Computing the Riemann Hypothesis

Volume II

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Cauchy Deconvolution
and Certified Computation

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Computing the Riemann Hypothesis, Volume II: Cauchy Deconvolution and Certified Computation

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Digital Object Identifier (DOI): `10.5281/zenodo.23861976`

Independently published

Second edition, in two volumes, November 2026

Cover: the backward Cauchy flow, from the level one density down to the zeros.

Source, code and data: `https://github.com/VadimSokolov/computing_rh`

Version 2.0.0, typeset by the authors in LaTeX from commit `5bc8c18` of the source (1 November 2026).

Preface

The Riemann Hypothesis can be stated as the positivity of a measure. The reciprocal Xi clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution exactly when every nontrivial zero lies on the critical line, and then its Thorin measure is the discrete measure $\sum_{\gamma>0}\delta_{\gamma^2}$ on the squared ordinates. This book asks what computation can say about that positivity condition.

Its answer has three parts. First, the Thorin measure is computable at every positive resolution, with explicit error bounds, from the Pólya kernel alone, with no primes and no list of zeros. Second, the distance between the computed measure and the measure demanded by RH can be measured exactly, and the signature an off line zero would leave can be calibrated. Third, a finite computation could refute RH by certifying a zero off the line, but neither a computation of the zeros up to a finite height nor a finite set of positivity tests on the heat trace can prove it. A last question is how much of the positivity can be proved. Moved from the centre of the critical strip to the basepoint one, the clock has a positive Thorin measure with no hypothesis. Chapters 12 to 14 follow each construction of the book back across the strip as far as it holds unconditionally; below the basepoint one, zero density estimates still show that the zeros to the right of the basepoint, the only possible source of negativity, carry a vanishing fraction of the mass of the density.

The book grew out of the first author's programme on generalised gamma convolutions and the Pólya density, collected in the monograph *On Hilbert's 8th Problem* [235], and out of a joint paper computing the Thorin measure directly [242]. The first author's arXiv paper of the same title is a separate work, which we cite by version: [233] is its version 24, of 2026, and [234] its version 5, of 2017. Chapters 10 and 11 incorporate Part IX of that monograph [235], the first author's work on the reciprocal zeta function, written as the manuscript [238], and Chapters 12 to 14 a paper of the authors on the reciprocal Xi function at the basepoint one and across the critical strip, together with notes on Riesz's criterion and on almost generalised gamma convolutions. Sections 15.1 and 15.2 draw on a manuscript of the first author on the Thorin measures of the Dedekind eta function [236], Chapter 20 on a note of the first author on ferromagnetic random variables, Chapter 19 on a manuscript of the first author on the Pólya density as a maximum order statistic [237], Chapter 6 on a working report of

the authors [243], Chapter 22 on a chapter of the first author on Pólya and the zeros of entire functions, and Chapters 15.12 to 16 on five further chapters of the first author, on Yoshida's Hermitian forms, on the compact windows of Zhu [279], on three families of generalised gamma convolutions, on deconvolution as a damped Weil criterion, and on Dirichlet L functions and the Linnik constant, and Chapters 18 and 23 on two more chapters of the first author, on friable approximants of Ξ after Miyatake and the cosh ladder and on the work of Haseo Ki read through the Thorin programme, and Chapter 5 on a chapter and two notes of Nick Polson on hitting times and the reciprocal ξ function [140, 152, 151], and Chapter 19.1.7 on a chapter of Nick Polson on extreme values, hitting times and a path to RH [133], Chapter 15 on a chapter of Nick Polson on Weil positivity and the Thorin condition [149], Chapter 20 on a note of Nick Polson on the identity of Comtet and Majumdar [131], Chapter 17.1.4 on two notes of Nick Polson on the Thorin truncation of $L(s, \chi_4)$ and on the Hermite coefficients of Ξ [146, 139], and Chapter 17 on two notes of Nick Polson on the secant dilates and the Möbius function, Chapter 10 on a manuscript of Nick Polson on the level one measure and the backward Cauchy flow [142], Chapter 11 on his notes on the complexity and acceleration of that flow, Chapter 12 on a manuscript of his on the explicit formula as a deconvolution [132], Chapter 7 on his notes on Cauchy deconvolution, Chapter 13 on his note on the deconvolution from level one to the critical line, and Chapter 14 on his note on the criterion of Balazard, Saias and Yor. The core evaluator runs in ball arithmetic. The code and data of every computation are in the book's GitHub repository. No proof of RH is claimed.

Chicago and Fairfax, September 2026

Contents

Preface v

I What computation can decide 1

1 What computation can and cannot decide 3
1.1 The path to RH, seen from the computation 3
1.2 What a finite computation can decide 4
1.3 What computation can reach, and what the sine squared lemma adds 4

1.4 What finite information cannot decide 8
1.5 A short history of computations on the zeros 16

2 Evaluating the Ξ function in ball arithmetic 19
2.1 The Pólya route 19
2.2 Classical evaluators, Turing's method and the Pólya route 25
2.3 A worked computation from first principles 29
2.4 Exercises 32

3 The Thorin measure at finite resolution 35
3.1 Distances to the discrete measure implied by RH 35
3.2 What an off line zero would do 39
3.3 Counting zeros with the Thorin measure 41
3.4 Exercises 45

4 The de Bruijn Newman flow computed from the Pólya density 47
4.1 The de Bruijn Newman flow from the Pólya density 47

5 The barrier method in Thorin form 81
5.1 The barrier argument 82
5.2 The barrier in Thorin form 84
5.3 A computed barrier at a modest height 87
5.4 What the Thorin form adds, and why it cannot reach $t = 0$ 90
5.5 Open problems 92

6 The Thorin phase diagram 93
6.1 Which computations see past the verified height 93
6.2 Clocks along the flow at every basepoint 95
6.3 The Thorin phase diagram 102
6.4 What a Thorin computation sees at positive time 104
6.5 A certificate for one point of the diagram 106
6.6 Where the uniform step could come from 110
6.7 Exercises 118

II Cauchy deconvolution 121

7 The Cauchy deconvolution operator 123
7.1 The operator 125
7.2 Positivity 126
7.3 Stability 128
7.4 Statistics 131
7.5 Atomic sources 133
7.6 What is known and what is not 134

8 Three families of generalised gamma convolutions and the deconvolution obstruction 137
8.1 Three families, and the tilt . . . 137
8.2 The basepoint family: a Poisson semigroup to the right of the zeros 140
8.3 What a single basepoint sees . . . 143
8.4 The heat flow: a stable step that cannot carry real zeros . . . 150

9 Deconvolution as a damped Weil criterion 153
9.1 Weil's distribution and the damped criterion . . . 153
9.2 The ladder . . . 156
9.3 What can be done and what is open . . . 162

10 The level one measure and the backward Cauchy flow 165
10.1 The level one measure . . . 167
10.2 The Fourier side: primes . . . 169
10.3 The backward flow . . . 171
10.4 The backward flow as a Bayesian inverse problem . . . 174
10.5 Generative Bayesian computation for the backward flow . . . 178
10.6 Is this a path to RH? . . . 181
10.7 Open problems . . . 182
10.8 The inverse problem: recovering the atoms by nonnegative deconvolution . . . 183

11 Complexity and acceleration of the backward flow 187
11.1 The deconvolution problem, precisely . . . 190
11.2 Three layers of cost . . . 192
11.3 EM as majorise minimise . . . 194
11.4 Acceleration . . . 199
11.5 Von Neumann's Monte Carlo inversion . . . 202
11.6 Cauchy as a scale mixture . . . 205
11.7 What is being computed . . . 207
11.8 Thorin measures at level 7/8 . . . 207
11.9 Pictures of the Thorin measures and the decision of positivity . . . 209
11.10 The limits of the computation . . . 218
11.11 Open problems worked . . . 221
11.12 Open problems . . . 241

12 The explicit formula as a deconvolution 243
12.1 The deconvolution at level one half . . . 246
12.2 Sparse recovery . . . 249
12.3 The computation . . . 252
12.4 Posterior concentration, and what it cannot do . . . 253
12.5 The functional equation makes the problem singular . . . 256

12.6 An adaptive prior, and a ReLU network . . . 263
12.7 What this does and does not do . . . 266
12.8 Questions taken up . . . 267
12.9 Open problems . . . 273

13 From level one to the critical line: particle filters and nested sampling 275
13.1 What the book already proves . . . 275
13.2 Two test functions . . . 277
13.3 A particle filter along the Cauchy flow . . . 279
13.4 Optimising the filter with generative Bayesian computation . . . 282
13.5 Concentration of the particle approximation . . . 287
13.6 From $\mathfrak{U}_1$ to $\mathfrak{U}_{3/4}$ . . . 293
13.7 Negative mass and converse results . . . 295
13.8 Zero free regions from positivity . . . 301
13.9 Nested sampling over heights . . . 310
13.10 What would close the gap . . . 313

14 Balazard, Saias and Yor: Monte Carlo along vertical lines 317
14.1 Thorin measures along vertical lines . . . 319
14.2 The Balazard, Saias and Yor criterion at every level and height . . 319
14.3 The dictionary: one derivative in σ . . . 321
14.4 The 2 sin squared lemma, EGGC and Davenport and Heilbronn . 323
14.5 Brownian motion and a probabilistic reading of the criterion . . . 328
14.6 The Monte Carlo solution . . . 333
14.7 Two dimensions: the smoothed Riesz identity . . . 346
14.8 Limits . . . 355
14.9 Open problems . . . 356

III Weil positivity in windows 359

15 Weil positivity and Yoshida's Hermitian forms: certificates and barriers 361
15.1 Weil's explicit formula and the heat trace . . . 363
15.2 Weil positivity is the Thorin condition . . . 364
15.3 The archimedean place is a Gumbel Lévy measure . . . 365
15.4 Three regimes . . . 367
15.5 Zeros from primes by Prony's method . . . 369
15.6 The prime point process: first and second order . . . 370
15.7 Does the extreme value approach offer insight? . . . 371
15.8 Windows, primes and rigidity . . . 372
15.9 Certificates, smoothing, energy and moments . . . 376
15.10 The sine squared lemma and a single heat time . . . 382
15.11 Is there a path to RH? . . . 385

15.12Open problems . . . 386
15.13Yoshida's paper . . . 387
15.14The ladder is the diagonal of Weil's form . . . 389
15.15Every large rung fails . . . 390
15.16Compact support and the signed Thorin formula . . . 392
15.17Localisation: a modulated heat trace . . . 393
15.18The tent and the sine squared lemma . . . 394
15.19The sine squared lemma off the diagonal . . . 399
15.20Damped forms and sharp coercivity . . . 401
15.21An uncertainty barrier . . . 408
15.22The question of a path to RH . . . 410
15.23Open problems . . . 411

16 Zhu's compact windows and the Landau Widom horizon 413
16.1 What Zhu proves . . . 413
16.2 The Thorin dictionary . . . 415
16.3 Zhu's test function through a truncated reciprocal Xi transform . . 418
16.4 The Landau Widom horizon . . . 419
16.5 The comb barrier, basepoints and damping . . . 424
16.6 What window certificates reach . . . 428
16.7 Open problems . . . 428

IV Approximants and finite tests 431

17 Truncations: the Biane Pitman Yor lattice, the beta lattice and Hermite coefficients 433
17.1 Truncations, finite conditions and the road to RH . . . 433
17.2 The beta lattice and its truncations . . . 455
17.3 The drift of the truncated zeros . . . 457
17.4 The side of approach is a Gram parity . . . 460
17.5 Transient zeros . . . 463
17.6 Symmetrised truncations . . . 464
17.7 Where this leaves the route . . . 466
17.8 Hermite coefficients of Ξ . . . 467
17.9 The turning point asymptotic . . . 469
17.10The first negative coefficient . . . 471
17.11Counting zeros from Φ alone . . . 473
17.12The best scale and the run lengths . . . 475
17.13The Thorin shadow . . . 476
17.14Notes and open problems . . . 477

18 Friable approximants of Ξ after Miyatake and the cosh ladder 479

18.1 Riemann's representation, the friable approximant and the kink . . 480
18.2 Approximation, the kink constant and no finite proxy 483
18.3 The deficit: positivity and Lorentzian form 486
18.4 Humps, pairs and the count of real zeros 488
18.5 A survey of the breakdown height up to height 21,000 491
18.6 Where the lost zeros go 492
18.7 Miyatake's theorem and the remainder R 493
18.8 The pole term, the secant and reweighting 495
18.9 The $2\sin^2$ lemma and the Thorin programme 496
18.10 The cosh ladder 498
18.11 Miyatake's identity, read probabilistically 499
18.12 The transition at $b = 2$ 501
18.13 The Pólya mixture of ψ_{M} 504
18.14 The ladder and the quasi Riemann abscissa 507
18.15 Open problems 516

19 Ferromagnetic and extreme value approximants with certified real zeros 519
19.1 Ferromagnetic approximations of the Pólya law 519
19.2 Is there a path to RH? 536
19.3 Work on the open problems 538
19.4 Second round on the open problems 543
19.5 Devroye's series, the mean excess function and detection 547
19.6 Remaining open problems 553

20 The Comtet Majumdar family: damped kernels and a solved model 555
20.1 The identity decoded 557
20.2 GGC reciprocals and the Laguerre Pólya class 559
20.3 The shifted Comtet Majumdar family 559
20.4 The uniform law and the $2\sin^2$ lemma 562
20.5 Damping the kernel 567
20.6 The de Bruijn Newman family and what the identity contributes . 570

21 Approximating clocks: Jensen, Euler product and boundary hitting times 573
21.1 Jensen hitting times 573
21.2 Extreme value hitting times 578
21.3 The prime side hitting times have a universal limit 581
21.4 Thorin rigidity 588
21.5 What the prime side limit would have to be 589
21.6 Diffusion approximations 590
21.7 Curved boundaries: the Roberts and Shortland view 590
21.8 Marc Yor's probabilistic Riemann function 596
21.9 The Davenport Heilbronn test 601
21.10 The coefficient ladder 601

21.11 The angle theorem: what the ladders can and cannot see 603
21.12 What a hitting time proof would need 607
21.13 Open problems 607
21.14 Expansions of the reciprocal Xi function 608

V Conditional results and synthesis 611

22 The quasi Riemann hypothesis and the Thorin programme 613
22.1 The claims and their status 614
22.2 The structure of the argument 615
22.3 What a half plane gives in the Thorin programme 619
22.4 Does the $2\sin^2$ lemma add anything? 621
22.5 From θ to $\frac{1}{2}$: the descent in level 622
22.6 Open problems 623

23 Conditional certificates under a quasi Riemann hypothesis at 7/8 625
23.1 Consequences of a quasi Riemann hypothesis at 7/8 625

24 What would close the gap: open problems of the computational programme 637
24.1 Problems on the descent itself 637
24.2 Problems on windows of the explicit formula 638
24.3 Problems on approximants 638
24.4 Problems on cost and certification 638

A The Pólya evaluators 639
A.1 The Pólya evaluators 639

B Notation 641

List of Figures 651

List of Tables 657

Part I

What computation can decide

Chapter 1

What computation can and cannot decide

This volume computes the objects that Volume I defines, and its centre is the Cauchy deconvolution approach to the Riemann Hypothesis. The conventions are those of Volume I (Section 1.3): every statement is proved, computed, certified, conditional, heuristic or open, and the hypothesis of a conditional statement is named. This volume writes the level θ and the density $u_\theta(v) = \pi^{-1}\,\mathrm{Re}(\xi'/\xi)(\theta + iv)$, which Volume I writes as v_α/π with $\alpha = \theta$.

1.1 The path to RH, seen from the computation

Volume I (Section 1.4) argues that the clearest path to RH is a descent in level: the Thorin measure at level θ is positive exactly when ζ has no zero with $\mathrm{Re}\, s > \theta$ (Theorem 11.9.1), its global minimum sits at the origin at level one with no hypothesis (Proposition 12.1.11 of Volume I), and RH is the statement that it stays there at every level down to $\frac{1}{2}$ (Theorem 12.1.18 of Volume I). In this volume the same statement is a deconvolution. Under RH the level one density is the Cauchy smoothing of the zeros, $u_1 = P_{1/2} * \mathfrak{U}_{1/2}$, and in general a positive deconvolution of u_1 down to level $\frac{1}{2} + \varepsilon$ exists exactly when no zero has real part above $\frac{1}{2} + \varepsilon$ (Theorem 8.2.2). The level one density is computable from the Euler product; RH is the statement that the Cauchy flow can be run backwards from it to the critical line through positive measures.

Part II of this volume does that computation as well as it can be done, and the results show both its power and its limit.

1. The operator (Chapter 7). Cauchy deconvolution is Hadamard's ill posed problem: unique when solvable, Hölder stable between levels and only logarithmically stable at the boundary, with a critical width beyond which no positive source exists. For the zeta family that width is one minus the supremum of the real parts of the zeros.

2. The backward flow (Chapters 10 and 11). From samples of the level one density alone, the flow run to level 0.75 places a peak within 0.035 of each

of the 109 zeros tested between heights 100 and 300; the EM algorithm of Shepp and Vardi does it without ringing, and a posteriori bounds certify window masses in ball arithmetic up to height 1419.9.

3. The bottom of the flow (Chapter 12). At level $\frac{1}{2}$ the deconvolution is Weil's explicit formula. For a general measure it is exponentially ill posed, heuristically requiring the primes up to $e^{\pi/\delta}$ for resolution δ; for a finite sum of atoms it is Prony's problem and only polynomially ill posed, and the matrix pencil recovers ordinates and real parts from level one data alone. But it reads the real parts; it does not constrain them.

4. The descent itself (Chapters 13 and 14). Particle filters run the flow in level, nested sampling scans heights, and the criterion of Balazard, Saias and Yor is solved by Monte Carlo at every level. Each works in a window of heights.

The limit is the one Volume I names. RH is a statement for all heights, and a perturbation of ξ that agrees with it below any computed height passes the finite tests of the heat trace listed there (Theorem 1.4.1, Chapter 1). So the computations of this volume can refute RH by certifying a zero off the line, can certify that windows are clean, and can measure how ill posed each step of the descent is; they cannot take the step. What would close the gap is an analytic positivity argument that uses the Euler product at level one and carries it to the line, and the last chapter of this volume (Chapter 24) collects what such an argument would need. Statements that assume the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026, which has not been refereed, are marked where they occur, in particular in Chapters 22 and 23.

1.2 What a finite computation can decide

The rest of this chapter makes the limit precise: what a computation of the Thorin measure can reach, what a finite set of positivity tests cannot decide, and how computations of the zeros reached their present height.

1.3 What computation can reach, and what the sine squared lemma adds

Two facts govern what computation can establish about the Thorin condition and which evaluator it should use (Sections 1.3.1 and 1.3.2). The evaluator here and in Chapters 9 and 8 is the Pólya density.

1.3.1 What a finite computation can measure

Below height 200 the Pólya route finds 79 sign changes of Ξ and an argument principle count $N(200) = 79.0000000000004$ (Section 2.1.2). So, numerically and in

agreement with [9, 232], the Thorin functional $U = \sum_k \delta_{a_k}$ of Section 8.2 coincides with U_{RH} on $(0, 200^2]$. The matched total variation distance of Section 3.1, computed against a model of the zero list, is $3.7 \times 10^{-9}\,\varepsilon$ at every resolution computed, the numerical floor of the comparison.

None of this constrains the zeros above the window. Platt and Trudgian [232] have verified every zero below height 3×10^{12}, yet the perturbation of ξ in Theorem 1.4.1 [235, Chapter 11] keeps the order, type and symmetry of $\xi(\frac{1}{2} + s)$, puts zeros off the line just above any large enough height R for which RH has been verified up to $R + 50$, and still passes the finite tests of the heat trace, from Hankel blocks and derivative signs (to order 10^{17}) to the small time expansion. So a computation from the zeros verified up to a given height, or from finitely many tests of the heat trace, cannot by itself measure a scalar distance to RH, although a finite computation could refute RH by certifying a zero off the line. What it can measure is a margin, the minimum of ρ_ε, at a stated resolution and height; positivity on one line is weaker than a zero free half plane (Remark 2.1.4).

Resolution is the first limitation. On $0 \le \theta \le 200$ the computed minimum is $\min \rho_\varepsilon = 0.0147\,\varepsilon$ at each of the eight resolutions of Section 2.1.2, attained at $\theta = 0$, where $\rho_\varepsilon(0) = \pi^{-1}\xi'(\frac{1}{2} + \varepsilon)/\xi(\frac{1}{2} + \varepsilon) = 2\varepsilon\pi^{-1}\sum_k a_k^{-1} + O(\varepsilon^3)$, because $(\xi'/\xi)(\frac{1}{2} + \varepsilon) = \sum_k 2\varepsilon/(\varepsilon^2 + a_k)$ and $\sum_k a_k^{-1} = 0.0231050$ (Section 2.1). So $\inf_\theta \rho_\varepsilon(\theta) \le \rho_\varepsilon(0) \to 0$ as $\varepsilon \downarrow 0$: near the critical line positivity is never won by a fixed margin. On the verified window the origin is the minimum: for every $\varepsilon \in (0, 6.85]$ the minimum of ρ_ε over $0 \le \theta \le 3 \times 10^{12} - 1.21$ is $\rho_\varepsilon(0)$, attained only at the origin, and that this holds at every height for every $\varepsilon \in (0, \frac{1}{2})$ is equivalent to RH (Theorem 12.1.18 and Remark 12.1.19).

Height is the second limitation. Even on the tilted contour of Section 2.1.3 the number of nodes, and with it the cost per point, grows linearly in the height, so the Pólya route is not a competitor for zero verification (Section 2.2.3.2). Its value is accuracy at small ε on moderate windows, where the Thorin measure is sharp and the prime side is ill conditioned.

1.3.2 The sine squared decomposition of the Thorin bracket

The sine squared lemma (2.1.9) converts the Lévy description of $\log \xi$, in which the primes are atoms, into the Thorin description of the reciprocal. At a basepoint $\alpha > 1$, applying the kernel $2\sin^2(x\theta/2) = 1 - \cos(x\theta)$ to the canonical measure of [235, Theorem 8.1] writes the bracket $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ as a sum of four

α	θ	drift	archimedean	pole	primes	$v_\alpha(\theta)$	$\kappa_\alpha(\theta)$
3.0	2	0.1144	0.0533	-0.2500	0.1988	0.1165	3.85
3.0	50	0.1144	1.2594	-0.4992	0.2464	1.1210	0.67
2.0	2	0.0691	0.0859	-0.8000	0.7154	0.0704	21.5
2.0	50	0.0691	1.3989	-0.9996	0.7561	1.2245	1.43
1.5	2	0.0461	0.1137	-1.8824	1.7683	0.0470	77.6
1.5	50	0.0461	1.4863	-1.9998	1.8527	1.3867	2.78

Table 1.1: The four pieces of the sine squared decomposition (1.3.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond 2×10^6 (1.3×10^{-3} at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece.

explicit pieces:

$$v_\alpha(\theta) = \underbrace{\frac{\xi'}{\xi}(\alpha)}_{\text{drift}} + \underbrace{\int_0^\infty (1-\cos\theta x)\frac{e^{-\alpha x}}{e^{2x}-1}\,dx}_{\text{archimedean } \ge 0} - \underbrace{\frac{\theta^2}{(\alpha-1)\big((\alpha-1)^2+\theta^2\big)}}_{\text{pole } \ge 0} + \underbrace{\sum_{n\ge 2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big)}_{\text{primes } \ge 0}. \quad (1.3.1)$$

The pole piece is the closed form of $\int_0^\infty (1-\cos\theta x)e^{-(\alpha-1)x}dx$, and the prime piece equals $-\frac{\zeta'}{\zeta}(\alpha) + \operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+i\theta)$.

The drift is the linear coefficient $c = \xi'/\xi(\alpha)$ of the Lévy representation of Section 2.1, not the drift d of a GGC. The archimedean piece, from $\Gamma(s/2)$ and the factor s of $\frac{1}{2}s(s-1)$, is smooth in θ and grows like $\frac{1}{2}\log\theta$, which accounts for the average density of zeros. The prime piece is a sum of oscillations with frequencies $\log n$ and amplitudes $\Lambda(n)n^{-\alpha}$, one for each prime power. The drift is positive.

Near $\alpha = 1$ the pole piece is almost cancelled by the prime piece (Table 1.1). Both pieces grow like $1/(\alpha-1)$, the pole explicitly and the primes through the pole of $-\zeta'/\zeta$ at one. At each fixed $\theta \neq 0$ their difference stays bounded as $\alpha \downarrow 1$ because ζ has no zero on $\operatorname{Re} s = 1$: the analytic input of the prime number theorem appears in the Thorin description as this cancellation, which [235, Section 8.2] identifies with the collapse onto the zeros. Measured by the condition number

$$\kappa_\alpha(\theta) = \frac{|\text{pole}| + |\text{primes}|}{|v_\alpha(\theta)|},$$

the prime side amplifies relative errors by 77.6 at $\alpha = 1.5$ and $\theta = 2$.

The lemma also prices the prime side. By the prime number theorem the truncation tail satisfies, for fixed $\theta \neq 0$,

$$\sum_{n>N} \frac{\Lambda(n)}{n^{\alpha}}\big(1-\cos(\theta \log n)\big) \asymp \frac{N^{1-\alpha}}{\alpha-1},$$

which is 1.4×10^{-3} at $\alpha = 1.5$ with $N = 2\times10^{6}$, the size of the errors in Table 2.2; an absolute error of 10^{-10} at $\alpha = 1.5$ would need a cutoff $N \approx 4\times10^{20}$. For $\alpha \leq 1$ the series diverges, and the lemma must be read through the regularised pairing of [235, Chapter 7] and the explicit formula. At the basepoint one the regularisation is explicit: the pole and prime pieces can be compensated against each other, and Section 12.1 converts the resulting Lévy form of $\xi(1+s)$ term by term into the Thorin density. Near the critical line Selberg's smoothed sum over the primes up to x^2 takes the place of the series, and at resolution ε, if no zero lies to the right of the critical line, its error at an ordinate is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that k correct digits need $x \approx 10^{k/\varepsilon}$ (Section 5.1.1). The Pólya route has none of these costs: its cost per point is polynomial in the height and independent of ε, and at height 200 it reaches $\varepsilon = 0.005$ with about 60 digits (Section 2.1.1).

1.3.3 Consequences for computation

The bracket is the derivative of an argument, $v_{\alpha}(\theta) = \partial_{\theta} \arg \xi(\alpha + i\theta)$, because $\partial_{\theta} \log \xi(\alpha+i\theta) = i(\xi'/\xi)(\alpha+i\theta)$. So the cumulative Thorin mass is an argument,

$$\mu_{\varepsilon}\big((0,\theta^2]\big) = \frac{1}{\pi} \arg \xi\Big(\frac{1}{2}+\varepsilon+i\theta\Big),$$

continued along the line from the real axis, where $\xi > 0$. It is computed without quadrature, by accumulating the arguments of quotients of consecutive values of ξ, with steps chosen so that each increment can be read from its principal value, a condition checked after the fact, not guaranteed (Section 2.1.2). Section 3.1 uses it for masses, cell masses, cumulative distances and the mass defect of Proposition 3.1.2, and Section 3.3 for the zero count.

The lemma is linear with a nonnegative kernel, so a nonnegative Lévy input gives a nonnegative Thorin output, and positivity certificates can be posed as linear programs in the sine squared basis [235, Sections 18.2.8 and 18.2.9]. For ξ the only negative part of the input $xL(dx)$ is the pole density $-e^{-(\alpha-1)x}dx$, so a certificate must show that the positive pieces dominate the pole piece at every θ. For $\alpha > 1$ they do (Theorem 2.1.2), though the proof goes through the zeros, not the input. For $\frac{1}{2} < \alpha \leq 1$, where the lemma is read through the regularised pairing, domination at every basepoint is Lagarias's criterion, that is RH, and Section 5.1.2 states RH as a domination of the same kind at the centre, Weil's criterion on variograms. Beurling generalised prime systems, in which the sign of the input is controlled by construction, are among the open computations of Section 25.2.3.

The lemma also gives an exact consistency check at $\alpha = \frac{1}{2}$, where the constant parts of the pole and prime pieces diverge and are continued analytically. The constant part $1/(\alpha-1)$ of the pole piece continues to -2 and enters with a minus sign; the constant part $-\zeta'/\zeta(\alpha)$ of the prime piece continues to $-\zeta'/\zeta(\frac{1}{2})$; and the gamma factor contributes the archimedean constant of [235, Section 6.8], $C_\Gamma = \frac{1}{2}\log\pi - \frac{1}{2}\psi(\frac{1}{4}) - 2 = 0.6860917096$, which is minus the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ at $s = \frac{1}{2}$. They sum to

$$C_\Gamma + 2 - \frac{\zeta'}{\zeta}\Big(\frac{1}{2}\Big) = 0,$$

which is equivalent to $\xi'(\frac{1}{2}) = 0$, because $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$ and $1/s + 1/(s-1)$ vanishes at $s = \frac{1}{2}$.

1.4 What finite information cannot decide

Every computation in this book is finite: a finite window, finitely many resolutions, finitely many derivatives or matrix blocks. Theorem 1.4.1 replaces two close zeros of ξ on the line by a quadruple off it [235, Chapter 11] and obtains an entire function whose heat trace passes the finite tests of Section 8.1.

1.4.1 A perturbation that passes the finite tests of the heat trace

Theorem 1.4.1 ([233, Theorem D]). (a) *There are ordinates* $1000 < A < B \le 2000$ *of zeros on the critical line, with* $B - A \le 2$*, such that, with* $g = (A+B)/2$ *and* $w = \frac{1}{4} + ig$*,*

$$F(s) = \frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})}\,\frac{(1 - s^2/w^2)(1 - s^2/\bar{w}^2)}{(1 + s^2/A^2)(1 + s^2/B^2)}$$

is an even entire function with the order, type and reflection symmetry of $\xi(\frac{1}{2}+s)$ *and with four zeros off the critical line, and the following hold. The reciprocal clock* $1/F(\sqrt{u})$ *is the Laplace transform of an infinitely divisible, self decomposable law, and its heat trace*

$$W_F(t) = W(t) + 2\,\mathrm{Re}\,e^{-at} - e^{-A^2t} - e^{-B^2t}, \qquad a = -w^2 = g^2 - \tfrac{1}{16} - \tfrac{ig}{2},$$

which is t *times its Lévy density, is strictly log convex.* W_F *has the same small time expansion as* W *through the constant term, and* $1/F$ *has the same nearest poles as the reciprocal Xi function.*

(b) *Let* $N \ge 1$ *and* $R \ge T_N$*, where* $T_N = 20000N\sqrt{1+\log N}$ *as in Section 8.1, and suppose that every zero of* ξ *with ordinate in* $(0, R+50]$ *lies on the critical line. Then the pair can instead be chosen with* $R < A \le B \le R+50$ *and* $B - A < 2$ *(if* $A = B$*, a multiple zero supplies both), so that* F *keeps every property listed in (a)*

and the leading $N \times N$ Hankel blocks of W_F remain positive definite at every $t > 0$. If moreover $R \geq 10^{12}$, the derivative signs of Theorem 8.1.6 are also preserved: $(-1)^k W_F^{(k)}(t) > 0$ for every $t > 0$ and every integer $0 \leq k \leq 10^{17}$. For every $N \leq 3 \times 10^7$ the verification to height 3×10^{12} [232] establishes the hypothesis on the zeros for $R = \max(T_N, 10^{12})$, so that such an F with both properties exists unconditionally.

The construction moves the zeros at heights A and B, with their conjugates, to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$. The heat trace changes by the oscillating term $W_F - W$, whose envelope is bounded as in Section 8.1 [233, Proposition 11.1], and at the heights of (b) this change is smaller than the margin by which each of its tests holds. The pair of (b), with the properties of (a) and the Hankel blocks, comes from [233, Theorem 12.5], the derivative signs from [233, Proposition 12.7], which holds for any pair of zeros on the line with $A > 10^{12}$ and $B - A \leq 2$, and the unconditional range from [233, Corollary 12.6], since $T_{3\times 10^7} + 50 < 2.6 \times 10^{12}$. Figure 1.1 shows the change for a pair in the range of (a); the pair it moves, near height 1977.22, is the closest pair among the first 1700 zeros, whether or not the spacings are normalised (Section 8.3.2).

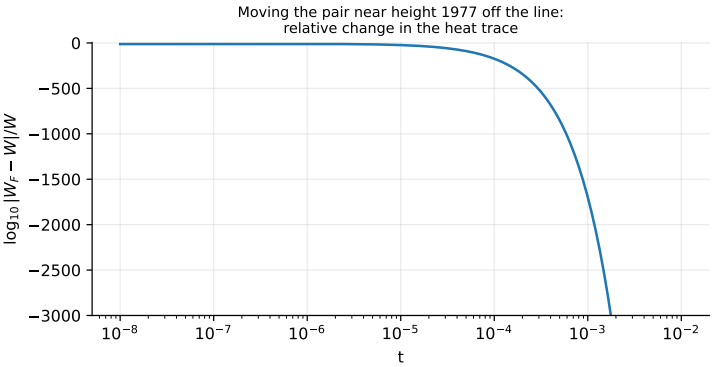


Figure 1.1: Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with W replaced by its sum over the first 1700 zeros. The change is at most 4×10^{-11}, near $t = 7 \times 10^{-7}$, and falls below 10^{-1000} by $t = 10^{-3}$. The sum over 1700 zeros is smaller than W, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with W from the explicit formula (2.1.5) the maximum is 3.66×10^{-11}, at $t = 7.2 \times 10^{-7}$.

Not every finite test is passed. Below any height $T > B$ the argument count of F equals that of ξ, since F loses the zeros at heights A and B and gains two at height g; but these are off the line, so F has two zeros fewer on the line, counted

with multiplicity; when $A < B$ and both zeros are simple, as for the pair of Figure 1.1, it has two sign changes fewer. Turing's comparison of the two counts, the first test of Table 1.3, then detects the perturbation once it passes height B, and the zero density identity at any height $T > g$ returns $\nu(\frac{1}{2}+\varepsilon, T) \geq 1$ for every $\varepsilon < \frac{1}{4}$, whatever the multiplicities.

The pair of Figure 1.1 lies far below the heights of (b), and there the derivative signs of F fail at a computable finite order. On the ray $t = k/u^2$ the normalised value $e^{-f_0} c_k^F(t)$ of Proposition 8.1.10, whose proof applies to F because F has at most one zero more than ξ below any height, is a sum of positive terms, one for each zero on the line, the oscillating term of the quadruple and terms, negligible here, from any zeros off the line above 3×10^{12}, and only the zeros within a few units of u matter. A scan over u and over the orders, at every order up to 500, then on a grid of ratio 1.002 and at every order of the last grid interval, finds that the first order at which $c_k^F(t) < 0$ for some $t > 0$ is $k = 6\,683\,655$, at $t = 1.7113393$. The grid loses nothing, because $c_{k-1}^F(t) = \int_t^\infty c_k^F(s)\, ds$: if $c_k^F \geq 0$ on $(0, \infty)$ then so is c_{k-1}^F, so the failing orders are all those from the first on. The negative value is certified in ball arithmetic, with explicit bounds for the zeros below $g - 40$, which lie on the line, and for those above $g + 40$, wherever the ones above 3×10^{12} lie; the positivity at the lower orders is computed, not certified. Table 1.2 repeats the computation for the closest pair of consecutive zeros in five other windows of height.

window	zeros moved	g	δ	k^*	t^*	$c_{k^*}^F(t^*)$, normalised
$(100, 200)$	71, 72	185.24	$\frac{1}{4}$	40 018	1.18413	-6.37×10^{-7}
$(200, 400)$	186, 187	376.08	$\frac{1}{4}$	219 169	1.54072	-6.70×10^{-8}
$(400, 700)$	363, 364	630.64	$\frac{1}{4}$	624 921	1.576635	-1.68×10^{-7}
$(700, 1000)$	453, 454	750.81	$\frac{1}{4}$	792 746	1.410725	-9.26×10^{-8}
$(1000, 1500)$	922, 923	1329.12	$\frac{1}{4}$	4 433 220	2.5121193	-3.37×10^{-8}
$(1500, 2000)$	1496, 1497	1977.22	$\frac{1}{4}$	6 683 655	1.7113393	-7.12×10^{-9}
$(1500, 2000)$	1496, 1497	1977.22	0.1	15 722 219	4.0257254	-6.71×10^{-11}

Table 1.2: The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, g the midpoint of the pair. k^* is the least order at which $c_k^F(t) < 0$ for some $t > 0$, found by the scan above (computed, not certified); $c_{k^*}^F(t^*) < 0$ is certified in ball arithmetic. The last column is $c_{k^*}^F(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$.

The first failing order is $k^* = t^* u_*^2$, where $u_* = \sqrt{k^*/t^*}$ is the height of the failure, between 0.69 and 1.40 from g, so k^* is within 1.6 per cent of $t^* g^2$, with t^* between 1.18 and 2.51 for $\delta = \frac{1}{4}$: it grows roughly with the square of the height, as Proposition 8.1.10 explains. Along the ray, $e^{-f_0} c_k^F(t)$ is, up to relative errors of order $t|\tilde{\gamma} - u|^3/u$ in the terms near u, small at these heights and times

(Section 8.1.5), the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros of F, in which the quadruple contributes $2e^{2t(\delta^2-(g-u)^2)}\cos(4t\delta(g-u))$. This term is negative only where $4t\delta|g-u|$ lies in $(\frac{\pi}{2},\frac{3\pi}{2})$ modulo 2π, hence at distance at least $\pi/(8t\delta)$ from g, where its modulus is at most $2e^{2t\delta^2-\pi^2/(32t\delta^2)}$, while a zero on the line at distance d from u contributes e^{-2td^2}. The first failure time is set by δ and by the gaps between the line zeros near the moved pair, which its removal widens; the mean spacing $2\pi/\log(g/2\pi)$ falls from 1.86 to 1.09 between the first and last rows, but the gaps near the moved pairs vary more, which is why t^* shows no trend. The first t at which $\min_{|u-g|\le 8}\mathcal{G}(u,t)<0$, computed in floating point, predicts k^* to 0.21 per cent at height 185 and to 0.08 per cent or better at the other heights, and the minimum lies just beyond the distance $\pi/(8t\delta)$ at which the first negative lobe begins. These failure times of $\mathcal{G}$ within 8 of g are certified in ball arithmetic, with the zeros from Arb, for all seven rows of Table 1.2: $\mathcal{G}(u,t)>0$ at every centre u within 8 of g for every t up to the failure time rounded down to two decimals, and $\mathcal{G}<0$ at one centre at the failure time rounded up to four decimals. Far from g the bracket says nothing: near or above 3×10^{12}, zeros off the line could make $\mathcal{G}$ negative earlier. One time suffices for the first statement, since $\sqrt{2t/\pi}\,\mathcal{G}(\cdot,t)$ at a smaller t is its value at a larger one convolved with a centred normal density: the sum over the zeros below 3×10^{12} is positive on the whole line at $t=1.71$, by an interval cover within 8 of g and by domination beyond, so it has a uniform positive lower bound at every smaller t within 8 of g, and the zeros above 3×10^{12}, wherever they lie, can lower $\mathcal{G}$ there by less than $10^{-2\times10^{12}}$; what they add may be large. Unlike the Hankel condition of Theorem 8.1.3 at one time, positivity of $\mathcal{G}$ at one resolution does not decide whether the zeros are on the line. Moving the pair near 1977.22 to distance 0.1 raises t^* from 1.71 to 4.03: the closer the quadruple to the line, the finer the resolution needed to see it.

The same mechanism puts the first failure for a quadruple just above the verified height far beyond the orders of (b). There $g^2\approx10^{25}$, so the first failing order is about $10^{25}t^*$, and t^* depends on the gaps around the quadruple, which a computation at height 3×10^{12} would have to supply. The mean spacing there is 0.23, and for $\delta=\frac14$ a failure at $t\le1$ would need a stretch of length 2.9, about twelve mean spacings, free of zeros on the line other than the moved pair. A model with the line zeros spread at their mean density gives $t^*\approx10$ there, but the same model gives 5.9 at height 1977 against the computed 1.71, because it ignores the gaps, so it gives the order of magnitude only. Theorem 8.1.13 certifies the signs of ξ to every order up to $1.29(H-23)^2>1.16\times10^{25}$, with $H=3\times10^{12}$, so a quadruple of ξ just above H could make a sign fail only at a time $t^*>1.29$, at an order of about t^*H^2: the orders certified at the time t^* reach that size, and by the model the orders certified at every time fall short of it by a factor of about eight. This is the entry of Table 1.3 on derivative signs.

1.4.2 What finite tests establish, and the local visibility condition

Each entry of the middle column of Table 1.3, made rigorous, is an unconditional theorem on its window; the right column is why no accumulation of such entries proves RH.

test	establishes	cannot rule out
sign changes and argument count to height T (Sections 2.1 and 2.2)	all zeros below T on the line	anything above T
$\rho_\varepsilon \geq 0$ on $0 < \theta \leq \Theta$ (Section 1.3)	no zero in the window at distance $d > \varepsilon$ from the line with $2/d^2 > \sum_j \delta_j^{-2}$ (to first order in ε)	zeros within ε of the line; zeros farther from it than about 0.78 of the mean spacing at their height; zeros above the window
zero density scan at T (Section 3.3)	$\nu(\frac{1}{2}+\varepsilon, T) = 0$ for the ε computed	zeros within ε of the line; zeros above T
Hankel blocks of size N (Section 8.1)	Weil positivity on a finite space	off line zeros above T_N
derivative signs to order K (Section 8.1)	finite order Williamson positivity	off line zeros above a height of about $\sqrt{K/1.29}$
HCM derivative signs (Section 9.1)	finitely many CM conditions	failure at other u or orders
matched total variation at resolution ε (Section 3.1)	agreement with the Poisson smoothing of the list	zeros outside the window
Li coefficients λ_n, $n \leq N$ (Section 23.6)	finitely many of Li's inequalities	off line zeros $\frac{1}{2} \pm d + i\gamma$ with Nd/γ^2 small

Table 1.3: What finite tests establish and what they cannot exclude.

The second row rests on a local computation. Suppose that ξ has a pair of zeros $\frac{1}{2} \pm d + i\gamma$ with $d > \varepsilon$, and that the other zeros near height γ lie on the line at the ordinates $\gamma + \delta_j$. A zero ρ contributes $\mathrm{Re}(s-\rho)^{-1}$ to $\pi\rho_\varepsilon(\theta)$, where $s = \frac{1}{2} + \varepsilon + i\theta$, so at the ordinate, ignoring distant zeros and those at negative heights,

$$\pi\rho_\varepsilon(\gamma) = \varepsilon\Big(\sum_j \frac{1}{\delta_j^2 + \varepsilon^2} - \frac{2}{d^2 - \varepsilon^2}\Big).$$

To first order in ε the off line zero makes ρ_ε negative at its ordinate if and only if $2/d^2 > \sum_j \delta_j^{-2}$, the local visibility condition. For neighbours at the mean spacing $\Delta = 2\pi/\log(\gamma/2\pi)$, that is $\delta_j = j\Delta$ with $j \neq 0$, the sum is $\pi^2/(3\Delta^2)$, and the condition becomes $d < (\sqrt{6}/\pi)\Delta = \sqrt{24}/\log(\gamma/2\pi)$, about 0.78 of the mean spacing. A zero farther from the line leaves ρ_ε positive at its ordinate at every small resolution and makes it negative there only at resolutions comparable to d.

The condition concerns the ordinate itself. For evenly spaced neighbours the neighbour sum is smallest and the pair term most negative there, so the condition also decides whether ρ_ε is negative near γ; in general it does not, because away from the ordinate the neighbour sum can fall faster than the pair term grows. With $d = 0.3$, $\varepsilon = 0.005$ and neighbours at $\gamma + 0.2 + m$ and $\gamma - 4 - m$, $m = 0, 1, 2, \ldots$, the sum $\sum_j \delta_j^{-2} = 26.55$ exceeds $2/d^2 = 22.2$ and $\rho_\varepsilon(\gamma) = 0.0069$, yet $\rho_\varepsilon(\gamma - 0.079) = -0.0060$ (Exercise 8.14). A search for negative values of ρ_ε must therefore cover every point of the window, not only the ordinates, and every resolution.

A zero on the line at the ordinate of the pair, $\delta_j = 0$, makes the neighbour sum infinite, and the pair is then invisible to first order at every distance. Exactly:

PROPOSITION 1.4.2 (A pair at the height of a line zero). *Let $d > 0$, $\varepsilon > 0$ with $\varepsilon \neq d$, and $u = (\theta - \gamma)^2$. The zeros $\frac{1}{2} + i\gamma$ and $\frac{1}{2} \pm d + i\gamma$ contribute to $\pi\rho_\varepsilon(\theta)$*

$$\frac{\varepsilon\big(3u^2 + 6\varepsilon^2 u + (3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)\big)}{(\varepsilon^2 + u)\big((d-\varepsilon)^2 + u\big)\big((d+\varepsilon)^2 + u\big)},$$

which is positive at every θ if and only if $\varepsilon < d/\sqrt{3}$ or $\varepsilon > d$. Hence, if every zero off the line of an entire function with the order, symmetry and reality of ξ shares its ordinate with a zero on the line of at least its multiplicity, and the distances d have a positive infimum d_0, its density is positive at every height on every line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ with $0 < \varepsilon < d_0/\sqrt{3}$, although zeros lie to the right of these lines.

Proof. The line zero contributes $\varepsilon/(\varepsilon^2 + u)$ and the pair $2\varepsilon(\varepsilon^2 - d^2 + u)/\big(((\varepsilon - d)^2 + u)((\varepsilon + d)^2 + u)\big)$ (proof of Corollary 14.3.7); over the common denominator the numerator is the one displayed. Its coefficients of u^2 and u are positive, so it is positive for every $u \geq 0$ exactly when its value $(3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)$ at $u = 0$ is, that is when $\varepsilon^2 < d^2/3$ or $\varepsilon^2 > d^2$. For the second statement group each pair, and its conjugate at $-\gamma$, with its line zero; every group and every remaining line zero contributes positively, and the sum converges absolutely, as in the proof of Corollary 14.3.7. □

To first order in ε the contribution is $\varepsilon(3u^2 + d^4)/(u(u + d^2)^2) > 0$ for every d: the polynomial $P(x) = x(x^2 + d^2)$ satisfies the Laguerre inequality $P'^2 - PP'' = 3x^4 + d^4 > 0$ on the whole real line without having only real zeros. Exercise 1.7 gives one such function at one line; the proposition gives a band of lines whose width is fixed by the nearest off line pair, so the Hadamard structure and the functional equation alone cannot make positivity on one line detect zeros to its right.

The zero density identity of Proposition 3.3.2 has no such blind spot: one evaluation at resolution ε returns $\nu(\frac{1}{2} + \varepsilon, T)$, the number of zeros with real part greater than $\frac{1}{2} + \varepsilon$ and ordinate in $(0, T]$, however far from the line they lie. The same count controls the negative part of ρ_ε: the negative kernels in its Poisson

form, which come from the zeros to the right of $\frac{1}{2}+\varepsilon$, have total mass on $(0,T]$ at most the number of such zeros below height $2T$ plus a tail that tends to zero as $T\to\infty$ when that number is $O(T^c)$ for some $c<1$ (Lemma 14.3.1). The zero density estimates of Ingham [176] and of Guth and Maynard [92] give such a bound with no hypothesis and make this mass a vanishing fraction of the positive mass (Theorem 14.3.2).

1.4.3 Refuting sufficient routes with one certificate

Not every route to RH through the Pólya density is a positivity question on the clock. Konstantopoulos, Patie and Sarkar [183] introduced a class $\mathcal{D}_P$ of solutions of the van Dantzig problem, the entire functions $\sum_{n\ge0}(-1)^n t^{2n}/(\Psi(1)\cdots\Psi(n))$ in which Ψ is the Laplace exponent of a spectrally negative Lévy process whose largest zero q is at most $\frac{1}{2}$; then $\Psi(u)=(u-q)\varphi(u)$ with φ a Bernstein function [183, Proposition 5.2]. If $q=0$ and φ is a Bernstein Pick function whose zeros and poles satisfy a separation condition, every zero of the entire function is real [183, Theorem 4.4], so such a representation of $\Xi/\Xi(0)$ would prove RH; this is the Bernstein Pick route. Let X have the Pólya density and put $M(s)=\mathbb{E}|X|^{2s}$. Since $\Xi(t)/\Xi(0)=\sum_{n\ge0}(-1)^n M(n)t^{2n}/(2n)!$, such a representation needs $\Psi(n)=n\varphi_I(n)$ for every $n\ge1$, where $\varphi_I(s)=4(s-\frac{1}{2})M(s-1)/M(s)$ is the canonical interpolant of the moment ratios; that is, φ must interpolate $n\varphi_I(n)/(n-q)$ at the integers. The function M is analytic for $\operatorname{Re}s>-\frac{1}{2}$, and for $-\frac{1}{2}<\operatorname{Re}s<0$ the classical formula for negative absolute moments gives

$$M(s)=\frac{2\,\Gamma(2s+1)\sin(-\pi s)}{\pi\,\Xi(0)}\int_0^\infty \Xi(\theta)\,\theta^{-2s-1}\,d\theta,$$

whose right side continues M meromorphically to $\operatorname{Re}s<0$.

Theorem 1.4.3 ([233, Theorem E]). *M has exactly one zero in the disc $|s-s_1|<10^{-8}$ with centre $s_1=-8.494182256992559+2.937767068567584i$; the zero is simple, and $M(s-1)$ does not vanish in the disc. Hence φ_I has a nonreal pole, and every Bernstein interpolant of the moment ratios agrees with φ_I on $(1,\infty)$ and cannot be continued through that pole, so none is a complete Bernstein function. For every $0<q<1$ the sequence $n\varphi_I(n)/(n-q)$ has no Bernstein interpolant either.*

Since the Bernstein Pick functions are the complete Bernstein functions, the theorem closes the Bernstein Pick route, and its last statement excludes $\Xi/\Xi(0)$ from $\mathcal{D}_P$ with $0<q\le\frac{1}{2}$, the case $q>0$ of the question raised in [183, Section 4.1]. The theorem belongs to the arcsine route of [235, Chapter 12], where its last statement is proved by a beta multiplier obstruction, and at the single remaining value $q=0$ membership in $\mathcal{D}_P$ reduces to a Hausdorff moment criterion.

1.4.4 A local model for the first failing order

Once the resolution $1/(2\sqrt{t})$ of the Gaussian smoothing (Section 8.1.5) is below the gaps between the line zeros near the moved pair, $c_k^F(t)$ near the quadruple is governed by the quadruple and its two neighbours on the line, and the first failing order becomes explicit.

Proposition 1.4.4. *Let F be obtained as in Table 1.2 by replacing two consecutive zeros of ξ on the line by the quadruple $\frac{1}{2} \pm \delta \pm ig$, g their midpoint, let $g - \Delta_-$ and $g + \Delta_+$ be the ordinates of the nearest zeros of F on the line below and above g, and put $\Delta = \max(\Delta_-, \Delta_+)$. Assume $\delta \ll \Delta \ll g$ and write $\omega = 2k\delta\Delta/g^2$, $\kappa = 2k\Delta^2/g^2$. Restricted to the quadruple and these two zeros, $c_k^F(t) < 0$ for some $t > 0$ if and only if, to leading order,*

$$\min_{0 \le \eta \le 1} \Big[2\cos(\omega\eta) + e^{-\kappa(1-\eta)} \Big] < 0,$$

and the first failing order is

$$k^* = \frac{\omega^* g^2}{2\delta\Delta}, \qquad \omega^* = \frac{\pi}{2} + \frac{\delta}{\Delta}\Big(1 + \log\frac{\Delta}{2\delta}\Big) + O\Big(\frac{\delta^2}{\Delta^2} + \frac{\Delta}{g}\Big),$$

so that $k^* = \frac{\pi}{4}\, g^2/(\delta\Delta)\,(1 + o(1))$ as $\delta/\Delta \to 0$ and $\Delta/g \to 0$. The failure is at the time $t^* = k^*/g^2\,(1 + O(\Delta/g)) = \omega^*/(2\delta\Delta)\,(1 + O(\Delta/g))$, which involves δ and Δ only.

Proof. Put $t = k(1+v)/g^2$ and $a = (g - i\delta)^2$. The quadruple contributes $2\operatorname{Re} a^k e^{-at} = 2|a|^k e^{-t\operatorname{Re} a}\cos\big(2\delta g t - 2k\arctan(\delta/g)\big)$ to $c_k^F(t)$, with $|a|^k e^{-t\operatorname{Re} a} = g^{2k} e^{-tg^2} e^{O(k\delta^2/g^2)}$ and phase $2\delta g k(1+v)/g^2 - 2k\arctan(\delta/g) = 2k\delta v/g + O(k\delta^3/g^3)$. The zero at $g+\Delta_+$ contributes $(g+\Delta_+)^{2k} e^{-t(g+\Delta_+)^2}$, and the logarithm of the ratio of its weight to $g^{2k}e^{-tg^2}$ is $2k\log(1+\Delta_+/g) - t(2g\Delta_+ + \Delta_+^2) = -(2k\Delta_+/g)(\Delta_+/g + v) + O(k\Delta_+^3/g^3)$. Up to a common factor these are, to this order, the quadruple term and the term of a zero on the line of the Gaussian smoothing of Section 8.1.5, at $u = g(1+v)^{-1/2}$. With $v = -\eta\Delta_+/g$ the phase is $-\omega_+\eta$ and the ratio is $e^{-\kappa_+(1-\eta)}$, the subscripted parameters being built from Δ_+; the other neighbour is reached with $v > 0$ and the parameters built from Δ_-. For $\eta > 1$ the neighbour dominates exponentially, so the minimum is attained on $[0, 1]$, and on the side of the larger gap, where a given phase is reached at a smaller neighbour weight. Since $k \asymp g^2/(\delta\Delta)$ in the relevant range, the neglected terms are a relative error $O(\delta/\Delta)$ in the weight of the quadruple, $O(\delta^2/(g\Delta))$ in its phase and $O(\kappa\Delta/g)$ in the exponent of the neighbour. With $\kappa/\omega = \Delta/\delta \gg 1$ the minimum is attained at $\eta = 1 - \sigma$ with ω just above $\pi/2$. Writing $\omega = \pi/2 + \theta$, the condition is $2(\theta - \omega\sigma) > e^{-\kappa\sigma}$; optimising $\sigma = \kappa^{-1}\log(\kappa/2\omega)$ gives $\theta > (\omega/\kappa)(1 + \log(\kappa/2\omega))$, the stated ω^*, with $\omega/\kappa = \delta/\Delta$. Because $e^{-\kappa\sigma} = 2\delta/\Delta$ at the optimum, a relative error $O(\delta/\Delta)$ in the weight of the quadruple moves θ by $O(\delta^2/\Delta^2)$ only, an error

$O(\kappa\Delta/g)$ in the exponent of the neighbour by $O(\Delta/g)$, and an error in the phase by that error, $O(\delta^2/(g\Delta))$. Finally $t = k(1+v)/g^2$ with $|v| \le \Delta/g$, which gives the time. □

The table below compares $t^*_{\mathrm{loc}} = \omega^*/(2\delta\Delta)$, the time of Proposition 1.4.4 without its remainder, with the certified failure times of Table 1.2, Δ being the larger gap from g to the zeros next to the moved pair; its last column is $100\,(t^*_{\mathrm{loc}}/t^* - 1)$. The local model is within 2.8 per cent of the certified time in every row, and the order $\omega^* g^2/(2\delta\Delta)$ within 2.9 per cent of k^*. Keeping only the quadruple and its two neighbours, with their exact weights, gives the first failing order of Table 1.2 exactly in four rows and to within 0.03 per cent in the other three (computed, not certified), so the differences come from the expansion, not from the zeros it omits. They are positive because the expansion drops the factor $e^{-2t\delta^2}$ in the weight of the neighbour relative to the quadruple, the amplification $e^{2t\delta^2}$ of the quadruple in Section 8.1.5; with that factor kept, the seven times are within 0.25 per cent of the certified ones. The model and the sum over the quadruple and its two neighbours are evaluated numerically from the zeros.

g	δ	Δ	t^*_{loc}	t^*	difference, per cent
185.24	$\frac{1}{4}$	3.03	1.1896	1.18413	0.46
376.08	$\frac{1}{4}$	2.36	1.5591	1.54072	1.19
630.64	$\frac{1}{4}$	2.31	1.5941	1.576635	1.11
750.81	$\frac{1}{4}$	2.57	1.4230	1.410725	0.87
1329.12	$\frac{1}{4}$	1.49	2.5815	2.5121193	2.76
1977.22	$\frac{1}{4}$	2.14	1.7341	1.7113393	1.33
1977.22	0.1	2.14	4.0339	4.0257254	0.20

1.5 A short history of computations on the zeros

Every verification in Table 1.4 from Backlund's on rests on an evaluator of ζ on or near the critical line and a certificate that no zero has been missed (Section 2.2). The window on which Chapters 3 to 8 measure distances, the 79 zeros below height 200, is the range that Backlund verified by hand in 1914 [8].

Where Turing bounds an average of S to certify a count (Section 2.2.1), the zero density identity of Section 3.3 is exact, an integer at every resolution; at the resolution $\frac{1}{2}$, the basepoint one of Section 12.1, it is Backlund's method with the vertical side moved from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = 1$, and it gives the count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$ with no hypothesis, which Section 2.2.3.2 computes at height 1000. Lehmer's close pair near height 7005 [193] is computed from the Pólya kernel in Section 2.2.3.1 and followed under the de Bruijn Newman flow in Section 4.1. The Odlyzko Schönhage algorithm, with Odlyzko's zeros as the reference measure, is one of the evaluators Section 2.2.2 proposes for the Thorin comparison at large height.

year	authors	result
1859	Riemann	the first zeros, in unpublished notes later found by Siegel [257]
1903	Gram	the first 15 zeros; Gram points [88]
1914	Backlund	all 79 zeros below height 200 [8, 9]
1925	Hutchinson	all 138 zeros below height 300 [174]
1936	Titchmarsh	the first 1041 zeros, all below height 1468, by the Riemann Siegel formula, computed with the help of Comrie [263]
1953	Turing	all 1104 zeros below height 1540, on the Manchester computer; Turing's method [270]
1956	Lehmer	the first 25000 zeros; the close pair near height 7005 [193, 194]
1969	Rosser, Yohe, Schoenfeld	the first 3.5×10^6 zeros; Rosser's rule [252]
1986	van de Lune, te Riele, Winter	the first 1.5×10^9 zeros [271]
1987	Odlyzko	10^5 zeros near the 10^{12}th zero, and the statistics of their spacings [220]
1988	Odlyzko, Schönhage	fast evaluation of ζ at many points [221]
2004	Gourdon	the first 10^{13} zeros, not interval certified [87]
2021	Platt, Trudgian	RH verified rigorously below height 3×10^{12} [232]

Table 1.4: Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [71, Chapters 6 to 8].

Chapter 2

Evaluating the Ξ function in ball arithmetic

2.1 The Pólya route

Pólya's integral (2.1.4) evaluates ξ in the complex plane from the kernel alone, without primes or zeros. In this chapter δ denotes the distance from the line of integration to the edge of the strip of analyticity of Φ, not the displacement of an off line zero.

2.1.1 The trapezoid rule on Pólya's integral

The Pólya kernel (2.1.3) in Titchmarsh's normalisation is [264, Section 10.1],

$$\Phi(u) = 2\sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right) \exp(-\pi n^2 e^{2u}), \tag{2.1.1}$$

even and positive on the real line, analytic in the strip $|\operatorname{Im} u| < \pi/4$ (Section 2.1) and doubly exponentially decaying in every closed substrip (proof of Proposition 2.1.1). With the Pólya variable X of density $p = \Phi/\Xi(0)$, the moment generating function (2.1.4) and its derivative under the integral sign give, for every complex s,

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad \frac{\xi'(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \mathbb{E}\, X e^{sX}, \qquad X \sim p, \tag{2.1.2}$$

so the Thorin density (2.1.7) is $\pi^{-1}\operatorname{Re}\{\mathbb{E}Xe^{sX}/\mathbb{E}e^{sX}\}$ at $s = \varepsilon + i\theta$. Throughout, log is the natural logarithm.

Analyticity and decay in the strip make the trapezoid rule with nodes $u_j = jh$ converge geometrically in $1/h$ [265, Theorem 5.1] (Exercise 2.1): for $s = \varepsilon + i\theta$,

$$\Big| \int_{\mathbb{R}} \Phi(u) e^{su}\, du - h\sum_{j\in\mathbb{Z}} \Phi(jh) e^{sjh} \Big| \leq \frac{2K_\eta(\varepsilon)\, e^{\eta|\theta|}}{e^{2\pi\eta/h} - 1}, \qquad 0 < \eta < \pi/4,$$

where $K_\eta(\varepsilon) = \int_{\mathbb{R}} |\Phi(x \pm i\eta)|\, e^{\varepsilon|x|}\, dx$, the same for both signs because Φ is real on $\mathbb{R}$, and $K_\eta = K_\eta(0)$. Since $|\xi(\frac{1}{2} + \varepsilon + i\theta)|$ decays like $e^{-\pi\theta/4}$, the relative error is at most of order $\exp\{(\eta + \pi/4)\theta - 2\pi\eta/h\}$. With $h = 1/200$ the bound is best for η close to $\pi/4$, and at $\theta = 200$ it is about 10^{-289}; halving h changes the computed values by less than 10^{-64}.

The practical limit is the cancellation in (2.1.2): the integrand is of order one and the integral of order $e^{-\pi\theta/4}$, so at b bits the rounding error is of order $2^{-b}e^{\pi\theta/4}$ relative to $|\xi|$, a loss of $\pi\theta/(4\log 10) = 0.341\,\theta$ decimal digits. Discretisation, rounding and the truncation of the u axis at $|u| \le L$ can be budgeted separately. To reach D correct digits at height θ one needs

$$b \;\ge\; \Big(D + \frac{\pi\theta}{4\log 10}\Big)\log_2 10, \qquad h \;\le\; \frac{2\pi\eta}{(D + 0.682\,\theta)\log 10},$$

where the condition on h uses $\eta + \pi/4 \le \pi/2$. The truncation needs $\pi e^{2L} \ge (D + 0.341\,\theta)\log 10 + |\varepsilon|L$, up to lower order terms.

We take $h = 1/200$ and $L = 3$, beyond which Φ is below 10^{-500}, and work at 420 bits (126 decimal digits): at height 200, where $|\xi|$ is about 10^{-63}, the cancellation consumes about half of them and leaves about 60. Truncating the sum at $|j| \le J = L/h$ and folding it with the evenness of Φ gives, with $E = e^{sh}$,

$$\xi(\tfrac{1}{2} + s) \approx P(E) + P(E^{-1}), \qquad \xi'(\tfrac{1}{2} + s) \approx Q(E) - Q(E^{-1}),$$

$$P(x) = \tfrac{h}{2}\Phi_0 + h\sum_{j=1}^{J} \Phi_j x^j, \quad Q(x) = h^2 \sum_{j=1}^{J} j\Phi_j x^j,$$

with $\Phi_j = \Phi(jh)$ and $J = 600$. The node values, the only place where the theta series is summed, are computed once for a given step, truncation and precision and serve every resolution and height; one evaluation of ξ and ξ' then costs four polynomial evaluations in ball arithmetic [179], about 0.7 ms at 420 bits. The ball radii certify only the rounding error, and overestimate it: at height 200 they certify about 38 digits of ξ and, at $\varepsilon = 0.005$, about 36 of $\mathrm{Re}(\xi'/\xi)$, while comparison with `mpmath` at 90 digits shows that the values themselves carry about 63.

2.1.2 Zeros and Thorin masses below height 200

With the parameters of Section 2.1.1, the zeros and the Thorin masses below height $\Theta = 200$ are computed at the eight resolutions $\varepsilon = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, 0.005$ in five steps.

1. Compute Φ_j, $0 \le j \le J$, from (2.1.1) at 420 bits.

2. For each resolution ε, evaluate $\rho_\varepsilon(\theta_k)$ and $M_\varepsilon(\theta_k) = \pi^{-1}\arg\xi(\frac{1}{2} + \varepsilon + i\theta_k)$, with the branch of the argument that vanishes at $\theta = 0$, on the grid $\theta_k = k\varepsilon/8$ ($k\varepsilon/4$ at $\varepsilon = 0.005$), $0 \le \theta_k \le \Theta$. Since $\partial_\theta \arg\xi(\alpha + i\theta) = v_\alpha(\theta)$ with $\alpha = \frac{1}{2} + \varepsilon$, $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ is exactly the cumulative Thorin mass.

3. Locate the line zeros as sign changes of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid of step 0.05, reading each sign from a ball that excludes zero, and refine them by Newton's method using $\Xi'(\theta) = -\operatorname{Im} \xi'(\frac{1}{2} + i\theta)$.

4. Count all zeros with $0 < \operatorname{Im} \rho \le \Theta$ by the argument principle, $N(\Theta) = \pi^{-1} \Delta \arg \xi(s)$ along $2 \to 2 + i\Theta \to \frac{1}{2} + i\Theta$ [264, Section 9.3].

5. Compare with U_{RH} in the metrics of Section 3.1.

Steps 3 and 4, which use the same two polynomials, together bound the number of off line zeros: each sign change of Ξ on the grid encloses at least one zero on the line, so $N(\Theta)$ minus the number of sign changes is an upper bound for the number of off line zeros with $0 < \operatorname{Im} \rho \le \Theta$. When the two numbers agree, every zero below height Θ is on the line and simple.

In step 2 the argument is accumulated from the increments $\arg(\xi(s_{k+1})/\xi(s_k))$ along $\operatorname{Re} s = \frac{1}{2} + \varepsilon$. The principal value is the true increment only if $\arg \xi$ changes by less than π between consecutive points, which the samples cannot show; the grids are chosen so that every observed increment is below $\pi/2$ in modulus, a margin checked after the fact, not guaranteed; the largest observed was 0.29. At large height ξ underflows double precision ($|\xi|$ is about 10^{-335} at height 1000), so the quotient must be formed in ball arithmetic before the argument is taken.

With $\Theta = 200$ there are 79 sign changes, and at none of the 4001 grid points does the ball for Ξ contain zero. The argument principle gives $N(200) = 79.0000000000004$, with a largest increment of 0.02 radians along the contour, and $\Theta = 200$ lies between $\gamma_{79} = 198.0153$ and $\gamma_{80} = 201.2648$, so it is not an ordinate. The Newton refined roots agree to 5.0×10^{-48} with the ordinates computed independently by `mpmath` [213] at 50 digits. The computation recovers, with no input but (2.1.1), the classical fact that the 79 zeros below height 200 lie on the line and are simple [9, 232]. It is not yet a proof: the argument increments are summed in double precision from samples, and the analytic error bounds are not added to the radii (Section 25.2.2 lists what a certified version requires). Figure 2.1 shows the computed density.

2.1.3 Tilting the contour into the strip

Since Φ is analytic in $|\operatorname{Im} u| < \pi/4$, the integrals behind (2.1.2) can be taken along any horizontal line in the strip.

Proposition 2.1.1. *Let $0 < \delta < \pi/4$ and $\eta = \pi/4 - \delta$. For every complex s,*

$$\xi\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} \Phi(x + i\eta)\, e^{sx}\, dx,$$
$$\xi'\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} (x + i\eta)\, \Phi(x + i\eta)\, e^{sx}\, dx.$$

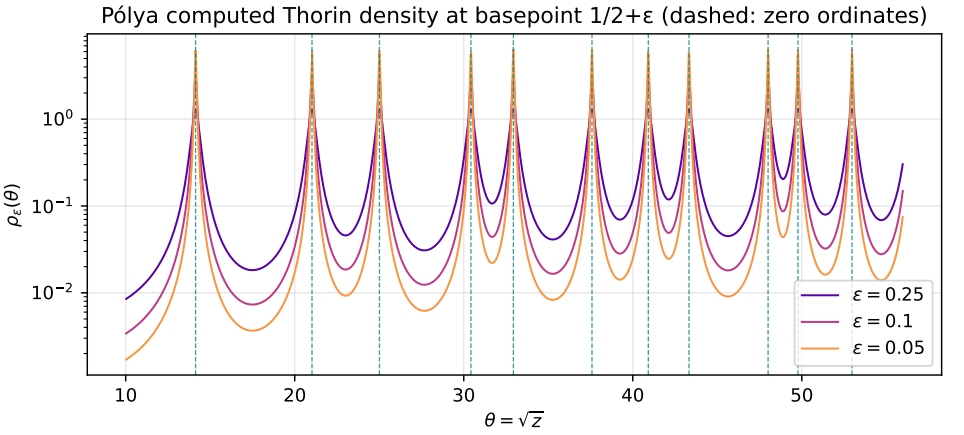


Figure 2.1: The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2}+\varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 2.1.3).

For real x, $|\Phi(x+i\eta)| \le C_\delta$ and $|(x+i\eta)\Phi(x+i\eta)| \le C_\delta$, with $C_\delta = O(\delta^{-5/2})$ as $\delta \to 0$.

Proof. By Section 2.1, Φ is analytic in $|\operatorname{Im} u| < \pi/4$, and by the identity theorem its evenness $\Phi(u) = \Phi(-u)$ extends from the real line to the strip. For $u = x + iy$ with $|y| \le \eta$ and $x \ge 0$ we have $\operatorname{Re} e^{2u} = e^{2x}\cos 2y \ge e^{2x}\sin 2\delta$, so

$$|\Phi(x+iy)| \le 2\sum_{n\ge 1}\left(2\pi^2 n^4 e^{9x/2} + 3\pi n^2 e^{5x/2}\right)\exp(-\pi n^2 e^{2x}\sin 2\delta), \tag{2.1.3}$$

and Φ decays doubly exponentially as $x \to +\infty$, uniformly in $|y| \le \eta$, and by evenness as $x \to -\infty$. Cauchy's theorem therefore moves the line of integration in $\xi(\frac{1}{2}+s) = \int_{\mathbb{R}} \Phi(u)e^{su}\,du$, which is (2.1.2), from $\operatorname{Im} u = 0$ to $\operatorname{Im} u = \eta$, and the second identity follows in the same way from $\xi'(\frac{1}{2}+s) = \int_{\mathbb{R}} u\Phi(u)e^{su}\,du$. Finally, $\sum_{n\ge1} n^k e^{-bn^2} = O(b^{-(k+1)/2})$ uniformly in $b > 0$, so the majorant (2.1.3) is $O(e^{-x/2}\delta^{-5/2})$ for $x \ge 0$, of order $\delta^{-5/2}$ at $x = 0$; with evenness this gives both bounds. □

On the real axis the integrand $\Phi(u)e^{su}$, with $s = \varepsilon + i\theta$, oscillates with frequency θ, and its integral, of order $\theta^{7/4+\varepsilon/2}e^{-\pi\theta/4}|\zeta(\frac{1}{2}+\varepsilon+i\theta)|$ by Stirling's formula, is exponentially smaller than the integrand. On the line $\operatorname{Im} u = \eta$ the factor $e^{i\theta u}$ has constant modulus $e^{-\theta\eta}$ and leaves the integral as the prefactor $e^{i\eta s}$, which carries almost all of the decay of ξ; the integral that remains is of order $e^{-\delta\theta}$ up to the same factors, against an integrand of size at most $C_\delta e^{\varepsilon x}$. The cancellation

drops from $e^{-\pi\theta/4}$ to $e^{-\delta\theta}$. The price is a larger kernel, by the factor K_η/K_0 with $K_0 = \xi(\frac{1}{2}) = 0.497$, and a finer step, because the natural boundary at $\operatorname{Im} u = \pi/4$ [248, Chapter 11, Section 1.4, and Chapter 5, Section 3.2] is now only δ away.

On the tilted line the kernel constant measures the integrand and costs $\log_{10} K_\eta$ digits. By Table 2.1 it grows like $\delta^{-5/2}$, but $\log_{10} K_\eta$ stays under five even at $\delta = 0.01$, which is negligible against the saving $(\pi/4 - \delta)\theta/\log 10$ at any height where tilting is worthwhile.

$\delta = \pi/4 - \eta$	K_η	$\max_x \lvert\Phi(x + i\eta)\rvert$
0.6854	0.554	0.981
0.3927	2.52	3.24
0.2	19.0	15.4
0.1	126	76.5
0.05	774	415
0.02	7836	3938
0.01	43503	22207

Table 2.1: Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts.

The trapezoid rule on the tilted line converges geometrically under two conditions (Exercise 2.2). By Poisson summation its error is $\sum_{k\neq 0} I(\theta + 2\pi k/h)$, where $I(t) = \int_{\mathbb{R}} \Phi(x + i\eta) e^{\varepsilon x} e^{itx}\, dx = e^{\eta t - i\eta\varepsilon}\xi(\frac{1}{2} + \varepsilon + it)$. The integrand is analytic in a strip extending δ above the line and $\pi/2 - \delta$ below it, so $|I(t)|$ is of order $e^{-\delta t}$ as $t \to +\infty$ and $e^{-(\pi/2-\delta)|t|}$ as $t \to -\infty$, up to the factors of Stirling's formula. Relative to $I(\theta)$ the terms with $k \ge 1$ are of order $e^{-2\pi\delta k/h}$, and those with $k \le -1$ are small only when the step resolves the oscillation $e^{i\theta x}$ with a margin. For D correct digits this gives

$$\frac{1}{h} \;\ge\; \frac{D\log 10}{2\pi\delta}, \qquad \frac{2\pi}{h} \;\ge\; \theta + \frac{\delta\theta + D\log 10}{\pi/2 - \delta}, \tag{2.1.4}$$

up to terms of order $\log\theta + \log(1/h)$. The first asks for about $D\log 10/(2\pi\delta)$ nodes per unit length, 458 at $D = 25$ and $\delta = 0.02$, that is $h \le 2.2 \times 10^{-3}$. The second binds at large height: at $\delta = 0.02$ and 25 digits it is the stronger from about $\theta = 2800$ on, and it asks for $h \le 1.23 \times 10^{-3}$ at $\theta = 5000$ and $h \le 8.8 \times 10^{-4}$ at $\theta = 7005$.

Taking δ proportional to $1/\theta$ holds the loss $\delta\theta/\log 10$ fixed, so the working precision grows only through the $\log_{10} K_\eta$ digits of the kernel constant, about $\frac{5}{2}\log_{10}(1/\delta)$, while both conditions (2.1.4) make the number of nodes, and with it the cost per point, grow linearly in θ up to a logarithmic factor from the truncation.

δ	bits	nodes	θ	error of ξ'/ξ	ms per point	untilted digits needed
0.10	200	2001	200	$< 2 \times 10^{-45}$	7	68
0.10	200	2001	500	9.8×10^{-42}	7	171
0.10	200	2001	1000	1.8×10^{-20}	7	341
0.02	220	8401	1000	$< 6 \times 10^{-45}$	28	341
0.02	220	8401	2000	$< 4 \times 10^{-45}$	28	682
0.02	220	8401	5000	2.9×10^{-25}	28	1705

Table 2.2: The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of ξ'/ξ, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height.

2.1.4 Precision and cost of the tilted route

The tilt makes height 5000 accessible (Table 2.2).

On the tilted line the rounding term of Section 2.1.1 becomes $2^{-b}e^{\delta\theta}$ times the kernel constant, and the step obeys (2.1.4). Table 2.3 compares the working precision of the two routes. Its formula for the tilted bits is conservative: it ignores the factor $\theta^{7/4+\varepsilon/2}$ in $|\xi(\frac{1}{2}+\varepsilon+i\theta)|$, worth about six digits at $\theta = 5000$, and it adds five digits for the kernel constant. The run at $\theta = 5000$ in Table 2.2 used 220 bits, fewer than the 244 of Table 2.3, and returned 25 digits; the Lehmer pair of Section 2.2.3.1 was computed with 256 bits at $\delta = 0.01$, where the formula asks for 201.

height θ	target digits	bits (untilted)	bits (tilted, $\delta = 0.02$)	nodes (tilted)
100	30	213	119	8400
200	30	326	122	8400
1000	30	1233	145	8400
5000	25	5748	244	8400
7005	25	8020	302	8400

Table 2.3: Working precision needed by the untilted and the tilted Pólya route for D target digits. The untilted bits are $(D + \pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D + \delta\theta/\log 10 + 5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8 \times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices.

With the cancellation reduced to $e^{-\delta\theta}$, double precision suffices at moderate height. With $\delta = 0.15$ and the node values rounded to double, a vectorised evaluation agrees with the ball arithmetic values of $\rho_{0.05}$ to within an absolute

error of 2.8×10^{-12} for $\theta \le 50$, 1.3×10^{-9} for $\theta \le 100$ and 7.6×10^{-7} for $\theta \le 150$, at $93\,\mu$s per point. The untilted route fails in double precision above $\theta \approx 40$ (Exercise 2.5).

2.1.5 Ball arithmetic on the tilted line

Horner's rule, which serves the untilted evaluator well, fails in ball arithmetic on the tilted line. The tilted evaluator writes the trapezoid sum as one polynomial of degree $2J$ in $E = e^{sh}$, with coefficients $h\Phi(x_k + i\eta)$ for x_k running from $-L$ to L, times the prefactor $e^{i\eta s}E^{-J}$. With 2001 nodes, $\delta = 0.1$ and 200 bits, the radius of the Horner recursion grew by a factor of about 1.3 per step, and the final ball at $\theta = 100$ had radius 10^{109} although its midpoint was correct to many digits. Evaluating each exponential e^{sx_k} directly and summing gives balls of radius about 10^{-55} at the same point.

The cause is the wrapping effect of rectangular complex balls. Arb stores a complex number as a real ball and an imaginary ball [179], so a multiplication by $E = e^{\varepsilon h}e^{i\theta h}$ rotates the midpoint but multiplies the radii by $e^{\varepsilon h}(|\cos\theta h| + |\sin\theta h|)$, up to rounding; at $\theta h = 0.4$ this factor is 1.31. In Horner's rule the coefficient of E^k passes through k such multiplications. On the untilted line the effect is contained: the polynomials are folded at $x = 0$, so the coefficient at $x = jh$ passes through at most 600 multiplications, and the coefficients $h\Phi(jh)$ decay like $\exp(-\pi e^{2jh})$, so those that pass through many multiplications are negligible and the radii stay far below $|\xi|$ up to height 200. On the tilted line the coefficients decay only like $\exp(-\pi e^{2x}\sin 2\delta)$, and the coefficient at x passes through $(x+L)/h$ multiplications. With $h = 0.004$ and $L = 4$, which give 2001 nodes, the coefficients near $x = 2$ are still of order 10^{-12} at $\delta = 0.1$, and they pass through about 1500 multiplications, which enlarge their radii by a factor of about 10^{176}.

2.2 Classical evaluators, Turing's method and the Pólya route

The classical evaluators on the critical line pay for height differently and, at large height, far less than the Pólya route, tilted or not.

2.2.1 The Riemann Siegel formula, Gram points and certified counts

On the critical line $\zeta(\frac{1}{2}+it) = e^{-i\vartheta(t)}Z(t)$, where $\vartheta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann Siegel theta function and $Z(t)$ is real for real t. Write $m = \lfloor\sqrt{t/2\pi}\rfloor$ and $q = \sqrt{t/2\pi} - m$. For every $K \ge 0$,

$$Z(t) = 2\sum_{n\le m}\frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + (-1)^{m-1}\Big(\frac{t}{2\pi}\Big)^{-1/4}\sum_{j=0}^{K}C_j(q)\Big(\frac{t}{2\pi}\Big)^{-j/2} + R_K(t),$$

where $R_K(t) = O(t^{-(2K+3)/4})$ as $t \to \infty$, $C_0(q) = \cos(2\pi(q^2 - q - \frac{1}{16}))/\cos(2\pi q)$, and the higher C_j are given in [71, Section 7.5] and [78]. This asymptotic expansion was found in Riemann's unpublished notes and published by Siegel [257].

The main sum is the approximate functional equation $\zeta(s) = \sum_{n \le x} n^{-s} + \chi(s) \sum_{n \le y} n^{s-1} + \cdots$, where $\zeta(s) = \chi(s)\zeta(1-s)$ and $2\pi xy = t$ [264, Chapter 4], taken at $x = y = \sqrt{t/2\pi}$ and written for Z; the C_j correct for the abrupt truncation at $n = m$. The terms are bounded by 2 with no exponential cancellation, so double precision suffices at every height computed in this book; near height 10^{12} and above, the phases $\vartheta(t) - t \log n$ must be reduced modulo 2π in higher precision. Gabcke's thesis gives explicit bounds for the remainders R_K with $K \le 10$ [78], which make the formula usable in rigorous computation.

To evaluate Z at many points near height T, Odlyzko and Schönhage [221] compute the main sum on a grid by fast Fourier transforms and rational function evaluation, and interpolate between grid points by band limited interpolation. About $T^{1/2}$ values then cost $T^{1/2+o(1)}$ operations, that is $T^{o(1)}$ per point. Odlyzko compared the spacings of the zeros near zero number 10^{12} with the Gaussian unitary ensemble [220], and with the algorithm he later took such computations to zero numbers 10^{20} [222] and 10^{22} [224].

The classical searches for zeros on the line sample Z at the Gram points g_n, defined by $\vartheta(g_n) = n\pi$ [88, pp. 301 and 302]. There $Z(g_n)$ usually has the sign of $(-1)^n$, and each Gram interval $[g_{n-1}, g_n)$ usually contains exactly one zero. This is Gram's law, and it fails with positive frequency [267, Corollary 5.4]. Call g_n good when $(-1)^n Z(g_n) > 0$, and call an interval $[g_j, g_{j+k})$ between consecutive good Gram points a Gram block. Rosser's rule asserts that such a block contains exactly k zeros [252], [267, Section 7.3]. It holds far longer than Gram's law, since its first exception is the block that begins at $g_{13\,999\,525}$ [31, Table 3], but it too fails with positive frequency [267, Corollary 7.8]. Counting sign changes of Z bounds the number of zeros on the line from below; the difficulty is to show that none is missing, off the line or in a close pair that produces no sign change.

Turing's answer [270] uses the counting function $N(T) = \vartheta(T)/\pi + 1 + S(T)$, where, for T not an ordinate, $\pi S(T) = \arg \zeta(\frac{1}{2} + iT)$, the argument being defined by continuous variation along $2 \to 2 + iT \to \frac{1}{2} + iT$. The function S is small on average: Littlewood proved $\int_0^T S(t)\,dt = O(\log T)$ [264, Theorem 9.9(A)], and Turing gave explicit constants for $|\int_{t_1}^{t_2} S(t)\,dt|$, later sharpened by Trudgian [266]. If the sign changes found near height T were fewer than $N(T)$, then S would be displaced by at least 2 over an interval long enough to violate the integral bound. The count is certified by locating the sign changes in a range slightly beyond T and checking them against Turing's bound for $\int S$, without evaluating ζ off the line. Turing's published argument contained errors, corrected by Lehman [192], and the corrected method underlies every large verification since, including Platt and Trudgian's to height 3×10^{12} [232].

Before Turing, counts were certified by Backlund's method. Since $\operatorname{Re} \zeta(2+it) \ge$

	Riemann Siegel and Turing	Odlyzko Schönhage	Pólya route
object	sign changes of $Z(t)$	zeros in blocks, statistics	Thorin density ρ_ε, mass, count
where evaluated	critical line	critical line	line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ and contour
count certificate	Gram points and $\int S$	Turing's method	argument principle count equal to the number of sign changes; computed, not yet certified
cost per point	$\sqrt{t/2\pi}$ terms, double	amortised, $t^{o(1)}$	$0.341\,t$ extra digits and step below π/t (untilted); $\delta t/\log 10$ digits and step below $2\pi/t$ (tilted)
accuracy	asymptotic remainder	asymptotic remainder	spectral; rounding certified by balls
reach	$t \approx 1540$ in 1953	$t \approx 10^{20}$ and beyond [222, 224]	$t \approx 7005$ (Section 2.2.3.1); 25 digits at $t = 5000$ (Table 2.2)

Table 2.4: The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached.

$2 - \zeta(2) > 0$ for every t, it suffices to check that $\operatorname{Re}\zeta(\sigma + iT) \neq 0$ for $\frac{1}{2} \leq \sigma \leq 2$: then $\arg\zeta$ stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$ along the whole path, so $|S(T)| < \frac{1}{2}$ and $N(T)$ is the integer nearest to $\vartheta(T)/\pi + 1$. Backlund carried this out for $T = 200$ [9] and Hutchinson for $T = 300$ [174, pp. 54 to 57], [266].

2.2.2 The Pólya route beside the classical evaluators

The Pólya route evaluates ξ and ξ' on and off the line, at $\frac{1}{2} + \varepsilon + i\theta$, from the Pólya kernel, and so delivers ρ_ε, its mass and the zero count from one evaluator. Its count has the logic of Turing's certificate: when the sign changes of Ξ and the argument principle count $N(T)$ agree, every zero up to height T is on the line and simple (Section 2.1.2).

The methods differ most in how their cost grows with height (Table 2.4). The untilted Pólya route needs about $0.341\,t$ extra digits, and by the budget of Section 2.1.1 its step must be below π/t, so its number of nodes also grows linearly with t, because of aliasing. By Poisson summation the trapezoid sum at $s = \varepsilon + it$ equals $\sum_{k\in\mathbb{Z}} \xi(\frac{1}{2} + s - 2\pi ik/h)$, and the term $k = 1$, at height $t - 2\pi/h$, is exponentially larger than $\xi(\frac{1}{2} + s)$ once $t > \pi/h$. The tilted contour of Section 2.1.3 cuts the loss of digits to about $\delta t/\log 10$, but aliasing still requires $2\pi/h > t$, by the second condition in (2.1.4), so there too the number of nodes grows linearly with t.

The distances of Section 3.1 do not depend on how ξ is evaluated, which suggests a hybrid at large height. The approximate functional equation holds

off the critical line [264, Chapter 4], so Riemann Siegel type expansions of ζ [6, Theorems 3.1 and 4.2] and of ζ', and the Odlyzko Schönhage algorithm for blocks of points, apply at $\frac{1}{2}+\varepsilon+i\theta$ as well. With them as the evaluator and Odlyzko's zeros [225] as the RH reference, one could compute ρ_ε, the matched total variation and the positivity margin at the heights of Odlyzko's computations, from about 10^{11} to beyond 10^{20}, while the tilted Pólya route supplies values with explicitly bounded errors on low windows.

2.2.3 Two computations beyond the untilted route

2.2.3.1 A Lehmer pair near height 7005

Lehmer found two zeros that are unusually close, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ [193, pp. 297 and 298], at distance 0.0377, the classical example of a Lehmer pair; such pairs give lower bounds for the de Bruijn Newman constant [51] (Section 4.1.6). The untilted Pólya route cannot reach this height: it would need about 2400 digits and a step below 4.5×10^{-4}. The tilted route with $\delta = 0.01$, 256 bits, 13001 nodes, step 6×10^{-4} and truncation $|x|\le 3.9$ computes $\Xi(\theta)e^{\pi\theta/4}$ at 34 milliseconds per point. On a grid of step 0.005 it finds sign changes in $(7005.060, 7005.065)$ and $(7005.100, 7005.105)$, the two members of the pair (Figure 2.2).

The Thorin density shows which resolution separates them. A single zero on the line produces a Cauchy peak of height $1/(\pi\varepsilon)$, that is 6.4, 15.9 and 31.8 at $\varepsilon = 0.05$, 0.02 and 0.01. The largest values of ρ_ε on the grid are 11.18, 19.58 and 33.89. At $\varepsilon = 0.05$ the two peaks merge into one of nearly double height; at $\varepsilon = 0.02$ they are partly separated; at $\varepsilon = 0.01$ each has its own peak, lifted slightly by its neighbour, as the Cauchy kernel predicts: two kernels with half width at half maximum ε at distance g have a single maximum when $g \le 2\varepsilon/\sqrt{3}$ and two maxima when $g > 2\varepsilon/\sqrt{3}$. With $g = 0.0377$ the pair is resolved as two peaks for $\varepsilon < \sqrt{3}\,g/2 \approx 0.033$. The criterion concerns two isolated kernels; Exercise 2.11 proves it and asks how much the other zeros move the threshold.

2.2.3.2 The zero count at height 1000

Proposition 3.3.2 of Section 3.3 gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, where $M_{1/2}(T) = \pi^{-1}\arg\xi(1+iT)$ is the cumulative Thorin mass at resolution $\frac{1}{2}$, that is at the basepoint one of Section 12.1, and $\tau_{1/2}(T)$ is π^{-1} times the change of $\arg\xi$ along the segment from $1+iT$ to $\frac{1}{2}+iT$. With the tilted route ($\delta = 0.1$, 200 bits, step 0.004 and truncation $|x|\le 4.2$, that is 2101 nodes), 10000 steps of 0.1 along $\operatorname{Re} s = 1$ and 2000 steps along the segment, we find

$$M_{1/2}(1000) = 648.7562315041018, \qquad \tau_{1/2}(1000) = 0.2437684958980,$$

$$M_{1/2}(1000) + \tau_{1/2}(1000) = 648.9999999999999,$$

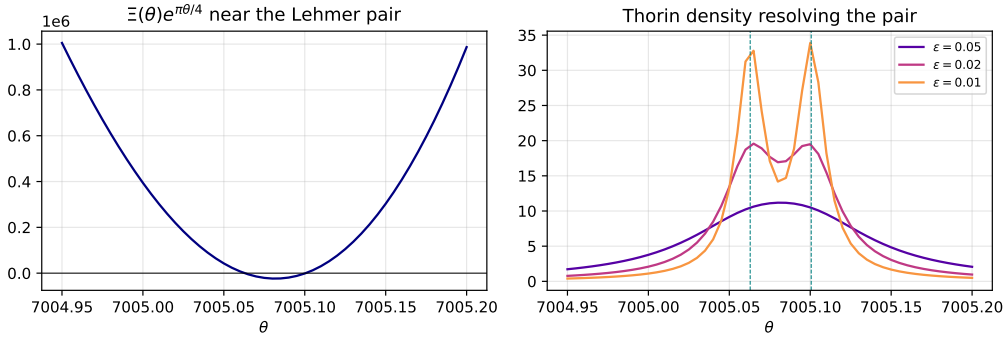


Figure 2.2: The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed.

so $N(1000) = 649$, with a largest argument increment of 0.43 radians. The computation took 77.5 seconds; `mpmath`'s `nzeros` returns 649 in 0.017 seconds. Where Backlund bounds the argument, this computation accumulates it from samples; ball arithmetic certifies the rounding in each sampled value but not the absence of winding between samples, so the count is computed, not certified. Certifying it would need, besides the analytic error bounds of Section 2.1, an enclosure of ξ between consecutive sample points s_k and s_{k+1}, for instance a check that $\operatorname{Re} \xi(s)/\xi(s_k) > 0$ on a ball containing the segment between them.

For counting and locating zeros at height the Riemann Siegel family with Turing's method is faster by orders of magnitude, and the Pólya route is no competitor. It reads the count as an integer from a smooth quantity, the Thorin mass, with no Gram point bookkeeping; it gives the zero density strip by strip (Section 3.3); and it evaluates objects that the classical methods, used on the critical line, do not compute: Thorin densities and their distances to U_{RH}.

2.3 A worked computation from first principles

We follow one small computation with the untilted route at low height, from the theta series to a Thorin mass and a zero.

The parameters are those of Section 2.1.1, step $h = 1/200$ and truncation $|u| \le 3$, and the working precision is 240 bits, about 72 decimal digits, the default of the code. As there, s is the displacement from $\frac{1}{2}$: with $E = e^{sh}$ and $\Phi_j = \Phi(jh)$, the evaluator returns $\xi(\frac{1}{2}+s) \approx P(E) + P(E^{-1})$ and $\xi'(\frac{1}{2}+s) \approx Q(E) - Q(E^{-1})$, from which $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$.

Step one: the kernel at a single point. At $u = 0$ the kernel (2.1.3) is

$$\Phi(0) = 2\sum_{n\geq 1} \bigl(2\pi^2 n^4 - 3\pi n^2\bigr) e^{-\pi n^2}.$$

The nth term is of order $n^4 e^{-\pi n^2}$ and the terms with $n \geq 8$ add up to less than 10^{-82}, so seven terms give more than eighty digits:

$$\Phi(0) = 0.89339380093424688817396933410941822644084873418965\ldots.$$

The ball returned by the arithmetic has radius 3×10^{-71}.

Step two: the coefficients. The truncation leaves 601 nodes $u_j = jh$, $0 \leq j \leq 600$. The coefficients of P are $\frac{h}{2}\Phi_0$ and $h\Phi_j$, $1 \leq j \leq 600$, and those of Q are $h^2 j\Phi_j$; the first is

$$\tfrac{h}{2}\Phi(0) = 0.00223348450233561722043492333527354556610\ldots.$$

The coefficients decay doubly exponentially: by the asymptotic formula of Section 2.1, $\Phi(2.7) \approx 4\pi^2 e^{12.15}\exp(-\pi e^{5.4}) \approx 6 \times 10^{-296}$, so the coefficients beyond $j \approx 540$ are below 10^{-297}.

Step three: $\xi(\frac{1}{2})$ as a trapezoid sum. At $s = 0$, $E = e^{sh} = 1$, and $\xi(\frac{1}{2}) = 2P(1)$ is the trapezoid sum of the kernel:

$$\xi(\tfrac{1}{2}) = 0.49712077818831410991277373968539771980729360955 77\ldots,$$

with ball radius 5×10^{-71}. `mpmath`'s evaluation from the definition, $\frac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ at $z = \frac{1}{2}$ in thirty digit arithmetic, gives

$$0.497120778188314109912773739685,$$

agreeing to every digit. By the bound of Section 2.1.1 the quadrature error is below 10^{-400}, so the seventy digits are set by the working precision, not by the rule.

Step four: the Thorin density at the first ordinate. At

$$s = 0.1 + 14.134725141734693790i,$$

that is at resolution $\varepsilon = 0.1$ and the first ordinate γ_1, the evaluator returns $\xi(\frac{1}{2}+s)$ and $\xi'(\frac{1}{2}+s)$, and

$$\rho_{0.1}(\gamma_1) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}\Bigl(\frac{1}{2}+s\Bigr) = 3.18525309376337225967538\ldots.$$

A single Cauchy kernel of width 0.1 contributes exactly $1/(\pi\varepsilon) = 3.18309886183791$ at its centre. By Proposition 2.1.3 the excess, 0.00215423, is the sum of the contributions of every other zero, $\sum_{j\neq 1}[P_{0.1}(\gamma_1-\gamma_j)+P_{0.1}(\gamma_1+\gamma_j)]$, plus the reflected kernel of γ_1 itself. The nearest neighbour, at distance 6.887, contributes 6.7×10^{-4}, the reflected kernel 4×10^{-5}, and the many distant zeros, whose contributions decay like the inverse square of the distance, the remaining 1.4×10^{-3}. Figure 2.3 shows the same decomposition along the whole stretch between the first two zeros.

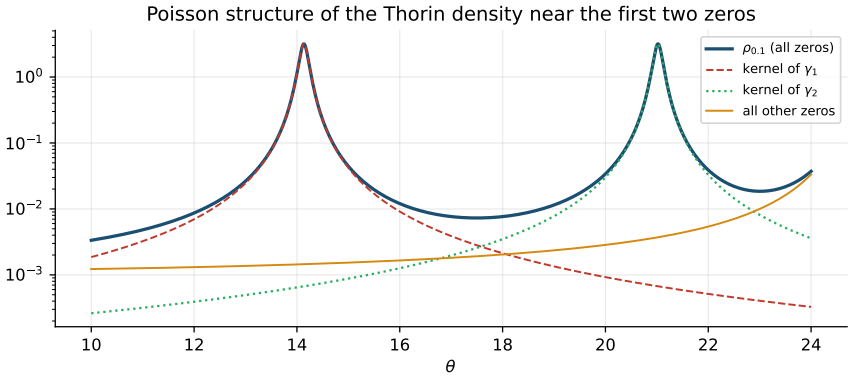


Figure 2.3: The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of γ_1, the kernel of γ_2 and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction.

Between zeros the density is small but strictly positive. At $\theta = 17$, between γ_1 and $\gamma_2 = 21.022$,

$$\rho_{0.1}(17) = 0.0076857707896205949783\ldots,$$

close to the minimum of $\rho_{0.1}$ between the two zeros (Figure 2.3). Lagarias's criterion [185] asks for this positivity at every θ and every resolution $\varepsilon > 0$.

Step five: the Thorin mass up to height 20. The cumulative Thorin mass of Section 2.1.2 is $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the argument continued from the real axis. Stepping along the line $\frac12+0.1+i\theta$ from $\theta=0$ to $\theta=20$ in 1600 steps of $\varepsilon/8 = 0.0125$, as in that section, and accumulating the arguments of the quotients $\xi(\frac12+s_{k+1})/\xi(\frac12+s_k)$, where $s_k = 0.1+i\theta_k$ and $\theta_k = 0.0125\,k$, gives

$$\mu_{0.1}\big((0,400]\big) = M_{0.1}(20) = 1.0509304\ldots,$$

in 0.77 seconds. There is exactly one zero below height 20. By Proposition 3.1.2 the excess 0.0509 over the count agrees to $O(\varepsilon^2)$ with the first order mass defect

$-(\varepsilon/\pi)\Xi'(20)/\Xi(20)$ (Exercise 2.16): it is the mass carried across the edge $\theta = 20$ by the Cauchy tails of the first zero and of the zeros above 20, while what crosses $\theta = 0$ is returned exactly by the reflected kernels. Adding $\tau_{0.1}(20)$, which is π^{-1} times the change of $\arg\xi$ along the segment from $\frac{1}{2} + 0.1 + 20i$ to $\frac{1}{2} + 20i$, returns the integer 1, as Proposition 3.3.2 requires.

Step six: locating the first zero. The ordinate γ_1 of step four would have appeared, up to a small shift from the other zeros, as the first peak of the density. To locate it from the Pólya route alone, one evaluates $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid, finds the sign change in (14.10, 14.15), and refines by Newton's method with $\Xi'(\theta) = -\operatorname{Im}\xi'(\frac{1}{2} + i\theta)$; six iterations give γ_1 to forty digits. Section 2.1.2 refines all 79 zeros below height 200 in the same way, at 420 bits.

2.4 Exercises

On Section 2.1.

2.1. Let f be analytic in a strip containing $|\operatorname{Im} u| \le \eta$, with $\int_{\mathbb{R}} |f(x \pm i\eta)|\,dx \le M$ and $f(u) \to 0$ uniformly as $|\operatorname{Re} u| \to \infty$ in $|\operatorname{Im} u| \le \eta$. Prove that $|\int_{\mathbb{R}} f(u)\,du - h\sum_j f(jh)| \le 2M/(e^{2\pi\eta/h} - 1)$ by writing the error as the sum of the integrals, from left to right, of $f(u)(e^{-2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = \eta$ and of $f(u)(e^{2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = -\eta$ [265, Theorem 5.1]. Deduce the bound of Section 2.1.1.

2.2. Let $g(x) = \Phi(x + i\eta)e^{\varepsilon x}$ with $\eta = \pi/4 - \delta$ and $\varepsilon \ge 0$, and $I(t) = \int_{\mathbb{R}} g(x)e^{itx}\,dx$. Show by Poisson summation that $h\sum_{j\in\mathbb{Z}} g(jh)e^{i\theta jh} - I(\theta) = \sum_{k\neq 0} I(\theta + 2\pi k/h)$. For $0 < a < \delta$, $0 < b < \pi/2 - \delta$ and $0 < \theta < 2\pi/h$, shift the line of integration up by a when $k \ge 1$ and down by b when $k \le -1$ to bound the error by $Ae^{-a\theta}/(e^{2\pi a/h} - 1) + Be^{b\theta}/(e^{2\pi b/h} - 1)$, where A and B are the suprema of $\int_{\mathbb{R}} |g(x + iy)|\,dx$ over $0 \le y \le a$ and over $-b \le y \le 0$. Derive (2.1.4) from $|I(t)| = e^{\eta t}|\xi(\frac{1}{2} + \varepsilon + it)|$ and Stirling's formula.

2.3. Using the formulas in the caption of Table 2.3, estimate the number of bits the untilted route needs at height 2000 for twenty digits, and compare with the tilted route at $\delta = 0.02$.

2.4. Show that on the tilted line $\operatorname{Im} u = \eta$ the factor e^{su} with $s = \varepsilon + i\theta$ has modulus $e^{\varepsilon x - \theta\eta}$. Explain why the prefactor $e^{i\eta s}$ carries all of the exponential decay of ξ except the factor $e^{-\delta\theta}$, and where that factor comes from.

2.5. Implement the two polynomial evaluator in double precision and find the height at which it first returns the wrong sign of Ξ.

2.6. Show that in midpoint radius arithmetic, with a complex number stored as a real and an imaginary ball, multiplication by an exact $w = |w|e^{i\varphi}$ rotates

the midpoint and maps real and imaginary radii r to $|w|(|\cos\varphi|+|\sin\varphi|)\,r$, up to rounding. Use this to explain why Horner's rule loses ball accuracy on the tilted line but not on the untilted line.

2.7. Use the tilted route to compute $\Xi(1000)$ and check its sign against $Z(1000)$ from the Riemann Siegel formula, with Z as in Section 2.2 and $\Xi(t) = -\frac{1}{2}\bigl(t^2+\frac{1}{4}\bigr)\pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{1}{2}it)|\,Z(t)$.

On Section 2.2.

2.8. Compute the Gram points $g_0,\dots,g_{20}$ and check Gram's law for them; find the first Gram interval that contains no zero (it lies beyond g_{100}).

2.9. Show that $N(T) = \vartheta(T)/\pi + 1 + S(T)$ follows from the argument principle applied to ξ on a rectangle.

2.10. Explain why a close pair of zeros that falls inside one Gram interval produces no sign change there, and how Turing's method detects that zeros are missing.

2.11. Prove that $P_\varepsilon(x-g/2)+P_\varepsilon(x+g/2)$ has a single maximum if $g \le 2\varepsilon/\sqrt{3}$ and two maxima if $g > 2\varepsilon/\sqrt{3}$, and estimate how the other zeros near height 7005 shift the threshold 0.033 for the Lehmer pair.

2.12. Estimate the cost of counting zeros to height 10^4 by the Thorin mass at resolution $\frac{1}{2}$ with the tilted route, including the growth of the number of nodes with height, and compare with the Riemann Siegel count.

On Section 2.3.

2.13. Compute $\Phi(0)$ with two, three, four and five terms of the series and explain the number of correct digits in each case.

2.14. Repeat step three with $h = 1/5$, $1/10$ and $1/20$ and plot $\log_{10}$ of the error against $1/h$. Show by Poisson summation that the error of the untruncated sum is exactly $2\sum_{k\ge1}\Xi(2\pi k/h)$, and confirm that it falls like $e^{-\pi^2/(2h)}$, the rate set by the strip half width $\pi/4$, up to slowly varying factors.

2.15. Compute $\rho_{0.1}(\gamma_1) - 1/(0.1\pi)$ directly as a sum over the first 1700 zeros with the smooth tail model of Section 3.1, and compare with step four.

2.16. Predict $M_{0.1}(20) - 1$ from the first order formula of Proposition 3.1.2, with $\Xi'(20)/\Xi(20)$ computed by the evaluator of step four, and compare with step five.

Chapter 3

The Thorin measure at finite resolution

The Thorin condition is RH (Proposition 2.1.1), and by Lagarias's criterion RH is the positivity of ρ_ε at every resolution (Remark 2.1.4). This chapter asks what finite computations of ρ_ε establish. On a window where the zeros are verified, the computed Thorin measure μ_ε converges to U_{RH}, the discrete measure demanded by RH, at explicit rates as $\varepsilon \downarrow 0$, and the rate depends on the metric (Section 3.1). An off line zero changes each of these distances in closed form and is most visible at resolutions comparable to its distance from the line (Section 3.2). Without any assumption on the zeros below the window, the argument principle turns the cumulative Thorin mass into an exact count of all zeros and of the zeros to the right of the line of evaluation (Section 3.3). Above the computed range every finite computation of these kinds remains consistent with a failure of the Thorin condition (Section 1.4).

3.1 Distances to the discrete measure implied by RH

On a window where the zeros are known, the Thorin measure μ_ε computed by the Pólya route can be compared with U_{RH} directly (Section 1.3.1), and how close the two are depends on the metric.

Fix $\Theta = 200$ and the window $E = (0, \Theta^2]$ in z, equivalently $(0, \Theta]$ in θ. The window contains 79 zeros, all simple and on the line, and Θ is not an ordinate, since $\gamma_{80} = 201.2648$ (Section 2.1.2). For a signed measure ν on E write $\|\nu\|_{\mathrm{TV}} = |\nu|(E)$, and $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ for the cumulative mass.

3.1.1 Total variation at matched resolution

A smooth approximation to a sum of point masses never converges to it in total variation.

Proposition 3.1.1. *If $\nu \geq 0$ is absolutely continuous and U is purely atomic with N unit atoms in E, then $\|\nu - U\|_{\mathrm{TV}} = \nu(E) + N$.*

Proof. Let A be the finite set of atoms. Then $\nu(A) = 0$, so $|\nu - U|(A) = U(A) = N$ and $|\nu - U|(E \setminus A) = \nu(E \setminus A) = \nu(E)$. □

The Thorin measure is positive on E, where its density has minimum $0.0147\,\varepsilon$ (Section 1.3.1), so $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}} = \mu_\varepsilon(E) + 79 \approx 158$ for every ε. Total variation becomes informative only after both measures are brought to the same resolution, or summed over cells.

By Proposition 2.1.3 the object to compare with μ_ε is the Poisson smoothing of the RH measure at the same resolution; write $d_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\|_{\mathrm{TV}}$ for the matched distance. Under RH the two measures coincide and $d_{\mathrm{match}}(\varepsilon) = 0$, so a computed matched distance measures its own error unless some zero is off the line.

A computation replaces $U^{\pm}_{\mathrm{RH}}$ by a model $\widetilde{U}$, built from the first 1700 ordinates, computed with `mpmath` [213], and from the smooth density $\pi^{-1}\vartheta'(t)\,dt$ of the Riemann Siegel theta function beyond the Gram point $T_* = g_{1699} = 2197.8796$ of Section 2.2.1, at which $\vartheta(T_*)/\pi + 1 = 1700$, each with its reflection. Write $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$. Since $\gamma_{1700} = 2197.2587 < T_* < \gamma_{1701} = 2198.6875$, we have $N(T_*) = \vartheta(T_*)/\pi + 1$, so the boundary term vanishes when $\int_{T_*}^{\infty} P_\varepsilon(\theta - t)\,d\big(N(t) - \vartheta(t)/\pi - 1\big)$ is integrated by parts, and the error of the tail model is at most of order ε/T_*^2 per unit length. The observed floor is far smaller: $\tilde{d}_{\mathrm{match}}(\varepsilon) = 3.66 \times 10^{-9}\,\varepsilon$ at every resolution from $\varepsilon = 1$ to $\varepsilon = 0.005$ (Table 3.1), that is $1.8 \times 10^{-11}\,\varepsilon$ per unit length. The exact proportionality to ε identifies the floor as the residual of the tail model. It is an observed value, not a bound for $d_{\mathrm{match}}(\varepsilon)$, which would need certified bounds on the error of the tail model and on the evaluation of μ_ε. The densities that enter the comparison are stored in double precision.

3.1.2 Cumulative distances and extrapolation in the resolution

The total mass is the simplest quantity to compare, since $U_{\mathrm{RH}}(E) = N(\Theta) = 79$. By Section 1.3.3 it is an argument, $\mu_\varepsilon(E) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\Theta)$, continued from $\arg \xi(\frac{1}{2} + \varepsilon) = 0$.

Proposition 3.1.2. *Suppose Θ is not an ordinate and every zero with $0 < \operatorname{Im} \rho \le \Theta$ lies on the line. Then*

$$\mu_\varepsilon\big((0, \Theta^2]\big) = N(\Theta) - \frac{\varepsilon}{\pi}\,\frac{\Xi'(\Theta)}{\Xi(\Theta)} + O(\varepsilon^2), \qquad \varepsilon \downarrow 0.$$

Proof. Let $\sigma_1 > 1$. By hypothesis, and because ξ is positive on the real axis, ξ has no zero in the closed rectangle with vertices $\frac{1}{2}+\varepsilon$, σ_1, $\sigma_1 + i\Theta$ and $\frac{1}{2}+\varepsilon+i\Theta$. So the change of $\arg \xi$ up its left side, which is $\pi\mu_\varepsilon((0, \Theta^2])$, equals the change along the path $\frac{1}{2} + \varepsilon \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + \varepsilon + i\Theta$. The classical formula $\pi N(\Theta) = \Delta \arg \xi$ along $\frac{1}{2} \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + i\Theta$ [264, Section 9.3] differs from it only by the

ε	mass	raw TV	matched TV	sup diff	cell TV	L^1 in θ	$W_\varepsilon(0.01)$
1	79.36738	158.37	3.66×10^{-9}	1.9×10^{-11}	4.992	59.23	0.29504
0.5	79.19417	158.19	1.83×10^{-9}	9.6×10^{-12}	2.655	41.86	0.22362
0.25	79.09887	158.10	9.14×10^{-10}	4.8×10^{-12}	1.351	28.77	0.18698
0.1	79.03977	158.04	3.66×10^{-10}	1.9×10^{-12}	0.5431	16.01	0.16469
0.05	79.01990	158.02	1.83×10^{-10}	9.6×10^{-13}	0.2718	9.737	0.15721
0.02	79.00796	158.01	7.31×10^{-11}	3.8×10^{-13}	0.1087	4.816	0.15270
0.01	79.00398	158.00	3.66×10^{-11}	2.0×10^{-13}	0.05437	2.756	0.15120
0.005	79.00199	158.00	1.83×10^{-11}	1.5×10^{-13}	0.02718	1.553	0.15045
RH	79	0	0	0	0	0	0.14970

Table 3.1: Distances between the Pólya computed Thorin measure μ_ε and U_{RH} on $0 < z \le 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.1.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \le 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, L^1 in θ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row.

segment from $\frac{1}{2} + \varepsilon + i\Theta$ to $\frac{1}{2} + i\Theta$, since $\arg \xi$ is constant on the real axis. Put $h(\varepsilon) = \xi(\frac{1}{2} + \varepsilon + i\Theta)$. Since $h(0) = \Xi(\Theta) \neq 0$, a branch of $\arg h$ is real analytic near $\varepsilon = 0$, and

$$\pi\mu_\varepsilon\big((0, \Theta^2]\big) = \pi N(\Theta) + \arg h(\varepsilon) - \arg h(0) = \pi N(\Theta) + \varepsilon \operatorname{Im} \frac{\xi'}{\xi}\Big(\frac{1}{2} + i\Theta\Big) + O(\varepsilon^2).$$

From $\frac{d}{d\theta}\Xi(\theta) = i\xi'(\frac{1}{2} + i\theta)$ we get $(\xi'/\xi)(\frac{1}{2} + i\Theta) = -i\,\Xi'(\Theta)/\Xi(\Theta)$, and Ξ'/Ξ is real, so the coefficient of ε is $-\Xi'(\Theta)/\Xi(\Theta)$. □

Remark 3.1.3. The remainder in Proposition 3.1.2 is in fact $O(\varepsilon^3)$: $\xi(s) = \xi(1-s)$ and the reality of ξ on the real axis give $h(-\varepsilon) = \overline{h(\varepsilon)}$, so $\arg h(\varepsilon) - \arg h(0)$ is odd in ε.

Choosing Θ midway between zeros makes the correction small, and adding it recovers $N(\Theta)$ up to $O(\varepsilon^3)$; Proposition 3.3.2 makes the recovery exact. At $\Theta = 200$ the Pólya route gives $\Xi'(200)/\Xi(200) = -1.2506879$, so the proposition predicts $(\mu_\varepsilon(E) - 79)/\varepsilon = -\Xi'(200)/(\pi\,\Xi(200)) + O(\varepsilon^2) = 0.3981063 + O(\varepsilon^2)$; the observed value is 0.398105 at $\varepsilon = 0.005$ and 0.398102 at $\varepsilon = 0.01$. Figure 3.1 compares M_ε with the counting function of U_{RH} across the window.

Summing over cells gives a distance that resolves the individual zeros. Let $b_0 = 0$, $b_k = (\gamma_k + \gamma_{k+1})/2$ for $1 \le k \le 78$ and $b_{79} = \Theta$, and let $C_k = (b_{k-1}, b_k]$. The cell masses $\mu_\varepsilon(C_k) = M_\varepsilon(b_k) - M_\varepsilon(b_{k-1})$ are differences of the argument, and

$$d_{\mathrm{cell}}(\varepsilon) = \sum_k |\mu_\varepsilon(C_k) - 1|$$

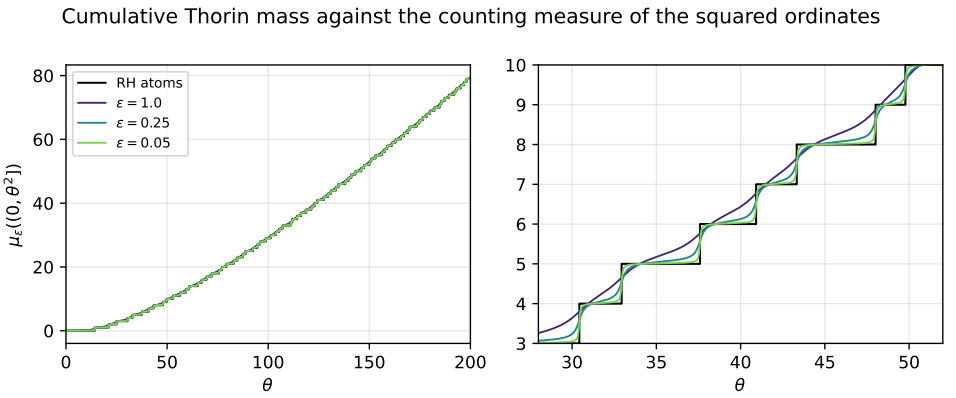


Figure 3.1: Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of U_{RH}, on the whole window (left) and on $28 \le \theta \le 52$ (right).

is the total variation distance between the images of μ_ε and U_{RH} under the map sending each cell to its atom. No b_k is an ordinate, so Proposition 3.1.2 with Remark 3.1.3 applies at both ends of every cell, with $\Xi'(0) = 0$ at b_0, and gives $\mu_\varepsilon(C_k) - 1 = -(\varepsilon/\pi)\big[\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\big] + O(\varepsilon^3)$. Hence $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$, where the cell constant $\kappa_{\mathrm{cell}} = \pi^{-1}\sum_k |\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})|$ of this window is 5.43665; every cell mass lies in $[0.99888, 1.00126]$ at $\varepsilon = 0.005$. Section 8.3 reads this constant as a statistic of the irregularity of the spacings.

The L^1 distance $\int_0^\Theta |M_\varepsilon(\theta) - N(\theta)|\,d\theta$ between the cumulative mass and the counting function is a transport distance, and because of the Cauchy tails it behaves like $\varepsilon\log(1/\varepsilon)$: it is 1.55 in θ and 360 in z at $\varepsilon = 0.005$, and Section 3.3.1 derives the rate. The heat trace $W(t) = \int e^{-tz}U_{\mathrm{RH}}(dz)$ is approximated by $W_\varepsilon(t) = \int_0^\infty e^{-t\theta^2}\rho_\varepsilon(\theta)\,d\theta = \int e^{-tz}\mu_\varepsilon(dz)$. At $t = 0.01$ the zeros give $W = 0.1496980112538$ (Table 2.1), while $W_\varepsilon(0.01) = 0.1504500280599$, 0.1512017825455 and 0.1527045050533 at $\varepsilon = 0.005$, 0.01 and 0.02, again with error linear in ε.

The mass and the heat trace are smooth in ε at $\varepsilon = 0$, so Richardson extrapolation applies to them: the mass by Proposition 3.1.2, and the heat trace because by Proposition 2.1.3 it is a sum over the zeros of Poisson integrals of the Gaussian $e^{-t\theta^2}$. The combination $2\mu_{0.005}(E) - \mu_{0.01}(E)$ equals $79 + 3.2\times10^{-8}$, against a raw defect of 2.0×10^{-3}. For the heat trace the error 7.5×10^{-4} of $W_{0.005}(0.01)$ falls to 2.6×10^{-7} after one Richardson step ($\varepsilon = 0.005$ and 0.01) and to 1.7×10^{-10} after two (adding $\varepsilon = 0.02$). Without extrapolation the same accuracy would need $\varepsilon \approx 8\times10^{-8}$ for the mass, and $\varepsilon \approx 2\times10^{-6}$ and 10^{-9} for the heat trace.

3.2 What an off line zero would do

By Proposition 2.1.3 an off line zero at distance δ from the line enters the Thorin density, together with its reflection in the line, only through the kernels $P_{\varepsilon+\delta}+P_{\varepsilon-\delta}$ at its ordinate, the second of which is negative once $\varepsilon < \delta$. So the effect on each distance of Section 3.1 can be computed in closed form. The three signatures of Proposition 3.2.1 show that an off line zero is most visible at resolutions comparable to its distance from the line, a theme that returns in Section 1.4.

3.2.1 Three signatures of an off line zero

Off line zeros come in pairs $\frac{1}{2} \pm \delta + i\gamma$ with $0 < \delta < \frac{1}{2}$, together with their conjugates, and near γ a pair contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ to ρ_ε where two line zeros would contribute $2P_\varepsilon$.

Proposition 3.2.1. *Let $\varepsilon > 0$ and $0 < \delta < \frac{1}{2}$; when $\delta = \varepsilon$, P_0 denotes the unit point mass at the origin.*

(i) (Displaced zeros, same count.) If two line zeros at height γ are replaced by the quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, the density near γ changes by $2g$, where $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$ is the change per displaced zero, and

$$\|g\|_{\mathrm{TV}} = \frac{3\sqrt{3}}{4\pi}\Big(\frac{\delta}{\varepsilon}\Big)^2 + O\Big(\frac{\delta^4}{\varepsilon^4}\Big), \qquad \delta < \varepsilon.$$

(ii) (Pair omitted from the list.) If a pair at $\frac{1}{2} \pm \delta + i\gamma$ exists but a list of line zeros omits it, the difference is $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ and

$$\|P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\|_{\mathrm{TV}} = \begin{cases} 2, & \delta \le \varepsilon, \\ \dfrac{8}{\pi} \arctan \sqrt{\dfrac{\delta+\varepsilon}{\delta-\varepsilon}} - 2, & \delta > \varepsilon. \end{cases}$$

(iii) (Negative mass.) For $\delta > \varepsilon$ the negative part of $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$, the density of one displaced zero, has mass

$$\frac{2}{\pi} \arctan \sqrt{\frac{\delta+\varepsilon}{\delta-\varepsilon}} - \frac{1}{2} = \frac{\varepsilon}{\pi\delta} + O\Big(\frac{\varepsilon^3}{\delta^3}\Big).$$

Proof. (i) Taylor expansion in δ gives $g = \frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$ [235, Proposition 19.7], with a remainder of L^1 norm $O(\delta^4/\varepsilon^4)$. Since $\partial_a^2 P_a(x) = (2a/\pi)(a^2 - 3x^2)/(a^2+x^2)^3$, the substitution $x = \varepsilon v$ gives $\partial_a^2 P_a|_{a=\varepsilon}\, dx = 2(1-3v^2)/(\pi\varepsilon^2(1+v^2)^3)\, dv$. The function $(1-3v^2)/(1+v^2)^3$ is the derivative of $v/(1+v^2)^2$, whose total variation is four times its maximum $9/(16\sqrt{3})$, that is $3\sqrt{3}/4$. So the L^1 norm of $\partial_a^2 P_a$ at $a = \varepsilon$ is $\frac{3\sqrt{3}}{2\pi\varepsilon^2}$, and multiplying by $\delta^2/2$ gives (i). (ii) For $\delta \le \varepsilon$

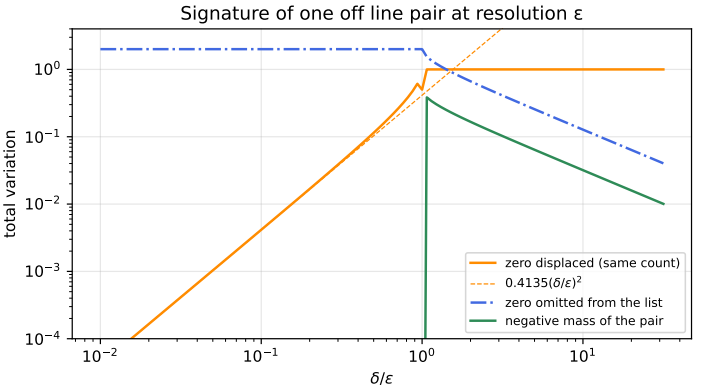


Figure 3.2: The three signatures of Proposition 3.2.1 as functions of δ/ε: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1.

both kernels are nonnegative with mass one. For $\delta > \varepsilon$ put $a = \delta + \varepsilon$, $b = \delta - \varepsilon$, so the difference is $P_a - P_b$, which is negative exactly on $|x| < \sqrt{ab}$. Integrating, $\|P_a - P_b\|_{\mathrm{TV}} = \frac{4}{\pi}[\arctan\sqrt{a/b} - \arctan\sqrt{b/a}]$, and $\arctan y + \arctan(1/y) = \pi/2$. (iii) $P_a - P_b$ has total mass zero, so its negative part is half its norm; halve again for the factor $\frac{1}{2}$. □

With $\varepsilon/\delta = \sin\phi$ one has $\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} = \pi/4 + \phi/2$, so for $\delta > \varepsilon$ the norm in (ii) is $(4/\pi)\arcsin(\varepsilon/\delta)$ and the mass in (iii) is $(1/\pi)\arcsin(\varepsilon/\delta)$. Figure 3.2 plots the three signatures.

3.2.2 Calibration of the matched distance

Proposition 3.2.1 converts the matched distance of Section 3.1 into detection thresholds. By part (ii), a computed matched distance $\tilde{d}_{\mathrm{match}}(\varepsilon)$ of at most $3.7 \times 10^{-9}\,\varepsilon$ excludes any omitted pair with ordinate in $(0, 200)$: such a pair contributes total variation $(4/\pi)\arcsin(\varepsilon/\delta)$, or 2 if $\delta \le \varepsilon$, less only the part of its kernels outside the window. By part (i), a displaced pair that keeps the count adds $2g$ to a difference whose norm is the floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, so by the triangle inequality the matched distance rises above the floor once $\|g\|_{\mathrm{TV}} > \tilde{d}_{\mathrm{match}}(\varepsilon)$, that is, to leading order, once δ exceeds

$$\delta_{\min}(\varepsilon) = \varepsilon\Big(\frac{4\pi\,\tilde{d}_{\mathrm{match}}(\varepsilon)}{3\sqrt{3}}\Big)^{1/2}. \tag{3.2.1}$$

With the floors of Table 3.1 this threshold is 3.3×10^{-5} at $\varepsilon = 0.5$, 1.1×10^{-6} at $\varepsilon = 0.05$ and 3.3×10^{-8} at $\varepsilon = 0.005$.

The threshold (3.2.1) is the square root law of [235, Proposition 19.7]: the signature grows like the square of the displacement, so the smallest detectable displacement falls like the square root of the achievable distance. The prime side calibration of [235, Section 19.3.6] obeys the same law and reaches $\delta \approx 0.05$ at $\varepsilon = 2$. What changes is the reach: the Pólya route evaluates ρ_ε at small ε to about 60 digits below height 200 (Section 2.1.1), so the floor is set by the tail model for the zero list (Section 3.1.1), not by the evaluator, whereas the prime side is limited by the truncation of Table 2.2. A positivity test sees less than the matched distance: only the negative part of the whole density, which by part (iii) is at most of order ε/δ per displaced zero and which the kernels of the neighbouring zeros can cancel entirely (Section 1.4).

These thresholds concern zeros below height 200 and calibrate the method above the verified window.

3.3 Counting zeros with the Thorin measure

Turing's method certifies the count $N(T)$ without evaluating ζ off the critical line, from the sign changes of Z and an explicit bound on the integral of S, where $\pi S(t) = \arg \zeta(\frac{1}{2} + it)$; Backlund's method, which it replaced, bounds the argument of ζ along a horizontal segment off the line (Section 2.2.1). Without the assumption of Proposition 3.1.2 that the zeros below the window lie on the line, the argument principle gives at every resolution an exact identity for $N(T)$ and for the number $\nu(\frac{1}{2}+\varepsilon, T)$ of zeros to the right of the line of evaluation, so that off line zeros are counted and not only detected. At $\varepsilon = \frac{1}{2}$ the identity is Backlund's count with the vertical side on $\operatorname{Re} s = 1$.

3.3.1 The cumulative Thorin mass as a smoothed counting function

Recall that $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2]) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\theta)$, with the argument continued along the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ from $\theta = 0$, where $\xi > 0$ (Sections 1.3.3 and 3.1). Since $\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the mass splits as

$$M_\varepsilon(\theta) = N_\varepsilon^{\mathrm{sm}}(\theta) + S_\varepsilon(\theta), \qquad S_\varepsilon(\theta) = \frac{1}{\pi} \arg \zeta\Big(\frac{1}{2} + \varepsilon + i\theta\Big),$$

where, with $s = \frac{1}{2} + \varepsilon + i\theta$ and $\theta > 0$, the smooth part $N_\varepsilon^{\mathrm{sm}}(\theta) = \pi^{-1}\big[\arg s + \arg(s-1) + \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{\varepsilon}{2} + \frac{i\theta}{2}) - \frac{\theta}{2} \log \pi\big]$ comes from $s(s-1)\pi^{-s/2}\Gamma(s/2)$, with the principal arguments of s and $s-1$, which lie in $(0, \pi)$, and the branch of $\log \Gamma$ that is real on the positive axis. Thus S_ε is $\pi^{-1} \arg \zeta$ continued along the vertical line from $S_\varepsilon(0+) = -1$ if $0 < \varepsilon < \frac{1}{2}$, when $\arg(s-1) \to \pi$ as $\theta \downarrow 0$, and from $S_\varepsilon(0+) = 0$ if $\varepsilon > \frac{1}{2}$.

As $\varepsilon \downarrow 0$, $N_\varepsilon^{\mathrm{sm}} \to \vartheta/\pi + 1$, where $\vartheta(\theta) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{i\theta}{2}) - \frac{\theta}{2} \log \pi$ is the Riemann Siegel theta function, and for θ not an ordinate Proposition 3.3.2 below gives $M_\varepsilon(\theta) \to N(\theta) - 2\nu(\frac{1}{2}, \theta)$, where $\nu(\sigma, \theta)$ counts the zeros with real part greater than σ and ordinate in $(0, \theta]$ (Section 1.1). When no zero below height θ lies to the right of the critical line, the limit is the Riemann von Mangoldt formula $N(\theta) = \vartheta(\theta)/\pi + 1 + S(\theta)$ [264, Section 9.3].

Integrating Proposition 2.1.3 in θ writes M_ε as a sum of Cauchy steps. Under RH, for $\theta \geq 0$,

$$M_\varepsilon(\theta) = \sum_{\gamma > 0} \big[F_\varepsilon(\theta - \gamma) - F_\varepsilon(-\gamma) + F_\varepsilon(\theta + \gamma) - F_\varepsilon(\gamma) \big],$$

$$F_\varepsilon(x) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{x}{\varepsilon},$$

with the sum over all positive ordinates. Since $F_\varepsilon(-\gamma) + F_\varepsilon(\gamma) = 1$, this says that $M_\varepsilon = P_\varepsilon * \tilde{N}$, where $\tilde{N}$ is the odd extension of N. As ϑ is odd, S_ε is then the Poisson average of the odd extension of S up to an explicit smooth function. If only the zeros below height $T + B$ are known to lie on the line, the display holds on $[0, T]$ up to an error of order $\varepsilon\delta^2/B^3$ for each off line pair above $T + B$ at distance δ from the line, counted as a double ordinate, because $P_{\varepsilon+\delta}(x) + P_{\varepsilon-\delta}(x) - 2P_\varepsilon(x) = -6\varepsilon\delta^2/(\pi x^4) + O(x^{-6})$.

For a single step,

$$\begin{aligned} \int_{-L}^{L} \big| F_\varepsilon(x) - \mathbf{1}_{x \geq 0} \big| \, dx &= \frac{2}{\pi} \Big[L \arctan \frac{\varepsilon}{L} + \frac{\varepsilon}{2} \log \Big(1 + \frac{L^2}{\varepsilon^2} \Big) \Big] \\ &= \frac{2\varepsilon}{\pi} \Big(\log \frac{L}{\varepsilon} + 1 \Big) + O\Big(\frac{\varepsilon^3}{L^2} \Big), \end{aligned}$$

which explains the $\varepsilon \log(1/\varepsilon)$ rate of the L^1 distance in Table 3.1. Taking half of it on each side of each of the 79 zeros below height 200, with L the half gap on that side, gives 1.61 at $\varepsilon = 0.005$, against the computed 1.55; the difference comes from the tails of the other steps.

3.3.2 A Littlewood identity and the zero density

Proposition 3.3.1. *Let $T > 0$. For every $\varepsilon > 0$ such that ξ has no zero on the segment from $\frac{1}{2} + \varepsilon$ to $\frac{1}{2} + \varepsilon + iT$,*

$$\pi M_\varepsilon(T) = \frac{\partial}{\partial \varepsilon} \int_0^T \log \Big| \xi\Big(\frac{1}{2} + \varepsilon + it \Big) \Big| \, dt,$$

and for every $\varepsilon > 0$

$$\int_0^T \log \Big| \xi\Big(\frac{1}{2} + \varepsilon + it \Big) \Big| \, dt = \int_0^T \log \Big| \xi\Big(\frac{1}{2} + it \Big) \Big| \, dt + \pi \int_0^\varepsilon M_{\varepsilon'}(T) \, d\varepsilon'.$$

Proof. Since $\partial_\sigma \log|\xi(\sigma+it)| = \mathrm{Re}(\xi'/\xi)(\sigma+it) = v_\sigma(t)$ and $v_\sigma(t) = \partial_t \arg\xi(\sigma+it)$, we have $M_\varepsilon(T) = \pi^{-1}\int_0^T v_{1/2+\varepsilon}(t)\,dt$, and when no zero lies on the segment, differentiation under the integral gives the first identity. The integral $I(\varepsilon)$ on the left of the second is continuous on $[0,\infty)$, because the logarithmic singularities at the zeros are integrable uniformly in ε, and continuously differentiable except at the finitely many ε' at which the segment meets a zero. There I' jumps by 2π for each zero met at an interior point of the segment, since $\int_{-c}^{c} a\,du/(a^2+u^2) \to \pm\pi$ as $a \to 0\pm$. So I is absolutely continuous, and $I(\varepsilon) - I(0) = \int_0^\varepsilon I'(\varepsilon')\,d\varepsilon'$ is the second identity. □

The jump of 2π in the slope $\pi M_\varepsilon(T)$ when the line crosses a zero is the term 2ν of Proposition 3.3.2 below. Littlewood's lemma [264, Section 9.9] is the same Cauchy Riemann identity with the argument taken along a path through $\mathrm{Re}\,s = 2$; the two arguments differ by $2\pi\nu(\frac12+\varepsilon,T)$, and Exercise 3.14 derives the lemma from the two propositions. Let $\tau_\varepsilon(T) = \pi^{-1}\Delta\arg\xi$ along the horizontal segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$.

PROPOSITION 3.3.2. *Let $\varepsilon > 0$ and $T > 0$, with T not an ordinate and no zero of ξ on the segment from $\frac12+\varepsilon$ to $\frac12+\varepsilon+iT$. Then*

$$M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\,\nu\Big(\frac12+\varepsilon, T\Big).$$

The segment condition holds at $\varepsilon = \frac12$, and $\nu(1,T) = 0$, which gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, so that $\nu(\frac12+\varepsilon,T) = \frac12\big[N(T) - M_\varepsilon(T) - \tau_\varepsilon(T)\big]$ for every ε satisfying the hypotheses.

Proof. Let $\sigma_1 > 1$. Write Δ_{right} for the change of $\arg\xi$ along the segment from σ_1 to σ_1+iT, and $\Delta_{\mathrm{top}}(a \to b)$ for its change along the segment from $a+iT$ to $b+iT$. The argument principle on the rectangle $[\frac12+\varepsilon,\sigma_1]\times[0,T]$, on whose bottom edge ξ is positive, gives $\Delta_{\mathrm{right}} + \Delta_{\mathrm{top}}(\sigma_1 \to \frac12+\varepsilon) - \pi M_\varepsilon(T) = 2\pi\nu(\frac12+\varepsilon,T)$. Subtracting this from the classical formula $\pi N(T) = \Delta_{\mathrm{right}} + \Delta_{\mathrm{top}}(\sigma_1 \to \frac12)$ [264, Section 9.3] leaves $\Delta_{\mathrm{top}}(\frac12+\varepsilon \to \frac12) + \pi M_\varepsilon(T) = \pi N(T) - 2\pi\nu(\frac12+\varepsilon,T)$, and $\Delta_{\mathrm{top}}(\frac12+\varepsilon \to \frac12) = \pi\tau_\varepsilon(T)$. At $\varepsilon = \frac12$ the segment lies on $\mathrm{Re}\,s = 1$, where ζ has no zeros, and $\nu(1,T) = 0$ because the Euler product has no zeros in $\mathrm{Re}\,s > 1$. □

Half the difference of $M_\varepsilon(T) + \tau_\varepsilon(T)$ at two resolutions counts the zeros in the strip between the two lines. The factor two comes from the functional equation: a zero $\frac12+\delta+i\gamma$ with $\delta > \varepsilon$ and its partner $\frac12-\delta+i\gamma$ add 2 to $N(T)$ and 1 to $\nu(\frac12+\varepsilon,T)$, hence nothing to $M_\varepsilon(T)+\tau_\varepsilon(T)$, whereas two zeros on the line at the same ordinate add 2. In the density the pair enters through the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of Proposition 2.1.3, whose masses cancel, as in the proof of Proposition 3.2.1(iii) (Exercise 3.12).

Zero density estimates such as Ingham's [176] bound the number $N(\sigma,T)$ of zeros with real part at least σ and ordinate in $(0,T]$, which is at least $\nu(\sigma,T)$, by

$T^{e(\sigma)+o(1)}$, with an exponent $e(\sigma) < 1$ for every $\sigma > \frac{1}{2}$. Section 14.3 turns these bounds, with no hypothesis, into a bound on the mass of the Cauchy kernels of the zeros to the right of the basepoint σ, the negative part of the split (14.3.1) (Theorem 14.3.2). Proposition 3.3.2 evaluates ν itself, as an integer, but only up to a finite height.

3.3.3 The count and the zero density below height 200

The right side of the identity is an integer, so a computation of the left side with error below one half determines it, provided no winding of the argument is missed. We follow the argument on a grid, summing the principal values of the arguments of quotients of successive values of ξ, with steps $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) up the vertical side from $\theta = 0$ and 4000 steps along the horizontal segment. A certificate would need, besides ball arithmetic at the grid points, a bound on the change of the argument between them. We do not supply that bound, and the conclusions below are computed, not certified.

Table 3.2 gives the identity at $T = 200$, evaluated by the Pólya route. Every row returns 79 to within 6×10^{-13}. The row $\varepsilon = \frac{1}{2}$ gives $N(200) = 79$, and with the 79 sign changes of Ξ on $(0, 200)$ (Section 2.1.2) this places every zero below height 200 on the line. Independently of the sign changes, the rows $\varepsilon < \frac{1}{2}$ give $\nu(\frac{1}{2} + \varepsilon, 200) = 0$ for every $\varepsilon \geq 0.005$, since ν is nonincreasing in σ, so that by the functional equation no zero below height 200 lies farther than 0.005 from the line.

ε	$M_\varepsilon(200)$	$\tau_\varepsilon(200)$	$M_\varepsilon + \tau_\varepsilon = N - 2\nu$
1	79.36737810665	−0.36737810665	79.00000000000003
0.5	79.19416875933	−0.19416875933	78.99999999999993
0.25	79.09886991193	−0.09886991193	79.00000000000024
0.1	79.03976768776	−0.03976768776	79.00000000000006
0.05	79.01989993142	−0.01989993142	78.99999999999967
0.02	79.00796178162	−0.00796178162	78.99999999999970
0.01	79.00398102016	−0.00398102016	78.99999999999993
0.005	79.00199052625	−0.00199052625	78.99999999999947

Table 3.2: Proposition 3.3.2 at $T = 200$ by the Pólya route (computed, not certified).

The identity also gives the mass defect of Proposition 3.1.2 exactly. If T is not an ordinate and every zero with ordinate in $(0, T]$ lies on the line, then $\nu(\frac{1}{2} + \varepsilon, T) = 0$ and $M_\varepsilon(T) = N(T) - \tau_\varepsilon(T)$ for every $\varepsilon > 0$. With that proposition and Remark 3.1.3 this gives $\tau_\varepsilon(T) = (\varepsilon/\pi)\,\Xi'(T)/\Xi(T) + O(\varepsilon^3)$. At $T = 200$ the first order term is $-0.3981063\,\varepsilon$, minus the mass defect found in Section 3.1.2, and for $\varepsilon \leq 0.1$ the column $\tau_\varepsilon(200)$ of Table 3.2 exceeds it by about $0.043\,\varepsilon^3$.

3.4 Exercises

On Section 3.1.

3.1. Prove Proposition 3.1.1 for a positive measure ν on a general measurable space that does not charge the atoms of U.

3.2. Show that the total variation distance between the unit point masses at a and b is 2 for $a \neq b$, and deduce that total variation cannot compare two atomic measures with slightly displaced atoms.

3.3. Compute the L^1 distance between the cumulative functions of a Cauchy kernel of width ε and a unit step on $[-L, L]$, derive the $\varepsilon \log(1/\varepsilon)$ rate, and compare with the formula for a single step in Section 3.3.1.

3.4. Derive the first order formula of Proposition 3.1.2 for the sinh clock of Section 4.1, where it reads $-(\varepsilon/\pi) f'/f$ with $f(\theta) = \sin(\pi\theta)/(\pi\theta)$, and compare it at $\Theta = m + \frac{1}{2}$ with the exact mass $m + \pi^{-1} \arctan(\varepsilon/\Theta)$ of Section 4.1.4; explain why the difference is of order ε^3.

3.5. Explain why the two point Richardson combination removes the linear term in ε and the three point combination the quadratic term, and reproduce the heat trace errors 2.6×10^{-7} and 1.7×10^{-10} from the values of $W_\varepsilon(0.01)$ given in Section 3.1.2.

On Section 3.2.

3.6. Verify the expansion $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon = \frac{\delta^2}{2} \partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$, show that the remainder has L^1 norm $O(\delta^4/\varepsilon^4)$, and compute the L^1 norm of $\partial_a^2 P_a$ at $a = \varepsilon$.

3.7. Show that for $\delta > \varepsilon$ the kernel $P_{\varepsilon+\delta} - P_{\delta-\varepsilon}$ is negative exactly on $|x| < \sqrt{\delta^2 - \varepsilon^2}$.

3.8. Compute the depth $2\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ of the negative dip of an off line pair at its centre, that is $\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ per displaced zero, compare it with the contribution $\varepsilon/(\pi(\varepsilon^2 + \delta_j^2)) \approx \varepsilon/(\pi\delta_j^2)$ of a line zero at distance δ_j, and derive the local visibility condition of Section 1.4.

3.9. Given the matched floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, derive the smallest displacement δ of a pair that keeps the count and would be detected, and evaluate it at $\varepsilon = 0.05$ using Table 3.1.

On Section 3.3.

3.10. Show that $I(\varepsilon) = \int_0^T \log|\xi(\frac{1}{2}+\varepsilon+it)|\,dt$ is continuous for $\varepsilon \geq 0$, that its derivative jumps by 2π when the line $\operatorname{Re} s = \frac{1}{2}+\varepsilon$ crosses a zero, and deduce the second identity of Proposition 3.3.1.

3.11. Prove Proposition 3.3.2 in detail, paying attention to the orientation of each side of the rectangle.

3.12. Show that the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of an off line pair at distance $\delta > \varepsilon$ from the line have total mass zero, and compute to leading order the mass that they and their mirror images at $-\gamma$ put on a window $(0, T]$ containing the ordinate γ. Explain how the factor two in Proposition 3.3.2 accounts for the pair.

3.13. Using the first order formula for $\tau_\varepsilon(T)$ in Section 3.3.3, predict $\tau_{0.05}(200)$ from $\Xi'(200)/\Xi(200) = -1.2506879$ and compare with Table 3.2; estimate the coefficient of the ε^3 term from the rows $\varepsilon = 0.01$ and 0.02.

3.14. Show that Littlewood's lemma follows from Proposition 3.3.1 and Proposition 3.3.2 by integrating in ε.

Chapter 4

The de Bruijn Newman flow computed from the Pólya density

4.1 The de Bruijn Newman flow from the Pólya density

The de Bruijn Newman flow reads RH from a deformation of the Pólya kernel: multiplying the kernel by a Gaussian factor turns Ξ into a family of entire functions H_t that evolve by the backward heat equation, and RH is the statement that no two zeros collide at positive time. The Pólya route computes H_t at the cost of Ξ. This chapter follows zeros backward to their collisions, reads the collisions through the Hankel test, brackets the collision times by Laguerre's quantity and Biane's theorem, compares them with the lower bounds for Λ of Csordas, Smith and Varga, and then turns to the tilted clock, the Laguerre expressions along the flow and the forward persistence of real zeros.

4.1.1 The flow, the motion of its zeros and its Thorin form

In the normalisation of Rodgers and Tao [250], which Theorem 17.1.16 also uses,

$$H_t(z) = \int_0^\infty e^{tu^2}\,\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du,$$
$$\Phi_{\mathrm{dBN}}(u) = \sum_{n\geq 1}\left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right)\exp(-\pi n^2 e^{4u}),$$

so that $H_0(z) = \frac{1}{8}\Xi(z/2)$. If $v = 2u$ denotes the variable of (2.1.3), then $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(v)$ and

$$8H_t(2\theta) = \int_{\mathbb{R}} \Phi(v)\,e^{tv^2/4}\cos(\theta v)\,dv,$$

so the flow multiplies the kernel of (2.1.3) by $e^{tv^2/4}$, and a zero z of H_t corresponds to the ordinate $\theta = z/2$.

De Bruijn showed that H_t has only real zeros for $t \geq \frac{1}{2}$ and that the property persists forward in time [61]. Newman showed that there is a constant $\Lambda > -\infty$,

at most $\frac{1}{2}$ by de Bruijn's theorem, such that H_t has only real zeros exactly for $t \geq \Lambda$, and conjectured $\Lambda \geq 0$ [215], which Rodgers and Tao proved [250]. Ki, Kim and Lee sharpened de Bruijn's bound to $\Lambda < \frac{1}{2}$ [182], the Polymath project proved $\Lambda \leq 0.22$ [247], and Platt and Trudgian improved this to $\Lambda \leq 0.2$ [232]. De Bruijn's bound comes from a theorem about strips: if the zeros of H_0 lie in $|\operatorname{Im} z| \leq \Delta$, then H_t has only real zeros for $t \geq \Delta^2/2$ [61, Theorem 13], and the critical strip gives $\Delta = 1$. A zero free half plane $\operatorname{Re} s > \alpha$ gives $\Delta = 2\alpha - 1$, hence $\Lambda \leq 2(\alpha - \frac{1}{2})^2$, which improves 0.2 only if $\alpha < \frac{1}{2} + \sqrt{0.1} \approx 0.816$. RH is the inequality $\Lambda \leq 0$, so by the theorem of Rodgers and Tao it is equivalent to $\Lambda = 0$.

While the zeros are simple they move according to

$$\frac{dz_k}{dt} = 2 \sum_{j \neq k} \frac{1}{z_k - z_j},$$

the sum running over all other zeros, positive and negative, with z_j and $-z_j$ taken together. Differentiating $H_t(z_k(t)) = 0$ in t and using the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ gives $dz_k/dt = \partial_z^2 H_t / \partial_z H_t$ at z_k, and the Hadamard product of the even function H_t turns this into the display (Exercise 18.15). The equation is the gradient flow of unit charges on the line with logarithmic interaction, so forward in time the zeros repel each other and spread. Backward in time neighbouring zeros approach each other, and a close pair collides and then leaves the real axis as a conjugate pair. A pair at distance D that is isolated from the other zeros obeys $dD/dt \approx 4/D$, so $D^2 \approx D_0^2 + 8t$ with $D_0 = D(0)$, and the pair collides at $t \approx -D_0^2/8$. RH is the statement that no two zeros collide at a positive time. Numerically the flow makes the first zeros more regular: unfolding the first 120 zeros of H_t by $\vartheta(z/2)/\pi$, the variance of the unfolded gaps is 0.1240, 0.1165, 0.1096 and 0.0975 at $t = 0$, 0.05, 0.1 and 0.2, and the smallest gap is 0.3868, 0.4092, 0.4300 and 0.4678, always between the 34th and 35th zeros (computed; [143]). This is an observation about these zeros and not a consequence of the repulsion: for the uniform law on $[-1, 1]$, whose transform $\sin\theta/\theta$ has the equally spaced zeros $k\pi$, the first order velocity of Remark 19.1.3 at the kth zero is $-2/(k\pi)$, so gaps that start equal become unequal.

At $t = 0$ the velocity has a closed form through Hardy's function Z of Section 2.2 [144].

Proposition 4.1.1. *Let γ_n be the nth positive zero of Z, and suppose it is simple, so that $z_n = 2\gamma_n$ is a simple zero of H_0 and moves smoothly with t near $t = 0$. Then*

$$\dot{z}_n(0) = \frac{1}{2}\Big[\frac{Z''(\gamma_n)}{Z'(\gamma_n)} + \frac{4\gamma_n}{\gamma_n^2 + \frac{1}{4}} - \operatorname{Im}\psi\big(\tfrac{1}{4} + \tfrac{i\gamma_n}{2}\big)\Big],$$

and the last two terms tend to $-\frac{\pi}{2}$ as $\gamma_n \to \infty$, a drift of $-\frac{\pi}{4}$ in $\dot{z}_n$.

Proof. Since $H_0(z) = \frac{1}{8}\Xi(z/2)$, the formula $dz_k/dt = \partial_z^2 H_t / \partial_z H_t$ gives $\dot{z}_n(0) = \Xi''(\gamma_n)/(2\Xi'(\gamma_n))$. Write $\Xi = gZ$ with $g(\theta) = -\frac{1}{2}(\theta^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\theta}{2})|$, which

is $\zeta(\frac12+i\theta)=e^{-i\vartheta(\theta)}Z(\theta)$ inserted into the definition of ξ. At a zero of Z, $\Xi'=gZ'$ and $\Xi''=2g'Z'+gZ''$, so $\Xi''/\Xi'=Z''/Z'+2g'/g$, and $2g'/g=4\theta/(\theta^2+\frac14)-\operatorname{Im}\psi(\frac14+\frac{i\theta}{2})$ because $\frac{d}{d\theta}\log|\Gamma(\frac14+\frac{i\theta}{2})|=-\frac12\operatorname{Im}\psi(\frac14+\frac{i\theta}{2})$. Finally $\psi(z)=\log z+O(1/z)$ as $z\to\infty$ in $\operatorname{Re}z>0$ [226, Section 5.11], so $\operatorname{Im}\psi(\frac14+\frac{i\theta}{2})\to\frac{\pi}{2}$. □

The rates of Section 4.1.8 are $\dot a_j(0)=\gamma_j\dot z_j(0)$, and the proposition reproduces $\dot a_1(0)$ there. The first 34 zeros move down at $t=0$, and the 35th, at $\gamma_{35}=111.874659177$, moves up, with $\dot z_{35}(0)=+0.0595012982$; all 35 signs are certified in ball arithmetic. Among the first 1500 zeros 286 move up, the velocities average -0.78280 against the drift $-\frac{\pi}{4}=-0.7854$, and the largest upward and downward velocities, $+9.16075$ and -11.21487, belong to the zeros $n=1497$ and $n=1496$, at distance 0.0975025 in γ (computed). In the equation of motion a zero at distance Δ below z_n adds $2/\Delta$ to $\dot z_n$, so the upper zero of a close pair is pushed up, the repulsion that Csordas, Smith and Varga [51] turned into lower bounds for Λ (Section 4.1.6); whether a given zero rises depends on all the others, not on one gap.

Let $R_t(u)=H_t(0)/H_t(2i\sqrt{u})$, where u is the variable of the clock (written s in Chapters 10 and 11), not the variable of integration in H_t, so that R_0 is the clock $1/G$. By Theorem 17.1.16, for every $t\ge\Lambda$, and so for every $t\ge0.2$, R_t is the Laplace transform of a GGC whose Thorin atoms move with the zeros of H_t. Put $\xi_t(s)=8H_t\big(-2i(s-\frac12)\big)$, so that $\xi_t(\frac12+i\theta)=8H_t(2\theta)$ and $\xi_0=\xi$, and

$$\rho_{\varepsilon,t}(\theta)=\pi^{-1}\operatorname{Re}(\xi_t'/\xi_t)\big(\tfrac12+\varepsilon+i\theta\big).$$

Each zero z of H_t gives the zero $\frac12+iz/2$ of ξ_t, and for every t the function $\rho_{\varepsilon,t}$ is the sum of the Poisson kernels $P_{\frac12+\varepsilon-\beta}(\theta-\gamma)$ over the zeros $\beta+i\gamma$ of ξ_t. For $t\ge\Lambda$ every β equals $\frac12$ and $\rho_{\varepsilon,t}$ is the Thorin density of R_t at resolution ε. As t decreases through a collision, $\rho_{\varepsilon,t}$ shows the off line signature of Section 3.2: two Poisson peaks merge, and after the collision the pair contributes the kernels $P_{\varepsilon+d(t)}+P_{\varepsilon-d(t)}$ at its ordinate, where $d(t)$, half the imaginary part of the pair in z, grows from zero; the second kernel is negative once $d(t)>\varepsilon$. The flow thus produces off line configurations of known geometry on which the diagnostics of Chapters 3 to 8 can be calibrated, an open computation listed in Section 25.2.3.

4.1.2 Two collisions computed from the Pólya kernel

The Pólya route evaluates H_t at any t by multiplying the node values $\Phi(v_j)$ by $e^{tv_j^2/4}$ and summing; the factor is entire, so the error analysis of Section 2.1 carries over. Among the 79 zeros below height 200 the closest pair is $\gamma_{71}=184.8745$, $\gamma_{72}=185.5988$, at distance 0.7243 in θ, that is $D_0=1.4486$ in z. We follow the pair by the sign changes of $H_t(2\theta)$ on a grid and refine each zero by Newton's method (Table 4.1).

t	zeros in $\theta = z/2$	sign changes
0.2	184.676, 185.590	2
0.1	184.771, 185.598	2
0.0	184.874, 185.599	2
−0.05	184.931, 185.595	2
−0.1	184.992, 185.587	2
−0.15	185.059, 185.573	2
−0.2	185.136, 185.550	2
−0.3	none	0

Table 4.1: The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; t is the de Bruijn Newman time.

Bisection on t for the disappearance of the two sign changes gives the collision time

$$t_* = -0.2880,$$

with a grid error of at most 6.1×10^{-5}. Near a collision the term $4/D$ dominates the motion of the pair, so $D^2 \approx 8(t - t_*)$ and the gap in θ is about $\sqrt{2(t - t_*)}$; on a grid of step h the sign changes therefore vanish at a time at most $h^2/2$ above t_*. The isolated pair estimate is $-D_0^2/8 = -0.262$: the neighbouring zeros, which the estimate ignores, delay the collision. Since nonreal zeros persist backward in time, H_t has nonreal zeros for every t below the collision time, so $\Lambda \geq t_* \approx -0.288$.

Lehmer's pair near height 7005, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ (Section 2.2.3.1), is much closer: $D_0 = 2 \times 0.0377 = 0.0754$, and its isolated pair estimate is $-D_0^2/8 = -7.1 \times 10^{-4}$. The tilted Pólya route evaluates H_t at that height at the same cost as Ξ. With the parameters of Section 2.2.3.1 we track the two sign changes of the pair on a grid of step 0.0025 in θ over $[7005.03, 7005.13]$. Both sign changes are present at $t = 0$ and absent at $t = -0.004$, and twelve bisection steps on this interval give -7.10×10^{-4}, which by the argument above lies above the collision time by at most $0.0025^2/2 \approx 3 \times 10^{-6}$. Newton's method for the double zero, $H_t(2\theta) = \partial_\theta H_t(2\theta) = 0$ in the unknowns θ and t, started from this value, gives

$$t_* = -7.11 \times 10^{-4}$$

at $\theta = 7005.0817$. For a pair this close the neighbours hardly matter and the isolated pair estimate is accurate; the bound $\Lambda \geq t_*$ is much sharper than the one from the pair below height 200.

Pólya's function $K_{i\theta/2}(2\pi)$ behaves differently under the same flow. Down to $t = -60$ none of its real zeros below height 90 collide, and as t decreases from 0 to -0.5 they rise almost as a rigid block, at speeds within 2% of the mean speed $\pi/8$ that the two functions share, while the zeros of Ξ rise at speeds between 0.004

and 0.735; yet by Newman's theorem the de Bruijn Newman constant of Pólya's function is finite (Section 18.1.6).

4.1.3 The Hankel test along the flow

The Stieltjes moment test of Section 8.1 (Table 8.2) needs only the Taylor coefficients of Ξ at the centre, and along the flow these are moments of the reweighted kernel [239]. Putting $\theta = i\sqrt{u}$ in the formula for $8H_t(2\theta)$ of Section 4.1.1 gives

$$G_t(u) = \frac{H_t(2i\sqrt{u})}{H_t(0)} = \sum_{k\ge 0} \frac{M_k(t)\,u^k}{(2k)!\,M_0(t)}, \qquad M_k(t) = \int_{\mathbb{R}} v^{2k} e^{tv^2/4}\Phi(v)\,dv,$$

and G_0 is the function G of Section 8.1. Write $\pm\theta_j(t)$ for the zeros of $\theta \mapsto H_t(2\theta)$, so that $\theta_j(0) = \gamma_j$ while the zeros are on the line. Since H_t is even and entire of order one, G_t has order $\frac{1}{2}$ and $G_t(u) = \prod_j (1 + u/a_j(t))$ with $a_j(t) = \theta_j(t)^2$, so the Newton identities turn the $M_k(t)$ into the moments $m_n(t) = \sum_j a_j(t)^{-n-1}$, exactly as at $t = 0$. If every $\theta_j(t)$ is real, the $m_n(t)$ are the moments of the positive measure $\nu_t = \sum_j a_j(t)^{-1}\delta_{1/a_j(t)}$, and every Hankel matrix $\mathcal{H}_N(t) = [m_{i+j}(t)]_{i,j<N}$ is positive definite. The converse is proved as at $t = 0$: if every $\mathcal{H}_N(t)$ is positive definite, every $a_j(t)$ is real, and it is positive because H_t has no zeros on the imaginary axis: $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, the kernel (2.1.3) being strictly positive. So all $\mathcal{H}_N(t)$ are positive definite exactly when H_t has only real zeros. The Cholesky factor gives the Jacobi matrix $J_N(t)$, whose eigenvalues return the lowest zeros of H_t as J_N returns the γ_j at $t = 0$ (Table 4.2).

t	$\theta_1(t)$	$\theta_2(t)$	$\theta_3(t)$	$\theta_5(t)$	$\theta_{10}(t)$
0.2	14.07681	20.94669	24.95221	32.88801	49.72988
0	14.13473	21.02204	25.01086	32.93506	49.77383
−0.4	14.25083	21.17365	25.12774	33.02595	49.85213
−1.0	14.42563	21.40341	25.30189	33.15277	49.93350

Table 4.2: Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \dots, m_{59}(t)$; at $t = 0$ they are γ_1, γ_2, γ_3, γ_5 and γ_{10}. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \dots, \theta_{12}(t)$ to within 3.1×10^{-7} of the zeros that Newton's method finds on H_t, and $\theta_{13}(t)$ only to within 5.2×10^{-4}.

Going back from $t = 0$, the matrix $\mathcal{H}_N(t)$ stays positive definite until a time t_N^* at which it ceases to be. Since $\mathcal{H}_N(t)$ is the leading block of $\mathcal{H}_{N+1}(t)$, $t_N^* \le t_{N+1}^*$. Under RH every t_N^* is negative, because $\mathcal{H}_N(0)$ is positive definite, and $t_N^* \to 0$: for each $t < 0$ the function H_t has nonreal zeros by the theorem of Rodgers and Tao, so some $\mathcal{H}_N(t)$ is not positive definite, and then $t_{N'}^* \ge t$ for every $N' \ge N$. With the trapezoid rule of step $1/600$ at $100 + 16N$ digits, the grid $t = 0, -0.05, \dots, -4$

and bisection to 10^{-7} give no failure of order 10 down to $t = -4$, and

$$t_{20}^* = -1.8991865, \qquad t_{30}^* = -1.2853067,$$
$$t_{40}^* = -1.2760788, \qquad t_{50}^* = -1.0325048;$$

for $N = 30$ a second discretisation (step 1/550, 650 digits) gives a bracket that overlaps the first in $[-1.28530669, -1.28530664]$.

Each failure registers a collision. We follow the real zeros of H_t that start at $2\gamma_j$, $j \le 64$, back to $t = -2.05$ by the equation of motion of Section 4.1.1, on the trapezoid sum for H_t, and locate each collision as a double zero, $H_t(z) = \partial_z H_t(z) = 0$, by Newton's method in the two unknowns z and t; the backward heat equation, which holds exactly for the trapezoid sum, gives the derivatives in t. Thirteen of the pairs below index 59 collide in this time range, among them those that start at (γ_k, γ_{k+1}) for $k = 9$, 13, 19, 24, 27 and 34, at

$$t = -1.87791,\ -1.28527,\ -1.27554,\ -1.14387,\ -0.88108,\ -0.38456$$

and the heights $\theta = 49.664$, 60.491, 77.027, 88.466, 95.540 and 111.638; the other seven start at $k = 40$, 43, 45, 48, 51, 53 and 57. To see which collision a failure registers, we continue each double zero into the complex plane, starting from $z_c \pm i\sqrt{2(t_c - t)}$ near the collision, and remove the contribution $2\,\mathrm{Re}\,\theta^{-2n-2}$ of the conjugate pair $\theta = z/2$, $\bar\theta$ from $m_n(t)$. At $t_N^* - \delta$, with $\delta = 0.01$ (for $N = 40$, $\delta = 0.0046$, half the time to the collision of $(\gamma_{13}, \gamma_{14})$), the only one of the thirteen pairs whose removal makes $\mathcal{H}_N(t)$ positive definite again is the one that starts at (γ_9, γ_{10}) for $N = 20$, at $(\gamma_{13}, \gamma_{14})$ for $N = 30$, at $(\gamma_{19}, \gamma_{20})$ for $N = 40$ and at $(\gamma_{27}, \gamma_{28})$ for $N = 50$. The same holds at $t_N^* - 10^{-5}$, except that there removing $(\gamma_{34}, \gamma_{35})$ also restores the order 50 matrix. The failures thus come 0.021, 3.7×10^{-5}, 5.3×10^{-4} and 0.151 after the collisions they register.

A Hankel test of finite order is a window on the low part of the spectrum. The order 30 matrix is positive definite at $t = -1.276, -1.278, \ldots, -1.284$ and -1.2852, after the collision of $(\gamma_{19}, \gamma_{20})$ at -1.27554 and before that of $(\gamma_{13}, \gamma_{14})$ at -1.28527: it passes the pair that is too high and too light for its order and registers the lower one. A conjugate pair at height θ enters $m_n(t)$ with weight of order θ^{-2n-2}, against $\theta_1(t)^{-2n-2}$ for the lowest zero, as at $t = 0$, and increasing the order widens the window.

The whole curve $N \mapsto t_N^*$ up to $N = 50$ follows from one elimination per time. Let $n_{\mathrm{fail}}(t)$ be the index of the first nonpositive pivot of Gaussian elimination on $\mathcal{H}_{50}(t)$, that is, the least $N \le 50$ for which $\mathcal{H}_N(t)$ is not positive definite, so that $t_N^* = \sup\{t < 0 : n_{\mathrm{fail}}(t) \le N\}$. On the grid of step 0.01 it is a staircase [239] (Table 4.3). To see what each step registers, the table gives, at one time on each step, the least order at which each complex pair breaks the test alone, that is, with the contributions of all the other complex pairs removed from the moments. A pair alone breaks the test at a definite order, a higher pair at a higher order,

and n_{fail} is close to the least of these orders. The order at which a pair is seen falls as the pair moves away from the axis: $(\gamma_{27}, \gamma_{28})$, which collides at -0.881, breaks order 50 at $t = -1.09$ and order 39 at -1.29, and $(\gamma_{34}, \gamma_{35})$, which collides earlier, at -0.385, but lies higher, breaks the test only from $t = -1.28$ on, at order 50 and then 47. Removing the pair with the least order moves n_{fail} up to about the next one: at $t = -1.30$ from 27 to 35, when $(\gamma_{13}, \gamma_{14})$ is removed, at -1.28 from 39 to 41, when $(\gamma_{19}, \gamma_{20})$ is removed, and at -1.09 from 50 to no failure, when $(\gamma_{27}, \gamma_{28})$ is removed. With all thirteen pairs removed, no order up to 50 fails at any of these times.

			order broken by (γ_k, γ_{k+1}) alone				
t	$n_{\text{fail}}(t)$	at $t =$	$k = 13$	19	24	27	34
$[-1.03, 0]$	none						
$[-1.14, -1.04]$	50	-1.09	real	real	real	50	> 50
$[-1.18, -1.15]$	46	-1.16	real	real	44	45	> 50
$[-1.27, -1.19]$	45	-1.23	real	real	44	45	> 50
-1.28	39	-1.28	real	38	40	41	50
-1.29	28	-1.29	28	35	37	39	47
-1.30	27	-1.30	27	35	37	39	47

Table 4.3: The first failing order $n_{\text{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01 and, at one time on each step, the least order at which the pair that starts at (γ_k, γ_{k+1}) breaks the test alone (real: the pair has not yet collided; > 50: no order up to 50 fails). The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone.

A rule suggested in [239] is that the collision of (γ_k, γ_{k+1}) is registered from order about $2k + 2$ on, so that $t_N^* \approx \max\{t_{\text{coll}}(k) : 2k + 2 \le N\}$, where $t_{\text{coll}}(k)$ is the collision time of the pair. It holds for $N = 30$ and 40 ($k = 13$ and 19), roughly for $N = 20$ ($k = 9$, delay 0.021), and fails at $N = 50$: it gives the collision of $(\gamma_{24}, \gamma_{25})$ at -1.144, while the order 50 matrix registers $(\gamma_{27}, \gamma_{28})$, for which $2k + 2 = 56$, at -1.0325, before $(\gamma_{24}, \gamma_{25})$ has collided. Table 4.3 shows why: a pair beyond the window of the rule enters it after a delay, and t_N^* is the latest of the collision times of the pairs inside the window and of the times at which pairs above it have moved far enough from the axis.

The curve rises to 0 through the collision times of ever closer pairs. Following every real zero that starts below index 215 in the same way (200 digits, step 1/200), 33 of the first 120 gaps close on $[-2.05, 0]$, 28 of them after $t = -1.6$, and 50 of the first 200 gaps close after $t = -1.6$. The latest collisions among the first 200 gaps are those of $(\gamma_{186}, \gamma_{187})$, at distance 0.498, at $t = -0.1310$, of $(\gamma_{159}, \gamma_{160})$ at -0.1695 and of $(\gamma_{135}, \gamma_{136})$ at -0.2015, and below height 230 those of $(\gamma_{91}, \gamma_{92})$ at -0.2821 and of the closest pair below height 200, $(\gamma_{71}, \gamma_{72})$, at -0.2880, the

value of Section 4.1.2. A collision close to $t = 0$ needs a pair as close as Lehmer's (Section 4.1.2), far beyond the reach of a Hankel test at the centre.

Further back in time the low pairs leave the axis one after another, and Theorem 8.1.7 turns the test into a count. Its proof applies at every time to the zeros of H_t, none of which is purely imaginary: the numbers $\iota_N(t)$ and $\iota'_N(t)$ of negative eigenvalues of $\mathcal{H}_N(t)$ and of $\mathcal{H}'_N(t) = [m_{i+j+1}(t)]_{i,j<N}$ do not decrease with N, never exceed the number of distinct conjugate pairs $\theta, \bar{\theta}$ of zeros with $\operatorname{Re}\theta > 0$, and tend to it, so that each negative eigenvalue certifies a pair. The signs of the pivots of one Gaussian elimination give $\iota_1(t), \dots, \iota_{30}(t)$, and $n'_{\text{fail}}(t)$ is to $\mathcal{H}'_N(t)$ what $n_{\text{fail}}(t)$ is to $\mathcal{H}_N(t)$. We follow the real zeros below $\theta = 88$, those that start at $\gamma_1, \dots, \gamma_{24}$, back to $t = -20$ and locate each collision as a double zero as above. Below $\theta = 88$ the collisions found above are those of (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$, and after them come those of $(\gamma_{22}, \gamma_{23})$, $(\gamma_{15}, \gamma_{16})$, (γ_4, γ_5), (γ_7, γ_8), $(\gamma_{17}, \gamma_{18})$ and (γ_2, γ_3), at

$$t = -2.32414,\ -2.75904,\ -3.95479,\ -4.03382,\ -7.31401,\ -12.49451$$

and the heights $\theta = 84.875$, 67.406, 33.099, 43.403, 72.746 and 27.321. Each pair is followed into the complex plane as above (Table 4.4). Removing the contribution of one pair from the moments shows what the first failure registers: at $t = -2$ and -3 the pair that starts at (γ_9, γ_{10}), whose removal moves n_{fail} from 19 to 21 and from 18 to 21, and from $t = -4$ to -12 the pair that starts at (γ_4, γ_5), whose removal raises n_{fail} by two or three, from 11 to 14 at $t = -4$; at these times the removal of any other pair that starts below $\theta = 88$ leaves n_{fail} unchanged. Both pairs are registered soon after their collisions, at the orders 19 and 11, close to $2k + 2 = 20$ and 10. At $t = -2$ the three pairs that start at (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$ account for every failure up to order 30: with them removed none remains, and alone they break the test at the orders 19, 21 and 30. From $t = -3$ on, pairs that start above $\theta = 88$ enter as well, since with all the pairs that start below $\theta = 88$ removed the test still fails, from order 24 at $t = -3$, 21 at -4, 15 at -8 and 11 at -15 and -20. At $t = -15$ and -20 the lowest pair is no longer the first to be seen: no single removal raises n_{fail}, and the pair that starts at (γ_4, γ_5) alone breaks the test as early as the lowest pair alone or earlier, at order 4 against 5 at $t = -20$. The two discretisations of the moments, step 1/300 at 600 digits and step 1/450 at 900 digits, give the same counts and the same removal tests, and at the orders 10 and 30 the eigenvalues of the diagonally scaled matrices give the same counts as the pivots.

4.1.4 Biane's theorem and Laguerre's quantity

The collision times of Section 4.1.3 can be predicted from H_0 alone through a quantity that free probability singles out. This section is based on [239, Section 8]; Corollaries 4.1.13 and 4.1.15 and the halved constant of Remark 4.1.11 are not in that note, and the numbers are recomputed here from H_t.

t	pair	θ	real	collided	n_{fail}	n'_{fail}	$(\iota_{10}, \iota'_{10})$	$(\iota_{30}, \iota'_{30})$
-2	$(9,10)$	$49.715 \pm 0.249i$	8	3	19	19	$(0,0)$	$(2,2)$
-3	$(9,10)$	$50.139 \pm 0.792i$	8	5	18	17	$(0,0)$	$(3,3)$
-4	$(4,5)$	$33.115 \pm 0.151i$	3	6	11	10	$(0,1)$	$(5,5)$
-5	$(4,5)$	$33.478 \pm 0.745i$	3	7	9	9	$(1,1)$	$(6,6)$
-6	$(4,5)$	$33.841 \pm 1.071i$	3	7	9	8	$(1,1)$	$(6,6)$
-8	$(4,5)$	$34.572 \pm 1.592i$	3	8	8	8	$(1,1)$	$(6,6)$
-10	$(4,5)$	$35.306 \pm 2.052i$	3	8	8	8	$(1,1)$	$(6,7)$
-12	$(4,5)$	$36.043 \pm 2.492i$	3	8	8	8	$(1,1)$	$(7,7)$
-15	$(2,3)$	$28.202 \pm 1.166i$	1	9	6	5	$(2,3)$	$(8,8)$
-20	$(2,3)$	$29.967 \pm 2.178i$	1	9	5	5	$(3,3)$	$(8,8)$

Table 4.4: The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros γ_k, γ_{k+1} it starts from, and its position θ; the number of real zeros below it; the number of collisions so far among $\gamma_1, \dots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\text{fail}}(t)$ and $n'_{\text{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30.

Let μ be a probability measure on $\mathbb{R}$, $G_\mu(z) = \int \mu(dx)/(z-x)$ its Cauchy transform, and σ_t the semicircle law of variance $t > 0$, with density $(2\pi t)^{-1}(4t - x^2)^{1/2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$. The free convolution $\mu \boxplus \sigma_t$ is the law of $A + S$ for free random variables A and S with laws μ and σ_t. Put

$$F_\mu(u) = \int \frac{\mu(dx)}{(u-x)^2} \in (0, \infty], \qquad v_t(u) = \inf\Big\{v \geq 0 : \int \frac{\mu(dx)}{(u-x)^2 + v^2} \leq \frac{1}{t}\Big\},$$

$$\psi_t(u) = u + t \int \frac{(u-x)\,\mu(dx)}{(u-x)^2 + v_t(u)^2}.$$

Theorem 4.1.2 (Biane [24]). *Let μ be a probability measure on $\mathbb{R}$ and $t > 0$.*

(a) ψ_t is a homeomorphism of $\mathbb{R}$, and $\mu \boxplus \sigma_t$ has a continuous density p_t with $p_t(\psi_t(u)) = v_t(u)/(\pi t)$ for every real u. For $\operatorname{Im} z > 0$, $G_{\mu\boxplus\sigma_t}(z) = G_\mu(\omega_t(z))$, where $\omega_t(z) = z - tG_{\mu\boxplus\sigma_t}(z)$, and ω_t is the inverse of the map $w \mapsto w + tG_\mu(w)$ on the region $\{u + iv : v > v_t(u)\}$, which that map sends onto the upper half plane.

(b) Let $U_t = \{u : F_\mu(u) > 1/t\}$. The interior of the support of $\mu \boxplus \sigma_t$ is $\psi_t(U_t)$, the support is its closure, the density is real analytic on $\psi_t(U_t)$, and the number of components of U_t does not increase with t.

(c) Let (a, b) be a bounded component of $\mathbb{R} \setminus \operatorname{supp} \mu$ with $\mu(\{a\}) > 0$ and $\mu(\{b\}) > 0$. Then F_μ is strictly convex on (a, b) and tends to $+\infty$ at both ends; let u_0 be its minimiser and

$$t_0 = \frac{1}{F_\mu(u_0)} = \Big(\min_{a<u<b} \int \frac{\mu(dx)}{(u-x)^2}\Big)^{-1}. \tag{4.1.1}$$

For $t < t_0$ the set $\{u \in (a,b) : F_\mu(u) \le 1/t\}$ is an interval $[a(t), b(t)]$ with $a(t) < b(t)$, and ψ_t maps $(a(t), b(t))$ onto a component of the complement of the support of $\mu \boxplus \sigma_t$. As t increases to t_0, $a(t)$ increases and $b(t)$ decreases to u_0, and at t_0 the gap closes.

Proof. Part (a) is Lemma 4, Proposition 2 and Corollaries 2 and 3 of [24], where σ_t has R transform tz as here; part (b) is Proposition 3, its proof, and Corollary 4 there. For (c), μ does not charge (a,b), so $F''_\mu(u) = 6\int (u-x)^{-4}\mu(dx) > 0$ there, and the atoms make F_μ tend to $+\infty$ at a and b. The sublevel sets of a strictly convex function are intervals; $\{F_\mu \le 1/t\} \cap (a,b)$ contains u_0 in its interior when $1/t > F_\mu(u_0)$, shrinks as t grows, and reduces to u_0 at $t = t_0$. Its interior is a component of $\mathbb{R} \setminus \overline{U_t}$, because the points of (a,b) just outside it lie in U_t, and ψ_t, a homeomorphism, maps it onto a component of the complement of $\overline{\psi_t(U_t)}$, which is the support by (b). This is the discussion following Corollary 5 of [24]. □

Call (4.1.1) Biane's threshold. For a real entire function f write

$$\mathcal{L}_f(x) = -\frac{d^2}{dx^2}\log|f(x)| = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2},$$

Laguerre's quantity. If f has genus at most one and zeros z_j, then $\mathcal{L}_f(x) = \sum_j (x - z_j)^{-2}$. For the normalised counting measure $\mu_p = n^{-1}\sum_j \delta_{z_j}$ of the zeros of a polynomial p of degree n with only real simple zeros, $F_{\mu_p} = \mathcal{L}_p/n$, and by Theorem 4.1.2(c) the gap (z_k, z_{k+1}) survives in $\mu_p \boxplus \sigma_t$ until $t_0 = n/\min_{(z_k,z_{k+1})} \mathcal{L}_p$.

For Ξ, whose zeros are $\pm\gamma_j$, and for $H_0(z) = \frac{1}{8}\Xi(z/2)$,

$$\mathcal{L}_{H_0}(z) = \tfrac{1}{4}\mathcal{L}_\Xi(z/2), \qquad \mathcal{L}_\Xi(x) = \sum_j \Big[\frac{1}{(x-\gamma_j)^2} + \frac{1}{(x+\gamma_j)^2}\Big] = \sum_j \frac{2(x^2+\gamma_j^2)}{(x^2-\gamma_j^2)^2}.$$

Both are computed from Ξ and its first two derivatives, without the zeros. Since $\Xi(x) = \Xi(0)G(-x^2)$, with $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.1,

$$\mathcal{L}_\Xi(x) = 2\frac{G'}{G}(-x^2) - 4x^2\Big(\frac{G'}{G}\Big)'(-x^2) = 2\frac{d}{dx}\Big(x\,\frac{G'}{G}(-x^2)\Big),$$
$$\frac{G'}{G}(u) = \sum_j \frac{1}{u+\gamma_j^2},$$

and G'/G is the Stieltjes transform of the Thorin measure $\sum_j \delta_{\gamma_j^2}$ of the clock $1/G$ (under RH; the identities hold in general with γ_j^2 replaced by $a_j = -(\rho_j - \frac{1}{2})^2$).

Lemma 4.1.3. *Let p be a real polynomial of degree $n \ge 2$ with only real simple zeros and critical points $c_1 < \cdots < c_{n-1}$, and let s be real with $s\,\mathcal{L}_p(c_i) < 1$ for every i. Then $p + sp''$ has only real simple zeros, one in each of the intervals $(-\infty, c_1)$, (c_1, c_2), …, (c_{n-1}, ∞).*

Proof. At a critical point $p'(c_i)=0$, so $p''(c_i)/p(c_i)=(\log|p|)''(c_i)=-\mathcal{L}_p(c_i)$, and $(p+sp'')(c_i)=p(c_i)\big(1-s\mathcal{L}_p(c_i)\big)$ has the sign of $p(c_i)$. There is exactly one critical point between consecutive zeros of p, so the signs of $p(c_1),\dots,p(c_{n-1})$ alternate. The polynomial $p+sp''$ has degree n and the sign of p at $\pm\infty$, which is opposite to the sign of $p(c_1)$ at $-\infty$ and to that of $p(c_{n-1})$ at $+\infty$, since p has one zero to the left of c_1 and one to the right of c_{n-1}. So $p+sp''$ changes sign in each of the n intervals and has n real zeros there. □

Since $e^{sD^2}=\lim_N(1+\frac{s}{N}D^2)^N$, the lemma suggests, without proving it, that the gap (z_k,z_{k+1}) of H_t stays open while $|t|<1/\mathcal{L}_{H_0}(c_k)$, where c_k is the critical point of H_0 in the gap. With $z_k=2\gamma_k$ and u_k the minimiser of $\mathcal{L}_{H_0}$ over the gap, this gives two predictions of the collision time from H_0 alone, the critical point form and Biane's form,

$$\tau_c=-\frac{1}{\mathcal{L}_{H_0}(c_k)}=-\frac{4}{\mathcal{L}_{\Xi}(c_k/2)},$$
$$\tau_B=-\frac{1}{m(0)},\qquad m(0)=\min_{(z_k,z_{k+1})}\mathcal{L}_{H_0}=\mathcal{L}_{H_0}(u_k).$$

For an isolated pair, $\mathcal{L}=8/D_0^2$ at the midpoint of the gap $D_0=z_{k+1}-z_k$, and both equal the isolated pair estimate $-D_0^2/8$ of Section 4.1.1, which is the exact collision time of z^2-a^2 under e^{sD^2}, since $e^{sD^2}(z^2-a^2)=z^2-a^2+2s$. Section 4.1.5 turns both predictions into theorems for functions with only real zeros.

Remark 4.1.4. Biane's theorem and the collisions concern opposite directions of the heat flow. For polynomials of degree n the finite free convolution of Marcus, Spielman and Srivastava [210] is $p\boxplus_n q=\hat{q}(D)p$ when $q=\hat{q}(D)x^n$, so $e^{-aD^2}p=p\boxplus_n e^{-aD^2}x^n$. The polynomial $e^{-aD^2}x^n$ is a rescaled Hermite polynomial [210] and has zeros with mean square $2a(n-1)$, and for large n their distribution is close to the semicircle law σ_{2an}; in the limit finite free convolution becomes free convolution [210]. So $\mu\mapsto\mu\boxplus\sigma_t$ is the limit of the forward flow e^{-aD^2} with $a\approx t/(2n)$, that is of H_t with t increasing: the zeros repel, and a gap between two groups of zeros fills in as the groups spread, which is what Theorem 4.1.2(c) describes. Zeros collide only in the other direction, under e^{sD^2} with $s=-t>0$. What carries over is the functional. Biane's threshold for μ_p is $n/\min\mathcal{L}_p$, and Theorem 4.1.9 below shows that under e^{sD^2} a gap of a function with only real zeros cannot close before $s=1/\min\mathcal{L}$ while all the zeros stay real. For the pair $\pm a$ the two times are a^2 and $a^2/2$, and the factor 2 is the n of the normalisation of μ_p.

Table 4.5 compares the two predictions with the collision times of the census. Every collision of the census, all 58 among the first 200 gaps on $[-2.05,0]$, comes later in the backward flow than both predictions and the isolated pair estimate: $t_{\mathrm{coll}}/\tau_B\geq 1.083$, $t_{\mathrm{coll}}/\tau_c\geq 1.104$ and $t_{\mathrm{coll}}/(-D_0^2/8)\geq 1.056$, all three for

$(186, 187)$, which collides at $t = -0.1310$. Over the 28 pairs of the table, t_{coll}/τ_B lies in $[1.114, 1.719]$, t_{coll}/τ_c in $[1.181, 1.951]$ and $t_{\mathrm{coll}}/(-D_0^2/8)$ in $[1.077, 1.518]$, and $|\tau_B|$ falls short of $|t_{\mathrm{coll}}|$ by 25.5% on average. The ratio t_{coll}/τ_B is close to 1 when the two zeros dominate their gap, and it is very nearly an increasing function of the share $6Q(0)/m(0)$ of the other zeros in Biane's functional (Section 4.1.5): ordered by this share, the ratios in the table rise with only small inversions, while the gap alone orders them much less well (compare $(114, 115)$ and $(13, 14)$ in the table). Real zeros outside the gap pull the two zeros apart while they fall together. Over all 58 collisions t_{coll}/τ_B lies in $[1.083, 2.329]$, and the largest ratios, 2.042 to 2.329, belong to the five collisions with $6Q(0)/m(0) > 0.9$, all at $t < -1.89$.

pair	$\gamma_{k+1} - \gamma_k$	t_{coll}	$-D_0^2/8$	τ_c	τ_B	t_{coll}/τ_B	$6Q/m$	frozen
$(91, 92)$	0.7158	−0.2821	−0.2562	−0.2369	−0.2455	1.149	0.250	−0.2826
$(71, 72)$	0.7243	−0.2880	−0.2623	−0.2305	−0.2517	1.144	0.242	−0.2883
$(97, 98)$	0.7370	−0.3040	−0.2716	−0.2454	−0.2587	1.175	0.286	−0.3047
$(63, 64)$	0.8175	−0.3686	−0.3341	−0.3122	−0.3199	1.152	0.256	−0.3694
$(34, 35)$	0.8451	−0.3846	−0.3571	−0.3115	−0.3452	1.114	0.200	−0.3852
$(53, 54)$	0.8717	−0.4231	−0.3800	−0.3361	−0.3624	1.167	0.277	−0.4244
$(114, 115)$	0.9307	−0.5380	−0.4331	−0.3933	−0.3997	1.346	0.462	−0.5364
$(93, 94)$	0.9763	−0.5533	−0.4766	−0.4179	−0.4471	1.237	0.371	−0.5589
$(106, 107)$	0.9650	−0.5691	−0.4656	−0.3620	−0.4306	1.321	0.451	−0.5724
$(75, 76)$	1.0531	−0.6572	−0.5545	−0.4537	−0.5166	1.272	0.410	−0.6648
$(111, 112)$	1.0745	−0.7102	−0.5772	−0.4715	−0.5310	1.338	0.481	−0.7240
$(117, 118)$	1.0571	−0.7653	−0.5588	−0.4318	−0.5029	1.522	0.599	−0.7676
$(78, 79)$	1.1388	−0.8235	−0.6485	−0.5695	−0.5930	1.389	0.513	−0.8318
$(27, 28)$	1.2193	−0.8811	−0.7433	−0.6313	−0.6937	1.270	0.400	−0.8861
$(45, 46)$	1.2588	−0.9600	−0.7923	−0.6731	−0.7325	1.311	0.452	−0.9751
$(82, 83)$	1.2050	−0.9734	−0.7260	−0.6170	−0.6533	1.490	0.601	−0.9985
$(80, 81)$	1.2288	−1.0044	−0.7550	−0.5459	−0.6808	1.475	0.589	−1.0283
$(87, 88)$	1.1991	−1.0343	−0.7190	−0.5805	−0.6390	1.619	0.667	−1.0540
$(100, 101)$	1.2456	−1.0435	−0.7757	−0.5739	−0.6963	1.499	0.614	−1.0797
$(104, 105)$	1.2476	−1.0485	−0.7783	−0.6591	−0.6983	1.502	0.617	−1.0858
$(57, 58)$	1.2524	−1.1004	−0.7842	−0.6484	−0.7031	1.565	0.621	−1.0984
$(24, 25)$	1.3838	−1.1439	−0.9575	−0.7980	−0.8898	1.286	0.424	−1.1578
$(40, 41)$	1.3100	−1.1519	−0.8580	−0.7345	−0.7771	1.482	0.566	−1.1458
$(65, 66)$	1.3427	−1.2447	−0.9014	−0.6381	−0.8066	1.543	0.631	−1.2737
$(19, 20)$	1.4401	−1.2755	−1.0370	−0.7719	−0.9573	1.332	0.461	−1.2837
$(13, 14)$	1.4847	−1.2853	−1.1022	−0.9064	−1.0334	1.244	0.374	−1.2947
$(51, 52)$	1.4218	−1.3190	−1.0107	−0.7548	−0.9126	1.445	0.582	−1.3684
$(48, 49)$	1.3875	−1.4611	−0.9626	−0.7573	−0.8497	1.719	0.703	−1.4679

Table 4.5: The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time t_{coll} from the census of Section 4.1.3, against predictions computed from H_0: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms τ_c and τ_B, and the frozen predictor of Section 4.1.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of H_t.

4.1.5 Collision brackets from Laguerre's quantity

Write $s = -t$ for the backward time, so that $f_s = H_{-s}$ solves $\partial_s f_s = f_s''$; for a real polynomial p, $f_s = e^{sD^2}p = \sum_{k\geq 0} s^k p^{(2k)}/k!$ solves the same equation. In this subsection f_s, $s \geq 0$, is a family of real entire functions of genus at most one, smooth in (z, s), with $\partial_s f_s = f_s''$, and $(z_k(s), z_{k+1}(s))$ is a gap between two adjacent simple real zeros, which move continuously until they meet, if they do. On the gap put $q = f_s'/f_s$ and $\mathcal{L} = \mathcal{L}_{f_s} = -q'$. For a point x of the gap write $x_j = 1/(x - z_j)$, where z_j runs over the zeros of f_s, and $S_k = \sum_j x_j^k$ for $k \geq 2$, a real and absolutely convergent sum; then

$$\mathcal{L}(x) = S_2, \qquad \mathcal{L}'(x) = -2S_3, \qquad \mathcal{L}''(x) = 6S_4.$$

When all zeros are real, q falls from $+\infty$ to $-\infty$ across the gap and vanishes at a single critical point $c(s)$; we write $\ell(s) = \mathcal{L}(c(s))$. In all cases $\mathcal{L} \to +\infty$ at both ends of the gap, and we write $m(s)$ for its minimum over the gap, attained at a minimiser $u(s)$, where $\mathcal{L}'(u) = 0$, that is $S_3 = 0$.

Lemma 4.1.5. *On the gap $\partial_s \mathcal{L} = \mathcal{L}'' + 2q\mathcal{L}' - 2\mathcal{L}^2$. If all zeros of f_s are real and simple, then $c(s)$ is differentiable with $\dot{c} = -\mathcal{L}'(c)/\mathcal{L}(c)$, and*

$$\dot{\ell} = 6S_4 - 2S_2^2 - \frac{4S_3^2}{S_2} \qquad \text{at } x = c(s).$$

Proof. With $w = \log f_s$ the heat equation gives $\partial_s w = f_s''/f_s = w'' + w'^2$, hence $\partial_s q = q'' + 2qq'$ and, with $\mathcal{L} = -q'$, $\partial_s \mathcal{L} = -q''' - 2q'^2 - 2qq'' = \mathcal{L}'' - 2\mathcal{L}^2 + 2q\mathcal{L}'$. Since $\partial_z q = -\mathcal{L} < 0$ at c, the implicit function theorem applied to $q(c(s), s) = 0$ gives $\dot{c} = -\partial_s q/\partial_z q = (q'' + 2qq')/\mathcal{L} = -\mathcal{L}'/\mathcal{L}$ at c, where $q = 0$. Hence $\dot{\ell} = \mathcal{L}'\dot{c} + \partial_s \mathcal{L} = \mathcal{L}'' - 2\mathcal{L}^2 - \mathcal{L}'^2/\mathcal{L}$ at c, and $\mathcal{L} = S_2$, $\mathcal{L}' = -2S_3$, $\mathcal{L}'' = 6S_4$. □

Lemma 4.1.6. *For every real square summable sequence (x_j) with power sums $S_k = \sum_j x_j^k$,*

$$E = 3S_2^3 - 6S_2S_4 + 4S_3^2 \geq 0,$$

and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.

Proof. By homogeneity take $S_2 = 1$. Let $S_k^+ = \sum_{x_j>0} x_j^k$ and $S_k^- = \sum_{x_j<0} |x_j|^k$, and $\alpha = (S_2^+)^{1/2}$, $\beta = (S_2^-)^{1/2}$, so that $\alpha^2 + \beta^2 = 1$, $S_3 = S_3^+ - S_3^-$ and $S_4 = S_4^+ + S_4^-$. Every positive term is at most α, so $S_4^+ \leq \alpha S_3^+$ and $S_3^+ \leq \alpha S_2^+ = \alpha^3$; likewise $S_4^- \leq \beta S_3^-$ and $S_3^- \leq \beta^3$. Therefore $E \geq g(S_3^+, S_3^-)$ with

$$g(p, n) = 3 - 6(\alpha p + \beta n) + 4(p - n)^2 \qquad \text{on the box } [0, \alpha^3] \times [0, \beta^3].$$

The function g is convex, its Hessian being $\begin{pmatrix} 8 & -8 \\ -8 & 8 \end{pmatrix}$, and its gradient $\big(-6\alpha + 8(p-n), -6\beta - 8(p-n)\big)$, whose components add up to $-6(\alpha + \beta) < 0$, never

vanishes; so its minimum over the box lies on the boundary. On the edge $p = 0$, $g = 3 - 6\beta n + 4n^2 \geq 3 - \frac{9}{4}\beta^2 \geq \frac{3}{4}$, and the edge $n = 0$ is symmetric. At the corner (α^3, β^3), using $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$ and $1 - 4\alpha^2\beta^2 = (\alpha^2 - \beta^2)^2$,

$$g = 4(\alpha^3 - \beta^3)^2 - 3(\alpha^2 - \beta^2)^2 = (\alpha - \beta)^2\big[4(\alpha^2 + \alpha\beta + \beta^2)^2 - 3(\alpha + \beta)^2\big],$$

and with $\sigma = \alpha+\beta$, $\alpha\beta = (\sigma^2-1)/2$, the bracket is $(\sigma^2+1)^2-3\sigma^2 = \sigma^4-\sigma^2+1 > 0$. On the edge $p = \alpha^3$, g is a convex quadratic in n with vertex at $n = \alpha^3 + \frac{3}{4}\beta$. If the vertex lies at or beyond β^3, the minimum over the edge is at the corner. Otherwise $\beta^3 - \frac{3}{4}\beta > \alpha^3 \geq 0$ forces $\beta^2 > \frac{3}{4}$ and $\alpha < \frac{1}{2}$, and the vertex value is

$$\tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3\beta \geq h(\alpha) = \tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3;$$

$h'(\alpha) = \alpha(\frac{9}{2} - 18\alpha - 24\alpha^2)$, so h increases up to $\alpha \approx 0.198$ and decreases after it, and on $[0, \frac{1}{2}]$ it is at least $\min(h(0), h(\frac{1}{2})) = \min(\frac{3}{4}, \frac{3}{16}) > 0$. The edge $n = \beta^3$ is symmetric. Hence $E \geq 0$. Equality forces the corner with $\alpha = \beta$, and $S_4^+ = \alpha S_3^+$ with $S_3^+ = \alpha^3$, so every positive term equals α and their squares add up to α^2: there is exactly one positive term, α, and likewise exactly one negative term, $-\alpha$. Conversely $E = 0$ for $\{a, -a\}$. □

Theorem 4.1.7 (critical point bound). *If all zeros of f_σ are real and simple for $0 \leq \sigma < s$, then $\dot\ell \leq \ell^2$ on $[0, s)$ and*

$$\ell(\sigma) \leq \frac{\ell(0)}{1 - \sigma\ell(0)} \qquad \big(0 \leq \sigma < \min(s, 1/\ell(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \geq 1/\mathcal{L}_p(c_k)$. The bound is attained by $p(z) = z^2 - a^2$.

Proof. By Lemma 4.1.5, $\dot\ell \leq \ell^2$ reads $6S_4 - 2S_2^2 - 4S_3^2/S_2 \leq S_2^2$, which is $E \geq 0$ divided by $S_2 > 0$, Lemma 4.1.6. Then $\frac{d}{d\sigma}(1/\ell) = -\dot\ell/\ell^2 \geq -1$, so $1/\ell(\sigma) \geq 1/\ell(0) - \sigma$, which is the bound. For the polynomial, all zeros are real and simple on $[0, s_1)$, and $\ell \geq (c - z_k)^{-2} + (z_{k+1} - c)^{-2} \geq 8/g^2$, where g is the length of the gap. If $s_1 < 1/\ell(0)$, then ℓ stays bounded on $[0, s_1)$, so g stays bounded below and the two zeros cannot meet at s_1. For $z^2 - a^2$, $e^{sD^2}p = z^2 - (a^2 - 2s)$, $c = 0$ and $\ell(s) = 2/(a^2 - 2s) = \ell(0)/(1 - s\ell(0))$, with the collision at $s = a^2/2 = 1/\ell(0)$. □

The theorem applies to every real rooted polynomial, in particular to every truncation $\prod_{j\leq J}(1-z^2/z_j^2)$ of H_0. For H_t itself the hypothesis fails at every $t < 0$: by the theorem of Rodgers and Tao [250], H_t has nonreal zeros for every $t < 0$ (Section 4.1.1). The minimum of $\mathcal{L}$ over the gap gives the sharper Biane form; at the minimiser $S_3 = 0$, while $q(u)$ need not vanish.

LEMMA 4.1.8. *For every real square summable sequence with $S_3 = 0$, $2S_4 \le S_2^2$, and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.*

Proof. In the notation of the proof of Lemma 4.1.6, without the normalisation, $S_3 = 0$ means $S_3^+ = S_3^- = \mu$, and for a nonzero sequence both α and β are then positive. Now $\mu \le \min(\alpha^3, \beta^3)$ and $S_4 \le \alpha S_3^+ + \beta S_3^- = (\alpha + \beta)\mu$. Let $\lambda = \min(\alpha, \beta)$ and $\max(\alpha, \beta) = r\lambda$ with $r \ge 1$. Then $S_4 \le (1 + r)\lambda^4$ and $S_2^2 = (1 + r^2)^2\lambda^4$, and $2(1 + r) \le (1 + r^2)^2$ for $r \ge 1$, with equality only at $r = 1$, since both sides equal 4 there and the right side has derivative $4r(1+r^2) \ge 8 > 2$. Equality in $2S_4 \le S_2^2$ forces $r = 1$, $\mu = \alpha^3$ and $S_4^+ = \alpha S_3^+$, hence one positive term α and one negative term $-\alpha$. □

THEOREM 4.1.9 (Biane form). *If all zeros of f_σ are real and simple for $0 \le \sigma < s$, then the upper right Dini derivative of m satisfies $D^+m \le m^2$ on $[0, s)$, and*

$$m(\sigma) \le \frac{m(0)}{1 - \sigma m(0)} \qquad \big(0 \le \sigma < \min(s, 1/m(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \ge 1/\min_{(z_k, z_{k+1})} \mathcal{L}_p$. This bound on s_1 contains that of Theorem 4.1.7, since $m(0) \le \ell(0)$, and it is attained by $z^2 - a^2$.

Proof. Since $\mathcal{L} \to +\infty$ at both ends of the gap, the minimisers are interior and m is continuous. If u is a minimiser at time σ, then $m(\sigma + h) \le \mathcal{L}(u, \sigma + h)$ for small $h > 0$, so by Lemma 4.1.5, with $\mathcal{L}'(u) = 0$,

$$D^+m(\sigma) \le \partial_\sigma \mathcal{L}(u, \sigma) = \mathcal{L}''(u) - 2\mathcal{L}(u)^2 = 6S_4 - 2S_2^2 \le S_2^2 = m(\sigma)^2,$$

the last step by Lemma 4.1.8, as $S_3 = 0$ at u. The lower right Dini derivative of the continuous function $1/m(\sigma) + \sigma$ is therefore nonnegative, so the function does not decrease, and $1/m(\sigma) \ge 1/m(0) - \sigma$. A collision forces $m \ge 8/g^2 \to \infty$, as in the proof of Theorem 4.1.7. For $z^2 - a^2$ the minimiser is the critical point 0, so $m = \ell$. □

As a numerical test, we draw 2400 polynomials of degree 3 to 12 with only real zeros, 600 in each of four families (zeros uniform on an interval; Gaussian; uniform with one added zero at 2 to 30% of the mean spacing from another, a Lehmer type pair; and a perturbed lattice), follow the zeros of $e^{sD^2}p$ until at most one real zero is left, and locate each collision as a double zero by Newton's method in (z, s) at 40 digits. For the 2398 polynomials it completed (two failed in the location of a collision), the first collision satisfied $s_1\ell(0) \ge 1.0000136$ and $s_1 m(0) \ge 1.0000055$, both for a Lehmer type pair, with medians 1.061 and 1.039.

After the first collision Theorem 4.1.9 no longer applies, yet each of the 4574 later collisions of two zeros adjacent at $s = 0$ satisfied $s_c m(0) \geq 1.0000326$, although the quantity δ of Corollary 4.1.13 below was positive at some step before 806 of them.

For H_t the hypothesis of Theorem 4.1.9 fails, but the argument survives with a correction. At the minimiser u, replace every nonreal pair $a \pm ib$ by the double real point a, its shadow.

LEMMA 4.1.10. *(i) For every real square summable sequence,* $S_4 \leq \frac{1}{2}S_2^2 + S_2^{1/2}|S_3|$.

(ii) If $w = 1/(d - ib)$ *with* $d \neq 0$ *and* b *real, then* $\left|2\operatorname{Re} w^k - 2d^{-k}\right| \leq k(k+1)\, b^2/|d|^{k+2}$ *for every* $k \geq 1$.

Proof. (i) Replacing the sequence by its negative if necessary, let $S_3 = \varepsilon \geq 0$, so $S_3^+ = S_3^- + \varepsilon$ in the notation of the proof of Lemma 4.1.6. Then $S_4 \leq \alpha S_3^+ + \beta S_3^- = (\alpha+\beta)S_3^- + \alpha\varepsilon$, where $S_3^- \leq \beta^3$ and $S_3^- = S_3^+ - \varepsilon \leq \alpha^3$. As in the proof of Lemma 4.1.8, $(\alpha+\beta)\min(\alpha,\beta)^3 \leq \frac{1}{2}(\alpha^2+\beta^2)^2 = \frac{1}{2}S_2^2$, and $\alpha\varepsilon \leq S_2^{1/2}|S_3|$. (ii) With $\epsilon = b/d$ and $\theta = \arctan\epsilon$, $w^k = d^{-k}(1-i\epsilon)^{-k}$ and $\operatorname{Re}(1-i\epsilon)^{-k} = A\cos k\theta$ with $A = (1+\epsilon^2)^{-k/2}$. Now $1 - A\cos k\theta = (1-A) + A(1-\cos k\theta)$, where $0 \leq 1-A \leq \frac{k}{2}\epsilon^2$ and $0 \leq 1-\cos k\theta \leq \frac{1}{2}k^2\theta^2 \leq \frac{1}{2}k^2\epsilon^2$, so $|\operatorname{Re}(1-i\epsilon)^{-k} - 1| \leq \frac{1}{2}(k+k^2)\epsilon^2$; multiply by $2|d|^{-k}$. □

REMARK 4.1.11. The coefficient of the second term in (i) can be halved, and $\frac{1}{2}$ is then best possible, since a single nonzero term gives $S_4 - \frac{1}{2}S_2^2 = \frac{1}{2}S_2^{1/2}|S_3|$. Normalise $S_2 = 1$ and $S_3 = \varepsilon \geq 0$ and put $\mu = S_3^-$, so that $S_4 \leq \alpha(\mu+\varepsilon) + \beta\mu$ with $0 \leq \mu \leq \min(\beta^3, \alpha^3 - \varepsilon)$; the bound increases with μ. For $\varepsilon \leq \alpha^3 - \beta^3$ it is at most $(\alpha+\beta)\beta^3 + \alpha\varepsilon$, which is linear in ε, and for $\varepsilon \geq \max(0, \alpha^3 - \beta^3)$ it is at most $\alpha^4 + \alpha^3\beta - \beta\varepsilon$, which decreases. So the bound minus $\frac{1}{2}(1+\varepsilon)$ is largest at $\varepsilon = 0$ or, when $\alpha > \beta$, at $\varepsilon = \alpha^3 - \beta^3$. At $\varepsilon = 0$ the bound is $(\alpha+\beta)\min(\alpha,\beta)^3 \leq \frac{1}{2}$, as in the proof of Lemma 4.1.8. At $\varepsilon = \alpha^3 - \beta^3$ it is $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$, and $\alpha^4 + \beta^4 - \frac{1}{2} \leq \frac{1}{2}(\alpha^3 - \beta^3)$ reads $(\alpha^2-\beta^2)^2 \leq \alpha^3 - \beta^3$, that is $(\alpha-\beta)(\alpha+\beta)^2 \leq 1 + \alpha\beta$; with $\nu = 2\alpha\beta \in [0,1]$ this is $(1-\nu)^{1/2}(1+\nu) \leq 1 + \frac{\nu}{2}$, which after squaring becomes $-\nu^2 - \nu^3 \leq \frac{1}{4}\nu^2$. Hence

$$S_4 \leq \tfrac{1}{2}S_2^2 + \tfrac{1}{2}S_2^{1/2}|S_3|,$$

and the coefficient 72 in Theorem 4.1.12 below may be replaced by 36; we keep the form of [239], which the computations below use. The bound (ii) is sharp as $b/d \to 0$, since $\operatorname{Re}(1-i\epsilon)^{-k} = 1 - \frac{1}{2}k(k+1)\epsilon^2 + O(\epsilon^4)$.

THEOREM 4.1.12 (local bound). *Let* u *be a minimiser of* $\mathcal{L}$ *over the gap at time* s, *with* $m = \mathcal{L}(u) > 0$, *and let* $B_p = \sum b^2|u-a|^{-p}$, *summed over the conjugate pairs* $a \pm ib$, $b > 0$, *of nonreal zeros of* f_s, *each pair once, be finite for* $p = 4, 5, 6$.

Then

$$\mathcal{L}''(u) - 3m^2 \le \eta\, m^2, \qquad \eta = \frac{36\, mB_4 + 108\, B_4^2 + 72\,(m + 6B_4)^{1/2} B_5 + 120\, B_6}{m^2},$$

and $D^+ m \le (1+\eta)m^2$. *Hence, if the two zeros bounding the gap meet at* s_c, *and* $m > 0$ *and* $\eta \le \bar\eta$ *on* $[0, s_c)$, *then* $s_c \ge 1/\big((1+\bar\eta)m(0)\big)$.

Proof. Let S'_k be the power sums of the real sequence in which every nonreal pair $a \pm ib$ is replaced by its shadow, two terms $1/(u-a)$, and let $D_k = S_k - S'_k$. By Lemma 4.1.10(ii) with $d = u - a$, $|D_2| \le 6B_4$, $|D_3| \le 12B_5$ and $|D_4| \le 20B_6$. Since $S_3 = 0$ at u, $S'_3 = -D_3$, and Lemma 4.1.10(i) for the shadow sequence gives $S'_4 \le \frac12 S_2'^2 + S_2'^{1/2}|D_3|$. With $S_4 = S'_4 + D_4$, $0 \le S'_2 = m - D_2 \le m + 6B_4$ and $S_2'^2 - m^2 = -2mD_2 + D_2^2$,

$$\begin{aligned} 6S_4 - 3m^2 &\le 3(S_2'^2 - m^2) + 6S_2'^{1/2}|D_3| + 6|D_4| \\ &\le 6m|D_2| + 3D_2^2 + 6S_2'^{1/2}|D_3| + 6|D_4| \le \eta m^2, \end{aligned}$$

and $\mathcal{L}''(u) = 6S_4$. As in the proof of Theorem 4.1.9, $D^+ m \le \mathcal{L}''(u) - 2m^2 \le (1+\eta)m^2$, so the function $1/m(\sigma) + (1+\bar\eta)\sigma$ does not decrease on $[0, s_c)$. At the collision $m \to \infty$: the share of the two meeting zeros tends to $+\infty$, while that of the other zeros, which stay away from the point where the two meet, remains bounded. No nonreal zero tends to that point: if z_c is a zero of multiplicity k of f_{s_c}, then $f_{s_c-\sigma} = e^{-\sigma D^2} f_{s_c}$, so $\sigma^{-k/2} f_{s_c-\sigma}(z_c + \sqrt{\sigma}\, w)$ tends to a nonzero multiple of $H_k(w/\sqrt{2})$ as $\sigma \downarrow 0$, uniformly on compact sets, and by Hurwitz's theorem the k zeros of $f_{s_c-\sigma}$ near z_c lie close to $\sqrt{2\sigma}$ times the zeros of H_k, which are real and simple; so they are real, and real zeros only add to m. □

Corollary 4.1.13. *In the setting of Theorem 4.1.12 put* $\delta = \mathcal{L}''(u)/m^2 - 3$, *computed from* f_s *and its first four derivatives at* u. *Then* $D^+ m \le (1+\delta)m^2$ *and* $\delta \le \eta$, *and* $\delta \le 0$ *when all zeros of* f_s *are real. Hence, if the two zeros bounding the gap meet at* s_c, *and* $m > 0$ *and* $\delta \le \bar\delta$ *on* $[0, s_c)$ *with* $\bar\delta > -1$, *then* $s_c \ge 1/\big((1+\bar\delta)m(0)\big)$; *if* $\delta \le 0$ *throughout, the collision cannot come before Biane's time* $1/m(0)$, *exactly as for functions with only real zeros.*

Proof. The first display of the proof of Theorem 4.1.9 gives $D^+ m \le \mathcal{L}''(u) - 2m^2 = (1+\delta)m^2$, Theorem 4.1.12 gives $\delta \le \eta$, and Lemma 4.1.8 gives $\delta \le 0$ for real zeros. The rest is the comparison argument of the proof of Theorem 4.1.12. □

Write $m = P + Q$, where $P = (u - z_k)^{-2} + (u - z_{k+1})^{-2}$ is the share of the two zeros bounding the gap and Q the share of all the others, and let $W = \sum |u - z|^{-4}$ over the nonreal zeros z of f_s.

THEOREM 4.1.14 (a deadline). *While the two zeros bounding the gap are real, the right derivative $m'(s^+)$ exists and is the least value of $\mathcal{L}''(u) - 2m^2$ over the minimisers u, and at every minimiser*

$$\mathcal{L}''(u) - 2m^2 \geq m^2 - 6mQ + 3Q^2 - 6W,$$

with Q and W taken at u. Suppose that $Q \leq \rho$ and $W \leq \bar{W}$ at every minimiser for as long as the two zeros bounding the gap are real, and that $m(0) > r_1$, where $r_{1,2} = 3\rho \pm (9\rho^2 + 6\bar{W})^{1/2}$. Then the gap closes, one of the two zeros bounding it meeting another zero, at a time

$$s_c \leq s^* = \frac{1}{r_1 - r_2} \log \frac{m(0) - r_2}{m(0) - r_1}.$$

If all zeros are real, $\bar{W} = 0$ and, for $\rho > 0$, $s^* = (6\rho)^{-1} \log\big(m(0)/(m(0) - 6\rho)\big) = 1/m(0) + 3\rho/m(0)^2 + 12\rho^2/m(0)^3 + \cdots$.

Proof. Near a given time the minimisers lie in a fixed compact subinterval of the gap, and by Danskin's theorem the right derivative of the minimum exists and equals the least value of $\partial_s \mathcal{L}$ over the minimisers, which is $\mathcal{L}''(u) - 2m^2 = 6S_4 - 2m^2$ by Lemma 4.1.5. In S_4 the two zeros bounding the gap contribute $x_k^4 + x_{k+1}^4 \geq \frac{1}{2}(x_k^2 + x_{k+1}^2)^2 = \frac{1}{2}P^2$, the other real zeros contribute nonnegative terms, and each nonreal zero z contributes $\mathrm{Re}(u - z)^{-4} \geq -|u - z|^{-4}$. So $\mathcal{L}''(u) - 2m^2 \geq 3P^2 - 6W - 2m^2$, and $P = m - Q$ gives the first inequality. While $m > 0$, $Q \leq \rho$ and $W \leq \bar{W}$, it gives $m'(s^+) \geq m^2 - 6\rho m - 6\bar{W} = (m - r_1)(m - r_2)$, so m stays above the solution y of $y' = (y - r_1)(y - r_2)$, $y(0) = m(0) > r_1 \geq 0$, which increases and tends to $+\infty$ as $s \to s^*$. If the gap were still open on $[0, s^*]$, its two zeros real, simple and apart, then m, being at most $\mathcal{L}$ at the midpoint of the gap, would be bounded on $[0, s^*]$, a contradiction. The first collision of either zero need not be with the other: one of them may first meet the zero beyond it, and the theorem then bounds the time of that collision. For a pair that is known to collide, as in the census, s^* bounds the collision time of the pair. □

COROLLARY 4.1.15. *Put $\kappa = \mathcal{L}''(u) - 3P^2$, computed from f_s at u and the two zeros bounding the gap, so that*

$$\kappa = 3\big(x_k^2 - x_{k+1}^2\big)^2 + 6 \sum_{j \neq k, k+1} x_j^4 \geq -6W,$$

with $\kappa \geq 0$ when all zeros are real. Theorem 4.1.14 holds with $6\bar{W}$ replaced by $\bar{\kappa} = \max\big(0, \sup(-\kappa)\big)$, the supremum over the minimisers at the times before the collision, and this form is never weaker.

Proof. The identity is $6(x_k^4 + x_{k+1}^4) - 3(x_k^2 + x_{k+1}^2)^2 = 3(x_k^2 - x_{k+1}^2)^2$, and the inequality is that of the proof of Theorem 4.1.14. That proof gives $m'(s^+) = \kappa + 3P^2 - 2m^2 \geq m^2 - 6mQ + 3Q^2 - \bar{\kappa}$, and $\bar{\kappa} \leq 6\bar{W}$. □

The inequalities behind Theorems 4.1.12 and 4.1.14 were also tested on random configurations. For 29620 configurations of 3 to 11 real zeros and 1 to 7 conjugate pairs, at the critical point u of $\mathcal{L}$ on a random gap, with $\mathcal{L}(u) > 0$, the quantity $6S_4 - 3S_2^2$ never exceeded $0.016\,\eta S_2^2$, and $6S_4 \geq 3P^2 - 6W$ held up to rounding. On the polynomials of the test above, with ρ the largest value of Q at the minimisers before the collision, the deadline s^* of Theorem 4.1.14 is defined for 2308 of the 2398 first collisions, with $\bar{W} = 0$, and for 3322 of the 4574 later ones (3345 in the form of Corollary 4.1.15), and no collision came after it: the largest ratio s_c/s^* is $1 - 1.5 \times 10^{-11}$ for the first collisions and $1 - 3.1 \times 10^{-10}$ for the later ones.

Corollaries 4.1.13 and 4.1.15 bracket every collision time by quantities along the flow that need only H_t near the gap, while Theorems 4.1.12 and 4.1.14 need the nonreal zeros. A computation gives both kinds for each of the 58 collisions of the census of Section 4.1.3. It evaluates H_t and its first four z derivatives in ball arithmetic at 800 bits, refines the double zero $H_t = \partial_z H_t = 0$ by Newton's method in (z, t), and on a grid of about 140 times s in $[0, |t_{\mathrm{coll}}|)$ it follows the two zeros by Newton's method and computes the minimiser u, the critical point c and the quantities ℓ, m, Q, δ and κ, the last three in ball arithmetic, since near the collision m and P are large and nearly equal. For η and W it also follows, by Newton's method in the complex plane, the conjugate pairs born in the census collisions before the time considered (60 pairs, two of them from collisions among the zeros 203 to 206), with their mirror images $-a \pm ib$, and adds a model of the zeros above the census, $z = 2\gamma_j$ for $207 \leq j \leq 2000$, in which every such zero is half of a pair $a \pm ib$ with $b^2 = 4|t|$. Newton's method reproduces the census collision times to 5×10^{-12}.

First, $\delta < 0$ at every point of the grid on all 58 trajectories; the largest value is -3.1×10^{-8}, for $(186, 187)$. So the hypothesis of Corollary 4.1.13 with $\bar{\delta} = 0$ holds on the grid in spite of the nonreal zeros, and its conclusion, the lower bound $1/m(0)$, holds for all 58 collisions of the census, as the direct comparison of Section 4.1.4 shows ($t_{\mathrm{coll}}/\tau_B \geq 1.083$; Table 4.5). Likewise $\ell(s) \leq \ell(0)/(1 - s\ell(0))$ for $s < 1/\ell(0)$ and $m(s) \leq m(0)/(1 - sm(0))$ for $s < 1/m(0)$ at every point of the grid.

Second, $\kappa \geq 0$ at every point of the grid except on the trajectories of $(108, 109)$ and $(153, 154)$, where $\bar{\kappa} = 0.029$ and 0.071. The upper bound s^* of Corollary 4.1.15, with ρ the largest value of Q on the grid, is defined, $m(0) > r_1$, for 55 of the 58 collisions; it is not for $(153, 154)$, $(178, 179)$ and $(181, 182)$, for which $6\rho/m(0)$ is 0.986, 1.068 and 1.103. Every collision time of the census lies in its bracket $[1/m(0), s^*]$, with $s^* = \infty$ for those three, and Table 4.7 shows the seven pairs of the note [239]. The upper bound is tight when the pair dominates its gap: $s^*/|t_{\mathrm{coll}}|$ ranges from 1.0010, for $(186, 187)$ with $6\rho/m(0) = 0.151$, to 2.021, for $(48, 49)$ with $6\rho/m(0) = 0.965$.

Third, the correction η of Theorem 4.1.12, computed from the tracked pairs and the model above the census, is at most 3.25×10^{-5} for the seven pairs of Table 4.7,

but it reaches 0.53 for $(153, 154)$ and exceeds 10^{-2} for four more collisions, all at $t < -1.89$. With $\bar{\eta}$ in place of $\bar{\delta}$ the lower bound still holds for all 58, the closest case being again $(186, 187)$, whose collision time exceeds the bound by the factor 1.083. For $(34, 35)$, $(71, 72)$, $(91, 92)$ and five other late collisions more than half of B_4 at the worst point comes from the model, so $\bar{\eta}$ depends on it, while δ and κ depend on no model. The model is not pessimistic in every respect: the tracked pairs leave the axis faster than an isolated pair would, with b^2 up to 3.39 times $2(t_c - t)$.

$s = -t$	0	0.2	0.4	0.6	0.8	1.0	1.2
$(13, 14)$: $\ell(s)$	1.1033	1.2462	1.4539	1.7833	2.3846	3.8293	12.051
$\ell(0)/(1 - s\ell(0))$	1.1033	1.4156	1.9748	3.2638	9.3992		
$m(s)$	0.9676	1.1099	1.3170	1.6457	2.2461	3.6898	11.911
$m(0)/(1 - sm(0))$	0.9676	1.1998	1.5787	2.3071	4.2837	29.903	
$(19, 20)$: $\ell(s)$	1.2955	1.4520	1.6768	2.0293	2.6688	4.2153	13.848
$\ell(0)/(1 - s\ell(0))$	1.2955	1.7485	2.6889	5.8171			
$m(s)$	1.0446	1.1904	1.4029	1.7413	2.3644	3.8914	13.500
$m(0)/(1 - sm(0))$	1.0446	1.3205	1.7944	2.7990	6.3583		

Table 4.6: Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13, 14)$ and $(19, 20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 4.1.7 and 4.1.9; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of H_t.

pair	$m(0)$	$Q(0)$	ρ	$10^8\bar{\delta}$	$\bar{\eta}$	pairs	$1/m(0)$	s^*	$\lvert t_{\text{coll}}\rvert$
$(13, 14)$	0.9676	0.0604	0.0610	-9.0	2.8×10^{-7}	49	1.0334	1.2982	1.2853
$(19, 20)$	1.0446	0.0803	0.0879	-12.8	2.4×10^{-6}	48	0.9573	1.3325	1.2755
$(24, 25)$	1.1238	0.0794	0.0794	-10.2	3.2×10^{-5}	44	0.8898	1.1578	1.1439
$(27, 28)$	1.4415	0.0962	0.1012	-10.2	4.0×10^{-6}	36	0.6937	0.9006	0.8811
$(34, 35)$	2.8970	0.0968	0.0969	-4.3	1.7×10^{-9}	11	0.3452	0.3853	0.3846
$(71, 72)$	3.9727	0.1605	0.1678	-5.5	2.0×10^{-9}	4	0.2517	0.2903	0.2880
$(91, 92)$	4.0736	0.1701	0.1757	-5.7	3.1×10^{-9}	3	0.2455	0.2841	0.2821

Table 4.7: Two sided brackets for the collision time of seven pairs, in the units of H_t: the lower bound $1/m(0)$ of Corollary 4.1.13, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound s^* of Corollary 4.1.15, with ρ the largest value of Q and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 4.1.12 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision.

Freezing Q at its value at $s = 0$, that is, taking $\rho = Q(0)$ and $\bar{\kappa} = 0$ in

Corollary 4.1.15, gives a predictor of the collision time from H_0 alone,

$$t_{\mathrm{coll}} \approx -\frac{1}{6Q(0)} \log \frac{m(0)}{m(0) - 6Q(0)},$$

the last column of Table 4.5. It is not a bound, since Q grows along the flow ($\rho > Q(0)$ for 41 of the 58 collisions) and the predictor falls on both sides of t_{coll}, but it is accurate when the pair dominates its gap. Over the 28 collisions of Table 4.5 its relative error is at most 0.32% for the six with $6Q(0)/m(0) < 0.3$ and 3.75% in all, for $(51, 52)$; over the 52 collisions of the census with $6Q(0)/m(0) < 0.8$ it is at most 5.75%, for $(130, 131)$. It fails when the other zeros carry most of $m(0)$: the six collisions with $6Q(0)/m(0) \geq 0.8$, all at $t < -1.87$, have errors from 10.4%, for $(68, 69)$, to 46.6%, for $(188, 189)$. To first order in $Q(0)$ the predictor is $1/m(0) + 3Q(0)/m(0)^2$ in absolute value, and if the two zeros sit symmetrically about u, so that $P(0) = 8/D_0^2$, it is

$$\frac{D_0^2}{8} + \frac{D_0^4 Q(0)}{32} + O\big(Q(0)^2\big).$$

The bound Λ_k of Csordas, Smith and Varga [51] in Section 4.1.6 has the expansion $-\Delta_k^2/8 - \Delta_k^4 g_k/64 - \cdots$ with $\Delta_k = D_0$: the same two terms with g_k in place of $2Q(0)$, and indeed $g_k \approx 2Q(0)$, since g_k adds up, over the other zeros, the inverse squares of their distances to each of the two zeros of the pair. For $(71, 72)$, $g_{71} = 0.33734$ and $2Q(0) = 0.3210$, and the frozen predictor gives -0.2883, against $t_{\mathrm{coll}} = -0.2880$ and $\Lambda_{71} = -0.3048$.

Both brackets rest on hypotheses along the flow, on δ or η for the lower bound and on Q and κ or W for the upper bound, which the computation checks on the grid of times but which cannot yet be guaranteed in advance.

Problem 4.1. Find a bound for how far the nonreal zeros of H_t, $t < 0$, travel, and for how close to a gap between real zeros they come. Such control would make the brackets unconditional statements about H_t, computed from H_0 alone.

Biane's work meets ξ in two more places. With Pitman and Yor [26], following Williams [275], he wrote $2\xi(s)$ as the Mellin transform of a sum of two Bessel hitting times, the identity behind the Pólya variable of Section 2.1. And in [25] he revisits the paper of Pólya [245], treated in full in Section 18.1.

4.1.6 Lehmer pairs and lower bounds for Λ

Every collision time is a lower bound for Λ, because nonreal zeros persist backward in time, and Lehmer pairs, pairs with unusually small D_0, collide at times close to zero. Csordas, Smith and Varga [51] turned this into a theorem that uses only the zeros of H_0. Write $x_j = 2\gamma_j$ for the positive zeros of H_0, $x_{-j} = -x_j$,

$\Delta_k = x_{k+1} - x_k$ and

$$g_k = \sum_{j \neq k,k+1}{}' \Big[\frac{1}{(x_k - x_j)^2} + \frac{1}{(x_{k+1} - x_j)^2} \Big],$$

where the prime restricts the sum to nonzero integers j, of either sign. The pair (x_k, x_{k+1}) is called a Lehmer pair if $\Delta_k^2 g_k < \frac{4}{5}$, and then [51, Theorem 1]

$$\Lambda \geq \Lambda_k = \frac{\big(1 - \frac{5}{4}\Delta_k^2 g_k\big)^{4/5} - 1}{8 g_k} = -\frac{\Delta_k^2}{8} - \frac{\Delta_k^4 g_k}{64} - \cdots .$$

The leading term is the isolated pair collision time $-D_0^2/8$ with $D_0 = \Delta_k$. The binomial series of $(1-x)^{4/5}$ has only negative coefficients after the constant term, so every further term is negative and decreases in g_k, and so does Λ_k.

For the closest pair below height 200, $\Delta_{71} = 1.44863$ and, from the first 1700 zeros, $g_{71} = 0.33734$, so $\Delta^2 g = 0.708 < \frac{4}{5}$: the pair is a Lehmer pair in this sense, and it gives $\Lambda_{71} = -0.3048$. The omitted zeros above height 2197 increase g by less than 10^{-3}, which lowers Λ_{71} by less than 3×10^{-4}, so with the tail the bound is -0.305. The theorem is proved, but this evaluation of it is a floating point computation with zeros computed by `mpmath` [213]. The bound lies below the computed collision time -0.288, as expected. The theorem itself does not order the bound and the collision time: its proof, by contradiction, assumes that H_t has only real zeros for $\Lambda_k \leq t \leq 0$, which fails for every $t < 0$ because $\Lambda \geq 0$.

Table 4.8 lists the published bounds. The last four lower bounds before 2020, among them that of Csordas, Odlyzko, Smith and Varga [50], came from this theorem applied to Lehmer pairs of increasingly high quality, found at heights far greater than 7005. Rodgers and Tao proved $\Lambda \geq 0$ by a different argument, showing that $\Lambda < 0$ would force the zeros into a local equilibrium incompatible with Montgomery's pair correlation estimates [211].

4.1.7 The shielding deficit

Let $a < b$ be adjacent simple real zeros of H_t with gap $g = b - a$, and put

$$S_{a,b} = \sum_{\rho \neq a,b} \frac{1}{(a-\rho)(b-\rho)}, \qquad \mathrm{Def} = 2 - g^2 S_{a,b},$$

the sum running over all the other zeros ρ (conjugate pairs, if any, give a real sum). Subtracting the two equations of motion of Section 4.1.1 gives $\dot g = 4/g - 2gS_{a,b}$, that is

$$\frac{d(g^2)}{dt} = 4\,\mathrm{Def}. \tag{4.1.2}$$

For an isolated pair $S_{a,b} = 0$ and $\mathrm{Def} = 2$, which is the law $g^2 = g(0)^2 + 8t$ of Section 4.1.1. For real zeros $S_{a,b} > 0$, and $2xy \leq x^2 + y^2$ gives $S_{a,b} \leq g_k/2$

bound	authors	year
$\Lambda > -\infty$	Newman	1976
$\Lambda > -50$	Csordas, Norfolk, Varga	1988
$\Lambda > -5$	te Riele	1991
$\Lambda > -0.385$	Norfolk, Ruttan, Varga	1992
$\Lambda > -0.0991$	Csordas, Ruttan, Varga	1991
$\Lambda > -4.379 \times 10^{-6}$	Csordas, Smith, Varga	1994
$\Lambda > -5.895 \times 10^{-9}$	Csordas, Odlyzko, Smith, Varga	1993
$\Lambda \geq -2.63 \times 10^{-9}$	Odlyzko	2000
$\Lambda > -1.15 \times 10^{-11}$	Saouter, Gourdon, Demichel	2011
$\Lambda \geq 0$	Rodgers, Tao	2020
$\Lambda \leq \frac{1}{2}$	de Bruijn	1950
$\Lambda < \frac{1}{2}$	Ki, Kim, Lee	2009
$\Lambda \leq 0.22$	Polymath	2019
$\Lambda \leq 0.2$	Platt, Trudgian	2021

Table 4.8: Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [250, Table 1], upper bounds from [61], [182], [247] and [232]. Odlyzko's bound is the one computed in [223, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication.

with the g_k of Csordas, Smith and Varga (computed here from the zeros of H_t) in Section 4.1.6, so $\mathrm{Def} \geq 2 - \Delta_k^2 g_k/2$. A regular lattice of gap g shields exactly: each side contributes $\sum_{m\geq 1} 1/(m(m+1)) = 1$, the factor 2 being what the other zeros must supply. On each side number the exterior gaps outward from the pair, write them $g(1+\varepsilon_m)$ with $\varepsilon_m > -1$, and put $D_0 = 0$, $D_m = \sum_{j\leq m} \varepsilon_j$. The mth exterior zero on that side is at distance $g(m + D_m)$ from the nearer end of the pair.

THEOREM 4.1.16 (Newton). *On one side, for every $N \geq 1$,*

$$1 - \sum_{m=1}^{N} \frac{1}{(m+D_m)(m+1+D_m)} = \sum_{m=1}^{N} \frac{\varepsilon_m}{(m+D_{m-1})(m+D_m)} + \frac{1}{N+1+D_N}.$$

If all zeros are real and the exterior zeros tend to infinity on both sides, then $\mathrm{Def} = \sum_{\text{sides}} \sum_{m\geq 1} \varepsilon_m / \big((m+D_{m-1})(m+D_m)\big)$.

Proof. Put $y_m = m + D_{m-1}$ and $z_m = m + D_m$, so $z_m - y_m = \varepsilon_m$, $y_1 = 1$ and $y_{m+1} = z_m + 1$. The mth exterior zero contributes $1/\big(z_m(z_m+1)\big)$ to $g^2 S_{a,b}$, because $a - \rho = -g(z_m+1)$ and $b - \rho = -g z_m$. Now $\frac{1}{z_m(z_m+1)} + \frac{z_m - y_m}{y_m z_m} = \frac{1}{y_m} - \frac{1}{z_m+1}$, and summing over $m \leq N$ gives $1/y_1 - 1/(z_N+1) = 1 - 1/(N+1+D_N)$. The last term tends to zero when the zeros tend to infinity; add the two sides. □

COROLLARY 4.1.17. *(a) If all $\varepsilon_m = 0$ then* $\mathrm{Def} = 0$ *and g is constant; the lattice of zeros of $e^{\kappa^2 t}\sin \kappa z$ is an example. (b) Keeping R exterior zeros on each side of*

a regular lattice leaves $\mathrm{Def} = 2/(R+1)$*, the size of the omitted tail. (c) A single enlarged exterior gap* $g(1+\delta)$ *at position* j *contributes exactly* $\delta/(j(j+\delta))$*. (d) With* Def *as above and* $d_{\nu,j}$ *the exterior gaps on the sides* $\nu = L, R$*,*

$$[\mathrm{Def}]_+ \le g^{-1} \sum_{j\ge 1} j^{-2} \big[(d_{L,j}-g)_+ + (d_{R,j}-g)_+\big],$$

and by (c) the constant j^{-2} *is optimal to first order. (e) If the configuration has a smallest gap* $g > 0$ *(a finite configuration or a lattice) and* g *is that gap, then every* $\varepsilon_m \ge 0$*, so* $\mathrm{Def} \ge 0$ *and the smallest gap does not shrink forward in time, with equality only for the lattice.*

Proof. (a) to (c) are read off from the theorem; in (b), $2(1-R/(R+1)) = 2/(R+1)$. For (d) let $P_m = \sum_{i\le m}(\varepsilon_i)_+$. Then $z_m \le m + P_m$ and $1/(z(z+1))$ decreases, so by the theorem applied to the sequence $(\varepsilon_i)_+$, one side contributes at most $1 - \sum_m \frac{1}{(m+P_m)(m+P_m+1)} = \sum_m \frac{(\varepsilon_m)_+}{(m+P_{m-1})(m+P_m)} \le \sum_m (\varepsilon_m)_+/m^2$. (e) is the theorem with $\varepsilon_m \ge 0$ and (4.1.2). □

Part (e) says nothing about H_t: its mean gap at height T is of order $1/\log T$, so there is no smallest gap. Being minimal among the neighbours is not enough either: take $g = 1$ and exterior zeros at distances 1, 1.01, 1.02 from the pair on both sides, so that the nearest exterior gaps equal g and the next ones are 0.01. Then $g^2 S_{a,b} \ge 1 + \frac{20000}{20301} + \frac{20000}{20604} = \frac{340039}{115039} > 2$, so $\mathrm{Def} < 0$ and this gap shrinks forward. Finally, (4.1.2) integrates to an identity, $4\int \mathrm{Def}\, dt = \Delta(g^2)$, so a bound on the shielding budget that already contains the moving gap is circular; an estimate has to bound Def from the exterior zeros alone.

4.1.8 The Thorin atoms of the tilted clock

Multiplying the kernel by e^{tu^2} tilts the clock. For $t \ge \Lambda$ the function R_t of Section 4.1.1 is the Laplace transform of a GGC T_t, the tilted clock, with the Thorin atoms $a_j(t) = z_j(t)^2/4 = \theta_j(t)^2$ (Theorem 17.1.16), and $T_0 = T$ is the clock of Section 2.1, which exists unconditionally (Theorem 8.1.1).

Proposition 4.1.18. *For* $t > 0$*, with* Z *a standard normal variable,*

$$H_t(z) = \mathbb{E}\, H_0\big(z + i\sqrt{2t}\, Z\big), \qquad z \in \mathbb{C},$$
$$\frac{1}{R_t(\sigma^2)} = \frac{\mathbb{E}\, \xi\big(\frac12 + \sigma + \sqrt{t/2}\, Z\big)}{\mathbb{E}\, \xi\big(\frac12 + \sqrt{t/2}\, Z\big)}, \qquad \sigma \ge 0.$$

Proof. With $c = \sqrt{2t}$ we have

$$\mathbb{E}\cos\big((z + icZ)u\big) = \cos(zu)\, \mathbb{E}\cosh(cuZ) = \cos(zu)\, e^{tu^2},$$

because the part odd in Z has mean zero. Integrating against Φ_{dBN} gives the first identity, Fubini applying because $\Phi_{\mathrm{dBN}}(u)$ decays like $\exp(-\pi e^{4u})$; the same computation with $e^{tu^2}\Phi_{\mathrm{dBN}}$ in place of Φ_{dBN} gives $H_{t+s}(z) = \mathbb{E}\, H_t\big(z + i\sqrt{2s}\, Z\big)$ for real t and $s > 0$. For the second identity, $8H_0(\pm 2i\sigma) = \xi(\frac{1}{2} + \sigma)$ and Z is symmetric. We checked the first identity at $t = 0.3$ and $z = -4i$ to 40 digits. $\square$

So for $t > 0$ the tilt is Gaussian convolution, along the real axis of σ, of the even positive function $\sigma \mapsto \xi(\frac{1}{2} + \sigma)$, with variance $t/2$. Theorem 8.4.1(b) is the smoothing along the real axis of z, which lowers t.

The Newton identities of Section 4.1.3 give the mean and variance of T_t from the moments $M_k(t)$ of the reweighted kernel,

$$\mathbb{E}T_t = \sum_j \frac{1}{a_j(t)} = \frac{M_1(t)}{2M_0(t)}, \qquad \operatorname{Var} T_t = \sum_j \frac{1}{a_j(t)^2} = \frac{M_1(t)^2}{4M_0(t)^2} - \frac{M_2(t)}{12M_0(t)},$$

and in the variable a_j the motion of the zeros in Section 4.1.1 reads

$$\frac{da_j}{dt} = \frac{1}{2} + 2\sum_{k \neq j} \frac{a_j}{a_j - a_k}, \tag{4.1.3}$$

the term $\frac{1}{2}$ coming from the mirror zero $-z_j$ and the others from the pairs $\pm z_k$. The table gives the first four atoms at $t = 0$ and $t = 0.2$, with $\mathbb{E}T_t$ and $\operatorname{Var} T_t$, the difference quotient $\Delta a_j/\Delta t = (a_j(0.2) - a_j(0))/0.2$ and the rate $\dot{a}_j(0)$ from (4.1.3).

	$t = 0$	$t = 0.2$	$\Delta a_j/\Delta t$	$\dot{a}_j(0)$
a_1	199.7905	198.1565	-8.170	-8.193
a_2	441.9262	438.7638	-15.812	-15.872
a_3	625.5430	622.6126	-14.652	-14.652
a_4	925.6731	919.9043	-28.844	-29.069
$\mathbb{E}T_t$	0.0231049931	0.0232009711		
$\operatorname{Var} T_t$	3.71726×10^{-5}	3.77203×10^{-5}		

At $t = 0$ the a_j are γ_j^2, the Thorin atoms of T if RH holds (Proposition 2.1.1), and $\mathbb{E}T_0$ and $\operatorname{Var} T_0$ are the constants of Section 2.1. The rate is the sum (4.1.3) over the first 8000 zeros completed by the tail $-2a_j(\log(\gamma_{8000}/2\pi) + 1)/(2\pi\gamma_{8000})$; a central difference of the zeros of H_t at $t = \pm 0.01$ and ± 0.005, with Richardson extrapolation, reproduces it to 2×10^{-5}, the order of the error of the tail.

Since $M_k'(t) = \frac{1}{4}M_{k+1}(t)$, the mean of the tilted clock increases along the flow:

$$\frac{d}{dt}\mathbb{E}T_t = \frac{M_0(t)M_2(t) - M_1(t)^2}{8M_0(t)^2} > 0$$

by the Cauchy Schwarz inequality, at every t at which T_t exists [144]. The atoms need not all decrease. Take the common coupling $T_t = \sum_j E_j/a_j(t)$ with the same independent standard exponential variables E_j for every t, which exists for $t \geq 0.2$ because every zero is real there [232]. The equation of motion gives $\dot{z}_{64}(0.2) = +0.26897$ and $\dot{z}_{92}(0.2) = +0.13385$, the only two rising zeros among the first 100, so a_{64} and a_{92} increase (computed, not certified: Gauss Hermite quadrature in double precision, the two velocities unchanged to 3×10^{-17} between 60 and 90 nodes). Put $b_j(t) = 1/a_j(t)$ and choose $h > 0$ so small that $c = b_{64}(0.2+h) - b_{64}(0.2) < 0$. Then $T_{0.2+h} - T_{0.2} = cE_{64} + Y$ with $Y = \sum_{j\neq 64} \big(b_j(0.2+h) - b_j(0.2)\big)E_j$, which converges absolutely almost surely, since $|Y| \leq T_{0.2+h} + T_{0.2}$, and is independent of E_{64}. Given any value of Y, the unbounded variable E_{64} makes $cE_{64} + Y < 0$ with positive probability, so $T_{0.2+h} < T_{0.2}$ with positive probability, and this coupling is not almost surely monotone on any right neighbourhood of $t = 0.2$ (computed). At $t = 0$ the same argument with $\dot{z}_{35}(0) > 0$ of Proposition 4.1.1 (certified) gives the statement on every right neighbourhood of 0, conditional on RH, which the coupling at $t = 0$ needs. The increase of the mean therefore does not come from monotone atoms.

Problem 4.2. Is the tilted clock stochastically increasing in t on $t \geq \Lambda$ (so, unconditionally, on $t \geq 0.2$), that is, is $P(T_t \leq x)$, whose Laplace transform in x is $R_t(u)/u$, nonincreasing in t for every $x > 0$? The failure of the coupling above does not decide it, since another coupling may be monotone, and an increasing mean is necessary but not sufficient.

For large t the tilt concentrates the kernel. Under the probability density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $(0,\infty)$, with expectation $\mathbb{E}_t$, we have $1/R_t(s) = \mathbb{E}_t \cosh(2\sqrt{s}\,u)$ and $H_t(z)/H_t(0) = \mathbb{E}_t \cos(zu)$.

Proposition 4.1.19. *As $t \to \infty$ let u_t be the mode of $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$, so that $e^{4u_t} \sim tu_t/(2\pi)$. Then*

$$R_t\big(s/u_t^2\big) \to \frac{1}{\cosh(2\sqrt{s})}, \qquad \frac{T_t}{u_t^2} \Rightarrow 2C, \qquad u_t\, z_j(t) \to \pi\big(j - \tfrac{1}{2}\big),$$

where C is the exit time of Brownian motion from $(-1,1)$, whose Laplace transform is $1/\cosh\sqrt{2\theta}$.

Proof. Put $g(u) = tu^2 + \log\Phi_{\mathrm{dBN}}(u) = tu^2 + 9u + \log 2\pi^2 - \pi e^{4u} + o(1)$. The stationary point satisfies $e^{4u_t} = (2tu_t + 9)/(4\pi) + o(1)$ and $g''(u_t) = -8tu_t + O(t)$, so under $\mathbb{E}_t$ the variable u has mean $u_t + o(1)$ and standard deviation of order $(tu_t)^{-1/2} \to 0$, with Gaussian tails in $(u - u_t)\sqrt{tu_t}$. Hence $u/u_t \to 1$ in probability with $\mathbb{E}_t e^{cu/u_t}$ bounded for each c, and $\mathbb{E}_t \cosh(2\sqrt{s}\,u/u_t) \to \cosh(2\sqrt{s})$. The Laplace transform $1/\cosh(2\sqrt{s})$ is that of $2C$, and the zeros follow from Hurwitz's theorem applied to $z \mapsto H_t(z/u_t)/H_t(0) \to \cos z$, uniformly on compacts. □

The convergence is fast: at $t = 300$, 1000 and 3000 the mode is $u_t = 0.9602$, 1.3418, 1.6707, and $u_t z_1(t) = 1.57334$, 1.57112, 1.57086 against $\pi/2 = 1.57080$. The path of clocks T_t thus runs from the Brownian exit time $2C$, after the scaling by u_t^2, at $t = \infty$ to T at $t = 0$, and consists of GGCs for $t \ge \Lambda$ (Theorem 17.1.16). The limit shape is a fixed shape of the flow: the zeros of $e^{\kappa^2 t}\cos\kappa z$ are the equally spaced points $\pi(j - \frac{1}{2})/\kappa$ at all times [217].

Gaussian smoothing in the variable $\sigma = \sqrt{s}$ preserves reciprocal GGC transforms in general, not only for the tilt.

THEOREM 4.1.20 (Gaussian smoothing of reciprocal GGC transforms). *Let $a_j > 0$ with $\sum_j a_j^{-1} < \infty$, and let $M(\sigma) = \prod_j (1 + \sigma^2/a_j)$, the reciprocal of the Laplace transform, at $s = \sigma^2$, of the GGC with the discrete Thorin measure $\sum_j \delta_{a_j}$ and no drift; M is an even entire function of genus zero in σ^2. For $c > 0$ put*

$$M_c(\sigma) = \frac{\mathbb{E}\, M(\sigma + cZ)}{\mathbb{E}\, M(cZ)}.$$

Then $M_c(\sigma) = \prod_j (1 + \sigma^2/a_j(c))$ with real $a_j(c) > 0$, so that $1/M_c(\sqrt{s})$ is the Laplace transform of a GGC S_λ with the Thorin measure $\sum_j \delta_{a_j(c)}$, where $\lambda = 2c^2$. The atoms obey (4.1.3) *with t replaced by λ while they are simple, and*

$$\frac{d}{d\lambda}\mathbb{E}S_\lambda = (\mathbb{E}S_\lambda)^2 - \frac{3}{2}\operatorname{Var} S_\lambda.$$

Proof. Write $\lambda = 2c^2$, $D = d/d\sigma$ and $u(\lambda, \sigma) = \mathbb{E}M(\sigma + cZ) = e^{\lambda D^2/4} M(\sigma)$, so that $\partial_\lambda u = \frac{1}{4}\partial_\sigma^2 u$, the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ in the variable $\sigma = iz/2$. The function $f(\tau) = M(i\tau) = \prod_j (1 - \tau^2/a_j)$ is an even real entire function of genus at most one with only real zeros, and $u(\lambda, i\tau) = \mathbb{E}f(\tau + icZ) = e^{-\lambda D_\tau^2/4} f(\tau)$ with $D_\tau = d/d\tau$. By Pólya's theorem that e^{-aD^2}, $a > 0$, preserves the class of real entire functions of genus at most one with only real zeros [246] (for H_t it is Theorem 8.4.1(c)), $u(\lambda, i\tau)$ has only real zeros in τ. It is even and $u(\lambda, 0) = \mathbb{E}M(cZ) \ge 1$, so $M_c = u/u(\lambda, 0)$ has the product form with real $a_j(c) > 0$. For a single atom, $\mathbb{E}\big(1 + (\sigma + cZ)^2/a\big) = (1 + c^2/a)\big(1 + \sigma^2/(a + c^2)\big)$: the atom moves from a to $a + \lambda/2$, the drift $\frac{1}{2}$ of (4.1.3). In general the atoms obey (4.1.3) by the computation of Section 4.1.1, which uses only the heat equation, the product form above and the simplicity of the zeros.

For the mean write $u = c_0 + c_2\sigma^2 + c_4\sigma^4 + \cdots$, so that $\dot{c}_0 = \frac{1}{2}c_2$ and $\dot{c}_2 = 3c_4$. Since $u/c_0 = \prod_j (1 + \sigma^2/a_j(\lambda)) = 1 + m\sigma^2 + \frac{1}{2}(m^2 - p)\sigma^4 + \cdots$ with $m = \sum_j a_j^{-1}$ and $p = \sum_j a_j^{-2}$, we have $c_2 = mc_0$ and $c_4 = \frac{1}{2}(m^2 - p)c_0$, and

$$\dot{m} = \frac{\dot{c}_2}{c_0} - m\,\frac{\dot{c}_0}{c_0} = \frac{3}{2}(m^2 - p) - \frac{1}{2}m^2 = m^2 - \frac{3}{2}p.$$

Finally $\mathbb{E}S_\lambda = m$ and $\operatorname{Var} S_\lambda = p$. □

Take $M(\sigma) = 1/R_t(\sigma^2)$. The proof of Proposition 4.1.18 gives

$$M_c(\sigma) = \frac{1}{R_{t+2c^2}(\sigma^2)},$$

so for $t \geq \Lambda$ the identity becomes

$$\frac{d}{dt}\mathbb{E}T_t = (\mathbb{E}T_t)^2 - \frac{3}{2}\operatorname{Var} T_t.$$

For the tilt it also follows without the zeros from $dM_k/dt = M_{k+1}/4$ and the formulas for $\mathbb{E}T_t$ and $\operatorname{Var} T_t$ above. At $t = 0$ both sides equal 4.7808×10^{-4} and at $t = 0.2$ they equal 4.8170×10^{-4}.

The clock $T_{0.2}$ is an unconditional GGC (Theorem 17.1.16). By Proposition 4.1.18 its Laplace transform is the reciprocal of the Gaussian smoothing of $\xi(\frac{1}{2} + \sigma)$ with variance 0.1 in σ, normalised at $\sigma = 0$. It is close to T: the means differ by 9.6×10^{-5} on 0.0231, and $R_{0.2}(s)/R_0(s) - 1$ is -9.6×10^{-5} at $s = 1$, -9.3×10^{-4} at $s = 10$ and -7.6×10^{-3} at $s = 100$. In this sense T is a GGC up to a smoothing of variance 0.1, and RH is the statement that the variance can be taken to be zero.

4.1.9 High order Laguerre expressions along the flow

This subsection and Section 4.1.7 report results of T. Newton [217]; the Lambert form of Proposition 4.1.22 is the authors' own. Put $\nu_t(u) = \frac{1}{2}e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $\mathbb{R}$, so that $H_t(z) = \int_{\mathbb{R}} \nu_t(u)e^{izu}\,du$, and define the Laguerre expressions $L_{n,t}(x)$ by

$$H_t(x + iy)H_t(x - iy) = \sum_{n\geq 0} L_{n,t}(x)\, y^{2n}.$$

Then $L_{1,t} = H_t'^2 - H_t H_t'' = H_t^2 \mathcal{L}_{H_t}$ with Laguerre's quantity of Section 4.1.4. With $w = u + v$ and $b = u - v$ the odd powers of b cancel under $(u, v) \mapsto (v, u)$, and

$$L_{n,t}(x) = \frac{1}{(2n)!}\iint b^{2n}\,\nu_t(u)\nu_t(v)\cos(xw)\,du\,dv. \tag{4.1.4}$$

Since H_t is even, $H_t(iy) = H_t(0)/R_t(y^2/4)$, so $\sum_n L_{n,t}(0)y^{2n} = H_t(0)^2 R_t(y^2/4)^{-2}$: the $L_{n,t}(0)$ are the Taylor coefficients of the square of the reciprocal tilted clock. By (4.1.4), $L_{n,t}(0) > 0$. Let $\mathbb{P}_{n,t}$ be the probability law on $\mathbb{R}^2$ with density proportional to $b^{2n}\nu_t(u)\nu_t(v)$, with expectation $\mathbb{E}_{n,t}$. Then $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, and with $k_n = (n + 1)(2n + 1)$ and $k_n(2n)!/(2n + 2)! = \frac{1}{2}$, the central ratio is

$$c_n(t) = \frac{k_n L_{n+1,t}(0)}{L_{n,t}(0)} = \tfrac{1}{2}\,\mathbb{E}_{n,t}[b^2].$$

Theorem 4.1.21 (Newton). $\lim_{n\to\infty} \sup_{0\leq t\leq 1/2} \mathbb{E}_{n,t}[w^2] = 0$. *Consequently, for every* $X < \infty$ *there is* N_X *with* $L_{n,t}(x) > 0$ *for all* $n \geq N_X$, $|x| \leq X$ *and* $0 \leq t \leq \frac{1}{2}$.

Proof. Let $V = -\log \Phi_{\mathrm{dBN}}$, an even smooth function with $V''(u) \sim 16\pi e^{4u}$ as $u \to \infty$ (the first theta summand dominates). So V'' is continuous, bounded below by $-K$ and tends to infinity; Newman's monotonicity of V'' [216] is not needed. Given $b = d$, the density of w is proportional to $e^{-Q(w)}$ with $Q(w) = V(\frac{w+d}{2}) + V(\frac{w-d}{2}) - \frac{t}{2}(w^2 + d^2)$ and $Q'' = \frac{1}{4}[V''(\frac{w+d}{2}) + V''(\frac{w-d}{2})] - t$. One of the two arguments has modulus at least $|d|/2$, so for $|d| \geq D_0$ and $t \leq \frac{1}{2}$ we get $Q'' \geq \kappa(D_0) := \frac{1}{4}[\inf_{|x| \geq D_0/2} V'' - K] - \frac{1}{2}$, which tends to infinity with D_0. Since V is even, Q is even, $Q'(0) = 0$ and $wQ'(w) \geq \kappa w^2$; integration by parts gives $\mathbb{E}[wQ'(w) \mid b = d] = 1$, so $\mathbb{E}[w^2 \mid b = d] \leq \kappa^{-1}$. For $|d| \leq D_0$ the conditional second moment is at most a constant $B(D_0)$ uniform in t, because $1 \leq e^{t(u^2+v^2)} \leq e^{(u^2+v^2)/2}$ and both integrals are continuous in d on a compact set. Finally $\nu_0 \leq \nu_t \leq \nu_{1/2}$, and comparing the weight b^{2n} on $\{|b| \leq D_0\}$ with that on $\{D_0+1 \leq |b| \leq D_0+2\}$ gives $\mathbb{P}_{n,t}(|b| \leq D_0) \leq C(D_0)\big(D_0/(D_0+1)\big)^{2n}$ uniformly in t. Hence $\mathbb{E}_{n,t}[w^2] \leq B(D_0)C(D_0)(\frac{D_0}{D_0+1})^{2n} + \kappa(D_0)^{-1}$; let $n \to \infty$ and then $D_0 \to \infty$. For the second claim, $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw) \geq 1 - \frac{x^2}{2}\mathbb{E}_{n,t}[w^2]$. □

Newton states that $c_n(t) = (\frac{1}{8} + o(1))(\log n)^2$ uniformly in $t \in [0, \frac{1}{2}]$. The Laplace calculation behind it gives the better centring $\frac{1}{8}\mathcal{W}(n/\pi)^2$, with $\mathcal{W}$ the Lambert function of Section 14.4.3.

Proposition 4.1.22. *Uniformly in* $t \in [0, \frac{1}{2}]$, $c_n(t) = \frac{1}{8}\,\mathcal{W}(n/\pi)^2 + o(\log n)$.

Sketch. For large $|b|$ only the first theta summand matters, and $\log \nu_t(u) + \log \nu_t(v) = -2\pi e^{2b}\cosh 2w + O(b^2)$. Integrating out w (a Gaussian of width of order e^{-b}) leaves a marginal $-2\pi e^{2b} + O(b^2)$, uniformly in t. The law of b has log density $2n\log b - 2\pi e^{2b} + O(b^2)$, maximal where $2b\,e^{2b} = n/\pi$ up to an $O(b^2/n)$ shift, that is at $b_n = \frac{1}{2}\mathcal{W}(n/\pi) + o(1)$, with width $O(\sqrt{b_n/n})$. So $\mathbb{E}_{n,t}[b^2] = b_n^2 + o(\log n)$. □

Since $\mathcal{W}(y) = \log y - \log\log y + o(1)$, the proposition is a statement about the second order term, and the relative error of $\frac{1}{8}(\log n)^2$ is of order $\log\log n/\log n$. The numerical values below, computed from the moment convolution $L_{n,t}(0) = \frac{1}{4}\sum_j M_j M_{n-j}/((2j)!(2n-2j)!)$ with $M_k = \int u^{2k} e^{tu^2}\Phi_{\mathrm{dBN}}(u)\,du$, suggest that the error is $o(1)$.

n	$c_n(\frac{1}{2})$	$c_n(0)$	$\frac{1}{8}\mathcal{W}(n/\pi)^2$
10^2	0.82596	0.82413	0.80112
10^3	2.32317	2.32216	2.31510
10^4	4.86185	4.86152	4.85998
10^5	8.52286	8.52278	8.52250
10^6	13.35216	13.35215	13.35210

The difference $c_n(\frac{1}{2}) - \frac{1}{8}\mathcal{W}(n/\pi)^2$ is 0.025, 0.0081, 0.0019, 0.00035 and 0.00006, and $c_n(0)$ differs from $c_n(\frac{1}{2})$ by 3×10^{-4} at $n = 10^4$. At $n = 10^5$, $\frac{1}{8}(\log n)^2 = 16.6$

against $c_n = 8.52$, so an effective version of Theorem 4.1.23 must use the Lambert centring.

THEOREM 4.1.23 (Newton). *For every $X < \infty$, uniformly in $0 \le t \le \frac{1}{2}$ and $|x| \le X$,*

$$\frac{k_n L_{n+1,t}(x)}{L_{n,t}(x)} = c_n(t)\,(1 + o(1)) = \tfrac{1}{8}(\log n)^2(1 + o(1)).$$

Proof. With $r_n(t,x) = L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, Theorem 4.1.21 gives $r_n \to 1$ uniformly, and $k_n L_{n+1,t}(x)/L_{n,t}(x) = c_n(t)\,r_{n+1}/r_n$; the second equality is Proposition 4.1.22 with $\mathcal{W}(n/\pi) \sim \log n$, used as stated. □

The result has the shape of the theorem of Griffin, Ono, Rolen and Zagier [89] (Problem 22.4): all large orders at each fixed window, for the Laguerre expressions along the whole flow. The index N_X is not effective, and effectivity in X is the open problem here: it would turn the positivity of all $L_{n,t}$ on $|x| \le X$ into a finite computation in $n < N_X$.

4.1.10 Forward persistence of real zeros

The motion $\dot z_j = 2\sum_{k\neq j} 1/(z_j - z_k)$ of the zeros of H_t (Section 4.1.1) says nothing at a collision, where the sum diverges. The lemma below and the barrier argument after it come from a discussion with a collaborator.

LEMMA 4.1.24. *Let c be a real zero of multiplicity $m \ge 1$ of H_{t^*}. For $t < t^*$ close to t^*, the m zeros of H_t near c are $c + i\sqrt{2(t^*-t)}\,x_k + o(\sqrt{t^*-t})$, $k = 1,\dots,m$, with x_k the roots of He_m, so at most one of them is real, and for $t > t^*$ close to t^* all m of them are real and simple. Consequently real zeros of H_t never collide as t increases; a real zero stays real, and a simple real zero is simple except at times when zeros arriving from the complex plane meet it; and nonreal zeros of H_τ cannot accumulate at a real point as τ decreases to a limit τ_1.*

Proof. Put $t = t^* + s$ and expand $H_t(c+w) = \sum a_{jk} w^j s^k$ (a convergent series, since $H_t(z)$ is an entire function of $(z,t) \in \mathbb{C}^2$, by the decay of Φ), and give w weight 1 and s weight 2. The lowest weighted degree part P satisfies $\partial_s P = -\partial_w^2 P$, because $\partial_t H_t = -\partial_z^2 H_t$ and the two sides have the same weighted degree after the shift, so $P(w,s) = e^{-s\partial_w^2}P(w,0)$. The part of P with s^0 is $a\,w^d$, and $a \neq 0$, since otherwise $P = 0$; as $H_{t^*}(c+w) = a_{m0}w^m + \cdots$ with $a_{m0} \neq 0$, we get $d = m$ and $P = a\,h_m$ with $h_m(w,s) = e^{-s\partial_w^2}w^m$. For $s = \mp\sigma$ the substitution $w = \sqrt{2\sigma}\,x$ turns h_m into $(2\sigma)^{m/2}e^{\pm\partial_x^2/2}x^m$. For $s > 0$ this is $(2s)^{m/2}\mathrm{He}_m(x)$, with $\mathrm{He}_m = e^{-\partial_x^2/2}x^m$ the Hermite polynomial, whose m roots are real and simple. For $s < 0$ put $w = i\sqrt{2|s|}\,x$ instead: $h_m(w,s) = (i\sqrt{2|s|})^m\mathrm{He}_m(x)$, so the zeros of h_m are $i\sqrt{2|s|}$ times the roots of He_m, simple and purely imaginary, with $w = 0$ among them exactly for odd m. Now write $w = \sqrt{|s|}\,\omega$ (respectively $\sqrt{s}\,\omega$): every monomial $w^j s^k$ becomes $(\operatorname{sgn} s)^k|s|^{(j+2k)/2}\omega^j$, so $|s|^{-m/2}H_t(c + \sqrt{|s|}\,\omega) \to$

$a\,h_m(\omega,\pm1)$ uniformly on compact sets as $s\to 0$, and the limit has m simple zeros. By Rouché's theorem, in small discs around the zeros of the limit the function has exactly one zero each; by Hurwitz's theorem applied to $H_t(c+w)$ on a fixed circle $|w|=r$, there are exactly m zeros in $|w|<r$, so these are all of them. For $s<0$ the discs are disjoint, and all lie off the real axis except the one at $\omega=0$ (odd m), which contains exactly one zero; for $s>0$ the zeros are real because they are simple, near a real simple zero of the limit, and the zero set is symmetric under conjugation. The consequences follow: two real zeros meeting at c at time t^* would give two real zeros near c for $t<t^*$; a real zero stays real by the symmetry (for a multiple one, by the case $t>t^*$), since a simple zero near a real simple zero is unique; and nonreal zeros of H_{τ_k}, $\tau_k\downarrow\tau_1$, with a limit point $c\in\mathbb{R}$ would give nonreal zeros near c just after τ_1, against the case $s>0$. □

Corollary 4.1.25. *If $\Lambda>0$, then H_Λ has a multiple real zero. Hence RH holds if and only if H_t has only simple zeros for every $t\in(0,0.2]$ [143].*

Proof. The set of t for which H_t has only real zeros is $[\Lambda,\infty)$, closed by Hurwitz's theorem, so the zeros of H_Λ are real. Suppose $\Lambda>0$ and that they are all simple. By the theorem of the Polymath project that for an absolute constant C and every $0<t\le\frac12$ every zero $x+iy$ of H_t with $|x|\ge e^{C/t}$ is real and simple [247, Theorem 1.5], there is $X\ge e^{2C/\Lambda}$, with no zero of H_Λ near $\pm X$, such that for $t\in[\frac{\Lambda}{2},\Lambda]\subset(0,\frac12]$ every zero of H_t with $|\operatorname{Re} z|>X$ is real. The zeros with $|\operatorname{Re} z|\le X$ lie in a bounded set [61]. As $t\uparrow\Lambda$, Hurwitz's theorem puts exactly one zero of H_t near each zero of H_Λ in $[-X,X]$ and no others in the box $[-X,X]\times[-1,1]$ (the strip of de Bruijn holds all zeros), and the zero near a real simple zero is real, since H_t is real on the real line and nonreal zeros come in conjugate pairs. So for $t<\Lambda$ close to Λ all zeros of H_t are real, against the definition of Λ. If RH fails, then H_0 has a zero off the real line, so $0<\Lambda\le 0.2$ [232], and H_Λ has a multiple zero. If RH holds, then $\Lambda\le 0$, and a multiple zero of H_{t^*} with $t^*>0$ would give, by Lemma 4.1.24, zeros of H_t off the real line for $t<t^*$ close to t^*, although $t>\Lambda$. □

For a height X let $0=t_0<t_1<\cdots<t_M=0.2$ and let $Q_i=[u_i,v_i]\times[-1,1]$ be rectangles in the z plane with $u_i<X<v_i$. The hypothesis $B(X)$ is:

(B0) every zero of H_0 with $|\operatorname{Re} z|\le u_0$ is real and simple (Platt and Trudgian [232] verified this to 6×10^{12} in z);

(B1) for each i, H_{t_i} has no zero on ∂Q_i, and the number of zeros of H_{t_i} in Q_i (counted with multiplicity by the argument principle) is at most the number of sign changes of H_{t_i} observed on $[u_i,v_i]$;

(B2) for each i and every $t\in[t_i,t_{i+1}]$, $|H_t-H_{t_i}|<|H_{t_i}|$ on ∂Q_i.

PROPOSITION 4.1.26. *Assume $B(X)$. Then for every $\lambda \in [0, 0.2]$ every zero of H_λ with $|\operatorname{Re} z| \le X$ is real and simple.*

Proof. Fix i. By (B2) and Rouché's theorem, H_t has no zero on ∂Q_i and the same number N_i of zeros in Q_i for all $t \in [t_i, t_{i+1}]$. At t_i a sign change is a real zero of odd multiplicity, so by (B1) the N_i zeros are real and simple. Let A be the set of $t \in [t_i, t_{i+1}]$ at which the N_i zeros in Q_i are real and simple. It contains t_i, and it is open in $[t_i, t_{i+1}]$, since a simple real zero persists as a simple zero, real by symmetry. Let τ be the supremum of the t with $[t_i, t] \subset A$ and take $t'_k \uparrow \tau$ in A: the zeros of H_τ in Q_i are limits of real zeros (Hurwitz's theorem, with no zero on the boundary), hence real, and two of them cannot coincide, since the N_i distinct real zeros of $H_{t'_k}$ near the common point would contradict Lemma 4.1.24; so there are N_i distinct real zeros, all simple, and $\tau \in A$. As A is open, $\tau = t_{i+1}$, so $A = [t_i, t_{i+1}]$, and every zero of H_t in Q_i is real and simple for $t \in [t_i, t_{i+1}]$. The same holds for the mirror image $-Q_i$.

Now let $\lambda \in [0, 0.2]$ and suppose H_λ has a nonreal zero z_0 with $|\operatorname{Re} z_0| \le X$. By de Bruijn's strip theorem [61, Theorem 13], the zeros of H_t satisfy $|\operatorname{Im} z| \le \sqrt{1 - 2t}$ for $0 \le t \le \frac{1}{2}$ (and $|\operatorname{Im} z| < 1$ at $t = 0$), so no zero meets the lines $\operatorname{Im} z = \pm 1$ and every zero of H_τ with $|\operatorname{Re} z| \le X$ lies in the compact set $K = \{|\operatorname{Re} z| \le X,\ |\operatorname{Im} z| \le 1\}$. Let F be the set of $\tau \in [0, \lambda]$ at which H_τ has a nonreal zero in K; then $\lambda \in F$. For $\tau \in [t_i, t_{i+1}]$ the first part shows that such a zero has $|\operatorname{Re} z| < u_i < X$ (otherwise it would lie in Q_i or $-Q_i$), and $|\operatorname{Im} z| < 1$, so it is an interior point of K and F is open in $[0, \lambda]$ by Hurwitz's theorem. If $\tau_k \downarrow \tau$ with $\tau_k \in F$, a limit point of the nonreal zeros is a zero of H_τ in K, and it is nonreal by the case $s > 0$ of Lemma 4.1.24. Hence $\tau_0 = \inf\{\tau : [\tau, \lambda] \subset F\}$ lies in F, openness forces $\tau_0 = 0$, and $0 \in F$ gives a nonreal zero of H_0 with $|\operatorname{Re} z| \le X$; by (B0) it has $|\operatorname{Re} z| > u_0$, so it lies in $\pm Q_0$, against the first part. Hence every zero of H_λ with $|\operatorname{Re} z| \le X$ is real. The constants in $B(X)$ do not change if X is replaced by $X' \in (X, \min_i v_i)$, so the same holds for $|\operatorname{Re} z| \le X'$. For simplicity at $\lambda = 0$ use (B0) and the first part. For $\lambda > 0$, a real zero c of multiplicity $m \ge 2$ with $|c| \le X$ would by Lemma 4.1.24 have at least $m - 1 \ge 1$ nonreal zeros of H_t near c for $t < \lambda$ close to λ, with $|\operatorname{Re} z| \le X'$, which is impossible. □

The hypothesis $B(X)$ has not been verified at any height X, and no computation is claimed. The intended verification is the count barrier of the collaborator: evaluate H_{t_i} on the boundary of Q_i and on its real diameter at the grid times, and check (B1) and (B2) there. The status of the problem is recorded in Remark 19.15.1.

Two results of Newton [217] bear on the verification of $B(X)$. The first is the local form of (B2). Let $t_* \ge 0$ and let $a < b$ be adjacent real simple zeros of H_{t_*} with $g_* = b - a$; let $0 < r < g_*/2$ and let Ω be a conjugation invariant Jordan domain containing the closed discs D_a, D_b of radius r about a, b and the segment

between them, with no other zero of H_{t_*} in $\overline{\Omega}$. Let $\Gamma = \partial D_a \cup \partial D_b \cup \partial\Omega$, $Y = \max_\Gamma |\operatorname{Im} z|$, $m = \min_\Gamma |H_{t_*}|$ and $\psi_Y(\sigma) = \int_0^\infty \Phi_{\mathrm{dBN}}(u) e^{t_* u^2 + Yu}(1 - e^{-\sigma u^2})\, du$.

Proposition 4.1.27 (Newton). *If $\psi_Y(\sigma) < m$, then for $0 \le \tau \le \sigma$ the function $H_{t_*-\tau}$ has exactly one zero in each of D_a, D_b, both real and simple, and no other zero in Ω; so the gap satisfies $g \ge g_* - 2r$ and, by* (4.1.2), *$4 \int \mathrm{Def} \le g_*^2 - (g_* - 2r)^2$ over the interval.*

Proof. Since $|\cos(zu)| \le e^{|\operatorname{Im} z| u}$, on Γ we have $|H_{t_*-\tau} - H_{t_*}| \le \psi_Y(\tau) \le \psi_Y(\sigma) < m \le |H_{t_*}|$. By Rouché's theorem $H_{t_*-\tau}$ has as many zeros in D_a, D_b and Ω as H_{t_*}, namely 1, 1 and 2. A single zero in a disc centred on the real axis is real by conjugation symmetry and is simple, and Ω has no further zero. □

This is the mechanism of (B2) in backward time, with a rectangle replaced by a pair of discs. The second result shows what it cannot do. The decay of H_t along the real axis is that of Ξ: for $0 < \delta < \pi/8$, Φ_{dBN} is analytic in $|\operatorname{Im} u| \le \delta$, where $|\exp(-\pi n^2 e^{4(u+i\eta)})| = \exp(-\pi n^2 e^{4u} \cos 4\eta)$ and, Φ_{dBN} being even, $\Phi_{\mathrm{dBN}}(u+i\eta) = \Phi_{\mathrm{dBN}}(-u - i\eta)$ decays as fast as $u \to -\infty$, and shifting the contour $u \mapsto u + i\delta$ in $H_t(z) = \int_{\mathbb{R}} \nu_t(u) e^{izu}\, du$ gives, for $\operatorname{Re} z = x > 0$ and $|\operatorname{Im} z| \le 1$,

$$|H_t(z)| \le C(\delta, t)\, e^{-\delta x}, \qquad C(\delta, t) = \tfrac{1}{2} \int_{\mathbb{R}} |\Phi_{\mathrm{dBN}}(u + i\delta)|\, e^{t(u^2 - \delta^2) + |u|}\, du < \infty.$$

On the other hand $\psi_Y(\sigma) \ge \psi_0(\sigma)$ for every $Y \ge 0$, and $\psi_0(\sigma) > 0$ does not depend on the height.

Proposition 4.1.28 (Newton, with the mechanism corrected). *Fix $t_* \ge 0$, $\sigma > 0$ and $0 < \delta < \pi/8$. For every real zero $a \ge 1$ of H_{t_*} and every contour Γ containing a point $a + i\eta$ with $0 < \eta \le 1$, we have $\min_\Gamma |H_{t_*}| \le C(\delta, t_*) e^{-\delta a} < \psi_0(\sigma) \le \psi_Y(\sigma)$ as soon as $a > \delta^{-1} \log\big(C(\delta, t_*)/\psi_0(\sigma)\big)$. So no criterion of the form $\psi_Y(\sigma) < m$ certifies a fixed step $\sigma > 0$ at every height, and the step it can certify at height a is $O(e^{-\delta a})$.*

Proof. $|H_{t_*}(a + i\eta)| \le C(\delta, t_*) e^{-\delta a}$ by the display, and $\psi_Y \ge \psi_0 > 0$. For $\sigma \le 1$, $1 - e^{-x} \ge x/(1+x)$ gives $\psi_0(\sigma) \ge \sigma \int_0^\infty \Phi_{\mathrm{dBN}}(u) e^{t_* u^2} u^2 (1+u^2)^{-1}\, du$, so a certified step is $O(e^{-\delta a})$. □

Newton proved this for pairs with gaps tending to 0, but the obstruction is the decay $e^{-\pi x/8}$ of H_t and not the gaps: it holds at every zero. It concerns bounds through the absolute value of the integrand $\Phi_{\mathrm{dBN}} e^{tu^2}(1 - e^{-\sigma u^2}) \cos(zu)$; bounds on $H_{t_*-\tau} - H_{t_*}$ that use H itself are not excluded. Jensen's inequality for the convex map $x \mapsto e^{-\sigma x}$ gives $\psi_Y(\sigma) \le A(1 - e^{-\sigma M/A}) \le 2A\sigma M/(2A + \sigma M)$, with A and M the mass and first moment of the image of $\Phi_{\mathrm{dBN}}(u) e^{t_* u^2 + Yu}\, du$ under $x = u^2$ (the second step is $2 \operatorname{artanh}(x/2) \ge x$); it improves the linear bound σM only when σM is comparable to A, and it bounds backward steps only: the

forward profile $\int (e^{\tau x} - 1)\, \mu(dx)$ of (B2) is convex in x, and Jensen then gives a lower bound.

The practical consequence concerns (B2). On ∂Q_i the minimum of $|H_{t_i}|$ is at most $C(\delta, t_i) e^{-\delta v_i}$, with $v_i > X$, while an absolute value bound for $|H_t - H_{t_i}|$ is at least $\psi_0(t - t_i)$, of order $t - t_i$ by $1 - e^{-x} \geq x/(1 + x)$. A step certified that way must therefore be $O(e^{-\delta X})$ for every $\delta < \pi/8$, about 10^{-3400} at $X = 2 \times 10^4$, and (B2) cannot be checked by it at an appreciable height. The perturbation has to be bounded through H: by Taylor's formula in t with $\partial_t^k H_t = (-1)^k H_t^{(2k)}$ and Cauchy estimates from $|H_t|$ on a larger contour at the same height, in ball arithmetic, or by a Riemann Siegel type approximation with relative error, as in the Polymath project [247].

Chapter 5

The barrier method in Thorin form

This chapter states the barrier method of Polymath15 [247], as used by Platt and Trudgian [232], translates it into the language of Thorin measures, and runs a small barrier check of our own. The method proves upper bounds for the de Bruijn Newman constant Λ by combining three things: a verification of RH to a height, an analytic zero free region for H_{t_0} at large heights, and a barrier, a window of heights through which no zero of H_t can pass at any time $t \in [0, t_0]$. With it Polymath15 proved $\Lambda \leq 0.22$ and Platt and Trudgian $\Lambda \leq 0.2$ (Section 4.1 and Table 4.8).

The main results are these. The barrier criterion is stated precisely as Polymath15 state it, with the role of each hypothesis (Theorem 5.1.1, cited). In Thorin language (Section 5.2) its conclusion is that the shifted clock φ_{t_0,α_0} with $\alpha_0 = \frac{1}{2} + \frac{y_0}{2}$ is a GGC transform, so that by Theorem 17.1.16 the reciprocal R_t of H_t is a GGC transform for every $t \geq t_0 + \frac{1}{2}y_0^2$ (Proposition 5.2.1, proved); the barrier becomes the statement that no atom of the Thorin measure of R_t, in the chart of squares, crosses a curvilinear window bounded by four confocal parabolas with focus at the origin (Lemma 5.2.2 and Proposition 5.2.3, proved); and a positive density $v_t = \mathrm{Re}(\xi_t'/\xi_t)$ on the barrier rectangle at every time is a sufficient certificate (Proposition 5.2.4, proved). Section 5.3 runs the barrier at the modest height $X/2 = 497.4$ for $t \in [0, 0.2]$ and $y_0 = 0.2$ in floating point, and certifies Proposition 5.3.1 in ball arithmetic for every $t \in [0, 0.2]$. Section 5.4 explains why the method, in either language, cannot reach $t = 0$.

For the path to RH of Section 1.4 the barrier method is the source we know of an analytic tail below the basepoint one, and the overture names its limit: what separates $\Lambda \leq 0.2$ from RH is the existence of an analytic region beyond the barrier at positive time and its absence at time zero. The Thorin form makes that statement exact but does not move it. Every point the method certifies lies on a level set $t + 2(\alpha - \frac{1}{2})^2 = c$ with $c > 0$, and the corner $(0, \frac{1}{2})$ of the phase diagram (Theorem 6.3.2) is reached only in the limit, where the analytic hypothesis becomes a zero free region for ζ itself.

5.1 The barrier argument

We use the normalisation of Section 4.1.1: $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$, $\xi_t(s) = 8H_t(-2i(s-\frac12))$, and a zero $x+iy$ of H_t is the zero $\frac12 - \frac{y}{2} + \frac{ix}{2}$ of ξ_t. The zeros of H_t are symmetric under $z \mapsto -z$ and $z \mapsto \bar z$, so it suffices to consider $x > 0$, $y \ge 0$. By de Bruijn's strip theorem (Theorem 8.4.1(c) with $r = 1$) every zero of H_t satisfies $y^2 \le \max(1-2t, 0)$ for $t \ge 0$.

5.1.1 De Bruijn's theorem and the canopy

The method rests on one consequence of the strip theorem. If at a time $t_0 \ge 0$ no zero of H_{t_0} satisfies $y > y_0$, then all zeros of H_{t_0} lie in $|\operatorname{Im} z| \le y_0$, and by the strip theorem started at t_0 the zeros of H_t satisfy $y^2 \le \max(y_0^2 - 2(t-t_0), 0)$, so they are real for $t \ge t_0 + \frac12 y_0^2$. Hence

$$\Lambda \le t_0 + \tfrac12 y_0^2 \quad \text{whenever } H_{t_0} \text{ has no zero with } y > y_0. \tag{5.1.1}$$

This is [61, Theorem 13] in the form of [247], and in the language of the phase diagram it is Theorem 6.3.2(ii) at the point $(t_0, \frac12 + \frac{y_0}{2})$. Since $y \le \sqrt{1-2t_0}$ in any case, the hypothesis of (5.1.1) is that the *canopy*

$$\mathcal{C}(t_0, y_0) = \{x + iy : \ x \ge 0,\ y_0 < y \le \sqrt{1-2t_0}\}$$

contains no zero of H_{t_0}. The canopy is unbounded in x, and no finite computation can clear it. The barrier method splits it at a height X into a part to the left, cleared by the verification of RH and by the dynamics of the zeros, and a part to the right, cleared by an analytic estimate.

5.1.2 The criterion of Polymath15

We state the criterion as it is printed in [247] (Theorem 1.2 there, read in the arXiv version and its HTML rendering), with ζ and H_t in our notation.

Theorem 5.1.1 (Polymath15, upper bound criterion [247]). *Let $0 < t_0 \le \frac12$, $X > 0$ and $0 < y_0 \le 1$, and suppose:*

(i) (numerical verification of RH at time 0) there is no zero $\zeta(\sigma + iT) = 0$ with $\frac{1+y_0}{2} \le \sigma \le 1$ and $0 \le T \le X/2$;

(ii) (asymptotic zero free region at time t_0) there is no zero $H_{t_0}(x+iy) = 0$ with $x \ge X + \sqrt{1-y_0^2}$ and $y_0 \le y \le \sqrt{1-2t_0}$;

(iii) (barrier at intermediate times) there is no zero $H_t(x+iy) = 0$ with $X \le x \le X + \sqrt{1-y_0^2}$, $\sqrt{y_0^2 + 2(t_0 - t)} \le y \le \sqrt{1-2t}$ and $0 \le t \le t_0$.

Then $\Lambda \le t_0 + \frac12 y_0^2$.

Polymath15 describe the roles of the hypotheses as follows: (i) prevents zeros from reaching the canopy from an initial position to the left of the barrier, (ii) prevents zeros from lying in the canopy to the right of the barrier, and (iii) prevents zeros from starting to the right of the barrier and reaching the canopy to the left of it. The proof combines (5.1.1) with the continuity of the zeros in t and the strip theorem applied on subintervals; we do not reproduce it. In practice [247] replace the region of (iii) by the larger box

$$X \le x \le X+1, \qquad y_0 \le y \le 1, \qquad 0 \le t \le t_0, \tag{5.1.2}$$

and they take $t_0 = y_0 = 0.2$ and $X = 6\times 10^{10} + 83952 - 0.5$, a value near the limit of the verifications of RH available to them, shifted so that a partial Euler product over the primes up to 11 is large on the barrier and $|H_t|$ stays away from zero there. This gives $\Lambda \le 0.22$ (Theorem 1.1 of [247]). By their account (ii) is the expensive hypothesis, verified with an effective Riemann Siegel type approximation of H_t in the region $0 < t \le \frac{1}{2}$, $0 \le y \le 1$, $x \ge 200$, and (iii) is verified numerically and quickly by comparison. Platt and Trudgian take the second row of the table of conditional bounds of [247], which needs RH to height 2.51×10^{12}, and their verification to 3×10^{12} gives $\Lambda \le 0.2$ [232]; Section 6.6.3 records the cost of further rows. The barrier idea itself was proposed in the seventh thread of the project, as a region $0 \le t \le t_0$, $X \le x \le X+1$, $y \ge y_0$ that prevents zeros from flowing into the canopy at time t_0, assuming they were not already there at time 0 [280].

Remark 5.1.2. The mechanism behind (iii) is the dynamics of Section 4.1.1. Forward in time the zeros repel, $\dot z_k = 2\sum_{j\ne k}(z_k - z_j)^{-1}$, a complex zero is pulled towards the real axis by the strip theorem, and zeros are continuous in t (Hurwitz's theorem; at a collision Lemma 4.1.24 describes the splitting). Heuristically, a zero that lies in the canopy at time t_0 and to the left of the barrier was higher at earlier times, as the bound on the highest zero suggests: the height $y(t)$ of the highest zero satisfies $y(t)^2 \ge y(t_0)^2 + 2(t_0 - t)$ by the strip theorem applied on $[t, t_0]$; so it was either to the left of the barrier at time 0, which (i) excludes, or it crossed the window $X \le x \le X+1$ at some intermediate time at a height above $\sqrt{y_0^2 + 2(t_0 - t)}$, which (iii) excludes. The width $\sqrt{1 - y_0^2} \le 1$ of the window is what prevents a zero from jumping over it: by the strip theorem a zero cannot move far horizontally without passing through it, a point that [247] make precise in their proof.

5.1.3 The rectangle in the variable of ξ_t

The box (5.1.2) becomes a rectangle in s. A zero $x + iy$ of H_t with $y \ge 0$ gives the zero $\frac{1}{2} - \frac{y}{2} + \frac{ix}{2}$ of ξ_t, and by $\xi_t(1-s) = \xi_t(s)$ (Lemma 6.2.1) and conjugation also $\frac{1}{2} + \frac{y}{2} + \frac{ix}{2}$. So the box at time t is free of zeros exactly when ξ_t has no zero in

$$\mathcal{B}(X, y_0) = \left[\tfrac{1}{2} + \tfrac{y_0}{2},\, 1\right] \times \left[\tfrac{X}{2},\, \tfrac{X}{2} + \tfrac{1}{2}\right]. \tag{5.1.3}$$

The right edge $\operatorname{Re} s = 1$ carries no zero for $t \geq 0$: at $t = 0$ because $\zeta(1 + i\theta) \neq 0$, and for $t > 0$ because every zero has $|\beta - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1 - 2t} < \frac{1}{2}$ (Lemma 6.5.2). The barrier at a given time is therefore the vanishing of the winding number of ξ_t around $\partial\mathcal{B}$, together with the absence of zeros on $\partial\mathcal{B}$; and since zeros move continuously, a winding number of 0 at the times of a grid together with a Rouché bound between grid times (hypothesis (B2) of Section 4.1.10) gives (iii) for all $t \in [0, t_0]$.

5.2 The barrier in Thorin form

5.2.1 The conclusion as a GGC statement

Proposition 5.2.1. *Under the hypotheses of Theorem 5.1.1, let* $\alpha_0 = \frac{1}{2} + \frac{y_0}{2}$. *Then:*

(a) the shifted clock $\varphi_{t_0,\alpha_0}(u) = \xi_{t_0}(\alpha_0)/\xi_{t_0}(\alpha_0 + \sqrt{u})$ *is the Laplace transform of a GGC, whose Thorin measure in the chart* $\theta = \sqrt{z}$ *is the Poisson sum of Theorem 6.2.3;*

(b) $(t, \alpha) \in \mathcal{G}$ *for every* $t \geq t_0$ *and* $\alpha \geq \frac{1}{2}$ *with* $(\alpha - \frac{1}{2})^2 + \frac{t}{2} \geq \frac{1}{4}y_0^2 + \frac{t_0}{2}$;

(c) for every $t \geq t_0 + \frac{1}{2}y_0^2$ *the reciprocal* $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ *is the Laplace transform of a GGC with Thorin measure* $\sum_k \delta_{(z_k(t)/2)^2}$, $z_k(t) > 0$ *the zeros of* H_t.

Proof. Theorem 5.1.1 is proved in [247, Proposition 3.3] through (5.1.1), that is, by showing that H_{t_0} has no zero with $y > y_0$; in the variable of ξ_{t_0} this says that every zero has $\operatorname{Re}\rho \leq \frac{1}{2} + \frac{y_0}{2} = \alpha_0$. Theorem 6.2.3 then gives (a). Part (b) is Theorem 6.3.2(i) from the point (t_0, α_0), and (c) is the case $\alpha = \frac{1}{2}$ of (b) combined with Theorem 17.1.16, since $\Lambda \leq t_0 + \frac{1}{2}y_0^2$. □

With $t_0 = y_0 = 0.2$ this is Corollary 6.3.3(b). The proposition only renames the conclusion; what this section adds is the reading of the hypotheses.

5.2.2 Atoms in the chart of squares

Let $t \geq 0$. By Lemma 23.1.1, applied to $F = H_t$ (whose kernel $e^{tu^2}\Phi_{\mathrm{dBN}}$ satisfies part (b) of the lemma), the function R_t is an infinite product of factors $(1 + 4u/w_k^2)^{-1}$ over one representative w_k of each pair $\pm w_k$ of zeros of H_t, and we choose w_k with $\operatorname{Re} w_k > 0$, or $\operatorname{Im} w_k > 0$ if $\operatorname{Re} w_k = 0$ (which does not occur, since $H_t > 0$ on the imaginary axis). We call

$$\mu_t = \sum_k \delta_{w_k(t)^2/4}$$

the *formal Thorin measure* of R_t. It is a positive measure on $(0,\infty)$, and then the Thorin measure of R_t, exactly when all zeros are real, that is when $t \geq \Lambda$; a nonreal zero gives a pair of atoms $w^2/4$ and $\bar{w}^2/4$ off the half line, and then R_t is not a GGC transform (Lemma 23.1.1(a)). In the language of μ_t, a collision of two real zeros as t decreases is a collision of two atoms on $(0,\infty)$ after which a conjugate pair of atoms leaves the half line, and RH is the statement that no atom of μ_t leaves $(0,\infty)$ for any $t > 0$.

The map $w \mapsto w^2$ carries the rectangular coordinates of $w = x + iy$ to the parabolic coordinates of w^2. Lemma 23.1.2 treats the horizontal lines; the vertical lines behave in the same way.

LEMMA 5.2.2 (The barrier as a parabolic quadrilateral). *For $b > 0$ let $\mathcal{P}_b = \{\xi + i\eta :\ \xi = \eta^2/(4b^2) - b^2\}$, the right opening parabola of Lemma 23.1.2, and for $a > 0$ let $\mathcal{Q}_a = \{\xi + i\eta :\ \xi = a^2 - \eta^2/(4a^2)\}$, a left opening parabola. All of them have their focus at the origin, and $\mathcal{P}_b$ meets $\mathcal{Q}_a$ at right angles at the points $(a \pm ib)^2$. The map $w \mapsto w^2$ is a bijection from the box $\{x+iy :\ a_1 \leq x \leq a_2,\ b_1 \leq y \leq b_2\}$, $0 < a_1 < a_2$, $0 < b_1 < b_2$, onto the closed curvilinear quadrilateral in the upper half plane bounded by arcs of $\mathcal{P}_{b_1}$, $\mathcal{P}_{b_2}$, $\mathcal{Q}_{a_1}$ and $\mathcal{Q}_{a_2}$. In particular the box* (5.1.2) *goes to the quadrilateral $\mathcal{W}(X, y_0)$ bounded by $\mathcal{P}_{y_0}$, $\mathcal{P}_1$, $\mathcal{Q}_X$ and $\mathcal{Q}_{X+1}$, and the canopy $\mathcal{C}(t_0, y_0)$ goes to the region of the closed upper half plane strictly to the left of $\mathcal{P}_{y_0}$ and on or to the right of $\mathcal{P}_{\sqrt{1-2t_0}}$.*

Proof. For $w = a + iy$ with $a > 0$ fixed, $w^2 = (a^2 - y^2) + 2iay$, so $\eta = 2ay$ and $\xi = a^2 - \eta^2/(4a^2)$: the vertical line goes to $\mathcal{Q}_a$. The horizontal line $w = x + ib$ goes to $\mathcal{P}_b$ by the proof of Lemma 23.1.2. A parabola $\xi = \eta^2/(4p) - p$ has its focus at $(0,0)$ for either sign of p, which gives the common focus. The map is conformal on $\operatorname{Re} w > 0$, so the images of the orthogonal lines $x = a$ and $y = b$ meet at right angles at $(a+ib)^2$, and at $(a-ib)^2$ by symmetry. The map is injective on the open right half plane, where it is a conformal bijection onto $\mathbb{C} \setminus (-\infty, 0]$, so the box goes onto the region bounded by the images of its four sides, which lies in the upper half plane since $\eta = 2xy > 0$ there. The statement for the canopy follows in the same way from the horizontal lines alone, together with $x \geq 0$, which is $\eta \geq 0$. □

The atoms of μ_t live in the scaled chart $w^2/4$, and scaling by $\frac{1}{4}$ maps confocal parabolas to confocal parabolas; we keep the chart w^2 for the geometry, as Lemma 23.1.2 does. In this chart de Bruijn's theorem reads: if at time t_0 every atom lies on or to the right of $\mathcal{P}_{y_0}$ (or on the half line), then at time t every atom lies on or to the right of $\mathcal{P}_{\sqrt{y_0^2 - 2(t-t_0)}}$, and the parabolas collapse onto $[0,\infty)$ at $t = t_0 + \frac{1}{2}y_0^2$ (Corollary 23.2.2 is the case $t_0 = 0$, $y_0 = 1$).

PROPOSITION 5.2.3 (The barrier as a window). *Write $\mathcal{W}_t$ for the image under $w \mapsto w^2$ of the region of hypothesis (iii) of Theorem 5.1.1 at time t, the quadrilateral bounded by $\mathcal{Q}_X$, $\mathcal{Q}_{X+\sqrt{1-y_0^2}}$, $\mathcal{P}_{\sqrt{y_0^2+2(t_0-t)}}$ and $\mathcal{P}_{\sqrt{1-2t}}$. Then the hypotheses of*

Theorem 5.1.1 are equivalent to the following statements about the formal Thorin measures:

(i′) μ_0 *has no atom in the image of* $\{0 \le x \le X,\ y_0 \le y \le 1\}$*, the part of the closed region between* $\mathcal{P}_{y_0}$ *and* $\mathcal{P}_1$ *to the left of* $\mathcal{Q}_X$*, scaled by* $\frac{1}{4}$*;*

(ii′) μ_{t_0} *has no atom in the part of the closed region between* $\mathcal{P}_{y_0}$ *and* $\mathcal{P}_{\sqrt{1-2t_0}}$ *on or to the right of* $\mathcal{Q}_{X+\sqrt{1-y_0^2}}$*, scaled by* $\frac{1}{4}$*;*

(iii′) for every $t \in [0, t_0]$*,* μ_t *has no atom in* $\frac{1}{4}\mathcal{W}_t$*.*

Under (i′) to (iii′), no atom of μ_{t_0} *lies strictly to the left of* $\mathcal{P}_{y_0}$ *off the half line, and* μ_t *is a positive measure on* $(0, \infty)$ *for* $t \ge t_0 + \frac{1}{2}y_0^2$*.*

Proof. The atoms of μ_t are the points $w^2/4$ with w a zero of H_t in the open right half plane; since $\bar{w}$ is then also such a zero, the set of atoms is invariant under conjugation, and it suffices to consider the atoms with $\eta \ge 0$, which come from the zeros with $x > 0$, $y \ge 0$. Lemma 5.2.2 maps each region of Theorem 5.1.1 bijectively onto the region named. For (i′), hypothesis (i) is the absence of zeros $\frac{1}{2} + \frac{y}{2} + \frac{ix}{2}$ of ξ with $y_0 \le y \le 1$ and $0 \le x \le X$, which is the absence of zeros of H_0 in that box. The last sentence is Theorem 5.1.1 with (5.1.1) and the definition of μ_t. □

The proposition says what the barrier certifies in Thorin language. The formal Thorin measure has its atoms on the positive half line, up to an unknown set of conjugate pairs which de Bruijn's theorem confines between the parabola $\mathcal{P}_{\sqrt{1-2t}}$ and the half line; the canopy is the parabolic band $\mathcal{P}_{\sqrt{1-2t}} \le \cdot < \mathcal{P}_{y_0}$; and the barrier is a window of heights $\eta = 2xy$ in that band, cut out by two confocal left opening parabolas, through which no complex atom passes during $[0, t_0]$. It is not a statement that atoms do not leave the half line through the window: an atom that leaves the half line at a collision does so at a point of $(0, \infty)$, to the right of $\mathcal{P}_{y_0}$, and it is the subsequent motion into the band that the window watches. What the barrier certifies is therefore that the off axis part of μ_t to the left of the parabola $\mathcal{P}_{y_0}$ is not fed through a window of heights $\eta \approx 2Xy$ during $[0, t_0]$, so that its fate is decided by (i′) on the left and by the analytic estimate (ii′) on the right.

5.2.3 The density certificate

In the chart $\theta = \sqrt{z}$, with the basepoint shift of Theorem 6.2.3, the formal Thorin measure has the density $\pi^{-1}v_t(\sigma, \theta)$, where

$$v_t(\sigma, \theta) = \operatorname{Re} \frac{\xi_t'}{\xi_t}(\sigma + i\theta),$$

the function of Section 6.2.1 and $\pi\rho_{\sigma-1/2,t}(\theta)$ of Section 4.1.1. It is a positive Thorin density at the basepoint σ when no zero of ξ_t lies to the right of σ (Theorem 6.2.3); at a general σ it is the Poisson sum of Proposition 2.1.3 for ξ_t, with negative kernels from the zeros to the right of σ. The barrier rectangle has a pointwise certificate in terms of this density.

Proposition 5.2.4 (The barrier in density form). *Let $0 \le t \le t_0$, $\frac{1}{2} < \alpha < \sigma_1$ and $\theta_1 < \theta_2$, and suppose $v_t(\sigma,\theta) > 0$ at every point of $[\alpha,\sigma_1] \times [\theta_1,\theta_2]$ at which $\xi_t(\sigma+i\theta) \neq 0$. Then ξ_t has no zero $\beta+i\gamma$ with $\alpha < \beta \le \sigma_1$ and $\theta_1 \le \gamma \le \theta_2$. In particular, if $v_t > 0$ on $\mathcal{B}(X,y_0)$ off the zeros of ξ_t for every $t \in [0,t_0]$, then ξ_t has no zero in $\mathcal{B}(X,y_0)$ with $\beta > \frac{1}{2} + \frac{y_0}{2}$ for any such t.*

Proof. Near a zero ρ of multiplicity m, $\xi_t'/\xi_t(s) = m/(s-\rho)+O(1)$, so $v_t(\beta-\delta,\gamma) = -m/\delta + O(1) < 0$ for small $\delta > 0$, and these points lie in the rectangle when $\alpha < \beta \le \sigma_1$ and $\theta_1 \le \gamma \le \theta_2$. This is the proof of Proposition 6.5.7 with the bottom edge $\theta = 0$ replaced by θ_1. □

Zeros on the left edge $\operatorname{Re} s = \alpha$ are not excluded by the proposition; they are the zeros with $y = y_0$, which hypothesis (iii) in the form (5.1.2) excludes and which the winding number of Section 5.1 also detects. The density certificate is a two dimensional condition on the whole rectangle, because positivity on one line does not exclude zeros to its right (Remark 2.1.4). It is stronger than the absence of zeros, since v_t can be negative at points of the rectangle where no zero lies to their right within it, so it can fail where the winding test succeeds. What it offers is a margin: the minimum of v_t over $\mathcal{B}$ is a number, comparable across times and heights, and Section 5.3 measures it.

5.3 A computed barrier at a modest height

We ran the box (5.1.2) of the barrier with $t_0 = y_0 = 0.2$, as in [247], at a height that the Pólya route reaches in minutes, and with both certificates: the winding number of ξ_t around $\partial\mathcal{B}$ and the minimum of v_t on $\mathcal{B}$. The run is a check of the method and of the evaluator, not a contribution to any bound on Λ: a barrier is useful only at a height up to which RH has been verified and beyond which (ii) is known, and at $X \approx 10^3$ hypothesis (ii) is not available.

5.3.1 What is computed

The check does the following.

1. The window. A scan of $\min|\zeta|$ over $\sigma \in \{0.6, 0.8, 1\}$ and five heights in each window $[\theta, \theta+\frac{1}{2}]$, $495 \le \theta < 505$ in steps of 0.05, at 20 digits, chose $X/2 = 497.4$, where the minimum is 2.2472. The window $[497.4, 497.9]$ lies between the zeros $\gamma_{268} = 496.4297$ and $\gamma_{269} = 498.5808$ of ζ. So $X = 994.8$,

and the rectangle is $\mathcal{B} = [0.6, 1] \times [497.4, 497.9]$. This imitates, on a small scale, the choice of X in [247] that makes $|H_t|$ large on the barrier.

2. The evaluator. $\xi_t(s)$, $\xi_t'(s)$ and the derivatives $\partial_t^k \xi_t(s) = 8 \int_0^\infty u^{2k} e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cosh(2(s - \frac{1}{2})u)\, du$ for $k \le 8$ are computed by the trapezoid rule of Section 2.1.1 with the weight e^{tu^2}, on $[0, 3]$ with step $h = 0.002361$ (1273 nodes), at a working precision of 214 digits; h is chosen from the Poisson summation bound with the strip $|\operatorname{Im} u| < 0.3$, as in Section 6.5.2, for 30 correct digits relative to $|\xi_t|$, which is about 4×10^{-165} on $\mathcal{B}$. At the four corners of $\mathcal{B}$ the evaluator at $t = 0$ agrees with $\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(\frac{s}{2}) \zeta(s)$ computed by `mpmath` [213] to a relative error between 2.1×10^{-51} and 4.0×10^{-51}, and at $t = 0.2$ a second rule with step $0.8h$ agrees to at most 3.6×10^{-51}.

3. The winding numbers. For each $t_i = 0, 0.01, \dots, 0.2$ the boundary of $\mathcal{B}$ is walked counterclockwise with an adaptive step keeping $|\xi_{t_i}(s_{j+1}) - \xi_{t_i}(s_j)| < \frac{1}{2} |\xi_{t_i}(s_j)|$, as in Lemma 6.5.5 but tested only at the points of the walk, and the winding number is the sum of the principal arguments of $\xi_{t_i}(s_{j+1}) / \xi_{t_i}(s_j)$ divided by 2π.

4. Between grid times. At every point s_j of the walk at time t_i we evaluate the relative Taylor sum $\sum_{k=1}^{8} \frac{D^k}{k!} |\partial_t^k \xi_{t_i}(s_j)| / |\xi_{t_i}(s_j)|$ with $D = t_{i+1} - t_i$, the form of (B2) of Section 4.1.10 that Proposition 4.1.28 forces: bounds through the absolute value of the integrand cannot certify any step at an appreciable height. The rest of the Taylor series beyond order 8 is bounded by moving the contour to $\operatorname{Im} u = \eta$ as in Lemma 6.5.3:

$$\Big| \sum_{k>8} \frac{D^k}{k!} \partial_t^k \xi_t(s) \Big| \le 8 e^{-2\eta\theta} \int_0^\infty e^{t(u^2 - \eta^2)} \cdot |\Phi_{\mathrm{dBN}}(u + i\eta)| \cosh(2xu) \frac{r^9 e^r}{9!}\, du, \qquad r = D|u + i\eta|^2,$$

for $s = \frac{1}{2} + x + i\theta$, by $|(u + i\eta)^{2k}| = |u + i\eta|^{2k}$ and $\sum_{k>8} r^k/k! \le r^9 e^r / 9!$; the integral is computed by Gauss Legendre quadrature at $x = \frac{1}{2}$, $\theta = 497.4$ and $\eta = 0.38$, which bounds it on $\mathcal{B}$ because the integrand increases with x and $e^{-2\eta\theta}$ decreases with θ.

5. The density. The minimum of v_t over a grid of 9×11 points of $\mathcal{B}$ (steps 0.05 in σ and in θ) at each t_i.

6. A control. At $t = 0$ the walk is repeated for $\xi(s)\,(w^2 - w_0^2)(w^2 - \bar{w}_0^2)$, $w = s - \frac{1}{2}$, $w_0 = \frac{1}{4} + i(X/2 + \frac{1}{4})$, which has exactly one zero, $0.75 + 497.65\, i$, in $\mathcal{B}$.

5.3.2 The results

t	winding	points	$\min\lvert\xi_t\rvert$ on the walk	Taylor ratio, $D = 0.01$	$\min_{\mathcal{B}} v_t$
0	-2.6×10^{-217}	35	4.087×10^{-165}	0.00773	0.25392
0.05	-1.3×10^{-216}	35	4.123×10^{-165}	0.00781	0.25815
0.10	4.6×10^{-217}	35	4.155×10^{-165}	0.00788	0.26252
0.15	-1.5×10^{-216}	35	4.184×10^{-165}	0.00795	0.26705
0.19	5.8×10^{-218}	35	4.204×10^{-165}	0.00800	0.27064
0.20	-1.9×10^{-216}	35	4.208×10^{-165}		0.27145

Table 5.1: The barrier $\mathcal{B} = [0.6, 1] \times [497.4, 497.9]$, $X = 994.8$, $y_0 = 0.2$, at six of the 21 grid times (computed, floating point at 214 digits; the rows $t = 0.05$ to 0.19 are not recomputed in ball arithmetic). Taylor ratio: the largest over the walk of $\sum_{k \le 8} \frac{D^k}{k!} |\partial_t^k \xi_t| / |\xi_t|$.

Proposition 5.3.1 (Computed, and certified in ball arithmetic for all $t \in [0, 0.2]$). *With $t_0 = y_0 = 0.2$ and the rectangle $\mathcal{B} = [0.6, 1] \times [497.4, 497.9]$:*

(a) at each of the 21 times $t_i = 0, 0.01, \ldots, 0.2$ the computed winding number of ξ_{t_i} around $\partial\mathcal{B}$ is 0, the sums of arguments being below 3×10^{-216} in absolute value, with 35 points per walk and the ratio $|\Delta\xi|/|\xi|$ at most 0.169 along the walks; the control function returns winding number 1.0 in 69 points;

(b) at every point of every walk the relative Taylor sum of order 8 over the step $D = 0.01$ is at most 0.00801, and the tilted tail beyond order 8 is at most 7.4×10^{-186}, that is 1.8×10^{-21} relative to the smallest $|\xi_t|$ met; one Taylor step from $t = 0$ over the whole interval, $D = 0.2$, gives the relative sum 0.171 and a relative tail below 9.9×10^{-10};

(c) the minimum of v_t over the 9×11 grid of $\mathcal{B}$ increases with t, from 0.25392 at $t = 0$ to 0.27145 at $t = 0.2$, and it is attained on the left edge $\sigma = 0.6$, at $\theta = 497.5$ for $t \le 0.17$ and at $\theta = 497.45$ for $t \ge 0.18$.

The floating point values above agree with a ball arithmetic computation, which certifies for every $t \in [0, 0.2]$ (not only at the grid times): the winding number of ξ_0 around $\partial\mathcal{B}$ is 0 (and the control returns 1); one Taylor step of order 8 from $t = 0$ over $D = 0.2$ gives $|\xi_t - \xi_0| \le 0.198|\xi_0|$ on all of $\partial\mathcal{B}$ for all $t \in [0, 0.2]$, hence ξ_t has no zero on $\partial\mathcal{B}$, the winding number stays 0, and there is no zero of ξ_t in $\mathcal{B}$ for $t \in [0, 0.2]$; and $v_t \ge 0.1587$ on all of $\mathcal{B}$ for all $t \in [0, 0.2]$. Still floating point or taken from elsewhere: the table rows at $t = 0.05$ to 0.19, the untilted bound below, the monotone increase of $\min v_t$ in t in part (c) (the certified bound is 0.1587, below the grid values because the cells are discs), and the Polymath15 barrier hypotheses (i) and (ii) of Theorem 5.1.1 and their use at $X = 994.8$.

Read with Section 5.1, part (a) says that at the grid times no zero of ξ_t lies in $\mathcal{B}$, and part (b) says that between grid times $|\xi_t - \xi_{t_i}| < \frac{1}{2}|\xi_{t_i}|$ at the points of the walks, which by Rouché's theorem would carry the winding number 0 across each interval if it held on the whole of $\partial\mathcal{B}$. Part (c) gives the same conclusion pointwise through Proposition 5.2.4, again only on a grid. The untilted bound for the same tail, the absolute value bound of Proposition 4.1.28, is 1.4×10^{-23} for $D = 0.2$, some 10^{141} times $|\xi_t|$ on $\mathcal{B}$, so the tilt is essential even at this height.

Three gaps separated the floating point proposition from a certified barrier at this height, and the ball arithmetic computation closes each by the methods of Section 6.5: the evaluator runs in ball arithmetic with the truncation and discretisation errors in the radius (Theorem 6.5.8); the winding at $t = 0$ is verified on the segments between walk points through a derivative bound (Lemmas 6.5.4 and 6.5.5); and the Taylor criterion is verified on all of $\partial\mathcal{B}$ by the same derivative bound applied to $\partial_t^k \xi_t$, with the tail integral enclosed by interval subdivision, as the constants $\bar{K}_\eta$ are in Section 6.5.2. The certified bounds are weaker than the point values (the one step ratio is 0.198 against 0.171, the supremum over discs against the maximum over the walk), and the threshold 1 of Rouché's theorem is met with a factor above 5 to spare.

Two features of the numbers are worth recording. First, the barrier is cheap in t: at this height a single Taylor step covers $[0, 0.2]$ with a factor of almost 3 to spare against the threshold $\frac{1}{2}$ used in the walks (Rouché's theorem needs only 1), whereas [247] needed many time values at $X \approx 6 \times 10^{10}$; heuristically, the relative size of $\partial_t \xi_t = \frac{1}{4}\partial_w^2 \xi_t$ is of order $(\log\theta)^2$ near zeros and of order the inverse squared distance to the nearest zero, so the number of time steps grows with the height and with the closeness of the zeros to the window. Second, $\min v_t$ on $\mathcal{B}$ moves by only 7% between $t = 0$ and $t = 0.2$, in line with Proposition 6.4.1(iii): the density certificate does not see the difference between the time at which the clock is known to be a GGC and the time at which it is not known to be one.

5.4 What the Thorin form adds, and why it cannot reach $t = 0$

5.4.1 What it adds

Three things. First, a single statement that joins the method to the rest of the book: the output of the barrier method is a point $(t_0, \frac{1}{2} + \frac{y_0}{2})$ of the GGC region $\mathcal{G}$ (Proposition 5.2.1), and every later improvement is a point further down the phase diagram of Chapter 6. Second, a geometry: the canopy, the barrier and the strip theorem become a parabolic band, a window between confocal parabolas, and a family of right opening parabolas collapsing onto the half line (Lemma 5.2.2, Proposition 5.2.3), the same picture as the shrinking parabolas of Corollary 23.2.2. Third, a quantitative certificate: the density v_t on the barrier rectangle (Proposition 5.2.4) gives a margin that can be compared across times and heights, which the winding number, an integer, does not.

5.4.2 What it does not add

It adds no analytic input. Hypothesis (ii) is the heart of the method, and the Thorin form restates it as (ii′) without proving it; Polymath15 prove it through an effective Riemann Siegel approximation of H_t, and the Thorin side has no substitute. Lemma 6.2.6 writes ξ_t as a Gaussian average of ζ shifted to the right by about $\frac{t}{4}\log\frac{\theta}{2\pi}$, which is one way to see why a positive time helps: the average samples ζ at a real part that grows with the height, where the Euler product dominates. The density certificate is also weaker than the winding test in the sense of Section 5.2: it can fail where no zero lies, and its margin at basepoints near $\frac{1}{2}$ tends to zero (Remark 12.1.19), so it can only be used with y_0 bounded away from 0.

5.4.3 Why the method cannot reach $t = 0$

The bound of Theorem 5.1.1 is $t_0 + \frac{1}{2}y_0^2$, and RH is $\Lambda \le 0$, so a proof of RH by the method needs $t_0 \to 0$ and $y_0 \to 0$. Each limit fails for a separate reason.

As $t_0 \to 0$, hypothesis (ii) becomes a zero free region for ζ itself: no zero with $\operatorname{Re} s \ge \frac{1}{2} + \frac{y_0}{2}$ above height $X/2$. For $y_0 < 1$ no such region is known; the regions of Vinogradov and Korobov narrow towards $\operatorname{Re} s = 1$, and the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 and not refereed [227, 228, 198] would give it only for $y_0 \ge \frac{3}{4}$, that is the point $(0, \frac{7}{8})$ of Corollary 6.3.3(d) and the bound $\Lambda \le \frac{9}{32}$, worse than 0.2. The effective approximation that proves (ii) at positive time depends on the time: heuristically the flow at time t makes the zeros near height x real once $t \log x$ exceeds a constant, and the theorem of [247] cited in the proof of Corollary 4.1.25 says that for an absolute constant C all zeros of H_t with $|x| \ge e^{C/t}$ are real and simple (we read only its statement as the book cites it, not its proof). The required height X therefore grows like e^{c/t_0}, and the table of [247] fits $\log X \approx 4.24/\Lambda + 8.28$ (Proposition 6.6.2), which needs verification of RH to about 10^{40} for $\Lambda \le 0.05$. In the limit $t_0 = 0$ the barrier height is infinite, and the method has reduced RH to RH.

As $y_0 \to 0$, the barrier rectangle $\mathcal{B}$ widens to the left edge $\operatorname{Re} s = \frac{1}{2}$, on which ξ_t has its real zeros for $t \ge \Lambda$, and the band of Lemma 5.2.2 closes onto the half line, where the atoms of μ_t live. The winding test on a rectangle touching the critical line needs a window free of zeros on the line at every time in $[0, t_0]$, which moving zeros make hard but not impossible (the box barrier of Proposition 4.1.26, not carried out; Remark 19.15.1), and the density margin vanishes (Remark 12.1.19). More fundamentally, a statement at $y_0 = 0$ and $t_0 > 0$ would only give $\Lambda \le t_0$.

Finally, the Thorin form cannot make up the difference, for the reason given in Section 1.4: a perturbation of ξ with the same zeros below any height passes the finite tests of the heat trace listed in Theorem 1.4.1, so a finite barrier computation together with an analytic tail at $t_0 > 0$ can certify points of $\mathcal{G}$ at positive time but never the corner $(0, \frac{1}{2})$. By Rodgers and Tao, $\Lambda \ge 0$ [250], so the corner is on

the boundary of the region the method can reach, and the step to it is a closed condition at $t = 0$ that must come from an analytic argument which uses the Euler product (Proposition 23.2.3 rules out arguments that survive a Gaussian tilt).

5.5 Open problems

Problem 5.1. Proposition 5.3.1 is certified in ball arithmetic at $X = 994.8$ (Section 5.3). Determine, as a function of the height X, the number of time steps that a certified barrier of width one at $t_0 = y_0 = 0.2$ needs. Is it of order $(\log X)^2$, as the size of $\partial_t \xi_t / \xi_t$ near the zeros suggests?

Problem 5.2. Measure $\min_{\mathcal{B}} v_t$ on barrier rectangles at heights up to 10^{10} with a Riemann Siegel type evaluator (step 1 of Section 6.6.4), at the windows chosen by the Euler product heuristic of [247]. Does the density certificate succeed wherever the winding test does, for $y_0 = 0.2$, and how does its margin depend on the distance from the window to the nearest zeros?

Problem 5.3. In the chart of squares, the barrier is a window between $\mathcal{Q}_X$ and $\mathcal{Q}_{X+1}$. Is there a version of Theorem 5.1.1 in which the window is replaced by a condition on the Thorin measure alone, for instance a bound on the mass of μ_t near $\mathcal{Q}_X$ along the flow, that is equivalent to (iii) and can be checked from moments of the Pólya kernel tilted by e^{tu^2}?

Problem 5.4. Find an analytic statement of type (ii) at $t_0 = 0$ and some $y_0 < \frac{3}{4}$ (the announced half plane would give $y_0 = \frac{3}{4}$ conditionally) that uses the Euler product: a zero free region for ζ to the right of $\operatorname{Re} s = \frac{1}{2} + \frac{y_0}{2}$ above a computable height. With the barrier and the verification of RH it would give $\Lambda \le \frac{1}{2} y_0^2$; it is a quasi Riemann hypothesis, and the problem is the hard core identified in Section 1.4.

Does the barrier method give a path to RH?

No. It gives a monotone sequence of theorems $\Lambda \le c$ with $c > 0$, each of them a point of the GGC region at positive time, and its cost grows like $e^{c'/\Lambda}$ in the height to which RH must be verified. The Thorin form states each step as a GGC theorem and adds a parabolic geometry and a density margin, but it supplies no analytic input below the basepoint one, and the limit $t_0 \to 0$ turns the analytic hypothesis of the method into a zero free region for ζ, which is the open step of the path to RH and not a way around it.

Chapter 6

The Thorin phase diagram

This chapter sets out the most promising computational path to RH that the book has found, and says exactly where it stops.

The sine squared lemma and the Thorin side of Chapter 3 offer no computational alternative to the de Bruijn Newman flow, only the right language for it. The Thorin side is the object to compute, prime free and well defined at every basepoint, and the flow is the only known source of an analytic tail below the basepoint $\frac{7}{8}$. What separates the theorems $\Lambda \le 0.22$ of Polymath15 [247] and $\Lambda \le 0.2$ of Platt and Trudgian [232] from RH is not anything a computed density shows; it is the existence of the analytic tail at $t = 0.2$ and its absence at $t = 0$. The chapter supersedes the authors' working report [243]. We use the normalisation of Section 4.1: $H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du$, $\xi_t(s) = 8H_t(-2i(s - \frac{1}{2}))$, a zero $x + iy$ of H_t is the zero $\frac{1}{2} - \frac{y}{2} + \frac{ix}{2}$ of ξ_t, and $P_a(x) = a/(\pi(a^2 + x^2))$ is the Cauchy density of scale $a > 0$.

6.1 Which computations see past the verified height

Every route to RH in this book replaces the hypothesis by an infinite family of statements, each certifiable by a finite computation. What separates the routes is what they say above the height the computation reaches. If nothing, the route is finite information in the sense of Section 1.4: by Theorem 1.4.1 a perturbation with the same zeros below that height and zeros off the line above it passes the same test, so the test certifies no more than a verification of the zeros up to a height determined by its size.

Definition 6.1.1. A route *reaches every height* if there are a finite computation C and an analytic statement A valid above a barrier X such that C and A together imply a statement about all zeros of ξ, or of a fixed approximant or deformation of ξ, that is not implied by the zeros below X alone.

6.1.1 The finite information routes

Most of the book's tests are finite information, and each buys a measurable height. The heat trace has its derivative signs to order 1.16×10^{25} from the verification to 3×10^{12} (Theorem 8.1.13), and the order is a height in disguise: an off line zero at height γ first breaks the alternation at an order of about $t\gamma^2$, with t the parameter of the heat trace and not the time of the flow. The Bosch function $\varphi(x^p)$ of Section 9.3 is HCM for $|p| \leq 1 - 1/(\pi T_0)$, and $1 - p \approx 1/(\pi T)$ is again a height (Corollary 9.3.7). Li coefficients see an off line zero at height γ and distance d from the line only from an order of about γ^2/d, while the Hankel test of Section 4.1.3 often registers a pair at an order close to $2k + 2$, set by its index k, at a cost in precision that grows linearly with the order (Section 14.4). The Cauchy flow ρ_ε is positive on the verified window for every ε, but its minimum is attained at the origin and tends to zero with ε (Theorem 12.1.18 and Remark 12.1.19), so positivity is never won by a fixed margin, and positivity on one line is not known to exclude zeros to its right (Remark 2.1.4). The ferromagnetic distance of Section 19.1 is below 1.932×10^{-6}, and it is zero exactly under RH, but the approximating laws keep only the first K zeros, so it sees nothing above them. The Weil Gram matrices and the arithmetic forms of Section 6.6.1 below have the same property, with a resolution in height set by the support of the test functions.

6.1.2 The three routes that reach every height

Three routes reach every height in the sense of Definition 6.1.1.

The Euler product: basepoints $\alpha \geq 1$. At the basepoint one, the Thorin density of the shifted clock is positive at every height, and the zeros below 2000 in ball arithmetic and explicit bounds for $S(T)$ [100, 9] certify that it is smallest only at the origin (Propositions 12.1.11 and 12.1.12). Above the barrier, the analytic input is the explicit bound for $S(T)$, and positivity itself comes from the Euler product, through the absence of zeros in $\operatorname{Re} s \geq 1$. Below $\alpha = 1$ the same structure gives a statement at every height only together with a zero free half plane (Proposition 12.1.13); the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [227] would extend the route to $\alpha \geq \frac{7}{8}$ (Section 25.1.2), and below $\frac{7}{8}$ nothing is known.

The functional equation: symmetrised truncations. The symmetrised truncations of the Biane, Pitman and Yor lattice of Section 17.1 have all their zeros in the strip on the line, at every height, for $n \leq 24$ and $n = 32, 36$, proved above explicit thresholds and computed below them (Theorem 17.1.7 and Corollaries 17.1.10 and 17.1.12). Above the threshold the analytic input is the symmetry of the functional equation. RH would follow from infinitely many n (Problem 17.1), and no argument uniform in n is known.

The heat flow: positive time. At time $t > 0$ and for a suitable $y_0 > 0$, the function H_t has no zeros with $\operatorname{Im} z \geq y_0$ above a barrier, by an analytic argument, the effective Riemann Siegel approximation of Polymath15 [247]; a computation of winding numbers keeps zeros from entering through a strip of width one at the barrier for the times up to t, and the verification of RH covers the zeros below it. Together they give an upper bound for Λ, since then $\Lambda \leq t + y_0^2/2$ by de Bruijn's strip theorem [61, Theorem 13] (Polymath15 take $t = y_0 = 0.2$): $\Lambda \leq 0.22$ [247] and, from the verification to 3×10^{12}, $\Lambda \leq 0.2$ [232].

route	analytic input above the barrier	what is certified at every height	what is missing
basepoint $\alpha \geq 1$	Euler product, explicit bounds for $S(T)$	the shifted clock is a GGC, its density minimal at the origin	any basepoint below one (below $\frac{7}{8}$ if the announced half plane is confirmed)
symmetrised truncations	functional equation symmetry	zeros of K_n in the strip on the line, $n \leq 24$ and $n = 32, 36$ (floating point counts below the thresholds)	infinitely many n
heat flow	Riemann Siegel at $t > 0$	$\Lambda \leq 0.2$; GGC at points (t, α)	a tail at $t = 0$

Table 6.1: The three routes of the book that reach every height. All other routes are finite information.

The first and the third routes are the two coordinates of the phase diagram of Section 6.3. The second route has no coordinate on the diagram and stands apart; Problem 6.3 asks whether it can be given one.

6.2 Clocks along the flow at every basepoint

Theorem 2.1.2 decides when the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is a generalised gamma convolution, and Theorem 17.1.16 with Corollary 9.2.5 decides it for the centre clock R_t of the flow. This section extends both to every time $t \geq 0$ and every basepoint $\alpha \geq \frac{1}{2}$.

Lemma 6.2.1 (Structure of ξ_t). *For every $t \geq 0$: (i) ξ_t is entire of order at most one, real on the real axis, and $\xi_t(1 - s) = \xi_t(s)$; (ii) $\xi_t(\frac{1}{2} + x) > 0$ for real x; (iii) with $w = s - \frac{1}{2}$ and the zeros written $\rho = \frac{1}{2} + \tilde{\rho}$,*

$$\xi_t(\tfrac{1}{2} + w) = \xi_t(\tfrac{1}{2}) \prod \Big(1 - \frac{w^2}{\tilde{\rho}^2}\Big),$$

the product running over one representative of each pair $\{\tilde\rho, -\tilde\rho\}$*, with multiplicity, and* $\sum_\rho |\tilde\rho|^{-2} < \infty$*.*

Proof. Since $\Phi_{\mathrm{dBN}}(u) \le Ce^{9u-\pi e^{4u}}$ for $u \ge 0$ and $|\cos(zu)| \le e^{|z|u}$,

$$|H_t(z)| \le C\int_0^\infty e^{tu^2+|z|u+9u-\pi e^{4u}}\,du \le \exp(C'|z|\log|z|)$$

for large $|z|$; so H_t and ξ_t are entire of order at most one. Evenness of H_t gives the reflection symmetry, and reality of the integrand on the imaginary axis gives ξ_t real on the real axis, with

$$\xi_t(\tfrac12 + x) = 8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(2xu)\,du > 0,$$

since $\Phi_{\mathrm{dBN}} > 0$. Hadamard's theorem for order at most one gives $H_t(z) = e^{a+bz}\prod_k E_1(z/z_k)$ over the zeros, none of which is zero since $H_t(0) > 0$. Pairing z_k with $-z_k$, a zero by evenness, gives $E_1(z/z_k)E_1(-z/z_k) = 1 - z^2/z_k^2$, and $\sum|z_k|^{-2} < \infty$ since the convergence exponent is at most one. Evenness forces $b = 0$, and the substitution $z = -2iw$, $\tilde\rho = iz_k/2$ gives (iii). □

Lemma 6.2.2 (One conjugate pair). *Let* $d, \gamma \in \mathbb{R}$*, not both zero, and* $\varphi(u) = (d^2+\gamma^2)/((\sqrt u + d)^2 + \gamma^2)$ *for* $u \ge 0$*.*

(a) If $d > 0$*, then* φ *is the Laplace transform of a GGC whose Thorin measure, in the chart* $\theta = \sqrt z$*, is* $[P_d(\theta-\gamma) + P_d(\theta+\gamma)]\,d\theta$ *on* $\theta > 0$ *(compare Proposition 2.1.3):*

$$-\log\varphi(u) = \int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big[P_d(\theta-\gamma)+P_d(\theta+\gamma)\big]\,d\theta.$$

(b) If $d = 0$*, then* $\varphi(u) = (1+u/\gamma^2)^{-1}$*, an exponential law, with Thorin measure* δ_{γ^2} *in the variable* z*, that is,* δ_γ *in the chart* $\theta = \sqrt z$*.*

(c) If $d < 0$*, then* φ *has a pole at* $u_0 = (-d+i\gamma)^2 \in \mathbb{C}\setminus(-\infty, 0]$ *and is not the Laplace transform of a GGC.*

Proof. (a) Both sides vanish at $u = 0$, so it suffices to compare derivatives in $u > 0$. On the left, with $F(u) = (\sqrt u + d)^2 + \gamma^2$, $F'/F = (\sqrt u + d)/(\sqrt u\, F)$. On the right, differentiating under the integral and using the symmetry $\theta \mapsto -\theta$ of the second kernel,

$$\begin{aligned}\int_{-\infty}^\infty \frac{P_d(\theta-\gamma)}{\theta^2+u}\,d\theta &= \frac{\pi}{\sqrt u}\int_{-\infty}^\infty P_d(\theta-\gamma)P_{\sqrt u}(\theta)\,d\theta\\ &= \frac{\pi}{\sqrt u}P_{d+\sqrt u}(\gamma) = \frac{d+\sqrt u}{\sqrt u\,((d+\sqrt u)^2+\gamma^2)},\end{aligned}$$

by the semigroup property $P_a * P_b = P_{a+b}$. The integral converges, since $\log(1 + u/\theta^2)$ is integrable at 0 and $O(\theta^{-2})$ at infinity, and the measure is positive since $d > 0$. (b) is immediate. (c) The equation $\sqrt{u} + d - i\gamma = 0$ has the solution $\sqrt{u} = -d + i\gamma$, whose real part is positive, so it is the principal root and u_0 lies off $(-\infty, 0]$ (if $\gamma = 0$ then $u_0 = d^2 > 0$); the Laplace transform of a GGC is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$ [30]. □

THEOREM 6.2.3 (Threshold along the flow). *Let $t \geq 0$ and $\alpha \geq \frac{1}{2}$. The shifted time t clock*

$$\varphi_{t,\alpha}(u) = \frac{\xi_t(\alpha)}{\xi_t(\alpha + \sqrt{u})}$$

is the Laplace transform of a GGC if and only if $\alpha \geq \sup_\rho \operatorname{Re}\rho$ over the zeros ρ of ξ_t. In that case its Thorin measure, in the chart $\theta = \sqrt{z}$, is

$$\sum_{\substack{\rho=\beta+i\gamma \\ \gamma>0,\ \beta<\alpha}} \left[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\right] d\theta \ + \sum_{\substack{\rho=\alpha+i\gamma \\ \gamma>0}} m_\rho\, \delta_\gamma,$$

where the first sum counts each zero with its multiplicity and m_ρ is the multiplicity of ρ; this is the Poisson form of Proposition 2.1.3 for ξ_t.

Proof. By Lemma 6.2.1, with $a = \alpha - \frac{1}{2} \geq 0$,

$$\varphi_{t,\alpha}(u) = \prod \frac{1 - a^2/\tilde{\rho}^2}{1 - (a + \sqrt{u})^2/\tilde{\rho}^2} = \prod_\rho \Big(1 - \frac{\sqrt{u}}{\rho - \alpha}\Big)^{-1},$$

where the last product runs over all zeros and is grouped so that each finite partial product contains, with every zero, its partner under $\tilde{\rho} \mapsto -\tilde{\rho}$; the denominators do not vanish since $\xi_t(\alpha) > 0$. Because ξ_t is real on the real axis, the zeros are also closed under conjugation, and regrouping into conjugate pairs is legitimate: both groupings give absolutely convergent products, the conjugate pairing because $(1-\sqrt{u}/(\rho-\alpha))(1-\sqrt{u}/(\bar{\rho}-\alpha)) = 1+(2d\sqrt{u}+u)/(d^2+\gamma^2)$ with $d = \alpha-\beta$ bounded (the zeros of H_t lie in $|\operatorname{Im} z| \leq 1$ by de Bruijn's strip theorem [61, Theorem 13], as in Section 4.1.1; and $0 \leq d \leq 2\alpha - 1$ directly when $\alpha \geq \sup \operatorname{Re}\rho$, the only case in which the pairing is used, by the symmetry $\rho \mapsto 1 - \rho$) and $\sum |\rho - \alpha|^{-2} < \infty$, and the partial products over $|\gamma| \leq T$ agree. Hence

$$\varphi_{t,\alpha}(u) = \prod_{\{\rho,\bar{\rho}\}} \frac{d^2 + \gamma^2}{(\sqrt{u} + d)^2 + \gamma^2},$$

a product of the functions of Lemma 6.2.2, one for each conjugate pair, every pair being genuine because ξ_t has no real zeros by Lemma 6.2.1(ii).

If $\alpha \geq \sup \operatorname{Re}\rho$, every factor is a GGC transform by Lemma 6.2.2(a) and (b). Finite products of GGC transforms are GGC transforms with Thorin measures

adding, the partial products converge pointwise to $\varphi_{t,\alpha}$, which is continuous with $\varphi_{t,\alpha}(0) = 1$, so by the continuity theorem and the closure of the class under weak limits ([30, Theorem 3.1.5], Section 9.1.1) the limit is a GGC transform. Its Thorin measure is the full sum μ displayed in the theorem, and there is no drift, by the following monotone convergence argument. Write μ_T for the sum of the kernels and atoms over the zeros with $0 < \gamma \le T$, so that $\mu_T \uparrow \mu$ as $T \to \infty$, and φ_T for the partial product over the same zeros. By Lemma 6.2.2, $-\log \varphi_T(u) = \int \log(1 + u/\theta^2)\, \mu_T(d\theta)$ for $u \ge 0$, and the integrand is nonnegative, so by monotone convergence $-\log \varphi_T(u)$ increases to $\int \log(1 + u/\theta^2)\, \mu(d\theta)$. Since $\varphi_T(u) \to \varphi_{t,\alpha}(u) > 0$, this gives $-\log \varphi_{t,\alpha}(u) = \int \log(1 + u/\theta^2)\, \mu(d\theta) < \infty$ for every $u \ge 0$. If some zero has $\beta > \alpha$, then $\xi_t(\alpha + \sqrt{u_0}) = 0$ at $u_0 = (\rho - \alpha)^2$, whose principal root is $\rho - \alpha$ because $\operatorname{Re}(\rho - \alpha) > 0$, so $\varphi_{t,\alpha}$ has a pole in $\mathbb{C} \setminus (-\infty, 0]$ and is not a GGC transform. □

Remark 6.2.4. The proof uses nothing about ξ_t beyond Lemma 6.2.1, so it applies to any even real entire function of order at most one with a genus zero product in w^2, positive on the real axis, in particular to the perturbation of Theorem 1.4.1, whose shifted clocks are therefore not GGC transforms at any basepoint below the real part of the displaced zeros. The Thorin density of $\varphi_{t,\alpha}$ in the chart θ is $\pi^{-1} \operatorname{Re}(\xi_t'/\xi_t)(\alpha + i\theta)$ away from atoms, which is $\rho_{\alpha - 1/2, t}(\theta)$ of Section 4.1.1. The shifted time t clock is a function of u only: where the threshold condition fails the theorem claims no probability law behind it, and for $\alpha = \frac{1}{2}$ and $0 < t < 0.2$ it is open, without RH, whether R_t is the transform of a probability law at all (Problem 19.3); the basepoint family at $t = 0$ and the flow at $\alpha = \frac{1}{2}$ are the two one parameter families of laws compared in Section 19.7.

6.2.1 The minimum at the origin along the flow

For $t \ge 0$ and $\alpha > \frac{1}{2}$ put $v_t(\alpha, \theta) = \operatorname{Re}(\xi_t'/\xi_t)(\alpha + i\theta)$, $b_t(\alpha) = v_t(\alpha, 0)$ and $V_t = (\log \xi_t)''(\frac{1}{2})$. For $\alpha \ge \sup \operatorname{Re} \rho$, $v_t(\alpha, \cdot)/\pi$ is the Thorin density of the shifted clock $\varphi_{t,\alpha}$ in the chart θ (Remark 6.2.4), and since $\int_0^\infty \log(1 + u/\theta^2)\, d\theta = \pi\sqrt{u}$,

$$\tilde{\varphi}_{t,\alpha}(u) = \varphi_{t,\alpha}(u)\, e^{b_t(\alpha)\sqrt{u}} = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big) \frac{v_t(\alpha, \theta) - b_t(\alpha)}{\pi}\, d\theta \Big\}.$$

So $\tilde{\varphi}_{t,\alpha}$, the shifted clock with the constant part of its Thorin density removed, is the Laplace transform of a GGC exactly when $v_t(\alpha, \theta) \ge b_t(\alpha)$ for every θ, that is when the minimum of $v_t(\alpha, \cdot)$ is at the origin. At $t = 0$ this is the question of Theorem 12.1.18. For $t \ge \Lambda$, unconditionally for $t \ge 0.2$, the zeros of ξ_t are $\frac{1}{2} \pm i\theta_k(t)$ with $0 < \theta_1(t) \le \theta_2(t) \le \cdots$, and $V_t = 2 \sum_k \theta_k(t)^{-2} = 2\mathbb{E}T_t$ (Section 4.1.8).

Proposition 6.2.5 (The minimum at the origin along the flow). *Let $t \ge \Lambda$, $\alpha > \frac{1}{2}$, $a = \alpha - \frac{1}{2}$, $\theta_k = \theta_k(t)$, $v = v_t(\alpha, \cdot)$ and $b = b_t(\alpha)$.*

(i) $b \le aV_t$.

(ii) *If* $a^2 \le \frac{3}{5}\theta_1^2$, *then* $v(\theta) \ge b$ *for* $|\theta| \le \theta_1$. *If* $a \le \frac12$ *and* $V_t < 4$, *then* $v(\theta) > b$ *at every* θ *within* $D_t = (1/V_t - \frac14)^{1/2}$ *of some* θ_k.

(iii) *Let* $a \le \frac12$ *and* $V_t < 4$. *If every* $\theta \in [\theta_1, T_*]$ *is within* D_t *of some* θ_k, *and for every* $\theta > T_*$ *some* $w > 0$ *has* $\#\{k : |\theta_k - \theta| \le w\} > V_t(w^2 + \frac14)$, *then* $v \ge b$.

(iv) *Let* $\alpha' > \frac12$ *and* $a' = \alpha' - \frac12$ *with* $a'^2 \le \frac35\theta_1^2$. *If* $v_t(\alpha', \theta) \ge a'V_t$ *for every* $\theta \ge \theta_1$, *then* $v_t(\alpha, \cdot) \ge b_t(\alpha)$ *for every* $\alpha \in (\frac12, \alpha']$.

(v) *As* $\theta \to 0$, $(v(\theta) - b)/\theta^2 \to \sum_k 2a(3\theta_k^2 - a^2)/(a^2 + \theta_k^2)^3 = 6a \operatorname{Var} T_t + O(a^3)$.

Under the hypotheses of (iii), respectively (iv), $\tilde\varphi_{t,\alpha}$ *is the Laplace transform of a GGC for every* $\alpha \in [\frac12, 1]$, *respectively* $[\frac12, \alpha']$, *the case* $\alpha = \frac12$ *being Theorem 6.2.3; there* $b_t(\frac12) = \operatorname{Re}(\xi_t'/\xi_t)(\frac12) = 0$ *by the symmetry* $\xi_t(1-s) = \xi_t(s)$, *so that* $\tilde\varphi_{t,1/2} = \varphi_{t,1/2}$.

Proof. By Lemma 6.2.1, with $q(u) = a/(a^2+u^2)$, $v(\theta) = \sum_k [q(\theta - \theta_k) + q(\theta + \theta_k)]$ and $b = \sum_k 2q(\theta_k)$. (i) $2q(\theta_k) \le 2a\theta_k^{-2}$. (ii) For $0 \le \theta \le \theta_k$ put $p_k(\theta) = q(\theta_k - \theta) + q(\theta_k + \theta)$. Since $u \mapsto u/(a^2+u^2)^2$ decreases on $u \ge a/\sqrt3$, p_k increases where $\theta_k - \theta \ge a/\sqrt3$, and elsewhere $p_k \ge q(a/\sqrt3) = \frac{3}{4a} \ge 2q(\theta_k)$, because $\theta_k^2 \ge \frac53 a^2$. So $p_k(\theta) \ge p_k(0)$ for $|\theta| \le \theta_1$, and summing gives the first claim. If $|\theta - \theta_k| = d$, then $v(\theta) \ge q(d) > aV_t \ge b$ as soon as $a^2 + d^2 < 1/V_t$, which holds for $d < D_t$ when $a \le \frac12$. (iii) Since $V_t \ge 2\theta_1^{-2}$, $\frac35\theta_1^2 > \frac14 \ge a^2$, so (ii) gives $v \ge b$ on $[0, T_*]$; above T_* the more than $V_t(w^2 + \frac14)$ zeros of the window give $v(\theta) > aV_t(w^2+\frac14)/(a^2+w^2) \ge aV_t \ge b$. (iv) Each $q(u)/a = 1/(a^2+u^2)$ decreases in a, so for $a \le a'$ and $|\theta| \ge \theta_1$, $v_t(\alpha, \theta) \ge \frac{a}{a'} v_t(\alpha', \theta) \ge aV_t \ge b_t(\alpha)$, as in Theorem 12.1.15; for $|\theta| \le \theta_1$ apply (ii), since $a^2 \le a'^2 \le \frac35\theta_1^2$. (v) $v(\theta) - b = \sum_k [p_k(\theta) - p_k(0)]$, $p_k''(0) = 2q''(\theta_k)$ with $q''(u) = 2a(3u^2 - a^2)/(a^2+u^2)^3$, the series of second derivatives converges uniformly near 0, and $\sum_k \theta_k^{-4} = \operatorname{Var} T_t$ (Section 4.1.8). The last assertion is the identity displayed before the proposition. □

At $t = 0.2$, with $\operatorname{Var} T_{0.2} = 3.77203 \times 10^{-5}$ from the table of Section 4.1.8, the coefficient in (v) is $2.2632 \times 10^{-4} a$ to first order in a, against $2.2304 \times 10^{-4} a$ at $t = 0$ under RH: the flow raises the curvature of v at its minimum slightly. From the same table, $V_{0.2} = 0.0464019$, $D_{0.2} = 4.6153$ and $\theta_1(0.2) = 14.0768$.

Lemma 6.2.6 (The heat average as a shifted Gaussian average). *Write* $\xi(s) = g(s)\zeta(s)$ *with* $g(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, *and* $\ell = g'/g$. *For* $t \ge 0$ *and* $\operatorname{Im} s > 0$,

$$\xi_t(s) = g(s)\, e^{t\ell(s)^2/4}\, \zeta_t(s), \qquad \zeta_t(s) = \mathbb{E}\, \Psi\Big(s, \frac{t}{2}\ell(s) + \sqrt{\tfrac{t}{2}}\, Z\Big),$$

$$\Psi(s,h) = \frac{\xi(s+h)}{g(s)\, e^{h\ell(s)}},$$

with Z standard normal. Moreover $\Psi(s,h) = \zeta(s+h)e^{r(s,h)}$ with $r(s,h) = h^2\int_0^1(1-u)\,\ell'(s+uh)\,du$, and if $Y = \min\big(\operatorname{Im} s, \operatorname{Im} s + \frac{t}{2}\operatorname{Im}\ell(s)\big) > 0$, then $|r(s,h)| \le |h|^2\big(\frac{1}{4Y} + \frac{7}{4Y^2}\big)$ for every $h \in \frac{t}{2}\ell(s) + \mathbb{R}$.

Proof. By Proposition 4.1.18, since $\xi_t(\frac12 + i\theta) = 8H_t(2\theta)$ and Z is symmetric, $\xi_t(s) = \mathbb{E}\,\xi(s+\tau Z)$ with $\tau = \sqrt{t/2}$. With ϕ_{N} the normal density, $u \mapsto \xi(s+\tau u)\phi_{\mathrm{N}}(u)$ is entire, and because ξ has order one the Gaussian factor lets the line of integration move to $u = v + \tau\ell(s)$. Since $\phi_{\mathrm{N}}(v+\tau\ell) = \phi_{\mathrm{N}}(v)e^{-\tau\ell v - t\ell^2/4}$, this gives $\xi_t(s) = e^{-t\ell^2/4}\mathbb{E}[\xi(s+h)e^{-\tau\ell Z}]$ with $h = \frac{t}{2}\ell + \tau Z$, and $\xi(s+h) = g(s)e^{h\ell}\Psi(s,h)$ with $h\ell - \tau\ell Z = \frac{t}{2}\ell^2$ gives the identity. Off the real axis g has no zeros or poles, so $\log g(s+h) - \log g(s) - h\ell(s) = r(s,h)$ along the segment from s to $s+h$, by Taylor's formula. For $h \in \frac{t}{2}\ell(s) + \mathbb{R}$ the imaginary part of $s+uh$ is linear in u, hence at least Y on the segment. Euler summation gives $\psi'(z) = \frac{1}{z} + \frac{1}{2z^2} - 2\int_0^\infty(\{x\} - \frac12)(z+x)^{-3}\,dx$, whose last term is at most $2/(\operatorname{Im} z)^2$ in absolute value; with $\ell'(v) = -(v-1)^{-2} + \frac14\psi'(\frac{v}{2}+1)$, which follows from $\psi'(z) = z^{-2} + \psi'(z+1)$, this gives $|\ell'(v)| \le \frac{1}{2y} + \frac{7}{2y^2}$ for $\operatorname{Im} v = y > 0$, and the bound on r follows. □

Proposition 6.2.7 (The zero count of ξ_t). *Let $t \ge 0$ and let $N_t(T)$ be the number of zeros ρ of ξ_t with $0 < \operatorname{Im}\rho \le T$, counted with multiplicity whether or not they lie on the critical line. For $T > 0$ not an ordinate let $\pi S_t(T)$ be the change of a continuous argument of $\zeta_t = \xi_t/(g\,e^{t\ell^2/4})$ along the segments from 2 to $2+iT$ and from $2+iT$ to $\frac12 + iT$, starting from $\arg\zeta_t(2) = 0$, and at ordinates take right limits.*

(i) $N_t(T) = 1 + \vartheta(T)/\pi + \frac{t}{4\pi}\operatorname{Im}\ell(\frac12 + iT)^2 + S_t(T)$, *and* $S_0 = S$.

(ii) The main term is $\frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac78 + \frac{t}{16}\log\frac{T}{2\pi} + O\big((1+t\log T)/T\big)$; *for* $0 < t \le \frac12$ *a count of this form, with error terms, is [247, Theorem 1.5].*

(iii) Let $T > \frac52$. If $\operatorname{Re}\zeta_t(2+iy) > 0$ for $0 \le y \le T$, and $M(\omega) \ge \max\big(|\zeta_t(2 + iT + \frac52 e^{i\omega})|, |\zeta_t(2 + iT + \frac52 e^{-i\omega})|\big)$ for $0 \le \omega \le 2\pi$, then

$$|S_t(T)| \le \frac12 + \frac{1}{\log\frac53}\Big(\frac{1}{2\pi}\int_0^{2\pi}\log M(\omega)\,d\omega - \log\operatorname{Re}\zeta_t(2+iT)\Big).$$

Proof. (i) By Lemma 6.2.1, ξ_t is positive on the real axis and $\xi_t(1-s) = \xi_t(s) = \overline{\xi_t(\bar s)}$; by de Bruijn's strip theorem, as in the proof of Theorem 6.3.2, its zeros satisfy $(\operatorname{Re} s - \frac12)^2 \le \max(\frac14 - \frac{t}{2}, 0)$, so for $t > 0$ they lie in $0 < \operatorname{Re} s < 1$, as they do for $t = 0$ because ζ has no zeros on $\operatorname{Re} s = 1$. The argument principle on the rectangle with corners -1, 2, $2+iT$, $-1+iT$ and the two symmetries give $\pi N_t(T)$ as the change of $\arg\xi_t$ along the two segments. There $\xi_t = g\,e^{t\ell^2/4}\zeta_t$ with each factor continuous and nonzero, and $g(2) = 1/\pi$ and $\ell(2)$ are real. The argument of g changes by $\vartheta(T) + \pi$: $s(s-1) = (s-\frac12)^2 - \frac14$ stays in the closed

upper half plane and ends on the negative axis, and $\pi^{-s/2}\Gamma(\frac{s}{2})$ has no zeros in $\operatorname{Re} s > 0$. The factor $e^{t\ell^2/4}$ contributes $\frac{t}{4}\operatorname{Im}\ell(\frac12+iT)^2$. (ii) Stirling's series gives $\operatorname{Re}\ell(\frac12+iT) = \frac12\log\frac{T}{2\pi} + O(T^{-2})$, $\operatorname{Im}\ell(\frac12+iT) = \frac{\pi}{4} - \frac{7}{4T} + O(T^{-3})$ and $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi e} - \frac{\pi}{8} + O(T^{-1})$. (iii) This is Backlund's argument [9]. Along the vertical segment $|\arg\zeta_t| < \frac{\pi}{2}$, and along the horizontal one the argument moves by at most π between consecutive zeros of $\operatorname{Re}\zeta_t(\sigma+iT)$, so $|S_t(T)| \le m+\frac12$ with m the number of these zeros on $[\frac12, 2]$. They are zeros of $f(z) = \frac12[\zeta_t(z+iT) + \overline{\zeta_t(\bar z + iT)}]$, which is analytic on $|z-2| \le \frac52$, has $f(2) = \operatorname{Re}\zeta_t(2+iT)$ and $|f| \le M$ on the circle, and they lie in $|z-2| \le \frac32$; Jensen's formula on the circle of radius $\frac52$ gives $m\log\frac53 \le \frac{1}{2\pi}\int_0^{2\pi}\log M(\omega)\,d\omega - \log f(2)$. □

The count includes zeros off the line, so (iii) applies below Λ too. Ki, Kim and Lee [182, Theorem 1.4] give $S_t(T) = O_t(1)$ for $t > 0$, and the Polymath project [247, Theorem 1.5] gives $S_t(T) = O(\log T)$ uniformly in $0 < t \le \frac12$, with constants that are not numerical; an explicit bound needs the two hypotheses of (iii) with an explicit M (Problem 6.6).

Corollary 6.2.8 (Conditional; the finite range computed). *Suppose that $v_{0.2}(10,\theta) \ge \frac{19}{2}V_{0.2}$ for every $\theta \ge \theta_1(0.2)$. Then $\tilde\varphi_{0.2,\alpha}$ is the Laplace transform of a GGC for every $\alpha \in [\frac12, 10]$. For $\theta_1(0.2) \le \theta \le 10^4$ the hypothesis is computed, not certified: on a grid of step 0.05, with a Gauss Hermite quadrature whose error is not bounded, the least margin is $v_{0.2}(10,\theta) - \frac{19}{2}V_{0.2} = 0.1163$, at $\theta = 14.0768$.*

Proof. $\Lambda \le 0.2$ [232], and Proposition 6.2.5(iv) applies with $\alpha' = 10$, since $(\frac{19}{2})^2 \le \frac35\theta_1(0.2)^2$. This needs only $\theta_1(0.2) \ge (\operatorname{Var} T_{0.2})^{-1/4} = 12.76$, which follows from $\operatorname{Var} T_{0.2} = \sum_k \theta_k(0.2)^{-4} \ge \theta_1(0.2)^{-4}$ and gives $\frac35\theta_1(0.2)^2 \ge 97.69 \ge 90.25$. □

Remark 6.2.9. At $t = 0$ the hypothesis of the corollary, with ξ in place of $\xi_{0.2}$, is a statement about the Gamma factor alone, up to an error below 10^{-3}, since $|\operatorname{Re}(\zeta'/\zeta)(10+i\theta)| \le \sum_n \Lambda(n)n^{-10} < 10^{-3}$; but there the comparison in (iv) needs the zeros on the line. Along the flow the zeros are on the line and the Euler product is lost: by Lemma 6.2.6, $v_t(\alpha',\theta) = \operatorname{Re}\big[\ell + \frac{t}{2}\ell\ell' + \zeta_t'/\zeta_t\big](\alpha'+i\theta)$, and what is missing is a bound for ζ_t'/ζ_t on the line $\operatorname{Re} s = 10$ that is uniform in the height (Problem 6.6(a)). The route of the note [138] is (iii): it needs the gaps between consecutive zeros of $\xi_{0.2}$ below a height T_* to be below $2D_{0.2} = 9.2306$, and an explicit count above T_* (Problem 6.6(b)). On our scan of $\xi_{0.2}(\frac12+i\theta)$ through Lemma 6.2.6, for $0 \le \theta \le 4200$, the largest gap between consecutive sign changes is 6.87, at $\theta = 14.0768$, and the number of sign changes below T differs from the main term of Proposition 6.2.7 by less than 0.65 at $T = 500, 1000, \dots, 4000$ (computed).

6.3 The Thorin phase diagram

Theorem 6.2.3 gives every time t its own threshold basepoint, $\alpha^*(t) = \max(\frac{1}{2}, \sup \operatorname{Re}\rho)$ over the zeros of ξ_t.

Definition 6.3.1. The *GGC region* $\mathcal{G}$ is the set of points (t,α) with $t \geq 0$ and $\alpha \geq \frac{1}{2}$ at which the shifted clock $\varphi_{t,\alpha}$ is the Laplace transform of a GGC, where at $\alpha = \frac{1}{2}$ the clock is R_t of Section 4.1.

By Theorem 6.2.3, $(t,\alpha) \in \mathcal{G}$ exactly when every zero $x+iy$ of H_t has $|y| \leq 2(\alpha - \frac{1}{2})$; at $\alpha = \frac{1}{2}$ this is the statement that the zeros are real, which is the GGC property of R_t. The slice $t = 0$ of $\mathcal{G}$ concerns the basepoint family of Section 19.8, a GGC for $\alpha \geq \sup \operatorname{Re}\rho$ and unconditionally for $\alpha \geq 1$, and the slice $\alpha = \frac{1}{2}$ the flow family R_t of Section 4.1, known to be a GGC unconditionally only for $t \geq 0.2$ (Problem 19.3); the diagram joins these two one parameter families. The region is closed upward in α, since the threshold condition is monotone.

Theorem 6.3.2 (de Bruijn's strip theorem in Thorin form). *(i) If* $(t_0,\alpha_0) \in \mathcal{G}$, *then* $(t,\alpha) \in \mathcal{G}$ *for every* $t \geq t_0$ *and every* $\alpha \geq \frac{1}{2}$ *with*

$$(\alpha - \tfrac{1}{2})^2 + \tfrac{t}{2} \;\geq\; (\alpha_0 - \tfrac{1}{2})^2 + \tfrac{t_0}{2}.$$

(ii) $\Lambda = \min\{\, t + 2(\alpha - \frac{1}{2})^2 : (t,\alpha) \in \mathcal{G} \,\}$.

(iii) RH holds if and only if $(0, \frac{1}{2}) \in \mathcal{G}$, *and if and only if* $(t, \frac{1}{2} + \sqrt{\delta - t}/\sqrt{2}) \in \mathcal{G}$ *for every* $\delta > 0$ *and some* $t \in [0,\delta]$.

Proof. De Bruijn's theorem on strips [61, Theorem 13], in the form used by Polymath15 [247], says that if the zeros of H_{t_0} lie in $|\operatorname{Im} z| \leq \Delta$, then the zeros of H_t lie in $(\operatorname{Im} z)^2 \leq \max(\Delta^2 - 2(t - t_0), 0)$ for $t \geq t_0$. With $\Delta = 2(\alpha_0 - \frac{1}{2})$ the zeros of H_t then satisfy $|\operatorname{Im} z| \leq 2(\alpha - \frac{1}{2})$ whenever $4(\alpha - \frac{1}{2})^2 \geq 4(\alpha_0 - \frac{1}{2})^2 - 2(t - t_0)$, which is (i). For (ii), a point $(t,\alpha) \in \mathcal{G}$ puts the zeros of H_t in $|\operatorname{Im} z| \leq 2(\alpha - \frac{1}{2})$, so by (i) those of $H_{t'}$ are real for $t' = t + 2(\alpha - \frac{1}{2})^2$, and $\Lambda \leq t + 2(\alpha - \frac{1}{2})^2$. Conversely $\Lambda \geq 0$ [250], the zeros of H_Λ are real, and so $(\Lambda, \frac{1}{2}) \in \mathcal{G}$, where the quantity equals Λ. (iii) The first equivalence is (ii) with $\Lambda \geq 0$. For the second, the points named lie on the level set $t + 2(\alpha - \frac{1}{2})^2 = \delta$, so their membership gives $\Lambda \leq \delta$ for every $\delta > 0$, and conversely RH puts $(0, \frac{1}{2})$ in $\mathcal{G}$, from which (i) gives every point of every level set. □

Corollary 6.3.3 (The known region). *The following points lie in* $\mathcal{G}$, *and with them every point that Theorem 6.3.2(i) carries them to; the bounds on* Λ *in (a) to (c) are those of Table 4.8.*

(a) $(0,1)$, *by the critical strip; it gives de Bruijn's* $\Lambda \leq \frac{1}{2}$.

(b) $(0.2, 0.6)$, *by the certificate of Polymath15 [247]; it gives* $\Lambda \leq 0.22$.

(c) *A point on the level set* $t + 2(\alpha - \frac{1}{2})^2 = 0.2$*, by the certificate of Polymath15 with the verification of Platt and Trudgian [232]; it gives* $\Lambda \le 0.2$.

(d) *Conditionally on the half plane announced in October 2026 [227],* $(0, \frac{7}{8})$*; it gives* $\Lambda \le \frac{9}{32}$.

Proof. (a) is the absence of zeros of ζ in $\operatorname{Re} s \ge 1$ and the functional equation. (b) is the statement proved in [247]: $H_{0.2}$ has no zeros with $\operatorname{Im} z > 0.2$, which is $(0.2, 0.6) \in \mathcal{G}$. (c) is [232], which takes a row of the table of [247] whose hypothesis their verification satisfies. (d) is immediate from [227]. The bounds on Λ are Theorem 6.3.2(ii). □

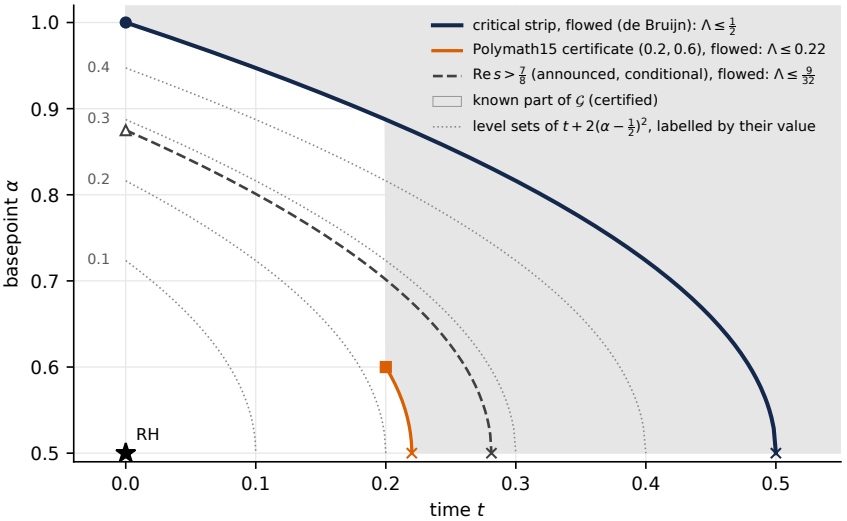


Figure 6.1: The Thorin phase diagram. Solid curves are the boundaries of certified GGC regions carried along the flow by Theorem 6.3.2(i), and the dashed curve is that of the region conditional on the half plane of Corollary 6.3.3(d); the dotted curves are the level sets of $t + 2(\alpha - \frac{1}{2})^2$, labelled by the bound on Λ that a point on them gives. RH is the point $(0, \frac{1}{2})$.

A half plane at time zero improves the current bound only below $\alpha = \frac{1}{2} + \sqrt{0.1} \approx 0.816$ (Section 4.1); Figure 6.1 shows the region.

Remark 6.3.4 (Two semigroups). The two coordinates of the diagram carry compatible monotonicities. In α the Cauchy densities form a semigroup, so when no zero of ξ_t lies to the right of α the Thorin density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the one at α, and positivity propagates to the right; going left is deconvolution, which does not preserve positivity (Section 14.3.7). In t the heat flow carries real zeros to real zeros and narrows strips, so membership

propagates forward in time. RH asks for the corner where both coordinates are smallest, and neither semigroup can be run backward.

6.4 What a Thorin computation sees at positive time

The Pólya route computes ξ_t at the cost of ξ, since the flow only reweights the kernel (Section 4.1.1), and, when $(t,\alpha)\in\mathcal{G}$, the Thorin density of the shifted time t clock is $\pi^{-1}v_t(\alpha,\theta)$ with

$$v_t(\alpha,\theta)=\operatorname{Re}\frac{\xi_t'}{\xi_t}(\alpha+i\theta),$$
$$\xi_t(s)=8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(2wu)\,du,$$
$$\xi_t'(s)=8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\,2u\sinh(2wu)\,du,$$

$w=s-\frac12$, by Remark 6.2.4; where membership in $\mathcal{G}$ is not known, as at $t=0$ with $\alpha=0.6$, the function v_t/π is still defined. We computed $v_t(\alpha,\theta)$ at the times $t=0$ and $t=0.2$ and the two basepoints $\alpha=0.6$ and 0.75, for $0\le\theta\le100$ in steps of 0.25, with no primes and no list of zeros. The integrals were evaluated by the trapezoid rule of Section 2.1.1 with step $1/160$ on $[0,1.6]$ (the tail beyond 1.6 is below 10^{-812}) at $\lfloor 0.36\,\theta\rfloor+35$ digits, the term $0.36\,\theta$ absorbing the cancellation $|\xi_t(\alpha+i\theta)|\approx e^{-\pi\theta/4}$ that the Pólya route always carries (Section 2.1). A second discretisation agrees with the first in v_t at all 1604 points of the four grids to a relative 2×10^{-35}; at $t=0$ the integrals, evaluated with 10 more digits, agree with the closed forms of ξ and ξ'/ξ through Γ, ψ, ζ and ζ' to at least 43 digits at both basepoints and at $\theta=0$, 17.5, 50 and 100, and at $t=0.2$ the integral for ξ_t' agrees with a central difference of ξ_t to 30 digits at two points.

Proposition 6.4.1 (Computed). *On $0\le\theta\le100$:*

(i) At both times and both basepoints the minimum of $v_t(\alpha,\cdot)$ is at the origin, $v_t(\alpha,0)=\xi_t'/\xi_t(\alpha)$, in agreement with Theorem 12.1.18 at $t=0$, and $v_t(\alpha,\cdot)$ is positive.

(ii) Between consecutive zeros the valleys are set by the gaps relative to the kernel width $\alpha-\frac12$, and they scale roughly in proportion to it: at $t=0$ and $\alpha=0.6$ they range from 0.023 near $\theta=17.5$ to 0.56 near $\theta=95$, and at $\alpha=0.75$ they are 2.2 to 2.5 times larger.

(iii) The 28 valleys above $\theta=10$, each taken at its minimum, change between $t=0$ and $t=0.2$ by a median of 2.3% and at most 14.2% at $\alpha=0.6$ (2.2% and 12.6% at $\alpha=0.75$), the largest change being that of the valley near $\theta=95$; the zeros below 60 move by 0.044 to 0.118, with mean 0.075, the first from 14.135 to 14.077 (Section 6.5.2).

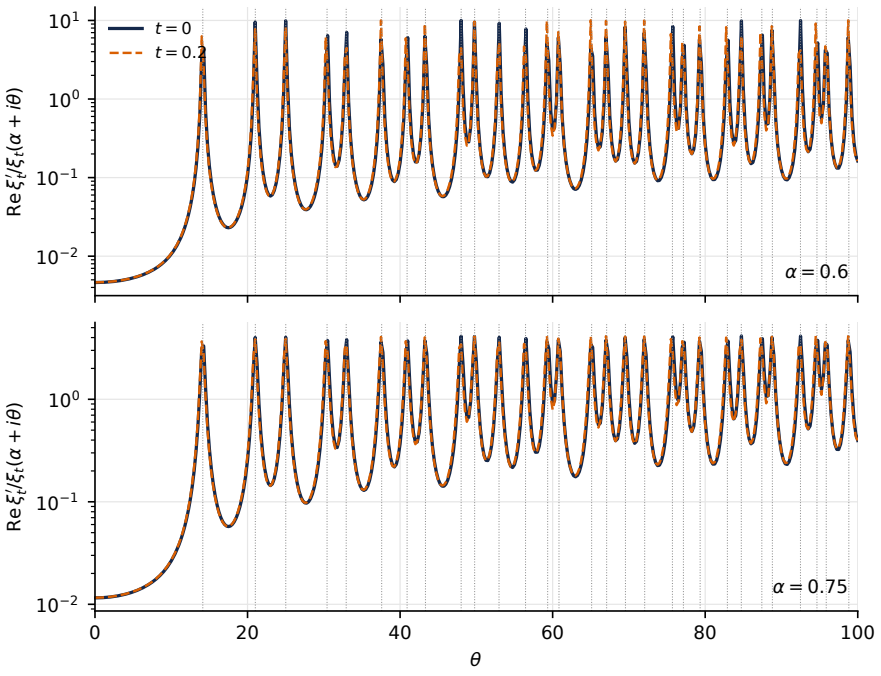


Figure 6.2: $v_t(\alpha, \theta) = \mathrm{Re}\ \xi_t'/\xi_t(\alpha + i\theta)$, which is π times the Thorin density of the shifted time t clock when $(t, \alpha) \in \mathcal{G}$, on the lines $\mathrm{Re}\, s = 0.6$ and 0.75, at $t = 0$ (solid) and $t = 0.2$ (dashed), from the Pólya integral. Dotted lines mark the ordinates of the zeros of ζ.

Part (ii) corrects a natural expectation: averaged over a window wider than the spacing of the zeros, $v_t(\alpha, \theta)$ is close to $\frac{1}{2}\log(\theta/2\pi)$, which suggests a margin of that size. At $t = 0$ Remark 12.1.17 proves a margin that does grow with the height, $v_\alpha(\theta)/\pi \geq c \log\theta$ for every $c < (\alpha - \frac{1}{2})/(\pi(2\alpha+1))$ and all large θ outside a set of density zero, but its constant is proportional to $\alpha - \frac{1}{2}$: at $\alpha = 0.6$ it is below $0.015 \log\theta$, against the average $(2\pi)^{-1}\log(\theta/2\pi)$ of v_α/π, and its exceptional set contains the window of every zero to the right of $\mathrm{Re}\, s = \alpha$. Below height 100 the computed margin is smaller still, set by the largest gaps between consecutive zeros, because the kernel width $\alpha - \frac{1}{2}$ is small against the mean spacing $2\pi/\log(\theta/2\pi)$.

On the windows computed here, below height 100, the clock of $\xi_{0.2}$, which is a GGC at the basepoint 0.6 by Corollary 6.3.3(b), and the clock of ξ, which is a GGC at the basepoint 0.6 only if there is no zero with $\beta > 0.6$ anywhere, are described by the positive functions $v_{0.2}/\pi$ and v_0/π, whose valleys differ by at most 14.2% (Proposition 6.4.1(iii)). Nothing in the computed density distinguishes the theorem from the conjecture. The Thorin side offers not a cheaper finite computation than the flow but the same one in the language of this book, and the

	$\alpha = 0.6$			$\alpha = 0.75$		
θ	$t = 0$	$t = 0.2$	change	$t = 0$	$t = 0.2$	change
0	0.004621	0.004640	+0.4%	0.011551	0.011599	+0.4%
17.50	0.0230	0.0231	+0.4%	0.0573	0.0575	+0.4%
23.01	0.0582	0.0578	−0.7%	0.1439	0.1429	−0.7%
27.67	0.0389	0.0393	+0.9%	0.0968	0.0976	+0.9%
35.27	0.0520	0.0526	+1.1%	0.1291	0.1304	+1.1%
42.12	0.1539	0.1496	−2.8%	0.3729	0.3627	−2.7%
45.63	0.0569	0.0579	+1.8%	0.1412	0.1438	+1.8%
95.26	0.5625	0.4829	−14.2%	1.2474	1.0906	−12.6%

Table 6.2: The value at the origin and at the minimum of a selection of the valleys of $v_t(\alpha, \cdot)$ between consecutive zeros. The ordinate θ is that of the minimum at $t = 0$ and $\alpha = 0.6$; at $t = 0.2$ the minimum lies 0.06 to 0.08 to the left, and at $\alpha = 0.75$ it moves by less than 0.002.

whole question is where the tail comes from. The same comparison at $t = 0.2$ and $t = 0$ above the barrier of Polymath15, at heights near 3×10^{10}, would require a Riemann Siegel type evaluator of $H_{0.2}$ such as the effective approximation of [247] (Section 2.2 gives the one for ζ); Problem 6.1 asks for it.

6.5 A certificate for one point of the diagram

This section writes the finite part of the statement $(0.2, 0.6) \in \mathcal{G}$, that $\xi_{0.2}$ has no zero with real part above 0.6, as a certified computation with explicit constants, and runs it in ball arithmetic below height 160. Given a barrier X (in the variable of $H_{0.2}$, so that it sits at height $X/2$ for $\xi_{0.2}$; see Theorem 6.5.6) beyond which this is known [247], it is the vanishing of the winding number of $\xi_{0.2}$ around the rectangle $Q = [0.6, 0.9] \times [0, X/2]$. Throughout, $t = 0.2$.

6.5.1 The lemmas

Lemma 6.5.1 (Bottom edge). *$\xi_t(\sigma) > 0$ for every real σ and every $t \geq 0$.*

Proof. Lemma 6.2.1(ii). □

Lemma 6.5.2 (Right edge). *For $0 \leq t \leq \frac{1}{2}$ every zero of ξ_t satisfies $|\beta - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1 - 2t}$. In particular for $t = 0.2$ every zero has $\beta \leq 0.8873$, so $\xi_{0.2}$ has no zero on $\operatorname{Re} s = 0.9$.*

Proof. De Bruijn's theorem on strips, as in the proof of Theorem 6.3.2, with $\Delta = 1$ from the critical strip and $\beta - \frac{1}{2} = \frac{1}{2} \operatorname{Im} z$ up to sign. □

LEMMA 6.5.3 (Tilted bound). *Let $0 < \eta < \pi/8$, $t \geq 0$ and $s = \frac{1}{2} + x + i\theta$ with $\theta \geq 0$. Then*

$$|\xi_t(s)| \leq K_\eta(t,x)\, e^{-2\eta\theta}, \qquad K_\eta(t,x) = 4\int_{-\infty}^{\infty} \left|e^{t(u+i\eta)^2}\right| |\Phi_{\mathrm{dBN}}(u+i\eta)|\, e^{2xu}\, du,$$

and $K_\eta(t,x)$ is finite and even in x. Write $\bar{K}_\eta(t,R) = \max_{|x|\leq R} K_\eta(t,x)$.

Proof. Since Φ_{dBN} is even, $\xi_t(s) = 4\int_{\mathbb{R}} e^{tu^2}\Phi_{\mathrm{dBN}}(u)e^{2(x+i\theta)u}\, du$. The kernel extends analytically to $|\operatorname{Im} u| < \pi/8$ with the same doubly exponential decay, which is Proposition 2.1.1 in the variable $v = 2u$, so the contour may be moved to $\operatorname{Im} u = \eta$, and $|e^{2i\theta(u+i\eta)}| = e^{-2\eta\theta}$. □

LEMMA 6.5.4 (Derivative bound). *With the notation of Lemma 6.5.3, for $r > 0$, $|\xi_t'(s)| \leq r^{-1}\bar{K}_\eta(t, |x|+r)\, e^{-2\eta(\theta-r)}$, and the factor $e^{-2\eta(\theta-r)}$ may be replaced by 1 when $\theta \leq r$.*

Proof. Cauchy's estimate on the disc of radius r about s. A point z of the disc has $|\operatorname{Re} z - \frac{1}{2}| \leq |x| + r$ and $\operatorname{Im} z \geq \theta - r$. If $\operatorname{Im} z \geq 0$, Lemma 6.5.3 bounds $|\xi_t(z)|$ by $\bar{K}_\eta(t,|x|+r)e^{-2\eta \operatorname{Im} z}$, which is at most $\bar{K}_\eta(t,|x|+r)e^{-2\eta(\theta-r)}$ and at most $\bar{K}_\eta(t,|x|+r)$. If $\operatorname{Im} z < 0$, which occurs only when $\theta < r$, then $\xi_t(\bar{z}) = \overline{\xi_t(z)}$ gives the bound $\bar{K}_\eta(t,|x|+r)$ for $|\xi_t(z)|$. So on the disc $|\xi_t| \leq \bar{K}_\eta(t,|x|+r)e^{-2\eta(\theta-r)}$, and $|\xi_t| \leq \bar{K}_\eta(t,|x|+r)$ when $\theta \leq r$. □

LEMMA 6.5.5 (Grid). *Let s_0, s_1 be points of a segment on which ξ_t has no zero, and suppose $|s_1 - s_0| \cdot \sup_{[s_0,s_1]} |\xi_t'| < |\xi_t(s_0)|$. Then the change of $\arg \xi_t$ along the segment equals the principal value $\operatorname{Arg}(\xi_t(s_1)/\xi_t(s_0)) \in (-\pi, \pi)$.*

Proof. The image of the segment lies in the open disc of radius $|\xi_t(s_0)|$ about $\xi_t(s_0)$, which does not contain 0, and on that disc a continuous branch of the argument varies by less than π from its value at the centre. □

THEOREM 6.5.6 (The certificate). *Let X be a barrier for $(0.2, 0.6)$ in the variable of $H_{0.2}$, as Polymath15 provide one [247] (not checked in this book), that is, a number such that $H_{0.2}$ has no zero z with $\operatorname{Re} z \geq X$ and $\operatorname{Im} z > 0.2$, so that $\xi_{0.2}$ has no zero with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \geq X/2$. Let $s_0, \ldots, s_M$ be points of the left, top and right edges of Q, in order along ∂Q from one bottom corner, s_0, to the other, s_M, and including the two top corners, and let B_j be ball enclosures of $\xi_{0.2}(s_j)$ with radii ε_j. Suppose that for every j there are $r > 0$ and $\eta \in (0, \frac{\pi}{8})$ such that*

$$|s_{j+1} - s_j|\, \frac{\bar{K}_\eta(0.2,\, 0.4 + r)}{r}\, e^{-2\eta(\theta_j - r)} + \varepsilon_j + \varepsilon_{j+1} < \min |B_j|,$$
$$\theta_j = \min(\operatorname{Im} s_j, \operatorname{Im} s_{j+1}),$$

where the factor $e^{-2\eta(\theta_j - r)}$ *may be replaced by* 1 *when* $\theta_j \le r$, *and that* $\sum_j \operatorname{Arg}(B_{j+1}/B_j)$, *evaluated in ball arithmetic, lies within* $(-\pi, \pi)$. *Then* $\xi_{0.2}$ *has no zero in* Q, *so* $(0.2, 0.6) \in \mathcal{G}$ *and* $\Lambda \le 0.22$.

Proof. By Lemma 6.5.1, $\xi_{0.2}$ is real and positive on the bottom edge, so $\arg \xi_{0.2}$ does not change along it, and by Lemma 6.5.2 the right edge is zero free. The hypothesis gives $|\xi_{0.2}(s_j)| \ge \min |B_j|$ at the grid points, and Lemma 6.5.4 with $|x| \le 0.4$ and $\operatorname{Im} z \ge \theta_j$ on the segment from s_j to s_{j+1} bounds the derivative between them, so Lemma 6.5.5 applies to every step, and the winding number is the integer nearest to $\frac{1}{2\pi} \sum_j \operatorname{Arg}(B_{j+1}/B_j)$, which is 0. So $\xi_{0.2}$ has no zero in Q; it has none with $\operatorname{Re} s > 0.9$ by Lemma 6.5.2, none with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \ge X/2$ by the barrier, and its zeros in $\operatorname{Im} s < 0$ are the conjugates of those in $\operatorname{Im} s > 0$. Hence it has no zero with $\operatorname{Re} s > 0.6$, which is $(0.2, 0.6) \in \mathcal{G}$, and the last clause is Theorem 6.3.2. □

The number of grid points is of order $(X/2) \sup_j K_\eta e^{2\eta r}/(r\,|\xi_t(s_j)| e^{2\eta\theta_j})$. The decay of ξ_t, of order $e^{-\pi\theta/4}$ away from its zeros, cancels against the decay $e^{-2\eta\theta}$ of the derivative bound only at $\eta = \pi/8$, which Lemma 6.5.3 excludes, and K_η grows without bound as $\eta \uparrow \pi/8$ (Table 6.3). With a tilt $\eta < \pi/8$ the count grows like $e^{(\pi/4 - 2\eta)\theta}$. We have no proof that the count is $O(X \log X)$, or polynomial in X, when the tilt is chosen at each height. In the runs below the left edge needs on average about 45 steps per unit height up to height 60, 73 up to 120 and 94 up to 160 (Theorem 6.5.8).

Proposition 6.5.7 (The certificate in Thorin form). *Let* $t \ge 0$ *and* $\frac{1}{2} < \alpha < \sigma_1$, *and suppose* $v_t(\sigma, \theta) > 0$ *for every* $(\sigma, \theta) \in [\alpha, \sigma_1] \times [0, T]$ *at which* $\xi_t(\sigma + i\theta) \ne 0$. *Then* ξ_t *has no zero* $\beta + i\gamma$ *with* $\alpha < \beta \le \sigma_1$ *and* $0 < \gamma < T$.

Proof. Near a zero ρ of multiplicity m, $\xi_t'/\xi_t(s) = m/(s-\rho) + O(1)$, so at $s = \rho - \delta$ with small $\delta > 0$ the real part is $-m/\delta + O(1) < 0$. If $\alpha < \beta \le \sigma_1$, these points lie in the rectangle for small δ. This is the argument of Remark 2.1.4, restricted to a rectangle of finite height. □

The proposition asks for the whole rectangle because positivity on the single line $\operatorname{Re} s = \alpha$ is not enough (Remark 2.1.4). It is the pointwise counterpart of Theorem 6.5.6: the same evaluations of ξ_t and ξ_t' certify it, with a two dimensional grid and a bound on the derivative of v_t in place of the argument increments, and the margin is the one measured in Proposition 6.4.1.

6.5.2 Constants and the march

Table 6.3 gives the constants of Lemma 6.5.3 at $t = 0.2$ and the values of $\xi_{0.2}$ on the real axis that fix the scale.

A floating point march of Lemma 6.5.5 around $Q_{60} = [0.6, 0.9] \times [0, 60]$, with the Pólya route evaluator (the trapezoid rule of Section 2.1.1 with the weight e^{tu^2})

η	decay $e^{-2\eta\theta}$	$K_\eta(0.2, 0.4)$	$K_\eta(0.2, 0.9)$
0.30	0.60	23.67	25.31
0.35	0.70	199.2	231.1
0.38	0.76	4847	6776

σ	$\xi_{0.2}(\sigma)$
0.5	0.49827
0.6	0.49839
0.9	0.50013
1.0	0.50117
1.5	0.50996

Table 6.3: Constants for the certificate at $t = 0.2$, in floating point.

at 40 digits (working precision 72 digits) and an adaptive step keeping $|\xi_t(s_{j+1}) - \xi_t(s_j)| < \frac{1}{2}|\xi_t(s_j)|$, returned the winding -2.9×10^{-72} in 910 evaluations. The same evaluator located the zeros of $\xi_{0.2}$ on the critical line below 60,

$$14.077,\ 20.947,\ 24.952,\ 30.330,\ 32.888,\ 37.503,$$
$$40.833,\ 43.282,\ 47.887,\ 49.730,\ 52.905,\ 56.358,\ 59.237,$$

thirteen zeros, as for ξ below 60, each moved slightly by the flow (five of them are in Table 4.2). For the control function that moves the pair at 30.330 and 32.888 to the quadruple $\frac{1}{2} \pm \frac{1}{4} + 31.609i$ and its conjugates, as in Theorem 1.4.1 (there for ξ and ordinates $1000 < A < B \le 2000$), the computed winding was 1 to within 10^{-35} in 905 evaluations: exactly one of the four new zeros, $0.75 + 31.609i$, lies in Q_{60}.

The finite part of Theorem 6.5.6 was then run in ball arithmetic, adding every truncation error to the radius of the enclosure. The evaluator is the same trapezoid rule, with $w = s - \frac{1}{2} = x + i\theta$ and $u_j = jh$: $\xi_t(s) = 8h \sum_{j=0}^{J} \omega_j e^{tu_j^2} \Phi_{\mathrm{dBN}}(u_j) \cosh(2wu_j) + E$, where $Jh \ge 3$, $\omega_0 = \frac{1}{2}$, $\omega_j = 1$ for $j \ge 1$, and $|E| \le E_{\mathrm{cut}} + E_{\mathrm{trap}}$. For $u \ge 3$ the kernel satisfies $|\Phi_{\mathrm{dBN}}(u)| \le 60\, e^{9u} e^{-\pi e^{4u}}$, so that for $|x| \le 1$ the summand is at most $8h\, G(u_j)$ with $G(u) = 60 \exp(tu^2 + 11u - \pi e^{4u})$, which decreases with $-G'/G \ge \lambda = 4\pi e^{4Jh} - 2tJh - 11$ for $u \ge Jh$, and the nodes beyond Jh contribute at most $E_{\mathrm{cut}} = 8G(Jh)/\lambda < 10^{-229000}$. The error E_{trap} of the rule is bounded by Poisson summation on the strip $|\operatorname{Im} u| < a$, $a = 0.30 < \pi/8$, in which the integrand of Lemma 6.5.3 is analytic: $E_{\mathrm{trap}} \le 2K_{0.30}(0.2, x) \cosh(2a\theta)/(e^{2\pi a/h} - 1)$, with $K_{0.30}(0.2, x) \le 23.72$ for $|x| \le 0.4$, bounded in the way described below, and h is taken so that this is below $5 \times 10^{-4} e^{-\pi T/4}$ at the top of the rectangle of height T: $h = 0.0199$, 0.0106 and 0.008 at $T = 60$, 120 and 160, at the working precisions 219, 288 and 334 bits. The series for the kernel is summed until its tail is below $2^{-\mathrm{prec}-30}$, the tail after N terms being enclosed by

$$\sum_{n>N} \left|(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}\right|$$
$$\le |e^{9u}| \Big(2\pi^2 + \frac{3\pi}{\kappa}\Big) \Big(\frac{4}{\pi\kappa}\Big)^2 e^{-2}\, \frac{e^{-\pi(N+1)^2\kappa/2}}{1 - e^{-\pi(N+1)\kappa}}, \qquad \kappa = \operatorname{Re}(e^{4u}),$$

using $|e^{5u}| \le |e^{9u}|/\kappa$, $n^4 e^{-\pi n^2 \kappa/2} \le (4/\pi\kappa)^2 e^{-2}$ and $n^2 \ge (N+1)^2 + 2(N+1)(n - N-1)$ for the geometric sum, for every N and every $\kappa > 0$. At a real node $\kappa = e^{4u}$; at a complex u κ is a lower bound of $\mathrm{Re}(e^{4u})$ over the input ball. The constants $\bar{K}_\eta(0.2, 0.4 + r)$ were bounded above by interval subdivision of $[0, 3]$ into $3 \cdot 2^{14}$ cells, plus the tail $u \ge 3$, below e^{-25000} since $\mathrm{Re}\, e^{4(u+i\eta)} \ge e^{12} \cos(4\eta) > 8000$ there, for the three tilts $\eta = 0.30$, 0.35 and 0.38 and the three radii $r = \frac{1}{4}$, $\frac{3}{8}$ and $\frac{1}{2}$: 24.37, 24.82 and 25.36, then 212.33, 221.11 and 231.71, then 5633.6, 6170.8 and 6841.2, each rounded up. The ordinates of the grid points are exact dyadic numbers, and the abscissae 0.6 and 0.9 are balls of radius $2^{-\mathrm{prec}}$ enclosing them, so that the radius of $\xi_t(s_j)$ is that of the evaluator. At each step the tilt η and the radius r giving the smallest derivative bound are used, the step is $0.8\,|\xi_t(s_j)|_{\mathrm{low}}$ divided by that bound, capped at 0.5, and the inequality of Theorem 6.5.6, with the exact length of the step and its exact lowest height, is then verified in ball arithmetic, the step being halved when it fails.

Theorem 6.5.8. *$\xi_{0.2}$ has no zero s with $0.6 \le \mathrm{Re}\, s \le 0.9$ and $0 \le \mathrm{Im}\, s \le 160$.*

Proof. For $T = 60$, 120 and 160 the computation walks the boundary of $Q_T = [0.6, 0.9] \times [0, T]$ counterclockwise, up the right edge, along the top and down the left edge, with the steps of Theorem 6.5.6. The bottom edge is not walked, since $\xi_{0.2}$ is real and positive on it (Lemma 6.5.1). The run took 4483 steps for $T = 60$ (1737 up the right edge, 31 along the top and 2715 down the left edge), 13,679 for $T = 120$ (4955, 11 and 8713) and 23,219 for $T = 160$ (8142, 20 and 15,057), and returned the winding number enclosures $[\pm 3.94 \times 10^{-5}]$, $[\pm 1.24 \times 10^{-6}]$ and $[\pm 3.47 \times 10^{-7}]$. In every run the inequality of Theorem 6.5.6 was verified in ball arithmetic at every step and no step failed, and the sum of the arguments lies within π of 0, so by Lemmas 6.5.1 to 6.5.5 the winding number is 0 and $\xi_{0.2}$ has no zero in Q_T, which for $T = 160$ is the theorem. □

With Lemma 6.5.2 this says that every zero of $\xi_{0.2}$ below height 160 has $|\beta - \frac{1}{2}| < 0.1$; as a control, the same walk around $[0.4, 0.6] \times [0, 60]$, which contains the thirteen zeros on the line, returns the winding number enclosure $[13.0000 \pm 5.04 \times 10^{-5}]$ in 4841 steps. The barrier of Polymath15 is near $X = 6 \times 10^{10}$, height 3×10^{10} for $\xi_{0.2}$, and reaching it in this framework requires the evaluator of the programme below.

6.6 Where the uniform step could come from

Section 25.1.2 names three possible sources of the missing uniform step: a positive approximating family whose positivity is proved independently, a Hilbert Pólya operator, or a direct Weil inequality.

6.6.1 The Weil Gram matrix on compactly supported test functions

For h smooth enough, even or not (each term depends on h only through its even part h_e, defined below), with $\hat{h}(r) = \int h(x)e^{irx}\,dx$, the explicit formula reads

$$\sum_{\rho} \hat{h}(\gamma_\rho) = \hat{h}(\tfrac{i}{2}) + \hat{h}(-\tfrac{i}{2}) - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\big(h(\log n) + h(-\log n)\big) + \frac{1}{2\pi}\int_{\mathbb{R}} \hat{h}(r)\ \mathrm{Re}\,\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\,dr - h(0)\log\pi,$$

with $\rho = \frac{1}{2} + i\gamma_\rho$ and $\psi = \Gamma'/\Gamma$. For $h = g * \tilde{g}$ with $\tilde{g}(x) = \overline{g(-x)}$ and g supported in $[-L/2, L/2]$, the left side is $\sum_\rho |\hat{g}(\gamma_\rho)|^2 \ge 0$ under RH, which is Weil's criterion [273], h is supported in $[-L, L]$, and the prime sum runs over prime powers $n \le e^L$. The archimedean term was evaluated in x space,

$$\frac{1}{2\pi}\int \hat{h}(r)\ \mathrm{Re}\,\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\,dr = \int_0^\infty \Big[\frac{h(0)e^{-2x}}{x} - \frac{2h_e(x)e^{-x/2}}{1-e^{-2x}}\Big]dx,$$
$$h_e(x) = \tfrac{1}{2}\big(h(x) + h(-x)\big),$$

which is far better conditioned than the r integral for compactly supported h: truncated at $r = 400$, the r integral carries an error of 6×10^{-6} in the archimedean term of the diagonal entry at $w = 0.1$, more than two thousand times the smallest eigenvalue 2.5×10^{-9} of Table 6.4. With hat functions $g_j(x) = \max(0, 1-|x-c_j|/w)$ at nodes spaced w apart inside $[-L/2, L/2]$, $N = L/w - 1$ functions in all, $h_{jk} = g_j * \tilde{g}_k$ is an explicit piecewise cubic, and the Gram matrix $G_{jk} = W(h_{jk})$ is Hermitian, with $W(g*\tilde{g}) \ge 0$ on the span exactly when G is positive semidefinite.

w	L	N	$\lambda_{\min}$	$\lambda_{\max}$	pole, −prime, archimedean
0.25	1	3	4.22×10^{-4}	0.034	0.3468, −0.0141, −0.3322
0.25	2	7	2.22×10^{-5}	0.050	0.8410, −0.2472, −0.5938
0.25	4	15	4.70×10^{-7}	0.096	1.6494, −0.8550, −0.7944
0.25	6	23	3.56×10^{-8}	0.144	2.7255, −1.8067, −0.9187
0.1	1	9	1.84×10^{-5}	0.083	0.1287, −0.0019, −0.1268
0.1	2	19	1.27×10^{-6}	0.149	0.2900, −0.0676, −0.2224
0.1	4	39	5.70×10^{-8}	0.290	0.7281, −0.3981, −0.3300
0.1	6	59	2.48×10^{-9}	0.431	1.1050, −0.7353, −0.3697

Table 6.4: The Weil Gram matrix on hat functions. The last column splits $\lambda_{\min} = v^*Gv$ for the minimising unit vector into the pole, prime (with its sign) and archimedean pieces.

Three findings (Tables 6.4 and 6.5, Figure 6.3). First, $\lambda_{\min} > 0$ throughout and it falls roughly geometrically in L, by a factor between about 2.5 and 20 per unit of L up to $L = 6$, as it must, since the nodes are nested (for the values of L used),

w	L	$\lambda_{\min}$ (arithmetic)	$\sum_\rho \lvert\hat{g}(\gamma_\rho)\rvert^2$ (zeros)	zeros carrying most of the sum
0.25	1	4.2184×10^{-4}	4.2184×10^{-4}	#2 (56%), #4 (20%), #5 (11%)
0.25	3	1.8238×10^{-6}	1.8237×10^{-6}	#9 (44%), #11 (24%), #20 (5%)
0.25	6	3.5562×10^{-8}	3.5560×10^{-8}	#10 (40%), #20 (13%), #30 (9%)
0.1	2	1.2691×10^{-6}	1.2691×10^{-6}	#13 (37%), #15 (15%), #16 (15%)
0.1	4	5.7008×10^{-8}	5.7004×10^{-8}	#14 (36%), #13 (23%), #16 (7%)
0.1	6	2.4846×10^{-9}	2.4841×10^{-9}	#41 (32%), #15 (17%), #14 (8%)

Table 6.5: The smallest eigenvalue from the arithmetic side against the sum of squares of $\hat{g}_{\min}$ over the first 1500 zeros, with the zeros (by index) that carry the largest shares.

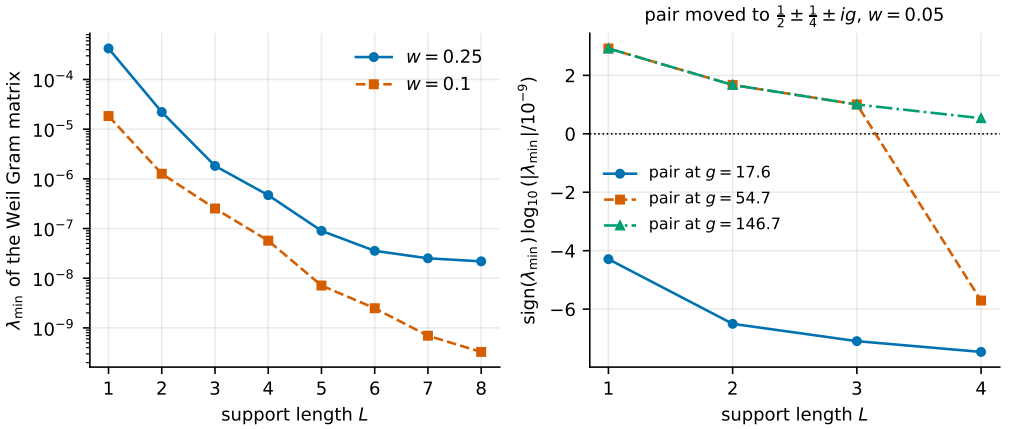


Figure 6.3: Left: $\lambda_{\min}$ of the Weil Gram matrix against the support length L for two resolutions. Right: a pair of zeros at height g moved to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$, signed logarithm of $\lambda_{\min}$ for $w = 0.05$. The pair at $g \approx 17.6$ is detected at every L, the pair at $g \approx 54.7$ only at $L = 4$, and the pair at $g \approx 146.7$ not at all within $L \leq 4$.

so that the matrix for a shorter support is a principal submatrix of the matrix for a longer one, and the smallest eigenvalue of the Weil form on $[-L/2, L/2]$ tends to zero as L grows: a long enough support can make $\hat{g}$ small at every low ordinate. Second, the two sides of Weil's identity agree to a relative 2×10^{-4} at every (w, L), down to $\lambda_{\min} = 2.5 \times 10^{-9}$; and the margin is carried by a handful of zeros, zero 41 at height 124.3 carrying 32% at $w = 0.1$, $L = 6$. Third, at $w = 0.05$ a pair displaced at height $g \approx 17.6$ makes $\lambda_{\min}$ negative at every $L \leq 4$, one at $g \approx 54.7$ only at $L = 4$ and one at $g \approx 146.7$ at no $L \leq 4$, an instance of the finite information principle of Section 6.1.1. How fine w must be to see a pair at height g is not fixed by one resolution: at $w = 0.2$ the pair at $g \approx 17.6$ is already seen at $L = 2$, with $wg \approx 3.5$.

The sign structure is the reverse of the Lévy picture of Section 12.1, where the pole of ζ is a negative input to $\log\xi$; in the Weil form the pole, on the minimising vectors of Table 6.4, contributes positively. A certificate for the Weil form would therefore have to show, for every L, that the three terms together are nonnegative on the whole span, although the pole term alone is not (for real odd g it is $-2G(\frac{1}{2})^2$, with $G(a)=\int g(x)e^{ax}\,dx$), and on the minimising vectors the pole term outweighs the prime and archimedean terms only by a margin that tends to zero. No uniformly positive quadratic form can do that. What could is a factorisation $W(g*\tilde{g})=\|Ag\|^2$ with A built from the arithmetic and exact positivity, the shape of the Hilbert Pólya mechanism, and Problem 6.4 asks for the first experiment in that direction.

6.6.2 The zero gas along the flow

In the variable $z=2\gamma$ the zeros of H_t move, while simple, by $dz_k/dt=2\sum_{j\neq k}(z_k-z_j)^{-1}$ (Section 4.1.1, with five of the zeros at four times in Table 4.2). We integrated this system for the first 600 zeros, heights up to 939.02, with a frozen tail for the omitted zeros (the zeros 601 to 1700 held at their positions at $t=0$ and a continuum of density $\frac{1}{4\pi}\log(z/4\pi)$ beyond them, whose drift on z_k is $-4z_k\sum_{j>600}z_j^{-2}$ to leading order), backward from $t=0$ until the first gap fell below 0.02 and forward to $t=0.3$. Along the trajectory we recorded the minimum normalised gap $s_{\min}(t)=\min_k(z_{k+1}-z_k)\frac{1}{4\pi}\log(z_{k+1}/4\pi)$ and the Csordas, Smith and Varga quantity $\min_k\Delta_k^2g_k$ over the first 400 pairs, with $\Delta_k=z_{k+1}-z_k$ and $g_k=\sum_{j\neq k,k+1}[(z_k-z_j)^{-2}+(z_{k+1}-z_j)^{-2}]$, where j runs over all the other zeros $\pm z_j$ of the model, the frozen tail and the continuum included; a pair with $\Delta_k^2g_k<\frac{4}{5}$ is a Lehmer pair in the sense of [51] and Section 4.1.6.

Proposition 6.6.1 (Computed; the upper end of (iii) certified). *(i) The backward integration stops at $t=-0.0493$, when the pair at ordinates 750.656 and 750.966, the 453rd and 454th zeros, collides; the isolated pair estimate is $-D_0^2/8=-0.048$. (ii) At the 171 times sampled in $[-0.049,0.30]$ both $s_{\min}(t)$ and $\min_k\Delta_k^2g_k$ are strictly increasing, from 0.0076 to 0.502 and from 0.035 to 0.977; at $t=0$ the minimiser of $\Delta_k^2g_k$ is the pair at 564.16 and 564.51, with value 0.238. (iii) For the pair near 185, the pair $(\gamma_{71},\gamma_{72})$ of Section 4.1.2, certified signs of $\xi_t(\frac{1}{2}+i\theta)$ in Arb at 640 bits (the trapezoid sum of Section 2.1.1 with an enclosure of its discretisation error) at 131 points of $[184.6,185.9]$, with bisection in t on the number of sign changes, show that the pair is still real at $t=-0.287979$, and with the pair law below place the collision in $[-0.28803,-0.287979]$, in agreement with the value -0.2880 of Section 4.1 and inside the bracket of Section 4.1.5. Newton's method on the same sum puts the double zero at $z_*=370.78087$, $t_*=-0.287979837009$; a Krawczyk test certifies a real double zero of the trapezoid sum itself in a box of half width 2^{-400} about this point (a statement about the sum, not about H_t), and at $t=t_*+1.2\times10^{-12}$ the signs $+,-,+$ of H_t at $z_*-4\times10^{-6}$, z_* and*

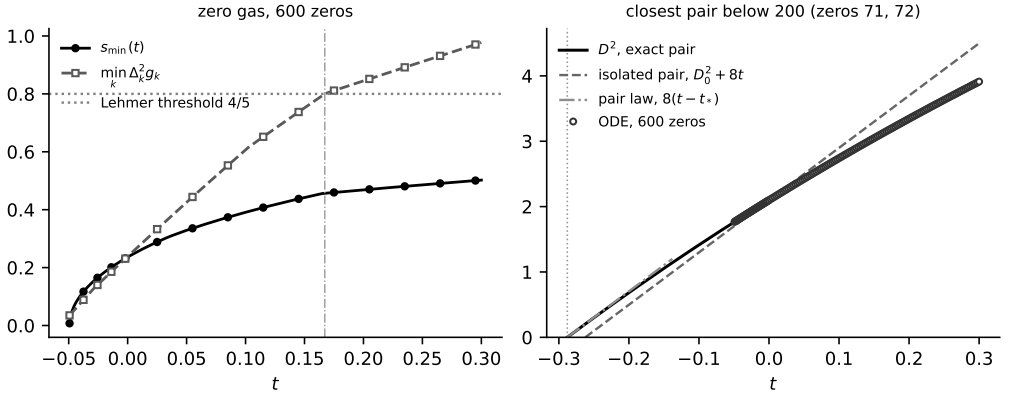


Figure 6.4: Left: the minimum normalised gap and $\min_k \Delta_k^2 g_k$ along the flow, both monotone on the computed range; the second crosses the Lehmer threshold $\frac{4}{5}$ near $t = 0.17$. Right: D^2 for the closest pair below height 200, at ordinates 184.87 and 185.60, against the isolated pair law $D_0^2 + 8t$.

$z_* + 4 \times 10^{-6}$ *show that the pair is still real, which certifies that the collision lies below* -0.287979837008*: the upper end of the bracket is certified and the lower end is not. The extrapolation of the integrated* D^2 *was* 7% *early and the isolated pair estimate* -0.262 *was* 9% *late.*

The bracket in (iii) uses that a pair at distance D in z obeys $D^2 \approx 8(t - t_*)$ near the collision (Section 4.1.2), so sign changes on a grid of step 0.01 in θ vanish at most $0.01^2/2 = 5 \times 10^{-5}$ above t_*. At $t = -0.287981$ the grid finds none, which does not exclude two real zeros less than 0.01 apart, and the lower end rests on the pair law, an asymptotic statement and not an enclosure.

The question is whether a gap functional, monotone as in (ii), can carry a positive lower bound from $t > 0$ down to $t = 0$ uniformly in the height; the data suggest not. Both are minimised at Lehmer pairs, whose normalised gaps are expected, on the GUE conjecture, to be arbitrarily small at large height, so their infimum over all heights would be zero at $t = 0$. A Lyapunov functional for RH must be global, insensitive to individual close pairs, in the manner of the pair correlation that Rodgers and Tao used to prove the reverse inequality [250]; we did not find a candidate. Two rigorous companions of the data are in Sections 4.1.7 and 4.1.10: an exact formula for the drift of a gap, and a proof that no bound through the absolute value of the integrand gives a step uniform in the height [217].

6.6.3 The cost of the flow route

The published points of the phase diagram give the cost of descending along it (Table 4.8 lists the bounds on Λ). Polymath15 proved $\Lambda \le 0.22$ with the barrier $X \approx 6\times10^{10}$, and Table 1 of their paper lists, for each bound from $\Lambda \le 0.21$ down to $\Lambda \le 0.1$, the barrier X that the same method needs, with RH verified to height $X/2$ [247]: $X \approx 2\times10^{12}$ for 0.21, 5×10^{12} for 0.2, 2×10^{13} for 0.19, 7×10^{15} for 0.15 and 9×10^{21} for 0.1. Platt and Trudgian obtained $\Lambda \le 0.2$ from the second row, which by their account needs verification to 2.51×10^{12}, and the third row, 0.19, requires verification slightly above 10^{13} [232]. The product $\Lambda \log X$ is 5.46 at 0.22 and falls along the table from 5.95 at 0.21 to 5.06 at 0.1.

Proposition 6.6.2 (Computed). *The least squares fit of $\Lambda = c/\log X$ in $1/\log X$ to the bound $\Lambda \le 0.22$ and the twelve rows of the table gives $c = 5.59$, with residuals below 0.013 in Λ, which along the table are errors of up to a factor of about 200 in X; on this law a bound $\Lambda \le \delta$ requires a barrier X of order $e^{c/\delta}$, that is verification to height $X/2$: $X = 10^{48.6}$ for $\delta = 0.05$ and $10^{242.9}$ for $\delta = 0.01$. Since $\Lambda \log X$ falls along the table, the law $\log X = a/\Lambda + b$ fits its twelve rows better, with $a = 4.24$, $b = 8.28$ and residuals below 0.003 in Λ, within a factor 1.6 in X, and gives $X = 10^{40.4}$ for $\delta = 0.05$ and $10^{187.7}$ for $\delta = 0.01$.*

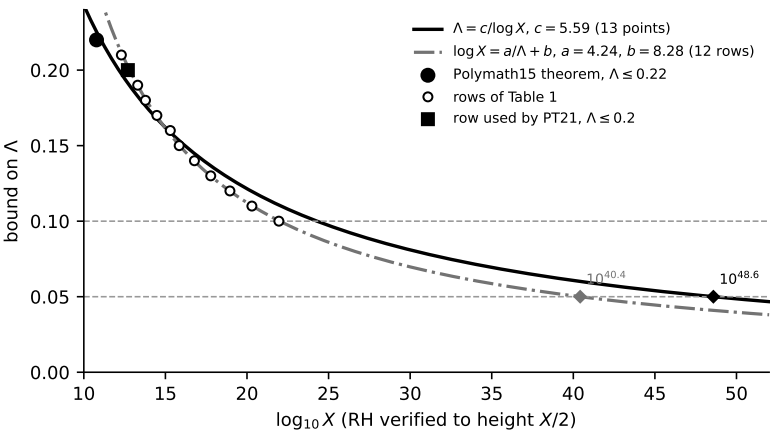


Figure 6.5: The bound $\Lambda \le 0.22$ of Polymath15 and the twelve rows of their table of conditional bounds [247] against the barrier X (verification height $X/2$), with the fits $\Lambda = c/\log X$ and $\log X = a/\Lambda + b$. Dashed lines mark $\Lambda = 0.1$ and 0.05.

Beyond the table the two laws are orders of magnitude. Their form agrees with the heuristic that the flow at time t makes the zeros near height x real once $t \log x$ exceeds a constant, which is how the Riemann Siegel type approximation of H_t enters [247]. Along the table, halving the bound from 0.2 to 0.1 raises the required barrier from 5×10^{12} to 9×10^{21}, to about its power 1.7, and the power

is 2 at every bound on the law $c/\log X$ and tends to 2 as the bound decreases on the law $\log X = a/\Lambda + b$. By the table itself the bound $\Lambda \le 0.15$ needs verification to a height of about 3.5×10^{15}, some 1200 times the record 3×10^{12} [232], and $\Lambda \le 0.1$ about 4.5×10^{21}; no conceivable computation reaches 0.05. The flow route produces a theorem at every stage, a monotone sequence of upper bounds for Λ, which is 0 exactly when RH holds [250], as no other route does, and it cannot reach the corner of the diagram by computation.

6.6.4 A programme

The phase diagram suggests five steps; the first three are engineering.

1. An evaluator: a Riemann Siegel type form of ξ_t and ξ_t', such as the effective approximation of H_t in [247], in ball arithmetic, at a cost of about $\sqrt{\theta/2\pi}$ terms per point (Table 2.4), replacing the Pólya integral, whose required precision grows like $0.341\,\theta$ digits (Section 2.1.1); see Section 25.2.2 for $t = 0$. Lemma 6.2.6 gives another evaluator: at $s = \frac{1}{2} + i\theta$, $\xi_t(s)$ is $g(s)e^{t\ell(s)^2/4}$ times a Gaussian average of ζe^r centred at $s + \frac{t}{2}\ell(s)$, close to $\frac{1}{2} + \frac{t}{4}\log\frac{\theta}{2\pi} + i(\theta + \frac{\pi t}{8})$, with variance $\frac{t}{2}$ in the real direction, which Gauss Hermite quadrature computes from ordinary evaluations of ζ and Γ; at $t = 0.2$ the relative differences between the rules with 40 and 60 nodes are 6.8×10^{-56} at $\theta = 1000.3$ and 2.4×10^{-47} at $\theta = 4000.3$, and an error bound for the quadrature is open.

2. Certificates in Thorin form: Theorem 6.5.6 or Proposition 6.5.7 at a target (t_0, α_0) up to the barrier.

3. The tail in the same framework: the barrier and the analytic region beyond it [247], so that the points of Corollary 6.3.3(b) and (c) are certified end to end and stated as GGC theorems.

4. A measurement: the optimal (t_0, α_0) on each level set as a function of the verification height, and the constant c of Proposition 6.6.2 as a function of the height.

5. The structural step: a tail at $t = 0$ below the basepoint one (below $\frac{7}{8}$ if the announced half plane is confirmed). Every certified point is a theorem, and by Proposition 6.4.1 no computed window distinguishes the corner from them; the step is the existence of the tail, and it must come from one of the mechanisms of Section 25.1.2.

6.6.5 Open problems

Problem 6.1. Compute $v_t(\alpha, \theta)$ at $t = 0$ and $t = 0.2$ near the Polymath barrier, heights near 3×10^{10}, with a Riemann Siegel type evaluator of $H_{0.2}$ such as the

effective approximation of [247] (Section 2.2 gives the one for ζ), and measure how the difference between the functions $v_{0.2}/\pi$ and v_0/π grows with the height. Proposition 6.4.1(iii) measures it below 100. For $t = 0$ see work package 2 of Section 25.2.2 and Section 25.1.3.

Problem 6.2. Describe the boundary $\alpha^*(t)$ of $\mathcal{G}$ for $0 \le t \le 0.2$ as far as computation and the known zero free regions allow. Is α^* continuous, and is $(\alpha^*(t)-\frac{1}{2})^2+\frac{t}{2}$ strictly decreasing in t where $\alpha^*(t) > \frac{1}{2}$? Theorem 6.3.2(i) shows that it is nonincreasing there. Problem 19.3 asks the neighbouring question for the flow clock T_t, $0 < t < 0.2$.

Problem 6.3. Give the symmetrised truncations of Section 17.1 a coordinate on the phase diagram: for some (t, α) outside the region of Corollary 6.3.3, is there a family of explicit truncations of ξ_t, built from finitely many terms of the Pólya integral or of a Dirichlet series and making no use of the zeros, as the symmetrised lattice truncations K_n of Subsection 17.1.1 are, that converges to ξ_t on compact sets, together with heights $h_n \to \infty$ such that for infinitely many n the truncation has no zero with $\operatorname{Re} s > \alpha$ and $|\operatorname{Im} s| < h_n$? Hurwitz's theorem then makes (t, α) a new point of $\mathcal{G}$, as in Corollary 19.12.2(c); at $t = 0$ and $\alpha = \frac{1}{2}$ the question for the K_n is Conjecture S (Theorem 17.1.7), and for the stages of Chapter 19 it is Problem 19.1.

Problem 6.4. Compute the Cholesky factor of the Weil Gram matrix of Section 6.6.1 for decreasing w and increasing L, and decide whether its entries converge to an object built from the arithmetic, the first test of a factorisation $W(g * \tilde{g}) = \|Ag\|^2$.

Problem 6.5. Find a global functional of the zeros of H_t, insensitive to individual close pairs, that is monotone along the flow and whose positivity at $t = 0$ excludes collisions at positive time. Biane's gap functional of Section 4.1.4 is local, the minimum over one gap of Laguerre's quantity, and Problems 4.1 and 25.5 ask for the control of the nonreal zeros that would make its brackets unconditional; the functional asked for here should instead be one quantity for all the zeros.

Problem 6.6. (a) Prove that $v_{0.2}(10, \theta) \ge \frac{19}{2} V_{0.2}$ for every $\theta \ge \theta_1(0.2)$, which by Corollary 6.2.8 makes $\tilde{\varphi}_{0.2,\alpha}$ a GGC transform for every $\alpha \in [\frac{1}{2}, 10]$. By Lemma 6.2.6 and Cauchy's estimate it suffices to bound $|\zeta_{0.2} - 1|$ on the discs of radius 5 about $10 + i\theta$ by a small constant uniform in θ; on these discs the shifted Gaussian average of the lemma reaches $\operatorname{Re} s \le 2$ with probability below 10^{-20} at every height, and at the four heights sampled the largest of 256 sampled values of $|\zeta_{0.2} - 1|$ on each boundary circle is at most 0.0326 (computed, not a bound on the discs). (b) For $0 < t \le 0.2$ prove an explicit bound $|S_t(T)| \le c_1 \log T + c_2$ for $T \ge T_1$, for instance by proving the two hypotheses of Proposition 6.2.7(iii) with an explicit M, from Lemma 6.2.6, Rademacher's convexity bound for $-\frac{1}{2} \le \operatorname{Re} s \le \frac{3}{2}$ [161] and the functional equation, with its Gamma factor kept, for $\operatorname{Re} s < -\frac{1}{2}$.

The minimum of $\operatorname{Re}\zeta_{0.2}(2+iy)$ over $0 \le y \le 200$ is 0.69006, near $y = 14.015$ (computed). At $t = 0$ much sharper explicit bounds are known [100].

Notes

The phase diagram theorem is de Bruijn's strip theorem [61, Theorem 13] read through Theorem 6.2.3; its case $t = 0$ is in Section 4.1. Theorem 6.2.3, the certificate of Section 6.5 with Theorem 6.5.8, and the experiments of Sections 6.6.1 and 6.6.2 are from the authors' working report [243]. The bounds on Λ are those of de Bruijn [61], Ki, Kim and Lee [182], Polymath15 [247] and Platt and Trudgian [232], and the lower bound is Rodgers and Tao [250]. The half plane $\operatorname{Re} s > \frac{7}{8}$ is the announcement [227], discussed in Section 25.1.2. The description of the Thorin measure as one object with a prime face and a zero face, joined by the sine squared lemma, is developed for automorphic L functions in [244]. Section 6.2.1, except part (iv) of Proposition 6.2.5 and Corollary 6.2.8, and part (b) of Problem 6.6 are based on [138].

6.7 Exercises

On Sections 6.2 and 6.3.

6.1. Prove the semigroup identity $P_a * P_b = P_{a+b}$ from the characteristic function $e^{-a|\xi|}$ of P_a, and deduce the identity $\int_{\mathbb{R}} P_d(\theta-\gamma)(\theta^2+u)^{-1}d\theta = \pi P_{d+\sqrt{u}}(\gamma)/\sqrt{u}$ used in Lemma 6.2.2.

6.2. Show that the Thorin density of Lemma 6.2.2(a), in the chart $z = \theta^2$, has total mass one, and that it converges vaguely to δ_{γ^2} as $d \downarrow 0$.

6.3. Show that the level sets of $t + 2(\alpha - \frac{1}{2})^2$ are the parabolas along which Theorem 6.3.2(i) carries the boundary of a certified region, and that the parabola through (t_0, α_0) meets the axis $\alpha = \frac{1}{2}$ at $t_0 + 2(\alpha_0 - \frac{1}{2})^2$.

6.4. Using Theorem 6.3.2(ii) and Corollary 6.3.3(d), find the α for which a zero free half plane $\operatorname{Re} s > \alpha$ at time zero would improve the bound $\Lambda \le 0.2$, and the smallest t for which the flowed half plane $\operatorname{Re} s > \frac{7}{8}$ gives a point of $\mathcal{G}$ at the basepoint 0.6.

6.5. Let F be the perturbation of Theorem 1.4.1, with a quadruple of zeros at $\frac{1}{2} \pm \frac{1}{4} \pm ig$. Assuming RH, show that the shifted clock of F at the basepoint α is a GGC exactly for $\alpha \ge \frac{3}{4}$, and compute its Thorin density near $\theta = g$ for $\alpha = 0.8$.

On Sections 6.4 and 6.5.

6.6. Assuming RH, show that the average of $v_0(\alpha,\theta)$ over a window of length H about θ is $\frac{1}{2}\log(\theta/2\pi)+O(\log\theta/H)$ when H is large compared with $\alpha-\frac{1}{2}$, and explain why the minimum over the window can be much smaller than the average when $\alpha-\frac{1}{2}$ is small compared with the mean spacing.

6.7. Show that $v_t(\alpha,0)=\xi_t'/\xi_t(\alpha)$ is positive for every $t\ge 0$ and $\alpha>\frac{1}{2}$, using Lemma 6.2.1(ii), and compute it at $t=0.2$, $\alpha=0.6$ from the Pólya integral.

6.8. Prove Lemma 6.5.5 and show that it remains true with $\le$ in place of $<$, the change of argument then lying in $(-\frac{\pi}{2},\frac{\pi}{2})$.

6.9. Explain why the number of grid points in Theorem 6.5.6 grows like $e^{(\pi/4-2\eta)\theta}$ for a tilt $\eta<\pi/8$.

6.10. Show that positivity of v_t on the boundary of a rectangle does not by itself exclude a zero inside it, and explain why Proposition 6.5.7 asks for the whole rectangle.

On Section 6.6.

6.11. Show that for g supported in $[-L/2,L/2]$ the function $h=g*\tilde g$ is supported in $[-L,L]$, and that the prime sum in Weil's explicit formula runs over $n\le e^L$.

6.12. Derive the x space form of the archimedean term of Section 6.6.1 from the integral representation $\psi(z)=\int_0^\infty(e^{-t}/t-e^{-zt}/(1-e^{-t}))\,dt$.

6.13. For an isolated pair of zeros of H_t at distance D, show from the equation of motion that $D^2\approx D_0^2+8t$ (compare Exercises 18.15 and 18.16). Explain why two certified sign changes of $\xi_t(\frac{1}{2}+i\theta)$ on a grid of step 0.01 show that the pair of Proposition 6.6.1(iii) is still real at t, while their absence does not show that it has left the line, and why the lower end of the bracket there rests on the law $D^2\approx 8(t-t_*)$.

6.14. Fit $\Lambda=c/\log X$ to the thirteen points of Proposition 6.6.2 by least squares in $1/\log X$ and check the value $c=5.59$; then fit $\Lambda=c'\log\log X/\log X$ to the same points and compare the barrier each law gives for $\Lambda=0.1$ with the last row of the table, $X\approx 9\times 10^{21}$.

Part II

Cauchy deconvolution

Chapter 7

The Cauchy deconvolution operator

Every computation of this part inverts one operator: convolution with the Cauchy density $P_a(x) = (a/\pi)/(a^2+x^2)$. This chapter studies the operator in its own right, proves the facts the part uses, and places them in the literature; the chapters that follow apply it. The Thorin measure at level θ' is the solution, when a positive one exists, of $P_{1-\theta'} * \mu = u_1$ (Chapter 8); the explicit formula is that equation at $\theta' = \frac{1}{2}$, in measures under RH (Theorem 8.2.2(b)) and in distributions unconditionally (Theorem 9.1.1); the backward flow and the EM inversion solve it numerically (Chapter 10), and the matrix pencil solves it for atoms (Chapters 11 and 12).

The facts are these. Convolution with P_a is harmonic extension to height a in the upper half plane, so Cauchy deconvolution is the Cauchy problem for Laplace's equation, Hadamard's original example of an ill posed problem [93]. It is unique when it is solvable. The forward map preserves positivity and the inverse does not: positive data can come from a signed source, and whether a positive source exists is a statement about positive harmonic functions, which Herglotz and Riesz characterised. Positivity of the deconvolution is monotone in the width, so every positive datum has a critical width beyond which no positive source exists; for the Thorin family it is 1 minus the supremum of the real parts of the zeros. Stability is Hölder between levels, with the exponent of the three lines theorem, and logarithmic at the boundary, and both rates are attained. In statistics Cauchy noise is supersmooth of order one and the minimax rates of density deconvolution are logarithmic; Bayesian posteriors contract at logarithmic rates in the cases treated (Section 7.4.2). For a sum of atoms the problem becomes Prony's: the error is Lipschitz in the data error, with a constant that grows polynomially as atoms approach, except near merging atoms, where it is the square root.

The chapter develops a survey chapter on Cauchy deconvolution by Nick Polson, together with the checks made for this book. New here are the generality of the statements (any nonnegative datum, any source in the class $\int (1+x^2)^{-1} d|\mu| < \infty$), the example of Section 7.2, the three lines bound with its sharpness, the bound for finite sources (Proposition 7.3.2), the positive two level bound (Corollary 7.3.3), the logarithmic bound and the survey of the statistical, algorithmic

and physical literature. The specialisation to the Thorin family, its uniqueness argument and the unboundedness of the deconvolution are in Chapters 8 and 9.

Notation. A level θ is the real part of $s = \theta + iv$, and $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$; in Chapter 8, which extends equation (2.1.7), this is $\rho_{\theta-1/2}(v)$. The measure Cauchy(γ, a) of Chapter 17 is $P_a(\cdot - \gamma)\,dx$, and the Thorin measure $\mathfrak{U}_\theta = \sum_\rho \text{Cauchy}(\gamma_\rho, \theta - \sigma_\rho)$ has density u_θ; $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$, with $\widetilde{U}_\theta$ the measure of Theorem 17.14.4, which has the factor $\frac{1}{2}$. The harmonic extension of a measure is h, and $\sigma_{\max} = \sup \operatorname{Re}\rho$. Widths are $a, c, d, w \geq 0$, except that b may be negative with $b > -w$ (Theorem 7.3.1, Proposition 7.3.2); the example of Section 7.2 has its own letters q, r and m, and η is the linear coefficient of Theorem 7.2.3. A measure μ is admissible when $\int (1+x^2)^{-1} d|\mu|(x) < \infty$; the zero counting measure $\sum_\gamma \delta_\gamma$ is the example that matters.

Statements from the literature are marked cited; the notes of the last section name the entry the citation verifier has not confirmed and the statements not checked against the papers. The results are these.

1. Uniqueness of Cauchy deconvolution, the range of P_a* on L^2 and the failure of the naïve description of it (Propositions 7.1.1 and 7.1.2, proved).

2. A positive datum with a signed source and the closed form of its critical width (Example 7.2.1, proved; the numbers are computed); the critical width of a general positive datum, attained and possibly infinite (Proposition 7.2.2, proved); $c^*(u_1) = 1 - \sigma_{\max}$ (Theorem 8.2.2(a), cited); the Herglotz characterisation of the data with a positive source (Theorem 7.2.3, proved), with Lagarias's criterion as its case $c = \frac{1}{2}$ (cited).

3. Hölder stability between levels, sharp (Theorem 7.3.1, proved); the same for finite signed sources with the sharp exponent $(b + w)/(a + w)$ (Proposition 7.3.2, proved); a two level bound for positive sources on the line (Corollary 7.3.3, proved); the logarithmic bound at the boundary, sharp in order and in the constant (Theorem 7.3.4, proved). Why $\widetilde{U}_{3/4}$ is recovered better than $\widetilde{U}_{1/2}$ is explained heuristically, and the exponent is tested on Lorentzians (Table 7.1, computed).

4. A survey of density deconvolution, Bayesian inverse problems and positivity preserving algorithms (cited), and a polynomial upper rate for sources analytic in a strip (Proposition 7.4.1, proved).

5. Atomic sources, merging atoms and the analytic continuation problem of many body physics (cited, with the proofs of Chapters 11 and 12), and three open problems.

7.1 The operator

7.1.1 Semigroup and harmonic extension

With $\widehat{f}(\omega) = \int f(x)e^{-i\omega x}dx$, the transform of P_a is $e^{-a|\omega|}$, so

$$P_a * P_b = P_{a+b}, \qquad P_a = e^{-a|D|}, \quad |D| = (-\partial_x^2)^{1/2} :$$

the Cauchy densities form the Poisson semigroup, the symmetric stable semigroup of index one (for the clock of ξ see Theorem 2.1.2 and Proposition 2.1.3, Section 2.1; for the Thorin measures, Theorem 17.14.4). For an admissible measure μ the function

$$h(x,y) = (P_y * \mu)(x), \qquad y > 0,$$

is harmonic in the upper half plane, being the real part of $\pi^{-1}i\int\big((z-t)^{-1} + t(1+t^2)^{-1}\big)d\mu(t)$ at $z = x + iy$. The integral converges absolutely because $(z - t)^{-1} + t(1 + t^2)^{-1} = (1 + tz)/((z - t)(1 + t^2)) = O_z(t^{-2})$, and the added term is real, so after multiplication by i it changes only the imaginary part. (Without the added term the integral diverges for Lebesgue measure and for the zero counting measure, since $\sum_\gamma 1/|\gamma| = \infty$.) The boundary values of h are μ in the vague sense, so $P_a * \mu$ is the harmonic extension of μ restricted to the line $y = a$, and

$$P_a * \mu = u \tag{7.1.1}$$

asks for the boundary values of a harmonic function known on a parallel line. A second description, used in Chapter 11 (Proposition 11.6.1) and in Section 12.2.3, is probabilistic: P_a is the law of planar Brownian motion started at height a when it first hits the real axis, equivalently a Gaussian with a random variance distributed as the first passage time of one dimensional Brownian motion to level a.

7.1.2 Uniqueness and the range

Proposition 7.1.1 (uniqueness). *If μ_1, μ_2 are admissible measures and $P_a * \mu_1 = P_a * \mu_2$, then $\mu_1 = \mu_2$.*

Proof. Let $\nu = \mu_1 - \mu_2$ and $g(x,y) = (P_y * \nu)(x)$, harmonic on $y > 0$. For $y > a$, $g(\cdot, y) = P_{y-a} * g(\cdot, a) = 0$; the semigroup property holds for the signed ν because $|g(\cdot, a)| \le P_a * |\nu|$ and $P_{y-a} * (P_a * |\nu|) = P_y * |\nu| < \infty$ by Tonelli, so Fubini applies. So g vanishes on an open set and hence on the whole half plane. An admissible measure is the vague boundary value of its Poisson integral, so $\nu = 0$. □

This is the uniqueness part of Theorem 8.2.2, for a general admissible measure.

The inverse is formally multiplication by $e^{a|\omega|}$, which is not the Fourier transform of a tempered distribution, so the inverse is not a convolution and is defined

only on the range of P_a*. If $u = P_a * \mu$, the Poisson extension of u continues harmonically to the depth a below the line, as $P_{y+a} * \mu$. The converse fails without conditions at depth a (part (c) below). For positive sources Theorem 7.2.3 characterises the range exactly. On $L^2(\mathbb{R})$ the inverse is unbounded in every Sobolev norm, which is the source of everything that follows; Chapter 8 makes the same remark for the Thorin density (after Theorem 8.2.2).

PROPOSITION 7.1.2 (the range). *(a) For $f \in L^2(\mathbb{R})$, $\int e^{2a|\omega|}|\widehat{P_a * f}|^2 d\omega = \int |\widehat{f}|^2 d\omega$, and every u with $\int e^{2a|\omega|}|\widehat{u}|^2 < \infty$ is $P_a * f$ for an $f \in L^2$. (b) For all real s, t, $\sup \|f\|_{H^t}/\|P_a * f\|_{H^s} = \infty$. (c) The function $u = 2 + \cos(x^2)$ is not $P_a * \mu$ for any $a > 0$ and any admissible μ, although its Poisson extension continues to an entire harmonic function.*

Proof. (a) $\widehat{P_a * f} = e^{-a|\omega|}\widehat{f}$, and conversely $\widehat{f} = e^{a|\omega|}\widehat{u}$ defines $f \in L^2$ with $P_a * f = u$. (b) Let f_N be real with $\widehat{f_N}$ the indicator of $[N, N+1] \cup [-N-1, -N]$. Then $\|P_a * f_N\|_{H^s} \le e^{-aN}\|f_N\|_{H^s}$, and $\|f_N\|_{H^t}/\|f_N\|_{H^s}$ lies between two powers of N, so $\|f_N\|_{H^t}/\|P_a * f_N\|_{H^s} \ge cN^{-|t-s|}e^{aN} \to \infty$. (c) The Poisson extension $h = P_y * u$ is continuous on $y \ge 0$, and $g(x, y) = 2 + \operatorname{Re}\cos((x + iy)^2)$ is an entire harmonic function with $g(x, 0) = u(x)$. So $h - g$ is harmonic on $y > 0$, continuous on $y \ge 0$ and zero on $y = 0$; by the Schwarz reflection principle it extends to an entire harmonic function, and so does $h = g + (h - g)$. Suppose $u = P_a * \mu$ with μ admissible. Then $u''' = (\partial_x^3 P_a) * \mu$ and $|\partial_x^3 P_a(x - t)| \le C_a(1 + (x - t)^2)^{-1} \le 2C_a(1+x^2)(1+t^2)^{-1}$, because $1+t^2 \le 2(1+x^2)(1+(x-t)^2)$. So $|u'''(x)| \le C(1+x^2)$. But $u'''(x) = 8x^3 \sin(x^2) - 12x\cos(x^2)$ is not $O(1 + x^2)$. □

7.2 Positivity

7.2.1 The forward map preserves it, the inverse does not

P_a is a probability density, so $\mu \ge 0$ implies $P_a * \mu \ge 0$, and the semigroup property carries positivity up the levels: if $P_c * \mu \ge 0$ for some c then $P_{c'} * \mu \ge 0$ for every $c' \ge c$. The converse fails, and the failure is quantitative.

EXAMPLE 7.2.1 (positive data, signed source). Let $d > 0$, $0 < q < 1$ and $\mu = \delta_0 - q\delta_d$. Then $P_a * \mu = P_a - qP_a(\cdot - d) \ge 0$ on $\mathbb{R}$ if and only if

$$q \;\le\; \frac{r - d}{r + d}, \qquad r = \sqrt{d^2 + 4a^2}.$$

Proof. Positivity is $q \le \inf_x f(x)$ with $f(x) = (a^2 + (x - d)^2)/(a^2 + x^2)$. Setting $f' = 0$ gives $x^2 - dx - a^2 = 0$, and at the root $x^* = (d+r)/2$, using $x^{*2} = dx^* + a^2$, the numerator is $r(r - d)/2$ and the denominator $r(r + d)/2$. The other root is a maximum, and $f \to 1$ at infinity. □

By Proposition 7.1.1 the signed measure is the only source of these data at width a, so a positive datum can have no positive source at width a. The example has the structure of the Thorin family. Deconvolving $u = P_a * \mu$ only part of the way, by P_c with $c < a$, gives $P_{a-c} * \mu = P_{a-c} - qP_{a-c}(\cdot - d)$, which is positive if and only if $q \le (r_c - d)/(r_c + d)$ with $r_c = \sqrt{d^2 + 4(a-c)^2}$; that is, if and only if

$$c \;\le\; c^* = a - \tfrac{1}{2}\sqrt{m^2 - d^2}, \qquad m = \frac{1+q}{1-q}\, d,$$

and $c^* \ge 0$ if and only if the datum is positive: positivity holds up to a critical width and is lost beyond it. For $q = 9 - 4\sqrt{5} \approx 0.0557$, $a = \frac{1}{2}$ and $d = 1$ one has $m = \sqrt{5}/2$ and $c^* = \frac{1}{4}$ exactly: the datum is positive (the bound of the example is $3 - 2\sqrt{2} \approx 0.1716$), its partial deconvolutions are positive for $c < \frac{1}{4}$, touch zero at $c = \frac{1}{4}$ and are negative beyond. (Computed: with $q = 0.0557$ the critical width is 0.25007, so the value $\frac{1}{4}$ needs $q = 9 - 4\sqrt{5}$.)

Proposition 7.2.2 (the critical width). *For $u \ge 0$ with $\int u(x)(1+x^2)^{-1}dx < \infty$ let $c^*(u) = \sup\{c \ge 0 :\ u = P_c * \nu$ for some measure $\nu \ge 0\} \in [0, \infty]$, where $P_0 * \nu = \nu$, so that $c = 0$ always qualifies. Then a positive source at width c exists for every finite $c \le c^*(u)$ and for no $c > c^*(u)$; in particular the supremum is attained when it is finite.*

Proof. If $u = P_c * \nu$ with $\nu \ge 0$ and $0 \le c' < c$, then $u = P_{c'} * (P_{c-c'} * \nu)$ with $P_{c-c'} * \nu \ge 0$, by Tonelli and the semigroup property, so the admissible widths form an interval containing 0. Let $0 < c^*(u) < \infty$ and let $0 < c_n \uparrow c^*(u)$ be admissible. Then $u = P_{c_1} * \nu_1$ is continuous, and by Theorem 7.2.3 the harmonic extension h of u extends to a nonnegative harmonic function V_n on $y > -c_n$. Since V_n and V_m agree with h on $y > 0$, they agree on the connected set $y > -\min(c_n, c_m)$, so together they define a nonnegative harmonic function on $y > -c^*(u)$ that extends h. Theorem 7.2.3 with $c = c^*(u)$ gives $u = P_{c^*(u)} * \nu$ with $\nu \ge 0$. □

The critical width is not capped at the width of a given deconvolution, so it does not presuppose a source at that width. For the Thorin family, let $u_1 = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + iv)$, the density of level one. By Theorem 8.2.2(a), with $\varepsilon_0 = \frac{1}{2}$ and $\varepsilon_1 = \frac{1}{2} - c$, a positive source at width $c \in (0, \frac{1}{2}]$ exists if and only if ζ has no zeros in $\operatorname{Re} s > 1 - c$, zeros on $\operatorname{Re} s = 1 - c$ contributing atoms; the source is then $u_{1-c}\, dv$ (the Thorin measure $\mathfrak{U}_{1-c}$), which is a function when $c < 1 - \sigma_{\max}$ and has atoms when $c = 1 - \sigma_{\max}$ and some zero has real part $\sigma_{\max}$. For $c > \frac{1}{2}$ there is no source at all, since its Poisson integral would be harmonic on $y > -c$ and equal to $\pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + y + iv)$ off the zeros, which has poles at the zeros on the critical line (compare Theorems 17.14.2, 17.14.4 and 10.3.1). So

$$c^*(u_1) = 1 - \sigma_{\max}, \qquad \sigma_{\max} = \sup\{\operatorname{Re}\rho\},$$

and the Riemann hypothesis is the statement $c^*(u_1) = \frac{1}{2}$: the critical width of the level one density is as large as the functional equation allows (Theorem 8.2.2,

where $\sigma_{\max} - \frac{1}{2}$ is called the depth to which the Thorin density can be deconvolved into a positive measure); Remark 2.1.4 says why positivity of the density on the single line is weaker.

7.2.2 When does a positive source exist

Theorem 7.2.3 (existence of a positive source). *Let $u \geq 0$ be continuous with $\int u(x)(1+x^2)^{-1}dx < \infty$, and let $h(x,y) = (P_y * u)(x)$ be its harmonic extension, $y > 0$. For $c > 0$ the following are equivalent.*

(i) $u = P_c * \nu$ for a measure $\nu \geq 0$.

(ii) h extends to a nonnegative harmonic function on the half plane $y > -c$.

Proof. If (i) holds, $V(x,y) = (P_{y+c} * \nu)(x)$ is harmonic and nonnegative on $y > -c$ and equals h for $y \geq 0$ by the semigroup property. Conversely, a nonnegative harmonic function V on $y > -c$ is, by the Herglotz and Riesz representation for the half plane (see [1]), of the form $V(x,y) = \eta(y+c) + (P_{y+c} * \nu)(x)$ with $\eta \geq 0$, $\nu \geq 0$ and $\int (1+t^2)^{-1} d\nu < \infty$, and $\eta = 0$: at $y = 0$ the representation says $u = \eta c + P_c * \nu$, since $V = h$ on $y \geq 0$ and u is continuous, so for $y > 0$ we have both $V(\cdot, y) = \eta(y+c) + P_{y+c} * \nu$ and $h(\cdot, y) = P_y * u = \eta c + P_{y+c} * \nu$, whence $\eta y = 0$. Thus $u = P_c * \nu$. □

For the level one density the harmonic extension is explicit. Smoothing raises the level, $P_y * u_1 = u_{1+y}$, and the formula continues below the line: u_θ is harmonic in (θ, v) wherever ξ has no zeros. So condition (ii) for u_1 at width c says that ξ has no zeros on $\operatorname{Re} s > 1 - c$ and $\operatorname{Re}(\xi'/\xi)(s) \geq 0$ there, and at $c = \frac{1}{2}$ it is Lagarias's criterion [185]: the Riemann hypothesis is equivalent to $\operatorname{Re}(\xi'/\xi)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$ (the conditions ≥ 0 and > 0 agree by the minimum principle). In the language of analytic continuation, (ii) says that h is the real part of an analytic function of positive real part on $\operatorname{Im} z > -c$, a Pick function after a rotation.

7.3 Stability

7.3.1 Hölder between levels

Theorem 7.3.1 (three lines). *Let $b_0 < b < a$ and let μ be a signed measure with $P_{b_0} * \mu \in L^2$, where for $b_0 < 0$ this means $\mu = P_{|b_0|} * \lambda$ with $\lambda \in L^2$, and $P_{b_0} * \mu$ stands for λ. Then*

$$\|P_b * \mu\|_2 \;\leq\; \|P_a * \mu\|_2^{\frac{b-b_0}{a-b_0}} \; \|P_{b_0} * \mu\|_2^{\frac{a-b}{a-b_0}}.$$

In particular, if two sources agree in their data at width a to within ε in L^2 and both are bounded by M at width b_0, they agree at width b to within $\varepsilon^\kappa (2M)^{1-\kappa}$, $\kappa = (b-b_0)/(a-b_0)$. The exponent is attained in rate.

Proof. Let F be the Fourier transform of $P_{b_0} * \mu \in L^2$ (of λ when $b_0 < 0$). For $y > b_0$, $P_y * \mu = P_{y-b_0} * (P_{b_0} * \mu)$, so $\|P_y * \mu\|_2^2 = (2\pi)^{-1} \int e^{-2(y-b_0)|\omega|} |F|^2 d\omega$, and with $\kappa = (b - b_0)/(a - b_0)$,

$$e^{-2(b-b_0)|\omega|}|F|^2 = \big(e^{-2(a-b_0)|\omega|}|F|^2\big)^{\kappa}\big(|F|^2\big)^{1-\kappa},$$

so Hölder's inequality with exponents $1/\kappa$ and $1/(1-\kappa)$ gives the bound. Equivalently, $y \mapsto \log \|P_y * \mu\|_2$ is convex, the three lines theorem for the harmonic extension. For sharpness take F concentrated near a frequency Ω with $e^{-(a-b_0)\Omega} = \varepsilon/M$: both sides are then equal up to constants. □

The exponent depends on how far below the target an a priori bound is available. For the Thorin family under the Riemann hypothesis the data are at level 1, which is width $a = \frac{1}{2}$ above the zero counting measure, and the level θ' measure is at width $b = \theta' - \frac{1}{2}$. The zero measure is atomic, so Theorem 7.3.1 admits no level below it ($b_0 < 0$ would require $\sum_\gamma \delta_\gamma = P_{|b_0|} * \lambda$ with $\lambda \in L^2$, a smooth function), and it has infinite mass. For the zeros in a window, a finite measure, Proposition 7.3.2 below with $w = 0$, $a = \frac{1}{2}$ and $b = \theta' - \frac{1}{2}$ gives the exponent $2\theta' - 1$: about $\frac{1}{2}$ at $\theta' = \frac{3}{4}$, falling to 0 at $\theta' = \frac{1}{2}$. The application is heuristic, since a window does not commute with P_y and the zeros outside it carry infinite mass; heuristically, it explains why $\widetilde{U}_{3/4}$ was recovered so much better than $\widetilde{U}_{1/2}$ in Chapters 10 and 11. Table 7.1 shows the exponent $(b + w)/(a + w)$ of Proposition 7.3.2 on a test source of Lorentzians of width w, to within 0.01.

Proposition 7.3.2 (finite sources). *Let ν be a finite signed measure of total variation at most B, let $w \geq 0$ and $h = P_w * \nu$ (with $P_0 * \nu = \nu$, and $P_b * h$ meaning $P_{b+w} * \nu$ when $b < 0$), and let $-w < b < a$. Then*

$$\|P_b * h\|_2 \leq \sqrt{2}\, D^{1-\kappa}\, \|P_a * h\|_2^{\kappa}, \qquad \kappa = \frac{b+w}{a+w}, \quad D = \frac{B}{\sqrt{2\pi(b+w)}},$$

and no larger exponent holds uniformly over this class.

Proof. The transform of h satisfies $|\widehat{h}(\omega)| \leq Be^{-w|\omega|}$. Let $\varepsilon = \|P_a * h\|_2$ and $\Omega \geq 0$. For $|\omega| \leq \Omega$, $e^{-2b|\omega|} = e^{2(a-b)|\omega|} e^{-2a|\omega|} \leq e^{2(a-b)\Omega} e^{-2a|\omega|}$, so by Plancherel

$$\begin{aligned}
\|P_b * h\|_2^2 &= \frac{1}{2\pi} \int e^{-2b|\omega|} |\widehat{h}(\omega)|^2 d\omega \\
&\leq e^{2(a-b)\Omega}\varepsilon^2 + \frac{B^2}{2\pi} \int_{|\omega|>\Omega} e^{-2(b+w)|\omega|} d\omega \\
&= e^{2(a-b)\Omega}\varepsilon^2 + D^2 e^{-2(b+w)\Omega}.
\end{aligned}$$

If $\varepsilon < D$, take $e^{(a+w)\Omega} = D/\varepsilon$: both terms equal $D^{2(1-\kappa)}\varepsilon^{2\kappa}$. If $\varepsilon \geq D$, take $\Omega = 0$: then $\|P_b * h\|_2 \leq D \leq D^{1-\kappa}\varepsilon^{\kappa}$. For sharpness let $\phi(x) = (2\pi)^{-1}\big(\sin(x/2)/(x/2)\big)^2$,

so that $\phi \ge 0$, $\int \phi = 1$ and $\widehat{\phi}(\omega) = (1-|\omega|)_+$, and let $\nu_\Omega = B\phi(x)\cos(\Omega x)\,dx$ with $\Omega \ge 2$, of total variation at most B, and $h_\Omega = P_w * \nu_\Omega$. Then $\widehat{\nu_\Omega} = \frac{B}{2}\big(\widehat{\phi}(\cdot-\Omega)+\widehat{\phi}(\cdot+\Omega)\big)$ has two disjoint bumps, and for every $y > -w$, $\|P_y * h_\Omega\|_2 = Be^{-(y+w)\Omega}c_\Omega(y)$ with $(6\pi)^{-1/2}e^{-(y+w)} \le c_\Omega(y) \le (6\pi)^{-1/2}e^{y+w}$. So $\|P_b * h_\Omega\|_2/\|P_a * h_\Omega\|_2^\kappa$ lies between two positive multiples of $B^{1-\kappa}$, independent of Ω, while $\|P_a * h_\Omega\|_2 \to 0$ as $\Omega \to \infty$; this excludes every exponent above κ. □

With $w = 0$ this bounds a finite signed measure from data at width a; with $b = 0$ and $w > 0$ it gives the exponent $w/(a+w)$ for recovering $h = P_w * \nu$ from the data at width a: the improvement comes from the analytic smoothing and the mass bound, not from positivity, and if $w \to 0$ or the mass bound is removed no uniform positive exponent follows. It is stated here for a finite mass; in Theorem 7.3.1 the mass is replaced by an L^2 bound at a level below the target.

Table 7.1: Stability between levels, computed. The source is a sum of 40 Lorentzians of width $w = 0.05$ on a period 400, the data are at width $a = \frac{1}{2}$ with white noise of size ε at every frequency, and $P_b * \mu$ is recovered by an oracle spectral cutoff, with ε from 10^{-12} to 10^{-3}. The fitted slope is that of log of the error in $P_b * \mu$ against $\log\varepsilon$; the predicted slope is the exponent $(b+w)/(a+w)$ of Proposition 7.3.2. Nineteen values of ε (the error does not depend on the noise draw). All four fitted slopes are within 0.004 of the prediction. Status: computed, illustration.

level b	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	0
predicted $(b+w)/(a+w)$	0.7727	0.5455	0.3182	0.0909
fitted slope	0.769	0.545	0.318	0.091

If the datum and both sources are nonnegative, the mass in a window is controlled by the data alone, and the three lines theorem needs no a priori bound.

Corollary 7.3.3 (positive sources). *If $\mu_1, \mu_2 \ge 0$, $u_j = P_a * \mu_j \in L^2$ and $\|u_1 - u_2\|_2 \le \varepsilon$, then for $0 < d < b < a$*

$$\|P_b * (\mu_1 - \mu_2)\|_2 \le \varepsilon^\kappa\big((a/d)(\|u_1\|_2 + \|u_2\|_2)\big)^{1-\kappa}, \qquad \kappa = \frac{b-d}{a-d}.$$

Proof. For $0 < d < a$, $P_d/P_a = (d/a)(a^2+x^2)/(d^2+x^2) \le a/d$ pointwise, so $P_d * \mu_j \le (a/d)P_a * \mu_j$ pointwise, and $\|P_d * (\mu_1 - \mu_2)\|_2 \le (a/d)(\|u_1\|_2 + \|u_2\|_2)$. Theorem 7.3.1 applies with $b_0 = d$. □

This needs data in L^2 on all of $\mathbb{R}$; for data on a window the question is open (Problem 7.3).

7.3.2 Logarithmic at the boundary

Theorem 7.3.4 (the boundary). *Suppose $\|\mu\|_{H^s} \le M$ for some $s > 0$ and $\|P_a * \mu\|_2 \le \varepsilon$, with ε/M small enough, depending on a and s. Then*

$$\|\mu\|_2 \;\le\; C_s\, M\Big(\frac{2a}{\log(M/\varepsilon)}\Big)^s,$$

and no bound of smaller order holds uniformly over the class: the modulus of continuity of the inverse is of this order.

Proof. Split the frequencies at Ω. Below, $|\widehat{\mu}| \le e^{a\Omega}|\widehat{P_a * \mu}|$, contributing at most $e^{a\Omega}\varepsilon$ to the norm; above, the H^s bound gives at most $M\Omega^{-s}$. With $\Omega = (2a)^{-1}\log(M/\varepsilon)$ the first term is $(M\varepsilon)^{1/2}$, of smaller order, and the second is the bound. For the lower bound, a source with transform concentrated near the frequency $\Omega = a^{-1}\log(M/\varepsilon)$ and H^s norm M has L^2 norm of order $M\Omega^{-s}$ and data of size at most ε (of size $o(\varepsilon)$ in fact, $\varepsilon\Omega^{-s}$). □

The cutoff at $2a$ is a convenient one. The sharp modulus $\sup\{\|\mu\|_2 : \|\mu\|_{H^s} \le M,\ \|P_a * \mu\|_2 \le \varepsilon\}$ equals $(1 + o(1))\, M\big(a/\log(M/\varepsilon)\big)^s$ as $\varepsilon/M \to 0$: in the upper bound take $\Omega = a^{-1}\big(L - (s+1)\log L\big)$ with $L = \log(M/\varepsilon)$, which makes the low frequency term ML^{-s-1} and the high frequency term $(1+o(1))M(a/L)^s$; the lower bound is the one given.

At the boundary the inverse loses all but a logarithm. Together the two theorems describe the backward Cauchy flow of the Thorin programme: heuristically, Hölder from level 1 to any level above $\frac{1}{2}$ for the zeros in a window, with the exponent $2\theta' - 1$ of Proposition 7.3.2, and logarithmic at $\frac{1}{2}$ for a source in a Sobolev ball (Theorem 7.3.4), which the zero measure is not.

7.4 Statistics

7.4.1 Supersmooth noise

In density deconvolution one observes $Y_i = X_i + E_i$, $i \le n$, with E_i of known law, and estimates the density f of X_i; the difficulty is set by the decay of the characteristic function ϕ_E of the noise. Cauchy noise has $\phi_E(\omega) = e^{-a|\omega|}$ and is supersmooth of order one in the classification of Fan [73], who showed that for supersmooth noise the minimax rate is logarithmic in n (cited). Carroll and Hall [35] found this for Gaussian noise before Fan. Stefanski and Carroll [173] introduced the deconvoluting kernel estimators, which attain these rates [35, 73]; the estimator has a Fourier cutoff at a frequency of order $\log n$. Heuristically, the logarithm is the statistical form of Theorem 7.3.4 with $\varepsilon = n^{-1/2}$.

The rates improve when the target is smooth in the strong sense. Butucea and Tsybakov [21, 22] constructed sharp asymptotic minimax estimators when the target is itself supersmooth, with rates much faster than the logarithmic ones

(cited). Cauchy noise with a target analytic in a strip is simpler, a bound of the form of Proposition 7.3.2, stated here without attribution.

PROPOSITION 7.4.1 (a strip). *Let E_i be Cauchy noise of width a and let f satisfy $\int e^{2w|\omega|}|\widehat{f}(\omega)|^2 d\omega \le 2\pi M^2$ with $w > 0$. The spectral cutoff estimator $\widehat{f_n}(x) = (2\pi)^{-1}\int_{|\omega|\le\Omega} e^{a|\omega|}\widehat{\phi}_n(\omega)e^{i\omega x}d\omega$, where $\widehat{\phi}_n(\omega) = n^{-1}\sum_{j\le n} e^{-i\omega Y_j}$ is the empirical transform of $Y_1,\dots,Y_n$ in the convention of Section 7.1, with $\Omega = \log n/(2(a+w))$, has root mean integrated squared error $O(n^{-w/(2(a+w))})$. This is an upper bound.*

Proof. The transform of the law of Y is $\widehat{f}(\omega)e^{-a|\omega|}$ and $\mathbb{E}|\widehat{\phi}_n(\omega) - \widehat{f}(\omega)e^{-a|\omega|}|^2 \le 1/n$. By Plancherel, the integrated variance is at most $(2\pi n)^{-1}\int_{|\omega|\le\Omega} e^{2a|\omega|}d\omega \le e^{2a\Omega}/(2\pi an)$, and the integrated squared bias is $(2\pi)^{-1}\int_{|\omega|>\Omega}|\widehat{f}|^2 \le M^2 e^{-2w\Omega}$. With $e^{2(a+w)\Omega} = n$ both are of order $n^{-w/(a+w)}$. □

A source of Lorentzians of width w (the test source of Table 7.1) is analytic in every strip of width less than w, and the exponent $(b+w)/(a+w)$ of Proposition 7.3.2 is positive even at $b = 0$. Meister's book [117] collects the theory.

7.4.2 Bayesian posteriors

For linear inverse problems with Gaussian priors, Knapik, van der Vaart and van Zanten [108] computed the posterior contraction rates in terms of the decay of the singular values, for polynomially decaying singular values only, the mildly ill posed case (cited). Cauchy deconvolution is severely ill posed: on a periodic domain its singular values are $e^{-a|k|}$ (convolution with P_a on $L^2(\mathbb{R})$ is not compact, being bounded below on the functions with spectrum in $[-1,1]$, so a periodic or windowed setting is needed). The severely ill posed case was treated by the same authors for the recovery of the initial condition of the heat equation [109], the Gaussian analogue of the backward Cauchy flow. For mixtures, where the unknown is a mixing measure seen through a known kernel, Nguyen [125] gave rates in Wasserstein distance for mixing measures with compact support: polynomial for ordinary smooth kernels and logarithmic for supersmooth ones, of order $(\log n)^{-1/\beta}$ for a kernel of supersmooth order β, $(\log n)^{-1/2}$ for the Gaussian kernel; for the Cauchy kernel, $\beta = 1$, the hypotheses of the paper are not checked here. Gao and van der Vaart [79] treated Dirichlet process mixtures of Laplace kernels (the noise is Laplace, not Cauchy). No posterior theorem for Cauchy deconvolution is cited. By analogy the Bayesian picture is the deterministic one with ε the noise level; this is a heuristic, since prior mass, testing and support conditions enter.

7.4.3 Positivity in the algorithms

Spectral cutoff, Tikhonov regularisation and Landweber iteration are linear and do not: they return signed estimates even when a positive source exists. The positivity preserving methods are nonlinear: nonnegative least squares (Section 10.8.1),

maximum entropy, and the EM algorithm of Richardson and Lucy [164, 115], studied for positive linear inverse problems by Vardi and Lee [200], whose multiplicative update keeps every iterate positive from a positive start (a zero coordinate stays zero) and on a finite grid converges to the Kullback Leibler projection of the data onto the cone of blurred positive measures. Chapters 10 and 11 use EM, and its accelerated form in the logarithmic domain (Section 11.4), for this reason. When no positive source exists, which by Proposition 7.2.2 is the case beyond the critical width, these methods still return positive estimates, each its own projection (least squares for nonnegative least squares, Kullback Leibler for EM), and the residual, compared with the data error, is what detects the failure (Proposition 10.4.1): positivity of the method is not evidence of positivity of the source.

7.5 Atomic sources

7.5.1 From logarithmic to polynomial

The rates above are for smoothness classes. For a finite sum of atoms, $\mu = \sum_k c_k \delta_{t_k}$, the problem is different in kind. The data are a sum of Lorentzians, $\sum_k c_k P_a(x - t_k)$, whose transform is the sum of exponentials $\sum_k c_k e^{-a|\omega| - i\omega t_k}$, and after a taper, as in Theorem 12.2.1, the samples on an arithmetic progression of frequencies are a sum of damped complex exponentials. Recovering their poles is Prony's problem, solved stably by Hankel matrix methods: the matrix pencil of Hua and Sarkar [101], and the Hankel singular value decomposition used in nuclear magnetic resonance spectroscopy since Barkhuijsen, de Beer and van Ormondt [11], where Lorentzian line shapes and their damped exponential transforms are the daily object. For a fixed number of separated atoms with amplitudes bounded below and no aliasing, the error is Lipschitz in the data error, with a constant that grows polynomially as atoms approach each other, except near merging atoms, where it is the square root; this is super resolution, studied by Donoho [67], Candès and Fernandez Granda [33], and, for positive measures without a separation condition, Schiebinger, Robeva and Recht [168], with sharp constants for close atoms by Li and Liao [113] (cited). Chapter 12 compares EM on a grid with the pencil on the zeros of ζ.

7.5.2 Merging atoms

The polynomial rate has one exception that matters for ζ. If two atoms approach each other symmetrically, the data depend on their separation only through its square, the Fisher information for the separation vanishes, and the rate for the separation is the square root of the data error; this is the case of two atoms merging symmetrically (with more merging atoms the exponent can be smaller). It is the singular rate of finite mixtures (Chen [39]; Nguyen [125]) in its simplest case, and Chapter 12 met it in the form forced by the functional equation: a zero off the line

comes with its partner at the same ordinate, their widths at level θ are $\theta - \frac{1}{2} \mp \delta$, the two poles merge as $\delta \to 0$, and a data error τ determines δ only to order $\tau^{1/2}$ (Lemma 12.5.1 and Theorem 12.5.2).

7.5.3 Analytic continuation in physics

A related problem is solved daily in many body physics, where a spectral function $A(\omega) \geq 0$ is to be recovered from its Cauchy transform $G(z) = \int A(\omega)(z-\omega)^{-1} d\omega$ known at points away from the real axis, the Matsubara frequencies. Its imaginary part at height y is $-\pi(P_y * A)$, but the observation operator differs from the one above: the data sample the Poisson integral at $x = 0$ and several heights, not along a line, so uniqueness and stability do not transfer (finite data are not unique). It is severely ill posed, unconstrained methods produce negative spectral weight, and the standard tools are maximum entropy (Jarrell and Gubernatis [102]) and, more recently, Nevanlinna interpolation (Fei, Yeh and Gull [74]), which builds positivity in by parametrising the Pick functions of Theorem 7.2.3 directly. Section 11.11.11 gives the identity that links the two: the Thorin density is $u_\theta(v) = \pi^{-1} \operatorname{Tr}\big[a((H-v)^2 + a^2)^{-1}\big]$ with $a = \theta - \frac{1}{2}$, for a hypothetical self adjoint H whose spectrum is the set of ordinates (so under RH); the resolvent of H itself is not trace class.

7.6 What is known and what is not

Known: uniqueness; the failure of positivity under the inverse and its characterisation by positive harmonic extension; Hölder stability between levels with the three lines exponent and logarithmic stability at the boundary, both sharp; logarithmic minimax rates for smooth sources and polynomial upper rates for sources analytic in a strip; Lipschitz error for separated atomic sources, with the square root rate for merging atoms; and a range of positivity preserving algorithms whose limits are projections on a finite grid.

Two features are specific to the book's use of the operator. The first is the systematic use of the whole family of partial deconvolutions, the Thorin family $\mathfrak{U}_\theta = P_{\theta-\theta'} * \mathfrak{U}_{\theta'}$, with positivity at each level, and its critical width $c^*(u_1) = 1 - \sigma_{\max}$ (Theorem 8.2.2(a)). The second is the identification of the full deconvolution of the level one density with Weil's explicit formula (Theorem 9.1.1 and Chapter 12), which is classical in substance.

Problem 7.1. Estimating the critical width. Given noisy data u + noise with $u \geq 0$ and $\int u(x)(1+x^2)^{-1} dx < \infty$, estimate $c^*(u)$ of Proposition 7.2.2. For atomic sources with positive weights it is the smallest width among the poles, estimable by the pencil on a class with bounded order, weights bounded below, separation and no aliasing; without weights bounded below there is no uniform rate, since c^* is discontinuous ($(1-\delta)P_A + \delta P_r$ has $c^* = r < A$ for every $\delta > 0$).

For general sources it is a boundary of a convex cone in an infinite dimensional space. Status: open.

Problem 7.2. Positivity and rates. Positivity does not improve the logarithmic minimax rate over smoothness classes of densities in the worst case (the lower bounds of Fan [73] are over such classes), and sparsity improves it to a polynomial one. Imposing a prescribed number of sign changes of a real Fourier transform does not help: two positive densities whose transforms are $e^{-|\omega|}$ and that function plus two nonnegative bumps near $\pm\Omega$ both have no sign change, and keep the logarithmic obstruction. Positive sources of the form $P_w * \nu$ with $\nu \geq 0$ of mass at most B have the polynomial exponent $w/(a+w)$ of Proposition 7.3.2 at $b = 0$, but that comes from the smoothing and the mass bound, not from positivity, and the zero counting measure has infinite mass. Open: is the Thorin family in a class on which positivity alone gives a polynomial rate?

Problem 7.3. Three lines with positivity on a window. Theorem 7.3.1 needs an a priori bound at a level below the target, and Corollary 7.3.3 removes it for data in L^2 on the whole line. For data known only on a window, the mass in a window of a positive source is controlled by the data alone, as in the a posteriori bound of Proposition 11.11.5. Does positivity give a Hölder bound between levels from data on a window, without any a priori bound below, with an exponent determined by the data? Status: open.

Notes

The chapter develops the survey chapter on Cauchy deconvolution by Nick Polson, with the corrections and additions of the book's reviews. The bibliography entries that are new for this chapter have been checked against CrossRef or OpenAlex, except [93], and the statements about the contents of four of the papers (Knapik et al. 2011 and 2013, Butucea and Tsybakov, Fan, Gao and van der Vaart) are those of a web check made for this chapter and about the others those of the reviews, not a reading of the papers here (in particular the supersmooth order β of Nguyen [125], and the form of the rates of [109], which is not stated).

Chapter 8

Three families of generalised gamma convolutions and the deconvolution obstruction

Three one parameter families of generalised gamma convolutions (GGCs) contain the clock of Riemann's ξ as a boundary case: the Esscher tilt, the basepoint family, which smooths the zero counting measure by Cauchy kernels, and the de Bruijn Newman heat flow; in each the Thorin property propagates away from the centre, and RH asks for it at the centre.

The word obstruction in the title names what the theorems on the three families have in common, not a theorem of its own. The chapter's own numbers are computed in floating point at the precision stated, and a sentence that reads a theorem as saying what a route needs is a heuristic and is marked as one.

8.1 Three families, and the tilt

Let ξ be Riemann's function, with zeros $\rho = \beta + i\gamma$, and $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$. The centre clock is the variable $T \geq 0$ with

$$\mathbb{E}\big[e^{-uT}\big] = \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})}, \qquad u \geq 0,$$

which exists with no hypothesis (Theorem 8.1.1) and is a GGC exactly under RH, with Thorin measure $\sum_{\gamma>0} \delta_{\gamma^2}$ (Proposition 2.1.1). Heights are written h in this chapter. A basepoint $\alpha = \frac{1}{2} + \varepsilon$ is also called the level ε, the resolution of Section 2.1. Put

$$\sigma_{\max} = \sup_{\rho} \operatorname{Re}\rho \in [\tfrac{1}{2}, 1],$$

so that RH is $\sigma_{\max} = \frac{1}{2}$. Since $\beta < 1$, every level $\varepsilon \geq \frac{1}{2}$ lies strictly to the right of every zero, $\varepsilon > \beta - \frac{1}{2}$, whether or not $\sigma_{\max} < 1$; the latter is the quasi Riemann hypothesis, and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [227] (Section 25.1.2), not yet refereed, would give $\sigma_{\max} \leq \frac{7}{8}$.

The clock is the boundary case $p = 0$ of three one parameter families:

(A) the *tilt* $u_0 \geq 0$, the Esscher transform $\xi(\frac{1}{2}+\sqrt{u_0})/\xi(\frac{1}{2}+\sqrt{u_0+u})$ (Theorem 12.2.2, Corollary 12.2.3 and Proposition 23.6.2 at $u_0 = \frac{1}{4}$);

(B) the *basepoint* $\alpha = \frac{1}{2}+\varepsilon$, the shifted clock $\xi(\alpha)/\xi(\alpha+\sqrt{u})$, whose Thorin density is a sum of Cauchy kernels over the zeros (Theorem 2.1.2 and Proposition 2.1.3);

(C) the *heat time* $t \geq 0$, the clock of the de Bruijn Newman function H_t (Section 4.1.1 and Theorem 17.1.16).

This chapter adds the converse half of the Poisson semigroup and the deconvolution form of RH (Theorems 8.2.1(b) and 8.2.2), the measured exponents of Table 8.1 for the smoothed prime sums, and, in Section 8.3, the visibility curve at every level with its closed forms in three lattice models and its level count.

PROPOSITION 8.1.1 (Invertible families do not localise the difficulty). *Let $\{S_p\}_{p\geq 0}$ be maps on a set $\mathcal{D}$ of Laplace transforms, with S_0 the identity. If each S_p is injective on $\mathcal{D}$, maps GGC Laplace transforms in $\mathcal{D}$ to GGC Laplace transforms, and has an inverse on $S_p(\mathcal{D})$ that does the same, then for every $\varphi \in \mathcal{D}$ and every p, $S_p\varphi$ is a GGC Laplace transform exactly when φ is.*

Proof. If φ is a GGC transform so is $S_p\varphi$, and if $S_p\varphi$ is one so is $S_p^{-1}S_p\varphi = \varphi$. □

In such a family the Thorin property is equally hard at every parameter: knowing it at some $p > 0$ would prove RH. The tilt is such a family, with no unconditional point known. The basepoint and heat families are not of this kind: the property is known at $\varepsilon \geq \frac{1}{2}$ and at $t \geq 0.2$, and their inverses do not preserve it. The two fail in opposite ways. The inverse of the Cauchy smoothing is an unbounded operator, with Fourier multiplier $e^{a|x|}$, so the step towards RH is an ill posed computation. The inverse of the heat flow is bounded, a Gaussian average, so the step towards RH is a stable computation; but by Corollary 8.4.2 a property that Gaussian averaging preserves and that implies real zeros holds for no H_t.

8.1.1 The tilt: invertible and uninformative

Write $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2})$ as in Section 2.1, an entire function of u, and for $u_0 \geq 0$

$$\widetilde{\varphi}_{u_0}(u) = \frac{G(u_0)}{G(u_0+u)}, \qquad u \geq 0,$$

the Laplace transform of the clock tilted by $e^{-u_0 T}$, that is of the variable $\widetilde{T}_{u_0}$ with law $e^{-u_0 t}P(T \in dt)/\mathbb{E}e^{-u_0 T}$ (the $\widetilde{T}$ of Theorem 12.2.2 for $u_0 = \frac{1}{4}$). Each pair $\{\rho, 1-\rho\}$ of zeros gives the number $a_\rho = -(\rho-\frac{1}{2})^2 = \gamma_\rho^2 = \gamma^2-(\beta-\frac{1}{2})^2-2i\gamma(\beta-\frac{1}{2})$, the a_k of (2.1.1), where $\gamma_\rho = -i(\rho-\frac{1}{2}) = \gamma-i(\beta-\frac{1}{2})$ is the complex ordinate of ρ,

and Hadamard's product (2.1.1) gives $G(u) = \prod(1+u/a_\rho)$, absolutely convergent since $\sum |a_\rho|^{-1} < \infty$. On the line $a_\rho = \gamma^2$.

THEOREM 8.1.2 (The tilt family). *(a) For $u_1 \geq 0$,*

$$\widetilde{\varphi}_{u_0+u_1}(u) = \frac{\widetilde{\varphi}_{u_0}(u_1+u)}{\widetilde{\varphi}_{u_0}(u_1)},$$

so the forward tilt is the Esscher transform by $e^{-u_1\widetilde{T}_{u_0}}$, and it maps GGCs to GGCs, moving the Thorin measure up by u_1. Backward, if $\widetilde{\varphi}_{u_0+u_1}$ is a GGC transform with Thorin measure in (u_1,∞), then

$$\widetilde{\varphi}_{u_0}(u) = \frac{\widetilde{\varphi}_{u_0+u_1}(u-u_1)}{\widetilde{\varphi}_{u_0+u_1}(-u_1)}$$

is a GGC transform, with the Thorin measure moved down by u_1; the denominator is $G(u_0+u_1)/G(u_0)$, finite and positive.

(b) $\widetilde{\varphi}_{u_0}$ is the Laplace transform of a GGC exactly when every u_0+a_ρ is real and positive. In particular $\operatorname{Im}(u_0+a_\rho) = -2\gamma(\beta-\frac{1}{2})$ does not depend on u_0.

(c) For ξ, the Thorin property at one tilt is equivalent to the property at every tilt and to RH.

Proof. (a) The identity is the definition. If $\varphi(u) = \exp(-du - \int \log(1+u/z)\,U(dz))$, then $\varphi(u_1+u)/\varphi(u_1) = \exp(-du - \int \log(1+u/(z+u_1))\,U(dz))$, a GGC with the same drift and the Thorin measure moved up by u_1. Backward, if U lies in (u_1,∞) and $\varphi(-u_1)$ is finite, as it is here, then $\varphi(u-u_1)/\varphi(-u_1) = \exp(-du - \int \log(1+u/(z-u_1))\,U(dz))$, a GGC with the measure moved down. (b) If every u_0+a_ρ is real and positive, $\widetilde{\varphi}_{u_0}(u) = \prod(u_0+a_\rho)/(u_0+a_\rho+u)$ is the transform of $\sum_\rho e_\rho/(u_0+a_\rho)$ with independent unit exponentials e_ρ, a GGC. Conversely a GGC transform extends to an analytic zero free function on $\mathbb{C}\setminus(-\infty,0]$, while $\widetilde{\varphi}_{u_0}$ has poles at $u = -(u_0+a_\rho)$, so each of these lies in $(-\infty,0)$. (c) The zeta function has no real zeros in $(0,1)$, so $\gamma \neq 0$ for every zero, and $\xi(\frac{1}{2}) \neq 0$. By (b) the property at any u_0 says that every a_ρ is real, that is $\beta = \frac{1}{2}$; conversely, under RH every $a_\rho = \gamma^2 > 0$, and (b) gives the property at every $u_0 \geq 0$. □

Theorem 8.1.2 is the case $\mathbb{Q}$ of Theorem 16.2.3, which states the equivalence for the Dedekind zeta function of every number field and every $u_0 \geq 0$.

In the chart $\theta = \sqrt{z}$ of the Thorin measure the tilt is the deformation $\theta \mapsto \sqrt{\theta^2+u_0}$ of the atoms, not a smoothing. What the tilt can move is real zeros: for the Dedekind zeta function ζ_K of a quadratic field K with $\zeta_K(\frac{1}{2}) \neq 0$, a real zero $\beta \in (\frac{1}{2},1)$ of ζ_K, which is a real zero of $L(s,\chi)$ for the quadratic character χ of K, gives a real $a_\rho = -(\beta-\frac{1}{2})^2 < 0$, admissible exactly when $u_0 > (\beta-\frac{1}{2})^2$.

8.2 The basepoint family: a Poisson semigroup to the right of the zeros

For real ε and θ such that $\frac{1}{2}+\varepsilon+i\theta$ is not a zero of ξ, let

$$\rho_\varepsilon(\theta) = \frac{1}{\pi}\,\mathrm{Re}\,\frac{\xi'}{\xi}\big(\tfrac{1}{2}+\varepsilon+i\theta\big)$$

be the density of (2.1.7), now for every real ε, and let $P_a(x) = a/(\pi(a^2+x^2))$ for real $a \neq 0$ be the Poisson kernel of Section 2.1, with $P_a * P_b = P_{a+b}$ for $a, b > 0$. For $\varepsilon > 0$ the shifted clock at the basepoint $\frac{1}{2}+\varepsilon$ has Thorin density ρ_ε in the chart $\theta = \sqrt{z}$ whenever it is a GGC and no zero lies on the line $\mathrm{Re}\,s = \frac{1}{2}+\varepsilon$ (Theorem 2.1.2). The function $(\theta,\varepsilon) \mapsto \rho_\varepsilon(\theta)$ is harmonic off the zeros, odd in ε by the functional equation, and Hadamard's product gives the absolutely convergent expansion

$$\rho_\varepsilon(\theta) = \sum_\rho P_{\varepsilon-(\beta-\frac{1}{2})}(\theta-\gamma), \qquad \varepsilon \neq \beta-\tfrac{1}{2}, \tag{8.2.1}$$

over all zeros with multiplicity (Proposition 2.1.3). A zero with $\beta-\frac{1}{2} < \varepsilon$ contributes a positive Cauchy density, one with $\beta-\frac{1}{2} > \varepsilon$ a negative one.

Theorem 8.2.1 (Where the Poisson semigroup holds). *(a) If $\varepsilon_0 > \beta-\frac{1}{2}$ for every zero, then $\rho_{\varepsilon_0+a} = P_a * \rho_{\varepsilon_0}$ for every $a > 0$, and $\rho_\varepsilon > 0$ for every $\varepsilon \geq \varepsilon_0$. This holds for every $\varepsilon_0 \geq \frac{1}{2}$, with no hypothesis.*

(b) If $\varepsilon_0 \geq 0$ lies at positive distance from the set $\{\beta-\frac{1}{2}\}$ and some zero has $\beta-\frac{1}{2} > \varepsilon_0$, then $\rho_{\varepsilon_0+a} \neq P_a * \rho_{\varepsilon_0}$ for some $a > 0$.

Proof. (a) Every term of (8.2.1) at ε_0 is a Cauchy density $P_c(\theta-\gamma)$ with $c = \varepsilon_0-(\beta-\frac{1}{2}) > 0$, and $P_a * P_c = P_{a+c}$. The terms are positive, so the convolution passes through the sum by monotone convergence. For $\varepsilon_0 \geq \frac{1}{2}$, $c \geq 1-\beta > 0$. (b) Now $|c| \geq c_0 > 0$ for every term, and $|c| < 1$ because $0 \leq \varepsilon_0 < \frac{1}{2}$, so $|P_c(\theta-\gamma)| \leq \min\big((\pi c_0)^{-1}, (\pi(\theta-\gamma)^2)^{-1}\big)$, and by the Riemann von Mangoldt formula the sum is $O(\log(2+|\theta|))$. Hence $w(\theta,a) = (P_a * \rho_{\varepsilon_0})(\theta)$ is defined and harmonic in $a > 0$. If $w(\cdot,a) = \rho_{\varepsilon_0+a}$ for every $a > 0$, then off the zeros $\rho_\varepsilon(\theta)$ would agree with a function harmonic on the whole half plane $\{\varepsilon > \varepsilon_0\}$, and in particular bounded near every point $(\gamma, \beta-\frac{1}{2})$ with $\beta-\frac{1}{2} > \varepsilon_0$. But near a zero, $\xi'/\xi(s) = m_\rho/(s-\rho)+O(1)$, so $\rho_\varepsilon(\theta)$ is unbounded there. □

Part (a) is the semigroup of Remark 6.3.4, and (b) is its converse (a rightmost zero need not exist).

Theorem 8.2.2 (Deconvolution form of RH). *Let $\varepsilon_0 > \beta-\frac{1}{2}$ for every zero (for instance $\varepsilon_0 \geq \frac{1}{2}$), and let $0 \leq \varepsilon_1 < \varepsilon_0$.*

(a) *There is a positive measure ν on $\mathbb{R}$ with $\int(1+x^2)^{-1}\nu(dx)<\infty$ and $\rho_{\varepsilon_0}=P_{\varepsilon_0-\varepsilon_1}*\nu$ if and only if $\beta-\frac12\le\varepsilon_1$ for every zero. When it exists, ν is unique: it is $\rho_{\varepsilon_1}\,d\theta$ if $\beta-\frac12<\varepsilon_1$ for every zero, and otherwise the vague limit of $\rho_\varepsilon\,d\theta$ as $\varepsilon\downarrow\varepsilon_1$, which carries point masses $m_\rho\delta_\gamma$ at the zeros with $\beta-\frac12=\varepsilon_1$.*

(b) *In particular RH holds if and only if $\rho_{\varepsilon_0}=P_{\varepsilon_0}*\mu$ for a positive measure μ (with $\int(1+x^2)^{-1}\mu(dx)<\infty$, which is automatic since $P_{\varepsilon_0}*\mu=\rho_{\varepsilon_0}$ is finite), and then $\mu=\sum_\gamma\delta_\gamma$, the sum over the ordinates of all zeros, of both signs, with multiplicity.*

Proof. If every zero has $\beta-\frac12<\varepsilon_1$, Theorem 8.2.1(a) gives $\nu=\rho_{\varepsilon_1}\,d\theta$. If every zero has $\beta-\frac12\le\varepsilon_1$ with equality for some, write $\rho_{\varepsilon_0}=P_{\varepsilon_0-\varepsilon}*\rho_\varepsilon$ for $\varepsilon_1<\varepsilon<\varepsilon_0$ and let $\varepsilon\downarrow\varepsilon_1$: by (8.2.1) the terms of the zeros with $\beta-\frac12<\varepsilon_1$ converge to Cauchy densities and those with $\beta-\frac12=\varepsilon_1$ to point masses at their ordinates, and the sum over the zeros passes to the limit by the domination of the discussion after Remark 2.1.4, so $\rho_\varepsilon\,d\theta$ converges vaguely to a positive measure ν, the sum of these limits. Then $P_{\varepsilon_0-\varepsilon_1}*\nu=\rho_{\varepsilon_0}$ term by term, because $P_{\varepsilon_0-\varepsilon_1}*P_{\varepsilon_1-(\beta-\frac12)}=P_{\varepsilon_0-(\beta-\frac12)}$ and $P_{\varepsilon_0-\varepsilon_1}*\delta_\gamma=P_{\varepsilon_0-(\beta-\frac12)}(\cdot-\gamma)$ when $\beta-\frac12=\varepsilon_1$, all terms being positive; in particular $\int(1+x^2)^{-1}\nu(dx)<\infty$. Conversely, suppose ν exists and some zero has $\beta-\frac12>\varepsilon_1$. The Poisson integral $u(\theta,\varepsilon)=(P_{\varepsilon-\varepsilon_1}*\nu)(\theta)$ is harmonic on $\{\varepsilon>\varepsilon_1\}$, and for $\varepsilon>\varepsilon_0$ it equals $P_{\varepsilon-\varepsilon_0}*\rho_{\varepsilon_0}=\rho_\varepsilon(\theta)$ by Theorem 8.2.1(a). On the connected set $\{\varepsilon>\varepsilon_1\}$ minus the zeros, $u-\rho_\varepsilon$ is harmonic and vanishes for $\varepsilon>\varepsilon_0$, so $u=\rho_\varepsilon$ there. Then $\rho_\varepsilon(\theta)$ is bounded near a zero with $\beta-\frac12>\varepsilon_1$, which contradicts the pole of ξ'/ξ. For uniqueness, let ν and ν' both qualify and put $\sigma=\nu-\nu'$, a signed measure with $\int(1+x^2)^{-1}|\sigma|(dx)<\infty$. For $\varepsilon>\varepsilon_0$ the Poisson integrals of ν and ν' at height $\varepsilon-\varepsilon_1$ both equal $P_{\varepsilon-\varepsilon_0}*\rho_{\varepsilon_0}$, so the harmonic function $(P_{\varepsilon-\varepsilon_1}*\sigma)(\theta)$ on $\{\varepsilon>\varepsilon_1\}$ vanishes for $\varepsilon>\varepsilon_0$, hence everywhere by real analyticity, and letting $\varepsilon\downarrow\varepsilon_1$ gives $\sigma=0$, since such a Poisson integral converges vaguely to σ. Part (b) is the case $\varepsilon_1=0$: $\beta-\frac12\le0$ for every zero means $\beta=\frac12$ for every zero by the symmetry $\rho\mapsto1-\rho$, and (8.2.1) at $\beta=\frac12$ gives μ. □

The *if* direction of (b), and the atoms at a boundary line in (a), are Theorem 2.1.2, Proposition 2.1.3 and Theorem 6.2.3; what this chapter adds is the *only if* direction and the uniqueness. Theorem 8.2.2 makes the obstruction of Section 14.3.7 exact: $\sigma_{\max}-\frac12$ is the depth to which the Thorin density at a basepoint beyond the zeros can be deconvolved into a positive measure. Two remarks qualify it. First, positivity of ρ_{ε_1} is necessary for deconvolvability to ε_1 but not sufficient: below $\sigma_{\max}-\frac12$ the function ρ_{ε_1} may well be positive (Remark 2.1.4 and Section 8.3), and still $P_{\varepsilon_0-\varepsilon_1}*\rho_{\varepsilon_1}\neq\rho_{\varepsilon_0}$. Second, the deconvolution is unbounded. On the Fourier side $\widehat{P_a}(x)=e^{-a|x|}$, and in the explicit formula the frequency $x=\log n$ carries the weight $\Lambda(n)n^{-\alpha}$ at basepoint α; deconvolving by P_a multiplies it by n^a, so the prime sums converge absolutely only for $\alpha>1$,

and below the basepoint one they need cancellation, which is a zero free region (Proposition 12.1.13).

PROPOSITION 8.2.3 (The smoothed prime sums at a basepoint). *Let χ be a primitive nonprincipal character, $N > 0$, and $s = \alpha + ib$ such that $s - k$ is not a zero of $L(s, \chi)$ for any integer $k \geq 0$. Put*

$$E(N, s) = \sum_{n \geq 1} \chi(n)\Lambda(n)n^{-s}e^{-n/N} + \frac{L'}{L}(s, \chi),$$

the difference between the smoothed prime sum and $-L'/L(s, \chi)$. Then

$$\begin{aligned} E(N, s) = & -\sum_{\rho} m_\rho \, \Gamma(\rho - s)N^{\rho - s} \\ & + \sum_{k \geq 1} \frac{(-1)^k}{k!}\Big(-\frac{L'}{L}\Big)(s - k, \chi)N^{-k}, \end{aligned} \tag{8.2.2}$$

where ρ runs over all zeros of $L(s, \chi)$, trivial ones included, with multiplicity m_ρ. Hence, for fixed s, $E(N, s) = O(N^{\sigma_{\max}(\chi) - \alpha}) + O(N^{-1})$ as $N \to \infty$, where $\sigma_{\max}(\chi)$ is the supremum of β over the nontrivial zeros $\beta + i\gamma$ of $L(s, \chi)$. Under GRH the sum over the nontrivial zeros is $N^{1/2-\alpha}F_s(\log N)$, where $F_s(x) = -\sum m_\rho \Gamma(\rho - s)e^{i(\gamma - b)x}$, summed over the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$, is an almost periodic function none of whose coefficients vanishes; so $F_s(\log N)$ does not tend to zero, and for $\alpha < \frac{3}{2}$ the error $E(N, s)$ is not $o(N^{1/2-\alpha})$.

Proof. This is the explicit formula for the smoothed prime sum (compare [54, Chapter 19]), derived here for the order of the error. Insert $e^{-x} = \frac{1}{2\pi i}\int_{(c)} \Gamma(w)x^{-w}dw$ with $c > \max(0, 1 - \alpha)$ and sum: the smoothed prime sum is $\frac{1}{2\pi i}\int_{(c)}(-L'/L)(s + w, \chi)\Gamma(w)N^w \, dw$. Move the contour to $\operatorname{Re} w = -R$ with $R \to \infty$ along a sequence avoiding the poles; $\Gamma(w)$ decays like $e^{-\pi|\operatorname{Im} w|/2}$ and L'/L grows at most polynomially on suitable horizontal segments and vertical lines [54, Chapters 16 and 19], so the integrals vanish in the limit. The poles are simple by the hypothesis on s. The residues are $(-L'/L)(s, \chi)$ at $w = 0$, which is moved to the left side, $-m_\rho\Gamma(\rho - s)N^{\rho - s}$ at $w = \rho - s$, since $-L'/L$ has residue $-m_\rho$ at ρ, and $(-1)^k(k!)^{-1}(-L'/L)(s - k, \chi)N^{-k}$ at $w = -k$. The sum over ρ converges absolutely and is dominated by the zeros with $|\gamma - b|$ small, because $|\Gamma(\rho - s)|$ decays exponentially in $|\gamma - b|$; the nontrivial zeros contribute $O(N^{\sigma_{\max}(\chi) - \alpha})$, the trivial zeros $O(N^{-\alpha})$, and $O(N^{-1-\alpha})$ for odd χ, and the last sum is $O(N^{-1})$. Under GRH each nontrivial term is $|\Gamma(\rho - s)|N^{1/2-\alpha}$ times a phase, Γ has no zeros, and the series defining F_s converges absolutely, so F_s is almost periodic in the sense of Bohr with the nonzero coefficients $-m_\rho\Gamma(\rho - s)$, and its mean square is the sum of the squares of their moduli, which is positive. □

So recovering the Thorin density at basepoint α to accuracy η from the primes with this smoothing needs primes up to about $N \approx \eta^{-1/(\alpha-\sigma_{\max}(\chi))}$, which is unbounded as $\alpha \downarrow \sigma_{\max}(\chi)$. This is a heuristic reading of the proposition as the deconvolution of Theorem 8.2.2 paid for in primes: the identity is for the weight $e^{-n/N}$ and does not bound every method that uses the primes up to N. Proposition 12.1.13 gives the same exponent for the compensated prime sums of ξ (the book writes Θ for $\sigma_{\max}$ there), and Section 15.2.2 treats $L(s, \chi_{12})$, where no pole needs compensating. Table 8.1 shows the exponent for χ_4, for which GRH is verified to height 2.5×10^7 [231]: the measured exponent matches $\frac{1}{2} - \alpha$ to within 0.006 for $0.6 \le \alpha \le 1.2$, while at $\alpha = 1.5$ the $O(N^{-1})$ term is of the same size as the zero sum, between 1.1 and 2.3 times it.

α	$N = 10^3$	10^4	10^5	2.5×10^5	slope	$\frac{1}{2} - \alpha$
1.5	3.83×10^{-3}	3.97×10^{-4}	4.72×10^{-5}	2.00×10^{-5}	-0.949	-1.00
1.2	1.56×10^{-2}	3.03×10^{-3}	6.24×10^{-4}	3.36×10^{-4}	-0.695	-0.70
1.0	5.86×10^{-2}	1.83×10^{-2}	5.73×10^{-3}	3.78×10^{-3}	-0.499	-0.50
0.9	1.21×10^{-1}	4.78×10^{-2}	1.87×10^{-2}	1.35×10^{-2}	-0.400	-0.40
0.8	2.60×10^{-1}	1.29×10^{-1}	6.36×10^{-2}	4.95×10^{-2}	-0.302	-0.30
0.7	5.66×10^{-1}	3.57×10^{-1}	2.22×10^{-1}	1.86×10^{-1}	-0.203	-0.20
0.6	1.23×10^{0}	9.78×10^{-1}	7.69×10^{-1}	6.93×10^{-1}	-0.103	-0.10

Table 8.1: Mean modulus of $E(N, s)$ of (8.2.2) for χ_4 over $s = \alpha + 10i$ and $s = \alpha + 25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10} N$ fitted by least squares on the four values of N. Four values of N and two values of b illustrate the GRH statement of the proposition and do not verify it.

The identity (8.2.2) itself was checked in the 56 cases of the table: $E(N, s)$ and the right side, built from the zeros of $L(s, \chi_4)$ of height below 62, the trivial zeros $-1, -3, \dots, -15$ and the terms $m \le 9$, agree to 5.4×10^{-14} in absolute value.

The book's route through basepoints, positivity at every height for $\alpha \ge 1$ (Section 6.1.2), stops at $\alpha = 1$, the edge of the known zero free half plane; the cost exponent and the zero free half plane are two faces of one fact (Proposition 12.1.13), not cause and effect.

8.3 What a single basepoint sees

Theorem 8.2.2, like Lagarias's criterion [185] (Remark 2.1.4), needs every level. A test at one level ε sees less; Section 1.4.2, with Table 1.3 and Exercise 8.14, sets out what it sees in a lattice model. Two facts frame it. A zero at distance $\delta = \beta - \frac{1}{2} > 0$ from the line contributes positively at every level $\varepsilon > \delta$, so those levels cannot see it. As $\varepsilon \uparrow \delta$ its contribution at $\theta = \gamma$ is $-1/(\pi(\delta - \varepsilon)) \to -\infty$, so the levels just below δ always see it. Near the line the test reduces to the sign of Laguerre's quantity $\mathcal{L}_\Xi = -(\log|\Xi|)''$.

A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ with $\delta > 0$ is added by hand to ξ'/ξ and $\mathcal{L}_\Xi$ computed from the actual zeros, as Section 14.3.4 inserts quadruples into the computed density; nothing below says that ξ has such zeros, and the tables, the figures and Corollary 8.3.3 are statements about a model. A quadruple puts two zeros in the upper half plane, placed among the actual zeros in one of three ways, which differ in the number of zeros up to a given height:

(i) the quadruple replaces one on line zero, at the ordinate γ^* of that zero, as in the lattice of Section 1.4.2; the count of zeros up to a given height above γ^* grows by one;

(ii) the quadruple is inserted at the midpoint γ^* of a gap between consecutive on line zeros; the count grows by two;

(iii) the quadruple replaces the two on line zeros that bound a gap and sits at the midpoint γ^* of the gap, as in the perturbation of Theorem 1.4.1; the count is unchanged.

Proposition 8.3.1 (The test near the line). *Let θ be real and not the ordinate of a zero on the line, and let $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$, defined by the same formula for every real ε. Then*

$$\rho_\varepsilon(\theta) = \frac{\varepsilon}{\pi}\, \mathcal{L}_\Xi(\theta) + O(\varepsilon^3) \quad (\varepsilon \to 0),$$

where

$$\mathcal{L}_\Xi(\theta) = -\frac{d^2}{d\theta^2} \log |\Xi(\theta)| = \sum_\rho \frac{1}{(\theta - \gamma_\rho)^2},$$

$\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ are the zeros of Ξ, and the sum is real, since the zeros of Ξ come in conjugate pairs. Under RH, $\mathcal{L}_\Xi(\theta) = \sum_\gamma (\theta - \gamma)^{-2} > 0$. A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ contributes

$$2 \operatorname{Re} \frac{1}{(\theta - \gamma^* + i\delta)^2} + 2 \operatorname{Re} \frac{1}{(\theta + \gamma^* + i\delta)^2},$$

which equals $-2/\delta^2 + O(\gamma^{*-2})$ at $\theta = \gamma^*$.

Proof. As a function of (θ, ε), $\rho_\varepsilon(\theta)$ is harmonic off the zeros and odd in ε by the functional equation, so near $(\theta, 0)$ it equals $\varepsilon\, \partial_\varepsilon \rho_\varepsilon(\theta)|_{\varepsilon=0} + O(\varepsilon^3)$. With $F = \xi'/\xi$ and $\Xi'(\theta) = iF(\frac{1}{2} + i\theta)\Xi(\theta)$, differentiating gives $F'(\frac{1}{2} + i\theta) = -(\Xi'/\Xi)'(\theta)$, which is real, and $\partial_\varepsilon \rho_\varepsilon = \pi^{-1} \operatorname{Re} F'$. Hadamard's product for Ξ, with the zeros $\pm\gamma_\rho$ paired, gives $\Xi'/\Xi = \sum (\theta - \gamma_\rho)^{-1}$, and differentiating once more gives the sum. For the quadruple the zeros of Ξ are $\gamma^* \pm i\delta$ and $-\gamma^* \pm i\delta$, and each conjugate pair contributes $2 \operatorname{Re}(\theta \mp \gamma^* + i\delta)^{-2}$; at $\theta = \gamma^*$ the first is $2 \operatorname{Re}(i\delta)^{-2} = -2/\delta^2$. □

At first order the test at γ^* sees the quadruple exactly when

$$\delta < \delta_c(\gamma^*) = \sqrt{2/\mathcal{L}_0(\gamma^*)},$$

with $\mathcal{L}_0(\gamma^*)$ the contribution of the other zeros to $\mathcal{L}_\Xi(\gamma^*)$; this is the local visibility condition of Section 1.4.2, written with Laguerre's quantity. In (i) $\mathcal{L}_0$ omits the replaced zero; with neighbours at the mean spacing $\Delta = 2\pi/\log(h/2\pi)$ this gives $\mathcal{L}_0 = \pi^2/(3\Delta^2)$ and $\delta_c = (\sqrt{6}/\pi)\Delta = 0.780\,\Delta$. In (ii), midway between two on line zeros, $\mathcal{L}_0 = \pi^2/\Delta^2$ and $\delta_c = (\sqrt{2}/\pi)\Delta = 0.450\,\Delta$. In (iii) the two zeros at $\gamma^* \pm \frac{1}{2}\Delta$ are removed from the lattice of (ii), $\mathcal{L}_0 = (\pi^2 - 8)/\Delta^2$ and $\delta_c = \sqrt{2/(\pi^2-8)}\,\Delta = 1.034\,\Delta$. A distance $\delta_c = c\Delta$ is below $\frac{1}{2}$ exactly when $h > 2\pi e^{4\pi c}$, that is above $2\pi e^{4\sqrt{6}} = 1.131 \times 10^5$ in (i), above $2\pi e^{4\sqrt{2}} = 1799$ in (ii) and above 2.8×10^6 in (iii). Table 8.2 replaces the lattice by the actual zeros.

		(i) replacing a zero			(ii) at a midpoint			(iii) count preserving		
height	mean gap	median	least	ratio	median	least	ratio	median	least	ratio
14	3.177	2.065	1.503	0.716	1.339	0.692	0.450	2.987	2.364	1.035
50	2.557	1.824	1.335	0.719	1.128	0.589	0.447	2.551	2.187	1.000
237	1.679	1.201	0.951	0.733	0.674	0.447	0.454	1.691	1.508	1.093
1419	1.178	0.799	0.655	0.711	0.469	0.297	0.452	1.167	0.975	1.062
9878	0.848	0.579	0.419	0.701	0.375	0.164	0.444	0.856	0.715	0.973
74921	0.661	0.446	0.235	0.662	0.296	0.088	0.437	0.651	0.527	0.902

Table 8.2: The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate γ_{n_0} of the n_0th zero. Replacing a zero: the 20 zeros γ_n, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of γ_n and γ_{n+1}, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros γ_n and γ_{n+1} are removed as well. Median and least are over the 20 values of δ_c, and the ratio is the median over the 20 points of δ_c divided by the local gap.

The ratio δ_c/gap is stable across heights in each configuration: about 0.45 at midpoints, as in the lattice; somewhat below the lattice value 0.78 when a zero is replaced; around the lattice value 1.034 when the count is preserved. The median of δ_c falls below $\frac{1}{2}$, the largest possible distance of a zero from the line, already by height 1419 at midpoints and by height 74921 when replacing a zero, below the lattice estimate 1.1×10^5 above. Above these heights an off line zero at distance between δ_c and $\frac{1}{2}$ passes the first order test at its own ordinate, unseen, wherever $\delta_c < \frac{1}{2}$, which is so at half or more of the sampled points.

Table 8.3 and Figure 8.1 show what happens beyond δ_c. We insert a hypothetical quadruple at the midpoint γ^* of a gap, configuration (ii), add its Cauchy

terms exactly to ρ_ε computed from ξ'/ξ, and record the levels $\varepsilon \in (0, \delta)$, on the grid $\varepsilon = \delta k/40$, $k = 1, \dots, 39$, with $0.999\,\delta$, at which ρ_ε is negative at some of 41 equally spaced points of the gap. At height 9878, where $\delta_c = 0.381$, the quadruple with $\delta = 0.45$ is seen only for $0.19125 \le \varepsilon < 0.45$; at every level of the grid below 0.19, the first order test included, ρ_ε stays positive at the 41 points of the gap. The level at which the sign changes at the ordinate is $\varepsilon = 0.1895$ (Proposition 8.3.2 below), and the minimum over the gap, which lies about 0.004 to the left of the midpoint, changes sign at $\varepsilon = 0.1893$; the grid level 0.19125 is the first above both.

γ^*	δ_c	$\delta = 0.05$	$\delta = 0.2$	$\delta = 0.3$	$\delta = 0.45$
100.07	1.109	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$
9878.22	0.381	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$	$0.19125 \le \varepsilon < 0.45$

Table 8.3: Levels ε at which ρ_ε is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)).

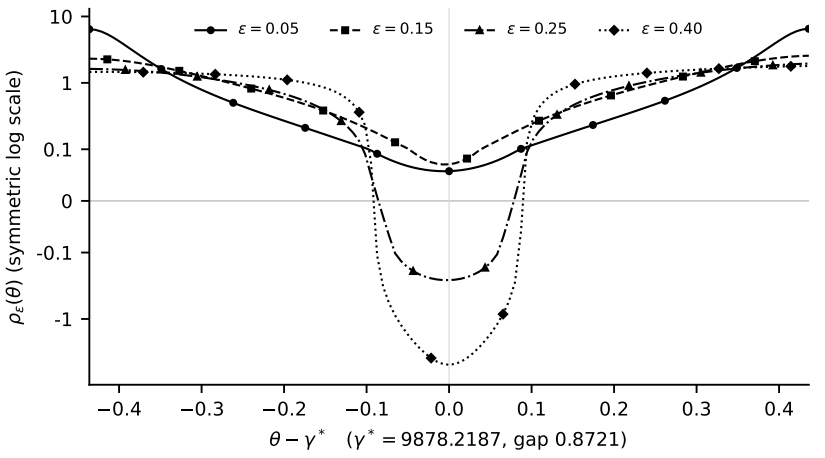


Figure 8.1: The Thorin density ρ_ε across the gap at height 9878 (gap 0.872) with a hypothetical quadruple at distance $\delta = 0.45$ inserted at its midpoint, at four levels $\varepsilon = 0.05$, 0.15, 0.25 and 0.4. It is positive at $\varepsilon = 0.05$ and $\varepsilon = 0.15$ and negative near the midpoint at $\varepsilon = 0.25$ and $\varepsilon = 0.4$. The vertical scale is symmetric logarithmic, linear within ± 0.1.

8.3.1 The visibility curve

Proposition 8.3.2 (The visibility curve). *Let ρ_ε^0 be the Thorin density of all zeros other than the two zeros $\frac{1}{2} \pm \delta + i\gamma^*$ of a quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$, so that the*

terms of the quadruple's images at $-\gamma^*$ are included in it (and $\delta_{\det}$ below depends on δ through them, by 10^{-9} in the computations of this section), and suppose $\rho_\varepsilon^0(\gamma^*) > 0$. For $0 < \varepsilon < \delta$, $\rho_\varepsilon(\gamma^*) < 0$ if and only if

$$\delta < \delta_{\det}(\varepsilon), \qquad \delta_{\det}(\varepsilon)^2 = \varepsilon^2 + \frac{2\varepsilon}{\pi\rho_\varepsilon^0(\gamma^*)}.$$

In the lattice models the other zeros are those of the lattice alone, and the images at $-\gamma^*$ and the distant zeros are neglected, as in Section 1.4.2. Let the other zeros lie on the line at the points $\gamma^* \pm (k+\frac12)\Delta$ with $k \ge 0$ in configuration (ii), $\gamma^* \pm k\Delta$ with $k \ge 1$ in (i), and $\gamma^* \pm (k+\frac12)\Delta$ with $k \ge 1$ in (iii). Then

$$\delta_{\det}(\varepsilon)^2 = \varepsilon^2 + \begin{cases} \dfrac{2\varepsilon\Delta}{\pi\tanh(\pi\varepsilon/\Delta)} & \text{in (ii),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\coth\frac{\pi\varepsilon}{\Delta} - \frac{1}{\varepsilon}} & \text{in (i),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\tanh\frac{\pi\varepsilon}{\Delta} - \frac{2\varepsilon}{\varepsilon^2+\Delta^2/4}} & \text{in (iii).} \end{cases}$$

As $\varepsilon \to 0$ these tend to $\delta_c^2 = 2\Delta^2/\pi^2$, $6\Delta^2/\pi^2$ and $2\Delta^2/(\pi^2-8)$, and for $\varepsilon \ge \Delta$ all three satisfy $\delta_{\det}(\varepsilon)^2 - \varepsilon^2 = 2\varepsilon\Delta/\pi + O(\Delta^2)$, that is $\delta_{\det}(\varepsilon) - \varepsilon = (\Delta/\pi)\big(1 + O(\Delta/\varepsilon)\big)$. In all three, $\delta_{\det}$ is increasing.

Proof. The two zeros at height γ^* contribute $\pi^{-1}\big((\varepsilon-\delta)^{-1} + (\varepsilon+\delta)^{-1}\big) = -2\varepsilon/(\pi(\delta^2-\varepsilon^2))$ at $\theta = \gamma^*$, so $\rho_\varepsilon(\gamma^*) < 0$ exactly when $\pi\rho_\varepsilon^0(\gamma^*) < 2\varepsilon/(\delta^2-\varepsilon^2)$, which rearranges to the stated inequality. A zero on the line at distance δ_j from γ^* contributes $\varepsilon/(\pi(\varepsilon^2+\delta_j^2))$ to $\rho_\varepsilon^0(\gamma^*)$ (Proposition 2.1.3), so in the lattice models $\pi\rho_\varepsilon^0(\gamma^*) = \varepsilon\sum_j(\varepsilon^2+\delta_j^2)^{-1}$, which is positive. The sum decreases with ε, and $\delta_{\det}(\varepsilon)^2 = \varepsilon^2 + 2\big/\sum_j(\varepsilon^2+\delta_j^2)^{-1}$ increases. For the lattices, $\sum_{k\in\mathbb{Z}}\varepsilon/(\varepsilon^2+(k+\frac12)^2\Delta^2) = (\pi/\Delta)\tanh(\pi\varepsilon/\Delta)$ and $\sum_{k\neq0}\varepsilon/(\varepsilon^2+k^2\Delta^2) = (\pi/\Delta)\coth(\pi\varepsilon/\Delta) - 1/\varepsilon$, by the partial fractions of tanh and coth, and removing the two zeros at $\gamma^* \pm \frac12\Delta$ subtracts $2\varepsilon/(\varepsilon^2+\Delta^2/4)$ from the first. The limits follow from $\tanh x = x + O(x^3)$ and $\coth x = 1/x + x/3 + O(x^3)$, which give $\pi\rho_\varepsilon^0(\gamma^*)/\varepsilon \to \pi^2/\Delta^2$, $\pi^2/(3\Delta^2)$ and $(\pi^2-8)/\Delta^2$. For $\varepsilon \ge \Delta$, $\tanh(\pi\varepsilon/\Delta)$ and $\coth(\pi\varepsilon/\Delta)$ equal 1 up to $O(e^{-2\pi\varepsilon/\Delta})$, so that $\pi\rho_\varepsilon^0(\gamma^*) = \pi/\Delta + O(1/\varepsilon)$ in all three, whence $\delta_{\det}^2 - \varepsilon^2 = 2\varepsilon\Delta/\pi + O(\Delta^2)$; dividing by $\delta_{\det} + \varepsilon = 2\varepsilon + O(\Delta)$ gives $\delta_{\det} - \varepsilon = (\Delta/\pi)(1 + O(\Delta/\varepsilon))$. □

The first display is the formula for $\pi\rho_\varepsilon(\gamma)$ at the ordinate in Section 1.4.2, solved for the distance, with all the other zeros in ρ_ε^0. So a level ε sees, at the ordinate, exactly the zeros at distances in $(\varepsilon, \delta_{\det}(\varepsilon))$: a band that starts at width δ_c near the line and has width close to $\Delta/\pi = 2/\log(h/2\pi)$ well above it, the scale $1/\log\gamma$ of Section 1.4.2 with its constant. The width $\delta_{\det}(\varepsilon) - \varepsilon$ is $\Delta/\pi + c\,\Delta^2/\varepsilon + \cdots$

with $c = \frac{1}{2\pi^2}$, $-\frac{1}{2\pi^2}$ and $\frac{3}{2\pi^2}$ in (i), (ii) and (iii), so that for large ε it is above Δ/π in (i) and (iii) and below it in (ii); in (i) and (iii) it falls monotonically to Δ/π (computed), while in (ii) it falls from $0.450\,\Delta$ to a least value $0.272\,\Delta$ at $\varepsilon = 0.581\,\Delta$ and then rises to Δ/π from below (computed). The curve explains Table 8.3. At the midpoint of the 10000th gap, $\pi\rho^0_\varepsilon(\gamma^*)$ computed from ξ'/ξ gives $\delta_{\det}(0.19) = 0.4503$, and solving $\delta_{\det}(\varepsilon) = 0.45$ gives $\varepsilon = 0.1895$. Figure 8.2 draws the region for the actual zeros and for the lattice with the same gap, which lies within 0.015 of it.

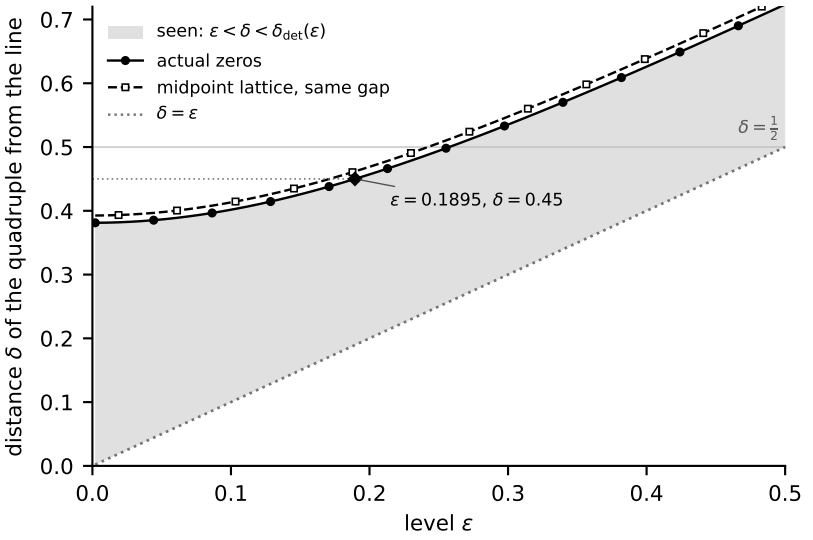


Figure 8.2: The visibility region $\varepsilon < \delta < \delta_{\det}(\varepsilon)$ (shaded band) at the midpoint of the 10000th gap. Solid: $\delta_{\det}$ from the actual zeros through ξ'/ξ; dashed: the midpoint lattice of Proposition 8.3.2 with the same gap; dotted: $\delta = \varepsilon$, the lower edge of the band. The marked point is $\varepsilon = 0.1895$, $\delta = 0.45$.

Corollary 8.3.3 (How many levels, in the lattice model). *Take any of the three lattice models, with gap $\Delta = 2\pi/\log(h/2\pi)$, and put $\varepsilon_1 = \delta_c$ and $\varepsilon_{k+1} = \delta_{\det}(\varepsilon_k)$. Let K be the least k with $\varepsilon_{k+1} > \frac{1}{2}$, so that $K = 0$ if $\delta_c > \frac{1}{2}$.*

(a) The sign of $\mathcal{L}_\Xi(\gamma^*)$, together with positivity of $\rho_\varepsilon(\gamma^*)$ at the levels $\varepsilon_1, \dots, \varepsilon_K$, each moved down slightly so that consecutive bands overlap, detects a single off line quadruple at every distance in $(0, \frac{1}{2}]$.

(b) No test made at the ordinate, that is the sign of $\mathcal{L}_\Xi(\gamma^*)$ together with the sign of $\rho_\varepsilon(\gamma^*)$ at finitely many levels, that detects a single off line quadruple at every distance in $(0, \frac{1}{2}]$ uses fewer than K levels.

(c) $K = \frac{1}{4}\log(h/2\pi) + O(\log\log h)$.

Proof. By Proposition 8.3.1 the first order test detects the distances in $(0, \delta_c)$, and by Proposition 8.3.2 the level ε detects exactly those in $(\varepsilon, \delta_{\det}(\varepsilon))$, since a zero at distance $\delta \le \varepsilon$ contributes positively and $\rho^0_\varepsilon(\gamma^*) > 0$. Write $f = \delta_{\det}$, so that $\varepsilon_{k+1} = f(\varepsilon_k)$; f is continuous and increasing with $f(\varepsilon) > \varepsilon$.

(a) The bands $(0, \varepsilon_1)$ and $(\varepsilon_k, \varepsilon_{k+1})$, $1 \le k \le K$, cover $(0, \varepsilon_{K+1})$ except the points $\varepsilon_1, \dots, \varepsilon_K$. Moving the levels down by suitably small amounts, which is possible because f is continuous and $\varepsilon_{K+1} = f(\varepsilon_K) > \frac{1}{2}$, makes consecutive bands overlap and still reach beyond $\frac{1}{2}$.

(b) We claim that if $j \ge 1$ bands $(\varepsilon', f(\varepsilon'))$ cover $[\ell, x)$ with $\ell > 0$, then $x < f^{(j)}(\ell)$, the jth iterate. A band containing ℓ has $\varepsilon' < \ell$, so its right end is $r = f(\varepsilon') < f(\ell)$. For $j = 1$ this gives $x \le r < f(\ell)$. For $j > 1$, either $x \le r < f(\ell) \le f^{(j)}(\ell)$, or the other $j-1$ bands cover $[r, x)$ and the claim for $j-1$ gives $x < f^{(j-1)}(r) < f^{(j-1)}(f(\ell)) = f^{(j)}(\ell)$. Now suppose that m levels, with the first order test, detect every distance in $(0, \frac{1}{2}]$. The point δ_c is not in $(0, \delta_c)$, so unless $\delta_c > \frac{1}{2}$ (and then $K = 0$) some bands cover $[\delta_c, \frac{1}{2}]$, and the claim with $\ell = \delta_c$ and $x = \frac{1}{2}$ gives $\frac{1}{2} < f^{(m)}(\delta_c) = \varepsilon_{m+1}$. By the definition of K, $m \ge K$.

(c) The bands advance by $a(\varepsilon) = \delta_{\det}(\varepsilon) - \varepsilon = \big(\delta_{\det}(\varepsilon)^2 - \varepsilon^2\big)/\big(\delta_{\det}(\varepsilon) + \varepsilon\big)$, which by the formulas of Proposition 8.3.2 is $\frac{\Delta}{\pi}\big(1 + O(\Delta/\varepsilon)\big)$ for $\varepsilon \ge \Delta$ and, for $\varepsilon \le \Delta$, lies between two positive constants times Δ, since $a(\varepsilon)/\Delta$ is a positive continuous function of ε/Δ. The number of steps from δ_c to $\frac{1}{2}$ is therefore $\int_\Delta^{1/2} \frac{d\varepsilon}{a(\varepsilon)} + O(\log(1/\Delta)) = \frac{\pi}{\Delta}\big(\frac{1}{2} + O(\Delta \log(1/\Delta))\big)$, and $\pi/(2\Delta) = \frac{1}{4}\log(h/2\pi)$, while $\log(1/\Delta) = O(\log\log h)$. □

h	$\frac{1}{4}\log(h/2\pi)$	(i) replacing	(ii) midpoint	(iii) count preserving
10^5	2.4	0	2	0
10^8	4.1	2	4	1
3×10^{12}	6.7	4	6	3
10^{22}	12.2	9	12	7

Table 8.4: The number K of greedy levels of Corollary 8.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$.

The count preserving lattice has the larger $\delta_c = 1.034\,\Delta$, which stays above $\frac{1}{2}$ up to $h = 2.8 \times 10^6$, so it uses fewer levels (Table 8.4), and at 10^5 the $\delta_c = 0.506$ of the replacement lattice already covers every distance. At the height 3×10^{12} to which RH has been verified [232] the three models use between three and six levels. The corollary is a statement about a model with a single quadruple and regularly spaced neighbours, and about tests at the ordinate: a test that looks for negative values anywhere in the window can see more (Section 1.4.2, Exercise 8.14), and for it no lower bound is proved here. Turning it into a certificate for the actual zeros is Problem 8.1.

8.4 The heat flow: a stable step that cannot carry real zeros

With the de Bruijn Newman kernel Φ_{dBN} of Section 4.1.1,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\,\Xi(z/2),$$

the function H_t is defined for every real t, since the kernel decays like $e^{9u}\exp(-\pi e^{4u})$, and the zeros of H_0 are the points $2\gamma_\rho$, with imaginary parts $-2(\beta-\frac{1}{2})$. The de Bruijn Newman constant Λ satisfies $0 \le \Lambda \le 0.2$ [250, 232].

Theorem 8.4.1 (The heat family). *(a) $\partial_t H_t = -\partial_z^2 H_t$, so $H_t = e^{-tD^2}H_0$ with $D = d/dz$.*

(b) For real t, $s > 0$ and complex z, with Z a standard normal variable,

$$H_{t-s}(z) = \mathbb{E}\big[H_t\big(z + \sqrt{2s}\,Z\big)\big].$$

(c) Forward in t real zeros persist, and if the zeros of H_{t_0} lie in $|\operatorname{Im} z| \le r$, then those of H_t lie in $(\operatorname{Im} z)^2 \le \max(r^2 - 2(t-t_0), 0)$ for $t \ge t_0$ [61, Theorem 13].

(d) Backward in t real zeros are not preserved: $f(z) = z^2 - c$, $c > 0$, has real zeros, and $\mathbb{E}f(z+\sqrt{2s}Z) = z^2 - c + 2s$ has none once $s > c/2$. For H itself, H_t has nonreal zeros for every $t < 0$ [250].

Proof. (a) Differentiate under the integral: ∂_t brings down u^2 and ∂_z^2 brings down $-u^2$. (b) $\mathbb{E}\cos((z+\sigma Z)u) = \cos(zu)e^{-\sigma^2u^2/2}$ for complex z, and with $\sigma^2 = 2s$ the factor e^{-su^2} turns the kernel $e^{tu^2}\Phi_{\mathrm{dBN}}$ into $e^{(t-s)u^2}\Phi_{\mathrm{dBN}}$; Fubini applies because Φ_{dBN} decays like $e^{9u}\exp(-\pi e^{4u})$, for every real t. (c) is de Bruijn's theorem, in the form used by Polymath15 [247] and in the proof of Theorem 6.3.2. (d) is the computation shown and [250]. □

Corollary 8.4.2 (No averaging stable property carries real zeros). *Let $\mathcal{P}$ be a property of entire functions that is preserved by $f \mapsto \mathbb{E}f(\cdot + \sqrt{2s}Z)$ for some $s > 0$. If $\mathcal{P}$ implies that all zeros are real, then H_t has $\mathcal{P}$ for no real t.*

Proof. If H_{t_0} has $\mathcal{P}$, then by (b), applied k times, so does H_{t_0-ks} for every integer $k \ge 1$, and for $t_0 - ks < 0$ it has only real zeros, against (d). The proof uses only that some H_t has a nonreal zero, which is Newman's theorem $\Lambda > -\infty$ [215]. □

The corollary is the heat flow form of the criticality filter of Section 14.4.5: Proposition 14.4.12 shows that no property of the kernel Φ_t that holds at some $t < 0$ can force real zeros. The step towards RH, from $t = 0.2$ down to 0, is an average of $H_{0.2}$ against a Gaussian of variance 0.4: a contraction, stable to compute.

A Gauss Legendre rule with 200 nodes in u on $[0, 1.5]$, beyond which $\Phi_{\mathrm{dBN}} < 10^{-540}$, and a Gauss Hermite rule with 80 nodes reproduce $H_0(z)$ from $H_{0.2}$ at $z = 0$, 10 and 28.27 to a relative error below 10^{-35} in 40 digit arithmetic. The step away from RH, e^{-tD^2}, preserves real zeros (Laguerre Pólya theory) and is a deconvolution in z. In the basepoint family the roles are reversed: the positivity preserving step is a convolution and the step towards RH an unbounded deconvolution.

The two informative families are tied by the conserved quantity of Theorem 6.3.2: a basepoint offset ε is worth heat time $2\varepsilon^2$, and $\Lambda \le 2(\sigma_{\max} - \frac{1}{2})^2$. The critical strip, $\sigma_{\max} \le 1$, gives de Bruijn's $\Lambda \le \frac{1}{2}$; the half plane announced in October 2026 [227], not yet refereed, would give $\sigma_{\max} \le \frac{7}{8}$ and, conditionally on it, $\Lambda \le \frac{9}{32}$, weaker than the known 0.2; and $\Lambda \le 0.2$ corresponds to the basepoint $\frac{1}{2} + \sqrt{0.1} = 0.816$ (Corollary 6.3.3). In the language of Theorem 8.2.2, whose *if* direction holds for ξ_t by the product of Lemma 6.2.1, the bound $\Lambda \le 0.2$ of Platt and Trudgian, obtained through the Polymath15 framework, says that the Thorin density of $\xi_{0.2}$ deconvolves to a positive measure all the way to the line; RH asks for the same at $t = 0$, and by Corollary 8.4.2 no property preserved by the Gaussian average joining the two times can carry it.

What would get past the obstruction. This paragraph is heuristic. In the basepoint family a proof needs a bound, uniform in the height, for the deconvolution of ρ_{ε_0}; from the primes that means cancellation in $\sum \Lambda(n) n^{-\alpha - ib}$ below $\alpha = 1$, which is a zero free region by Proposition 12.1.13, and the half plane announced in October 2026 [227], not yet refereed, would move the end of the range to $\alpha = \frac{7}{8}$. In the heat family a proof needs a property of H_0 that implies real zeros and that fails for every H_t with $t < 0$, since those have nonreal zeros (Theorem 8.4.1(d)): a property that sees the boundary $t = \Lambda$ itself. In the tilt family nothing can be gained. And since $\Lambda \ge 0$, any such statement must hold at the boundary with no slack (Remark 6.3.4).

8.4.1 Open problems

Problem 8.1 (How many levels, for the actual zeros). Corollary 8.3.3 counts levels in a model with one quadruple among regularly spaced zeros, and only at its ordinate. Turn it into a certificate: find levels $\varepsilon_1 < \cdots < \varepsilon_K$ such that positivity of every ρ_{ε_k} on $[0, h]$, together with positivity of $\mathcal{L}_\Xi$, excludes all off line zeros up to height h, allowing several off line zeros, irregular spacing, and minima off the ordinate (Exercise 8.14). Is K still of order $\log h$? Such a certificate is finite information in the sense of Section 1.4 and Theorem 1.4.1; its interest is as a verification of RH up to height h from ξ on lines to the right of the critical line rather than from sign changes on it.

Problem 8.2 (Stability with an a priori bound). On a strip $\frac{1}{2} + \varepsilon_1 \le \operatorname{Re} s \le \frac{1}{2} + \varepsilon_0$

free of zeros, Hadamard's three lines theorem bounds the error of an approximation g to ξ'/ξ at level ε by $\eta^{\lambda} M^{1-\lambda}$, $\lambda = (\varepsilon - \varepsilon_1)/(\varepsilon_0 - \varepsilon_1)$, where η is the error at ε_0 and M a bound at ε_1. The bound M presupposes the zero free strip. Is there a formulation in which the a priori bound comes from the Thorin structure itself, for instance from positivity of ρ_ε at the levels already certified?

Problem 8.3 (Seeing the boundary). By Theorem 8.4.1(d), a property of H_0 that implies real zeros fails for H_t at every $t < 0$. Which of the positivity statements of the book, for instance the GGC property of the clock, whose Laplace transform is R_0, the case $t = 0$ of $R_t(s) = H_t(0)/H_t(2i\sqrt{s})$ in Theorem 17.1.16, or the Pólya frequency property of the density of $Z\sqrt{2T}$ (Proposition 22.5.1, where the clock is written Y), can be shown to fail for $t < 0$ by a mechanism visible at finite height, as Lehmer pairs make Λ close to 0 visible [51]?

Problem 8.4 (The window below δ). For a quadruple at distance $\delta > \delta_c$ the level test detects it on a set of levels contained in $(0, \delta)$, an interval (ε_c, δ) in our examples. At the ordinate ε_c is the inverse of the visibility curve of Proposition 8.3.2. Describe the distribution of δ_{det}, and of the onset $\varepsilon_c(\delta, \gamma^*)$, over heights in terms of the local zero statistics, and compare it with the prediction of the sine kernel; Table 8.2 gives its lower end, δ_c.

Notes

The ordinate formula for $\pi\rho_\varepsilon(\gamma)$, the first order visibility condition and the lattice threshold $(\sqrt{6}/\pi)\Delta$ of Section 8.3 are those of Section 1.4.2, and inserting hypothetical quadruples into the density computed from the zeros is the device of Section 14.3.4; the closed forms of Proposition 8.3.2, the level count of Corollary 8.3.3 and the computations at the actual zeros are in this chapter. The bound $\Lambda \leq 0.2$ is that of Platt and Trudgian [232], and the visibility of a Lehmer pair through Λ is the work of Csordas, Smith and Varga [51].

Chapter 9

Deconvolution as a damped Weil criterion

The half plane $\operatorname{Re} s > \frac{7}{8}$ used below has not been refereed. The eigenvalues of the ladder are variational upper bounds, and the converged values are extrapolations of such bounds with an error estimate; none is a certificate.

The question. Let ξ have zeros $\rho = \beta + i\gamma$, let $\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ be the zeros of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$, and put $\sigma_{\max} = \sup_\rho \operatorname{Re}\rho \in [\frac{1}{2}, 1]$. We say that a level $\varepsilon > 0$ is *to the right of the zeros* if $\beta < \frac{1}{2} + \varepsilon$ for every zero. Every $\varepsilon \geq \frac{1}{2}$ is, since $\beta < 1$. For real α write $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\theta)$ and $b_\alpha = v_\alpha(0) = \frac{\xi'}{\xi}(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (2.1.7): for $\varepsilon > 0$ the Poisson smoothing of the zeros (Proposition 2.1.3) and, when ε is to the right of the zeros, the Thorin density of the shifted clock at basepoint $\frac{1}{2}+\varepsilon$ (Theorem 2.1.2). With $P_a(x) = a/(\pi(a^2+x^2))$, Theorem 8.2.2 says that, for ε_0 to the right of the zeros and $0 \leq \varepsilon < \varepsilon_0$, there is a positive measure ν with $\int (1+\theta^2)^{-1}\nu(d\theta) < \infty$ and $\rho_{\varepsilon_0} = P_{\varepsilon_0 - \varepsilon} * \nu$ if and only if no zero has $\beta > \frac{1}{2} + \varepsilon$. The proof runs a Poisson integral of ν down from ε_0 and uses harmonic continuation: a positive ν would make $\rho_\varepsilon(\theta)$ harmonic in (θ, ε) near a zero with $\beta - \frac{1}{2} > \varepsilon$, where ξ'/ξ has a pole.

Asking whether the deconvolution *can be done* has two meanings. One is constructive: can $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ be produced, from data we can compute, without assuming where the zeros are? The other is the theorem itself: can it be shown to be a positive measure? We answer the first completely (Section 9.1.1) and reduce the second to a two parameter family of finite problems (Sections 9.1.2 and 9.2), whose behaviour we partly prove and compute.

9.1 Weil's distribution and the damped criterion

9.1.1 The deconvolved object exists: it is Weil's distribution

With $\hat{f}(t) = \int f(x)e^{itx}dx$ the Fourier transform of P_a is $e^{-a|x|}$, so deconvolution by P_a multiplies Fourier transforms by $e^{a|x|}$. Let $\mathcal{B}$ be the space of even functions $\varphi(\theta) = \frac{1}{2\pi}\int \hat{\varphi}(x)e^{-i\theta x}dx$ with $\hat{\varphi}$ of compact support, Lipschitz, and with $\hat{\varphi}'$ of

bounded variation. Then φ is entire of exponential type and $\varphi(\theta+is)=O(\theta^{-2})$ uniformly for $|s|\le 1$, since $\hat\varphi(x)e^{sx}$ has the same properties. For $a\ge 0$ let $P_a^{-1}\varphi\in\mathcal{B}$ have Fourier transform $\hat\varphi(x)e^{a|x|}$; the corner of $e^{a|x|}$ at 0 keeps it in $\mathcal{B}$.

THEOREM 9.1.1 (The deconvolved functional). *Let ε_0 be to the right of the zeros. For $\varphi\in\mathcal{B}$ put $D(\varphi)=\int \rho_{\varepsilon_0}(\theta)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$. Then D does not depend on ε_0, and with $g(x)=\hat\varphi(x)/2\pi$, so that $\varphi(\theta)=\int g(x)e^{i\theta x}dx=\hat g(\theta)$,*

$$D(\varphi)=\sum_\rho \varphi(\gamma_\rho)=\hat g\big(\tfrac{i}{2}\big)+\hat g\big(-\tfrac{i}{2}\big)-(\gamma_E+\log\pi)\,g(0)$$
$$+\int_0^\infty \frac{2\big(e^{-2x}g(0)-e^{-x/2}g(x)\big)}{1-e^{-2x}}\,dx$$
$$-2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n),$$

where the sum over ρ runs over all zeros with multiplicity, in conjugate pairs, and converges absolutely. Through the x form of its archimedean term, the Thorin functional U of Section 8.2.1 extends to $\mathcal{B}$, and then $D(\varphi)=2\langle U,\varphi(\sqrt{\cdot})\rangle$.

Proof. By the expansion of ρ_{ε_0} over the zeros (Proposition 2.1.3, summed over all zeros $\rho=\beta+i\gamma$, with γ of both signs), $D(\varphi)=\sum_\rho\int P_{c_\rho}(\theta-\gamma)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$ with $c_\rho=\varepsilon_0-(\beta-\frac12)>0$; the interchange is justified because $\rho_{\varepsilon_0}=O(\log(2+|\theta|))$ and $P_{\varepsilon_0}^{-1}\varphi=O(\theta^{-2})$. By Parseval each term is $\frac{1}{2\pi}\int\hat\varphi(x)e^{(\varepsilon_0-c_\rho)|x|}e^{-i\gamma x}\,dx$. A zero ρ off the line and its partner $1-\bar\rho$ have the same γ and opposite $\beta-\frac12=\pm\delta$, so the pair gives $\frac{1}{2\pi}\int\hat\varphi(x)e^{-i\gamma x}\,2\cosh(\delta x)\,dx=\varphi(\gamma-i\delta)+\varphi(\gamma+i\delta)$, because $e^{\delta|x|}+e^{-\delta|x|}=e^{\delta x}+e^{-\delta x}$; a zero on the line is its own partner and gives the single term $\varphi(\gamma)$. This is $\sum_\rho\varphi(\gamma_\rho)$, absolutely convergent by the uniform $O(\theta^{-2})$ bound, and it does not involve ε_0. The second expression is the explicit formula of Weil [273], [177, Theorem 5.12], with the archimedean term written in the variable x by $\operatorname{Re}\psi(\frac14+\frac{it}{2})=-\gamma_E+\int_0^\infty\frac{e^{-u}-e^{-u/4}\cos(tu/2)}{1-e^{-u}}\,du$; this integral converges because g is Lipschitz at 0, and the formula extends from the usual class to $\mathcal{B}$ by approximation: mollifying g gives functions of the usual class supported in a fixed interval, with uniformly bounded Lipschitz constants, and both sides are continuous along such sequences, the zero side by the uniform $O(\theta^{-2})$ bound. The comparison with (8.2.1), whose functional sums once over each pair $\{\rho,1-\rho\}$, gives the factor 2. □

For g even, Lipschitz and of compact support, with g' of bounded variation, so that $\hat g\in\mathcal{B}$, write $\langle D,g\rangle=D(\hat g)$, and for g not even apply it to the even part of g. By the theorem $\langle D,g\rangle=\sum_\rho\hat g(\gamma_\rho)$, which is the action on g of Weil's distribution $D(x)=\sum_\rho e^{(\rho-\frac12)x}$ of Proposition 17.1.14.

The distribution $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ on $\mathcal{B}$ exists with no hypothesis, and is a positive measure exactly when RH holds: under RH it is $\sum_\gamma\delta_\gamma$, and if $D(\varphi)=\int\varphi\,d\mu$ on

$\mathcal{B}$ with $\mu \geq 0$, then $\int \rho_{\varepsilon_0}\psi = D(P_{\varepsilon_0} * \psi) = \int (P_{\varepsilon_0} * \psi)\, d\mu$ for every $\psi \in \mathcal{B}$, which gives $\int (1+\theta^2)^{-1} d\mu < \infty$ (take $\psi \geq 0$ with $P_{\varepsilon_0} * \psi \geq c(1+\theta^2)^{-1}$) and $\rho_{\varepsilon_0} = P_{\varepsilon_0} * \mu$, so that RH follows from Theorem 8.2.2. On the functions $\varphi = |\hat{h}|^2$ the statement $D(\varphi) \geq 0$ is Weil's criterion [273], in the form the book uses in Section 5.1.2. We checked the second expression against $2\sum_{\gamma>0} \hat{g}(\gamma)$ for the Gaussian $g(x) = e^{-(x/c)^2}$: at $c = \frac{1}{2}$ both sides are $6.69239857682 \times 10^{-6}$ and they differ by less than 10^{-69} at seventy digits of working precision.

9.1.2 The damped Weil criterion

Partial deconvolution, to depth ε instead of to the line, corresponds to damping. For bounded h of bounded variation and compact support put $\tilde{h}(x) = \overline{h(-x)}$, so that $h * \tilde{h}$ is Lipschitz and Hermitian, and

$$Q_\varepsilon(h) = \big\langle D,\, e^{-\varepsilon|x|}\,(h * \tilde{h}) \big\rangle .$$

The Fourier transform of $e^{-\varepsilon|x|}$ is $2\pi P_\varepsilon$ and that of $h * \tilde{h}$ is $|\hat{h}|^2$, so

$$\widehat{e^{-\varepsilon|\cdot|}(h * \tilde{h})} = P_\varepsilon * |\hat{h}|^2,$$

a nonnegative function. Real h suffice in what follows, since $Q_\varepsilon(h) = Q_\varepsilon(\operatorname{Re} h) + Q_\varepsilon(\operatorname{Im} h)$.

Theorem 9.1.2 (Damped Weil criterion). *Let $0 \leq \varepsilon \leq \frac{1}{2}$. Then $Q_\varepsilon(h) \geq 0$ for every $h \in C_c^\infty(\mathbb{R})$ if and only if no nontrivial zero of ζ has $\beta > \frac{1}{2} + \varepsilon$. When ε is to the right of the zeros,*

$$Q_\varepsilon(h) = \int_{\mathbb{R}} |\hat{h}(\theta)|^2\, \rho_\varepsilon(\theta)\, d\theta \tag{9.1.1}$$

for every bounded h of bounded variation and compact support. In particular: at $\varepsilon = 0$ this is Weil's criterion for RH; at $\varepsilon = \frac{1}{2}$ it holds with no hypothesis; and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [227], which has not been refereed, is equivalent to $Q_{3/8} \geq 0$.

Proof. Formula (9.1.1) follows from Theorem 9.1.1. Let h be real, bounded, of bounded variation and of compact support. Then $F = h * \tilde{h}$ is Lipschitz and of compact support, and its derivative $h * d\tilde{h}$, the convolution of a bounded function of bounded variation with a finite measure, is of bounded variation; so $e^{-\varepsilon|x|}F$ is Lipschitz, of compact support, with a derivative of bounded variation, and $\varphi = P_\varepsilon * |\hat{h}|^2$, the Fourier transform of $e^{-\varepsilon|x|}F$, and $|\hat{h}|^2$, that of F, lie in $\mathcal{B}$. As $P_\varepsilon^{-1}\varphi = |\hat{h}|^2$, the first identity of Theorem 9.1.1 with $\varepsilon_0 = \varepsilon$ gives $Q_\varepsilon(h) = D(\varphi) = \int |\hat{h}|^2 \rho_\varepsilon\, d\theta$; for complex h add the two parts, the cross terms of $|\hat{h}|^2$ being odd and ρ_ε even. For bounded piecewise C^1 functions h this is Proposition 15.20.1 of Chapter 15.12 at $\alpha = \frac{1}{2} + \varepsilon$, where the form $\langle \varphi, \varphi \rangle_\alpha$ is $Q_\varepsilon(\varphi)$. The *if* direction

is Theorem 15.20.3(i): if no zero has $\beta > \frac12 + \varepsilon$, then either ε is to the right of the zeros and $\rho_\varepsilon > 0$, so that $Q_\varepsilon \ge 0$ by (9.1.1), or some zero has $\beta = \frac12 + \varepsilon$, and then $Q_\varepsilon(h) = \lim_{\varepsilon' \downarrow \varepsilon} Q_{\varepsilon'}(h) \ge 0$, the limit holding because $Q_\varepsilon(h)$ is a finite combination of integrals continuous in ε.

Conversely, suppose $Q_\varepsilon \ge 0$. The map $T_\varepsilon : g \mapsto \langle D, e^{-\varepsilon|x|} g\rangle$ is a distribution of order at most one, the corner of $e^{-\varepsilon|x|}$ at 0 being harmless in the archimedean integral, and $Q_\varepsilon(h) = T_\varepsilon(h * \tilde h) \ge 0$ says that it is positive definite. By the Bochner Schwartz theorem [80, Chapter II] it is tempered and $T_\varepsilon(g) = \int \hat g \, d\nu_\varepsilon$ for a positive tempered measure ν_ε and every Schwartz g. Fix $\varepsilon_0 > \varepsilon$ with $\varepsilon_0 \ge \frac12$, put $c = \varepsilon_0 - \varepsilon$, and let $k_\eta(x) = e^{-c\sqrt{x^2+\eta^2}}$, smooth, positive definite and of exponential decay, with $\hat k_\eta \ge 0$, $\hat k_\eta \to 2\pi P_c$ pointwise and $\hat k_\eta \le C(1+\theta^2)^{-1}$ uniformly in $\eta \le 1$. For $h \in C_c^\infty$ and $g = h * \tilde h$, $T_\varepsilon(k_\eta g) = \frac{1}{2\pi} \int (\hat k_\eta * |\hat h|^2)\, d\nu_\varepsilon$, and as $\eta \to 0$ the left side tends to $T_\varepsilon(e^{-c|x|} g) = Q_{\varepsilon_0}(h) = \int |\hat h|^2 \rho_{\varepsilon_0}$, because $k_\eta g \to e^{-c|x|} g$ uniformly, with supports in a fixed interval and uniformly bounded Lipschitz constants, which is enough for dominated convergence in the archimedean integral and in the other terms of $\langle D, \cdot\rangle$. By Fatou, $\int (P_c * |\hat h|^2)\, d\nu_\varepsilon \le \int |\hat h|^2 \rho_{\varepsilon_0} < \infty$; choosing h with $|\hat h|^2 > 0$ near 0 gives $P_c * |\hat h|^2 \ge c'(1+\theta^2)^{-1}$, so $\int (1+\theta^2)^{-1} d\nu_\varepsilon < \infty$, and dominated convergence then gives $\int (P_c * |\hat h|^2)\, d\nu_\varepsilon = \int |\hat h|^2 \rho_{\varepsilon_0}$, that is $\int |\hat h|^2 \, (P_c * \nu_\varepsilon) = \int |\hat h|^2 \rho_{\varepsilon_0}$. Polarising, the same holds with $|\hat h|^2$ replaced by $\hat h_1 \overline{\hat h_2}$, and these products span a dense set, so $\rho_{\varepsilon_0} = P_{\varepsilon_0 - \varepsilon} * \nu_\varepsilon$. Theorem 8.2.2 gives $\beta \le \frac12 + \varepsilon$ for every zero. □

The theorem is the Fourier dual of the deconvolution theorem; at $\varepsilon = 0$ it proves Weil's criterion by the harmonic continuation of Theorem 8.2.2. The damping explains why $\varepsilon = \frac12$ is free: the primes enter with weight $\Lambda(n) n^{-1/2-\varepsilon}$, absolutely summable exactly for $\varepsilon > \frac12$: the damped criterion at $\varepsilon = \frac12$ sits at the edge of the Euler product, the basepoint one of Chapter 12.

9.2 The ladder

Restrict the support of h. For $a > 0$ and $\varepsilon \ge 0$ let

$$\lambda_{\min}\big(\tfrac12 + \varepsilon, a\big) = \inf \Big\{ \frac{Q_\varepsilon(h)}{\|h\|_2^2} : h \text{ bounded, real, of bounded variation, supported in } [-a,a],\ h \ne 0 \Big\}.$$

This is the least eigenvalue $\lambda_{\min}(\alpha, a)$, $\alpha = \frac12 + \varepsilon$, of Chapter 15.12. The function $e^{-\varepsilon|x|}(h * \tilde h)$ is supported in $[-2a, 2a]$, so Q_ε involves only the prime powers $n \le e^{2a}$: each $\lambda_{\min}(\frac12 + \varepsilon, a)$ is the bottom of the spectrum of an explicit form. RH is $\lambda_{\min}(\frac12, a) \ge 0$ for every a. Yoshida [278, Theorem 1] proved positivity of Weil's form for $a \le \frac12 \log 2$, the range in which no prime enters, Bombieri [29] treated other test functions of small support, and Connes and Consani reproved Yoshida's

range from the semilocal trace formula [44]. Zhu certifies the lower bound 8.9×10^{-18} for $\lambda_{\min}(\frac{1}{2}, a)$ at $a = 0.8$ (Corollary 6.3 of [279]), and reports an empirical law for the decay of $\lambda_{\min}(\frac{1}{2}, a)$ [279] (Chapter 16), and Connes, Consani and Moscovici find that the Fourier transform of the lowest eigenvector of a truncated form reproduces the low zeros to high accuracy [45]. The last two papers are cited as preprints.

THEOREM 9.2.1 (The ladder at depth $\varepsilon > 0$). *Let $\varepsilon > 0$ be to the right of the zeros, and suppose that $v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$ for $\theta \neq 0$, with $v''_{1/2+\varepsilon}(0) > 0$ and $\liminf_{|\theta|\to\infty} v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$.*

(a) $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ is nonincreasing in a, and $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \geq 2b_{1/2+\varepsilon} > 0$ for every a.

(b) $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \to 2b_{1/2+\varepsilon}$ as $a \to \infty$, and there are constants $c, a_0 > 0$ with

$$\frac{c}{a^2} \leq \lambda_{\min}\big(\tfrac{1}{2}+\varepsilon, a\big) - 2b_{1/2+\varepsilon} \leq \frac{\pi^2 v''_{1/2+\varepsilon}(0)/4 + o(1)}{a^2} \qquad (a \geq a_0).$$

(c) At $\varepsilon = 0$, with none of the hypotheses above, under RH, $\lambda_{\min}(\frac{1}{2}, a) \geq 0$ and $\lambda_{\min}(\frac{1}{2}, a) = O(a^{-N})$ for every N, so $\lambda_{\min}(\frac{1}{2}, a) \to 0$; if RH fails, $\lambda_{\min}(\frac{1}{2}, a) < 0$ for all large a.

Proof. (a) is Theorem 15.20.5(i) of Chapter 15.12 at $\alpha = \frac{1}{2} + \varepsilon$, whose $m_\alpha = \inf_\theta v_\alpha(\theta)$ is b_α here, the minimum being at the origin; and $b_{1/2+\varepsilon} = \sum_\rho \mathrm{Re}(\frac{1}{2} + \varepsilon - \rho)^{-1} > 0$ because every term is positive. We use the first step of that proof: by (9.1.1) and Parseval, $\|\hat{h}\|_2^2 = 2\pi\|h\|_2^2$, so $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$, the average of $v_{1/2+\varepsilon}$ against the probability density $|\hat{h}|^2/\|\hat{h}\|_2^2$, which is at least $2\min v_{1/2+\varepsilon} = 2b_{1/2+\varepsilon}$.

(b) Write $v = v_{1/2+\varepsilon}$. Upper bound: take $h(u) = \cos(\pi u/2a)$ on $[-a, a]$. Then $\langle\theta^2\rangle_h = \|h'\|_2^2/\|h\|_2^2 = \pi^2/4a^2$, and the normalised density satisfies $|\hat{h}(\theta)|^2/\|\hat{h}\|_2^2 = O(a^{-3}\theta^{-4})$ for $|\theta| \geq 1/a$. Write $v(\theta) = v(0) + \frac{1}{2}v''(0)\theta^2 + R(\theta)$, with $|R(\theta)| \leq C\theta^4$ for $|\theta| \leq \eta$ and $|R(\theta)| \leq C\theta^2\log(2+|\theta|)$ for all θ. Splitting at $|\theta| = \eta$ gives $\langle R\rangle_h = O(\eta a^{-3}) + O(a^{-3}\eta^{-1})$, and with $\eta = a^{-1/2}$ this is $o(a^{-2})$. Hence $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \leq 2v(0) + v''(0)\pi^2/4a^2 + o(a^{-2})$. Lower bound: by the hypotheses there is $c_1 > 0$ with $v(\theta) - v(0) \geq c_1 \min(\theta^2, 1)$. For h supported in $[-a, a]$ the fraction of $\|\hat{h}\|_2^2$ carried by $|\theta| \leq \kappa/a$ is at most the largest eigenvalue $\ell_0(\kappa)$ of the time and band limiting operator with product κ, which is less than 1 for every finite κ and tends to 0 with κ [258]. So $\langle\min(\theta^2, 1)\rangle_h \geq (\kappa/a)^2(1 - \ell_0(\kappa))$ once $\kappa \leq a$, which is the lower bound with $c = 2c_1\kappa^2(1 - \ell_0(\kappa))$. Convergence $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \to 2b_{1/2+\varepsilon}$ follows from the two bounds.

(c) Under RH, $Q_0(h) = \sum_\gamma |\hat{h}(\gamma)|^2 \geq 0$ by Theorem 9.1.1. Fix a real $h_1 \in C_c^\infty[-1, 1]$, $h_1 \neq 0$, and put $h_a(u) = h_1(u/a)$. Then $\hat{h}_a(\theta) = a\hat{h}_1(a\theta)$ with $|\hat{h}_1(t)| \leq$

$C_N|t|^{-N}$, so $Q_0(h_a)/\|h_a\|_2^2 \le C'_N a^{1-2N}\sum_\gamma \gamma^{-2N}$, every $|\gamma|$ being larger than 14. If RH fails, Weil's criterion gives $h \in C_c^\infty$, which may be taken real, with $Q_0(h) < 0$; it is supported in some $[-a_0, a_0]$, and by monotonicity $\lambda_{\min}(\frac12, a) \le \lambda_{\min}(\frac12, a_0) < 0$ for $a \ge a_0$. □

At $\varepsilon = \frac12$ the hypotheses of the theorem hold with no assumption on the zeros: the level is to the right of the zeros because ζ has no zeros on $\operatorname{Re} s = 1$, and Proposition 12.1.11 gives the minimum at the origin, with $v_1''(0) > 0$, and in its proof $v_1 > 0.05 > b_1$ for $|\theta| \ge 2000$, which is the third condition. For $0 < \varepsilon < \frac12$ the level is to the right of the zeros exactly when the closed half plane $\operatorname{Re} s \ge \frac12 + \varepsilon$ is free of zeros, which the announced half plane gives only for $\varepsilon > \frac38$; the minimum at the origin is certified for $0 < |\theta| \le H_0 - \frac{1+\sqrt2}{2}$, $H_0 = 3 \times 10^{12}$, by Theorem 12.1.18(ii) (Table 12.4), and its holding on every line with $\frac12 < \alpha < 1$ jointly is equivalent to RH (part (vi) there). The third condition is not known for $0 < \varepsilon < \frac12$ without RH; under RH it holds, because $v_{1/2+\varepsilon}$ is then a sum of Poisson kernels of the zeros and tends to infinity.

Remark 9.2.2. At $\varepsilon = \frac12$ the hypotheses of Theorem 9.2.1(a) and (b) hold, as just said, and $2b_1 = 2\frac{\xi'}{\xi}(1) = 2\big(1 + \frac{\gamma_E}{2} - \frac12\log 4\pi\big) = 2 + \gamma_E - \log 4\pi = 0.0461914\ldots$. So for every $a > 0$ and every bounded real h of bounded variation supported in $[-a, a]$,

$$Q_{1/2}(h) \;\ge\; \big(2 + \gamma_E - \log 4\pi\big)\,\|h\|_2^2 = 0.0461914\ldots\,\|h\|_2^2,$$

and by (9.1.1) this is Proposition 12.1.11 read through Plancherel, and Theorem 15.20.5(i) at $\alpha = 1$, which is Corollary 15.20.8 of Chapter 15.12; what the ladder adds is the rate in a of Theorem 9.2.1(b).

9.2.1 Numerical results

We computed $\lambda_{\min}(\frac12+\varepsilon, a)$ variationally, with h in the span of the first M Legendre polynomials $q_0, \ldots, q_{M-1}$ on $[-a, a]$, normalised in L^2: the correlations $q_j * \tilde q_k$ by Gauss Legendre quadrature, which is exact for polynomials; the archimedean and pole integrals by Gauss Legendre quadrature on $[0, 2a]$, plus the exact tail of the $g(0)$ term; and the primes exactly. We repeat this for $M = 24$ to 128 and compute the same eigenvalue on a second basis, the normalised Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, for $N = 24$ to 1536, from the matrices of Chapter 15.12. Every computed eigenvalue is a variational upper bound for $\lambda_{\min}$ and needs no hypothesis on the zeros, because Q_ε is an explicit finite form; the hypotheses of Theorem 9.2.1 enter only in what the numbers are compared with. Rounding is not the limit: double precision agrees with thirty digit arithmetic to 1.4×10^{-14} at most, and 1.5 times the quadrature nodes changes the eigenvalue by at most 2×10^{-15}. The limit is the size of the basis.

Between $M = 24$ and $M = 32$ the Legendre bounds change by at most 2.6×10^{-6}, but they are not converged at that level: for $\varepsilon > 0$ the $M = 32$ values lie

above the converged values defined next by 0.6×10^{-7} to 1.8×10^{-6}, so the change between two sizes understates the error. The Fourier sections are nested, and their eigenvalues converge like a power, $\lambda_N - \lambda_\infty \propto N^{-p}$ with p between 1.18 and 1.69 over the whole grid. The *converged value* is the power law extrapolation from the three largest sizes $N = 384$, 768, 1536, and its uncertainty is the shift of that value when the largest size is dropped (sizes 192, 384, 768): at most 2.7×10^{-8} at $a = 0.25$, 3.0×10^{-9} at $a = 0.5$ and 1.7×10^{-9} for $a \geq 0.75$. This is an error estimate, not a bound, and two checks support it. The Legendre bound with $M = 128$ lies above the converged value in all 40 entries with $\varepsilon > 0$, by 1.8×10^{-8} to 1.5×10^{-7}, as an upper bound must. And the converged values of Table 15.10 of Chapter 15.12, obtained there by Richardson extrapolation of the sizes $N = 96$, 192, 384, agree with the values computed here for the same cells to 2.0×10^{-8} at most over its twenty entries with $\alpha > \frac{1}{2}$ and to 4.1×10^{-9} for $a \geq 0.8$.

Table 9.1 shows the converged values for $\varepsilon > 0$ and, at $\varepsilon = 0$, for $a = 0.25$ and 0.5, and Table 9.2 the rung at $\varepsilon = 0$ for $0.25 \leq a \leq 0.5$. Table 15.10 of Chapter 15.12 and Table 16.1 of Chapter 16 list eigenvalues of the same kind for other windows and levels; the level 0.375 of Table 9.1 is that of the announced half plane. Figure 9.1 plots the eigenvalue on a logarithmic scale.

a	$\varepsilon = 0.5$	0.375	0.2	0.1	0.05	0
0.25	0.0894856	0.0756372	0.0560458	0.0447429	0.0390618	0.0333609
0.5	0.0495266	0.0372606	0.0200130	0.0100860	0.0050779	9.33×10^{-7}
0.75	0.0474318	0.0355980	0.0190070	0.0095123	0.0047593	
1	0.0468396	0.0351417	0.0187506	0.0093779	0.0046899	
1.5	0.0464620	0.0348541	0.0185931	0.0092974	0.0046489	
2	0.0463394	0.0347613	0.0185428	0.0092720	0.0046361	
3	0.0462555	0.0346979	0.0185087	0.0092548	0.0046275	
4	0.0462270	0.0346764	0.0184971	0.0092490	0.0046246	
$2b_{1/2+\varepsilon}$	0.0461914	0.0346497	0.0184828	0.0092418	0.0046210	0

Table 9.1: The ladder $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \leq N$, by power law extrapolation from $N = 384$, 768, 1536. The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2}+\varepsilon)$ of Theorem 9.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed. At $\varepsilon = 0$ the entries for $a \geq 0.75$ are omitted: in double precision they are below the rounding level of about 10^{-13}, and the upper bounds from extended precision are given in the text.

For every level in Table 9.1 the converged values decrease to $2b_{1/2+\varepsilon}$ and never cross it (nor does any computed upper bound, in either basis), as Theorem 9.2.1(a) predicts when its hypotheses hold; at $\varepsilon = \frac{1}{2}$ this is Remark 9.2.2, and at $\varepsilon = \frac{3}{8}$

a	$M = 32$	converged	uncertainty
0.25	3.3363×10^{-2}	3.3361×10^{-2}	2.7×10^{-8}
0.3	7.5722×10^{-3}	7.5715×10^{-3}	1.4×10^{-8}
$\frac{1}{2}\log 2$	1.3294×10^{-3}	1.3293×10^{-3}	3.4×10^{-9}
0.35	1.1938×10^{-3}	1.1935×10^{-3}	1.3×10^{-9}
0.4	1.8173×10^{-4}	1.8126×10^{-4}	1.1×10^{-9}
0.45	1.6232×10^{-5}	1.6149×10^{-5}	1.7×10^{-10}
0.5	9.4039×10^{-7}	9.3317×10^{-7}	1.4×10^{-11}

Table 9.2: The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is 1.3293×10^{-3}.

the announced half plane gives $\lambda_{\min}(\frac{7}{8}, a) \ge 0$ (Theorem 9.1.2), while $\lambda_{\min} \ge 2b_{7/8}$ needs in addition the minimum of $v_{7/8}$ at the origin, which Theorem 12.1.18(ii) certifies only up to height 3×10^{12}. The margin is linear in ε. With no hypothesis $2b_{1/2+\varepsilon} \le 2V_\xi\,\varepsilon$, where $V_\xi = (\log\xi)''(\frac{1}{2}) = 0.04620998623$ is the constant of Theorem 12.1.18(i), and $2b_{1/2+\varepsilon}/\varepsilon \to 2V_\xi = 0.0924200$ as $\varepsilon \downarrow 0$; under RH $V_\xi = 2p_1$ with $p_1 = \sum_{\gamma>0}\gamma^{-2} = 0.0231050$.

The rate of approach changes character at the line. Table 9.3 compares the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ of the converged values with the upper constant $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 9.2.1(b): the ratio falls towards 1 as a grows. Upper bounds approaching the constant from above cannot tell whether the limit is that constant or a smaller one, so we fit the converged values: a least squares fit on $2 \le a \le 7$ (prime powers to e^{14} at $a = 7$) of $a^2(\lambda_{\min} - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$ gives $C_0 = 1.0002$, 1.0003, 1.0005 at $\varepsilon = \frac{1}{2}$, 0.2, 0.1, with an uncertainty from the errors of the data of 0.00004, 0.0001, 0.0002, and over the choice of the window ($a \ge 1.5$, 2 or 3) and of the degree (2 or 3) C_0 ranges over 0.9999 to 1.0004, 0.9995 to 1.0010 and 0.9991 to 1.0016. The data support the limit $\bar{c}_\varepsilon$ to within 0.04 to 0.2 percent, with a positive correction $C_1 = 0.139$, 0.159, 0.166 and $C_2 = 0.037$, 0.045, 0.051 and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$; the term in a^{-2} is needed, a fit by $C_0 + C_1/a$ alone being rejected by its residuals. This is numerical evidence and not a proof (Problem 9.2). Numerically, $\bar{c}_\varepsilon$ is not a bound at finite a: the scaled excess exceeds it at every computed a ($1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$).

At $\varepsilon = 0$ and $a = 0.35$ the 1.20×10^{-3} of Table 15.10 is the finite section with $N = 24$, an upper bound for the value 1.19×10^{-3} of Table 9.2. From $a = 0.75$ on double precision returns only rounding noise, below about 10^{-13}, and extended precision gives the upper bounds $\lambda_{\min}(\frac{1}{2}, a) < 4.4 \times 10^{-15}$, 1.92×10^{-17} and 7.3×10^{-30} at $a = 0.75$, 0.8 and 1 (Fourier modes $|n| \le 48$, 48 and 60, at

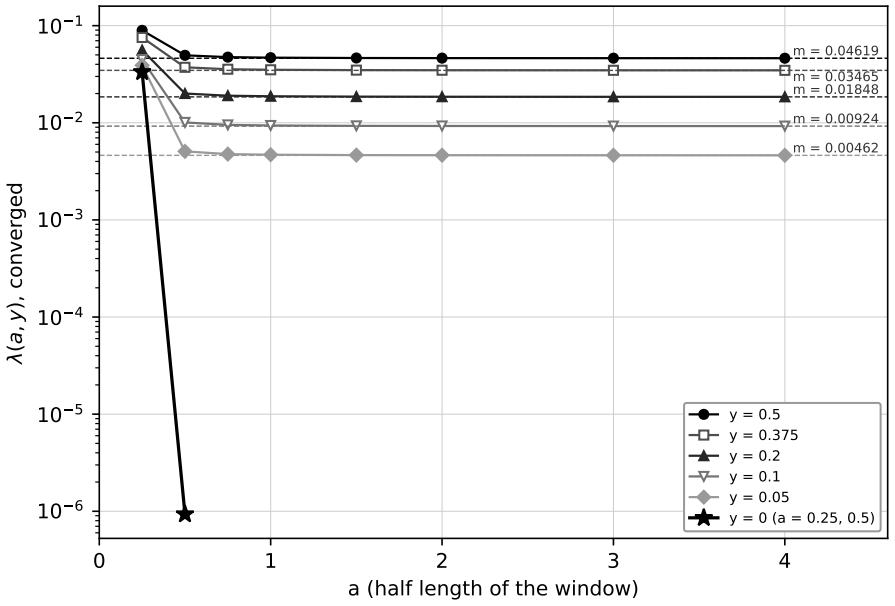


Figure 9.1: The converged ladder on a logarithmic scale: $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ against the window half length a at $a = 0.25$, 0.5, 0.75, 1, 1.5, 2, 3, 4, for $\varepsilon = 0.5$, 0.375, 0.2, 0.1, 0.05 (five marker styles in black and grey), each with the value $2b_{1/2+\varepsilon}$, the limit where the hypotheses of Theorem 9.2.1 hold, as a dashed line labelled m, and for $\varepsilon = 0$ at $a = 0.25$ and 0.5 only (heavy line with stars). The values are those of Table 9.1; the legend and the axis write y and $\lambda(a, y)$ for ε and $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$.

60, 60 and 80 digits). They fall steeply with the size of the section and depend on the basis (the Legendre bounds with $M = 24$ are 2.7×10^{-13}, 2.7×10^{-14} and 2.1×10^{-16}, and with $M = 96$ they are 3.7×10^{-15}, 1.66×10^{-17} and 6.3×10^{-30}, each of these three below the Fourier bound above), so they are upper bounds and not the order of the eigenvalue. Zhu's certified lower bound at $a = 0.8$ is 8.9×10^{-18} (Corollary 6.3 of [279], Chapter 16), and the bounds 1.92×10^{-17} and 1.66×10^{-17} lie above it, the second within a factor 1.9.

The decay of the rung at $\varepsilon = 0$ is stated four ways in this book: Section 6.6.1 observes that the least eigenvalue on hat functions falls roughly geometrically in the support length $L = 2a$, up to $L = 8$; Chapter 15.12 observes that the finite sections follow $\exp(-2c\,ae^{2a})$ with $4.1 \le c \le 5.1$ for $0.25 \le a \le 0.8$, and does not prove it; Zhu proves, under RH and for large a, that the least eigenvalue over real even test functions, and so $\lambda_{\min}(\frac{1}{2}, a)$, is at most $\exp(-ae^a)$ (Theorem 1.3 of [279], Chapter 16, a preprint); and Theorem 9.2.1(c) proves, under RH, that the rung is $O(a^{-N})$ for every N. The four do not conflict: each fall is faster than any power of a, and Theorem 9.2.1(c) is the weakest conditional bound and the only

one proved in the text.

ε	$\pi^2 v''_{1/2+\varepsilon}(0)/4$	$a=1$	$a=2$	$a=3$	$a=4$
0.5	5.485	6.48	5.92	5.76	5.69
0.2	2.200	2.68	2.40	2.33	2.29
0.1	1.100	1.36	1.21	1.17	1.15

Table 9.3: The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of 10^{-4}, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 9.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 9.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times a^2, is at most 4×10^{-5} in these units, an error estimate and not a bound.

Table 9.4 runs the proof of Theorem 9.2.1(b) with numbers. Write $\psi(\theta) = v_{1/2+\varepsilon}(\theta) - v_{1/2+\varepsilon}(0)$. The infimum c_1 of $\psi(\theta)/\min(\theta^2, 1)$ is the limit $v''_{1/2+\varepsilon}(0)/2$ at the origin, on a grid of 600 points to height 40 and with the assumption that ψ stays above its minimum on $[20, 40]$ beyond 40, and $\max_\kappa \kappa^2(1-\ell_0(\kappa)) = 0.5348$ at $\kappa = 1.54$, where $\ell_0(1) = 0.5726$ and $\ell_0(2) = 0.8806$. So the lower constant of the theorem is $c = 2c_1 \cdot 0.5348 = 0.2167\,\bar{c}_\varepsilon$ for $a \geq 1.54$. Using at every radius at once the bound $P(|\theta| \leq r) \leq \ell_0(ar)$ for the probability density $|\hat{h}|^2/\|\hat{h}\|_2^2$ gives the layer cake bound $\lambda_{\min} - 2b_{1/2+\varepsilon} \geq 2\int_0^\infty \tilde{\psi}'(r)\,(1-\ell_0(ar))\,dr$, with $\tilde{\psi}$ the nondecreasing minorant of ψ in $|\theta|$; it is $0.666\,\bar{c}_\varepsilon$ at $a=4$, $\varepsilon = \frac{1}{2}$ (3.66×10^{-4}), and tends to $8I_1/\pi^2\,\bar{c}_\varepsilon = 0.683\,\bar{c}_\varepsilon$ as $a\to\infty$, with $I_1 = \int_0^\infty \kappa\,(1-\ell_0(\kappa))\,d\kappa = 0.8425$. These lower bounds are floating point computations on a grid, not certificates, and for $\varepsilon < \frac{1}{2}$ the identity $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$ behind them presupposes that the level is to the right of the zeros.

The cosine trial function of the proof gives upper bounds, computed from the prime side (Table 9.4, last column). At $\varepsilon = \frac{1}{2}$ the ratio of its scaled excess to $\bar{c}_\varepsilon$ is 1.52, 1.37, 1.17 and 1.11 at $a = 1$, 2, 4 and 6, and a times its remainder over $\bar{c}_\varepsilon$ is 0.65 to 0.68 for $3 \leq a \leq 8$, a positive correction of order a^{-1}. At $\varepsilon = 0.2$ and 0.1 the remainder oscillates in a (a times it over $\bar{c}_\varepsilon$ lies between 1.4 and 2.2 at $\varepsilon = 0.2$ and between 2.3 and 4.8 at $\varepsilon = 0.1$ for $1.5 \leq a \leq 8$), because the tail of $|\hat{h}|^2$, of order $\cos^2(a\theta)/\theta^4$, meets the peaks of $v_{1/2+\varepsilon}$ at the zeros, of height $1/\varepsilon$, and the cosine is a poor trial function for small ε.

9.3 What can be done and what is open

9.3.1 What can be done

Partly, and exactly so: the deconvolution of the Thorin density always exists, as Weil's distribution computed from the primes with no hypothesis (Theorem 9.1.1), and its positivity to depth ε is a zero free half plane (Theorem 9.1.2), known for $\varepsilon = \frac{1}{2}$ and announced for $\varepsilon = \frac{3}{8}$, each rung $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \geq 0$ being a finite problem

ε	a	lower (κ)	lower (layer cake)	converged	upper (trial)
0.5	1	0.950 (1.00)	3.834	6.482	8.353
0.5	2	1.189 (1.54)	3.692	5.919	7.493
0.5	4	1.189 (1.54)	3.656	5.690	6.413
0.5	6	1.189 (1.54)	3.661	5.619	6.095
0.2	1	0.381 (1.00)	1.538	2.678	4.291
0.2	2	0.477 (1.54)	1.481	2.401	4.584
0.2	4	0.477 (1.54)	1.466	2.295	3.258
0.2	6	0.477 (1.54)	1.468	2.262	2.857
0.1	1	0.191 (1.00)	0.769	1.361	2.813
0.1	2	0.239 (1.54)	0.741	1.206	3.745
0.1	4	0.239 (1.54)	0.733	1.150	2.331
0.1	6	0.239 (1.54)	0.734	1.133	1.863

Table 9.4: Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of 10^{-4}: the lower bound of the proof of Theorem 9.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1-\ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 9.3, extended to $a = 6$), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound.

in the primes up to e^{2a} whose limit, under the hypotheses of Theorem 9.2.1, is a margin $2b_{1/2+\varepsilon}$ of about $0.092\,\varepsilon$, approached at a rate of order a^{-2}.

A rung is finite information in the sense of Section 1.4.2 (Theorem 1.4.1 is the rigorous form of this for the heat trace), so heuristically no finite set of rungs sees every height, and at $\varepsilon = 0$ the margin is 0 and, under RH, the approach to it is faster than any power of a (Theorem 9.2.1(c)): this is the deconvolution obstruction, consistent with the theorem of Rodgers and Tao that RH, if true, holds with no slack [250].

9.3.2 Open problems

Problem 9.1 (Certified rungs). Certify $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) > 0$ in interval arithmetic for $\varepsilon \in \{0.05, 0.1, 0.2, 0.375\}$ and $a = 4$, which by monotonicity in a covers every $a \le 4$, and determine how far in a a certificate can be pushed at fixed ε. Each certificate is unconditional; at $\varepsilon = \frac{3}{8}$ it agrees with the $\lambda_{\min} \ge 0$ that the announced half plane gives (Theorem 9.1.2) and checks it independently.

Problem 9.2 (The exact rate). Prove that, under the hypotheses of Theorem 9.2.1, $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) \to \pi^2 v''_{1/2+\varepsilon}(0)/4$, and find the next term. The converged values of Table 9.3 exceed the constant by a relative amount that falls like a^{-1}: a least squares fit on $2 \le a \le 7$ gives $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$, with $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$,

$C_0 = 1.0002,\ 1.0003,\ 1.0005$, $C_1 = 0.139,\ 0.159,\ 0.166$ and $C_2 = 0.037,\ 0.045,\ 0.051$ at $\varepsilon = \frac{1}{2},\ 0.2,\ 0.1$, and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$. Numerically, the correction is positive and $\bar{c}_\varepsilon$ is not a bound at finite a (the scaled excess is $1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$); the fit is numerical evidence for the limit, to within 0.04 to 0.2 percent, and for the order a^{-1} of the correction, and neither is proved.

Problem 9.3 (The transition at the line). Describe $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ jointly as $a \to \infty$ and $\varepsilon \to 0$. For fixed $\varepsilon > 0$, under the hypotheses of Theorem 9.2.1, the limit is $2b_{1/2+\varepsilon}$, which is at most $2V_\xi\,\varepsilon$ and asymptotic to it as $\varepsilon \downarrow 0$, and the rate is algebraic; at $\varepsilon = 0$, under RH, the decay is faster than any power. On which curves $\varepsilon = \varepsilon(a) \to 0$ does $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$ stay comparable to $2b_{1/2+\varepsilon}$, and does the crossover occur at $\varepsilon \approx$ the mean spacing of the zeros at height e^{2a}?

Problem 9.4 (Uniformity from structure). Find a structural reason, valid for all a, why $Q_\varepsilon \geq 0$ on $L^2[-a, a]$ for some fixed $\varepsilon < \frac{1}{2}$, using properties of the archimedean and pole terms that dominate the primes up to e^{2a} uniformly. By Theorem 9.1.2 any such argument proves a zero free half plane $\operatorname{Re} s > \frac{1}{2} + \varepsilon$.

Chapter 10

The level one measure and the backward Cauchy flow

Chapter 17 shows that for $\theta \geq 1$ the function $\xi(\theta)/\xi(\theta+\sqrt{s})$ is, unconditionally, the Laplace transform of a generalised gamma convolution, that its Thorin measure is the Cauchy smoothing of the zero measure at width $\theta - \operatorname{Re}\rho$, and that the family $\theta \mapsto U_\theta$ is a Cauchy semigroup above the zero free abscissa (Theorems 17.14.2 and 17.14.4). It asks whether the semigroup can be run backward from a level one measure computed without the zeros (Problem 17.5). This chapter applies the operator of Chapter 7 to that question in three ways.

First, the level one measure in closed form. Its density is $\pi^{-1}\operatorname{Re}(\xi'/\xi)(1+iv)$, a smooth gamma part minus the prime sum $\pi^{-1}\sum_{n\geq 2}\Lambda(n)n^{-1}\cos(v\log n)$, read as a compensated limit because the series itself diverges. The Fourier transform of the measure is, apart from the transform of the smooth part and a constant 1 from the pole of ζ, a sum of point masses $-\Lambda(n)/n$ at $\omega = \pm\log n$: the level one measure carries the primes in its spectrum and the zeros as the centres of its Cauchy components, and the identity between the two descriptions is a Weil type explicit formula for one analytic test function evaluated at the zeros.

Second, the backward flow. On the Fourier side the semigroup multiplies by $e^{-(\theta-\theta')|\omega|}$, so going from level 1 to level θ' multiplies by $e^{(1-\theta')|\omega|}$. The band limited backward flow converges to the level θ' measure if and only if ζ has no zeros to the right of θ'; a zero to the right makes the limit fail to be a measure μ with $\int |\mu|(dv)/(1+v^2) < \infty$. Numerically, from samples of the level one density on the line $\operatorname{Re} s = 1$ only, the flow places a peak within 0.035 of each of the 109 zeros tested, in $100 < v < 300$.

Third, the backward flow as a positive linear inverse problem of Vardi and Lee type with the Cauchy kernel as the blur. The EM algorithm of Shepp and Vardi, an information geometric analogue of von Neumann's alternating projection (Csiszár and Tusnády), gives a positivity preserving backward flow with a diagnostic, the total Kullback Leibler residual; its data augmentation Gibbs sampler is its Monte Carlo version; and generative Bayesian computation amortises a

posterior under a GUE prior, trained on simulations. The EM inversion locates the 109 zeros without ringing, and a redesigned injection test separates an injected off line quartet from a control; the GBC implementation, rerun here with a corrected forward model, counts the zeros in unit bins almost exactly.

None of this is a path to the Riemann hypothesis. It is a dictionary in which RH reads as the statement that the backward flow runs all the way to $\theta = \frac{1}{2}$, and in which the exponential amplification $e^{(1-\theta')|\omega|}$ is the classical error term $\psi(x) - x = O(x^{\theta'+\varepsilon})$ in Fourier clothing.

This chapter is based on [142]. Its numbers are recomputed, with the sampler of u_1 and the EM iteration of Chapter 11, unless marked as the author's or the review's; where the recomputation differed from the source the recomputed value is printed.

Notation. $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, a zero is $\rho = \sigma_\rho + i\gamma_\rho$, and $P_c(v) = (c/\pi)/(c^2+v^2)$ is the Cauchy kernel of width $c > 0$, with $P_0 = \delta_0$ (the Poisson kernel of Section 2.1, with the same index). The Thorin measure $\mathfrak{U}_\theta$ (Section 10.1) carries no factor $\frac{1}{2}$, so it is twice the measure $\widetilde{U}_\theta$ of Theorem 17.14.4; its density is $\rho_{\theta-1/2}$ of Chapter 8. The digamma function is written Γ'/Γ, because $\psi(x) = \sum_{n \le x} \Lambda(n)$ is Chebyshev's function here; the grid spacing of the numerics is h and the test function of the explicit formula is φ, the Fourier transform being $\widehat{f}(\omega) = \int e^{-i\omega v} f(v)\, dv$.

Nothing in this chapter is certified in ball arithmetic except the sampler of u_1, evaluated with `python-flint` balls at 128 bits and checked at 192. The results are these.

1. The level one Thorin measure of ξ has the explicit real analytic density (10.1.3), for every real v including $v = 0$, where the prime sum is a compensated limit with the value $u_1(0) = (2 + \gamma_E - \log 4\pi)/(2\pi)$ (Theorem 10.1.1, proved; for $v \neq 0$ Propositions 12.1.2 and 12.1.3). Its positivity is automatic from the Euler product; the nonvanishing of ζ on $\operatorname{Re} s = 1$ is equivalent to the absence of atoms (Section 10.6, proved).

2. The Fourier transform of the measure is the sum of prime atoms, the transform of the smooth part and the constant 1, and the identity of the two descriptions is (10.2.2), an explicit formula for one analytic test function (Proposition 10.2.1, proved). The spectral peaks lie at $\log n$ to 10^{-4} (Table 10.1, computed).

3. The band limited backward flow converges at level θ' if and only if ζ has no zeros in $\operatorname{Re} s > \theta'$, and RH is the convergence at every $\theta' > \frac{1}{2}$ (Theorem 10.3.1, proved). The flow recovers 109 zeros to 0.035 at $\theta' = \frac{3}{4}$ and to 0.027 at $\theta' = \frac{1}{2}$; the sharp cutoff rings and the Fejér window does not (Section 10.3.1, computed).

4. The existence of a positive solution of the deconvolution $P_{1-\theta'} * x = u_1$ is

the zero free half plane $\operatorname{Re} s > \theta'$, and the total Kullback Leibler residual is bounded away from zero as the grid is refined exactly when a zero lies to the right of θ' (Proposition 10.4.1, proved, with a counterexample to the residual per unit length). The EM runs and a quartet injection test with controls are computed (Section 10.4.4).

5. The data augmentation sampler needs an exposure and pseudo counts; it is a method and carries no result. Generative Bayesian computation under a GUE prior counts the zeros in unit bins with 98 per cent exact bins on the 200 test bins, for a corrected forward model (Section 10.5, computed; the formulation is heuristic).

6. None of this is a path to RH (Section 10.6); four open problems close the chapter (Section 10.7).

10.1 The level one measure

Throughout, $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, zeros are $\rho = \sigma_\rho + i\gamma_\rho$ counted with multiplicity, and $\theta_0 = \sup \sigma_\rho \le 1$. For every $\theta \ge \theta_0$, that is for every θ such that ζ has no zeros in $\operatorname{Re} s > \theta$ (in particular for every $\theta \ge 1$, unconditionally), let

$$\mathfrak{U}_\theta = \sum_\rho \operatorname{Cauchy}(\gamma_\rho, \theta - \sigma_\rho), \qquad \operatorname{Cauchy}(\gamma, 0) = \delta_\gamma, \tag{10.1.1}$$

the sum over the zeros of both half planes, a positive measure on $\mathbb{R}$ in the variable v with $\int \mathfrak{U}_\theta(dv)/(1+v^2) < \infty$. It is twice the symmetric measure $\widetilde{U}_\theta$ of Theorem 17.14.4, and the image of its restriction to $v > 0$ under $v \mapsto v^2$ is the Thorin measure U_θ, each conjugate pair of zeros contributing one term.

Theorem 10.1.1 (density of the Thorin measure; at $\theta = 1$, $v \ne 0$, Propositions 12.1.2 and 12.1.3). *If $\theta > \sigma_\rho$ for every ρ (for instance if $\theta > \theta_0$, and for every $\theta \ge 1$ because $\zeta(1+it) \ne 0$), the measure $\mathfrak{U}_\theta$ is absolutely continuous with real analytic density*

$$u_\theta(v) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\theta + iv) = \frac{1}{\pi}\, \partial_\sigma \log |\xi(\sigma + iv)|\Big|_{\sigma=\theta}, \tag{10.1.2}$$

and $\mathfrak{U}_\theta((0,V]) = \frac{1}{\pi}\arg \xi(\theta + iV)$ for the branch with $\arg \xi(\theta) = 0$. At $\theta = 1$,

$$\begin{aligned} u_1(v) = \frac{1}{\pi}\Big[&\frac{1}{1+v^2} - \frac{\log \pi}{2} + \frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma}\Big(\frac{1+iv}{2}\Big) \\ &- \lim_{X\to\infty} \Big(\sum_{n \le X} \frac{\Lambda(n)}{n} \cos(v \log n) - \frac{\sin(v \log X)}{v} \Big) \Big] \end{aligned} \tag{10.1.3}$$

for every real v, where $\sin(v \log X)/v$ means $\log X$ at $v = 0$. The limit exists, locally uniformly in v, and equals $\operatorname{Re} C(1+iv)$ with $C(s) = -\zeta'(s)/\zeta(s) - 1/(s-1)$:

it is $-\operatorname{Re}(\zeta'/\zeta)(1+iv)$ *for* $v \neq 0$ *and minus Euler's constant at* $v = 0$*, while the cosine series itself has no limit for any real* v *(compare Proposition 12.1.3). For* $\theta > 1$*,*

$$u_\theta(v) = \frac{1}{\pi}\Big[\frac{\theta}{\theta^2+v^2} + \frac{\theta-1}{(\theta-1)^2+v^2} - \frac{\log\pi}{2} + \frac{1}{2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac{\theta+iv}{2}\Big) - \sum_{n\geq 2}\frac{\Lambda(n)}{n^\theta}\cos(v\log n)\Big],$$

the prime sum converging absolutely.

Proof. Each term of (10.1.1) is a probability measure, and for $0 \leq c \leq \theta$

$$\int \frac{P_c(v-\gamma)}{1+v^2}\,dv = \pi P_{1+c}(\gamma) = \frac{1+c}{(1+c)^2+\gamma^2} \leq \frac{1+\theta}{1+\gamma^2};$$

since $\sum_\rho (1+\gamma_\rho^2)^{-1} < \infty$, the sum converges to a positive measure with $\int \mathfrak{U}_\theta(dv)/(1+v^2) < \infty$. If every width $\theta - \sigma_\rho$ is positive, fix $R > 0$. The finitely many zeros with $|\gamma_\rho| \leq R + 2\theta + 2$ give Cauchy densities of positive width, which are real analytic; for the others the densities, continued to $|\operatorname{Re} z| \leq R$, $|\operatorname{Im} z| \leq \frac{1}{2}$, are holomorphic and bounded by $2\theta/(\pi(|\gamma_\rho| - R)^2)$, so their sum is holomorphic there. Hence $\mathfrak{U}_\theta$ has a real analytic density.

By the Hadamard product with conjugate pairing, $(\xi'/\xi)(s) = \sum_\rho 1/(s-\rho)$, so

$$\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\theta+iv) = \frac{1}{\pi}\sum_\rho \frac{\theta-\sigma_\rho}{(\theta-\sigma_\rho)^2+(v-\gamma_\rho)^2},$$

which is the density of (10.1.1). The second form in (10.1.2) is the Cauchy Riemann equation for $\log\xi$, and integrating $\partial_v \arg\xi = \partial_\sigma \log|\xi|$ gives the mass formula. For (10.1.3), $\xi'/\xi = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\Gamma'/\Gamma(s/2) + \zeta'/\zeta$, so $u_1 = g_1 - \pi^{-1}\operatorname{Re} C(1+iv)$, where g_1 collects the first three terms of (10.1.3) and $C(s) = -\zeta'(s)/\zeta(s) - 1/(s-1)$ is analytic on a neighbourhood of $\operatorname{Re} s \geq 1$, because $\zeta(1+it) \neq 0$ and the pole of ζ cancels; at $v = 0$ this is the value of the continuous function u_1. Write $\sum_{n\leq t}\Lambda(n) = t + R(t)$ with $R(t) = O(te^{-c\sqrt{\log t}})$. Partial summation gives, for $\operatorname{Re} s \geq 1$,

$$\sum_{n\leq X}\frac{\Lambda(n)}{n^s} - \int_1^X \frac{dt}{t^s} = \frac{R(X)}{X^s} - R(1) + s\int_1^X \frac{R(t)}{t^{s+1}}\,dt,$$

and as $X \to \infty$ the right side converges, uniformly for s in compact subsets of $\operatorname{Re} s \geq 1$, because $\int^\infty e^{-c\sqrt{\log t}}\,dt/t < \infty$. For $\operatorname{Re} s > 1$ the left side tends to $C(s)$, so by continuity the limit is $C(s)$ on $\operatorname{Re} s = 1$ too. At $s = 1+iv$ the real part of $\int_1^X t^{-s}\,dt$ is $\sin(v\log X)/v$, and $\log X$ at $v = 0$, which has no limit as $X \to \infty$; this gives the compensated limit in (10.1.3) and the divergence of the cosine series. For $\theta > 1$ the Dirichlet series converges absolutely, and $\operatorname{Re} 1/(s-1) = (\theta-1)/((\theta-1)^2+v^2)$ at $s = \theta + iv$. □

At $v=0$ the compensated limit is $\lim(\sum_{n\le X}\Lambda(n)/n-\log X)=-\gamma_E$, and (10.1.3) gives $u_1(0)=(1+\gamma_E/2-\log 2-\frac12\log\pi)/\pi=(2+\gamma_E-\log 4\pi)/(2\pi)=0.00735159249234$, the value of the continuous function u_1 at the origin (computed, to 30 digits; the values $u_1(10^{-3})-u_1(0)=3.5\times10^{-11}$ and $u_1(0.05)-u_1(0)=8.8\times10^{-8}$ confirm it). It is also $(2\pi)^{-1}\sum_\rho 1/(\rho(1-\rho))$.

So the level one Thorin measure of the reciprocal xi function is an explicit function of the primes, with no reference to zeros, and a positive measure by Theorem 17.14.2. Positivity of (10.1.3) is itself a statement about primes: it says that the limit in (10.1.3) is at most $1/(1+v^2)-\frac12\log\pi+\frac12\operatorname{Re}\Gamma'/\Gamma((1+iv)/2)$, the right side being $\frac12\log(v/2\pi)+O(1/v^2)$. It follows from the Euler product (Proposition 12.1.2). Numerically, the mass of u_1 on $[0,400]$ is 201.67 and there are 202 zeros below height 400 (computed: Simpson on step 0.05 gives 201.66682, and the formula below over the first 1700 ordinates plus the smooth tail gives 201.66681), the difference coming from the Cauchy smoothing of the zero count,

$$\mathfrak{U}_1((0,V])=\frac1\pi\sum_{\gamma_\rho>0}\Big[\arctan\frac{V-\gamma_\rho}{1-\sigma_\rho}+\arctan\frac{V+\gamma_\rho}{1-\sigma_\rho}\Big].$$

10.2 The Fourier side: primes

Write $\widehat{\mu}(\omega)=\int e^{-i\omega v}\mu(dv)$ for the Fourier transform of a tempered measure.

Proposition 10.2.1 (spectrum of the level one measure). *For $\theta\ge1$ let $g_\theta(v)=\frac1\pi\big[\theta/(\theta^2+v^2)-\frac12\log\pi+\frac12\operatorname{Re}\frac{\Gamma'}{\Gamma}((\theta+iv)/2)\big]$. As tempered distributions,*

$$\widehat{\mathfrak{U}_1}(\omega)=\widehat{g_1}(\omega)+1-\sum_{n\ge2}\frac{\Lambda(n)}{n}\big(\delta_{\log n}+\delta_{-\log n}\big)(\omega),\tag{10.2.1}$$

the constant 1 being the transform of the atom δ_0 that the pole of ζ at $s=1$ contributes in the limit $\theta\downarrow1$; on the zero side $\widehat{\mathfrak{U}_1}(\omega)=\sum_\rho e^{-(1-\sigma_\rho)|\omega|}e^{-i\omega\gamma_\rho}$, the limit of the sums over $|\gamma_\rho|<T$. Equivalently, for every $\varphi\in\mathcal{S}(\mathbb{R})$,

$$\sum_\rho\big(P_{1-\sigma_\rho}*\varphi\big)(\gamma_\rho)=\int g_1\varphi\,dv+\varphi(0)-\frac{1}{2\pi}\sum_{n\ge2}\frac{\Lambda(n)}{n}\big(\widehat{\varphi}(\log n)+\widehat{\varphi}(-\log n)\big),\tag{10.2.2}$$

with P_c the Cauchy kernel of width c. For $\theta\ge1$ the semigroup $\mathfrak{U}_\theta=P_{\theta-1}*\mathfrak{U}_1$ reads, since $g_\theta=P_{\theta-1}*g_1$, $\widehat{\mathfrak{U}_\theta}(\omega)=\widehat{P_{\theta-1}*\mathfrak{U}_1}(\omega)=e^{-(\theta-1)|\omega|}\widehat{\mathfrak{U}_1}(\omega)=\widehat{g_\theta}(\omega)+e^{-(\theta-1)|\omega|}-\sum_{n\ge2}\Lambda(n)n^{-\theta}(\delta_{\log n}+\delta_{-\log n})(\omega)$, which on the prime side is the replacement of $\Lambda(n)/n$ by $\Lambda(n)n^{-\theta}$.

Proof. For $1<\theta\le2$ the formula of Theorem 10.1.1 reads

$$u_\theta=g_\theta+P_{\theta-1}-\pi^{-1}\sum_n\Lambda(n)n^{-\theta}\cos(v\log n),$$

absolutely convergent. Integrating against $\varphi \in \mathcal{S}(\mathbb{R})$ and using

$$\int \varphi(v)\cos(v\log n)\,dv = \tfrac{1}{2}\big(\widehat{\varphi}(\log n) + \widehat{\varphi}(-\log n)\big),$$

we get

$$\sum_{\rho}(P_{\theta-\sigma_\rho} * \varphi)(\gamma_\rho) = \int g_\theta \varphi\,dv + (P_{\theta-1} * \varphi)(0) - \frac{1}{2\pi}\sum_{n\ge 2}\frac{\Lambda(n)}{n^{\theta}}\big(\widehat{\varphi}(\log n) + \widehat{\varphi}(-\log n)\big).$$

Let $\theta \downarrow 1$. On the left, splitting the integral at $|v-\gamma| = |\gamma|/2$ gives $|(P_c * \varphi)(\gamma)| \le C_\varphi/(1+\gamma^2)$ uniformly for $0 \le c \le 2$; each term is continuous in c and $\sum_\rho (1+\gamma_\rho^2)^{-1} < \infty$, so dominated convergence over the zeros applies. On the right, $g_\theta \to g_1$ pointwise with $|g_\theta(v)| \le C\log(2+|v|)$ for $1 \le \theta \le 2$, $(P_{\theta-1} * \varphi)(0) \to \varphi(0)$, and $\sum_n \Lambda(n)n^{-1}|\widehat{\varphi}(\pm\log n)| < \infty$ because $\widehat{\varphi}(\omega) = O((1+|\omega|)^{-3})$ and $\sum_{n\le x}\Lambda(n)/n = \log x + O(1)$. This proves (10.2.2). Its left side is $\int \varphi\, d\mathfrak{U}_1$, and $\int f\varphi\,dv = \frac{1}{2\pi}\int \widehat{f}(\omega)\widehat{\varphi}(-\omega)\,d\omega$, so (10.2.2) for all $\varphi \in \mathcal{S}(\mathbb{R})$ is (10.2.1). The zero side is the transform of (10.1.1) term by term, $\widehat{\text{Cauchy}(\gamma,c)}(\omega) = e^{-c|\omega|}e^{-i\omega\gamma}$, the sums over $|\gamma_\rho| < T$ converging in $\mathcal{S}'$ by the same bound. For $\theta > 1$ the computation without the limit gives the last formula, and $\mathfrak{U}_\theta = P_{\theta-1} * \mathfrak{U}_1$ holds term by term because $P_{\theta-1} * P_c = P_{\theta-1+c}$. □

The identity (10.2.2) is an explicit formula of Weil's type [273]. For a conjugate pair of zeros $\rho, \bar{\rho}$, with

$$H_\varphi(z) = \frac{1}{2\pi}\int_0^\infty \big(\widehat{\varphi}(t) + \widehat{\varphi}(-t)\big)e^{(z-1)t}\,dt,$$

one has $(P_{1-\sigma} * \varphi)(\gamma) + (P_{1-\sigma} * \varphi)(-\gamma) = H_\varphi(\rho) + H_\varphi(\bar{\rho})$, so the left side of (10.2.2) is $\sum_\rho H_\varphi(\rho)$, summed over conjugate pairs: one test function evaluated at the zeros. Under RH it is the classical explicit formula tested against $P_{1/2} * \varphi$ (compare Theorem 9.1.1). With Theorem 9.1.1, it identifies the Thorin measure with the object of the explicit formula. As a check, for $\varphi(v) = e^{-v^2/2}$ the two sides of (10.2.2) agree to thirteen digits in the form $\int (u_1 - g_1)\varphi\,dv = \varphi(0) - (2\pi)^{-1}\sum_n \Lambda(n)n^{-1}(\widehat{\varphi}(\log n) + \widehat{\varphi}(-\log n))$, both equal to 0.4267700651131 (computed in the review of the source with 120 point Gauss and Legendre quadrature and 20 digit arithmetic, not rerun here).

The spectral peaks of u_1 sit at $\log 2, \log 3, \log 4, \log 5, \log 7, \ldots$ with heights roughly proportional to $\Lambda(n)/n$. Table 10.1 shows them recovered from a Hann windowed Fourier transform of u_1 sampled on $[-400, 400]$ at spacing 0.05 (by symmetry), padded to a frequency step of 3×10^{-5}.

n	peak ω	e^{ω}	relative height	$\Lambda(n)/n$ relative to $n=3$
2	0.6931	2.000	0.950	0.946
3	1.0986	3.000	1.000	1.000
4	1.3863	4.000	0.449	0.473
5	1.6095	5.000	0.872	0.879
7	1.9459	7.000	0.747	0.759
8	2.0794	8.000	0.202	0.237
9	2.1972	9.000	0.303	0.333
11	2.3979	11.000	0.576	0.595
13	2.5650	13.000	0.517	0.539
17	2.8332	17.000	0.430	0.455
19	2.9444	19.000	0.397	0.423
23	3.1355	23.000	0.343	0.372
29	3.3673	29.000	0.286	0.317
31	3.4340	31.001	0.271	0.302

Table 10.1: Spectral peaks of the level one Thorin measure (computed). Every peak lies within 10^{-4} of the logarithm of a prime or prime power. The heights, relative to that of $n=3$, depend on the window and follow the relative value of $\Lambda(n)/n$ to within 0.04.

10.3 The backward flow

Fix $\frac{1}{2} \le \theta' < 1$, put $a = 1-\theta'$, and let $\Delta_\Omega(\omega) = (1-|\omega|/\Omega)_+$ be the Fejér window. Let $K_{\theta',\Omega}$ be the inverse Fourier transform of $\Delta_\Omega(\omega)e^{a|\omega|}$; this multiplier is even, continuous, compactly supported and has a derivative of bounded variation, so $K_{\theta',\Omega}$ is smooth and $|K_{\theta',\Omega}(v)| \le C_{\Omega,a}/(1+v^2)$. The band limited backward flow from level one to level θ' is

$$B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_1 * K_{\theta',\Omega}, \tag{10.3.1}$$

a smooth function of growth $O(\log(2+|v|))$, since $\mathfrak{U}_1([t,t+1]) = O(\log(2+|t|))$; formally its transform is $\Delta_\Omega(\omega)e^{a|\omega|}\widehat{\mathfrak{U}_1}(\omega)$. Let $\Phi_\Omega(v) = (1-\cos\Omega v)/(\pi\Omega v^2)$ be the Fejér kernel, the inverse transform of Δ_Ω.

Theorem 10.3.1 (dichotomy for the backward flow). *Let $\frac{1}{2} \le \theta' < 1$ and $a = 1-\theta'$.*

(i) If ζ has no zeros in $\operatorname{Re} s > \theta'$, then $B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * \Phi_\Omega$, with $\mathfrak{U}_{\theta'}$ as in (10.1.1), which has atoms at the zeros with $\sigma_\rho = \theta'$. As $\Omega \to \infty$, $B_{\theta',\Omega}\mathfrak{U}_1 \to \mathfrak{U}_{\theta'}$ vaguely, and $\int B_{\theta',\Omega}\mathfrak{U}_1(v)\,dv/(v^2+s) \to \int \mathfrak{U}_{\theta'}(dv)/(v^2+s)$ for every $s>0$. If $\theta' > \sigma_\rho$ for every ρ, the convergence is locally uniform as densities.

(ii) If ζ has a zero ρ with $\sigma_\rho > \theta'$, there is no signed measure μ with $\int |\mu|(dv)/(1+v^2) < \infty$ such that $\int B_{\theta',\Omega}\mathfrak{U}_1(v)\,dv/(v^2+s) \to \int \mu(dv)/(v^2+s)$ as $\Omega \to \infty$ for all sufficiently large s.

Proof. Write $f_s(v) = 1/(v^2+s)$, so that $f_s = (\pi/\sqrt{s})P_{\sqrt{s}}$ and $\widehat{f_s}(\omega) = (\pi/\sqrt{s})e^{-\sqrt{s}|\omega|}$. The measures below are positive or satisfy $\int |\mu|/(1+v^2) < \infty$, and the kernels are $O(1/(1+v^2))$, so every use of Fubini's theorem is absolute.

(i) Every zero has $\sigma_\rho \le \theta'$, so $\mathfrak{U}_{\theta'}$ is defined and, term by term, $\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * P_a$, because $P_c * P_a = P_{c+a}$ for $c \ge 0$ (Theorem 17.14.4). The integrable function $P_a * K_{\theta',\Omega}$ has transform $e^{-a|\omega|}\Delta_\Omega(\omega)e^{a|\omega|} = \Delta_\Omega(\omega)$, so $P_a * K_{\theta',\Omega} = \Phi_\Omega$ and $B_{\theta',\Omega}\mathfrak{U}_1 = \mathfrak{U}_{\theta'} * P_a * K_{\theta',\Omega} = \mathfrak{U}_{\theta'} * \Phi_\Omega$. The kernel Φ_Ω is nonnegative, has integral one and satisfies $\Phi_\Omega(u) \le 2/(\pi\Omega u^2)$. For f continuous with support in $[-R,R]$, $\Phi_\Omega * f \to f$ uniformly and $|\Phi_\Omega * f(v)| \le 8\|f\|_1/(\pi\Omega v^2)$ for $|v| \ge 2R$, so $|\Phi_\Omega * f - f| \le \eta_\Omega/(1+v^2)$ with $\eta_\Omega \to 0$, and $\int f\, d(\mathfrak{U}_{\theta'} * \Phi_\Omega) = \int (\Phi_\Omega * f)\, d\mathfrak{U}_{\theta'} \to \int f\, d\mathfrak{U}_{\theta'}$. For $f = f_s$, $\Phi_\Omega * f_s \to f_s$ pointwise and $\Phi_\Omega * f_s \le C_s/(1+v^2)$ for $\Omega \ge 1$, and dominated convergence gives the second statement. If $\theta' > \sigma_\rho$ for every ρ, $\mathfrak{U}_{\theta'}$ has a continuous density $u_{\theta'}$ (Theorem 10.1.1). For $|v| \le R$ and $0 < \delta \le 1$, $|\Phi_\Omega * u_{\theta'}(v) - u_{\theta'}(v)|$ is at most the largest change of $u_{\theta'}$ over distances δ on $[-R-1, R+1]$, plus $(2/(\pi\Omega)) \int_{|t|\ge\delta} u_{\theta'}(v-t)t^{-2}\, dt + (4/(\pi\Omega\delta)) \max_{[-R,R]} u_{\theta'}$, and the last two terms are $O_{R,\delta}(1/\Omega)$.

(ii) Let $s > a^2$ and $b = \sqrt{s} - a > 0$. The integrable function $K_{\theta',\Omega} * f_s$ has transform $(\pi/\sqrt{s})\Delta_\Omega(\omega)e^{-b|\omega|}$, so $K_{\theta',\Omega} * f_s = (\pi/\sqrt{s})\Phi_\Omega * P_b = (b/\sqrt{s})\Phi_\Omega * f_{b^2}$ and

$$\int B_{\theta',\Omega}\mathfrak{U}_1(v) f_s(v)\, dv = \int (K_{\theta',\Omega} * f_s)\, d\mathfrak{U}_1 = \frac{b}{\sqrt{s}} \int (\Phi_\Omega * f_{b^2})\, d\mathfrak{U}_1$$
$$\longrightarrow \frac{b}{\sqrt{s}} \int \frac{\mathfrak{U}_1(dv)}{v^2+b^2}$$

as $\Omega \to \infty$, by the domination used in (i). Since $\int P_c(v-\gamma) f_{b^2}(v)\, dv = (\pi/b)P_{b+c}(\gamma) = b^{-1}\operatorname{Re}(b+c-i\gamma)^{-1}$, summing over the zeros with $c = 1-\sigma_\rho$ gives

$$\int \frac{\mathfrak{U}_1(dv)}{v^2+b^2} = \frac{1}{b}\sum_\rho \operatorname{Re}\frac{1}{1+b-\rho} = \frac{1}{b}\,\frac{\xi'}{\xi}(1+b),$$

so the limit is $(\xi'/\xi)(\theta' + \sqrt{s})/\sqrt{s}$. If a measure μ as in (ii) existed, then for all large $s > 0$

$$G(s) := \int \frac{\mu(dv)}{v^2+s} = \frac{1}{\sqrt{s}}\,\frac{\xi'}{\xi}(\theta' + \sqrt{s}).$$

On a compact subset of $\mathbb{C}\setminus(-\infty,0]$ one has $|v^2+s| \ge c(1+v^2)$, so G is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. The right side, with the principal square root, is meromorphic on $\mathbb{C} \setminus (-\infty, 0]$, and the identity theorem on the connected set $\mathbb{C} \setminus (-\infty, 0]$ minus its poles makes G equal to it there. Squaring maps $\operatorname{Re} w > 0$ onto $\mathbb{C} \setminus (-\infty, 0]$, so $s_0 = (\rho - \theta')^2 \in \mathbb{C} \setminus (-\infty, 0]$ with $\sqrt{s_0} = \rho - \theta'$, and near s_0 the right side is $2m_\rho/(s-s_0) + O(1)$, where $m_\rho \ge 1$ is the multiplicity of ρ, while G is bounded near s_0. This contradiction proves (ii). □

The theorem is the Thorin criterion of Chapter 17 in dynamical form, with part (ii) proved by the pole argument of Theorem 17.14.2: RH is the convergence of the backward flow from the explicit level one measure at every level $\theta' > \frac{1}{2}$. The flow relocates the difficulty into the multiplier $e^{(1-\theta')|\omega|}$, which amplifies whatever is not known about $\widehat{\mathfrak{U}_1}$ at high frequency. On the prime side, high frequency means large n: the content of $\widehat{\mathfrak{U}_1}$ on $|\omega| \le \Omega$ is $\sum_{n \le e^{\Omega}} \Lambda(n)/n$, and the backward flow to level θ' heuristically asks for $\sum_{n \le x} \Lambda(n) n^{-\theta'}$ to behave, which is the statement $\psi(x) - x = O(x^{\theta'+\varepsilon})$ for every $\varepsilon > 0$ (the classical equivalence with the zero free half plane $\operatorname{Re} s > \theta'$).

10.3.1 Numerical backward flow

We sampled u_1 on $[0, 400]$ at spacing 0.05 from (10.1.2) with $\theta = 1$, evaluating ζ and ζ' on the line $\operatorname{Re} s = 1$ only (`python-flint`, 128 bits), with the analytic value $u_1(0)$ at $v = 0$, extended by symmetry, applied a Hann window and the multiplier of (10.3.1) with a sharp cutoff at Ω, not the Fejér window of (10.3.1), inverted and divided by the window. A peak is a local maximum of the output in $100 < v < 300$ of height at least a quarter of the maximum, and the peaks are compared with the 109 zeros of ζ in that range ($N(300) - N(100) = 138 - 29$). The Fejér variant multiplies the sharp multiplier by $1 - |\omega|/\Omega$.

θ'	Ω	cutoff	peaks	zeros	max error	min value
0.75	40	sharp	109	109	0.035	+0.075
0.50	40	sharp	109	109	0.027	−3.2
0.50	60	sharp	213	109	0.042	−16.5
0.50	60	Fejér	109	109	0.025	−0.020

Table 10.2: The Fourier backward flow from samples of u_1 on $\operatorname{Re} s = 1$ (computed). The error is the largest distance between a zero in $100 < v < 300$ and the nearest peak. At $\Omega = 60$ the sharp cutoff has 213 peaks of height at least a quarter of the maximum, 109 of them at the zeros, and the maximum is +26.6.

At $\theta' = 0.75$ the flow produces a positive function with 109 peaks, each within 0.035 of a zero, as Theorem 10.3.1(i) predicts if ζ has no zeros in $\operatorname{Re} s > 3/4$: the level $\frac{3}{4}$ measure is then a smooth positive density whose peaks lie near the zeros. At $\theta' = 0.5$ the peaks lie within 0.027 of the zeros, but the function oscillates below zero between them, because of the sharp cutoff: its kernel $\sin(\Omega v)/(\pi v)$ has negative lobes, the first of depth about $0.22\,\Omega/\pi \approx 2.8$ at $\Omega = 40$, whereas under RH the Fejér approximant $\mathfrak{U}_{1/2} * \Phi_\Omega$ of Theorem 10.3.1(i) is nonnegative, because $\Phi_\Omega \ge 0$. Theorem 10.3.1 does not cover the sharp cutoff. The Fejér run at $\Omega = 60$ is clean: 109 peaks, error 0.025 and minimum −0.020 (the small negative values come from the Hann window and its division); at $\Omega = 40$ its minimum is −0.011.

The cutoff is limited by the accuracy of the samples: at level $\frac{1}{2}$ the multiplier

at the cutoff is $e^{\Omega/2}$, about 10^{13} at $\Omega = 60$, so an error of 10^{-7} in the single sample at $v = 0$ already produces oscillations of order 10^2 in $100 < v < 300$. This is what the source printed: its $\Omega = 60$ row, with 1022 peaks, error 1.9 and minimum -430, has its minimum reproduced only if the sample at $v = 0$, the pole of ζ, is replaced by the sample at $v = 0.05$ (an error of 8.8×10^{-8}; minimum -430.2, maximum $+430.9$; the peak count and the error are not reproduced, the source giving no peak rule), and the source read it as amplified rounding error. With the analytic $u_1(0)$ the output of the sharp cutoff at $\Omega = 60$ is not noise but the ringing of the cutoff: each atom gives a sinc of height $\Omega/\pi \approx 19$ and the sidelobes of 213 peaks follow; double precision rounding alone would give oscillations of order 3×10^{-4}. The line $\operatorname{Re} s = 1$ determines the zeros, and the 109 zeros tested are read off it to 0.025 to 0.035 (two decimals, not the three decimals and the first two hundred zeros of the source), but at level $\frac{1}{2}$ the precision required of the level one data grows like $e^{\Omega/2}$ with the cutoff.

10.4 The backward flow as a Bayesian inverse problem

10.4.1 Positive linear inverse problem

Discretise. Let $y_i = u_1(v_i)$ on a grid v_i of spacing h covering the data window, let $x_j \geq 0$ be the unknown densities of $\mathfrak{U}_{\theta'}$ on a grid w_j of the same spacing covering a larger support (cell masses hx_j), and let

$$y = Ax, \qquad A_{ij} = hP_{1-\theta'}(v_i - w_j). \tag{10.4.1}$$

Every entry of A is positive, and a column centred well inside the data window has $\sum_i A_{ij} \approx 1$. This is a positive linear inverse problem in the sense of Vardi and Lee [200], the structure of emission tomography, image deblurring and the portfolio problem of their title, with the Cauchy kernel as the blur. Two features are unusual: the data are exact, so the problem is one of existence rather than estimation, and the existence question is the zero free half plane $\operatorname{Re} s > \theta'$, the Riemann hypothesis when $\theta' = \frac{1}{2}$: by Proposition 10.4.1(a), a positive solution of the continuous problem exists if and only if ζ has no zeros to the right of θ'.

10.4.2 Likelihood, prior, posterior

The data y_i are exact real numbers, so a Poisson model is a device. Read hy_i as the expected count in cell i of a Poisson process whose intensity is the Cauchy blur of an unknown positive measure: the complete data are counts z_{ij} of mass originating in cell j and observed in cell i, Poisson with means $hA_{ij}x_j$, and the observed counts are $\sum_j z_{ij}$. Replacing these counts by the data hy_i, the log likelihood of x is, up to constants and the factor h,

$$\ell(x) = \sum_i \big(y_i \log (Ax)_i - (Ax)_i \big),$$

concave in x, and maximising it over $x \geq 0$ minimises the generalised Kullback Leibler divergence $\mathrm{KL}(y \,\|\, Ax) = \sum_i [y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i]$, which is defined for real data. The EM algorithm with the z_{ij} as missing data is the Shepp and Vardi iteration [169]

$$x_j^{(k+1)} = \frac{x_j^{(k)}}{\sum_i A_{ij}} \sum_i A_{ij} \frac{y_i}{(Ax^{(k)})_i}, \tag{10.4.2}$$

which is also the Richardson and Lucy deblurring algorithm [164, 115]. It preserves positivity at every step, increases the likelihood monotonically, and, from a strictly positive start, converges to a maximum likelihood solution [206]. Csiszár and Tusnády [52] identified it as alternating I projection between the set of blurred positive measures and the set of distributions consistent with the data, an information theoretic analogue of von Neumann's alternating projection theorem for subspaces of a Hilbert space.

A Bayesian completes the model with a prior on x, the conjugate choice being a product of gamma laws. A sampler needs integer counts, which exact data do not supply: fix an exposure $\tau > 0$ and let N_i be the integer nearest to $\tau h y_i$, modelled as $N_i = \sum_j Z_{ij}$ with independent Z_{ij}, Poisson with means $\tau h A_{ij} x_j$. Given x and N, the vector $(Z_{ij})_j$ is multinomial with total N_i and probabilities $A_{ij}x_j/(Ax)_i$; given Z, with independent $\mathrm{Gamma}(\alpha_j, \beta_j)$ priors, x_j is gamma with shape $\alpha_j + \sum_i Z_{ij}$ and rate $\beta_j + \tau h \sum_i A_{ij}$. Alternating the two steps is a data augmentation Gibbs sampler, the Monte Carlo counterpart of (10.4.2) with the E step replaced by simulation: the EM fixed point becomes a posterior over positive measures, whose spread depends on the exposure τ, which exact data do not determine. The sampler is a method; no run of it is reported here (Problem 10.1).

The generative Bayesian computation of Section 10.5 amortises the posterior of a different model: instead of a Gibbs chain for each data set, one simulates pairs from a prior on configurations of zeros and the forward map, trains a quantile network to represent conditional quantiles of chosen functionals given the data, and evaluates it at the observed data. The three methods share the forward map but not the prior or the unknown: EM gives the posterior mode under a flat prior on the grid masses, data augmentation samples the posterior under a gamma prior on the same masses, and GBC regresses functionals of a GUE configuration of zeros on simulated data.

10.4.3 What positivity buys

The Fourier backward flow of Section 10.3 ignores positivity, which is why the sharp cutoff rings below zero at $\theta' = \frac{1}{2}$ and why its error is amplified by $e^{(1-\theta')\Omega}$. The EM inversion enforces positivity and uses the whole frequency content of the data; its total residual is the diagnostic of Proposition 10.4.1.

PROPOSITION 10.4.1 (the EM residual as a diagnostic). *Let $\theta' \in [\frac12, 1)$ and $a = 1-\theta'$.*

(a) There is a positive measure x on $\mathbb{R}$ with $\int x(dv)/(1+v^2) < \infty$ and $P_a * x = u_1$ if and only if ζ has no zeros in $\operatorname{Re} s > \theta'$. It is then unique and equals $\mathfrak{U}_{\theta'}$.

(b) Let $h_k \to 0$ and $L_k \to \infty$ with $h_k^2 L_k \log L_k \to 0$. For positive measures $x = \sum_j x_j \delta_{jh_k}$ on the grid $h_k\mathbb{Z}$ with $\sum_j x_j/(1+j^2h_k^2) < \infty$, let $m_i(x) = (P_a * x)(v_i)$ at the data points $v_i \in h_k\mathbb{Z} \cap [-L_k, L_k]$, let $y_i = u_1(v_i)$, and let

$$D_k = \inf_x \, h_k \sum_i \Big(y_i \log \frac{y_i}{m_i(x)} - y_i + m_i(x) \Big)$$

be the total, not normalised, Kullback Leibler residual. Then $D_k \to 0$ if ζ has no zeros in $\operatorname{Re} s > \theta'$, and $\liminf_k D_k > 0$ if it has one.

Proof. (a) is Theorem 8.2.2(a) with $\varepsilon_0 = \frac12$ and $\varepsilon_1 = \theta' - \frac12$.

Write $d(y,m) = y\log(y/m) - y + m$; then $(\sqrt{y} - \sqrt{m})^2 \le d(y,m) \le (y-m)^2/m$ for $y, m > 0$. Also $|\partial_t \log P_a(t)| = 2|t|/(a^2+t^2) \le 1/a$, so $P_a(t')$ lies between $P_a(t)e^{-|t-t'|/a}$ and $P_a(t)e^{|t-t'|/a}$.

(b), no zeros in $\operatorname{Re} s > \theta'$. Let $x_j = \mathfrak{U}_{\theta'}([jh_k - h_k/2, jh_k + h_k/2))$, which satisfies the summability condition because $\mathfrak{U}_{\theta'}$ does. By (a), $y_i = \int P_a(v_i - w)\, \mathfrak{U}_{\theta'}(dw)$, and moving each w to the centre of its cell changes $P_a(v_i - w)$ by a factor in $[e^{-h_k/2a}, e^{h_k/2a}]$, so $|m_i(x) - y_i| \le (e^{h_k/2a} - 1) y_i$ and $d(y_i, m_i(x)) \le C h_k^2 y_i$ for $h_k \le a$. As $u_1 = P_a * \mathfrak{U}_{\theta'}$ gives $|u_1'| \le u_1/a$, we have $h_k \sum_i y_i \le e^{h_k/2a} \int_{-L_k-1}^{L_k+1} u_1\, dv$, which is $O(L_k \log L_k)$ by the Riemann von Mangoldt formula: the zeros with $|\gamma_\rho| \le 2L_k + 2$ contribute at most their number, and the others $O(\log L_k)$. Hence $D_k = O(h_k^2 L_k \log L_k) \to 0$. Atoms of $\mathfrak{U}_{\theta'}$, at zeros with $\sigma_\rho = \theta'$, are allowed.

(b), a zero in $\operatorname{Re} s > \theta'$. Suppose that $D_k \to 0$ along a subsequence, and choose grid measures x_k with total residual at most $D_k + 1/k$. The function $F_k = P_a * x_k$ is smooth and positive with $|F_k'| \le F_k/a$, and $h_k \sum_{|v_i| \le R} (\sqrt{y_i} - \sqrt{F_k(v_i)})^2 \to 0$ for every $R > 0$. Since $F_k \ge e^{-1/a} F_k(0)$ on $[-1,1]$, the case $R = 1$ shows that $F_k(0) = (a/\pi) \int x_k(dw)/(a^2+w^2)$ is bounded. So the measures $\nu_k(dw) = x_k(dw)/(a^2+w^2)$ have bounded mass, and along a further subsequence they converge weakly on the compactification $\mathbb{R} \cup \{\infty\}$ to a finite measure ν. Now $F_k(v) = \int \kappa_v \, d\nu_k$ with $\kappa_v(w) = (a/\pi)(a^2+w^2)/(a^2+(v-w)^2)$, which is jointly continuous on $[-R,R] \times (\mathbb{R} \cup \{\infty\})$ with $\kappa_v(\infty) = a/\pi$. Hence $F_k \to F$ uniformly on $[-R,R]$, where $F = P_a * x + c$ with $x = (a^2+w^2)\nu$ on $\mathbb{R}$ and $c = (a/\pi)\nu(\{\infty\}) \ge 0$, and Riemann sums give $\int_{-R}^{R} (\sqrt{u_1} - \sqrt{F})^2\, dv = 0$. As R is arbitrary and $P_a * (c\, dv) = c$, the positive measure $x + c\, dv$ has $\int (x + c\, dv)/(1+v^2) < \infty$ and $P_a * (x + c\, dv) = u_1$, which (a) excludes. So $\liminf_k D_k > 0$. □

The total residual cannot be replaced by the residual per unit length of the data window: when only finitely many zeros lie to the right of θ', the latter tends

to zero, since an atom of mass m_ρ at each of their ordinates leaves the defect $\sum_\rho (P_{1-\sigma_\rho} - P_a)(v - \gamma_\rho)$, whose total residual is finite. The proof of (b) also shows that the residual restricted to some fixed window $[-R, R]$ stays bounded away from zero, and the same argument works for every $R > 0$ since u_1 and $P_a * x + c$ are real analytic, but the proposition gives no rate. The runs below, with a fixed data window and a finite unknown support, illustrate it; no finite computation certifies either alternative.

10.4.4 Numerical EM inversion

The runs use the EM iteration (10.4.2) (the convolution by FFT agrees with the dense product to 2×10^{-15}). Data are u_1 on $[100, 300]$, the unknown is supported on $[80, 320]$, spacing 0.05, flat start with the total mass matched. The residual printed is the plain sum $\sum_i [y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i]$ over the 4001 data points, that is $\mathrm{KL}(y \,\|\, Ax)$ of Proposition 10.4.1 divided by the spacing h, not per point and not per unit length; it changes by 1.6 per cent between spacings 0.05 and 0.1, so two digits at most are meaningful. A peak is a local maximum of prominence at least 1 per cent of the maximum, and $\min x$ is the smallest x_j.

θ'	steps	KL residual	peaks	max error	$\min x$
0.75	300	2.4×10^{-5}	109 of 109	0.040	7.7×10^{-2}
0.50	300	4.9×10^{-2}	109 of 109	0.085	2.7×10^{-5}
0.50	3000	4.0×10^{-3}	109 of 109	0.030	9.9×10^{-12}

Table 10.3: EM inversion of u_1 on $[100, 300]$ (computed). The source printed 4.1×10^{-3} in the last row; the recomputation gives 4.05×10^{-3}, and the grid sensitivity of the residual is of that size.

At $\theta' = 0.75$ the iteration converges quickly to a strictly positive density with one peak per zero; at $\theta' = 0.5$ the target is atomic, the iteration sharpens slowly, as EM does when the truth lies on the boundary of the positive cone, and after 3000 steps the mass has concentrated at the zeros to 0.03 with the density between them driven to 10^{-11}. There is no ringing.

With an off line pair injected at $\sigma = 0.75$ and $\theta' = 0.75$, the source reports a residual after 3000 steps of 1.6×10^{-5} against 2.3×10^{-7} for the real u_1, a factor of about 70; the recomputation gives 1.62×10^{-5} and 2.27×10^{-7} (ratio 71, and 53 on the data window $[150, 250]$) for a quartet at $\sigma = \theta'$ that keeps the total mass, and the source's local misfit near the injected zero could not be reproduced, because its definition is not given. This zero lies on the level, not to its right, so a positive solution still exists: at level one the pair adds $P_{1/4}(v - \gamma) + P_{3/4}(v - \gamma)$, which is $P_{1/4} * (\delta_\gamma + P_{1/2}(\cdot - \gamma))$. The larger residual measures the slow concentration of EM onto an atom, as in the $\theta' = 0.5$ rows, and does not illustrate Proposition 10.4.1(b), which needs $\sigma > \theta'$.

The test was redesigned. The two consecutive line zeros nearest the centre of the window ($\gamma = 198.0153$ and 201.2648, midpoint 199.6400) are removed and replaced by the quartet $\rho, 1-\bar{\rho}, \bar{\rho}, 1-\rho$ with real part σ at the midpoint (widths $1-\sigma$ and σ at the midpoint and at its mirror image, total mass unchanged to 10^{-4}). The control puts two line zeros at the midpoint. Table 10.4 gives the total residual at 10000 steps and the ratio of quartet to control at 300, 1000, 3000 and 10000 steps, on the data windows $[100, 300]$ (unknown $[80, 320]$) and $[150, 250]$ (unknown $[130, 270]$).

θ', σ	window	real u_1	control	quartet	ratio at $300, \dots, 10^4$ steps
0.50, 0.75	$[100, 300]$	1.09×10^{-3}	1.08×10^{-3}	1.85×10^{-2}	1.4, 2.4, 5.4, 17
0.50, 0.75	$[150, 250]$	5.0×10^{-4}	5.0×10^{-4}	1.78×10^{-2}	1.8, 3.8, 10, 36
0.75, 0.85	$[100, 300]$	3.2×10^{-8}	3.3×10^{-8}	7.03×10^{-3}	366, 4000, 31000, 220000
0.75, 0.85	$[150, 250]$	5.9×10^{-8}	5.9×10^{-8}	7.02×10^{-3}	627, 4500, 23000, 120000
0.75, 0.75	$[100, 300]$	3.2×10^{-8}	3.3×10^{-8}	4.4×10^{-6}	9.0, 30, 72, 136

Table 10.4: Quartet injection test with controls (computed); total residual after 10000 EM steps. The last row is the boundary case $\sigma = \theta'$.

The redesigned test separates the situations. At $\theta' = 0.75$ and $\sigma = 0.85 > \theta'$ the quartet residual stops at 7.03×10^{-3}, the same for both windows to 0.3 per cent: a floor, with no positive solution, as Proposition 10.4.1(b) predicts for a zero to the right of θ', while the control and the real data decay below 10^{-7}. At $\theta' = 0.5$ and $\sigma = 0.75$ the control falls like $1/k$ (4.9×10^{-2}, 1.3×10^{-2}, 4.0×10^{-3}, 1.1×10^{-3} at 300, 1000, 3000, 10000 steps) while the quartet leaves 6.9×10^{-2}, 3.1×10^{-2}, 2.2×10^{-2}, 1.85×10^{-2}; the floor is not reached within 10000 steps and the ratio keeps growing, so this row shows slower decay than the control and not a plateau. The boundary case $\sigma = \theta'$ keeps decaying, like $1/k$, but more slowly than the control (ratio 9.0 at 300 steps, 136 at 10000), because a positive solution exists and EM is slow onto an atom. The plain sum is $1/h$ times the total residual of the proposition, a fixed factor at fixed spacing; a finite run certifies neither alternative.

10.5 Generative Bayesian computation for the backward flow

The backward flow is a Bayesian inverse problem once a prior is given for the unknown, the the configuration $Z = \{\gamma\}$ of zeros in a window; the forward map is the Cauchy semigroup,

$$Z \mapsto \sum_{\gamma \in Z} \mathrm{Cauchy}(\gamma, \tfrac{1}{2})$$

on the window, plus the contribution of the zeros outside the window, and nothing else: by Theorem 10.1.1 the zero sum is the whole of u_1, and the smooth part g_1

belongs to the prime side decomposition (10.1.3). The data are the values of u_1 on a grid, exact to rounding. A prior for Z is supplied by Montgomery's pair correlation [211] and Odlyzko's computations [220]: locally the zeros look like the eigenvalues of a large GUE matrix, unfolded to density $\frac{1}{2\pi}\log\frac{v}{2\pi}$.

Generative Bayesian computation [156, 157] needs no likelihood or Markov chain. One simulates (Z_i, y_i) from the prior and the forward map, trains a deep quantile network $q_\tau(y)$ to predict functionals of Z from y at quantile levels τ, and evaluates the trained network at the observed y. The network is meant as an amortised posterior: by the noise outsourcing theorem [104] any conditional law is the push forward of a uniform variable through a measurable map, and the quantile network learns that map for the functionals of interest; in practice it estimates marginal conditional quantiles of those functionals, not their joint law. Here the functionals are the counts of zeros in eight unit bins at the centre of a window of length 16, which determine the zero positions to within the bin width and are what a deconvolution has to get right. The formulation is heuristic; what follows is a computation.

10.5.1 Implementation

The run uses two seeds. Training set: 24,000 windows. For each window a density d is drawn uniformly from $[0.45, 0.8]$, which covers the heights $150 \le v \le 350$, where the zero density $(2\pi)^{-1}\log(v/2\pi)$ runs from 0.50 to 0.64 (and in fact the heights 106 to 958); a GUE matrix $H = (A + A^*)/2$ of size $M = 400$, with $A = X + iY$ and X, Y independent standard normal matrices, so that the off diagonal entries have variance 1 and the spectral edge is $2\sqrt{M} = 40$; the eigenvalues x are unfolded to unit spacing by the semicircle law, $e = MF(x/(2\sqrt{M}))$ with F the semicircle distribution function, and rescaled to density d (the measured number of zeros per unit length divided by d is 0.998 over the 24,000 windows); the forward map is evaluated on the grid of spacing 0.1 as the sum of $P_{1/2}(t - z)$ over the simulated zeros within $150/d$ of the centre, plus the Cauchy smoothing of the mean density d for the zeros beyond; the bin counts are the targets. Dividing the eigenvalues by $\sqrt{2M}$ instead of $2\sqrt{M}$ would give the density $d/\sqrt{2}$, which does not cover the real density above $v = 220$. Network: two hidden layers of 256 rectified units, 24 outputs (eight bins at $\tau = 0.1, 0.5, 0.9$), pinball loss, Adam, 25 epochs, plain numpy. Test set: windows of the real u_1 centred at $v = 150, 158, \ldots, 342$ (25 windows, 200 bins), with the true counts from the zeros. The forward map with the real zeros within $150/d$, $d = (2\pi)^{-1}\log(c/2\pi)$ at the window centre c, and the mean density beyond differs from u_1 by at most 4.4×10^{-4} on these windows (mean 1.9×10^{-4}, against $\max u_1 = 1.07$): the modelling error of the outside zeros, the mirror zeros and the density gradient.

10.5.2 Results

Values are for seed 1 and seed 2. The pinball loss at epoch 25 is 0.0157 and 0.0166 on the training simulations, and 0.0161 and 0.0175 on held out simulations. On the 200 real bins the rounded posterior median equals the true count in 196 and 197 bins (98.0 and 98.5 per cent) and is within one of it in all 200; the errors are 3 undercounts and 1 overcount, and 3 undercounts, and about half of them fall in the five bins that hold two or more zeros, where two Cauchy bumps of width $\frac{1}{2}$ merge into one. On held out simulations the quantile levels are calibrated to within about 0.05 for integer counts: $P(N \le q_{0.1}) = 0.086$ and 0.095, $P(N \le q_{0.5}) = 0.486$ and 0.449, $P(N \le q_{0.9}) = 0.892$ and 0.891, and the interval $[q_{0.1}, q_{0.9}]$ covers 0.806 and 0.796 (nominal 0.8); on the real windows it covers 0.805 and 0.775. The first run of the source reported 68 per cent exact bins, 99 per cent within one, and uncalibrated quantiles, with training inputs and test inputs that did not match (the smooth part g_1 was counted twice); with the corrected model the network counts almost exactly, and the earlier numbers are not those of this model.

A misfit statistic, the root mean square distance over $|t| \le 4$ between y and the forward map of the reconstruction that places the rounded median counts at the bin centres, is 0.173 (standard deviation 0.031) on the real windows and 0.331 (0.154) and 0.307 (0.132) on the same windows with an injected quartet at $\sigma = 0.75$ (total mass kept), for the two seeds; with two line zeros placed at the midpoint instead it is 0.190 and 0.186. The injected misfit exceeds the real one in 20 and 21 of the 25 windows. The source's statistic, which did not separate real from injected windows (0.31 against 0.34), is not defined in the source and was not reproduced; the statistic here is a plug in misfit and not a posterior predictive check proper.

This first implementation counts the zeros in unit bins and does not locate them. The EM inversion of Section 10.4.4, with the same positivity and no prior, locates them to 0.03; the two are not compared on one metric. A GBC inversion with zero positions as continuous outputs and the EM reconstruction as a feature is left open (Problem 10.1).

10.5.3 What the Bayesian view adds

The GUE prior encodes the critical line: every simulated configuration has all its zeros at real part $\frac{1}{2}$, so every simulated y is a sum of Cauchy bumps of width exactly $\frac{1}{2}$. A zero at $\sigma \neq \frac{1}{2}$ produces, with its mirror, bumps of widths $1 - \sigma$ and σ in place of the two bumps of width $\frac{1}{2}$ of two zeros on the line. Such a window is atypical under the prior, and the misfit statistic is meant to flag it; above it separates injected from real windows on average, and a posterior predictive check proper would simulate from the posterior rather than plug in its median. The method can do two things that the Fourier deconvolution does not: regularise the ill posed inversion with a prior rather than with a frequency cutoff, and report

uncertainty and misfit, to the extent that its quantiles are calibrated. It cannot establish the absence of off line zeros: the prior assumes them away, and the misfit statistic can only flag a window where the data disagree with that assumption, at the resolution the window allows.

In the language of the book on Hilbert's sixth problem [158], the forward map is the Poisson semigroup, the subordination of the heat semigroup by the stable one half law, and Cauchy(γ, a) is the law of $\gamma + B(T_a)$ for a Brownian motion B and the hitting time T_a of level a by an independent Brownian motion. Under RH the level θ Thorin measure is thus the zero measure run through Brownian motion to the hitting time of level $\theta - \frac{1}{2}$, and the backward flow is the unstopping of that Brownian motion; without RH the zeros carry widths $\theta - \sigma_\rho$ and the unconditional version is Section 12.2.3. That book reads this in thermodynamic terms; for the infinite measures here entropy is not defined, and the reading is heuristic, not an independent handle on RH.

10.6 Is this a path to RH?

No; the chapter produces several objects that look like handles, and we say precisely why they are not.

The level one measure is explicit and unconditional. Its positivity, which is what the Thorin theory gives for free, is automatic: by the Euler product every zero has $\sigma_\rho \leq 1$, so every term of (10.1.1) at $\theta = 1$ is a positive measure. The nonvanishing of ζ on $\operatorname{Re} s = 1$ is equivalent to the absence of atoms: $\mathfrak{U}_1$ has no atoms, and then has a continuous density, if and only if $\zeta(1+it) \neq 0$ for every real t. Indeed a zero $1 + i\gamma_0$ would contribute the atom Cauchy$(\gamma_0, 0) = \delta_{\gamma_0}$, which the other terms, all positive, cannot cancel, and if every width $1 - \sigma_\rho$ is positive the proof of Theorem 10.1.1 gives a continuous density. Everything to the left of the line enters only through the backward flow.

The backward flow is a deconvolution by $e^{(1-\theta')|\omega|}$. Its convergence at level θ' is equivalent, by Theorem 10.3.1, to the zero free half plane $\operatorname{Re} s > \theta'$, and the proof of that equivalence is the pole argument of Chapter 17. On the prime side the flow is the passage from the weights $\Lambda(n)/n$ to $\Lambda(n)n^{-\theta'}$, and its convergence is equivalent to the classical bound $\psi(x) - x = O(x^{\theta'+\varepsilon})$ for every $\varepsilon > 0$. The dictionary is exact and nothing is lost, but nothing is gained either: the error term in the prime number theorem has become the growth of a Fourier multiplier.

The Bayesian inversion, by EM, data augmentation or generative Bayesian computation, is a numerical method. Its residual is a diagnostic equivalent to the zero free half plane (Proposition 10.4.1), but equivalent is the operative word: the total residual stays bounded away from zero as the discretisation is refined exactly when there is a zero to the right of θ', and showing that it tends to zero at every $\theta' > \frac{1}{2}$ is RH. The GUE prior of the GBC version assumes the critical line outright. These are tools for computation and verification, not for proof.

What the chapter contributes is narrower: the prime formula of Propositions 12.1.2 and 12.1.3 extended to every level $\theta \geq 1$ and to $v = 0$; the explicit formula of Theorem 9.1.1 read at level one, for one analytic test function; a dichotomy theorem for the backward flow; the total residual theorem for the discretised inversion; and a computation in which 109 zeros are read off the line $\operatorname{Re} s = 1$ by deconvolution to 0.03, with a first GBC implementation that recovers bin counts. The question that remains is Problem 17.5, now sharpened: find a property of the explicit measure (10.1.3), visible from the primes, that is preserved by the backward Poisson flow and forces it to remain a measure. Positivity is not such a property, since it is not preserved backward. We do not know one.

10.7 Open problems

Problem 10.1. Implement the data augmentation sampler of Section 10.4 for the real u_1 and compare its posterior spread at $\theta' = 0.75$ and $\theta' = 0.5$ with the EM point estimate; then build the GBC inversion with the EM reconstruction as an input feature and zero positions as continuous outputs, and calibrate its quantiles on held out simulations.

Problem 10.2. Characterise, without reference to zeros, the functions of the form (10.1.3): which densities $\frac{1}{\pi}[g(v) - \lim_X(\sum_{n\leq X} a_n \cos(v\log n) - \sin(v\log X)/v)]$ with $a_n \geq 0$ supported on prime powers are Thorin densities at level one of a function $\Xi(1)/\Xi(1+\sqrt{s})$ with Ξ in the Selberg class? The Thorin conditions constrain the growth and the integrability near $u = 0$ but say nothing about the arithmetic of the a_n. (The wording is the author's; the Selberg class needs the completed function, the gamma factors, the conductor and the poles to be specified, and the sign of the coefficients of $-L'/L$.)

Problem 10.3. Quantify the backward flow. For $\theta' \in (\theta_0, 1)$ give the rate of locally uniform convergence of $B_{\theta',\Omega}\mathfrak{U}_1$ to $\mathfrak{U}_{\theta'}$ in terms of Ω and $\theta' - \theta_0$, and give the matching lower bound in case (ii): with $\sigma_{\max} = \sup \sigma_\rho > \theta'$ attained by a zero, show that $\sup_{|v|\leq V} |B_{\theta',\Omega}\mathfrak{U}_1(v)|$ grows like $e^{(\sigma_{\max}-\theta')\Omega}$ up to polynomial factors. For one zero with $d = \sigma - \theta' > 0$ the Fejér inverse at distance t from its ordinate is $(\pi\Omega)^{-1}\operatorname{Re}[(e^{z\Omega} - 1 - z\Omega)/z^2]$ with $z = d + it$, which at $t = 0$ is asymptotic to $e^{d\Omega}/(\pi d^2\Omega)$; with infinitely many zeros the supremum need not be attained and the cancellations among them are part of the problem.

Problem 10.4. Replace the GUE prior by a prior that allows a fraction of zeros off the line, for instance pairs at $\frac{1}{2} \pm \delta + i\gamma$ with δ drawn from a prior on $[0, \frac{1}{2})$, and train the quantile network to report the posterior of δ for each detected zero. The posterior of δ at the real u_1 is then a window by window numerical diagnostic at the resolution the data allow; being conditional on the prior, it is not a certificate.

10.8 The inverse problem: recovering the atoms by nonnegative deconvolution

The inverse problem is to recover the zeros of ζ from the Thorin density alone, by the positivity preserving deconvolution of Section 10.4.4 with nonnegative least squares in place of EM. Under RH the density is the Poisson smoothing of a positive measure with an atom at each zero, so the recovery is a deconvolution under a positivity constraint, and RH on a window becomes a feasibility question. The monograph [235, Section 19.3] poses it as a Bayesian problem with a positivity prior and a posterior predictive check, not run there because ρ_ε was available only from the primes, at resolutions above one half. The Pólya route of Section 2.1 supplies ρ_ε at small resolution to about 60 digits below height 200 (Section 2.1.1); the experiment uses it in double precision.

10.8.1 Recovering the zeros by nonnegative least squares

Write $P_\varepsilon(x) = \varepsilon/(\pi(\varepsilon^2+x^2))$ and $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$. Under RH, $\rho_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ on $\theta > 0$ (Proposition 2.1.3); since P_ε has Fourier transform $e^{-\varepsilon|t|}$, recovering $U^{\pm}_{\mathrm{RH}}$ is severely ill posed. An off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta < \varepsilon$ contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at γ, and $P_{\varepsilon-\delta}$ is sharper than the Poisson smoothing at level ε of any positive measure, since $P_{\varepsilon-\delta} = P_\varepsilon * \lambda$ would give λ the exponentially growing Fourier transform $e^{\delta|t|}$. Positivity of the recovered measure is the test.

Fix a data window $[\theta_a, \theta_b]$ with sample points $\theta_1 < \cdots < \theta_n$, and candidate atom locations $x_1 < \cdots < x_m$ on a grid over a candidate window $[x_a, x_b]$ that contains $[\theta_a, \theta_b]$. With $A_{ij} = P_\varepsilon(\theta_i - x_j) + P_\varepsilon(\theta_i + x_j)$ and data $y_i = \rho_\varepsilon(\theta_i) - b_i$, where b_i is the contribution of the zeros outside the candidate window, the problem is

$$\min_{w \geq 0} \ \|Aw - y\|_2.$$

Under a Gaussian likelihood with a flat prior on the positive orthant its solution is the maximum a posteriori estimate, and the minimal residual is the discrepancy statistic of the posterior predictive check: up to the error of the grid, it vanishes when the data are the Poisson smoothing of a positive measure. The active set algorithm of Lawson and Hanson [190] solves the problem in finitely many steps.

We took $\varepsilon = 0.1$, the Pólya computed $\rho_{0.1}$ on $[15, 55]$ at 1601 points, and candidates on $[5, 65)$ at spacing 0.02. The background b_i is the Poisson smoothing of the zeros outside $[5, 65]$ among the first 1700 ordinates, the list of Section 3.1 without its smooth tail, which would add about 3×10^{-5} to each sample. The nonnegative solution concentrates on clusters, runs of adjacent grid points with weights above 10^{-3}, one at each zero in the data window (Table 10.5 and Figure 10.1, left): every ordinate is recovered to within 10^{-5}, with mass one to within half a per cent. The remaining mass, 0.071 of the total 10.094, lies in small clusters at the edges of the data window (0.025), which absorb the tails of the zeros at 14.13 and 56.45 just

outside it, and in weights below 10^{-3} (0.046). The residual norm is 0.106. It is set by the grid: the true atoms lie between grid points, and a positive combination of two neighbouring Cauchy kernels reproduces an off grid kernel only to second order in the grid spacing.

k	γ_k	c_k	$10^6(c_k-\gamma_k)$	mass
2	21.022040	21.022034	−5.5	1.0013
3	25.010858	25.010859	1.6	1.0027
4	30.424876	30.424867	−8.8	1.0020
5	32.935062	32.935070	8.7	1.0020
6	37.586178	37.586171	−7.4	1.0023
7	40.918719	40.918723	3.8	1.0008
8	43.327073	43.327068	−5.5	1.0025
9	48.005151	48.005145	−6.1	1.0024
10	49.773832	49.773840	7.9	1.0028
11	52.970321	52.970330	8.8	1.0042

Table 10.5: The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero γ_k in the data window, the centre of mass c_k of its cluster, the error of the centre and the total mass of the cluster.

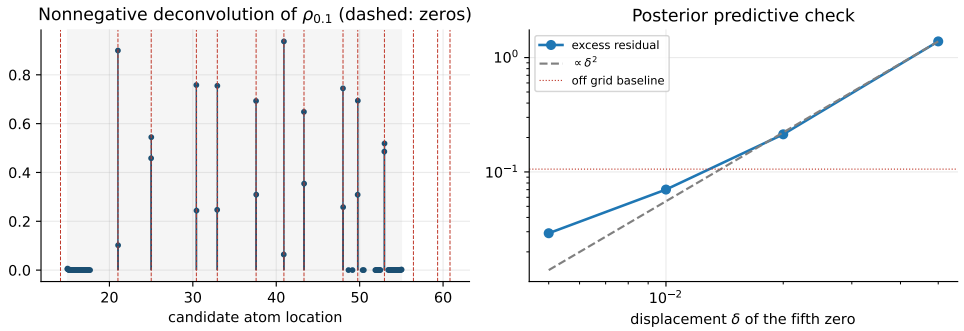


Figure 10.1: Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. The zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement δ of the fifth zero, the δ^2 law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$.

10.8.2 A displaced zero and the threshold of detection

To calibrate the check we displaced the fifth zero, $\gamma_5 = 32.935$, in the sense of Proposition 3.2.1(i): we added to the data the change per displaced zero, $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$, centred at γ_5, and refitted. Table 10.6 gives the residuals.

δ	residual $r = \|Aw - y\|$	excess over baseline	total mass
0	0.1058	0	10.0942
0.005	0.1098	0.029	10.0947
0.01	0.1269	0.070	10.0963
0.02	0.2380	0.213	10.1031
0.05	1.3891	1.385	10.1566

Table 10.6: Nonnegative deconvolution of $\rho_{0.1}$ with the signature g of a displaced fifth zero added to the data. Here r is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$.

For $\delta \geq 0.02$ the excess residual grows roughly like δ^2, as the signature does (Proposition 3.2.1(i) and [235, Proposition 19.7]): it rises from 0.21 at $\delta = 0.02$ to 1.39 at $\delta = 0.05$, a factor 6.5 for a factor 2.5 in δ, against $2.5^2 = 6.25$. At the smallest displacements it grows more slowly, by a factor 2.4 from $\delta = 0.005$ to 0.01, because there it is dominated by the cross term between the signature and the off grid misfit at γ_5, which has the same shape. The off grid baseline limits detection: the excess reaches it between $\delta = 0.01$ and 0.02 (Figure 10.1, right). This is the square root law that (3.2.1) states for the matched distance (Section 3.2.2), whatever limits the residual.

For $\delta \geq \varepsilon$ the check is unreliable. At $\delta = \varepsilon$ the zero $\frac{1}{2} + \delta + i\gamma_5$ lies on the line of evaluation, where its kernel is a point mass that the samples miss, and the residual, 0.099, falls below the baseline. For $\delta > \varepsilon$ the atom at γ_5 leaves the data and is replaced by the signed term $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ of total mass zero, whose negative part, of mass $\varepsilon/(\pi\delta) + O(\varepsilon^3/\delta^3)$ by Proposition 3.2.1(iii), fades as δ grows. The unit of mass that leaves the data is what the zero density identity of Proposition 3.3.2 counts, and zeros farther from the line than ε are left to that identity.

As [235, Section 19.3.7] emphasises, this is a model check at finite height, not a route to a proof. The Pólya route adds the small resolution, where the signature $(\delta/\varepsilon)^2$ of Proposition 3.2.1(i) is largest for a given δ. The recovery and the calibration are computed, at one resolution and on one window.

This is the hard direction: Poisson smoothing, which moves the basepoint to the right, preserves positivity, while deconvolution, which moves it back towards the critical line, does not (Section 14.3.7).

Chapter 11

Complexity and acceleration of the backward flow

Chapter 10 turned the backward Cauchy flow into a positive linear inverse problem and solved it by the EM algorithm of Vardi and Lee. This chapter asks three questions about that computation. What does it cost, and what does log concave sampling in the sense of Applegate, Kannan and Polson say about it? Can EM be accelerated, by Nesterov momentum or otherwise, and what does the accelerated iteration compute? Does the representation of the Cauchy kernel as a scale mixture of normals help?

The short answers. The cost splits into three layers. Evaluating the level one measure needs ζ on the line $\operatorname{Re} s = 1$, for which the best algorithms are polynomial in the height T and not in $\log T$; this is the bottleneck, and the Thorin formulation does nothing to it. The inversion, given the data, is a convex minimisation whose Bayesian version is log concave sampling; the ill posedness appears as a precision requirement linear in the resolution. Verification is Turing's method and costs $T^{1+\varepsilon}$. EM is majorise minimise in the Kullback Leibler geometry; Nesterov momentum in the logarithmic domain keeps positivity, is an order of magnitude faster at $\theta' = 1/2$ in the runs, and in every run reached the reverse information projection of the data onto the cone of Cauchy blurred positive measures, which is the Thorin measure at level θ' when it exists. The scale mixture of normals turns the Cauchy blur into a heat flow at a first passage time and identifies the Thorin measure with the subordinated heat trace of the zeros; it changes nothing in the complexity classification or the arithmetic.

The chapter then turns to the Thorin measures themselves: the levels 1, 7/8, 3/4 and 1/2, what positivity means, an integer valued mass deficit that decides positivity up to any finite height, a certificate in ball arithmetic up to height 1419.9, the limits of the computation, and the global minimum of the Thorin density at the origin, unconditionally at level one (Proposition 12.1.11) and, for every $\theta \in (\frac{1}{2}, 1]$, equivalent at all levels in $[\theta, 1]$ to the zero free half plane $\operatorname{Re} s > \theta$ (Theorem 12.1.18). Its last part works open problems: an a posteriori certificate for the backward flow, a smoothed prime sum that evaluates u_1 to fixed accuracy in

time polynomial in $\log v$, given that the zeros near that height lie on the critical line (verified below 3×10^{12}; at every height conditionally on the announced half plane $\operatorname{Re} s > 7/8$), and the heuristic reason, from the explicit formula read backwards, why $\mathfrak{U}_{1/2}$ (equal to $2\widetilde{U}_{1/2}$ of Theorem 17.14.4) cannot be computed efficiently by deconvolution.

The chapter develops a source chapter on complexity and acceleration of the backward flow by Nick Polson, together with the computations done for this book. The numbers were recomputed for this book (Section 11.4, Theorem 11.9.6, the global minimum certificates and Proposition 11.11.4); where a recomputation differed from the source the recomputed value is printed, and numbers not recomputed are marked as the author's.

Notation. A level θ is the real part of $s = \theta + iv$, and $u_\theta(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta + iv)$; in the notation of Chapter 8 this is $\rho_{\theta - 1/2}(v)$. The Cauchy density is $P_a(x) = (a/\pi)/(a^2 + x^2)$, with $P_0 = \delta_0$. A zero is $\rho = \sigma_\rho + i\gamma_\rho$. The symmetric Thorin measure $\mathfrak{U}_\theta$ of equation (10.1.1) has density $\sum_\rho P_{\theta - \sigma_\rho}(v - \gamma_\rho)$ with no factor $\frac{1}{2}$, so $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with the $\widetilde{U}_\theta$ of Theorem 17.14.4. Certified means computed in ball arithmetic, with the sentence saying what the certificate covers. The announced zero free half plane $\operatorname{Re} s > 7/8$ (OpenAI [227, 228], formalised in Lean, and Liu [198]; not yet refereed) is never used as a theorem: a result that uses it is stated as conditional on it. The results and their status are these.

1. The matrix A of the discretised deconvolution, with cell densities as unknowns, has the properties (a) to (e) of Section 11.1 (proved), including a kernel of dimension exactly $2(N - M)$ when the unknown window is larger. The level one data must be supplied to absolute precision $e^{-(1-\theta')\Omega}$ for the linear inverse (Proposition 11.2.2, proved). The Poisson posterior is log concave (Theorem 11.2.1); the sampling clause is a cited theorem, conditional on a bound for the ratio of radii not proved here.

2. EM has the rate $D(x^* \| x^{(0)})/k$ in the consistent case (Theorem 11.3.1, proved, with the repaired step) and local linear convergence in the interior (Theorem 11.3.2, proved; its injectivity hypothesis fails in the runs). Under three growth conditions the EM limits converge vaguely to $\mathfrak{U}_{\theta'}$ when ζ has no zeros in $\operatorname{Re} s > \theta'$, and the residual stays bounded away from zero when it has one (Proposition 11.3.4, proved).

3. Log domain Nesterov momentum with restart and SQUAREM are an order of magnitude faster than EM at $\theta' = 1/2$ on a window; on the full range (202 zeros below height 400) and on the first thousand zeros (994 of 995 zeros below 1415 found as peaks) they locate the zeros to about 10^{-2} (Section 11.4, computed; the rates $O(1/k)$ and $O(1/k^2)$ and the identification of the limit are observed, not proved).

4. The von Neumann and Ulam estimator for the backward flow has infinite

variance whenever $h < 0.5221\,\pi(1-\theta')$ (Proposition 11.5.1, proved); the positive reverse walk is the EM step in expectation. The Cauchy kernel is a first passage mixture of Gaussians, giving the subordinated heat trace of the zeros and an inverse gamma data augmentation sampler (Proposition 11.6.1, proved; the sampler comparison is computed).

5. The level 7/8 Thorin measure is computed by three routes; its identification as a Thorin measure is conditional on the announced half plane. The positivity theorem (Theorem 11.9.1), the mass deficit identity (Proposition 11.9.2) and the level mass function (Corollary 11.9.3) are proved; the deficit tables, the $U_{3/4}$ and $U_{1/2}$ tables and Turing's check in Thorin bookkeeping are computed. Decidability by level is proved for finite heights (Section 11.10).

6. Certified positivity (Theorem 11.9.6, ball arithmetic at 96 and 160 bits): $u_{3/4}$ and $u_{7/8}$ are positive on $[0, 1420]$, every certified lower bound exceeding 4×10^{-5}; the argument change of ξ gives exactly 1000 zeros below height 1419.9, all on the critical line and simple, which is known (Platt and Trudgian [232]); the positivity is what the certificate adds.

7. The global minimum of the Thorin density is at the origin: at level 1 this is Proposition 12.1.11 (Theorem 11.10.2(i) gives an independent computed proof); at level 7/8 it is conditional on the announced half plane and computed, the least certified bound on $[13.5, 1420]$ being 0.0360 at $\theta = 1$ and 0.0272 at $\theta = 7/8$, both at $v = 17.49$ (Theorem 11.10.2); the Riemann hypothesis form is Theorem 12.1.18(vi) (Theorem 11.10.3). The uniform law and the lower bound $W \geq w^*$ for the excess (Theorems 11.11.1 and 11.11.3) are computed, and conditional on the announced half plane at level 7/8.

8. The slope of the level one density has certified infima 0.1758 to 0.1793 (Proposition 11.11.4, computed); under the Riemann hypothesis the limiting bracket X has its minimum in the first gap at $v^* = 17.4979 \pm 0.0001$, with $0.1838985 \leq X(v^*) \leq 0.1838988$ (computed, conditional on RH); that this is its infimum over $13.5 \leq |v| \leq 1418.9$ is the author's computation, not rerun.

9. The a posteriori bound for window masses and centroids of the backward flow is proved (Proposition 11.11.5); the certificate of Theorem 11.11.8 is the author's computation, not rerun, and is conditional on the absence of zeros to the right of θ' (the Riemann hypothesis at $\theta' = \frac{1}{2}$); it certifies Gaussian weighted masses and centroids, and single zeros only through Corollary 11.11.9 (proved).

10. The smoothed prime sum for u_1 has an exact identity (proved) and an error bound proved under the local hypothesis on the zeros, including the pole term the source omitted (Theorem 11.11.10); conditionally on the announced

half plane it holds at every height with X of order $(\log v/\varepsilon)^8$; the nonuniform FFT experiment is the author's, not rerun.

11. The explicit formula read backwards and the efficiency of the backward flow (Section 11.11.10), and the reading of the Hilbert and Pólya conjecture as a resolvent trace (Section 11.11.11), are heuristic.

11.1 The deconvolution problem, precisely

11.1.1 Why the Thorin measure is a deconvolution

Fix $\frac{1}{2} \le \theta' < \theta$ and write $P_a(x) = (a/\pi)/(a^2 + x^2)$ for the density of the Cauchy law of scale a. Theorem 17.14.4 says that if ζ has no zeros in $\operatorname{Re} s > \theta'$ then

$$\mathfrak{U}_\theta = P_{\theta-\theta'} * \mathfrak{U}_{\theta'}. \tag{11.1.1}$$

At $\theta = 1$ the left side is known without any information about zeros, with density $u_1(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + iv)$ (Theorem 10.1.1). So the Thorin measure at level θ' is the solution μ of

$$P_{1-\theta'} * \mu = \mathfrak{U}_1, \qquad \mu \ge 0, \tag{11.1.2}$$

a deconvolution. Three facts fix its nature (Theorem 8.2.2).

Uniqueness. $\widehat{P_a} = e^{-a|\omega|}$ never vanishes, so $\mu \mapsto P_a * \mu$ is injective on signed measures with $\int (1 + v^2)^{-1} d|\mu| < \infty$, and (11.1.2) has at most one solution.

Existence. A positive solution exists if and only if ζ has no zeros in $\operatorname{Re} s > \theta'$, and it is then $\mathfrak{U}_{\theta'}$. The deconvolution is an existence problem, not an estimation problem: existence at θ' is the half plane $\operatorname{Re} s > \theta'$, and at all $\theta' > \frac{1}{2}$ together it is the Riemann hypothesis.

Ill posedness. The inverse has Fourier symbol $e^{a|\omega|}$: a perturbation of $\mathfrak{U}_1$ of size δ at frequency ω becomes $\delta e^{(1-\theta')|\omega|}$ in μ, exponential rather than polynomial growth.

At its own level the Thorin measure of $F_\theta(s) = \xi(\theta)/\xi(\theta + \sqrt{s})$ is obtained by Stieltjes inversion and is immediate once ξ'/ξ is available there. The passage between levels is the hard part: the Cauchy kernel is the Poisson kernel of the half plane $\operatorname{Re} s > \theta'$, so (11.1.1) is the Poisson integral formula for the harmonic function $\partial_\sigma \log|\xi|$, and deconvolution is harmonic continuation backward, into the region where a zero may destroy harmonicity.

11.1.2 The matrix A

Discretise (11.1.2) with a common spacing h. The data grid is $v_i = ih$, $|i| \le M$, covering $[-V, V]$ with $V = Mh$, and the unknown grid is $w_j = jh$, $|j| \le N$, covering $[-W, W]$ with $W = Nh > V$. The unknown is the vector of cell densities

$x_j = h^{-1}\mu([w_j - \frac{h}{2}, w_j + \frac{h}{2}))$, so that $\mu \approx \sum_j x_j h\, \delta_{w_j}$, and the data vector is $y_i = u_1(v_i)$. Then

$$A_{ij} = h\, P_{1-\theta'}(v_i - w_j) = \frac{h(1-\theta')/\pi}{(1-\theta')^2 + (i-j)^2 h^2}, \tag{11.1.3}$$

so $A \in \mathbb{R}^{(2M+1)\times(2N+1)}$ and $(Ax)_i$ is the midpoint rule for the density of $P_{1-\theta'} * \mu$ at v_i. (Chapter 10 uses the cell masses hx_j as unknowns and the matrix $P_{1-\theta'}(v_i - w_j)$; the residuals of both chapters include the factor h.) The chapter uses the following properties.

(a) Every entry is positive, so A maps the orthant into the orthant and $y = Ax$ is a positive linear inverse problem in the sense of Vardi and Lee [200].

(b) A_{ij} depends only on $i - j$: A is the restriction to $|i| \le M$, $|j| \le N$ of the symmetric Toeplitz operator with kernel $k_m = hP_{1-\theta'}(mh)$, $m \in \mathbb{Z}$. Hence Ax and $A^{\mathsf{T}}r$ are restricted convolutions with k, computed by FFT after circulant embedding in length $L \ge 2(2N+1)$ at cost $O(L \log L)$ per product, which makes the full range feasible; the dense form has $(2M+1)(2N+1)$ entries, 4001×4801 for the window runs below.

(c) The column sums $c_j = \sum_{|i|\le M} A_{ij} = h \sum_{|i|\le M} P_{1-\theta'}((i-j)h)$ equal 1 up to the discretisation error and the Cauchy tail outside the data window: for $|w_j| < V$,

$$\begin{aligned} c_j = 1 + O\big(e^{-2\pi(1-\theta')/h}\big) - \frac{1-\theta'}{\pi}\Big(\frac{1}{V - w_j} + \frac{1}{V + w_j}\Big) \\ + O\big(h(V - |w_j|)^{-2} + (V - |w_j|)^{-3}\big), \end{aligned}$$

because $h \sum_{m\in\mathbb{Z}} P_a(mh) = \coth(\pi a/h)$ and each tail is $\pi^{-1} \arctan\big(a/(V \mp w_j + \frac{h}{2})\big)$ up to the midpoint error. EM divides by them, which accounts for mass leaving the data window. This is also why $W > V$: zeros just outside $[-V, V]$ contribute Cauchy tails to the data inside.

(d) The symbol of the bi infinite Toeplitz operator is, by Poisson summation,

$$\begin{aligned} \phi(\xi) &= \sum_{m\in\mathbb{Z}} k_m e^{im\xi} = \sum_{\ell\in\mathbb{Z}} e^{-(1-\theta')|\xi + 2\pi\ell|/h} \\ &= \frac{\cosh\big(r(\pi - |\xi|)\big)}{\sinh(\pi r)}, \qquad |\xi| \le \pi, \quad r = \frac{1-\theta'}{h}, \end{aligned}$$

so the nonzero singular values of A lie between $1/\sinh(\pi r)$ and $\coth(\pi r)$, and the condition number of the bi infinite operator is $\cosh(\pi r)$, of order $e^{(1-\theta')\pi/h}$, which is e^{78} for $h = 0.02$ and $\theta' = 1/2$. With $\omega = \xi/h$, the frequency of the multiplier (10.3.1), the singular value at ω is $e^{-(1-\theta')|\omega|}$.

(e) If $N \le M$, A is injective. If $N > M$, A has rank $2M+1$ and a kernel of dimension exactly $2(N-M)$, for every h. Since $A > 0$ the kernel vectors are signed, and when $2(N-M) \le M+1$ (the unknown window exceeds the data window by at most about half its length) none of them vanishes on the data window. The minimisers of $\mathrm{KL}(y\|Ax)$ over $x \ge 0$ share one image $z^* = Ax$ and form the polytope $\{x \ge 0 : Ax = z^*\}$; the EM limit is a point of it that may depend on the start. For fixed windows the continuous operator $\mu \mapsto (P_{1-\theta'} * \mu)|_{[-V,V]}$ on measures carried by $[-W,W]$ is injective, $P_{1-\theta'} * \mu$ being real analytic, with unbounded inverse. The uniqueness clause of Theorem 11.3.1 and the hypothesis of Theorem 11.3.2 fail in the runs below, where $N > M$, while the bound on f in Theorem 11.3.1 holds.

Proof of (c) to (e). For (c) and (d), Poisson summation gives $\sum_m hP_a(mh)e^{im\xi} = \sum_\ell e^{-a|\xi+2\pi\ell|/h}$, which sums to the displayed hyperbolic quotient, and at $\xi = 0$ to $\coth(\pi r)$. For (e), with $r = (1-\theta')/h$ every finite section B of the Toeplitz operator satisfies $z^{\mathsf T}Bz = \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi(\xi)\big|\sum_j z_j e^{ij\xi}\big|^2 d\xi \ge \phi(\pi)|z|^2$, and $\phi(\pi) = 1/\sinh(\pi r) > 0$. If $N \le M$ the rows $|i| \le N$ of A form such a section; if $N > M$ the central $(2M+1)$ block is one, so A has full row rank and nullity $(2N+1)-(2M+1)$. When $2(N-M) \le M+1$, a kernel vector carried by $|j| > M$ gives $\sum_j z_j(a^2 + (v-w_j)^2)^{-1} = 0$ at the $2M+1$ points v_i; this rational function has distinct poles $w_j \pm ia$ and a numerator of degree at most $4(N-M)-2 < 2M+1$, so all $z_j = 0$. The function $\mathrm{KL}(y\|z)$ is strictly convex in z for $y > 0$. For μ carried by $[-W,W]$, $P_a * \mu$ extends holomorphically to $|\operatorname{Im} v| < a$, so vanishing on $[-V,V]$ forces $P_a * \mu = 0$ and $\mu = 0$. □

11.1.3 Three inverses of the same matrix

The three computations of Chapter 10 are three inverses of (11.1.3). The Fourier backward flow is the truncated singular value inverse: discarding singular values below $e^{-(1-\theta')\Omega}$ gives the multiplier $e^{(1-\theta')|\omega|}\mathbf{1}_{|\omega|\le\Omega}$, which is $B_{\theta',\Omega}$ of (10.3.1) with a sharp window; it is linear, ignores positivity, and rings. EM is the maximum likelihood inverse under the Poisson model, the minimiser of $\mathrm{KL}(y\|Ax)$ over $x \ge 0$; it uses every singular value and enforces positivity (Theorem 11.3.1). The Bayesian inverse is the posterior over $x \ge 0$ under the Poisson likelihood and a gamma prior, sampled by data augmentation or by the random walk of Theorem 11.2.1; with a flat prior its mode is the EM solution.

11.2 Three layers of cost

Fix a height T, a level $\theta' \in [\frac{1}{2}, 1)$, a grid spacing h, and an accuracy ε.

11.2.1 The data

The level one density $u_1(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(1+iv)$ requires $\zeta(1+iv)$ and $\zeta'(1+iv)$. The Riemann Siegel formula costs $v^{1/2+\varepsilon}$ per evaluation; Hiary's algorithms cost $v^{1/3+\varepsilon}$ [97] and $v^{4/13+\varepsilon}$ (the exponents are the source's, unchecked); the Odlyzko and Schönhage method [221] evaluates ζ at all heights up to T on a grid of spacing $T^{-1/2}$ in time $T^{1+\varepsilon}$, which is T^{ε} per point. All are exponential in the bit length of T; no algorithm polynomial in $\log T$ is known for ζ at a single point of a vertical line. So the data layer costs $T^{1+\varepsilon}$ for the measure up to height T, whatever follows.

11.2.2 The inversion

The discretised problem is $y = Ax$ with n unknowns $x_j \geq 0$ and the Poisson log likelihood

$$\ell(x) = \sum_i \bigl(y_i \log (Ax)_i - (Ax)_i \bigr). \tag{11.2.1}$$

Theorem 11.2.1 (log concavity). *The function ℓ is concave on the open orthant $x > 0$. With independent gamma priors $x_j \sim \operatorname{Gamma}(\alpha_j, \beta_j)$, $\alpha_j \geq 1$, the posterior density $\pi(x) \propto e^{\ell(x)} \prod_j x_j^{\alpha_j - 1} e^{-\beta_j x_j}$ is log concave on the orthant. Consequently, for every $\varepsilon > 0$, a sample from a distribution within total variation ε of π can be drawn by a random walk in time polynomial in n, $\log(1/\varepsilon)$ and the ratio of the radii of a ball containing most of the mass of π and a ball on which π is bounded below, given an oracle for ℓ.*

Proof. Each $(Ax)_i$ is a positive linear form, log is concave and increasing, and $y_i \geq 0$, so $y_i \log (Ax)_i$ is concave; $-(Ax)_i$ is linear. The log prior $(\alpha_j - 1)\log x_j - \beta_j x_j$ is concave for $\alpha_j \geq 1$. The orthant is convex. The sampling statement is the random polynomial time theorem for log concave distributions of Applegate, Kannan and Polson [3] and Applegate and Kannan [2], in the form given by Lovász and Vempala [116]; the condition on radii is their roundness condition, and it is not verified here. □

The EM limit minimises a convex function and is computed to accuracy ε in function value in deterministic polynomial time. We expect the ill posedness to appear not in the running time but in the precision of the data.

Proposition 11.2.2 (precision). *Let Ω be the largest frequency to be resolved in the level θ' measure. Then, for the linear inverse, the level one data must be supplied to absolute precision, on the scale of u_1, $e^{-(1-\theta')\Omega}$, that is, to $(1-\theta')\Omega/\log 2$ binary digits, and this is sharp for the linear inverse: an error of size δ at frequency ω in $\widehat{u_1}$ produces an error of size $\delta e^{(1-\theta')|\omega|}$ in the inverted measure.*

Proof. The backward multiplier is $e^{(1-\theta')|\omega|}$ (equation (10.3.1)). □

The number of bits is linear in Ω. Resolving the zeros near height T, of mean spacing $2\pi/\log T$, needs Ω of order $\log T$ (close pairs need more), hence $(1-\theta')\log_2 T$ bits plus a constant. The precision is cheap; evaluating ζ to that precision at height T is not.

11.2.3 Verification

Certifying that no zero lies to the right of θ' below height T is Turing's method [270], an application of the argument principle to ξ on a rectangle, and costs $T^{1+\varepsilon}$. The EM residual of Proposition 11.3.4(b) is a diagnostic for the same statement; heuristically, a residual computed to precision ε on a window resolved to frequency Ω detects a zero to the right of θ' only beyond a distance set by ε and Ω (no bound is proved here), and obtaining that precision costs the data layer.

Remark 11.2.3. The three layers do not place the Riemann hypothesis in a complexity class. What they give is conditional: given ζ on $\operatorname{Re} s = 1$ to the required precision, the EM limit above the zero free abscissa is a convex minimisation, and its posterior can be sampled in the time of Theorem 11.2.1, whose ratio of radii is not bounded here.

11.3 EM as majorise minimise

Write $f(x) = -\ell(x) = \sum_i \big((Ax)_i - y_i \log (Ax)_i\big)$, convex on the orthant, and $c_j = \sum_i A_{ij}$. The EM iteration of Chapter 10 is

$$x_j^{+} = x_j \, \frac{1}{c_j} \sum_i A_{ij} \frac{y_i}{(Ax)_i} = x_j \Big(1 - \frac{\partial_j f(x)}{c_j}\Big). \tag{11.3.1}$$

The bound of Theorem 11.3.1 below and the Jacobian of Theorem 11.3.2 are classical for EM with a Poisson likelihood [200, 52]; the proofs are given to fix the constants and the hypotheses. Three readings of (11.3.1) are useful.

It is an MM algorithm. Jensen's inequality for $-\log (Ax)_i$, with weights proportional to $A_{ij} x_j^{(k)}$, majorises f at $x^{(k)}$ by a separable surrogate, and (11.3.1) is the exact minimiser of the surrogate. Hence $f(x^{(k+1)}) \le f(x^{(k)})$, with equality only at a fixed point, and every fixed point with $x > 0$ is a minimiser of the convex f.

It is related to mirror descent. With the entropy mirror map, the mirror descent step of size η is

$$x_j^{+} = x_j \exp(-\eta \, \partial_j f(x));$$

(11.3.1) is its first order expansion with $\eta = 1/c_j$. In the language of relative smoothness [14], f is $\|y\|_1$ smooth relative to the Burg entropy $-\sum_j \log x_j$ (constant unchecked), which places EM near the Bregman proximal methods and their sublinear guarantees, though EM is not literally the Burg Bregman step. A direct argument gives the rate with an explicit constant.

THEOREM 11.3.1 (rate of EM in the consistent case). *Normalise the columns so that $c_j = 1$, and suppose the problem is consistent: $y = Ax^*$ for some $x^* \geq 0$. Then for every $x^{(0)} > 0$ and $k \geq 1$ the EM iterates satisfy*

$$f(x^{(k)}) - f(x^*) \leq \frac{D(x^* \,\|\, x^{(0)})}{k}, \quad D(x^*\|x) = \sum_j \Big(x_j^* \log \frac{x_j^*}{x_j} - x_j^* + x_j \Big).$$

If moreover $y > 0$ and A is injective, the minimiser is unique and $x^{(k)} \to x^*$.

Proof. For every $x > 0$, $\sum_i (Ax)_i = \sum_j x_j$ because $c_j = 1$, and $Ax^* = y$ gives $f(x^*) = \sum_i y_i - \sum_i y_i \log y_i$, so

$$\begin{aligned} f(x) - f(x^*) &= \sum_i \Big(y_i \log \frac{y_i}{(Ax)_i} - y_i + (Ax)_i \Big) \\ &= \sum_i y_i \log \frac{y_i}{(Ax)_i} + \sum_j x_j - \sum_i y_i \;\geq 0. \end{aligned}$$

Summing (11.3.1) over j with $c_j = 1$ gives $\sum_j x_j^+ = \sum_i y_i$. Now

$$\begin{aligned} D(x^*\|x^+) - D(x^*\|x) &= \sum_j x_j^* \log \frac{x_j}{x_j^+} + \sum_j x_j^+ - \sum_j x_j \\ &\leq -\sum_j x_j^* \sum_i A_{ij} \log \frac{y_i}{(Ax)_i} + \sum_i y_i - \sum_j x_j \end{aligned}$$

by Jensen's inequality with the weights A_{ij}, $\sum_i A_{ij} = 1$, applied to $x_j^+/x_j = \sum_i A_{ij} y_i/(Ax)_i$. The first term on the right is $-\sum_i (Ax^*)_i \log(y_i/(Ax)_i) = -\sum_i y_i \log(y_i/(Ax)_i)$, so the right side is $-(f(x) - f(x^*))$. Thus $D(x^*\|x^{(j+1)}) \leq D(x^*\|x^{(j)}) - (f(x^{(j)}) - f(x^*))$ for every $j \geq 0$. Summing over $j < k$ and using that $f(x^{(j)})$ is nonincreasing, $k\,(f(x^{(k)}) - f(x^*)) \leq \sum_{j<k} (f(x^{(j)}) - f(x^*)) \leq D(x^*\|x^{(0)})$. If $y > 0$ and A is injective, f is strictly convex, since $t \mapsto -y_i \log t$ is strictly convex and $x \mapsto Ax$ is one to one, so the minimiser is unique; the iterates are bounded by the mass identity, and since $f(x^{(k)}) \to f(x^*)$ every limit point is a minimiser, hence equals x^*. □

The constant explains the slow convergence at the boundary. For an atomic target, x^* puts mass about one on a few cells around each zero, and for a flat start $D(x^*\|x^{(0)})$ is heuristically of order $N \log(1/(hd))$, with N the number of zeros and d their density: the entropy of the target relative to the start. In the interior, where x^* is a smooth positive density, convergence is linear with a computable rate.

THEOREM 11.3.2 (local linear convergence in the interior). *In the setting of Theorem 11.3.1, with $c_j = 1$, suppose $x^* > 0$ componentwise and A injective. Let $B = \operatorname{diag}(y)^{-1/2} A \operatorname{diag}(x^*)^{1/2}$ and $S = B^{\mathsf{T}} B$. Then the eigenvalues of S lie in*

$(0,1]$, *with* 1 *attained, and the Jacobian of the EM map at* x^* *is similar to* $I - S$, *so its spectrum lies in* $[0,1)$ *and*

$$\|x^{(k)} - x^*\| = O\big((1 - \lambda_{\min}(S))^k\big)$$

for $x^{(0)}$ *near* x^*. *The rate* $1 - \lambda_{\min}(S)$ *is that of the linearised iteration along the eigenvector* $\operatorname{diag}(x^*)^{1/2} s$ *of the Jacobian,* s *an eigenvector of* S *for* $\lambda_{\min}$, *and* $\lambda_{\min}(S) \geq \min_j x_j^* \cdot \sigma_{\min}(A)^2 / \max_i y_i$.

Proof. Write $M(x)_j = x_j \sum_i A_{ij} y_i / (Ax)_i$ for the EM map. Differentiating,

$$\frac{\partial M_j}{\partial x_k} = \delta_{jk} \sum_i A_{ij} \frac{y_i}{(Ax)_i} - x_j \sum_i A_{ij} A_{ik} \frac{y_i}{(Ax)_i^2}.$$

At x^* the first sum is $c_j = 1$ because $Ax^* = y$, and the second is $(\operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1} A)_{jk}$. So $J = I - \operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1} A = I - \operatorname{diag}(x^*)^{1/2} S \operatorname{diag}(x^*)^{-1/2}$, which is similar to $I - S$. S is symmetric positive semidefinite and positive definite when A is injective and $x^* > 0$. For the upper bound on its spectrum, $BB^{\mathsf{T}} = \operatorname{diag}(y)^{-1/2} A \operatorname{diag}(x^*) A^{\mathsf{T}} \operatorname{diag}(y)^{-1/2}$ has the same nonzero eigenvalues as S, and $P = A \operatorname{diag}(x^*) A^{\mathsf{T}}$ is a positive matrix with row sums $(P\mathbf{1})_i = \sum_j A_{ij} x_j^* c_j = y_i$. Hence $\operatorname{diag}(y)^{-1} P$ is a positive stochastic matrix, with Perron root 1 and all eigenvalues of modulus at most 1, and BB^{T} is similar to it. So the spectrum of S lies in $(0,1]$, that of J in $[0,1)$, and the local linear rate follows from the contraction of the linearisation. The lower bound on $\lambda_{\min}(S)$ is $\min_j x_j^* \, \sigma_{\min}(A)^2 / \max_i y_i$ from the definition of B. □

For the linearised iteration, the spectrum of J is nonnegative, so EM does not oscillate near x^*; and the rate is bounded through $\sigma_{\min}(A)^2 \geq 1/\sinh^2(\pi r)$, of order $e^{-2(1-\theta')\pi/h}$ (Section 11.1 (d)), so the rate constant can be exponentially close to one for the highest modes, the ill posedness again. At the boundary level $\theta' = 1/2$ the hypothesis $x^* > 0$ fails between the zeros, and there the inactive components converge sublinearly, at the rate of Theorem 11.3.1 (an observation); the full range runs, where A has a kernel and the theorem does not apply, behave in the same way.

Remark 11.3.3 (a limit on acceleration). Dragomir, Taylor, d'Aspremont and Bolte [68] showed that for problems that are merely smooth relative to a Bregman distance, no Bregman first order method can improve the $O(1/k)$ rate in general. Any acceleration of EM here must exploit the structure of the problem, and we prove none; the $O(1/k^2)$ behaviour below is observed.

It is an information projection. The minimisers of f are the minimisers of $\mathrm{KL}(y \,\|\, Ax)$, so the EM limit is the reverse I projection of the data onto the convex cone $\{Ax : x \geq 0\}$, and the iteration is the alternating minimisation of Csiszár and Tusnády [52] between that cone and the data, the information geometric form of von Neumann's alternating projections.

PROPOSITION 11.3.4 (what EM computes). *Let $\frac12 \le \theta' < 1$ and $a = 1-\theta'$. For a grid of spacing h with data window $[-V,V]$ and unknown window $[-W,W]$, $W > V$, let*

$$R_h(x) = h \sum_{|i|\le M} \Big(y_i \log \frac{y_i}{(Ax)_i} - y_i + (Ax)_i \Big), \qquad R_h^* = \min_{x\ge 0} R_h(x),$$

the residual quoted in the tables below (it is not divided by the length of the data window), let x_h be any minimiser, for instance the limit of EM from a positive start, and let $\mu_h = \sum_j x_{h,j} h\, \delta_{w_j}$. Consider a sequence of grids with $h \to 0$ and $V \to \infty$.

(a) If ζ has no zeros in $\operatorname{Re} s > \theta'$ and

$$\frac{(\log W)^2}{W-V} \to 0, \qquad \frac{(W-V)^2 \log V}{V^2} \to 0, \qquad h^2 V \log^2 V \to 0,$$

then $R_h^* \to 0$ and $\mu_h \to \mathfrak{U}_{\theta'}$ vaguely.

(b) If ζ has a zero with real part greater than θ', then $\liminf R_h^* > 0$, whatever the choice of $W > V$.

Proof. Constants C depend on θ' only. For $y, z > 0$,

$$(\sqrt{y} - \sqrt{z})^2 \;\le\; y \log \frac{y}{z} - y + z \;\le\; \frac{(y-z)^2}{z}, \tag{11.3.2}$$

which with $t = y/z$ reduces to $s \log s - s + 1 \ge 0$, $s = \sqrt{t}$, on the left and to $t - 1 - \log t \ge 0$ on the right. Since $|P_a'(x)| = \frac{2|x|}{a^2+x^2} P_a(x) \le P_a(x)/a$, every positive measure ν with $g = P_a * \nu$ finite satisfies

$$g(v) \le e^{|v-u|/a} g(u) \qquad (u, v \in \mathbb{R}), \tag{11.3.3}$$

and so does P_a itself. We use $u_1(0) \le u_1(v) \le C \log(|v|+3)$ (Proposition 12.1.11 and $\zeta'/\zeta(1+iv) = O(\log|v|)$), and that a measure $\sum_\rho P_{a_\rho}(\cdot - \gamma_\rho)$ with $0 \le a_\rho \le \frac12$ gives mass at most $C\log(|t|+3)$ to $[t, t+1]$, by the zero count.

Step 1 (comparison point; for (a) only). Put $x_j^* = h^{-1} \mathfrak{U}_{\theta'}([w_j - \frac{h}{2}, w_j + \frac{h}{2}))$ and $z^* = Ax^*$. Since $y_i = \int P_a(v_i - w)\, d\mathfrak{U}_{\theta'}(w)$ and $|P_a(x+s) - P_a(x)| \le \frac{|s|}{a} e^{|s|/a} P_a(x)$ by (11.3.3), for $h \le a$

$$|z_i^* - y_i + \tau_i| \le \frac{h}{a} y_i, \qquad \tau_i = \int_{|w| \ge W + h/2} P_a(v_i - w)\, d\mathfrak{U}_{\theta'}(w) \le \frac{C \log W}{W - |v_i|}.$$

So $z_i^* \ge u_1(0)/2$ for all i once h is small and $W - V$ is large, and by the right side of (11.3.2)

$$R_h^* \le R_h(x^*) \le \frac{4}{u_1(0)}\, h \sum_i \Big(\frac{h^2 y_i^2}{a^2} + \tau_i^2 \Big) \le C \Big(h^2 V \log^2 V + \frac{(\log W)^2}{W-V} \Big) \to 0.$$

Step 2 (local bounds; needs only $R_h^* \to 0$). Write $z = Ax_h$ and $g_h = P_a * \mu_h$, so that $z_i = g_h(v_i)$ exactly. By the left side of (11.3.2), $h\sum_i(\sqrt{y_i}-\sqrt{z_i})^2 \le R_h^*$; with $z_i \le 2y_i + 2(\sqrt{z_i}-\sqrt{y_i})^2$ and Cauchy and Schwarz, for every interval $J \subset [-\varrho,\varrho]$, $\varrho \le V$,

$$h\sum_{v_i\in J} z_i \le C(|J|+h)\log(\varrho+3) + 2R_h^*,$$

$$h\sum_{|v_i|\le\varrho} |z_i - y_i| \le (R_h^*)^{1/2}\Big(h\sum_{|v_i|\le\varrho}(\sqrt{z_i}+\sqrt{y_i})^2\Big)^{1/2} \to 0.$$

By (11.3.3), $\int_J g_h \le e^{h/(2a)} h\sum_{v_i\in J+[-h,h]} z_i$, and with the smoothness of u_1 this gives $g_h \to u_1$ in $L^1_{\mathrm{loc}}(\mathbb{R})$. If $K+[-2,2] \subset [-V,V]$, then $\mu_h(K)\,P_a([-1,1]) \le \int_{K+[-1,1]} g_h$ with $P_a([-1,1]) = \frac{2}{\pi}\arctan\frac{1}{a} > \frac{7}{10}$, so

$$\mu_h([t,t+1]) \le C\log(|t|+3) + 3R_h^* \qquad (|t| \le V-3).$$

Hence $\{\mu_h\}$ is vaguely precompact, every vague limit μ satisfies $\mu([t,t+1]) \le C\log(|t|+3)$, so $\int(1+w^2)^{-1}d\mu < \infty$, and $g_h(0) \le e^{1/a}\,\frac12\int_{-1}^{1} g_h \le C$.

Step 3 (the limit equation). Let $\mu_h \to \mu$ vaguely along a subsequence, fix ϱ_0 and take $R \ge 2\varrho_0$ with $\mu(\{\pm R\}) = 0$. For $|v| \le \varrho_0$ split $g_h(v)$ over $|w| \le R$, $R < |w| \le V-3$ and $V-3 < |w| \le W$. The first part tends to $\int_{|w|\le R} P_a(v-w)\,d\mu(w)$ uniformly in $|v| \le \varrho_0$. By Step 2 the second is at most $\sum_{k\ge0}\mu_h(2^kR < |w| \le 2^{k+1}R)\,\frac{4a}{\pi(2^kR)^2} \le C\frac{\log R}{R}$, uniformly in h. In the third, $\big|\frac{a^2+w^2}{a^2+(v-w)^2} - 1\big| \le C\varrho_0/V$, so it equals $\kappa_h(1+O(\varrho_0/V))$ with $\kappa_h = \int_{V-3<|w|\le W} P_a(w)\,d\mu_h(w) \le g_h(0) \le C$. Along a further subsequence $\kappa_h \to \kappa \ge 0$; letting $h \to 0$ and then $R \to \infty$, $g_h \to P_a * \mu + \kappa$ uniformly on compact sets, and with Step 2

$$P_a * \mu + \kappa = u_1 \ \text{ on } \mathbb{R}, \qquad \mu \ge 0, \quad \kappa \ge 0. \tag{11.3.4}$$

Step 4 (no constant; for (a) only). For $v \in [V-5, V-4]$ and $V-3 < w \le W$, $P_a(v-w) \ge \frac{a/\pi}{a^2+(W-V+5)^2}$; integrating over $v \in [V-5,V-4]$ and using Step 2, $\mu_h((V-3,W]) \le C(W-V+5)^2\log V$, and likewise on the negative side. Hence

$$\kappa_h \le \frac{a/\pi}{(V-3)^2}\,\mu_h(V-3<|w|\le W) \le C\,\frac{(W-V+5)^2\log V}{V^2} \to 0,$$

and $\kappa = 0$.

Proof of (a). By Step 1, $R_h^* \to 0$, so Steps 2 to 4 apply, and every subsequential vague limit satisfies $P_a * \mu = u_1 = P_a * \mathfrak{U}_{\theta'}$. For the signed measure $\nu = \mu - \mathfrak{U}_{\theta'}$, which has $\int(1+w^2)^{-1}d|\nu| < \infty$, the Poisson extension $F(t,v) = (P_t * \nu)(v)$ is harmonic in $t > 0$ and vanishes at $t = a$, hence for $t \ge a$ because $P_t = P_{t-a} * P_a$, hence for all $t > 0$ by real analyticity; so $\nu = \lim_{t\downarrow0} F(t,\cdot) = 0$. Every subsequence of (μ_h) thus has a further subsequence converging to $\mathfrak{U}_{\theta'}$, and $\mu_h \to \mathfrak{U}_{\theta'}$ vaguely.

Proof of (b). Suppose $R_h^* \to 0$ along a subsequence. Steps 2 and 3 use neither the hypothesis nor the growth conditions of (a), so (11.3.4) holds for some $\mu \geq 0$ and $\kappa \geq 0$. For $\sigma > \theta'$ put $H(\sigma + iv) = \pi\big((P_{\sigma-\theta'} * \mu)(v) + \kappa\big)$, a harmonic function on $\operatorname{Re} s > \theta'$. For $\sigma \geq 1$, $P_{\sigma-\theta'} * \mu = P_{\sigma-1} * (u_1 - \kappa)$ and $\pi P_{\sigma-1} * u_1(v) = \operatorname{Re}(\xi'/\xi)(\sigma+iv)$ (Theorem 17.14.4 at levels at least 1), so $H = \operatorname{Re} \xi'/\xi$ on $\operatorname{Re} s \geq 1$. Both are harmonic on the connected open set $\{\operatorname{Re} s > \theta'\}$ minus the zeros of ξ, so they agree there. Near a zero ρ_0 of multiplicity m with $\operatorname{Re}\rho_0 > \theta'$, $\operatorname{Re}(\xi'/\xi)(s) = \operatorname{Re}\frac{m}{s-\rho_0} + O(1)$ is unbounded while H is continuous at ρ_0, a contradiction. Hence $\liminf R_h^* > 0$. □

Remark 11.3.5. (1) Part (b) is qualitative: it gives no lower bound for $\liminf R_h^*$ in terms of the distance of the zero from the line $\operatorname{Re} s = \theta'$, which is why the residual is a diagnostic and not a certificate. (2) The residual must not be divided by the length $2V$ of the data window: a defect confined to bounded heights costs a bounded total residual, which the division sends to 0. For instance, with $0 < b < a$, $0 < \epsilon < \pi b$ and data $y = 1 - \epsilon P_b > 0$, the measure dv has bounded total residual, so the normalised residual tends to 0, yet no $\mu \geq 0$, $\kappa \geq 0$ satisfy $P_a * \mu + \kappa = y$: away from $\omega = 0$ the Fourier transform of μ would be $-\epsilon e^{(a-b)|\omega|}$, which is not tempered. (3) Without the second growth condition in (a), the argument does not exclude a constant $\kappa > 0$ in (11.3.4), left by mass beyond the data window; then $\mu = \mathfrak{U}_{\theta'} - \kappa\, dv$, a positive measure whenever $\kappa \leq \inf u_{\theta'}$, so uniqueness of the continuous equation alone does not identify the limit.

Where the exact backward flow fails to be a measure, EM does not diverge: it returns a positive measure whose residual stays bounded away from zero, which is what one wants from a diagnostic.

11.4 Acceleration

EM is slow when the solution lies on the boundary of the cone, as at $\theta' = 1/2$ where the target is atomic: 3000 steps gave a residual of 4×10^{-3} in Chapter 10. This section was recomputed (Cauchy products as FFT convolutions, data u_1 from ξ'/ξ in ball arithmetic at 128 bits). Conventions: the KL residual is $R = h\sum_i \big(y_i \log(y_i/(Ax)_i) - y_i + (Ax)_i\big)$, as in Proposition 11.3.4; the start is flat, $x_j = \sum_i y_i / \sum_{i,j} A_{ij}$; a peak is a local maximum of x of prominence at least 0.05, located at its grid point unless refined by a parabola; the basin mass of a zero is $h\sum x_j$ between the midpoints to its neighbours (the first basin starts at $v = 0$, the last ends at $v = 400$); steps are EM evaluations; timings are orders of magnitude only.

11.4.1 Nesterov momentum in the logarithmic domain

Momentum in the variable x would leave the orthant. In the mirror geometry the natural momentum is geometric: with $z = \log x$,

$$z_{\mathrm{ext}}^{(k)} = z^{(k)} + \beta_k\big(z^{(k)} - z^{(k-1)}\big), \qquad x^{(k+1)} = \mathrm{EM}\big(\exp z_{\mathrm{ext}}^{(k)}\big), \tag{11.4.1}$$

with $\beta_k = (t_k - 1)/t_{k+1}$, $t_{k+1} = (1 + \sqrt{1 + 4t_k^2})/2$, the Nesterov sequence [124], and adaptive restart [126]: if f increases, take a plain EM step and reset t to 1. Positivity is automatic. This is related to accelerated mirror descent in the sense of Krichene, Bayen and Bartlett [106], with the EM surrogate as the proximal step.

11.4.2 SQUAREM

The SQUAREM of Varadhan and Roland [205] takes two EM steps, forms the differences $r = x_1 - x_0$ and $q = (x_2 - x_1) - r$, and extrapolates $x_0 - 2\alpha r + \alpha^2 q$ with $\alpha = -\|r\|/\|q\|$, followed by one more EM step. It is a quasi Newton extrapolation costing three EM evaluations per iteration. The step counts depend on two safeguards: the extrapolated point is clipped below at 10^{-30} to keep positivity, and the iteration falls back to x_2 if f at the stabilised point is not smaller than at x_2.

11.4.3 Results on a window

Levels are values of the KL residual; data on $[100, 300]$ at spacing 0.05, unknown on $[80, 320]$, dense matrix.

θ'	method	steps to 10^{-2}	to $4 \cdot 10^{-3}$	to 10^{-3}	to $3 \cdot 10^{-4}$
0.75	EM	25	36	64	104
0.75	Nesterov	11	14	20	28
0.75	SQUAREM	9	15	21	27
0.50	EM	1305	never	never	never
0.50	Nesterov	85	131	250	775
0.50	SQUAREM	114	249	993	never

("Never" means within 3000 steps.) After 3000 EM evaluations at $\theta' = 1/2$ the residuals are 4.05×10^{-3} for EM, 5.7×10^{-5} for Nesterov and 8.2×10^{-4} for SQUAREM. All three recover the 109 zeros in $(100, 300)$ to 0.03 at worst (maximum distance 0.022, 0.010 and 0.014 at $\theta' = 1/2$ with peaks refined by a parabola, 0.017, 0.027 and 0.020 at $\theta' = 3/4$; at $\theta' = 1/2$ Nesterov and SQUAREM also show 115 and 113 peaks, the extra ones small) and keep every x_j positive, the smallest being of order 10^{-52}.

11.4.4 Results on the full range by FFT

With the FFT products of (b) we took the data u_1 on $[-400, 400]$ by symmetry at spacing 0.02 and the unknown on $[-440, 440]$ (44,001 cells), and compared the 202 zeros below height 400 with the peaks; each basin mass should be one.

θ'	method	steps	KL	peaks	err (max/mean)	mass
0.75	EM	400	$4.4 \cdot 10^{-5}$	202	0.045 / 0.0067	0.999 ± 0.043
0.75	Nest.	400	$6.9 \cdot 10^{-8}$	202	0.045 / 0.0063	0.999 ± 0.042
0.50	EM	3000	$1.6 \cdot 10^{-2}$	201	$2.07^{\dagger}$ / 0.013	1.001 ± 0.038
0.50	Nest.	3000	$1.8 \cdot 10^{-4}$	205	0.010 / 0.0055	1.001 ± 0.031

The source printed $6.5{\cdot}10^{-5}$ and $2.6{\cdot}10^{-5}$ at $\theta' = 3/4$ and $1.5{\cdot}10^{-2}$ and $1.2{\cdot}10^{-4}$ at $\theta' = 1/2$; the Nesterov value at 3/4 differs by a factor of 400 and is the least stable entry (it depends on the restart details). Errors are for grid point peaks, except † (peaks refined by a parabola, one zero without a peak: with 202 grid point peaks the maximum is 0.029 and the value depends on the peak rule); the 205 peaks of the last row include three small spurious ones; the mass spreads are dominated by the two edge basins (first 1.054, last 0.517 at $\theta' = 3/4$) and are 0.026 in the interior at 3/4 and 0.006 for accelerated EM at 1/2. A run takes of the order of a minute on one cluster core (57 s for EM and 82 s for Nesterov at $h = 0.02$). At $\theta' = 1/2$ plain EM has not separated one close pair after 3000 steps, hence its large maximum error; accelerated EM separates every pair and gives each zero basin mass one with standard deviation 3.1 per cent, the largest deviations at the two edge basins. So the first two hundred zeros, with multiplicity one (Theorem 11.9.6), are read off the line $\operatorname{Re} s = 1$ by a positivity preserving iteration in a minute.

Against Theorem 11.3.1: at $\theta' = 1/2$, with R^* the least residual observed, $k\,(R(x^{(k)}) - R^*)$ for plain EM is 54.7, 51.9, 49.5 and 48.1 at $k = 500$, 1000, 2000, 3000 (the source printed 52.9, 49.4, 46.1, 44.0), bounded and slowly decreasing, and the same on grids of spacing 0.04, 0.02 and 0.01 (Section 11.11.1). For accelerated EM the product $k^2(R(x^{(k)}) - R^*)$ is 518, 355 and 416 at $k = 500$, 1000, 2000 (the source: 411, 295, 313): bounded, the empirical $O(1/k^2)$, with a sawtooth caused by the restarts. At $\theta' = 0.75$ the product $k\,(R(x^{(k)}) - R^*)$ for EM falls from 0.32 at $k = 50$ to 0.018 at $k = 400$, faster than $1/k$: the interior linear regime.

11.4.5 The first thousand zeros

The same code run to height 1450, with $h = 0.025$, the unknown on $[-1490, 1490]$ (119,201 cells) and 4000 accelerated steps, gives the following (reproduced by the recomputation). Below height 1415 there are 995 zeros; the inversion finds 994 peaks. The one failure is the closest pair in the range, 1329.0435 and 1329.2050, gap 0.16, which the iteration has not yet separated and reports as a single peak at 1329.125 carrying mass two (2.0012); the next closest pair, gap 0.22 at 1054.78 and 1055.00, is resolved. Over the 995 zeros the distance from each zero to the nearest

grid point peak has mean 0.0065, median 0.0063, 99th percentile 0.015 (with the peaks refined by a parabola, 0.0015, 0.0011 and 0.0095), and exceeds 0.03 only for the two members of the merged pair. The basin masses are 1.0007 ± 0.0053, every zero simple. The residual after 4000 steps is 3.8×10^{-4}; more steps are expected to separate the last pair, as the window runs at $\theta' = 1/2$ showed for gaps of this size.

The source reports 1375 seconds for u_1 on $[0, 1450]$ at spacing 0.025 (58,000 points) against 151 seconds for the inversion; with the evaluator of this book (`python-flint`, 128 bits, eight processes) the data took about ten seconds and the inversion a few hundred. Which layer dominates at height 1450 depends on the evaluator; what does not is that the data layer grows like $T^{1+\varepsilon}$ and the inversion like $n \log n$.

11.4.6 What the accelerated iteration computes

Restart falls back to EM, so accepted steps never increase f; if the accelerated iterates converge to a positive limit, the limit is an EM fixed point with positive coordinates, hence a minimiser of f, and in every run the accelerated iteration reached the same object as EM, the reverse I projection of Proposition 11.3.4, hence $\mathfrak{U}_{\theta'}$ when it exists. Convergence and the observed rate are not proved. What changes is the path. Su, Boyd and Candès [167] showed that Nesterov's sequence is a discretisation of the damped second order equation $\ddot{z} + (3/t)\dot{z} + \nabla f = 0$, a Euclidean analogue of the accelerated flow here. With noisy data, where the EM iteration count acts as a regularisation parameter in the manner of Landweber iteration, acceleration reaches the overfitting regime correspondingly sooner, and the restart rule becomes the stopping rule.

11.5 Von Neumann's Monte Carlo inversion

Before Markov chain Monte Carlo there was von Neumann's Monte Carlo, invented with Ulam at Los Alamos to invert matrices by random walks and written up by Forsythe and Leibler [76]. It applies to the backward flow directly; why it fails there and the Bayesian inversion succeeds is instructive.

11.5.1 The von Neumann and Ulam estimator

To solve $x = Hx + b$ when the Neumann series $x = \sum_{k \ge 0} H^k b$ converges, choose a transition matrix $p_{jk} > 0$ wherever $H_{jk} \neq 0$ and a stopping probability $q_j > 0$. Start a walk at $j_0 = j$, move $j_0 \to j_1 \to \cdots$ according to p, stop after each step with probability q (the walk is at state j_τ when it stops), and accumulate the weights $w_k = \prod_{l<k} H_{j_l j_{l+1}} / (p_{j_l j_{l+1}} (1 - q_{j_l}))$. The estimator

$$\widehat{x}_j = w_\tau \, \frac{b_{j_\tau}}{q_{j_\tau}}$$

has expectation $\sum_{k\ge0}(H^k b)_j$ term by term, because the probability of a path of length k that stops at its end is $\prod_{l<k}(1-q_{j_l})p_{j_l j_{l+1}}\cdot q_{j_k}$; this is a statement about an absolutely convergent series, which needs $\rho(|H|)<1$. Its second moment from the state j is $\sum_{m\ge0}(Q^m g)_j$ with $Q_{jk}=H_{jk}^2/((1-q_j)p_{jk})$ and $g_j=b_j^2/q_j$, so it is finite when $\rho(Q)<1$; and by the Cauchy and Schwarz step in the proof of Proposition 11.5.1 it is infinite for every admissible p,q when $b>0$ and $\rho(|H|)>1$. That $\rho(|H|)<1$ is also sufficient for suitable p,q is a standard fact; for row sums of $|H|$ below one it is shown at the end of the proof of Proposition 11.5.1.

11.5.2 Applied to the backward flow

The deconvolution $y=Ax$ is the fixed point equation $x=(I-A)x+y$ with A square ($M=N$), so $H=I-A$, $b=y$, and the Neumann series is the Richardson iteration $x^{(k+1)}=x^{(k)}+(y-Ax^{(k)})$ with unit step, the linear cousin of EM. Under $p\propto|H|$ a step is a Cauchy jump, so the estimator is a Cauchy random walk with alternating signs in the weights, started at v and evaluating y at its positions. Its variance is not finite.

Proposition 11.5.1 (infinite variance). *Let $M=N$, so that A is the square section of order $2N+1$ of the Toeplitz matrix* (11.1.3) *and $x=(I-A)x+y$ is defined, and put $t=h/(\pi(1-\theta'))$. If $t<t_0=0.52219\ldots$, the root in $(0,1)$ of $\coth(1/t)=2t$, in particular if $h<\pi(1-\theta')/2$, then the bi infinite operator $|I-A|$ has spectral radius $1-2t+\coth(1/t)>1$, and so do its sections for all $N\ge N_0(h,\theta')$. For such a section consider the von Neumann and Ulam estimator of $\sum_{k\ge0}(I-A)^k y$ which, at state j, stops with probability $q_j\in(0,1)$ and otherwise moves to k with probability $(1-q_j)p_{jk}$, $p_{jk}>0$ wherever $(I-A)_{jk}\ne0$, and returns the product of the weights $(I-A)_{jk}/((1-q_j)p_{jk})$ along the path times y_j/q_j at the stopping state j. Its second moment is infinite from every starting state. If $t_0<t\le1$, the spectral radius of $|I-A|$ is below one for the bi infinite operator and all its sections, and the conclusion fails.*

Proof. For $t\le1$ the diagonal entry is $A_{jj}=t$, so $|I-A|=(1-2t)I+A$ entrywise, with symbol $1-2t+\phi(\xi)$ and $\phi(\xi)=\cosh((\pi-|\xi|)/(\pi t))/\sinh(1/t)$, the symbol of (11.1.3). For a Laurent operator with nonnegative summable entries the spectral radius on $\ell^2(\mathbb{Z})$ is the maximum of the symbol, here its value $1-2t+\coth(1/t)$ at $\xi=0$, which is also the row sum. The function $\coth(1/t)-2t$ decreases on $(0,1]$, its derivative being $(x/\sinh x)^2-2<0$ with $x=1/t$; it is positive near 0 and equals $\coth 1-2<0$ at $t=1$, so the spectral radius exceeds one exactly when $t<t_0$. For a section, a symmetric nonnegative matrix, the spectral radius is the largest Rayleigh quotient; the normalised constant vector has quotient tending to the row sum as $N\to\infty$, which gives N_0. If $t_0<t\le1$, every row sum of every section is at most $1-2t+\coth(1/t)<1$. With $r_j=\sum_k|H_{jk}|$ the row sums of $H=I-A$, the choice $q_j=1-r_j$ and $p_{jk}=|H_{jk}|/r_j$ gives $(1-q_j)p_{jk}=|H_{jk}|$

and $Q_{jk} = |H_{jk}|$, so $\rho(Q) \le \max_j r_j < 1$ and the second moment $\sum_m (Q^m g)_j$ is finite: for this p, q the conclusion fails.

Let $H = I - A$ be a section with $\lambda = \rho(|H|) > 1$ and Perron vector $w > 0$ (all entries of $|H|$ are positive). With $Q_{jk} = H_{jk}^2/((1-q_j)p_{jk})$, summing over the paths gives the second moment $\sum_{m\ge 0}(Q^m g)_{j_0}$ from the state j_0, where $g_j = y_j^2/q_j > 0$ since $y = u_1 > 0$. By Cauchy and Schwarz,

$$\begin{aligned}\lambda w_j = \sum_k |H_{jk}| w_k &\le \Big(\sum_k Q_{jk} w_k^2\Big)^{1/2} \Big(\sum_k (1-q_j)p_{jk}\Big)^{1/2} \\ &\le \big(Q w^{(2)}\big)_j^{1/2}, \qquad w^{(2)} = (w_k^2)_k,\end{aligned}$$

so $Qw^{(2)} \ge \lambda^2 w^{(2)}$ and $Q^m w^{(2)} \ge \lambda^{2m} w^{(2)}$. As $g \ge c\, w^{(2)}$ for some $c > 0$, the series diverges. (The same argument without squares shows that the estimator is not even integrable: $E|\widehat{x}_j| = \sum_k (|H|^k y)_j = \infty$.) □

All grids of this chapter are in the first case ($t = 0.0127$ for $h = 0.02$ and $\theta' = \frac{1}{2}$). So the oldest Monte Carlo inversion exists on paper and is useless in practice, and the reason is sign: $I - A$ has entries of both signs, and the cancellation the Neumann series relies on is what a sample cannot reproduce. This is the usual fate of von Neumann and Ulam on ill conditioned systems, and the condition number here is of order $e^{(1-\theta')\pi/h}$.

11.5.3 The positive walk

The Bayesian inversion is also a Monte Carlo of a random walk, but a positive one. Under the current x, read A as a Markov kernel from sources w_j to detectors v_i: a unit of mass at w_j lands at v_i with probability A_{ij}/c_j, a Cauchy jump. Bayes' rule reverses it: a unit of mass observed at v_i came from w_j with probability

$$\pi_{ij}(x) = \frac{A_{ij} x_j}{(Ax)_i},$$

a reverse Cauchy jump weighted by the current estimate of the sources. The EM step is the expectation of this reverse walk,

$$x_j^+ = \frac{1}{c_j} \sum_i y_i\, \pi_{ij}(x)\,:$$

every unit of observed mass sent one step backward in expectation. The data augmentation sampler of Chapter 10 replaces the expectation by a sample: draw the source of each unit of observed mass from $\pi_{ij}(x)$ (the y_i are densities, so the units are pseudo counts and the sampler is a computational device, not an exact likelihood), then draw x from its gamma conditional. The weights are probabilities, so the sampled step has multinomial variance and no more: not the

Neumann series sampled by a signed walk, which diverges, but the EM fixed point sampled by a positive reverse walk, a Gibbs sampler. The nonlinearity of the EM map, x appearing in $\pi_{ij}(x)$, is what removes the signs.

Remark 11.5.2. Von Neumann's alternating projection theorem is, through Csiszár and Tusnády, the structure of the EM iteration itself (a reading, not used as a theorem), and the signed random walk shows what goes wrong without positivity.

11.6 The Cauchy kernel as a scale mixture of normals

The Cauchy law is a scale mixture of normals, a fact used for robust regression since Andrews and Mallows [4] and in Bayesian computation since Carlin and Polson [47]. Here the mixture has a probabilistic meaning that ties the chapter to the hitting time chapters.

Proposition 11.6.1 (subordination). *Let B be a standard Brownian motion and T_a the first passage time of an independent Brownian motion to level $a > 0$, with density $\ell_a(t) = a(2\pi)^{-1/2} t^{-3/2} e^{-a^2/(2t)}$. Then $\gamma + B(T_a)$ has the Cauchy law of centre γ and scale a, that is*

$$\frac{a/\pi}{a^2 + (v-\gamma)^2} = \int_0^\infty \frac{e^{-(v-\gamma)^2/(2t)}}{\sqrt{2\pi t}}\, \ell_a(t)\, dt, \tag{11.6.1}$$

and on the Fourier side $e^{-a|\omega|} = \int_0^\infty e^{-t\omega^2/2} \ell_a(t)\, dt$. Consequently, if ζ has no zeros in $\operatorname{Re} s > \theta_0$ and $\theta > \theta_0$,

$$\mathfrak{U}_\theta = \sum_\rho \int_0^\infty \left(e^{t\Delta/2} \delta_{\gamma_\rho}\right) \ell_{\theta - \sigma_\rho}(t)\, dt, \tag{11.6.2}$$

the Thorin measure at level θ is the sum over the zeros of the heat flow of δ_{γ_ρ} run to the random time $T_{\theta-\sigma_\rho}$, and under RH it is the heat flow of $\sum_\rho \delta_{\gamma_\rho}$ to the single random time $T_{\theta-1/2}$.

Proof. The Fourier identity is the Laplace transform of the first passage law, $E e^{-\lambda T_a} = e^{-a\sqrt{2\lambda}}$ at $\lambda = \omega^2/2$; the density identity is its inverse transform; (11.6.2) is equation (10.1.1) (the sum of Cauchy terms $P_{\theta-\sigma_\rho}(\cdot - \gamma_\rho)$, a sum of positive measures) with each Cauchy term written as the mixture. □

Formula (11.6.2) connects the Thorin measure to the heat trace of the zeros, the sum of $e^{-t\gamma^2/2}$ over the zeros (compare Section 12.2.3): under RH the Fourier transform of $\mathfrak{U}_\theta$ is

$$\int_0^\infty \Big(\sum_\gamma e^{-i\omega\gamma}\Big) e^{-t\omega^2/2}\, \ell_{\theta-1/2}(t)\, dt,$$

the Fourier transform of the zero measure multiplied by the subordinated Gaussian. It also explains why the backward Cauchy flow is less ill posed than a backward heat flow: the forward multiplier $e^{-(1-\theta')|\omega|} = \int_0^\infty e^{-t\omega^2/2}\ell_{1-\theta'}(t)\,dt$ is an average of heat multipliers, carried for large $|\omega|$ by small t, where $\ell_a(t)$ behaves like $e^{-a^2/(2t)}$. The backward multiplier is the reciprocal of this average, not the average of the reciprocals $e^{t\omega^2/2}$ (whose integral against ℓ_a diverges for every $\omega \neq 0$, as $\ell_a(t) \sim a(2\pi)^{-1/2}t^{-3/2}$ at infinity), so it grows exponentially and not like a Gaussian.

11.6.1 Data augmentation with latent times

The mixture gives the Bayesian inversion a conditionally Gaussian structure. Augment the complete data of Chapter 10: each unit of mass observed at v_i (a pseudo count, as above) carries one latent time t_i with prior ℓ_a, $a = 1 - \theta'$, and given t_i and its source w_j its displacement is $N(0, t_i)$. Then

$$\begin{aligned} p(t_i \mid v_i, w_j) &\propto t_i^{-1/2} e^{-(v_i - w_j)^2/(2t_i)}\, t_i^{-3/2} e^{-a^2/(2t_i)} \\ &= t_i^{-2} \exp\Big(-\frac{a^2 + (v_i - w_j)^2}{2t_i}\Big), \end{aligned}$$

an inverse gamma law with shape 1 and scale $\frac{1}{2}(a^2 + (v_i - w_j)^2)$, explicit and cheap to sample. Given the times, the allocation of the observed mass at v_i to the sources is multinomial with Gaussian weights

$$x_j \exp\big(-(v_i - w_j)^2/(2t_i)\big),$$

normalised over j, and given the allocations the x_j are gamma as before. The three block Gibbs sampler alternates these steps; the allocation step is dense, since each observation is allocated among all the sources, so a sweep costs of the order of $(2M + 1)(2N + 1)$ operations. This is the Carlin and Polson augmentation [47] applied to the forward map rather than to an error law, and it is the Monte Carlo inversion of Chapter 10 with one extra block.

11.6.2 Does the mixture help?

For the complexity classification, no. The marginal posterior in x is the log concave law of Theorem 11.2.1 with or without the latent times, and the augmented posterior in (x, t) is not jointly log concave, the inverse gamma factors being log convex in t on part of their range; the random walk guarantee applies to the marginal, which is what one samples. For the data layer, no: the latent times do not touch the evaluation of ζ. For the sampler, yes in the usual way: the conditionally Gaussian blocks mix well, and each conditional is a standard law. For understanding, yes: (11.6.2) says that the Thorin semigroup is Brownian motion run to a first passage time, that the levels θ index the level of the barrier, and that

RH is the statement that one barrier, at distance $\theta - \frac{1}{2}$, serves every zero. The backward flow is the un stopping of that Brownian motion, and its ill posedness is the usual one for the backward heat equation, tamed by the first passage mixture to an exponential rather than Gaussian amplification.

11.7 What is being computed

Collecting the chapter: given ζ on the line $\operatorname{Re} s = 1$ to an absolute precision on the scale of u_1, which for the linear inverse costs $(1 - \theta') \log_2 T$ bits near height T (Proposition 11.2.2), which is the expensive part and the only part that grows faster than the output, the EM iteration and its accelerated forms compute the information projection of the level one Thorin measure onto the cone of Cauchy blurred positive measures at level θ'. When ζ has no zeros to the right of θ' the limit is the Thorin measure $U_{\theta'}$ of $\xi(\theta')/\xi(\theta' + \sqrt{s})$ (Proposition 11.3.4(a), under its conditions on the windows). Under the Riemann hypothesis this is the Brownian image of the zero measure stopped at the first passage of level $\theta' - \frac{1}{2}$, and the zeros are its peaks. When there is a zero to the right of θ', the residual of the grid problems stays bounded away from zero as the grid is refined (Proposition 11.3.4(b)); the residual is a diagnostic, not a certificate. The posterior over the grid problem is log concave (Theorem 11.2.1). None of this reaches the arithmetic, which lives in the evaluation of ζ and costs polynomial time in T by every known method.

In the rest of this part, Theorems 11.9.6 and 11.10.2 and the EM runs of Section 11.9.6 were recomputed; the three routes of Section 11.8, the mass deficit and level mass tables, the tables of $U_{3/4}$ and the timings are the author's, not rerun. The author's figures are not reproduced; the text says what they show.

11.8 Thorin measures at level 7/8

11.8.1 Thorin measures, not primes

What the backward flow computes well is Thorin measures. At every level $\theta \geq 1$ the Thorin measure of $\xi(\theta)/\xi(\theta+\sqrt{s})$ has the explicit density $\pi^{-1} \operatorname{Re}(\xi'/\xi)(\theta+iv)$, with the prime sum $-\sum \Lambda(n) n^{-\theta} \cos(v \log n)$ inside it (absolutely convergent for $\theta > 1$; at $\theta = 1$ it stands for the value of ζ'/ζ on the line), computable on one line with no zeros and no inversion (Chapter 10, Theorem 17.14.2). Below level one, above the zero free abscissa, the Thorin measure is the EM limit from the level one data, and at $\theta' = \frac{1}{2}$ it is resolved down to the atoms: the thousand zero run of Section 11.4.5 computes the level one half Thorin measure of the reciprocal ξ function from the Euler product side. Under the Riemann hypothesis the measure at level $\theta > \frac{1}{2}$ is the Cauchy smoothing of the zero measure at width $\theta - \frac{1}{2}$, so a computed Thorin measure is to be compared with that smoothed density, not

with the atomic measure in total variation, where the two are mutually singular whatever the truth.

What the flow does not compute is primes. They are the spectrum of the level one measure, point masses at $\pm\log n$ of weight proportional to $-\Lambda(n)/n$, put in through the Euler product when u_1 is evaluated; reading them off (Chapter 10) recovers what was supplied. The reverse passage, from zeros to primes, is the classical explicit formula for $\psi(x)$, which nothing here changes.

11.8.2 The level 7/8 measure

The quasi Riemann hypothesis (Section 17.7) is the zero free half plane $\operatorname{Re} s > 7/8$, announced in October 2026 by OpenAI, Liu and Thorner with a formalisation in Lean and not yet refereed [227, 228, 198]. It makes $\theta = 7/8$ the lowest level at which the Thorin representation is currently claimed; every statement below that rests on it is conditional on it, as in Chapter 17:

$$\frac{\xi(7/8)}{\xi(7/8+\sqrt{s})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{u}\Big)U_{7/8}(du)\Big), \tag{11.8.1}$$

with $U_{7/8}$ a positive Thorin measure (Theorem 17.14.2). We compute it three ways, the agreement being the check. Throughout, $u(v)$ denotes the density of the symmetric measure $\mathfrak{U}_{7/8}$ in the variable v; the Thorin measure $U_{7/8}$ in the variable $u = v^2$ is the image of the restriction of $\mathfrak{U}_{7/8}$ to $v > 0$, so that $\int g(u)U_{7/8}(du) = \int_0^\infty g(v^2)u(v)\,dv$, and its density is $u(\sqrt{u})/(2\sqrt{u})$.

Route (a), direct. $u(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{7}{8}+iv)$, with ξ'/ξ from its gamma factor and ζ'/ζ (Section 11.9.8), needs ζ and ζ' inside the critical strip, which any multiprecision library supplies; on $[0,400]$ at spacing 0.05 it took 100 seconds in the author's run. The formula defines a signed density unconditionally; the announced half plane is what makes it the Thorin measure of (11.8.1). Its minimum on $[0,400]$ is $0.0055 > 0$ and its mass on $[0,400]$ is 201.63 against 202 zeros, the deficit being the Cauchy tails of width $3/8$ at the ends.

Route (b), deconvolution. Run the accelerated EM of Section 11.4 on the level one data with the Cauchy kernel of width $1-\frac{7}{8} = \frac{1}{8}$. The condition number $e^{\pi/(8h)}$ is $e^{7.9}$ at $h = 0.05$, mild, and 400 steps reach a residual of 2×10^{-10} (the author's run). This route uses ζ on the line $\operatorname{Re} s = 1$ only, never inside the strip. Away from the edges of the data window it agrees with route (a) to 6×10^{-4} absolutely and 0.33 per cent relatively.

Route (c), zeros. Under the Riemann hypothesis, which is verified far beyond height 400 [232],

$$u(v) = \sum_\rho \frac{(3/8)/\pi}{(3/8)^2 + (v-\gamma_\rho)^2},$$

summed over all zeros with both signs of γ. With the first 1000 zeros and a tail from the zero density $\frac{1}{2\pi}\log\frac{v}{2\pi}$ above the last one, this agrees with route (a) to

9×10^{-5} absolutely and 1.6 per cent relatively, the relative discrepancy being the crude tail at the top of the range.

Finally the identity (11.8.1) itself, with route (a) on the right. Minus the logarithm of the left side is $\log \xi(\frac{7}{8}+\sqrt{s}) - \log \xi(\frac{7}{8})$ and the right side is $\int_0^\infty \log(1+ s/v^2)\, u(v)\, dv$. At $s = 1$ the left side is 0.040368, the integral over $[0, 400]$ is 0.038361 and the tail from the zero density 0.002051, total 0.040412; at $s = 4$ the three numbers are 0.126495, 0.118335, 0.008202, total 0.126537. The agreement is to the accuracy of the tail.

11.8.3 What this gives the programme

Route (b) is the one to notice. It produces the Thorin measure at level 7/8 from data on the line $\operatorname{Re} s = 1$, where the Euler product is in force, by a positivity preserving iteration with no reference to zeros and no continuation of ζ into the strip. The same computation runs at any $\theta' \in (1/2, 7/8)$; the half plane adds at 7/8 the guarantee that the limit is a Thorin measure rather than an information projection with a residual. Watching the residual at $\theta' < 7/8$ is therefore the computational form of the question whether the zero free half plane extends, a diagnostic and not a certificate, since Proposition 11.3.4(b) is qualitative.

11.9 Pictures of the Thorin measures and the decision of positivity

11.9.1 Pictures

The densities u_θ on $[0, 60]$ at $\theta = 0.55, 0.6, 0.75, 7/8, 1$, each computed directly on its own line, show the Cauchy semigroup (Proposition 2.1.3) from the side: as θ decreases to $\frac{1}{2}$ the peaks at the zeros sharpen like $1/(\pi(\theta - \frac{1}{2}))$ and the troughs between them deepen, and at $\theta = \frac{1}{2}$ the picture becomes the zero measure.

The level 7/8 measure on $[0, 400]$ oscillates about $(2\pi)^{-1} \log(v/2\pi)$, the density of the zeros, as the average of a sum of Cauchy bumps must. In the variable $u = v^2$ the Thorin measure $U_{7/8}$ has density $g(u) = u_{7/8}(\sqrt{u})/(2\sqrt{u})$, and its two asymptotes make the Thorin conditions visible: $g(u) \sim u_{7/8}(0)/(2\sqrt{u})$ as $u \to 0$, so $\int_0^1 |\log u|\, U_{7/8}(du) < \infty$, and $g(u) \approx \log(\sqrt{u}/2\pi)/(4\pi\sqrt{u})$ on average as $u \to \infty$, so $\int_1^\infty u^{-1} U_{7/8}(du) < \infty$. The total mass is infinite, consistent with $F_{7/8}(s) = \xi(7/8)/\xi(7/8 + \sqrt{s})$ decaying faster than any power of s, in fact like $\exp(-\frac{1}{2}\sqrt{s}\log\sqrt{s})$.

A third figure compares routes (b) and (c) with route (a): both differences are smooth and below 10^{-3} on the whole range.

11.9.2 What positivity means

Theorem 11.9.1 (positivity). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent.*

(i) U_θ *is a positive Thorin measure, that is,* $F_\theta(s) = \xi(\theta)/\xi(\theta + \sqrt{s})$ *is the Laplace transform of a generalised gamma convolution.*

(ii) ζ *has no zeros in* $\operatorname{Re} s > \theta$.

(iii) $\operatorname{Re}(\xi'/\xi)(s) > 0$ *for* $\operatorname{Re} s > \theta$*; that is,* $u_\sigma(v) > 0$ *for every* $\sigma > \theta$ *and every* v.

For $\theta = \frac{1}{2}$ *the equivalence of (ii) and (iii) is Lagarias's theorem [185].*

Proof. (i) and (ii) are equivalent by Theorem 17.14.2. (ii) implies (iii) term by term, since $\operatorname{Re}(\xi'/\xi)(s) = \sum_\rho (\sigma - \sigma_\rho)/|s - \rho|^2$, the sum taken over the zeros in pairs ρ, $1 - \bar\rho$, and every $\sigma - \sigma_\rho$ is positive. For (iii) implies (ii), let $\rho_0 = \sigma_0 + i\gamma_0$ be a zero of multiplicity m with $\sigma_0 > \theta$. Near ρ_0, $\xi'/\xi(s) = m/(s - \rho_0) + h(s)$ with h analytic, so $\operatorname{Re}(\xi'/\xi)(\sigma_0 - \varepsilon + i\gamma_0) = -m/\varepsilon + O(1)$, which is negative for small ε, and $\sigma_0 - \varepsilon > \theta$. □

Positivity on the single line $\operatorname{Re} s = \theta$, $u_\theta \geq 0$, is necessary for (ii) but not sufficient. An off line zero ρ_0 puts a negative bump of height $1/(\pi(\sigma_0 - \theta))$ at $v = \gamma_0$, and its mirror $1 - \bar\rho_0$ a positive bump of height $1/(\pi(\theta + \sigma_0 - 1))$. The remaining zeros add a background which, once $\theta - \frac{1}{2}$ exceeds the mean spacing of the zeros near γ_0, is about the zero density

$$d(\gamma_0) = \frac{1}{2\pi} \log \frac{\gamma_0}{2\pi}.$$

Heuristically, since the background grows with the height, a zero high enough up is invisible to pointwise positivity on a line to its left (an open question, Remark 2.1.4); at height 60 the opposite holds (last column of Table 11.1).

11.9.3 The mass deficit

The invariant that does not mask is the mass.

Proposition 11.9.2 (mass deficit). *Let* $\frac{1}{2} < \theta < 1$ *and* $T > 0$*, and suppose no zero lies on the segment* $\operatorname{Re} s = \theta$*,* $0 \leq \operatorname{Im} s \leq T$*, nor on the segment* $\operatorname{Im} s = T$*,* $\theta \leq \operatorname{Re} s \leq 1$*. Let* $M_\sigma(T) = \int_0^T u_\sigma(v)\,dv$ *and let* $N(\theta, T)$ *be the number of zeros with* $\operatorname{Re}\rho > \theta$ *and* $0 < \operatorname{Im}\rho < T$*. Then*

$$2N(\theta, T) = M_1(T) - M_\theta(T) + \frac{1}{\pi}\Delta_{\mathrm{top}}(\theta, T), \tag{11.9.1}$$

where $\Delta_{\mathrm{top}}(\theta, T) = -\int_\theta^1 \operatorname{Im}(\xi'/\xi)(\sigma + iT)\,d\sigma$ *is the change of* $\arg\xi$ *along the top side.*

Proof. Apply the argument principle to ξ on the rectangle with corners θ, 1, $1+iT$, $\theta+iT$, traversed counterclockwise. There are no zeros on $\operatorname{Re} s = 1$, and ξ is real and positive on $[\theta, 1]$, so the bottom side contributes nothing. By the Cauchy Riemann equations $\partial_v \arg \xi(\sigma+iv) = \operatorname{Re}(\xi'/\xi)(\sigma+iv) = \pi u_\sigma(v)$, so the right side contributes $\pi M_1(T)$ and the left side, traversed downward, $-\pi M_\theta(T)$. The top side contributes Δ_{top}, since $\partial_\sigma \arg \xi = \operatorname{Im}(\xi'/\xi)$. The total is $2\pi N(\theta, T)$. □

So, after the correction along the top, the Thorin measure at level θ carries two units less mass up to height T than at level one for each zero to its right. The deficit is an integer, decided by computing it to accuracy better than one half. The identity is the argument principle of Backlund and Turing; the Thorin picture supplies the bookkeeping, not a cheaper computation. As θ decreases to $\frac{1}{2}$ with no zeros off the line below T, $M_\theta(T)$ tends to the number of zeros on the critical line below T, and (11.9.1) becomes the classical formula $N(T) = M_1(T) + \pi^{-1}\Delta_{\text{top}}(\frac{1}{2}, T)$, which is Backlund's count and holds with $N(T)$ the number of all zeros below T, with no hypothesis, when T is not an ordinate.

Table 11.1 evaluates (11.9.1) at $T = 100$ for the real ξ, and for a synthetic $\tilde{\xi}$ in which the consecutive zeros $\frac{1}{2} + 59.347i$ and $\frac{1}{2} + 60.832i$ and their conjugates are replaced by the off line quadruple $0.85 \pm 60.089i$, $0.15 \pm 60.089i$. The factor is a ratio of products of two quadratics, with the exact zeros in the denominator, symmetric under $s \mapsto 1-s$ and positive on the real axis, so $\tilde{\xi}$ is entire of order one with the functional equation of ξ. Arguments were tracked on a grid of spacing 0.02 in v and 0.0005 in σ (the author's computation, not rerun).

θ	$M_\theta(100)$	Δ_{top}/π	$2N$ for ξ	$2N$ for $\tilde{\xi}$	$\min \tilde{u}_\theta$
0.55	29.0109	−0.1002	0.0000	2.0000	−0.26
0.60	29.0218	−0.0893	0.0000	2.0000	−0.55
0.65	29.0328	−0.0784	0.0000	2.0000	−0.94
0.70	29.0438	−0.0674	0.0000	2.0000	−1.51
0.75	29.0548	−0.0563	0.0000	2.0000	−2.57
0.80	29.0659	−0.0452	0.0000	2.0000	−5.38
0.875	29.0827	−0.0284	0.0000	0.0000	0.006
0.90	29.0884	−0.0228	0.0000	0.0000	0.006
0.95	29.0997	−0.0114	0.0000	0.0000	0.007
1.00	29.1112	0	0.0000	0.0000	0.007

Table 11.1: The mass deficit at $T = 100$, where $M_1(100) = 29.1112$ and $N(100) = 29$ (the author's computation, not rerun).

For the real ξ the deficit vanishes to four decimals at every level of the table, and $M_\theta(100)$ decreases almost linearly, with slope 0.223 per unit of θ, towards 29.000 as $\theta \to \frac{1}{2}$: the smoothed zero count becomes the zero count. For $\tilde{\xi}$ the deficit is 2 exactly when $\theta < 0.85$ and 0 above, as the table shows; at this height the synthetic density is also visibly negative near 60 at every level below 0.85.

Corollary 11.9.3 (the level mass function). *Let $T > 0$ not be the ordinate of a zero. For $\theta \in (\frac{1}{2}, 1]$ not the real part of a zero below height T,*

$$M_\theta(T) = M_1(T) + \frac{1}{\pi}\Delta_{\mathrm{top}}(\theta, T) - 2N(\theta, T).$$

The first two terms form a real analytic function of θ, since ξ has no zero on the line $\operatorname{Im} s = T$; its derivative is

$$\frac{1}{\pi} \operatorname{Im} \frac{\xi'}{\xi}(\theta + iT).$$

The last term is a step function which drops by $2m$ as θ decreases through $\operatorname{Re}\rho$, for each zero ρ of multiplicity m with $\operatorname{Re}\rho > \frac{1}{2}$ and $0 < \operatorname{Im}\rho < T$. Hence $\theta \mapsto M_\theta(T)$ extends continuously to $(\frac{1}{2}, 1]$ if and only if every zero below height T lies on the critical line, and its jumps occur exactly at the real parts of the off line zeros, of size twice the total multiplicity of the zeros with that real part and ordinate below T.

Run θ down from 1, and a zero off the line shows itself by a jump of exactly 2 at its real part. At $T = 100$, for ξ the masses on the ten vertical lines of the table agree with the smooth part, computed from one horizontal sweep, to 2.3×10^{-5}, and the smooth part has slope 0.223 per unit of θ; for $\tilde{\xi}$ they lie exactly 2.0000 below it for $\theta < 0.85$ and on it above (the author's computation, not rerun). The corollary does not locate real parts more cheaply than the argument principle, since the jumps are seen only by computing along vertical lines.

11.9.4 Positivity of $U_{7/8}$

Given the zero free half plane $\operatorname{Re} s > 7/8$ (announced, not refereed, as above), Theorem 11.9.1 makes $U_{7/8}$ positive: every term of $u_{7/8} = \sum_\rho P_{7/8-\sigma_\rho}(v - \gamma_\rho)$, with P_a the Cauchy density of width a and $P_0 = \delta_0$, is nonnegative, a zero on the line $\operatorname{Re} s = 7/8$, if there were one, contributing an atom. The computation displays the margin. On $[0, 400]$ the minimum of $u_{7/8}$ is 0.00551, attained at $v = 0$, where $u_{7/8}(0) = \pi^{-1}(\xi'/\xi)(7/8) = 0.005514663528$; this value is positive unconditionally, because $\xi(\frac{1}{2} + t) = \int \Phi(x)\cosh(tx)\,dx$ with Riemann's kernel Φ positive, so ξ increases on $(\frac{1}{2}, \infty)$. Away from the origin the minimum on $[15, 400]$ is 0.0272, at $v = 17.5$ between the first two zeros; with the second difference estimate $\max|u''| \approx 11.8$ on the grid of spacing 0.05, the grid minimum less $h^2/8 \cdot \max|u''|$ is 0.0235. Theorem 11.9.6 certifies the positivity of $u_{7/8}$ in ball arithmetic on $[0, 1420]$, and Theorem 11.10.2 certifies the lower bound 0.0272 for $u_{7/8}$ on $[13.5, 1420]$, the least certified bound lying at $v = 17.49$. The mass deficit at $\theta = 7/8$, $T = 100$, is zero.

11.9.5 The measure $U_{3/4}$ below level 7/8

At every height the positivity of $U_{3/4}$ is, by Theorem 11.9.1, the open zero free half plane $\operatorname{Re} s > 3/4$. We computed it as at 7/8, the direct route on a grid of spacing 0.02 (20,001 points) since the peaks have width 1/4. The table is the author's computation, not rerun apart from the value at the origin.

quantity	value
$\min u_{3/4}$ on $[0, 400]$	0.00368 at $v = 0$
$\min u_{3/4}$ on $[15, 400]$	0.0182 at $v = 17.5$
$\max \lvert u'' \rvert$; margin	40.5; 0.0162
mass on $[0, 400]$	201.60 against $N(400) = 202$
zeros plus tail minus direct	5.7×10^{-5}
EM: residual; relative error	1.5×10^{-9}; 1.1 per cent
EM density: minimum	4.0×10^{-3}
$2N(\frac{3}{4}, 400.9)$	$202.1778 - 202.0891 - 0.0887 = 0.0000$

The EM run used the Cauchy kernel of width 1/4, condition number $e^{\pi/(4h)} = e^{15.7}$ at $h = 0.05$, and 1500 accelerated steps; the height $T = 400.9$ for the deficit lies between the zeros 399.985 and 401.839. The minimum at the origin equals $\pi^{-1}(\xi'/\xi)(3/4) = 0.003676904191$, and the margin is the grid minimum on $[15, 400]$ less $\frac{h^2}{8} \max \lvert u'' \rvert$. So $u_{3/4}$ is positive on $[0, 400]$, by a margin of 0.0037 at the origin and 0.016 elsewhere, and its mass deficit to height 400.9 is zero: no zero lies to the right of 3/4 below that height, as was known far higher [232]; Theorem 11.9.6 certifies this on $[0, 1420]$. Against $U_{7/8}$, the author's figure shows the peaks higher and the troughs deeper, the semigroup run one eighth further towards the zeros.

11.9.6 All the way down: $U_{1/2}$

At $\theta = \frac{1}{2}$ the densities collapse. Since $\xi(\frac{1}{2} + iv) = \Xi(v)$ is real, $\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + iv) = 0$ away from the zeros, and the measure is purely atomic: $\pi\, \mathfrak{U}_{1/2} = \partial_v \arg \xi(\frac{1}{2} + iv)$, which jumps by πm at a zero of multiplicity m on the line. Under the Riemann hypothesis $\mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho}$ and $U_{1/2} = \sum_{\gamma > 0} m_\gamma \delta_{\gamma^2}$, the Thorin measure of the reciprocal ξ function around which the book is organised. It can be computed three ways; the author's last figure shows the cumulative masses $M_\theta(v)$ at $\theta = 1, 0.75, 0.55$ converging to the zero staircase $N(v)$, and the EM backward flow at $\theta' = \frac{1}{2}$ on a logarithmic scale with the sign changes of Z marked.

Directly at level $\frac{1}{2}$, the atoms of odd multiplicity are the sign changes of Hardy's function $Z(v) = e^{i\vartheta(v)} \zeta(\frac{1}{2} + iv)$, which is real with the sign of $\Xi(v)$ up to a fixed sign. A scan of $[1, 400.5]$ at spacing 0.02, 19,976 evaluations of Z in 122 seconds (the author's run), followed by root finding, gives 202 atoms below height 400, agreeing with the tabulated zeros to machine precision, the classical computation, included for comparison.

By the backward flow at $\theta' = \frac{1}{2}$ from level one, the runs of Section 11.4 apply: on $[-400, 400]$ with 3000 accelerated steps the peaks lie within 0.010 of the zeros, 0.0055 on average, with basin masses 1.0013 ± 0.0313 (± 0.0061 for the interior basins), and to height 1415 the basin masses are 1.0007 ± 0.0053, the unresolved pair at 1329.125 carrying mass 2.001. The author's centroid statements (the basin centroids lie on average 0.0014 and at most 0.081 from the zeros, the density between the atoms falls to 10^{-12}) were not rerun.

As the limit of the cumulative masses, $M_\theta(v)$ approaches $N(v)$: $M_1(100) = 29.1112$, $M_{0.75}(100) = 29.0548$, $M_{0.55}(100) = 29.0109$, and $N(100) = 29$.

Is $U_{1/2}$ positive? That is the Riemann hypothesis (Theorem 17.14.2). To certify positivity up to height T, the sign changes, which count zeros of odd multiplicity on the line, need the total count $N(T)$, which Backlund's count supplies from level one data:

$$N(T) = M_1(T) + \frac{1}{\pi}\Delta_{\text{top}}(\tfrac{1}{2}, T).$$

At $T = 400.9$ this gives $202.1778 - 0.1778 = 202.0000$, and there are 202 sign changes of Z below 400.9. Every zero below height 400.9 is therefore on the line and simple, which Theorem 11.9.6 certifies to height 1419.9, and $\mathfrak{U}_{1/2}$ restricted to $|v| < 400.9$ is the sum of unit atoms at the ordinates. This is Turing's verification [270] in Thorin bookkeeping: the level one mass counts the zeros, the level one half atoms locate them on the line, and the counts agree.

11.9.7 Can positivity be decided?

By level. For $\theta \geq 1$ positivity is an unconditional theorem, and for $\theta \geq 7/8$ it holds conditionally on the announced half plane; the computation only displays it.

For $\frac{1}{2} < \theta < 7/8$ and finite T, positivity up to height T, $N(\theta, T) = 0$, is decidable: compute the deficit (11.9.1) to accuracy better than one half in interval arithmetic, provided no zero lies on the boundary of the rectangle, which is known within the verified range. For $\theta = \frac{1}{2}$ Backlund's count and the sign changes decide it. Theorem 11.9.6 does this in ball arithmetic at $\theta = 3/4$ and $T = 1419.9$.

For $\frac{1}{2} \leq \theta < 7/8$ and all heights, positivity is the zero free half plane $\operatorname{Re} s > \theta$, and for $\theta = \frac{1}{2}$ the Riemann hypothesis. If false it is refutable by a finite computation: an offending zero lies in a finite rectangle, over which the deficit is a positive even integer. If true it is not confirmable by finitely many heights without an argument for the rest. That is why the half plane at 7/8 is claimed by a proof, and why no computation in this book settles the positivity of U_θ below 7/8.

11.9.8 Computational details

The author's computations used Python with `mpmath` [213] at 15 significant digits for ζ, ζ', $\log\Gamma$, ψ and Hardy's Z, `numpy` for the fast Fourier transforms and

the EM iterations, and `matplotlib` for the figures. The first 1000 zeros were taken from `mpmath.zetazero`. The recomputations of this book used Arb through `python-flint` [179, 160] at 96 to 160 bits.

Densities. $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$ was evaluated from

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log\pi}{2} + \frac{1}{2}\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s).$$

The grid spacing was chosen so that each Cauchy peak of width $\theta - \frac{1}{2}$ is covered by at least ten points: 0.05 at $\theta = 7/8$ and 1, 0.02 at $\theta = 3/4$, 0.01 at $\theta = 0.55$ and 0.6.

Arguments. The masses $M_\theta(T)$ and the top correction were obtained by continuous variation of $\operatorname{Im}\log\xi$, computed as $\operatorname{Im}[\log s + \log(s-1) - \frac{s}{2}\log\pi + \log\Gamma(s/2) + \log\zeta(s)]$ and unwrapped along the path. The unwrapping is valid when the phase increment per step is below π; since $\partial_v \arg\xi = \pi u_\theta$ and $\pi u_\theta \le 1/(\theta - \frac{1}{2})$ plus a background of order $\log v$, a step of 0.02 gives increments below 0.5 radians at $\theta = 0.55$. The horizontal sweeps used step 0.0005 in σ. The integer deficits came out within 10^{-4} of integers in every case.

Deconvolution. The matrix is (11.1.3) with cell densities as unknowns, the data grid $[-400, 400]$ by evenness of u_1, the unknown grid $[-440, 440]$, Toeplitz products by FFT, and the conventions for residuals (with the factor h; without it every value is $1/h$ times larger), steps, peaks and basins are those of Section 11.4.

The author's timings on one core, with `mpmath` for the data, orders of magnitude only (with Arb the data layer does not dominate; Section 11.4.5):

computation	seconds
u_1 on $[0, 400]$, 8001 points	103
$u_{7/8}$ on $[0, 400]$, 8001 points	100
$u_{3/4}$ on $[0, 400]$, 20,001 points	239
u_θ on $[0, 60]$ at five levels	60
arguments on ten vertical lines to $T = 100$ and one sweep	148
arguments to $T = 400.9$ at $\theta = 3/4$ and 1, two sweeps	207
EM at width 1/8, 400 steps, 17,601 unknowns	3
EM at width 1/4, 1500 steps	11
sign changes of Z on $[1, 400.5]$	122
u_1 on $[0, 1450]$, 58,001 points; EM to height 1450	1375; 151

The positivity of $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$ and the zero counts above are certified in ball arithmetic in Section 11.9.9; the other numbers above are floating point.

11.9.9 Certification in ball arithmetic

The positivity statements and integer counts above were obtained in floating point. Here they are repeated in the ball arithmetic of Arb [179], where each quantity

is an interval guaranteed to contain the true value, including all rounding and truncation errors. Two lemmas reduce a line or segment to finitely many ball evaluations.

LEMMA 11.9.4 (cells). *Let $L = \xi'/\xi$, $u(v) = \pi^{-1} \operatorname{Re} L(\theta + iv)$, and let ξ have no zero in the closed cell $\{\theta + iv : |v - m| \le r\}$. Then for $|t| \le r$*

$$u(m + t) \;\ge\; u(m) - r\,|u'(m)| - \frac{r^2}{2\pi} \sup_{|v-m|\le r} |L''(\theta + iv)|,$$

where $u'(m) = -\pi^{-1} \operatorname{Im} L'(\theta + im)$.

Proof. Taylor's theorem with the Lagrange remainder, using $\partial_v L(\theta + iv) = iL'$ and $\partial_v^2 L(\theta + iv) = -L''$, so that $|u''| \le |L''|/\pi$. □

In practice $u(m)$ and $u'(m)$ are evaluated at the exact midpoint, where the balls are tight, and the supremum is bounded by evaluating the power series of L on the whole cell as a ball, which overestimates but is multiplied by $r^2/2$. A cell whose bound is not certifiably positive, or on which the ball enclosure of ξ contains 0, is bisected.

LEMMA 11.9.5 (segment). *Let $[a, b]$ be a segment of length d with*

$$d \sup_{s\in[a,b]} |\xi'(s)| < |\xi(a)|.$$

Then ξ has no zero on the segment, and the continuous change of $\arg \xi$ along it is the principal value $\operatorname{Arg}(\xi(b)/\xi(a))$, which lies in $(-\pi/2, \pi/2)$.

Proof. For $s \in [a, b]$, $|\xi(s) - \xi(a)| \le d \sup |\xi'| < |\xi(a)|$, so $\xi(s)/\xi(a)$ lies in the open disk of radius less than one about 1, which lies in the half plane $\operatorname{Re} w > 0$. The continuous argument of $\xi(s)/\xi(a)$ therefore stays in $(-\pi/2, \pi/2)$ along the segment and ends at the principal value. □

The cells have exact dyadic endpoints, the segments exact dyadic endpoints except for the corner $T^* = 14199/10$, which is one ball shared by all paths, so that the contours close and every enclosure contains the value at the exact corner. A version that accumulated endpoints in floating point produced an enclosure of the deficit that excluded its true value 0 by 10^{-17}, which is how the closure requirement was noticed.

THEOREM 11.9.6 (certified). *Let $T^* = 1419.9$. Then:*

(i) $u_{3/4}(v) > 0$ and $u_{7/8}(v) > 0$ for $0 \le v \le 1420$;

(ii) $N(T^*) = 1000$;

(iii) $N(\frac{3}{4}, T^*) = 0$, with $M_{3/4}(T^*) = 1000.01952570266\ldots$ to within 10^{-20};

(iv) Ξ *has* 1000 *certified sign changes in* $(0, T^*)$.

Consequently every zero of ζ *with* $0 < \operatorname{Im}\rho < T^*$ *lies on the critical line and is simple; the measure* $\pi^{-1}\partial_v \arg\xi(\frac{1}{2}+iv)$ *of Section 11.9.6 restricted to* $0 \le v < T^*$, *which under the Riemann hypothesis is the Thorin measure* $\mathfrak{U}_{1/2}$, *is the sum of* 1000 *unit atoms at the ordinates; and* $u_{3/4}$ *and* $u_{7/8}$ *have positive densities there, by (i).*

Proof. Computed, computer assisted, in the ball arithmetic of Arb at 96 bits and again at 160 bits; two standard facts are used and not certified, that $\xi > 0$ on $(\frac{1}{2}, \infty)$ and Backlund's relation $\pi N(T) = \Delta \arg\xi$ along $2 \to 2+iT \to \frac{1}{2}+iT$. For (i), Lemma 11.9.4 on the dyadic cells of width 2^{-7} over $[0, 1420]$, bisected once where necessary, gives 182,409 cells for $\theta = 3/4$ and 181,781 for $\theta = 7/8$ at 96 bits (182,512 and 181,783 at 160 bits; the counts belong to the method and the grid, not to u); every certified lower bound is positive, and in both runs and at both levels it is at least 4×10^{-5}. The least bound of the certificate depends on the grid and the precision and is not the minimum of u, which for $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$ is attained near 0, where $u_{3/4}(0) = \pi^{-1}(\xi'/\xi)(3/4) = 0.003676904191347$ and $u_{7/8}(0) = 0.005514663527790$ (balls of radius below 10^{-17}). For (ii), Lemma 11.9.5 along the path $2 \to 2+iT^* \to \frac{1}{2}+iT^*$, which gives $\pi N(T^*)$ as the change of argument because ξ is real and positive on $[\frac{1}{2}, 2]$ and real on the critical line: the enclosure contains 1000 and has width below 10^{-20} (11,470 segments on $\operatorname{Re} s = 2$ and 96 pieces of length $1/64$ on the top side at 96 bits). For (iii), Lemma 11.9.5 around the rectangle $[\frac{3}{4}, 2] \times [0, T^*]$, using Proposition 11.9.2 with the right side moved from $\operatorname{Re} s = 1$ to $\operatorname{Re} s = 2$, which changes nothing since there are no zeros with $\operatorname{Re} s \ge 1$ and keeps the balls away from the pole of ζ: the enclosure of $2N(\frac{3}{4}, T^*)$ has radius below 10^{-20} about 0 and contains the single integer 0 (32,387 segments on $\operatorname{Re} s = \frac{3}{4}$), and $M_{3/4}(T^*) = 1000.0195257026609355$ with the same twenty digits at both precisions. For (iv), the signs of $\Xi(v) = \xi(\frac{1}{2}+iv)$ at the 70,996 points $v = k/50$, $k = 0, \ldots, 70995$, are certified, none undetermined, with $\Xi(0) > 0$, and change 1000 times. Since the sign changes give at least 1000 distinct zeros of odd multiplicity on the line and (ii) gives exactly 1000 zeros in all, each zero is on the line and simple. The statement about $\pi^{-1}\partial_v \arg\xi$ is (ii) with (iv); the positivity statement is (i); (iii) is implied by (ii) and (iv) and is an independent check.

□

The author's run to height 400.9 certifies $N(400.9) = 202$, an enclosure of radius below 10^{-16} for $2N(\frac{3}{4}, 400.9)$, 202 sign changes, and both positivity statements on $[0, 400]$, in under two minutes (not rerun).

Parts (ii) to (iv) are known [232]; the theorem adds part (i), the positivity of the Thorin densities at levels 3/4 and 7/8 on $[0, 1420]$ as a computer assisted statement, fast enough for a laptop.

certificate	enclosure at 96 bits (160 bits)	size
(i) $u_{3/4} > 0$ on $[0, 1420]$	least bound at least 4×10^{-5}	182,409 cells
(i) $u_{7/8} > 0$ on $[0, 1420]$	least bound at least 4×10^{-5}	181,781 cells
(ii) $N(1419.9)$	$1000 \pm 4 \times 10^{-22}$ ($\pm 2 \times 10^{-41}$)	11,470 + 96 segments
(iii) $2N(\frac{3}{4}, 1419.9)$	$0 \pm 1.4 \times 10^{-21}$ ($\pm 7 \times 10^{-41}$)	32,387 segments
(iv) signs of Ξ	1000 changes, none undetermined	70,996 points

Table 11.2: The certificates of Theorem 11.9.6, as recomputed.

11.10 The limits of the computation

Can these computations be pushed past what is known? For ζ, no.

Height. Theorem 11.9.6 reaches height 1419.9 in minutes. Platt and Trudgian verified the Riemann hypothesis rigorously to height 3×10^{12} [232], about 1.2×10^{13} zeros, combining Turing's method with the fast multi evaluation of Odlyzko and Schönhage [221], a few operations per zero. The mass deficit is Turing's method in Thorin masses, so its known cost is the same, of order $T^{1+\varepsilon}$; matching the record would mean rebuilding that machinery.

All heights. Below 7/8, positivity of U_θ at every height is the zero free half plane $\operatorname{Re} s > \theta$, which a computation stopping at a finite height H cannot confirm (Section 11.9.7); above H an argument is needed, and that is the open problem.

Combining the two gains nothing either. Below $3 \times 10^{12} - \frac{1}{2}$ every zero is on the line, so U_θ restricted to that range is known to be positive for every $\theta > \frac{1}{2}$ (Corollary 14.3.7). Above it, the announced half plane gives positivity for $\theta \geq 7/8$, conditionally, and nothing below. Every computation in these chapters lies inside what is already known. The frontier that moves the half plane is analytic: zero density estimates such as those of Guth and Maynard [92], and the announced, not yet refereed, half plane $\operatorname{Re} s > 7/8$.

What computation can still add is narrower: two results about the Thorin densities themselves, one unconditional (Proposition 12.1.11) and one conditional on the announced half plane, whose Riemann hypothesis form is Theorem 12.1.18(vi). They rest on a lemma.

Lemma 11.10.1 (pairing). *Let $\nu = \sum_\rho \delta_{\gamma_\rho}$ be the counting measure of the ordinates of the nontrivial zeros, with multiplicity and both signs, and P_a the Cauchy density of width a, with $P_0 = \delta_0$. If ζ has no zeros in $\operatorname{Re} s > \theta_0$, where $\frac{1}{2} \leq \theta_0 \leq 1$, then for every $\theta \geq \theta_0$, as measures,*

$$\mathfrak{U}_\theta \;\geq\; \tfrac{1}{2}\, P_{2\theta-1} * \nu.$$

In particular the bound holds unconditionally for $\theta \geq 1$, and for $\theta \geq 7/8$ it holds conditionally on the announced half plane. Under the Riemann hypothesis $\mathfrak{U}_\theta = P_{\theta-1/2} * \nu$ exactly.

Proof. Group the zeros into orbits under $\rho \mapsto 1-\bar{\rho}$, which preserves the ordinate. An orbit of two zeros $\sigma+i\gamma$, $1-\sigma+i\gamma$ contributes P_a+P_b at the point γ, with widths $a=\theta-\sigma$ and $b=\theta-1+\sigma$, both nonnegative because $\theta \ge \theta_0 \ge \max(\sigma, 1-\sigma)$, and $a+b=2\theta-1$. For fixed $x \ne 0$ write $\pi P_a(x) = a\,h(a)$ with $h(a) = 1/(a^2+x^2)$, which is decreasing in $a \ge 0$. Then

$$a\,h(a) + b\,h(b) \;\ge\; a\,h(a+b) + b\,h(a+b) = (a+b)\,h(a+b),$$

that is $P_a(x) + P_b(x) \ge P_{a+b}(x) = \frac{1}{2}\cdot 2P_{2\theta-1}(x)$, which is the claim for the two zeros of the orbit (the two zeros count twice in ν); at $x=0$ the inequality holds as measures because a width zero term is an atom. An orbit of one zero, on the critical line, contributes $P_{\theta-1/2}$, and $P_{\theta-1/2}(x) \ge \frac{1}{2}P_{2\theta-1}(x)$ by the same monotonicity with $a=\theta-\frac{1}{2}$, $a+b=2a$. Summing over orbits gives the lemma. □

Even when off line pairs move to the edges of the allowed strip, the density keeps at least half of its smoothing under the Riemann hypothesis, at twice the width. With the cell certificate and an explicit zero count, the lemma proves the following.

THEOREM 11.10.2 (global minimum). *(i) Unconditionally, for every real $v \ne 0$,*

$$u_1(v) \;>\; u_1(0) = \frac{1}{\pi}\Big(1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi\Big) = 0.0073515924923\ldots,$$

where γ_E is Euler's constant; this is Proposition 12.1.11. (ii) If ζ has no zeros in $\operatorname{Re} s > 7/8$ (the announced half plane), then, as measures, $\mathfrak{U}_{7/8} \ge u_{7/8}(0)\,dv$ with $u_{7/8}(0) = \pi^{-1}(\xi'/\xi)(7/8) = 0.0055146635278\ldots$, and $u_{7/8}(v) > u_{7/8}(0)$ wherever $v \ne 0$ is not the ordinate of a zero on the line $\operatorname{Re} s = 7/8$.

Proof. Part (i) is Proposition 12.1.11; the argument below is the same in three ranges and gives (ii) as well. It is computed, computer assisted. Write θ for 1 or 7/8; both functions are even. Three ranges.

$|v| \le 13.5$. Here $u_\theta = \sum_\rho P_{\theta-\sigma_\rho}(v-\gamma_\rho)$ with every width in $[0,1)$ for $\theta=1$, and in $[0,\frac{3}{4}]$ for $\theta = 7/8$ under the hypothesis. The Cauchy density $P_a(x)$ is convex for $|x| \ge a/\sqrt{3}$, and every zero has $|\gamma_\rho| \ge 14.1347$ (Theorem 11.9.6), so $|v-\gamma_\rho| \ge 0.63 > 1/\sqrt{3}$. Hence u_θ is a convergent sum of convex functions, strictly convex and even on $[-13.5, 13.5]$, with its unique minimum at 0. The value $u_1(0) = \pi^{-1}(\xi'/\xi)(1) = \pi^{-1}\sum_\rho 1/\rho$ is π^{-1} times the classical constant $1 + \gamma_E/2 - \frac{1}{2}\log 4\pi$.

$13.5 \le |v| \le 1420$. Lemma 11.9.4 with the threshold $u_\theta(0)$ in place of 0: on 175,813 cells of radius 0.004 for each of $\theta = 1$ and $\theta = 7/8$, with no bisection, every certified lower bound exceeds the upper end of the ball for $u_\theta(0)$, the smallest being 0.0360 at $v = 17.49$ for $\theta = 1$ and 0.0272 at $v = 17.49$ for $\theta = 7/8$ (the margins over $u_\theta(0)$ are 0.0287 and 0.0217). The computation evaluates ξ'/ξ on the line and uses no hypothesis.

$|v| \ge 1420$. By Lemma 11.10.1, unconditional at $\theta = 1$ and conditional at $7/8$, $u_\theta(v) \ge \frac12 \sum_\gamma P_{2\theta-1}(v-\gamma) \ge \frac12\, n(v)\, P_{2\theta-1}(8)$, where $n(v)$ is the number of ordinates in $(v-8, v+8]$ (positive ordinates, with multiplicity). Hasanalizade, Shen and Wong [100] (the form used in Proposition 12.1.11) give $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \le E(T)$ with $E(T) = 2.5167 + 1/(50T)$ for $e \le T \le X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + 1/(50T)$ for $T > X_0$. Hence $n(v) \ge D(v) - E(v+8) - E(v-8)$ with $D(v) = M(v+8) - M(v-8)$ and $M(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. The right side is $8.7678\ldots$ at $v = 1418.9$, so more than 8.767, and it increases with v as long as $v + 8 \le X_0$, because $M'(T) = \log(T/2\pi)/(2\pi)$ increases, so D increases, and E decreases; for larger v it is at least 44 at $X_0 - 8$ and increases (a numerical check of the derivative, in the recomputation). So $n(v) \ge 9$ for $v \ge 1420$, since n is an integer, and $u_\theta(v) \ge 4.5\, P_{2\theta-1}(8)$, which is 0.0220 for $\theta = 1$ and 0.0166 for $\theta = 7/8$, more than twice $u_\theta(0)$ in both cases. The conclusion would survive a loss of five ordinates at $\theta = 1$ and of six at $\theta = 7/8$. □

The same argument, with the Riemann hypothesis in place of the zero free half plane, covers every level at once, and the conclusion turns out to be equivalent to the hypothesis.

THEOREM 11.10.3 (a Thorin form of the Riemann hypothesis). *The Riemann hypothesis holds if and only if, for every $\theta \in (\frac12, 1]$, the Thorin density u_θ attains its global minimum at $v = 0$; that is,*

$$\operatorname{Re}\frac{\xi'}{\xi}(\theta + it) \;\ge\; \frac{\xi'}{\xi}(\theta) \qquad \text{for all real } t \text{ and } \tfrac12 < \theta \le 1.$$

Under the hypothesis the minimum is attained only at $t = 0$.

Proof. This is the equivalence of Theorem 12.1.18(vi), $v_\alpha(\theta) > b_\alpha$ for $\alpha \in (\frac12, 1)$, $\theta \ne 0$, in the form with the weak inequality; the proof here uses the cell certificate of Theorem 11.10.2 and an explicit zero count, and the forward direction is computer assisted. Suppose the hypothesis, and let $a = \theta - \frac12 \in (0, \frac12]$. Then $u_\theta = \sum_\rho P_a(v - \gamma_\rho)$, so

$$\frac{u_\theta(v)}{a} = \frac{1}{\pi}\sum_\rho \frac{1}{a^2 + (v-\gamma_\rho)^2} \;\ge\; \frac{1}{\pi}\sum_\rho \frac{1}{\frac14 + (v-\gamma_\rho)^2} = 2u_1(v),$$

$$\frac{u_\theta(0)}{a} \;\le\; \frac{1}{\pi}\sum_\rho \frac{1}{\gamma_\rho^2}.$$

Since $\gamma_\rho^2 \ge \gamma_1^2$ for every zero, the last sum is at most

$$\Big(1 + \frac{1}{4\gamma_1^2}\Big)\frac{1}{\pi}\sum_\rho \frac{1}{\frac14 + \gamma_\rho^2} = \Big(1 + \frac{1}{4\gamma_1^2}\Big)\frac{2 + \gamma_E - \log 4\pi}{\pi} \le 0.01473.$$

For $13.5 \le |v| \le 1420$, Theorem 11.10.2 gives $2u_1(v) \ge 0.0720$ (twice the certified 0.0360). Beyond 1420, under the hypothesis, $u_1(v) = \sum_\gamma P_{1/2}(v-\gamma) \ge 9P_{1/2}(8)$ by the count of that proof, so $2u_1(v) \ge 18P_{1/2}(8) \ge 0.0445$. Both exceed 0.01473, so $u_\theta(v) > u_\theta(0)$ there. For $|v| \le 13.5$ the convexity argument applies with every width equal to $a \le \frac{1}{2}$. Conversely, if a zero $\rho_0 = \sigma_0 + i\gamma_0$ has $\sigma_0 > \frac{1}{2}$, take $\theta = \sigma_0 - \varepsilon$; then $\operatorname{Re}(\xi'/\xi)(\theta + i\gamma_0) = -m/\varepsilon + O(1) < 0$, while $(\xi'/\xi)(\theta) > 0$ unconditionally, because $\xi(\frac{1}{2}+x) = \int \Phi(y)\cosh(xy)\,dy$ with Riemann's kernel Φ positive increases for $x > 0$. So the minimum is not at 0. □

Theorem 11.11.1 below extends Theorem 11.10.2(ii) to every level in $[7/8, 1]$. Theorem 11.10.3 sits beside the reformulations of Lagarias [185], $\operatorname{Re}(\xi'/\xi) > 0$ on $\operatorname{Re} s > \frac{1}{2}$, and of Sondow and Dumitrescu [259], that $|\xi(\sigma + it)|$ increases in σ for $\sigma > \frac{1}{2}$, and strengthens both under the hypothesis: $\partial_\sigma \log|\xi(\sigma+it)|$ is not merely positive but smallest on the real axis, at every level in $(\frac{1}{2}, 1]$. Its forward direction is computer assisted, and it is a reformulation, not progress towards the hypothesis.

The certificates transfer to any completed L function with a ball arithmetic evaluator (Dirichlet L functions, including the Dirichlet beta function, that of Ramanujan's Δ, Dedekind zeta functions); whether they would add anything to the published verifications is not examined here. The Thorin measures at levels 7/8 and 3/4 and their margins are tables of numbers, not theorems about zeros.

11.11 Open problems worked

Notation of this section, as in the rest of the chapter: θ is the level and v the height, $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$, $P_a(x) = a/(\pi(a^2+x^2))$, and a zero is $\rho = \sigma_\rho + i\gamma_\rho$ with σ_ρ and γ_ρ real (so γ_ρ is the height of the zero, which is the book's ordinate of ρ only under the Riemann hypothesis). The symmetric Thorin measure $\mathfrak{U}_\theta$ in the variable v has density $\sum_\rho P_{\theta-\sigma_\rho}(v-\gamma_\rho)$ with no factor $\frac{1}{2}$, so it is twice the measure $\widetilde{U}_\theta$ of Theorem 17.14.4; it exists as a positive measure when ζ has no zeros in $\operatorname{Re} s > \theta$, and then $P_a * \mathfrak{U}_\theta = \mathfrak{U}_{\theta+a}$, in particular $P_{1-\theta} * \mathfrak{U}_\theta = u_1\,dv$ (Theorem 17.14.4). The zero free half plane $\operatorname{Re} s > \frac{7}{8}$ is announced, formalised in Lean and not refereed [227, 228, 198], as described in Section 17.7: every statement below that uses it is conditional on it.

Three open problems were posed for this chapter: the rate of EM at the boundary (answered by experiment, observed and not proved), the value of the latent Brownian times in the sampler (a comparison and an experiment), and a fast algorithm for u_1 (answered at fixed accuracy). The global minimum of Section 11.10 is extended to every level in $[\frac{7}{8}, 1]$ and then to every level under the matching hypothesis, with the slope of the excess, and the a posteriori bound for the backward flow is worked up to a certificate in ball arithmetic at the levels $\frac{1}{2}$ and $\frac{3}{4}$.

11.11.1 The rate at the boundary

The question was whether the factor of about 15 by which acceleration beat EM at $\theta' = \frac{1}{2}$ is generic, and how the constants depend on the grid. The runs show a grid independent constant, an EM exponent close to 1 for the first few thousand steps that then rises, and an accelerated scheme whose restarts make the objective a sawtooth with no single exponent; all observed, not proved. The runs use the conventions of Section 11.4.

Grid independence. The full range problem at $\theta' = \frac{1}{2}$ was run for 3000 steps at $h = 0.04$, 0.02 and 0.01. With R^* the smallest residual reached by either method, $k(R_k - R^*)$ for EM is the same on the three grids to four digits (the values of Section 11.4), and the ratio of EM steps to accelerated steps needed to reach EM's residual at 3000 steps is 23.1 (3000/130) on all three. We read the iteration as a discretisation of the continuous Richardson and Lucy iteration for measures, whose iterates stay resolved by the grid over thousands of steps. The bound $D(x^* \| x^{(0)})$ of Theorem 11.3.1, which grows like $\log(1/h)$ at an atomic target, is therefore not the active constraint, and the constant of order 50 appears to belong to the continuous problem.

The exponent. A reference value R^* taken from a finite run biases every exponent estimated from $R_k - R^*$, so we use the three point exponent

$$p(k) = \log_2 \frac{R_k - R_{2k}}{R_{2k} - R_{4k}},$$

which equals p exactly when $R_k - R^* = Ck^{-p}$ and involves no reference value. For EM it is 1.080, 1.067, 1.062, 1.054 at $k = 500$, 1000, 2000, 4000 at $h = 0.08$, equal to about 10^{-3} on the grids 0.04 and 0.02; it decreases slowly and stays close to 1. After about 4000 steps it rises, to 1.10 at 8000, 1.22 at 16,000, 1.53 at 32,000 and 1.84 at 50,000 in a run of 200,000 steps at $h = 0.08$. The residual itself is 9.0×10^{-4}, 5.8×10^{-4} and 5.0×10^{-4} at 5×10^4, 10^5 and 2×10^5 steps. A geometric rate $R_k - R^* \approx Ce^{-k/K}$ is not claimed: no estimator of K is fixed, and the estimate from three equally spaced differences gives $K \approx 32{,}000$ at $k = 5 \times 10^4$ only. The interpretation consistent with the theory of EM on a cone is a sublinear phase at rate $1/k$, while the mass away from the atoms is driven to zero, followed by the local linear convergence of Theorem 11.3.2 on the identified support. What is observed is that the constant of order 50 describes the first few thousand steps and that the longest run turns up after 4000 steps; no exponent is claimed to be generic.

Acceleration. For the logarithmic domain scheme with function restart the three point exponent reads 2.8, 0.6, 1.2, 1.2, 1.3 at $k = 500$, 1000, 2000, 4000, 5000, a sawtooth whose single values depend on the restart instants; its running value at $k = 2000$, 3000, 5000 is 1.2, 1.4, 1.3. The speedup over EM, the ratio of step counts to a common residual at $h = 0.04$, is 13.5, 18.9, 29.6, 41.7 and 39.1 at the residuals EM reaches after 1000, 2000, 5000, 10,000 and 20,000 steps: it grows

and saturates near 40. Five restart rules were compared at $h = 0.04$: function restart, no restart, a gradient restart (here the step against the momentum in the logarithmic variables, an adaptation of the rule of O'Donoghue and Candès [126]), a fixed restart every 500 steps, and a momentum capped at 0.9. They reach EM's residual at 20,000 steps in 512, 344, 344, 344 and 1991 steps, with final values 2.0, 1.7, 1.5, 2.8 and 18×10^{-5} after 20,000 steps. The capped momentum, a heavy ball iteration, is about four times slower than function restart. The exponent without restart is erratic (between 0.2 and 0.3 at $k = 4000$ and 5000, not monotone earlier). No restart rule is uniformly best: function restart is the safe one, never increasing f, at about one and a half times the steps of the others. There is no generic speedup factor, and the accelerated scheme is the one to run.

11.11.2 The latent times do not help the sampler

The question was whether the three block sampler of Section 11.6 mixes better than the two block sampler near $\theta' = \frac{1}{2}$. It does not, and for this scan there is a general reason: the two block sampler is the three block sampler with the latent times integrated out analytically. Let Z be the allocations, X the masses and T the times, with X and T independent given Z. The allocation marginal of the three block scan is data augmentation through (X, T), that of the collapsed scan is data augmentation through X, and for centred $g(Z)$ their reversible kernels satisfy, by conditional Jensen,

$$\langle g, K_3 g \rangle = \mathbb{E}\big[\mathbb{E}(g \mid X, T)^2\big] \;\geq\; \mathbb{E}\big[\mathbb{E}(g \mid X)^2\big] = \langle g, K_2 g \rangle,$$

so the additional augmentation cannot improve the L^2 spectral contraction of the allocation marginal. This is the comparison of Liu, Wong and Kong [114], in the scope proved here: it orders neither wall clock efficiency nor finite run autocorrelations for other scans and observables.

Both samplers were run on real data (the author's runs, not repeated). The level one density on the window $100 \leq v \leq 120$, sampled at spacing $h = 0.05$, was turned into Poisson pseudo counts with means

$$200\, h\, u_1(v_i).$$

The unknown was on $[98, 122]$, the prior Gamma$(1, 0.01)$ per cell, the start the EM solution, and each run had 500 burn in and 3000 recorded sweeps. Integrated autocorrelation times were estimated by the initial positive sequence of Geyer [81], for the mass within 0.5 of a zero, the mass in the trough between two zeros, and the log likelihood.

θ'	sampler	near a zero	trough	log likelihood	seconds
0.75	two block	4.3	7.1	1.8	22
0.75	three block	4.9	9.2	2.9	65
0.55	two block	5.4	7.8	2.5	23
0.55	three block	6.4	10.4	2.5	59

The latent times lengthen the autocorrelation of the two masses by 14 to 19 per cent (near a zero) and 30 to 33 per cent (trough), and that of the log likelihood by 61 per cent at $\theta' = 0.75$ and not at all at $\theta' = 0.55$; they multiply the cost per sweep by 2.6 to 3 at both levels, including the one near $\frac{1}{2}$ where they were expected to help. The posterior means agree (0.80 for the mass near the zero). The scale mixture keeps its value as an interpretation and a source of proposals, not as a device for speed.

11.11.3 The global minimum at every level between 7/8 and 1

The next problem asked for Theorem 11.10.2(ii) at every level $\theta \in [\frac{7}{8}, 1]$. On the middle range a certificate uniform in θ seemed to need a two dimensional search; a monotonicity argument removes the second dimension. The theorem is conditional on the announced half plane.

Theorem 11.11.1 (global minimum, uniform in the level; conditional). *Suppose ζ has no zeros in $\operatorname{Re} s > \frac{7}{8}$. Then for every $\theta \in [\frac{7}{8}, 1]$ and every real $v \neq 0$ that is not the height of a zero on $\operatorname{Re} s = \theta$,*

$$u_\theta(v) \;>\; u_\theta(0) = \frac{1}{\pi}\frac{\xi'}{\xi}(\theta),$$

and the singular part of $\mathfrak{U}_\theta$ consists of positive atoms, which occur only if $\theta = \frac{7}{8}$ and a zero lies on $\operatorname{Re} s = \frac{7}{8}$; so $\mathfrak{U}_\theta - u_\theta(0)\,dv$ is a positive measure. Unconditionally, $\theta \mapsto u_\theta(0)$ is increasing on $[\frac{1}{2}, \infty)$.

Proof. Write $g(\theta, v) = u_\theta(v) - u_\theta(0)$ and $a_\rho = \theta - \sigma_\rho \in [\theta - \frac{7}{8}, \theta - \frac{1}{8}] \subset [0, \frac{7}{8}]$, which uses the hypothesis and the functional equation. Monotonicity of $u_\theta(0)$: $\xi(\frac{1}{2} + x)$ is a positive multiple of $\int_0^\infty \Phi(y)\cosh(xy)\,dy$ with the positive density Φ of (2.1.3), and the logarithm of the Laplace transform of a positive measure is strictly convex (Hölder), so $(\xi'/\xi)(\frac{1}{2} + x)$ is strictly increasing in x. In particular $u_\theta(0) \le u_1(0) = 0.00736$ on the range.

$|v| \le 13.5$: every term $P_{a_\rho}(v - \gamma_\rho)$ with $a_\rho > 0$ is convex, since $|v - \gamma_\rho| \ge 0.63 > \frac{7}{8}/\sqrt{3}$; the sum is strictly convex and even, with its minimum at 0. This is the argument of Theorem 11.10.2, valid for all $\theta \le 1$ at once.

$|v| \ge 1418.9$: Lemma 11.10.1 and the zero count of [100] give at least seven ordinates in $(v-8, v+8]$ (the explicit bound is 8.76786 at $v = 1418.9$, so there are at least nine, and it increases with v, the derivative of its main term $(2\pi)^{-1}\log\frac{v+8}{v-8} \ge 8/(\pi v)$ dominating the derivatives of the error terms, which are $O(1/v)$; this was checked numerically in the computation of Section 11.11.3). Seven suffice: $u_\theta(v) \ge \frac{7}{2}P_{2\theta-1}(8) \ge \frac{7}{2}P_{3/4}(8) = 0.01294 > u_1(0) \ge u_\theta(0)$, using that $P_c(8)$ increases in c on $[\frac{3}{4}, 1]$.

$13.5 \le |v| \le 1418.9$: by Theorem 11.9.6 every zero with $|\gamma| < 1419.9$ lies on the critical line and is one of the 1000 certified ordinates (and their negatives), so

its width is exactly $a = \theta - \frac{1}{2} \le \frac{1}{2}$, and the zeros with $|\gamma| \ge 1419.9$ are at distance at least 1 from v. Two cases.

If some certified zero has $|v - \gamma| \le 1.05$, then $u_\theta(v) \ge P_{\theta-1/2}(1.05) \ge P_{3/8}(1.05) = 0.0960 > u_\theta(0)$, since $P_c(1.05)$ increases in c on $[\frac{3}{8}, \frac{1}{2}]$.

If every certified zero has $|v - \gamma| \ge 0.95$, differentiate in θ:

$$\partial_\theta u_\theta(v) = \frac{1}{\pi} \sum_\rho f\big(|v - \gamma_\rho|, a_\rho\big), \qquad f(d, a) = \frac{d^2 - a^2}{(d^2 + a^2)^2}.$$

For the far zeros, $d \ge 1 > \frac{7}{8} \ge a$, so their terms are positive. For the certified zeros, $a \le \frac{1}{2}$, and f is decreasing in a for $d > a/\sqrt{3}$ and decreasing in d for $d > \sqrt{3}\, a$, both satisfied when $d \ge 0.95$ and $a \le \frac{1}{2}$. Hence

$$\partial_\theta u_\theta(v) \;\ge\; \frac{1}{\pi} \sum_{|\gamma| < 1419.9} f\big(|v - \gamma|, \tfrac{1}{2}\big) =: S(v),$$

a function of the certified ordinates alone, independent of θ. On the other side, $\partial_\theta u_\theta(0) = \pi^{-1} \sum_\rho (\gamma_\rho^2 - a_\rho^2)/(a_\rho^2 + \gamma_\rho^2)^2 \le \pi^{-1} \sum_\rho \gamma_\rho^{-2}$. The sum over the certified ordinates, both signs, is at most 0.044804, and the sum over $|\gamma| \ge 1419.9$ is at most 0.001442 (the computed value is 0.0014417) by partial summation against the bound of [100] for $N(T)$, so $\partial_\theta u_\theta(0) \le c^* = 0.0147225$ for every θ. The ball arithmetic computation described below certifies $S(v) > c^*$ on every cell of the second case. Therefore $\partial_\theta g \ge 0$ on $[\frac{7}{8}, 1]$, and $g(\theta, v) \ge g(\frac{7}{8}, v) > 0$ by Theorem 11.10.2(ii).

Atoms: a zero contributes an atom to $\mathfrak{U}_\theta$ only if its width $a_\rho = 0$, that is $\sigma_\rho = \theta$, and $\sigma_\rho \le \frac{7}{8} \le \theta$ forces $\theta = \frac{7}{8}$. □

The computation partitions $[13.5, 1418.9]$ into cells of radius 0.1, bisected where neither case applies to the whole cell. The ordinates enter as the certified sign change enclosures. A cell is of the first kind if for one certified ordinate the largest distance between a point of the cell and a point of its enclosure is at most 1.05, and of the second kind if the smallest distance between the cell and every enclosure is at least 0.95; then S is bounded below by evaluating each $f(\cdot, \frac{1}{2})$ at the largest such distance, since f decreases in d. Of the 7,245 cells, 6,733 are of the first kind and 512 of the second, every cell is certified, and the least certified value of S on a cell of the second kind is 0.066 against c^*. The run takes seconds. The level enters only through the two monotonicities, in a for the near zeros and in sign for the far ones.

COROLLARY 11.11.2. *Given the zero free half plane* $\operatorname{Re} s > \frac{7}{8}$, *for every* $\theta \in [\frac{7}{8}, 1]$ *the Thorin measure* $\mathfrak{U}_\theta$ *dominates the multiple* $u_\theta(0)\, dv$ *of Lebesgue measure, with* $u_\theta(0)$ *increasing from* 0.0055147 *at* $\theta = \frac{7}{8}$ *to* 0.0073516 *at* $\theta = 1$.

11.11.4 Below 7/8: a level by level theorem

The extension below $\frac{7}{8}$, under the corresponding hypothesis, gives a cleaner statement and proof than either theorem it extends. The case $\theta = 1$ is Proposition 12.1.11 and the case $\theta \downarrow \frac{1}{2}$ is Theorem 12.1.18(vi).

Theorem 11.11.3 (level by level). *Let $\frac{1}{2} < \theta \le 1$. The following are equivalent.*

(i) ζ has no zeros in $\operatorname{Re} s > \theta$.

(ii) For every $\theta' \in [\theta, 1]$ the function $u_{\theta'}$ attains its global minimum at the origin: $u_{\theta'}(v) \ge u_{\theta'}(0)$ for all real v that are not heights of zeros on $\operatorname{Re} s = \theta'$.

When they hold, $\mathfrak{U}_{\theta'} \ge u_{\theta'}(0)\,dv$ as measures for $\theta' \in [\theta, 1]$, and for every $\theta' \in [\theta, 1]$, with $c = 2\theta' - 1$ and the sum over all zeros,

$$u_{\theta'}(v) - u_{\theta'}(0) \;\ge\; \frac{c}{2\pi}\Big(\sum_{\rho} \frac{1}{c^2 + (v - \gamma_\rho)^2} - \pi w^*\Big), \tag{11.11.1}$$

with $w^* = 0.0294450$, and the bracket is at least $\pi(0.06174 - w^*) > 0.1014$ for $13.5 \le |v| \le 1418.9$ and at least $7/65 - \pi w^* > 0.0151$ for $|v| \ge 1418.9$.

Proof. (ii) implies (i) exactly as in Theorem 11.10.3: a zero ρ_0 of multiplicity m with $\sigma_0 > \theta$ makes $u_{\theta'}(\gamma_0) = -m/\varepsilon + O(1)$ at the level $\theta' = \sigma_0 - \varepsilon \in [\theta, 1]$, which is negative, while $u_{\theta'}(0) = \pi^{-1}(\xi'/\xi)(\theta') > 0$.

(i) implies (ii). Fix $\theta' \in [\theta, 1]$, write $c = 2\theta' - 1 \in (0, 1]$, and let $a_\rho = \theta' - \sigma_\rho \ge 0$. Two bounds, one on each side.

At the origin, grouping the zeros into orbits under $\rho \mapsto 1 - \bar\rho$, the widths in an orbit sum to c (a pair) or equal $c/2$ (a zero on the line), so

$$u_{\theta'}(0) = \frac{1}{\pi}\sum_{\rho} \frac{a_\rho}{a_\rho^2 + \gamma_\rho^2} \;\le\; \frac{1}{\pi}\sum_{\text{orbits}} \frac{c}{\gamma^2} \;\le\; \frac{c}{\pi}\sum_{\rho} \frac{1}{\gamma_\rho^2} \;\le\; c\,c^*,$$

with $c^* = 0.0147225$, the certified value of Theorem 11.11.1: the certified ordinates below 1419.9 contribute at most 0.044804 to $\sum_\rho \gamma_\rho^{-2}$ and the rest at most 0.001442 by [100]. (An orbit contributes at most $\frac{c}{2}\sum \gamma^{-2}$ over its zeros, which would halve the bound; the weaker form is what the printed constants use.)

Away from the origin, Lemma 11.10.1 gives $u_{\theta'}(v) \ge \frac{1}{2}\sum_\rho P_c(v - \gamma_\rho) = \frac{c}{2\pi}\sum_\rho \frac{1}{c^2 + (v-\gamma_\rho)^2} \ge \frac{c}{2\pi}\sum_\rho \frac{1}{1 + (v-\gamma_\rho)^2}$, since $c \le 1$. So (11.11.1) holds with $w^* = 2c^*$, and (ii) at v follows from

$$W(v) = \frac{1}{\pi}\sum_{\rho} \frac{1}{1 + (v - \gamma_\rho)^2} \;\ge\; 2c^* = w^*, \tag{11.11.2}$$

a condition on the ordinates alone, with no θ' in it. Dropping the zeros above height 1419.9 and those of negative height only lowers W. Ball arithmetic certifies

(11.11.2) on $[13.5, 1418.9]$ with the 1000 certified positive ordinates, 7027 cells of radius 0.1, each term evaluated at the farthest point of the cell; the smallest certified cell bound is $W \geq 0.06174$, in the cell about $v = 17.4$, between the first two zeros, and the bracket is at least $\pi(0.06174 - w^*) = 0.10146$. With the negative ordinates included the same cells give 0.0653. For $|v| \geq 1418.9$ the seven ordinates within distance 8 give $W \geq 7/(65\pi) > 0.03427 > w^*$ and the bracket is at least $7/65 - \pi w^* = 0.01518$ (with the nine ordinates of the count of [100] it is $9/65 - \pi w^* = 0.0459$). For $|v| \leq 13.5$ every term $P_{a_\rho}(v - \gamma_\rho)$ is convex, because $a_\rho \leq \theta' - (1 - \theta) \leq 1$ and $|v - \gamma_\rho| \geq 0.63 > 1/\sqrt{3}$, so $u_{\theta'}$ is strictly convex and even there and its minimum is at 0. The excess (11.11.1) is the difference of the two bounds. The positivity of the atoms gives the measure statement. □

Theorem 11.11.3 contains the weak form of Theorem 11.10.2(i) ($\theta = 1$), Theorem 11.11.1 ($\theta = \frac{7}{8}$) and Theorem 11.10.3 ($\theta \downarrow \frac{1}{2}$). Its proof is shorter because the pairing lower bound and the orbit upper bound both scale linearly in c, so the level drops out and one certified function of the ordinates, W, decides every level; the monotonicity in θ is not used.

As $\theta' \downarrow \frac{1}{2}$ the excess over the value at the origin has a lower bound linear in $c = 2\theta' - 1$, with slope at least $0.1014/(2\pi) = 0.0161$ on the certified range and $0.0151/(2\pi) = 0.0024$ beyond it. The slope is not sharp, since the pairing lemma halves the density and W was bounded with $c = 1$, but the order is right: under the Riemann hypothesis, at the midpoint between two consecutive zeros with gap G, $u_{\theta'}(v) = \sum P_{c/2}(v - \gamma) \approx \frac{c}{2\pi} \cdot \frac{8}{G^2}$ for small c, and $u_{\theta'}(0) \approx c\,c^*/2$, so the excess is of order c at fixed v (of order $1/c$ at a zero).

11.11.5 The slope of the excess

The losses in (11.11.1) are avoidable for the zeros whose position is known. By Theorem 11.9.6 every zero with $|\gamma| < 1419.9$ lies on the critical line, so at level θ' its width is exactly $c/2$ and its contribution to $u_{\theta'}$ is exactly $P_{c/2}(v - \gamma)$, with no pairing. Only the zeros above height 1419.9, whose real parts are unknown, need Lemma 11.10.1 and the orbit bound. Writing $V_c(v) = \sum_{|\gamma|<1419.9} \big(\frac{c^2}{4} + (v-\gamma)^2\big)^{-1}$, both signs of γ included, the two sides become

$$u_{\theta'}(v) \;\geq\; \frac{c}{2\pi}\Big(V_c(v) + \sum_{|\gamma|\geq 1419.9} \frac{1}{c^2 + (v-\gamma)^2}\Big),$$

$$u_{\theta'}(0) \;\leq\; \frac{c}{2\pi}\Big(V_c(0) + 2\sum_{|\gamma|\geq 1419.9} \frac{1}{\gamma^2}\Big),$$

and dropping the positive far sum on the left,

$$u_{\theta'}(v) - u_{\theta'}(0) \;\geq\; \frac{c}{2\pi}\,B_c(v), \qquad B_c(v) = V_c(v) - V_c(0) - 2\sum_{|\gamma|\geq 1419.9} \frac{1}{\gamma^2}, \quad (11.11.3)$$

where the last sum is at most 0.001442 by [100], so that $B_c(v) \geq V_c(v) - V_c(0) - 0.0029$ (the far sum in the display below is written with c^2 in place of $\frac{c^2}{4}$, which is the weaker bound). On $13.5 \leq |v| \leq 1418.9$ the function B_c decreases in c (its derivative in c is $\frac{c}{2}\sum_\gamma\big[(\frac{c^2}{4}+\gamma^2)^{-2} - (\frac{c^2}{4}+(v-\gamma)^2)^{-2}\big]$, in which the first sum is below $\sum\gamma^{-4} < 7.5\times10^{-5}$ while the nearest certified ordinate, within 3.45 of v, alone gives more than $(\frac{1}{4}+3.45^2)^{-2} > 6.7\times10^{-3}$ to the second; single terms need not decrease), so its value at $c=0$, which involves only $\sum(v-\gamma)^{-2}$ over the certified ordinates, is a lower bound for the slope as $\theta' \downarrow \frac{1}{2}$.

Proposition 11.11.4 (the slope). *Under the hypothesis of Theorem 11.11.3, for every $\theta' \in [\theta, 1]$ and $c = 2\theta' - 1$,*

$$\inf_{13.5\leq|v|\leq1418.9} B_c(v) \;\geq\; \begin{cases} 0.17575 & c = 1, \\ 0.17838 & c = \frac{1}{2}, \\ 0.17904 & c = \frac{1}{4}, \\ 0.17927 & c \downarrow 0, \end{cases}$$

each infimum being approached at $v = 17.49$ to 17.50, between the first two zeros; and, for $|v| \geq 1418.9$,

$$B_c(v) + \sum_{|\gamma|\geq1419.9} \frac{1}{c^2+(v-\gamma)^2} \;\geq\; \frac{7}{c^2+64} - V_c(0) - 0.0029 \;\geq\; 0.0599,$$

while $B_c(v)$ itself tends to $-V_c(0) - 2\sum_{|\gamma|\geq1419.9}\gamma^{-2} < 0$ as $|v| \to \infty$. Under the Riemann hypothesis the limiting bracket

$$X(v) = \lim_{c\downarrow0} \frac{2\pi}{c}\big(u_{\theta'}(v) - u_{\theta'}(0)\big) = \sum_\rho \Big(\frac{1}{(v-\gamma_\rho)^2} - \frac{1}{\gamma_\rho^2}\Big)$$

has, in the first gap between the zeros 14.1347 and 21.0220, a unique minimiser $v^* \in [17.4978, 17.4980]$ (below the midpoint 17.5784 of the gap), and

$$0.1838985448 \;\leq\; X(v^*) \;\leq\; 0.1838987563;$$

on the rest of $13.5 \leq |v| \leq 1418.9$, $V_0(v) - V_0(0) \geq 0.184074$ (the author's computation, not rerun).

Proof. First clause, ball arithmetic as for (11.11.2): the interval $[13.5, 1418.9]$ is cut into cells of radius 0.1, V_c is bounded below on a cell by evaluating each term at the farthest point of the cell from the enclosure of γ, $V_c(0)$ is bounded above using the lower endpoints of the enclosures, and the cells whose lower bound falls below the best upper bound plus 0.005 are bisected. The printed values are valid lower bounds; a rerun with bisection to radius 2^{-14} and the enclosures of width 0.02 gives 0.175808, 0.178430, 0.179098, 0.179322 at $v = 17.493$ to 17.498.

Far range: seven ordinates lie within distance 8 of v; each contributes at least $(c^2+64)^{-1}$, through $V_c(v)$ if it is certified and through the far sum of the first display otherwise. The minimum over $c \in [0,1]$ of $7/(c^2+64) - V_c(0) - 0.0029$ is 0.0600 (at $c=1$, using $V_1(0) \le 0.044785$), so 0.0599 holds.

The last statement. Under the Riemann hypothesis $X(v) = [V(v) - V(0)] + F(v)$ with $V = V_0$ over the certified ordinates and $F(v) = \sum_{\gamma \ge 1419.9} f_v(\gamma)$, where

$$f_v(\gamma) = \frac{1}{(\gamma - v)^2} + \frac{1}{(\gamma + v)^2} - \frac{2}{\gamma^2} = \frac{2v^2(3\gamma^2 - v^2)}{\gamma^2(\gamma^2 - v^2)^2} \;\ge\; 0 \qquad (\gamma > v),$$

so the uncertified zeros only raise X, and $X \ge V - V(0)$ on the whole range. Under the Riemann hypothesis $X''(v) = 6\sum_\rho (v - \gamma_\rho)^{-4} > 0$ between consecutive ordinates, so X is strictly convex on $(14.1347, 21.0220)$ and has a unique minimiser v^* there; on $[13.5, 14.1347)$ it is increasing, being even and convex on $(-\gamma_1, \gamma_1)$, and $X(13.5) > 2$, the nearest ordinate alone contributing 2.48 while $\sum_\rho \gamma_\rho^{-2} = 0.0462$. For $0 < v < \gamma$, $\partial_v f_v(\gamma) = 4v(3\gamma^2 + v^2)/(\gamma^2 - v^2)^3 > 0$, so $F' \ge 0$. With the 1000 certified ordinates (`python-flint`, 160 and 320 bits) the derivative Q' of $Q = V - V(0)$ is enclosed by $Q'(17.4978) \in [-1.0391209, -1.0367081] \times 10^{-5}$ and $Q'(17.4980) \in [7.2326504, 7.2567776] \times 10^{-6}$, including a rigorous bound $0 \le F' \le 2.41 \times 10^{-8}$ for the tail (partial summation against [100]). So X' changes sign exactly once in $[17.4978, 17.4980]$. For the value, tangent lines of Q at the two endpoints give the lower bound, and $Q(17.4979)$ plus the tail bound 2.11×10^{-7} the upper. The statement for the rest of the range is the cell computation of the first clause for $V - V(0)$ with the enclosures of the author, not rerun: the rerun of the first clause gives $V - V(0) \ge B_0 \ge 0.17927$ only. □

The certified values for the four levels differ by two per cent, so the slope is essentially independent of the level: the bracket of (11.11.1) is at least 0.1757 rather than 0.1014 on the middle range and 0.0599 rather than 0.0151 beyond, a slope per unit c of at least $0.1757/(2\pi) = 0.0279$. By the author's computation (the last clause of Proposition 11.11.4, not rerun) the infimum over $13.5 \le v \le 1418.9$ is attained between the first two zeros, at v^*, pushed below their midpoint 17.578 by the zeros above; its value as $\theta' \downarrow \frac{1}{2}$ is 0.18390 to five digits under the Riemann hypothesis, the uncertified zeros contributing less than 3×10^{-7} because their paired terms $f_v(\gamma)$ are of order v^2/γ^4. The two neighbours alone give $8/G^2 = 0.1687$ with $G = 6.887$ at the midpoint of the gap (0.1689 at v^*), the other 1998 certified ordinates of both signs add 0.060, and $V(0)$ takes away 0.0448. The unconditional certificate (11.11.3) is weaker than this by the orbit bound at the origin, $2\sum_{\text{far}} \gamma^{-2} \le 0.0029$, which is the price of not assuming where the far zeros lie.

11.11.6 A posteriori bounds for the backward flow

The next problem asked for a bound on the distance between the EM output and the Thorin measure in terms of the residual, so that quantities read from the output become certified. The bound is a duality.

Proposition 11.11.5 (a posteriori window bound). *Let $\theta' \in [\frac12, 1)$, $a = 1-\theta'$, and suppose ζ has no zeros in $\operatorname{Re} s > \theta'$, so that $\mathfrak{U}_{\theta'}$ exists and $P_a * \mathfrak{U}_{\theta'} = u_1\,dv$. Let μ be a finite positive measure, for instance $\mu = \sum_j x_j h\,\delta_{w_j}$ built from an EM iterate, and let $r = P_a * \mu - u_1$ on $\mathbb{R}$. For a test function φ with $\int e^{a|\omega|}|\widehat{\varphi}(\omega)|\,d\omega < \infty$ put $\psi = \mathcal{F}^{-1}[e^{a|\omega|}\widehat{\varphi}]$, and assume that $\psi\cdot(P_a*\mu+u_1)$ is integrable (this holds for the Gaussian windows below, by the tails of Lemma 11.11.6). Then*

$$\Big|\int \varphi\,d\mu - \int \varphi\,d\mathfrak{U}_{\theta'}\Big| \;\le\; \int_{\mathbb{R}} |r(v)|\,|\psi(v)|\,dv.$$

Proof. $\widehat{P_a} = e^{-a|\omega|}$, so $P_a * \psi = \varphi$. By Fubini, justified by the integrability hypothesis applied to the absolute integrand, $\int \varphi\,d\mu = \int \psi\,(P_a*\mu)\,dv$ and $\int \varphi\,d\mathfrak{U}_{\theta'} = \int \psi\,u_1\,dv$. Subtract. □

The residual is needed on all of $\mathbb{R}$: inside the data window it is the EM residual, between the data window and the edge of the grid carrying μ it is computable from u_1, and beyond that grid $r = P_a*\mu - u_1$ with $P_a*\mu(v) \le a\|\mu\|/(\pi(a^2+(|v|-W)^2))$ for $\operatorname{supp}\mu \subset [-W,W]$ and $|v| > W$, against the decay of ψ, which is $|\psi(v)| \sim a\sigma\sqrt{2\pi}/(\pi(v-c)^2)$ for the Gaussian window $\varphi = e^{-(v-c)^2/2\sigma^2}$, from the kink of $e^{a|\omega|}$ at the origin. The data must be known to certified accuracy on the computation grid: in the author's experience, interpolating u_1 from a coarser grid introduces errors of 10^{-3}, larger than the EM residual, and the bound fails. And the tail must be measured from the window centre, not from the origin. Centroid certificates come from the odd test function $\varphi_1(v) = (v-c)\varphi(v)$: with certified mass $M \pm b_0$ and certified first moment $m_1 \pm b_1$, the φ weighted centroid of $\mathfrak{U}_{\theta'}$ lies within $(|m_1| + b_1)/(M - b_0)$ of c. An interval for a single zero needs in addition a bound on the other zeros (Corollary 11.11.9).

In floating point, with exact data at spacing 0.02 on $[-400, 400]$, the unknown on $[-440, 440]$ and 4000 accelerated steps, the author found the bound to hold for all 198 windows centred at the zeros in $(5, 395)$, for masses and moments, at $\theta' = \frac12$ and $\frac34$ and $\sigma = 0.5$ and 0.3 (not repeated here); the certified figures below supersede these.

11.11.7 The certificate

Two things stood in the way of a certificate: ψ came from a floating point transform, and $\int |r||\psi|$ was a Riemann sum. Both are removed by replacing the Gaussian φ with the function the grid actually sees.

LEMMA 11.11.6 (the dual function in closed form). *For* $\varphi(v) = e^{-(v-c)^2/2\sigma^2}$ *and* $a > 0$, *with* $q = a + i(v-c)$ *and*

$$E(q) = e^{q^2/2\sigma^2}\operatorname{erfc}\Big(-\frac{q}{\sigma\sqrt{2}}\Big),$$

the dual functions of φ *and of* $\varphi_1(v) = (v-c)\varphi(v)$ *are*

$$\psi(v) = \operatorname{Re} E(q), \qquad \psi_1(v) = \operatorname{Im}\big(q\,E(q)\big) = -\sigma^2\psi'(v).$$

Both extend to entire functions of v: *with* $q_\pm = a \pm i(z-c)$, $\Psi(z) = \frac12[E(q_+) + E(q_-)]$ *and* $\Psi_1(z) = \frac{1}{2i}[q_+E(q_+) - q_-E(q_-)]$. *For every* $x = v - c \neq 0$,

$$|\psi(v)| \le \frac{\sigma\sqrt{2\pi}}{\pi}\Big(\frac{a}{x^2} + \frac{B_4}{x^4}\Big), \qquad |\psi_1(v)| \le \frac{\sigma^3\sqrt{2\pi}}{\pi}\Big(\frac{2a}{|x|^3} + \frac{B_5}{|x|^5}\Big),$$

where, with $g(\omega) = e^{a\omega - \sigma^2\omega^2/2}$ *and* $G(\omega) = \omega g(\omega)$, $B_4 = |a^3 - 3\sigma^2 a| + \int_0^\infty |g^{(4)}|$ *and* $B_5 = 4|a^3 - 3\sigma^2 a| + \int_0^\infty |G^{(5)}|$.

Proof. By definition,

$$\psi(v) = \frac{1}{2\pi}\int e^{a|\omega|}\widehat{\varphi}(\omega)e^{i\omega v}d\omega = \frac{\sigma\sqrt{2\pi}}{\pi}\int_0^\infty g(\omega)\cos(\omega x)\,d\omega,$$
$$\int_0^\infty e^{-p\omega^2+q\omega}d\omega = \tfrac12\sqrt{\pi/p}\,e^{q^2/4p}\operatorname{erfc}\big(-q/2\sqrt{p}\big),$$

and the second, classical, integral with $p = \sigma^2/2$ and $q = a + ix$ gives the first formula; at $a = 0$ it reduces to φ, since erfc of an imaginary argument has real part 1. Since $\overline{E(q)} = E(\bar q)$, the real part is $\frac12[E(q_+) + E(q_-)]$ on the real axis, which is entire in z. Differentiating, $E'(q) = (q/\sigma^2)E(q) + \sqrt{2/\pi}/\sigma$, so $\psi'(v) = \operatorname{Re}(iE'(q)) = -\sigma^{-2}\operatorname{Im}(qE(q))$; and $\widehat{\varphi_1} = -i\sigma^2\omega\widehat{\varphi}$ gives $\psi_1 = -\sigma^2\psi'$. The tails come from integrating by parts:

$$\int_0^\infty g\cos(\omega x)\,d\omega = -\frac{g'(0)}{x^2} + \frac{g'''(0)}{x^4} + \frac{1}{x^4}\int_0^\infty g^{(4)}\cos(\omega x)\,d\omega,$$
$$\int_0^\infty G\sin(\omega x)\,d\omega = -\frac{G''(0)}{x^3} + \frac{G''''(0)}{x^5} + \frac{1}{x^5}\int_0^\infty G^{(5)}\cos(\omega x)\,d\omega,$$

with $g'(0) = a$, $g'''(0) = a^3 - 3\sigma^2 a$, $G''(0) = 2a$ and $G''''(0) = 4(a^3 - 3\sigma^2 a)$; the second integral is the one for $\psi_1 = \frac{\sigma^3\sqrt{2\pi}}{\pi}\int_0^\infty G\sin(\omega x)\,d\omega$. □

The integrals $\int|g^{(4)}|$ and $\int|G^{(5)}|$ are enclosed by writing $g^{(k)} = H_k g$ with $H_{k+1} = H_k' + (a - \sigma^2\omega)H_k$, taking absolute values of the coefficients and integrating in balls. The author reports $B_4 = 38.0$ and $B_5 = 550.6$ at $a = \frac12$, $\sigma = 0.3$ (not

rerun); generous, but they multiply x^{-4} and x^{-5} and are used only at distances above 440.

Now fix the grid $v_i = ih$, $|v_i| \le 880$, and let $\psi_h = \sum_i \psi(v_i) h\, \delta_{v_i}$ and $\varphi_h = P_a * \psi_h$. For any finite positive μ with $P_a * \mu$ finite at the grid points, and $U = \mathfrak{U}_{\theta'}$,

$$\int \varphi_h \, d\mu - \int \varphi_h \, d\mathfrak{U}_{\theta'} = \sum_i \psi(v_i)\, h\, r_i, \qquad r_i = (P_a * \mu)(v_i) - u_1(v_i), \quad (11.11.4)$$

exactly, by Fubini over a finite sum and $P_a * \mathfrak{U}_{\theta'} = u_1$. The right side is a finite sum of balls. For $\mu = \sum_j x_j h \delta_{w_j}$, $(P_a * \mu)(v_i)$ is a polynomial product, which Arb [179] multiplies with rigorous error bounds in a fraction of a second, the x_j being the double precision EM output taken as exact rationals. The identity does not depend on EM having converged. It remains to compare φ_h with φ:

$$\int \varphi \, d\mu - \int \varphi \, d\mathfrak{U}_{\theta'} = \sum_i \psi(v_i) h r_i + \int (\varphi - \varphi_h)\, d\mu - \int (\varphi - \varphi_h)\, d\mathfrak{U}_{\theta'}.$$

Write φ_h^{∞} for the same sum over the whole lattice $h\mathbb{Z}$. Then $\varphi - \varphi_h = (\varphi - \varphi_h^{\infty}) + \sum_{|v_i|>880} \psi(v_i) h P_a(\cdot - v_i)$, the first term the trapezoid error of an analytic integrand, the second a truncation controlled by the tails of Lemma 11.11.6.

Lemma 11.11.7 (trapezoid error). *Let Ψ be the entire extension of ψ, $f_w(v) = \Psi(v) P_a(w - v)$, and assume $h \le \pi\sigma^2/(2a)$. For every real w,*

$$|\varphi(w) - \varphi_h^{\infty}(w)| \;\le\; \min\Big(\frac{4 S_\Psi}{e^{\pi a/h} - 1},\; \frac{E_{\mathrm{dec}}}{(w-c)^2} \Big),$$

where $S_\Psi = 2e^{9a^2/8\sigma^2} + 1$ bounds $|\Psi|$ on the strip $|\operatorname{Im} v| \le a/2$, and

$$E_{\mathrm{dec}} = \frac{\sigma\sqrt{2\pi}}{2\pi}\,(4a + J)\,\frac{2e^{-2\pi a/h}}{1 - e^{-2\pi a/h}},$$
$$J = \int_{\mathbb{R}} \big((2a + \sigma^2|\omega|)^2 + \sigma^2 \big) e^{2a|\omega| - \sigma^2\omega^2/2}\, d\omega;$$

at $w = c$ the second term of the minimum is $+\infty$. The same holds for ψ_1 and φ_1 with $b = \frac{3a}{2}$, $S_1 = \sigma^3\sqrt{2\pi}\,\pi^{-1}\int_0^\infty t e^{bt - \sigma^2 t^2/2} dt$ in place of S_Ψ with numerator $4S_1$, $C_1 = \sup_{t \ge 0} t e^{2at - \sigma^2 t^2/2}$, $J_1 = \int_{\mathbb{R}} \big[2(2a + \sigma^2|\omega|) + |\omega| \big((2a + \sigma^2|\omega|)^2 + \sigma^2 \big) \big] e^{2a|\omega| - \sigma^2\omega^2/2} d\omega$, and

$$E_{\mathrm{dec},1} = \frac{\sigma^3\sqrt{2\pi}}{2\pi}\,(2aC_1 + J_1)\,\frac{2e^{-2\pi a/h}}{1 - e^{-2\pi a/h}}.$$

Proof. The uniform bound is the theorem of Trefethen and Weideman [265] for the trapezoid rule on $\mathbb{R}$: if f is analytic in $|\operatorname{Im} v| < d$, tends to zero there, and $\int |f(v + ib)| dv \le M$ for $|b| < d$, then

$$\Big| h \sum_k f(kh) - \int f \Big| \le \frac{2M}{e^{2\pi d/h} - 1}.$$

Take $d = a/2$, inside the poles of $P_a(w - \cdot)$ at $w \pm ia$. On the strip,

$$|a^2 + (t - ib)^2| = |t - i(b + a)|\,|t - i(b - a)| \geq t^2 + \tfrac{a^2}{4},$$

so $\int |P_a(w - v - ib)|\,dv \leq (a/\pi)(2\pi/a) = 2$ and $M \leq 2 \sup |\Psi|$. For

$$\Psi(v + ib) = \tfrac{1}{2}\big[E(a - b + i(v - c)) + E(a + b - i(v - c))\big]$$

write $E(q) = 2e^{z^2} - w(iz)$ with $z = q/\sigma\sqrt{2}$ and w the Faddeeva function, which satisfies $|w(\zeta)| \leq 1$ for $\operatorname{Im}\zeta \geq 0$, that is for $\operatorname{Re} q > 0$; and $|e^{z^2}| = e^{((\operatorname{Re} q)^2 - (\operatorname{Im} q)^2)/2\sigma^2} \leq e^{9a^2/8\sigma^2}$. Hence S_Ψ. The decaying bound is Poisson summation: $\varphi_h^\infty(w) - \varphi(w) = \sum_{k \neq 0} \widehat{f_w}(2\pi k/h)$ and

$$\widehat{f_w}(\omega_0) = \frac{e^{-i\omega_0 w}}{2\pi} \int \widetilde{G}(\omega) e^{i\omega(w-c)} d\omega,$$
$$\widetilde{G}(\omega) = \sigma\sqrt{2\pi}\, e^{a|\omega| - a|\omega - \omega_0| - \sigma^2\omega^2/2},$$

which is smooth except for kinks at 0 and ω_0. Two integrations by parts give $(w - c)^{-2}$ times the sum of the jumps of $\widetilde{G}'$, each at most $2a\sigma\sqrt{2\pi}e^{-a|\omega_0|}$ since $e^{2a|\omega_0| - \sigma^2\omega_0^2/2} \leq 1$ for $|\omega_0| \geq 4a/\sigma^2$, which holds for every sampling frequency $|\omega_0| \geq 2\pi/h$ exactly when $h \leq \pi\sigma^2/(2a)$; plus $\int |\widetilde{G}''| \leq \sigma\sqrt{2\pi}e^{-a|\omega_0|}J$, using $e^{a|\omega| - a|\omega - \omega_0|} \leq e^{-a|\omega_0| + 2a|\omega|}$. Summing the geometric series over k gives E_{dec}. For ψ_1 the transform carries the factor $-i\sigma^2\omega$, the strip bound is the Fourier integral of Ψ_1 on the strip, and the second derivative of $\omega\,\widetilde{G}$ is bounded in the same way, which gives the constants S_1, C_1, J_1 (derived in the review of the source, not rerun). □

With $h = 0.02$ the condition $h \leq \pi\sigma^2/(2a)$ holds for all four parameter pairs of the table, $e^{\pi a/h}$ is $e^{78.5}$ at $a = \frac{1}{2}$ and $e^{39.3}$ at $a = \frac{1}{4}$, and the constants are below 10^{-30} and 10^{-15} respectively; they are carried through the computation, and are why the grid spacing was never allowed to grow. The last ingredient is a bound on the mass of $\mathfrak{U}_{\theta'}$, which is unknown but dominated through the forward equation: for $L > 0$,

$$\mathfrak{U}_{\theta'}([-W, W]) \;\leq\; \frac{2M_1(W + L)}{1 - 2P_a((L, \infty))},$$
$$M_1(T) = \int_0^T u_1(v)\,dv = \pi^{-1} \operatorname{Im} \log \xi(1 + iT),$$

since $P_a * \mathfrak{U}_{\theta'} \geq P_a * (\mathfrak{U}_{\theta'}|_{[-W,W]})$ and the latter integrates to at least $P_a([-L, L])\,\mathfrak{U}_{\theta'}([-W, W])$ over $[-W - L, W + L]$. The function M_1 is explicit up to $\arg\zeta(1 + iT)$ (continuous branches of the logarithms, continued horizontally from $\sigma = +\infty$):

$$\pi M_1(T) = \arctan T + \tfrac{\pi}{2} - \tfrac{T}{2}\log\pi + \operatorname{Im}\log\Gamma\big(\tfrac{1+iT}{2}\big) + \arg\zeta(1 + iT),$$

and $|\arg\zeta(1+iT)| \le 24.303 + \log\log T + 0.498$ for $T \ge 13$, by integrating the explicit bound $|\zeta'/\zeta(\sigma+it)| \le 24.303\log t$ on $\sigma \ge 1$ from 1 to $1+1/\log T$, the bound $-\zeta'/\zeta(\sigma) \le 1/(\sigma-1)$ from there to 2, and $\log\zeta(2) = 0.4977$ beyond. The explicit bound is Leong's [111]; the constant 24.303 is the author's, and the corollary of that paper is reported (unchecked) to give $18.062\log t$ for $\sigma \ge 1$, $t \ge 5$, so 24.303 is conservative. So M_1 is a ball, and the mass of $\mathfrak{U}_{\theta'}$ on $[-W, W]$ is bounded by an increasing explicit function $\overline{U}(W)$, with $\overline{U}(880) = 1286.4$ at $a = \frac{1}{2}$ (the author's value, not rerun).

THEOREM 11.11.8 (certified windows; conditional, computed by the author). *Let $\theta' \in \{\frac{1}{2}, \frac{3}{4}\}$, $a = 1-\theta'$, $\sigma \in \{0.3, 0.5\}$, suppose ζ has no zeros in $\operatorname{Re} s > \theta'$ (at $\theta' = \frac{1}{2}$ this is the Riemann hypothesis; at $\theta' = \frac{3}{4}$ it is Theorem 11.9.6 up to height 1419.9 and an assumption beyond), and let μ be the accelerated EM output after 4000 steps on the grid of spacing $h = 0.02$ on $[-440, 440]$. For each of the 198 windows φ of width σ centred at the grid point c nearest a zero in $(5, 395)$, with $\varphi_1 = (v-c)\varphi$,*

$$\Big|\int \varphi\,d\mu - \int \varphi\,d\mathfrak{U}_{\theta'}\Big| \le B_0(c), \qquad \Big|\int \varphi_1\,d\mu - \int \varphi_1\,d\mathfrak{U}_{\theta'}\Big| \le B_1(c),$$

$$B_0(c) = h\sum_{|v_i|\le 880} |\psi(v_i)||r_i| + \|\mu\|\,\varepsilon_\mu + \overline{U}(880)\,\varepsilon_U + T(c),$$

$$T(c) = \int_{|w|>880} D(|w| - |c|)\,d\mathfrak{U}_{\theta'}(w),$$

and B_1 likewise with ψ_1, where $\varepsilon_\mu, \varepsilon_U$ are the uniform bounds on $|\varphi - \varphi_h|$ over the support of μ and over $[-880, 880]$ given by Lemma 11.11.7 and the tails of Lemma 11.11.6, D is the explicit decreasing majorant

$$D(d) = \frac{E_{\mathrm{dec}}}{d^2} + \frac{K_{\mathrm{far}}\,a}{\pi(1-\lambda)^2 d^2} + \Big(1 + \frac{h}{\pi a}\Big)\bar{k}(\lambda d), \qquad \lambda = 0.9,$$

with $\bar{k}$ the tail bound of Lemma 11.11.6 and K_{far} a bound on $\sum_{|v_i|>880} |\psi(v_i)|h$, and the last integral is bounded by Abel summation against $\overline{U}$ over blocks $W_k = 880\cdot 1.005^k$ up to 10^7 and by $\overline{U}(w) \le w\log w$ beyond. Every quantity is a ball. The 792 pairs of bounds (mass and moment, for 198 windows, two levels and two widths) were compared with floating point reference values and agree. The certified figures are in the table below.

The finite computations are the author's, not rerun; given them, the proof is the identity (11.11.4), Lemma 11.11.7 and the mass bound. The majorant D is valid after Lemma 11.11.7: split the omitted sample points into $|v_i - c| \le \lambda d$, where $P_a(w - v_i) \le a/[\pi(1-\lambda)^2 d^2]$, and the complement, where $|\psi(v_i)| \le \bar{k}(\lambda d)$ and $h\sum_i P_a(w - v_i) \le 1 + h/(\pi a)$. The Abel summation is an upper sum: if D is decreasing, integration by parts gives $\int_{|w|>V} D(|w| - |c|)\,d\mathfrak{U} \le \int_V^\infty [-D'(R -$

$|c|)]\,\overline{U}(R)\,dR$, and on a block $[W_k, W_{k+1}]$ the integral is at most $\overline{U}(W_{k+1})[D(W_k - |c|) - D(W_{k+1} - |c|)]$.

θ'	σ	mass bound B_0			moment B_1		centroid bound	
		max	mean	far tail	max	mean	median	max
1/2	0.5	2.2×10^{-2}	5.4×10^{-3}	1.7×10^{-3}	1.8×10^{-3}	1.1×10^{-3}	0.027	0.20
1/2	0.3	1.4×10^{-2}	5.1×10^{-3}	1.1×10^{-3}	2.0×10^{-3}	1.2×10^{-3}	0.0072	0.11
3/4	0.5	5.7×10^{-3}	1.7×10^{-3}	8.3×10^{-4}	2.2×10^{-4}	1.1×10^{-4}	0.042	0.14
3/4	0.3	3.4×10^{-3}	1.0×10^{-3}	4.9×10^{-4}	3.6×10^{-5}	1.2×10^{-5}	0.0068	0.071

The far tail column is the mean of the last term of B_0, the price of knowing $\mathfrak{U}_{\theta'}$ beyond 880 only through its mass bound; it is a fifth to a half of the mean bound and would shrink on a longer grid. The main term, the finite sum in (11.11.4), has mean 3.6×10^{-3} at $\theta' = \frac{1}{2}$, $\sigma = 0.3$, against 4.5×10^{-3} for the floating point bound with its own tail estimate. At $\theta' = \frac{1}{2}$ every window has certified mass at least 0.9929, so, since an interval of length one below height 400 holds at most $n(400) < 9.34$ ordinates (the count of Theorem 11.11.10), each window has a zero within 0.73 of its centre at $\sigma = 0.3$ and within 1.22 at $\sigma = 0.5$: if no zero lay within r of the centre, the Gaussian mass would be at most $2n \sum_{k \geq 0} e^{-(r+k)^2/(2\sigma^2)}$, which is 0.967 and 0.952 at these radii, below 0.9929 (the critical radii are 0.7267 and 1.2115). The certified centroid lies within 0.05 of the window centre for 192 of the 198 windows at $\sigma = 0.3$ (as reported by the author), the six exceptions being the close pairs, where a neighbouring atom at distance d moves the centroid by an amount of the order of $d\,e^{-d^2/2\sigma^2}$. At $\theta' = \frac{3}{4}$ the target is the Thorin density $\sum_\rho P_{3/4-\sigma_\rho}(v - \gamma_\rho)$, which is $\sum P_{1/4}(v - \gamma)$ when the zeros are on the line, the window mass is about 0.7, the bounds are five times tighter because the multiplier $e^{|\omega|/4}$ is milder, and the certified quantities are masses and centroids of the Thorin measure at level $\frac{3}{4}$ itself. The serialisation of the EM vector and of the data balls must be lossless for the bound to be a certificate; this was not repeated here.

Corollary 11.11.9 (from centroids to zeros). *Assume the Riemann hypothesis, so that* $\mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho}$. *Let* $\varphi(v) = e^{-(v-c)^2/(2\sigma^2)}$, *let* $\int \varphi\, d\mathfrak{U}_{1/2} \in [\widehat{M} - B_0, \widehat{M} + B_0]$ *with* $\widehat{M} > B_0$ *and* $\int (v-c)\varphi\, d\mathfrak{U}_{1/2} \in [\widehat{m} - B_1, \widehat{m} + B_1]$ *be certified, let* γ_* *be an ordinate with* $|\gamma_* - c| \leq R$, *and let* $\Theta \geq \sum_{\gamma \neq \gamma_*} (|\gamma - c| + R)\varphi(\gamma)$, *the sum over the other ordinates with multiplicity. Then*

$$|\gamma_* - c| \;\leq\; \frac{|\widehat{m}| + B_1 + \Theta}{\widehat{M} - B_0}.$$

Proof. With $M = \int \varphi\, d\mathfrak{U}_{1/2}$, $\int (v-c)\varphi\, d\mathfrak{U}_{1/2} = (\gamma_* - c)M + \sum_{\gamma \neq \gamma_*} (\gamma - \gamma_*)\varphi(\gamma)$ and $|\gamma - \gamma_*| \leq |\gamma - c| + R$. Hence $|\gamma_* - c|\, M \leq |\widehat{m}| + B_1 + \Theta$, and $M \geq \widehat{M} - B_0 > 0$. □

The windows are centred at grid points, within 0.01 of the zeros, so that $\psi(v_i - c)$ is a shift of one table; the centroid certificate reports the offset from c

and is unaffected. Below height 1419.9 the ordinates are already certified (Theorem 11.9.6), so Corollary 11.11.9 locates nothing new there. The hypothesis on the zeros is used only to know that $\mathfrak{U}_{\theta'}$ exists and satisfies $P_a * \mathfrak{U}_{\theta'} = u_1$. The bound is a statement about $\mathfrak{U}_{\theta'}$: at $\theta' = \frac{3}{4}$ about the Thorin density itself, at $\theta' = \frac{1}{2}$, under the Riemann hypothesis, about Gaussian weighted sums over the zeros, and about single zeros only through Corollary 11.11.9.

11.11.8 An iteration adapted to the certificate

The certificate's main term $h \sum_i |\psi_c(v_i)||r_i|$ is a weighted ℓ^1 norm of the residual; does an iteration aimed at it beat EM? We compared, at $\theta' = \frac{1}{2}$ on the grid of spacing 0.02, with $\sigma = 0.3$ and the weights $w_i = \sum_c |\psi_c(v_i)|$ summed over the 198 windows: accelerated EM; FISTA with nonnegativity on the weighted least squares $\frac{1}{2} \sum_i w_i (Ax - y)_i^2$, which bounds the sum over the windows of the weighted ℓ^1 norms by Cauchy and Schwarz, $h \sum_i w_i |r_i| \le h (\sum_i w_i)^{1/2} (\sum_i w_i r_i^2)^{1/2}$ (an aggregate bound, not a bound for each window); FISTA on the plain least squares; and accelerated EM followed by a weighted FISTA polish. The figure of merit is the mean main term over the windows. The experiment is the author's, not repeated.

iteration (steps)	main term	KL
accelerated EM (4000)	5.32×10^{-5}	1.2×10^{-4}
weighted FISTA (4000)	7.38×10^{-5}	0.95
plain FISTA (4000)	5.26×10^{-5}	1.4×10^{-4}
EM (4000), weighted FISTA polish (1000)	4.12×10^{-5}	4.3×10^{-4}
accelerated EM (5000)	4.73×10^{-5}	9.8×10^{-5}

The weighted objective from a cold start is worse on its own figure of merit, and much worse in divergence: the weights concentrate on the peaks of u_1, where the deconvolution is most ill conditioned, and the iteration spends its steps there. Plain least squares and EM are indistinguishable. The polish gains thirteen per cent over the same number of EM steps. Our reading is that the main term is dominated by the residual within a unit of each zero, where every objective fits equally badly, the limit being the conditioning rather than the loss. In this experiment the certificate does not reward an iteration of its own; a tighter certificate needs more accelerated steps or a finer grid near the zeros. One experiment does not show that no other objective helps.

11.11.9 A fast algorithm for u_1 at fixed accuracy

The third problem asked for an algorithm for $u_1(v)$ polynomial in $\log v$. With the accuracy entering through $\log(1/\varepsilon)$ it is open (Problem 11.3). With the accuracy fixed the explicit formula answers it: given that the zeros near the height v lie

on the critical line (verified below 3×10^{12} [232]; at every height conditionally on the announced half plane $\operatorname{Re} s>\frac78$, with a larger exponent), the primes below X determine u_1 at that height to accuracy of order $X^{-1/2}\log v$, so the cost of one value is $O(X)$ whatever the height, and a nonuniform fast Fourier transform gives a whole grid of heights for the price of one.

THEOREM 11.11.10 (smoothed prime sum). *Let $v\ge2$, $X\ge2$, $s=1+iv$, $L=\zeta'/\zeta$, and*

$$u_1^X(v)=\frac1\pi\Big[\frac{1}{1+v^2}-\frac{\log\pi}{2}+\frac12\operatorname{Re}\psi\Big(\frac{1+iv}{2}\Big)\\ -\sum_{n\ge2}\frac{\Lambda(n)}{n}\,e^{-n/X}\cos(v\log n)\Big].$$

Then, with $R_X(s)=(2\pi i)^{-1}\int_{\operatorname{Re}w=-3/2}(-L(s+w))\Gamma(w)X^w\,dw$,

$$\pi\big(u_1(v)-u_1^X(v)\big)=-\operatorname{Re}\sum_\rho\Gamma(\rho-s)X^{\rho-s}+\operatorname{Re}\,\Gamma(1-s)X^{1-s}\\ +\operatorname{Re}\big[X^{-1}L(s-1)\big]+\operatorname{Re}R_X(s),$$

with $|R_X(s)|\le2.5X^{-3/2}\log(v+2)$, and if every zero with $|\gamma-v|\le Y$, $Y\ge1$, lies on the critical line,

$$|u_1(v)-u_1^X(v)|\le\frac1\pi\Big[c_\Gamma\,n(v)\,X^{-1/2}+2.6\,c'_\Gamma\,n(v+Y+1)\,e^{-\pi Y/2}\\ +\sqrt{\frac{\pi}{v\sinh\pi v}}+\frac{27\log v}{X}+\frac{2.5\log(v+2)}{X^{3/2}}\Big],$$

where $n(t)=0.39\log t+7$ bounds the number of ordinates in any interval of length one below height t, by [100, (1.7), (2.7) and Corollary 1.4]; $c_\Gamma=\sup_{v\ge2}\sum_{j\in\mathbb Z}\sup_{j\le y<j+1}|\Gamma(-\frac12+iy)|\,\frac{n(v+|j|+1)}{n(v)}\le8.34$ (without the factor $n(v+|j|+1)/n(v)$ the sum is 8.14376); and $c'_\Gamma=\sup\{|\Gamma(x+iy)|e^{\pi|y|/2}:-1<x<0,\ |y|\ge1\}\le2.51$.

Proof. $\sum_n\Lambda(n)n^{-s}e^{-n/X}=(2\pi i)^{-1}\int_{(2)}(-L)(s+w)\Gamma(w)X^w\,dw$ by Mellin inversion of e^{-y}. Shift the contour to $\operatorname{Re}w=-\frac32$. This crosses the pole of Γ at $w=0$, with residue $-L(s)$, the zeros $w=\rho-s$ (real part in $(-1,0)$), with residues $-\Gamma(\rho-s)X^{\rho-s}$ times the multiplicity, the pole $w=1-s$ of L, with residue $\Gamma(1-s)X^{1-s}$, and the pole $w=-1$ of Γ, with residue $X^{-1}L(s-1)$; it crosses none of the trivial zeros ($\operatorname{Re}w=-2k-1\le-3$) or of the later poles of Γ. The horizontal segments vanish along heights that avoid the zeros, by the exponential decay of Γ. Taking real parts and using $\pi u_1(v)=\operatorname{Re}[1/s-\frac12\log\pi+\frac12\psi(s/2)]+\operatorname{Re}L(s)$ gives the identity. On $\operatorname{Re}w=-\frac32$, $\operatorname{Re}(s+w)=-\frac12$; by the functional equation $L(z)=\chi'/\chi(z)-L(1-z)$ with $\operatorname{Re}(1-z)=\frac32$, where the Dirichlet series converges absolutely, so $|L(1-z)|\le-\zeta'/\zeta(\frac32)<1.5053$. Here $\chi'/\chi(z)=$

$\log 2\pi + \frac{\pi}{2}\cot\frac{\pi z}{2} - \psi(1-z)$ with $|\cot\frac{\pi z}{2}| = 1$ on $\operatorname{Re} z = -\frac12$, and Binet's formula $\psi(a) = \log a - \frac{1}{2a} - 2\int_0^\infty \frac{t\,dt}{(t^2+a^2)(e^{2\pi t}-1)}$, with $|t^2+a^2| \ge \frac94$ for $\operatorname{Re} a = \frac32$, gives $|\psi(\frac32 + i\tau)| \le \log(|\tau|+2) + \frac{\pi}{2} + \frac13 + \frac{1}{27}$. For $w = -\frac32 + iy$, as $\log(|v+y|+2) \le \log(v+2) + \log(1+|y|)$, this gives $|L(s+w)| \le \log(v+2) + \log(1+|y|) + 6.856$, and with $\int_{\mathbb R} |\Gamma(-\frac32+iy)|\,dy < 2.527$ and $\int_{\mathbb R} |\Gamma(-\frac32+iy)|\log(1+|y|)\,dy < 0.798$ (computed), $|R_X(s)| \le X^{-3/2}\big(0.41\log(v+2) + 2.89\big) \le 2.5X^{-3/2}\log(v+2)$, since $\log(v+2) \ge \log 4$.

For the bound: the pole term is $\operatorname{Re}\Gamma(-iv)X^{-iv}$, of modulus at most $|\Gamma(-iv)| = (\pi/(v\sinh\pi v))^{1/2}$ since $|X^{-iv}| = 1$. For $v \ge 5$, $|L(iv)| \le 27\log v$ by the functional equation, $|\chi'/\chi(iv)| \le \log v$ and the explicit bound of [111] for $|L(1-iv)|$ on $\operatorname{Re} s = 1$; for $2 \le v \le 5$ it is a direct evaluation (the maximum of $|L(iv)|/\log v$ on $[2, 60]$ is 1.014). For the zeros on the line, $|\Gamma(-\frac12+iy)|^2 = \pi/(\cosh(\pi y)(\frac14+y^2))$ and $|X^{\rho-s}| = X^{-1/2}$; the zeros are counted in the unit intervals $\gamma - v \in [j, j+1)$, each holding at most $n(v+|j|+1)$ ordinates, and the resulting sum is $c_\Gamma n(v) X^{-1/2}$. A zero at distance more than Y contributes at most $c'_\Gamma e^{-\pi|\gamma-v|/2}$ ($|X^{\rho-s}| \le 1$ and $|\Gamma(\sigma_\rho - 1 + i(\gamma-v))| \le c'_\Gamma e^{-\pi|\gamma-v|/2}$), and $\sum_{k\ge0} 2e^{-\pi k/2} n(v+Y+k+1) \le 2.6\, n(v+Y+1)$. The constants 8.3388, 8.14376 and 2.509, and the count $n(t)$ against [100], were recomputed. □

The pole term cannot be dropped. At $v = 2$ every zero has $|\gamma - 2| \ge 12$ and lies on the critical line below 3×10^{12}; taking $Y = 10^{12}$, the bound without the pole term tends to 0 as $X \to \infty$. But $\operatorname{Re}\Gamma(-2i)X^{-2i} = |\Gamma(-2i)|\cos(\arg\Gamma(-2i) - 2\log X)$ equals $|\Gamma(-2i)| = 0.0766$ at $X_k = \exp((\arg\Gamma(-2i) + 2\pi k)/2)$, and the other terms tend to 0, so $\limsup_{X\to\infty} |u_1(2) - u_1^X(2)| \ge 0.0766/\pi = 0.0244$. At large heights the pole term is exponentially small and does not affect the experiments below.

The hypothesis on the zeros within Y of v is verified for $v + Y \le 3 \times 10^{12}$ with Y of a few dozen [232], and $e^{-\pi Y/2}$ is then negligible. Beyond, under the zero free region alone, $|X^{\rho-s}| \le X^{-c/\log v}$ and the first term decays only when $\log X \gg \log v$. Conditionally on the announced half plane $\operatorname{Re} s > \frac78$ [227, 228, 198], every zero has $\frac18 \le \beta \le \frac78$, so $|X^{\rho-s}| \le X^{-1/8}$ and $\operatorname{Re}(\rho - s)$ stays in $[-\frac78, -\frac18]$, away from the poles of Γ; the same count gives $|u_1(v) - u_1^X(v) - \pi^{-1}\operatorname{Re}\Gamma(1-s)X^{1-s}| \le C\log(v+2)(X^{-1/8} + X^{-1})$ at every height, so that a fixed accuracy ε costs X of order $(\log v/\varepsilon)^8$. The sum can be cut at $n = 30X$ with a dropped tail of about $e^{-30}\log X$ (at about $X\log(1/\delta)$ for a tolerance δ), so one value costs $O(X)$ operations after a sieve of $O(X\log\log X)$, independent of v; the bit cost adds the phases $v\log n$, which need $\log n$ to about $\log(v/\varepsilon)$ bits. For accuracy ε take $X = (c_\Gamma n(v)/\varepsilon)^2$: the cost is $O(\varepsilon^{-2}\log^2 v)$ operations, polynomial in $\log v$ and in $1/\varepsilon$.

On a grid $v_k = kh$ the sum is

$$\operatorname{Re}\sum_n a_n e^{ik(h\log n)}, \qquad a_n = \frac{\Lambda(n)}{n}\, e^{-n/X},$$

a type one nonuniform discrete Fourier transform with the frequencies $h \log n$, and a nonuniform FFT evaluates it at K heights with M retained prime powers in $O(K \log K + M \log(1/\delta))$ operations for a transform accuracy δ [75], to which the coefficients, the sieve and the smooth gamma factor part are to be added. The experiment is the author's, not rerun. With $X = 10^6$ and 1.86 million prime powers below 3×10^7, the error $|u_1 - u_1^X|$ at the heights 10^3, 10^4, 10^5, 10^6 and 10^7 is reported as 9×10^{-4}, 4×10^{-4}, 8×10^{-4}, 1.1×10^{-3} and 1.4×10^{-3}, against Arb [179]; averaged over fifty one heights near 10^6 the error is $0.05 \log v \, X^{-1/2}$ for $X = 10^4$, 10^5 and 10^6, which is the law of the theorem with a constant about twenty times smaller than the leading coefficient of the proved bound ($c_\Gamma \cdot 0.39/\pi = 1.0$ per $\log v$), the zero sum being oscillatory. The nonuniform FFT (`finufft`) is reported to give u_1^X at 1,048,576 heights at spacing 0.02, up to height 20,972, in 0.4 seconds for the prime sums alone (the smooth part is not in the timing and the thread count is not fixed), agreeing with the direct sums to 10^{-8}; Arb needs about two milliseconds per height, two thousand seconds for the same grid, but gives twenty digits.

So for the data layer: twenty digits at one height cost of order $v^{1/2}$ by the formula of Riemann and Siegel [71, 257] and of order $v^{1/3}$ by Hiary [97]; three digits at a million heights cost a second by the primes; the deconvolution needs four or five digits on a grid, which by the observed law above the smoothed sum delivers with $X = 10^8$ to 10^{10} (the proved bound asks for more), at the edge of a laptop, and Arb delivers in minutes up to the heights of this chapter.

11.11.10 Is $\mathfrak{U}_{1/2}$ computed efficiently?

The Cauchy deconvolution at the bottom level is the equation

$$P_{1/2} * \mathfrak{U}_{1/2} = u_1, \qquad \mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho} \text{ under the Riemann hypothesis,}$$

that is, $u_1(v) = \sum_\rho P_{1/2}(v - \gamma_\rho)$: the level one density is the Poisson integral of the counting measure of the zeros at distance $\frac{1}{2}$ from the line. To compute $\mathfrak{U}_{1/2}$ is to undo that Poisson integral. In Fourier variables the equation reads $e^{-|\omega|/2}\, \widehat{\mathfrak{U}_{1/2}}(\omega) = \widehat{u_1}(\omega)$, and by the prime sum of Theorem 11.11.10 the spectrum $\widehat{u_1}$ consists of atoms of weight $-\Lambda(n)/n$ at $\omega = \pm \log n$ and a smooth part from the Gamma factor (the prime part of u_1 is $-\pi^{-1} \sum \Lambda(n) n^{-1} \cos(v \log n)$ and $\widehat{\cos} = \pi(\delta_{\log n} + \delta_{-\log n})$). So the deconvolution multiplies the atom at $\log n$ by $e^{(\log n)/2} = \sqrt{n}$:

$$\widehat{\mathfrak{U}_{1/2}}: \quad -\frac{\Lambda(n)}{n} \;\longmapsto\; -\frac{\Lambda(n)}{\sqrt{n}} \qquad \text{at } \omega = \pm \log n,$$

which is the Fourier side of Weil's explicit formula,

$$\sum_\rho \widehat{g}(\gamma_\rho) = -\sum_n \frac{\Lambda(n)}{\sqrt{n}} \big(g(\log n) + g(-\log n) \big) + \cdots .$$

The backward Cauchy flow from level 1 to level $\frac{1}{2}$ is the explicit formula: the multiplier $e^{|\omega|/2}$ turns $\sum \Lambda(n)/n^{1+iv}$, convergent on $\operatorname{Re} s = 1$ only in a smoothed sense, into $\sum \Lambda(n)/n^{1/2+iv}$, meaningful only as a distribution.

This is the heuristic answer on efficiency. To resolve $\mathfrak{U}_{1/2}$ at scale δ one needs its spectrum to frequency $\Omega \approx \pi/\delta$, hence the primes to e^{Ω}, exponentially many; equivalently, the data u_1 must be known to accuracy $e^{-\Omega/2}$, since that is the factor by which errors at frequency Ω are amplified. The problem is exponentially ill posed in the resolution, and no linear band limited inversion of u_1 escapes this; EM shows no sign of escaping it. This is worst case stability, not a lower bound for every method: for a single unit atom with blur $y(v) = P_a(v - \gamma)$, $\gamma = a\,[1/y(-t) - 1/y(t)]/(4\pi t)$ for any $t \neq 0$, so extra structure changes the inverse problem. The backward flow with 4000 accelerated steps on a grid of spacing 0.02 certifies Gaussian weighted centroids with an uncertainty of about 10^{-3} (the moment bound B_1 of Theorem 11.11.8, divided by the mass), not the atoms (Corollary 11.11.9), and further digits cost further steps, at the observed rates of Section 11.11.1, or a finer grid with data accurate to about $e^{-\pi/2h}$. Each FFT step costs $O(T \log T)$ on a grid to height T, and the number of steps for a fixed resolution was observed to be grid independent; no bound on it uniform in T, the least gap or the accuracy is proved, so the total $T^{1+\varepsilon}$ of Odlyzko and Schönhage [127] is not claimed for the deconvolution. In the sense that matters it is not efficient: direct zero finding gives γ far more accurately in less time, because the functional equation carries the spectrum beyond any Ω in closed form, where the deconvolution has only the data below Ω. The efficient algorithm for $\mathfrak{U}_{1/2}$ is Turing's [270]; the deconvolution contributes the certificate, a bound on the measure from the residual alone, and the levels above $\frac{1}{2}$, where the Thorin measure is a density that no zero finder produces.

11.11.11 A von Neumann representation of $\mathfrak{U}_{1/2}$

Two things could be meant. In the sense of the von Neumann and Ulam estimator of Section 11.5, $\mathfrak{U}_{1/2} = (I - (I - A))^{-1} u_1$ with $A = P_{1/2}*$ has the Neumann series $\sum_{k \geq 0} (I - A)^k u_1$, which converges at every frequency, since the symbol $1 - e^{-|\omega|/2}$ lies in $[0, 1)$, but to the unbounded operator with symbol $e^{|\omega|/2}$. Its partial sums are the Richardson iterates with unit step: the sum to K terms applied to u_1 has symbol $[1 - (1 - e^{-|\omega|/2})^{K+1}]e^{|\omega|/2}$, so relative to the target the effective filter is $1 - (1 - e^{-|\omega|/2})^{K+1}$, which is near 1 for $|\omega| \leq 2 \log K$ and near 0 beyond. The truncated Neumann series is therefore a soft band limited inverse with resolution of order $\pi/(2 \log K)$, and the resolution 0.1 costs $K = e^{16}$ terms. The random walk that samples this series has kernel $|I - A| = \delta + P_{1/2}$, of total mass 2, and the estimator with random termination has infinite second moment (Proposition 11.5.1 for the grid version below the threshold $h < 0.52\pi(1 - \theta')$); finitely truncated walks have finite second moments. That representation is exact, convergent and useless,

for the reason of the previous subsection.

In the sense of the spectral theorem, the question is whether $\mathfrak{U}_{1/2}$ is the spectral measure of a natural self adjoint operator, the conjecture of Hilbert and Pólya. Under the Riemann hypothesis a diagonal operator with eigenvalues γ_ρ exists trivially, so existence alone is not the conjecture. What is unconditional is the shape of the answer at the levels above. If H is self adjoint with spectral measure $\mu_H = \sum \delta_\gamma$, then for $a > 0$ the resolvent is not trace class, since $N(T) \asymp T \log T$, but its imaginary part is, and

$$\frac{1}{\pi} \operatorname{Tr} \operatorname{Im}(H - v - ia)^{-1} = \sum_\gamma \frac{a/\pi}{(\gamma - v)^2 + a^2} = (P_a * \mu_H)(v),$$

so the Thorin density at level $\theta = \frac{1}{2} + a$ is this resolvent trace at height a, and the backward flow $\theta \downarrow \frac{1}{2}$ is Stieltjes inversion. Unconditionally the chapter has computed the function that would be that trace:

$$u_\theta(v) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\theta + iv) = \frac{1}{\pi} \sum_\rho \operatorname{Re} \frac{1}{\theta + iv - \rho} = \frac{1}{\pi} \sum_\rho \frac{a_\rho}{a_\rho^2 + (v - \gamma_\rho)^2},$$

with $a_\rho = \theta - \sigma_\rho$, which is that trace exactly when all a_ρ equal a. So the densities u_θ, $\theta > \frac{1}{2}$, are the traces $\pi^{-1} \operatorname{Tr} \operatorname{Im}(H - v - i(\theta - \frac{1}{2}))^{-1}$ of the one operator H with spectrum $\{\gamma_\rho\}$ if and only if the Riemann hypothesis holds. (If a zero has $\sigma_0 > \frac{1}{2}$, then at $v = \gamma_0$ the density u_θ is unbounded as $\theta \downarrow \sigma_0$, while the trace of Im of the resolvent is finite for every $\theta > \frac{1}{2}$.) The conjecture of Hilbert and Pólya asks for a natural H. The Thorin measures at levels $\theta > \frac{1}{2}$ provide the resolvent trace without the operator: ξ'/ξ plays the trace of the resolvent at $z = v + i(\theta - \frac{1}{2})$, and Theorem 11.11.8 bounds the boundary measure from the resolvent at height $\frac{1}{2}$, the Stieltjes inversion made quantitative. This does not produce H; it says what H would have to do.

11.12 Open problems

Problem 11.1. Prove the two phase picture of Section 11.11.1 for the continuous Cauchy deconvolution at an atomic target: a sublinear phase with $k(R_k - R^*)$ near a constant of order 50, lasting of the order of 10^4 steps on this data, followed by faster convergence on the support of the atoms with a grid independent rate; and compute that rate from the curvature of the divergence at the atoms. For the accelerated scheme with restart, explain the observed three point exponents, which oscillate between 0.6 and 2.8 with a running value near 1.3.

Problem 11.2. Proposition 11.11.4 gives the limiting slope under the Riemann hypothesis to six digits, 0.183899 at $v^* = 17.4979$. Is there a closed form? By the Hadamard product, under the Riemann hypothesis, $\log|\xi(\frac{1}{2} + iv)| = \log \xi(\frac{1}{2}) + \sum_{\gamma>0} \log|1 - v^2/\gamma^2|$, so with $\mathcal{C}(v) = \partial_v^2 \log|\xi(\frac{1}{2} + iv)|$ the bracket is $X(v) =$

$\mathcal{C}(0)-\mathcal{C}(v)$: the slope of the excess is the drop of the second logarithmic derivative of ξ along the critical line from the origin to v, and v^* is the zero of $\mathcal{C}'$ between the first two zeros. Whether $\mathcal{C}(v^*)$ has an expression in known constants is not clear.

Problem 11.3. Theorem 11.11.10 evaluates u_1 to accuracy ε in time polynomial in $\log v$ and in $1/\varepsilon$. Is there an algorithm polynomial in $\log v$ and $\log(1/\varepsilon)$? Such an algorithm for ζ and ζ' on $\operatorname{Re} s = 1$ would give one; the converse is not known, and the question is open. The smoothed sum cannot be the route, since the zeros themselves contribute $X^{-1/2}$.

Problem 11.4. The certificate's main term is set by the residual within a unit of each zero, which no objective reduces faster than the conditioning allows (as observed in Section 11.11.8). Quantify this: for a single atom at γ blurred by P_a and observed exactly, what is the best achievable $h\sum_i |\psi_\gamma(v_i)||r_i|$ after k steps of any first order method on the cone, as a function of k, h and a? The answer would say how far Theorem 11.11.8 can be pushed at a given cost.

Chapter 12

The explicit formula as a deconvolution

Chapters 10 and 11 write the Thorin measure at level θ as the Cauchy smoothing of the zeros at width $\theta - \frac{1}{2}$ (under the Riemann hypothesis; in general the widths are $\theta - \sigma_\rho$), turn the passage from level 1 to level θ into a deconvolution, and solve it by EM to about 10^{-3} in the positions of the zeros, with a certificate. This chapter looks at the bottom of that flow. At level $\frac{1}{2}$, under the Riemann hypothesis, the Thorin measure is the counting measure of the zeros, the data are $u_1(v) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(1 + iv)$, and the deconvolution, in Fourier variables, is Weil's explicit formula: the multiplier $e^{|\omega|/2}$ that undoes the Cauchy kernel is the factor $\sqrt{n}$ that turns the weights $\Lambda(n)/n$ of the level one spectrum into the weights $\Lambda(n)/\sqrt{n}$ of the formula. This says how ill posed the problem is. For a general measure the resolution δ costs, heuristically, the primes to $e^{\pi/\delta}$, or data accurate to $e^{-\pi/2\delta}$. For the measure at hand, a finite sum of atoms on every window, it costs much less: the Fourier transform of a sum of Lorentzians is a sum of damped exponentials, which Prony's method recovers from its samples with an error that, for a fixed configuration of atoms, is proportional to the data error. The matrix pencil applied to the tapered Fourier transform of u_1 on a window of width 60 returns, in the computation reported by the author, the ordinates and real parts of the central zeros to within 8×10^{-10}, from level one data alone, in seconds; a synthetic pair of zeros off the line at $0.85 \pm 60.089i$ is returned with real parts 0.85 and 0.15 to seven digits. EM solves the problem on a dense grid, where it is exponentially ill posed; the matrix pencil solves it in the model where it is polynomially ill posed. A posterior concentration theorem makes the recovery Bayesian, and the functional equation makes it singular: off line zeros come in pairs at $\frac{1}{2} \pm \delta$, the data depend on δ only through δ^2, the real part of a pair is resolved at the rate $\tau^{1/2}$ when the data are accurate to τ, and Bayes factors for the line against the pair grow only like $\tau^{-1/2}$. None of this is a path to the hypothesis: the method reads real parts, it does not constrain them.

This chapter is based on [132]. Its numbers were recomputed here from the mathematics, except those marked as computed by the author, which were not rerun; where a recomputation differed from the source the recomputed value is

printed. The Lorentzian representation of u_1 (Proposition 2.1.3), the positivity criterion for the deconvolution (Theorem 8.2.2), the explicit formula for even test functions (Theorem 9.1.1, Section 9.1.1), the EM, grid and certificate methods (Chapters 10 and 11) and the survey of Cauchy deconvolution (Chapter 7) are cited, not restated.

Notation. A level θ is the real part of $s = \theta + iv$, and $u_\theta(v) = \pi^{-1}\,\mathrm{Re}(\xi'/\xi)(\theta + iv)$; in the notation of Chapter 8 (equation (2.1.7)) it is $\rho_{\theta-1/2}(v)$. For a zero $\rho = \sigma_\rho + i\gamma_\rho$ the width at level 1 is $a_\rho = 1 - \sigma_\rho \in (0,1)$. The Cauchy density is $P_a(x) = (a/\pi)/(a^2 + x^2)$, and $\mathfrak{U}_\theta = \sum_\rho \mathrm{Cauchy}(\gamma_\rho, \theta - \sigma_\rho)$ is the Thorin measure with no factor $\frac{1}{2}$, as in Chapter 11, so that $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ with $\widetilde{U}_\theta$ of Theorem 17.14.4; γ_E is Euler's constant and Λ the von Mangoldt function. In the Bayesian sections τ is the standard deviation of each real coordinate of the noise. Certified is used only for the sign changes of Hardy's function Z computed in ball arithmetic in Sections 12.8.1, 12.8.6 and 12.8.7; everything else done in floating point is computed. The results are these.

1. The explicit formula of Guinand and Weil is the Fourier side of the deconvolution from level 1 to level $\frac{1}{2}$: for every test function in $C_c^\infty(\mathbb{R})$, unconditionally, the zero side $\sum_\rho \int g\, e^{(\sigma_\rho - 1/2)|\omega|} e^{-i\omega\gamma_\rho} d\omega$ equals the prime side with the pole term and the finite part of the Gamma factor; for even g this is Theorem 9.1.1, and under the Riemann hypothesis it reads $\sum_\rho e^{-i\omega\gamma_\rho} = e^{|\omega|/2}(1 + \widehat{G}) - \sum \Lambda(n) n^{-1/2}(\delta_{\log n} + \delta_{-\log n})$ (Theorem 12.1.1, proved; the reading is conditional). A Gaussian perturbation of the measure of total variation 1 that data of accuracy ε cannot see exists as soon as $0.72\, e^{-\Omega/2} \le 2\varepsilon$ (Proposition 12.1.2, proved); that resolution δ needs the primes to $e^{\pi/\delta}$ is heuristic (Remark 12.1.3).

2. The tapered Fourier transform of u_1 on a window is a sum of damped exponentials, one per zero, the pole $e^{\Delta(-a_\rho + i(\gamma_\rho - c))}$ carrying both coordinates of the zero, plus a remainder of order $e^{-\ell^2\omega^2/2}$, with no Gamma terms (Theorem 12.2.1, proved). The matrix pencil recovers the poles of an exact sum (Lemma 12.2.2, proved); its error analysis is heuristic. At height 200 the eleven zeros within 10 of the centre are placed to within 8×10^{-10} in both coordinates, and a synthetic pair off the line at 60.089 is read to seven digits (Section 12.3, computed by the author, not rerun).

3. Posterior concentration on a window holds at the rate τ for zeros with separated ordinates, under explicit hypotheses (Theorem 12.4.1, proved); the hypotheses fail at the settings of the computations (Remark 12.4.2, computed), and concentration cannot prove the hypothesis, since a finite computation settles finitely many zeros.

4. The functional equation makes the model singular: with one amplitude common to its two members, a pair at $\frac{1}{2} \pm \delta$ enters the data through δ^2

(Lemma 12.5.1, proved). The posterior of δ^2 concentrates at the rate τ, so the real part is resolved at the rate $\tau^{1/2}$ when the line is the truth; no method, neural networks included, separates $\delta = 0$ from $\delta = c\,(\tau/\|t_\perp\|)^{1/2}$, $t_\perp$ the projected tangent of the pair, with error probabilities summing to less than $1 - c^2/2 - O(\tau)$; and the matched filter and the likelihood ratio test attain the scale $\tau^{1/2}$ (Theorem 12.5.2, proved for data simulated from the model in a window without aliasing, not for the runs on the zeros of ζ).

5. Bayes factors with the right alternative, the pair and not a single zero: under the line the Bayes factor of the line against the pair is of order $\tau^{-1/2}$ in probability, under a fixed pair it is exponentially small, and against a single zero alternative it is of order τ^{-1} (Theorem 12.5.4, proved in the common amplitude model). For the zero at 201.264752, at an assumed noise level and on the assumed design of the detection experiment, the computed $\log_{10}$ of the Bayes factor is 6.1 against the pair and 11.8 against the single zero alternative: the functional equation takes roughly the square root of the evidence (computed, an illustration).

6. An adaptive prior, the sine kernel process on the distinct ordinates at the Riemann and von Mangoldt density and a spike and slab on the pairs for the real parts, is a heuristic choice resting on a conjecture about the ordinates; the properties used here are proved (Remarks 12.5.3 and 12.6.1). A detection experiment compares the pencil and a plug in matched filter with the benchmark of Theorem 12.5.2(iii) (computed).

7. The open problems of the source are worked: certification of the pencil's output by sign changes and a winding number (logic proved, eleven zeros near height 200 computed), close pairs, the levels $\frac{3}{4}$ and 0.6 and the signed level θ transform (Proposition 12.8.1, proved), the level as a cost parameter, and many windows (173 zeros on $[100, 400]$ from 31 windows, computed by the author).

8. A window at $c = 3\times10^{12}+100$ lies inside the range $0 < \gamma \le 3\,000\,175\,332\,800$ where Platt and Trudgian [232] verify the hypothesis, so the run is a demonstration near the limit of the verification: the data are values of ξ'/ξ at $\operatorname{Re} s = 0.6$ computed in ball arithmetic, no primes are used and no new zero is found. The pencil returns 52 components; an independent scan of Hardy's function finds exactly 52 certified sign changes there, and the one by one comparison with the pencil's ordinates has not been made. The selection of components in the source's runs presumes the real part $\frac{1}{2}$ (Section 12.8.6, computed).

9. A window at $c = 10^{13} + 100$, above the verified height by a factor 3.3, is run with the book's own pencil, whose selection uses the width of each pole:

53 components, real parts within 5×10^{-9} of $\frac{1}{2}$ (computed, an uncertified estimate), every ordinate within 10^{-7} of one of 53 certified sign changes of Z and no certified sign change without a pencil ordinate. The count of zeros in the window and the real parts are not certified, and nothing is proved about any zero outside the 53 (Section 12.8.7).

The open problems close the chapter.

12.1 The deconvolution at level one half

12.1.1 The forward equation

Write $s = 1+iv$ and, for a zero $\rho = \sigma_\rho + i\gamma_\rho$, $a_\rho = 1-\sigma_\rho \in (0,1)$. Proposition 2.1.3 and the Hadamard product give, with the zeros paired as $\rho, \bar\rho$ so that the sum converges,

$$u_1(v) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(1+iv) = \frac{1}{\pi}\sum_\rho \frac{a_\rho}{a_\rho^2 + (v-\gamma_\rho)^2} = \sum_\rho P_{a_\rho}(v-\gamma_\rho), \tag{12.1.1}$$

a sum of Cauchy densities of widths a_ρ centred at the ordinates. Under the Riemann hypothesis every width is $\frac{1}{2}$ and

$$u_1 = P_{1/2} * \mathfrak{U}_{1/2}, \qquad \mathfrak{U}_{1/2} = \sum_\rho \delta_{\gamma_\rho} : \tag{12.1.2}$$

the Poisson integral of the zero counting measure at distance $\frac{1}{2}$. In general (12.1.1) smooths each atom of $\sum_\rho \delta_{(\gamma_\rho, a_\rho)}$ at its own distance a_ρ, and by Theorem 8.2.2 a positive solution μ of $P_a * \mu = u_1$ exists for a given $a \in (0, \frac{1}{2}]$ if and only if no zero has $a_\rho < a$.

12.1.2 The Fourier side

With $\widehat{f}(\omega) = \int f(v)e^{-i\omega v}dv$ and $\widehat{P_a}(\omega) = e^{-a|\omega|}$, termwise

$$\widehat{u_1}(\omega) = \sum_\rho e^{-a_\rho|\omega|}e^{-i\omega\gamma_\rho}, \tag{12.1.3}$$

and under the Riemann hypothesis $\widehat{u_1} = e^{-|\omega|/2}\,\widehat{\mathfrak{U}_{1/2}}$ with $\widehat{\mathfrak{U}_{1/2}}(\omega) = \sum_\rho e^{-i\omega\gamma_\rho}$, a tempered distribution. On the other hand, from the Dirichlet series, as tempered distributions,

$$\widehat{u_1} = 1 + \widehat{G} - \sum_{n\ge 2}\frac{\Lambda(n)}{n}\big(\delta_{\log n} + \delta_{-\log n}\big), \tag{12.1.4}$$

where the 1 is the transform of δ_0, the boundary value on $\operatorname{Re} s = 1$ of $\pi^{-1}\operatorname{Re}(s-1)^{-1}$, which the prime series does not contain, and $G(v) = \pi^{-1}[1/(1+v^2) - \frac12 \log\pi + \frac12 \operatorname{Re}\psi(\frac{1+iv}{2})]$ is the Gamma part, whose transform is the distribution

$$\langle \widehat{G}, g\rangle = (1 - \log\pi - \gamma_E)\, g(0) + \sum_{k\ge1}\Big[\frac{g(0)}{k+1} - \int_{\mathbb{R}} g(\omega)\, e^{-(2k+1)|\omega|}\, d\omega\Big]. \quad (12.1.5)$$

Away from the origin $\widehat{G}(\omega) = -e^{-3|\omega|}/(1-e^{-2|\omega|}) = -1/(2|\omega|) + 1 + O(|\omega|)$, the transform of the growth $(2\pi)^{-1}\log|v|$ of G: a singularity that is not integrable, and (12.1.5) is its finite part regularisation. Multiplying (12.1.4) by $e^{|\omega|/2}$ multiplies the atom at $\pm\log n$ by $\sqrt{n}$.

THEOREM 12.1.1 (the explicit formula is the deconvolution). *For every $g \in C_c^\infty(\mathbb{R})$, with the sum over ρ absolutely convergent,*

$$\sum_\rho \int_{\mathbb{R}} g(\omega)\, e^{(\sigma_\rho - \frac12)|\omega|}\, e^{-i\omega\gamma_\rho}\, d\omega = \int_{\mathbb{R}} g(\omega)\, e^{|\omega|/2}\, d\omega$$
$$- \sum_{n\ge2} \frac{\Lambda(n)}{\sqrt{n}}\big(g(\log n) + g(-\log n)\big) + (1 - \log\pi - \gamma_E)\, g(0)$$
$$+ \sum_{k\ge1}\Big[\frac{g(0)}{k+1} - \int_{\mathbb{R}} g(\omega)\, e^{-(2k+\frac12)|\omega|}\, d\omega\Big]. \quad (12.1.6)$$

For even g this is Theorem 9.1.1, the explicit formula of Guinand [91] and Weil [273]; for odd g both sides vanish. Under the Riemann hypothesis the left side is $\sum_\rho \widehat{g}(\gamma_\rho)$, and (12.1.6) *says that on $C_c^\infty(\mathbb{R})$*

$$\sum_\rho e^{-i\omega\gamma_\rho} = e^{|\omega|/2}\big(1 + \widehat{G}(\omega)\big) - \sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\big(\delta_{\log n} + \delta_{-\log n}\big),$$

where $e^{|\omega|/2}\widehat{G}$ is the functional $g \mapsto \langle \widehat{G}, e^{|\cdot|/2} g\rangle$ given by (12.1.5). *The left side is tempered; the right side is tempered only as a whole, since the prime measure has mass about $2e^{X/2}$ on $[0, X]$.*

Proof. (a) For $\sigma > 1$, $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac12\log\pi + \frac12\psi(s/2) + \zeta'/\zeta(s)$. Let $\sigma \downarrow 1$ with $s = \sigma + iv$. The left side tends to πu_1 in $\mathcal{S}'(\mathbb{R})$, since ξ'/ξ is analytic across $\operatorname{Re} s = 1$ and grows like $\log|v|$. On the right, $\operatorname{Re} 1/s \to 1/(1+v^2)$, $\frac12\operatorname{Re}\psi(s/2) \to \frac12\operatorname{Re}\psi(\frac{1+iv}{2})$, $\pi^{-1}\operatorname{Re}(s-1)^{-1} = P_{\sigma-1}(v) \to \delta_0$, and $\operatorname{Re}\zeta'/\zeta(s) = -\sum_n \Lambda(n) n^{-\sigma}\cos(v\log n)$, whose transform $-\pi\sum_n \Lambda(n)n^{-\sigma}(\delta_{\log n} + \delta_{-\log n})$ converges in $\mathcal{S}'$ as $\sigma \downarrow 1$ by dominated convergence, the measure $\sum_n \Lambda(n) n^{-1}\delta_{\log n}$ having mass $O(X)$ on $[0, X]$. This gives (12.1.4). The digamma series gives $G(v) = (1 - \log\pi - \gamma_E)/2\pi + \sum_{k\ge1}[1/(2\pi(k+1)) - P_{2k+1}(v)]$, the $1/(1+v^2)$ cancelling the term $k = 0$, and transforming termwise gives (12.1.5); the series converges because $\int g(\omega) e^{-b|\omega|} d\omega = 2g(0)/b + O(b^{-2})$ and $1/(k+1) - 2/(2k+1) = O(k^{-2})$.

(b) By (12.1.1) the partial sums of $\sum_\rho P_{a_\rho}(v-\gamma_\rho)$ are positive and increase to $u_1 = O(\log|v|)$, so the series converges in $\mathcal{S}'$ and $\widehat{u_1} = \sum_\rho e^{-a_\rho|\omega|}e^{-i\omega\gamma_\rho}$.

(c) Let $g \in C_c^\infty(\mathbb{R})$ and $\varphi = e^{|\omega|/2}g$, which is Lipschitz with compact support and has φ' of bounded variation. Let $\varphi_\epsilon = \varphi * \eta_\epsilon$ with a mollifier η_ϵ: then $\varphi_\epsilon \in C_c^\infty$ with supports in a fixed compact set, $\varphi_\epsilon \to \varphi$ uniformly, and $\|\varphi_\epsilon'\|_\infty$ and the total variation of φ_ϵ' are bounded uniformly in ϵ. Pair (a) and (b) with φ_ϵ. On the zero side the term of ρ is the transform of $e^{-a_\rho|\omega|}\varphi_\epsilon$ at γ_ρ; two integrations by parts, the second against the measure $d\big(e^{-a_\rho|\omega|}\varphi_\epsilon\big)'$ of bounded mass, bound it by $C(1+\gamma_\rho^2)^{-1}$ uniformly in ϵ and ρ, and $\sum_\rho(1+\gamma_\rho^2)^{-1} < \infty$, so by dominated convergence the zero side tends to $\sum_\rho \int g(\omega)e^{(1/2-a_\rho)|\omega|}e^{-i\omega\gamma_\rho}d\omega$, the left side of (12.1.6), since $\frac12 - a_\rho = \sigma_\rho - \frac12$. On the other side the prime sum is finite on the support, $\int \varphi_\epsilon \to \int g\, e^{|\omega|/2}$, and $\langle \widehat{G}, \varphi_\epsilon\rangle \to \langle \widehat{G}, \varphi\rangle$, because each term of (12.1.5) converges and $|\varphi_\epsilon(\omega) - \varphi_\epsilon(0)| \le C|\omega|$ bounds the k-th term by C/k^2 uniformly. This is (12.1.6). Under the Riemann hypothesis $a_\rho = \frac12$, and $N(T) = O(T\log T)$ makes $\sum_\rho e^{-i\omega\gamma_\rho}$ tempered. □

The theorem is classical, and that is the point (Section 9.1.1): the backward Cauchy flow from level 1 to level $\frac12$, computed by EM in Chapters 10 and 11, is the explicit formula, and the statement of Theorem 8.2.2 that the Thorin measure at level θ solves $P_{1-\theta} * \mu = u_1$ is, at $\theta = \frac12$ and under the Riemann hypothesis, the statement that the zeros are the Fourier transform of the primes. The unbounded multiplier $e^{|\omega|/2}$ is the passage from $\Lambda(n)/n$ to $\Lambda(n)/\sqrt{n}$, from a tempered measure of mass about X on $[0,X]$ to one of mass about $2e^{X/2}$, which pairs only with test functions of compact support. The conditioning of the grid matrix in Chapter 11 is a statement about this passage.

12.1.3 What the ill posedness costs

Proposition 12.1.2 (an unresolvable perturbation). *Let $\Omega > 0$, $L \ge 1$, $\chi_L(v) = (2\pi L^2)^{-1/2}e^{-v^2/2L^2}$, $N = \int_{\mathbb{R}} |\cos(\Omega v)|\,\chi_L(v)\,dv$ and $\nu = N^{-1}\cos(\Omega v)\,\chi_L(v)\,dv$. Then $\|\nu\|_{TV} = 1$, $\widehat{\nu}$ is concentrated within $O(1/L)$ of $\pm\Omega$, and*

$$\|P_{1/2} * \nu\|_\infty \le \sqrt{\pi/8}\; L^{-1}e^{1/8L^2}\, e^{-\Omega/2} \le 0.72\, e^{-\Omega/2}.$$

Consequently, if $0.72\, e^{-\Omega/2} \le 2\varepsilon$ and μ is a positive measure with $\mu \ge N^{-1}\chi_L\,dv$, then μ and $\mu+\nu$ are positive measures at total variation distance 1 whose data $P_{1/2}*\mu$ and $P_{1/2}*(\mu+\nu)$ lie within ε in L^∞ of one function: data of accuracy ε do not distinguish them.

Proof. $\widehat{\chi_L}(\omega) = e^{-L^2\omega^2/2}$, so $\widehat{\nu}(\omega) = (2N)^{-1}\big(e^{-L^2(\omega-\Omega)^2/2} + e^{-L^2(\omega+\Omega)^2/2}\big)$. By

Fourier inversion, the evenness of $e^{-|\omega|/2}$ and $e^{-|\omega|/2} \le e^{-\omega/2}$,

$$\begin{aligned}\|P_{1/2} * \nu\|_\infty &\le \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\omega|/2}|\widehat{\nu}(\omega)|\,d\omega \\ &\le \frac{1}{2\pi N}\int_{\mathbb{R}} e^{-\omega/2}e^{-L^2(\omega-\Omega)^2/2}\,d\omega = \frac{e^{1/8L^2}}{\sqrt{2\pi}\,L\,N}\,e^{-\Omega/2}.\end{aligned}$$

From $|\cos x| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{k\ge1}(-1)^{k+1}\frac{\cos 2kx}{4k^2-1}$ and $\int \cos(2k\Omega v)\chi_L(v)\,dv = e^{-2k^2L^2\Omega^2}$,

$$N = \frac{2}{\pi} + \frac{4}{\pi}\sum_{k\ge1}(-1)^{k+1}\frac{e^{-2k^2L^2\Omega^2}}{4k^2-1} \ge \frac{2}{\pi},$$

the series being alternating with decreasing terms. With $N \ge 2/\pi$ this is the bound, and $\sqrt{\pi/8}\,e^{1/8} = 0.7101$. For the last statement, $\nu \ge -N^{-1}\chi_L$, so $\mu+\nu \ge 0$, and the midpoint of the two data lies within $\frac{1}{2}\|P_{1/2} * \nu\|_\infty \le \varepsilon$ of each. □

Remark 12.1.3 (heuristic). Identifying resolution at scale δ with frequency $\Omega = \pi/\delta$, Proposition 12.1.2 suggests that a general density cannot be resolved at scale δ from data less accurate than about $\frac{1}{2}e^{-\pi/2\delta}$, and, since in (12.1.6) a test function supported in $[-\Omega,\Omega]$ involves $\Lambda(n)$ only for $n \le e^{\Omega}$, that the primes to $e^{\pi/\delta}$ are the matching arithmetic input. Neither is a theorem: no loss function is fixed, no method is excluded, and both fail for measures with few atoms, as the computations below show.

For $\delta = 0.1$ the heuristic gives $\varepsilon \approx 10^{-7}$ and primes to $e^{31} \approx 10^{13}$; for $\delta = 0.01$ it is 3×10^{-69} and 10^{136}. The EM computation of Chapter 11 reached $\delta \approx 10^{-3}$ in the positions of well separated atoms from data accurate to 10^{-20}, better than the heuristic allows for a general measure, because the target is not a general measure: a positive sum of a few atoms per unit length has far fewer degrees of freedom than a density, and the question is whether a method can use that.

12.2 Sparse recovery

12.2.1 Positivity and sparsity

Two facts from super resolution apply. First, Candès and Fernandez Granda [33] proved that a measure $\mu = \sum_k c_k\delta_{t_k}$ on $[0,1]$ is recovered exactly from its Fourier coefficients $\widehat{\mu}(\omega)$, $|\omega| \le \Omega$, by minimising the total variation subject to the data, provided the atoms are separated by at least $2\cdot 2\pi/\Omega$; with noise ε the recovery error is $O(\varepsilon)$ times a super resolution factor, polynomial in $\Omega\delta$ rather than exponential [32]. Second, for positive measures the separation is not needed at all: de Castro and Gamboa [56], and Schiebinger, Robeva and Recht [168], showed that a positive measure with k atoms is the unique positive measure with its moments of orders 0 to $2k$, and is recovered by a linear program. Our measure

is positive, and on a window of length W at height v it has about $W\log(v/2\pi)/2\pi$ atoms separated on average by $2\pi/\log(v/2\pi)$, with occasional pairs much closer. So $P_{1/2} * \mu = u_1$, $\mu \ge 0$, restricted to a window and to the frequencies the data support, is of the kind these theorems treat.

The gridded convex methods of Chapter 11 respect positivity, and nonnegative least squares has sparse vertex solutions; but on a grid of spacing h the model has W/h degrees of freedom, the unknown is a density, and the conditioning is that of Proposition 12.1.2. Sparsity pays only when the model is the set of sums of atoms, and for that the natural tool is on the Fourier side.

12.2.2 Lorentzians become exponentials

Theorem 12.2.1 (tapered transform). *Let* $w(v) = e^{-(v-c)^2/2\ell^2}$, $\ell > 0$, *and*

$$F(\omega) = \int_{\mathbb{R}} u_1(v)\, w(v)\, e^{i\omega(v-c)}\, dv.$$

Then for $\omega\ell^2 > 1$

$$F(\omega) = \sum_{\rho} w(\gamma_\rho + ia_\rho)\, e^{-a_\rho\omega} e^{i\omega(\gamma_\rho - c)} + R(\omega), \tag{12.2.1}$$

where $w(x+iy) = w(x)e^{y^2/2\ell^2}e^{-iy(x-c)/\ell^2}$ *is the entire extension of the taper, the series converges absolutely, and, with* $Y = \omega\ell^2$,

$$|R(\omega)| \le \frac{1}{\pi}\, e^{-\ell^2\omega^2/2} \sum_{\rho} a_\rho \int_{\mathbb{R}} \frac{e^{-(x-c)^2/2\ell^2}\, dx}{(x-\gamma_\rho)^2 + (Y - a_\rho)^2},$$

$$\int_{\mathbb{R}} \frac{e^{-(x-c)^2/2\ell^2}\, dx}{(x-\gamma)^2 + B^2} \le \min\Big(\frac{\pi}{B}, \frac{\sqrt{2\pi}\,\ell}{B^2}\Big),$$

so that, heuristically, R *is of the order* $e^{-\ell^2\omega^2/2}$ *times the number of zeros per unit length at height* c, *divided by* $\omega\ell$. *In particular, for* $\omega_j = \omega_0 + j\Delta$ *with* $\omega_0\ell^2 > 1$,

$$F(\omega_j) = \sum_{\rho} A_\rho\, z_\rho^{\,j} + R(\omega_j), \qquad |R(\omega_j)| \le C_0\, e^{-\ell^2\omega_j^2/2},$$

$$A_\rho = w(\gamma_\rho + ia_\rho) e^{-a_\rho\omega_0} e^{i\omega_0(\gamma_\rho - c)}, \qquad z_\rho = e^{\Delta(-a_\rho + i(\gamma_\rho - c))},$$

where C_0 *is the bound on* $|R(\omega_0)|$ *divided by* $e^{-\ell^2\omega_0^2/2}$. *The Gamma factor does not enter:* (12.1.1) *is all of* u_1, *and the Lorentzians* P_{2k+1} *centred at* 0, *which the trivial zeros* $-2k$ *contribute to* $\operatorname{Re}\zeta'/\zeta$, *are cancelled in* ξ'/ξ *by the poles of* $\Gamma(s/2)$.

Proof. By (12.1.1), $u_1 = \sum_\rho P_{a_\rho}(\cdot - \gamma_\rho)$ exactly, with positive terms, and $\int u_1 w < \infty$; so $F(\omega) = \sum_\rho \int P_{a_\rho}(v-\gamma_\rho)\phi(v)\, dv$ with $\phi(v) = w(v)e^{i\omega(v-c)}$, which is entire with

$$|\phi(x+iy)| = e^{-(x-c)^2/2\ell^2}\, e^{y^2/2\ell^2 - \omega y}.$$

Write $P_a(v-\gamma) = (2\pi i)^{-1}[(v-\gamma-ia)^{-1} - (v-\gamma+ia)^{-1}]$ and move the integral over $\mathbb{R}$ to the line $\operatorname{Im} v = Y$, $Y = \omega\ell^2$, where $|\phi|$ is smallest. Since $0 < a < 1 < Y$, the first term picks up the residue at $v = \gamma + ia$, the second has no pole in between, the vertical sides contribute nothing by the Gaussian decay in x, and

$$\int_{\mathbb{R}} P_a(v-\gamma)\phi(v)\,dv = \phi(\gamma+ia) + \frac{1}{2\pi i}\int_{\operatorname{Im} v = Y} \frac{2ia\,\phi(v)}{(v-\gamma)^2 + a^2}\,dv.$$

On the line, $|\phi| = e^{-(x-c)^2/2\ell^2} e^{-\ell^2\omega^2/2}$ and $|(v-\gamma)^2+a^2| = |v-\gamma-ia|\,|v-\gamma+ia| \geq (x-\gamma)^2+(Y-a)^2$, which gives the term of the bound attached to ρ; the two bounds for the integral follow from $\int((x-\gamma)^2 + B^2)^{-1}dx = \pi/B$ and $\int e^{-(x-c)^2/2\ell^2}dx = \sqrt{2\pi}\,\ell$. The sums over ρ converge: $|w(\gamma+ia)| \leq e^{1/2\ell^2} e^{-(\gamma-c)^2/2\ell^2}$, the integrals are $O((\gamma-c)^{-2})$ uniformly because $Y - a \geq Y - 1 > 0$, and $N(T) = O(T\log T)$. Finally $\phi(\gamma+ia) = w(\gamma+ia)e^{i\omega(\gamma-c)}e^{-a\omega}$, and the bound at ω_j holds because the bound on $|R(\omega)|$ divided by $e^{-\ell^2\omega^2/2}$ decreases in ω. □

Some hypothesis $\omega\ell^2 > a_\rho$ is needed, since otherwise the line $\operatorname{Im} v = \omega\ell^2$ does not reach the pole; $\omega\ell^2 > 1$ suffices. The settings of the computations below ($\omega_0\ell^2 = 500$ at height 200, 20 at height 3×10^{12}) satisfy it.

The content is in the last display: the samples are a finite sum of damped complex exponentials in j, one per zero that the taper sees, plus a remainder, and the modulus of z_ρ gives $a_\rho = 1 - \sigma_\rho$, its argument $\gamma_\rho - c$ modulo $2\pi/\Delta$. Nothing is assumed about the real parts; (12.2.1) is (12.1.1) and nothing else.

12.2.3 Prony and the matrix pencil

Recovering the poles of a sum of exponentials from J equally spaced samples is Prony's problem; the matrix pencil of Hua and Sarkar [101] solves it stably. Form the Hankel matrix $H_{ik} = F(\omega_{i+k})$, $0 \leq i < J - L$, $0 \leq k < L$, with $L \approx J/2$, and its two submatrices H_0, H_1 obtained by deleting the last and the first row. If the samples are an exact sum of M exponentials with distinct nonzero poles and nonzero amplitudes, and $M \leq \min(L, J-L-1)$, then H_0 and H_1 have rank M and the values of z at which $H_1 - zH_0$ drops rank are the poles z_ρ (Lemma 12.2.2). With noise, take the rank M truncated singular value decomposition $H_0 \approx U_M \Sigma_M V_M^*$, chooses M by the gap in the singular values, and computes the eigenvalues of $U_M^* H_1 V_M \Sigma_M^{-1}$. The amplitudes A_ρ follow by least squares against the Vandermonde matrix of the recovered poles, and a component is accepted as a zero when the modulus of its amplitude matches the prediction $|w(\gamma_\rho + ia_\rho)|e^{-a_\rho\omega_0}$ of Theorem 12.2.1, a diagnostic and not a certificate (in the runs at levels $\theta < 1$ below, the prediction is evaluated at the width $\theta - \frac{1}{2}$ of a zero on the line); spurious components, which absorb the zeros at the edge of the taper and the wrapped poles of zeros farther away, have amplitudes orders of magnitude off the profile.

LEMMA 12.2.2 (exact matrix pencil). *Let* $f_j = \sum_{m=1}^{M} A_m z_m^j$, $0 \le j < J$, *with distinct nonzero* z_m, *nonzero* A_m *and* $M \le \min(L, J-L-1)$, *and let* $H_0 = (f_{i+k})$ *and* $H_1 = (f_{i+k+1})$, $0 \le i \le J-L-2$, $0 \le k < L$. *Then* $\operatorname{rank} H_0 = \operatorname{rank} H_1 = M$, $\operatorname{rank}(H_1 - zH_0) < M$ *exactly when* $z \in \{z_1, \dots, z_M\}$, *and if* $H_0 = U_M \Sigma_M V_M^*$ *is a reduced singular value decomposition then* $U_M^* H_1 V_M \Sigma_M^{-1}$ *is similar to* $\operatorname{diag}(z_1, \dots, z_M)$.

Proof. With $V = (z_m^i)$ of size $(J-L-1) \times M$, $W = (z_m^k)$ of size $L \times M$, $D = \operatorname{diag}(A_m)$ and $Z = \operatorname{diag}(z_m)$, we have $H_0 = VDW^T$ and $H_1 = VDZW^T$. V and W are Vandermonde with distinct nodes and at least M rows, so both have rank M; hence $\operatorname{rank} H_0 = \operatorname{rank} H_1 = M$ and $\operatorname{rank}(H_1 - zH_0) = \operatorname{rank}(Z - zI) = M - \#\{m : z_m = z\}$. The columns of U_M span the column space of V, so $P = U_M^* V$ is invertible; from $H_0 V_M = U_M \Sigma_M$ we get $PD(W^T V_M) = \Sigma_M$, so $W^T V_M = D^{-1} P^{-1} \Sigma_M$ and $U_M^* H_1 V_M \Sigma_M^{-1} = PDZD^{-1}P^{-1} = PZP^{-1}$. □

Heuristically, by the standard error analysis of the pencil, if the samples carry an error η and the exponentials are separated, the poles move by $O(\eta / |A_{\min}|)$ times a condition number polynomial in J and in the separation, not exponential in the bandwidth. The bandwidth enters through $|w(\gamma_\rho + ia_\rho)| e^{-a_\rho \omega_{J-1}}$: at the largest frequency used, the signal of a zero is $e^{-\Omega/2}$ times its amplitude, and the data error at that frequency is η, so the effective noise is $\eta e^{\Omega/2}$. This is the $e^{\Omega/2}$ of Proposition 12.1.2, but it now multiplies the data error, not the resolution: with data exact to 10^{-20} and $\Omega = 45$ the effective noise is 10^{-10}, and with double precision transforms, $\eta = 10^{-16}$, it is 10^{-6} at the top frequency and far smaller at the frequencies that carry most of the signal.

12.3 The computation

The numbers of this section are the author's, computed by the author and not rerun. The data are u_1 on $[150, 250]$ at spacing 0.02, enclosed in balls of radius below 5×10^{-21} by the method of Chapter 11; the taper is centred at $c = 200$ with $\ell = 10$, so that the zeros within 15 of the centre have taper weight at least 0.3 and those beyond 30 weight below 0.012. The transform $F(\omega_j)$ is computed as a Riemann sum over the data in ball arithmetic, at $\omega_j = 5 + 0.1j$, $j < 400$, so $\Omega = 45$, and converted to double precision. The balls enclose the finite sum, not F: the aliasing error of the sum is of the order $e^{-a(2\pi/h - \Omega)} = e^{-135}$ with $a = \frac{1}{2}$ and $h = 0.02$, but the truncation of the data at $c \pm 5\ell$, where the taper is $e^{-12.5} = 3.7 \times 10^{-6}$, is not bounded. The Hankel matrix is 200×200; the singular values fall from 1 to 10^{-9} over the first sixty and the model order is taken there. At $\Delta = 0.1$ the ordinates are determined modulo $2\pi/\Delta = 62.8$, so zeros farther than 31.4 from the centre appear wrapped into the window, with taper weights below $e^{-4.9}$; the amplitude test sets them aside, and only ordinates within 25 of the centre are reported.

$\lvert\gamma - c\rvert \le$	zeros	position error max	position error median	error in σ_ρ max	error in σ_ρ median
10	11	4.5×10^{-10}	1.8×10^{-10}	7.9×10^{-10}	2.3×10^{-10}
15	17	5.0×10^{-7}	2.5×10^{-10}	8.4×10^{-7}	3.8×10^{-10}
20	21	5.0×10^{-7}	4.5×10^{-10}	8.4×10^{-7}	7.9×10^{-10}
25	28	4.1×10^{-5}	2.7×10^{-9}	4.0×10^{-5}	1.1×10^{-8}

Table 12.1: The matrix pencil at height 200, level one data, taper $\ell = 10$: errors in the position and in the real part of the recovered zeros, by distance from the centre (computed by the author, not rerun).

The eleven zeros within 10 of the centre are placed to within 8×10^{-10} in both coordinates; the error grows toward the edge of the taper as the amplitudes fall, as the heuristic error analysis predicts. The backward flow of Chapter 11 on the same data, with 4000 accelerated EM steps and the certificate, placed the centroids to about 10^{-3}: a gain of seven orders of magnitude at the centre, from the same u_1. The modulus of the tapered transform $F(\omega)$ follows the line $e^{-\omega/2}$, so that the signal falls by e^{-20} over the frequencies used (the author's figure is not reproduced).

The real parts are the new element. Every recovered pole has $\lvert z_\rho\rvert = e^{-\Delta a_\rho}$ with $a_\rho = 0.5000000$ to the digits shown; the fit does not constrain the real parts. To make the point we added to u_1 the Lorentzians of a fictitious pair of zeros at $0.85 \pm 60.089i$ and $0.15 \pm 60.089i$, which is what a zero off the line and its reflection would contribute, and ran the pencil on a window centred at 60. It returned two components at $\gamma = 60.08900000$ and 60.08899905 with $a = 0.15000000$ and 0.84999991, beside the genuine zeros 59.347044 and 60.831779 with $a = 0.500000$. The pencil reads the real parts of the zeros off the level one data, to seven digits, without any assumption.

12.4 Posterior concentration, and what it cannot do

A posterior concentration result for the recovery of $\mathfrak{U}_{1/2}$ holds under hypotheses; a proof of the Riemann hypothesis from it does not, and the theorem shows why.

Fix a window, centre c and taper ℓ, a sampling step Δ, the samples $\omega_j = \omega_0 + j\Delta$, $0 \le j < J$, with $\omega_0 \ell^2 > 1$, and a radius R with $2R\Delta < 2\pi$, so that $\gamma \mapsto e^{i\Delta(\gamma - c)}$ is one to one on $[c - R, c + R]$. By Theorem 12.2.1 the data vector $F = (F(\omega_j))_j$ is

$$F = \Phi(\vartheta^0) + b, \qquad \Phi(\vartheta)_j = \sum_{m=1}^{M} A_m \, z_m^j, \qquad z_m = e^{\Delta(-a_m + i(\gamma_m - c))},$$

where ϑ^0 collects the zeros with $\lvert\gamma_m - c\rvert \le R$ and b is the deterministic bias from the zeros beyond R and the remainder of Theorem 12.2.1; its size is about

$2n(c)(\ell^2/R)e^{-R^2/2\ell^2}$ per sample, with $n(c)$ the density of zeros at height c. To make a Bayesian statement we model the observation as $Y = F + \tau\xi$, where $\xi \in \mathbb{C}^J$ has independent standard normal real and imaginary parts, and put a prior on ϑ.

THEOREM 12.4.1 (posterior concentration on a window). *Fix $s > 0$, $0 < a_- < a_+ < 1$, $0 < \alpha_- < A_+$ and an integer $M_{\max}$ with $J \geq 2M_{\max}$. For $0 \leq M \leq M_{\max}$ let K_M be the compact set of $\vartheta = (\gamma_m, a_m, A_m)_{m\leq M}$ with $c - R \leq \gamma_1 < \cdots < \gamma_M \leq c + R$, $\gamma_{m+1} - \gamma_m \geq s$, $a_m \in [a_-, a_+]$ and $\alpha_- \leq |A_m| \leq A_+$. Let the prior be $\Pi = \sum_M p_M \Pi_M$, where Π_M has a density on K_M with values between $\pi_- > 0$ and π_+. Let ϑ^0 be an interior point of K_{M^0} with $p_{M^0} > 0$, and let $Y = \Phi(\vartheta^0) + b + \tau\xi$ with $\|b\|_2 \leq B\tau$. Put $d(\vartheta, \vartheta^0) = \max_m\big(|\gamma_m - \gamma_m^0| + |a_m - a_m^0| + |A_m - A_m^0|\big)$ if $M = M^0$ and $d = \infty$ otherwise. Then there are $\tau_0, r_0, K, \kappa_0 > 0$, depending on ϑ^0, the sets K_M, the prior, J, Δ and B, such that for $\tau \leq \tau_0$ and $K\tau \leq \varepsilon \leq r_0$, with probability at least $1 - e^{-\kappa_0\varepsilon^2/\tau^2}$,*

$$\Pi\big(d(\vartheta, \vartheta^0) \leq \varepsilon \,\big|\, Y\big) \;\geq\; 1 - e^{-\kappa_0\varepsilon^2/\tau^2}.$$

In particular, for fixed $\varepsilon_1 > 0$ and data with $\|b\|_2 \leq B\tau$, as $\tau \to 0$ the posterior probability that every component has $|\sigma_m - \frac12| \leq \varepsilon_1$ tends to 1 in probability if every zero of ϑ^0 has $|\sigma_m^0 - \frac12| < \varepsilon_1$, and to 0 if one has $|\sigma_m^0 - \frac12| > \varepsilon_1$.

Proof. Identifiability. On each K_M the poles z_m are distinct and nonzero, and $(\gamma, a) \mapsto z$ is one to one on $[c - R, c + R] \times (0, 1)$ because $2R\Delta < 2\pi$. If $\Phi(\vartheta) = \Phi(\vartheta^0)$ with $\vartheta \in K_M$, the difference is a combination of at most $M + M^0 \leq J$ distinct exponentials vanishing at J consecutive indices; the Vandermonde matrix of the distinct poles has full column rank, so after merging equal poles every coefficient vanishes, and since all amplitudes are nonzero each pole of ϑ is a pole of ϑ^0 with the same amplitude and conversely. Hence $M = M^0$ and, the labels being fixed by the order of the γ_m, $\vartheta = \vartheta^0$. By continuity and compactness $g = \min_{M\neq M^0} \min_{K_M} \|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 > 0$. In the $4M^0$ real coordinates the derivative of Φ at ϑ^0 is injective: with respect to (z_m, A_m) it is the $J \times 2M^0$ matrix with columns $(z_m^0)^j$ and $jA_m^0(z_m^0)^{j-1}$, whose first $2M^0$ rows have determinant $\pm\prod_m A_m^0 \prod_{m<n}(z_n^0 - z_m^0)^4 \neq 0$, and $dz = z\Delta(-da + i\,d\gamma)$ with $z \neq 0$. So $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \geq \lambda_0 d$ for $d \leq r_1$, and on the compact set $\{d \geq r_1\} \cap K_{M^0}$ the left side has a positive minimum; hence $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \geq \lambda d(\vartheta, \vartheta^0)$ on K_{M^0} for some $\lambda > 0$, and $\|\Phi(\vartheta) - \Phi(\vartheta^0)\|_2 \leq L_\Phi d$ on the d-ball of radius r_1 about ϑ^0, which lies in K_{M^0} if r_1 is small.

Noise. Put $u = \Phi(\vartheta) - \Phi(\vartheta^0)$ and $e = b + \tau\xi$. The log likelihood ratio of ϑ to ϑ^0 is $\big(2\,\mathrm{Re}\langle u, e\rangle - \|u\|^2\big)/2\tau^2$. On the event $E_t = \{\|\xi\|_2 \leq \sqrt{2J} + \sqrt{2t}\}$, of probability at least $1 - e^{-t}$ by Gaussian concentration, $\|e\| \leq q\tau$ with $q = B + \sqrt{2J} + \sqrt{2t}$, so the ratio lies between $e^{-\|u\|^2/2\tau^2 - q\|u\|/\tau}$ and $e^{-\|u\|^2/4\tau^2 + q^2}$, by $-x^2/2 + qx \leq -x^2/4 + q^2$.

Denominator. On $\{M = M^0,\ d \leq \tau\}$, $\tau \leq r_1$, we have $\|u\| \leq L_\Phi\tau$, and this set has prior mass at least $p_{M^0}\pi_- v_0\tau^D$, $D = 4M^0$, where $v_0\tau^D$ is the vol-

ume of the d-ball of radius τ. So the integral of the ratio against Π is at least $p_{M^0}\pi_- v_0 \tau^D e^{-L_\Phi^2/2 - L_\Phi q}$.

Numerator. Since $e^{-\lambda^2 d^2/4\tau^2} \le e^{-\lambda^2\varepsilon^2/8\tau^2} e^{-\lambda^2 d^2/8\tau^2}$ for $d > \varepsilon$, the integral of the ratio over $\{M = M^0,\ d > \varepsilon\}$ is at most

$$p_{M^0}\pi_+ e^{q^2} e^{-\lambda^2\varepsilon^2/8\tau^2} \int_{\mathbb{R}^D} e^{-\lambda^2 d(\vartheta,\vartheta^0)^2/8\tau^2}\, d\vartheta = p_{M^0}\pi_+ C_D \Big(\frac{\tau}{\lambda}\Big)^D e^{q^2} e^{-\lambda^2\varepsilon^2/8\tau^2},$$

and the integral over $\{M \ne M^0\}$ is at most $e^{q^2} e^{-g^2/4\tau^2}$.

Conclusion. The powers τ^D cancel:

$$\Pi(d > \varepsilon \mid Y) \le C_1 e^{q^2 + L_\Phi q}\big(e^{-\lambda^2\varepsilon^2/8\tau^2} + \tau^{-D} e^{-g^2/4\tau^2}\big).$$

For $\varepsilon \le r_0 := g/\lambda$ and $\tau \le \tau_0$, with $\tau_0 \le r_1$ so small that $\tau^{-D} e^{-g^2/8\tau^2} \le 1$, the second term is at most the first. Since $q^2 + L_\Phi q \le 2q^2 + L_\Phi^2 \le C_2 + 12t$, on E_t we get $\Pi(d > \varepsilon \mid Y) \le C_3 e^{12t} e^{-\lambda^2\varepsilon^2/8\tau^2}$. Take $t = \lambda^2\varepsilon^2/192\tau^2$: then $\Pi(d > \varepsilon \mid Y) \le C_3 e^{-\lambda^2\varepsilon^2/16\tau^2} \le e^{-\lambda^2\varepsilon^2/32\tau^2}$ once $\varepsilon \ge K\tau$ with $K^2\lambda^2 \ge 32 \log\max(C_3, 1)$, on an event of probability at least $1 - e^{-\lambda^2\varepsilon^2/192\tau^2}$. This is the claim with $\kappa_0 = \lambda^2/192$. For the last statement, $|\sigma_m - \sigma_m^0| = |a_m - a_m^0| \le d$; apply the bound with the fixed radius $\varepsilon = \varepsilon_1 - \max_m |\sigma_m^0 - \frac12|$ in the first case and $\varepsilon = \frac12(|\sigma_m^0 - \frac12| - \varepsilon_1)$ in the second, reduced to r_0 if necessary. □

Remark 12.4.2. The theorem needs the bias below the noise, $\|b\|_2 \le B\tau$ with B of order one, and no aliasing, $R < \pi/\Delta$. At the settings of the computations of this chapter these are incompatible. With $\ell = 10$, $\Delta = 0.1$ and $\tau = 10^{-16}$ at height 200, the bias falls below τ only for $R \ge 86$, while $\pi/\Delta = 31.4$; with $\ell = 2$ and $\Delta = 0.2$ at height 3×10^{12} it needs $R \ge 17.4$, while $\pi/\Delta = 15.7$. With R just below π/Δ the bias is about 2×10^{-3} per sample at height 200, and the theorem then holds only with B of the order of $\|b\|_2/\tau$, that is at radii of the order of the bias divided by λ, which is no information. The theorem describes a computation with $\Delta < \pi/86 = 0.036$ at $\ell = 10$, about 1100 samples on $[5, 45]$, or one in which the amplitudes are tied to (γ_m, a_m) through the taper, which removes the aliasing; that variant is not proved here.

The separation $s > 0$ excludes two components at one ordinate, and so the pairs $\frac12 \pm \delta$ that the functional equation imposes on zeros off the line: such a configuration is not an interior point of any K_M, and the rate τ of the theorem does not apply to it. Near the line the two poles of such a pair merge; Section 12.5 treats that singular case.

Under its hypotheses the posterior for the zeros in a window, real parts included, contracts to the truth at the rate of the data accuracy: the Bayesian form of Theorem 12.2.1. The hypotheses fail at the settings of the computations above (Remark 12.4.2), so the errors of 10^{-10} to 10^{-7} that the pencil reaches from data with $\tau = 10^{-16}$ are an observation, not an instance of the theorem. The constants

depend on the separation of the poles, and this is not a technicality: the functional equation forces off line zeros into pairs whose poles merge at the line, where the rate for the real part of a pair is $\tau^{1/2}$, not τ (Theorem 12.5.2).

The theorem concentrates the posterior at ϑ^0, whatever ϑ^0 is: in the limit of exact data the data decide the real parts of finitely many zeros to any finite precision, but the theorem does not say what they decide. If the hypothesis holds in the window, the posterior probability that its zeros lie within ε of the line tends to one; if not, it tends to zero; consistency is a statement about the mechanism, not about the parameter. A proof would need, first, $\varepsilon = 0$, which a posterior at a positive noise level does not give (sign changes of Z with a count of zeros prove $\sigma = \frac{1}{2}$ exactly for the zeros they cover), and second, all windows at once, the obstacle that the verification to height 3×10^{12} also meets: a finite computation settles finitely many zeros. Nor is there a loophole in the limit $\tau \to 0$: the data u_1 are not random, the noise is a modelling device, and the limiting posterior is the point mass at the true zeros, where the problem started. Posterior concentration describes what the pencil computation does; the step from it to a theorem about the parameter is one no statistical argument can make.

What the theorem adds is a precise sense in which $\mathfrak{U}_{1/2}$ is recoverable: under its hypotheses the atomic model is identifiable from level one data on a window without aliasing, the posterior contracts at the rate of the data accuracy, and the real parts are recovered with the ordinates. That is the sparse answer to the deconvolution, and it is as far as the data go.

12.5 The functional equation makes the problem singular

12.5.1 Pairs, not zeros

Theorem 12.4.1 treats each zero as a free parameter (γ, a). They are not free: $\xi(s) = \xi(1-s)$, so a zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\delta > 0$ comes with its partner $1 - \bar{\rho} = \frac{1}{2} - \delta + i\gamma$ at the same ordinate. At level θ the pair has widths $a_0 \mp \delta$ with $a_0 = \theta - \frac{1}{2}$, and its two poles $e^{\Delta(-(a_0 \mp \delta) + i(\gamma - c))}$ merge as $\delta \to 0$ into one pole of doubled amplitude, the contribution of a double zero on the line. The pair enters the samples as $A\,[f(a_0 - \delta) + f(a_0 + \delta)]$ with one complex amplitude A common to its members, and $\delta = 0$ is a zero on the line of free amplitude $2A$. The right alternative to "this zero is on the line" is not "this zero has width $a_0 + \varepsilon$", which the functional equation forbids, but "this zero is a pair at $\frac{1}{2} \pm \delta$", and the two behave very differently.

Lemma 12.5.1 (evenness). *In the notation of Theorem 12.2.1 at level θ, with $0 \le \delta < a_0$, write the contribution of one zero of width a at ordinate γ to the samples as $f(a)_j = \exp(-q^2/2\ell^2 + i\omega_j q)$ with $q = (\gamma - c) + ia$. The contribution of the pair at $\frac{1}{2} \pm \delta$ is $f(a_0 - \delta) + f(a_0 + \delta)$, an even entire function of δ, hence*

an entire function $g(\eta)$ of $\eta = \delta^2$, with

$$g(\eta) = 2f(a_0) + \eta\, t + O(\eta^2),$$
$$t_j = f''(a_0)_j = f(a_0)_j \Big[\Big(\omega_j + \frac{i q_0}{\ell^2} \Big)^2 + \frac{1}{\ell^2} \Big], \quad q_0 = (\gamma - c) + i a_0.$$

Proof. Swapping the two members of the pair changes δ to $-\delta$ and leaves the sum unchanged, so the sum is even; it is entire in δ because f is entire in a. An even entire function is an entire function of δ^2. The coefficient of δ^2 is $f''(a_0)$, and $\partial_a \log f = -(\omega_j + iq/\ell^2)$, $\partial_a^2 \log f = 1/\ell^2$, which gives t. □

With the amplitudes of Theorem 12.2.1, the limit $\delta \to 0$ of a pair is a double zero, $2f(a_0)$, so a simple zero, $f(a_0)$, is not nested in the pair model. The singular analysis therefore compares, in the common amplitude model, a zero of free amplitude with a pair of one free common amplitude; for a simple zero $A^0 = \frac{1}{2}$ and $t = f''(a_0)/2$. Freeing the two amplitudes independently discards the multiplicity: the contribution is then not even in δ, and the joint Jacobian is rank deficient at $\delta = 0$. We use the common amplitude model throughout this section; statements about "a zero" below are in the free amplitude model.

The data thus depend on δ only through δ^2: the Fisher information for δ vanishes at $\delta = 0$, and the model is singular there, in the way a mixture with two merging components is singular [39, 125]. In the parameter $\eta \geq 0$ it is regular, one sided at $\eta = 0$, when $t_\perp \neq 0$ (Theorem 12.5.2).

THEOREM 12.5.2 (the rate for the real part). *Fix a window, a level $\theta > \frac{1}{2}$ with $a_0 = \theta - \frac{1}{2}$, the samples $\omega_j = \omega_0 + j\Delta$, $0 \leq j < J$, and f as in Lemma 12.5.1. Let one ordinate γ carry either a zero on the line or a pair at $\frac{1}{2} \pm \delta$, $0 \leq \delta \leq \delta_+ < a_0$, entering the samples as $A\, g(\eta;\gamma)$, where $g(\eta;\gamma) = f(a_0 - \sqrt{\eta}) + f(a_0 + \sqrt{\eta})$, $\eta = \delta^2$, and A is one complex amplitude common to the two members; let the other M zeros of the window enter as $\sum_{m \leq M} A_m z_m^j$ as in Theorem 12.4.1, labelled by increasing ordinate. Write $\vartheta' = (\gamma, A, (\gamma_m, a_m, A_m)_{m \leq M})$, of real dimension $D' = 3 + 4M$, and $\Phi(\eta, \vartheta') \in \mathbb{C}^J = \mathbb{R}^{2J}$ for the noiseless samples, and assume:*

(H1) $Y = \Phi(\eta^0, \vartheta'^0) + b + \tau\xi$, where $\xi \in \mathbb{C}^J$ has independent standard normal real and imaginary parts, as in Theorem 12.4.1, so that τ is the standard deviation of each real coordinate, and $b \in \mathbb{C}^J$ is deterministic with $\|b\|_2 \leq B\tau$;

(H2) the parameter set is $\Theta = [0, \eta_+] \times K'$ with $\eta_+ = \delta_+^2$ and K' compact, $\eta^0 \in [0, \eta_+)$, ϑ'^0 is interior to K', and Φ is injective on Θ;

(H3) the real $2J \times D'$ Jacobian T' of $\vartheta' \mapsto \Phi(\eta^0, \vartheta')$ at ϑ'^0 has rank D'; with $t = \partial_\eta \Phi(\eta^0, \vartheta'^0)$, which is $A^0 f''(a_0)$ when $\eta^0 = 0$, P' the orthogonal projection of $\mathbb{R}^{2J}$ onto the range of T' and $t_\perp = (I - P')t$, we have $\|t_\perp\| > 0$;

(H4) the prior puts mass $1-p \in [0,1)$ *at* $\eta = 0$ *and has density* $p\,\pi_\eta(\eta)$ *on* $(0,\eta_+]$, $\pi_\eta(\eta) = \pi(\sqrt{\eta})/2\sqrt{\eta}$ *with* π *continuous and positive on* $[0,\delta_+]$, *and, independently,* ϑ' *has a density* p' *bounded on* K', *continuous and positive at* ϑ'^0.

Let $r_\tau = \tau/\|t_\perp\|$ *and let* $\tau \to 0$ *with* B *and* ϑ'^0 *fixed, and with* η^0 *fixed in (i) and (ii). Then:*

(i) for every $M_\tau \to \infty$, $\Pi(|\eta-\eta^0| > M_\tau r_\tau \mid Y) \to 0$ *in probability; hence* $\Pi(\delta > (M_\tau r_\tau)^{1/2} \mid Y) \to 0$ *if* $\delta^0 = 0$, *and* $\Pi(|\delta-\delta^0| > M_\tau r_\tau/\delta^0 \mid Y) \to 0$ *if* $\delta^0 > 0$;

(ii) let $\eta^0 = 0$ *and, for* $0 < \delta_1 \le \delta_+$, *let* $D(\delta_1) = \min_{\vartheta' \in K'} \|\Phi(\delta_1^2, \vartheta'^0) - \Phi(0,\vartheta')\|$, *attained at* ϑ'^*; *then* $D(\delta_1) = \delta_1^2\|t_\perp\| + O(\delta_1^4)$, *the Kullback Leibler divergence of the law of* Y *under* $(\delta_1^2, \vartheta'^0)$ *from the law under* $(0,\vartheta'^*)$ *is* $D(\delta_1)^2/2\tau^2$, *and for* $\delta_1 = c\,r_\tau^{1/2}$ *the two error probabilities of any test of* $\delta = 0$ *against* $\delta = \delta_1$, *whatever the method, sum to at least* $1 - c^2/2 - O(\tau)$;

(iii) let $\eta^0 = v\,r_\tau$ *with* $v \ge 0$ *fixed (*$v = 0$ *is the null), let* $\|b\|_2 = o(\tau)$, *take* $t_\perp$ *at* $\eta = 0$, *and let* $Z = \langle t_\perp, r\rangle/(\tau\|t_\perp\|)$ *with* r *the residual of the least squares fit of* $\Phi(0,\cdot)$ *over* K'; *then* $Z = v + \zeta_\perp + o_P(1)$ *with* $\zeta_\perp$ *standard Gaussian, so the one sided matched filter* $Z > z_\alpha$ *has asymptotic level* α *and asymptotic power* $1 - N(z_\alpha - v)$, N *the standard normal distribution function, which is* $\frac{1}{2}$ *at* $\delta^* = (z_\alpha r_\tau)^{1/2}$; *and twice the logarithm of the generalised likelihood ratio of* $\eta \ge 0$ *against* $\eta = 0$ *is* $\max(Z,0)^2 + o_P(1)$, *so that for* $\alpha < \frac{1}{2}$ *the generalised likelihood ratio test at level* α *is asymptotically the matched filter.*

Remark 12.5.3. (H2) and (H3) hold, and $\|t_\perp\| > 0$ is automatic when $\eta^0 = 0$, if $J \ge 2M+4$, $2R\Delta < 2\pi$ as in Theorem 12.4.1, the amplitudes in K', A included, are bounded away from 0, the ordinate γ ranges over an interval I of length less than $2\pi/\Delta$, and the poles z_m keep a distance at least $s > 0$ from one another and from the set $S = \{e^{\Delta(-a+i(\gamma'-c))} : |a-a_0| \le \delta_+,\ \gamma' \in I\}$ that contains the poles of the pair. Indeed $f(a)_j = f(a)_0\, z(a)^j$ with $z(a) = e^{\Delta(-a+i(\gamma-c))}$ and $f(a)_0 \ne 0$, so two parameters with the same samples give two exponential sums whose difference has at most $2M+4 \le J$ distinct poles and vanishes for $0 \le j < J$; the Vandermonde matrix of these poles has full column rank, so the two sums have the same poles with the same coefficients. The poles in S are those of the pair: one pole means $\eta = 0$, two poles of moduli $e^{-\Delta(a_0 \mp \sqrt{\eta})}$ give $\eta > 0$, their common argument gives γ, and the coefficients divided by $f(a_0 \mp \sqrt{\eta})_0$ give A; the remaining poles give the (γ_m, a_m, A_m), the ordinates by $2R\Delta < 2\pi$ and the labels by their order. For (H3) at $\eta^0 = 0$, the real columns of T' lie in the complex span of $(z_m^j)_j$, $(jz_m^j)_j$, $m \le M$, and $(z_0^j)_j$, $(jz_0^j)_j$, $z_0 = z(a_0)$: four real columns per zero, z_m^j, iz_m^j, $iA_m\Delta j z_m^j$, $-A_m\Delta j z_m^j$, and three for the pair, independent over $\mathbb{R}$ because the confluent Vandermonde matrix of these $2M+2$ complex columns has full

rank for $J \geq 2M+2$. Finally $t = A^0 f''(a_0)$ and $\partial_a^2 z(a)^j = \Delta^2 j^2 z(a)^j$, so t has the component $A^0\Delta^2 f(a_0)_0\,(j^2 z_0^j)_j$ with a nonzero coefficient, and the confluent Vandermonde matrix with the further column $(j^2 z_0^j)_j$ has full rank for $J \geq 2M+3$; so t is not in the complex span of the columns of T', a fortiori not in their real span, and $t_\perp \neq 0$. With a prior on M as in Theorem 12.4.1 (amplitudes bounded below, $M \leq M_{\max}$) and $J \geq 2M_{\max}+4$, Step 1 of the proof extends across M by the Vandermonde argument of that theorem, and (i) holds unchanged. Like Theorem 12.4.1, the theorem needs the bias below the noise and no aliasing, so by Remark 12.4.2 it does not cover the runs on the zeros of ζ in this chapter; it covers data simulated from the model with $b = 0$ in a window of radius below π/Δ.

Proof. Write $\vartheta = (\eta, \vartheta')$, $\vartheta^0 = (\eta^0, \vartheta'^0)$, $u(\vartheta) = \Phi(\vartheta) - \Phi(\vartheta^0)$ and $n = b + \tau\xi$, so that the log likelihood ratio of ϑ to ϑ^0 is $\ell(\vartheta) = (2\langle u, n\rangle - \|u\|^2)/2\tau^2$, with $\langle\cdot,\cdot\rangle$ the real inner product of $\mathbb{R}^{2J}$. By Lemma 12.5.1, Φ is real analytic on a neighbourhood of Θ, including $\eta < 0$, so there are L and C_2 with $\|u(\vartheta)\| \leq L|\vartheta - \vartheta^0|$, $\|J_0\| \leq L$ and $\|u(\vartheta) - J_0(\vartheta - \vartheta^0)\| \leq C_2|\vartheta - \vartheta^0|^2$ on Θ, where $J_0 = [t, T']$.

Step 1 (separation). By (H3), t is not in the range of T', so J_0 has full column rank and $\lambda = \sigma_{\min}(J_0) > 0$; hence $\|u(\vartheta)\| \geq \lambda|\vartheta - \vartheta^0|/2$ for $|\vartheta - \vartheta^0| \leq r_1 = \lambda/2C_2$. By (H2), continuity and compactness, $g_1 = \min\{\|u(\vartheta)\| : \vartheta \in \Theta,\ |\vartheta - \vartheta^0| \geq r_1\} > 0$. With $\lambda_1 = \min(\lambda/2, g_1/r_1)$, $\|u(\vartheta)\| \geq \lambda_1 \min(|\vartheta - \vartheta^0|, r_1)$ on Θ; note that $\lambda_1 \leq \lambda/2 \leq \sigma_{\min}(T')/2$.

Step 2 (localisation). Let $E_\tau = \{\|\xi\| \leq \sqrt{2J} + 2(\log 1/\tau)^{1/2}\}$, of probability at least $1 - \tau^2$ by Gaussian concentration. On E_τ, $\|n\| \leq \tau\Lambda_\tau$ with $\Lambda_\tau = B + \sqrt{2J} + 2(\log 1/\tau)^{1/2}$, hence $\ell \leq (2\|u\|\tau\Lambda_\tau - \|u\|^2)/2\tau^2$ everywhere and $\ell \geq -L\Lambda_\tau - L^2/2$ on $B_\tau = \{\vartheta \in \Theta : |\vartheta - \vartheta^0| \leq \tau\}$. By (H4) the prior has a density bounded below by a positive constant on Θ near ϑ^0, and B_τ contains at least half a ball, so $\Pi(B_\tau) \geq c_0\tau^{1+D'}$ and $\int e^\ell\, d\Pi \geq c_0\tau^{1+D'}e^{-L\Lambda_\tau - L^2/2}$. Let $\varepsilon_\tau = K\tau(\log 1/\tau)^{1/2}$ with $\lambda_1 K > \max(8, 2(1+D')^{1/2})$. For small τ, $\varepsilon_\tau \leq r_1$ and $\lambda_1\varepsilon_\tau \geq 4\tau\Lambda_\tau$; as $2x\tau\Lambda_\tau - x^2 \leq -x^2/2$ for $x \geq 4\tau\Lambda_\tau$, $\ell \leq -\lambda_1^2\varepsilon_\tau^2/4\tau^2$ on $\{|\vartheta - \vartheta^0| > \varepsilon_\tau\}$. Hence, on E_τ,

$$\Pi\big(|\vartheta - \vartheta^0| > \varepsilon_\tau \;\big|\; Y\big) \leq c_0^{-1} \exp\Big(-\Big(\frac{\lambda_1^2 K^2}{4} - 1 - D'\Big)\log\frac{1}{\tau} + L\Lambda_\tau + \frac{L^2}{2}\Big) \to 0.$$

The radius ε_τ only localises the posterior; the rate comes from Steps 3 and 4.

Step 3 (linearisation). On $U_\tau = \{\vartheta \in \Theta : |\vartheta - \vartheta^0| \leq \varepsilon_\tau\}$ write $\vartheta - \vartheta^0 = (h_\eta, h')$ and $k = h' + h_\eta w$ with $w = (T'^{\mathsf{T}}T')^{-1}T'^{\mathsf{T}}t$, so that $J_0(\vartheta - \vartheta^0) = h_\eta t_\perp + T'k$, the two terms being orthogonal. Then $\ell = \ell_1(h_\eta) + \ell_2(k) + \rho$ with

$$\ell_1(x) = \frac{2x\langle t_\perp, n\rangle - x^2\|t_\perp\|^2}{2\tau^2}, \qquad \ell_2(k) = \frac{2\langle T'k, n\rangle - \|T'k\|^2}{2\tau^2},$$

and, on E_τ, $|\rho| \leq (2C_2\varepsilon_\tau^2\tau\Lambda_\tau + 2LC_2\varepsilon_\tau^3 + C_2^2\varepsilon_\tau^4)/2\tau^2 = O(\tau(\log 1/\tau)^{3/2})$ uniformly on U_τ.

Step 4 (the marginal of η). Let $W = \langle t_\perp, n\rangle/(\tau\|t_\perp\|) = \zeta_\perp + \langle t_\perp, b\rangle/(\tau\|t_\perp\|)$, where $\zeta_\perp = \langle t_\perp, \xi\rangle/\|t_\perp\|$ is standard Gaussian and the second term is at most B in modulus; so $W = O_P(1)$ and $\ell_1(vr_\tau) = Wv - v^2/2$. Let $I_2 = \int_{\mathbb{R}^{D'}} e^{\ell_2(k)}\,dk$, a Gaussian integral centred at $\hat{k} = (T'^{\mathsf{T}}T')^{-1}T'^{\mathsf{T}}n$ with covariance $\tau^2(T'^{\mathsf{T}}T')^{-1}$; on E_τ, $|\hat{k}| \le \tau\Lambda_\tau/\sigma_{\min}(T') \le \varepsilon_\tau/4$ for small τ, because $\sigma_{\min}(T')K \ge 2\lambda_1 K > 16$ and $\Lambda_\tau \le 4(\log 1/\tau)^{1/2}$ for small τ. Put $S_\tau = U_\tau \cap \{|h_\eta| > M_\tau r_\tau\}$ and $S'_\tau = U_\tau \cap \{|h_\eta| \le r_\tau\}$. The change of variables $(h_\eta, h') \mapsto (h_\eta, k)$ has Jacobian 1. Bounding each k integral by I_2,

$$\int_{S_\tau} e^{\ell}\,d\Pi \le e^{o(1)} \sup_{K'} p'\; I_2 \int_{|x|>M_\tau r_\tau} e^{\ell_1(x)}\, p\,\pi_\eta(\eta^0 + x)\,dx.$$

For $|h_\eta| \le r_\tau$ the k section of U_τ contains the ball of radius $\varepsilon_\tau/2$ about $\hat{k}$, since there $|h'| \le |\hat{k}| + \varepsilon_\tau/2 + r_\tau|w| \le (\frac{3}{4} + o(1))\varepsilon_\tau$, and the Gaussian gives this ball the mass $(1 - o(1))I_2$, its radius being large against $\tau/\sigma_{\min}(T')$; so

$$\int_{S'_\tau} e^{\ell}\,d\Pi \ge e^{-o(1)}(1 - o(1)) \inf_{U_\tau} p'\; I_2 \int_{|x|\le r_\tau} e^{\ell_1(x)}\, p\,\pi_\eta(\eta^0 + x)\,dx.$$

If $\eta^0 > 0$, then $\pi_\eta(\eta^0 + x) = \pi_\eta(\eta^0)(1 + o(1))$ for $|x| \le \varepsilon_\tau$, and with $x = vr_\tau$ the ratio of the right sides is at most $(1 + o(1)) \int_{|v|>M_\tau} e^{Wv - v^2/2}dv \big/ \int_{|v|\le 1} e^{Wv - v^2/2}dv$, which tends to 0 in probability because $W = O_P(1)$; the atom at $\eta = 0$ lies outside U_τ and is covered by Step 2. If $\eta^0 = 0$, then $x = \eta \ge 0$, $\pi_\eta(\eta) = \pi(0)(1 + o(1))/2\sqrt{\eta}$ on $(0, \varepsilon_\tau]$, the factor $\pi(0)r_\tau^{1/2}/2$ cancels, and the ratio is at most $(1 + o(1)) \int_{M_\tau}^{\infty} v^{-1/2} e^{Wv - v^2/2}dv \big/ \int_0^1 v^{-1/2} e^{Wv - v^2/2}dv \to 0$; the atom at $\eta = 0$ lies in S'_τ and only increases the denominator. As $\Pi(|\eta - \eta^0| > M_\tau r_\tau \mid Y) \le \Pi(U_\tau^c \mid Y) + \Pi(S_\tau \mid Y)/\Pi(S'_\tau \mid Y)$, this proves (i) for η; the statements for δ follow from $|\sqrt{\eta} - \sqrt{\eta^0}| \le |\eta - \eta^0|^{1/2}$ and $|\sqrt{\eta} - \sqrt{\eta^0}| \le |\eta - \eta^0|/\sqrt{\eta^0}$.

(ii). The two laws are Gaussian with covariance $\tau^2 I_{2J}$ and means differing by $\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta'^*)$, b cancelling, so the divergence is $D(\delta_1)^2/2\tau^2$. The choice $\vartheta' = \vartheta'^0 + \delta_1^2 w$, in K' for small δ_1, gives $\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta') = \delta_1^2 t_\perp + O(\delta_1^4)$, so $D(\delta_1) \le \delta_1^2\|t_\perp\| + O(\delta_1^4)$. Conversely, let $r_2 = \min(r_1, \sigma_{\min}(T')/2C_2)$. If $|\vartheta' - \vartheta'^0| \ge r_2$, Step 1 and the triangle inequality give $\|\Phi(\delta_1^2, \vartheta'^0) - \Phi(0, \vartheta')\| \ge \lambda_1 r_2 - L\delta_1^2 > \delta_1^2\|t_\perp\| + O(\delta_1^4)$ for small δ_1. If $h' = \vartheta' - \vartheta'^0$ has $|h'| < r_2$, the difference is $\delta_1^2 t - T'h' + e$ with $\|e\| \le C_2(\delta_1^4 + |h'|^2)$; its norm is at least $\sigma_{\min}(T')|h'|/2 - \delta_1^2\|t\| - C_2\delta_1^4$, which exceeds the upper bound for $D(\delta_1)$ unless $|h'| = O(\delta_1^2)$, and at least $\|(I - P')(\delta_1^2 t - T'h')\| - \|e\| = \delta_1^2\|t_\perp\| - C_2(\delta_1^4 + |h'|^2)$. So $D(\delta_1) = \delta_1^2\|t_\perp\| + O(\delta_1^4)$. By Le Cam's two point lemma and Pinsker's inequality, the two error probabilities of any test sum to at least $1 - \mathrm{TV} \ge 1 - (\mathrm{KL}/2)^{1/2} = 1 - D(\delta_1)/2\tau$, which is $1 - c^2/2 - O(\tau)$ at $\delta_1 = c\,r_\tau^{1/2}$.

(iii). Let $\hat{\vartheta}'$ minimise $\|Y - \Phi(0, \vartheta')\|$ over K'. Since $\|Y - \Phi(0, \vartheta'^0)\| \le \|n\| + L\eta^0 = O_P(\tau)$, also $\|\Phi(0, \hat{\vartheta}') - \Phi(0, \vartheta'^0)\| = O_P(\tau)$, and Step 1 with the base point $(0, \vartheta'^0)$ gives $|\hat{\vartheta}' - \vartheta'^0| = O_P(\tau)$; so with probability tending to one $\hat{\vartheta}'$ is

interior and satisfies the normal equations. Expanding, $Y - \Phi(0, \hat{\vartheta}') = n + \eta^0 t - T'(\hat{\vartheta}' - \vartheta'^0) + O_P(\tau^2)$ and $T'^{\mathsf{T}}(Y - \Phi(0, \hat{\vartheta}')) = O_P(\tau^2)$, so $r = (I - P')(n + \eta^0 t) + O_P(\tau^2)$ and $Z = \zeta_\perp + \langle t_\perp, b\rangle/(\tau\|t_\perp\|) + v + O_P(\tau)$, the middle term being at most $\|b\|_2/\tau \to 0$. The level and the power follow, and the power is $\frac{1}{2}$ when $v = z_\alpha$, that is $\eta^0 = z_\alpha r_\tau$. The least squares fit over $\eta \geq 0$ and ϑ' is consistent in the same way, and in the linearised model the minimum of the residual sum of squares is $\|r\|^2 - \max(\langle t_\perp, r\rangle, 0)^2/\|t_\perp\|^2 + O_P(\tau^3)$; so twice the log likelihood ratio is $\max(Z, 0)^2 + o_P(1)$, and for $\alpha < \frac{1}{2}$ the events $\max(Z, 0)^2 > z_\alpha^2$ and $Z > z_\alpha$ coincide. □

Part (i) holds under the conventions of Theorem 12.4.1 (the noise, $\|b\|_2 \leq B\tau$, separation and no aliasing), with no logarithmic factor in the rate; its second rate, $\tau/(\delta_0\|t_\perp\|)$, is claimed only for fixed $\delta_0 > 0$, since under the spike of Theorem 12.5.4 it fails in the intermediate range. Part (ii) shows that no method beats the rate $\tau^{1/2}$ for the real part, and (iii) that the matched filter attains it.

The scale is visible in the data. Take the window about 60 at level 0.6 with taper $\ell = 3$ and the samples $\omega_j = 2 + 0.2j$, $j < 100$ (the design of the detection experiment below, assumed because the source states none for this window), with a pair at $\frac{1}{2} \pm \delta$ added at 60.089 to the zeros, the pair fitted with one free common amplitude, and the noise $\tau = 1.7 \times 10^{-11}$ of the source, an estimate from a residual, taken as the standard deviation per real coordinate. In this design $\|t_\perp\| = 323.8$ for the free amplitude and 406.5 for an amplitude fixed by the model, the matched filter has $\delta^* = (1.645\,\tau/\|t_\perp\|)^{1/2} = 2.9 \times 10^{-7}$, and the Bayes factor, which pays an Occam factor, crosses one at about 3×10^{-7} (computed, one noise draw per δ: a pair at 10^{-6} is already detected, with a profile likelihood gain of 380 in $2\log$). So data accurate to 10^{-11} resolve a real part to a few times 10^{-7} in this design. The comparison with Theorem 12.5.2 is with its scale only: the error of the run is deterministic, one noise draw, and the theorem does not cover the reported runs (Remark 12.5.3).

12.5.2 Bayes factors with the right alternative

Theorem 12.5.4 (Bayes factors). *In the setting of Theorem 12.5.2, put the spike and slab prior on the pair: $\delta = 0$ with probability $1 - p$, and otherwise δ with a density π continuous and positive on $[0, \delta_+]$. Let $s = \|t_\perp\|/\tau$, $G(Z) = \int_0^\infty u^{-1/2} e^{Zu - u^2/2}\,du$, and Z the statistic of Theorem 12.5.2(iii).*

(a) If $\eta^0 = 0$, or $\eta^0 = v\,r_\tau$ with v fixed, and $\|b\|_2 = o(\tau)$, and ϑ' has the same prior under the line and under the pair, the Bayes factor for the line against the pair is

$$\mathrm{BF}_{01} = \frac{2\,s^{1/2}}{\pi(0)\,G(Z)}\,\big(1 + o_P(1)\big);$$

under the null Z is asymptotically standard Gaussian and $\mathrm{BF}_{01} \asymp \tau^{-1/2}$ *in probability.*

(b) Under a pair with fixed $\delta \in (0, \delta_+)$, if p' is bounded below on K' and every point of K' has prior mass at least $c\tau^{D'}$ within distance τ, then $\log \mathrm{BF}_{01} = -D(\delta)^2(1 + o_P(1))/2\tau^2$, *with $D(\delta) = \delta^2\|t_\perp\| + O(\delta^4)$ as in Theorem 12.5.2(ii).*

(c) Against the alternative of a single zero of width $a_0 + \varepsilon$, with ε having a density positive and continuous at 0 and $\|(I - P')\partial_\varepsilon\Phi\| > 0$, the same data give $\mathrm{BF}_{01} \asymp \tau^{-1}$ *in probability under the null.*

Proof. (a) $\mathrm{BF}_{10} = m_1/m_0$ with $m_1 = \iint_{\eta>0} e^{\ell(\eta,\vartheta')}\pi_\eta(\eta)\,p'(\vartheta')\,d\eta\,d\vartheta'$ and $m_0 = \int e^{\ell(0,\vartheta')}p'(\vartheta')\,d\vartheta'$, ℓ the log likelihood ratio to $(0, \vartheta'^0)$. The offset $\Phi(\eta^0, \vartheta'^0) - \Phi(0, \vartheta'^0) = \eta^0 t + O(\tau^2)$ is $O(\tau)$ and can be carried in b in Steps 2 and 3 of the proof of Theorem 12.5.2, which then apply to m_0 and to m_1: the parts outside U_τ are negligible, and on U_τ the integrand factorises as in Step 3 with W replaced by $W_0 = \langle t_\perp, n + \eta^0 t\rangle/(\tau\|t_\perp\|) = Z + o_P(1)$, by the proof of (iii). The k integrals give $p'(\vartheta'^0)I_2(1+o_P(1))$ in both (in m_1 it suffices, as in Step 4, to control the k sections for $\eta \le M_\tau r_\tau$, the rest being negligible), and, with $\eta = u\,r_\tau$ and $\varepsilon_\tau/r_\tau \to \infty$,

$$\int_0^{\varepsilon_\tau} e^{W_0\eta/r_\tau - \eta^2/2r_\tau^2}\,\frac{\pi(\sqrt{\eta})}{2\sqrt{\eta}}\,d\eta = \frac{\pi(0)}{2}\,r_\tau^{1/2}\,G(W_0)\,(1 + o(1)).$$

Since $r_\tau^{1/2} = s^{-1/2}$ and G is continuous and positive, $\mathrm{BF}_{10} = \frac{1}{2}\pi(0)s^{-1/2}G(Z)(1 + o_P(1))$. Under the null $Z = \zeta_\perp + o_P(1)$, so $G(Z)$ is bounded above and away from 0 in probability.

(b) Take ℓ relative to the truth (δ^2, ϑ'^0) and work on E_τ. For the slab, $\ell \le \sup_x(2x\tau\Lambda_\tau - x^2)/2\tau^2 = \Lambda_\tau^2/2$ and Step 2 bounds m_1 below, so $|\log m_1| = O(\log 1/\tau)$. For the spike, $\|u(0, \vartheta')\| \ge D(\delta)$ for every ϑ', so $\ell(0, \vartheta') \le -(D^2 - 2D\tau\Lambda_\tau)/2\tau^2$ once $D \ge \tau\Lambda_\tau$; and within distance τ of the minimiser ϑ'^*, $\|u(0, \vartheta')\| \le D + L\tau$ and $\ell(0, \vartheta') \ge -(D+L\tau)^2/2\tau^2 - (D+L\tau)\Lambda_\tau/\tau$, on a set of prior mass at least $c\tau^{D'}$. Hence $\log m_0 = -D^2/2\tau^2 + O(\Lambda_\tau/\tau) + O(\log 1/\tau)$, and $\log \mathrm{BF}_{01} = \log m_0 - \log m_1 = -D^2(1 + o(1))/2\tau^2$.

(c) As in (a), with the regular coordinate ε in place of η: the ε integral is $\pi_\varepsilon(0)\,\tau\,\|(I - P')\partial_\varepsilon\Phi\|^{-1}\int e^{Z_\varepsilon v - v^2/2}\,dv\,(1 + o_P(1))$, over the half line or the line allowed by the sign of ε, which is of order τ; so $\mathrm{BF}_{10} \asymp \tau$. □

Using the formula on data needs Z and s; for the real zero at 201.264752 it is an illustration under assumptions, since the source gives no noise level for that window. With π uniform on $(0, \frac{1}{2})$, the design of the detection experiment below (level 0.6, $\ell = 3$, $\omega_j = 2 + 0.2j$, an assumption) and $\tau = 1.7 \times 10^{-11}$ assumed, the simple zero, of amplitude $A^0 = \frac{1}{2}$, has $\|t_\perp\| = 148.6$ and $s = \|t_\perp\|/\tau = 8.74 \times 10^{12}$; the formula of Theorem 12.5.4 gives $\log_{10} \mathrm{BF}_{01} = 6.14$ at $Z = 0$ and 5.51 at

$Z = 1.7$, and the integral computed exactly gives a median 6.10 over 40 noise draws, with range 5.55 to 6.48. The single zero alternative gives 11.83 at $Z_1 = 0$ and 10.92 at $Z_1 = 1.7$, and 9.66 to 11.78 for the other ten central zeros at $Z_1 = 0$, the spread arising from the norm of the projected tangent, which varies from 0.2 to 29 over the central zeros (computed). Under these assumptions one window of data at this accuracy is worth a factor of about 10^6 per zero in favour of the line against the pair, against about 10^{12} against the single zero alternative: the functional equation takes the square root of the evidence (computed in the free amplitude model; heuristic beyond it).

12.6 An adaptive prior, and a ReLU network

12.6.1 Where to look: the ordinates

The prior of Theorem 12.4.1 puts the ordinates uniformly on the window. A prior on where the zeros are can use two pieces of information. The first is unconditional: the zeros have density $d(t) = \log(t/2\pi)/2\pi$ at height t, with an explicit error, so the number in a window of fixed length W is $Wd(c) + O(\log c)$ (proved, Riemann and von Mangoldt). The second is conjectural (the pair correlation theorem of Montgomery [211], under RH and for restricted support, and the computations of Odlyzko [220]) the local statistics of the ordinates are those of the sine kernel determinantal process. Take the prior on the ordinates in a window to be the determinantal point process with kernel

$$K(x, y) = \frac{\sin \pi d(c)(x - y)}{\pi(x - y)},$$

which adapts to the height through $d(c)$ and repels: its Janossy density vanishes like $\prod_{i<j}(x_i - x_j)^2$ at coincidences. Three consequences. First, at a configuration of distinct ordinates the rates of Theorems 12.4.1 and 12.5.2 are unchanged, the prior density being continuous and positive there, since the prior mass of a ball of radius τ about it is still of order $\tau^{\dim}$; only the constants move. Second, the posterior on the number M of zeros in the window is pulled toward $Wd(c)$, a prior tendency and the Bayesian form of the count that Section 12.8.6 could not certify at 3×10^{12} (heuristic). Third, spurious components that sit on top of genuine ones are penalised, a priori and by the two point correlation, by the factor $(\pi d\,|x_i - x_j|)^2/3$, which is 1×10^{-4} at the density $d = 0.569$ of height 224 and a separation 0.0097, and which may help the many window scheme remove them (heuristic). The repulsion is a conjecture about the zeros and enters only as a prior. A determinantal process has almost surely no two points at one ordinate, so the process is put on the distinct ordinates, at the intensity $d(c)$, a pair at $\frac{1}{2} \pm \delta$ counting as one point marked by δ; put on the zeros themselves it would give prior probability zero to every off line pair, and so decide the real parts before the data. With this convention the ordinate prior is the same under the line and under the

pair, and to leading order as $\tau \to 0$ it leaves the Bayes factor of Theorem 12.5.4 unchanged (Remark 12.6.1); at finite τ it can still move the posterior of the real parts, as any prior on parameters that the likelihood couples with them can.

Remark 12.6.1. For a determinantal process with continuous kernel K and intensity d the second correlation function is $\rho_2(x,y) = d^2 - K(x,y)^2 \le d^2$, so the expected number of ordered pairs of distinct points of a window of length W at distance less than ϵ is at most $2\epsilon W d^2$, which tends to 0 with ϵ: almost surely no two points coincide, while a pair at $\frac{1}{2} \pm \delta$, and a double zero, put two zeros at one ordinate. On the distinct ordinates the prior density p' of ϑ' in Theorem 12.5.2 (the Janossy density of the ordinates times the priors of the amplitudes and widths) does not depend on η, and it is continuous and positive at distinct ordinates; in the proof of Theorem 12.5.4(a) the integrals over ϑ' give $p'(\vartheta'^0) I_2 (1 + o_P(1))$ in both marginal likelihoods, so p' cancels from BF_{01} to leading order. At finite noise it does not cancel from the posterior of a real part: if $Y_1 = \gamma + a + \sigma\epsilon_1$ and $Y_2 = \gamma + \sigma\epsilon_2$, with ϵ_1, ϵ_2 independent standard normal and independent priors $\gamma \sim N(m, v_\gamma)$, $a \sim N(0, v_a)$, then

$$E[a \mid Y_1, Y_2] = \frac{v_a (Y_1 - m_2)}{v_a + v_2 + \sigma^2}, \qquad m_2 = \frac{\sigma^2 m + v_\gamma Y_2}{v_\gamma + \sigma^2}, \quad v_2 = \frac{v_\gamma \sigma^2}{v_\gamma + \sigma^2},$$

which depends on the prior mean m of the ordinate whenever $\sigma > 0$ (given Y_2, γ is $N(m_2, v_2)$, and then Y_1 given a is $N(m_2 + a, v_2 + \sigma^2)$).

12.6.2 What to believe: the real parts

For the real parts the adaptive prior is the spike and slab on pairs of Theorem 12.5.4: the functional equation supplies the structure, the spike at $\delta = 0$ the shrinkage, and the slab keeps the alternative honest. A computation like that of Section 12.8.6 should be read under this prior, and Theorem 12.5.4 says what it delivers: under the line, Bayes factors of order $\tau^{-1/2}$ in probability per pair, the other pairs held on the line, in the free amplitude model (the posterior odds are $(1-p)\,\mathrm{BF}_{01}/p$), and a resolution of $(\tau / \|t_\perp\|)^{1/2}$ in the real part.

12.6.3 A ReLU network

The question was whether a ReLU network can serve as the adaptive prior or regulariser. Polson and Ročková [153] proved that deep ReLU networks with a spike and slab prior on their weights have posteriors that contract at the near minimax rate for Hölder functions, adapting to the unknown smoothness. That theorem is about nonparametric regression and gives no rate for the window model, and what it suggests here is heuristic; part (ii) of Theorem 12.5.2 bounds every method, networks included. Where a network can help is in amortisation: it learns a classifier of the line against the pair, which at the population optimum is the

posterior probability of the pair under the training law, and costs one pencil fit and one forward pass per window. In the author's experiment the training law is a recentred circular unitary ensemble of 40 points truncated to the window, with a log uniform slab on $[10^{-7.5}, 0.09]$ for the displacement of the pair, which is neither the sine kernel process nor the spike and slab of Theorem 12.5.4. That experiment, with two ReLU detectors (a network with two hidden layers of 256 units on the raw transform and the same network on the residual of the fit on the line), was run by the author and is unchecked: its two rows were not reproduced here and are not printed.

The pencil and a plug in matched filter were rerun on the window model of Theorem 12.2.1 at level 0.6 with taper $\ell = 3$ and 100 samples $\omega_j = 2 + 0.2j$ (the sampling that the author appears to use; the chapter does not give it), ordinates drawn from the circular unitary ensemble scaled to the density at height 200, and the zero nearest the centre either a double zero on the line or a pair at $\frac{1}{2} \pm \delta$; the noise has complex root mean square τ, that is standard deviation $\tau/\sqrt{2}$ per real coordinate. Two detectors, each calibrated to five per cent false alarms on 1200 null windows: the pencil statistic $\max|\hat{a} - a_0|$ over the components within 0.5 of the centre, and a plug in version of the matched filter of Theorem 12.5.2(iii) in which the tangent t is that of the central zero at the pencil's ordinate and $t_\perp$ is orthogonal to the tangent columns (amplitude and pole) of every pencil component (this is our reading of "built from the pencil's components"; the source does not define it, and no efficiency claim is made for any plug in filter). The quantity δ^* is the smallest displacement detected with power one half.

	detectable δ^* at noise $\tau =$				
detector	10^{-10}	10^{-8}	10^{-6}	10^{-4}	10^{-2}
benchmark $0.0537\,\tau^{1/2}$	5.4×10^{-7}	5.4×10^{-6}	5.4×10^{-5}	5.4×10^{-4}	5.4×10^{-3}
pencil	8.5×10^{-7}	7.0×10^{-6}	6.1×10^{-5}	9.1×10^{-4}	7.2×10^{-3}
plug in filter	1.0×10^{-6}	1.3×10^{-5}	1.3×10^{-4}	1.0×10^{-3}	1.2×10^{-2}

The benchmark is Theorem 12.5.2(iii) at the half power point, $\delta^* = (1.645\,\tau'/\|t_\perp\|)^{1/2}$, with the median 403 of $\|t_\perp\|$ over 300 windows (10th and 90th percentiles 382 and 407) and $\tau' = \tau/\sqrt{2}$, the standard deviation per real coordinate; with τ itself as the standard deviation per real coordinate the constant is 0.0639 and the row is 6.4×10^{-7} to 6.4×10^{-3}. It is the half power point at the median, a benchmark and not a bound (a law of $\|t_\perp\|$ with two values can have a smaller average power at it). The pencil and plug in rows are the means of two seeds (150 windows per δ, 16 values of δ; computed).

First, the law $\delta^* \propto \tau^{1/2}$ of Theorem 12.5.2 holds over eight decades of noise for the pencil, with $\delta^*/\tau^{1/2}$ between 0.057 and 0.093, within a factor 1.7 of the benchmark; a data accuracy of 10^{-10} resolves a real part to about 10^{-6} in this design. Second, the plug in filter is 1.1 to 2.1 times worse than the pencil in our implementation. The source's rows were not reproduced; they depend on the

pencil's order rule and on the definition of the plug in filter, which the source does not give.

So a ReLU network does not change what can be found: the bound of Theorem 12.5.2(ii) holds for it, and a network trained on residuals is a learned test that cannot beat it; whether it is more robust than the pencil is a matter for the author's unchecked experiment. The regularisation that matters is in the prior: the sine kernel for where the zeros are, the spike and slab on pairs for their real parts. None of this reaches the hypothesis: each window at accuracy τ gives finitely many zeros at finite precision, with the Bayes factors and resolution stated above.

12.7 What this does and does not do

Sparse methods solve the deconvolution. The problem $P_{1/2} * \mu = u_1$ is exponentially ill posed for a general μ and polynomially ill posed for a sum of atoms, and the matrix pencil, which works in the second model, recovers the atoms to ten digits where EM, which works in the first, reaches three. The structural reason is Theorem 12.2.1: the explicit formula, read as a deconvolution and tapered, is a sum of damped exponentials, Prony's problem, and the level one data determine the poles. Two further things follow. The pencil does not need to assume a width: it returns a_ρ and so places each zero in the strip, which the backward flow at a fixed level cannot do, since that flow presupposes a width. And the certificate question of Chapter 11 changes character: a certified version of the pencil would need interval versions of the singular value decomposition and the eigenvalue problem, which exist, and a bound linking the matrix model to ξ (the remainder of Theorem 12.2.1 and the far zeros), which does not, and would then give certified zeros from the line $\operatorname{Re} s = 1$; we have not done it, and it is the first open problem below.

What it does not do is also clear. The data u_1 to twenty digits are $\operatorname{Re}(\xi'/\xi)$ on $\operatorname{Re} s = 1$ to twenty digits, which is computed from ζ by the same algorithms that find zeros directly; the pencil is a different route from the same information, not a cheaper one. It reads the real parts; it does not constrain them. The one thing it offers the Thorin programme is a sharper statement of what the programme computes: the backward flow at level θ is the resolvent trace $\pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$, whose Fourier transform is $\sum_\rho \operatorname{sgn}(\theta - \sigma_\rho) e^{-|\theta-\sigma_\rho||\omega|} e^{-i\omega\gamma_\rho}$ (Proposition 12.8.1), and the Riemann hypothesis is the statement that every damping rate $|\theta - \sigma_\rho|$ at level θ is $\theta - \frac{1}{2}$, which is what the pencil measures and what no measurement proves.

The two models are two priors on the same unknown (an analogy, since no posterior was computed here). The grid prior of Chapter 11, a product of gamma laws on the cells, is a prior on densities, and the EM solution is a point estimate under that model, with its 10^{-3}. A spike prior, a point process of atoms with amplitudes bounded away from zero, is a prior on sums of atoms; in the setting

of Theorem 12.4.1 its posterior concentrates at the true atoms at the rate of the noise, and the matrix pencil, which is exact up to the remainder of Theorem 12.2.1 when the data come from finitely many atoms, reaches the same atoms without computing a posterior. The deconvolution is the same; what changed is the model, and in an exponentially ill posed problem the model is everything.

12.8 Questions taken up

This section takes up certification of the pencil's output, close pairs, levels above one half, the choice of level, combining windows, and large heights. Its tables are the authors' computations at heights near 200, not rerun, except where a sentence says otherwise.

12.8.1 Certifying the pencil

The pencil proposes; ball arithmetic disposes. For each of the eleven zeros within 10 of the centre $c = 200$, with the pencil's value γ, two checks were made in Arb at 80 bits. First, $\xi(\frac{1}{2} + it)$ changes sign on $[\gamma - 2 \times 10^{-9}, \gamma + 2 \times 10^{-9}]$, which certifies a zero on the line within 2×10^{-9} of γ. Second, the winding number of ξ around the circle of radius 10^{-3} about $\frac{1}{2} + i\gamma$, computed as $(2\pi i)^{-1} \oint \xi'/\xi$ by Arb's rigorous integrator, is the ball 1.0000 ± 10^{-9}, hence exactly 1: there is one zero in the disc, it is simple, and it is the one on the line. All eleven pass. This is not Rump's verified eigenvalue bounds [166] for the pencil, which would enclose the eigenvalues of the matrix model and still need a bound linking that model to ξ; it is what one wants in practice: the pencil locates, the argument principle certifies, and together they give certified zeros from level one data in the time the pencil takes plus a few seconds per zero. For a zero off the line the same winding number around a circle of radius less than $|\sigma - \frac{1}{2}|$ about the pencil's $\sigma + i\gamma$ would certify it with the same rigour, which is more than a sign change can do.

12.8.2 Close pairs

The pencil's resolution of a close pair was tested on synthetic data built from Theorem 12.2.1: the Lorentzians of the true zeros within 15 of the centre, with those within 1.5 of the centre removed and a pair at $c \pm d/2$ put in their place, for d from 1 down to 0.01, with Gaussian noise $\eta = 10^{-16}$ or 10^{-12} on the samples and bandwidths $\Omega = 25$, 45, 65. The error in the recovered separation is below 10^{-12} at $\eta = 10^{-16}$ and below 5×10^{-9} at $\eta = 10^{-12}$, for every d down to 0.01 and every Ω: the loss from closeness is a factor of a few hundred at most over two decades of separation, a polynomial loss consistent with the bounds of Donoho and of Li and Liao [67, 113], not the exponential one of Proposition 12.1.2. The closest real pair below height 1420 reported by the authors, at 1329.043518 and 1329.205019 with gap 0.1615 (not checked here), is recovered from the real data to about 2×10^{-8}

at level 1 and to about 3×10^{-11} at level 0.6. Down to a separation of 0.01, close pairs are not the limitation of the method.

12.8.3 Levels above one half

The pencil was run on u_θ for $\theta = 1$, $\frac{3}{4}$ and 0.6 on the same window. The zeros enter with a sign, and the sign matters for a zero to the right of θ.

Proposition 12.8.1 (levels below the zeros). *Let $\frac{1}{2} < \theta \le 1$ with $\theta \neq \sigma_\rho$ for every ρ, and $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$. Then*

$$u_\theta(v) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, P_{|\theta-\sigma_\rho|}(v - \gamma_\rho),$$
$$\widehat{u_\theta}(\omega) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, e^{-|\theta-\sigma_\rho||\omega|} e^{-i\omega\gamma_\rho},$$

and for $\omega\ell^2 > 1$ the tapered transform $F_\theta(\omega) = \int u_\theta(v) w(v) e^{i\omega(v-c)} dv$ is

$$F_\theta(\omega) = \sum_\rho \operatorname{sgn}(\theta - \sigma_\rho)\, w(\gamma_\rho + i|\theta - \sigma_\rho|)\, e^{-|\theta-\sigma_\rho|\omega} e^{i\omega(\gamma_\rho - c)} + R_\theta(\omega),$$

with R_θ bounded as in Theorem 12.2.1 with a_ρ replaced by $|\theta - \sigma_\rho|$. A zero with $\sigma_\rho > \theta$ therefore gives the pole $e^{\Delta(-(\sigma_\rho-\theta)+i(\gamma_\rho-c))}$, inside the unit circle on the ring of a zero at $2\theta - \sigma_\rho$, with an amplitude of sign opposite to the taper profile, while its partner $1 - \sigma_\rho < \theta$ gives a pole at the same ordinate with the profile's sign. The Riemann hypothesis holds if and only if every damping rate $|\theta - \sigma_\rho|$ equals $\theta - \frac{1}{2}$.

Proof. With the zeros paired, $\xi'/\xi(s) = \sum_\rho (s-\rho)^{-1}$, and $\operatorname{Re}(\theta + iv - \rho)^{-1} = (\theta - \sigma_\rho)/((\theta-\sigma_\rho)^2 + (v-\gamma_\rho)^2) = \pi \operatorname{sgn}(\theta - \sigma_\rho) P_{|\theta-\sigma_\rho|}(v - \gamma_\rho)$; the series converges absolutely, its terms being $O(\gamma_\rho^{-2})$. Transform termwise with $\widehat{P_b}(\omega) = e^{-b|\omega|}$. For F_θ apply the proof of Theorem 12.2.1 to each $P_{|\theta-\sigma_\rho|}$, whose width lies in $(0, 1)$, below $\omega\ell^2$. A pole of modulus $e^{-\Delta b}$ is read as the width b, that is as the real part $\theta - b$, which for $b = \sigma_\rho - \theta$ is $2\theta - \sigma_\rho$. For the last statement: if $|\theta - \sigma_\rho| = \theta - \frac{1}{2}$ for every ρ, each σ_ρ is $\frac{1}{2}$ or $2\theta - \frac{1}{2}$; but if $\sigma_\rho = 2\theta - \frac{1}{2}$ then $1 - \sigma_\rho = \frac{3}{2} - 2\theta$ is also the real part of a zero, with rate $3(\theta - \frac{1}{2}) \neq \theta - \frac{1}{2}$. □

For $\sigma_\rho < \theta$ the poles are $e^{\Delta(-(\theta-\sigma_\rho)+i(\gamma_\rho-c))}$ (equation (12.1.1) with $a_\rho = \theta - \sigma_\rho$ and Proposition 12.8.1). The authors' run gives three findings.

θ	position error, max	median	error in σ_ρ, max	median
1	4.5×10^{-10}	1.8×10^{-10}	7.9×10^{-10}	2.3×10^{-10}
3/4	3.3×10^{-11}	1.5×10^{-11}	5.3×10^{-11}	1.4×10^{-11}
0.6	2.4×10^{-11}	8.0×10^{-12}	2.3×10^{-11}	7.2×10^{-12}

First, the real parts agree across levels: $\sigma_\rho = \theta - a_\rho$ is 0.50000000 for each of the eleven central zeros at all three levels, a consistency check on the computation of u_θ. Second, lower levels are better conditioned, by an order of magnitude from 1 to $\frac{3}{4}$ and a little more to 0.6: the damping $e^{-(\theta-1/2)\omega}$ is milder, so the signal at the top of the band is $e^{-4.5}$ at $\theta = 0.6$ against $e^{-22.5}$ at $\theta = 1$, and the pencil's noise amplification falls accordingly. Third, the point of the exercise: nothing in the method needs the level to be above the zeros: u_θ is defined for every θ that is not the real part of a zero, and by Proposition 12.8.1 a zero to the right of θ appears as a pole of modulus $e^{-\Delta(\sigma_\rho-\theta)}$, read at $2\theta - \sigma_\rho$, with amplitude of sign opposite to the taper profile; the sign of the amplitude is the test, a test that the amplitude filter used in these runs, which compares moduli, does not perform. The pencil at level θ could therefore be a test for zeros in $\operatorname{Re} s > \theta$ on a window through the sign of the amplitude (not implemented in the runs reported here, open), with a certificate available by the winding number of Section 12.8.1. The recovered poles lie on three concentric rings, one per level, all at $\sigma_\rho = \frac{1}{2}$. The selection of components in these runs compares the amplitude with the taper profile at the width $\theta - \frac{1}{2}$ of a zero on the line (two independent readings of the authors' computation agree), so that these runs presume the real part $\frac{1}{2}$ in the selection step.

12.8.4 The level as a cost parameter

In a separate run the pencil was run at $\theta = 1$, $\frac{3}{4}$, 0.6, 0.55, 0.52 and 0.51 on the window about 200, with the grid spacing reduced as the level drops so that the leading aliasing error $e^{-(\theta-1/2)(2\pi/h-\Omega)}$ of the transform stays below 3×10^{-12}: $h = 0.02$ down to 0.6, then 0.01, 0.004 and 0.002.

θ	h	data points	position error, median	error in σ_ρ, median
1	0.02	5001	1.8×10^{-10}	2.3×10^{-10}
3/4	0.02	5001	1.5×10^{-11}	1.3×10^{-11}
0.6	0.02	5001	8.0×10^{-12}	7.2×10^{-12}
0.55	0.01	10001	3.7×10^{-12}	4.2×10^{-12}
0.52	0.004	25001	2.8×10^{-12}	3.9×10^{-12}
0.51	0.002	50001	9.9×10^{-13}	1.4×10^{-12}

The error falls by a factor twenty from level 1 to level 0.6 and then only slowly, by another factor eight down to 0.51, while the number of data points, and with it the cost of the data, grows like $1/(\theta - \frac{1}{2})$. The gain presumably saturates because the pencil's error is set by the lower frequencies, where the signal is large at every level, and not by the top of the band where the damping $e^{-(\theta-1/2)\Omega}$ differs most (heuristic). The level is a weak parameter: anything in $[0.55, 0.75]$ is within a factor of twenty of the best, and $\theta = 0.6$ with the grid 0.02 is a good default, costing the same data as level 1 and giving twenty times the accuracy. There is

no level below which the data become as expensive as finding zeros directly, since Arb evaluates ζ at a point of the strip at about the cost of a point on the line at the same height; what grows is the number of points, and only linearly.

12.8.5 Many windows

Windows are combined by the simplest rule: tapers of width $\ell = 10$ centred every 10 units, each zero taken from the window whose centre is nearest to it, with half a unit of overlap. On $[100, 400]$ at level 0.6 this is 31 windows, and all 173 zeros in the range are recovered (the count agrees with $N(400) - N(100) = 173$), with median position error 2.4×10^{-12}, maximum 2.6×10^{-11}, and similar figures for the real parts; the pencils take four seconds, the data on $[40, 460]$ at spacing 0.02 seven. Seven spurious components survive the amplitude filter, all at the edges of windows and all with amplitude at the lower limit of the filter; a stricter filter, or the agreement of two overlapping windows, should remove them. The cost to height T is about T/h evaluations of ξ'/ξ off the line, fifty per unit height at $h = 0.02$, each about as expensive as an evaluation of ζ at that height, plus $T/10$ pencils whose order grows like $\log T$ with the density of zeros; a sign change scan needs a few evaluations of Z per zero, on the line. Both counts grow like T up to logarithms at about the same cost per evaluation, so the constants decide, and they favour the direct methods by a wide margin; the pencil's advantage is the real parts, which the sign changes never see.

12.8.6 A window near the limit of the verified range

Platt and Trudgian [232] verify the hypothesis for $0 < \gamma \le 3\,000\,175\,332\,800$, and we ran the pencil on the window centred at $c = 3 \times 10^{12} + 100$, inside that range and near its limit, at level $\theta = 0.6$ with taper $\ell = 2$, chosen because the zeros there are 4.3 per unit height and a window of ± 6 holds about fifty of them. No prime sum is used: the data are values of ξ'/ξ at $\operatorname{Re} s = 0.6$. They are $u_{0.6}$ at 1001 points at spacing 0.02, each from the first two Taylor coefficients of ξ at $0.6 + iv$ in Arb at 160 bits, since the Riemann and Siegel sum has 7×10^5 terms there and the Gamma factor costs 46 bits of the working precision at this height. The pencil used $\omega \in [5, 85]$ at spacing 0.2, a Hankel matrix of order 200, and model order 90 by the singular value gap. This is the authors' computation in the source, not rerun.

It returned 52 components with amplitudes on the taper's profile within 6 of the centre, at mean spacing 0.2325 between the extreme ordinates, the density $\log(c/2\pi)/2\pi = 4.28$ per unit predicting 51 (the smooth count is 51.36). Every one of them has

$$\sigma_\rho = 0.5000000 \pm 1.4 \times 10^{-8},$$

the real part read from the modulus of the pole, which the fit does not constrain. The components are selected by comparing the moduli of their amplitudes with

the taper profile computed at the width $\theta - \frac{1}{2}$ of a zero on the line, so a zero far from the line would have been discarded by the selection; that the selection discarded no zero on the line is shown by the count of sign changes below. The ordinates are then certified on the line by sign changes of Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, computed in balls with ϑ from $\log\Gamma$ at 160 bits, on the intervals $[\gamma - 10^{-7}, \gamma + 10^{-7}]$: all 52 of the 52 intervals contain a sign change, which places a zero on the line within 10^{-7} of each pencil ordinate (the authors' certificate, not rerun). An independent scan of Hardy's function in ball arithmetic at 160 bits finds exactly 52 certified sign changes in $[c - 6, c + 6]$, at ordinates from $c - 5.91$ to $c + 5.95$, with brackets of width below 10^{-10}; the comparison with the pencil's ordinates, one by one, has not been made (the pencil's list is not available), and a sign scan cannot see two zeros inside one step, so the count 52 is not a proof of completeness. The winding numbers that closed the certificate near height 200 did not complete here. Arb's rigorous integrator, fed ξ'/ξ on a circle of radius 0.03 about the pencil's zero nearest the centre, ran for over an hour at 3.8 seconds an evaluation without converging, and the alternative of enclosing ξ on arcs of the circle by ball evaluation failed because Arb's enclosure of ζ at a ball argument of radius 10^{-4} at this height already contains zero: the derivative bounds it uses are not tuned to the Riemann and Siegel range. So at 3×10^{12} a zero on the line within 10^{-7} of each of the 52 pencil ordinates is certified; that these are all the zeros in the window is not, and would follow from a winding number or from Turing's method [270] with the explicit bounds for $S(T)$ [266], and which the pencil's amplitude profile supports without proving. A rigorous argument principle at large height needs certified nonzero values of ζ on a mesh of the circle together with an explicit bound for $|\zeta'|$ on a small disc, available from the explicit convexity bounds, and is the natural next step.

This is a demonstration that the pencil runs at a height near the limit of the rigorous verification, recovers the zeros of a window from data on the line $\operatorname{Re} s = 0.6$, and reports their real parts, which the fit does not constrain, although the selection of components presumes the real part $\frac{1}{2}$. It is not a finding about the zeros, and the window lies inside the verified range. Zeros at and far above this height have been computed before: Gourdon's unrigorous computation of the first 10^{13} zeros [87] reaches about 2.4×10^{12}, and Odlyzko's computations near the 10^{20}th and higher zeros [222, 224] located long runs of zeros on the line by the same sign change test used here (these heights and attributions are the source's and were not checked here). Nor is reading a real part a stronger test than counting: an off line pair shows up in Turing's method as two missing sign changes against the count from $N(T)$, and the pencil adds nothing to that test. The real parts $0.5000000 \pm 1.4 \times 10^{-8}$ come from a floating point pencil and are an estimate, not a certificate; what is certified is a sign change of Z near each ordinate, the standard verification. Since the count in the window is not certified, an off line pair the pencil missed is not excluded by anything computed here.

What the run establishes is narrower: the deconvolution route of this chapter, from data on $\operatorname{Re} s = 0.6$ to positions and real parts, works at height 3×10^{12}, with real parts within 1.4×10^{-8} of $\frac{1}{2}$, at a cost of half an hour per window, and its output agrees with the authors' sign change certificate for every zero it reports.

12.8.7 Above the verified range

The verified range of Platt and Trudgian ends at $3\,000\,175\,332\,800$, and the window centred at $c = 10^{13} + 100$ lies above it, by a factor 3.33 in height. We repeated the computation there, generating the data and running the pencil, with an independent scan of Hardy's function. The setting is that of the previous subsection: level 0.6, taper $\ell = 2$, 1001 values $u_{0.6}(v_k)$ at $v_k = c - 10 + 0.02k$, each a ball at 160 bits from the first two Taylor coefficients of ζ at $0.6 + iv_k$ (the largest radius is 1.6×10^{-27}, and evaluating the first point of every task again at 200 bits changes it by at most 1.7×10^{-47}), at 2.6 seconds a point on one core, 2634 core seconds in all; the pencil uses $\omega \in [5, 85]$ at spacing 0.2 (401 samples) and a Hankel matrix of order 200 in double precision, in 0.7 seconds. The model order is 132, the number of singular values above ten times the plateau of the spectrum: the spectrum decays gradually, a gap rule is not usable, and a smaller order loses zeros.

The pencil returns 53 components with amplitude on the taper's profile within 6 of the centre, at ordinates from $c-5.7$ to $c+5.8$, the smooth count of the window being 53.66; the ratio of the amplitude to the profile is 1 to within 1.6×10^{-7} for all 53. The real part of each pole is read from its modulus, and the filter that selects the components compares the amplitude with the taper profile computed at the pole's own width, so that no real part is assumed (this is the difference from the runs of the source, whose selection uses the width $\theta - \frac{1}{2}$). The largest $|\sigma_\rho - \frac{1}{2}|$ over the 53 poles is 4.7×10^{-9} (median 7.1×10^{-11}), an estimate from a double precision pencil and not a certificate.

The scan finds 53 certified sign changes of Hardy's function in $[c - 6, c + 6]$: each step of 0.02 has opposite signs of Z at its ends in ball arithmetic at 160 bits, and the bisection brackets of width 10^{-10} were certified at both ends again at 192 bits. Every pencil ordinate lies within 10^{-7} of the midpoint of a certified bracket, and every certified bracket has a pencil ordinate: all 53 are matched, no pencil component and no sign change is left over, and the largest ordinate difference is 3.9×10^{-9} (median 5.7×10^{-11}). The result is stable: with the model order changed by ± 5, 53 components remain, the ordinates move by at most 3.1×10^{-9} and the real parts by at most 2.1×10^{-9}. Against the scan, the largest ordinate difference over the model order is 1.1×10^{-3} at order 70, 1.9×10^{-4} at 80, 4.8×10^{-6} at 90, 5.4×10^{-9} at 95 and 4.9×10^{-9} at 100, and below order 65 the pencil finds only between 8 and 49 of the 53 zeros. The same pencil was validated on known zeros, at $c = 200$ (seven zeros) and $c = 1000$ (ten zeros): all are matched, the largest ordinate difference is 7.1×10^{-15} and 1.4×10^{-14}, and the largest $|\sigma - \frac{1}{2}|$ is 1.7×10^{-14} and 2.3×10^{-14}.

Status. Computed: the 53 components, their ordinates and real parts. Certified: 53 zeros of ζ lie on the critical line in $[c-6, c+6]$, each located to 10^{-10} (this is a certified computation above the height verified in the literature, but it certifies only these 53 zeros on the line). Not certified: the count (a sign scan cannot see two zeros inside one step of 0.02, the smallest gap between consecutive zeros found is 0.0326, so a pair closer than the step is not excluded, and no Turing method or winding number was run; the pencil independently returns the same 53 and no further pole on the amplitude profile, which supports the count without proving it), and the real parts (the 4.7×10^{-9} is an estimate; it says that no component of the window is off the line by more than the pencil's error, and it is not a proof of the Riemann hypothesis in the window: a zero outside the window, or hidden in a component of the profile, is not excluded). Nothing is certified off the line. The data enter the pencil as midpoints in double precision, an error of about 10^{-16}, and the accuracy falls at the edge of the taper: the further 14 poles kept at 6 to 7.5 from the centre, outside the window and not compared with the scan, have $|\sigma - \frac{1}{2}|$ up to 2.6×10^{-8}. No priority is claimed: unrigorous computations of zeros at and far above this height exist [87, 222, 224], and the sign change scan is the standard method; what this subsection records is that the deconvolution route of this chapter, from data on $\operatorname{Re} s = 0.6$, reproduces the certified zeros of a window of this height with real parts read within 5×10^{-9} of $\frac{1}{2}$.

12.9 Open problems

Problem 12.1. Rigorous pencil. The certification of Section 12.8.1 goes through the argument principle after the fact. A verified pencil in the sense of Rump would need, besides verified linear algebra on ball data, a perturbation bound for the pencil under the model error (the far zeros and the remainder of Theorem 12.2.1), which makes the Hankel matrix of exact rank only in the limit. State and prove that bound, and compare the width of the resulting enclosures with the winding number route. Status: open.

Problem 12.2. Spurious components. At the edges of overlapping windows a few components pass the amplitude filter without being zeros. Characterise them: they should be the pencil's representation of the taper's tail of the next zeros outside the window, with poles close to the genuine ones but amplitudes off the profile. A rule based on the agreement of two windows, or on the phase of the amplitude, which Theorem 12.2.1 predicts as $e^{i(\omega_0 - a/\ell^2)(\gamma - c)}$, should remove all of them; under explicit hypotheses on the separation and the amplitudes, prove that it does. Status: open.

Problem 12.3. Higher and wider. The runs near and above the verified range covered one window each, of fifty two and fifty three zeros. The cost per point is the Riemann and Siegel sum, $t^{1/2}$ terms, and Hiary's method [97] would bring it

to $t^{1/3+o(1)}$; the number of points per zero is fixed by the taper and the spacing. Make the argument principle rigorous at large height, with an explicit bound for $|\zeta'|$ on a small disc from the explicit convexity bounds, so that the count in the window is certified and not only the existence on the line. Estimate the height at which a window of the pencil, with its full certificate, is feasible within a day, and run it; and determine whether the many window scheme at a fixed height, say 10^3 zeros above $3\,000\,175\,332\,800$, is competitive with Turing's method for that range when the real parts are wanted as output and not only tested. Status: open.

Problem 12.4. The adaptive prior in full. Place the sine kernel process on marked distinct ordinates, a pair counting as one point marked by its displacement, and implement the posterior on the number, the positions and the marks of the zeros in a window, with the spike and slab on the marks, by reversible jump or by a determinantal birth and death sampler started at the pencil; measure on the many window run whether it removes the spurious components. State the count with its multiplicity (a double zero counts twice), and report the posterior probability of each count, conditional on the model, separately from the argument principle certificate, which it does not replace. Status: open.

Problem 12.5. The constant in the rate. Theorem 12.5.2 gives $\delta^* = (z_\alpha \tau / \|t_\perp\|)^{1/2}$, with τ the standard deviation per real coordinate, and $\|t_\perp\|$ depends on the taper, the bandwidth and the level. Find the design that maximises $\|t_\perp\|/\tau$, the information and not $\|t_\perp\|$ alone, for a given cost, and determine how it scales with the height, through the gaps to the neighbouring zeros, which enter $t_\perp$ through the projection, and not through the mean density alone. Status: open.

Chapter 13

From level one to the critical line: particle filters and nested sampling

Under the Riemann hypothesis the level one density of Chapters 10 to 7 is the Cauchy smoothing of the zero counting measure, $u_1 = P_{1/2} * \mathfrak{U}_{1/2}$ (equation (12.1.2)), and $\mathfrak{U}_\theta$ at every level $\theta \in (\frac{1}{2}, 1)$ is the same smoothing at scale $\theta - \frac{1}{2}$. Can the deconvolution from $\mathfrak{U}_1$, which the Euler product makes accessible, to $\mathfrak{U}_{1/2}$, the object of the Riemann hypothesis, be carried out, and can the particle filters of [158], accelerated by generative Bayesian computation (GBC) [156], make progress on it? We recall the identities that relate the two measures (Section 13.1), build a particle filter whose time is the level, running the Cauchy flow backwards from $\theta = 1$ towards $\theta = \frac{1}{2}$ in a window of heights, with a GBC screen that replaces most of its work by one pass of a network, and record the gap between local computation and a statement for all heights (Section 13.10).

This chapter is based on [130]. The zeros and counts of the Davenport and Heilbronn function, the contour moments, the values of the density and the negative masses, the information numbers, the linear programs and polynomial tables and the nested sampling scores were recomputed for this book; where they differ from the note the recomputed value is printed. The runs of the particle filter, the GBC networks, the concentration experiments and the nested sampler are the note's and were not rerun; they are marked computed by the author. Statements at the level $\frac{7}{8}$ are conditional on QRH($\frac{7}{8}$), announced by OpenAI [227] and formalised in Lean but not refereed (Section 17.7).

13.1 What the book already proves

The notation is that of Section 10.1: zeros $\rho = \sigma_\rho + i\gamma_\rho$ with multiplicity, levels θ, and $u_\theta(v) = \pi^{-1}\,\mathrm{Re}(\xi'/\xi)(\theta + iv) = \pi^{-1}\partial_\sigma \log|\xi(\sigma + iv)||_{\sigma=\theta}$ the density of Theorem 10.1.1. The measure $\mathfrak{U}_\theta$ of equation (10.1.1) lives on the line in the variable v, with $\mathfrak{U}_\theta = 2\widetilde{U}_\theta$ (Theorem 17.14.4); the symbol U_θ is kept for the Thorin measure in $x = v^2$ (Section 17.14). With $P_a(x) = (a/\pi)/(a^2 + x^2)$ for

$a > 0$ we write $Q_a(x) = (a/\pi)/(a^2 + x^2)$ for every real $a \neq 0$, so $Q_a = -P_{|a|}$ when $a < 0$, and $Q_0(x) = 0$ for $x \neq 0$. For θ not the real part of a zero at the height in question the Hadamard sum (equation (12.1.1), and the proof of Theorem 11.9.1) reads

$$u_\theta(v) = \sum_\rho Q_{\theta-\sigma_\rho}(v - \gamma_\rho), \tag{13.1.1}$$

absolutely convergent because $\sum_\rho |\gamma_\rho|^{-2} < \infty$. Positivity of u_θ at every level $\theta > \sigma_0$ is the monotone increase of $|\xi(\sigma + iv)|$ in $\sigma > \sigma_0$ studied in [259, 121].

Remark 13.1.1 (a level through a zero). At a level $\theta = \sigma_\rho$ the term $Q_{\theta-\sigma_\rho}(v-\gamma_\rho)$ of (13.1.1) has no meaning at $v = \gamma_\rho$. For every $\theta > \frac{1}{2}$ we take $\mathfrak{U}_\theta$ to be the vague limit of $u_{\theta'}(v)\,dv$ as $\theta' \downarrow \theta$. Each term converges vaguely, and the far terms are dominated by $|\gamma_\rho|^{-2}$ uniformly for θ' in a compact set, so the limit exists; it is the measure of equation (10.1.1) with an atom at the ordinate of every zero with $\sigma_\rho = \theta$ and a negative Cauchy term $Q_{\theta-\sigma_\rho}$ for every zero with $\sigma_\rho > \theta$. At $\theta = \frac{1}{2}$ it is $\mathfrak{U}_{1/2} = \sum_{\sigma_\rho = 1/2} \delta_{\gamma_\rho}$ plus the contribution of the zeros off the line, which vanishes (Proposition 13.2.1).

Proposition 13.1.2 (Cauchy transport). *For $\theta > \frac{1}{2}$ and $\theta + iv$ not a zero,*

$$u_\theta = P_{\theta-1/2} * \mathfrak{U}_{1/2} + R_\theta, \qquad R_\theta(v) = \sum_{\sigma_\rho > 1/2} \big[Q_{\theta-\sigma_\rho} + Q_{\theta-1+\sigma_\rho}\big](v - \gamma_\rho), \tag{13.1.2}$$

the sum over the zeros off the critical line, each taken together with its reflection $1 - \bar\rho$, which has the same ordinate. In particular $u_1 = P_{1/2} * \mathfrak{U}_{1/2} + R_1$: the density at level one is the Cauchy smoothing of $\mathfrak{U}_{1/2}$ at scale $\frac{1}{2}$ plus one Cauchy pair $P_{1-\sigma_\rho} + P_{\sigma_\rho}$ for each such zero. The pair is positive at $\theta = 1$ and is a signed term only for $\theta < \sigma_\rho$.

Proof. Split the sum (13.1.1) into the zeros on the line, each contributing $P_{\theta-1/2} * \delta_{\gamma_\rho}$, and the pairs ρ, $1 - \bar\rho$ off it. □

This is Theorem 17.14.4 above the abscissa of the last zero, and equation (12.1.2) under RH, where $R_\theta = 0$. In Fourier terms a zero on the line contributes $e^{-(\theta-1/2)|\omega|}e^{i\gamma\omega}$ and a pair at distance $\delta = \sigma_\rho - \frac{1}{2}$ contributes $2\cosh(\delta\omega)\, e^{-(\theta-1/2)|\omega|}e^{i\gamma\omega}$ for $\theta > \sigma_\rho$; deconvolving by $P_{1/2}$ from $\theta = 1$ leaves $2\cosh(\delta\omega)$, which is not the Fourier transform of a tempered distribution (Chapter 7). We use without proof the deconvolution criterion (Theorem 8.2.2(b), Proposition 7.2.2), Thorin positivity as a zero free half plane (Theorem 11.9.1, after Lagarias [185] and Sondow and Dumitrescu [259]; positivity at a single level is necessary and not known to be sufficient, Remark 2.1.4), unconditional positivity for $\theta \geq 1$ (Theorem 10.1.1, Corollary 17.14.3), and the pairing of Proposition 3.2.1(ii) (Corollary 14.3.7(1), Lemma 11.10.1). The pairing also gives a local statement for zero free regions that narrow with the height, to which Theorem 11.9.1 does not apply.

Corollary 13.1.3 (local positivity from a zero free region). *Let $\eta : [0,\infty) \to (0, \frac{1}{2}]$ be nonincreasing and suppose that every zero satisfies $\sigma_\rho \le 1 - \eta(|\gamma_\rho|)$. Then $u_\theta(v) > 0$ whenever $\theta > \frac{1}{2}$, $\theta \ge 1 - \eta(|v| + \frac{1}{2})$ and $\xi(\theta + iv) \ne 0$.*

Proof. Group the zeros in (12.1.1) into orbits of $\rho \mapsto 1 - \bar\rho$. As in the proof of Corollary 14.3.7(1), a zero on the critical line contributes $P_{\theta-1/2}(v - \gamma_\rho) > 0$, an orbit with $\sigma_\rho \le \theta$ a nonnegative amount, and an orbit with $\sigma_\rho > \theta$ is negative only where $|v-\gamma_\rho| < \sqrt{(\sigma_\rho - \theta)(\sigma_\rho + \theta - 1)} < \frac{1}{2}$. Such a zero would have $|\gamma_\rho| < |v| + \frac{1}{2}$, hence $\sigma_\rho \le 1 - \eta(|\gamma_\rho|) \le 1 - \eta(|v| + \frac{1}{2}) \le \theta$, a contradiction. So every orbit contributes a nonnegative amount, and the zero $\frac{1}{2} + i\gamma_1$ a positive one. □

With the region of de la Vallée Poussin ($\eta(T)$ a constant over $\log T$, Proposition 12.1.4) or of Vinogradov and Korobov ($\eta(T) = c(\log T)^{-2/3}(\log\log T)^{-1/3}$ [272]), taken as $\frac{1}{2}$ at small heights, this is positivity of u_θ at height v for θ within that width of $\operatorname{Re} s = 1$, the width taken at height $|v| + \frac{1}{2}$. It is local in v, as it must be.

The logarithmic form of the transport is the formula of Balazard, Saias and Yor [20], proved at every level in Theorem 14.2.1; here it enters only through Proposition 13.8.3. Since Cauchy(γ, a) is the harmonic measure of a half plane (Sections 10.5 and 12.2.3), the forward flow is a Markov semigroup; the backward flow has no positive kernel, and the natural particle method for it is a Bayesian filter, not a forward simulation.

13.2 Two test functions

Proposition 13.2.1 (the two ends). *(a) (cited) For $\theta \ge 1$, unconditionally, $u_\theta(v) = \sum_\rho P_{\theta-\sigma_\rho}(v - \gamma_\rho) > 0$; in particular $\mathfrak{U}_1 = \sum_\rho P_{1-\sigma_\rho}(\cdot - \gamma_\rho)$ is a positive mixture of Cauchy laws centred at the ordinates, with scales in $(0,1)$. This is Theorem 10.1.1 and Corollary 17.14.3, and it holds for ζ because $\zeta(1+it) \ne 0$.*

(b) (proved) As $\theta \downarrow \frac{1}{2}$, in the sense of distributions, $u_\theta \to \mathfrak{U}_{1/2} = \sum_{\sigma_\rho = 1/2} \delta_{\gamma_\rho}$. Zeros off the line are invisible in $\mathfrak{U}_{1/2}$: a pair σ_ρ, $1-\sigma_\rho$ contributes $Q_{\theta-\sigma_\rho} + Q_{\theta-1+\sigma_\rho} \to Q_{1/2-\sigma_\rho} + Q_{\sigma_\rho-1/2} = 0$.

Proof. (b) For a zero on the line $P_{\theta-1/2} \to \delta_0$. For a pair, $\theta - \sigma_\rho$ and $\theta - 1 + \sigma_\rho$ tend to $\frac{1}{2} - \sigma_\rho \ne 0$ and $\sigma_\rho - \frac{1}{2} \ne 0$, and $a \mapsto Q_a$ is continuous from $\mathbb{R} \setminus \{0\}$ to $L^1(\mathbb{R})$, so the pair terms converge to $-P_{\sigma_\rho-1/2} + P_{\sigma_\rho-1/2} = 0$. The far terms are bounded by $|\gamma_\rho|^{-2}$ uniformly for $\theta \in [\frac{1}{2}, 1]$, which gives dominated convergence on compact sets. □

In the variable $x = v^2$, $U_{1/2} = \sum_\gamma \delta_{\gamma^2}$ is the discrete measure of the RH programme.

All computations of this chapter use ζ and the Davenport and Heilbronn function, which has no Euler product and has zeros off the critical line. With $\kappa = (\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1) = 0.28407904384\ldots$ it is [60, 264]

$$f(s) = 5^{-s}\Big[\zeta(s,\tfrac{1}{5}) + \kappa\,\zeta(s,\tfrac{2}{5}) - \kappa\,\zeta(s,\tfrac{3}{5}) - \zeta(s,\tfrac{4}{5})\Big],$$

and the completed function $\Lambda(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$ is entire and satisfies $\Lambda(s) = \Lambda(1-s)$, the identity of Davenport and Heilbronn [60] (see [264]); as a diagnostic the relative difference $|\Lambda(s) - \Lambda(1-s)|/|\Lambda(s)|$ is at most 1.1×10^{-87} at four points in ball arithmetic. It is the control (Section 21.11.2). The zeros of f off the line below height 200 were found by Spira [261]; refined by Newton's method in ball arithmetic they are

$$0.8085171825 + 85.6993484854\,i, \qquad 0.6508300806 + 114.1633427308\,i,$$
$$0.5743560505 + 166.4793059132\,i, \qquad 0.7242576946 + 176.7024612429\,i,$$

together with their reflections in the critical line, each certified to be the only zero in a square of half width 10^{-16} about the centre, so that all ten printed decimals are correct. These are all the zeros off the line below height 200 (computed, certified in ball arithmetic): Λ has 130 zeros with $0 < \gamma < 200$ by the argument principle on a rectangle, and $\Lambda(\frac{1}{2} + it)$ has 122 sign changes on the segment, with certified signs on a grid of step $\frac{1}{50}$; the difference $8 = 2 \cdot 4$ accounts exactly for these four zeros and their reflections, so no zero on the line is missed or multiple and there is no other zero off the line.

For Λ the density is $u_\theta = \pi^{-1}\operatorname{Re}\Lambda'/\Lambda(\theta + iv)$. The Hadamard sum (13.1.1), Proposition 13.1.2 and the deconvolution criterion with the base line $\operatorname{Re} s = 1$ replaced by $\operatorname{Re} s = 2$ apply to it ($|f(s) - 1| \le \zeta(2) - 1 < 1$ there, as $|\kappa| < 1$); Proposition 13.2.1(a) does not, since f has infinitely many zeros in $\operatorname{Re} s > 1$ [60] (Proposition 13.8.3). A method that detects zeros off the line of ξ must detect them for Λ (Section 21.11.2).

For ζ the densities u_θ on the window $[60, 120]$ stay positive at every level computed (the zeros of ζ at these heights are on the line [232]) and sharpen into spikes at the ordinates as θ decreases ($R_\theta = 0$). For Λ the densities at $\theta = 1$ and $\theta = 0.9$ are positive on this window (computed; the zeros of f in the window have $\sigma_\rho \le 0.8085$). They turn negative near $v = 85.70$ once $\theta < 0.8085$ and near $v = 114.16$ once $\theta < 0.6508$: at the ordinates the minima of u_θ over $[60, 120]$ are -36.8 at $\theta = 0.8$ and -4.96 at $\theta = 0.6$ (computed; the grid values -36.6 and -4.6 of the note are grid minima). This is the signed defect R_θ of Proposition 13.1.2.

13.3 A particle filter along the Cauchy flow

13.3.1 The state space model

Fix a window of heights. The parameter is the local configuration $z = \{(\gamma_j, \sigma_j)\}_{j=1}^{K}$ of the zeros with ordinates in a slightly larger window, with K fixed by the argument principle (a contour integral of ξ'/ξ, computed once). The observation at level θ_k is the vector $y_k = (u_{\theta_k}(v_i))_i$ on a grid, computed to high precision. Equation (13.1.1) gives the observation equation

$$y_k(v) = \sum_{j=1}^{K} Q_{\theta_k - \sigma_j}(v - \gamma_j) + b_k(v) + \varepsilon_k(v),$$

where b_k is the contribution of the zeros outside the window. For a distant zero the pair terms equal $2(\theta - \frac{1}{2})/(\pi(v-\gamma)^2)$ to leading order, so we model $b_k(v) = (\theta_k - \frac{1}{2})(c_0 + c_1 w + c_2 w^2 + c_3 w^3) + (\theta_k - \frac{1}{2})^3 c_4$ with w the centred height. The coefficients c enter linearly and are integrated out in closed form under a Gaussian prior, by the Woodbury identity, so the particles carry only (γ_j, σ_j). The error $\varepsilon = 0.01$ is a tolerance, not noise: the data are deterministic, and ε absorbs the background approximation. Every statement below about posteriors, information and coverage refers to this Gaussian tolerance likelihood.

The levels $1 = \theta_1 > \theta_2 > \cdots > \theta_n$ play the role of time. With the background coefficients shared by the levels and integrated out jointly, the filtering distributions are

$$\pi_k(z) \;\propto\; \pi_0(z)\, p(y_1, \ldots, y_k \mid z),$$

the Feynman Kac flow of [158] with potentials $G_k = p(y_k \mid y_1, \ldots, y_{k-1}, z)$ and Markov moves invariant for π_k. The prior is uniform: ordinates uniform in the window and every σ_j uniform on $(0,1)$, so that the functional equation is not imposed on the particles (Section 13.3.4).

13.3.2 Algorithm

1. Draw N particles from the prior.
2. For $k = 1, \ldots, n$: move from π_{k-1} to π_k through the tempered targets $\pi_{k-1}^{1-\lambda}\pi_k^{\lambda}$, choosing each increment of λ by bisection so that the effective sample size of the incremental weights is $N/2$; resample systematically; rejuvenate by componentwise random walk Metropolis moves with scales set from the current particle cloud.
3. Report the posterior of every σ_j, and of $\delta^* = \max_j |\sigma_j - \frac{1}{2}|$ over the ordinates in the interior of the window.

This is the sequential Monte Carlo sampler of [40, 55] with the Cauchy level as time (compare Sections 12.6, 10.4 and 10.6).

13.3.3 Results for ζ

The runs of this section and the next are the author's (not rerun). We used $N = 1000$ particles, the window $[78, 94]$ with 401 grid points, $K = 10$ zeros in $[74, 98]$, and levels $\theta \in \{1, 0.9, 0.8, 0.7, 0.65, 0.6\}$. Table 13.1 gives the posterior for the zeros in the data window and Table 13.2 the posterior of δ^* as levels are added. The ordinates are recovered to within 10^{-4} and every σ_j is pinned at $\frac{1}{2}$ to within 4×10^{-4}. The bound on δ^* falls by a factor of fourteen between $\theta = 1$ and $\theta = 0.6$. Each σ_j is a free parameter, so the information is first order, $\int (\partial_a P_a)^2 \, dx = 1/(4\pi a^3)$, and $(\sum_l a_l^{-3}/a_1^{-3})^{1/2} = 13.6$ for these levels (Section 13.3.4); Proposition 13.3.1, which concerns a symmetric pair, would give 7.8. Zeros just outside the data window are poorly identified nuisance parameters.

posterior median γ	posterior median σ_j	95% interval for σ_j
79.3373	0.49999	[0.49964, 0.50036]
82.9104	0.50000	[0.49966, 0.50035]
84.7355	0.50002	[0.49967, 0.50038]
87.4253	0.49999	[0.49964, 0.50034]
88.8091	0.49999	[0.49961, 0.50036]
92.4919	0.50001	[0.49967, 0.50037]

Table 13.1: Particle filter posterior for the zeros of ζ in the data window $[78, 94]$ after all levels down to $\theta = 0.6$ (computed by the author). The true ordinates are 79.3374, 82.9104, 84.7355, 87.4253, 88.8091 and 92.4919.

levels absorbed down to $\theta =$	1	0.9	0.8	0.7	0.65	0.6
median of δ^* ($\times 10^{-3}$)	3.65	2.15	1.31	0.73	0.46	0.26
95% quantile of δ^* ($\times 10^{-3}$)	6.43	3.69	2.33	1.34	0.85	0.46
likelihood evaluations ($\times 10^6$)	1.86	2.46	2.95	3.43	3.79	4.15

Table 13.2: The backward Cauchy flow as a filter: the posterior of $\delta^* = \max|\sigma_j - \frac{1}{2}|$ over the zeros of ζ in $[80, 92]$ as levels are added, with cumulative cost (computed by the author). The bounds measure the displacement of one unpaired zero from the line, which the functional equation forbids.

For a pair $\frac{1}{2} \pm \delta$ the same data bound δ only at about 4.4×10^{-3} (one sided 95%, second order Gaussian approximation, six levels; 0.0345 after level one alone), ten times weaker than the bound on an unpaired displacement (computed). A rerun with the pair parametrisation of Section 12.5 (Lemma 12.5.1, Theorem 12.5.2), each ordinate on the line or a pair $\frac{1}{2} \pm \delta$ with $\eta = \delta^2 \geq 0$, has not been done.

The cost is the obstacle: 4.15×10^6 likelihood evaluations for a window of length 16 with six zeros. For the Davenport and Heilbronn window, with seventeen zeros

and a uniform prior, the tempering stalled at $\lambda \approx 0.003$ after several hundred thousand evaluations: the posterior is so concentrated relative to the prior that the bridge needs thousands of steps. GBC addresses this (Section 13.4).

13.3.4 How much information each level carries

Proposition 13.3.1 (detectability). *Let $a = \theta - \frac{1}{2} > \delta > 0$. At the centre $v = \gamma$, a pair of zeros at $\frac{1}{2} \pm \delta + i\gamma$ exceeds a double zero on the line at $\frac{1}{2} + i\gamma$ by*

$$\Delta(a,\delta) = \frac{1}{\pi}\Big(\frac{1}{a-\delta} + \frac{1}{a+\delta}\Big) - \frac{2}{\pi a} = \frac{2\delta^2}{\pi a(a^2-\delta^2)}.$$

In the parameter $\eta = \delta^2$ the pair differs from the double zero by $\eta\,\partial_a^2 P_a + O(\eta^2)$, and

$$\int_{\mathbb{R}} \big(\partial_a^2 P_a(x)\big)^2 dx = \frac{3}{4\pi a^5}.$$

Proof. Direct computation, with $\partial_a^2 P_a(x) = 2a(a^2-3x^2)/\big(\pi(a^2+x^2)^3\big)$ and the scaling $P_a(x) = a^{-1}P_1(x/a)$. The integral was also checked by quadrature at 40 digits. □

For a symmetric pair (Section 12.5) the information about δ is second order, the Fisher information at $\delta = 0$ vanishing, so one works with $\eta = \delta^2$; and it grows like a^{-5} as the level approaches the line. With observation error ε on a grid of spacing Δv, the level a resolves δ once roughly $\delta^4 \gtrsim (4\pi/3)\,\varepsilon^2 \Delta v\, a^5$ (heuristic, for the pair model). Coarse levels are well posed and uninformative, fine levels informative and ill posed, a structure sequential Monte Carlo handles well: each new level sharpens the likelihood of a pair in η by a factor of order $((\theta_{k-1}-\frac{1}{2})/(\theta_k-\frac{1}{2}))^5$. The filter of this section does not impose the pairing, so its information is first order.

13.3.5 Information rates

For a single zero with free real part σ (no pairing imposed), $\partial_\sigma Q_{\theta-\sigma} = -\partial_a P_a$ and $\int (\partial_a P_a)^2 dx = 1/(4\pi a^3)$. With error ε on a grid of spacing Δv, the Fisher information after levels $a_1, \dots, a_k$, with $a_l = \theta_l - \frac{1}{2}$, is about $(4\pi\varepsilon^2\Delta v)^{-1} \sum_{l \le k} a_l^{-3}$, so the Bernstein and von Mises posterior standard deviation of the Gaussian tolerance model is

$$\mathrm{sd}(\sigma_j) \approx \varepsilon\sqrt{4\pi\Delta v}\,\Big(\sum_{l\le k} a_l^{-3}\Big)^{-1/2} \qquad \text{(heuristic)}.$$

Table 13.3 compares this with the filter: within 8% at the first level and exact at the last, so the filter matches the first order Fisher information of the model with free real parts; a deterministic tolerance likelihood gives no frequentist coverage, and the efficient information for the full multizero model with unknown

background is not that of an isolated zero, so no more is claimed. The ratios of the median bound of Table 13.2 to its value at level one (observed 1.70 to 1.74, 2.76 to 2.79, 4.80 to 5.00, 7.56 to 7.93, 13.98 to 14.04) follow the first order law, $(\sum_l a_l^{-3}/a_1^{-3})^{1/2} = 1.72,\ 2.75,\ 4.82,\ 7.76,\ 13.61$ for the levels 0.9 to 0.6, and not the law of a pair, $(\sum_l a_l^{-5}/a_1^{-5})^{1/4} = 1.42,\ 2.03,\ 3.27,\ 4.79,\ 7.77$ (computed).

levels absorbed down to $\theta =$	1	0.9	0.8	0.7	0.65	0.6
predicted sd of σ_j ($\times 10^{-4}$)	25.1	14.6	9.1	5.2	3.2	1.84
particle filter, mean over six zeros	27.1	15.3	9.4	5.3	3.3	1.84

Table 13.3: Posterior standard deviation of σ_j for the six zeros of ζ in $[78, 94]$, against the first order Fisher information formula ($\varepsilon = 0.01$, $\Delta v = 0.04$). The predicted row is computed; the filter row is the author's run.

When the functional equation is imposed, a zero on the line is compared with a symmetric pair and the information is second order (Proposition 13.3.1): the relevant quantity is $\sum_l a_l^{-5}$ and the resolvable δ scales like $a_{\min}^{5/4}$ (heuristic). Either way information comes from the levels nearest the line: the levels $\theta \geq 0.9$ supply 1.6% of $\sum a^{-3}$ and 0.1% of $\sum a^{-5}$ in this schedule.

Problem 13.1. What is the best possible resolution of the pair parameter $\eta = \delta^2$ from data at levels $\theta \geq \theta_{\min}$, and does the particle filter with the pair parametrisation of Section 12.5 reach it? (open)

13.4 Optimising the filter with generative Bayesian computation

13.4.1 Contour moments

Proposition 13.4.1 (contour moments). *Let $R = [\theta_0, \theta_1] \times [T_1, T_2]$ with $\xi \neq 0$ on ∂R, let $c \in \mathbb{C}$ and*

$$M_k(R) = \frac{1}{2\pi i} \oint_{\partial R} (s - c)^k \, \frac{\xi'}{\xi}(s)\, ds \qquad (k \geq 0).$$

Then $M_k(R) = \sum_{\rho \in R} (\rho - c)^k$, with multiplicity. In particular $K = M_0(R)$ is the number of zeros in R, and $M_1(R), \dots, M_K(R)$ determine them: with $e_0 = 1$ and $j e_j = \sum_{r=1}^{j} (-1)^{r-1} e_{j-r} M_r(R)$ for $1 \leq j \leq K$, the numbers $\rho - c$, $\rho \in R$, are the roots of $z^K - e_1 z^{K-1} + \cdots + (-1)^K e_K$.

Proof. The function $(s-c)^k \xi'(s)/\xi(s)$ is meromorphic inside R, with a simple pole at each zero ρ of multiplicity m, of residue $m(\rho - c)^k$; the residue theorem gives the first claim. The power sums $M_1, \dots, M_K$ of the K numbers $\rho - c$ determine their elementary symmetric functions $e_1, \dots, e_K$ by Newton's identities, which is

the recursion displayed, and these are the coefficients of the monic polynomial with those roots. □

Remark 13.4.2. This is the method of Delves and Lyness [64]. The moments use ξ'/ξ on the whole of ∂R, horizontal edges included, and give the zeros in R exactly; they are not functions of the filter's grid values, so not a sufficient statistic of its data. The real part of an analytic function on the two vertical edges does not determine it: with $a = \theta_0$, $b = \theta_1$ and $T_1 < t_0 < T_2$, the entire function $g(s) = \cos\big(\pi(s - a - it_0)/(b - a)\big)$ is real and nonzero on both vertical lines, g'/g is purely imaginary there, and g vanishes at $\frac{a+b}{2} + it_0 \in R$; so ξ and ξg have the same density on both vertical edges and different zeros in R. Within the class of functions like ξ, of order one with zeros in the strip, the density on one segment of the line $\operatorname{Re} s = 1$ does determine ξ'/ξ (Proposition 13.6.1(a)), but only through analytic continuation, which no finite grid of values carries.

function	rectangle	M_0	recovered zeros right of θ_0
ζ	$[0.6, 3] \times [80, 92]$	0 (radius 3×10^{-48})	none
ζ	$[0.55, 3] \times [60, 120]$	0 (radius 7×10^{-46})	none
Davenport and Heilbronn	$[0.6, 3] \times [80, 92]$	1 (radius 3×10^{-48})	$0.808517182457 + 85.699348485378\,i$

Table 13.4: Contour moments, computed in ball arithmetic at 128 and 192 bits; the two precisions agree, and the integrals are enclosed, so that each M_0 is an enclosure containing one integer. The note printed 7×10^{-22} and 0.006 for the first two entries, floating point quadrature residuals (the second from zeros at distance 0.05 from the edge); the exact values are 0. The zero of the last row is the Newton root with $K = 1$ and M_1 as an interval.

So in the noise free limit the deconvolution in a window is a handful of contour integrals, to twelve digits for the zero of the last row. The difficulty is that there are infinitely many windows. GBC aims to make each window nearly free, with quantified uncertainty, when the data are a grid of noisy values.

13.4.2 GBC: learned summary and quantile network

Following [156] and the semi automatic summaries of Fearnhead and Prangle [77] (Section 10.5), we simulate from the joint law of parameters and data, learn the conditional mean as a summary, and then learn the posterior quantile function given the summary.

Simulator. A window of half width 4 around a centre T; ordinates from a rigid point process (Gamma(4) spacings, density uniform on $[0.35, 0.8]$ per unit; the density of ζ is 0.35 at height 56.8 and lower below, so the windows of ζ centred at 30 to 55 lie outside the training range); with probability $\frac{1}{2}$ one zero in the central interval $|\gamma - T| \le 3$ is replaced by a symmetric pair at distance δ, mostly uniform on $[0.02, 0.45]$; with probability 0.2 a decoy pair outside the central interval; a random quadratic background proportional to $\theta - \frac{1}{2}$; observations u_θ on 161 points at $\theta \in \{1, 0.85, 0.7, 0.6\}$ with error 0.003. The target is $\delta^* = \max\{|\sigma_\rho - \frac{1}{2}| : |\gamma_\rho - T| \le 3\}$, equal to 0 when the central zeros lie on the line.

Stage one: summary. A network S (two hidden layers, 256 and 128 units; the code's input has 652 entries for the 644 grid values, and the note does not say what the other eight are) is trained by squared error to $S(y) \approx \mathbb{E}[(\delta^*, \mathbf{1}\{\delta^* > 0\}, \gamma^*) \mid y]$, with γ^* the ordinate of the displaced zero. The location loss is masked by $A = \mathbf{1}\{\delta^* > 0\}$, so the third output estimates $\mathbb{E}[A\gamma^* \mid y]/\mathbb{E}[A \mid y]$, and the second network receives the three outputs with eight hand built minima of the data, an eleven dimensional conditioning vector. By the argument of Fearnhead and Prangle, the conditional mean, when the network equals it, is the optimal summary for these targets under quadratic loss.

Stage two: posterior by noise outsourcing. A second network $Q(\tau, S(y))$ is trained by the pinball loss $\rho_\tau(\delta^* - Q)$ with $\tau \sim U(0,1)$ as an input. By noise outsourcing [104], $\delta^* \stackrel{d}{=} Q(U, S(y))$ with U uniform and independent, for the exact conditional quantile function, so a posterior draw costs one forward pass; the network approximates that function, and the posterior given $S(y)$ is that given y only if S is sufficient. The network is not constrained to be monotone in τ.

Training and calibration. We use 120,000 training and 20,000 test simulations. The root mean square error of the first output of S for δ^* is 0.016. Calling a window flagged when the posterior median exceeds 0.03 gives a true positive rate of 0.978 and a false positive rate of 0.002; detection is essentially perfect for $\delta^* \ge 0.05$ and falls to 0.67 for $\delta^* \in [0.02, 0.05]$ (computed by the author). Coverage of the posterior quantiles is in Table 13.5.

nominal τ	0.05	0.25	0.50	0.75	0.95
empirical	0.049	0.182	0.467	0.725	0.917

Table 13.5: Coverage of the GBC posterior quantiles of δ^* on held out simulations (computed by the author). The middle quantiles undercover, an approximation and calibration error of the quantile network; at an atom exact quantiles overcover. The upper tail is slightly short and is the place to add conformal calibration.

13.4.3 Screening real windows

We applied the networks to 35 windows centred at $T = 30, 35, \dots, 200$ for each function. Each window costs 644 evaluations of ξ'/ξ (of Λ'/Λ for the Davenport and Heilbronn function) and then 6.7 milliseconds of network time. Table 13.6 lists the flagged windows.

window centre	GBC location	true zero	GBC median δ^*	true δ
85	85.68	$0.8085 + 85.6993\,i$	0.314	0.3085
115	114.08	$0.6508 + 114.1633\,i$	0.143	0.1508
165	166.47	$0.5744 + 166.4793\,i$	0.086	0.0744
175	176.72	$0.7243 + 176.7025\,i$	0.211	0.2243
180	176.58	same zero, at the edge	0.361	0.2243

Table 13.6: Davenport and Heilbronn windows flagged by GBC (computed by the author; the zeros are those of Section 13.2). The four zeros off the line below height 200 are flagged, and located to within 0.1 in the four windows that target them. The window at 180 sees the zero at 176.70 near its edge, outside its target interval $[177, 183]$, and overstates δ: a boundary false alarm. No other window of either function is flagged; for all 35 windows of ζ the posterior median is 0 and the 95% quantile is at most 0.007.

13.4.4 GBC inside the particle filter

The GBC posterior is fast but approximate; the particle filter targets the model posterior but is slow. We use the learned posterior as the starting distribution of the particle system.

1. Screen every window with GBC.

2. In a flagged window, build a proposal q: ordinates near the peaks of u_{θ_n} at the lowest level for zeros on the line, a pair near each negative dip with δ drawn from the GBC posterior by noise outsourcing, and diffuse components for the rest.

3. Run the tempered sampler from q to the posterior, through the bridge $q^{1-\lambda}(\pi_0 L)^{\lambda}$, whose potentials are powers of $\pi_0 L/q$ and so involve $\log \pi_0 - \log q$, with the same resample and move steps as in Section 13.3.

Step 3 works far better when the proposal is first moved to the posterior mode by a Gauss Newton fit started from the GBC output, with q Gaussian with covariance four times the inverse Gauss Newton Hessian.

For ζ the four zeros inside $[80, 92]$ are recovered as before, with 95% intervals for σ_j of half width 4×10^{-4} around $\frac{1}{2}$. For the Davenport and Heilbronn function the

sampler	window	steps	evaluations	seconds
plain filter, uniform prior, ζ	$[78, 94]$	many	4.15×10^6	minutes
GBC, Gauss Newton, filter, ζ	$[80, 92]$	7	3.6×10^4	3.6
plain filter, uniform prior, Davenport and Heilbronn	$[78, 94]$	stalled	n/a	stopped
GBC, Gauss Newton, filter, Davenport and Heilbronn	$[80, 92]$	1020	5.1×10^6	645

Table 13.7: Costs of reported runs (computed by the author). The two runs for ζ differ in window ($[78, 94]$ with six zeros against $[80, 92]$ with four), in zeros and in data, so their ratio of about 115 in evaluations is a comparison of two different runs and not a matched speedup. For Davenport and Heilbronn the GBC initialised filter succeeds where the plain filter stalls (at tempering exponent $\lambda \approx 0.003$), but the posterior is strongly non Gaussian because the level $\theta = 0.8$ lies within 0.0085 of $\sigma = 0.8085$, so the bridge from the Gaussian proposal still needs many steps.

filter finds the pair at $\gamma = 85.6993$ with $\sigma = 0.80851$ (interval $[0.80851, 0.80852]$) and $\sigma = 0.18994$ (interval $[0.18867, 0.19084]$). The first agrees with the true 0.808517 to 10^{-5}. The second misses the true 0.191483 by 0.0015, and the intervals for several zeros on the line exclude $\frac{1}{2}$ by up to 6×10^{-4}. Both are symptoms of misspecification: with 12 zeros in a window of length 18 the cubic background is too crude for $\varepsilon = 0.01$, and the posterior is overconfident. A larger zero window removes this for these data (end of the next subsection).

13.4.5 Three computational questions

Sine kernel statistics in the simulator. We replaced the rigid Gamma(4) spacing of the GBC simulator by eigenphases of Haar unitary matrices (the circular unitary ensemble), rescaled to the local density, whose local statistics approximate the sine kernel (heuristic; the matrix size is not stated in the note); the empirical spacing variance is 0.16 against 0.25 before. Retraining both networks took 285 seconds. On held out simulations of the new simulator the 95% quantile has coverage 0.952, against 0.917 for the rigid networks on theirs. On the real windows the screen again flags exactly the four Davenport and Heilbronn zeros, plus the edge window at 180, and nothing for ζ (largest posterior median 0.001). The estimates of δ improve about threefold (Table 13.8, computed by the author).

Conformal calibration. For the rigid networks, split conformal calibration on 10,000 fresh simulations shifts the 95% upper bound by an order statistic of the calibration residuals, 0.0015, and raises its coverage on 20,000 test simulations from 0.915 to 0.948. Conditional on an off line pair being present, coverage rises only from 0.885 to 0.906: marginal conformal calibration does not repair condi-

zero	0.8085 +85.70 i	0.6508 +114.16 i	0.5744 +166.48 i	0.7243 +176.70 i
true δ	0.3085	0.1508	0.0744	0.2243
rigid simulator	0.314	0.143	0.086	0.211
sine kernel simulator	0.312	0.149	0.079	0.228

Table 13.8: GBC posterior medians of δ. Mean absolute error 0.0096 with the rigid simulator and 0.0034 with the sine kernel simulator.

tional coverage. A Mondrian version would have to partition by the network's own flag, since $\mathbf{1}\{\delta^* > 0\}$ is not observed at prediction time (open). For the sine kernel networks the shift is -0.0002 and no correction is needed.

Removing the misspecification. Widening the zero window of Section 13.4.4 to $[74, 98]$ with $K = 17$ zeros, counted by the argument principle, and data on $[78, 94]$ at the five levels other than $\theta = 0.8$, the Gauss Newton fit started from the GBC output converged in 82 evaluations and 4.7 seconds, with essentially zero residual. With Laplace standard errors it gives

$$\sigma = 0.80852 \pm 0.00015, \qquad 1 - \sigma = 0.19149 \pm 0.00129, \qquad \gamma = 85.6993 \pm 0.0001,$$

against the true 0.808517, 0.191483 and 85.699348. Every zero on the line in the data window comes out as 0.50000 ± 0.00019. The six components outside the data window are unidentified and drift to the boundary of the prior box; for the same reason the full tempered sampler on this window did not finish in forty minutes, the posterior in those coordinates being flat and truncated. Absorbing the unobserved zeros into a background built from the Riemann von Mangoldt density has not been tried.

13.5 Concentration of the particle approximation

We prove a non asymptotic concentration inequality for the particle approximation, in the style of Del Moral [58], for the system defined below: multinomial resampling, with potentials G_k and mutation kernels M_k fixed in advance. A tempering bridge is also a Feynman Kac flow, so the plain filter of Section 13.3 with a fixed schedule and fixed kernels is covered, and so is the GBC initialised filter when in addition $\log \pi_0 - \log q$ has finite oscillation B, since its potentials are powers of $\pi_0 L / q$ (in Corollary 13.5.4 replace Ω_ς by $\Omega_\varsigma + B$). The samplers run in Sections 13.3 and 13.4 resample systematically, choose their tempering increments and move scales from the particles, and move by random walk Metropolis kernels; the theorem does not cover them (Lemma 13.5.3), and the $N^{-1/2}$ behaviour in Table 13.9 is computed. Unlike Theorem 12.4.1, which concerns contraction of the posterior as the noise tends to zero, the posterior here (that of the Gaussian

tolerance likelihood of Section 13.3) is fixed, and the question is the Monte Carlo error. Theorem 13.5.1, Remark 13.5.2, Lemma 13.5.3, the stability bound and Corollary 13.5.4 are proved; the two closing remarks are heuristic. The symbols ϵ (a minorisation constant) and ε (the tolerance) are different.

13.5.1 Setting

Let η_0 be a probability measure on a measurable space E, let $G_0, \dots, G_{n-1}$ be positive bounded potentials with $\sup G_k / \inf G_k < \infty$, and let $M_1, \dots, M_n$ be Markov kernels. Write $\Psi_G(\mu)(dx) = G(x)\mu(dx)/\mu(G)$ and

$$\eta_{k+1} = \Phi_{k+1}(\eta_k), \qquad \Phi_{k+1}(\mu) = \Psi_{G_k}(\mu) M_{k+1}.$$

In the filter, G_k is the incremental likelihood (or its tempered power) and M_{k+1} the Metropolis kernel that leaves η_{k+1} invariant. The particle system draws $X_0^1, \dots, X_0^N$ independently from η_0 and, given generation k, draws $X_{k+1}^1, \dots, X_{k+1}^N$ independently from $\Phi_{k+1}(\eta_k^N)$, where $\eta_k^N = N^{-1}\sum_i \delta_{X_k^i}$. This is multinomial resampling followed by mutation.

Put $\mathcal{Q}_{k+1}(x, dy) = G_k(x) M_{k+1}(x, dy)$, $\mathcal{Q}_{k,n} = \mathcal{Q}_{k+1} \cdots \mathcal{Q}_n$ (the identity when $k = n$),

$$G_{k,n} = \mathcal{Q}_{k,n}(1), \qquad \mathcal{P}_{k,n}(f) = \frac{\mathcal{Q}_{k,n}(f)}{\mathcal{Q}_{k,n}(1)}, \qquad \Phi_{k,n}(\mu)(f) = \frac{\mu\mathcal{Q}_{k,n}(f)}{\mu\mathcal{Q}_{k,n}(1)},$$

so that $\eta_n = \Phi_{k,n}(\eta_k)$. Define the oscillation ratio and the Dobrushin coefficient

$$g_{k,n} = \sup_{x,y} \frac{G_{k,n}(x)}{G_{k,n}(y)}, \qquad \varkappa_{k,n} = \sup_{x,y} \big\| \mathcal{P}_{k,n}(x,\cdot) - \mathcal{P}_{k,n}(y,\cdot) \big\|_{\mathrm{TV}},$$

$$S_n = \sum_{k=0}^{n} g_{k,n}\, \varkappa_{k,n}.$$

Theorem 13.5.1 (concentration). *For every measurable f with $\operatorname{osc}(f) \le 1$, every N and every $x > 0$,*

$$\mathbb{P}\big(|\eta_n^N(f) - \eta_n(f)| \ge x \big) \;\le\; 2(n+1) \exp\Big(-\frac{N x^2}{2 S_n^2} \Big),$$

$$\mathbb{E}\big[(\eta_n^N(f) - \eta_n(f))^2 \big]^{1/2} \le \frac{S_n}{\sqrt{N}}.$$

Proof. Since $\Phi_{k,n} \circ \Phi_k = \Phi_{k-1,n}$, with the convention $\Phi_0(\eta_{-1}^N) = \eta_0$ the error telescopes:

$$\eta_n^N(f) - \eta_n(f) = \sum_{k=0}^{n} \Big[\Phi_{k,n}(\eta_k^N)(f) - \Phi_{k,n}\big(\Phi_k(\eta_{k-1}^N)\big)(f) \Big].$$

For probability measures μ, ν, using $\nu\mathcal{Q}_{k,n}\big(f - \Phi_{k,n}(\nu)f\big) = 0$,

$$\Phi_{k,n}(\mu)(f) - \Phi_{k,n}(\nu)(f) = \frac{(\mu-\nu)\big(G_{k,n}\,h\big)}{\mu(G_{k,n})}, \qquad h = \mathcal{P}_{k,n}f - \Phi_{k,n}(\nu)(f).$$

Because $\Phi_{k,n}(\nu)(f)$ is a weighted average of the values of $\mathcal{P}_{k,n}f$, we have $|h| \le \operatorname{osc}(\mathcal{P}_{k,n}f) \le \varkappa_{k,n}$. Write $G_{k,n}h/\mu(G_{k,n}) = \tilde{h}\cdot\|G_{k,n}\|_\infty/\mu(G_{k,n})$ with $\tilde{h} = G_{k,n}h/\|G_{k,n}\|_\infty$, so that $|\tilde{h}| \le \varkappa_{k,n}$ and $\|G_{k,n}\|_\infty/\mu(G_{k,n}) \le g_{k,n}$. Take $\mu = \eta_k^N$ and $\nu = \Phi_k(\eta_{k-1}^N)$. Then $\tilde{h} = \tilde{h}_k$ is measurable with respect to generation $k-1$, and conditionally on that generation $\eta_k^N(\tilde{h}_k)$ is a mean of N independent variables with values in an interval of length $2\varkappa_{k,n}$ and mean $\nu(\tilde{h}_k)$. Hoeffding's inequality gives

$$\mathbb{P}\big(|Z_k| \ge u\big) \le 2\exp\big(-Nu^2/(2\varkappa_{k,n}^2)\big), \qquad Z_k = (\eta_k^N - \nu)(\tilde{h}_k),$$

and the kth term is at most $g_{k,n}|Z_k|$ in absolute value. Allocate x in proportion to $g_{k,n}\varkappa_{k,n}$: if $\sum_k g_{k,n}|Z_k| \ge x$ then $|Z_k| \ge x\varkappa_{k,n}/S_n$ for some k, and the union bound gives the tail estimate. For the second moment, Minkowski's inequality and $\mathbb{E}[Z_k^2] \le \varkappa_{k,n}^2/N$ give $\|\sum_k g_{k,n}|Z_k|\|_2 \le S_n/\sqrt{N}$. □

Remark 13.5.2 (a tail bound without the factor $n+1$). Under the hypotheses of Theorem 13.5.1, for every f with $\operatorname{osc}(f) \le 1$, every N and every $x > 0$,

$$\mathbb{P}\big(|\eta_n^N(f) - \eta_n(f)| \ge x\big) \le 2\exp\Big(-\frac{Nx^2}{2S_n^2}\Big).$$

Proof. Terms with $\varkappa_{k,n} = 0$ vanish; for the others put $w_k = g_{k,n}\varkappa_{k,n}/S_n$ and $V_k = |Z_k|/\varkappa_{k,n}$, with Z_k as in the proof of Theorem 13.5.1. Conditionally on generation $k-1$ (on nothing for $k=0$), $Z_k/\varkappa_{k,n}$ is a mean of N independent centred variables with values in intervals of length at most 2, so Hoeffding's lemma gives $\mathbb{E}\big[e^{\pm\lambda Z_k/\varkappa_{k,n}} \mid \text{generation } k-1\big] \le e^{\lambda^2/(2N)}$, hence $\mathbb{E}\,e^{\lambda V_k} \le 2e^{\lambda^2/(2N)}$ for $\lambda \ge 0$. The proof of Theorem 13.5.1 gives $|\eta_n^N(f) - \eta_n(f)|/S_n \le \sum_k w_k V_k$ with $\sum_k w_k = 1$, so by convexity of the exponential

$$\mathbb{E}\exp\Big(\lambda\,\frac{|\eta_n^N(f) - \eta_n(f)|}{S_n}\Big) \le \sum_k w_k\,\mathbb{E}\,e^{\lambda V_k} \le 2e^{\lambda^2/(2N)}.$$

Markov's inequality with $\lambda = Nx/S_n$ gives the bound. □

13.5.2 Stability gives bounds uniform in the number of steps

S_n can grow with n, but not when the mutation kernels mix. Suppose, as in Del Moral [58], that

$$M_k(x,\cdot) \ge \epsilon\, M_k(y,\cdot) \text{ for all } x, y, \qquad \sup G_k \le e^c \inf G_k \text{ for all } k.$$

Then $g_{k,n} \le e^c/\epsilon$, because $G_{k,n} = G_k\, M_{k+1}(G_{k+1,n})$ and the minorisation bounds the ratio of the second factor by $1/\epsilon$. The one step kernels of $\mathcal{P}_{k,n}$, namely

$$R_j(x, dy) = \frac{M_j(x, dy)\, G_{j,n}(y)}{M_j(G_{j,n})(x)},$$

satisfy $R_j(x,\cdot) \ge \epsilon^2 R_j(y,\cdot)$ (the ratio of the densities of $R_j(x,\cdot)$ and $R_j(y,\cdot)$ is $\big[M_j(x,dz)/M_j(y,dz)\big] \cdot \big[M_j(G_{j,n})(y)/M_j(G_{j,n})(x)\big]$, and each factor is at least ϵ), so $\varkappa_{k,n} \le (1-\epsilon^2)^{n-k}$, since $\mathcal{P}_{k,n} = R_{k+1} \cdots R_n$. Hence $S_n \le 1 + e^c(1-\epsilon^2)\epsilon^{-3} \le e^c\epsilon^{-3}$ for every n.

The hypothesis is strong, and the kernels of the filter do not satisfy it.

Lemma 13.5.3 (Metropolis kernels are not ratio minorised). *Let M be a Markov kernel on E, and suppose there are points x, y and a measurable set B with $M(y,B) > 0 = M(x,B)$. Then no $\epsilon > 0$ satisfies $M(x',\cdot) \ge \epsilon M(y',\cdot)$ for all x', y'. This is the case for a random walk Metropolis kernel whose proposals have a density, at any y with positive rejection probability ($B = \{y\}$, $x \ne y$), for its powers, and for componentwise Metropolis sweeps ($B = \{z : z_1 = y_1\}$, $x_1 \ne y_1$, when rejection of the first coordinate has positive probability at y).*

Proof. $M(x,B) = 0 < \epsilon M(y,B)$. From x the chain reaches the point y, or the hyperplane $z_1 = y_1$, only by accepting a proposal that lands there, an event of probability zero, in one step and in m steps; from y a rejection keeps it in B with positive probability. □

If M_k has a density $m_k(x,z)$ with respect to a common reference measure with $\alpha_k h_k(z) \le m_k(x,z) \le \alpha_k' h_k(z)$ for all x, z, then $M_k(x,\cdot) \ge (\alpha_k/\alpha_k')M_k(y,\cdot)$; exact Gibbs sweeps are an example when the target density is bounded above and below on Θ_ς (defined below). With $S_n \le e^c\epsilon^{-3}$, Remark 13.5.2 gives $\mathbb{P}(|\eta_n^N(f) - \eta_n(f)| \ge x) \le 2\exp(-\frac{1}{2}Nx^2\epsilon^6 e^{-2c})$ for every n. The number of steps does not enter the bound.

13.5.3 The Cauchy flow

Corollary 13.5.4 (concentration along the Cauchy flow). *Fix $\varsigma > 0$ and restrict the parameter space to $\Theta_\varsigma = \{z : |\theta_l - \sigma_j| \ge \varsigma$ for all levels θ_l and zeros $j\}$, with ordinates in the window. Then the total log likelihood has finite oscillation*

$$\operatorname{osc}(\ell) \le \Omega_\varsigma = \frac{1}{2\varepsilon^2} \sum_{l,i} \Big(|y_l(v_i)| + \frac{K}{\pi\varsigma} \Big)^2,$$

and a tempering schedule with $n = \max(1, \lceil \Omega_\varsigma / c \rceil)$ equal increments has potentials with $\sup G / \inf G \le e^c$. If the mutation kernels satisfy the ratio minorisation with constant ϵ (by Lemma 13.5.3 this needs kernels with densities, as in the

condition above), then for every posterior functional with oscillation at most one, in particular for the posterior probability that $\max_j |\sigma_j - \frac{1}{2}|$ *exceeds any threshold,*

$$\mathbb{P}\big(|\eta_n^N(f) - \eta_n(f)| \geq x\big) \leq 2\exp\Big(-\tfrac{1}{2}Nx^2\epsilon^6 e^{-2c}\Big) \qquad \text{for every } N \text{ and } x > 0.$$

Without the minorisation, Theorem 13.5.1 holds with its finite S_n*, which is not uniform in* n*.*

Proof. On Θ_ς each term $|Q_{\theta_l-\sigma_j}(v)|$ is at most $1/(\pi\varsigma)$. The marginal likelihood after integrating out the background is $\exp(-\frac{1}{2}r^\top\Sigma^{-1}r)$ with $\Sigma \succeq \varepsilon^2 I$, so its logarithm lies between $-\|r\|^2/(2\varepsilon^2)$ and 0, and $\|r\|^2$ is at most the sum in the display, since each entry of r is at most $|y_l(v_i)| + K/(\pi\varsigma)$ in absolute value. A potential $L^{\Delta\lambda}$ then has $\sup/\inf \leq e^{\Delta\lambda\,\Omega_\varsigma} \leq e^c$ for $\Delta\lambda = 1/n \leq c/\Omega_\varsigma$. The rest is Theorem 13.5.1, Remark 13.5.2 and the stability bounds. □

Two remarks. The restriction to Θ_ς is needed: when a candidate zero approaches a grid point (θ_l, v_i), that is, when $\sigma_j \to \theta_l$ with $\gamma_j \to v_i$, the term $Q_{\theta_l-\sigma_j}$ blows up, the likelihood tends to zero, and $\inf G = 0$. It is the singularity that makes the deconvolution informative. Under QRH(θ) the truncation is automatic for levels above $\theta + \varsigma$, provided the prior is restricted to $1 - \theta \leq \sigma_j \leq \theta$; the sampler's prior is uniform on $(0, 1)$. Second, the constants are explicit but pessimistic: Ω_ς is of order 10^6 for the grids of the source note (not recomputed here), and the minorisation constant enters as ϵ^6. The theorem guarantees the $N^{-1/2}$ rate and exponential tails for the system it covers, not the constant observed.

13.5.4 The rate in practice

The source note reran the GBC initialised filter for ζ twelve times at each of $N \in \{250, 500, 1000, 2000\}$ and measured the spread across runs of three posterior functionals for the zero at 84.7355: the posterior means of σ and of γ, and the posterior probability that $|\sigma - \frac{1}{2}| > 2 \times 10^{-4}$, which has oscillation one (Table 13.9, computed by the author, not rerun). The sampler is the systematic and adaptive one that Theorem 13.5.1 does not cover.

N	250	500	1000	2000
sd of posterior mean of σ ($\times 10^{-5}$)	6.5	5.5	3.2	2.8
sd of posterior mean of γ ($\times 10^{-5}$)	6.9	6.7	2.5	2.3
sd of posterior probability	0.104	0.062	0.048	0.038

Table 13.9: Monte Carlo spread over twelve runs, computed in the source note. Least squares log log slopes of the printed values are -0.44, -0.62 and -0.47, against $-1/2$ from Theorem 13.5.1; with twelve runs per cell each standard deviation has a relative error of about 20%, so the slopes carry about ± 0.15.

For the probability functional, Theorem 13.5.1 gives root mean square error at most $S_n/\sqrt{N}$. The observed 0.038 at $N = 2000$ corresponds to an effective $S_n \approx 1.7$, far below the worst case constants of Corollary 13.5.4; it is a lower estimate, since the theorem bounds the root mean square error, bias included, and the spread across runs measures less. That the adaptive rule (effective sample size $N/2$) controls the constant in practice is a heuristic; the adaptive version remains to be analysed.

13.5.5 Two open problems

Remark 13.5.5 (the continuous level limit; heuristic, open). Let $\eta_\lambda \propto q^{1-\lambda}(\pi_0 L)^\lambda$ be the tempered flow, with $V = \log(\pi_0 L) - \log q$. Exactly,

$$\frac{d}{d\lambda}\eta_\lambda(f) = \mathrm{Cov}_{\eta_\lambda}(V, f),$$

and Metropolis moves that leave η_λ invariant do not change $\eta_\lambda(f)$. The particle version is a Moran system: each particle moves by the generator $\mathcal{L}_\lambda$ of the moves and, at rate $(V(X^j) - V(X^i))_+/N$, particle i is replaced by a copy of particle j; its mean drift for the empirical measure is $\mathrm{Cov}_{\eta^N_\lambda}(V, f)$. As the level spacing tends to zero one expects convergence at rate $N^{-1/2}$ and a Gaussian limit for the fluctuations, by the martingale problem method (Ethier and Kurtz [72], as used by Del Moral and Miclo [65]), with an asymptotic variance that in discrete time is $\sum_k \eta_k\big[(G_{k,n}/\eta_k G_{k,n})^2(\mathcal{P}_{k,n}f - \eta_n f)^2\big]$ (the formula of the source, quoted from the literature and unchecked here, see [58]). This is heuristic: no scaling of the likelihood increments, tolerance, mutation times and resampling rates has been fixed, and with the same ε at every new level the log likelihood of a mesh h is of order $1/h$ and need not converge. On a finite state space with bounded potentials and mutation generators the limit is the corresponding ordinary differential equation and the Moran approximation has error $O(N^{-1/2})$; for the Cauchy flow V is unbounded below near the levels, as in Corollary 13.5.4, and on Θ_ς it is bounded. A continuous time limit theorem for the singular potentials, with a Lyapunov function that controls how often particles approach the singular set $\{\sigma_j = \theta_l\}$, and the corresponding result for the adaptive schedule (Beskos, Jasra, Kantas and Thiery [15]), are open problems.

Remark 13.5.6 (concentration under adaptive tempering; heuristic, open). Theorem 13.5.1 assumes a fixed schedule. The filter of Section 13.4 chooses each tempering increment so that the effective sample size of the incremental weights is $N/2$, which says that the empirical relative second moment $\eta^N_k(G_k^2)/\eta^N_k(G_k)^2$ equals 2. The proof of Theorem 13.5.1 uses the oscillation ratio $g_{k,n}$ only to bound $\|G_{k,n}\|_\infty/\eta^N_k(G_{k,n})$. A natural idea is to replace that sup norm bound by a second moment bound, so that by Cauchy Schwarz each term of the telescoping sum is controlled by $\eta_k(G^2_{k,n})/\eta_k(G_{k,n})^2$, the quantity that the adaptive rule keeps near

2 for one step. The non asymptotic variance theorem of Cérou, Del Moral and Guyader [37] is reported to give the resulting bound for a fixed schedule with such relative moments, and Beskos, Jasra, Kantas and Thiery [15] show that the adaptive schedule converges to its deterministic limit (both statements are the source's and the papers were not read for this book). Combined, with the singular potentials kept away from the levels as in Corollary 13.5.4, they would give a version of Theorem 13.5.1 with constant set by the effective sample size threshold rather than $\sup G/\inf G$. The sketch is heuristic: control of the one step relative second moment says nothing about $\eta_k(G_{k,n}^2)/\eta_k(G_{k,n})^2$ for $k < n-1$, because $G_{k,n}$ involves all later potentials, so a multistep moment bound does not follow. That version is open.

13.6 From $\mathfrak{U}_1$ to $\mathfrak{U}_{3/4}$

The level $\theta = \frac{3}{4}$ is halfway between the two ends of the deconvolution and is the level of QRH(3/4). There are two ways to obtain $\mathfrak{U}_{3/4}$.

Directly. $\mathfrak{U}_{3/4}\big((0,V]\big) = \int_0^V u_{3/4}(v)\,dv$ with $u_{3/4}(v) = \pi^{-1}\,\mathrm{Re}\,\xi'/\xi(\frac{3}{4}+iv)$, evaluated to high precision. This forward computation needs no filter. On $[78,94]$, on a grid of spacing 0.02, we find the following (computed).

- For ζ, $u_{3/4}$ is positive on the grid, with minimum 0.0734 at $v = 90.68$ between the zeros 88.81 and 92.49, and $\mathfrak{U}_{3/4}\big((78,94]\big) = 6.162$, close to the six zeros in the window spread by Cauchy kernels of scale 1/4. Positivity for every $\theta > \frac{1}{2}$ at these heights is Corollary 14.3.7, not a consequence of the grid values.
- For Davenport and Heilbronn, $u_{3/4}$ has a negative spike of depth -4.80 at $v = 85.70$, and the negative part of $\mathfrak{U}_{3/4}$ on the window has mass 0.578. Its total increment is 8.763. The spike comes from the zero $0.8085 + 85.699i$, whose real part exceeds $\frac{3}{4}$: it contributes the negative Cauchy term $Q_{3/4-0.8085}$ of scale 0.0585, partly cancelled by the positive terms (Theorem 13.7.1).

By the backward flow. The test of the method is to predict $u_{3/4}$ from a fit to the two levels $\theta \in \{1, 0.9\}$: fit the filter to these levels and push each posterior particle forward to $\theta = \frac{3}{4}$ through the Hadamard sum (12.1.1),

$$u_{3/4}(v \mid z) = \sum_j Q_{3/4-\sigma_j}(v-\gamma_j) + b_{3/4}(v \mid z),$$

with the background coefficients refitted given z (to their conditional mean, so that the band below is the band of the conditional mean prediction and omits

the uncertainty of those coefficients). The data are on lines to the right and the prediction is to the left. Since $Q_{3/4-\sigma}$ changes sign as σ crosses $\frac{3}{4}$, the prediction is only as good as the posterior for each σ_j relative to $\frac{3}{4}$ (compare Proposition 13.3.1).

The source note ran the GBC initialised filter of Section 13.4 with $N = 1000$ particles (computed by the author, not rerun; the source's figure is not reproduced).

- For ζ, the filter needed 7 tempering steps and 3.6×10^4 likelihood evaluations (one second). The predicted density of $\mathfrak{U}_{3/4}$ on $[80, 92]$ has median absolute error 0.0012 and maximum error 0.008 against the direct computation, and the 95% band covers the truth at every grid point.

- For Davenport and Heilbronn, again 7 steps and two seconds. From the fit to the levels 1 and 0.9 the filter places the pair at $\gamma = 85.699$ with real parts $\sigma = 0.8082$ (interval $[0.8080, 0.8084]$, which excludes the true value 0.80852, so the posterior is overconfident, as in Section 13.4) and $\sigma = 0.1987$, against the true 0.8085 and 0.1915. It then predicts the negative spike of $\mathfrak{U}_{3/4}$: median absolute error 0.0019 and maximum error 0.042 against a spike of depth 4.8, with the band covering 80% of grid points. The misses are at the bottom of the spike, where a bias of 3×10^{-4} in σ is amplified by a factor of order $1/(\pi(\sigma - \frac{3}{4})^2) \approx 90$.

So, within the window model of Section 13.3, the filter predicts $u_{3/4}$ on the window from the levels 1 and 0.9, to about two digits at the bottom of the spike and three for the median error, including the failure of positivity caused by a zero off the line between the data and the target. The initial draws came from the GBC screen, which uses the levels 1, 0.85, 0.7 and 0.6, so this is not a computation from $\theta \geq 0.9$ alone; a rerun with a screen trained on $\{1, 0.9\}$ only has not been done. Exact data on a window determine $u_{3/4}$ only through unstable analytic continuation (Proposition 13.6.1), and positivity of $\mathfrak{U}_\tau$ at all heights for every $\tau > \frac{3}{4}$, which is QRH(3/4) (Theorem 11.9.1), is beyond any finite computation.

Proposition 13.6.1 (what one level determines). *(a) Let F_1, F_2 be real entire functions of order at most one with all zeros in $0 < \operatorname{Re} s < 1$, and let $u_\theta^{(j)}(v) = \pi^{-1} \operatorname{Re}(F_j'/F_j)(\theta + iv)$. If $u_1^{(1)} = u_1^{(2)}$ on an interval, then $F_1'/F_1 = F_2'/F_2$; in particular $u_\theta^{(1)} = u_\theta^{(2)}$ for every θ.*

(b) The order hypothesis cannot be dropped: $F = \xi e^G$ with $G(s) = -(10\pi)^{-1} \cos(10\pi(s - 1))$ is real entire with the zeros of ξ, its densities at the levels 1 and 0.9 equal u_1 and $u_{0.9}$ on the whole line, and its density at $\frac{3}{4}$ is $u_{3/4}(v) - \pi^{-1} \cosh(10\pi v)$.

(c) If no zero has $\sigma_\rho \geq \frac{3}{4}$ (in particular under RH), then $u_1 = P_{1/4} * u_{3/4}$: recovering $u_{3/4}$ from u_1 inverts a Poisson smoothing, whose Fourier multiplier is $e^{-|\omega|/4}$, so an error at frequency ω grows by the factor $e^{|\omega|/4}$.

Proof. (a) Neither F_j vanishes on $\operatorname{Re} s = 1$, so $v \mapsto u_1^{(j)}(v)$ is real analytic and the two densities agree on $\mathbb{R}$. By Hadamard's factorisation, $\pi u_\theta^{(j)}(v) = c_j + \sum_\rho (\theta - \sigma_\rho)/((\theta-\sigma_\rho)^2 + (v-\gamma_\rho)^2)$, with a real constant c_j and an absolutely convergent sum over the zeros $\rho = \sigma_\rho + i\gamma_\rho$ of F_j, because $\sum_\rho |\rho|^{-2} < \infty$. For $\theta > 1$ every term of the sum is positive and $P_{\theta-1} * P_{1-\sigma_\rho}(\cdot - \gamma_\rho) = P_{\theta-\sigma_\rho}(\cdot - \gamma_\rho)$, so by Tonelli $u_\theta^{(j)} = P_{\theta-1} * u_1^{(j)}$. Hence $\operatorname{Re}(F_1'/F_1 - F_2'/F_2) = 0$ on $\operatorname{Re} s > 1$, so $F_1'/F_1 - F_2'/F_2$ is an imaginary constant there, which is 0 because both functions are real on the real axis; by the identity theorem they agree on $\mathbb{C}$.

(b) $G'(s) = g(s) = \sin(10\pi(s-1))$ and $F'/F = \xi'/\xi + g$. On $\operatorname{Re} s = 1$, $g(1 + iv) = i\sinh(10\pi v)$; on $\operatorname{Re} s = 0.9$, $g(0.9 + iv) = \sin(-\pi + 10\pi i v) = -i\sinh(10\pi v)$; both are purely imaginary. On $\operatorname{Re} s = \frac34$, $g(\frac34 + iv) = \sin(-\frac{5\pi}{2} + 10\pi i v) = -\cosh(10\pi v)$.

(c) By the Hadamard sum (12.1.1), every zero then contributes $P_{3/4-\sigma_\rho}(\cdot - \gamma_\rho)$ to $u_{3/4}$, with $\frac34 - \sigma_\rho > 0$, and $P_{1/4} * P_{3/4-\sigma_\rho} = P_{1-\sigma_\rho}$; sum by Tonelli. Finally $\int e^{i\omega x} P_a(x)\,dx = e^{-a|\omega|}$. □

13.7 Negative mass and converse results

In the Hadamard sum (13.1.1) every term is nonnegative for $\theta \geq \sup \sigma_\rho$; for smaller θ the terms with $\sigma_\rho > \theta$ are negative, and we ask how much they can cost. Zero density bounds the negative mass (Theorem 13.7.1, proved), and a computed example for the Davenport and Heilbronn function shows the bound at work (Table 13.10, computed). A converse, in which negative mass forces zeros, is proved for separated zeros and in a weighted form for all zeros (Proposition 13.7.2, Corollary 13.7.3, Theorem 13.7.4). Under $\mathrm{QRH}(\theta_0)$ the density has a margin (Proposition 13.7.5, conditional on $\mathrm{QRH}(\theta_0)$).

13.7.1 How negative the density can be

Problem 13.2. How negative can $\mathfrak{U}_\theta$ be, for $\frac12 < \theta < 1$, without assuming RH?

Write $N(\theta, T)$ for the number of zeros with $\sigma_\rho > \theta$ and $0 < \gamma_\rho \leq T$, counted with multiplicity, and $u_\theta^- = \max(-u_\theta, 0)$.

Theorem 13.7.1 (zero density controls the negative Thorin mass). *Let $\frac12 < \theta < 1$ and $T \geq 2$.*

(a) $\displaystyle\int_0^T u_\theta^-(v)\,dv \leq N\big(\theta, T + \tfrac12\big).$

(b) If T is not an ordinate and no zero with $0 < \gamma_\rho < T$ has real part θ, then

$$\mathfrak{U}_\theta\big((0,T]\big) = \int_0^T u_\theta(v)\,dv = \frac{T}{2\pi}\log\frac{T}{2\pi e} - 2N(\theta,T) + O(\log T).$$

(c) *Consequently, for T as in (b), Selberg's density theorem $N(\theta,T) \ll T^{1-(\theta-\frac12)/4}\log T$ [256] shows that the negative part of $\mathfrak{U}_\theta$ on $(0,T]$ is a fraction $O\big(N(\theta,T+\frac12)/(T\log T)\big) = O\big(T^{-(\theta-\frac12)/4}\big)$ of its mass.*

Proof. (a) Group the zeros in the Hadamard sum into orbits of $\rho \mapsto 1-\bar\rho$, which preserves ordinates. A zero on the critical line contributes $P_{\theta-1/2}(v-\gamma_\rho) > 0$, and an orbit with $\frac12 < \sigma_\rho \le \theta$ contributes $Q_{\theta+\sigma_\rho-1}(v-\gamma_\rho) + Q_{\theta-\sigma_\rho}(v-\gamma_\rho) \ge 0$, with $Q_0 = 0$ off the ordinate. An orbit with $\sigma_\rho > \theta$ contributes $P_a(v-\gamma_\rho) - P_b(v-\gamma_\rho)$ with $a = \theta+\sigma_\rho-1 > b = \sigma_\rho-\theta > 0$, and

$$\pi\big(P_a(x)-P_b(x)\big) = \frac{(a-b)(x^2-ab)}{(a^2+x^2)(b^2+x^2)},$$

so its negative part is at most $P_b(x)$ for $x^2 < ab$ and vanishes for $x^2 \ge ab$, where $ab < \big(\frac{a+b}{2}\big)^2 = (\sigma_\rho-\frac12)^2 < \frac14$ (as in the proof of Proposition 3.2.1(ii)). Since the negative part of a sum is at most the sum of the negative parts, $u_\theta^-(v) \le \sum_{\sigma_\rho>\theta} P_{\sigma_\rho-\theta}(v-\gamma_\rho)\,\mathbf{1}\{|v-\gamma_\rho| < \frac12\}$, the sum over zeros of both signs of γ_ρ. On $[0,T]$ only zeros with $-\frac12 < \gamma_\rho < T+\frac12$ contribute, each at most $\int P_b = 1$, and no zero has $|\gamma_\rho| < \frac12$; this gives (a).

(b) Apply the argument principle to ξ on the rectangle with vertices θ, 2, $2+iT$, $\theta+iT$, positively oriented. It contains the $N(\theta,T)$ zeros with $\sigma_\rho > \theta$ and $0 < \gamma_\rho < T$, and ξ is real and positive on $[\theta,2]$. Writing Δ_I for the continuous change of $\arg\xi$ along a segment I,

$$\Delta_{[\theta,\theta+iT]} = \Delta_{[2,2+iT]} + \Delta_{[2+iT,\theta+iT]} - 2\pi N(\theta,T).$$

The left side is $\pi\int_0^T u_\theta$ (Theorem 10.1.1). As in the proof of the Riemann von Mangoldt formula, $\Delta_{[2,2+iT]} + \Delta_{[2+iT,\frac12+iT]} = \pi N(T)$, and the change along the horizontal segment from $\frac12+iT$ to $\theta+iT$ is $O(\log T)$, by Stirling's formula for the Gamma factor and, for ζ, by the bound that gives $S(T) = O(\log T)$. With $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T)$ this is (b); it is Proposition 11.9.2 in the present notation.

(c) By (b) and $N(\theta,T) = o(T\log T)$, $\mathfrak{U}_\theta((0,T]) = \frac{T}{2\pi}\log T\,(1+o(1))$; divide (a) by it and apply Selberg's bound to $N(\theta,T+\frac12)$. □

The display in (b) without the term $-2N(\theta,T)$ holds under RH and is equivalent to $N(\theta,T) = O(\log T)$, which is open. Theorem 14.3.2 gives better exponents (Ingham below $\frac{7}{10}$, Guth and Maynard above), so (c) follows from it; what (a) adds is the constant, with no tail term. Ingham's bound $N(\theta,T) \ll T^{3(1-\theta)/(2-\theta)}\log^5 T$ is better than Selberg's near $\theta = 1$, the density hypothesis $N(\theta,T) \ll T^{2(1-\theta)+\epsilon}$ would make the fraction $O(T^{1-2\theta+\epsilon})$, and RH makes it zero. Define the Thorin defect exponent, with the convention $\log 0 = -\infty$,

$$\operatorname{def}(\theta) = \limsup_{T\to\infty} \frac{\log\int_0^T u_\theta^-(v)\,dv}{\log T}.$$

By Theorem 13.7.1(a), $\mathrm{def}(\theta)$ is at most the density exponent $\limsup \log N(\theta,T)/\log T$ (proved). It equals $-\infty$ under RH.

For the Davenport and Heilbronn function f (Section 13.2) the bound is seen at work. Its density u_θ is the same expression with Λ_{DH} of Proposition 13.8.3 in place of ξ, and the pairing argument of (a) applies to its zeros with $\frac{1}{2} < \sigma_\rho < 1$. Table 13.10 gives the negative mass of u_θ on intervals of length 3 centred at the two lowest zeros off the line, $0.8085\ldots+85.6993\ldots i$ and $0.6508\ldots+114.1633\ldots i$ (certified, Section 13.2). Within each interval only the zero at its centre can make u_θ negative, by the pairing, and it contributes at most one when $\sigma_\rho > \theta$ and nothing when $\sigma_\rho \le \theta$. The mass is largest just below σ_ρ and decays further left, as the negative Cauchy term spreads and is absorbed by the positive background.

level θ	0.9	0.79	0.7	0.6
negative mass near 85.70 ($\sigma_\rho = 0.8085$)	0	0.763	0.424	0.195
bound for this interval	0	1	1	1
negative mass near 114.16 ($\sigma_\rho = 0.6508$)	0	0	0	0.454
bound for this interval	0	0	0	1

Table 13.10: Negative mass of u_θ for the Davenport and Heilbronn function on intervals of length 3 centred at its two lowest zeros off the line (computed). The bound is the number of zeros with $\sigma_\rho > \theta$ at the centre.

Problem 13.3 (converse). Does $\mathrm{def}(\theta) \le A(1-\theta)$ for all θ imply $N(\theta,T) \ll T^{A(1-\theta)+\epsilon}$? A zero with $\sigma_\rho > \theta$ forces negative mass at levels just below σ_ρ (Theorem 11.9.1), but nearby zeros with σ_ρ just below the level contribute large positive terms, and the cancellation is not controlled. A converse would make the negative Thorin mass a proxy for zero density. This is open; the next subsection settles separated zeros and bounds the general case from below.

13.7.2 A partial converse

For a zero ρ, of multiplicity $k(\rho)$, let $d(\rho)$ be the distance from ρ to the nearest other zero, and

$$\mathrm{cr}(\rho) = \sum_{\rho' \ne \rho,\ |\gamma_{\rho'}-\gamma_\rho| \le 2} \frac{k(\rho')}{|\rho'-\rho|},$$

the sum over distinct zeros, the reflected zero $1-\bar\rho$ included, weighted by multiplicity; $d(\rho)$ is the distance to a distinct zero location.

Proposition 13.7.2 (a separated zero forces negative Thorin mass). *Let $\rho_0 = \sigma_0 + i\gamma_0$ be a zero of multiplicity k with $\sigma_0 > \frac{1}{2}$ and $\gamma_0 \ge 3$. There is an absolute*

constant $C_1 \geq 1$ such that if

$$r \leq \min\Big(1,\ \sigma_0 - \tfrac{1}{2},\ \tfrac{1}{2}d(\rho_0),\ \frac{k}{\pi\big(\mathrm{cr}(\rho_0) + C_1 \log \gamma_0\big)}\Big),$$

then

$$\int_{\sigma_0 - r}^{\sigma_0} \int_{\gamma_0 - r}^{\gamma_0 + r} u_\theta^-(v)\, dv\, d\theta \ \geq\ \frac{k\, r}{\pi}.$$

Proof. Let D^- be the half disc $\{|s - \rho_0| < r,\ \mathrm{Re}\, s < \sigma_0\}$, which lies in $\{\mathrm{Re}\, s > \frac{1}{2}\}$ and in the rectangle of the claim. Since $u_\theta^- \geq -u_\theta$, the integral is at least $-\iint_{D^-} u\, dA$, with $u(\theta + iv) = u_\theta(v)$. Split u by the Hadamard sum. In polar coordinates $s - \rho_0 = ue^{i\phi}$ with $\phi \in (\frac{\pi}{2}, \frac{3\pi}{2})$ the own term is

$$\frac{k}{\pi} \iint_{D^-} \mathrm{Re}\, \frac{1}{s - \rho_0}\, dA = \frac{k}{\pi} \int_0^r \int_{\pi/2}^{3\pi/2} \cos\phi\, d\phi\, du = -\frac{2kr}{\pi}.$$

A zero ρ' with $|\gamma_{\rho'} - \gamma_0| \leq 2$ is at distance at least $2r$ from ρ_0, so $|s - \rho'| \geq |\rho' - \rho_0|/2$ on D^- and its term is at most $\pi^{-1}k(\rho')\big(\frac{\pi r^2}{2}\big)\big(\frac{2}{|\rho' - \rho_0|}\big)$ in absolute value; these sum to at most $r^2 \mathrm{cr}(\rho_0)$. For $|\gamma_{\rho'} - \gamma_0| > 2$ we have $|\theta - \sigma_{\rho'}| \leq 1$ and $|v - \gamma_{\rho'}| \geq |\gamma_{\rho'} - \gamma_0|/2$, so $|\mathrm{Re}(s - \rho')^{-1}| \leq 4/(\gamma_{\rho'} - \gamma_0)^2$, and $\sum_{|\gamma' - \gamma_0| > 2} (\gamma' - \gamma_0)^{-2} \ll \log \gamma_0$; these contribute at most $\frac{1}{2}C_1 r^2 \log \gamma_0$. Hence

$$-\iint_{D^-} u\, dA \ \geq\ \frac{2kr}{\pi} - r^2\big(\mathrm{cr}(\rho_0) + \tfrac{1}{2}C_1 \log \gamma_0\big) \ \geq\ \frac{kr}{\pi}$$

by the choice of r. □

Corollary 13.7.3 (two sided control for separated zeros). *Let $\frac{1}{2} < \theta_0 < 1$, $T \geq 3$ and $E(\theta_0, T) = \int_{\theta_0}^1 \int_0^{T+1} u_\theta^-(v)\, dv\, d\theta$. Let $N_{\mathrm{sep}}(\theta_0, T; c, \delta)$ count the zeros with $\sigma_\rho > \theta_0$, $3 \leq \gamma_\rho \leq T$, $\mathrm{cr}(\rho) \leq c$ and distance at least 2δ from every other zero. With $r = \min\big(\delta,\ 1/(\pi(c + C_1 \log T))\big)$,*

$$\begin{aligned} \frac{r}{\pi} N_{\mathrm{sep}}(\theta_0 + r, T; c, \delta) \ &\leq\ E(\theta_0, T) \ \leq\ \int_{\theta_0}^1 N\big(\theta, T + \tfrac{3}{2}\big)\, d\theta \\ &= \sum_{\sigma_\rho > \theta_0,\ 0 < \gamma_\rho \leq T + \frac{3}{2}} (\sigma_\rho - \theta_0). \end{aligned}$$

Proof. The half discs of distinct separated zeros are disjoint, and for $\sigma_\rho > \theta_0 + r$ they lie in $\{\theta_0 < \theta < 1,\ 0 < v < T+1\}$, so the lower bound adds Proposition 13.7.2 over them (with $r \leq \sigma_\rho - \frac{1}{2}$ because $\theta_0 > \frac{1}{2}$, and $r < 1$). The upper bound integrates Theorem 13.7.1(a), applied with $T + 1$ in place of T, over θ. □

The lower bound counts only separated zeros and the upper bound counts every zero, weighted by its distance to the level θ_0, so the two bounds meet only when the zeros to the right of θ_0 are separated at scale $\delta \gg 1/\log T$ with crowding $c = O(\log T)$; then r is of order $1/\log T$ and the two sides agree up to powers of $\log T$. Clusters are the open case: crowded zeros off the line can cancel each other's Cauchy terms, so the negative mass undercounts them; Theorem 13.7.4 quantifies this.

THEOREM 13.7.4 (every zero off the line forces negative Thorin mass). *Let $\theta_0 > \frac{1}{2}$ and, for a zero ρ with $\sigma_\rho > \theta_0$,*

$$r(\rho) = \min\Big(1,\ \sigma_\rho - \theta_0,\ \tfrac{1}{2}d(\rho),\ \frac{k(\rho)}{\pi\,\big(\mathrm{cr}(\rho) + C_1 \log \gamma_\rho\big)}\Big),$$

with C_1 the constant of Proposition 13.7.2. Then for $T \geq 3$, with E as in Corollary 13.7.3,

$$E(\theta_0, T) \ \geq\ \frac{1}{\pi} \sum_{\sigma_\rho > \theta_0,\ 3 \leq \gamma_\rho \leq T} k(\rho)\, r(\rho),$$

the sum running over distinct zeros. Moreover, for an absolute $c > 0$,

$$r(\rho) \geq c \min\big(\sigma_\rho - \theta_0,\ \min(d(\rho), 1)/\log \gamma_\rho\big),$$

and so

$$E(\theta_0, T) \ \gg\ \frac{1}{\log T} \sum_{\sigma_\rho > \theta_0,\ 3 \leq \gamma_\rho \leq T} \min\big((\sigma_\rho - \theta_0)\log T,\ d(\rho),\ 1\big).$$

Proof. The proof of Proposition 13.7.2 applies to each zero with its own radius $r(\rho)$, and the half discs $D^-(\rho)$ are pairwise disjoint because $r(\rho) \leq d(\rho)/2$. They lie in $\{\theta_0 < \theta < 1,\ 0 < v < T + 1\}$, so their contributions add. For the second statement, at most $C_2 \log \gamma_\rho$ zeros, counted with multiplicity, satisfy $|\gamma' - \gamma_\rho| \leq 2$, each at distance at least $d(\rho)$, so $\mathrm{cr}(\rho) \leq C_2 \log \gamma_\rho / d(\rho)$. Replacing $d(\rho)$ by $\min(d(\rho), 1)$ in this bound gives $k/(\pi(\mathrm{cr} + C_1 \log \gamma_\rho)) \geq \min(d(\rho), 1)/(\pi(C_2 + C_1)\log \gamma_\rho)$, and, enlarging C_1 and C_2 if necessary, $\frac{1}{2}d(\rho)$ is larger than this for $\gamma_\rho \geq 3$. The last display follows from $k(\rho) \geq 1$ and $\log \gamma_\rho \leq \log T$. □

So every zero off the line pays, in negative Thorin mass over levels and heights, at least its isolation distance over $\log T$, or its distance from θ_0 if smaller; a cluster of M zeros at mutual distances at least $d \leq 1$, each at least $d/\log T$ to the right of θ_0, pays at least about $Md/\log T$. Suppose, as a hypothesis (conditional, and heuristic since no zero off the line is known), that every zero with $\sigma_\rho > \theta_0$ has $d(\rho) \geq c'/\log \gamma_\rho$, which is the typical isolation for zeros with the sine kernel statistics of the zeros of ζ. Then Theorem 13.7.4 and Corollary 13.7.3 give, for every $\epsilon > 0$,

$$\frac{N(\theta_0 + \epsilon, T)}{(\log T)^2} \ \ll_\epsilon\ E(\theta_0, T) \ \ll\ \int_{\theta_0}^{1} N\big(\theta, T + \tfrac{3}{2}\big)\, d\theta,$$

a two sided equivalence between negative Thorin mass and zero density up to powers of $\log T$ (conditional on that hypothesis). The step from isolation of a typical zero to every zero is not proved.

13.7.3 A check on the Davenport and Heilbronn function

Table 13.11 integrates u_θ^- numerically over the left half disc of radius r around each zero off the line of the Davenport and Heilbronn function. The negative mass lies between r/π and the own term $2r/\pi$; the gap measures the cancellation from the other zeros. This illustrates the mechanism of Proposition 13.7.2 and does not check it, since C_1, and so the admissible r, is not given.

zero	r	negative mass on D^-	r/π	$2r/\pi$
$0.8085 + 85.6993\,i$	0.05	0.0296	0.0159	0.0318
$0.8085 + 85.6993\,i$	0.10	0.0545	0.0318	0.0637
$0.8085 + 85.6993\,i$	0.14	0.0710	0.0446	0.0891
$0.6508 + 114.1633\,i$	0.05	0.0275	0.0159	0.0318
$0.6508 + 114.1633\,i$	0.10	0.0457	0.0318	0.0637
$0.6508 + 114.1633\,i$	0.14	0.0530	0.0446	0.0891

Table 13.11: Negative mass of u_θ on the left half disc around the zeros of the Davenport and Heilbronn function off the line, on a 40×81 grid (computed by the author, not recomputed).

13.7.4 Positivity at 7/8 and what it gives

Problem 13.4. What does $\mathrm{QRH}(\theta_0)$, the absence of zeros with $\sigma_\rho > \theta_0$, give in this framework, for instance at $\theta_0 = \frac{7}{8}$?

A zero free half plane $\operatorname{Re} s > \frac{7}{8}$ is announced for Dirichlet L functions in a preprint of OpenAI with a Lean formalisation, with a small improvement claimed by Liu and a related preprint attributed to Thorner (unchecked), none of them refereed (Section 17.7); every statement below that uses $\mathrm{QRH}(\frac{7}{8})$ is conditional on it. By Theorem 11.9.1, $\mathrm{QRH}(\theta_0)$ is equivalent to positivity of $\mathfrak{U}_\tau$ for every $\tau > \theta_0$. The Balazard, Saias and Yor criterion for $\mathrm{QRH}(\theta_0)$, with computations at $\theta_0 = \frac{7}{8}$, is Theorem 14.2.1. Here is the margin.

Proposition 13.7.5 (positivity margin, conditional on $\mathrm{QRH}(\theta_0)$). *Assume* $\mathrm{QRH}(\theta_0)$ *with* $\frac{1}{2} \le \theta_0 < 1$. *There are constants* $c > 0$ *and* v_0 *such that* $u_\theta(v) \ge c\,(\theta - \theta_0)\log v$ *for* $\theta_0 < \theta \le 1$ *and* $v \ge v_0$.

Proof. Every zero has $1 - \theta_0 \le \sigma_\rho \le \theta_0$, so every term $P_{\theta-\sigma_\rho}(v - \gamma_\rho)$ of the Hadamard sum has width $\theta - \sigma_\rho \in [\theta - \theta_0, 1)$ and is positive. For $|v - \gamma_\rho| \le 1$

it is at least $(\theta-\theta_0)/(2\pi)$, because $a^2+x^2\le 2$. By the Riemann von Mangoldt formula, the number of zeros with $|\gamma_\rho - v|\le 1$ is $\pi^{-1}\log(v/2\pi)+O(1)+S(v+1)-S(v-1)$, and the explicit bound $|S(T)|\le 0.1095\log T+0.2042\log\log T+3.0305$ of Hasanalizade, Shen and Wong [100] (as used in Proposition 12.1.11) leaves at least $\big(\pi^{-1}-0.219-o(1)\big)\log v\ge 0.09\log v$ of them for large v, since $\pi^{-1}=0.3183\ldots$. Summing gives the claim with $c=0.09/(2\pi)$. □

The particle filter can start at the level $\theta=\frac{7}{8}$ without any hypothesis, since the levels at or above 0.9 carry only 1.6% of $\sum a_l^{-3}$ and 0.1% of $\sum a_l^{-5}$ in the schedule of Section 13.3 (computed). Conditionally on QRH($\frac{7}{8}$) the prior can also be truncated to real parts in $[\frac{1}{8},\frac{7}{8}]$. This helps little, since neither touches the informative levels between $\frac{1}{2}$ and $\frac{7}{8}$.

13.7.5 Certificates

Problem 13.5. Turn the screen of Section 13.4 into a proof in each window.

The screen is calibrated on simulations, so it is evidence, not proof. A certificate in a window needs the contour moment M_0 of Proposition 13.4.1 computed in interval arithmetic and shown to be less than $\frac{1}{2}$; this is the argument principle, as in Section 11.10 (Theorem 11.9.6 and Section 11.9.7 treat certified counts). A certificate is valid whatever its cost; the screen would move the verified height 3×10^{12} [232] only if it were cheaper per zero than the existing methods, which we do not expect. The role of the screen is to choose where to spend the expensive rigorous integrations, and how close to the line the rectangle can be placed.

13.8 Zero free regions from positivity

Here the Euler product enters, as an upper bound on a positive combination of values of u_θ at several heights and, in the multilevel form, at several levels. We characterise the admissible combinations, optimise the zero free constant they give, and prove that no member of the class gives more than a region of width $c/\log\gamma$ with $c<1$ (Proposition 13.8.7). The linear programming values, the Stechkin constant and the optimised constants are computed. The de la Vallée Poussin region itself is Proposition 12.1.4; what is new is the general form of the inequality.

13.8.1 The Euler product as an inequality between Cauchy data

From $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, for $s = \theta + iv$,

$$\begin{aligned} \pi u_\theta(v) &= g_\theta(v) - \Pi_\theta(v), \\ g_\theta(v) &= \operatorname{Re}\Big[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\big(\tfrac{s}{2}\big)\Big], \\ \Pi_\theta(v) &= -\operatorname{Re}\frac{\zeta'}{\zeta}(\theta + iv). \end{aligned} \tag{13.8.1}$$

The gamma part g_θ is explicit and carries no arithmetic. The arithmetic is in $\Pi_\theta(v) = \sum_{n\ge 2}\Lambda(n)n^{-\theta}\cos(v\log n)$, which for $\theta > 1$ is a cosine series with non-negative coefficients (Λ the von Mangoldt function); that is all of the Euler product used below.

Theorem 13.8.1 (de la Vallée Poussin in Thorin form, general version; the first display is Proposition 12.1.4(i) in the present notation). *For $\theta > 1$ and every real v,*

$$3u_\theta(0) + 4u_\theta(v) + u_\theta(2v) \;\le\; \frac{1}{\pi}\big[3g_\theta(0) + 4g_\theta(v) + g_\theta(2v)\big].$$

More generally, for every cosine polynomial $p(\omega) = \sum_{k=0}^{K} a_k\cos k\omega$ with $p \ge 0$ and $a_k \ge 0$, $\sum_k a_k u_\theta(kv) \le \pi^{-1}\sum_k a_k g_\theta(kv)$.

Proof. $\sum_k a_k\Pi_\theta(kv) = \sum_n \Lambda(n)n^{-\theta}p(v\log n) \ge 0$, and $\Pi_\theta = g_\theta - \pi u_\theta$ by (13.8.1). The first display is $p = 3 + 4\cos\omega + \cos 2\omega = 2(1+\cos\omega)^2$. □

Corollary 13.8.2 (zero free region). *Let $c_* = (14 - 8\sqrt{3})/5 = 0.028719\ldots$ For every $c < c_*$ there is γ_c such that every zero with $\gamma_\rho \ge \gamma_c$ has $1 - \sigma_\rho \ge c/\log\gamma_\rho$.*

Proof. Take $v = \gamma_0$ and $\theta = 1 + a/\log\gamma_0$. For $\theta > 1$ every term of the Hadamard sum is positive, so $\pi u_\theta(\gamma_0) \ge 1/(\theta - \sigma_0)$, and $u_\theta(0), u_\theta(2\gamma_0) \ge 0$. Since $g_\theta(0) = 1/(\theta-1) + O(1)$ and $g_\theta(v) = \frac{1}{2}\log v + O(1)$, Theorem 13.8.1 gives $4/(\theta - \sigma_0) \le (3/a + \frac{5}{2} + o(1))\log\gamma_0$. Rearranging, $(1-\sigma_0)\log\gamma_0 \ge 4a/(3 + \frac{5}{2}a) - a - o(1)$. The maximum over $a > 0$ is attained at $a = (4\sqrt{3} - 6)/5 = 0.18564\ldots$, where $3 + \frac{5}{2}a = 2\sqrt{3}$, and equals $a(2/\sqrt{3} - 1) = c_*$. The $o(1)$ is why only $c < c_*$ follows. □

The choice $a = \frac{1}{4}$ gives the constant $\frac{3}{116}$ of Proposition 12.1.4. The constant c_* is the crude one: sharper cosine polynomials and smoothing give $1/5.69693$ (Kadiri [103]), $1/5.573412$ (Mossinghoff and Trudgian [122]) and $1/5.558691$ (Mossinghoff, Trudgian and Yang [123], for $|t| \ge 2$), constants as quoted in those papers and not checked here, and exponential sum estimates give the Vinogradov and Korobov shape [272]. None of this reaches a fixed $\theta < 1$. Through Corollary 13.1.3 these regions give positivity of u_θ only locally in v.

13.8.2 The Davenport and Heilbronn barrier

The function f of Davenport and Heilbronn has no Euler product, and the inequality of Theorem 13.8.1 is what fails; the rest of the framework survives.

Proposition 13.8.3 (the Davenport and Heilbronn barrier). *Let* $\Lambda_{\mathrm{DH}}(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$*, which is entire of order one, real on the real axis and satisfies* $\Lambda_{\mathrm{DH}}(s) = \Lambda_{\mathrm{DH}}(1-s)$*. The Hadamard sum, the identity* (13.1.2)*, Theorem 11.9.1 and Proposition 13.3.1 hold for* Λ_{DH} *with the same proofs; so does Theorem 14.2.1 with* $h = f$ *and the sum over all zeros of* f *to the right of* θ*; and so does Theorem 8.2.2 with the line* $\operatorname{Re} s = 1$ *replaced by* $\operatorname{Re} s = 2$*, to the right of all zeros since* $|f(s) - 1| \le \zeta(2) - 1 < 1$ *there. The function* f *has infinitely many zeros with* $\sigma_\rho > 1$ *[60], so* Λ_{DH} *violates* $\mathrm{QRH}(\theta)$ *for every* $\theta < 1$*; its lowest zeros off the line,* $0.8085\ldots + 85.6993\ldots i$ *and* $0.6508\ldots + 114.1633\ldots i$ *[261], already violate* $\mathrm{QRH}(\frac{3}{4})$ *and* $\mathrm{QRH}(\frac{5}{8})$*.*

The step from (ii) to (i) in Theorem 8.2.2 uses that ξ'/ξ is analytic in $\operatorname{Re} s > 1$, which fails for Λ_{DH}; positivity on the line $\operatorname{Re} s = 1$ (Theorem 10.1.1) and the constant of Proposition 13.7.5 are specific to ζ. So these properties do not prove $\mathrm{QRH}(\theta)$ for any $\theta < 1$; a further property is needed, naturally the Euler product. It enters through (13.8.1): for $\theta > 1$ the prime sum has nonnegative cosine coefficients, so a nonnegative cosine polynomial applied to it bounds a combination of u_θ at heights $0, v, 2v$ from above by explicit gamma terms, and a zero close to $1 + iv$ would make $u_\theta(v)$ too large. For f the corresponding combination has no sign: the source note reports that $3A_\theta(0) + 4A_\theta(v) + A_\theta(2v)$, with $A_\theta = -\operatorname{Re} f'/f$, is positive for ζ on $1 \le v \le 200$ (minima 142.2, 23.3 and 5.0 at $\theta = 1.02$, 1.1 and 1.3) and negative for f on more than half of that interval at every level, with its minimum near $v = 85.72$, the height of the zero at 0.8085 (values -17.5, -12.2, -6.3); the same note reports that the optimised curvature kernel of Table 13.12 (level $x = 0.382$, degree 4 polynomial), on 700 heights in $[10, 200]$, is positive for ζ (minimum 10.84) and negative for f on 55% of the heights (minimum -5.97 at $v = 85.84$). These values are the author's, not recomputed, and the source's figure is not reproduced.

13.8.3 Multilevel inequalities

Fix $v \ge 2$, write $L = \log v$, and parametrise levels to the right of the line $\operatorname{Re} s = 1$ by $\theta = 1 + x/L$ with $x > 0$. A level kernel is a finite signed measure w with compact support in $(0, \infty)$; it acts on the density by $w[u](y) = \int u_{1+x/L}(y)\, w(dx)$, and similarly on g and Π. Its Laplace transform is $\phi_w(\lambda) = \int e^{-x\lambda}\, w(dx)$.

Theorem 13.8.4 (multilevel inequality). *Let* $p(\omega) = \sum_{k=0}^{K} a_k \cos k\omega \ge 0$ *with* $a_k \ge 0$*, and let* w *be a level kernel with* $\phi_w(\lambda) \ge 0$ *for* $\lambda > 0$*. Then for every*

$v \ge 2$,

$$\sum_{k=0}^{K} a_k\, w[u](kv) \;\le\; \frac{1}{\pi} \sum_{k=0}^{K} a_k\, w[g](kv).$$

Proof. By (13.8.1), $\pi\, w[u] - w[g] = -w[\Pi]$, and, the support of w being compact in $(0,\infty)$ so that the series and the integral may be interchanged,

$$\sum_{k} a_k\, w[\Pi](kv) = \sum_{n\ge 2} \Lambda(n)\, n^{-1}\, p(v \log n)\, \phi_w(\log n/L) \;\ge 0. \qquad \square$$

The single level case $w = \delta_x$, with $\phi_w(\lambda) = e^{-x\lambda}$, is Theorem 13.8.1. The second ingredient controls the zeros.

LEMMA 13.8.5 (admissible kernels). *Let w be a level kernel supported in $[x_0, x_1] \subset (0,\infty)$, and let $F_w(z) = \int w(dx)/(z+x)$. The following are equivalent.*

(i) $\int w(dx)\, P_{x+y}(\tau) \ge 0$ *for all $y \ge 0$ and all real τ; that is, every zero with $\sigma_\rho \le 1$, seen through the kernel, contributes a nonnegative amount (widths and distances in units of $1/L$).*

(ii) $\operatorname{Re} F_w(i\tau) = \int w(dx)\, x/(x^2+\tau^2) \ge 0$ *for all real τ.*

(iii) $\int_0^\infty \cos(\tau\lambda)\, \phi_w(\lambda)\, d\lambda \;\ge\; 0$ *for all τ; that is, the even extension of ϕ_w is positive definite.*

When they hold, $|\phi_w(\lambda)| \le \phi_w(0)$ for all λ.

Proof. $\pi \int w(dx) P_{x+y}(\tau) = \operatorname{Re} F_w(y+i\tau)$ is harmonic in $\operatorname{Re} z > -x_0$, tends to 0 at infinity, and equals the left side of (ii) on the imaginary axis; the minimum principle on the right half plane gives (ii)⇒(i), and (i)⇒(ii) is the case $y = 0$. Since $F_w(z) = \int_0^\infty e^{-z\lambda} \phi_w(\lambda)\, d\lambda$, (ii) and (iii) are the same statement. The last claim is Bochner's theorem. $\square$

Call (p, w) admissible if p is as in Theorem 13.8.4 and w satisfies $\phi_w \ge 0$ and the conditions of Lemma 13.8.5. For two levels, $w = \delta_{x_1} - \kappa\, \delta_{x_2}$ with $x_1 < x_2$, admissibility is exactly $\kappa \le x_1/x_2$ (the cosine transform of w is $x_1/(x_1^2+\tau^2) - \kappa x_2/(x_2^2+\tau^2)$, which is nonnegative for all τ exactly then). Stechkin's device, which pairs each zero with its reflection, uses a two level kernel with κ above this bound, so it lies outside the class (Proposition 13.8.10).

COROLLARY 13.8.6 (zero free constant of an admissible pair). *Let (p, w) be admissible, with w of compact support in $(0,\infty)$ and $\phi_w(0) > 0$; put $I = \int_0^\infty \phi_w$, $J(y) = \int_0^\infty e^{-y\lambda} \phi_w(\lambda)\, d\lambda$ and $A = a_1 + \cdots + a_K$. If $(a_1 - a_0) I > \frac{A}{2} \phi_w(0)$, the equation $a_1 J(y) = a_0 I + \frac{A}{2}\phi_w(0)$ has a unique root $y^*(p,w) > 0$, and every zero with γ_ρ large has $(1-\sigma_\rho)\log\gamma_\rho \ge y^*(p,w) - o(1)$. Otherwise put $y^*(p,w) = 0$: the scheme gives no positive constant.*

Proof. J is continuous and strictly decreasing on $[0,\infty)$, since $\phi_w \geq 0$ is positive near 0, with $J(0) = I$ and $J(y) \to 0$; the hypothesis puts the right side strictly between $a_1 J(0)$ and 0, which gives a unique root. Apply Theorem 13.8.4 at $v = \gamma_0$. By Lemma 13.8.5 every zero contributes a nonnegative amount to the left side, so it is at least a_1 times the contribution of ρ_0 at height γ_0, which is $(L/\pi)J(y)$ with $L = \log\gamma_0$ and $y = (1-\sigma_0)L$. On the right, $g_{1+x/L}(0) = L/x + O(1)$ and $g_{1+x/L}(k\gamma_0) = \frac{1}{2}L + O(1)$ for $k \geq 1$, uniformly on the support of w, and $\int w(dx)/x = I$. Dividing by L/π gives $a_1 J(y) \leq a_0 I + \frac{A}{2}\phi_w(0) + o(1)$. For $\epsilon > 0$, $a_1 J(y^* - \epsilon)$ exceeds the right side by a fixed amount, so $y < y^* - \epsilon$ fails for γ_0 large. □

13.8.4 Optimising the constant

We computed y^* numerically, over cosine polynomials $p = |\sum_j b_j e^{ij\omega}|^2$ with $b_j \geq 0$, which makes every $a_k \geq 0$, and three families of kernels. The first is a single level, for which $y^* = 2(\sqrt{a_1} - \sqrt{a_0})^2/A$ at the best level. The second is two levels with $\kappa = x_1/x_2$, where the optimum pushes $x_2 \to x_1$. In that limit the kernel becomes

$$w[u] \;\longrightarrow\; u_\theta - (\theta-1)\,\partial_\theta u_\theta = u_\theta + \frac{\theta-1}{\pi}\,\partial_v^2 \log|\xi(\theta+iv)|,$$

the density corrected by the curvature of $\log|\xi|$ in v, using harmonicity, after division by $1 - x_1/x_2$ (the derivative kernels are limits of finite measure kernels and not covered by Theorem 13.8.4 itself). Its Laplace transform is $\phi(\lambda) = e^{-x\lambda}(1 + x\lambda)$. The third family is a single level with derivatives, $\phi(\lambda) = e^{-x\lambda}q(\lambda)$ with q a polynomial, constrained by $q \geq 0$ and by $\sum_j q_j\, j!\,\mathrm{Re}(x+i\tau)^{-j-1} \geq 0$, both imposed on finite grids.

The curvature correction improves every polynomial by about 30%, and derivatives of the level kernel add 13% at degree 4 and a little more at degree 5. All of this sits below the cap of the next proposition, which is not sharp; the last subsection puts the best kernel within about 5% of the linear programming values (computed).

13.8.5 The $1/\log\gamma$ barrier

Proposition 13.8.7 (no admissible pair beats $1/\log\gamma$). *Let (p,w) be admissible with $a_1 > a_0$, normalised by $\phi_w(0) = 1$, and let $y > 0$ satisfy $a_1 \int_0^\infty e^{-y\lambda}\phi_w(\lambda)\,d\lambda \geq a_0 \int_0^\infty \phi_w(\lambda)\,d\lambda + \frac{A}{2}$. Then*

$$y \;\leq\; \frac{2}{A}\Big(a_1 - a_0 - a_0\log\frac{a_1}{a_0}\Big) \;<\; \frac{2(a_1-a_0)}{A} \;\leq\; 1.$$

In particular $y^*(p,w)$ of Corollary 13.8.6 obeys these bounds; the first is 0.0548 *for* $3 + 4\cos\omega + \cos 2\omega$ *and* 0.1051 *for the degree* 4 *polynomial of Table 13.12.*

kernel	cosine polynomial	$c = y^*$	$1/c$
single level	$3 + 4\cos\omega + \cos 2\omega$	0.0287	34.8
single level	best of degree 2	0.0376	26.6
single level	best of degree 3	0.0542	18.5
single level	best of degree 4 to 8	0.0573	17.4
two levels, curvature limit	$3 + 4\cos\omega + \cos 2\omega$	0.0377	26.5
two levels, curvature limit	best of degree 3	0.0704	14.2
two levels, curvature limit	best of degree 4	0.0742	13.5
level with derivatives, $\deg q = 2$	best of degree 4	0.0800	12.5
level with derivatives, $\deg q = 3$	best of degree 4	0.0826	12.1
level with derivatives, $\deg q = 4$	best of degree 4	0.0841	11.9
level with derivatives, $\deg q = 5$	best of degree 4	0.0849	11.8
cap of Proposition 13.8.7	best of degree 4	≤ 0.1051	≥ 9.5
one far level paired (Proposition 13.8.10), outside the class	best of degree 4	0.10367	9.646

Table 13.12: Zero free constants $1 - \sigma_\rho \ge c/\log\gamma_\rho$ from admissible pairs, in the scheme of Corollary 13.8.6 (computed). The optimal polynomial of degree 4 has $a \approx (1, 1.705, 1.044, 0.425, 0.089)$, and the single level rows to degree 8 are over squares with $b_j \ge 0$ (with only $p \ge 0$ and $a_k \ge 0$ the value grows by about 1% at degree 8); the best level for the derivative family at degree 4 is $x = 0.89$. The rows with derivatives run to $\deg q = 5$ and are optima on discrete constraint grids (computed); at $\deg q = 4$ and 5 the kernels are not certified admissible (at $\deg q = 5$ a dense audit of positive definiteness gives -1.6×10^{-6}). The cap of the source note, 0.432 for the degree 4 polynomial, is replaced by the sharper cap of Proposition 13.8.7. The last row is outside the class of Lemma 13.8.5.

Proof. A nonzero nonnegative cosine polynomial has $a_0 > 0$. By Lemma 13.8.5, $0 \le \phi_w \le \phi_w(0) = 1$. The function $h(\lambda) = a_1 e^{-y\lambda} - a_0$ is positive exactly for $\lambda < \lambda_* = y^{-1}\log(a_1/a_0)$, so

$$\frac{A}{2} \le \int_0^\infty h\,\phi_w \le \int_0^{\lambda_*} h = \frac{a_1 - a_0 - a_0\log(a_1/a_0)}{y},$$

which is the first inequality. The second holds because $\log(a_1/a_0) > 0$. Finally $a_1 \le A$, and $a_1 = \pi^{-1}\int_0^{2\pi} p(\omega)\cos\omega\,d\omega \le \pi^{-1}\int_0^{2\pi} p = 2a_0$, so $2(a_1 - a_0)/A \le 2(1 - a_0/a_1) \le 1$. □

Within this class the scheme gives $1 - \sigma_\rho \ge c/\log\gamma_\rho$ with $c < 1$, never a region of fixed width. The proof uses $\phi_w \le \phi_w(0)$, which kernels that pair each zero

with its reflection violate (Proposition 13.8.10), so it says nothing about them. The term $\frac{A}{2}\phi_w(0)\log v$ is the Cauchy smoothing of the Riemann von Mangoldt density, entering through g, while positive definiteness caps the kernel's weight on the zero itself at $\phi_w(0)/y$, so the zero cannot outweigh the density once y is of order one.

Remark 13.8.8 (what goes beyond; heuristic). The Vinogradov and Korobov region beats $1/\log v$ by replacing the gamma bound $g_\theta(v) \approx \frac{1}{2}\log v$ with an upper bound for $\log|\zeta|$ near $\theta = 1$ from exponential sums. Through Jensen's formula this replaces the density term $\frac{1}{2}\log v$ by a quantity of order $(\log v)^{2/3}(\log\log v)^{1/3}$ in a disc of radius about $(\log v)^{-2/3}$ around $1+iv$; in the Thorin language that is a local upper bound on the mass of $\mathfrak{U}_\theta$ near height v, sharper than the density. A fixed zero free strip would need a uniform bound of the type $\log|\zeta(\theta+iv)| = o(\log v)$ for θ in a fixed neighbourhood of 1, together with an inequality that converts it into exclusion of zeros. Even the Lindelöf hypothesis is not known to suffice. The suggestion that the method alone yields only regions of width $(\log v)^{-2/3}$ up to logarithms is a heuristic and unchecked.

13.8.6 The computed ceiling, and Stechkin's pairing

The kernels of Table 13.12 reach 0.085 while Proposition 13.8.7 caps them at 0.105. We computed the best constant attainable in the scheme of Corollary 13.8.6 by any smoothing in a larger formal class: every ϕ with $0 \le \phi \le 1$ and $\phi(0) = 1$ whose even extension is positive definite, not only Laplace transforms of level kernels. This is the class in which the proof of Proposition 13.8.7 still runs; whether each such ϕ comes from a level kernel is not claimed. For a fixed cosine polynomial and a fixed y, whether some ϕ certifies y is a linear program:

$$\max_{\phi}\Big\{a_1\int_0^{\lambda_{\max}} e^{-y\lambda}\phi - a_0\int_0^{\lambda_{\max}}\phi\Big\} \quad \text{subject to}$$

$$0 \le \phi \le 1, \qquad \phi(0) = 1, \qquad \int_0^{\lambda_{\max}} \cos(\tau\lambda)\phi(\lambda)\,d\lambda \ge 0 \text{ for all } \tau,$$

and y is certified exactly when the maximum is at least $A/2$. For a fixed ϕ, optimising the polynomial (with $p \ge 0$ imposed on a grid of angles and $a_k \ge 0$) is again a linear program. We alternate the two and bisect on y; the joint problem is not linear, so this is a search and not a global optimisation. We discretise with $\lambda_{\max} = 12$ and 300 points in λ and impose positive definiteness at finitely many frequencies τ.

In this class, where every zero contributes nonnegatively (Lemma 13.8.5(i)), the discretised linear program finds about $1/(11.3\log\gamma)$ for polynomials of degree at most 8 (Table 13.13). The values are computed, not certified: positive definiteness is imposed at finitely many frequencies, the support is truncated at $\lambda_{\max}$, and the polynomial and the smoothing are optimised alternately. For $3+4\cos\omega+\cos 2\omega$

problem	ceiling c	$1/c$
$p = 3 + 4\cos\omega + \cos 2\omega$, smoothing optimised; $\lambda_{\max} = 12$, 300 points, 460 uniform frequencies to 400	0.0460	21.7
same, 699 frequencies concentrated below 25	0.0450	22.2
same, $\lambda_{\max} = 20$, 600 points, frequencies concentrated below 25	0.0451	22.2
degree 4 polynomial of Table 13.12, smoothing optimised	0.0874	11.4
polynomial of degree ≤ 4 and smoothing both optimised	0.0874	11.4
polynomial of degree ≤ 8 and smoothing both optimised	0.0881	11.35
best level kernel (Table 13.12)	0.0849	11.8
cap of Proposition 13.8.7, degree 4	0.1051	9.5
one far level paired (Proposition 13.8.10), degree 4, outside the class	0.10367	9.646

Table 13.13: Ceiling of the scheme of Corollary 13.8.6 over nonnegative positive definite smoothings, by discretised linear programming (computed, not certified). Refining the frequency grid near the origin moves the value for $3+4\cos\omega+\cos 2\omega$ from 0.0460 to 0.0450; on the stated uniform grid, changing the number of points or $\lambda_{\max}$ moves the third digit. The rows with the optimised polynomial are lower estimates, since the alternating searches end at local optima (the degree 4 gain over the fixed polynomial is below 10^{-4}).

Proposition 13.8.7 proves the bound 0.0548, against the computed 0.0450. On grids that resolve the origin the best level kernel, 0.0849, is about 4% below the computed value 0.0881 for the whole class at degree at most 8; raising the degree of the polynomial beyond 4 adds about 1%.

Pairing each zero with its reflection $1-\bar\rho$ leaves the class. The kernel of the pairing, $\delta_x - \kappa\delta_{x_1}$ with $x_1 = (\theta_1 - 1)L$ of order L, has the cosine transform $x/(x^2+\tau^2) - \kappa x_1/(x_1^2+\tau^2)$, which is negative for large τ because $x < \kappa x_1$, so it is not admissible and there is no contradiction with the linear program. The following lemma is the pair inequality behind Stechkin's device [172], proved here by direct algebra.

Lemma 13.8.9 (Stechkin's pair inequality). *Let $\theta > 1$, $\theta_1 = \frac{1}{2}\bigl(1+\sqrt{1+4\theta^2}\bigr)$ and $\kappa = (2\theta-1)/(2\theta_1-1)$. For $0 \le b \le 1$ and real τ put*

$$F_\theta(b,\tau) = \operatorname{Re}\Bigl[\frac{1}{\theta - b + i\tau} + \frac{1}{\theta - 1 + b + i\tau}\Bigr].$$

Then $F_\theta(b,\tau) \ge \kappa\, F_{\theta_1}(b,\tau)$.

Proof. Put $c = \theta - \frac12$, $c_1 = \theta_1 - \frac12$, $e = (b-\frac12)^2 \le \frac14$, $q = \tau^2$ and $K = 4eq$. With $p = \theta - b$ and $p' = \theta - 1 + b$, both positive,

$$F_\theta = \frac{p}{p^2+q} + \frac{p'}{p'^2+q} = \frac{(p+p')(pp'+q)}{(p^2+q)(p'^2+q)} = \frac{2cX}{X^2+K}, \qquad X = c^2 - e + q,$$

because $p + p' = 2c$, $pp' = c^2 - e$ and $(p^2+q)(p'^2+q) = (c^2-e+q)^2 + 4eq$. In the same way $F_{\theta_1} = 2c_1X_1/(X_1^2+K)$ with $X_1 = X + D$ and $D = c_1^2 - c^2 = \theta > 0$. Since $\kappa = c/c_1$,

$$F_\theta - \kappa F_{\theta_1} = 2c\Big[\frac{X}{X^2+K} - \frac{X+D}{(X+D)^2+K}\Big] = \frac{2cD\,\big(X(X+D)-K\big)}{(X^2+K)\big((X+D)^2+K\big)},$$

and

$$X(X+D) - K = q^2 + (c^2 + c_1^2 - 6e)\,q + (c^2-e)(c_1^2-e).$$

Here $c^2 \ge \frac14 \ge e$, $c_1^2 > c^2$ and $c^2 + c_1^2 = 2\theta^2 - \theta + \frac12 \ge \frac32 \ge 6e$, so every coefficient is nonnegative. □

Proposition 13.8.10 (one far level and no smoothing). *Let* $p(\omega) = \sum_{k=0}^K a_k \cos k\omega \ge 0$ *with* $a_k \ge 0$ *and* $a_1 > a_0$, *let* $A = a_1 + \cdots + a_K$ *and* $\kappa_* = 1/\sqrt5$. *Every zero* $\rho_0 = \sigma_0 + i\gamma_0$ *with* γ_0 *large has*

$$(1-\sigma_0)\log\gamma_0 \;\ge\; \frac{2(\sqrt{a_1}-\sqrt{a_0})^2}{(1-\kappa_*)A} - o(1).$$

For $p = 3 + 4\cos\omega + \cos2\omega$ *the constant is* 0.05195*; for the degree* 4 *polynomial of Table 13.12,* $a \approx (1, 1.705, 1.044, 0.425, 0.089)$, *it is* $0.10367 = 1/9.646$ *(computed values; the proposition is proved).*

Proof. Let $L = \log\gamma_0$, $\theta = 1 + x/L$ with $x > 0$, and θ_1, κ as in Lemma 13.8.9, so that $\kappa = \kappa_* + O(1/L)$. For $\theta' > 1$ write $\pi u_{\theta'} = g_{\theta'} - \Pi_{\theta'}$ as in (13.8.1). Since $n^{-\theta} - \kappa n^{-\theta_1} > 0$ for $n \ge 2$,

$$\sum_k a_k\big[\Pi_\theta(k\gamma_0) - \kappa\Pi_{\theta_1}(k\gamma_0)\big] = \sum_{n\ge2}\Lambda(n)\big(n^{-\theta} - \kappa n^{-\theta_1}\big)p(\gamma_0\log n) \ge 0,$$

so $\sum_k a_k[u_\theta - \kappa u_{\theta_1}](k\gamma_0) \le \pi^{-1}\sum_k a_k[g_\theta - \kappa g_{\theta_1}](k\gamma_0)$. By the Hadamard sum, grouping the zeros into orbits of $\rho \mapsto 1-\bar\rho$, which preserves ordinates, $\pi[u_\theta - \kappa u_{\theta_1}](y)$ is a sum over orbits of $F_\theta(\sigma_\rho, y-\gamma_\rho) - \kappa F_{\theta_1}(\sigma_\rho, y-\gamma_\rho)$ (half of it for a zero on the critical line), each nonnegative by Lemma 13.8.9. Keep only the orbit of ρ_0 in the term $k = 1$; at $y = \gamma_0$ it contributes at least $1/(\theta-\sigma_0) - O(1)$. On the right, $g_\theta(0) - \kappa g_{\theta_1}(0) = L/x + O(1)$, from the pole of ζ, and $g_\theta(k\gamma_0) - \kappa g_{\theta_1}(k\gamma_0) = (1-\kappa)\frac{L}{2} + O(1)$ for $k \ge 1$, by Stirling's formula. With $y = (1-\sigma_0)L$, so that $\theta - \sigma_0 = (x+y)/L$, division by L/π gives

$$\frac{a_1}{x+y} \le \frac{a_0}{x} + \frac{(1-\kappa_*)A}{2} + o(1),$$

and the choice $x = 2(\sqrt{a_0a_1} - a_0)/((1-\kappa_*)A)$ gives the bound. □

The Stechkin constant over polynomials is the single level optimum divided by $1-1/\sqrt{5}$: the best polynomials of degree 2, 3 and 4 give $0.06809 = 1/14.687$, $0.09800 = 1/10.204$ and $0.10367 = 1/9.646$ (computed). The last is the constant 9.646 of Rosser and Schoenfeld [165], an attribution recalled from the literature and not verified here. The constants $1/(5.573412\log\gamma)$ [122] and $1/(5.558691\log\gamma)$ [123] combine the pairing with smoothing and a finer accounting. The gap between the computed ceiling of the class and the constants that use pairing is open.

13.9 Nested sampling over heights

The earlier computations live below height 200. Since ξ'/ξ costs about 0.07 seconds even at height 10^7 (Riemann and Siegel in mpmath, a timing reported by the source), the question is how to spend evaluations at large heights. Nested sampling [170] concentrates them on the most suspicious heights and, under idealised assumptions (Subsection 13.9.1), estimates the prior volume of the set of near failures of Thorin positivity. Heights and scores below are recomputed in ball arithmetic; the run parameters, the volumes $\log X$ and the run times are the author's and were not rerun.

13.9.1 The score and the algorithm

At height t define the normalised Thorin density

$$s(t) = \frac{u_{\theta_0}(t)}{\varrho(t)}, \qquad \theta_0 = 0.6, \qquad \varrho(t) = \frac{1}{2\pi}\log\frac{qt}{2\pi},$$

with $q = 1$ for ζ and $q = 5$ for the Davenport and Heilbronn function, so that s has mean about one. A negative value of $s(t)$ forces a zero ρ with $\sigma_\rho > \theta_0$ and $|t-\gamma_\rho| < \sqrt{(\sigma_\rho-\theta_0)(\sigma_\rho+\theta_0-1)} < \frac{1}{2}$, by the pairing in the proof of Proposition 3.2.1(ii). A zero with $\sigma_\rho > \theta_0$ makes u_θ negative near its height at levels θ just below σ_ρ (Theorem 11.9.1); at the fixed level θ_0 it need not (Lemma 14.3.9, Remark 2.1.4), and Theorem 13.7.4, which integrates over levels, does not bound s. For ζ the positivity of s below height $3\times10^{12}-\frac{1}{2}$ is Corollary 14.3.7(1) with [232]; the computation measures the margin, and the lower tail of s is governed by the largest gaps between consecutive zeros. In unfolded coordinates $x = \varrho t$, and keeping only the zeros near t, $s \approx \sum_j P_b(x-x_j)$ with $b = (\theta_0-\frac{1}{2})\varrho(t)$.

Take the prior $t \sim U[T_1, T_2]$ and the likelihood $-s(t)$. Nested sampling keeps N live heights. At iteration k it removes the live height with the largest s, records the threshold ℓ_k, and replaces it by a copy of another live height moved by M Metropolis steps for the uniform law on $\{s < \ell_k\}$. Each step is a local Gaussian move of a quarter of the mean spacing with probability 0.9, or a global uniform proposal with probability 0.1. Under exact independent replacements and a continuous score distribution, the prior volume $X_k = P(s(t) < \ell_k)$ satisfies that $-\log X_k$ is

Gamma distributed with shape k and rate N, so $\log X_k = -k/N \pm \sqrt{k}/N$ in the sense of mean and standard deviation [170, 41], and the error of the log volume is about $\sqrt{\log(1/X)/N}$, which does not vanish for rare sets. The implemented clones are not independent, so these statements are the ideal model and not an error certificate. The method is related to adaptive multilevel splitting [36]; the concentration theorem of Section 13.5 is not shown to cover it, since its thresholds are random and its potentials vanish.

13.9.2 Results

Davenport and Heilbronn On $t \in [1000, 1100]$, with $N = 50$ live heights, $M = 4$ moves and 300 iterations (1250 evaluations of Λ'/Λ, twelve minutes), no point of the initial uniform sample had $s < 0$. Nested sampling crossed $s = 0$ at iteration 239, so the set of heights with $s < 0$ has estimated prior volume $X = e^{-4.78 \pm 0.31} \approx 0.008$, about 0.8 units of height. Its negative points form two clusters, near 1012.0 and 1018.8. Newton's method started there gives two zeros of f off the line, computed in ball arithmetic and certified (one zero in a square of half width 10^{-18} around each centre, at 256 and 384 bits),

$$\rho = 0.691404808984 + 1012.019981489088\, i,$$
$$\rho = 0.698097971267 + 1018.795762313197\, i,$$

together with their reflections. Both zeros are in the table of Balanzario and Sánchez-Ortiz [17], and zeros off the line of this function are classical [60, 261, 17]. The argument principle in interval arithmetic, with $f \neq 0$ for $\operatorname{Re} s \geq 2$ since $|f(s) - 1| \leq \zeta(2) - 1 < 1$ there, gives 0 zeros in $[0.6, 2] \times [1000, 1010]$ and 2 in $[0.6, 2] \times [1010, 1020]$ (computed, certified). So nested sampling found every zero with $\beta > 0.6$ in $[1000, 1020]$, and the two it found are all of them there. The rest of the band $[1020, 1100]$ was not counted.

ζ at heights 10^6 to 10^{10} The source ran nested sampling on bands of length 10^4 starting at 10^6, 10^7, 10^9 and 10^{10}. Each band contains between 19,000 and 34,000 zeros.

For the most suspicious heights of the bands at 10^6 and 10^7 the source reports argument principle counts of zeros with $\beta > 0.6$ in $[0.6, 2] \times [t - \frac{1}{2}, t + \frac{1}{2}]$ equal to 0 (floating point quadrature, not recomputed), as they must be inside the range verified in [232]. At 10^9 and 10^{10} the runs are shorter and the live points collapse into a single gap, so those minima (0.252 and 0.292) are upper bounds on the band minimum. The source reports that the sine kernel predicts the 10^{-4} quantile of s at 0.231 for 10^9 and the 10^{-3} quantile at 0.287 for 10^{10}, at the matching $b = 0.30$ and 0.34 (not recomputed).

band	N, M, K	evals	$\log X$	min s	height, normalised gap
10^6	100, 8, 1200	9700	-12	0.175	1009611.61, gap 2.44
10^7	100, 8, 1200	9700	-12	0.182	10005411.18, gap 2.62
10^9	100, 8, 1000	8100	-10	0.252	1000004287.61, gap 2.54
10^{10}	50, 4, 400	1650	-8	0.292	10000009878.55, gap 2.38

Table 13.14: Nested sampling for ζ on bands of length 10^4, with N live heights, M moves per replacement and K iterations (computed). In every band the score stays positive. The column min s is the smallest value found by the run, not the band minimum. The scores at the printed heights and the normalised gaps, measured between consecutive zeros found as sign changes of Hardy's Z function, are recomputed with enclosure radius below 6×10^{-12}; the run parameters and $\log X$ are the source's. The smallest values sit in the largest normalised gaps.

Checking the volume estimates, and a failure mode A plain Monte Carlo sample of 5000 uniform heights in the band at 10^6 gives $P(s < 0.399) = 0.130$, $P(s < 0.235) = 0.0104$ and $P(s < 0.2005) = 0.0018$. Nested sampling put these levels at $X = 0.10$, 0.010 and 0.0010. The two agree to within the stated error down to $X \approx 10^{-2}$ in this one comparison, and nested sampling underestimates by a factor of about two at 10^{-3}. The Monte Carlo sample also found a point with $s = 0.142$, in a gap of normalised length 2.87 at height 1004606.8 (both values recomputed: $s = 0.14185$, gap 2.865), deeper than anything nested sampling reached. The set $\{s < \ell\}$ is a union of thousands of short intervals, one inside each large gap, and local moves cannot travel between them; once global proposals are rarely accepted, the live points collapse into a few intervals. No reliability threshold in the volume is established; beyond about 10^{-2} nested sampling is a search heuristic, not an exhaustive search. The remedy, not yet tried, is a gap guided form that proposes new heights at the large gaps directly, located cheaply by sign changes of Hardy's Z function.

Comparison with the sine kernel The source compares the tails with the prediction of the sine kernel process, modelled by circular unitary matrices of size 600 at the matching value of b, a finite unitary ensemble and not the infinite sine process; the uncertainty of the extreme tail, which comes from few matrices, is not estimated. At volumes 10^{-1} and 10^{-2} the band at 10^7 gives 0.394 against 0.410 and 0.258 against 0.259, and the band at 10^6 agrees to about 10% (not recomputed). Deeper in the tail the zeros of ζ give fewer small values of s than the sine kernel model, both in nested sampling and in plain Monte Carlo (0.0018 against about 0.004 at $s < 0.2005$). This is the direction of the finite height deficit of large gaps, which Bogomolny, Bohigas, Leboeuf and Monastra describe by a unitary ensemble of effective size $\log(t/2\pi)/\sqrt{12\Lambda_0}$ with $\Lambda_0 = 1.57314\ldots$ [28]. The coefficient is 0.2302 (not $0.2887 = 1/\sqrt{12}$), and the sizes are 2.76 at 10^6, 3.29 at 10^7, 4.35 at

10^9 and 4.88 at 10^{10} (computed). Their comparison with the finite height data is heuristic and has not been checked quantitatively.

13.9.3 What this buys

Nested sampling does not verify RH at new heights; exhaustive verification by Turing's method already reaches 3×10^{12} [232]. It gives a cheap, scale free measure of how close ζ comes to violating positivity of u_{θ_0}: the smallest margins found are between 0.14 and 0.29 of the mean density at heights 10^6 to 10^{10}, always in the largest gaps, with a lower tail that agrees with the sine kernel model down to volumes of about 10^{-2} and has fewer extreme gaps beyond (computed, heuristic as a statement about the tail). As a heuristic search it found two zeros off the line of the Davenport and Heilbronn function in $[1010, 1020]$. Lifting it to a certificate needs the gap guided proposals above and an exhaustive count, which brings it back to Turing's method; the certificates that the book has are in Section 11.9.7, Theorem 11.9.6, Theorem 11.10.2 and Section 11.10.

13.10 What would close the gap

The proved statements of this chapter are Theorems 13.7.1, 13.5.1, 13.8.1, 13.8.4 and 13.7.4, Corollaries 13.1.3, 13.8.2, 13.7.3 and 13.8.6, Lemmas 13.8.5 and 13.8.9, and Propositions 13.2.1, 13.3.1, 13.4.1, 13.6.1, 13.7.2, 13.8.7, 13.8.10 and 13.8.3; Proposition 13.7.5 is conditional on QRH(θ), and the formula of Balazard, Saias and Yor is Theorem 14.2.1. With Section 13.1 they place the reciprocal Xi programme as follows.

1. $\mathfrak{U}_1$ is unconditionally a positive Cauchy mixture, and $\mathfrak{U}_{1/2}$ is the counting measure of the critical zeros. The passage between them is one step of the Cauchy semigroup, and RH is exactly the statement that this step can be inverted within positive measures (Theorem 8.2.2).

2. Positivity of $\mathfrak{U}_\tau$ for every $\tau > \sigma$ is QRH(σ), a zero free half plane (Theorem 11.9.1), while positivity at the single level σ is necessary and not known to be sufficient (Remark 2.1.4). There is no slack: the Thorin route to RH is the zero free region problem in another language.

3. Unconditionally, $\mathfrak{U}_\theta$ is positive up to a fraction $O(T^{-(\theta-1/2)/4})$ of its mass on $(0, T]$ (Theorem 13.7.1, proved with Selberg's density theorem). This is the unconditional content of the Thorin picture inside the strip.

4. The Davenport and Heilbronn function satisfies the Hadamard sum, the identity (13.1.2), Theorem 11.9.1, Proposition 13.3.1, Theorem 14.2.1 and Theorem 8.2.2 (with the line $\operatorname{Re} s = 2$), and it has zeros off the line (Proposition 13.8.3). Its zeros with $\beta > 1$ show that these properties do not prove

QRH(θ) for any $\theta < 1$: some further property is needed, the Euler product being the natural one. It enters as an upper bound on positive combinations of the Thorin density at several heights (Theorem 13.8.1), which ζ satisfies and the Davenport and Heilbronn function violates, most strongly at the height of its zero off the line. With the Cauchy structure this yields positivity within $c/\log t$ of $\theta = 1$ (Corollary 13.8.2, with the local form of Corollary 13.1.3). Combining levels as well as heights improves the best single level constant by about 48% (computed, Table 13.12), but in the class in which every zero contributes nonnegatively no inequality gives more than $1 - \beta \geq c/\log\gamma$ with a constant $c < 1$ (Proposition 13.8.7): the density of the zeros, entering through the gamma factor, caps it. The ceiling of that class, computed by a discretised linear program and not certified, is about $1/(11.3\log\gamma)$ (Table 13.13). Pairing each zero with its reflection leaves the class (Proposition 13.8.10), so the barrier says nothing about it.

5. Every zero off the line forces negative Thorin mass at least its isolation distance over $\log T$ (Theorem 13.7.4); for separated zeros the negative mass and the zero count control each other up to a factor $\log T$ when the isolation is at least of order $1/\log T$ and the counts are of order $\log T$ or smaller (Corollary 13.7.3).

The computational statements are local; nested sampling (Section 13.9) samples heights up to 10^{10} and certifies only the windows counted in interval arithmetic. The particle filter recovers the zeros of ζ in a window to four decimals and bounds their distance from the line by 5×10^{-4} at the 95% level under the Gaussian tolerance model, and it matches the first order Fisher information of the model with free real parts (heuristic; Section 13.3). The GBC screen scans a window in milliseconds and found all four zeros off the line of the Davenport and Heilbronn function below height 200, with one boundary false alarm (the window centred at 180, Table 13.6) and none for ζ. The contour moments recover a zero off the line to ten digits, computed in ball arithmetic (Table 13.4). None of this extends past the verified height 3×10^{12}, and none of it is a proof.

A proof along these lines needs one of two things. The first is an arithmetic input, coming from the Euler product, that controls u_θ uniformly in t for some $\theta < 1$ beyond the Vinogradov and Korobov boundary. Section 13.8 shows that positivity inequalities of de la Vallée Poussin type, in which every zero contributes nonnegatively, cannot supply it: they are capped at $c/\log t$ with $c < 1$, and in the computed basic form at about $1/(11.3\log t)$. The input could instead be a growth bound for ζ near $\theta = 1$, as in Vinogradov and Korobov, but uniform (heuristic). The second is a density estimate strong enough to force the negative Thorin mass to vanish. Theorem 13.7.4 shows this direction is not lost to clustering: every zero off the line pays negative mass in proportion to its isolation. But the density estimate needed is itself equivalent to the absence of such zeros. Both routes are

open.

Chapter 14

Balazard, Saias and Yor: Monte Carlo along vertical lines

Balazard, Saias and Yor [20] proved in 1999 that the Riemann Hypothesis is equivalent to a single Cauchy average of $\log|\zeta|$ on the critical line being zero. The Thorin programme of this book studies the measures $\mathfrak{U}_\sigma$ with density $m_\sigma = \pi^{-1}\operatorname{Re}\xi'/\xi(\sigma+it)$ and their images U_σ, the Thorin measures of $\xi(\sigma)/\xi(\sigma+\sqrt{s})$ (equation (10.1.1) and Chapter 10). This chapter shows that the two are the same harmonic analysis, one derivative in σ and one archimedean term apart, and then solves the resulting Cauchy averages numerically by quadrature, by plain Monte Carlo and by a vertical likelihood sampler after Polson and Scott (Section 14.6; computed, where the plain sampler has the smaller standard error). Chapter 13 is the companion: it cites Theorem 14.2.1 for the criterion, and its particle filter and nested sampling over heights supply the scans of Section 14.6.

The chapter is based on [129]. The zeros and counts behind Theorem 14.4.3 are certified in ball arithmetic; the other tables were recomputed in floating point, and where the recomputation differed from the note the recomputed value is printed.

Every value at the level $\frac{7}{8}$ is conditional on the quasi Riemann hypothesis $\mathrm{QRH}(\frac{7}{8})$, announced by OpenAI [227], formalised in Lean and not refereed (Section 17.7). The notation is that of Chapters 10 to 13: $\mathfrak{U}_\sigma$ is the measure on $\mathbb{R}$ without the factor $\frac{1}{2}$; $P_a(x) = a/(\pi(a^2+x^2))$ is the Cauchy density of scale a and, for $a<0$, the same formula, so that $P_{-a} = -P_a$ as in Proposition 2.1.3 (Chapter 13 writes Q_a for this signed kernel); and $\mathrm{QRH}(\theta)$ is the statement that ζ has no zero with $\beta > \theta$.

The chapter contains:

1. a dictionary between the logarithmic level (Balazard, Saias and Yor) and the Thorin level: the Poisson integral becomes a Cauchy smoothing of $\mathfrak{U}_{1/2}$, and the Green potential of each zero off the line a signed Cauchy defect, with an archimedean term that the logarithmic level does not see (Section 14.3);
2. the criterion at every level $\theta \in [\frac{1}{2}, 1)$ and every height, as the Nevanlinna formula of the half plane for $\zeta(s)(s-1)/(s-2\theta+1)$, which characterises

QRH(θ); the case $\theta = \frac{1}{2}$, $t_0 = 0$ is the theorem of Balazard, Saias and Yor (Section 14.2);

3. the reading of the 2 $\sin^2$ lemma (equation (2.1.9)): the Lévy density of the reciprocal Xi function is positive with no hypothesis, and RH is equivalent to that reciprocal being an extended generalised gamma convolution (EGGC) with the Thorin measure $\mathfrak{U}_{1/2}$ (Section 14.4);
4. a probabilistic reading of the criterion by Brownian motion and the Riesz decomposition, in the spirit of Yor (Section 14.5), and its complex form: under QRH(θ) the boundary values of $\log h_\theta$ are the stopped value of a time changed complex Brownian motion, so $\log|\zeta|$ and $\arg h_\theta$ have equal variances and zero covariance under harmonic measure at every height; without the hypothesis the defects of these moments are integrals along horizontal cuts from the zeros to the line (Proposition 14.5.4);
5. a proof that the 2 $\sin^2$ Lévy density of the Davenport and Heilbronn function is positive although it has zeros off the line, the three finite zero computations behind it being certified in ball arithmetic, so infinite divisibility of the reciprocal does not force the zeros onto the line (Section 14.4);
6. a vertical (layer cake) form of the criterion, in which RH says that the upper and lower tails of $\log|\zeta|$ under harmonic measure balance, solved by quadrature, by plain and vertical Monte Carlo, and by the weighted Monte Carlo of Yakowitz, Krimmel and Szidarovszky with an exact Dirichlet polynomial control; an exact local form of the criterion on a half disc, for which the sorted trapezoid rule has mean square error $O(N^{-4})$, and $O(N^{-5})$ after a change of variable (proved); and the truncated whole line average, whose tail is $O(T^{-2}\log T)$ under RH [16] (computed in Table 14.6; Section 14.6);
7. a smoothed Riesz identity in the plane, with the Cauchy kernel replaced by a compactly supported bump. The integrand is then smooth away from the zeros, the Voronoi and Delaunay forms of the weighted rule have root mean square error $O(N^{-3/2})$ at independent uniform nodes (proved, Theorem 14.7.3, which gives $O(N^{-1/2-2/d})$ in dimension d; computed at jittered nodes), a shifted grid reaches machine precision in a box free of zeros, and deflation handles a zero in the box; the same count is a linear functional of the horizontal logarithmic derivative alone (Section 14.7).

Assuming RH, Bui, Lester and Milinovich [16] bound the tail of the truncated Balazard, Saias and Yor integral and disprove a conjecture of Borwein, Bradley and Crandall [13] on its rate of decay; Section 14.6.6 uses their results. Except for the positivity of $\mathfrak{U}_\sigma$ for $\sigma \geq 1$ and Theorem 14.5.2, every identity of Sections 14.1 to 14.5 holds, with the changes listed in Section 14.8, for the Davenport and Heilbronn function, which has zeros off the line; Section 14.8 records what that implies.

14.1 Thorin measures along vertical lines

Write $\rho = \beta + i\gamma$ for the zeros of ζ (the zeros of ξ), counted with multiplicity. For $\sigma > \frac{1}{2}$ that is not the real part of a zero let $\mathfrak{U}_\sigma$ be the measure on $\mathbb{R}$ with density $m_\sigma(t) = \pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it)$ in the variable t. For $\sigma \geq \theta_0 = \sup \beta$ this is the positive measure of equation (10.1.1), with $m_\sigma = u_\sigma$ of Theorem 10.1.1; for $\frac{1}{2} < \sigma < \theta_0$ it is a signed measure. At a level $\sigma = \beta$ through a zero the density has a simple pole, and $\mathfrak{U}_\sigma$ is the right limit as in Remark 13.1.1. The Thorin measure U_σ is the image of the restriction of $\mathfrak{U}_\sigma$ to $t > 0$ under $t \mapsto t^2$, so that $\pi U_\sigma\big((0, t^2]\big) = \arg \xi(\sigma + it)$ with the argument followed continuously up the vertical line. When $\xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a generalised gamma convolution (GGC) this is its Thorin measure; for $\sigma > 1$ it always is (Theorem 2.1.2, Corollary 17.14.3).

Lemma 14.1.1 (Cauchy and Riemann). *The density of $\mathfrak{U}_\sigma$ in the variable t is*

$$m_\sigma(t) = \frac{1}{\pi}\, \partial_t \arg \xi(\sigma + it) = \frac{1}{\pi}\, \partial_\sigma \log |\xi(\sigma + it)| = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it),$$

a harmonic function of (σ, t) away from the zeros.

This is the Cauchy and Riemann equations for $\log \xi$, as in the proof of Theorem 10.1.1 and Proposition 3.3.1. Hadamard's product gives, absolutely convergently (Proposition 2.1.3),

$$m_\sigma(t) = \sum_\rho P_{\sigma - \beta}(t - \gamma), \tag{14.1.1}$$

the sum over all zeros, with the signed P_a of Proposition 2.1.3. Hence $\mathfrak{U}_1 = \sum_\rho P_{1-\beta}(\cdot - \gamma)$ is unconditionally a positive Cauchy mixture, and as $\sigma \downarrow \frac{1}{2}$ the measure tends vaguely (Theorem 17.14.4; Proposition 13.2.1) to $\mathfrak{U}_{1/2} = \sum_{\beta = 1/2} \delta_\gamma$, the counting measure of the critical zeros. Pairs off the line cancel in this limit, since $P_{1/2-\beta} + P_{\beta-1/2} = 0$.

14.2 The Balazard, Saias and Yor criterion at every level and height

Fix $\theta \in [\frac{1}{2}, 1)$ and a height t_0. The factor $(s-1)/(s - 2\theta + 1)$ below has modulus one on the line $\operatorname{Re} s = \theta$, because $2\theta - 1$ is the reflection of 1 in that line. For a zero ρ with $\beta > \theta$ let

$$G_\theta(s, \rho) = \log \left| \frac{s - 2\theta + \bar\rho}{s - \rho} \right| \geq 0$$

be the Green function of the half plane $\{\operatorname{Re} s > \theta\}$ with pole at ρ.

THEOREM 14.2.1 (Balazard, Saias and Yor at level θ). *Let $\frac{1}{2} \le \theta < 1$, $H = \{\operatorname{Re} s > \theta\}$ and $h_\theta(s) = \zeta(s)\,(s-1)/(s-2\theta+1)$. For every $s = \sigma + it \in H$ with $h_\theta(s) \neq 0$,*

$$\int_{-\infty}^{\infty} \log|\zeta(\theta+i\tau)|\, P_{\sigma-\theta}(t-\tau)\, d\tau - \log|h_\theta(s)| = \sum_{\beta>\theta} G_\theta(s,\rho) \;\ge 0, \qquad (14.2.1)$$

the integral absolutely convergent and the sum, over the zeros with $\beta > \theta$ counted with multiplicity, convergent. In particular

$$D_\theta(t_0) := \int_{-\infty}^{\infty} \log|\zeta(\theta+i\tau)|\, P_{1-\theta}(t_0-\tau)\, d\tau - \log|h_\theta(1+it_0)| = \sum_{\beta>\theta} G_\theta(1+it_0,\rho),$$

and QRH(θ) *holds if and only if $D_\theta(t_0) = 0$ for one (equivalently every) t_0. At $\theta = \frac{1}{2}$, $t_0 = 0$ this is the theorem of Balazard, Saias and Yor [20]:*

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\log|\zeta(\frac{1}{2}+it)|}{\frac{1}{4}+t^2}\, dt = \sum_{\beta>1/2} \log\left|\frac{\rho}{1-\rho}\right|.$$

Proof. Write $F = h_\theta$. Since $2\theta - 1 < \theta$, F is analytic on a neighbourhood of $\overline{H}$ (the pole of ζ at 1 is cancelled by $s-1$); its zeros in H are those of ζ with $\beta > \theta$, with the same multiplicities, and $|F| = |\zeta|$ on ∂H. Every $s \in \overline{H}$ is at least as close to 1 as to $2\theta - 1$, so $|F(s)| \le |\zeta(s)|$ for $|s-1| \ge 1$, while F is bounded on $\overline{H} \cap \{|s-1| \le 1\}$. The convexity bound for ζ in $\theta \le \sigma \le 2$ [264, Chapter V] and $|\zeta(s)| \le \zeta(2)$ for $\sigma \ge 2$ give a constant A with $\log|F(s)| \le A\log(2+|s|)$ on $\overline{H}$. Hence F is of exponential type in $\overline{H}$ and $\int \log^+|F(\theta+i\tau)|\,(1+\tau^2)^{-1}d\tau < \infty$, and Nevanlinna's formula for a half plane [112] gives, for $s \in H$ not a zero,

$$\log|F(s)| = \int_{-\infty}^{\infty} \log|F(\theta+i\tau)|\, P_{\sigma-\theta}(t-\tau)\, d\tau - \sum_{\beta>\theta} G_\theta(s,\rho) + \kappa\,(\sigma-\theta),$$

$$\kappa = \limsup_{r\to\infty} \frac{\log|F(\theta+r)|}{r},$$

with the integral absolutely convergent and the sum convergent. Since $F(\theta+r) \to 1$ as $r \to \infty$, $\kappa = 0$, which is (14.2.1). Every term of the sum is positive, because $|s-2\theta+\bar\rho|^2 - |s-\rho|^2 = 4(\sigma-\theta)(\beta-\theta) > 0$. At $s = 1+it_0$, which is never a zero, this is the formula for $D_\theta(t_0)$; the sum is empty exactly under QRH(θ), and otherwise positive at every t_0. The term $\kappa(\sigma-\theta)$ cannot be dismissed by finite order: $F(s) = e^{-c(s-\theta)}$ with $c > 0$ has order one, no zeros and $|F| = 1$ on ∂H, but $\log|F(1+it_0)| = -c(1-\theta)$; for it $\kappa = -c$. □

The theorem is the Riesz decomposition of the subharmonic function $\log|h_\theta|$ on the half plane (the formula is cited from [112] without a page). Two consequences

are used below. The first is the case $\theta = \frac{1}{2}$ at a general point $\sigma + it$,

$$\log|h_{1/2}(\sigma+it)| = \big(P_{\sigma-1/2} * \log|\zeta(\tfrac{1}{2}+i\,\cdot)|\big)(t) - \sum_{\beta>1/2} G_{1/2}(\sigma+it,\rho). \quad (14.2.2)$$

The second is the form at $\theta = \frac{7}{8}$, $t_0 = 0$, where $h_{7/8}(1) = \frac{1}{2-2\theta} = 4$: the Cauchy average of $\log|\zeta(\frac{7}{8}+i\tau)|$ at scale $\frac{1}{8}$ equals $\log 4$ if and only if QRH($\frac{7}{8}$) holds. That equivalence is unconditional; that the average does equal $\log 4$ is conditional on QRH($\frac{7}{8}$), announced and not refereed (Section 17.7). Table 14.3 lists the defects computed at the levels $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{7}{8}$.

Remark 14.2.2 (local versus global). A zero at height γ with $\beta = \theta+\delta$ contributes about $2\delta(1-\theta)/\gamma^2$ to $D_\theta(0)$. That is below 10^{-25} beyond the verified height 3×10^{12} [232], so the global criterion is numerically blind: summed over all possible zeros above that height, $D_\theta(0)$ is below 7.4×10^{-13} at $\theta=\frac{1}{2}$, 1.85×10^{-13} at $\theta=\frac{3}{4}$ and 4.7×10^{-14} at $\theta=\frac{7}{8}$ (computed from the zero counting bound of Hasanalizade, Shen and Wong [100]; the last bound is 4.62×10^{-14} to three digits). Centred at $t_0=\gamma$, the same zero contributes $\log\frac{1-\theta+\delta}{1-\theta-\delta}\approx\frac{2\delta}{1-\theta}$, which is of order δ itself. For the Davenport and Heilbronn function at $\theta=\frac{3}{4}$ the zero $0.8085\ldots+85.6993\ldots i$ gives $\log\frac{0.308517}{0.191483}=0.4770$ at its own height (as computed in Table 14.2) and about 4×10^{-6} seen from $t_0=0$. So the kernel should be centred where the zeros are. Computing $D_{7/8}(t_0)$ on a grid of heights is a practical local test of QRH($\frac{7}{8}$) at those heights, and the generative screen of Section 13.4 tells where to centre it; being local, it is not a proof.

14.3 The dictionary: one derivative in σ

Differentiating (14.2.2) in σ gives the Thorin statement in two steps: the transport of $\mathfrak{U}_{1/2}$ to $\mathfrak{U}_\sigma$, and the archimedean term.

Theorem 14.3.1 (transport). *For $\sigma > \frac{1}{2}$,*

$$m_\sigma = P_{\sigma-1/2} * \mathfrak{U}_{1/2} + \sum_{\beta>1/2} \big[P_{\sigma-\beta} + P_{\sigma-1+\beta}\big](\cdot-\gamma).$$

Proof. The formula is (14.1.1) split into zeros on the line and zeros off it, a zero ρ with $\beta>\frac{1}{2}$ being paired with $1-\bar\rho$ at the same height γ. □

Remark 14.3.2 (zero by zero). A zero on the line enters (14.2.2) through the logarithmic singularity of the boundary values, whose Poisson extension has σ derivative $\pi P_{\sigma-1/2}(\cdot-\gamma)$. A pair $\rho, 1-\bar\rho$ off the line enters through two terms. The Poisson extension of $\log|(\cdot-\rho)(\cdot-1+\bar\rho)|$ from the line is $2\log|s-1+\bar\rho|$, with σ derivative $2\pi P_{\sigma-1+\beta}$. The Green term $-G_{1/2}(s,\rho)$ has σ derivative $\pi(P_{\sigma-\beta}-P_{\sigma-1+\beta})$. The sum is $\pi(P_{\sigma-\beta}+P_{\sigma-1+\beta})$. The derivatives are direct:

$\partial_\sigma \log|\sigma+it-c| = \mathrm{Re}(s-c)^{-1} = \pi P_{\sigma-\mathrm{Re}\,c}(t-\mathrm{Im}\,c)$, and $2\log|s-1+\bar\rho|$ is harmonic on $\{\mathrm{Re}\,s>\frac12\}$ (its singularity $1-\bar\rho$ lies to the left of the line) with the right boundary values.

Differentiating (14.2.2) in σ does not give m_σ directly, because $\pi m_\sigma = \partial_\sigma \log|\xi|$ while (14.2.2) concerns $\log|h_{1/2}|$, and $\xi = C\,h_{1/2}$ with

$$C(s) = \frac{\xi(s)}{h_{1/2}(s)} = \tfrac12 s^2 \pi^{-s/2}\Gamma(\tfrac{s}{2}),$$

analytic and zero free on $\{\mathrm{Re}\,s>0\}$. With $a_\sigma(t) = \pi^{-1}\,\mathrm{Re}\,(C'/C)(\sigma+it) = \pi^{-1}\,\mathrm{Re}\big[\frac{2}{s} - \frac12\log\pi + \frac12\psi(\frac{s}{2})\big]$, for $\sigma>\frac12$,

$$m_\sigma(t) = \frac{1}{\pi}\,\partial_\sigma\big(P_{\sigma-1/2} * \log|\zeta(\tfrac12+i\,\cdot)|\big)(t) + a_\sigma(t) + \sum_{\beta>1/2}\big[P_{\sigma-\beta} - P_{\sigma-1+\beta}\big](t-\gamma), \tag{14.3.1}$$

equivalently, by Theorem 14.3.1, $\frac{1}{\pi}\partial_\sigma(P_{\sigma-1/2} * \log|\zeta(\frac12+i\cdot)|) = P_{\sigma-1/2} * \mathfrak{U}_{1/2} + \sum_{\beta>1/2} 2P_{\sigma-1+\beta}(\cdot-\gamma) - a_\sigma$. The term a_σ cannot be absorbed into the Poisson integral: $\log|C(\frac12+it)| = -\frac{\pi}{4}|t| + O(\log|t|)$, whose Cauchy average diverges, which is why the criterion is stated for ζ and not for ξ.

Proof. On $\{\mathrm{Re}\,s>\frac12\}$, $\log|\xi| = \log|h_{1/2}| + \log|C|$, so $\pi m_\sigma = \partial_\sigma \log|h_{1/2}| + \pi a_\sigma$ by Lemma 14.1.1. Differentiate (14.2.2): the Poisson integral is harmonic, and by the derivative of the preceding remark,

$$\partial_\sigma G_{1/2}(s,\rho) = \partial_\sigma \log|s-1+\bar\rho| - \partial_\sigma\log|s-\rho| = \pi\big[P_{\sigma-1+\beta} - P_{\sigma-\beta}\big](t-\gamma),$$

the differentiated series converging locally uniformly off the zeros. This gives (14.3.1); the second form follows by substituting Theorem 14.3.1. The asymptotics of $\log|C|$ are Stirling's formula. □

logarithmic level	Thorin level
$\log\|\xi\|$	$\pi\, m_\sigma = \partial_\sigma \log\|\xi\|$
Poisson integral of $\log\|\zeta(\frac12+i\cdot)\|$	$P_{\sigma-1/2} * \mathfrak{U}_{1/2}$ plus $2P_{\sigma-1+\beta}$ per zero with $\beta>\frac12$, minus a_σ
minus the Green potential, $-G(\cdot,\rho)\le 0$	signed defect $P_{\sigma-\beta} - P_{\sigma-1+\beta}$
average equals $\log\|h(1)\|$	$\mathfrak{U}_\sigma \ge 0$ for $\sigma>\frac12$
$D_\theta \equiv 0$	$\mathfrak{U}_\sigma \ge 0$ for $\sigma>\theta$
D of order δ at $t_0=\gamma$	negative spike of m_σ at γ

Table 14.1: The dictionary between the two levels.

COROLLARY 14.3.3 (equivalent forms of RH). *The following are equivalent: (i) RH; (ii) $D_{1/2}(t_0) = 0$ for some t_0; (iii) $\mathfrak{U}_\sigma$ is positive for every $\sigma \in (\frac12, 1)$; (iv) $\mathfrak{U}_1 = P_{1/2} * \nu$ for a positive measure ν on $\mathbb{R}$ with $\int (1+t^2)^{-1}\nu(dt) < \infty$, and then $\nu = \mathfrak{U}_{1/2}$. More generally, for $\theta \in [\frac12, 1)$, QRH(θ) holds if and only if $D_\theta \equiv 0$, if and only if $\mathfrak{U}_\sigma \geq 0$ for all $\sigma > \theta$.*

(i)⇔(ii) is Theorem 14.2.1. Statement (iii) and its form for QRH(θ) are Lagarias's argument (Remark 2.1.4); (iv) is Theorem 8.2.2(a).

Proof. Let ρ_0 be a zero with $\beta_0 > \theta$. By (14.1.1), $m_\sigma(\gamma_0) \to -\infty$ as $\sigma \uparrow \beta_0$, because the term of ρ_0 is $P_{\sigma-\beta_0}(0) = 1/(\pi(\sigma-\beta_0))$ and the other terms stay bounded near γ_0; so m_σ is negative on a neighbourhood of γ_0 for $\sigma \in (\theta, \beta_0)$ close to β_0, and $\mathfrak{U}_\sigma$ is not positive. This gives (iii)⇒(i), and its form for QRH(θ). Conversely, under QRH(θ) every term of (14.1.1) with $\sigma > \theta$ is positive. For (iv), if RH holds, Theorem 14.3.1 at $\sigma = 1$ gives $\nu = \mathfrak{U}_{1/2}$. Conversely a positive ν makes $P_{\sigma-1/2} * \nu$ a positive harmonic function on $\{\sigma > \frac12\}$ equal to m_σ for $\sigma \geq 1$ (the two agree at $\sigma = 1$ and are harmonic with the growth of Poisson integrals; this is the argument of Theorem 8.2.2(a)). By unique continuation they agree off the zeros, which is impossible near a zero with $\beta > \frac12$, where m_σ is unbounded of both signs. So RH holds, and then $P_{\sigma-1/2} * \nu = P_{\sigma-1/2} * \mathfrak{U}_{1/2}$ gives $\nu = \mathfrak{U}_{1/2}$. □

14.4 The 2 sin² lemma, the extended gamma convolution and the Davenport and Heilbronn trace

The $2\sin^2$ lemma (equation (2.1.9)) is the Lévy and Khintchine representation written with $1 - \cos(xu) = 2\sin^2(xu/2)$. Applied to the factors of the reciprocal Xi function along the real direction, with Frullani's integral $\log(1 + x^2/c^2) = \int (1 - \cos xu)\, e^{-c|u|}\, du/|u|$ for $\operatorname{Re} c > 0$, it gives for real x

$$\log \frac{\xi(\frac12 + x)}{\xi(\frac12)} = \int_{-\infty}^{\infty} 2\sin^2\Big(\frac{xu}{2}\Big)\, \frac{K(u)}{|u|}\, du, \qquad K(u) = \sum_{\gamma>0} e^{-\gamma|u|} \cos\big((\beta - \tfrac12)u\big), \tag{14.4.1}$$

the sum over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$, with multiplicity. Under RH, $K(u) = \sum_{\gamma>0} e^{-\gamma|u|}$ is the Poisson trace $K_{1/2}$ of Section 13.1.4, and the left side is $\sum_{\gamma>0} \log(1 + x^2/\gamma^2)$.

PROPOSITION 14.4.1 (the Lévy density is positive with no hypothesis). *$K(u) > 0$ for every $u \neq 0$. So $x \mapsto \xi(\frac12)/\xi(\frac12 + x)$ is, unconditionally, the characteristic function of an infinitely divisible law with Lévy density $K(u)/|u|$ (Theorem 8.1.1). Under RH it is moreover an extended generalised gamma convolution (EGGC) $\exp\big(-\int_{(0,\infty)} \log(1 + x^2/y^2)\, U(dy)\big)$ with $U = \sum_{\gamma>0} \delta_\gamma$, the restriction of $\mathfrak{U}_{1/2}$ to $(0, \infty)$.*

Proof. Write $\delta = \beta - \frac{1}{2}$, so $|\delta| < \frac{1}{2}$. For $0 < |u| \le \pi$, $|\delta u| < \frac{\pi}{2}$ and every term of $K(u)$ is positive. Let $|u| > \pi$. The zeros with $0 < \gamma < 100$ lie on the line [232], the lowest at $\gamma_1 = 14.1347\ldots$, and a zero off the line contributes at least $-e^{-\gamma|u|}$, so

$$e^{\gamma_1|u|}K(u) \ge 1 - \sum_{\gamma \ge 100} e^{-(\gamma-\gamma_1)\pi}.$$

Pairing ρ with $1-\bar\rho$, $\operatorname{Re}\frac{1}{\rho} + \operatorname{Re}\frac{1}{1-\bar\rho} = \frac{\beta}{|\rho|^2} + \frac{1-\beta}{|1-\rho|^2} \ge \frac{1}{1+\gamma^2}$, hence $\sum_\rho (1+\gamma^2)^{-1} \le 2\sum_\rho \operatorname{Re}\frac{1}{\rho} = 2 + \gamma_E - \log 4\pi < 0.05$ [54, Chapter 12]. Since $(1+\gamma^2)e^{-\pi\gamma}$ decreases for $\gamma \ge 1$, the last sum is at most $0.05\,(1+10^4)\,e^{-\pi(100-\gamma_1)} < 10^{-113}$, and $K(u) > 0$. The display follows from the Hadamard product of $\xi(\frac{1}{2}+x)$ over pairs and quadruples of zeros and Frullani's integral with $c = \gamma - i\delta$; positivity of K and the Lévy and Khintchine formula give infinite divisibility. Under RH, $\exp(-\sum_{\gamma>0}\log(1+x^2/\gamma^2)) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+x)$. □

So positivity of the Lévy density carries no information on RH: it is proved with no hypothesis (the numbers in the proof are $2 + \gamma_E - \log 4\pi = 0.046191$ and $10001\,e^{-\pi(100-\gamma_1)} = 7.0 \times 10^{-114}$, computed). The book has a stronger form: with no hypothesis the law is self decomposable and K decreases on $(0,\infty)$ (Section 4.1.2, whose k is this K because $\sqrt{a_j} = \gamma - i(\beta - \frac{1}{2})$; Theorem 8.1.1); the proof above is direct. The EGGC property is what sees RH.

Proposition 14.4.2 (RH as an EGGC statement). *RH holds if and only if $\xi(\frac{1}{2})/\xi(\frac{1}{2}+x)$ is the characteristic function of an EGGC. When it is, its Thorin measure is the restriction of $\mathfrak{U}_{1/2}$ to $(0,\infty)$, the measure that the Cauchy deconvolution of Corollary 14.3.3(iv) recovers from $\mathfrak{U}_1$.*

Proof. An EGGC characteristic function $\exp\bigl(-\int \log(1+x^2/y^2)\,U(dy)\bigr)$ is analytic off the imaginary axis; a drift or a Gaussian part would change the exponent by an entire function and would not affect this. A zero $\frac{1}{2}+\delta+i\gamma$ with $\delta \ne 0$ produces a pole of $1/\xi(\frac{1}{2}+x)$ at $x = \delta + i\gamma$, off that axis. Conversely under RH the product formula above exhibits the EGGC. □

This is Theorem 13.1.2 at the centre (see also Section 4.1.2).

The same computation goes through with zeros off the line. Let Λ be entire, with $\Lambda(\frac{1}{2}+x)$ even, real on the real axis, of order one, with $\Lambda(\frac{1}{2}) \ne 0$ and no real zeros, whose zeros are exactly, with multiplicity, the pairs $\pm i\gamma'$ on the imaginary axis and the quadruples $\pm\delta \pm i\gamma$ with $\gamma > 0$ and $\delta \ne 0$ bounded. Frullani's integral with $c = \gamma - i\delta$, $\operatorname{Re} c > 0$, gives

$$\log\frac{\Lambda(\frac{1}{2}+x)}{\Lambda(\frac{1}{2})} = \int (1-\cos xu)\,\frac{K_\Lambda(u)}{|u|}\,du,$$

$$K_\Lambda(u) = \sum_{\gamma'>0} e^{-\gamma'|u|} + 2\sum_{\text{quadruples}} e^{-\gamma|u|}\cos(\delta u),$$

and by uniqueness of the Lévy and Khintchine representation, $\Lambda(\frac12)/\Lambda(\frac12+x)$ is an infinitely divisible characteristic function exactly when $K_\Lambda \ge 0$. For the converse, suppose $\Lambda(\frac12)/\Lambda(\frac12+x)$ is the characteristic function of an infinitely divisible law, with Gaussian coefficient $c \ge 0$ and Lévy measure ν. The function is analytic near 0, so the law has a finite second moment and $\int u^2\nu(du) < \infty$. Differentiating both exponents twice shows that the finite measures $c\,\delta_0 + u^2\nu(du)$ and $|u|K_\Lambda(u)\,du$ (finite, since $\int |u||K_\Lambda(u)|\,du \le 2\sum \gamma^{-2} < \infty$) have the same Fourier transform; so $c = 0$, $\nu(du) = K_\Lambda(u)\,du/|u|$ and $K_\Lambda \ge 0$, everywhere since K_Λ is continuous on $u \ne 0$. Off line zeros make terms of K_Λ change sign, but need not make K_Λ negative (Remark 18.9.2; for the function of Exercise 1.7, $K_\Lambda(u) \ge (e^{|u|}-1)^{-1} - 2e^{-10.5|u|} > 0$, and by the argument of Proposition 14.4.2 its reciprocal is not an EGGC).

THEOREM 14.4.3 (infinite divisibility without the EGGC property). *Let f be the Davenport and Heilbronn function, $\Lambda(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f(s)$, so that $\Lambda(s) = \Lambda(1-s)$, and let K_{DH} be its trace in the display above. Then $K_{\mathrm{DH}}(u) > 0$ for every $u \ne 0$, and $\Lambda(\frac12)/\Lambda(\frac12+x)$ is the characteristic function of an infinitely divisible law. That law is not an EGGC, because Λ has a zero off the line, near $0.8085 + 85.6993\,i$ [261, 17]. The proof rests on three finite computations, certified in ball arithmetic: (a) Λ has a zero $\frac12 + ig$ with $5.09 < g < 5.10$; (b) Λ has no zero $\beta + i\gamma$ with $\beta \ne \frac12$ and $0 < \gamma < 85.69$; (c) Λ has exactly one zero in a square of half width 10^{-16} about $0.8085171824566374 + 85.6993484853775922\,i$.*

Proof. Real zeros. With $\kappa = (\sqrt{10-2\sqrt5}-2)/(\sqrt5-1) \in (0,1)$ the coefficients of f are $1, \kappa, -\kappa, -1, 0$ modulo 5, and for real $\sigma > 0$

$$f(\sigma) = \sum_{k\ge0}\Big[(5k+1)^{-\sigma} - (5k+4)^{-\sigma} + \kappa\big((5k+2)^{-\sigma} - (5k+3)^{-\sigma}\big)\Big] > 0,$$

each bracket being positive and $O(k^{-\sigma-1})$; the grouped series is analytic in $\sigma > 0$ and equals f for $\sigma > 1$. So $\Lambda > 0$ on $[\frac12, \infty)$ and, by the functional equation, on $\mathbb{R}$.

A strip. For $\sigma \ge 2$, $|f(s)-1| \le \zeta(2) - 1 < 1$, so every zero of Λ has $|\beta - \frac12| < \frac32$.

A weighted bound. Write the zeros of $F(x) = \Lambda(\frac12 + x)$ with positive imaginary part (the upper zeros) as $\delta + i\gamma$, and $c = \gamma - i\delta$, so that $F(x)/F(0) = \prod(1 + x^2/c^2)$ over the upper zeros with multiplicity. At $x = \frac52$, $\frac{\Lambda'}{\Lambda}(3) = \sum 5\ \mathrm{Re}(x^2+c^2)^{-1}$, and with $A = \gamma^2 + x^2 - \delta^2 \ge \gamma^2$, $\mathrm{Re}(x^2+c^2)^{-1} = (A + 4\gamma^2\delta^2/A)^{-1} \ge (\gamma^2 + \frac{61}{4})^{-1}$. Since $\frac{\Lambda'}{\Lambda}(3) = \frac12\log\frac{5}{\pi} + \frac12\psi(2) + \frac{f'}{f}(3)$, $|f(3)-1| \le \zeta(3) - 1 < \frac14$ and $|f'(3)| \le -\zeta'(3) < \frac14$, we get $\frac{\Lambda'}{\Lambda}(3) < \frac14 + \frac12 + \frac13 < 2$, so

$$\sum_{\text{upper zeros}} \Big(\gamma^2 + \frac{61}{4}\Big)^{-1} < \frac25.$$

Positivity. $K_{\mathrm{DH}}(u) = \sum_{\text{upper zeros}} e^{-\gamma|u|}\cos(\delta u)$. For $0 < u \le \pi/3$ every term is positive, since $|\delta u| < \frac{\pi}{2}$. For $u \ge \pi/3$ keep the term of the zero in (a), drop the

other terms of zeros on the line, and bound each term of a zero off the line below by $-e^{-\gamma u}$. By (b), the weighted bound and the decrease of $(t^2+\frac{61}{4})e^{-(t-5.1)\pi/3}$ for $t \ge 85.69$,

$$e^{gu}K_{\mathrm{DH}}(u) \ge 1 - \sum_{\text{off line, upper}} e^{-(\gamma-5.1)\pi/3}$$
$$\ge 1 - \frac{2}{5}\Big(85.69^2+\frac{61}{4}\Big)e^{-(85.69-5.1)\pi/3} > 1-10^{-33}.$$

So $K_{\mathrm{DH}} > 0$ on $(0,\infty)$, and the representation above gives infinite divisibility. By (c) Λ has a zero off the line, so $1/\Lambda(\frac{1}{2}+x)$ has a pole off the imaginary axis and the law is not an EGGC (proof of Proposition 14.4.2).

The computations. The computation works in Arb balls at two precisions, with every count and every zero agreeing at both. (a) $\operatorname{Re}\Lambda(\frac{1}{2}+5.09i) = 3.564\times10^{-4}$ and $\operatorname{Re}\Lambda(\frac{1}{2}+5.10i) = -4.974\times10^{-4}$, each with a certified sign, so the real function $\Lambda(\frac{1}{2}+it)$ changes sign on $[5.09, 5.10]$; the zero is $\gamma_1 = 5.09416\ldots$. (b) The argument principle on the rectangle $[-2,3]\times[-85.69,85.69]$ gives 86 zeros, so 43 in the upper half; the certified sign changes of $\Lambda(\frac{1}{2}+it)$ on $(0,85.69)$ number 43, Λ is certified nonzero on the real axis, and so all 43 upper zeros are on the line and simple, and none lies off it. (c) The square contains exactly one zero, certified by the argument principle, at distance 0.3085 to the right of the line. These certified computations make the theorem proved. The floating point zero list to height 300 (204 sign changes of $\Lambda(\frac{1}{2}+it)$ and 5 quadruples, at heights 85.6993, 114.1633, 166.4793, 176.7025 and 240.4047, and $204+2\cdot5=214$ by the argument principle) is computed evidence only; the proof uses just the part below 85.69.

□

Remark 14.4.4. If a narrower strip is wanted, $\sigma_1 = 1.4$ gives $|\beta-\frac{1}{2}| < 0.9$ and the threshold $\pi/1.8 = 1.7453$ for the first range of u (the sum $\sum_{n\ge2}|a_n|n^{-\sigma}$ equals $0.98468 < 1$ at $\sigma = 1.4$ and $1.00428 > 1$ at 1.3938; the root of the equation sum $= 1$ lies in $[1.39492, 1.39531]$). The theorem does not use it.

Figure 14.1 shows K_{DH} computed two ways. One is the sum over the zeros up to height 300. The other is the explicit formula, which uses no zeros:

$$K_{\mathrm{DH}}(u) = \frac{1}{\pi}\operatorname{Re}\big[e^{i\alpha u}A(u)\big] + \frac{1}{\pi}\int_0^\alpha \frac{\Lambda'}{\Lambda}\Big(\tfrac{1}{2}+v\Big)\sin(vu)\,dv,$$

$$A(u) = \int_0^\infty \frac{\Lambda'}{\Lambda}(c+it)\,e^{-tu}\,dt,$$

with $c=3$, $\alpha = c-\frac{1}{2}$, and f'/f expanded as a Dirichlet series. It follows from the residue theorem on the rectangles $1-c \le \operatorname{Re} s \le c$, $0 \le \operatorname{Im} s \le T$, with $T\to\infty$ through heights away from the zeros, the functional equation and the reality of Λ'/Λ on the real axis. The two computations agree to a relative error of 4×10^{-13}

for $0.1 \le u \le 0.5$ and 10^{-9} for $0.1 \le u \le 2$, at the sampled points (computed, floating point), and the off line terms of the zeros below height 300 stay below 0.02 percent of K_{DH} there; the share reaches 4 percent only at $u = 0.01$, outside the sampled range, where the zero sum cut at height 300 has not converged. So the $2\sin^2$ Lévy density can stay positive with zeros off the line: RH is the EGGC property, which infinite divisibility alone does not see.

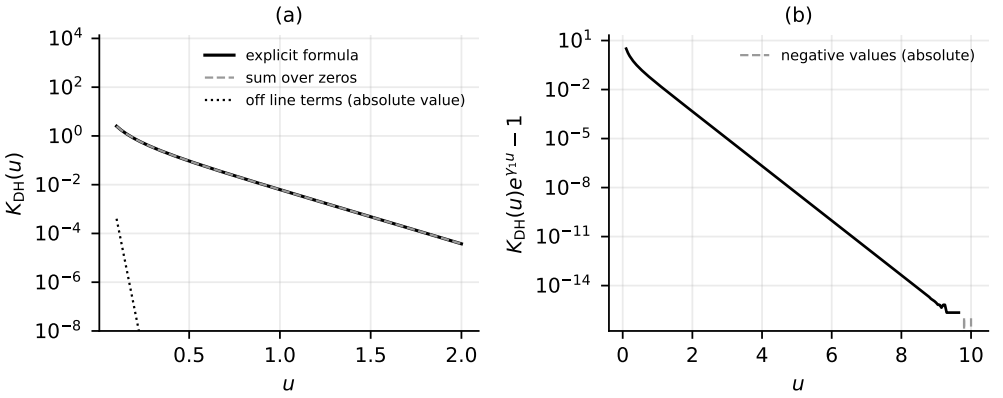


Figure 14.1: The Davenport and Heilbronn Lévy density. Left: K_{DH} by the explicit formula and by the sum over zeros, with the off line terms in absolute value. Right: $K_{\mathrm{DH}}(u)e^{\gamma_1 u} - 1$ on a logarithmic scale; the quantity $K_{\mathrm{DH}}(u)e^{\gamma_1 u}$ tends to 1, from above for large u since the first zero is simple and the second lies on the line. Beyond $u = 9$ the right panel shows rounding noise.

Positivity of the trace is not automatic for Dirichlet series with a functional equation.

Remark 14.4.5 (a negative trace). Let $B(s) = L(s, \chi_4)$, $\Lambda_B(s) = (4/\pi)^{s/2}\Gamma(\frac{s+1}{2})B(s)$, let $p < q < 2p$ be odd primes, and put $a = \frac{1}{2}\log(pq)$, $b = \frac{1}{2}\log(p/q)$ and $P(s) = 1 + 2\sqrt{p}\,p^{-s} + 2\sqrt{q}\,q^{-s} + \sqrt{pq}\,(pq)^{-s}$. Then

$$\Big(\frac{4pq}{\pi}\Big)^{s/2}\Gamma\Big(\frac{s+1}{2}\Big)P(s)B(s) = 2(pq)^{1/4}\Lambda_B(s)\big[\cosh(a(s-\tfrac{1}{2})) + 2\cosh(b(s-\tfrac{1}{2}))\big],$$

which is entire of order one, real and positive on $\mathbb{R}$, and invariant under $s \mapsto 1 - s$. With $B_p = B\,(1 - \chi_4(p)p^{-s})$, the L function of the character modulo $4p$ induced by χ_4, $p^{-s}B = \chi_4(p)(B - B_p)$, and similarly for q and pq, so PB is a real linear combination of L functions of characters modulo 4, $4p$, $4q$ and $4pq$. With $z = a(s - \frac{1}{2})$ and $\epsilon = b/a$, at a zero of $H_\epsilon(z) = \cosh z + 2\cosh(\epsilon z)$ we have $|\sinh x| \le 2\cosh(x/2)$, so $|x| < 3$; by Rouché's theorem against $\cosh z + 2$, for $|\epsilon|$ small H_ϵ has exactly two zeros with $0 \le \operatorname{Im} z \le 2\pi$, near $\pm\operatorname{arcosh} 2 + i\pi$. For p large they give the lowest zeros of the completed function above the real

axis, a pair $\frac{1}{2} \pm \delta_0 + i\gamma_0$ with $\delta_0 > 0$ and γ_0 below the first zero of Λ_B; hence the trace satisfies $K(u) = 2e^{-\gamma_0 u}\cos(\delta_0 u) + O(e^{-(\gamma_0+\eta)u})$ for some $\eta > 0$, and $K((2k+1)\pi/\delta_0) < 0$ for all large k. So the reciprocal of this completed function is not infinitely divisible. For $p = 101$, $q = 103$: $\delta_0 = 0.28475$, $\gamma_0 = 0.67928$ and $K(\pi/\delta_0) = -1.11 \times 10^{-3}$ (computed in floating point from the zeros of H_ϵ with $m \le 5$ and the first five zeros of $L(s, \chi_4)$).

The primitive version of the question stays open (Open problem 14.6).

14.5 Brownian motion and a probabilistic reading of the criterion

The Cauchy kernel $P_{1-\theta}(t_0 - \cdot)$ is the law of the height at which planar Brownian motion B started at $1 + it_0$ first hits the line $\operatorname{Re} s = \theta$. As in Section 12.2.3, the Cauchy process is Brownian motion run along the first passage times of an independent Brownian motion, indexed by the level.

Proposition 14.5.1 (the criterion as optional stopping). *Let B be a planar Brownian motion started at $s_0 = 1 + it_0$, τ its hitting time of the line $\{\operatorname{Re} s = \theta\}$ and $H = \{\operatorname{Re} s > \theta\}$. Then $\log|h_\theta(B_\tau)|$ is integrable and*

$$\mathbb{E}\big[\log|h_\theta(B_\tau)|\big] = \log|h_\theta(s_0)| + \sum_{\beta>\theta} G_\theta(s_0, \rho).$$

The process $\log|h_\theta(B_{t\wedge\tau})|$ is a local martingale and a submartingale; it is a uniformly integrable martingale if and only if ζ has no zero with $\beta > \theta$. Each zero contributes its Riesz mass 2π, times $\frac{1}{2}$, times the value at ρ of $\pi^{-1}G_\theta(s_0, \cdot)$, the density of the expected occupation measure of B killed at τ. Planar Brownian motion almost surely never visits a given point, so no local time at the zeros is involved.

Proof. The law of B_τ is $P_{1-\theta}(t_0 - \tau)\,d\tau$ on the line (Section 12.2.3) and $|h_\theta| = |\zeta|$ there, so integrability and the identity are Theorem 14.2.1. The function $\log|h_\theta|$ is harmonic on H off the zeros, which B almost surely never visits; stopping at the exit times of the compact sets $\{|s| \le n,\ \operatorname{Re} s \ge \theta + \frac{1}{n},\ |s-\rho| \ge \frac{1}{n} \text{ for all } \rho\}$ shows that $\log|h_\theta(B_{t\wedge\tau})|$ is a local martingale. By (14.2.1), $\log|h_\theta| = v - \sum_{\beta>\theta} G_\theta(\cdot, \rho)$ on H, where $v(z) = \mathbb{E}_z \log|\zeta(B_\tau)|$; by the strong Markov property $v(B_{t\wedge\tau}) = \mathbb{E}[\log|\zeta(B_\tau)| \mid \mathcal{F}_{t\wedge\tau}]$ is a uniformly integrable martingale. Each $G_\theta(B_{t\wedge\tau}, \rho)$ is a positive local martingale, hence a supermartingale, and so is their sum, by monotone convergence and $\sum G_\theta(s_0, \rho) < \infty$. So $\log|h_\theta(B_{t\wedge\tau})|$ is a submartingale. If there is a zero with $\beta > \theta$, its terminal value has expectation strictly above its initial value, so it is not a uniformly integrable martingale; if there is none it equals $v(B_{t\wedge\tau})$. Finally $\pi^{-1}G_\theta(\cdot, z)$ is the Green function of H for the generator $\frac{1}{2}\Delta$, since $\frac{1}{2}\Delta\,\pi^{-1}G_\theta(\cdot, z) = -\delta_z$ and $G_\theta(\cdot, z) = 0$ on ∂H, which gives the occupation statement. $\square$

So RH says that $\log|h_{1/2}|$ is a fair game for Brownian motion stopped on the critical line, and each zero off the line is a source of drift. Under RH, $m_\sigma(t)$ is the hitting density of the critical line for Brownian motion from $\sigma+it$, summed over the critical zeros, and a zero off the line is the one thing that can make that density negative.

14.5.1 The complex logarithm: a conformal martingale

Proposition 14.5.1 uses only the real part of $\log h_\theta$. Under QRH(θ) the complex logarithm is analytic on the half plane, and conformal invariance fixes all its analytic moments on the line under harmonic measure; with an energy identity they give its second order structure.

Theorem 14.5.2 (moments of $\log h_\theta$ under harmonic measure). *Assume* QRH(θ), $\frac12 \le \theta < 1$. *Let L be the analytic logarithm of h_θ on $H=\{\operatorname{Re} s>\theta\}$ with $L(\sigma+it)\to 0$ as $\sigma\to+\infty$. For almost every τ the limit $Z(\tau)=X(\tau)+iY(\tau)=\lim_{\sigma\downarrow\theta} L(\sigma+i\tau)$ exists, $X(\tau)=\log|\zeta(\theta+i\tau)|$, and $Y(\tau)=\arg h_\theta(\theta+i\tau)$ continued along the horizontal path from $+\infty$. Let τ have law $P_{1-\theta}(t_0-\cdot)$ and $z_0=x_0+iy_0=L(1+it_0)$. Then $\mathbb{E}|Z|^p<\infty$ for every $p<\infty$, $\mathbb{E}[Z^k]=z_0^k$ for every $k\ge1$, and in particular $\mathbb{E}X=x_0$, $\mathbb{E}Y=y_0$, $\operatorname{Var}X=\operatorname{Var}Y$, $\operatorname{Cov}(X,Y)=0$, with*

$$\operatorname{Var}X=\operatorname{Var}Y=\frac{1}{\pi}\iint_H G_\theta(1+it_0,w)\left|\frac{h_\theta'}{h_\theta}(w)\right|^2 dA(w).$$

Proof. Under QRH(θ), h_θ is analytic and zero free on H ($h_\theta(1)=1/(2-2\theta)$), so L exists. By Theorem 14.2.1 with an empty sum, $\operatorname{Re}L=\log|h_\theta|$ is the Poisson integral of X, and X has every moment under $P_{1-\theta}(t_0-\cdot)$ by Lemma 14.6.1. The map $w=(s-1-it_0)/(s-2\theta+1-it_0)$ sends H onto the unit disc, $1+it_0$ to 0, and the harmonic measure to normalised arc length; put $F=L\circ s(w)$. Then $\operatorname{Re}F$ is the Poisson integral of a function in L^p of the circle for every $p<\infty$, so by M. Riesz's theorem on conjugate functions [69, Theorem 4.1] $F\in H^p$ for every $p<\infty$. Hence F has nontangential limits F^* almost everywhere, with $F^*=Z$: at boundary points where $h_\theta\neq0$, L is continuous up to the line and the horizontal convention gives the same value, and the boundary zeros form a null set. For $k\ge1$, $F^k\in H^1$, so $\mathbb{E}[Z^k]=\int (F^*)^k\frac{d\phi}{2\pi}=F(0)^k=z_0^k$. With $k=1,2$, $\mathbb{E}[(Z-z_0)^2]=0$, that is $\mathbb{E}(X-x_0)^2=\mathbb{E}(Y-y_0)^2$ and $\mathbb{E}(X-x_0)(Y-y_0)=0$. Finally, by the Littlewood and Paley identity, $\mathbb{E}|Z-z_0|^2=\|F-F(0)\|_{H^2}^2=\frac{2}{\pi}\iint_{\mathbb{D}}|F'(w)|^2\log\frac{1}{|w|}\,dA(w)$, and since $\log(1/|w(s)|)=G_\theta(1+it_0,s)$ and $|F'(w)|^2dA(w)=|L'(s)|^2dA(s)$, this is $\frac{2}{\pi}\iint_H G_\theta(1+it_0,s)|L'(s)|^2dA(s)$, half of it for each coordinate. □

Remark 14.5.3 (the Brownian reading). For planar Brownian motion B from $1+it_0$, stopped at its hitting time T of the line, $L(B_{t\wedge T})=z_0+W_{A_{t\wedge T}}$ with $A_t=\int_0^t|L'(B_r)|^2\,dr$ and a complex Brownian motion W (Lévy's theorem; the clock is adapted, not independent of W). The bound $|F|^p\le P[|F^*|^p]$ of the proof

is the harmonic majorant that makes the stopped powers uniformly integrable, so $\mathbb{E}[Z^k] = z_0^k$ is optional stopping. A zero of multiplicity q to the right of the line removes this argument and makes Y jump upwards by $2\pi q$ at its height; a zero on the line itself gives a jump πq already under QRH(θ).

The coordinate Y is $\arg h_\theta$, not the raw argument of ζ, which differs from it by an explicit term of the order of $2\arctan\frac{1-\theta}{t}$ and fails the identities of the theorem: at $\theta = \frac{1}{2}$, $t_0 = 0$ under RH the variance of $\arg\zeta$ exceeds that of $\log|\zeta|$ by more than $\pi^2/3$. Indeed, with $w = (s-1)/s$ the raw argument is $Y - \phi$, where $\phi = \operatorname{Arg} w(\frac{1}{2} + i\tau)$ is uniform on $(-\pi, \pi)$, so $\operatorname{Var}\phi = \pi^2/3$. Writing $F(w) = \sum_{n\geq 1} a_n w^n$ as in the proof of Theorem 14.5.2 and using $\mathbb{E}[\phi \sin n\phi] = (-1)^{n+1}/n$ gives $\operatorname{Cov}(Y, \phi) = -\int_0^1 F(-r)\, dr/r$, and $F(-r) = \log h_{1/2}(\frac{1}{1+r}) > 0$ because $h_{1/2}(s) = 1 + (1-s)\int_1^\infty \{x\} x^{-s-1}\, dx > 1$ for $\frac{1}{2} < s < 1$. Hence $\operatorname{Var}(Y - \phi) = \operatorname{Var} X + \frac{\pi^2}{3} - 2\operatorname{Cov}(Y, \phi) > \operatorname{Var} X + \frac{\pi^2}{3}$.

The moments do not determine the joint law of (X, Y) by themselves; the law is the image of the Cauchy law under L. What the theorem adds is the moment identities and the variance formula. Selberg's central limit theorem says that for t uniform on $[T, 2T]$, $\log|\zeta(\frac{1}{2} + it)|$ and $\arg\zeta(\frac{1}{2} + it)$ are asymptotically independent normals with the same variance $\frac{1}{2}\log\log T$. Theorem 14.5.2 is an exact finite height counterpart: under QRH(θ), at every height, the two coordinates have equal variances and zero covariance under harmonic measure. A zero to the right of the line removes this proof: the logarithm then has a branch point in the half plane, and Y jumps by $2\pi q$ at the height of a zero of multiplicity q.

	θ	t_0	$\mathbb{E}X - x_0$	$\mathbb{E}Y - y_0$	$\operatorname{Var} X$	$\operatorname{Var} Y$	$\operatorname{Cov}(X, Y)$
ζ	$\frac{1}{2}$	0	0.0000	0.0000	0.1415	0.1415	0.0000
ζ	$\frac{1}{2}$	85.7	0.0000	0.0000	0.7056	0.7056	0.0000
ζ	$\frac{1}{2}$	1000	0.0000	0.0000	1.0539	1.0539	0.0000
ζ	$\frac{1}{2}$	5000	0.0000	0.0000	1.1298	1.1298	0.0000
ζ	$\frac{3}{4}$	85.7	0.0000	0.0000	0.1341	0.1341	0.0000
DH	$\frac{1}{2}$	85.7	+1.4407	−0.0016	0.6410	4.4414	−0.0727
DH	$\frac{3}{4}$	85.7	+0.4770	−0.0003	1.0124	3.2682	−0.0162

Table 14.2: Moments of $Z = \log h_\theta(\theta + i\tau)$ under $P_{1-\theta}(t_0 - \cdot)$, computed in floating point: Poisson weighted panels between the zeros with tanh sinh nodes, midpoint cells beyond distance 1000 out to $|\tau - t_0| \leq 13000$, and the mass beyond the window (2.45×10^{-5} at $\theta = \frac{1}{2}$, 1.22×10^{-5} at $\theta = \frac{3}{4}$) estimated from the sample moments of the outer window. Entries are from the finer of two meshes; the larger of the difference between the meshes and half the tail correction is at most 2.1×10^{-5} in every entry. For ζ the equal variance and zero covariance identities of Theorem 14.5.2 hold to that accuracy; for Davenport and Heilbronn (DH), with $h = f$, they fail at both levels.

In Table 14.2, for ζ the variances of $\log|\zeta|$ and $\arg h_\theta$ agree and the covariance

vanishes to 2.1×10^{-5} in every row, the size of the mesh differences. For Davenport and Heilbronn the variance of the argument is 6.9 times that of the modulus at $\theta=\frac12$ and 3.2 times at $\theta=\frac34$, as the jump of Y at the height of the zero suggests (an observation, not a consequence of the jump), and the first moment shows the defect of Theorem 14.2.1. At the three positive heights shown the variance at $\theta=\frac12$ is close to $\frac12\log\log t_0$ (0.71 against 0.75 at height 85.7, 1.05 against 0.97 at height 1000, 1.13 against 1.07 at height 5000), the scale of Selberg's theorem, which concerns t uniform on $[T,2T]$ and implies nothing for a window of fixed width at a given centre.

14.5.2 The defect of the complex moments

Without QRH(θ) the moments of Z have explicit defects. Bui, Lester and Milinovich [16] move the line of integration across horizontal cuts from the zeros for the logarithm itself; we extend the argument to every polynomial moment, and the extension is proved here.

Proposition 14.5.4 (complex moments through the cuts). *Let $\frac12\le\theta<1$, $z=1+it_0$, $z^*=2\theta-1+it_0$ and $K(s)=\frac{1}{s-z^*}-\frac{1}{s-z}$. For each ordinate γ of a zero with $\beta>\theta$ let b_γ be the largest real part of the zeros with that ordinate, $C_\gamma=[\theta+i\gamma,\,b_\gamma+i\gamma]$, and for $x+i\gamma\in C_\gamma$ let $M_\gamma(x)$ be the number of zeros $\beta'+i\gamma$ with $\beta'>x$, counted with multiplicity. Let L be the logarithm of h_θ on $\{\operatorname{Re}s>\theta\}$ minus the cuts, continued along horizontal paths from $+\infty$, where it tends to 0; let $z_0=L(z)$ and let L_- be the boundary value from below on C_γ, so that the value above minus the value below is $2\pi i M_\gamma$. Let $Z(\tau)=\lim_{\sigma\downarrow\theta}L(\sigma+i\tau)$, which exists for almost every τ, and $\tau\sim P_{1-\theta}(t_0-\cdot)$. Then for every polynomial F*

$$\mathbb{E}F(Z)-F(z_0)=\sum_\gamma\frac{1}{2\pi i}\int_{C_\gamma}\bigl[F(L_-+2\pi iM_\gamma)-F(L_-)\bigr]K(s)\,ds,$$

each integral taken from left to right and the sum absolutely convergent. In particular

$$D:=\mathbb{E}Z-z_0=\sum_{\beta>\theta}\Bigl[\log\frac{\rho-z^*}{\rho-z}-\log\frac{\theta+i\gamma-z^*}{\theta+i\gamma-z}\Bigr],$$

the sum over the zeros with multiplicity and the logarithms continued along the cut, and

$$\mathbb{E}Z^2-z_0^2=2\sum_\gamma\int_{C_\gamma}M_\gamma\,(L_-+\pi iM_\gamma)\,K(s)\,ds.$$

The real part of the formula for D is Theorem 14.2.1 at $1+it_0$. The second order balance $\operatorname{Var}X=\operatorname{Var}Y$, $\operatorname{Cov}(X,Y)=0$ holds exactly when $2\sum_\gamma\int_{C_\gamma}M_\gamma\,(L_-+\pi iM_\gamma-z_0)\,K\,ds=D^2$. When the zeros with $\beta>\theta$ are simple with distinct ordinates, $M_\gamma=1$ on every cut and $L_-+\pi i$ is the mean of the two boundary values.

Proof. On the line $K(\theta+i\tau) = 2a/(a^2+(\tau-t_0)^2)$ with $a = 1-\theta$, so $K(s)\,ds/(2\pi i) = P_{1-\theta}(t_0-\tau)\,d\tau$ there and $\mathbb{E}F(Z) = \frac{1}{2\pi i}\int F(L)K\,ds$ along the line upwards. Take the rectangle $\theta \le \operatorname{Re} s \le X$, $|\operatorname{Im} s - t_0| \le T$, with $t_0 \pm T$ not ordinates of zeros; remove thin strips along the cuts, small discs about their right ends and about the zeros on them, and small half discs about the zeros on the line, and apply the residue theorem to $F(L)K$ on the rest. Its only pole there is z, with residue $-F(z_0)$. With the positive orientation the line is traversed downwards and the upper bank of each cut from left to right, so, as the strips, discs and half discs shrink, $\frac{1}{2\pi i}\int F(L)K\,ds$ along the line upwards equals $F(z_0)$ plus the cut terms of the statement plus $\frac{1}{2\pi i}$ times the integrals over the other three sides; the discs and half discs contribute $O(r(1+|\log r|)^{\deg F}) \to 0$. In a fixed vertical strip $L(s) = A(s)+\sum m_\rho \log(s-\rho)$, the sum over the $O(\log(|t|+2))$ zeros with $|\gamma-t| \le 1$, where $A = O(\log(|t|+2))$ (remove these factors in a disc and apply the Borel and Carathéodory lemma to the rest); so the integrals of $|L|^p$ over horizontal segments and over vertical segments of unit length are $O_p(\log^{c_p}(|t|+2))$. As $K = O(t^{-2})$ in the strip, the horizontal sides contribute $o(1)$ as $T \to \infty$, and the sum over the cuts converges absolutely, since there are $O(T\log T)$ zeros up to height T and every cut is shorter than $\frac{1}{2}$. On the right side $L(s) = O(2^{-\operatorname{Re} s}) + O(|s|^{-1})$, which contributes $o(1)$ as $X \to \infty$. For $F(w) = w$ the jump is $2\pi i M_\gamma$, and $\int_{C_\gamma} M_\gamma K\,ds$ is the sum over the zeros with ordinate γ of $\int_{\theta+i\gamma}^{\rho} K\,ds$; for $F(w) = w^2$ the jump is $4\pi i M_\gamma(L_- + \pi i M_\gamma)$. Finally $\mathbb{E}Z^2 - (\mathbb{E}Z)^2 = (\mathbb{E}Z^2 - z_0^2) - 2z_0 D - D^2$. □

The same holds for the Davenport and Heilbronn function with $h = f$, provided no zero has $\gamma = t_0$ and $\beta \ge 1$, so that z lies off the cuts and $f(z) \ne 0$; the proof uses the local factorisation and the Dirichlet series bounds of f in place of those of ζ.

We checked Proposition 14.5.4 on the Davenport and Heilbronn function at $t_0 = 85.7$, with the zeros with $\beta > \theta$ below height 700 and their conjugates (computed, floating point): L is continued along each cut, the cut integrals use Gauss and Legendre nodes, and the jump across every cut is $2\pi i$ to 7 digits. At $\theta = \frac{1}{2}$ the cut formula gives $\mathbb{E}Z - z_0 = 1.4407 - 0.0016\,i$ and the balance defect $\operatorname{Var} X - \operatorname{Var} Y + 2i\operatorname{Cov}(X,Y) = -3.8003 - 0.1455\,i$, against $1.4407 - 0.0016\,i$ and $-3.8004 - 0.1455\,i$ from the unrounded entries of Table 14.2; at $\theta = \frac{3}{4}$ it gives $0.4770 - 0.0003\,i$ and $-2.2558 - 0.0323\,i$, the same to the digits shown.

For one simple zero with $\beta > \theta$ the balance condition reads exactly $\bar{L} - z_0 = \frac{1}{2}D$ when $D \ne 0$, where $\bar{L} = D^{-1}\int_C (L_- + \pi i)K\,ds$ is a mean of $L_- + \pi i$ along its cut with the complex weight K/D. For a zero far from t_0 the weight is nearly constant along the short cut, so $\bar{L}$ nearly does not depend on t_0, while $D \to 0$; balance at every such t_0 would make $z_0 = \log h_\theta(1+it_0)$ nearly constant, which it is not. So the balance should fail for most t_0 when a zero lies to the right of the line (heuristic: the variation of $\bar{L}$ with t_0 has not been bounded). With several cuts the contributions can cancel, and no proof is known (Open problem 14.5).

14.6 The Monte Carlo solution

14.6.1 The vertical representation

Write $f(\tau) = \log|\zeta(\theta + i\tau)|$ (or $\log|h|$ for another function) and $P = P_{1-\theta}(t_0 - \cdot)$. The layer cake formula gives

$$\mathbb{E}_P[f] = \int_0^\infty P(f > \lambda)\, d\lambda - \int_0^\infty P(f < -\lambda)\, d\lambda =: A^+ - A^-.$$

In this vertical form Theorem 14.2.1 says that QRH(θ) holds exactly when the upper and lower tail areas of $\log|\zeta|$ under harmonic measure differ by $\log|h_\theta(1 + it_0)|$, for one (equivalently every) t_0. The lower tail $P(f < -\lambda)$ is the harmonic measure of the neighbourhoods $\{|\zeta| < e^{-\lambda}\}$ of the zeros on the line, which decays like $e^{-\lambda}$ when $\theta = \frac{1}{2}$ and the zeros are simple (heuristic on the whole line).

Lemma 14.6.1 (finite moments). *For $\frac{1}{2} \le \theta < 1$, every t_0 and every $p < \infty$, $\mathbb{E}_P[|f|^p] < \infty$. Hence plain Monte Carlo with exact Cauchy draws is unbiased, needs no truncation of the heavy tail, and has root mean square error $(\mathrm{Var}_P f/n)^{1/2}$.*

Proof. The density of P is $\ll_{t_0} (1+t^2)^{-1}$, $|\zeta(\theta - it)| = |\zeta(\theta + it)|$, and f has only logarithmic singularities on bounded intervals, so $\int_{-2}^{2} |f|^p\, dP < \infty$, and it suffices to show $\int_T^{T+1} |f(t)|^p\, dt \ll_p \log^p T$ for $T \ge 2$, since $\sum_T T^{-2}\log^p T < \infty$. By the convexity bound, $\log^+|\zeta(\theta + it)| \ll \log T$ on $[T, T+1]$. By the local formula $\log|\zeta(s)| = \sum_{|\gamma - t| \le 1} \log|s - \rho| + O(\log t)$, uniform in $-1 \le \sigma \le 2$ [264, Chapter IX], and $|\theta + it - \rho| \ge |t - \gamma|$,

$$\log^-|\zeta(\theta + it)| \le \sum_{|\gamma - t| \le 1} \log^-|t - \gamma| + O(\log T).$$

The number n_T of zeros with $|\gamma - T| \le 2$ is $O(\log T)$. By Hölder's inequality for the sum over n_T terms,

$$\int_T^{T+1} \Big(\sum_{|\gamma - t| \le 1} \log^-|t - \gamma| \Big)^p\, dt \le n_T^{p-1} \sum_{|\gamma - T| \le 2} \int_{\mathbb{R}} \big(\log^-|x|\big)^p\, dx \ll_p \log^p T,$$

since $\int_{\mathbb{R}} (\log^-|x|)^p\, dx = 2\,\Gamma(p+1)$. □

14.6.2 Three estimators

Quadrature Adaptive Gauss quadrature on $|\tau - t_0| \le W$, with the zeros on the line used as breakpoints so that the logarithmic singularities are endpoint singularities. The neglected Cauchy mass is $\frac{2}{\pi}\arctan\frac{1-\theta}{W} \approx \frac{2(1-\theta)}{\pi W}$; the neglected part of the integral needs its own bound (Cauchy and Schwarz with Lemma 14.6.1).

Plain Monte Carlo Draw $\tau_i = t_0 + (1-\theta)\tan\big(\pi(u_i - \frac{1}{2})\big)$ with u_i uniform, and average $f(\tau_i)$. Lemma 14.6.1 makes this unbiased with a finite standard error.

Vertical likelihood Monte Carlo Following Polson and Scott [202], sample instead from a law that reweights the levels of f,

$$P_w(d\tau) \;\propto\; P(d\tau)\, w\big(f(\tau)\big),$$

and recover $\mathbb{E}_P[f]$ and the vertical curves $P(f > \lambda)$, $P(f < -\lambda)$ by self normalised weights $1/w$. We use $w(f) = |f - c| + \epsilon$, with c a pilot estimate, a regularised form of the weight $|f - \mathbb{E}_P f|$, which is optimal for independent draws (Remark 14.6.2). We sample P_w by a Metropolis chain mixing Cauchy independence proposals with local moves. The weighting visits the deep levels $\{f < -\lambda\}$ near the zeros and the upper tail more often than plain sampling (computed, Table 14.3 and below).

Remark 14.6.2 (the optimal weight). Let $\mu = \mathbb{E}_P f$, $w > 0$, $Z = \mathbb{E}_P w$ and $q = Pw/Z$. For independent draws $\tau_i \sim q$ the self normalised estimator $\hat{\mu} = \sum_i f(\tau_i)/w(\tau_i) \big/ \sum_i 1/w(\tau_i)$ satisfies $n \operatorname{Var}(\hat{\mu}) \to Z\, \mathbb{E}_P[(f-\mu)^2/w]$, since $\hat{\mu} - \mu = Z\, n^{-1} \sum_i (f-\mu)(\tau_i)/w(\tau_i) + o_P(n^{-1/2})$ and $\mathbb{E}_q[(f-\mu)^2/w^2] = Z^{-1}\mathbb{E}_P[(f-\mu)^2/w]$. By the Cauchy and Schwarz inequality the limit is at least $(\mathbb{E}_P|f-\mu|)^2$, with equality exactly for $w \propto |f-\mu|$, and $(\mathbb{E}_P|f-\mu|)^2 \le \operatorname{Var}_P f$, with equality only if $|f-\mu|$ is constant. So for independent draws the weight $|f-\mu|$, the standard optimal proposal of self normalised importance sampling, strictly beats plain Monte Carlo for these integrands, and $|f-c|+\epsilon$ approximates it. For the integrands of Table 14.3 the factor $\operatorname{Var}_P f/(\mathbb{E}_P|f-\mu|)^2$ ranges over the nine rows from 1.35 (DH, $\theta = \frac{3}{4}$, $t_0 = 85.7$) to 2.46 (ζ, $\theta = \frac{7}{8}$, $t_0 = 85.7$; the nine values are 1.35, 1.41, 1.43, 1.48, 1.55, 1.86, 2.02, 2.13 and 2.46), with an error of up to 0.03 (computed by quadrature), so for independent draws no weight gains more than that. For a Markov chain the limit is $Z^2\big[\operatorname{Var}_q Y_0 + 2\sum_{j\ge 1} \operatorname{Cov}_q(Y_0, Y_j)\big]$ with $Y = (f-\mu)/w$, when a central limit theorem holds; negative autocorrelation can bring it below the value for independent draws, so the factor above does not bound Markov samplers, and the optimum changes.

14.6.3 Results

Table 14.3 reports the local defect $D_\theta(t_0)$ for ζ and for the Davenport and Heilbronn function (with $h = f$, which is entire), whose zero $0.808517182457 + 85.699348485378\,i$ lies off the line (certified in Section 13.2; see also [261, 17]). The prediction for ζ is zero, under RH for $\theta = \frac{1}{2}$ and under QRH(θ) for the other levels. For Davenport and Heilbronn it is the sum of the Green potentials of its zeros with $\beta > \theta$, at least that of this zero (the other zeros below height 200 add about 2×10^{-4} at $\theta = \frac{1}{2}$ and less than 10^{-5} at $\theta = \frac{3}{4}$): $\log(0.8085/0.1915) \approx 1.440$ at $\theta = \frac{1}{2}$ and $\log(0.3085/0.1915) \approx 0.477$ at $\theta = \frac{3}{4}$. The table prints the sum over the zeros below height 700.

(θ, t_0)	prediction	quad. $W = 60$	quad. $W = 200$	plain MC	vertical MC
ζ $(\frac{1}{2}, 0)$	0	-4.1×10^{-5}	5.0×10^{-6}	-0.0040 ± 0.0026	-0.0021 ± 0.0055
ζ $(\frac{7}{8}, 0)$	0	-7.9×10^{-6}	8.7×10^{-7}	-0.0032 ± 0.0049	-0.0173 ± 0.0132
ζ $(\frac{1}{2}, 85.7)$	0	1.79×10^{-4}	-1.5×10^{-6}	-0.0146 ± 0.0061	-0.0064 ± 0.0141
ζ $(\frac{3}{4}, 85.7)$	0	5.8×10^{-5}	-7.9×10^{-7}	-0.0022 ± 0.0026	-0.0074 ± 0.0055
ζ $(\frac{7}{8}, 85.7)$	0	2.2×10^{-5}	-3.8×10^{-7}	-0.0011 ± 0.0016	0.0037 ± 0.0032
DH $(\frac{1}{2}, 85.7)$	1.4407	1.4405	1.4406	1.4374 ± 0.0133	1.4727 ± 0.0229
DH $(\frac{3}{4}, 85.7)$	0.4770	0.47697	0.47698	0.4759 ± 0.0159	0.4740 ± 0.0312
DH $(\frac{3}{4}, 100)$	1.49×10^{-4}	1.16×10^{-4}	1.47×10^{-4}	0.0099 ± 0.0054	0.0139 ± 0.0130
DH $(\frac{7}{8}, 85.7)$	0	-3.8×10^{-6}	-3.6×10^{-7}	-0.0160 ± 0.0141	-0.0311 ± 0.0260

Table 14.3: Local defects $D_\theta(t_0)$ (computed, floating point). Davenport and Heilbronn is abbreviated DH. The quadrature columns are the integral over $|\tau - t_0| \le W$ minus $\log|h_\theta(1 + it_0)|$, with the zeros as breakpoints: they are realised truncation errors of the window, not accuracies, and change with W. Outside the window the Cauchy mass is 0.0053 at $\theta = \frac{1}{2}$, 0.0027 at $\theta = \frac{3}{4}$ and 0.0013 at $\theta = \frac{7}{8}$ for $W = 60$. Monte Carlo (computed) uses $n = 20{,}000$ evaluations for ζ and 4,000 for DH. Plain Monte Carlo uses exact Cauchy draws, with standard error $\mathrm{sd}/\sqrt{n}$. The vertical sampler takes a pilot mean c from $n/20$ draws, $\epsilon = 0.05$ times the standard deviation of the pilot, an even mixture of Cauchy independence proposals and Gaussian steps of scale $(1 - \theta)/2$, and a burn in of $n/20$ steps, with batch means standard errors (50 batches). The Davenport and Heilbronn prediction is the sum of the Green potentials of the zeros found below height 700; for $\theta = \frac{7}{8}$ none has $\beta > \frac{7}{8}$, but f has zeros with $\beta > 1$ [60], so the exact defect is positive.

The quadrature values agree with the predictions to the size of the neglected tail, which is larger than the printed values suggest. For ζ at $(\frac{1}{2}, 0)$ the Cauchy and Schwarz bound for the neglected part, over blocks of unit length, is 5.3×10^{-3} (computed up to height 1000) plus 1.1×10^{-3} (extrapolated, heuristic) at $W = 60$, and 1.4×10^{-3} plus 1.1×10^{-3} at $W = 200$; at $(\frac{7}{8}, 0)$ it is 6.3×10^{-4} plus 1.7×10^{-4} and 1.6×10^{-4} plus 1.7×10^{-4}; for Davenport and Heilbronn at $(\frac{3}{4}, 85.7)$ it is 1.0×10^{-3} plus 6.2×10^{-4} and 2.0×10^{-4} plus 6.2×10^{-4}. The two Monte Carlo columns agree with the quadrature values within 2.4 standard errors (plain Monte Carlo for ζ at $(\frac{1}{2}, 85.7)$ is 2.4 off, the other seventeen entries are within 2), and at height 100 neither sampler resolves the defect 1.49×10^{-4}, which is far below their standard errors 0.005 and 0.013. The local character of the criterion is visible in the last rows: centred at the zero the defect of Davenport and Heilbronn at $\theta = \frac{3}{4}$ is the order one number 0.477, and seen from height 100 the same zero gives 1.4×10^{-4}.

At $\theta = \frac{7}{8}$ and $t_0 = 0$ the criterion of Theorem 14.2.1 reads

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \log|\zeta(\tfrac{7}{8} + i\tau)| \, \frac{1/8}{1/64 + \tau^2} \, d\tau = \log 4 + D_{7/8}(0),$$

with $D_{7/8}(0) = 0$ under QRH($\frac{7}{8}$); the quadrature gives $\log 4$ to within the window error of the table. With no hypothesis, each zero with $\beta > \theta$ adds at most $2(1-\theta)(\beta-\theta)/\gamma^2$ to $D_\theta(0)$, and by Remark 14.2.2 all possible zeros above the verified height add less than 10^{-12} at every level (computed). The global criterion is therefore numerically blind above the verified range and useful only locally.

All three estimators agree with the predictions. Quadrature gives the smallest realised errors, with the caveat above. Plain Monte Carlo has the smaller standard error of the two samplers; the vertical sampler pays for its Markov dependence and its importance weights, and buys more visits to the deep levels: at $\theta = \frac{1}{2}$ it visits $\{f < -2\}$ more often (computed: 315 visits against 25 for ζ at $(\frac{1}{2}, 0)$, 2952 against 679 at $(\frac{1}{2}, 85.7)$, 76 against 43 for Davenport and Heilbronn), without reaching a deeper minimum of f. So it samples the lower tail A^-, the harmonic measure of the neighbourhoods of the zeros, more densely, at the price of a larger standard error.

Figure 14.2 shows the vertical picture. For ζ the running balance $A^+(\Lambda) - A^-(\Lambda) - \log|h(1+it_0)|$ settles at zero once the level cutoff passes about 4. For Davenport and Heilbronn it settles near the Green potential 1.44 of the zero off the line, an imbalance between the upper and lower tails.

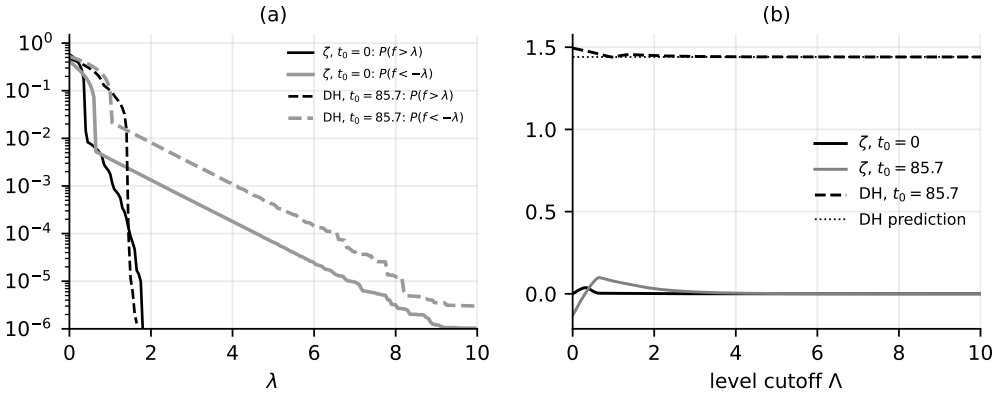


Figure 14.2: Vertical curves at $\theta = \frac{1}{2}$ (computed, floating point). Left: $P(f > \lambda)$ (black) and $P(f < -\lambda)$ (grey) under the Cauchy law centred at t_0, for ζ at $t_0 = 0$ (solid) and Davenport and Heilbronn at $t_0 = 85.7$ (dashed), $\lambda \le 10$, logarithmic vertical axis. Right: the running balance $\mathbb{E}[\mathrm{clip}(f, -L, L)] - \log|h(1+it_0)|$, which converges to the defect $D_{1/2}(t_0)$, for ζ at $t_0 = 0$ and 85.7 and for Davenport and Heilbronn at $t_0 = 85.7$, with the prediction 1.4407 dotted; window $|\tau - t_0| \le 200$, neglected Cauchy mass 0.0016.

14.6.4 Weighted Monte Carlo of Yakowitz, Krimmel and Szidarovszky

The map $\tau(u) = t_0 + a\tan\bigl(\pi(u-\tfrac12)\bigr)$, with $a = 1-\theta$, writes every Cauchy average as an integral over the unit interval, $\mathbb{E}_P[f] = \int_0^1 g(u)\,du$ with $g = f\circ\tau$. Yakowitz, Krimmel and Szidarovszky [277] proposed sorting uniform nodes $u_{(1)} < \cdots < u_{(N)}$ and applying the trapezoid rule:

$$T_N = \sum_{i=0}^{N} \bigl(u_{(i+1)} - u_{(i)}\bigr)\,\frac{g(u_{(i)}) + g(u_{(i+1)})}{2}, \qquad u_{(0)} = 0,\ u_{(N+1)} = 1.$$

We set $g(0) = g(1) = 0$, a convention: g has no limit at the ends, and 0 is the mean of $\log|\zeta|$ on the line $\operatorname{Re} s = \theta$ for $\theta > \tfrac12$, and for $\theta = \tfrac12$ under RH. Their theorem: if $g \in C^2[0,1]$ then $\mathbb{E}(T_N - \mathbb{E}_P f)^2 = O(N^{-4})$, against $O(N^{-1})$ for the sample mean. Lemma 14.6.8 below gives the leading bias. The same weights define a rule for any node set, so the states of the vertical likelihood chain can serve as nodes, the spacings replacing the weights $1/w$, although the theorem does not transfer to dependent nodes.

The spacings remove the variance that the sample mean spends on the smooth part of g, and the Euler product supplies a second, exact reduction.

Lemma 14.6.3 (exact Cauchy means). *For any Dirichlet polynomial* $D(s) = \sum_{n\le M} b_n n^{-s}$ *and* $a = 1-\theta$,

$$\mathbb{E}_P\bigl[\operatorname{Re} D(\theta + i\tau)\bigr] = \operatorname{Re} D(1 + it_0),$$
$$\mathbb{E}_P\Bigl[\log\Bigl|\frac{\theta+1+i\tau}{\theta-1+i\tau}\Bigr|\Bigr] = \log\Bigl|\frac{2+it_0}{2-2\theta+it_0}\Bigr|.$$

Proof. The characteristic function of the Cauchy law gives $\mathbb{E}_P[n^{-i\tau}] = n^{-it_0}e^{-a\log n}$, so each term $n^{-\theta-i\tau}$ averages to n^{-1-it_0}. For the second, $|\theta-1+i\tau| = |\theta-(2\theta-1)+i\tau|$ on the line, and $\log|s+1| - \log|s-2\theta+1|$ is harmonic on $\{\operatorname{Re} s > \theta\}$; its Poisson integral at $1+it_0$ is its value there. □

With D_M the Selberg smoothed truncation of $\log\zeta$, $b_n = \frac{\Lambda(n)}{\log n}\bigl(1 - \frac{\log n}{\log M}\bigr)$, we apply the trapezoid rule to $g - \operatorname{Re} D_M\circ\tau - \ell\circ\tau$, where ℓ is the pole term of Lemma 14.6.3, and add back the exact means. Table 14.4 and Figure 14.3 report root mean square errors over 400 replications, at $t_0 = 0$, against the values 0 and $\log 4$ that the criterion gives under RH and QRH($\frac78$); with no hypothesis they hold to within 7.4×10^{-13} and 4.7×10^{-14} (Remark 14.2.2).

At $\theta = \frac78$ the trapezoid weights alone reduce the error by a factor between 12 and 18, and with the control by a factor between 65 and 75, equivalent to between 4,000 and 6,000 times as many plain samples. At $\theta = \frac12$ the factor is between 3 and 4.3. With coefficient one the control alone makes the sample mean worse: over 400 replications its root mean square error is 3.3 times that of plain sampling

θ	N	plain	YKS	YKS with control	gain
$\frac{1}{2}$	500	1.7×10^{-2}	9.6×10^{-3}	5.3×10^{-3}	3.2
$\frac{1}{2}$	2000	8.4×10^{-3}	4.0×10^{-3}	2.6×10^{-3}	3.3
$\frac{1}{2}$	8000	4.2×10^{-3}	1.6×10^{-3}	1.1×10^{-3}	3.7
$\frac{1}{2}$	16000	3.1×10^{-3}	9.7×10^{-4}	7.2×10^{-4}	4.3
$\frac{7}{8}$	500	3.0×10^{-2}	2.3×10^{-3}	4.0×10^{-4}	74
$\frac{7}{8}$	2000	1.5×10^{-2}	1.3×10^{-3}	2.3×10^{-4}	65
$\frac{7}{8}$	8000	7.7×10^{-3}	4.8×10^{-4}	1.0×10^{-4}	75
$\frac{7}{8}$	16000	5.1×10^{-3}	2.9×10^{-4}	6.9×10^{-5}	75

Table 14.4: Root mean square error of the Cauchy average of $\log|\zeta(\theta+i\tau)|$, $t_0 = 0$, over 400 replications (computed, floating point; the sampling error of each entry is about 3.5% and larger for the heavy tailed plain estimator). The control uses $M = 100$ at $\theta = \frac{1}{2}$ and $M = 1000$ at $\theta = \frac{7}{8}$. Gain is the ratio of plain to YKS with control. Fitted slopes over $500 \le N \le 16{,}000$ at $\theta = \frac{1}{2}$ and $\frac{7}{8}$: plain -0.49 and -0.50, YKS -0.66 and -0.60, YKS with control -0.58 and -0.52.

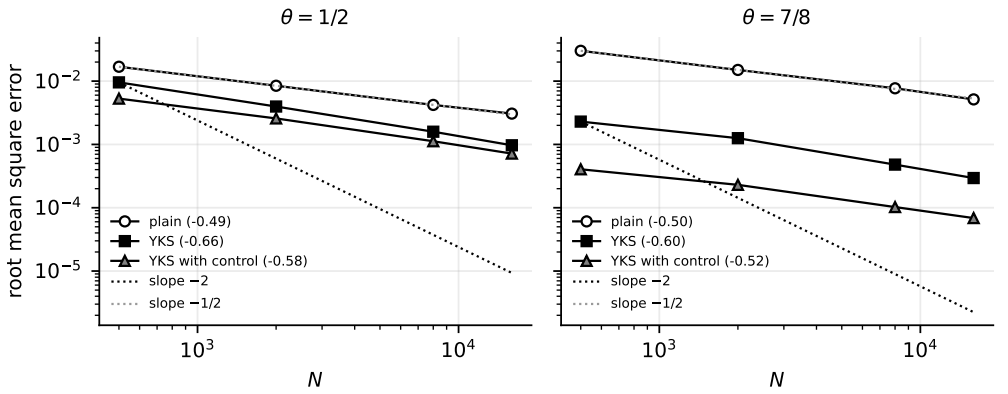


Figure 14.3: Root mean square error against N (from the 400 replications of Table 14.4), drawn for $\theta = \frac{1}{2}$ and $\frac{7}{8}$ for plain sampling, YKS and YKS with control, with the fitted slopes in the legend. The dotted lines have slope -2 (anchored at the first YKS point), the rate of the theorem of Yakowitz, Krimmel and Szidarovszky for integrands in $C^2[0,1]$, and slope $-\frac{1}{2}$ (anchored at the first plain point).

at $\theta = \frac{1}{2}$ and 1.04 times at $\theta = \frac{7}{8}$ (computed); it helps only once the trapezoid weights have removed the variance of the smooth part.

The rate, however, is not N^{-2}: the fitted slopes over $500 \le N \le 16{,}000$ lie between -0.49 and -0.66 for every method. The hypothesis $g \in C^2[0,1]$ fails, and indeed $g \notin C^1[0,1]$:

PROPOSITION 14.6.4 (why the N^{-4} bound does not apply). *Let $\frac{1}{2} \le \theta < 1$, $f(\tau) = \log|\zeta(\theta + i\tau)|$ and $g = f \circ \tau$. If $\theta = \frac{1}{2}$, g is unbounded near $\tau^{-1}(\gamma)$ for every ordinate γ of a zero on the line. If $\frac{1}{2} < \theta < 1$, g' is unbounded near $u = 0$ and near $u = 1$. In both cases $g \notin C^1[0,1]$.*

Proof. For $\theta = \frac{1}{2}$, $f(\tau) \to -\infty$ as $\tau \to \gamma$. Let $\frac{1}{2} < \theta < 1$ and suppose $|g'| \le C$ on $(1-\eta, 1)$. Then $g(u)$ has a limit as $u \to 1$, so $|\zeta(\theta + i\tau)| \to e^L$ for some L as $\tau \to \infty$. The mean value theorems for $\sigma > \frac{1}{2}$ [264, Chapter VII] give $T^{-1}\int_1^T |\zeta(\theta+it)|^2 dt \to \zeta(2\theta)$ and $T^{-1}\int_1^T |\zeta(\theta+it)|^4 dt \to \zeta(2\theta)^4/\zeta(4\theta)$, so $e^{2L} = \zeta(2\theta)$ and $e^{4L} = \zeta(2\theta)^4/\zeta(4\theta)$, whence $\zeta(2\theta)^2 = \zeta(4\theta)$. This is false, since $\zeta(4\theta) = \sum_{n=m^2} n^{-2\theta} < \sum_n d(n)\, n^{-2\theta} = \zeta(2\theta)^2$. The end $u = 0$ is the same, because $|\zeta(\theta - i\tau)| = |\zeta(\theta + i\tau)|$. □

REMARK 14.6.5 (a heuristic rate). For a single component $f(\tau) = \cos(k\tau)$ the following count suggests a mean square error of order $\sqrt{ak}\, N^{-3/2}$ for the sorted trapezoid rule, that is a root mean square error of order $N^{-3/4}$ (heuristic). Near $u = 1$, $\tau - t_0 \approx a/(\pi(1-u))$, and $\tau'(u) = \pi\big(a^2 + (\tau - t_0)^2\big)/a$. A cell of width $h \approx 1/N$ in u has width $\Delta \approx h\pi(\tau - t_0)^2/a$ in τ for large τ. Cells with $k\Delta \lesssim 1$ contribute variance of order $h^2(k\Delta)^4$; cells beyond $T_N \asymp (aN/k)^{1/2}$, where $k\Delta \gtrsim 1$, behave like independent samples and contribute of order h^2 each. There are about $Na/(\pi\tau^2)$ cells per unit of τ. The two sums are each of order $\sqrt{ak}\, N^{-3/2}$.

A test with $f(\tau) = \cos(\tau \log 2)$ gives the slope -0.74 for the trapezoid rule at both levels over $500 \le N \le 128{,}000$ (and -0.50 for plain sampling), close to the predicted $-\frac{3}{4}$. The Selberg smoothed polynomials $\operatorname{Re} D_{1000}$ give -0.67 at $\theta = \frac{7}{8}$ and -0.71 at $\theta = \frac{1}{2}$. The true integrands give -0.60 and -0.66 for the trapezoid rule alone, and only the plain slope is near $-\frac{1}{2}$. The likely cause (heuristic) is the features that the polynomial lacks: the logarithmic singularity at each zero on the line, and the dips of width $\theta - \frac{1}{2}$ at the heights of the zeros, whose number grows like $\log\tau$. Each of these adds error at the scale of one cell.

Nor does the N^{-4} rate transfer to the vertical scale. First, the vertical representation needs the masses $X(\lambda) = P(f < -\lambda)$ at the nodes. In nested sampling and in the vertical likelihood chain these are estimated, with a Monte Carlo error (of order $N^{-1/2}$ for independent draws) that a better quadrature rule alone does not remove (heuristic). Second, even computed exactly, the quantile function of f under P has a logarithmic singularity at 0 when $\theta = \frac{1}{2}$, because $P(f < -\lambda) \asymp e^{-\lambda}$ (heuristic, as above), and a flat point at every nondegenerate critical value of f, where the law of f has an inverse square root singularity, one for each gap between consecutive zeros. Whether the vertical chain's visited states, which concentrate nodes at the deep levels and leave wider gaps elsewhere, work as nodes of the trapezoid rule is not tested here; its error is governed by the widest gaps where g oscillates (heuristic).

In one dimension, quadrature with the zeros as breakpoints remains the

method of choice. The Yakowitz weights with the Dirichlet control are for averages that quadrature cannot reach cheaply, such as double integrals over level and height, where the trapezoid rule becomes the Voronoi and Delaunay rules (Section 14.7). On the line, Section 14.6.5 gives an exact local identity at the rates of the theory, and Section 14.6.6 a truncation whose tail is bounded under RH.

14.6.5 An exact identity on a half disc

Whether another parametrisation of the whole line gives an integrand smooth enough for the rates of Yakowitz, Krimmel and Szidarovszky is not settled here: for a positive integrand the distribution function of fP makes the transformed integrand constant, but it needs the integral itself. A change of domain does work. Cut the line at the heights $t_0 \pm R$ and close the contour by a semicircle in the half plane, where $\log|h|$ is harmonic away from the zeros; the Cauchy weight becomes the harmonic measure of a half disc, and the identity stays exact.

Theorem 14.6.6 (Balazard, Saias and Yor on a half disc). *Let $c = \theta + it_0$, $R > 0$, $H = \{|s-c| < R,\ \operatorname{Re} s > \theta\}$, $z \in H$, and let ω_z be the harmonic measure of ∂H seen from z. Let $h \not\equiv 0$ be analytic on a neighbourhood of $\overline{H}$, with no zero on the closed arc $\{|s-c| = R,\ \operatorname{Re} s \geq \theta\}$, with zeros $\rho_1, \dots, \rho_n$ of multiplicities $m_1, \dots, m_n$ on the open diameter, and with $h(z) \neq 0$. Then*

$$\int_{\partial H} \Big(\log|h| - \sum_j m_j \log|s-\rho_j| \Big)\, d\omega_z = \log|h(z)| - \sum_j m_j \log|z-\rho_j| + \sum_{\rho \in H} G_H(z,\rho),$$

the last sum over the zeros of h in H with multiplicity, and the integrand on the left is real analytic on each of the two closed pieces of ∂H. In the coordinate $w = (s-c)/R$, with $w_z = (z-c)/R$, $w^* = -\overline{w_z}$ the reflection of w_z in the diameter and $G_D(a,w) = \log|(1-\bar a w)/(w-a)|$,

$$G_H(z,\cdot) = G_D(w_z,\cdot) - G_D(w^*,\cdot),$$

$$\frac{d\omega_z}{d\phi} = \frac{1-|w_z|^2}{2\pi|e^{i\phi}-w_z|^2} - \frac{1-|w^*|^2}{2\pi|e^{i\phi}-w^*|^2}$$

on the arc $w = e^{i\phi}$, $|\phi| \leq \frac{\pi}{2}$, and on the diameter $w = iy$, $|y| \leq 1$, the density with respect to dy is $\frac{1}{2\pi}\partial_x \big[G_D(w_z, x+iy) - G_D(w^*, x+iy) \big]$ at $x = 0$. Both densities are real analytic up to the corners $w = \pm i$, where they vanish to first order. In particular h has no zero in H if and only if the identity holds without the Green sum.

Proof. Put $q(s) = h(s)\prod_j (s-\rho_j)^{-m_j}$. Then q is analytic on a neighbourhood of $\overline{H}$ with no zero on ∂H, so $\log|q| = \log|h| - \sum_j m_j \log|s-\rho_j|$ is real analytic on a neighbourhood of ∂H. Green's identity on H with small discs about the zeros of q removed (on the circle of radius r about a zero of multiplicity m the normal derivative of $\log|q|$ integrates to $2\pi m + O(r)$) gives the Riesz decomposition $\log|q(z)| = \int_{\partial H} \log|q|\, d\omega_z - \sum_{\rho\in H} G_H(z,\rho)$, which is the identity. The function $G_D(w_z,\cdot) - G_D(w^*,\cdot)$ vanishes on the unit circle and, because w^* is the reflection of w_z in the imaginary axis, on the diameter, and its only singularity in the half disc is the logarithmic pole at w_z; so it is the Green function of H. The harmonic measure density is its inner normal derivative divided by 2π, which gives the two formulas. They are rational in the boundary point, and at the right angled corners $w = \pm i$ each has a simple zero, as the reflection principle predicts ($r^{\pi/(\pi/2)-1} = r$ at distance r from a corner). Since $G_H > 0$ on H, the Green sum vanishes exactly when H contains no zero. □

Remark 14.6.7 (the half plane as a limit). Let $h = h_\theta$ and $z = 1 + it_0$. Each $\log|s-\rho_j|$ is the Poisson integral of its boundary values (it is the real part of $\log(s-\rho_j)$, which lies in H^p of the half plane for every $p < \infty$), so the identity also holds with the subtracted terms removed on both sides. As $R \to \infty$ the density on the diameter increases to the Cauchy density $P_{1-\theta}(t_0 - \cdot)$, since harmonic measure on the common boundary increases with the domain, and the Green sum increases to the sum of Theorem 14.2.1. On the arc the density is at most $CR^{-1}\cos\phi$, so the part of the arc in $\theta \le \operatorname{Re} s \le 2$ has harmonic measure $O(R^{-3})$; along radii $R_k \to \infty$ with $|t_0 \pm R_k - \gamma| \ge c/\log R_k$ for every zero, $|\log|h_\theta|| = O(\log^2 R_k)$ there, while $|\log|h_\theta(s)|| = O(2^{-\operatorname{Re} s} + R_k^{-1})$ on the rest of the arc. So the arc term tends to 0, and with Lemma 14.6.1 and dominated convergence on the diameter Theorem 14.6.6 becomes Theorem 14.2.1 at $1+it_0$. That the arc carries vanishing harmonic measure would not suffice by itself: for $h(s) = e^{s-\theta}$ the arc contributes $1-\theta$ for every R.

Wunderlich [207] gives other finite radius forms of the criterion, by Jensen's formula on discs with the trivial zeros removed explicitly, and a relative entropy form: if X has the density $6/(\pi(1+t^2)(4+t^2))$, which is twice the Cauchy$(0,1)$ density minus the Cauchy$(0,2)$ density, $\Xi(t) = \xi(\frac12 + it)$ is Riemann's Xi function, $\mathbb{E}\log|\Xi(X)|$ is finite, $W = |\Xi(X)/\Xi(0)|^2$, and ν is the law with density $W/\mathbb{E}W$ with respect to the law λ of X, then $D_{\mathrm{KL}}(\lambda\|\nu) = \log\mathbb{E}W - \mathbb{E}\log W$, with no hypothesis; in his formulation RH is the saturation of a separate bound. The half disc sits on the line and is local in height, which is what the quadrature rates below use.

Lemma 14.6.8 (the sorted trapezoid rule). *Let $g \in C^4[a,b]$ and let T_N be the sorted trapezoid rule at N independent uniform nodes on $[a,b]$ with the end points added, as in Section 14.6 (there g vanishes at the end points, here its end values*

are used). Then

$$\mathbb{E}T_N - \int_a^b g = \frac{(b-a)^2\big(g'(b)-g'(a)\big)}{2(N+1)(N+2)} + O(N^{-3}), \qquad \operatorname{Var} T_N = O(N^{-5}).$$

So the mean square error is $O(N^{-4})$, and $O(N^{-5})$ when $g'(a) = g'(b)$.

Proof. Take $[a,b] = [0,1]$ and write $T_N - \int_0^1 g = \sum_{i=0}^{N} e(u_{(i)}, u_{(i+1)})$, $u_{(0)} = 0$, $u_{(N+1)} = 1$, where $e(x, x+h) = \frac{h}{2}(g(x)+g(x+h)) - \int_x^{x+h} g = \frac{h^3}{12} g''(x+\frac{h}{2}) + O(h^5)$ and $|e(x,x+h)| \le \frac{h^3}{12}\|g''\|_\infty$. The inner intervals form a point process in (left end, length) of intensity $N(N-1)(1-h)^{N-2}$ on $\{x>0,\ h>0,\ x+h<1\}$, and the two outer ones have lengths of density $N(1-h)^{N-1}$. Since $\int_0^{1-h} g''(x+\frac{h}{2})\,dx = g'(1-\frac{h}{2}) - g'(\frac{h}{2}) = g'(1) - g'(0) + O(h)$ and $N(N-1)\int_0^1 (1-h)^{N-2} h^3\,dh = \frac{6}{(N+1)(N+2)}$, the inner intervals give the main term with error $O(N^2 \int_0^1 (1-h)^{N-2} h^4 dh) = O(N^{-3})$; the outer intervals give $O(N\int_0^1 (1-h)^{N-1} h^3 dh) = O(N^{-3})$, and the $O(h^5)$ terms give $O(\mathbb{E}\sum_i h_i^5) = O(N^{-4})$. Replacing one node by an independent copy merges its two adjacent intervals and splits the interval containing the copy, so at most six intervals change, each a spacing, a size biased spacing or a sum of two spacings, with sixth moment $O(N^{-6})$. Hence $\mathbb{E}(T_N - T'_N)^2 = O(N^{-6})$, and the Efron and Stein inequality gives $\operatorname{Var} T_N \le \frac{N}{2}\mathbb{E}(T_N - T'_N)^2 = O(N^{-5})$. □

Each piece of the boundary integral in Theorem 14.6.6 is a smooth integrand on a closed interval that vanishes at the ends. By Lemma 14.6.8 the sorted trapezoid rule with $N/2$ nodes on each piece has mean square error $O(N^{-4})$; its bias $\frac{2(b-a)^2(g'(b)-g'(a))}{(N+2)(N+4)} + O(N^{-3})$ vanishes to that order only if the end slopes agree. The change of variable $y = \Phi(v) = v + \sin(\pi v)/\pi$ on each piece, written on $[-1,1]$, makes the transformed integrand $g(\Phi(v))\Phi'(v)$ vanish to at least fifth order at the ends, because $\Phi'(\pm(1-r)) = \frac{\pi^2}{2} r^2 + O(r^4)$ and $1 - |\Phi(\pm(1-r))| = \frac{\pi^2}{6} r^3 + O(r^5)$; both end slopes are then zero, and by the same lemma the mean square error is $O(N^{-5})$. These are upper bounds (proved).

Table 14.5 and Figure 14.4 test this with $R = 1$, $t_0 = 85.7$ and $z = c + \frac{1}{2}R$, over 200 replications with $N/2$ nodes on each piece (computed, floating point), in three cases: ζ at $\theta = \frac{3}{4}$, where nothing is subtracted; ζ at $\theta = \frac{1}{2}$, where the zero $\frac{1}{2} + 84.7354\ldots i$ on the diameter is subtracted; and the Davenport and Heilbronn function at $\theta = \frac{1}{2}$, with its zero $0.8085\ldots + 85.6993\ldots i$ in the half disc, so that the right side includes $G_H(z,\rho) = 1.12940$; no zero of it lies on the diameter. Gauss and Legendre quadrature of the boundary integral agrees with the right side to 4.2×10^{-15}. The sorted trapezoid rule falls with slopes between -2.07 and -2.28, within the N^{-4} mean square bound, its error mostly bias (the bias term of Lemma 14.6.8 is within 8 percent of the measured mean error for $N \ge 4096$); with the change of variable the slopes are between -2.49 and -2.51, consistent with the N^{-5} bound; plain Monte Carlo falls as $N^{-1/2}$. At $N = 32{,}768$ the weighted rules

are between 1.5×10^5 and 1.5×10^6 times more accurate than plain sampling. The change of variable has the steeper slope, but at these N it is not the more accurate rule for ζ and DH at $\theta = \frac{1}{2}$ (whether the two rules cross beyond $N = 32{,}768$ was not computed).

case	target	plain	sorted trapezoid	with Φ
ζ, $\theta = \frac{3}{4}$	0.028322	1.2×10^{-3} (-0.49)	3.4×10^{-9} (-2.07)	2.7×10^{-9} (-2.49)
ζ, $\theta = \frac{1}{2}$	0.048774	2.7×10^{-3} (-0.52)	1.9×10^{-9} (-2.28)	5.1×10^{-9} (-2.49)
DH, $\theta = \frac{1}{2}$	-0.365254	2.1×10^{-3} (-0.50)	7.8×10^{-9} (-2.16)	1.4×10^{-8} (-2.51)

Table 14.5: Root mean square error at $N = 32{,}768$ for the half disc identity of Theorem 14.6.6, $R = 1$, $t_0 = 85.7$, 200 replications, with the slopes fitted for $N \geq 1024$ in brackets (computed, floating point). The target is the right side of the identity; for the Davenport and Heilbronn function (DH) it includes the Green term $G_H(z, \rho)$.

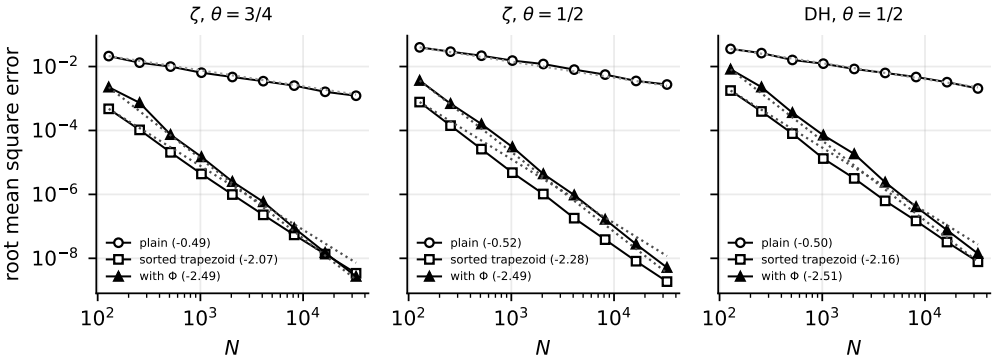


Figure 14.4: Root mean square error against N for the half disc identity (data of Table 14.5). The dotted lines have slopes $-\frac{1}{2}$, -2 and $-\frac{5}{2}$.

So the rates N^{-4} and N^{-5} are reached on the line by an exact local form of the criterion, which still detects zeros off the line through the Green function of the half disc. For the whole line no such rate is proved (Open problem 14.3).

14.6.6 Truncation on the whole line

On the line itself the Cauchy tail beyond a height can be bounded directly. Assume RH. Then

$$\int_T^t \log|\zeta(\tfrac{1}{2} + iu)|\, du = O\Big(\frac{\log t}{(\log\log t)^2}\Big) \qquad (3 \leq T \leq t),$$

the estimate behind Theorem 1.1 of Bui, Lester and Milinovich [16]. Let $A(t)$ be the left side with T replaced by $t_0 + T$ and $w = P_{1/2}(t_0 - \cdot)$. Integration by parts

gives $\int_{t_0+T}^{\infty} \log|\zeta(\frac12+i\tau)|\, w(\tau)\, d\tau = -\int_{t_0+T}^{\infty} A(\tau)w'(\tau)\, d\tau$, since $A(t)w(t) \to 0$, and $w'(\tau) = O_{t_0}(\tau^{-3})$. With the same bound below $t_0 - T$,

$$\int_{|\tau-t_0|>T} \log|\zeta(\tfrac12+i\tau)|\, P_{1/2}(t_0-\tau)\, d\tau = O_{t_0}\Big(\frac{\log T}{T^2(\log\log T)^2}\Big)$$

(conditional on RH). At $t_0 = 0$ this is [16, Theorem 1.1], where it is also shown that the tail is not $O(T^{-2})$. At a level $\theta \in (\frac12, 1)$, under RH, the integral of $\log\zeta$ around the rectangle with vertices $\theta+iT$, $\theta+it$ and $+\infty$ bounds the primitive of $\log|\zeta(\theta+iu)|$ by $O_\theta(\log t)$, and the tail by $O_\theta(T^{-2}\log T)$.

For the rule place $N = 2T/h$ nodes $\tau_j = t_0 - T + h(j+U)$, $0 \le j < N$, with one shift U uniform on $[0,1)$. At $\theta = \frac34$ nothing is deflated. At $\theta = \frac12$ deflate each zero $\rho_k = \frac12 + i\gamma_k$ with $|\gamma_k - t_0| \le T+1$ by

$$u_k(s) = \log\Big|\frac{(s-\rho_k)(s-\rho_k+2)}{(s-\rho_k+1)^2}\Big|,$$

which is harmonic on $\operatorname{Re} s > \frac12$ and $O(|s|^{-2})$; on the line it is $\frac12 \log\frac{x^2(4+x^2)}{(1+x^2)^2}$ with $x = \tau - \gamma_k$, about x^{-2} for large $|x|$, and its integral over $\mathbb{R}$ is 0. The estimate is $h\sum_j\big[(f - \sum_k u_k)P\big](\tau_j) + \sum_k \int_{|\tau-t_0|\le T} u_k P\, d\tau$, the window means of the u_k computed by quadrature. The shifted grid is unbiased for the integral over the window, so the mean error over U is exactly minus the Cauchy tail beyond T, and one shift adds a discretisation error of mean zero. Adding back instead the whole line means $u_k(1+it_0)$ would add the tails $\sum_k \int_{|\tau-t_0|>T} u_k P\, d\tau$, about the value of P at the edge times the sum of the reciprocal distances of the zeros to the edges; for this variant at $\theta = \frac12$ the measured $T^2 \cdot |\text{error}|$ grows from 0.15 at $T = 10$ to 1.24 at $T = 640$, more than 97 percent of the error at $T = 640$ being the deflation tails (computed), so it is not the tail of $\log|\zeta|$.

In Table 14.6 the measured error is the tail, which under RH is $O(T^{-2}\log T)$, that is $O(N^{-2}\log N)$ at a fixed step, and the spread over the shifts is at most 8.1×10^{-7} (at $T = 10$, $\theta = \frac12$) and below 1.0×10^{-8} for $T \ge 160$. The error changes sign with T (between $T = 40$ and 160 and again, at $\theta = \frac34$, between 640 and 1280), a bounded oscillation of $T^2 \cdot |\text{error}|$ consistent with T^{-2} up to logarithms in this range (computed). This is computed, not a theorem: at a fixed step the discretisation error of one shift need not tend to 0 as T grows (besides an endpoint term of order h times the integrand at the edges of the window, it contains an aliasing term, of order $e^{-2\pi b/h}$ for an integrand analytic in a strip of width b, which in general does not decay as T grows), so a rate in N needs bounds on the deflated integrand, or a step that shrinks with N (Open problem 14.3). The tail is bounded under RH, not corrected; only the half disc of Theorem 14.6.6 gives an exact identity at these rates.

With no hypothesis, Bui, Lester and Milinovich [16, Theorem 1.2] prove that

	$\theta = \frac{3}{4}$, $h = 0.02$			$\theta = \frac{1}{2}$, $h = 0.01$, zeros deflated			
T	error	spread	$T^2 \cdot \lvert\text{error}\rvert$	error	spread	$T^2 \cdot \lvert\text{error}\rvert$	zeros
10	-9.47×10^{-4}	5.5×10^{-8}	0.095	-2.09×10^{-3}	8.1×10^{-7}	0.209	10
40	-3.40×10^{-5}	2.2×10^{-8}	0.054	-5.37×10^{-5}	1.3×10^{-8}	0.086	33
160	$+4.36 \times 10^{-6}$	9.8×10^{-9}	0.112	$+1.05 \times 10^{-5}$	9.4×10^{-9}	0.268	123
640	$+6.47 \times 10^{-8}$	8.2×10^{-10}	0.027	$+7.84 \times 10^{-8}$	5.5×10^{-10}	0.032	742
1280	-1.36×10^{-7}	1.8×10^{-10}	0.223				

Table 14.6: Truncated Cauchy average of $\log|\zeta(\theta + i\tau)|$ over $|\tau - t_0| \le T$ at $t_0 = 85.7$, minus $\log|h_\theta(1 + it_0)|$; as in Remark 14.2.2, the zeros above 3×10^{12} move the whole line average away from $\log|h_\theta(1+it_0)|$ by less than 10^{-12}, with no hypothesis. Mean error and spread over 10 random shifts of the grid (computed, floating point). At $\theta = \frac{1}{2}$ the zeros with $|\gamma - t_0| \le T+1$ (the zero range convention, counted in the last column) are deflated and their window means added back (the unbiased rule). The shifted grid is unbiased for the window integral, so the mean error is minus the Cauchy tail beyond T, up to the spread. In this range $T^2 \cdot |\text{error}|$ stays below 0.27; the bounds of [16] allow slow growth.

for $T \ge 3$

$$I(T) = \int_{-T}^{T} \frac{\log|\zeta(\frac{1}{2} + it)|}{\frac{1}{4} + t^2}\, dt = 2\pi \sum_{\substack{|\gamma| \le T \\ \beta > 1/2}} \log\Big|\frac{\rho}{1-\rho}\Big| + O\Big(\frac{\log T}{T^2}\Big) :$$

up to that error the truncated integral is the sum over the zeros below height T, the global counterpart of centring the kernel at the height of interest. With Theorem 14.2.1 the tail is $I(\infty) - I(T) = 2\pi \sum_{|\gamma|>T,\, \beta>1/2} \log|\rho/(1-\rho)| + O(\log T/T^2)$, which is $O(\log T/T)$ with no hypothesis, since each term lies between 0 and $(2\beta - 1)/(2|1-\rho|^2) \le 1/(2\gamma^2)$ and there are $O(T \log T)$ zeros up to height T. Under RH their Omega theorem disproves the conjecture $I(T) = O(T^{-2})$ of Borwein, Bradley and Crandall [13], as stated in [16].

14.6.7 Scanning heights

The local criterion invites a scan over t_0. Chapter 13 provides the particle filter along the Cauchy flow (Section 13.3) and nested sampling over heights (Section 13.9), with the results of Tables 13.6 and 13.14. Once a suspicious height is found, the defect $D_\theta(t_0)$ of Theorem 14.2.1 gives a check, in floating point: centred at a zero at distance δ to the right of the line $\operatorname{Re} s = \theta$ it is at least $\log \frac{1-\theta+\delta}{1-\theta-\delta} \approx \frac{2\delta}{1-\theta}$.

14.7 Two dimensions: the smoothed Riesz identity

The one dimensional rule stalls between $N^{-1/2}$ and $N^{-2/3}$ because the Cauchy kernel has a heavy tail and $\log|\zeta|$ never stops oscillating. Both come from the boundary formulation: Theorem 14.2.1 tests $\log|h_\theta|$ against the Green function of a half plane, whose normal derivative on the line is 2π times the Cauchy density. Testing against a smooth bump of compact support instead gives an identity with no tail at all, at the price of one more dimension.

Theorem 14.7.1 (smoothed Riesz identity). *Let f be meromorphic on a neighbourhood of a compact set $K \subset \mathbb{C}$, not identically zero on any component of the neighbourhood, and let $\varphi \in C_c^\infty$ with support in K. Then*

$$I(\varphi) := \iint \log|f(\sigma+it)|\,\Delta\varphi(\sigma,t)\,d\sigma\,dt = 2\pi\Big(\sum_{f(\rho)=0}\varphi(\rho) - \sum_{f(p)=\infty}\varphi(p)\Big),$$

with zeros and poles counted with multiplicity. Equivalently, for $f = \xi$, $-\iint\big(\pi\, m_\sigma(t)\,\partial_\sigma\varphi + \partial_t\log|\xi|\,\partial_t\varphi\big)\,d\sigma\,dt = 2\pi\sum\varphi(\rho)$, so the Thorin density m_σ is the σ component of the field whose divergence counts the zeros.

Proof. In the sense of distributions $\Delta\log|f| = 2\pi\big(\sum\delta_\rho - \sum\delta_p\big)$, because $\log|f| = \log|g| + \sum\log|s-\rho| - \sum\log|s-p|$ with g analytic and zero free near K, $\log|g|$ harmonic, $\Delta\log|s-\rho| = 2\pi\delta_\rho$ and $\Delta\log|s-p|^{-1} = -2\pi\delta_p$ at a pole. Pair with φ and integrate by parts twice. The second form is one integration by parts, with $\partial_\sigma\log|\xi| = \pi m_\sigma$ (equation (10.1.1) and Theorem 10.1.1); $\nabla\log|\xi|$ is locally integrable since its singularities are of order $1/|s-\rho|$. □

Theorem 14.2.1 corresponds formally to taking for φ the Green function of $\{\operatorname{Re} s > \theta\}$ with pole at $1+it_0$, which is singular at the pole and not compactly supported; integrating out σ leaves the Cauchy average on the line. With a bump instead, both obstacles of the one dimensional rule disappear. If f has no zeros in the support, $g = \log|f|\,\Delta\varphi$ is in C_c^∞ and $I(\varphi) = 0$ exactly; a zero ρ in the support moves I by $2\pi\varphi(\rho)$, of order one unless ρ is near the edge. With a partition of unity $\sum_k\varphi_k = 1$ on a region, $\sum_k I(\varphi_k) = 2\pi$ times the number of zeros in the region, plus $2\pi\sum_k\varphi_k(\rho)$ for each zero ρ where $0 < \sum_k\varphi_k < 1$, so overlapping bumps leave no blind spots. This smoothed argument principle has a smooth, compactly supported integrand when no zero lies in the support (after deflation otherwise), the setting in which weighted Monte Carlo pays.

14.7.1 Weighted Monte Carlo in the plane

Map the box to $[-1,1]^2$ and take $\varphi(x,y) = b(x)b(y)$ with $b(x) = e^{-1/(1-x^2)}$ on $|x| < 1$. The rules we compare all use N nodes:

1. plain Monte Carlo, the sample mean at uniform nodes;

2. Voronoi weights: each uniform node is weighted by the area of its Voronoi cell, the two dimensional form of weighting by spacings;
3. Delaunay weights: piecewise linear interpolation on the Delaunay triangulation of the nodes and the four corners, the two dimensional trapezoid rule; at uniform nodes and at jittered nodes, one uniform point in each cell of a $\sqrt{N} \times \sqrt{N}$ grid;
4. the randomly shifted product trapezoid rule on a $\sqrt{N} \times \sqrt{N}$ grid;
5. deflation: replace $\log|f|$ by $\log|f(s)/(s-\hat{\rho})|$ for an estimate $\hat{\rho}$ of a zero, and add back $2\pi\varphi(\hat{\rho})$, which is exact by Theorem 14.7.1 for meromorphic f.

All of these are unbiased or asymptotically unbiased, and the spread over replications estimates their random error, not their bias.

PROPOSITION 14.7.2 (rates). *After an affine change of variables the box is the torus $[0,1)^2$; the shifted product trapezoid rule with $N = m^2$ nodes is $Q(\delta) = m^{-2}\sum_{j\in\{0,\dots,m-1\}^2} g(j/m+\delta)$, with δ uniform on $[0,1/m)^2$. (i) If g has a smooth periodic extension (for example the integrand $\log|f|\,\Delta\varphi$ with the flat bump of the text and no zero in the box), then $Q(\delta) - \int g = O(N^{-k})$ for every k, uniformly in δ, and $\mathbb{E}_\delta Q = \int g$. (ii) If $g = \psi \log|s-\rho|$ with $\psi \in C_c^\infty((0,1)^2)$ and $\rho \in (0,1)^2$, then $\mathbb{E}_\delta Q = \int g$ and $\left(\mathbb{E}_\delta|Q - \int g|^2\right)^{1/2} = O(N^{-1})$. For a fixed shift whose nodes all lie at distance at least c/m from ρ, $|Q - \int g| = O_c(N^{-1}\log N)$. If $\psi(\rho) \neq 0$, no bound uniform in the shift holds. (iii) On the box, for g as in Theorem 14.7.3 (for example the integrand of the text when f has no zero in the closed box), Voronoi weights and Delaunay weights at independent uniform nodes have root mean square error $O(N^{-3/2})$, against $N^{-1/2}$ for the sample mean (Theorem 14.7.3); at jittered nodes the same rate is computed, not proved.*

Proof. For g square integrable on the torus, Poisson summation gives, in $L^2(d\delta)$, $Q(\delta) - \int g = \sum_{\ell\neq 0} \hat{g}(m\ell)\, e^{2\pi i m\langle \ell,\delta\rangle}$ with $\hat{g}(k) = \int g(x) e^{-2\pi i\langle k,x\rangle} dx$. These exponentials are orthonormal for δ uniform on $[0,1/m)^2$, so $\mathbb{E}_\delta Q = \int g$ and $\mathbb{E}_\delta|Q - \int g|^2 = \sum_{\ell\neq 0} |\hat{g}(m\ell)|^2$. (i) $\hat{g}$ decays faster than every power, and the series converges absolutely, uniformly in δ. (ii) In the sense of distributions $\Delta g = 2\pi\psi(\rho)\delta_\rho + q$ with $q = 2\nabla\psi\cdot\nabla\log|s-\rho| + \Delta\psi\,\log|s-\rho| \in L^1$, so $4\pi^2|k|^2|\hat{g}(k)| \le 2\pi|\psi(\rho)| + \|q\|_1$ and $\mathbb{E}_\delta|Q-\int g|^2 \le Cm^{-4}\sum_{\ell\neq 0}|\ell|^{-4} = O(N^{-2})$. For the separated bound put $h = 1/m$: on a cell at distance $r \ge 2h$ from ρ the trapezoid error is $O(h^4 r^{-2})$, and the $O(r\,dr/h^2)$ cells at distance about r give $O(h^2\log(1/h))$ in total; the $O(1)$ cells within $2h$ of ρ contribute $O(h^2\log(1/h))$ to $\int g$ and, by the separation, $O(h^2\log(1/(ch)))$ to Q. If $\psi(\rho) \neq 0$, a node at distance η from ρ contributes $m^{-2}\psi(\rho)\log\eta + O(m^{-2})$, which is unbounded as $\eta \to 0$, so no bound uniform in δ holds. (iii) is Theorem 14.7.3 with $d = 2$. □

14.7.2 A proof of the $N^{-3/2}$ rate

The rate of Proposition 14.7.2(iii) holds at independent uniform nodes in every dimension, for the Voronoi rule and for the Delaunay rule with the corners of the box added. Write $x_{\mathrm{nn}}(x)$ for the node nearest to x, $d(x) = |x - x_{\mathrm{nn}}(x)|$ and $v(x) = x_{\mathrm{nn}}(x) - x$. Then $\frac{1}{2}d(x)^2 = \frac{1}{2}|x|^2 - \max_i\big(x \cdot X_i - \frac{1}{2}|X_i|^2\big)$ is Lipschitz with gradient $-v(x)$ almost everywhere, so the displacement to the nearest node is a gradient field, and the first order error of the Voronoi rule integrates by parts into a second order quantity.

Theorem 14.7.3 (weighted Monte Carlo for integrands flat at the boundary). *Let $Q = [-1,1]^d$, let $X_1, \dots, X_N$ be independent and uniform on Q, and let $g \in C^3(Q)$ satisfy (a) $g = 0$ within distance $\delta > 0$ of ∂Q, or (b) $g \in C^\infty(Q)$ with every derivative vanishing on ∂Q. Let $V_N = \sum_i |V_i \cap Q|\, g(X_i)$ be the Voronoi rule, with the cells clipped to Q, and $D_N = \int_Q I_N g$ the Delaunay rule, where $I_N g$ interpolates g linearly on the Delaunay triangulation of the nodes and the 2^d corners of Q. Then*

$$V_N - \int_Q g = \frac{1}{2}\int_Q d(x)^2\, \Delta g(x)\, dx + \frac{1}{2}\int_Q v^{\mathsf{T}} D^2 g(x)\, v\, dx + \int_Q r(x)\, dx,$$
$$|r(x)| \le \tfrac{1}{6}\|D^3 g\|_\infty |v|^3,$$

and for $A_N = V_N$ and for $A_N = D_N$

$$\Big|\mathbb{E}A_N - \int_Q g\Big| = \begin{cases} O(Ne^{-cN\delta^d}) & \text{in case (a),} \\ O(N^{-k}) \text{ for every } k & \text{in case (b),} \end{cases} \qquad \operatorname{Var} A_N = O(N^{-1-4/d}).$$

So the root mean square error is $O(N^{-1/2-2/d})$ in every dimension: $N^{-5/2}$ on the line and $N^{-3/2}$ in the plane. The constants depend on g, d and, in case (a), δ. Case (b) includes $g = \log|f|\,\Delta\varphi$ for the bump φ of Section 14.7, mapped to Q, when f is analytic and zero free on a neighbourhood of the closed box.

Proof. Identity. The Voronoi rule is $\int_Q g(x_{\mathrm{nn}}(x))\, dx$, so its error is $\int_Q [g(x+v) - g(x)]\, dx$; expand to third order. The first order term is $\int_Q \nabla g \cdot v = -\int_Q \nabla g \cdot \nabla(\frac{1}{2}d^2) = \frac{1}{2}\int_Q d^2 \Delta g$, with no boundary term because $\nabla g = 0$ on ∂Q in both cases.

Bias of the Voronoi rule. By exchangeability $\mathbb{E}V_N = \int_Q g(y)\, w_N(y)\, dy$ with

$$w_N(y) = \frac{N}{|Q|}\int_Q \Big(1 - \frac{|B(x, |x-y|) \cap Q|}{|Q|}\Big)^{N-1} dx,$$

which is $N/|Q|$ times the expected volume of the cell of a node placed at y. Let $r = \frac{1}{2}\mathrm{dist}(y, \partial Q)$ and $\kappa_d = |B(0,1)|$. For $|x-y| \le r$ the ball $B(x, |x-y|)$ lies in Q and the integrand is $(1 - \kappa_d |x-y|^d/|Q|)^{N-1}$. For $x \in Q$ and $\rho \le 2$,

$|B(x,\rho)\cap Q| \ge 4^{-d}\kappa_d\rho^d$ (for $\rho \le 1$ the ball contains its part in the orthant at x that points to the centre of Q, which lies in Q, and for $1 < \rho \le 2$ use the radius 1), so for $|x-y| > r$ the integrand is at most $e^{-c_1 N r^d}$. Since $\frac{N}{|Q|}\int_{\mathbb{R}^d}(1-\kappa_d|u|^d/|Q|)_+^{N-1}du = N\int_0^1(1-t)^{N-1}dt = 1$ and the two integrands agree for $|x-y| \le r$, $|w_N(y)-1| \le CNe^{-c_1Nr^d}$. Hence $|\mathbb{E}V_N - \int_Q g| \le CN\int_Q |g(y)|\, e^{-c_1N(\operatorname{dist}(y,\partial Q)/2)^d}\,dy$. In case (a) this is $O(Ne^{-c_1N(\delta/2)^d})$. In case (b) Taylor's formula at the nearest boundary point gives $|g(y)| \le C_m \operatorname{dist}(y,\partial Q)^m$ for every m, and the bound is $O(N^{1-(m+1)/d})$ for every m.

Variance of the Voronoi rule. Let R be the largest distance from a node to the points of its clipped cell. If $R > \rho$ for a node y, the convex cell contains a point x with $|x-y| = \rho$, and $B(x,\rho)$ holds no other node; one of a fixed set of C_d points x' on the sphere $|x'-y| = \rho$ has $|x-x'| \le \frac{\rho}{4}$, and then $B(x,\frac{\rho}{2}) \subset B(x',\frac{3}{4}\rho) \subset B(x,\rho)$, so $B(x',\frac{3}{4}\rho)$ holds no other node and, as $x \in Q$, meets Q in volume at least $4^{-d}\kappa_d(\rho/2)^d$. So $P(R > \rho \mid y) \le C_d e^{-cN\rho^d}$ for $\rho \le 4$, uniformly in y, and $\mathbb{E}R^p = O(N^{-p/d})$ for every p. The node nearest to y is at distance at most $2R$, because the midpoint lies in the cell of y; so removing y moves the nearest node of each point of its cell to distance at most $3R$. Write $V_N - \int_Q g = \int_Q \frac{1}{2}d^2\Delta g\,dx + \int_Q \mathcal{R}(x,v(x))\,dx$ with $\mathcal{R}(x,v) = g(x+v) - g(x) - \nabla g(x)\cdot v$, $|\mathcal{R}| \le \frac{1}{2}\|D^2g\|_\infty|v|^2$. Replacing X_1 by an independent copy X_1' changes the integrands only on the old cell of X_1 and the new cell of X_1', where d and $|v|$ are at most $3\max(R,R')$ before and after, R and R' being the two cell radii. So $|V_N - V_N'| \le C(R^{d+2} + R'^{d+2})$, $\mathbb{E}(V_N - V_N')^2 = O(N^{-2-4/d})$, and the Efron and Stein inequality gives $\operatorname{Var} V_N \le \frac{N}{2}\mathbb{E}(V_N - V_N')^2 = O(N^{-1-4/d})$.

The Delaunay rule. Linear interpolation has no first order error, since $\sum_j \lambda_j(x)(y_j - x) = 0$ for the barycentric coordinates of x in its simplex S, and $|I_N g - g| \le \frac{1}{2}\|D^2g\|_\infty \operatorname{diam}(S)^2$ on S, whatever the shape of S. In case (a) identify opposite faces of Q to make a flat torus, on which g extends smoothly. There the Delaunay weights $w_i = \int \lambda_i$ are nonnegative, sum to $|Q|$ and move with translations of the configuration, so $\mathbb{E}(w_i \mid X_i = x) = |Q|/N$ and the torus rule is exactly unbiased. Fix η small compared with δ. Outside an event of probability $O(e^{-cN\eta^d})$ every ball of radius η in a fixed finite net inside Q contains a node. On that event the empty circumball of a simplex that meets the support of g has radius $O(\eta)$, because it contains a ball inside Q of radius comparable to the smaller of its radius and δ; so the simplices that carry g, and the weights of the nodes in the support, are the same for the box with its corners and for the torus. Both rules are bounded by $|Q|\,\|g\|_\infty$, so the box rule has bias $O(e^{-cN\delta^d})$. Inserting a node changes the triangulation only in the union of the simplices whose circumballs contain it; on the same event the part of it that carries g lies within a radius R of the node with $P(R > \rho) \le Ce^{-cN\rho^d}$, by the net argument above, and by the value bound the rule changes by $O(R^{d+2})$; the Efron and Stein inequality gives $\operatorname{Var} D_N = O(N^{-1-4/d})$ as before. In case

(b) write $g = g\chi + g(1-\chi)$ with a cutoff χ equal to 1 at distance at least $2\delta_N$ from ∂Q and to 0 within δ_N, $\delta_N = N^{-1/(2d)}$, whose derivatives of order j are $O(\delta_N^{-j})$. The second part changes both the rule and the integral by at most $|Q| \sup |g(1-\chi)| = O(N^{-k})$ for every k, since the weights are nonnegative and sum to $|Q|$. The first part satisfies (a) with $\delta = \delta_N$, so its bias is $O(e^{-cN^{1/2}})$, and its second derivatives are bounded uniformly in N, because the derivatives of g within $2\delta_N$ of ∂Q are $O(\delta_N^m)$ for every m; so its variance is $O(N^{-1-4/d})$. □

Lemma 14.6.8 gives the rate of Yakowitz, Krimmel and Szidarovszky its exact form: their mean square N^{-4} is the squared bias $\frac{(b-a)^4(g'(b)-g'(a))^2}{4N^4}$ to leading order, and when $g'(a) = g'(b)$ the mean square error is $O(N^{-5})$ (for $g \in C^4$ by the lemma, and for g as in Theorem 14.7.3 with $d = 1$ by that theorem). The integrand of Section 14.7 satisfies (b) when f has no zero in the closed box, so for ζ in Figure 14.6 the Voronoi and Delaunay rules at independent uniform nodes have root mean square error $O(N^{-3/2})$ (proved; the fitted slopes are -1.43 and -1.40), and at jittered nodes the slope -1.54 is computed. With a zero in the box the integrand is not C^3, and the slopes between -1.0 and -1.3 are outside the theorem.

Remark 14.7.4. If g is smooth on Q but does not vanish near ∂Q, the integration by parts leaves the boundary term $-\frac{1}{2}\oint_{\partial Q} d^2\, \partial_n g$, and $\int_Q \Delta g = \oint_{\partial Q} \partial_n g$ need not vanish, so the bias of these rules is in general of order $N^{-2/d}$ (affine g is an exception); the exponential rows of Table 14.7 show it (computed).

integrand and rule	$N = 1000$	largest N	slope	bound
bump, $d = 1$, Voronoi	1.2×10^{-6}	3.5×10^{-14} (1,024,000)	-2.51	$-\frac{5}{2}$
e^x, $d = 1$, trapezoid	4.8×10^{-6}	4.5×10^{-12} (1,024,000)	-2.00	-2
bump, $d = 2$, Voronoi	1.4×10^{-4}	5.1×10^{-9} (1,024,000)	-1.46	$-\frac{3}{2}$
bump, $d = 2$, Delaunay	1.1×10^{-4}	4.2×10^{-9} (1,024,000)	-1.42	$-\frac{3}{2}$
$e^{x+y/2}$, $d = 2$, Delaunay	1.0×10^{-2}	1.2×10^{-5} (1,024,000)	-0.98	-1

Table 14.7: Root mean square error of the weighted rules at independent uniform nodes on $Q = [-1, 1]^d$, 200 replications in one dimension and 30 in two (about 5 percent noise in each root mean square error in one dimension and about 13 percent in two), the largest N in brackets, slopes fitted for $N \geq 4000$ (computed, floating point). The bump $\prod_i b(x_i/0.9)$ vanishes near ∂Q (case (a) of Theorem 14.7.3); the exponentials do not, and their error is the bias of Remark 14.7.4. The slopes are consistent with the last column, the exponent of Theorem 14.7.3, of Lemma 14.6.8 for the trapezoid row and of the Remark for the last row.

14.7.3 Results

The box is $0.6 \le \sigma \le 1.4$, $83.7 \le t \le 87.7$, centred on the Davenport and Heilbronn zero at height 85.70 (Figure 14.5). For ζ there are no zeros in the box, so $I(\varphi) = 0$. For Davenport and Heilbronn the box contains $\rho = 0.808517182457 + 85.699348485378\,i$, so $I(\varphi) = 2\pi\varphi(\rho) = 0.63165631$. Both functions are evaluated by a vectorised Euler and Maclaurin formula, accurate to about 10^{-12}, so a million evaluations take seconds. Table 14.8 and Figure 14.6 give root mean square errors over 20 replications (computed, floating point; the relative standard error of an entry is 16%).

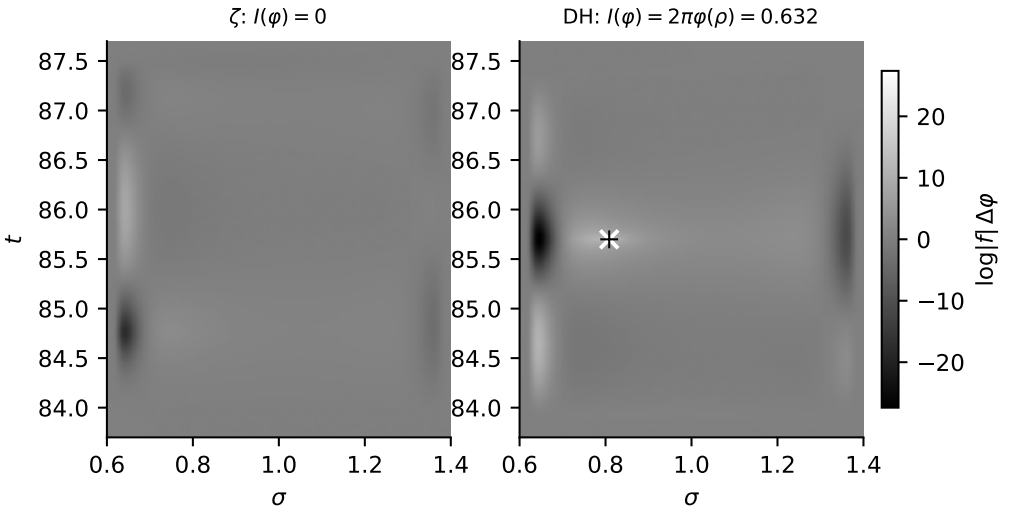


Figure 14.5: The integrand $\log|f|\,\Delta\varphi$ on the box (121×121 points, one grey scale for both functions). The cross marks the Davenport and Heilbronn zero. The large values are at the left and right edges of the box, where $\Delta\varphi$ is largest and $\log|f|$ is large; the picture does not show the zero, which is why the two look alike. Their integrals are 0 and 0.632.

The fitted slopes for $N \ge 4096$ are consistent with Proposition 14.7.2. For ζ, with no zero in the box, they are -0.51 for plain Monte Carlo, -1.43 for Voronoi weights, -1.40 for Delaunay weights and -1.54 for Delaunay at jittered nodes (the observed slopes lie between -1.4 and -1.55, so $N^{-3/2}$ is an upper estimate). The shifted grid falls faster than any power, to 2.2×10^{-14} at $N = 262{,}144$, the accuracy of the function values. So the spacings rule gains a factor N over plain sampling (proved at independent uniform nodes, Theorem 14.7.3; computed at jittered nodes) and the grid is limited only by arithmetic. This answers, in two dimensions, the question raised at the start of Section 14.6.5: the change that makes the integrand smooth is not a change of variable on the line but a move off

	N	plain	Voronoi	Delaunay	jittered	grid	deflated
ζ	1024	1.2×10^{-1}	4.3×10^{-2}	4.5×10^{-2}	2.0×10^{-2}	3.0×10^{-3}	
ζ	16384	3.9×10^{-2}	1.7×10^{-3}	1.5×10^{-3}	4.4×10^{-4}	1.9×10^{-6}	
ζ	262144	9.0×10^{-3}	3.2×10^{-5}	2.4×10^{-5}	5.6×10^{-6}	2.2×10^{-14}	
DH	1024	2.3×10^{-1}	5.8×10^{-2}	5.3×10^{-2}	3.3×10^{-2}	1.5×10^{-2}	2.7×10^{-2}
DH	16384	6.7×10^{-2}	3.0×10^{-3}	2.5×10^{-3}	1.0×10^{-3}	6.9×10^{-4}	1.5×10^{-5}
DH	262144	1.2×10^{-2}	9.5×10^{-5}	7.4×10^{-5}	6.1×10^{-5}	4.8×10^{-5}	3.5×10^{-6}

Table 14.8: Root mean square error of $I(\varphi)$ (computed, floating point, 20 replicates; the relative standard error of an entry is 16%). Jittered is Delaunay on jittered nodes; deflated grid uses an estimate of the zero that is off by 1.4×10^{-4}, along $(1+i)/\sqrt{2}$.

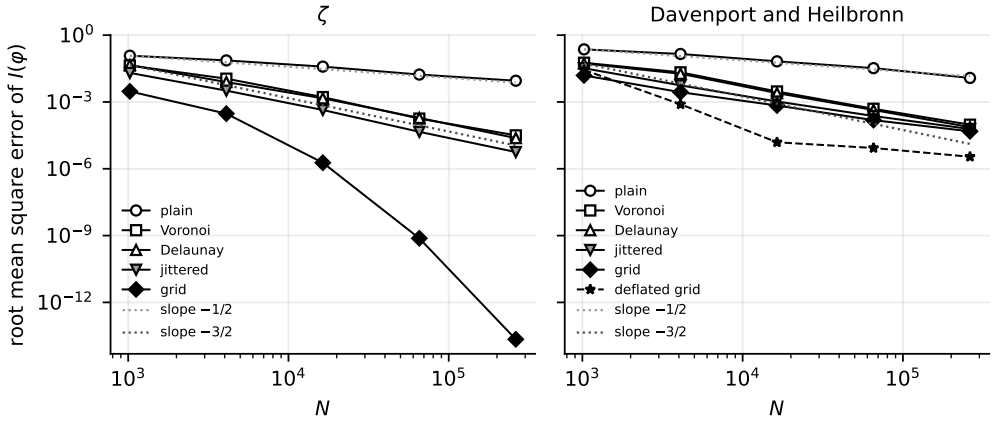


Figure 14.6: Root mean square error of $I(\varphi)$ against N in two dimensions, 20 replicates, for ζ (left) and Davenport and Heilbronn (right), with the plain, Voronoi, Delaunay, jittered, grid and (Davenport and Heilbronn) deflated grid rules, the deflating zero being off by 1.4×10^{-4}. The dotted lines have slopes $-\frac{1}{2}$ and $-\frac{3}{2}$.

it.

With a zero in the box the logarithmic singularity takes over at large N. All the weighted rules slow to slopes between -1.0 and -1.3; for the grid this is the rate of Proposition 14.7.2(ii). Deflation with the exact zero restores the smooth rate. With an estimate at distance ε the residual keeps two logarithmic singularities, and Proposition 14.7.5 gives two regimes: while the grid does not separate them ($\varepsilon\sqrt{N} \lesssim 1$) the error is at most of order $\varepsilon N^{-1/2}$ up to a logarithm, so at fixed N it is proportional to ε (about $0.03\,\varepsilon$ at $N = 262{,}144$ for $\varepsilon \leq 10^{-4}$ in Table 14.9, and the ratio is 0.014 at $N = 2^{20}$, so it is the bound $\varepsilon N^{-1/2}$ that holds and not a fixed constant, where $\hat{\rho} = \rho + \varepsilon(1+i)/\sqrt{2}$); once $\varepsilon\sqrt{N} \gtrsim 1$ it falls like N^{-1}. There is no floor. With the zero refined by Newton's method the error is of order

5×10^{-14} at $N = 262{,}144$ (Table 14.9, $\varepsilon = 0$). Deflated Delaunay at jittered nodes keeps the slope -1.52. The singularity is integrable and mild: untreated, uniform nodes already reach 10^{-4} at $N = 262{,}144$ (heuristic as a reason; concentrating nodes near the zero was not tested).

PROPOSITION 14.7.5 (approximate deflation). *Let $\psi \in C_c^\infty((0,1)^2)$, $\rho, \hat\rho \in (0,1)^2$ with $0 < \varepsilon = |\hat\rho - \rho| \le \frac12$, and $r = \psi\,(\log|s-\rho| - \log|s-\hat\rho|)$. The shifted product trapezoid rule of Proposition 14.7.2 with N nodes is unbiased for $\int r$, and its root mean square error over the shift is at most*

$$C\Big(\varepsilon N^{-1/2}\big(1+\log^+\tfrac{1}{\varepsilon\sqrt{N}}\big)^{1/2} + \varepsilon\log\tfrac{2}{\varepsilon}\,N^{-1}\Big) \quad \text{if } \varepsilon\sqrt{N}\le 1,$$
$$C\,N^{-1} \quad \text{if } \varepsilon\sqrt{N}\ge 1,$$

with C depending only on ψ. So for every fixed ε the error tends to zero: approximate deflation leaves no floor.

Proof. As in Proposition 14.7.2, the rule is unbiased and $\mathbb{E}_\delta|Q - \int r|^2 = \sum_{\ell\neq0}|\hat r(m\ell)|^2$, $m = \sqrt{N}$. In the sense of distributions $\Delta r = 2\pi\big(\psi(\rho)\delta_\rho - \psi(\hat\rho)\delta_{\hat\rho}\big) + q_\varepsilon$, where $q_\varepsilon = 2\nabla\psi\cdot\nabla\ell_\varepsilon + \Delta\psi\,\ell_\varepsilon$ and $\ell_\varepsilon = \log|s-\rho| - \log|s-\hat\rho|$. Since $|\nabla\ell_\varepsilon| = \varepsilon/(|s-\rho||s-\hat\rho|)$, and $|\ell_\varepsilon| \le 2\varepsilon/|s-\rho|$ for $|s-\rho|\ge 2\varepsilon$, $\|q_\varepsilon\|_1 \le C\varepsilon\log(2/\varepsilon)$. With $|\psi(\rho)e^{-2\pi i\langle k,\rho\rangle} - \psi(\hat\rho)e^{-2\pi i\langle k,\hat\rho\rangle}| \le |\psi(\rho)|\min(2, 2\pi|k|\varepsilon) + \|\nabla\psi\|_\infty\varepsilon$,

$$|\hat r(k)| \le C\,\frac{\min(1,|k|\varepsilon) + \varepsilon\log(2/\varepsilon)}{|k|^2}.$$

If $m\varepsilon \ge 1$ the sum is $O(m^{-4})$. If $m\varepsilon < 1$, split at $|\ell| = 1/(m\varepsilon)$: $\sum_{1\le|\ell|\le1/(m\varepsilon)}\varepsilon^2m^{-2}|\ell|^{-2} \le C\varepsilon^2m^{-2}(1+\log\frac{1}{m\varepsilon})$ and $\sum_{|\ell|>1/(m\varepsilon)}m^{-4}|\ell|^{-4} \le C\varepsilon^2m^{-2}$, while the second part of the bound on $\hat r$ contributes $C\varepsilon^2\log^2(2/\varepsilon)\,m^{-4}$. Take square roots. □

With the partition of unity, the smoothed identity is an argument principle count with C^∞ weights, so a box free of zeros is indicated, in floating point and without a certificate, to near machine precision with a few hundred thousand evaluations; Turing's method confirms that rigorously and far higher up. The gain is on the Monte Carlo side: the one dimensional Balazard, Saias and Yor average is a hard integral, while its localised Riesz form in two dimensions is smooth away from the zeros, and the Voronoi and Delaunay rules reach $N^{-3/2}$, proved at independent uniform nodes (Theorem 14.7.3) and computed at jittered nodes. Whether this extends to node sets produced by the particle filter or by nested sampling, which are dependent, is open (Open problem 14.4).

14.7.4 Counting with the horizontal logarithmic derivative

THEOREM 14.7.6 (zeros from the horizontal logarithmic derivative). *Let $f \not\equiv 0$ be meromorphic on a connected neighbourhood of $\{\operatorname{Re} s \ge \sigma_0\}$ and equal, for $\operatorname{Re} s$*

ε	$N = 1024$	4096	16,384	65,536	262,144	1,048,576
0	3.2×10^{-2}	1.0×10^{-3}	8.1×10^{-6}	4.0×10^{-9}	5.1×10^{-14}	1.2×10^{-15}
10^{-8}	3.2×10^{-2}	1.0×10^{-3}	8.1×10^{-6}	4.1×10^{-9}	3.1×10^{-10}	1.4×10^{-10}
10^{-6}	3.2×10^{-2}	1.0×10^{-3}	8.0×10^{-6}	5.8×10^{-8}	3.1×10^{-8}	1.4×10^{-8}
10^{-4}	3.2×10^{-2}	1.0×10^{-3}	1.1×10^{-5}	5.7×10^{-6}	3.0×10^{-6}	1.4×10^{-6}
10^{-3}	3.3×10^{-2}	9.3×10^{-4}	1.5×10^{-4}	4.4×10^{-5}	2.1×10^{-5}	1.0×10^{-5}
10^{-2}	3.4×10^{-2}	2.0×10^{-3}	9.6×10^{-4}	2.9×10^{-4}	3.0×10^{-5}	1.5×10^{-5}
none	1.7×10^{-2}	2.9×10^{-3}	8.2×10^{-4}	1.8×10^{-4}	4.2×10^{-5}	7.2×10^{-6}

Table 14.9: Deflated shifted grid for Davenport and Heilbronn, root mean square error over 10 shifts, as a function of the error ε in the deflating zero (computed, floating point; the relative standard error of an entry is 22%). The last row is the grid without deflation. A synthetic test of the two regimes of Proposition 14.7.5 shows the error falling to $N = 2^{22}$ for every $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, with slope near -0.6 while $\varepsilon\sqrt{N} \le 1$ and near -1.1 afterwards; the constant depends on the bump and the box: at $\varepsilon = 10^{-4}$, $N = 262{,}144$ the ratio of the error to ε is 0.030 here and 0.0050 in the synthetic test, a factor 6.

large, to an absolutely convergent Dirichlet series $1 + \sum_{n \ge 2} a_n n^{-s}$*, such as* ζ *or the Davenport and Heilbronn function. Let* $d_f(u,t) = \pi^{-1} \operatorname{Re}(f'/f)(u+it)$*, let* $\varphi \in C_c^\infty(\{\operatorname{Re} s > \sigma_0\})$*, and put* $\Phi(u,t) = \int_{-\infty}^{u} \varphi(\sigma,t)\, d\sigma$ *and* $K_\varphi = \partial_u \varphi + \partial_t^2 \Phi$*. Then*

$$\sum_{f(\rho)=0} \varphi(\rho) - \sum_{f(p)=\infty} \varphi(p) = -\frac{1}{2} \int_{-\infty}^{\infty} \int_{\sigma_0}^{\infty} d_f(u,t)\, K_\varphi(u,t)\, du\, dt,$$

zeros and poles counted with multiplicity, the double integral absolutely convergent. K_φ *is smooth; to the right of the support of* φ *it equals* $\partial_t^2 \Phi(\infty, t)$ *and does not depend on* u*.*

Proof. Theorem 14.7.1 for meromorphic f gives $2\pi J = \iint \log|f|\, \Delta\varphi$, with J the left side. Choose b with $\sum |a_n| \log n\, n^{-b} < \infty$. For $u \ge b+1$ large enough that $|f-1| \le \frac{1}{2}$, $|f'/f| \le C n_0^{-u}$ uniformly in t, where n_0 is the least $n \ge 2$ with $a_n \ne 0$ (if there is none, $f \equiv 1$ and both sides vanish). For almost every t the horizontal ray meets no zero or pole, and since $f(u+it) \to 1$, $\log|f(\sigma+it)| = -\pi \int_\sigma^\infty d_f(u,t)\, du$. Substitute and exchange the integrals: d_f is locally integrable (near a zero or pole a of order q it is $q \operatorname{Re}[(s-a)^{-1}]/\pi + O(1)$), it decays exponentially in u uniformly in t, and K_φ is bounded and vanishes for t outside the projection of the support. Finally $\int_{-\infty}^{u} \Delta\varphi(\sigma,t)\, d\sigma = \partial_u \varphi(u,t) + \partial_t^2 \Phi(u,t)$. □

Remark 14.7.7. (i) For $f = \zeta$, $d_\zeta = m_\sigma - \pi^{-1} \operatorname{Re}\big[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2})\big]$, where m_σ is the Thorin density of Section 14.1; so the Thorin density gives the count after this known term is subtracted, and m_σ itself cannot be substituted ($\log|\xi(u+it)|$ does not tend to 0). (ii) Local form: for every $\varphi \in C_c^\infty$,

$\iint m_\sigma(t)\,\Delta\varphi(\sigma,t)\,d\sigma\,dt = -2\sum_\rho \partial_\sigma\varphi(\rho)$, which uses m_σ only on the support of φ; the weights $-2\partial_\sigma\varphi$ integrate to zero along every horizontal line, so a region is counted with signed weights in its collar (for $\varphi = -\frac{1}{2}(\sigma - c)\chi$ with $\chi = 1$ near a region R, zeros in R have weight one and zeros where $0 < \chi < 1$ have weight $\chi + (\operatorname{Re}\rho - c)\partial_\sigma\chi$). (iii) Cutoff: if U lies to the right of the support and $A(t) = \Phi(\infty, t)$, then $J + \frac{1}{2}\int\int_{\sigma_0}^{U} d_f K_\varphi\,du\,dt = \frac{1}{2\pi}\int A''(t)\log|f(U+it)|\,dt$. (iv) Deflation by $R(s) = (s - \hat\rho + 1)/(s - \hat\rho)$: $J_f = J_{fR} + \varphi(\hat\rho) - \varphi(\hat\rho - 1)$.

For (ii): $m_\sigma = \pi^{-1}\partial_\sigma \log|\xi|$ is locally integrable and $\Delta m_\sigma = \pi^{-1}\partial_\sigma\Delta\log|\xi| = 2\sum_\rho \partial_\sigma\delta_\rho$ in the sense of distributions. For (iii): $K_\varphi = A''$ for $u \ge U$ and the integral of d_f from U to ∞ is $-\pi^{-1}\log|f(U+it)|$. For (iv): add the divisors of f and R; the formula of the theorem holds for fR, since its proof uses only $\log|fR(u+it)| \to 0$ and the absolute convergence of the double integral, and $|R'/R| = |(s-\hat\rho)(s-\hat\rho+1)|^{-1} = O(u^{-2})$ uniformly in t.

So the zero count is a linear functional of the horizontal component d_f alone, on the half strip to the right of the support (the integrand $d_f K_\varphi$ is not smooth at the zeros). On the box of Section 14.7, with Gauss and Legendre nodes in u and the periodic trapezoid rule in t, the functional with the tail of Remark (iii) added returns 0 for ζ to rounding error, below 10^{-14} (whose pole at 1 lies outside the box). For Davenport and Heilbronn it converges slowly and irregularly to $\varphi(\rho) = 0.1005312239$, with errors between 10^{-5} and 10^{-4} at 128 to 1024 nodes in t (-8.6×10^{-6} at 512, a coincidence of the grid), since the integrand has a singularity of order $1/|s-\rho|$. Deflated by $R(s) = (s-\hat\rho+1)/(s-\hat\rho)$ with $\hat\rho = \rho$, it agrees with $\varphi(\rho)$ to 4×10^{-16}. The factor R tends to 1 only like $1/u$, so its tail beyond the last node, $-\pi^{-1}\log|R(u_{\max}+it)|$, is added exactly; the zero of R at $\hat\rho - 1$ lies outside the support, so only $\varphi(\hat\rho)$ is added back. With an error ε in the deflating zero the result moves by 8×10^{-10} at $\varepsilon = 10^{-6}$ and 10^{-7} at $\varepsilon = 10^{-4}$ (512 nodes in t). These are floating point values with f'/f evaluated by the Euler and Maclaurin formula (computed); with the tail of R alone, and not the tail of f, the result needs $u_{\max}$ near 40 for an error of 10^{-14}.

14.8 Limits

Every statement of Sections 14.1 to 14.5 uses only Hadamard's product, the functional equation and order one growth, with two exceptions: the positivity of $\mathfrak{U}_\sigma$ for $\sigma \ge 1$ (Section 14.1 and Corollary 14.3.3(iv)) uses $\beta < 1$, the non-vanishing of ζ on $\operatorname{Re} s = 1$, which comes from the Euler product; and Theorem 14.5.2 assumes QRH(θ). The other statements hold for the completed Davenport and Heilbronn function [60], which has zeros off the line [261, 17], with three changes: $h_\theta = f$, since f is entire; the archimedean term in (14.3.1) becomes $\pi^{-1}\operatorname{Re}[\frac{1}{2}\log\frac{5}{\pi} + \frac{1}{2}\psi(\frac{s+1}{2})]$; and every sum over zeros with $\beta > \theta$ includes the zeros with $\beta \ge 1$. So the dictionary, the EGGC form and the Brownian reading are reformulations; none of them can prove RH.

The arithmetic enters only through the Euler product. In the Thorin language it appears as an upper bound on positive combinations of m_σ at several heights, for example $3m_\sigma(0) + 4m_\sigma(t) + m_\sigma(2t)$ for $\sigma > 1$ (Proposition 12.1.4), which the Davenport and Heilbronn function, having no Euler product, need not satisfy. Such inequalities give zero free regions of width $c/\log t$.

Remark 14.8.1 (the explicit formula at a Cauchy smoothed level). Let $F \geq 0$ have a Fourier transform supported in $[-L, L]$, Lipschitz, with derivative of bounded variation, as for the class $\mathcal{B}$ of Section 9.1.1; let $F_c = F(\cdot - c)$, and let $\frac{1}{2} < \sigma < 1$ not be the real part of a zero (at such a level take right limits in σ). The explicit formula of Theorem 9.1.1, applied to the even part of $P_{\sigma-1/2} * F_c$, has on its prime side only the prime powers $n \leq e^L$, besides the archimedean terms. On its zero side a zero with $\beta < \sigma$ contributes $\int F_c\, P_{\sigma-\beta}(\cdot - \gamma)$, its term in $\int F_c\, d\mathfrak{U}_\sigma$ by (14.1.1); so if no zero has $\beta > \sigma$ the formula gives $\int F_c\, d\mathfrak{U}_\sigma \geq 0$, which is the content of Theorem 9.1.2 for $F = |\hat{h}|^2$. A zero with $\beta > \sigma$ contributes instead the continuation in β of that term, which exceeds its signed term $-\int F_c\, P_{\beta-\sigma}(\cdot - \gamma)$ in $\int F_c\, d\mathfrak{U}_\sigma$ by $F_c(\gamma + i(\beta - \sigma)) + F_c(\gamma - i(\beta - \sigma))$, by Fourier inversion at complex arguments as in the proof of Theorem 9.1.1; these corrections need not be positive. So the primes compute the continued Weil functional, which equals the Thorin integral $\int F_c\, d\mathfrak{U}_\sigma$ in general only when no zero lies to the right of σ; positivity of $\mathfrak{U}_\sigma$ is not a finite computation with the primes.

The Monte Carlo side can show, in floating point, that the defect is zero to a given accuracy in a given window, but a zero at height γ moves the global average by at most $2(1-\theta)(\beta-\theta)/\gamma^2$.

14.9 Open problems

Problem 14.1. Find an implementable sampler for P_w with w close to $|f - \mathbb{E}_P f|$, or another weight, whose variance per unit of computing cost (pilot, burn in, autocorrelation and evaluations of f included) is below that of plain Monte Carlo for the defect $D_\theta(t_0)$, or show that the Metropolis chains of Section 14.6 cannot reach it. For independent draws no weight gains more than the factor $\mathrm{Var}_P f/(\mathbb{E}_P|f - \mathbb{E}_P f|)^2$ of Remark 14.6.2, which is between 1.35 and 2.46 for the integrands of Table 14.3 (computed); a Markov chain is not bound by it.

Problem 14.2. Section 14.7 replaces the negative Thorin mass $\iint m_\sigma(t)^-\, d\sigma\, dt$, a nonlinear functional with a kink along the curve $m_\sigma = 0$, by the linear functional $I(\varphi)$, and Theorem 14.7.6 writes the count through the horizontal logarithmic derivative alone. Near an undeflated zero m_σ^- is integrable but not square integrable, so independent uniform nodes give infinite variance there. Determine the root mean square rate of the Voronoi and Delaunay rules for $\iint_R m_\sigma^-$ on a bounded region R free of zeros, or after exact deflation of the zeros in R. Theorem 14.7.3 does not apply to this nonsmooth functional.

PROBLEM 14.3. On a half disc the sorted trapezoid rule reaches the mean square rates N^{-4} and N^{-5} for an exact local form of the criterion (Theorem 14.6.6 and Lemma 14.6.8). For the whole line: prove a rate in N for the truncated rule of Section 14.6.6 under RH, with a step that shrinks with N and with the discretisation and deflation errors bounded; find a correction for the tail beyond T that is finite, efficiently computable and certified, as the half disc gives for its arc. Are the empirical slopes -0.60 and -0.66 of the sorted trapezoid rule for ζ (Table 14.4) a preasymptotic effect of the zeros, or the true rate?

PROBLEM 14.4. Theorem 14.7.3 proves the rate $N^{-1/2-2/d}$ at independent uniform nodes. Extend it to dependent node sets, such as those of the particle filter along the Cauchy flow (Section 13.3) or of nested sampling, under explicit coverage and mixing hypotheses on the nodes (some hypothesis is needed: nodes that all coincide give one evaluation); for deterministic quasi uniform meshes the rate can be N^{-1}. For a box containing an undeflated zero, where plain Monte Carlo keeps the rate $N^{-1/2}$, is the root mean square error of the Voronoi and Delaunay rules at independent uniform nodes of order N^{-1} up to logarithms, as the slopes between -1.0 and -1.3 of Section 14.7 suggest?

PROBLEM 14.5. Fix $\theta \in [\frac{1}{2}, 1)$ and let Z_{t_0} be the boundary value of the horizontal branch of $\log h_\theta$ under $P_{1-\theta}(t_0 - \cdot)$. If $\mathbb{E}[Z_{t_0}^2] = (\mathbb{E}Z_{t_0})^2$ for every t_0, does $\mathrm{QRH}(\theta)$ follow? Proposition 14.5.4 reduces it to the identity $2\sum_\gamma \int_{C_\gamma} M_\gamma(L_- + \pi i M_\gamma - z_0)K\,ds = D^2$ for every t_0, with K the cut kernel there. Under RH, is $\operatorname{Var} \log|\zeta(\frac{1}{2} + i\tau)| = \frac{1}{2}\log\log t_0 + O(1)$ under $P_{1/2}(t_0 - \cdot)$, for every t_0 or for t_0 outside a set of density zero; and does $(Z_{t_0} - \mathbb{E}Z_{t_0})/(\frac{1}{2}\log\log t_0)^{1/2}$ converge in law to the standard normal law of $\mathbb{R}^2$ when t_0 is uniform on $[T, 2T]$?

PROBLEM 14.6. Is there a linear combination of $L(s, \chi)$ over primitive characters χ of one modulus, with a functional equation and zeros off the line, whose trace K is negative somewhere? With imprimitive characters allowed, Remark 14.4.5 gives one. If the lowest zero $\frac{1}{2} + i\gamma_1$ of such a function is on the line, the argument of the proof of Theorem 14.4.3, with $\pi/3$ replaced by $\pi/(2\delta_{\max})$, where $\delta_{\max}$ is the largest $|\beta - \frac{1}{2}|$, gives a positive trace whenever $\sum e^{-(\gamma-\gamma_1)\pi/(2\delta_{\max})} < 1$, the sum over the upper zeros off the line with multiplicity. So a negative trace needs the opposite inequality, and if there are finitely many, M, upper zeros off the line, one of them lies within $\frac{2\delta_{\max}}{\pi}\log M$ of γ_1.

Part III

Weil positivity in windows

Chapter 15

Weil positivity and Yoshida's Hermitian forms: certificates and barriers

Weil's criterion states RH as the positivity of a functional built from the explicit formula. On Gaussian test functions and their derivatives in the heat time this functional is the heat trace $W(t)$ of the zeros, and positivity becomes the Thorin condition of the reciprocal Ξ law: RH holds if and only if W is completely monotone. The explicit formula writes $W = \mathcal{A} - \mathcal{P}$ for every $t > 0$, with an archimedean part $\mathcal{A}$ and a prime part $\mathcal{P}$, and both have an extreme value form: $\mathcal{A}$ is u times the Lévy measure of a Gumbel type variable of order $\frac{1}{4}$, and $\mathcal{P}$ is a positive weighted series of Fréchet distribution functions, one for each prime power. The chapter reads the explicit formula in that language, maps the three regimes of the Gaussian family, recovers the zeros from the primes by Prony's method, and collects what can be proved and what can only be computed about windows of positivity, certificates from finitely many primes, rigidity of the primes, smoothing of the prime error, energy, Hankel positivity and a one heat time form of RH, saying where each route stops.

The chapter is based on [149]. Its numbers were recomputed from the mathematics alone, without the code of the source. Every number printed below is one of those recomputations unless it is marked unchecked, and where the recomputation differed from the source the recomputed value is printed. Statements of papers that were not read are marked unchecked.

Notation. The heat trace of the zeros is the $W(t)$ of Section 2.1; the $\Pi + \mathcal{A}$ of Table 8.4 is written $\mathcal{A}(t)$ here, and $\mathcal{P}(t)$ is the prime part. The Gaussian kernel is $g_t(u) = (4\pi t)^{-1/2} e^{-u^2/4t}$. Chebyshev's function is $\psi_{\mathrm{C}}(y) = \sum_{n \le y} \Lambda(n)$, and $\psi = \Gamma'/\Gamma$. The normalised prime error is $D(u) = (\psi_{\mathrm{C}}(e^u) - e^u)e^{-u/2}$. The shifted functional of Section 15.10 is $V(t, a)$. Zeros are indexed as in Section 2.1 (one zero of each pair, with multiplicity), with local symbols: $\rho_j = \beta_j + i\tau_j$ with $\tau_j > 0$ (the γ_k of Section 2.1); $\gamma_j = \tau_j + i(\frac{1}{2} - \beta_j)$ is complex, so that $\rho_j = \frac{1}{2} + i\gamma_j$ and γ_j is real exactly when ρ_j is on the critical line; and $\lambda_j = \gamma_j^2$ is the $a_k = -(\rho_k - \frac{1}{2})^2$ of the book. A sum over positive real γ is written only under RH. The first window

root is r_* with $\lambda_* = r_*^2$.

Results already in the book are cited, not restated: the heat trace, its Frullani and Weil forms and the equivalence of RH with complete monotonicity (Section 2.1, Proposition 2.1.1, Equation (2.1.5)), the digits of Tables 8.4 and 2.1, the theorems that $W > 0$ and $W' < 0$ for every $t > 0$ (Theorem 8.1.1), that W is completely monotone on $(0, 1.29]$ with no information on the zeros below 0.019 (Theorem 8.1.9) and that $(-1)^k W^{(k)} > 0$ for all $t > 0$ and $k \le 1.16 \times 10^{25}$ (Theorem 8.1.13), the tent version of the sine squared lemma (Theorem 15.18.2), the Gumbel kernels of fractional order (Section 19.17) and the Prony extraction of the atoms (Section 5.2.2). The results of the chapter are these.

1. RH holds if and only if W is completely monotone, if and only if the Thorin measure $\sum_j \delta_{\lambda_j}$ is supported on $(0, \infty)$, if and only if $\mathcal{A} - \mathcal{P}$ is completely monotone (Theorem 15.2.1, proved). For the converse, the Laplace transform of W is meromorphic with a nonzero residue at every $-\lambda_j$, and a nonreal λ_j cannot be the pole of a Stieltjes transform.

2. The archimedean density is u times the Lévy density of $-\frac{1}{2}\log G_{1/4}$, and the prime part is a weighted series of Fréchet distribution functions with scales $(\log n)^2/4$ (Section 15.3, proved).

3. W is computed to 110 digits from Γ and the primes; Prony's method on 16 such samples gives $\gamma_1, \gamma_2, \gamma_3$ to more than 30 digits and the eighth zero to 4 digits (Table 15.4, computed).

4. The archimedean part $\mathcal{A}$ has a spectral density with exactly one sign change, at $\lambda_* = 39.562 \approx 4\pi^2$, where r_* is the critical point of the Riemann Siegel theta function. The first sign change t_n of $(-1)^n \mathcal{A}^{(n)}$ satisfies $t_n = n/\lambda_* + \beta + \delta/n + O(n^{-2})$ with explicit β and δ (Propositions 15.8.2 and 15.9.1, proved; the table is computed). Complete monotonicity of order n of $\mathcal{A}$ holds only on $(0, t_1]$.

5. Each prime power added extends the order one window to about $\log q/2\gamma_1$ (computed, a heuristic balance). A one sided certificate from the primes up to N_0 is proved (Proposition 15.9.2); the windows tabulated from it are computed in floating point on a grid, not certified.

6. The variance of ψ_{C} in short intervals follows the GUE prediction and not the Poisson level (computed, with a heuristic prediction). RH is equivalent to finite discounted energy of D (Proposition 15.9.4, proved); the mean square constant needs simple zeros.

7. The first two cumulants of the reciprocal Ξ law are read off the primes, and RH is equivalent to the positive semidefiniteness of two Hankel families of the normalised cumulants (Section 15.9.5, proved).

8. RH holds if and only if $V(t,a) \geq 0$ for every $a \geq 0$ at one fixed heat time t (Theorem 15.10.1, proved by Landau's theorem for nonnegative functions).

The open problems close the chapter.

15.1 Weil's explicit formula and the heat trace

Write the nontrivial zeros as $\rho = \frac{1}{2} + i\gamma$, with the indexing of the introduction. For an even test function h, holomorphic in a closed strip wider than $|\operatorname{Im} r| \leq \frac{1}{2}$ and decaying like $O((1+|\operatorname{Re} r|)^{-2-\eta})$ there, with $g(u) = \frac{1}{2\pi}\int h(r)e^{-iru}\,dr$, the Guinand and Weil explicit formula [91, 273], [177, Theorem 5.12] is

$$\sum_{\rho} h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\,\operatorname{Re}\psi\big(\tfrac{1}{4}+\tfrac{ir}{2}\big)\,dr - 2\sum_{n\geq 1}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n), \quad (15.1.1)$$

the sum over all nontrivial zeros with multiplicity. The terms $h(\pm\frac{i}{2})$ come from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s=1$ and $s=0$, which the factor $s(s-1)$ of ξ removes. Weil's criterion [273, 29] states that RH holds if and only if the right side is nonnegative for every even part of a test function $F(r)\overline{F(\bar r)}$ in the admissible class. Under RH this is clear, since then $h(\gamma) = |F(\gamma)|^2$ for real γ; the content is the converse. For small support the positivity is known: Yoshida proved it for support in a short interval [278], and Connes and Consani showed that the archimedean place controls the sign for small support [44]. These results (Chapters 15.12 and 16) use no information on the zeros.

Gaussian test functions

Take $h_t(r) = e^{-tr^2}$ with $t > 0$. It is entire and decays faster than any power on every strip, so (15.1.1) holds for every $t > 0$, with $g_t(u) = (4\pi t)^{-1/2}e^{-u^2/4t}$. Pairing ρ with $1-\rho$ (the zeros with $\tau > 0$ are closed under $\rho \mapsto 1-\bar\rho$) gives $\sum_\rho h_t(\gamma_\rho) = 2W(t)$ with

$$W(t) = \sum_j e^{-\lambda_j t}, \qquad \operatorname{Re}\lambda_j = \tau_j^2 - (\tfrac{1}{2}-\beta_j)^2 \geq \tau_j^2 - \tfrac{1}{4}.$$

This is the unconditional heat trace of Section 2.1; the zero counting bound $N(T) = O(T\log T)$ gives absolute convergence for every $t > 0$ and after any number of differentiations, and W is real because the nonreal γ_j come with their

conjugates. Then (15.1.1) becomes $W(t) = \mathcal{A}(t) - \mathcal{P}(t)$, where

$$\mathcal{A}(t) = e^{t/4} - \tfrac{1}{2} g_t(0) \log \pi + \frac{1}{4\pi} \int_{\mathbb{R}} e^{-tr^2} \operatorname{Re} \psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\, dr, \tag{15.1.2}$$

$$\mathcal{P}(t) = \sum_{n \geq 2} \Lambda(n)\, n^{-1/2}\, g_t(\log n). \tag{15.1.3}$$

This is the identity (2.1.5) (with the polar term included in $\mathcal{A}$), whose digits are in Tables 2.1 and 8.4; Table 15.1 adds a few more. At 110 digits $\mathcal{A} - \mathcal{P}$ equals the sum over the first 1700 zeros to 10^{-70} or better at each t of the table (computed), reproducing the printed digits of the source.

t	$W(t)$ from the zeros	$\mathcal{A}(t) - \mathcal{P}(t)$ from Γ and the primes
0.005	0.5371919640910408957	0.5371919640910408957
0.02	0.018541291731319482254	0.018541291731319482254
0.05	$4.5878350485735742685 \times 10^{-5}$	$4.5878350485735742685 \times 10^{-5}$
0.1	$2.1047997958223512845 \times 10^{-9}$	$2.1047997958223512845 \times 10^{-9}$
0.2	$4.4301821802236448843 \times 10^{-18}$	$4.4301821802236448843 \times 10^{-18}$

Table 15.1: The explicit formula for the heat trace, agreeing to all printed digits (computed).

15.2 Weil positivity is the Thorin condition

Put $H(s) = \xi(\frac{1}{2} + \sqrt{s})/\xi(\frac{1}{2})$, which is entire in s by the functional equation. By Hadamard's factorisation (order $\frac{1}{2}$), $H(s) = \prod_j (1 + s/\lambda_j)$ with $\sum_j |\lambda_j|^{-1} < \infty$, and the candidate $L = 1/H$ has the form

$$-\log L(s) = \sum_j \log\Big(1 + \frac{s}{\lambda_j}\Big) = \int_0^\infty (1 - e^{-sx})\, \frac{W(x)}{x}\, dx,$$

by Frullani's integral $\log(1 + s/\lambda) = \int_0^\infty (1 - e^{-sx}) e^{-\lambda x}\, dx/x$ for $\operatorname{Re}\lambda > 0$ (the unconditional statement is [235, Theorem 4.3], quoted in Section 2.1). Under RH every $\lambda_j = \tau_j^2$ is positive, $L(s) = \mathbb{E} e^{-sT}$ for the clock variable $T = \sum_j E_j/\lambda_j$ of Section 2.1 (the reciprocal Ξ law, with E_j independent standard exponentials), its Lévy measure is $W(x)\, dx/x$ and its Thorin measure is $U = \sum_j \delta_{\lambda_j}$, with the multiplicities. Without RH only the analytic identities hold, and $W(x)/x$ is the only possible Lévy density (Theorem 8.1.1 proves $W > 0$ and $W' < 0$ unconditionally).

Theorem 15.2.1. *The following are equivalent.*

(i) RH.

(ii) Every λ_j lies in $(0, \infty)$, that is, the Thorin measure $U = \sum_j \delta_{\lambda_j}$ is supported on $(0, \infty)$ and T is a generalised gamma convolution.

(iii) W *is completely monotone on* $(0,\infty)$*:* $(-1)^n W^{(n)}(t) \geq 0$ *for all* $n \geq 0$ *and* $t > 0$*.*

(iv) $\mathcal{A} - \mathcal{P}$ *is completely monotone on* $(0,\infty)$*, with* $\mathcal{A}$ *and* $\mathcal{P}$ *as in* (15.1.2) *and* (15.1.3)*.*

Proof. (iii) and (iv) are the same statement, since $W = \mathcal{A} - \mathcal{P}$ for every $t > 0$. Under RH every $\lambda_j = \tau_j^2 > 0$, which gives (ii), and then $W = \int e^{-\lambda t} U(d\lambda)$ is completely monotone (termwise differentiation is justified by the convergence bound above), which gives (iii). For (iii) implies (i) define $S(s) = \sum_j (s + \lambda_j)^{-1}$. The series converges normally away from its poles, and for real $s > \frac{1}{4}$ Fubini applies because $\sum_j (s + \operatorname{Re}\lambda_j)^{-1} < \infty$, so $S(s) = \int_0^\infty e^{-sx} W(x)\, dx$. If W is completely monotone, Bernstein's theorem gives $W(x) = \int_{[0,\infty)} e^{-ux} \mu(du)$ with $\mu \geq 0$, and Tonelli gives $S(s) = \int (s+u)^{-1} \mu(du)$ for $s > \frac{1}{4}$. Finiteness at one such s gives $\int (1+u)^{-1} \mu(du) < \infty$, so the right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. It agrees with the meromorphic series on a real interval, hence on the connected set $\mathbb{C} \setminus (-\infty, 0]$ minus the poles of S. A pole $-\lambda_j$ outside $(-\infty, 0]$ would be removable, but its residue is the positive integer multiplicity of λ_j. So every $\lambda_j \in [0, \infty)$. Finally $\operatorname{Im}\lambda_j = 2\tau_j(\frac{1}{2} - \beta_j)$ and $\tau_j > 0$, so $\beta_j = \frac{1}{2}$ and $\lambda_j = \tau_j^2 > 0$. This gives (i); (ii) implies (i) by the same last step. □

This is Proposition 2.1.1 with the converse spelled out by the pole argument for the resolvent, in place of the uniqueness of exponents of a complex exponential series asserted in the source. Statement (iv) is Weil positivity restricted to the cone generated by the Gaussian test functions and their t derivatives, $h(r) = r^{2n} e^{-tr^2}$ for all $n \geq 0$ and $t > 0$ (these are $|r^n e^{-tr^2/2}|^2$ on the real line and equal $(-\partial_t)^n$ of h_t). Positivity of W alone is weaker, and is proved without RH (Theorem 8.1.1). The prime side converges for every $t > 0$, because $g_t(\log n)$ decays like $e^{-(\log n)^2/4t}$.

Remark 15.2.2 (The Stieltjes form and the parabola). The Laplace transform of W is the meromorphic function $S(s) = \sum_j (\lambda_j + s)^{-1} = H'(s)/H(s)$, a Stieltjes function if and only if RH holds. Its Weil kernel $h_s(r) = (r^2 + s)^{-1} = |r + i\sqrt{s}|^{-2}$ is admissible only when the pole $-i\sqrt{s}$ lies outside the strip, that is $\operatorname{Re}\sqrt{s} > \frac{1}{2}$, which is also where the prime sum $\sum \Lambda(n) n^{-1/2-\sqrt{s}}$ converges. In the s plane this is the outside of the parabola $\operatorname{Re}\sqrt{s} = \frac{1}{2}$. A pole $-\gamma^2$ of S with $\gamma = \gamma_r + i\beta$ lies inside the parabola, because $\operatorname{Re}(-\gamma^2) - \frac{1}{4} + (\operatorname{Im}\gamma^2)^2 = -(1 - 4\beta^2)(\gamma_r^2 + \frac{1}{4}) < 0$ for $|\beta| < \frac{1}{2}$, and for $\beta = 0$ the quantity is $-(\gamma_r^2 + \frac{1}{4}) < 0$ as well. So all the poles of S, violations or not, sit where the Euler product diverges. The Gaussian form of Theorem 15.2.1 avoids this by using test functions that are admissible everywhere.

15.3 The archimedean place is a Gumbel Lévy measure

Put $k(u) = e^{-u/2}/(1 - e^{-2u}) = \sum_{m \geq 0} e^{-(2m+\frac{1}{2})u}$. The digamma integral in (15.1.2) has an absolutely convergent form. Start from $\psi(z) = -\gamma_E + \int_0^\infty (e^{-s} - e^{-zs})(1 -$

$e^{-s})^{-1}\,ds$ with $z=\frac14+\frac{ir}{2}$ and $s=2u$, which gives

$$\frac{1}{4\pi}\int e^{-tr^2}\,\mathrm{Re}\,\psi\big(\tfrac14+\tfrac{ir}{2}\big)dr=-\frac{\gamma_E}{2}g_t(0)+\int_0^\infty\frac{g_t(0)e^{-2u}-e^{-u/2}g_t(u)}{1-e^{-2u}}\,du. \quad (15.3.1)$$

Splitting off $g_t(0)e^{-u/2}$ from the numerator and using $\int_0^\infty(e^{-u/2}-e^{-2u})(1-e^{-2u})^{-1}du=\frac12(\psi(1)-\psi(\frac14))$, the archimedean part is

$$\mathcal{A}(t)=e^{t/4}+\tfrac12\big[\psi(\tfrac14)-\log\pi\big]g_t(0)+\int_0^\infty\big[g_t(0)-g_t(u)\big]k(u)\,du. \quad (15.3.2)$$

The compensation $g_t(0)-g_t(u)=O(u^2)$ is essential: since $k(u)\sim 1/2u$ at the origin, $\int g_t(u)k(u)\,du$ and $\int g_t(0)k(u)\,du$ separately diverge, so the archimedean place cannot be set beside the primes as two finite integrals against g_t. The identity (15.3.1) was verified numerically to 10^{-71} in the source (unchecked here). The prime part is $\mathcal{P}(t)=\int_0^\infty e^{-u/2}g_t(u)\,d\psi_{\mathrm{C}}(e^u)$ with the atom at $u=0$ left out, and both measures have an infinitely divisible meaning.

- The density $k(u)$ is u times the Lévy density $e^{-u/2}/\big(u(1-e^{-2u})\big)$ of $-\frac12\log G_{1/4}$, where $G_{1/4}$ has the Gamma$(\frac14)$ law. More generally $-\frac12\log G_a$ has Lévy density $e^{-2au}/\big(u(1-e^{-2u})\big)$ on $u>0$, with compensation (the density behaves like $1/2u^2$ at the origin), by taking two derivatives of $\log\Gamma(a-i\theta/2)-\log\Gamma(a)$. The variable $-\log G_k$ is the kth largest point of a Poisson process with intensity $e^{-v}\,dv$ for an integer k (put $z=e^{-v}$ to get the arrival times of a unit rate process); for a fractional order a it is the log gamma extension of that family, the Gumbel kernel of fractional order of Section 19.17, not an ordinary ranked point. It is a real valued infinitely divisible variable, not a subordinator.

- The prime measure $\Lambda(n)n^{-1/2}\delta_{\log n}$ is u times the atomic measure $\sum_n\frac{\Lambda(n)}{\log n}n^{-\sigma}\delta_{\log n}$ at $\sigma=\frac12$. For $\sigma>1$ this is the Lévy measure $\sum_{p,m\ge1}p^{-m\sigma}\delta_{m\log p}/m$ of $\log N_\sigma$, with $\mathbb{P}(N_\sigma=n)=n^{-\sigma}/\zeta(\sigma)$, which is compound Poisson with total mass $\log\zeta(\sigma)$. At $\sigma=\frac12$ the total mass beyond any point is infinite ($\sum_p p^{-1/2}=\infty$), so the measure is locally finite and arithmetic but it is not a Lévy measure and the law of N_σ does not exist. The Gaussian test function is what makes the pairing $\int u\,g_t(u)\,\nu_{1/2}(du)=\mathcal{P}(t)$ finite.

So Weil positivity in Gaussian form asks that the compensated archimedean functional (15.3.2) minus the prime pairing be completely monotone in t; it is not a difference of two probability Lévy measures.

Fréchet form of the prime part

As a function of t, each prime power enters through

$$\sqrt{4\pi t}\,g_t(\log n)=\exp\Big(-\frac{(\log n)^2}{4t}\Big),$$

the Fréchet distribution function of index one with scale $(\log n)^2/4$, evaluated at $4t/(\log n)^2$ through $\Phi_1(y) = e^{-1/y}$. So

$$\mathcal{P}(t) = \frac{1}{\sqrt{4\pi t}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, \Phi_1\Big(\frac{4t}{(\log n)^2}\Big),$$

a positive weighted series of Fréchet distribution functions whose scales are the squared logarithms of the prime powers. The weights $\Lambda(n)n^{-1/2}$ are not summable ($\sum_{n\leq N} \Lambda(n)n^{-1/2} \sim 2\sqrt{N}$), so this is not a probability mixture, and a finite truncation is one only after division by its total weight. The smallest scale, $(\log 2)^2/4 = 0.120$, belongs to the prime 2, and for $t \downarrow 0$

$$\mathcal{P}(t) = \frac{\log 2}{\sqrt{8\pi t}} \exp\Big[-\frac{(\log 2)^2}{4t}\Big]\Big\{1 + O\Big(\exp\Big[-\frac{(\log 3)^2 - (\log 2)^2}{4t}\Big]\Big)\Big\},$$

proved by isolating $n = 2$ and factoring out the exponential of the term $n = 3$: the prime part is governed by one extreme atom as $t \downarrow 0$.

15.4 Three regimes

Table 15.2 shows $\mathcal{A}$, $\mathcal{P}$ and W at 110 digits (computed; a run at 130 digits differs by at most 2×10^{-110} in W at all 23 values of t tried). Here $\mathcal{A}$ is the sum of the columns Π and $\mathcal{A}$ of Table 8.4.

t	$\mathcal{A}(t)$	$\mathcal{P}(t)$	$W(t)$	digits cancelled	share of the prime 2 in $\mathcal{P}$
0.002	1.783	2.6×10^{-26}	1.783	0	1.0000
0.005	0.537	7.2×10^{-11}	0.537	0	1.0000
0.01	0.1497	8.4×10^{-6}	0.1497	0	1.0000
0.02	0.02095	0.00241	0.01854	0.05	0.9999
0.05	0.05796	0.05791	4.59×10^{-5}	3.1	0.966
0.1	0.16281	0.16281	2.10×10^{-9}	7.9	0.808
0.2	0.30152	0.30152	4.43×10^{-18}	16.8	0.562
0.3	0.39603	0.39603	9.32×10^{-27}	25.6	0.427

Table 15.2: Archimedean part, prime part and heat trace (computed). Digits cancelled is $\log_{10}(\mathcal{A}/W)$, which differs from the measure of the caption of Table 8.4.

1. Archimedean dominance, $t \leq 0.01$. Here $\mathcal{P}/W \leq 6 \times 10^{-5}$ (computed: 5.6×10^{-5} at $t = 0.01$), and all of $\mathcal{P}$ comes from the prime 2, through its Fréchet factor $e^{-(\log 2)^2/4t}$. Positivity of W is inherited from the archimedean part, and so are the derivative conditions of the orders up to 6 tested on the grid of t (unchecked as a computation). The book proves more: every order $k \geq 1$ for $t \leq 0.019$ with no information on the zeros (Theorem 8.1.9(a)), and all orders up to 1.16×10^{25} for every $t > 0$ (Theorem 8.1.13). A small $\mathcal{P}/W$

controls values, not derivatives, since the derivatives of $\mathcal{P}$ are not dominated by those of $\mathcal{A}$ uniformly in the order. This regime is the Gaussian counterpart of small support positivity: a Gaussian of variance $2t$ sees essentially nothing beyond $\log 2$.

2. Crossover, $t \approx 0.02$ to 0.05. The prime part becomes comparable to W (at $t = 0.05$ the ratio $\mathcal{P}/W$ is about 1260) but is still almost entirely the prime 2, so positivity compares one extreme atom with the archimedean density.

3. Cancellation, $t \gtrsim 0.05$. As $t \to \infty$ both $\mathcal{A}$ and $\mathcal{P}$ grow like $e^{t/4}$; on the range of the table $\mathcal{A}$ rises only from 0.02 to 0.40. While $W \approx e^{-\gamma_1^2 t}$ (for large t, if λ_1 is simple and separated), the two parts agree to $\log_{10}(\mathcal{A}/W) \approx \gamma_1^2 t/\ln 10 + \log_{10}\mathcal{A} \approx 87t$ digits ($\gamma_1^2/\ln 10 = 86.767$), and the prime 2's share falls as the bulk of the primes enters. The information about the zeros lives in this exponentially small difference, whose evaluation is very ill conditioned; this is a statement about conditioning, not a theorem that the information is confined to one interval of t.

15.4.1 The archimedean positivity window

How far does the archimedean part alone satisfy the Bernstein condition? Combining the archimedean terms of (15.1.2),

$$\mathcal{A}(t) = e^{t/4} + \int_0^\infty e^{-\lambda t} w(\lambda)\, d\lambda, \qquad w(\lambda) = \frac{\operatorname{Re}\psi(\frac{1}{4} + \frac{i}{2}\sqrt{\lambda}) - \log\pi}{4\pi\sqrt{\lambda}} \tag{15.4.1}$$

(substitute $\lambda = r^2$). So $\mathcal{A}$ is the Laplace transform of a signed measure: an atom of weight one at $\lambda = -\frac{1}{4}$, coming from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, and a density w that is negative for $\lambda < \lambda_* = r_*^2$ and positive above (Lemma 15.8.1). Here r_* solves $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) = \log\pi$. Since $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) - \log\pi = 2\vartheta'(r)$ for the Riemann Siegel theta function ϑ (Section 8.1), r_* is the critical point of ϑ (its minimum on $r > 0$). Numerically $r_* = 6.2898359888$ and $\lambda_* = 39.5620$ (computed; $r_* \approx 2\pi$ because $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) \approx \log(r/2)$). Only the digamma piece is a Gumbel Lévy density; the atom and the term in $\log\pi$ are not.

The nth derivative weights w by $\lambda^n e^{-\lambda t}$, which peaks at $\lambda = n/t$. So $(-1)^n\mathcal{A}^{(n)}(t) \geq 0$ should hold while n/t stays above λ_*, that is for $t \lesssim n/\lambda_*$ (heuristic, made precise in Proposition 15.8.2). Table 15.3 gives the first sign change t_n of $(-1)^n\mathcal{A}^{(n)}$, found by bisection at 35 and 45 digits (computed; the two runs agree). For $n \leq 6$ it is within 1 percent of n/λ_*.

The relative deviation decreases from $n = 2$ on; in absolute terms $t_n - n/\lambda_*$ is $0.55, 4.86, 3.05, 2.27, 1.77, 1.44$ in units of 10^{-4}. The table tests the single inequality of order n. Since $t_1 < t_2 < \cdots < t_6$, the conditions of all orders $k \leq n$ hold together only on $(0, t_1]$, so $\mathcal{A}$ is completely monotone of order n only up to t_1 and not up to t_n. For the full heat trace all six orders are positive at every t tested from

order n	first sign change t_n of $(-1)^n\mathcal{A}^{(n)}$	n/λ_*	relative deviation
1	0.025332	0.025277	0.22%
2	0.051040	0.050554	0.96%
3	0.076135	0.075830	0.40%
4	0.101334	0.101107	0.22%
5	0.126561	0.126384	0.14%
6	0.151805	0.151661	0.095%

Table 15.3: The first sign change t_n of $(-1)^n\mathcal{A}^{(n)}$ against n/λ_*, $\lambda_* = r_*^2 = 39.5620$ (computed).

0.002 to 0.2 (unchecked as a computation), as Theorem 8.1.13 proves for all t. The binding order is $n = 1$: the archimedean part alone is decreasing only up to $t_1 = 0.02533 \approx 1/(4\pi^2) = 0.025330$, and beyond that point W decreases only because of the prime part, which at t_1 is still essentially the prime 2 alone, through $e^{-(\log 2)^2/4t_1} = 0.0087$ (computed, 0.0087253).

Under RH this gives a spectral division of labour. The archimedean place supplies the positive spectral density above $\lambda_* \approx 4\pi^2$ and two defects: negative spectral mass below λ_* and the atom at $-\frac{1}{4}$. Subtracting the Laplace representations of $\mathcal{A}$ and $W = \int e^{-\lambda t}U(d\lambda)$ gives $\mathcal{P}(t) = \int e^{-\lambda t}\,[\delta_{-1/4} + w(\lambda)\,d\lambda - U(d\lambda)]$, so the prime part is not itself a positive Thorin measure; it cancels the defects and leaves the atoms of U at λ_j, all of which lie above λ_* since $\gamma_1^2 = 199.79$ (computed). Inside $t < t_n$ the single condition $(-1)^n\mathcal{A}^{(n)} \geq 0$ holds without the primes, which is a Gaussian analogue of the small support theorems.

15.5 Zeros from primes by Prony's method

In the cancellation regime $W(t) = \sum_j e^{-\lambda_j t}$ is a sum of decaying exponentials, and Prony's method applies as in Section 5.2.2 (Tables 19.5 and 19.8). Here W comes from (15.1.2) and (15.1.3), from the gamma function and the primes, with no evaluation of ζ and no zeros. We computed W at 110 digits at $t = 0.02 + 0.006j$, $j = 0, \ldots, 15$, and fitted eight exponentials. The rule that terms with $\log n > \sqrt{4t \cdot 110 \ln 10}$ are below the working precision keeps the prime powers up to 2767 at $t = 0.062$ (403 primes, 438 prime powers) but up to 38 436 at the largest sample $t = 0.11$. Cutting at $n \leq 2767$ at every sample leaves an absolute error 1.2×10^{-62} at $t = 0.11$ (against $W = 2.85 \times 10^{-10}$), and the fitted exponents $k \geq 4$ are unchanged by it, while the errors of $\gamma_1, \gamma_2, \gamma_3$ are then 4×10^{-52}, 5×10^{-43} and 2×10^{-34} (computed). So the recovery of Table 15.4 is valid from the 403 primes, and the 110 digit evaluation of W itself needs the larger cutoff at the larger t.

That the prime side of the explicit formula carries the zeros is classical [91, 273]. The zeros are the exponential rates of W, omitted prime atoms bias the fit, and the number of digits needed to see the kth zero grows like $(\gamma_k^2 - \gamma_1^2)t/\ln 10$ (heuristic,

k	$\widehat{\gamma}_k$ from primes	γ_k	error
1	14.13472514173469379	14.13472514173469379	1.3×10^{-52}
2	21.022039638771554993	21.022039638771554993	1.6×10^{-41}
3	25.010857580145688763	25.010857580145688763	2.9×10^{-34}
4	30.42487612585951321	30.42487612585951321	7.2×10^{-24}
5	32.935061587739189691	32.935061587739189691	3.0×10^{-19}
6	37.58617815884	37.58617815883	1.5×10^{-11}
7	40.9187208	40.9187190	1.8×10^{-6}
8	43.3279	43.3271	7.9×10^{-4}

Table 15.4: Zeros recovered by Prony's method from the heat trace computed from Γ and the primes (computed, 110 digits; the source printed errors of 10^{-30} for $k \le 3$, which was the floor of its reference, and the digits shown are those of the source).

as in Section 5.2.2); this does not count the cost of evaluating W itself.

15.6 The prime point process: first and second order

On the log scale, $u = \log y$, the Λ weighted prime powers form a point process with counting function $\psi_{\mathrm{C}}(e^u)$. By the prime number theorem its intensity is e^u, which is the intensity e^{-v} of the Gumbel point process with $v = -u$. In this heuristic reading the prime side is, to first order, the reflected Gumbel process, and the weight $e^{-u/2}$ in (15.1.1) is the square root of that intensity, the normalisation under which a Poisson process with intensity e^u has fluctuations of order one. This is an analogy for the cumulative counts and not an identification of the processes.

RH lives in the second order. By von Koch's theorem [203], RH holds if and only if $\psi_{\mathrm{C}}(y) = y + O(y^{1/2+\epsilon})$ for every $\epsilon > 0$, that is $(\psi_{\mathrm{C}}(y) - y)/y = O(y^{-1/2+\epsilon})$. The error changes sign infinitely often (it is $\Omega_{\pm}(\sqrt{y} \log\log\log y)$ by Littlewood), so this is a growth bound and not an index of regular variation. We compare the primes with the Cramér random model, which keeps n independently with probability $1/\log n$.

- For $10^3 < y \le 10^7$ the normalised fluctuation $|\psi_{\mathrm{C}}(y) - y|/\sqrt{y}$ of the primes is at most 0.777 at integers (attained at $n = 1423$) and 0.803 as a supremum over real y (computed, sieve to 2.1×10^7). The book quotes the explicit bound $0.94\sqrt{y}$ in Section 12.1.7.
- For the Cramér model the same maximum over 20 realisations lies between 5.7 and 12.6, with median 7.8 (computed); the source's single draw 11.1 is within this range.

The primes are far more rigid than a Poisson process with the same intensity at this scale. The size of $\psi_{\mathrm{C}}(y) - y$ at scale y is set under RH by the sum $\sum 1/|\rho|$

over the low zeros and by Cramér's mean square theorem (Proposition 15.9.4), not by repulsion between pairs of zeros; pair correlation governs the variance in short intervals, treated in Section 15.8.3. Calling the rigidity the prime side counterpart of the repulsion of the zeros is a heuristic.

15.7 Does the extreme value approach offer insight?

It offers four things, each checkable.

1. A common language for the two sides of the explicit formula. The archimedean place is u times the Lévy density of $-\frac{1}{2}\log G_{1/4}$, a log gamma variable of fractional order in the family of Section 19.17; the primes are the atomic measure $\sum \Lambda(n)n^{-1/2}\delta_{\log n}$, which is u times the formal Lévy measure of the zeta distribution on the critical line; and Weil positivity in Gaussian form is complete monotonicity of the compensated difference.

2. A map of where positivity is easy and where it is hard. For $t \leq 0.01$ it is inherited from the archimedean part, with the prime 2 a negligible extreme atom. For $t \gtrsim 0.05$ it is a statement about an exponentially small difference between two large quantities. This is consistent with the small support theorems of Yoshida and of Connes and Consani; with the verified zeros, the orders $n \leq 1.16 \times 10^{25}$ are known at every $t > 0$ (Theorems 8.1.1 and 8.1.13). The boundary of the archimedean part alone is computable: its nth derivative condition first fails at $t_n \approx n/\lambda_*$, fixed by where its spectral density changes sign.

3. A computational method. Prony's method on the heat trace computed from the primes recovers the zeros, with the precision cost made explicit (Section 15.5).

4. A heuristic point process reading of RH. To first order the primes on the log scale look like the reflected Gumbel process; RH is a second order statement that their fluctuations stay at square root scale; and numerically they are more rigid than a Poisson process.

It does not offer the missing estimate. In the cancellation regime positivity needs uniform control of $\psi_{\mathrm{C}}(y) - y$ at square root scale, which is RH in von Koch's form (heuristic): the extreme value view states the obstruction precisely but does not remove it. The two places where it could still help are the spectrum of complex indices of the second order term of ψ_{C} (Section 15.8.3) and the rigidity of the primes, which suggests looking for a structural reason, from the Euler product, why the prime process is less variable than Poisson.

15.8 Windows, primes and rigidity

15.8.1 The archimedean window: sign structure and asymptotics

By (15.4.1), $D_n(t) := (-1)^n \mathcal{A}^{(n)}(t) = (-\frac{1}{4})^n e^{t/4} + I_n(t)$ with $I_n(t) = \int_0^\infty \lambda^n e^{-\lambda t} w(\lambda)\, d\lambda$.

Lemma 15.8.1. *w has exactly one sign change on $(0,\infty)$, from negative to positive, at $\lambda_* = r_*^2$, $r_* = 6.28984$, and $w'(\lambda_*) > 0$.*

Proof. With $z = \frac{1}{4} + \frac{ir}{2}$, $\frac{d}{dr}\operatorname{Re}\psi(z) = -\frac{1}{2}\operatorname{Im}\psi'(z)$ and $\psi'(z) = \sum_{k\ge 0}(z+k)^{-2}$ give $\operatorname{Im}\psi'(z) = -\sum_k (\frac{1}{4}+k)\, r\, |z+k|^{-4} < 0$ for $r > 0$. So $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2})$ increases strictly in r, from $\psi(\frac{1}{4}) = -4.227$ to $+\infty$, and crosses $\log\pi$ exactly once, with positive slope (this is the monotonicity of $m = 2\vartheta'$ of Section 8.1). The sign of w is that of the numerator. □

Proposition 15.8.2 (Sign structure). *For every $n \ge 1$, $D_n(t) > 0$ for small $t > 0$. For odd n, D_n has exactly one sign change on $(0,\infty)$, at a point t_n, and is negative after it. For even n, D_n has at most two sign changes, and is positive for large t; if it has two, the first is t_n and the second lies where the atom $4^{-n}e^{t/4}$ overtakes the negative part of the density. As $n \to \infty$ a first sign change t_n exists, and $t_n = n/\lambda_* + O(1)$.*

Proof. The measure $\lambda^n \mu(d\lambda)$, $\mu = \delta_{-1/4} + w(\lambda)\, d\lambda$, read along increasing λ, has the sign pattern $(-1)^n$ at $\lambda = -\frac{1}{4}$, then $-$ on $(0,\lambda_*)$ by Lemma 15.8.1, then $+$. That is one sign change for odd n and two for even n. The Laplace kernel $e^{-\lambda t}$ is totally positive, so D_n has no more sign changes in t than the measure has in λ (Karlin [105, Chapter 5]). As $t \to 0$ the density $w(\lambda) \sim (\log\lambda)/(8\pi\sqrt{\lambda})$ at infinity makes $I_n(t) \to +\infty$, so $D_n > 0$. As $t \to \infty$, $I_n(t) = O(t^{-n-1/2})$ (the density behaves like $c_0\lambda^{-1/2}$ at zero) while the atom is $e^{t/4}4^{-n}$ in size, so for odd n the atom is negative and $D_n < 0$ for large t, which gives exactly one sign change, and for even n the atom is positive and $D_n > 0$ for large t, which gives zero or two changes. For the asymptotics, I_n has exactly one zero: write $I_n = B_n - C_n$ with B_n the integral over (λ_*,∞) and C_n the absolute value of the integral over $(0,\lambda_*)$; the ratio B_n/C_n has logarithmic derivative in t equal to minus the difference of two mean values of λ, the first larger than the second, so it decreases strictly from $+\infty$ to 0. The location of this zero is computed in Proposition 15.9.1: it is $n/\lambda_* + O(1)$, and the atom, of relative size $O(\sqrt{n}b^n)$ with $b < 1$ near $t = n/\lambda_*$, does not move it by more than an exponentially small amount. So for large n, D_n changes sign near n/λ_*, and this is the first change. □

So a first sign change for even n is proved for large n and computed for $n = 2, 4, 6$. A scan of D_n on $[0.005, 7]$ at 35 digits (computed) finds exactly one sign change for $n = 1$ and $n = 3$, at 0.025332 and 0.076135 (the entries of Table 15.3), and for $n = 2, 4, 6$ a second change at 1.3612, 3.2456 and 5.2669 (computed), as

the proposition allows. In Table 15.3 the offset $t_n - n/\lambda_*$ is below 5×10^{-4} for $n \le 6$, so the $O(1)$ term is small in practice.

15.8.2 Prime by prime: each extreme atom buys a window

Let $\mathcal{P}_S$ be the prime part restricted to the prime powers of a finite set of primes S (all powers of its primes included), and let $t_n(S)$ be the first point where $(-1)^n(\mathcal{A} - \mathcal{P}_S)^{(n)}$ fails to be positive. Adding primes in increasing order adds the atoms $\log 2, \log 4, \log 8, \log 3, \dots$ in groups, and $\mathcal{A} - \mathcal{P}_S = W + \mathcal{P}_{\text{rest}}$ differs from W by the omitted prime powers. The first omitted one, q, contributes about $\Lambda(q)q^{-1/2}g_t(\log q)$, of exponential size $e^{-(\log q)^2/4t}$, while W and its derivatives decay like $e^{-\gamma_1^2 t}$. The two meet at

$$t(S) \approx \frac{\log q}{2\gamma_1}, \qquad q = \text{the smallest prime power not generated by } S. \tag{15.8.1}$$

This is a heuristic balance of exponents: coefficients, derivative polynomials, the other zeros and the other omitted prime powers contribute corrections with no uniform bound, so (15.8.1) is an empirical law for this truncation scheme and not a theorem. Read as an uncertainty principle, a Gaussian of width $\sqrt{2t}$ in $u = \log n$ resolves the first missing atom when its dual width resolves the lowest zero.

Table 15.5 tests (15.8.1), with $t_n(S)$ found by an upward scan to the first failure followed by bisection (computed, 35 and 45 digits, scanned to 0.8).

primes S	first omitted q	$\log q/2\gamma_1$	$t_1(S)$	$t_2(S)$	$t_3(S)$
none	2	0.0245	0.0253	0.0510	0.0761
2	3	0.0389	0.0394	0.146	0.223
2, 3	5	0.0569	0.0575	0.476	0.0608
≤ 5	7	0.0688	0.0696	> 0.8	0.0720
≤ 7	11	0.0848	0.0856	> 0.8	0.0872
≤ 13	17	0.1002	0.1010	> 0.8	0.1022
≤ 31	37	0.1277	0.1289	> 0.8	0.1297
≤ 97	101	0.1633	0.1635	> 0.8	0.1641

Table 15.5: End of the positivity window of order n when only the primes in S are kept (computed). The entries > 0.8 mean no failure up to the end of the scan.

Three facts emerge.

1. The law (15.8.1) holds for the order one window. Its error, as a percentage of $t_1(S)$, is 3.2% for S empty ($q = 2$), between 0.8% and 1.3% for the next six rows (about 1% from $q = 5$ on), and 0.17% at $q = 101$.

2. Among the orders $1, 2, 3$ tested, order 1 always fails first. A mechanism for the sign: each omitted term $g_t(\log q)$ is increasing in

t for $t < (\log q)^2/2$, since $\partial_t \log g_t(u) = u^2/4t^2 - 1/2t$, so that $-W' \geq -(\mathcal{A} - \mathcal{P}_S)'$ there (Proposition 15.9.2(1)). For the higher derivatives $\partial_t^n g_t(u) = (-1)^n n!\, t^{-n} g_t(u) L_n^{(-1/2)}(u^2/4t)$ is positive once $u^2/4t$ exceeds the largest zero of the Laguerre polynomial $L_n^{(-1/2)}$, a threshold that grows with n; for a finite set of orders the omitted primes then subtract from the odd order conditions and add to the even ones on a small common interval. This accounts for the parity effect in the table ($t_2 > 0.8$ once the primes up to 5 are in) but not for a ranking of order 1 against every higher order, and for S empty or $\{2\}$ the table has $t_2 < t_3$, so only order 1 is first. There is no fixed $t > 0$ at which every t derivative of $g_t(u)$, $u > 0$, is nonnegative (otherwise its Taylor series at t, of radius t, would have a singularity at its positive real radius). That order 1 binds for all orders is only an observation on the computed orders.

3. Keeping the first factor $e^{-(\log q)^2/4t}$ below $e^{-\gamma_1^2 t}$ takes every prime power up to about $e^{2\gamma_1 t}$ in this balance, and each atom $\log q$ extends the order one window to about $\log q/2\gamma_1$. These are heuristic scales for this truncation scheme, not theorems of necessary information (a tail can sometimes be bounded without listing its atoms); the full condition on $(0, \infty)$ would need them all.

15.8.3 Rigidity of the prime point process

Goldston and Montgomery [84] proved that, under RH, a strong form of Montgomery's pair correlation conjecture for the zeros is equivalent to an asymptotic for the variance of ψ_{C} in short intervals. Montgomery and Soundararajan [120] refined the constant under a uniform Hardy and Littlewood prime tuples hypothesis, not under RH alone. The GUE prediction transported to the primes is

$$V_\delta = \frac{1}{b-a} \int_a^b \frac{[\psi_{\mathrm{C}}((1+\delta)y) - \psi_{\mathrm{C}}(y) - \delta y]^2}{\delta y}\, dy \approx \log \frac{1}{\delta} + 1 - \gamma_E - \log 2\pi,$$

with $[a, b] = [10^6, 10^7]$ here, a normalised mean square (the source's ratio of the variance of the short interval error to $h = \delta y$ with y random is this quantity). A Poisson process with the same intensity, such as the Cramér model, gives instead about $\log y - 1$, averaged over the range 14.37. The gap between $\log y$ and $\log(1/\delta)$ measures the rigidity. Table 15.6 compares them (computed: random points of $(10^6, 10^7)$ and a sieve to 2.1×10^7, two independent prime samples, standard errors 0.003 to 0.03 and the prime column the first sample; the Cramér column is one realisation with standard errors 0.03 to 0.05).

The primes follow the GUE prediction over four decades of δ: the difference is 0.03, −0.09, −0.17, −0.35, 0.03, 0.21 at the six scales printed (from the rounded table values). The Cramér model stays near the Poisson level. These finite numbers

δ	primes	GUE prediction	Cramér model (one realisation)	Poisson $\log y - 1$
10^{-5}	10.13	10.10	14.40	14.37
10^{-4}	7.71	7.80	14.53	14.37
10^{-3}	5.32	5.49	14.60	14.37
3×10^{-3}	4.04	4.39	14.81	14.37
10^{-2}	3.22	3.19	14.76	14.37
10^{-1}	1.10	0.89	14.55	14.37

Table 15.6: V_δ for the primes, the GUE prediction, the Cramér model and the Poisson level (computed). The source printed $10.19, 7.69, 5.34, 4.06, 3.23, 1.10$ for the primes (all within two standard errors of these) and Cramér values 14.19 to 15.22 with a last entry 10.09, which the recomputation does not reproduce.

are compared with a conjectural prediction; they do not establish GUE statistics for the prime point process, since a variance fixes second moments only, and they do not prove an unconditional variance asymptotic. A variance estimate of this kind is a statement under RH in its Goldston and Montgomery form, so by itself it cannot be cited to prove RH; a future unconditional variance bound strong enough could still contribute to a proof, and no impossibility theorem is claimed. That any proof of positivity in the cancellation regime would have to use this property is an opinion, not a result.

15.8.4 Second order regular variation

By the explicit formula, with symmetric truncation in the height, $\psi_{\mathrm{C}}(y) - y = -\sum_\rho y^\rho/\rho + O(\log y)$ for $y \geq 2$. Relative to the first order term y, the correction is a superposition of terms $y^{\rho-1}$ with complex indices $\rho - 1 = -\frac{1}{2} + \delta + i\gamma$, in the convention $\rho = \frac{1}{2} + \delta + i\gamma$ of Section 8.1.5. Classical second order regular variation (de Haan and Stadtmüller [57]) requires a limit of $(f(ty)/f(t) - y^\alpha)/A(t)$ with A regularly varying, and handles one real second order index. For $f = \psi_{\mathrm{C}}$ no such limit exists: under RH, $(\psi_{\mathrm{C}}(ty) - ty)/\sqrt{t}$ is an almost periodic function of $\log t$, and the second order term is an oscillating sum with a whole spectrum of complex indices. What is true is that $\limsup_{y\to\infty} \log|\psi_{\mathrm{C}}(y) - y|/\log y = \sup \mathrm{Re}\,\rho$ unconditionally, and RH says that this exponent is $\frac{1}{2}$, which is von Koch's criterion. A theory of second order regular variation with complex index spectra would turn RH into the statement that the second order spectrum lies on a vertical line; this is a reformulation only.

15.9 Certificates, smoothing, energy and moments

15.9.1 The window law with explicit constants

Carrying Laplace's method of Proposition 15.8.2 one order further gives the constants of t_n. Let w be smooth near its simple zero λ_* (Lemma 15.8.1), of polynomial growth at infinity and integrable at zero, and put, uniformly for c near λ_*,

$$J_n(c) = \frac{(n/c)^{n+1}}{n!}\int_0^\infty \lambda^n e^{-n\lambda/c} w(\lambda)\,d\lambda = \mathbb{E}\,w(cT_n),$$

where T_n has the gamma law of shape $n+1$ and rate n. The moments about one are $\mathbb{E}(T_n-1)=n^{-1}$, $\mathbb{E}(T_n-1)^2 = n^{-1}+2n^{-2}$, $\mathbb{E}(T_n-1)^3 = 5n^{-2}+O(n^{-3})$ and $\mathbb{E}(T_n-1)^4 = 3n^{-2}+O(n^{-3})$, and a Taylor expansion with gamma concentration for the remainder gives $J_n(c) = w(c) + A_1(c)/n + A_2(c)/n^2 + O(n^{-3})$ with

$$A_1(c) = c\,w'(c) + \tfrac{1}{2}c^2 w''(c), \qquad A_2(c) = c^2 w''(c) + \tfrac{5}{6}c^3 w'''(c) + \tfrac{1}{8}c^4 w^{(4)}(c).$$

Proposition 15.9.1 (Window law). *The first sign change of D_n satisfies $t_n = n/\lambda_* + \beta + \delta/n + O(n^{-2})$, with*

$$\beta = \frac{1}{\lambda_*}\Big(1 + \frac{\lambda_* w''(\lambda_*)}{2w'(\lambda_*)}\Big), \qquad \delta = \frac{\alpha^2}{\lambda_*^3} - \frac{\eta}{\lambda_*^2},$$

where $\alpha = -A_1(\lambda_*)/w'(\lambda_*)$ and $\eta = -\big[\tfrac{1}{2}w''(\lambda_*)\alpha^2 + A_1'(\lambda_*)\alpha + A_2(\lambda_*)\big]/w'(\lambda_*)$.

Proof. Put $t = n/c$. Then $D_n(n/c)$ is, up to a positive factor, $J_n(c)$ plus the atom term $(-\frac{1}{4})^n e^{n/4c}$ divided by $n!(c/n)^{n+1}$, which is $O(\sqrt{n}\,b(c)^n)$ with $b(c) = e^{1+1/4c}/4c$. Since $\operatorname{Re}\psi(\frac{1}{4}+\frac{i}{2}) \le \frac{1}{5} - \gamma_E < \log\pi$ (the value is about -0.88) gives $\lambda_* > 1$, one has $b(\lambda_*) < 1$, and near λ_* the atom, with its derivatives, is exponentially small. So the root $c_n = n/t_n = \lambda_* + \alpha/n + \eta/n^2 + O(n^{-3})$ is found by comparing powers of $1/n$ in $J_n(c_n) = 0$, which uses $w'(\lambda_*) > 0$ (Lemma 15.8.1) and the implicit function theorem for the remainder. Inverting $t_n = n/c_n$ gives $\beta = -\alpha/\lambda_*^2$ and the stated δ. □

At λ_* the computations give $w' = 1.60238\times10^{-4}$ and $w'' = -8.1092\times10^{-6}$, so $\beta = -2.70691\times10^{-5}$ (computed). With the offsets $t_n - n/\lambda_*$ of Section 15.4.1, $n(t_n - n/\lambda_* - \beta)$ for $n = 2,\dots,6$ is $1.026, 0.995, 1.015, 1.023, 1.027$ in units of 10^{-3} (computed). The formula gives $\delta = 1.0365\times10^{-3}$ (evaluated in the review of the source; not evaluated in the book). The finite order fit 1.02×10^{-3} is an effective coefficient and does not identify the asymptotic δ; the 2% difference is within what a further term would contribute. The case $n = 1$ lies outside the asymptotic regime ($n(t_1 - 1/\lambda_* - \beta) = 0.082\times10^{-3}$).

For even n the second sign change of D_n is at $t = 1.3612$, 3.2456 and 5.2669 for $n = 2, 4, 6$ (computed). Balancing the positive atom against the negative part

of the integral with $w(\lambda) \approx c_0 \lambda^{-1/2}$ near 0, $c_0 = (\psi(\frac{1}{4}) - \log \pi)/4\pi = -0.427505$, gives the equation $4^{-n} e^{t/4} = |c_0| \Gamma(n + \frac{1}{2}) t^{-n-1/2}$, with roots 1.98, 3.94 and 5.99 (computed). This captures the growth but overestimates, with relative error falling from 45.7% to 13.7%. For large n the second change is $a_* n + O(\log n)$ with $a_*/4$ the solution of $b + 1 + \log b = 0$, that is $a_* = 1.113858$ (unchecked here, but $4W(1/e) = 1.11386$): at $t = an$ Laplace's method gives the negative integral as $w(1/a)\, n!\, t^{-n-1}(1+O(1/n))$, and equating its exponential rate with that of $4^{-n} e^{t/4}$ gives $a/4 + 1 + \log(a/4) = 0$, with saddle $\lambda = 1/a_* = 0.8978$ where $w < 0$ (proved, as an asymptotic for the local balanced root; it does not replace the computed values for small n).

15.9.2 Certificates from finitely many primes

Proposition 15.9.2 (One sided certificates). *Let q be the smallest prime power omitted from $\mathcal{P}_S$, so that $W = (\mathcal{A} - \mathcal{P}_S) - \mathcal{P}_{\mathrm{rest}}$, where $\mathcal{P}_{\mathrm{rest}}$ collects the omitted prime powers, all at least q.*

(1) For $t < (\log q)^2/2$, $-W'(t) \geq -(\mathcal{A} - \mathcal{P}_S)'(t)$. Hence if $\mathcal{A} - \mathcal{P}_S$ is decreasing on $(0, X]$ with $X < (\log q)^2/2$, so is W, with no tail estimate needed.

(2) For even orders a tail bound is needed. Let S be all prime powers up to N_0 and let $f(y) = y^{-1/2} \partial_t^2 g_t(\log y)$ be nonnegative and decreasing on (N_0, ∞). Then, with $\psi_{\mathrm{C}}(y) < 1.03883\, y$ [251],

$$W''(t) \geq (\mathcal{A} - \mathcal{P}_{\leq N_0})''(t) - 1.03883 \Big(N_0 f(N_0) + \int_{\log N_0}^{\infty} e^{u/2} \partial_t^2 g_t(u)\, du \Big).$$

The hypotheses on f hold for every $0 < t \leq X$ as soon as $X \leq (\log N_0)^2/[2(5 + \sqrt{10})]$.

Proof. For (1), $\partial_t \log g_t(u) = u^2/4t^2 - 1/2t > 0$ when $t < u^2/2$, so every term of $\mathcal{P}_{\mathrm{rest}}$ has positive t derivative, and the differentiated prime sum converges locally uniformly. For (2), Stieltjes integration by parts gives for $f \geq 0$ decreasing and integrable with $yf(y) \to 0$ that $\sum_{m > N_0} \Lambda(m) f(m) = -\psi_{\mathrm{C}}(N_0) f(N_0) - \int_{N_0}^{\infty} \psi_{\mathrm{C}} f'\, dy \leq 1.03883 \big(N_0 f(N_0) + \int_{N_0}^{\infty} f\, dy \big)$, and $\int_{N_0}^{\infty} f(y)\, dy = \int_{\log N_0}^{\infty} e^{u/2} \partial_t^2 g_t(u)\, du$. For the hypotheses put $z = (\log y)^2/4t$ and $Q(z) = z^2 - 3z + \frac{3}{4}$, so that $f(y) = y^{-1/2} g_t(\log y) t^{-2} Q(z)$; the u derivative of f has the sign of $\frac{u}{2t}(Q'(z) - Q(z)) - \frac{1}{2} Q(z)$ with $Q' - Q = -z^2 + 5z - \frac{15}{4}$, and both $f \geq 0$ and $f' \leq 0$ hold when $z \geq (5 + \sqrt{10})/2$. □

We evaluated the lower bound in (2) on the geometric grid $0.01 \cdot 1.08^k$ of t, at 30 digits, checking that f is nonnegative and decreasing on a grid of u of step 0.01 (computed). The bound stays positive up to the points of Table 15.7; the first failing grid points, where it is negative, are 0.0799, 0.1597 and 0.2534.

prime powers kept	order 2 window from the bound	law $\log q/2\gamma_1$	order 1 window (Table 15.5)
≤ 10	0.0740	0.0848 ($q=11$)	0.0856
≤ 100	0.1479	0.1633 ($q=101$)	0.1635
≤ 1000	0.2346	0.2447 ($q=1009$)	

Table 15.7: Windows for $W''>0$ suggested by the bound of Proposition 15.9.2(2) from the primes up to N_0 with the Rosser and Schoenfeld tail (computed in floating point on a grid; not certified).

The windows follow the same law and fall short of it by 12.8%, 9.4% and 4.1%, the cost of the explicit tail. The inequalities are proved; the table is not a certificate, because floating point positivity at grid points proves neither the signs of the computed quantities nor positivity between the points, and it does not cover $0<t<0.01$. A continuous certificate would partition $[\varepsilon,X]$ into intervals $I_i=[a_i,b_i]$, enclose the lower bound $B_N(c_i)\ge L_i$ at a point $c_i\in I_i$ and a bound $|B_N'|\le M_i$ on I_i, and check $L_i>M_i\max(c_i-a_i,b_i-c_i)$; below ε, Theorem 8.1.9(a) covers $(0,0.019]$ for every order. For every order and every t the theorem is Theorem 8.1.13, which uses the zeros. On the method of Theorem 8.1.9: its bound for part (a) vanishes at $t=0.0193$ (Section 8.1.3), and a reader of that section reported that the sharp envelope of the prime terms in place of the proved one moves the root to about 0.022 (0.0218 with the supremum over $k\le 300$; the book does not reproduce it, unchecked), the limit then being set by E_1, which vanishes at 0.0261.

15.9.3 The heat trace as a Gaussian smoothing of the prime error

Let $D(u)=(\psi_{\mathrm{C}}(e^u)-e^u)e^{-u/2}$ be the normalised prime error on the log scale, so $D(0)=-1$.

Proposition 15.9.3. *For every $t>0$ (without RH),*

$$W(t)=\Sigma_\infty(t)+\int_0^\infty D(u)\,k_t(u)\,du,\qquad k_t=\partial_u g_t-\tfrac12 g_t, \tag{15.9.1}$$

with

$$\Sigma_\infty(t)=\int_0^\infty g_t(u)e^{-u/2}du-g_t(0)\big(1+\tfrac12\log\pi\big)+\frac{1}{4\pi}\int e^{-tr^2}\operatorname{Re}\psi\big(\tfrac14+\tfrac{ir}{2}\big)dr.$$

Proof. Write $d\psi_{\mathrm{C}}(e^u)=e^u\,du+d(D\,e^{u/2})$ in (15.1.3) and integrate the second part by parts: $\mathcal{P}=\int_0^\infty g_te^{u/2}du-g_t(0)D(0)-\int_0^\infty D\,(\partial_u g_t-\frac12 g_t)\,du$, the upper boundary vanishing by Gaussian decay and $\psi_{\mathrm{C}}(y)\ll y$. Since $\int_0^\infty g_t(e^{u/2}+e^{-u/2})\,du=\int_{\mathbb{R}}g_te^{u/2}du=e^{t/4}$, subtracting $\mathcal{P}$ from (15.1.2) gives the claim. □

The source verified (15.9.1) against the zeros to 18 digits at $t=0.01,0.05,0.1,0.2$, where Σ_∞ is of order one and negative: -3.20, -1.77,

-1.33, -0.99 (unchecked here). So the Gaussian derivative smoothing of D at scale $\sqrt{t}$ must cancel an explicit archimedean quantity to exponential accuracy. By the explicit formula $D(u) \approx -\sum_\rho e^{(\rho-1/2)u}/\rho$, and the bilateral Gaussian transform $\int_{\mathbb{R}} g_t(u)e^{qu}du = e^{tq^2}$, for $q = \rho - \frac{1}{2} = i\gamma$, is the factor $e^{-t\gamma^2}$ of the heat trace, complex off the critical line; this explains the heat factor, although the half line integral in (15.9.1) also carries a derivative, boundary and archimedean terms, so it is not a pure Fourier multiplier on the zero series. Under RH $W(t) \sim m_{\gamma_1} e^{-\gamma_1^2 t}$ as $t \to \infty$, and an off line zero, with growth $e^{\delta u}$ in D, would spoil complete monotonicity. Integrating $W = \mathcal{A} - \mathcal{P}$ term by term is valid on a finite interval $(0, X]$; it must not be read as separate integration over $(0, \infty)$, since $\int_0^\infty g_t(u)\,dt = \infty$ and $\mathcal{A}$ contains $e^{t/4}$, and the convergent full integral comes from the cancellation in $\mathcal{A} - \mathcal{P}$.

15.9.4 Variance, energy and RH

Proposition 15.9.4 (Energy form of RH). *RH holds if and only if* $E_\epsilon = \int_0^\infty D(u)^2 e^{-2\epsilon u}\,du < \infty$ *for every* $\epsilon > 0$. *Under RH,*

$$\lim_{U\to\infty} \frac{1}{U}\int_0^U D(u)^2\,du = 2\sum_{\gamma>0} \frac{m_\gamma^2}{\frac{1}{4}+\gamma^2},$$

the sum over distinct positive ordinates, with multiplicities m_γ. *This equals* $\sum_\rho |\rho|^{-2} = 2 + \gamma_E - \log 4\pi = 0.0461914179$ *if all zeros are simple, and equals* $2 + \gamma_E - \log 4\pi + 2\sum_{\gamma>0} m_\gamma(m_\gamma - 1)/(\frac{1}{4}+\gamma^2)$ *in general.*

Proof. For $\operatorname{Re} s > 1$, $\int_0^\infty D(u)e^{-zu}du = \int_1^\infty (\psi_{\mathrm{C}}(y) - y)y^{-s-1}dy = -\frac{\zeta'(s)}{s\zeta(s)} - \frac{1}{s-1}$ with $s = \frac{1}{2} + z$, by the substitution $y = e^u$ and the Dirichlet series of $-\zeta'/\zeta$; the singularity at $s = 1$ cancels, and at a zero ρ of multiplicity m_ρ the residue is $-m_\rho/\rho \neq 0$. If $E_\epsilon < \infty$, then for $\sigma > \epsilon$ the Cauchy and Schwarz inequality gives $\int |D|e^{-\sigma u}du \le E_\epsilon^{1/2}/\sqrt{2(\sigma-\epsilon)}$, and with powers of u the transform is holomorphic on $\operatorname{Re} z > \epsilon$, so no zero has $\operatorname{Re}\rho > \frac{1}{2} + \epsilon$; for every ϵ this gives RH by the functional equation. Conversely, under RH von Koch's bound $\psi_{\mathrm{C}}(y) - y = O(\sqrt{y}\log^2 y)$ gives $D(u) = O((1+u)^2)$ and $E_\epsilon < \infty$ for every $\epsilon > 0$; neither Paley and Wiener nor a mean square theorem is needed. For the limit, under RH $D(u) = -2\operatorname{Re}\sum_{\gamma>0} m_\gamma e^{i\gamma u}/(\frac{1}{2}+i\gamma) + o(1)$ in the B^2 sense, and averaging the square of a finite sum over distinct frequencies leaves the squared coefficients, which gives the formula (Cramér [48] for the case of simple zeros; the multiplicity sensitive form is in Akbary, Ng and Shahabi, arXiv 1306.1657, unchecked). The unconditional identity $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma_E - \log 4\pi$, with multiplicity, comes from the logarithmic derivative of the Hadamard product at 0 and at 1. □

Table 15.8 checks the energy constant from the primes up to 2×10^7, integrating D^2 exactly on each interval between integers (computed). The last column is the

sum under simple zeros, 0.046191 (computed from the identity); finite windows can be consistent with it without proving the limit or the simplicity of the zeros.

window in $u = \log y$	mean of D^2, primes	Cramér model (one realisation)	$\sum_\rho \lvert\rho\rvert^{-2}$
$[5, 10]$	0.0509	5.19	0.0462
$[8, 12]$	0.0442	3.43	0.0462
$[10, 14]$	0.0465	2.93	0.0462
$[12, 16.8]$	0.0468	5.14	0.0462

Table 15.8: Energy of the normalised prime error on the log scale (computed). The source printed 0.0506 for the first window and 7.54, 9.78, 9.14, 7.15 for a Cramér realisation; the Cramér values are heavy tailed and one realisation does not support a growth rate.

This settles the variance question raised in Section 15.8.3 in three parts.

1. The short interval variance of Goldston and Montgomery is a statement under RH, so it cannot be used by itself to prove RH.

2. A bound on the energy alone cannot prove $W(t) \geq 0$ at a given t when $\Sigma_\infty(t) < 0$. If $E_\epsilon \leq M^2$, weighted Cauchy and Schwarz gives $|\int D k_t| \leq M K_{t,\epsilon}$ with $K_{t,\epsilon}^2 = \int_0^\infty k_t^2 e^{2\epsilon u} du$, so $W(t) \in [\Sigma_\infty - M K_{t,\epsilon}, \Sigma_\infty + M K_{t,\epsilon}]$. A small enough norm bound could certify a negative sign, and the limitation is exact for norm information: the two functions $D_\pm = \pm M e^{2\epsilon u} k_t / K_{t,\epsilon}$ have the same discounted energy and opposite extreme pairings. The norm depends on t and is not uniformly of order one.

3. The global energy of D is equivalent to RH (Proposition 15.9.4).

In the language of the second order term, RH says that D has finite discounted energy on the log scale, and under RH its energy per unit of u is the sum of Proposition 15.9.4. The Cramér model has finite discounted energy too, almost surely for every $\epsilon > 0$ at once: with $\Psi_C(y) = \sum_{3 \leq n \leq y} B_n \log n$ and B_n Bernoulli with parameter $1/\log n$, one has $\mathbb{E}\Psi_C(y) = y + O(1)$ and $\operatorname{Var} \Psi_C(y) = y \log y - 2y + O(\log y)$, so $\mathbb{E} D_C(u)^2 = O(1 + u)$ and Tonelli gives $\mathbb{E} \int D_C^2 e^{-2\epsilon u} du < \infty$. What it lacks is a bounded mean square per unit of u, which grows like u in expectation. That this reflects the Euler product is a heuristic; the energy equivalence for the primes uses the zeta specific transform and is not a characterisation of Euler products.

15.9.5 The reciprocal Xi law from primes

Under RH the reciprocal Ξ law $T = \sum_j E_j / \lambda_j$ has Lévy measure $W(x)\, dx/x$ and Thorin measure $U = \sum_j \delta_{\lambda_j}$, so its cumulants are $\kappa_j = \int_0^\infty x^{j-1} W(x)\, dx = (j-1)!\, s_j$ with $s_j = \sum_k \lambda_k^{-j} = \sum_{\gamma > 0} \gamma^{-2j}$. (The $(j-1)!$ appears once:

$\int_0^\infty x^{j-1}e^{-\lambda x}dx = (j-1)!\lambda^{-j}$.) The integral identity $\int x^{j-1}W\,dx = (j-1)!\,s_j$ needs no RH: $\operatorname{Re}\lambda_j > 0$ follows from $|\operatorname{Im}\gamma| < \frac12$ and the verification of RH to a finite height [232], and Fubini applies since $\sum_\gamma \int x^{j-1}|e^{-\gamma^2 x}|dx = (j-1)!\sum(\operatorname{Re}\gamma^2)^{-j} < \infty$. So each s_j is computable from Γ and the primes alone. For a finite X, $\int_0^X g_t(u)\,dt = \sqrt{X/\pi}\,e^{-u^2/4X} - \frac{u}{2}\operatorname{erfc}(u/2\sqrt{X})$ for $u \ge 0$, and the archimedean part integrates in closed form against $e^{t/4}$ and $g_t(0)$, leaving one integral against $\operatorname{Re}\psi(\frac14 + \frac{ir}{2})$. The source cut at $X = 0.25$ and kept the prime powers up to 4000; at 35 digits it printed the following (unchecked, not recomputed).

	Γ and primes ≤ 4000	from ξ at $\frac12$
$s_1 = \sum\gamma^{-2}$	0.023104993115418970788932 79	0.023104993115418970788932 81
$s_2 = \sum\gamma^{-4}$	0.000037172599285269686164 61	0.000037172599285269686164 87

The right column uses $\xi(\frac12+z) = \xi(\frac12)\prod(1+z^2/\gamma^2)$, so $s_1 = \xi''(\frac12)/2\xi(\frac12)$ and $s_2 = s_1^2 - \xi^{(4)}(\frac12)/12\xi(\frac12)$; these are analytic coefficients, and calling them the mean and variance of T is conditional on RH. The columns differ by 1.02×10^{-24} and 2.6×10^{-25}, which are the omitted heat tails beyond $X = \frac14$: the first zero alone gives $\int_X^\infty W \ge e^{-\gamma_1^2 X}/\gamma_1^2 = 1.0173\times10^{-24}$, so the cut is not below 10^{-24} as the source stated, and for $j=2$ the first zero gives $e^{-\gamma_1^2 X}(X/\gamma_1^2 + 1/\gamma_1^4) = 2.594\times10^{-25}$. The relative differences are 4×10^{-23} and 7×10^{-21}, so the agreement is to 23 and 20 significant digits, not 24 for both. The omitted prime tail, bounded by the partial summation of Proposition 15.9.2 applied to $G_X(u) = \int_0^X g_t(u)\,dt$ (nonnegative and decreasing in u), is about 2×10^{-31} at $N = 4000$ (unchecked) and is far smaller.

This is a consistency check and not evidence for RH. What RH adds is that the atoms λ_j^{-1} are real and positive, which is a moment problem. Put $m_j = s_{j+1} = \Gamma(j+1)^{-1}\int_0^\infty x^j W\,dx = \kappa_{j+1}/j!$, the normalised cumulants, and $\nu = \sum_j \lambda_j^{-1}\delta_{\lambda_j^{-1}}$, so that $m_j = \int t^j\,\nu(dt)$ and $|m_j| \le SR^j$ with $S = \sum|\lambda_j|^{-1}$ and $R = \sup|\lambda_j|^{-1}$.

Proposition 15.9.5 (Hankel form of RH). *RH holds if and only if the Hankel matrices $H_d = (m_{i+j})_{0\le i,j\le d}$ and $H_d^{(1)} = (m_{i+j+1})_{0\le i,j\le d}$ are positive semidefinite for every $d \ge 0$.*

Proof. Under RH ν is a positive measure and the two quadratic forms are $\int|p|^2 d\nu$ and $\int t|p|^2 d\nu$. Conversely, Stieltjes' theorem gives a positive measure μ on $[0,\infty)$ with these moments; from $b^{2k}\mu([b,\infty)) \le m_{2k} \le SR^{2k}$ it is supported on $[0,R]$. Near $z=0$, $M(z) = \int(1-zt)^{-1}\mu(dt) = \sum_j m_j z^j = \sum_k 1/(\lambda_k - z)$. The integral is holomorphic off $[1/R,\infty)$, while the spectral series is meromorphic, converging normally away from its poles, each of which has residue minus its positive integer multiplicity, so no cancellation occurs. Analytic continuation forces every λ_k to be positive, and then $\operatorname{Re}\gamma_k > 0$ with $\gamma_k^2 > 0$ gives γ_k real, which is RH. □

Semidefiniteness means all quadratic forms, equivalently all principal minors; some leading determinants do not replace it. Each entry is a finite combination of integrals of $\mathcal{A}-\mathcal{P}$ against powers of t, convergent at the Gaussian rate in the primes. This is a criterion of the type of the Keiper and Li criteria and of Theorem 8.1.3, which has Hankel matrices at one heat time; here the entries are moments of the Thorin side. The extreme value structure of W near 0 (Section 15.3) makes the integrals converge rapidly.

15.10 The sine squared lemma and a single heat time

The Thorin condition of Theorem 15.2.1 asks for every derivative of W at every t. The sine squared lemma (2.1.9), in its Lévy and Khintchine form $1-\cos\theta = 2\sin^2(\theta/2)$ (valid for every complex θ), trades that ladder for a one parameter family at a single heat time. The tent version used by Yoshida's window is Theorem 15.18.2.

15.10.1 The shifted functional

For $t>0$ and $a\in\mathbb{R}$ put

$$f_t(a)=\sum_j e^{-\lambda_j t}\cos(a\gamma_j), \qquad V(t,a)=W(t)-f_t(a)=\sum_j 2\sin^2\!\big(\tfrac{a\gamma_j}{2}\big)e^{-\lambda_j t},$$

with the indexing of the introduction ($\operatorname{Re}\gamma_j>0$, multiplicities counted). For every complex z, $1-\cos az=\frac12(1-e^{iaz})(1-e^{-iaz})$. With $F(z)=(1-e^{iaz})e^{-tz^2/2}/\sqrt{2}$ and $F^*(z)=\overline{F(\bar z)}$, $F(z)F^*(z)=(1-\cos az)e^{-tz^2}$, which is $|F(z)|^2$ on the real axis. So V is the Weil functional of the even part of an admissible test function FF^* (it decays like a Gaussian on every strip). At a nonreal zero the summand need not be nonnegative (at $\gamma=ib$ the weight $2\sin^2(a\gamma/2)$ equals $-2\sinh^2(ab/2)<0$), so termwise nonnegativity holds only under RH. The explicit formula (15.1.1) with $h(r)=e^{-tr^2}\cos(ar)$, for which $g(u)=\frac12\big(g_t(u-a)+g_t(u+a)\big)$, gives

$$f_t(a)=e^{t/4}\cosh\tfrac{a}{2}-\tfrac12 g_t(a)\log\pi+\frac{1}{4\pi}\int_{\mathbb{R}} e^{-tr^2}\cos(ar)\,\operatorname{Re}\psi\big(\tfrac14+\tfrac{ir}{2}\big)\,dr \\ -\frac12\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\big(g_t(\log n-a)+g_t(\log n+a)\big). \quad (15.10.1)$$

At $a=0$ this is $W=\mathcal{A}-\mathcal{P}$. For $a>0$ the shift moves each prime power atom from $\log n$ to $\log n-a$, so the primes near e^a are brought to the centre of the Gaussian, while the pole term grows like $\frac12 e^{t/4}e^{a/2}$ and must be cancelled by them. That cancellation is the prime number theorem near $y=e^a$, read through a Gaussian of width $\sqrt{2t}$ in $\log y$.

Theorem 15.10.1 (One heat time suffices). *Fix $t>0$. RH holds if and only if $V(t,a)\ge0$ for every $a\ge0$.*

Proof. Under RH every γ_j is real and each term of V is nonnegative. Conversely let $V(t,\cdot) \geq 0$ and suppose a zero is off the line, so that some $\gamma_j = \tau_j + i\beta$ has $\beta \neq 0$; its conjugate occurs as well, so we may take $\beta > 0$. Since $|\operatorname{Im}\gamma| < \frac{1}{2}$, $|\cos(a\gamma)| \leq e^{a/2}$ for $a \geq 0$, and $\sum_j |e^{-\gamma_j^2 t}| < \infty$ by the zero counting bound, the Laplace transform

$$L(s) = \int_0^\infty e^{-sa} V(t,a)\, da = \frac{W(t)}{s} - \sum_j e^{-\gamma_j^2 t} \frac{s}{s^2 + \gamma_j^2}$$

holds for $\operatorname{Re} s > \frac{1}{2}$. On each compact set avoiding the candidate poles the series converges normally (the summands are $O(e^{-\tau^2 t}/\tau^2)$), so the right side is meromorphic on $\mathbb{C}$, with poles at $s = \pm i\gamma_j$, and a possible pole at $s = 0$. The residue at a pole is $-\frac{1}{2}$ times the multiplicity times $e^{-\gamma^2 t}$, which is nonzero, and distinct ordinates with the same pole coincide, so poles do not cancel. The point $s = -i\gamma_j = \beta - i\tau_j$ is a genuine pole of real part $\beta > 0$, so the abscissa of convergence σ_c of the transform is at least $\beta > 0$. Because $V \geq 0$, Landau's theorem says that L is singular at the real point σ_c: if L were analytic there, choose $s_0 = \sigma_c + d$ with $d > 0$ small, so that the Taylor disk at s_0 has radius larger than d; its coefficients $(-1)^k L^{(k)}(s_0)/k! = \int a^k e^{-s_0 a} V\, da/k!$ are nonnegative, and Tonelli sums the series at a real point $s_0 - h$ with $h > d$ inside the disk to $\int e^{-(s_0-h)a} V\, da < \infty$, contradicting the definition of σ_c. The only real candidates for singularities of the right side are $s = 0$ and the points $s = \pm i\gamma_j$ with γ_j purely imaginary. The latter cannot occur: $\operatorname{Re}\gamma_j > 0$ by the indexing, which is legitimate because ξ has no real zeros ($\xi(s) > 0$ for real s: for $0 < s < 1$ the alternating series gives $\eta(s) > 0$, hence $\zeta(s) < 0$ while $s(s-1) < 0$, for $s > 1$ the factors are positive, for $s < 0$ use the functional equation, and $\xi(0) = \xi(1) = \frac{1}{2}$). And $\sigma_c \geq \beta > 0$ excludes $s = 0$. This contradicts the existence of the real singularity, so every γ_j is real. □

Where complete monotonicity used every order at every t, Theorem 15.10.1 uses one t and every shift a; the shift plays the role of the order. By (15.9.1), $f_t(a)$ is the Gaussian smoothing at scale $\sqrt{t}$ of the prime error D near $u = a$, plus explicit archimedean terms, so the theorem is a one sided bound on a smoothed D that is uniform in a. That is the quantity between pointwise and energy bounds left open in Section 15.9. It is close in spirit to classical facts about smoothed sums over zeros and is presented as a reformulation. The tent version Theorem 15.18.2 is the compactly supported form of the same lemma.

15.10.2 Numerical check

The source took $t = 0.02$, the first 120 zeros and the prime powers up to 5000, at 30 digits (unchecked here; an independent calculation reproduced the six values to the printed digits from the first eight ordinates). The values from the zeros and from Γ and the primes agree at every printed digit.

a	$V(t,a)$ from zeros	$V(t,a)$ from Γ and primes, (15.10.1)
0.3	0.0267570868155	0.0267570868155
$\log 2$	0.0357321878430	0.0357321878430
$\log 3$	0.0367065917181	0.0367065917181
1.5	0.0313514051449	0.0313514051449
$\log 8$	0.0264484602913	0.0264484602913
3.0	0.0185315145626	0.0185315145626

The source reported a minimum of V/W of 4.5×10^{-4} on a grid of 1600 points in $0<a\le 8$ (unreproduced; on the grid $a=8k/1600$ an independent calculation gives about 5.6×10^{-4} at $a=0.89$, and other grids give other values, none a continuous minimum). At this t the lowest zero carries 99.2% of W (0.99198; the source's 99.9% is not possible, because the second zero alone brings the share below 0.99218). The local minima sit near $a\approx 2\pi k/\gamma_1 = 0.4445k$ and are deep for even k (ratio about 4.9×10^{-4} at $k=2$ and 4.2×10^{-4} at $k=4$, unchecked) and shallow for odd k (about 0.016, unchecked), since $2\gamma_2/\gamma_1 = 2.975$ is close to an integer. For smaller t at a fixed positive shift the prime power resonance $a_0=\log m$ gives a peak, not a dip: $V(t,a_0)=W(t)+\Lambda(m)/(4\sqrt{\pi m t})+O(1)$ as $t\downarrow 0$.

To see the criterion fail, the source moved γ_1 off the line, replacing it by the pair $\gamma_1\pm i\beta$ and keeping the other zeros. The pair contributes $2\,\mathrm{Re}[c(1-\cos(a(\gamma_1+i\beta)))]$ with $c=e^{-t(\gamma_1+i\beta)^2}$, whose leading oscillation at large a is $-e^{-t(\gamma_1^2-\beta^2)}e^{\beta a}\cos(a\gamma_1+2t\gamma_1\beta)$. For $\beta=0.05$ the minimum of V over $0<a\le 200$ is of the order of -4×10^2 (the source printed -391 on a coarse sample; the single shift $a\approx 199.588$ gives -396.8 in an independent calculation), and for $\beta=0.2$ it is about -4×10^{15}, both near $a=200$ (unchecked here). The off line pair grows like $e^{\beta a}$ while oscillating, and the sine squared weights cannot hold it. This is an experiment on one finite model, not a proof of a general domination.

15.10.3 What it costs

The prime side of $V(t,a)$ is governed by the prime powers near e^a in $\log n$. Write $V=V_N-\sum_{m>N}\Lambda(m)m^{-1/2}H_{t,a}(\log m)$ with $H_{t,a}(u)=g_t(u)-\frac12[g_t(u-a)+g_t(u+a)]=g_t(u)\big[1-e^{-a^2/4t}\cosh(au/2t)\big]$ and V_N the same expression with the sum truncated at N and the archimedean part written out. For $a>0$, $H_{t,a}(u)\le 0$ for $u\ge u_0(t,a)=(2t/a)\,\mathrm{arcosh}(e^{a^2/4t})$, so $\log N\ge u_0$ implies the rigorous one sided bound $V\ge V_N$; for $0\le a\le A$ the condition $\log N\ge A+\sqrt{2t}$ suffices, by convexity of g_t on $[\sqrt{2t},\infty)$. At $a=0$, $V=0$ exactly, so a positive uniform margin is impossible, and the quotient $V(t,a)/a^2$, whose value at 0 is $-W'(t)/2$, is what a certificate must enclose. This proved extension of the certificate argument needs the tail hypotheses checked as in Proposition 15.9.2; where the sign condition is unavailable an absolute tail majorant with partial summation is used. The number of primes needed is not simply "up to $e^{a+O(\sqrt{t})}$": to keep a fixed small

absolute tail η at large a, completing the square gives $\int_L^\infty e^{u/2} g_t(u-a)\,du = \frac{1}{2} e^{a/2+t/4} \operatorname{erfc}\big((L-a-t)/2\sqrt{t}\big)$ and so $L = \log N = a + t + O(\sqrt{t(a+\log(1/\eta))})$ in the bound of that type, and a one sided certificate that drops a helpful signed tail can use a different cutoff. For large a the condition is square root control of ψ_{C} near $y = e^a$ in a smoothed one sided sense, which is not literally von Koch's pointwise estimate. So the lemma collapses the derivative ladder into a positive Weil functional at one heat time. The obstruction moves from large t to large a but is not removed; the criterion is local in a, and a failure of RH must show up as a negative value of a single explicit function of one variable.

15.11 Is there a path to RH?

The chapter gives four exact forms of RH, each computable from Γ and the primes, and one identity.

1. complete monotonicity of $\mathcal{A} - \mathcal{P}$ (Theorem 15.2.1);
2. finite discounted energy of D on the log scale (Proposition 15.9.4);
3. positive semidefiniteness of the two Hankel families of the normalised cumulants of the reciprocal Ξ law (Proposition 15.9.5);
4. nonnegativity of the sine squared functional $V(t,\cdot)$ at a single heat time (Theorem 15.10.1).

The identity (15.9.1) writes W as a Gaussian smoothing of D and is a restatement of the first form; the source listed the nonnegativity of W as a fifth exact form, but $W > 0$ holds unconditionally (Theorem 8.1.1), and positivity of an exponential sum does not force its exponents to be real (the ordinates $1, 2 \pm \frac{i}{4}$ give $e^{-x} + 2e^{-63x/16}\cos x > 0$ for all $x > 0$). Complete monotonicity does give the converse, by the pole argument of Theorem 15.2.1.

The forms split by regime. For small t, the archimedean part alone is positive to the first sign changes $t_n \approx n/\lambda_*$ (Proposition 15.8.2, proved). This transfers to W at order 1 (Proposition 15.9.2(1), proved). The window obtained from finitely many primes follows the law $\log q/2\gamma_1$ (computed, a heuristic balance, Table 15.5), and Proposition 15.9.2(2) suggests windows at order 2 (computed on a grid, Table 15.7). For every order and every t the theorems are those of Chapter 8, Theorems 8.1.9 and 8.1.13, which use the verified zeros, except part (a) of the first. This is the Gaussian counterpart of the small support results. To prove RH by the prime side one would need the window for every order and every t, and the law suggests the cost: reaching t takes prime powers up to about $e^{2\gamma_1 t}$, which is infinite as $t \to \infty$ (heuristic). A finite prime side certificate covers a finite window and no more, whereas the zero side covers all t at orders up to 1.16×10^{25}.

In the cancellation regime the four forms are heuristically controlled by one estimate, $\psi_{\mathrm{C}}(y) - y$ at square root scale, in forms with different quantifiers: the energy form says it exactly, and a pointwise, a discounted square or a one sided smoothed bound are different statements that the zeta specific transforms relate. The smoothing identity shows that no bound of norm type alone can prove $W(t) > 0$ at a single t where $\Sigma_\infty(t) < 0$. The extreme value view maps the obstruction precisely but gives no way around it. Two structural leads remain. The first is the rigidity of the prime point process, which follows the GUE variance and not the Poisson level; whether the Euler product forces this rigidity is open, and a proof that it does would prove more than RH (heuristic). The second is the Hankel form, which turns RH into the positivity of minors built from the integrals $\int t^j(\mathcal{A} - \mathcal{P})\,dt$. Like the Keiper and Li criteria it is an equivalence and not a reduction, but its entries converge at the Gaussian rate in the primes and its early minors can be computed to high order.

15.12 Open problems

Problem 15.1. Make the windows of Table 15.7 rigorous by continuous interval enclosure, with an analytic bound below the first grid point, extend Proposition 15.9.2 to all orders up to some K, and push the certified window as far as the primes up to 10^6 allow, where the law predicts about 0.49.

Problem 15.2. Carry out the shifted certificate of Section 15.10 (the enclosure of $V(t, a)/a^2$ on $0 \le a \le \log N_0$ from the primes up to N_0), and find the best heat time t for such certificates.

Problem 15.3. Compute the principal minors of the two Hankel families of Proposition 15.9.5 from the primes to high order, and find which features of the prime point process control their sign. Do the same for Devroye's sign test and the mean excess of the reciprocal Ξ law (Section 19.5).

Problem 15.4. Give a structural reason, from the Euler product, for the GUE rigidity of Table 15.6, or a theory of second order regular variation with a spectrum of complex indices in which RH is the statement that the spectrum lies on a vertical line.

The first open problem of the source, the $1/n$ coefficient δ of the window law, is answered by Proposition 15.9.1.

Notes

Guinand and Weil are credited for the explicit formula, Bombieri for the quadratic functional, Yoshida and Connes and Consani for the small support positivity, Goldston and Montgomery and Montgomery and Soundararajan for the short interval variance, Cramér for the mean square, von Koch for the pointwise criterion,

Rosser and Schoenfeld for the bound on ψ_{C} and de Haan and Stadtmüller for second order regular variation. The converse in Theorem 15.2.1, Theorem 15.10.1 and Proposition 15.9.5 is by the pole argument for a resolvent and by Landau's theorem for nonnegative functions, in the form of the review of the source. The figures of the source are not reproduced.

Yoshida's Hermitian forms, Weil positivity and the Thorin programme

The Gaussian test functions above see every prime power at once, with weights that decay in $\log n$. Compactly supported test functions see none beyond the window, and this is the setting of Hiroyuki Yoshida's 1992 study of Weil's explicit formula as a Hermitian form on functions supported in $[-a, a]$ [278]. As the first author reads the paper, Yoshida proved the form positive definite for $a \leq \frac{1}{2}\log 2$, the range in which no prime power enters, and reduced RH to its nondegeneracy on a completed space for every a (Section 15.13). We read Yoshida against the Thorin programme (Section 1.1) and the sine squared lemma (2.1.9), and work the open problems that come out of the comparison.

Yoshida's functional $\mathcal{Y}(F) = \sum_\rho \Phi(\rho)$ is Weil's distribution, and the heat trace W is half its value on a Gaussian. The sine squared lemma is the hinge: its Brownian level x is Yoshida's support length $2a$. There is an uncertainty barrier for one coefficient: Proposition 15.21.1 bounds the off line part of the tent coefficient $D(x) = x\langle\chi_0, \chi_0\rangle$ at $a = x/2$ by an explicit $\varepsilon_H(x)$; against the lower bound for the on line part given by the first 1700 zeros (a double precision computation), this keeps D positive at levels below about 47 (Proposition 15.21.2) whatever the zeros above the verified height 3×10^{12} [232] do, so no computation of D at those levels can detect an off line zero.

Numbers are computed in floating point at the precision stated, or in ball arithmetic where the text says so. The barrier concerns the tent coefficient only. The account of Yoshida's paper in Section 15.13 is the first author's reading, used as context and not checked against the paper.

15.13 Yoshida's paper

For $F \in C_c^\infty(\mathbb{R})$ put

$$\Phi(s) = \int_{\mathbb{R}} F(y) e^{(s-1/2)y}\, dy, \qquad \widehat{F}(t) = \Phi(\tfrac{1}{2} + it) = \int_{\mathbb{R}} F(y) e^{ity}\, dy.$$

The letter Φ is local and is not the Pólya kernel of Section 2.1. As there, write each nontrivial zero as $\rho = \frac{1}{2} + i\gamma_\rho$, that is $\gamma_\rho = -i(\rho - \frac{1}{2})$. The number γ_ρ is complex for a zero off the line: if $\rho = \frac{1}{2} + \delta + i\gamma$ with γ and δ real, then $\gamma_\rho = \gamma - i\delta$,

whereas $\gamma = \operatorname{Im}\rho$ is the ordinate and $\beta_\rho = \operatorname{Re}\rho$ the real part. Then $\Phi(\rho) = \widehat{F}(\gamma_\rho)$, the entire extension of $\widehat{F}$, and Weil's explicit formula reads

$$\begin{aligned}\mathcal{Y}(F) := \sum_\rho \widehat{F}(\gamma_\rho) = {} & \int_{\mathbb{R}} F(y)\big(e^{y/2} + e^{-y/2}\big)dy - (\log\pi)F(0) \\ & - \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\big(F(\log n) + F(-\log n)\big) + V_1(F),\end{aligned} \tag{15.13.1}$$

summed over all nontrivial zeros with multiplicity, with the archimedean term in two forms,

$$\begin{aligned}V_1(F) &= \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{F}(t)\operatorname{Re}\psi\Big(\frac{1}{4} + \frac{it}{2}\Big)dt \\ &= -\tfrac{1}{2}\operatorname{Pf}\Big(\frac{1}{|y|}\Big)(F) - (\gamma_E + \log 2)F(0) - \int_{\mathbb{R}}\Big(\frac{e^{|y|/2}}{2\sinh|y|} - \frac{1}{2|y|}\Big)F(y)\,dy,\end{aligned} \tag{15.13.2}$$

where $\psi = \Gamma'/\Gamma$ and $\operatorname{Pf}(1/|y|)(F) = \lim_{\epsilon\downarrow 0}\big(\int_{|y|>\epsilon} F(y)\,|y|^{-1}dy + 2F(0)\log\epsilon\big)$ is the finite part of $1/|y|$, the integral cut at $|y| = 1$. Each term of (15.13.1) depends on F only through its even part, because the zeros are symmetric under $\rho \mapsto 1-\rho$, which negates γ_ρ. The formula holds for the Gaussians used below, whose transforms are entire and decrease rapidly in every strip (the class of Section 2.1), and, in Weil's original form [273, 29], for continuous, piecewise C^1 functions of compact support, the class of the tent and of the products $\chi_n * \widetilde{\chi_m}$ of Sections 15.18 and 15.19.

Remark 15.13.1 (The archimedean term in the real variable). The second line of (15.13.2) is the archimedean integral of Section 2.1 carried to the variable y, its divergence at $y = 0$ cut at $|y| = 1$; the forms of Theorem 9.1.1 and Section 6.6.1 differ from it only in where γ_E is placed.

The Hermitian form is $\langle\varphi_1, \varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ with $\widetilde{\varphi}(y) = \overline{\varphi(-y)}$. Since $\widehat{\varphi * \widetilde{\varphi}} = |\widehat{\varphi}|^2$ on $\mathbb{R}$, Weil's criterion [273, 29] says that RH holds if and only if $\langle\varphi,\varphi\rangle \geq 0$ for all $\varphi \in C_c^\infty(\mathbb{R})$ (Proposition 17.1.14 and Section 5.1.2). Let $C(a)$ be the space of the functions of $C_c^\infty(\mathbb{R})$ supported in $[-a,a]$, let $\chi_n(y) = (2a)^{-1/2}e^{\pi i n y/a}1_{[-a,a]}(y)$ for $n \in \mathbb{Z}$, an orthonormal basis of $L^2[-a,a]$, and let $K(a)$ be the space of the restrictions to $[-a,a]$ of smooth $2a$ periodic functions, that is the series $\sum c_n\chi_n$ with rapidly decreasing coefficients. Yoshida's results, as the first author reads the paper, are the following.

(Y1) Positivity on odd φ alone is equivalent to RH; on even φ alone, to RH up to real zeros. The proof is a localisation lemma that concentrates a test vector on one zero.

(Y2) The prime sum in $\langle\varphi_1,\varphi_2\rangle$ for $\varphi_1,\varphi_2\in C(a)$ runs over $\log n\le 2a$ only. The form is positive definite on $C(a)$ for small a, and on the complement of the low Fourier modes for any a.

(Y3) The form is nondegenerate on $C(a)$ for every a, unconditionally.

(Y4) There are closed formulas for the coefficients $\langle\chi_n,\chi_m\rangle$ on the basis of $K(a)$.

(Y5) Theorem 1: the form is positive definite on $K(a)$ for $a=\frac12\log 2$.

(Y6) Theorem 2: RH holds if and only if the form is nondegenerate on a completion of $K(a)$ for every a. If RH fails there is a first a_0 beyond which positivity is lost, and continuity in a forces degeneracy there.

(Y7) For $\mathbb{Q}$ the form is nondegenerate on $K(a)$ for every a.

The statements (Y1) to (Y7) were not checked against the paper, and the proofs of Sections 15.14, 15.15, 15.16 and 15.17 do not use (Y1) to (Y7).

Throughout, $W(x)=\sum_{\gamma>0}e^{(\rho-1/2)^2x}$ is the heat trace of Section 2.1, the sum over one zero from each pair $\{\rho,1-\rho\}$; $T_k=(-1)^kW^{(k)}$ is the ladder, the function written c_k in Section 8.1; U is the Thorin functional of Section 8.2.1, which under RH is the measure $\sum_{\gamma>0}\delta_{\gamma^2}$; and $p_1=\sum_{\gamma>0}\gamma_\rho^{-2}=\frac12(\log\xi)''(\frac12)=0.0231049931$, so that $2p_1$ is the number V_ξ of Remark 12.1.19. In this chart the sine squared lemma is

$$k_x(w):=e^{-wx}+wx-1=\int_0^\infty\log\Big(1+\frac{u}{z}\Big)\varpi_x(z)\,dz,\qquad u=\frac{w^2}{2},$$
$$\varpi_x(z)=\frac{x}{\pi\sqrt{2z}}\,2\sin^2\Big(x\sqrt{\tfrac{z}{2}}\Big). \tag{15.13.3}$$

It is (2.1.9) after the substitution $z\mapsto 2z$, and (22.2.3) of Chapter 22.

15.14 The ladder is the diagonal of Weil's form

Let $h_x(y)=(4\pi x)^{-1/2}e^{-y^2/(4x)}$, so $\widehat{h_x}(t)=e^{-t^2x}$ and $h_x=h_{x/2}*\widetilde{h_{x/2}}$.

Proposition 15.14.1. *For $k\ge 0$ and $x>0$ let $g_{k,x}=(i\,d/dy)^kh_{x/2}$, so that $\widehat{g_{k,x}}(t)=t^ke^{-t^2x/2}$. Then*

$$\langle g_{k,x},g_{k,x}\rangle=\sum_\rho\gamma_\rho^{2k}e^{-\gamma_\rho^2x}=2\,T_k(x),\qquad W(x)=\tfrac12\langle h_{x/2},h_{x/2}\rangle.$$

Proof. $F=g_{k,x}*\widetilde{g_{k,x}}$ has $\widehat F(t)=t^{2k}e^{-t^2x}$, so $\Phi(\rho)=\gamma_\rho^{2k}e^{-\gamma_\rho^2x}$. The summand is even in γ_ρ and $\rho\mapsto 1-\rho$ negates γ_ρ, so the sum over all zeros is twice the sum over the zeros with $\gamma>0$, which is $2T_k(x)$. □

Dividing the prime side of $\langle h_{x/2}, h_{x/2}\rangle$ by two gives the explicit formula (2.1.5) for W. Both sides give $W(0.003) = 1.09522012145284$ (the first 1700 zeros against the primes, agreeing to thirty digits). Since W is completely monotone exactly under RH (Proposition 2.1.1), RH is positivity of Weil's form on the diagonal of the two parameter family (k, x), with no cross terms.

15.15 Every large rung fails

The rungs $T_k(x) = \sum_{\gamma>0} w_\rho^k e^{-w_\rho x}$, with $w_\rho = \gamma_\rho^2$, satisfy $T_{k-1}' = -T_k$, and $T_{k-1}(x) \to 0$ as $x \to \infty$ because every exponent has real part $\gamma^2 - \delta^2 > \gamma_1^2 - \frac{1}{4} > 0$, where $\gamma_1 = 14.1347\ldots$ is the least ordinate. Hence

$$T_{k-1}(x) = \int_x^\infty T_k(s)\, ds, \qquad k \geq 1,\ x > 0. \tag{15.15.1}$$

So $T_k \geq 0$ on $(0,\infty)$ forces $T_{k-1} \geq 0$ there, the argument of Section 8.1.4 for the Williamson measures U_k, which are positive exactly when $T_k \geq 0$ on $(0,\infty)$.

Theorem 15.15.1. *(i) If $T_k \geq 0$ on $(0,\infty)$ then $T_j \geq 0$ on $(0,\infty)$ for every $j \leq k$, so the set of such orders is an initial segment of $\mathbb{N}$, and it is all of $\mathbb{N}$ if and only if RH holds. (ii) If RH fails there is K_0 such that for every $k \geq K_0$ there is $x > 0$ with $T_k(x) < 0$. Consequently, for any infinite set $\mathcal{K}$ of orders, RH holds if and only if $T_k \geq 0$ on $(0,\infty)$ for every $k \in \mathcal{K}$. In particular RH is equivalent to $\langle g_{k,x}, g_{k,x}\rangle \geq 0$ for all odd k and all $x > 0$, a positivity statement on odd Hermite Gaussians only.*

Proof. (i) The first claim is (15.15.1). All T_k are nonnegative on $(0,\infty)$ if and only if W is completely monotone, which holds if and only if RH holds (Proposition 2.1.1). (ii) If RH fails, some T_{k_0} takes a negative value, and then by (i) so does every T_k with $k \geq k_0$; let K_0 be the least such k_0. If $T_k \geq 0$ for every k in an infinite set $\mathcal{K}$, then by (i) it holds for every k, since $\mathcal{K}$ is unbounded, and RH holds; the converse is clear. The last statement is Proposition 15.14.1 for odd k. □

For infinite $\mathcal{K}$ this is the equivalence of Section 8.1.4 between RH and $U_k \geq 0$ for infinitely many k. New here are the reading of T_k as the diagonal of Yoshida's form, which makes the odd orders the Hermite counterpart of Yoshida's parity reduction (Y1), and the location of the failure in the next proposition, which the theorem does not need. With $y = u^2$ and $x = k/y$ the computation is the ray $t = k/u^2$ of Proposition 8.1.10: $e^{-k(\log y - 1)} T_k(k/y)$ is the quantity $e^{-f_0} c_k(t)$ there.

Proposition 15.15.2 (Where the rungs fail). *Suppose RH fails. For a zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\gamma > 0$ write $w_\rho = \gamma_\rho^2 = r_\rho e^{i\theta_\rho}$, so that $\theta_\rho = -2\arctan(\delta/\gamma)$ and $\sec\theta_\rho = (\gamma^2 + \delta^2)/(\gamma^2 - \delta^2)$. Then $S = \max \log \sec\theta_\rho > 0$ is attained, by finitely many zeros. Let $w_* = r_* e^{i\theta_*}$ be one maximiser, of multiplicity m, and*

$y_* = r_* \cos\theta_* = \gamma_*^2 - \delta_*^2$. *There are* $\eta > 0$ *and* K_1 *and, for every* $k \ge K_1$, *a point* y_k *with* $|y_k - y_*| = O(1/k)$ *such that*

$$e^{-k(\log y_k - 1)} T_k(k/y_k) = -2m\, e^{kS}\big(1 + O(1/k)\big) + O\big(e^{k(S-\eta)}\big) < 0.$$

Proof. Since $|\delta| < \frac{1}{2} < \gamma$, $\operatorname{Re} w_\rho = \gamma^2 - \delta^2 > 0$. The zeros ρ and $1 - \overline{\rho}$ give conjugate values of w, so T_k is real. Evaluate at $x = k/y$ and multiply by the positive factor $e^{-k(\log y - 1)}$:

$$e^{-k(\log y - 1)} T_k(k/y) = \sum_{\gamma > 0} e^{k g(w_\rho, y)}, \qquad g(w, y) = \log\frac{w}{y} - \frac{w}{y} + 1.$$

For fixed $w = re^{i\theta}$, $\operatorname{Re} g(w, y) = \log(r/y) - r\cos\theta/y + 1$ has a unique maximum in y, equal to $\log\sec\theta$, at $y = r\cos\theta$. On line zeros $\theta = 0$ and $\operatorname{Re} g \le 0$. Suppose an off line zero exists. Since $|\theta_\rho| < 1/\gamma \to 0$, only finitely many zeros have $|\theta_\rho|$ above any given positive bound, so $S = \max \log\sec\theta_\rho$ is attained, by finitely many zeros, and $S > 0$. Every zero other than w_* and $\overline{w_*}$ has, for y in a small neighbourhood J of y_*, $\operatorname{Re} g(w_\rho, y) \le S - \eta$ for some $\eta > 0$: on line zeros because $\operatorname{Re} g \le 0$, for zeros with $|\theta_\rho| < |\theta_*|$ because only finitely many have $\sec\theta_\rho$ near the maximum, and for the other maximisers because y_* is not their optimal point. Since $e^{ku} \le e^u$ for $u \le 0$ and $k \ge 1$, the remainder satisfies

$$\Big| \sum_{\rho \ne w_*, \overline{w_*}} e^{k g(w_\rho, y)} \Big| \le e^{k(S-\eta)} \sum_\rho e^{\operatorname{Re} g(w_\rho, y) - S + \eta} \le C\, e^{k(S-\eta)}, \qquad y \in J,$$

with C independent of k, because $\sum_\rho (r_\rho/y) e^{1 - r_\rho \cos\theta_\rho / y}$ converges uniformly on J. The main term is $2m\, e^{k \operatorname{Re} g(w_*, y)} \cos\big(k \operatorname{Im} g(w_*, y)\big)$ with $\operatorname{Im} g(w_*, y) = \theta_* - r_* \sin\theta_*/y$, whose derivative in y is $r_* \sin\theta_*/y^2 \ne 0$. Choose y_k with $k \operatorname{Im} g(w_*, y_k) \equiv \pi \pmod{2\pi}$ and $|y_k - y_*| = O(1/k)$. Then $\operatorname{Re} g(w_*, y_k) = S - O(1/k^2)$, the main term is $-2m\, e^{kS}(1 + O(1/k))$, and it dominates the remainder for $k \ge K_1$. Hence $T_k(k/y_k) < 0$. □

The proposition gives no explicit K_1, hence no explicit bound for K_0, because C depends on the zeros. Two constraints enter. The first is the phase condition: y_k lies at distance about $\pi y_*/(k \tan\theta_*)$ from y_*, which costs a factor $\exp(-\pi^2/(2k\tan^2\theta_*))$ in the main term, so the asymptotics start only once $k\theta_*^2$ is of order π. The second is the competition of the main term e^{kS}, with $S = \log\sec\theta_*$ and $\sec\theta_* - 1 = 2\delta_*^2/(\gamma_*^2 - \delta_*^2)$, with the nearest true zeros.

Remark 15.15.3 (A stress test). Add a hypothetical pair at $\gamma = 60.5$, $\delta = 0.45$ to the true zeros (only those between heights 40 and 85 matter at these orders: every excluded term is below $e^{-0.23k}$ and, relative to the main terms, below 10^{-420} for $k \ge 3349$), so that $\sec\theta_* - 1 = 1.1065 \times 10^{-4}$ and $y_* = 3660.05$, and test the sign of $T_k(k/y)$ on 3001 points of $|y/y_* - 1| \le 0.06$. The first failing order is $K_0 = 6349$. Every one of the 3000 orders in $[6349, 9348]$ fails, every one of 395 sampled orders

$6349 + 997j$ up to 399167 fails, and none of the orders 1 to 6348 fails on this grid. For the order 6348 the sign is positive at every point of two further grids (200001 points near y_*, with both local minima of the scale free margin polished, where the margin is at least 2.9×10^{-4}, and 18421 points of $y \in [1, 10^8]$ with all 1700 zeros), while at $y = 3584.65$, that is $x = 1.771$, arithmetic with 60 digits gives $T_{6348}(6348/y) = +1.289 \times 10^{19803}$ and $T_{6349}(6349/y) = -5.700 \times 10^{19805}$. By Theorem 15.15.1(i) positivity of T_{6348} gives that of every lower order, so 6349 is the first failing order of this configuration; all of this is computed, not proved. For this pair $K_0(\sec\theta_* - 1) = 0.7025$.

The first failing order found on a grid is an upper bound for K_0. For the pair $(\gamma, \delta) = (30.5, 0.45)$ the grid of the stress test gives 1064, while a grid of ± 30 percent gives 396, the negative lobe lying 27 percent from y_*; the other thirteen pairs of the next sentence give the same order on both grids. Over fourteen pairs, with γ from 30.5 to 250.5 and δ from 0.02 to 0.45, $K_0(\sec\theta_* - 1)$ lies between 0.03 and 0.82, and at fixed γ it is proportional to δ: the logarithmic slope of $K_0(\sec\theta_* - 1)$ against δ is 1.04 for $\gamma = 60.5$ and 1.05 for $\gamma = 120.5$, and the ratio $K_0(\sec\theta_* - 1)/\delta$ lies between 0.97 and 2.01 for thirteen of the fourteen pairs. So K_0 grows with γ^2 and, at fixed γ over the range tested, like $1/\delta$, not like $(\sec\theta_* - 1)^{-1} \approx \gamma^2/(2\delta^2)$: for $\gamma = 60.5$ and $\delta = 0.02$ the second quantity is 4.6×10^6 and the computed order is 128610. This is an observation from computation, not a theorem. A scale argument (the pair is a Gaussian envelope times a cosine in the variable $\sqrt{k}\log(y/y_*)$, and its negative lobe has to beat the true zeros that sit near it) suggests that K_0 is of order $\gamma^2/(\delta d)$, where d is the distance from γ to the nearest true ordinate, and that γ^2/δ^2 appears only when δ is of the size of d.

15.16 Compact support, the signed Thorin formula, and the first windows

Section 8.2.1 defines the Thorin functional U on the class $\mathcal{T}$ of the functions $\varphi(t) = h(\sqrt{t})$ by (8.2.1), and identifies it with $\varphi \mapsto \sum_k \varphi(a_k)$ through Weil's formula, with no hypothesis on the zeros. In the variable $t = r^2$ the same functional is the formal signed measure

$$U^{\pm}(dt) = \delta_{-1/4}(dt) + \frac{\operatorname{Re}\psi(\frac{1}{4} + \frac{i\sqrt{t}}{2}) - \log\pi}{4\pi\sqrt{t}}\,dt - \frac{1}{2\pi\sqrt{t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(\sqrt{t}\log n)\,dt, \tag{15.16.1}$$

that is $\langle U, \varphi\rangle = \int \varphi\, dU^{\pm}$: the atom is the first term of (8.2.1), the density is its integral term with $t = r^2$, and the third term is the prime sum. The prime series in (15.16.1) is not a tempered distribution in the ordinate $r = \sqrt{t}$. Let Z be the space of the $\widehat{F}$ with $F \in C_c^\infty(\mathbb{R})$, and Z' its dual.

Proposition 15.16.1. *In the ordinate $r = \sqrt{t}$ the prime series of* (15.16.1) *defines*

an element of Z', and for every even $F \in C_c^\infty(\mathbb{R})$ with support in $[-L,L]$,

$$\mathcal{Y}(F) = 2\int \widehat{F}(\sqrt{t})\,U^{\pm}(dt), \qquad \text{with } \widehat{F}(\sqrt{-1/4}) = \widehat{F}(i/2);$$

only the prime powers $n \le e^L$ contribute.

Proof. Pairing $\cos(r\log n)$ with $\widehat{F}(r)$ gives $\pi\big(F(\log n)+F(-\log n)\big)$, which vanishes for $\log n > L$; hence the series is a finite sum on every element of Z and defines an element of Z'. With $t = r^2$ the three parts of (15.16.1) give, after the factor 2, which is the passage from one zero of each pair $\{\rho, 1-\rho\}$ in (8.2.1) to all zeros in (15.13.1), the pole term $\widehat{F}(i/2)+\widehat{F}(-i/2)$, the term $V_1(F)-(\log\pi)F(0)$, and the prime sum of (15.13.1). The function $\varphi(t) = \widehat{F}(\sqrt{t})$ lies in $\mathcal{T}$, because $\widehat{F}$ is even, entire and $O((1+|r|)^{-N})$ in every strip. □

For $\varphi \in K(a)$ with $a = \frac12\log 2$, $F = \varphi * \widetilde{\varphi}$ is continuous, vanishes at $\pm\log 2$, and the only prime power level in its support is that endpoint. On Yoshida's window the form is the pole and archimedean part alone: his Theorem 1, as the first author reads it, uses only the gamma factor, the pole and the fact that Dirichlet series begin at $n = 2$. Connes and Consani reproved this window from the semilocal trace formula [44]. Beyond it the primes enter: the window of half length 0.8, of autocorrelation support 1.6, contains the prime powers 2, 3 and 4, and Zhu [279] has certified positivity of the form on it (Chapter 16). Windows are indexed by their half length a in the ladder of Section 9.2, each of them a finite problem in the prime powers up to e^{2a}.

15.17 Localisation: a modulated heat trace

Yoshida's localisation lemma (Y1) concentrates a test vector on one zero. Its Gaussian form is the smoothing $\mathcal{G}(u,t) = \sum_{\gamma>0} e^{-2t(\gamma_\rho - u)^2}$ of the zeros of Proposition 8.1.10, the sum over one zero of each pair $\{\rho, 1-\rho\}$ (there $\widetilde{\gamma}$ is the γ_ρ of this chapter). For real u and $x > 0$ let g be the vector with $\widehat{g}(t) = e^{-(t-u)^2x/2}$, so that $g * \widetilde{g} = h_x e^{-iuy}$ and $\widehat{g * \widetilde{g}}(t) = e^{-(t-u)^2x}$.

Proposition 15.17.1. *(i) $\langle g, g\rangle = \mathcal{Y}(h_x e^{-iuy}) = \sum_\rho e^{-(\gamma_\rho - u)^2 x} = \mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$, which is $2W(x)$ at $u = 0$. (ii) Unconditionally,*

$$\begin{aligned}\mathcal{G}(u,\tfrac{x}{2}) + \mathcal{G}(-u,\tfrac{x}{2}) = {}& 2e^{-(u^2-1/4)x}\cos(ux) - \frac{\log\pi}{\sqrt{4\pi x}} \\ & + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-(t-u)^2x}\,\mathrm{Re}\,\psi\Big(\frac14 + \frac{it}{2}\Big)dt \\ & - \frac{1}{\sqrt{\pi x}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(u\log n)\,e^{-(\log n)^2/(4x)}.\end{aligned}$$

(iii) Under RH it equals $\int \big[e^{-(\sqrt{t}-u)^2x} + e^{-(\sqrt{t}+u)^2x}\big]U(dt)$. *(iv) RH holds if and only if it is nonnegative for all real* u *and all* $x > 0$.

Proof. (i) The transform $e^{-(t-u)^2x}$ is entire, so (15.13.1) applies; the zero $1-\rho$ of the pair $\{\rho, 1-\rho\}$ has $\gamma_{1-\rho} = -\gamma_\rho$ and gives $\mathcal{G}(-u, \frac{x}{2})$, and at $u = 0$ the sum is $2W(x)$ by Proposition 15.14.1. (ii) Apply (15.13.1) to $F = h_x e^{-iuy}$: the pole term is $\widehat{F}(i/2) + \widehat{F}(-i/2) = e^{-(u+i/2)^2x} + e^{-(u-i/2)^2x} = 2e^{-(u^2-1/4)x}\cos(ux)$, the term $-(\log\pi)F(0)$ is $-(\log\pi)/\sqrt{4\pi x}$, $V_1(F)$ is the integral, and the prime sum is $-\sum \Lambda(n)n^{-1/2}h_x(\log n)\big(e^{-iu\log n} + e^{iu\log n}\big)$. This is (8.1.3) at $t = x/2$, term by term: its polar term $2e^{t/2-2tu^2}\cos 2tu$, its integral $A(u,t)$ (with $m(r) = \operatorname{Re}\psi(\frac{1}{4}+\frac{ir}{2}) - \log\pi$) and its prime term $P(u,t)$ are the four terms above, because $2t = x$. (iii) Group $\pm\gamma$ in $\sum_{\gamma>0}\big[e^{-x(\gamma-u)^2} + e^{-x(\gamma+u)^2}\big]$ and use $U = \sum_{\gamma>0}\delta_{\gamma^2}$. (iv) This is Proposition 8.1.11, in the form of the remark after its proof: the term $\mathcal{G}(-u,t)$, which comes from the zeros $1-\rho$, is too small there to change the sign. □

Table 8.3 computes the identity (ii). A vector modulated to a centre u within $\frac{\delta}{2}$ of the ordinate γ of an off line zero $\rho = \frac{1}{2} + \delta + i\gamma$, with finitely many exceptional centres, makes the form negative at arbitrarily large x (Proposition 8.1.11), and u is free: modulation reaches a zero above any verified height, which is why the barrier of Proposition 15.21.1 is stated for the tent coefficient only.

The ladder reaches the height u only at order $k \approx u^2x$, because the weight $r^{2k}e^{-r^2x}$ of $T_k(x)$ peaks at $r = \sqrt{k/x}$ and the power r^{2k} is its only way to move the window; Theorem 8.1.13 states the consequence, that verification to a height H reaches the orders up to about the square of H. Modulation moves it at no cost. Figure 15.1 shows the modulated heat trace resolving isolated zeros into bumps of height one at $x = 0.5$; nearby zeros overlap, and zeros less than 2 apart merge into one bump (computed).

15.18 The tent and the sine squared lemma

Yoshida's basic vector is $\chi_0 = (2a)^{-1/2}1_{[-a,a]}$, and $\chi_0 * \widetilde{\chi_0} = (2a)^{-1}(2a - |y|)_+$. Put $x = 2a$,

$$\Delta_x(y) = (x - |y|)_+, \qquad D(x) = \mathcal{Y}(\Delta_x) = x\,\langle \chi_0, \chi_0\rangle,$$

with χ_0 taken for $a = x/2$: Δ_x is the tent of height x on $[-x, x]$, and $D(x)$ is the coefficient $\langle\chi_0,\chi_0\rangle$ of Yoshida's matrix times the window length x.

Proposition 15.18.1. *Let* $k_x(w) = e^{-wx} + wx - 1$ *be the compensated exponential on the left of the sine squared lemma* (2.1.9)*, and let* $\gamma_\rho = -i(\rho - \frac{1}{2})$*, which is complex for an off line zero.*

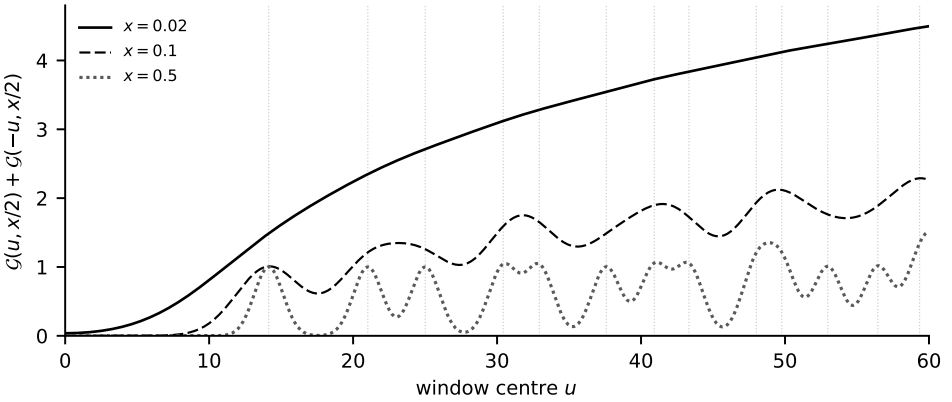


Figure 15.1: The modulated heat trace $\mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$ against the window centre u at $x = 0.02$ (solid), 0.1 (dashed) and 0.5 (dotted), from the first 1700 zeros; dotted verticals mark the ordinates of the zeros.

(a) For complex w and real t,

$$\int_{\mathbb{R}} \Delta_x(y) e^{wy}\, dy = \frac{k_x(w) + k_x(-w)}{w^2} = \frac{4\sinh^2(wx/2)}{w^2},$$

$$\widehat{\Delta_x}(t) = \frac{4\sin^2(xt/2)}{t^2} \geq 0.$$

(b) Unconditionally $D(x) = \sum_\rho 4\sin^2(\gamma_\rho x/2)/\gamma_\rho^2$, the sum over all nontrivial zeros. Under RH, with the Thorin measure $U = \sum_{\gamma>0} \delta_{\gamma^2}$ of Proposition 2.1.1,

$$D(x) = 8\int \frac{\sin^2(x\sqrt{t}/2)}{t}\, U(dt)$$
$$= \frac{8\pi}{x}\int \frac{x\sin^2(x\sqrt{t}/2)}{\pi\sqrt{t}}\, \frac{U(dt)}{\sqrt{t}},$$

so D samples the Thorin density of the sine squared lemma (2.1.9) *at level x on the Thorin atoms $t = \gamma^2$, with the weight $t^{-1/2}$.*

(c) Unconditionally,

$$D(x) = 32\sinh^2\frac{x}{4} - x\log\pi - 2\sum_{n\leq e^x} \frac{\Lambda(n)}{\sqrt{n}} \log\frac{e^x}{n} + V_1(\Delta_x),$$

$$V_1(\Delta_x) = x - x\log x - (\gamma_E + \log 2)x$$
$$- 2\int_0^x \Big(\frac{e^{y/2}}{2\sinh y} - \frac{1}{2y}\Big)(x-y)\, dy,$$

and the prime term is a Riesz mean of order one of $\Lambda(n)n^{-1/2}$.

(d) With ψ the digamma function, $V_1(\Delta_x)$ is also the residue series

$$V_1(\Delta_x) = x\,\psi(\tfrac14) + \tfrac12\psi'(\tfrac14) - 2\sum_{k\ge0}\frac{e^{-(2k+1/2)x}}{(2k+\frac12)^2}.$$

Proof. (a) Taylor's formula with integral remainder gives $k_x(w) = w^2\int_0^x (x-y)e^{-wy}dy$, so $\int \Delta_x e^{wy}dy = (k_x(w)+k_x(-w))/w^2$, and $k_x(w)+k_x(-w) = e^{wx} + e^{-wx} - 2 = 4\sinh^2(wx/2)$; put $w = it$.

(b) Evaluate $\widehat{\Delta_x}$ at γ_ρ and sum over the zeros; the series converges absolutely, since $|\sin^2(x\gamma_\rho/2)| \le \cosh^2(x/4)$ for $|\operatorname{Im}\gamma_\rho| < \frac12$ and $\sum|\gamma_\rho|^{-2} < \infty$. Under RH group $\pm\gamma$, which gives $D(x) = 2\sum_{\gamma>0} 4\sin^2(\gamma x/2)/\gamma^2$.

(c) Insert Δ_x in (15.13.1). By (a) at $w = \pm\frac12$ the first term is $2\cdot 4\sinh^2(x/4)/(1/4) = 32\sinh^2(x/4)$, the second is $-(\log\pi)\Delta_x(0) = -x\log\pi$, and $\Delta_x(\log n) + \Delta_x(-\log n) = 2(x-\log n)_+$. For V_1 use the second form of (15.13.2), with $\operatorname{Pf}(1/|y|)(\Delta_x) = 2(x\log x - x)$ for all $x > 0$, because $\int_{\epsilon<|y|<x}(x-|y|)\,dy/|y| = 2x\log(x/\epsilon) - 2x + 2\epsilon$. To see that (15.13.1) holds for the tent, apply it to $\Delta_x * h_\eta$, with $h_\eta(y) = (4\pi\eta)^{-1/2}e^{-y^2/(4\eta)}$, whose transform $\widehat{\Delta_x}(t)e^{-\eta t^2}$ decays rapidly in every strip, and let $\eta\downarrow 0$. The zero sum converges by dominated convergence, since $|e^{-\eta\gamma_\rho^2}| \le e^{\eta/4}$ and the bound of (b) holds; the Fourier form of V_1 converges because $\widehat{\Delta_x}(t) = O(t^{-2})$; and the prime sum converges because $\Delta_x * h_\eta \to \Delta_x$ uniformly with Gaussian tails.

(d) By the Fourier form of (15.13.2), $V_1(\Delta_x) = \frac{1}{2\pi}\int \widehat{\Delta_x}(t)\psi(\frac14+\frac{it}{2})\,dt$, because the imaginary part of $\psi(\frac14+\frac{it}{2})$ is odd in t and $\widehat{\Delta_x}$ is even. With $w = it$ and (a) this is

$$V_1(\Delta_x) = \frac{1}{2\pi i}\int_{\operatorname{Re} w=0}\frac{e^{wx}+e^{-wx}-2}{w^2}\,\psi\big(\tfrac14+\tfrac{w}{2}\big)\,dw.$$

The integrand is regular at $w = 0$, so the line may be moved to $\operatorname{Re} w = -\frac14$. Split the numerator into its three terms, and close the contour to the left for e^{wx} and to the right for e^{-wx} and for -2; the arcs, which pass between the poles of ψ, contribute nothing in the limit, because the integrands are $O(|w|^{-2}\log|w|)$ on them. On the left, $\psi(\frac14+\frac{w}{2})$ has simple poles at $w = -(2k+\frac12)$ with residue -2, which gives $-2\sum_k e^{-(2k+1/2)x}/(2k+\frac12)^2$. On the right the double pole at $w = 0$ gives, with the clockwise orientation, $x\psi(\frac14) - \frac12\psi'(\frac14)$ from e^{-wx} and $\psi'(\frac14)$ from -2. □

The kernel $K(w,x) = \int\Delta_x(y)e^{wy}dy = (k_x(w)+k_x(-w))/w^2$ of (a) has two readings. The sine squared lemma fixes w and sums over the level x against the canonical measure of Section 8.2, built from $\Lambda(n)n^{-1/2}$ at $x = \log n$, the gamma factor and the pole; Yoshida fixes the level x and sums over the zeros.

The tent in the variogram form. Put $\varphi_x(z) = \widehat{\Delta_x}(\sqrt{z}) = 4\sin^2(x\sqrt{z}/2)/z$. By (b) and (c), $D(x) = 2\langle U, \varphi_x\rangle$, where $\langle U, \cdot\rangle$ is the Thorin functional of Section 8.2.1, and (c) is twice (8.2.1) for φ_x. The tent lies just outside the class $\mathcal{T}$ of that section, because $\widehat{\Delta_x}(r) = O(r^{-2})$, and the limiting argument in the proof of (c) takes the place of the decay condition. In the variogram form of Weil's criterion (Section 5.1.2) the tent is the function $g = \Delta_x$ of (5.1.2): its transform $h = \widehat{\Delta_x}$ is nonnegative, and its variogram is $g(0) - g(u) = \min(u, x)$ for $u \geq 0$,

$$\int_0^\infty \varphi_x(z)\,\frac{2\sin^2(u\sqrt{z}/2)}{2\pi\sqrt{z}}\,dz = \min(u, x),$$

since both sides are $\pi^{-1}\int_0^\infty \widehat{\Delta_x}(r)(1-\cos ur)\,dr$, and (5.1.2) needs only $h \in L^1$. Theorem 15.18.2 is therefore Weil's criterion restricted to the one parameter family of tent variograms $\min(u, x)$, $x > 0$.

THEOREM 15.18.2. *RH holds if and only if $D(x) \geq 0$ for all $x > 0$, if and only if D is bounded below. Equivalently, RH holds if and only if the single coefficient $\langle \chi_0, \chi_0\rangle$ is nonnegative for every $a > 0$.*

Proof. Under RH every term in (b) is nonnegative. Conversely put $\omega(x) = \sum_\rho e^{(\rho-1/2)x}/(\rho-\frac{1}{2})^2$, which converges absolutely. Using $\rho \mapsto 1-\rho$ and $(\rho-\frac{1}{2})^2 = -\gamma_\rho^2$ we get $D = 2\omega + 4p_1$, where $p_1 = \sum_k a_k^{-1} = \frac{1}{2}\sum_\rho \gamma_\rho^{-2}$ is the mean of the clock (Section 2.1). For $\operatorname{Re} s > \frac{1}{2}$,

$$\int_0^\infty \omega(x)e^{-sx}dx = R(s) := \sum_\rho \frac{1}{(\rho-\frac{1}{2})^2\,(s-\rho+\frac{1}{2})},$$

which is meromorphic on $\mathbb{C}$ with poles only at the points $\rho - \frac{1}{2}$, and with the nonzero residues $m/(\rho-\frac{1}{2})^2$, m the multiplicity. If $D \geq -C$, then $f = \omega + 2p_1 + C/2 \geq 0$ has the Laplace transform $R(s) + (2p_1 + C/2)/s$. By Landau's theorem [274, Chapter II, Theorem 5b] this transform is singular at the real point of its abscissa of convergence σ_c. No $\rho - \frac{1}{2}$ is real, because ξ has no real zero, so the only real singularity is 0, $\sigma_c \leq 0$, and the transform is holomorphic in $\operatorname{Re} s > 0$; hence no zero has $\operatorname{Re}(\rho - \frac{1}{2}) > 0$, and by the symmetry $\rho \mapsto 1-\rho$ none has $\operatorname{Re}(\rho - \frac{1}{2}) < 0$. □

The equivalence is classical in substance (Landau's theorem applied to a Riesz mean, as in Ingham [175]); its interest is its place in Yoshida's matrix and in the lemma. Yoshida's Theorem 1 gives $D > 0$ on $(0, \log 2]$ (Section 15.13), where no prime power contributes to (c), and $\langle \chi_0, \chi_0\rangle = D(\log 2)/\log 2 = 0.18338$ at $a = \frac{1}{2}\log 2$. Computed from the prime side (c), which is explicit between consecutive prime powers, on a grid of step 10^{-4}, $D(x) \geq 0.0550$ on $[0.05, 16]$, with the minimum at $x = 8.04$, and the mean of D on $[2, 16]$ is 0.092409 against $4p_1 = 0.092420$. The difference -1.07×10^{-5} is the mean of the oscillating part

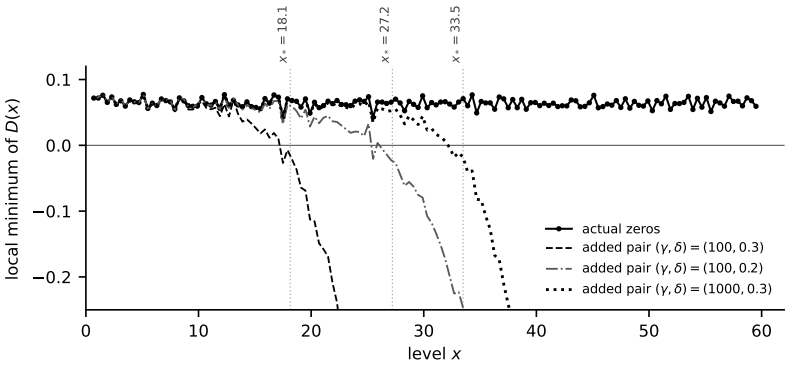


Figure 15.2: The least value of D over windows of length 0.4 on $[0, 60]$, for the actual zeros (solid, the first 1700 zeros with the smooth tail; the least window minimum is 0.0427, in the window centred at $x = 25.5$) and with one hypothetical off line pair added at height γ and displacement δ, $(\gamma, \delta) = (100, 0.3)$ (dashed), $(100, 0.2)$ (dash dotted) and $(1000, 0.3)$ (dotted). Vertical dotted lines mark the predicted levels $x_* = 18.1$, 27.2 and 33.5 of Remark 15.18.3.

2ω: it equals $-\frac{4}{14}\sum_{\gamma>0}(\sin 16\gamma - \sin 2\gamma)/\gamma^3$, which the first 1700 zeros give to the same digits (computed).

Remark 15.18.3 (Detection scale). A hypothetical off line zero $\rho = \frac{1}{2} + \delta + i\gamma$ with its three companions adds to D the term $4\,\mathrm{Re}\big[4\sin^2\big((\gamma - i\delta)x/2\big)/(\gamma - i\delta)^2\big]$, whose amplitude is about $4e^{\delta x}/\gamma^2$ with oscillating sign. Against the mean level $4p_1$ of D this makes D first turn negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$. For $(\gamma, \delta) = (100, 0.3)$, $(100, 0.2)$, $(1000, 0.3)$ the first negative values of D plus the pair term occur at $x = 17.3$, 25.4, 32.4, against the predictions 18.1, 27.2, 33.5 (Figure 15.2); the predictions are upper estimates. On a grid of step 0.01 the third value is 32.83 and not 32.4, because at $\gamma = 1000$ the pair term has period $2\pi/\gamma = 0.0063$, shorter than the step. A negative value of the single coefficient $D(2a)$ shows that the form is not positive on $C(a)$, so Yoshida's threshold a_0 of Section 15.13 is at most about $x_*/2$, logarithmic in the height. This is a bound, not an estimate, because the tent is one vector of the window: a vector that vanishes at the on line ordinates near γ sees the pair earlier, and Table 16.3 of Chapter 16 gives, for the full form on a window, detection at half lengths far below $x_*/2$ in the cases computed there (a heuristic experiment of that chapter, with its own cutoff).

Table 15.9 compares the two sides of Proposition 15.18.1. The prime side is (c) with the exact $\Lambda(n)$. The zero side is $\sum_{\gamma>0} 8\sin^2(\gamma x/2)/\gamma^2$ over the first 1500 zeros (to height 1980.91) plus the mean tail $2(\log(T/2\pi) + 1)/(\pi T)$ at $T = \gamma_{1500}$. The two sides agree to 3.6×10^{-6} except at $x = \log 2$ and $x = \log 5$, where they

differ by 1.6×10^{-4} and 2.3×10^{-4}. These are corners of the prime side, where its slope drops by $2\Lambda(n)/\sqrt{n}$, which a zero sum cut at T and completed by a mean tail cannot follow: by Landau's formula [187] for $\sum_{0<\gamma\le T} n^{i\gamma}$ the zeros above T contribute $2\Lambda(n)/(\pi\sqrt{n}\,T)$ at $x=\log n$, which is 1.575×10^{-4} and 2.313×10^{-4} at $T=\gamma_{1500}$, and with it the two corner rows agree to 1.3×10^{-6}. The last two columns agree to 10^{-29} in every row (eight decimals are printed).

x	prime side	zero side	$V_1(\Delta_x)$ by (c)	by (d)
0.2	0.1089996	0.1089981	0.25787887	0.25787887
0.5	0.0803252	0.0803270	0.15008051	0.15008051
$\log 2$	0.1271112	0.1269547	−0.04998528	−0.04998528
1.0	0.0880146	0.0880147	−0.50847597	−0.50847597
1.5	0.0898189	0.0898186	−1.52909231	−1.52909231
$\log 5$	0.1343264	0.1340964	−1.78866457	−1.78866457
2.0	0.1066822	0.1066850	−2.80144646	−2.80144646
3.0	0.0794186	0.0794176	−5.86891443	−5.86891443
4.0	0.0702819	0.0702855	−9.39384635	−9.39384635
6.0	0.0962464	0.0962437	−17.16435327	−17.16435327

Table 15.9: The tent value $D(x)$ from the primes (Proposition 15.18.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x=\log 2$ and $x=\log 5$ differ by Landau's term, see the text.

15.19 The sine squared lemma off the diagonal: Yoshida's matrix is a Löwner matrix

Recall Yoshida's basis $\chi_n=(2a)^{-1/2}e^{\pi iny/a}1_{[-a,a]}$ of $K(a)$ (Section 15.13, [278]), and write $\nu_n=\pi n/a$. The Fourier transform of the basis vector is

$$\widehat{\chi_n}(t)=(2a)^{-1/2}\int_{-a}^{a}e^{i(t+\nu_n)y}\,dy=(2a)^{-1/2}\,\frac{2(-1)^n\sin(at)}{t+\nu_n},$$

because $\sin(a(t+\nu_n))=(-1)^n\sin(at)$: every basis vector carries the factor $\sin(at)$, and only the pole moves. The Löwner matrix of a function g on the nodes ν_n has the entries $(g(\nu_n)-g(\nu_m))/(\nu_n-\nu_m)$ off the diagonal and $g'(\nu_n)$ on it, and a Pick function is a function holomorphic on the upper half plane with nonnegative imaginary part there.

Theorem 15.19.1. *(i) Unconditionally,*

$$\langle\chi_n,\chi_m\rangle=(-1)^{n+m}\,\frac{2}{a}\sum_{\rho}\frac{\sin^2(a\gamma_\rho)}{(\gamma_\rho+\nu_n)(\gamma_\rho+\nu_m)},$$

an absolutely convergent series.

(ii) Let $g_a(\nu)=\sum_\rho \sin^2(a\gamma_\rho)/(\gamma_\rho-\nu)$, *the sum over all zeros taken in symmetric pairs; unconditionally it is an odd meromorphic function of* ν. *Then for* $n\neq m$

$$\langle \chi_n,\chi_m\rangle=(-1)^{n+m}\,\frac{2}{a}\,\frac{g_a(\nu_n)-g_a(\nu_m)}{\nu_n-\nu_m},\qquad \langle \chi_n,\chi_n\rangle=\frac{2}{a}\,g_a'(\nu_n),$$

so Yoshida's matrix is, up to the sign change $\operatorname{diag}((-1)^n)$, $\frac{2}{a}$ *times the Löwner matrix of* g_a *on the lattice* $\frac{\pi}{a}\mathbb{Z}$.

(iii) Under RH, with the weighted Thorin measure $U_a^\sharp(dt)=2\sin^2(a\sqrt{t})\,U(dt)\geq 0$ *and its Stieltjes transform* $\mathcal{S}_a(s)=\int U_a^\sharp(dt)/(t+s)$,

$$g_a(\nu)=\nu\,\mathcal{S}_a(-\nu^2),$$
$$\langle \chi_n,\chi_m\rangle=(-1)^{n+m}\frac{2}{a}\int \frac{2\sin^2(a\sqrt{t})\,(t+\nu_n\nu_m)}{(t-\nu_n^2)(t-\nu_m^2)}\,U(dt),$$

and g_a *is an odd Pick function. By the elementary half of Löwner's theorem every finite Yoshida matrix is then positive semidefinite, for every* a.

(iv) For each fixed $a>0$, *RH holds if and only if* g_a *is a Pick function.*

Proof. (i) $F=\chi_n*\widetilde{\chi_m}$ has $\widehat{F}(t)=\widehat{\chi_n}(t)\overline{\widehat{\chi_m}(\bar t)}=(-1)^{n+m}\frac{2}{a}\sin^2(at)/((t+\nu_n)(t+\nu_m))$, entire because $\sin^2(at)$ vanishes at $-\nu_n$ and $-\nu_m$. It is $O(t^{-2})$ in every strip, and F is continuous, piecewise smooth and compactly supported, so (15.13.1) applies to it as it does to the tent (proof of Proposition 15.18.1(c)). Sum $\widehat{F}(\gamma_\rho)$.

(ii) Replace γ_ρ by $-\gamma_\rho$ (the zero set is symmetric), then use $\frac{1}{(\gamma-\nu_n)(\gamma-\nu_m)}=\frac{1}{\nu_n-\nu_m}\big(\frac{1}{\gamma-\nu_n}-\frac{1}{\gamma-\nu_m}\big)$; the diagonal is the limit. The series of g_a converges in symmetric pairs, since the pair $\pm\gamma_\rho$ contributes $2\nu\sin^2(a\gamma_\rho)/(\gamma_\rho^2-\nu^2)=O(|\nu|\,|\gamma_\rho|^{-2})$, and $g_a(-\nu)=-g_a(\nu)$ because the zero set is symmetric. A zero with $\gamma_\rho=\nu_n$ contributes nothing at the lattice, since $\sin^2(a\gamma_\rho)=0$.

(iii) Under RH group $\pm\gamma$: $\sin^2(a\gamma)\big(\frac{1}{\gamma-\nu}-\frac{1}{\gamma+\nu}\big)=\frac{2\nu\sin^2(a\gamma)}{\gamma^2-\nu^2}$, and with $t=\gamma^2$ this is $\nu\int U_a^\sharp(dt)/(t-\nu^2)$. In the same way the pair $\pm\gamma$ in (i) gives $2\sin^2(a\gamma)(\gamma^2+\nu_n\nu_m)/((\gamma^2-\nu_n^2)(\gamma^2-\nu_m^2))$, which is the second formula. Each atom $\nu/(t-\nu^2)$ has imaginary part $\operatorname{Im}\nu\,(t+|\nu|^2)/|t-\nu^2|^2>0$ for $\operatorname{Im}\nu>0$, so g_a is a Pick function. For a Pick function g and real nodes $\nu_1,\dots,\nu_N$, the matrix $[(g(\nu_i)-g(\nu_j))/(\nu_i-\nu_j)]$ is positive semidefinite, which is Löwner's easy direction [197, 66].

(iv) If RH fails, an off line pair gives the nonreal ordinates $\pm\gamma\pm i\delta$, two of them in the upper half plane, where g_a has a pole with residue $-m\sin^2(a\gamma_\rho)$, m the multiplicity. This residue is nonzero because sin vanishes only on the real axis, and a Pick function is holomorphic in the upper half plane. The converse is (iii). □

The mechanism of the theorem is that of the rigidity of the clock, Proposition 2.1.1: there G'/G is the Stieltjes transform of U, holomorphic off the negative real axis exactly when every zero is on the line, and $\mathcal{S}_a$ is the same transform with the atoms weighted by $2\sin^2(a\gamma)$, sampled on the cut at the squared modal frequencies ν_n^2 and differenced.

Numerically, the prime side of $\langle \chi_n, \chi_m \rangle$, computed from (15.13.1) and (15.13.2) by Gauss Legendre quadrature, gives $\langle \chi_0, \chi_0 \rangle = 0.18338$ at $a = \frac{1}{2}\log 2$, and at $a = 1$ it matches part (i) summed over the first 1500 zeros entry by entry; the tail of the smooth count from $T = 1982.0$ accounts for the difference up to 9×10^{-7}.

15.20 Basepoints: damped forms, multipliers and sharp coercivity

The Thorin programme leaves the critical line for a basepoint $\alpha > \frac{1}{2}$, where the Thorin density of the shifted clock is $\rho_{\alpha-1/2} = \pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau)$ (equation (2.1.7)). On Yoshida's side the same move damps the autocorrelation $\varphi * \widetilde{\varphi}$ by $e^{-(\alpha-1/2)|y|}$.

15.20.1 The multiplier identity

For a basepoint $\alpha \geq \frac{1}{2}$ put $\varepsilon = \alpha - \frac{1}{2}$ and define the damped Yoshida form

$$\langle \varphi_1, \varphi_2 \rangle_\alpha = \mathcal{Y}\big(e^{-\varepsilon|y|}\, \varphi_1 * \widetilde{\varphi_2}\big),$$

where $\mathcal{Y}$ is the functional of (15.13.1) and $\widetilde{\varphi}(y) = \overline{\varphi(-y)}$; at $\alpha = \frac{1}{2}$ it is the form of Section 15.13, and $C(a) \subset K(a)$ are the spaces of that section. On the prime side the damping replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$, and in the pole term $e^{y/2} + e^{-y/2}$ becomes $e^{(1-\alpha)|y|} + e^{-\alpha|y|}$; on $K(a)$ the prime sum is still finite, since only $\log n \leq 2a$ occurs. Let $\sigma_{\max} = \sup_\rho \operatorname{Re}\rho \in [\frac{1}{2}, 1]$, let $v_\alpha(\tau) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\tau)$ and $b_\alpha = v_\alpha(0) = \xi'(\alpha)/\xi(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (2.1.7), and write $P_w(\tau) = w/(\pi(w^2+\tau^2))$ for the Poisson kernel of width w, with $P_{-w} = -P_w$, as in Proposition 2.1.3.

Proposition 15.20.1. *Let $\alpha \geq \sigma_{\max}$, and suppose that no zero has real part α. Then ρ_ε is real analytic and positive, $(1+\tau^2)^{-1}\rho_\varepsilon$ is integrable over $\mathbb{R}$, and for every $a > 0$ and every $\varphi \in K(a)$*

$$\langle \varphi, \varphi \rangle_\alpha = \int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2\, \rho_\varepsilon(\tau)\, d\tau = \frac{1}{\pi}\int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2\, v_\alpha(\tau)\, d\tau.$$

Proof. For $w > 0$ the kernel P_w has Fourier transform $\int P_w(\tau) e^{i\tau y}\, d\tau = e^{-w|y|}$, so for a bounded F with compact support and $\gamma \in \mathbb{R}$, $\int \widehat{F}(\tau) P_w(\tau - \gamma)\, d\tau = \int F(y) e^{-w|y|} e^{i\gamma y}\, dy$. Let $F = \varphi * \widetilde{\varphi}$, which is continuous, piecewise smooth and

supported in $[-2a,2a]$, so that (15.13.1) applies to $e^{-\varepsilon|y|}F$ and

$$\mathcal{Y}(e^{-\varepsilon|y|}F)=\sum_{\rho}\int F(y)e^{-\varepsilon|y|}e^{i\gamma_\rho y}\,dy,$$

the series converging absolutely. Group the zeros in fours $\{\rho,\bar\rho,1-\rho,1-\bar\rho\}$, counted with multiplicity; for a zero on the critical line the group is $\{\rho,\bar\rho\}$. For the four zeros $\frac12\pm\delta\pm i\gamma$ with $\gamma>0$ and $\delta>0$ we have $e^{i\gamma_\rho y}=e^{\pm i\gamma y}e^{\pm\delta y}$, so their contributions add up to

$$\begin{aligned}&\int F(y)e^{-\varepsilon|y|}\big(e^{i\gamma y}+e^{-i\gamma y}\big)\big(e^{\delta y}+e^{-\delta y}\big)dy\\&\quad=\int F(y)\big(e^{-(\varepsilon-\delta)|y|}+e^{-(\varepsilon+\delta)|y|}\big)\big(e^{i\gamma y}+e^{-i\gamma y}\big)dy,\end{aligned}$$

because $e^{\delta y}+e^{-\delta y}$ is even. The widths $\varepsilon-\delta=\alpha-\beta$ and $\varepsilon+\delta=\alpha-(1-\beta)$, with $\beta=\frac12+\delta$, are the differences between α and the real parts of the zeros of the group, and they are positive, since every zero has real part less than α. By the identity for P_w the group contributes $\int\widehat F(\tau)\sum P_{\alpha-\beta_{\rho'}}(\tau-\operatorname{Im}\rho')\,d\tau$, the sum over its four members ρ' with real part $\beta_{\rho'}$; the same holds for a zero on the line, with the single width ε. Hence $\mathcal{Y}(e^{-\varepsilon|y|}F)=\sum_\rho\int\widehat F(\tau)P_{\alpha-\beta_\rho}(\tau-\operatorname{Im}\rho)\,d\tau$ over all zeros, where $\beta_\rho=\operatorname{Re}\rho$. Now $\widehat F=|\widehat\varphi|^2\ge0$ on $\mathbb{R}$, so every term is nonnegative and the sum and the integral may be interchanged. By Proposition 2.1.3, which is Hadamard's product, $\sum_\rho P_{\alpha-\beta_\rho}(\tau-\operatorname{Im}\rho)=\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)=\rho_\varepsilon(\tau)$, a sum of positive kernels; it is real analytic because no zero lies on the line $\operatorname{Re}s=\alpha$. Finally $\int P_w(\tau-t)(1+\tau^2)^{-1}d\tau=(1+w)/((1+w)^2+t^2)\le(1+\alpha)/(1+t^2)$ for $0<w\le\alpha$, so by Tonelli $\int(1+\tau^2)^{-1}\rho_\varepsilon\,d\tau\le(1+\alpha)\sum_\rho(1+(\operatorname{Im}\rho)^2)^{-1}<\infty$, because the number of zeros with $|\operatorname{Im}\rho|\le T$ is $O(T\log T)$. The proof used only that F is continuous, piecewise C^1 and supported in $[-2a,2a]$, so the identity holds for every bounded piecewise C^1 function φ supported in $[-a,a]$. □

For the Fejér taper $\varphi=e^{ity}\chi_0$ on the window $a=X/2$, $\varphi*\widetilde{\varphi}=e^{ity}(1-|y|/X)_+$, and since $\mathcal{Y}(F)$ depends only on the even part of F, $\langle\varphi,\varphi\rangle_\alpha=\mathcal{Y}\big(e^{-\varepsilon|y|}\cos(ty)(1-|y|/X)_+\big)=2A_X(\varepsilon,t)$ in the notation of Proposition 19.14.1. Its part (c), $A_X(\varepsilon,t)\to v_\alpha(t)$ as $X\to\infty$ when $\alpha>\sigma_{\max}$ (so that every zero has $|\beta-\frac12|<\varepsilon$), is the concentration argument of Theorem 15.20.5(i) below for this family of test functions.

Remark 15.20.2. For $\alpha<\sigma_{\max}$ the two sides of the identity differ: a zero with $\beta>\alpha$ contributes the growing factor $e^{(\beta-\alpha)|y|}$ to the damped Weil form, but the negative kernel $P_{\alpha-\beta}=-P_{\beta-\alpha}$ to the symbol. A computation checks the identity at $\alpha=1$, $a=0.8$ and $\varphi=\chi_0$.

Theorem 15.20.3. *(i) For $\alpha\ge\sigma_{\max}$ the damped form is positive semidefinite on $C_c^\infty(\mathbb{R})$; unconditionally this holds for every $\alpha\ge1$.*

(ii) RH holds if and only if the damped form is positive semidefinite for every $\alpha > \frac{1}{2}$.

(iii) If the damped form at α *is positive semidefinite on* $C(a)$*, so is the form at every* $\alpha' > \alpha$*; thus* $a_0(\alpha) = \sup\{a : \langle\cdot,\cdot\rangle_\alpha \geq 0 \text{ on } C(a)\}$ *is nondecreasing in* α.

Proof. (i) If no zero has real part α this is Proposition 15.20.1, because $\rho_\varepsilon \geq 0$ and every $\varphi \in C_c^\infty(\mathbb{R})$ lies in some $K(a)$. If a zero has real part α, then $\alpha = \sigma_{\max}$ and $\langle\varphi,\varphi\rangle_\alpha = \lim_{\alpha' \downarrow \alpha} \langle\varphi,\varphi\rangle_{\alpha'} \geq 0$, since for fixed φ each part of the form is continuous in ε: the pole term and the finite prime sum obviously, and $V_1(e^{-\varepsilon|y|}F)$ because $\widehat{e^{-\varepsilon|y|}F} = P_\varepsilon * \widehat{F}$ with $|\widehat{F}(t)| \leq C(1+t^2)^{-1}$, which gives a bound $C'(1+t^2)^{-1}$ uniform in $\varepsilon \leq 1$, while $\operatorname{Re}\psi(\frac{1}{4}+\frac{it}{2}) = O(\log(2+|t|))$, and dominated convergence applies to the Fourier form of V_1 in (15.13.2). Unconditionally $\sigma_{\max} \leq 1$ and no zero has real part 1, so every $\alpha \geq 1$ is covered.

(ii) Under RH, $\sigma_{\max} = \frac{1}{2}$ and (i) applies to every $\alpha > \frac{1}{2}$. Conversely, positivity for a sequence $\alpha_n \downarrow \frac{1}{2}$ gives $\langle\varphi,\varphi\rangle \geq 0$ for every $\varphi \in C_c^\infty(\mathbb{R})$ by the same continuity, which is Weil's criterion (Section 15.13).

(iii) Write $e^{-\varepsilon'|y|} = \int P_{\varepsilon'}(\lambda) e^{i\lambda y}\, d\lambda$ with $\varepsilon' = \alpha' - \alpha$. Since $(\varphi_\lambda * \widetilde{\varphi_\lambda})(y) = e^{i\lambda y}(\varphi * \widetilde{\varphi})(y)$ for $\varphi_\lambda = e^{i\lambda y}\varphi \in C(a)$, we get $\langle\varphi,\varphi\rangle_{\alpha'} = \int P_{\varepsilon'}(\lambda)\langle\varphi_\lambda,\varphi_\lambda\rangle_\alpha\, d\lambda$ (Fubini, as in (i)), which is nonnegative when the form at α is positive semidefinite on $C(a)$. □

Remark 15.20.4 (One basepoint). Part (ii) uses every $\alpha > \frac{1}{2}$. The converse of (i) holds at a single basepoint: if $\frac{1}{2} \leq \alpha \leq 1$ and $\langle\varphi,\varphi\rangle_\alpha \geq 0$ for every $\varphi \in C_c^\infty(\mathbb{R})$, then no zero has real part greater than α. This is Theorem 9.1.2 of Chapter 9. With (i), the damped form at $\alpha \in [\frac{1}{2}, 1]$ is positive semidefinite if and only if $\sigma_{\max} \leq \alpha$. A statement about the symbol on one line is weaker: positivity of v_α on the line $\operatorname{Re} s = \alpha$ is not known to exclude zeros to its right (Remark 2.1.4).

15.20.2 Sharp coercivity

For $a > 0$ let

$$\lambda_{\min}(\alpha, a) = \inf\Big\{ \frac{\langle\varphi,\varphi\rangle_\alpha}{\|\varphi\|^2} :\ 0 \neq \varphi \in K(a) \Big\},$$

the least eigenvalue of the damped form on the window $[-a, a]$, where $\|\cdot\|$ is the norm of $L^2(\mathbb{R})$ and the test functions are complex valued. The matrix of the form on the basis χ_n, $|n| \leq N$, is the form restricted to a subspace of $K(a)$, so its least eigenvalue is an upper bound for $\lambda_{\min}(\alpha, a)$: these are the finite sections of Table 15.10. Under the hypotheses of Proposition 15.20.1 the infimum over the smaller space $C(a)$ is the same. Indeed, let $\varphi \in K(a)$ and let $\psi_\delta \in C_c^\infty(-a, a)$ satisfy $0 \leq \psi_\delta \leq 1$ and $\psi_\delta = 1$ on $[-a+\delta, a-\delta]$, with ψ_δ monotone on each of the two end intervals. Then $\eta_\delta = \varphi(1 - \psi_\delta)$ lies in $K(a)$, has total variation bounded

independently of δ and L^1 norm at most $2\delta\|\varphi\|_\infty$, so that $|\widehat{\eta_\delta}(\tau)| \le C\min(\delta, |\tau|^{-1})$, and $\langle \eta_\delta, \eta_\delta\rangle_\alpha = \int |\widehat{\eta_\delta}|^2 \rho_\varepsilon \to 0$ as $\delta \downarrow 0$ by dominated convergence, since $(1+\tau^2)^{-1}\rho_\varepsilon$ is integrable. The form being positive semidefinite on $K(a)$, the Rayleigh quotients of $\varphi\psi_\delta \in C(a)$ tend to that of φ. The argument applies verbatim to the functions supported in $[-a,a]$ and smooth on $[-a,a]$, to which Proposition 15.20.1 extends; this class contains the Legendre polynomials on the window used in Chapter 9, whose finite sections are therefore upper bounds as well. Put $m_\alpha = \inf_{\tau\in\mathbb{R}} v_\alpha(\tau)$.

THEOREM 15.20.5. *Let $V_\xi = (\log\xi)''(\frac12)$ and $p_1 = \frac12 V_\xi = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and, under RH, let $p_2 = \sum_{\gamma>0}\gamma^{-4} = 3.71726\ldots\times 10^{-5}$.*

(i) If $\alpha \ge \sigma_{\max}$ and no zero has real part α, then $\lambda_{\min}(\alpha,a) \ge 2m_\alpha$ for every $a>0$, $\lambda_{\min}(\alpha,a)$ is nonincreasing in a, and $\lambda_{\min}(\alpha,a) \to 2m_\alpha$ as $a\to\infty$.

(ii) Under RH, $m_\alpha = b_\alpha = \xi'(\alpha)/\xi(\alpha)$ for every $\alpha > \frac12$, and $v_\alpha(\tau) > b_\alpha$ for $\tau \ne 0$: the minimum of $\operatorname{Re}\xi'/\xi$ on each vertical line lies on the real axis, and only there. For $\alpha \ge 1$ this holds with no hypothesis (Proposition 12.1.12, and Proposition 12.1.11 for $\alpha = 1$); for $\frac12 < \alpha < 1$ it holds with no hypothesis for $0 < |\tau| \le 3\times 10^{12} - 1.21$ (Theorem 12.1.18(ii)); and RH is equivalent to it on every line $\frac12 < \alpha < 1$ (Theorem 12.1.18(vi), which sharpens Lagarias's criterion [185]).

(iii) Hence RH holds if and only if

$$\langle\varphi,\varphi\rangle_\alpha \;\ge\; 2b_\alpha\,\|\varphi\|^2 \qquad \text{for all } \alpha\in(\tfrac12,1],\ a>0,\ \varphi\in K(a),$$

and under RH the constant is sharp: $\lambda_{\min}(\alpha,a)\downarrow 2b_\alpha$ as $a\to\infty$.

(iv) For every $\alpha > \frac12$, with no hypothesis, $2b_\alpha \le 4p_1(\alpha-\frac12)$; under RH also $4p_1(\alpha-\frac12) - 4p_2(\alpha-\frac12)^3 \le 2b_\alpha$. The coercivity constant vanishes linearly at $\alpha = \frac12$.

Proof. (i) By Proposition 15.20.1 and Plancherel, $\int|\widehat\varphi|^2 = 2\pi\|\varphi\|^2$, we have, for $\varphi\in K(a)$,

$$\langle\varphi,\varphi\rangle_\alpha = \pi^{-1}\int |\widehat\varphi|^2 v_\alpha \ge \pi^{-1}m_\alpha\int|\widehat\varphi|^2 = 2m_\alpha\|\varphi\|^2.$$

Monotonicity holds because $\lambda_{\min}(\alpha,a)$ is also the infimum over $C(a)$ and $C(a)\subset C(a')$ for $a<a'$. For the limit, fix τ_0 and $\psi\in C_c^\infty(-1,1)$ with $\|\psi\|=1$, and let $\varphi_a(y) = a^{-1/2}e^{-i\tau_0 y}\psi(y/a)\in C(a)$. Then $\|\varphi_a\|=1$, $|\widehat{\varphi_a}(\tau)|^2 = a\,|\widehat\psi(a(\tau-\tau_0))|^2$, and $|\widehat{\varphi_a}|^2/2\pi$ is a probability density that concentrates at τ_0 as $a\to\infty$. The function v_α is continuous and $(1+\tau^2)^{-1}v_\alpha$ is integrable (Proposition 15.20.1), and $\widehat\psi$ is a Schwartz function, so $\langle\varphi_a,\varphi_a\rangle_\alpha = \pi^{-1}\int|\widehat{\varphi_a}|^2 v_\alpha \to 2v_\alpha(\tau_0)$. Hence $\limsup_a \lambda_{\min}(\alpha,a) \le 2v_\alpha(\tau_0)$ for every τ_0, that is, $\le 2m_\alpha$.

(ii) This is in the book at the places cited.

(iii) Under RH, (i) and (ii) give the inequality and its sharpness. Conversely, $b_\alpha > 0$ for every $\alpha > \frac12$ with no hypothesis (Theorem 12.1.18(i)), so the inequality gives positive semidefiniteness for $\alpha \in (\frac12, 1]$, Theorem 15.20.3(i) covers $\alpha > 1$, and part (ii) of that theorem gives RH.

(iv) The upper bound is Theorem 12.1.18(i), $b_\alpha \le (\alpha - \frac12)V_\xi$, with $V_\xi = 2p_1$. Under RH, $b_\alpha = \sum_{\gamma>0} 2\varepsilon/(\varepsilon^2+\gamma^2)$ and $\gamma^{-2} - \varepsilon^2\gamma^{-4} \le (\varepsilon^2+\gamma^2)^{-1} \le \gamma^{-2}$. □

REMARK 15.20.6. At a single $\alpha \in (\frac12, 1)$ with no zero on the line, the inequality $\langle\varphi,\varphi\rangle_\alpha \ge 2b_\alpha\|\varphi\|^2$ for all a and φ holds if and only if $\sigma_{\max} \le \alpha$ and $v_\alpha \ge b_\alpha$ on the line: if it holds, the form is positive semidefinite (as $b_\alpha > 0$), so $\sigma_{\max} \le \alpha$ by Remark 15.20.4, and then (i) gives $2m_\alpha \ge 2b_\alpha$, while $m_\alpha \le v_\alpha(0) = b_\alpha$; the converse is (i).

LEMMA 15.20.7. *Consecutive ordinates of the zeros of ζ differ by at most* 6.8874. *Hence every real τ with $|\tau| \ge \gamma_1$ lies within* 3.4437 *of an ordinate.*

Proof. No hypothesis on the zeros is used. The first 1800 zeros of Proposition 12.1.11, computed in ball arithmetic, reach height 2304.3: consecutive ordinates among them differ by at most $\gamma_2 - \gamma_1 = 6.88731\ldots$, and the later gaps by at most 5.4141 (between $\gamma_3 = 25.0109$ and $\gamma_4 = 30.4249$). It remains to show that $\gamma' - \gamma \le h := 6.8874$ for consecutive ordinates $\gamma < \gamma'$ with $\gamma \ge 2304.3$. Take T with $\gamma < T < \gamma'$ and write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). The second derivative $\vartheta''(T) = \frac{T}{4}\sum_{k\ge0}(k+\frac14)\big((k+\frac14)^2 + \frac{T^2}{4}\big)^{-2}$ is positive, so ϑ is convex, $\vartheta(T+h) - \vartheta(T) \ge h\vartheta'(T)$, and

$$N(T+h) - N(T) \;\ge\; \frac{h\,\vartheta'(T)}{\pi} - |S(T)| - |S(T+h)|.$$

We use the bounds of [100] as they are quoted in the proof of Proposition 12.1.11: $|S(t)| \le 2.5167$ for $0 \le t \le X_0 = 30\,610\,046\,000$, and $|S(t)| \le B(t) = 0.1095\log t + 0.2042\log\log t + 3.0305$ for $t \ge e$, with B increasing. For $2304.3 \le T \le X_0 - h$ the first bound applies to both terms, and $\vartheta'(T) \ge \vartheta'(2304.3) = 2.9523$ gives $N(T+h) - N(T) \ge 6.4725 - 5.0334 > 1.4$. For $T > X_0 - h$ both $|S(T)|$ and $|S(T+h)|$ are at most $B(T+h)$, because $B(X_0) = 6.32 > 2.5167$, so $N(T+h) - N(T) \ge D(T) = h\vartheta'(T)/\pi - 2B(T+h)$. At $T = X_0 - h$, $h\vartheta'(T)/\pi = 24.452$ and $2B(X_0) = 12.649$, so $D(X_0 - h) = 11.80 > 0$; and D increases, because a comparison of the sum for ϑ'' with an integral gives $\vartheta''(T) \ge \frac{1}{2T} - \frac{1}{T^2}$, while $2T\,B'(T+h) \le 0.219 + 0.4084/\log T$, so that $T\,D'(T) \ge \frac{h}{2\pi}(1 - \frac{2}{T}) - 0.219 - \frac{0.4084}{\log T} > 0.8$ for $T \ge X_0 - h$. In every case $N(T+h) - N(T) > 0$, so some ordinate lies in $(T, T+h]$, and letting $T \downarrow \gamma$ gives $\gamma' - \gamma \le h$. Finally, a real τ with $|\tau| \ge \gamma_1$ lies between two consecutive ordinates, hence within half a gap of one of them. □

COROLLARY 15.20.8. *For every $a > 0$ and every $\varphi \in K(a)$, with no hypothesis on the zeros,*

$$\langle\varphi,\varphi\rangle_1 \;\ge\; 2b_1\,\|\varphi\|^2 = \big(2 + \gamma_E - \log 4\pi\big)\|\varphi\|^2 = 0.0461914\ldots\,\|\varphi\|^2,$$

and no larger constant works for every a: $\lambda_{\min}(1,a) \downarrow 2+\gamma_E-\log 4\pi$ *as* $a\to\infty$.

Proof. At $\alpha=1$ no zero has real part 1 and $\sigma_{\max}\le 1$, so Theorem 15.20.5(i) applies, and $m_1=b_1$ by Proposition 12.1.11. Finally $b_1=\xi'(1)/\xi(1)=1+\frac{\gamma_E}{2}-\frac{1}{2}\log 4\pi$, twice which is $2+\gamma_E-\log 4\pi=0.0461914179\ldots$. □

Remark 15.20.9 (A Laguerre type inequality). Laguerre's quantity is

$$\mathcal{L}_\Xi(\tau)=-(\log\Xi)''(\tau)=\sum_\rho(\tau-\gamma_\rho)^{-2},\qquad \Xi(\tau)=\xi(\tfrac12+i\tau),$$

and it is the limit of $v_{1/2+\varepsilon}(\tau)/\varepsilon$ as $\varepsilon\downarrow 0$, taken termwise: a zero on the line gives $(\tau-\gamma)^{-2}$, and a pair $\frac12\pm\delta+i\gamma$ gives $2\,\mathrm{Re}(\tau-\gamma+i\delta)^{-2}$. Dividing $v_\alpha(\tau)\ge b_\alpha$ by ε, part (ii) of Theorem 15.20.5 becomes, as $\varepsilon\downarrow 0$,

$$\mathcal{L}_\Xi(\tau)\ \ge\ \mathcal{L}_\Xi(0)=2p_1,$$

a sharpening of Laguerre's inequality $\Xi'^2-\Xi\Xi''\ge 0$, which says $\mathcal{L}_\Xi\ge 0$. The sharpened inequality holds under RH for every τ, and with no hypothesis for $|\tau|\le 3\times10^{12}-1.21$ (Theorem 12.1.18(ii)). Numerically $\mathcal{L}_\Xi(0)=0.0462100$ is the global minimum of $\mathcal{L}_\Xi$ on $[0,1800]$, and the next local minimum, 0.2301087 at $\tau=17.498$ between the first two ordinates, is 4.98 times larger.

Table 15.10 lists $\lambda_{\min}(\alpha,a)$ for five windows and four levels $\alpha>\frac12$, with the floor $2b_\alpha$ in the last row, and the finite section values at $\alpha=\frac12$. The matrices are assembled from the damped prime side of (15.13.1).

a	$\alpha=\frac12$	$\alpha=0.55$	$\alpha=0.6$	$\alpha=0.75$	$\alpha=1$
0.35	1.20×10^{-3}	0.0066141	0.0120197	0.0281496	0.0547526
0.8	2.28×10^{-17}	0.0047355	0.0094659	0.0236477	0.0472506
1.5	2.43×10^{-64}	0.0046489	0.0092974	0.0232400	0.0464620
2.5	1.95×10^{-145}	0.0046305	0.0092608	0.0231497	0.0462846
4.0	1.59×10^{-286}	0.0046246	0.0092490	0.0231206	0.0462270
$2b_\alpha$	0	0.0046210	0.0092418	0.0231027	0.0461914

Table 15.10: The least eigenvalue $\lambda_{\min}(\alpha,a)$ of the damped Yoshida form on the window $[-a,a]$, from the matrices of the form on the modes χ_n, $|n|\le N$. For $\alpha>\frac12$ the entries are converged values, from Richardson extrapolation of $N=96$, 192, 384, with an uncertainty of at most 3×10^{-7} at $a=0.35$ and 1.1×10^{-8} for $a\ge0.8$ (an independent Legendre basis agrees to 1.5×10^{-7} at most). For $\alpha=\frac12$ the entries are the finite section values at $N=\max(24,24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a=4$ ($N=96$) was computed with 640 digits. The last row is the floor $2b_\alpha=2\xi'(\alpha)/\xi(\alpha)$. Computed.

Every entry for $\alpha > \frac{1}{2}$ exceeds the floor and decreases with a, as Theorem 15.20.5 says (under RH for $\alpha < 1$). The excess over the floor falls roughly like a^{-2}, as in Theorem 9.2.1(b) of Chapter 9.

The column $\alpha = \frac{1}{2}$ is of another kind: its entries depend strongly on N (at $a = 0.8$ the value is 6.9×10^{-17} at $N = 12$, 2.3×10^{-17} at $N = 24$ and 1.9×10^{-17} at $N = 48$). Two other tables of the book compute the same eigenvalue differently: Table 16.1 of Chapter 16 at $\alpha = \frac{1}{2}$, to 60 digits, for even and for odd test functions separately, and Table 9.1 of Chapter 9, the rungs at more levels $\alpha > \frac{1}{2}$, as converged values.

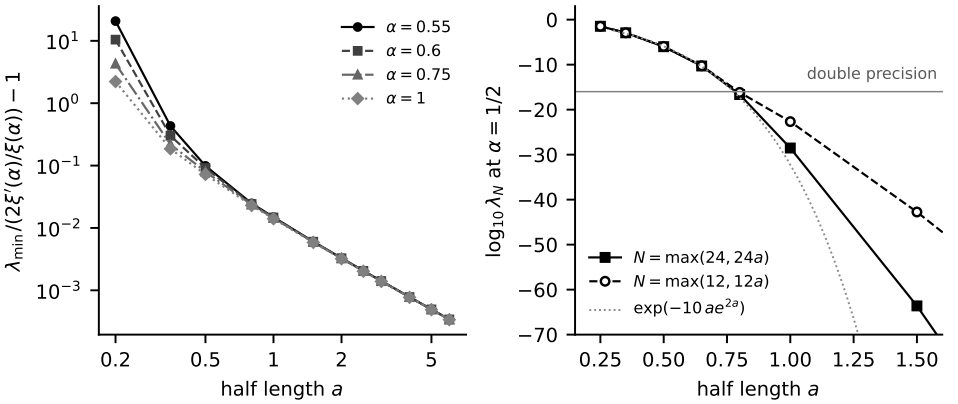


Figure 15.3: Left: the excess of the least eigenvalue of the damped Yoshida form over its floor, $\lambda_{\min}(\alpha, a)/(2b_\alpha) - 1$, against the window half length a, for $\alpha = 0.55$, 0.6, 0.75, 1 (finite sections of $\max(96, \lceil 24a \rceil)$ modes, logarithmic scales); it tends to zero, roughly like a^{-2}. Right: $\log_{10}$ of the finite section value at $\alpha = \frac{1}{2}$ for $N = \max(24, 24a)$ and $N = \max(12, 12a)$ modes, with the level of double precision and the curve $\exp(-10ae^{2a})$.

Figure 15.4 shows the symbol on four lines.

At $\alpha = \frac{1}{2}$ Theorem 15.20.5 does not apply. Under RH the symbol there is the atomic measure on the ordinates, the vague limit of ρ_ε as $\varepsilon \downarrow 0$; it vanishes between the atoms, which is why $\lambda_{\min}(\frac{1}{2}, a) \to 0$, and under RH the decay is faster than any power of a (Theorem 9.2.1(c) of Chapter 9). The speed of the collapse in Table 15.10 is a numerical observation: the finite section values at $\alpha = \frac{1}{2}$ follow $\exp(-2c\,ae^{2a})$ with $c = -\log\lambda/(2ae^{2a}) = 4.12$, 4.77, 5.10, 5.01, 4.86 at $a = 0.25$, 0.35, 0.5, 0.65, 0.8 (so $4.1 \le c \le 5.1$, at $N = 48$). Zhu proves, under RH and for all large a, only the bound $\exp(-ae^{a})$ for the least eigenvalue over real even test functions (Theorem 1.3 of [279], Chapter 16): it has e^{a} where the observed decay has e^{2a}. For $a \ge 1.5$ the finite sections saturate and c falls, so those values say nothing about the infimum. A heuristic for the form of the decay: $\widehat{\varphi}$ is an entire function of exponential type a and has at most about $2aH/\pi$ real zeros in

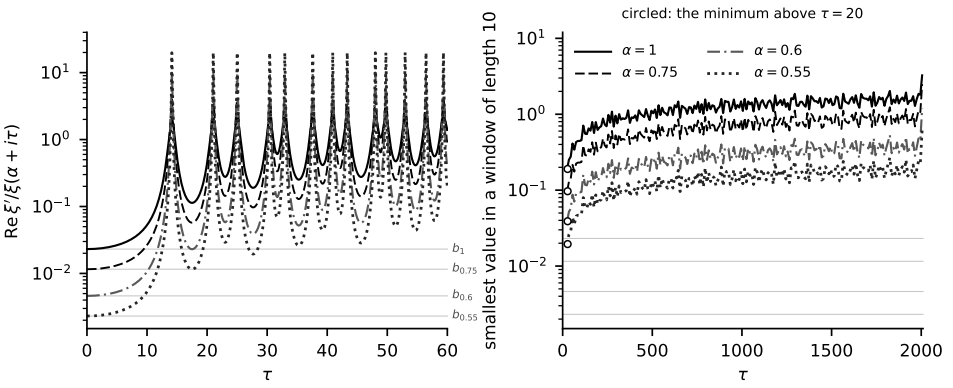


Figure 15.4: Left: the symbol $v_\alpha(\tau) = \operatorname{Re}\xi'/\xi(\alpha+i\tau)$ on $[0,60]$ for $\alpha = 1$, 0.75, 0.6, 0.55 (logarithmic scale), with b_α as thin horizontal lines. Right: the smallest value of v_α in windows of length 10 on $[0,2000]$, with the minimum above $\tau = 20$ circled. Above $\tau = 20$ the minima are 0.1901, 0.0968, 0.0389 and 0.0195 for $\alpha = 1$, 0.75, 0.6, 0.55, which are 8.2, 8.4, 8.4 and 8.4 times b_α, all at $\tau = 27.67$, between the third and the fourth ordinate (the largest gap above the second); the lowest local minimum other than the one at the origin is lower, $4.90\,b_1$ at $\tau = 17.49$ for $\alpha = 1$ and $4.98\,b_{0.55}$ for $\alpha = 0.55$, in the gap between the first two ordinates.

$[-H, H]$, against about $(H/\pi)\log(H/2\pi)$ ordinates there, so a vector of type a can vanish at every ordinate up to a height of about $2\pi e^{2a}$ and no further, which suggests a decay like $\exp(-c\,ae^{2a})$. This is not a proof.

15.21 An uncertainty barrier

Every prime side computation is limited in one of two ways. Compact support in the level variable keeps the prime sum finite but limits the level to $\log n \le X$. A finite basis limits the frequency, the ordinate $|\tau| \le T$. For the tent coefficient D of Proposition 15.18.1 the limit in level runs into zero verification, as the next proposition makes precise.

Proposition 15.21.1. *Suppose all zeros with $0 < \gamma \le H$ lie on the critical line, and let $D_{\mathrm{on}} \ge 0$ be the part of D from on line zeros. Then for every $x > 0$*

$$|D(x) - D_{\mathrm{on}}(x)| \le \varepsilon_H(x) := 8\cosh^2\frac{x}{4}\sum_{\gamma>H}\frac{1}{\gamma^2},$$

the sum being over all zeros above H, counted with multiplicity. With $H = 3\times10^{12}$ (Platt and Trudgian [232], the height H_0 of Theorem 12.1.18) the sum differs from $(\log(H/2\pi)+1)/(2\pi H) = 1.4797\times10^{-12}$ by less than 2×10^{-24} (Theorem 14.3.6), so that $\varepsilon_H(x) < 1.184\times10^{-11}\cosh^2(x/4)$.

Proof. Off line zeros have $|\gamma| > H$ and $|\delta| < \frac{1}{2}$, so $|\sin(\gamma_\rho x/2)|^2 \le \cosh^2(\delta x/2) \le \cosh^2(x/4)$ and $|\gamma_\rho|^2 \ge \gamma^2$. Each off line zero therefore contributes at most $4\cosh^2(x/4)/\gamma^2$ to D, and summing over all such zeros, two (a zero and its conjugate) for each ordinate $\gamma > H$ of the upper half plane, gives the bound. □

The proposition is of use only against a lower bound for D_{on}. Every term of D_{on} is nonnegative, so the first N zeros give one: $D_{\mathrm{on}}(x) \ge L_N(x) := \sum_{n\le N} 8\sin^2(\gamma_n x/2)/\gamma_n^2$. For the $N = 1700$ certified zeros of the book the least value of L_N on $[0.05, 47.1]$ is 0.040704, at $x = 25.383$, and 0.04070 is a lower bound for it computed by branch and bound; on $[0.05, 16]$ the least value is 0.0531.

Proposition 15.21.2 (Computed positivity of the tent coefficient). *Let L_N be as above with $N = 1700$, and let $H = 3 \times 10^{12}$. Since all zeros with $0 < \gamma \le H$ lie on the critical line [232], $D(x) \ge L_N(x) - \varepsilon_H(x)$ for every $x > 0$. Moreover $L_N(x) > \varepsilon_H(x)$ on $[0.05, 47.48]$, so $D > 0$ on $[0.05, 47.48]$ from the verification of RH up to H and the first 1700 zeros alone, with no prime sum. The first inequality is proved. The second is computed in double precision and not in ball arithmetic: for $x < 46.688$ it follows from the floor 0.04070 of L_N on $[0.05, 47.1]$, found by branch and bound, and for $46.688 \le x \le 47.48$ from a grid of step 5×10^{-4} and the Lipschitz constant 10.85 of L_N.*

The level 47.10 at which ε_H reaches 0.05 ($e^{47.10} = 2.85 \times 10^{20}$) is arithmetically right, but it is the barrier only if D_{on} stays above 0.05, which it does not; with the floor 0.0407 the bound reaches it at $x = 46.69$ ($e^{46.69} = 1.9 \times 10^{20}$), and pointwise at $x = 47.48$ ($e^{47.48} = 4.2 \times 10^{20}$). The barrier is at about 47 either way, because ε_H grows exponentially. A prime side computation of D below level 47.48, which needs all prime powers up to e^x, can only confirm a positive sign that the zeros already give; it cannot produce the negative value of D by which an off line zero would show. To see one through D a computation must reach levels beyond 47.48, with all prime powers up to 4×10^{20}, or heights beyond H, and the second is the zero verification itself.

In frequency, Proposition 15.20.1 makes the same point for a basepoint $\alpha \ge \sigma_{\max}$: a test vector with spectrum concentrated in $|\tau| \le T < H$ sees only the symbol there, and a zero $\beta + i\gamma$ with $\gamma > H$ contributes to the Hadamard sum for the symbol the term $(\alpha - \beta)/((\alpha - \beta)^2 + (\tau - \gamma)^2)$, of modulus at most $|\alpha - \beta|/(\gamma - T)^2$.

What the barrier does not say: Proposition 15.21.1 is proved for the single coefficient D and for no other vector. A normalised function supported in the window $[-a, a]$ and modulated to a height above H (Proposition 15.17.1) sees an off line zero there with a weight that does not decay with the height, and ε_H, which carries the factor γ^{-2} of the tent, does not bound the contribution of the zeros above H to the form of such a vector. Nor could a perturbation bound of size 10^{-11} settle the sign of a windowed form at $\alpha = \frac{1}{2}$, whose least eigenvalue

is below 10^{-15} from $a = 0.8$ on (Table 15.10). The tent is one vector of the window, so the level x_* of Remark 15.18.3 bounds Yoshida's threshold only from above, $a_0 \le x_*/2$, and the window experiment of Chapter 16 (Table 16.3) finds a hypothetical off line pair at half lengths far below $x_*/2$.

15.22 The question of a path to RH

The chapter does not give a path to RH. Every criterion proved here has the same structure. On the RH side it is a positivity statement: $D \ge 0$, $T_k \ge 0$ for infinitely many k, the modulated heat trace of Proposition 15.17.1 nonnegative, a Pick function, a coercive damped form. On the other side it is the presence of a pole off the real axis, or a negative Cauchy kernel, at an off line ordinate. The barrier of Section 15.21 says, for D only, that a computation at levels below 47 cannot go beyond what zero verification already knows. A proof needs an inequality valid at all levels or all heights at once, and it must use the Euler product: the function f_{DH} of Davenport and Heilbronn [60] (Section 14.4.5.4) shares the functional equation and has zeros off the line, so it violates the analogue of every criterion in this chapter.

What the chapter changes is the shape of the missing inequality, which Theorem 15.20.5 puts in its sharpest form:

$$\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau) \;\ge\; \frac{\xi'}{\xi}(\alpha) \qquad \text{for all } \tau,\ \alpha \in (\tfrac12, 1], \tag{15.22.1}$$

which we write $v_\alpha(\tau) \ge b_\alpha$ as in Theorem 12.1.18. Taken as a family over $\alpha \in (\frac12, 1]$, and not line by line, this is equivalent to the coercivity of every damped Yoshida form with the constant it has on functions of low frequency. Three facts about it:

1. It is true with no hypothesis for every $\alpha \ge 1$ and every τ (Propositions 12.1.11 and 12.1.12), and for $\frac12 < \alpha \le \alpha_{\max} = 7.35$ and $0 < |\tau| \le H - \frac{1+\sqrt2}{2}$ (Theorem 12.1.18(ii)); for $\frac12 < \alpha < 1$ and all τ it is equivalent to RH (Theorem 12.1.18(vi)). On the line $\alpha = 1$ the minimum is at $\tau = 0$, the next lowest local minimum of v_1, at $\tau = 17.49$, is 4.90 times b_1, and on $[20, 2000]$ the least value is $8.23\,b_1$, at $\tau = 27.67$ (computed).

2. If it holds, even in the weak form $v_{\alpha'}(\tau) > 0$, on every line $\operatorname{Re} s = \alpha'$ with $\alpha < \alpha' \le 1$, then $\sigma_{\max} \le \alpha$, since a zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 > \alpha$ sends $v_{\alpha'}(\gamma_0)$ to $-\infty$ as $\alpha' \uparrow \beta_0$; on all of $(\frac12, 1]$ it is RH. On a single line it is not known to exclude a zero to its right (Remark 2.1.4). Its $\alpha \downarrow \frac12$ limit, under RH, is the sharpened Laguerre inequality $-(\log \Xi)''(\tau) \ge 2p_1$.

3. Its failure mode is local. A zero with $\beta > \alpha$ would create a negative Cauchy kernel of height $1/(\pi(\beta - \alpha))$ at its ordinate, competing against a symbol of

size $\frac{1}{2\pi}\log\tau$ there. So the inequality at α asks for control of $-\operatorname{Re}\zeta'/\zeta(\alpha+i\tau)$ by $\frac{1}{2}\log(\tau/2\pi)$, a one sided bound on ζ'/ζ; the Fejér tapered form of the question is Proposition 19.14.1. No method known today proves it at any fixed $\alpha<1$ for all τ; the zero free half plane $\operatorname{Re}s>\frac{7}{8}$ announced in October 2026 (Section 25.1.2) has not been refereed and is not counted.

15.23 Open problems

Problem 15.5 (The symbol inequality below the line $\alpha=1$). The inequality $v_\alpha(\tau)\geq b_\alpha$ is a theorem of the book for every $\alpha\geq 1$ and every τ (Propositions 12.1.11 and 12.1.12), and for $\frac{1}{2}<\alpha\leq\alpha_{\max}$ up to height $3\times10^{12}-1.21$ (Theorem 12.1.18(ii)); for $\frac{1}{2}<\alpha<1$ and all τ it is equivalent to RH (Theorem 12.1.18(vi)). What is open is the region $\alpha<1$ above the verified height. Determine the part of it in which $v_\alpha(\tau)>b_\alpha$ can be proved with no hypothesis and with explicit constants; the Vinogradov Korobov region $\alpha\geq 1-c(\log|\tau|)^{-2/3}(\log\log|\tau|)^{-1/3}$ [272, 184] is the natural target. A proof on every line of a strip $\alpha_0<\alpha\leq1$ for all τ would give $\sigma_{\max}\leq\alpha_0$ (Section 15.22), which is why the target is a region that narrows with the height.

Problem 15.6 (A prime side proof at $\alpha=1$). The book proves $v_1(\tau)\geq b_1$ through the zeros (Proposition 12.1.11), by the Cauchy kernels of the ordinates nearest to τ and a bound on the gaps between them (compare Lemma 15.20.7). At $\alpha=1$ the statement can be written entirely in primes. Since $\xi'/\xi(1)=1+\frac{\gamma_E}{2}-\frac{1}{2}\log4\pi$ and $\operatorname{Re}\frac{1}{i\tau}=0$, it reads

$$\sum_{n\geq2}\frac{\Lambda(n)}{n}\cos(\tau\log n)\ \leq\ \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\tau}{2}\Big)-\frac{1}{2}\log\pi+\frac{1}{1+\tau^2}-\frac{\xi'}{\xi}(1)\qquad(\tau\neq0),$$

the left side being the conditionally convergent series for $-\operatorname{Re}\zeta'/\zeta(1+i\tau)$. Find a proof using only Mertens type information, in the spirit of the Fejér taper of Proposition 19.14.1; the right side grows like $\frac{1}{2}\log(\tau/2\pi)$ and the left side is $O(\log\log\tau)$ under RH.

Problem 15.7 (Löwner at a single window). By Theorem 15.19.1, positivity on $K(a)$ is positivity of the Löwner matrices of g_a on the lattice $\frac{\pi}{a}\mathbb{Z}$. Löwner's hard theorem [197, 66] recovers the Pick property from Löwner matrices on all finite subsets of an interval. What does positivity on the lattice alone, for one fixed a, imply about the location of the zeros? A positive answer that grew with a would quantify Yoshida's threshold a_0.

Problem 15.8 (The constants of the ladder and of the window). Determine the order of K_0 for the ladder and of a_0 for Yoshida's window, for a single off line pair, and compare with the displacement $\sigma_{\max}-\frac{1}{2}$ and with the de Bruijn Newman constant $\Lambda\leq2(\sigma_{\max}-\frac{1}{2})^2$ (de Bruijn [61], in the normalisation of Rodgers and

Tao [250] used in Section 4.1). For the ladder the computations of Section 15.15 give $K_0(\sec\theta_* - 1) \approx 0.70$ for the pair of the stress test (Remark 15.15.3) and, over fourteen pairs, a first failing order that grows like γ^2 and, at fixed γ, like $1/\delta$ and not like the heuristic γ^2/δ^2; its dependence on δ and on the spacing of the true ordinates is what is to be determined. For the window only $a_0 \le x_*/2$ is known (Remark 15.18.3), and the window experiment of Chapter 16 (Table 16.3) finds the threshold far below it.

Problem 15.9 (Zero density and the barrier). Proposition 15.21.1 allows every zero above H to have δ arbitrarily close to $\frac{1}{2}$. Zero density estimates make zeros with large δ rare. Sharpen the barrier level using an explicit $N(\sigma, T)$ bound, and determine whether any prime side computation of D at feasible levels can, even in principle, exclude off line zeros beyond the verified height. For windows no barrier is proved: decide whether a window of half length a below about 23.7 (level 47.48) can see an off line zero above H, for which Section 16.4 of Chapter 16 gives numerical evidence and a heuristic horizon, not a proof.

Notes

The forms of this chapter are Yoshida's [278], and the explicit formula is Weil's [273]. Lagarias's criterion [185] asks only for $v_\alpha > 0$; the minimum at the origin of Theorem 12.1.18(vi) sharpens it. In integrated form, under RH, the minimum at the origin says that $\sigma \mapsto |\xi(\sigma + i\tau)|/\xi(\sigma)$ is nondecreasing on $(\frac{1}{2}, \infty)$, a normalised version of the reformulation of Sondow and Dumitrescu [259] that $|\xi(\sigma + i\tau)|$ increases in σ. Theorem 15.18.2 is classical in substance.

Chapter 16

Zhu's compact windows and the Landau Widom horizon

Xuefeng Zhu's preprint on Weil positivity in compact windows [279] continues Yoshida's programme of Chapter 15.12 by certified computation.

The account of the preprint is the first author's reading of version 2 of 2 September 2026, checked statement by statement: Zhu's theorems are quoted with the status he gives them and are not reproved, and his certificates are not rechecked. Numbers are computed in floating point at the precision stated; the least eigenvalues of Table 16.1 are consistent with Zhu's certified intervals and carry no interval certificate of their own. The passage from a window to a height of the zeros (the horizon of Section 16.4) is Zhu's heuristic and is marked as one wherever it is used. The detection experiment of that section concerns a hypothetical configuration, a quadruple of zeros added to the actual zeros of ξ, and is not a statement about ξ.

16.1 What Zhu proves

Zhu [279] writes L for the half length of the window; here it is a, as in Chapter 15.12, where L is the support of an autocorrelation, so his $\lambda^*(L)$, A_L and $T_1(L)$ appear below as $\lambda_{\min}^{\mathrm{even}}(\frac12, a)$, A_a and $T_1(a)$. For real even continuous f supported in $[-a,a]$, with $F=\widehat{f}$, $\widehat{f}(t)=\int_{\mathbb{R}} f(y)e^{ity}\,dy$, Zhu writes Weil's quadratic form as

$$Q(f)=2F\big(\tfrac{i}{2}\big)^2+\frac{1}{2\pi}\int_{\mathbb{R}}|F(t)|^2\,\Psi_a(t)\,dt,\qquad \Psi_a(t)=\operatorname{Re}\psi\Big(\frac14+\frac{it}{2}\Big)-\log\pi-\mathcal{P}_a(t), \tag{16.1.1}$$

where $\mathcal{P}_a(t)=\sum_{\log n<2a}2\Lambda(n)n^{-1/2}\cos(t\log n)$ is the prime comb. He sets $\lambda_{\min}^{\mathrm{even}}(\frac12,a)=\inf Q(f)/\|f\|^2$ over the real even $f\neq0$ supported in $[-a,a]$. His results, with the status he gives them, are these.

(Z1) One stroke reduction (Theorem 1.1, unconditional). Let $A_a=\sup_t\mathcal{P}_a(t)=\sum_{\log n<2a}2\Lambda(n)n^{-1/2}$: the supremum is attained at $t=0$, and by Weyl

equidistribution of the phases $t \log p$ values arbitrarily close to it occur at arbitrarily large t (his Lemma 3.2). The envelope $\operatorname{Re}\psi(\frac{1}{4}+\frac{it}{2}) - \log\pi \geq \log\frac{t}{2\pi} - \frac{1}{t}$ for $t \geq \frac{15}{4}$ (Lemma 3.1) gives $\Psi_a(t) \geq \beta^* > 0$ for $t \geq T_\sharp$, where $\beta^* = \log\frac{T_\sharp}{2\pi} - \frac{1}{T_\sharp} - A_a$ and $\beta^* > 0$ holds for every $T_\sharp$ above a threshold that exceeds $T_1 := 2\pi e^{A_a}$ by the factor $1 + O(T_1^{-1})$ (120.1 against 119.1 at $a = 0.8$). Positivity on the window then reduces to the positive semidefiniteness of a finite matrix in an even Legendre basis, with explicit tails.

(Z2) Certificate (Theorem 1.2, a computer assisted theorem). At $a = 0.8$, with $T_\sharp = 200$, 200 even Legendre modes (orders up to 398), working precision 50 digits and a verified Cholesky factorisation, $Q(f) \geq 8.9\times10^{-18}\|f\|^2$ for every real even f supported in $[-0.8, 0.8]$. The autocorrelation then has support $[-1.6, 1.6]$, which is 2.3 times the range $2a \leq \log 2$ that Zhu attributes to Yoshida [278] and, by a different method, to Connes and Consani [44]. His variational upper bound, computed on the geometric side in interval arithmetic, is $\lambda_{\min}^{\text{even}}(\frac{1}{2}, 0.8) \leq 2.27 \times 10^{-17}$, which encloses the infimum within a factor 2.6.

(Z3) Under RH, $\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) \leq \exp(-ae^a)$ for a large (Theorem 1.3, conditional on RH), by an explicit test function: a canonical product over the ordinates up to $T = 2\pi e^a$ times a power of the sinc function. The construction works at half saturation, and with $T = 2\pi e^{2\theta a}$ it closes only for $\theta \lesssim 0.65$ (his Remark 13.6).

(Z4) Decay law (Conjecture 12.1, fitted to computed bounds). The certified variational upper bounds for a from 0.5 to 2, unconditional and computed on the geometric side, fall from 10^{-6} to 3.2×10^{-283} and follow $-\log\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) = C\,N(T_*)/\log N(T_*)\,(1 + o(1))$, where $T_* = 2\pi e^{2a}$, $N(T)$ is the number of ordinates up to T and $C = 20.13\ldots \approx 2\pi^2$ (the fit of his equation (5)). The constant is fitted and not derived: Zhu calibrates the law on $a \leq 1.2$ and tests it on $a = 1.4$ to 2, where it underpredicts by about 5 percent with no trend. That the law describes the infimum itself, with $2\pi^2$ in place of the fitted C, is his Conjecture 12.1.

(Z5) Barrier (Theorem 1.4, unconditional). Since $A_a = (4 + o(1))e^a$ by the prime number theorem, any application of (Z1) needs $T_\sharp > T_1(a) = 2\pi e^{A_a}$, about 1.0×10^7 at $a = 1.6$ and 2.4×10^{11} at $a = 2$ ($A_{1.6} = 14.323$, $A_2 = 24.383$), so that the size of the matrix grows doubly exponentially in a; and since $\sup_t \mathcal{P}_a(t) = A_a$ exactly, no pointwise bound on the comb lowers T_1. To improve on pointwise envelopes one would have to prove that the phases $t \log p$ cannot align, which Zhu calls an arithmetic statement about simultaneous Diophantine approximation of $\{\log p\}$ (his Remark 1.6).

(Z6) Parity (Lemma 6.1, Theorem 6.2 and Corollary 6.3). For real $f = f_e +$

f_o, $Q(f) = Q(f_e) + Q(f_o)$, with the pole term entering as $+2c^2$ on the even part and $-2s^2$ on the odd part, where $c = \int f_e(x)\cosh\frac{x}{2}\,dx$ and $s = \int f_o(x)\sinh\frac{x}{2}\,dx$. At $a = 0.8$ the even ground state is simple, with the certified bounds $8.9\times10^{-18} \le \lambda_1^{\mathrm{even}} \le 2.523\times10^{-16}$ (the upper bound 2.27×10^{-17} of (Z2) is the sharper), $\lambda_2^{\mathrm{even}} \ge 2.085\times10^{-12}$ and $8.206\times10^{-15} \le \lambda_1^{\mathrm{odd}} \le 2.347\times10^{-14}$, where $\lambda_1^{\mathrm{even}} \le \lambda_2^{\mathrm{even}}$ are the first two min max values of $Q(f)/\|f\|^2$ over real even f supported in $[-0.8, 0.8]$ and λ_1^{odd} is the infimum over real odd f. Hence $Q(f) \ge 8.9\times10^{-18}\|f\|^2$ for complex f as well. Zhu connects the simple even ground state with one of the two spectral steps that the programme of Connes, Consani and Moscovici [45] isolates as missing.

Zhu also retracts a claim of an earlier draft at autocorrelation support 2.38 ($a = 1.19$), keeping a computation that is consistent with positivity there and no certificate (his Section 7), and he warns that evaluating the geometric side of Q in double precision produces spurious negative eigenvalues (his Remark 2.1 and Section 10).

For complex test functions the least eigenvalue of Chapter 15.12 is $\lambda_{\min}(\frac12, a)$, the infimum of $\langle\varphi,\varphi\rangle/\|\varphi\|^2$ over $0 \ne \varphi \in C(a)$ (Section 15.20.2). Write $\lambda_{\min}^{\mathrm{odd}}(\frac12, a)$ for the infimum over real odd test functions. The form has no cross terms between the real and the imaginary part of φ, nor between its even and its odd part (Zhu's Lemma 6.1), so that

$$\lambda_{\min}(\tfrac12, a) = \min\bigl(\lambda_{\min}^{\mathrm{even}}(\tfrac12, a),\, \lambda_{\min}^{\mathrm{odd}}(\tfrac12, a)\bigr),$$

and $\lambda_{\min}(\frac12, a) = \lambda_{\min}^{\mathrm{even}}(\frac12, a)$ when the ground state is even, which Zhu's Theorem 6.2 shows at $a = 0.8$. Yoshida's parity reduction (Y1) concerns the same two sectors. The splitting holds for the damped forms of Section 15.20.1 at every α, because the damping $e^{-\varepsilon|y|}$ is even.

16.2 The Thorin dictionary

The functional $\mathcal{Y}$, the form $\langle\varphi_1,\varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ and the spaces $C(a)$ and $K(a)$ are those of Section 15.13. Call a real even f supported in $[-a, a]$ admissible if the explicit formula applies to $f * f$, as it does for every $f \in C_c^\infty(\mathbb{R})$. Let PW_a be the Paley Wiener space of the entire functions of exponential type at most a that are square integrable on $\mathbb{R}$, and PW_a^{even} its real even members, which contain the transforms $F = \widehat{f}$ of the admissible f. The measure $U = \sum_{\gamma>0}\delta_{\gamma^2}$ is the Thorin measure of the clock under RH (Section 2.1), each ordinate counted with its multiplicity.

Proposition 16.2.1. *For admissible f, with $F = \widehat{f}$, Zhu's form is Yoshida's form*

on the window $C(a)$: $Q(f)=\langle f,f\rangle=\mathcal{Y}(f*f)$. Under RH

$$Q(f)=\sum_{\rho}|F(\gamma_\rho)|^2=2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt),$$

$$\lambda_{\min}^{\mathrm{even}}(\tfrac12,a)=\inf_f\frac{2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt)}{\frac{1}{2\pi}\int_{\mathbb{R}}|F|^2},$$

so that $\lambda_{\min}^{\mathrm{even}}(\frac12,a)$ is the lower sampling constant of the Thorin measure U for the transforms $F\in PW_a^{\mathrm{even}}$ of the admissible f.

Proof. The autocorrelation $g=f*f$ is even, supported in $[-2a,2a]$ and has $\widehat{g}=F^2=|F|^2\geq 0$ on $\mathbb{R}$. By Proposition 15.16.1 (there $L=2a$; the proposition is stated for smooth f, and for an admissible f the identity is the term by term explicit formula that follows), with no hypothesis on the zeros, $\langle f,f\rangle=\mathcal{Y}(g)=2\int\widehat{g}(\sqrt{t})\,U^{\pm}(dt)$, and only the prime powers $n\leq e^{2a}$ enter; under RH the functional $\varphi\mapsto\int\varphi\,dU^{\pm}$ is the sum of φ over the atoms of U (Section 8.2.1). Term by term, the explicit formula (15.13.1) for g has the pole term $\widehat{g}(\frac{i}{2})+\widehat{g}(-\frac{i}{2})=2F(\frac{i}{2})^2$. Its prime term $-\sum_{n\geq2}\Lambda(n)n^{-1/2}\big(g(\log n)+g(-\log n)\big)$ equals $-\frac{1}{2\pi}\int|F|^2\mathcal{P}_a$, because $\widehat{g}=|F|^2$ and g vanishes beyond $2a$. Its archimedean term is $V_1(g)-g(0)\log\pi=\frac{1}{2\pi}\int|F|^2\big(\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi\big)dt$, by the first line of (15.13.2) and $g(0)=\frac{1}{2\pi}\int|F|^2$. The three terms add up to (16.1.1). Under RH every γ_ρ is real, and the zero side of (15.13.1) is $\sum_\rho\widehat{g}(\gamma_\rho)=\sum_\rho F(\gamma_\rho)^2=2\sum_{\gamma>0}F(\gamma)^2$, because F is even and the ordinates come in pairs $\pm\gamma$; this is $2\int F(\sqrt{t})^2\,U(dt)$. Finally $\|f\|^2=\frac{1}{2\pi}\int|F|^2$ by Plancherel. □

Two consequences follow. First, under RH the positivity of the infimum for every a is a sampling statement, and it is Zhu's Proposition 2.3. Littlewood's theorem on the gaps between consecutive ordinates [264, Section 9.12], which holds without RH although Zhu states it as a consequence of RH, says that the gaps tend to zero, so under RH the set $\{\pm\gamma\}$ contains, above some height, a separated subset Λ of lower uniform density larger than a/π (a maximal subset of points at least $\pi/(4a)$ apart), and Beurling's sampling theorem [255, Chapters 2 and 3] gives $c>0$ with $\sum_{\lambda\in\Lambda}|F(\lambda)|^2\geq c\int_{\mathbb{R}}|F|^2$ for all $F\in PW_a$. Hence $Q(f)\geq 2\pi c\|f\|^2$ and $\lambda_{\min}^{\mathrm{even}}(\frac12,a)>0$, with a constant that is not explicit (Zhu's Remark 2.4). Without RH the argument gives nothing (Zhu's Remark 2.4 counts the gap bound as a second use of RH and says that unconditionally the gaps are only known to be bounded). Second, in this picture the smallness of the infimum is the poor conditioning of sampling by a set that is sparse at low frequency: there is no ordinate in $(-14.13,14.13)$.

Zhu's parity lemma (Z6) has a Thorin reading. The pole is the atom of mass one of the signed Thorin measure $U^{\pm}$ at $t=-\frac14$ (equation (15.16.1)), that is at the imaginary ordinates $\pm\frac{i}{2}$, and it is paired with $F(\frac{i}{2})F(-\frac{i}{2})$. For even f this

a	modes N	$\lambda_{\min}^{\text{even}}$	$\lambda_{\min}^{\text{odd}}$	Zhu, even
0.5	8	1.336×10^{-6}	2.51×10^{-4}	
0.5	14	1.082×10^{-6}	2.39×10^{-4}	
0.5	20	1.005×10^{-6}	2.30×10^{-4}	
0.5	26	9.743×10^{-7}	2.24×10^{-4}	$\leq 1.05 \times 10^{-6}$
0.5	160	9.344×10^{-7}	1.997×10^{-4}	
0.8	8	5.49×10^{-15}	2.67×10^{-12}	
0.8	14	4.127×10^{-17}	3.86×10^{-14}	
0.8	20	2.651×10^{-17}	2.333×10^{-14}	
0.8	26	2.144×10^{-17}	2.157×10^{-14}	$[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$
0.8	160	1.673×10^{-17}	1.644×10^{-14}	

Table 16.1: Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order N shown. The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling.

is $c^2 \geq 0$ and for odd f it is $-s^2 \leq 0$ (Proposition 15.16.1): it helps the even sector and hurts the odd one. That the even ground state is nevertheless the smaller is explained by Zhu (Section 6.3) through the constraint $F_o(0) = 0$ on the odd sector, which denies odd test functions the zero free interval $[0, \gamma_1)$ in which even minimisers park most of their mass: the low frequency gap, and not the pole, controls the conditioning. At $a = 0.8$ the odd sector lies about three orders of magnitude above the even one and at $a = 0.5$ two orders (Table 16.1), and Zhu reports that the ratio grows almost geometrically in a over $[0.5, 1.2]$, as $10^{2.07a+1.32}$.

The least eigenvalues of the two sectors are computed from the prime side matrices of Yoshida's form, on the even basis $(2a)^{-1/2}$, $a^{-1/2}\cos(\pi jy/a)$ and the odd basis $a^{-1/2}\sin(\pi jy/a)$, $1 \leq j \leq N$, with N the number of modes in Table 16.1.

The values of Table 16.1 are consistent with Zhu's numbers, which come from another basis and another code: Fourier modes on $[-a, a]$ against his Legendre polynomials, and quadrature of the explicit formula in the variable y against his spherical Bessel integrals. At $a = 0.8$ both sectors lie inside his certified intervals at 26 modes. At 160 modes the even values 9.344×10^{-7} and 1.673×10^{-17} lie 0.04 and 1.4 percent above the reference floors 9.34×10^{-7} and 1.65×10^{-17} of Zhu's Table 1 (a converged exact assembly, which he gives as reference values and not as certificates), and the odd values 1.997×10^{-4} and 1.644×10^{-14} lie 2.9 and 4.7 percent above his 1.94×10^{-4} and 1.57×10^{-14} and are still falling. At $a = 0.8$ the second even eigenvalue is 8.42×10^{-12} at 160 modes (Zhu: 8.38×10^{-12} in his Table 1, and the certified bound $\lambda_2^{\text{even}} \geq 2.085 \times 10^{-12}$), about 5×10^5 times the

first, in agreement with the simple ground state of (Z6). Raising the precision to 90 digits, or the number of quadrature nodes by half, changes the least eigenvalues by a relative 2×10^{-58} at most (checked at 26 and 40 modes), so each value is an upper bound for the infimum over its sector.

As a check of the normalisation of Proposition 16.2.1, take the unit bottom eigenvector f of the matrices of 26 and 40 modes at $a = 0.5$ and 0.8, in both sectors. The partial sums $2\sum_{\gamma \le \gamma_K} F(\gamma)^2$ over the first K ordinates increase with K and stay below the eigenvalue, and with the mean tail $2\int_{\gamma_K}^{\infty} F(t)^2 \, \frac{1}{2\pi} \log \frac{t}{2\pi} \, dt$ of the missing ordinates they reproduce it to within 0.16 percent for $K = 300$, 400 and 1700 in all eight cases.

Three other tables of the book list the same eigenvalue at other levels: Table 15.10 of Chapter 15.12 for the windows $a = 0.35$ to 4 (its entry 2.28×10^{-17} at $a = 0.8$, from 24 modes, lies between our values at 20 and 26 modes), and Tables 9.1 and 9.2 of Chapter 9. Table 16.1 is at $\alpha = \frac{1}{2}$ only, with the sectors apart, in 60 digits, and shows the convergence in the number of modes without extrapolating.

16.3 Zhu's test function through a truncated reciprocal Xi transform

Zhu's proof of (Z3) takes $\widehat{f_0}(r) = B(r)h(r)$, with $B(r) = \prod_{0<\gamma_j \le T}(1 - r^2/\gamma_j^2)$ a finite product over the ordinates up to a truncation height T, and a window kernel h of type a. In the notation of Section 2.1, $G(u) = \prod_{\gamma>0}(1 + u/\gamma^2)$ is the reciprocal of the Laplace transform of the clock, which under RH is $\sum_{\gamma>0} e_\gamma/\gamma^2$. Truncate the product, $G_T(u) = \prod_{0<\gamma_j \le T}(1 + u/\gamma_j^2)$. Then $1/G_T$ is the Laplace transform of the finite gamma convolution $\sum_{\gamma_j \le T} e_{\gamma_j}/\gamma_j^2$ and

$$B(r) = G_T(-r^2),$$

the reciprocal of that Laplace transform continued to the cut $s = -r^2$: Zhu's test function is the window kernel divided by the truncated reciprocal Xi transform on the cut. In the variable $t = r^2$ it vanishes at the Thorin atoms $\gamma_j^2 \le T^2$ (and at the zeros of h), and under RH

$$Q(f_0) = 2\int_{t>T^2} h(\sqrt{t})^2 \, G_T(-t)^2 \, U(dt).$$

The optimisation is a trade: each atom killed costs a factor r^2/γ_j^2 of growth, which h must absorb with its limited type.

Table 16.2, a computation of this chapter and not of Zhu, evaluates this family with $h(r) = \big(\sin(ar/m)/(ar/m)\big)^m$, of type a, optimising over the number K of atoms killed and the even exponent $m \ge 2K + 2$. The quotient $Q(f_0)/\|f_0\|^2$ is evaluated on the zero side under RH, with the 1700 ordinates and an explicit

a	best bound	K	m	γ_K	$N(T_*)$	$T_* = 2\pi e^{2a}$
0.5	3.88×10^{-4}	2	6	21.0	1	17.1
0.8	4.03×10^{-9}	6	18	37.6	4	31.1
1.0	7.04×10^{-14}	9	26	48.0	8	46.4

Table 16.2: The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number K of atoms killed and the even exponent m, with the optimal K and m, the last ordinate γ_K killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 16.4, and T_* itself.

Riemann von Mangoldt tail [100], and on the geometric side by Zhu's formula (16.1.1), which needs no hypothesis. The two agree to 13 digits or more at $a = 0.8$ and 1.0 and to 3.5×10^{-7} relative at $a = 0.5$, where the tail beyond 1700 ordinates limits the comparison, so every entry is an unconditional upper bound for $\lambda_{\min}^{\text{even}}(\frac{1}{2}, a)$.

The product construction is crude. At $a = 0.8$ it gives 4.03×10^{-9} against about 1.7×10^{-17} for the window infimum (the entry at 160 modes of Table 16.1, an upper bound), and a general element of PW_a does much better. The number of atoms worth killing grows with a and exceeds the number $N(T_*)$ of ordinates below the horizon T_* of the next section: in each row the last atom killed lies above T_* and below eT_*. When K is restricted to $N(T_*)$ the bound is worse by the factors 1.27, 1.40 and 1.05. At $a = 0.35$ and $a = 1.2$ (not tabulated) the optimum is $K = 0$ and $K = 16$, equal to $N(T_*)$ in both cases. So atoms above the horizon are worth killing at the three tabulated half lengths; three rows with $a \le 1$ cannot settle the asymptotics (Zhu's Remark 13.6 in (Z3)).

16.4 The Landau Widom horizon

Zhu's resolution height comes from a comparison of densities (his Section 9). By Jensen's formula a real entire function of exponential type a has real zeros of average density at most a/π on long intervals $[-R, R]$ about the origin, which Zhu takes as a bound for the local density, while the ordinates have density $\nu(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$ at height t. The two meet at

$$\frac{a}{\pi} = \frac{1}{2\pi} \log \frac{T}{2\pi} \iff T = T_*(a) := 2\pi e^{2a}, \tag{16.4.1}$$

which we call the Landau Widom horizon, or the horizon. Zhu's argument is that below T_* a test function of type a can in principle interpolate the ordinates, placing a zero of F on each of them, and that above T_* the ordinates outnumber the available oscillations, so that $|F|^2$ must leak onto them. A count of degrees of freedom gives the same height up to a factor. An even function of PW_a concentrated on the frequencies $[0, T]$ has about aT/π degrees of freedom, since the

time frequency rectangle $[-a,a]\times[-T,T]$ has about $2aT/\pi$ [258, 188], halved for evenness, while $N(T)\approx\frac{T}{2\pi}\log\frac{T}{2\pi e}$ ordinates lie below T. Equating the two counts gives $2\pi e^{2a+1}=eT_*$.

Zhu's decay law (Z4) uses T_*. He models the cost of killing $N(T_*)$ ordinates at saturated budget as an excursion of order $N(T_*)$ eigenvalues beyond the Shannon number of the window, at the plunge rate $\lambda_{K+j}\approx\exp(-\pi^2 j/\log K)$ of the eigenvalues of the concentration operator of time and band limiting [189]. This gives $-\log\lambda_{\min}^{\mathrm{even}}(\frac12,a)\asymp\pi^2N(T_*)/\log N(T_*)$, with the constant supplied by the fit (his Section 9 and Conjecture 12.1): a model, fitted and not derived.

The passage from a window statement to a statement about the height of the zeros is a heuristic. Nothing in this book proves that positivity of Weil's form on every window of half length a is implied by, or implies, the verification of the zeros up to a height of order $T_*(a)$, and Problem 16.1 asks for the theorem that the heuristic would need.

A quadruple of off line zeros $\frac12\pm\delta\pm i\gamma$ contributes $4\,\mathrm{Re}\,F(\gamma-i\delta)^2$ to $Q(f)$ for real even f, and a negative value of Q needs a test function that nearly vanishes at the on line ordinates near γ while keeping its energy near $\gamma\pm i\delta$. Heuristically this is possible only when the window has more degrees of freedom near γ than the ordinates have, $a/\pi>\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, which predicts a detection threshold

$$a_{\mathrm{det}}(\gamma,\delta)\approx\tfrac12\log\frac{\gamma}{2\pi}+O(1),$$

with a weak dependence on δ for small δ. Once the test function vanishes at the neighbours and at γ itself, the contribution of the quadruple is $4\,\mathrm{Re}\,F(\gamma-i\delta)^2=-4\delta^2F'(\gamma)^2+O(\delta^4)$, which is negative for small $\delta>0$, and it must exceed the residual of the on line zeros.

We tested this on a hypothetical configuration: add to the on line zeros one quadruple $\frac12\pm\delta\pm i\gamma$, where γ is the midpoint of the gap between two consecutive ordinates (numbers 10 and 11, 79 and 80, 649 and 650, 1517 and 1518). The matrix is the zero side formula of Theorem 15.19.1 for Yoshida's form on $C(a)$, with the first 2100 ordinates (the last at 2620.28) and the modes $|n|\le\lceil a(\gamma+45)/\pi\rceil+2$, whose frequencies $\pi n/a$ reach 45 beyond the height of the quadruple. The on line zeros contribute a positive semidefinite Gram matrix and the quadruple a real symmetric matrix of rank at most four, so any negativity comes from the quadruple. Two criteria are used. In criterion (i) the entry is the smallest a on a grid of step 0.05 at which the least eigenvalue of the float64 matrix is below -10^{-11}. In criterion (ii) it is the smallest a at which the exact least eigenvalue is negative, which is decided in ball arithmetic: in each parity sector the quadruple is a perturbation of rank two of a positive definite matrix, so the sign is decided by a two by two determinant. At the heights 1000.57 and 1999.99 criterion (ii) is only an upper bound, from the principal submatrix on the modes with $|\pi n/a-\gamma|\le150$: by interlacing a detection there is a detection for the full matrix, and a failure to

γ	$\frac{1}{2}\log\frac{\gamma}{2\pi}$	$\delta = 0.01$	$\delta = 0.05$	$\delta = 0.2$	$\delta = 0.45$
Criterion (i): the float64 least eigenvalue falls below -10^{-11}					
51.37	1.051	1.20	1.15	1.15	1.10
199.64	1.729	1.85	1.80	1.75	1.75
1000.57	2.535	2.65	2.65	2.60	2.50
1999.99	2.882	3.05	3.00	2.95	2.45
Criterion (ii): the exact least eigenvalue is negative					
51.37	1.051	1.05	1.00	0.95	0.95
199.64	1.729	1.55	1.50	1.50	1.50
1000.57	2.535	≤ 2.65	≤ 2.65	≤ 2.60	≤ 2.50
1999.99	2.882	≤ 3.05	≤ 3.00	≤ 2.95	≤ 2.45

Table 16.3: Smallest window half length a at which Yoshida's form on $[-a, a]$ detects a hypothetical off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05. The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\leq$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90.

detect proves nothing. The entry of the table is the smaller of this band value and the entry of criterion (i).

The rounding error of the float64 eigenvalue is bounded a priori, from the rounding of the Gram matrix and a textbook bound for the eigensolver, by 4×10^{-12} to 5×10^{-12} at height 51, 9×10^{-12} at 200, about 2.6×10^{-11} at 1000 and 4×10^{-11} at 2000. The cut -10^{-11} of criterion (i) is therefore safe in double precision at the heights 51 and 200, whereas at 1000 and 2000 the bound exceeds the cut, so that there a detection is certified only when the float64 eigenvalue lies below minus the bound. At the printed value of a every one of the sixteen events is certified: in fourteen cells by the float64 eigenvalue and this bound, in the other two (at $(51.37, 0.01)$ and $(199.64, 0.2)$) by the exact test of the cut. The grid point below the printed value is certified as a non detection at the heights 51 and 200 only. Each entry of criterion (i) is an upper bound, for the model of the table, for the sign change in exact arithmetic.

For $\delta \leq 0.2$ every threshold of criterion (i) lies 0.02 to 0.17 above $\frac{1}{2}\log(\gamma/2\pi)$. For $\delta = 0.45$ it lies above it by 0.05 and 0.02 at the heights 51 and 200 and below it by 0.04 and 0.43 at the heights 1000 and 2000 (Figure 16.1). Refined by bisection to 0.005 the thresholds of criterion (ii) are 1.003, 0.975, 0.947, 0.925 at $\gamma = 51.37$

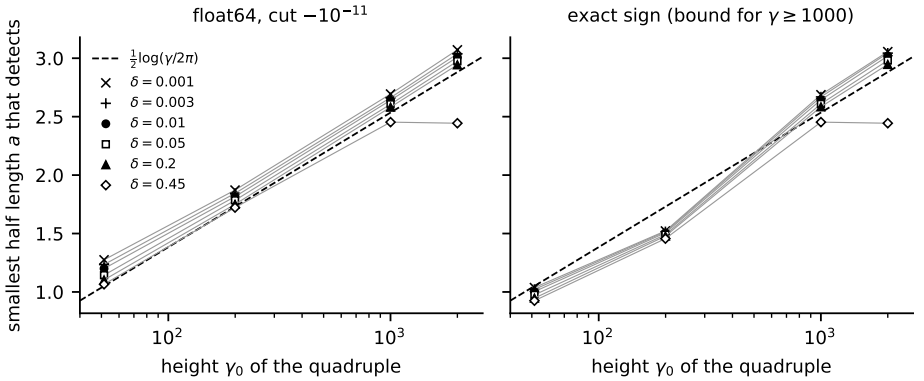


Figure 16.1: Detection thresholds of Table 16.3 and of the displacements $\delta = 0.001$ and 0.003 of the text against the height γ of the hypothetical quadruple (logarithmic axis), for $\delta = 0.001$, 0.003, 0.01, 0.05, 0.2, 0.45 (points of equal δ joined), with the horizon $\frac{1}{2}\log(\gamma/2\pi)$ dashed; the vertical axis is the half length a of the window. Left: criterion (i). Right: criterion (ii), with upper bounds at the heights 1000 and 2000. The points are the thresholds refined by bisection to 0.005, so that they lie up to 0.05 below the entries of the table.

and 1.506, 1.491, 1.478, 1.456 at $\gamma = 199.64$, for $\delta = 0.01$, 0.05, 0.2, 0.45, which lie below the horizon by 0.05 to 0.13 and by 0.22 to 0.27. So in these data the horizon is neither an upper nor a lower bound for the detection threshold. The table is set by the cut and by the model as much as by the horizon: near the cut the negative eigenvalue grows by a factor 2×10^2 to 2×10^4 per step 0.05 in a, so that any cut between 10^{-16} and 10^{-9} moves the thresholds by less than 0.15, but double precision cannot see the sign change itself, because between the exact sign change and the cut the float64 eigenvalue is rounding noise (between -2×10^{-15} and -5×10^{-15} for every a from 1.3 to 1.7 at height 200).

The dependence on δ is weak for small δ. At $\delta = 0.001$ the thresholds of criterion (i) are 1.30, 1.90, 2.70, 3.10 at the four heights, and at $\delta = 0.003$ they are 1.25, 1.85, 2.70, 3.05. From $\delta = 0.001$ to 0.2 they fall by 0.10 to 0.15, and they are nonincreasing in δ in every row. So a displacement of 0.01 is detected almost as early as one of 0.2, but one of 0.45 is detected at height 2000 much earlier than the horizon suggests. The weak dependence is a property of these computations, not a consequence of the expansion above. What lets a large displacement be seen early is the growth of a test function off the real axis: for $F \in PW_a$ bounded on $\mathbb{R}$, the Phragmén Lindelöf bound gives $|F(\gamma - i\delta)|^2 \le e^{2a\delta} \sup_{\mathbb{R}} |F|^2$, which is 9.1 at $a = 2.45$, $\delta = 0.45$. It is an upper bound and not a measured gain, and the data are consistent with the reading that at height 2000 the term of the quadruple, amplified by up to that factor, can beat on line neighbours spaced about 0.9 apart

(the gap that contains the quadruple is 0.889, the mean spacing 1.09) before the window resolves them individually. The experiment does not test this reading.

Three features of the model limit what the table says. First, the configuration is not count preserving. The variant that removes the two on line zeros flanking the gap is detected earlier, by 0.15 to 0.30 in each of the eight cells computed ($\delta = 0.01$ and 0.2 at the four heights), and every one of its thresholds lies at or below the horizon, by 0.03 to 0.24. The thresholds of the table are therefore those of a configuration with two extra zeros, not of one with the true zero count. Second, the truncation to 2100 ordinates can only advance detection, because the omitted zeros add positive semidefinite terms. In the eight cells that were also run with 4000 and with 8000 ordinates the float64 thresholds are unchanged, but the rounding bound grows with the number of zeros, and the entry $(1999.99, 0.2)$ is no longer certified at either size (the certified value is 3.00), nor is $(1000.57, 0.01)$ at 8000. Third, with the margin 90 in place of 45 in the number of modes the entry $(1999.99, 0.01)$ is 3.00 and not 3.05.

Compare the tent coefficient $D(x) = x\langle\chi_0, \chi_0\rangle$ of Chapter 15.12 at the level $x = 2a$. By Remark 15.18.3 it turns negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$, with $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231049931$. The tent is one vector of the window $C(a)$, so $x_*/2$ is only an upper bound for the half length at which the window detects the quadruple. For $\delta = 0.2$ the level at which the quadruple can first push D negative for some phase, computed from the first 8000 ordinates, gives $x_*/2 = 8.68$, 16.2, 24.2, 27.8 at the four heights, against the thresholds 1.15, 1.75, 2.60, 2.95 of Table 16.3. Over the 16 cells of the table the window thresholds are 4.0 to 177 times smaller than $x_*/2$, the ratio being roughly $2/\delta$ (144 to 177 at $\delta = 0.01$, 29 to 37 at 0.05, 7.5 to 9.4 at 0.2 and 4.0 to 5.1 at 0.45). The formula for x_* overestimates the computed level by 2 to 22 percent, and the window data do not support an order $\delta^{-1}\log\gamma$ for the window threshold itself. For the height 3×10^{12} the formula gives $x_*/2 = 26.85/\delta$, that is 59.7 at $\delta = 0.45$ and 134 at $\delta = 0.2$, against the half length 13.45 of the horizon, with the same ratio $2/\delta$. Proposition 15.21.1 (level 47) is proved for D only, and Table 16.3 shows that the window is far more sensitive than the coefficient, so 47 is not the level at which windows begin to see beyond the verification of the zeros, and the conversion from half lengths to levels, here and in Chapter 15.12, is a heuristic.

The horizon turns a window statement into a height statement only heuristically, and only for small displacements. Large displacements are seen below the horizon at the heights 1000 and 2000 (Table 16.3), and if the margin below the horizon keeps growing beyond the range tested, a window of half length near 13 could see a large displacement far above 3×10^{12}. Table 16.4 lists the heights $2\pi e^{2a}$ of some window statements.

Yoshida's window lies below the first ordinate: its height $4\pi = 12.57$ is less than $\gamma_1 = 14.13$. Zhu's certificate reaches about the fourth ordinate ($\gamma_4 = 30.42 < 31.12 < \gamma_5 = 32.94$). Under the heuristic, a statement about these windows speaks

window statement	half length a	height $2\pi e^{2a}$
Yoshida's Theorem 1, as Zhu reports it	0.347	12.6
Zhu's certificate (Z2)	0.8	31.1
practical limit of envelope certificates, Zhu's Remark 1.6	≈ 1.6	154
verification of the zeros to 3×10^{12} [232]	≈ 13.4	3×10^{12}
RH	∞	∞

Table 16.4: The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem.

about the zeros below the height shown, which the verification of the zeros covers far beyond.

16.5 The comb barrier, basepoints and damping

Damping to a basepoint $\alpha \geq \frac{1}{2}$ (Section 15.20) replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$ in the comb. Put

$$A_a(\alpha) = \sum_{\log n < 2a} \frac{2\Lambda(n)}{n^\alpha}, \qquad T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}, \tag{16.5.1}$$

so that $A_a(\frac{1}{2}) = A_a$ and $T_1(\frac{1}{2}, a) = T_1$. For $\alpha > \frac{1}{2}$ the height $T_1(\alpha, a)$ is a definition by analogy and not the cutoff of a theorem: Zhu's Theorem 1.1 is a statement about the undamped form, no damped analogue is proved here, and for $\frac{1}{2} < \alpha < 1$ the pole part of the damped form, with kernel $2e^{-(\alpha-1/2)|y|}\cosh(y/2)$, grows like $e^{(1-\alpha)|y|}$ and is not a positive rank one term. By the prime number theorem, as $a \to \infty$,

$$\begin{gathered} A_a(\alpha) \sim \frac{2e^{2(1-\alpha)a}}{1-\alpha} \quad (\tfrac{1}{2} \leq \alpha < 1), \qquad A_a(1) = 4a - 2\gamma_E + o(1), \\ A_a(\alpha) \text{ increases to } -2\frac{\zeta'}{\zeta}(\alpha) \quad (\alpha > 1). \end{gathered} \tag{16.5.2}$$

Table 16.5 lists the values, computed.

Corollary 16.5.1. *Let $\alpha \geq 1$, $a > 0$ and $f \in C(a)$, and write $F = \widehat{f}$ and $b_\alpha = \xi'(\alpha)/\xi(\alpha)$. With no hypothesis on the zeros,*

$$\langle f, f\rangle_\alpha = \frac{1}{\pi}\int_{\mathbb{R}} |F(t)|^2 \operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt \;\geq\; 2b_\alpha \|f\|^2, \qquad b_\alpha = \inf_{t\in\mathbb{R}} \operatorname{Re}\frac{\xi'}{\xi}(\alpha+it) > 0,$$

and the constant $2b_\alpha$ does not depend on a.

	$\alpha=\frac12$		$\alpha=0.75$		$\alpha=1$	
a	$A_a(\alpha)$	$\log_{10}T_1$	$A_a(\alpha)$	$\log_{10}T_1$	$A_a(\alpha)$	$\log_{10}T_1$
0.35	0.980	1.22	0.824	1.16	0.693	1.10
0.80	2.942	2.08	2.278	1.79	1.772	1.57
1.60	14.323	7.02	8.497	4.49	5.222	3.07
2.00	24.383	11.39	12.566	6.26	6.874	3.78
3.00	75.190	33.45	26.647	12.37	10.856	5.51
4.00	213.548	93.54	49.930	22.48	14.855	7.25
5.00	588.024	256.17	88.183	39.10	18.844	8.98

Table 16.5: The comb constant $A_a(\alpha)$ of (16.5.1) and the barrier height $T_1(\alpha,a)=2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha=\frac12$ is Zhu's [279], who prints $A_{0.8}(\frac12)=2.9420$ and $T_1=119$ at $a=0.8$, 1.0×10^7 at $a=1.6$ and 2.4×10^{11} at $a=2$; the digits here are 2.94197, 119, 1.04×10^7 and 2.44×10^{11}. For $\alpha>1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha)=18.88$, 3.01, 1.14 at $\alpha=1.1$, 1.5, 2.

First proof, $\alpha\ge1$. With $\sigma_{\max}=\sup_\rho\operatorname{Re}\rho$ (Section 15.20.1), $\sigma_{\max}\le1\le\alpha$ and no zero has real part α, so Proposition 15.20.1 gives the identity, and Theorem 15.20.5(i) gives the inequality with $m_\alpha=\inf_t\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$. By part (ii) of that theorem, which for $\alpha\ge1$ needs no hypothesis (Propositions 12.1.11 and 12.1.12), $m_\alpha=b_\alpha$, and $b_\alpha>0$ because the density $v_\alpha=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\cdot)$ is positive (Theorem 2.1.2). □

Second proof, $\alpha>1$. Let $g=f*\widetilde{f}$, which is supported in $[-2a,2a]$ and has $\widehat{g}=|F|^2$. By (15.13.1) with the damping $e^{-(\alpha-1/2)|y|}$, the form is the sum of a pole part, an archimedean part and the prime part $-\sum_{n\ge2}\Lambda(n)n^{-\alpha}\big(g(\log n)+g(-\log n)\big)$. Since $\int_{\mathbb R}|F(t)|^2\cos(t\log n)\,dt=\pi\big(g(\log n)+g(-\log n)\big)$, which vanishes for $\log n>2a$, and since for $\alpha>1$ the full comb $\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)$ converges absolutely and uniformly in t, the prime part equals $-\frac{1}{2\pi}\int|F|^2\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)\,dt$ with the full comb, inserted term by term. The pole part is $\int g(y)\big(e^{-(\alpha-1)|y|}+e^{-\alpha|y|}\big)dy$, which is $\frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\big(\frac1s+\frac{1}{s-1}\big)dt$ at $s=\alpha+it$, because $e^{-w|y|}$ has Fourier transform $2w/(w^2+t^2)$. The archimedean part is the Poisson smoothing by $P_{\alpha-1/2}$ of the multiplier $\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi$, and this is $\operatorname{Re}\psi(\frac s2)-\log\pi$ at $s=\alpha+it$, because $z\mapsto\psi(\frac14+\frac{iz}{2})$ is holomorphic for $\operatorname{Im}z<\frac12$. Now $\frac{\xi'}{\xi}(s)=\frac1s+\frac{1}{s-1}-\frac12\log\pi+\frac12\psi(\frac s2)+\frac{\zeta'}{\zeta}(s)$ and $-\frac{\zeta'}{\zeta}(s)=\sum\Lambda(n)n^{-s}$ for $\operatorname{Re}s>1$, so the three multipliers add up to $2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$, and $\langle f,f\rangle_\alpha=\frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt$. This symbol is positive (Theorem 2.1.2) and tends to infinity like $\frac12\log|t|$, because $\operatorname{Re}\frac{\zeta'}{\zeta}$ is bounded on the line $\operatorname{Re}s=\alpha$; its infimum is therefore positive and equals $2b_\alpha$ by Theorem 15.20.5(ii), and Plancherel, $\int|F|^2=2\pi\|f\|^2$, gives the bound. □

For $\alpha \geq 1$ no certificate is needed: the damped form is positive on every window (Theorem 15.20.3(i)), although the envelope height $T_1(1,a)$ still grows like e^{4a} (Table 16.5). The line $\operatorname{Re} s = 1$ is the same boundary as in the basepoint picture: the shifted clock $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is a GGC with Thorin density $v_\alpha > 0$ for every $\alpha \geq 1$ with no hypothesis, and for $\alpha < 1$ exactly when $\alpha \geq \sigma_{\max}$ (Theorem 2.1.2). The one sided bound on ζ'/ζ that corresponds to $v_\alpha > 0$ is the bound of Theorem 23.1.1(ii), which for $\alpha < 1$ holds under the hypothesis of that theorem.

For $\frac{1}{2} \leq \alpha < 1$ the full comb diverges, since $\sum \Lambda(n) n^{-\alpha}$ diverges, and the second proof is not available. The symbol $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + it)$ still exists by analytic continuation, and the damped form is its multiplier whenever $\alpha \geq \sigma_{\max}$ and no zero has real part α (Proposition 15.20.1). Positivity of the symbol on this one line follows from the absence of zeros with real part at least α and is not known to imply it (Remark 2.1.4), whereas positivity of the damped form on every window is equivalent to the absence of zeros with real part greater than α (Theorem 9.1.2, Remark 15.20.4). What the comb would have to supply there is the one sided bound for $-\operatorname{Re} \zeta'/\zeta(\alpha+it)$ of Section 15.22, that is cancellation in $\sum \Lambda(n) n^{-\alpha-it}$ beyond what absolute convergence gives, which is Problem 15.5 of Chapter 15.12.

16.5.1 Damping removes the collapse

Write $\lambda_{\min}(\alpha, a)$ for the least eigenvalue of the damped form of Section 15.20 on the window $[-a, a]$, over complex test functions in $C(a)$ (Section 15.20.2), and $\lambda_{\min}^{\text{even}}(\alpha, a)$ for the infimum of $\langle f, f\rangle_\alpha / \|f\|^2$ over real even $f \in C(a)$. By the splitting of Section 16.1, $\lambda_{\min}(\alpha, a) \leq \lambda_{\min}^{\text{even}}(\alpha, a)$.

Let $p_1 = \frac{1}{2}(\log \xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0} \gamma^{-2}$ under RH, and $p_2 = 3.7172599\ldots \times 10^{-5}$, which is $\sum_{\gamma>0} \gamma^{-4}$ under RH, as in Theorem 15.20.5.

Theorem 16.5.2 (Two parameter bound). *Let $a > 0$ and $\alpha > \frac{1}{2}$, and write $b_\alpha = \xi'(\alpha)/\xi(\alpha)$.*

(i) With no hypothesis, $\lambda_{\min}(\alpha, a) \geq \lambda_{\min}(\frac{1}{2}, a)$.

(ii) If RH holds, or if $\alpha \geq 1$, then $\lambda_{\min}(\alpha, a) \geq 2b_\alpha$. Here $2b_\alpha \leq 4p_1(\alpha - \frac{1}{2})$ with no hypothesis, and under RH $2b_\alpha \geq 4p_1(\alpha - \frac{1}{2}) - 4p_2(\alpha - \frac{1}{2})^3$.

Hence, under RH, $\lambda_{\min}(\alpha, a) \geq \max\big(\lambda_{\min}(\frac{1}{2}, a),\, 2b_\alpha\big)$ for every $\alpha > \frac{1}{2}$ and every $a > 0$.

Proof. (i) Put $\varepsilon = \alpha - \frac{1}{2}$ and recall $e^{-\varepsilon|y|} = \int_{\mathbb{R}} P_\varepsilon(\lambda) e^{i\lambda y} d\lambda$ with the Poisson kernel $P_\varepsilon(\lambda) = \varepsilon/(\pi(\varepsilon^2 + \lambda^2))$. For $f \in C(a)$ put $f_\lambda(y) = e^{i\lambda y} f(y) \in C(a)$, so that $f_\lambda * \widetilde{f_\lambda} = e^{i\lambda y}\,(f * \widetilde{f})$. The interchange of $\mathcal{Y}$ and the integral over λ is that of the

proof of Theorem 15.20.3(iii), taken from the basepoint $\frac{1}{2}$, and gives

$$\langle f,f\rangle_\alpha = \mathcal{Y}\big(e^{-\varepsilon|y|}f * \widetilde{f}\big) = \int_{\mathbb{R}} P_\varepsilon(\lambda)\langle f_\lambda, f_\lambda\rangle_{1/2}\, d\lambda$$
$$\geq \lambda_{\min}(\tfrac{1}{2},a)\int_{\mathbb{R}} P_\varepsilon(\lambda)\|f_\lambda\|^2 d\lambda = \lambda_{\min}(\tfrac{1}{2},a)\|f\|^2,$$

because $\|f_\lambda\| = \|f\|$ and $\int P_\varepsilon = 1$.

(ii) Under RH, $\sigma_{\max} = \frac{1}{2}$ and no zero has real part $\alpha > \frac{1}{2}$; for $\alpha \geq 1$, $\sigma_{\max} \leq 1 \leq \alpha$ and no zero has real part α. In both cases Theorem 15.20.5(i) gives $\lambda_{\min}(\alpha,a) \geq 2m_\alpha$, and part (ii) of that theorem gives $m_\alpha = b_\alpha$, under RH and with no hypothesis for $\alpha \geq 1$. The bounds on $2b_\alpha$ are Theorem 15.20.5(iv). □

At $a = 0.8$ the first bound is certified: Corollary 6.3 of [279] gives $\lambda_{\min}(\frac{1}{2},0.8) \geq 8.9\times10^{-18}$ for complex test functions, so by (i) $\langle f,f\rangle_\alpha \geq 8.9\times10^{-18}\|f\|^2$ for every $\alpha \geq \frac{1}{2}$ and every $f \in C(0.8)$, with no hypothesis. The second bound is not a multiple of $4p_1(\alpha-\frac{1}{2})$ but only comparable to it: $2b_\alpha$ lies between $4p_1(\alpha-\frac{1}{2}) - 4p_2(\alpha-\frac{1}{2})^3$ and $4p_1(\alpha-\frac{1}{2})$, so under RH a bound $\lambda_{\min}(\alpha,a) \geq 4p_1(\alpha-\frac{1}{2})$ would fail for large a, where $\lambda_{\min}(\alpha,a) \to 2b_\alpha$ (Theorem 15.20.5(i) and (ii)). Unconditionally the second bound is available for $\alpha \geq 1$ only (Corollary 16.5.1); at a single $\alpha < 1$ with no zero on the line the inequality $\lambda_{\min}(\alpha,a) \geq 2b_\alpha$ for all a is equivalent to $\sigma_{\max} \leq \alpha$ together with $v_\alpha \geq b_\alpha$ on the line (Remark 15.20.6).

Numerically, the matrices of Table 16.1 with the damping give upper bounds for $\lambda_{\min}^{\text{even}}(\alpha,a)$ (computed at 60 digits, for 26 and 40 modes). In all 44 values, for $a = 0.5$ and 0.8 and $\alpha - \frac{1}{2}$ between 10^{-17} and 10^{-3}, the finite section value lies above both bounds of the theorem: the least ratio to the floor $2b_\alpha$ is 1.030 at $a = 0.8$, $\alpha - \frac{1}{2} = 10^{-4}$ and 1.121 at $a = 0.5$, $\alpha - \frac{1}{2} = 10^{-3}$, and the least ratio to the undamped value of the same size is 1.011 at $a = 0.5$, $\alpha - \frac{1}{2} = 10^{-7}$ and 1.047 at $a = 0.8$, $\alpha - \frac{1}{2} = 10^{-17}$.

The damped value is the sum of the two terms and not their maximum. For $\alpha - \frac{1}{2} = \varepsilon$ the difference $\lambda_{\min}^{\text{even}}(\frac{1}{2}+\varepsilon,a) - \lambda_{\min}^{\text{even}}(\frac{1}{2},a)$ equals $s\cdot 4p_1\varepsilon$ with $s = 1.111$ at $a = 0.5$ for ε from 10^{-7} to 10^{-4} and $s = 1.082$ at $a = 0.8$ for ε from 10^{-17} to 10^{-14}, the same at 26 and 40 modes to the digits quoted. For larger ε at $a = 0.8$ the factor s falls, to 1.067, 1.043, 1.040 and 1.030 at $\varepsilon = 10^{-10}$, 10^{-8}, 10^{-6} and 10^{-4}, so that from $\varepsilon = 10^{-8}$ on the finite section value lies within 4.5 percent of the floor $2b_\alpha$; the fall begins at ε of order 10^{-10}, where ε times the slope is of the size of the gap of about 10^{-11} to the second even eigenvalue. Far above the crossover the least eigenvalue is close to the floor, as the converged entries of Table 15.10 show.

The two terms of the theorem are equal, to first order in the damping, at the crossover $\varepsilon_\times = \lambda_{\min}^{\text{even}}(\frac{1}{2},a)/(4p_1)$, at which the least eigenvalue is about twice $\lambda_{\min}^{\text{even}}(\frac{1}{2},a)$ (the damping at which it exactly doubles is 0.90 times $\varepsilon_\times$ at $a = 0.5$ and 0.92 times $\varepsilon_\times$ at $a = 0.8$, for 26 modes). Its value depends on the number used for $\lambda_{\min}^{\text{even}}(\frac{1}{2},a)$; Table 16.6 gives the choices (the 26 mode values are not converged).

a	value used for $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)$	λ	$\lambda/(4p_1)$
0.5	Zhu, certified upper bound	1.05×10^{-6}	1.14×10^{-5}
0.5	26 modes, Table 16.1	9.743×10^{-7}	1.05×10^{-5}
0.5	extrapolated from 256 to 400 modes	9.33×10^{-7}	1.01×10^{-5}
0.8	Zhu, certified lower bound	8.9×10^{-18}	9.6×10^{-17}
0.8	extrapolated from 256 to 400 modes	1.63×10^{-17}	1.77×10^{-16}
0.8	26 modes, Table 16.1	2.144×10^{-17}	2.32×10^{-16}
0.8	Zhu, certified upper bound	2.27×10^{-17}	2.46×10^{-16}

Table 16.6: The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 16.5.2 for several values of $\lambda = \lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices, not bounds; Zhu's reference values of his Table 1, 9.34×10^{-7} and 1.65×10^{-17}, give 1.01×10^{-5} and 1.79×10^{-16}.

The collapse is explained at the end of Section 15.20. A damping of the size of the crossover already doubles the finite section values (computed); for larger damping the least eigenvalue is, under RH, at least $2b_\alpha$, a margin of order $\alpha-\frac{1}{2}$ in place of $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)$. Whether certificates for the damped forms say more or less about the zeros than the certificates at $\alpha=\frac{1}{2}$ is not settled here (Problem 16.3).

16.6 What window certificates reach

A window of half length a resolves the zeros up to the horizon $2\pi e^{2a}$ (a heuristic, Section 16.4, after Section 9 of [279]), and whether the verification of the zeros up to a height H implies positivity on the window of half length $\frac{1}{2}\log(H/2\pi)-C$ is Problem 16.1; for large displacements it may fail.

16.7 Open problems

Problem 16.1 (The horizon as a theorem). Determine the largest half length $a(H,\delta_0)$ such that, if all zeros with $0<\gamma\le H$ lie on the line and every zero above H has $|\beta-\frac{1}{2}|\le\delta_0$, Weil's form is positive on the window $[-a,a]$ with $a=a(H,\delta_0)$. Is $a(H,\delta_0)\ge\frac{1}{2}\log\frac{H}{2\pi}-C(\delta_0)$, with a constant $C(\delta_0)$ that depends only on δ_0? Status: open. The heuristic of Section 16.4 suggests the bound for small δ_0. A window gains up to a factor $e^{2a\delta}$ on a zero at distance δ from the line, and in Table 16.3 the largest displacement is detected below the horizon at the largest heights, so a constant C that does not depend on δ_0 may not exist. A proof would need a quantitative form of the lower sampling inequality of Proposition 16.2.1 for the ordinates, with constants that depend only on the local density of the zeros. Compare Proposition 15.21.1, which is proved for the tent coefficient only.

Problem 16.2 (Decay law from the Thorin side). Derive the exponent of Zhu's decay law, $-\log\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)=C\,N(T_*)/\log N(T_*)\,(1+o(1))$ with $T_*=2\pi e^{2a}$ and

the fitted constant $C = 20.13\ldots$ of his equation (5) (his Conjecture 12.1 [279] states the law with $C = 2\pi^2$), from Zhu's test function $\widehat{f} = h\,G_T(-r^2)$ of Section 16.3, optimising T and h jointly, and explain the constant, which Zhu finds close to $2\pi^2 = 19.74\ldots$, through the Landau Widom plunge. Status: open. Zhu fits the law to certified upper bounds for $a \le 1.2$, tests it on $a = 1.4$ to 2, and states that the constant is fitted and not derived. The only proved decay beyond the bound $O(a^{-N})$ for every N of Theorem 9.2.1(c) of Chapter 9 is his bound $\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) \le \exp(-ae^a)$ for large a, under RH (Theorem 1.3), which kills the zeros only up to $2\pi e^a$, half the exponent of T_*, and is weaker than the law.

Problem 16.3 (Certified damped windows). Extend the certification of Zhu's Theorem 1.2 to the damped forms at $\alpha \in (\frac{1}{2}, 1)$, where under RH the margin is at least $2\xi'(\alpha)/\xi(\alpha)$, of order $\alpha - \frac{1}{2}$, in place of $\lambda_{\min}(\frac{1}{2}, a)$ (Theorem 16.5.2), and measure how far a can be pushed. Status: open. No damped analogue of Zhu's Theorem 1.1 is proved here, and below $\alpha = 1$ the damped pole part is not a positive rank one term (Section 16.5); for $\alpha \ge 1$ no certificate is needed (Corollary 16.5.1). The question for the rungs of Chapter 9 is Problem 9.1.

Problem 16.4 (Comb cancellation on average). Zhu's barrier concerns the supremum of the comb $\mathcal{P}_a(t) = \sum_{\log n < 2a} 2\Lambda(n) n^{-1/2} \cos(t \log n)$. The form sees the comb only through $\int |F|^2 \mathcal{P}_a$ with F in the Paley Wiener space PW_a of Proposition 16.2.1, an average over a frequency band of width about $1/a$. Determine the supremum of $\int_{|t| \ge T} |F|^2 \mathcal{P}_a \big/ \int_{|t| \ge T} |F|^2$ over $F \in PW_a$, as a function of T and a. Status: open. Zhu's Lemma 3.2 shows that the pointwise supremum of $\mathcal{P}_a$ is A_a, which bounds the ratio from above. The average over the whole line is smaller than A_a for every F, because $g = f * \widetilde{f}$ vanishes at $\pm 2a$ and damps the terms with $\log n$ near $2a$; how much smaller the restricted ratio is, uniformly in T, is not known, and a bound of smaller order than $A_a = (4 + o(1))e^a$ would be a statement about short sums of $\Lambda(n) n^{-1/2-it}$, the object of the Fejér tapered sums of Proposition 19.14.1.

Notes

Zhu's theorems and heuristics (the horizon and the plunge rate of his Section 9, the parity of the ground state) are from [279] and are credited where they are stated. Yoshida's form is [278], Weil's criterion is [273, 29], and the verification of the zeros to 3×10^{12} is [232]. The computations of the chapter are its own: the table of least eigenvalues in 60 digit arithmetic (Table 16.1), the optimisation of Zhu's product test function (Section 16.3), the comb table (Table 16.5), the detection experiment (Table 16.3 and Figure 16.1) and the damping numbers (Table 16.6).

Part IV

Approximants and finite tests

Chapter 17

Truncations: the Biane Pitman Yor lattice, the beta lattice and Hermite coefficients

17.1 Truncations, finite conditions and the road to RH

Cutting the series for S_2 in Theorem 11.1.21 after n terms gives variables whose Mellin transforms are a gamma factor times an exponential polynomial, with zeros converging to the points $\rho/2$; a condition on these zeros for infinitely many n implies RH. The one sided condition fails because the side from which the truncated zeros approach the line changes at zeros where $|S(\gamma)|$ is large, with $S(t) = \pi^{-1} \arg \zeta(\frac{1}{2} + it)$ (Appendix A). Symmetrising by the functional equation removes this mechanism.

17.1.1 Truncations of the Biane, Pitman and Yor lattice

In this subsection w is the complex variable of a Mellin transform and $u = w + 1$.

17.1.1.1 Truncated lattices and Conjecture T

Let e_k, e'_k be independent standard exponential variables, and put $\mathcal{S} = \sum_{k\geq 1} k^{-2}(e_k + e'_k) = \frac{\pi^2}{2} S_2$, with S_2 the variable of Theorem 11.1.21, and $\mathcal{S}_n = \sum_{k\leq n} k^{-2}(e_k + e'_k)$, a GGC with finitely many Thorin atoms. Put $F(w) = \mathbb{E}\mathcal{S}^w$ and $F_n(w) = \mathbb{E}\mathcal{S}_n^w$ for $\operatorname{Re} w > -2$. By Theorem 11.1.21,

$$F(w) = 2\pi^w \, \xi(2w),$$

so the zeros of F are the points $w = \rho/2$, and RH says they lie on $\operatorname{Re} w = \frac{1}{4}$. Each F_n is explicit. Define α_k and β_k, which depend on n, by the partial fraction expansion

$$\prod_{j\leq n} \Big(1 + \frac{\lambda}{j^2}\Big)^{-2} = \sum_{k\leq n} k^{-2} \Big\{ \alpha_k \Big(1 + \frac{\lambda}{k^2}\Big)^{-1} + \beta_k \Big(1 + \frac{\lambda}{k^2}\Big)^{-2} \Big\},$$

so that $\beta_k = k^2 \prod_{j\le n,\, j\ne k} (1 - k^2/j^2)^{-2}$ and $\alpha_k = -2\beta_k \sum_{j\le n,\, j\ne k} k^2/(j^2 - k^2)$; for $n = 2$ they are $\beta = (\frac{16}{9}, \frac{4}{9})$ and $\alpha = (-\frac{32}{27}, \frac{32}{27})$. The left side is the Laplace transform of $\mathcal{S}_n$, whose density is therefore $\sum_k (\alpha_k + \beta_k k^2 x) e^{-k^2 x}$, and taking moments gives, with $u = w + 1$, $F_n(u-1) = \Gamma(u) Q_n(u)$, where $Q_n(u) = \sum_{k\le n} k^{-2u}(\alpha_k + \beta_k u)$.

THEOREM 17.1.1. *If for infinitely many n the function F_n has no zeros in the half plane $\operatorname{Re} w > \frac14$, then the Riemann Hypothesis holds.*

Proof. For $\operatorname{Re} w > -2$ the variables $\mathcal{S}_n^w$ are dominated by an integrable variable, since $\mathcal{S}_n \ge e_1 + e_1'$ and $\mathcal{S}_n \le \mathcal{S}$, so $F_n \to F$ locally uniformly on $\operatorname{Re} w > -2$. If F had a zero w_0 with $\operatorname{Re} w_0 > \frac14$, Hurwitz's theorem would give zeros of F_n near w_0 for all large n. Hence every zero $\rho/2$ of F has $\operatorname{Re}\rho \le \frac12$, and the symmetry $\rho \mapsto 1 - \rho$ gives $\operatorname{Re}\rho = \frac12$. □

CONJECTURE T. For every $n \ge 1$, $F_n(w) = \mathbb{E}\big[\big(\sum_{k\le n} k^{-2}(e_k + e_k')\big)^w\big]$ has no zeros with $\operatorname{Re} w > \frac14$.

Conjecture T is a stability statement of Hermite Biehler or Lee and Yang type [191]. It is false (Corollary 17.1.4); we state it because the mechanism of its failure also defeats, for large n, the Hermite Biehler inequality on which the bounds of Theorems 17.1.9 and 17.1.11 for the symmetrised truncations rest.

At small n Table 17.1 finds no zero of Q_n with $\operatorname{Re} u > \frac54$ in any rectangle searched; zeros are counted by the argument principle and located by Newton's method from a grid, and the two agree.

n	region, in $u = w + 1$	zeros with $\operatorname{Re} u > \frac54$
$2, 3, 4, 5, 6, 8, 12, 16, 24, 32$	$\frac54 < \operatorname{Re} u < 6$, $0.3 < \operatorname{Im} u < 40$	none
$8, 16, 32$	$\frac54 < \operatorname{Re} u < 6$, $40 < \operatorname{Im} u < 100$	none
$48, 64$	$\frac54 < \operatorname{Re} u < 6$, $0.3 < \operatorname{Im} u < 30$	none
$16, 32, 64$	$6 < \operatorname{Re} u < 40$, $0.3 < \operatorname{Im} u < 40$	none

Table 17.1: Argument principle counts for Conjecture T: rectangles searched and zeros of Q_n found with $\operatorname{Re} u > \frac54$.

For fixed n and large $|\operatorname{Im} u|$ the zeros move far to the left, because β_1 and β_n differ by a factor of order 16^n, up to a power of n. Conjecture T is therefore at risk at intermediate heights, which Table 17.1 searches; the failure of Corollary 17.1.4 lies above them (Remark 17.1.5). An experiment shows that the absence of such zeros at small n is not a consequence of the GGC structure alone: replacing the weights k^{-2} by $k^{-2}(1 + \eta_k)$, with independent η_k uniform on $[-0.15, 0.15]$, one of two draws at $n = 16$ gives a zero with $\operatorname{Re} u > \frac54$.

17.1.1.2 How the truncated zeros reach the line, and why Conjecture T fails

PROPOSITION 17.1.2. *Let ρ be a simple zero of ξ and put*

$$\kappa(\rho) = \frac{\rho\,\xi(\rho-2)}{2\pi\,\xi'(\rho)} = \frac{\rho\,\xi(3-\rho)}{2\pi\,\xi'(\rho)}.$$

There is a constant C such that for all large n, F_n has exactly one zero $w_n(\rho)$ in the disc $|w - \frac{\rho}{2} - \frac{\kappa(\rho)}{n}| < C/n^2$; thus $w_n(\rho) = \frac{\rho}{2} + \frac{\kappa(\rho)}{n} + O(n^{-2})$.

Proof. Write $\mathcal{S} = \mathcal{S}_n + \mathcal{T}_n$ with $\mathcal{T}_n = \sum_{k>n} k^{-2}(e_k + e'_k)$, independent of $\mathcal{S}_n$, so that $\mathbb{E}\mathcal{T}_n = 2\sum_{k>n} k^{-2} = \frac{2}{n} + O(n^{-2})$ and $\mathbb{E}\mathcal{T}_n^2 = O(n^{-2})$. For $a > 0$, $b \ge 0$ and $0 < \operatorname{Re} w < 2$, Taylor's formula with integral remainder and $|(a+tb)^{w-2}| = (a+tb)^{\operatorname{Re} w-2} \le a^{\operatorname{Re} w-2}$ give

$$\bigl|(a+b)^w - a^w - wb\,a^{w-1}\bigr| \le \tfrac{1}{2}|w(w-1)|\,b^2 a^{\operatorname{Re} w-2},$$
$$\bigl|(a+b)^{w-1} - a^{w-1}\bigr| \le |w-1|\,b\,a^{\operatorname{Re} w-2}.$$

Since $\mathcal{S}_n \ge e_1 + e'_1$, a $\Gamma(2)$ variable, $\mathbb{E}\mathcal{S}_n^{\operatorname{Re} w-2} \le \Gamma(\operatorname{Re} w)$. Taking expectations with $a = \mathcal{S}_n$ and $b = \mathcal{T}_n$, and using independence,

$$F(w) = F_n(w) + w\,\mathbb{E}\mathcal{T}_n\,F_n(w-1) + O(n^{-2}), \qquad F_n(w-1) = F(w-1) + O(n^{-1}),$$

uniformly on compact subsets of $0 < \operatorname{Re} w < 2$. Hence $F_n(w) = F(w) - \frac{2}{n}wF(w-1) + O(n^{-2})$ uniformly there. Put $w_0 = \rho/2$, so $0 < \operatorname{Re} w_0 < \frac{1}{2}$, and recall $F(w_0) = 0$ and $F'(w_0) = 4\pi^{w_0}\xi'(\rho) \ne 0$. For $|w - w_0| = O(1/n)$, Taylor expansion of F and of $wF(w-1)$ about w_0 gives

$$F_n(w) = F'(w_0)\Bigl(w - w_0 - \frac{\kappa(\rho)}{n}\Bigr) + O(n^{-2}), \qquad \frac{\kappa(\rho)}{n} = \frac{2}{n}\,\frac{w_0 F(w_0-1)}{F'(w_0)},$$

and $F(w_0 - 1) = 2\pi^{w_0-1}\xi(\rho-2)$ turns the second identity into the stated κ. On the circle $|w - w_0 - \kappa/n| = C/n^2$ the linear term has modulus $|F'(w_0)|C/n^2$, which exceeds the $O(n^{-2})$ error once C is larger than its constant divided by $|F'(w_0)|$. Rouché's theorem gives exactly one zero inside for all large n. The second form of κ follows from $\xi(s) = \xi(1-s)$. □

If $\operatorname{Re}\kappa(\rho) > 0$ the zero $w_n(\rho)$ lies to the right of the critical line for all large n. This happens at zeros where $|S(\gamma)|$ is large, the first being γ_{213} (Theorem 17.1.3 and the computation after it), and is why Conjecture T fails.

For each of the first 100 zeros $\operatorname{Re}\kappa(\rho) < 0$, starting with $\kappa(\rho_1) = -4.762 - 1.944i$, $\kappa(\rho_2) = -8.673 - 0.627i$, $\kappa(\rho_3) = -10.149 - 3.872i$ and $\kappa(\rho_4) = -15.165 + 3.705i$. These truncated zeros approach the line from the left, at distance about $-\operatorname{Re}\kappa(\rho)/n$, which grows with the height. Figure 17.1 shows the first four, with the zeros of F_n located directly at each n, an independent check on Proposition 17.1.2 at low height.

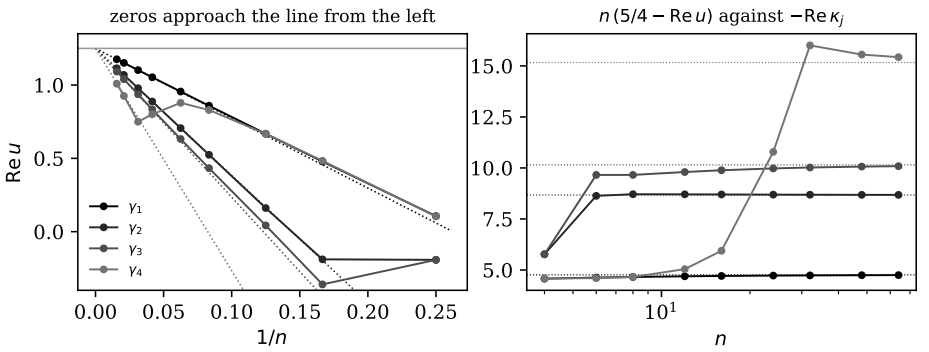


Figure 17.1: Left: real parts of the zeros of F_n near $\rho_j/2$, in the variable $u = w+1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against n, converging to $-\operatorname{Re}\kappa(\rho_j)$.

THEOREM 17.1.3. *Let $\rho = \frac{1}{2} + i\gamma$ be a simple zero on the critical line, let θ be the Riemann Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ Hardy's function, and put $a(\gamma) = e^{i\theta(\gamma)}\zeta(\frac{5}{2} + i\gamma)$. Then*

$$\operatorname{Re}\kappa(\rho) = \frac{(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma) - 4\gamma \operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)},$$
$$\operatorname{Im}\kappa(\rho) = \frac{4\gamma \operatorname{Im} a(\gamma) + (\gamma^2 - \frac{15}{4})\operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)}.$$

Proof. On the line $\xi(\frac{1}{2} + it) = \Xi(t)$ is real, so $\xi'(\rho) = -i\,\Xi'(\gamma)$, and $\Xi(t) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ gives $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|\overline{Z'(\gamma)}$ at a zero. Since $\xi(3-\rho) = \xi(\frac{5}{2} - i\gamma)$ and ξ is real on the real axis, $\xi(3-\rho) = \overline{\xi(\frac{5}{2} + i\gamma)}$, and $\Gamma(\frac{5}{4} + \frac{i\gamma}{2}) = (\frac{1}{4} + \frac{i\gamma}{2})\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$ with $\pi^{-i\gamma/2}\Gamma(\frac{1}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|e^{i\theta(\gamma)}$. Substituting, the moduli of Γ cancel, $(\frac{1}{2} + i\gamma)(\frac{1}{4} - \frac{i\gamma}{2}) = \frac{1}{2}(\gamma^2 + \frac{1}{4})$ cancels the remaining factor, and $\kappa = -i(\frac{5}{2} - i\gamma)(\frac{3}{2} - i\gamma)\overline{a(\gamma)}/(4\pi^2 Z'(\gamma))$, whose real and imaginary parts are as stated. □

For large γ the term $(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma)$ decides the sign of $\operatorname{Re}\kappa(\rho)$. Since $|\zeta(\frac{5}{2} + i\gamma) - 1| \le \zeta(\frac{5}{2}) - 1 < 0.35$, $a(\gamma)$ is a positive multiple of $e^{i\theta(\gamma)}$ up to a phase below 0.35 in modulus, and up to that phase the sign of $\operatorname{Re}\kappa(\rho)$ is the sign of $\sin\theta(\gamma)/Z'(\gamma)$. Let S_{mid} be the midpoint of the jump of S at the nth zero γ_n. Then $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\text{mid}})$ and the sign of $Z'(\gamma_n)$ is $(-1)^{n+1}$, so the sign of $\sin\theta(\gamma_n)/Z'(\gamma_n)$ is that of $-\cos(\pi S_{\text{mid}})$: negative when $|S_{\text{mid}}| < \frac{1}{2}$ and positive when $\frac{1}{2} < |S_{\text{mid}}| < \frac{3}{2}$. The phase of $\zeta(\frac{5}{2} + i\gamma)$ moves the threshold $\frac{1}{2}$ by at most

about 0.11 either way. For the first 460 zeros the sign of $\operatorname{Re}\kappa$ agrees with the sign of $\operatorname{Im} a(\gamma)/Z'(\gamma)$ in every case, and $\operatorname{Re}\kappa > 0$ occurs three times, at

$$\gamma_{213} = 415.4552, \quad \gamma_{289} = 527.9036, \quad \gamma_{379} = 650.6687,$$

with $\kappa = 233.96 - 2298.91i$, $604.16 + 3074.08i$, $1164.60 - 4296.91i$.

At these three zeros the midpoints of the jumps are 0.598, -0.645 and 0.664. Seven of the first 460 zeros have $S(\gamma^-) < -1$: besides γ_{289} these are γ_{127}, γ_{196}, γ_{233}, γ_{368}, γ_{380} and γ_{401}, with midpoints -0.506, -0.505, -0.537, -0.556, -0.538 and -0.514, and $\operatorname{Re}\kappa$ is negative at every one of them. Three further zeros have midpoints above $\frac{1}{2}$ and $\operatorname{Re}\kappa < 0$: γ_{136}, γ_{256} and γ_{399}, at 0.507, 0.510 and 0.539. The largest $|S_{\mathrm{mid}}|$ with $\operatorname{Re}\kappa < 0$ is 0.556, at γ_{368}, and the smallest with $\operatorname{Re}\kappa > 0$ is 0.598, at γ_{213}, so the threshold on the midpoint lies between about 0.56 and 0.60, inside the range $\frac{1}{2} \pm 0.11$ allowed by the phase of $\zeta(\frac{5}{2} + i\gamma)$.

Corollary 17.1.4. *Conjecture T is false: for all sufficiently large n, F_n has a zero with $\operatorname{Re} w > \frac{1}{4}$ near $\rho_{213}/2$.*

Proof. The first 1.5×10^9 zeros, ρ_{213} among them, lie on the critical line and are simple [271], and ball arithmetic applied to the formula of Theorem 17.1.3 gives $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ (Section 17.1.3). By Proposition 17.1.2, for all large n the function F_n has a zero w_n with $|w_n - \frac{\rho_{213}}{2} - \frac{\kappa(\rho_{213})}{n}| < C/n^2$, so $\operatorname{Re} w_n > \frac{1}{4} + \frac{233.95}{n} - \frac{C}{n^2} > \frac{1}{4}$ once $n > C/233.95$. □

Remark 17.1.5. The constants in Proposition 17.1.2 depend on ρ, and the first order regime requires n large compared with $|\kappa(\rho)|$. For the zero ρ_{213} of Corollary 17.1.4, $|\kappa(\rho_{213})| = 2311$, so the phenomenon begins at n of order 10^3 or beyond, larger than every n computed in this section.

The failure is intrinsic to the method, not to the lattice. Any truncation whose first order error is a multiple of $F(w-1)$ produces the displacement $\kappa(\rho)/n$, whose phase is governed by $S(\gamma)$. By Selberg's central limit theorem [256] S exceeds $\pm\frac{1}{2}$ for a positive proportion of heights, so one expects a positive proportion of zeros to be approached from the right, and one sided Hurwitz arguments cannot succeed for such truncations; a scheme is needed in which symmetry keeps the zeros on the line. The Dyck chain of Section 19.17 is an analogue in the variable $s = 2w$: its first order error is a multiple of $s\,\xi(s-1)$ where the lattice has $\xi(s-2)$, with a displacement of order $n^{-1/2}$, and the first zero approached from the right is ρ_{64} (the index computed in ball arithmetic; Propositions 19.17.3 and 19.17.4; conditional on Conjecture 19.17.2).

17.1.1.3 Symmetrised truncations and Conjecture S

The functional equation supplies it. Put

$$K_n(w) = \pi^{-w} F_n(w) + \pi^{w-\frac{1}{2}} F_n(\tfrac{1}{2} - w), \qquad Z_n(y) = K_n(\tfrac{1}{4} + iy).$$

PROPOSITION 17.1.6. *$K_n(\frac12 - w) = K_n(w)$, $K_n(\bar w) = \overline{K_n(w)}$, so Z_n is real for real y and the zeros of K_n off the line $\operatorname{Re} w = \frac14$ come in quadruples. As $n \to \infty$, $K_n \to 4\,\xi(2w)$ locally uniformly on $-\frac74 < \operatorname{Re} w < \frac94$. Every simple zero $\frac14 + i\frac{\gamma}{2}$ of $\xi(2w)$ on the line is, for all large n, the limit of a zero of K_n lying exactly on the line.*

Proof. The symmetries are immediate from the definition and the reality of F_n on the real axis. The limit follows from $F_n \to F = 2\pi^w \xi(2w)$ and $\xi(1-2w) = \xi(2w)$; since $F_n \to F$ on $\operatorname{Re} w > -2$, the convergence holds on $-2 < \operatorname{Re} w < \frac52$, which contains the strip of the statement. Near a simple zero y_0 of the real function $4\xi(\frac12 + 2iy)$, Z_n converges uniformly and so changes sign near y_0; by Hurwitz, K_n has exactly one zero near $\frac14 + iy_0$, which must be the real zero of Z_n, since an off line zero would bring its mirror image $\frac12 - \bar w$ with it. □

THEOREM 17.1.7. *If for infinitely many n all zeros of K_n in the strip $0 < \operatorname{Re} w < \frac12$ lie on the line $\operatorname{Re} w = \frac14$, then RH holds.*

Proof. A zero $\rho/2$ of $\xi(2w)$ off the line lies in the strip $0 < \operatorname{Re} w < \frac12$, since $0 < \operatorname{Re}\rho < 1$, and by Hurwitz it is the limit of zeros of K_n in the strip, which would be off the line for large n. □

CONJECTURE S. For every $n \ge 1$, all zeros of K_n in $0 < \operatorname{Re} w < \frac12$ lie on $\operatorname{Re} w = \frac14$.

The symmetrised zeros move along the line. The first order argument of Proposition 17.1.2, applied to $K_n = 2\operatorname{Re}[\pi^{-w} F_n(w)]$ on the line, gives

$$w_n^{\mathrm{sym}}(\rho) = \frac{\rho}{2} + \frac{i\,\operatorname{Im}\kappa(\rho)}{n} + O(n^{-2}) :$$

the symmetrisation removes the real part of the displacement and keeps the imaginary part.

For $n = 1$ the one off line quadruple of Table 17.2 lies at $\operatorname{Re} w = -2.001, 2.501$, $\operatorname{Im} w = \pm 2.471$, and for $n = 2$ at $\operatorname{Re} w = -2.949, 3.449$, $\operatorname{Im} w = \pm 3.261$; for $n = 3$ it is similar. These have $|\operatorname{Re} w - \frac14| > 2$, outside the strip $0 < \operatorname{Re} w < \frac12$, and do not bear on Conjecture S. For $n \ge 4$ every zero found lies on the line.

The window $195 < \operatorname{Im} w < 215$ contains $\gamma_{213}/2 = 207.73$, where the unsymmetrised truncations are forced off the line. There the count for K_8 exceeds that for ξ by one: the zeros of ξ with $390 < \gamma < 430$ are γ_{196} to γ_{221}, twenty six of them, the nearest excluded ordinates being $\gamma_{195} = 388.85$ and $\gamma_{222} = 430.33$. This is no contradiction: at this height and $n = 8$ the displacement $\operatorname{Im}\kappa(\rho)/n$ along the line is of order 300 for ρ_{213}, so K_8 is far from its limit and its zeros need not match those of ξ one to one. For Conjecture S what matters is that every zero found lies on the line.

n	window in $\operatorname{Im} w$	zeros in $-4<\operatorname{Re} w<4.5$: total, on the line	zeros of ξ
1	$(0.5,40)$, $(40,70)$	$23,21$; $27,27$	21; 27
2	$(0.5,40)$, $(40,70)$	$23,21$; $27,27$	21; 27
3	$(0.5,40)$, $(40,70)$	$23,21$; $27,27$	21; 27
$4,5,6,8,16$	$(0.5,40)$, $(40,70)$	$21,21$; $27,27$	21; 27
32	$(0.5,40)$	$21,21$	21
8	$(70,140)$, $(195,215)$	$78,78$; $27,27$	78; 26

Table 17.2: Zeros of the symmetrised truncations K_n in $-4<\operatorname{Re} w<4.5$: total found, how many lie on the line, and the number of zeros of ξ in the doubled window for comparison. In the window $195<\operatorname{Im} w<215$ the count for K_8 exceeds that for ξ by one; see the text.

The symmetrised truncations are the lattice analogue of the approximations $\zeta_N(s)=\sum_{m\le N} m^{-s}+\chi(s)\sum_{m\le N} m^{s-1}$ derived from the approximate functional equation, with χ the factor in $\zeta(s)=\chi(s)\zeta(1-s)$. Spira [260, Section 6] and Gonek and Montgomery [85] studied their zeros, and Gonek and Montgomery proved that for $N\le T^{o(1)}$ almost all zeros of ζ_N with ordinates in $[0,T]$ lie on the critical line and are simple. A proof of Conjecture S for infinitely many n would require controlling the zeros of Z_n uniformly in height, including the range of heights comparable with n, where K_n is neither close to $4\xi(2w)$ nor dominated by a single exponential term; this is the same difficulty as for ζ_N.

Proposition 17.1.8. *Write $s=\sigma+i\tau$.*

(i) All but finitely many zeros of ζ_2 in $0<\sigma<1$ lie on the critical line, and the number with $0<\tau\le T$ is $\vartheta(T)/\pi+O(1)$.

(ii) Let σ^* be the root of $2^{-\sigma}+3^{-\sigma}=1$. For every $\sigma_0\in(\frac12,\sigma^*)$ there are zeros $\sigma_k+i\tau_k$ of ζ_3 with $\tau_k\to\infty$ and $\sigma_k\to\sigma_0$. In particular ζ_3 has zeros off the critical line at arbitrarily large heights.

This follows a note of Nicholas Polson merged into the book [137]; compare [260, 85].

Sketch. (i) With $G(s)=\Gamma_{\mathbb R}(s)(1+2^{-s})$ we have $\Gamma_{\mathbb R}(s)\zeta_2(s)=G(s)+G(1-s)$, which is $2\operatorname{Re} G$ on the critical line. A zero needs $L(\sigma,\tau)=\log|G(\sigma+i\tau)|-\log|G(1-\sigma-i\tau)|=0$, and by Stirling's formula

$$L(\sigma,\tau)=(\sigma-\tfrac12)\log\frac{|\tau|}{2\pi}+\log|1+2^{-s}|-\log|1+2^{s-1}|+O(|\tau|^{-1}).$$

For $0<\sigma\le\frac12$ the last two terms are at most $\log 2-\log(1-2^{-1/2})$, so $L<0$ unless $\frac12-\sigma=O(1/\log|\tau|)$; on $[\frac14,\frac12)$ they have bounded σ derivative while that

of the first term is $\log(|\tau|/2\pi)$, and $L(\frac{1}{2},\tau)=0$, so $L<0$ there too once $|\tau|$ is large. The symmetry $s\mapsto 1-\bar{s}$ covers $\sigma>\frac{1}{2}$. On the line $G(\frac{1}{2}+i\tau)=|G|e^{i\Theta(\tau)}$ with $\Theta(\tau)=\vartheta(\tau)+\arg(1+2^{-\frac{1}{2}-i\tau})$; the second term has modulus at most $\frac{\pi}{4}$ and a bounded derivative, so Θ is eventually increasing and the count is $\Theta(T)/\pi+O(1)$. (ii) For $\sigma>\frac{1}{2}$, $|\chi(s)|\asymp(|\tau|/2\pi)^{\frac{1}{2}-\sigma}\to 0$, so near $\sigma_0+i\tau$ the function ζ_3 is $1+2^{-s}+3^{-s}+o(1)$. Since $2^{-\sigma_0}+3^{-\sigma_0}>1>|2^{-\sigma_0}-3^{-\sigma_0}|$ there are unit numbers u,v with $1+2^{-\sigma_0}u+3^{-\sigma_0}v=0$, and σ_0 is a simple zero of $w\mapsto 1+2^{-w}u+3^{-w}v$, because a double zero would force $2^{-\sigma_0}u=\log 3/(\log 2-\log 3)$, of modulus larger than 1. As $\log 2$ and $\log 3$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem gives $\tau_k\to\infty$ with $(2^{-i\tau_k},3^{-i\tau_k})\to(u,v)$, and Rouché's theorem, applied first to $1+2^{-s}+3^{-s}$ near $\sigma_0+i\tau_k$ and then to ζ_3, gives the zeros. □

17.1.1.4 A single term bound for small n

For small n, Conjecture S reduces to a finite computation, because at large heights the gamma factor makes $\log|A_n|$ increase across the strip. Here $A_n(w)=\pi^{-w}F_n(w)$, so that $K_n(w)=A_n(w)+A_n^*(w)$ with $A_n^*(w)=A_n(\frac{1}{2}-w)$ and $|A_n^*(\sigma+iy)|=|A_n(\frac{1}{2}-\sigma+iy)|$. If $\sigma\mapsto\log|A_n(\sigma+iy)|$ is increasing on $(0,\frac{1}{2})$, then $|A_n|>|A_n^*|$ for $\frac{1}{4}<\sigma<\frac{1}{2}$, and K_n has no zeros there at height y; by symmetry none to the left of the line either. The derivative is $\operatorname{Re}\frac{A_n'}{A_n}=-\log\pi+\operatorname{Re}\psi(w+1)+\operatorname{Re}\frac{Q_n'}{Q_n}(w+1)$, and for large y the digamma term dominates.

THEOREM 17.1.9. *Let $Q_n(u)=\sum_k k^{-2u}(\alpha_k+\beta_k u)$ as above and, for $y>|\alpha_1|/|\beta_1|$, let $\delta_n(y)=\sum_{k\ge 2}k^{-2}\frac{|\beta_k|y+|\alpha_k|}{|\beta_1|y-|\alpha_1|}$ and*

$$\delta_n'(y)=\sum_{k\ge 2}k^{-2}\frac{2\log k\,(|\beta_k|y+|\alpha_k|)+|\beta_k|}{|\beta_1|y-|\alpha_1|}+\delta_n(y)\,\frac{|\beta_1|}{|\beta_1|y-|\alpha_1|},$$

which bounds a derivative in the proof and is not the derivative of δ_n. If $\delta_n(y)<1$ and

$$\log\frac{y}{\pi}-\frac{1}{y}-\frac{|\beta_1|}{|\beta_1|y-|\alpha_1|}-\frac{\delta_n'(y)}{1-\delta_n(y)}>0,$$

then K_n has no zeros off the line in $0<\operatorname{Re}w<\frac{1}{2}$ at any height $\ge y$. The condition holds for all $y\ge Y_n$, with $Y_1=4.81$, $Y_2=5.85$, $Y_3=7.11$, $Y_4=9.08$, $Y_5=12.8$, $Y_6=20.8$, $Y_8=104.1$ and $Y_{10}=21232$.

Proof. $|\psi(z)-\log z|\le 1/|z|$ for $\operatorname{Re}z\ge 1$, so $\operatorname{Re}\psi(w+1)\ge\log y-\frac{1}{y}$ at height y. Write $Q_n(u)=(\alpha_1+\beta_1u)(1+\delta(u))$. On $1\le\operatorname{Re}u\le\frac{3}{2}$, $|u|\ge y$, we have $|k^{-2u}|\le k^{-2}$, so $|\delta(u)|\le\delta_n(y)$, and differentiating the quotient $\delta(u)$ term by term gives $|\delta'(u)|\le\delta_n'(y)$; both bounds decrease in y. Then $\operatorname{Re}\frac{Q_n'}{Q_n}\ge-\frac{|\beta_1|}{|\alpha_1+\beta_1u|}-\frac{|\delta'|}{1-|\delta|}$, and the condition makes $\operatorname{Re}(A_n'/A_n)>0$ on the strip at every height $\ge y$. The values of Y_n come from evaluating these bounds with the exact coefficients. □

Corollary 17.1.10. *Conjecture S holds for $n = 1, 2, 3, 4, 5, 6, 8$ above the heights Y_n of Theorem 17.1.9, and is verified numerically below them, by a floating point argument principle count and not by an enclosure.*

Proof. Above Y_n this is Theorem 17.1.9. Below it, the argument principle counts cover the strip $0 < \operatorname{Re} w < \frac{1}{2}$ for $0 < \operatorname{Im} w < 6, 7, 8$ when $n = 1, 2, 3$, and the windows of Table 17.2 cover $0 < \operatorname{Im} w < 70$ for $4 \le n \le 6$ and $0 < \operatorname{Im} w < 140$ for $n = 8$, with $0 < \operatorname{Im} w < 0.6$ checked separately and $K_n > 0$ on the real segment; in each case the total count in the strip equals the number of sign changes of Z_n on the line. □

The single term bound of Theorem 17.1.9 deteriorates quickly: the lattice coefficients $|\beta_k/\beta_1|$ grow with n, $Y_{10} = 21232$, and for $n \ge 12$ the bound fails because $\delta_n > 1$.

17.1.1.5 Torus bounds and where they stop

Write $u = x + iy$ and $Q_n(u) = u\,P_n(u) + R_n(u)$ with the Dirichlet polynomials $P_n(u) = \sum_{k\le n} \beta_k k^{-2u}$ and $R_n(u) = \sum_{k\le n} \alpha_k k^{-2u}$. At height y the values of P_n depend on y only through the prime phases $(p^{-2iy})_{p\le n}$, which lie on the torus $\mathbb{T}^{\pi(n)}$. Replacing them by independent phases $\omega = (\omega_p)_{p\le n} \in \mathbb{T}^{\pi(n)}$ gives the torus function

$$P_n(x,\omega) = \sum_{k\le n} \beta_k k^{-2x} \exp\Big(-i \sum_{p\le n} v_p(k)\,\omega_p\Big),$$

and in the same way the torus function $P_n'(x,\omega)$ of $P_n'(u) = -\sum_k 2\log k\,\beta_k k^{-2u}$. Their extremes over the torus bound the values at every height. Put $r(x) = \sum_k |\alpha_k| k^{-2x}$ and $r'(x) = \sum_k 2\log k\,|\alpha_k| k^{-2x}$.

Theorem 17.1.11. *Let $m_n = \min |P_n(x,\omega)|$, $M_n = \max\{-\operatorname{Re}(P_n'/P_n)(x,\omega)\}$ and $\ell_n = \max |P_n'/P_n|(x,\omega)$, over $\omega \in \mathbb{T}^{\pi(n)}$ and the band $1.2 \le x \le 1.3$, and let*

$$B_n(y) = \max_{0.3\le\sigma\le\frac12,\ \omega} \frac{\log\big(|P_n(\frac32-\sigma,\omega)| + r(\frac32-\sigma)/y\big) - \log\big(|P_n(1+\sigma,\omega)| - r(1+\sigma)/y\big)}{2\sigma - \frac12}.$$

If $m_n > r(1.2)/y$,

$$\log\frac{y}{\pi} - \frac{2}{y} - M_n - \frac{\ell_n r(1.2) + r'(1.2) + r(1.2)/y}{y\,(m_n - r(1.2)/y)} > 0 \quad \textit{and} \quad \log\frac{y}{\pi} - \frac{1}{y} > B_n(y),$$

then K_n has no zeros off the line in $0 < \operatorname{Re} w < \frac12$ at height y, and the same holds at every larger height.

Proof. For $\frac{1}{4} < \sigma \le 0.3$ both σ and $\frac{1}{2} - \sigma$ lie in $[0.2, 0.3]$, and the first condition makes $\operatorname{Re}(A_n'/A_n) > 0$ there: $\operatorname{Re}\psi(w+1) \ge \log y - \frac{1}{y}$, $\operatorname{Re}\frac{1}{u} > 0$, and $\frac{Q_n'}{Q_n} = \frac{1}{u} + \frac{P_n' + R_n'/u - R_n/u^2}{P_n + R_n/u}$ differs from $\frac{1}{u} + \frac{P_n'}{P_n}$ by at most the stated correction. So $\log|A_n|$ increases on $[\frac{1}{2} - \sigma, \sigma]$ and $|A_n| > |A_n^*|$. For $0.3 \le \sigma \le \frac{1}{2}$ compare directly: $\log|\Gamma(1+\sigma+iy)| - \log|\Gamma(\frac{3}{2} - \sigma + iy)| \ge (2\sigma - \frac{1}{2})(\log y - \frac{1}{y})$ by integrating $\operatorname{Re}\psi$, $|1+\sigma+iy| > |\frac{3}{2} - \sigma + iy|$, and $|P_n + R_n/u|$ is bounded below on the right by $|P_n| - r/y$ and above on the left by $|P_n| + r/y$; the second condition then gives $|A_n(\sigma+iy)| > |A_n(\frac{1}{2} - \sigma + iy)|$. Both conditions improve as y increases. □

Table 17.3 lists bounds for the torus extremes (Remark 17.1.13) and the smallest integer heights at which the first condition holds with them. For $n = 7$ and $9 \le n \le 15$ we use a single band variant: if $m_n > r(1)/y$ and the first condition of Theorem 17.1.11 holds with m_n, M_n, ℓ_n the extremes over the band $1 \le x \le \frac{3}{2}$ and with $r(1)$, $r'(1)$ in place of $r(1.2)$, $r'(1.2)$, then $\operatorname{Re}(A_n'/A_n) > 0$ on $0 < \operatorname{Re} w < \frac{1}{2}$ at height y, so $\log|A_n|$ increases across the strip and K_n has no zeros off the line there; the condition improves as y increases. The proof is the first part of the proof of Theorem 17.1.11 with the band widened, since r and r' decrease in x and $\operatorname{Re}\psi(w+1) \ge \log y - \frac{1}{y}$ on the whole strip.

n	16	17	18	19	20	21	22	23	24	32	36
$m_n \ge$	1.062	1.003	0.947	0.895	0.846	0.800	0.757	0.716	0.677	0.361	0.203
$M_n \le$	0.862	0.895	0.926	0.956	0.985	1.012	1.039	1.064	1.089	1.259	1.331
$\ell_n \le$	3.600	3.948	4.310	4.689	5.086	5.504	5.954	6.473	7.075	17.203	33.568
$r(1.2) \le$	7.522	7.776	8.009	8.223	8.420	8.600	8.834	9.062	9.280	10.675	11.313
$r'(1.2) \le$	4.648	5.019	5.361	5.675	5.961	6.222	6.650	7.078	7.489	10.181	11.577
threshold	30	34	37	41	46	50	56	62	69	213	558

Table 17.3: Bounds for the torus extremes over the band $1.2 \le x \le 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 17.1.11 holds with these values. The bounds are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 17.1.13); the programs are in `verification_reciprocal_zeta/torus/`.

Corollary 17.1.12. *Conjecture S holds for $n = 16, \dots, 24, 32, 36$ above the thresholds of Table 17.3 and for $n = 7$ and $9 \le n \le 15$ above those of Table 17.4, and for all these n it is verified numerically below the thresholds, by floating point argument principle counts. With Corollary 17.1.10 this covers every $n \le 24$ and $n = 32, 36$.*

Proof. Above the thresholds this is Theorem 17.1.11, with the bounds of Table 17.3 and the second condition of Remark 17.1.13, and its single band variant, with the bounds of Table 17.4. The hypothesis $m_n > r/y$ and the first condition, here and

n	7	9	10	11	12	13	14	15
$m_n \geq$	1.403	1.052	0.892	0.744	0.607	0.482	0.367	0.261
$M_n \leq$	0.576	0.715	0.777	0.834	0.887	0.937	0.984	1.028
$\ell_n \leq$	1.796	3.193	4.167	5.428	7.121	9.511	13.140	19.314
$r(1) \leq$	4.848	5.888	6.291	6.629	6.971	7.445	7.886	8.294
$r'(1) \leq$	2.014	3.096	3.520	3.869	4.282	5.032	5.746	6.418
threshold	14	24	31	41	57	84	131	229

Table 17.4: As Table 17.3, for the single band variant: bounds for the extremes over the band $1 \leq x \leq \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values.

in the variant, hold with the true values wherever they hold with these bounds, because both improve as m_n increases and as M_n, ℓ_n and the values of r and r' decrease. Below the thresholds the argument principle count in $0 < \operatorname{Re} w < \frac{1}{2}$ equals the number of sign changes of Z_n on the line, with the following heights and counts:

n	7	9	10	11	12	13	14	15	16	17
height	15	24	32	43	58	86	132	230	500	35
zeros	3	9	14	23	36	64	116	242	649	17
n	18	19	20	21	22	23	24	32	36	
height	250	43	47	51	57	62	110	215	570	
zeros	269	23	26	30	35	40	90	221	763	

Each height is at least the threshold for the same n, with equality for $n = 9$ and $n = 23$, so the counts and the theorem together cover all heights. For $n = 36$ the sampling step on the line is 0.02, cross checked by the argument principle. Every count equals the number of zeros of ξ with ordinate below twice the height, except for $n = 9$, where the ninth zero of K_9, at $\operatorname{Im} w = 23.82$, is the image of $\gamma_9/2 = 24.003$ displaced below height 24. □

Remark 17.1.13. The bounds of Tables 17.3 and 17.4 are proved by a branch and bound over the band and the half torus $0 \leq \omega_2 \leq \pi$, which suffices because $P_n(x, -\omega) = \overline{P_n(x, \omega)}$. On a box with centre (x_c, ω_c) and half widths h in x and h_p in ω_p, the kth term of P_n differs from its value at the centre by at most $\beta_k k^{-2x_c}\{(k^{2h}-1)+k^{2h}\min(2, \sum_p v_p(k)h_p)\}$, and the resulting enclosures of P_n, P_n' and their partial derivatives bound $|P_n|$ from below and $-\operatorname{Re}(P_n'/P_n)$ and $|P_n'/P_n|$ from above on the box; for M_n and ℓ_n the proved bound for m_n serves as a lower bound for $|P_n|$. Since $2\sigma - \frac{1}{2} > 0$, the second condition of Theorem 17.1.11 at

height y is equivalent to

$$\Big(|P_n(1+\sigma,\omega)| - \frac{r(1+\sigma)}{y}\Big)\exp\Big(\big(2\sigma-\tfrac12\big)\Big(\log\frac{y}{\pi}-\frac{1}{y}\Big)\Big) \\ > |P_n(\tfrac32-\sigma,\omega)| + \frac{r(\frac32-\sigma)}{y}$$

for $0.3 \le \sigma \le \frac12$ and all ω, which also gives the positivity of $|P_n(1+\sigma,\omega)|-r(1+\sigma)/y$ that $B_n(y)$ needs. The same branch and bound, applied to the difference of the two sides, proves this at the heights 29, 33, 36, 40, 45, 49, 55, 61, 68, 211, 555 for $n = 16,\dots,24,32,36$, each below the threshold, and so at every larger height. The searches run in double precision and only choose the boxes: every final box, about 1.1×10^8 in all, is proved again in ball arithmetic with Arb [179], with exact coefficients and on the exact box.

With the band $1.2 \le x \le 1.3$ the method stops between $n = 40$ and $n = 44$. The band minimum m_n falls to 0.20 at $n = 36$, 0.056 at $n = 40$ and 0 at $n = 44$, so at $n = 44$ the hypothesis $m_n > r(1.2)/y$ of Theorem 17.1.11 fails at every height. This reflects the functions, not only the bound: the torus function $P_n(x,\omega)$ then vanishes at a point of the band, and since by Kronecker's theorem the prime phases at height y come arbitrarily close to every point of the torus, A_n has zeros with real part near $[0.2, 0.3]$ at arbitrarily large heights, where $\log|A_n|$ cannot increase across the strip. On the line $x = \frac54$ itself the minimum is still 0.29 at $n = 44$ and 0.19 at $n = 48$, so these torus zeros lie off the line, and a narrower band may postpone the stop. A torus zero with $x > \frac54$ would end every comparison of this kind: it gives zeros of A_n with $\operatorname{Re} w > \frac14$ at arbitrarily large heights, where $|A_n(w)| > |A_n^*(w)|$ fails. On which side of the line the torus zeros at $n = 44$ lie is not determined here.

The monotonicity used in Theorem 17.1.9 implies the Hermite Biehler inequality $|A_n(w)| > |A_n(\frac12 - \bar w)|$ for $\frac14 < \operatorname{Re} w < \frac12$, and the inequality holds far beyond the proven range. On a grid of $\sigma \in \{0.26, 0.3, 0.4, 0.49\}$ and $y \le 100$ it holds for $n = 8, 16, 24$ at every height above 3.6, and it fails only below 3.6, where the strip contains no zeros. For $n = 8, 16$ it also holds for $200 \le y \le 212$, around $\gamma_{213}/2$. It cannot hold for all large n: by Corollary 17.1.4, for large n the function A_n has a zero w with $\frac14 < \operatorname{Re} w < \frac12$, where the inequality fails. Quantitatively, near a zero $\rho/2$ the first order margin is governed by the residue of $w\xi(2w-2)/\xi(2w)$, which is a positive multiple of $\kappa(\rho)$, so at zeros with $\operatorname{Re}\kappa(\rho) > 0$ the inequality reverses once n is large compared with $|\kappa(\rho)|$. A proof of Conjecture S for infinitely many n must establish real rootedness without a one sided inequality.

17.1.2 What is proved, and how far it reaches

In Table 17.5 every entry in the middle column is a positivity statement about a transform or about finitely many derivatives, and every entry on the left one

about all derivatives, equivalently about a measure. The question is whether the middle column can be pushed to the left. So far it reaches Weil positivity on the Bernstein cone and $(-1)^n W^{(n)} > 0$ up to a finite order, and neither the de Bruijn Newman flow nor a GGC multiplier closes the gap. Chapters 12 to 14 measure the same gap from the right of the centre: $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a GGC for every $\sigma \geq 1$ with no hypothesis (Theorem 11.1.14; Proposition 12.1.2 computes its Thorin measure at $\sigma = 1$), and for $\frac{1}{2} < \sigma < 1$ zero density estimates show that in the density $\rho_{\sigma-1/2}$ of (2.1.7) the Cauchy kernels of the zeros to the right of σ carry, below height T, a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$ (Theorem 14.3.2).

17.1.2.1 Weil's criterion in Thorin coordinates

Proposition 10.2.16 left open which signed primal measures have a positive sine squared image. For $\tilde{\mu}_\xi$ the answer is Weil's criterion.

PROPOSITION 17.1.14. *Let $D(x) = \sum_\rho e^{(\rho - \frac{1}{2})x}$ be Weil's distribution, whose action on a test function h is $\sum_\rho \hat{h}(\gamma_\rho)$ with $\gamma_\rho = -i(\rho - \frac{1}{2})$. Then R is the Laplace transform of a GGC, with Thorin measure the sine squared image of $\tilde{\mu}_\xi$, if and only if D is positive definite, that is, $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for every $h = g * \tilde{g}$ with g smooth and compactly supported. Both conditions are equivalent to RH.*

Proof. By Proposition 2.1.1, R is the Laplace transform of a GGC if and only if RH holds, and its Thorin measure is then $\sum_{\gamma>0} \delta_{\gamma^2}$, which is the sine squared image of $\tilde{\mu}_\xi$ by Proposition 10.2.16(ii). Weil's theorem [273, 29] states that RH holds if and only if $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for all $h = g * \tilde{g}$ with g smooth and compactly supported, that is, if and only if D is positive definite. □

A formal computation shows how the two formulations correspond. For a positive measure U on $(0, \infty)$ with $\int U(dt)/(1+t) < \infty$, the primal measure $\tilde{\mu}_U(dx) = -2\big(\int \cos(\sqrt{t}\,x)\, U(dt)\big)dx$ on $x > 0$, understood as a distribution, has sine squared image U: since $\int_0^\infty (1 - \cos x\sqrt{z})(-2\cos bx)\, dx = \pi\delta(\sqrt{z} - b)$, $\tilde{\mu}_U$ maps to $\int \delta(z-t)U(dt) = U$. The map of Lemma 10.2.5 is injective on the class of that lemma: if $\int_0^\infty \sin^2(x\sqrt{z}/2)\, \nu(dx) = 0$ for every $z > 0$, then $\psi(t) = \int (1 - \cos xt)\, \nu(dx)$ vanishes for all real t, and $\psi(t) - \frac{1}{2}\int_{t-1}^{t+1} \psi(s)\, ds = -\int \cos(xt)\big(1 - \frac{\sin x}{x}\big)\nu(dx)$ shows that the finite measure $(1 - \frac{\sin x}{x})\nu(dx)$ has vanishing cosine transform, so $\nu = 0$. Carried over to distributions, these two facts say that a primal measure has a positive sine squared image if and only if $-\tilde{\mu}$ is the restriction to $x > 0$ of an even positive definite distribution on $\mathbb{R}$, that is, by the Bochner Schwartz theorem, the Fourier transform of a positive measure. For ξ, $-\tilde{\mu}_\xi$ is the restriction to $x > 0$ of D, by (10.2.6); D is even because $\rho \mapsto 1 - \rho$ permutes the zeros, and the explicit formula (10.2.5) is its prime side. The computation is formal, for two reasons: Lemma 10.2.5 is proved only for primal measures with $\int (1 \wedge$

Equivalent form of RH	Proved with no unproved hypothesis	Computation the middle column uses
$R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is a GGC, Thorin measure $\sum \delta_{\gamma^2}$ [233, Proposition 3.2]	$e^{-c\sqrt{s}}/G$ is a GGC (Theorem 10.2.12); $e^{\sqrt{s}\zeta'/\zeta(\beta)}\zeta(\beta)/\zeta(\beta + \sqrt{s})$ is a GGC for $\beta > 1$ (Proposition 10.2.6)	none
$W(x) = \frac{1}{2}\sum_\rho e^{x(\rho-\frac{1}{2})^2}$ is completely monotone	$W > 0$ (proof of Theorem 17.1.15) and $(-1)^n W^{(n)} > 0$ for $n \le 1.16 \times 10^{25}$ (Theorem 8.1.13); $W + c/(2\sqrt{\pi x})$ is positive (Theorem 10.2.15) and n monotone for $n \le 10^{10}$ (Theorem 10.2.17)	positivity: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}
$e^{-c\sqrt{s}}R$ is a GGC	it is infinitely divisible (Theorem 10.2.15)	RH to 50 and enclosures
w is the Laplace transform of a real measure on $[0, \infty)$ (Theorem 10.1.15)	$w > 0$ (Theorem 11.1.1)	RH to 50 and enclosures
the sine squared image of $\tilde{\mu}_\xi$ is positive, equivalently Weil's distribution is positive definite (Proposition 17.1.14)	its subordination stage is positive; Weil positivity holds on the Bernstein cone (Theorem 17.1.15), against every Gaussian (its proof) and against $r^{2n}e^{-xr^2}$ for $n \le 1.16 \times 10^{25}$ (Theorem 8.1.13); the canonical function k of the floor is n monotone for every $n \le 10^{10}$ (Theorem 10.2.17), which is strictly weaker than n monotonicity of W	Bernstein cone: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}

Table 17.5: Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none.

$x)|\tilde{\mu}|(dx) < \infty$, and when RH fails D grows exponentially and is not tempered, so the Bochner Schwartz theorem applies only in Weil's own formulation, with compactly supported test functions. Riesz's criterion has a Thorin form of the same kind (Theorem 13.2.7 of Section 13.2).

The unconditional positivity results of Chapter 10, of this chapter and of Section 8.1 are partial statements of Weil type. The gamma part is a positive definite archimedean piece (Theorem 10.2.12). The heat trace results are Weil positivity along the Gaussian family: with h Gaussian, $\sum_\rho \hat{h}(\gamma_\rho) = 2W(x)$, and $W(x) > 0$ for every $x > 0$ unconditionally (Theorem 8.1.1 [235, Theorem A]). Weil positivity against $r^{2n}e^{-xr^2}$ is $(-1)^n W^{(n)} \geq 0$, which Theorem 8.1.6 [233, Theorem 7.10] proves for every $n \leq 10^{17}$ and Theorem 8.1.13 for every $n \leq 1.16 \times 10^{25}$. Theorem 10.2.17 proves the corresponding statement for the canonical function $k = W + c/(2\sqrt{\pi x})$ of the floor, for $n \leq 10^{10}$, and not for W itself. The known unconditional cases of Weil positivity, due to Yoshida [278, Theorem 1] and to Bombieri [29], concern test functions of small support; the Gaussian family, in which Theorems 8.1.6, 8.1.13 and 10.2.17 apply, is the opposite regime.

17.1.2.2 The Bernstein cone and the resolution of finite order tests

Proposition 17.1.14 identifies the Thorin positivity of R with Weil positivity for all $\hat{h} = |\hat{g}|^2$. A cone of another kind, completely monotone in r^2, is settled unconditionally.

THEOREM 17.1.15. *Let $\hat{h}(r) = \int_0^\infty e^{-yr^2} g(y)\, dy$ with $g \geq 0$, $g \not\equiv 0$ and $\int_0^1 g(y)\, y^{-1/2}\big(1 + \log(1/y)\big)\, dy + \int_1^\infty g(y)\, dy < \infty$, so that $\hat{h}$ is a completely monotone function of r^2. Then*

$$\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)\, g(y)\, dy > 0,$$

where $\gamma_\rho = -i(\rho - \frac{1}{2})$ and the sum is over all nontrivial zeros. In particular Weil's functional is positive for every test function of Thorin type $\hat{h}(r) = \prod_k (1 + r^2/t_k)^{-1}$, a nonempty product with $t_k > 0$ and $\sum_k 1/t_k < \infty$, for $\hat{h}(r) = \operatorname{sech}(cr)$ with $c > 0$, and for $\hat{h}(r) = (1 + r^2/a)^{-m}$ with $a > 0$ and $m > \frac{1}{2}$. The proof uses the verification of RH to height 50 and the interval enclosures of Theorem 11.1.1 together with the enclosures E5 and E6 of Section 17.1.3.

Proof. For each $y > 0$, $\sum_\rho e^{-y\gamma_\rho^2} = \sum_\rho e^{y(\rho - \frac{1}{2})^2} = 2W(y)$, and $W(y) > 0$ for every $y > 0$. For $y \leq 0.003$ this comes from the prime side: $W = w + W_0$ with $w > 0$ (Theorem 11.1.1) and $W_0 \geq 0$, because the function F of Lemma 10.1.12 satisfies $F(0.003) > 0.0329$ (enclosure E6) and is decreasing. For $y \geq 0.003$ it comes from the zeros below height 50, which lie on the line and contribute $e^{-y\gamma^2} > 0$: the first alone contributes more than $e^{-0.003 \cdot 14.14^2} > 0.548$ at $y = 0.003$, the off line tail above height 50 is below 0.0162 there (enclosure E5, with the bound $N(t+1) -$

$N(t) \le 2\log(t+1)$ of the proof of Theorem 11.1.1), and the ratio of the tail to the first term decreases as y grows. The interchange of sum and integral is justified by Fubini's theorem: $\sum_\rho |e^{y(\rho-\frac12)^2}| \le e^{y/4}\sum_\rho e^{-y(\operatorname{Im}\rho)^2}$, which is $O(y^{-1/2}(1+\log(1/y)))$ for $y \le 1$ by the Riemann von Mangoldt formula and bounded for $y \ge 1$, and the integrability condition makes this majorant integrable against g. The Thorin type functions are completely monotone in r^2 because $\prod(1+s/t_k)^{-1}$ and $\operatorname{sech}(c\sqrt{s})$ are GGC transforms. The examples satisfy the integrability condition. A nonempty product $\prod_k(1+s/t_k)^{-1}$ is the Laplace transform of $\sum_k e_k/t_k$, with e_k independent standard exponential variables, and the density g of this sum is at most t_1, the maximum of the density of e_1/t_1; the function $\operatorname{sech}(c\sqrt{s}) = \prod_{k\ge1}\bigl(1+4c^2s/((2k-1)^2\pi^2)\bigr)^{-1}$ is such a product; and for $(1+r^2/a)^{-m}$ the function $g(y) = a^m y^{m-1}e^{-ay}/\Gamma(m)$ satisfies the condition because $m > \frac12$. $\square$

The Bernstein cone consists of test functions that decrease in $|r|$ and so cannot concentrate near a given height; this is why positivity on it does not imply RH, and it matches the resolution analysis of Table 17.6: an off line zero at height γ and distance δ from the line is detected only by test functions able to localise at γ. Theorem 8.1.13 gives $(-1)^n W^{(n)} > 0$ for $n \le 1.16 \times 10^{25}$, which is Weil positivity for $r^{2n}e^{-yr^2}$ and, after integration against g where the integral converges, for the image of the Bernstein cone under multiplication by r^{2n}.

Complete monotonicity of W is Bernstein's condition, and its Post Widder form is

$$L_n[W](t) = \frac{(-1)^n}{n!}\Bigl(\frac{n}{t}\Bigr)^{n+1} W^{(n)}\Bigl(\frac{n}{t}\Bigr) \ge 0 \qquad \text{for all } n, t.$$

$L_n[W]$ is the Thorin measure $\sum \delta_{\gamma^2}$ seen through a kernel of relative width about $n^{-1/2}$, and an off line quadruple $\rho = \frac12 \pm \delta \pm i\gamma$ contributes $2\operatorname{Re}\{(n/t)^{n+1}z^n e^{-zn/t}/n!\}$ with $z = \gamma^2 - \delta^2 - 2i\delta\gamma$. In Table 17.6 the quadruple is inserted midway between two consecutive true ordinates among the first 300 zeros.

γ	$\delta = 0.4$	$\delta = 0.2$	$\delta = 0.1$	$\delta = 0.05$
27.7	724	1448	2896	5792
60.1	11585	23170	46340	92681
120.1	32768	65536	131072	262144

Table 17.6: Smallest order n^* at which an inserted off line quadruple at height γ and distance δ makes $\min_t L_n[W](t)$ negative.

In every row the detection order doubles when δ halves, within the resolution $\sqrt{2}$ of the scan, and across the rows it grows roughly like γ^2: the ratio $n^*\delta/\gamma^2$ lies between 0.38 and 1.3 in Table 17.6. An off line zero at height γ thus first becomes visible at order about γ^2/δ, and a zero above the verified height T_0 of Theorem 10.2.17 at order about T_0^2/δ. Theorem 8.1.13 reaches orders of this size,

every order up to $1.29(T_0 - 23)^2$ for $T_0 = 3 \times 10^{12}$ (Section 8.1.6), while the method of Theorem 10.2.17, whose Cauchy estimate loses a factor 2^n, certifies n monotonicity only up to orders proportional to T_0. Finite order tests therefore cannot prove RH: each order sees only a bounded height, and the height it sees tends to infinity only with the order.

17.1.2.3 Weak limits and the de Bruijn Newman flow

The class of GGC laws is closed under convolution and weak limits [30, Section 3.1], so RH is equivalent to exhibiting R as a limit of Laplace transforms of GGC laws, and positivity passes to the limit at every order at once. The de Bruijn Newman flow gives candidates. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$, with Φ the Pólya kernel (2.1.3); Φ_{dBN} is the kernel of Section 4.1, in the rescaled variable of Rodgers and Tao [250]. Let $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with Λ defined in this normalisation, and put

$$R_t(s) = \frac{H_t(0)}{H_t(2i\sqrt{s})}, \qquad R_0 = R.$$

THEOREM 17.1.16. *Let Λ be the de Bruijn Newman constant. For every $t \geq \Lambda$, R_t is the Laplace transform of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the zeros of H_t. In particular R_t is a GGC for every $t \geq 0.2$, unconditionally. RH holds if and only if R_t is a GGC for arbitrarily small $t > 0$.*

Proof. For $t \geq \Lambda$ all zeros of H_t are real by definition of Λ, and H_t is even and entire of order one with the Hadamard product $H_t(z) = H_t(0)\prod_k(1 - z^2/z_k^2)$, so $R_t(s) = \prod_k(1 + 4s/z_k^2)^{-1}$, an infinite convolution of exponential laws. The bound $\Lambda \leq 0.2$ is [232], improving $\Lambda \leq 0.22$ [247]. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t \to R$ pointwise on $(0, \infty)$; if each R_t is a GGC transform, so is the limit, and Proposition 10.1.9 gives RH. Conversely RH means $\Lambda \leq 0$, so every R_t with $t \geq 0$ is a GGC. By Rodgers and Tao [250], $\Lambda \geq 0$, so RH is equivalent to $\Lambda = 0$. □

Conversely, for $t < \Lambda$, and so for every $t < 0$ [250], R_t is not the Laplace transform of a GGC (Corollary 9.2.5), and Proposition 14.4.12 turns this into a test for the properties of the Pólya kernel that could imply RH.

The zeros $z_k(t)$ move by the backward heat flow, and the Thorin measure of R_t can cease to be positive only by two real zeros colliding and leaving the real line, which is the collision picture of Rodgers and Tao; Section 4.1 computes the flow numerically. What Chapter 10 and this chapter add concerns $t = 0$: any failure there comes from zeros above the verified height. For $t \neq 0$ no prime side is known for the Thorin measure of R_t, and changing the test function does not give one. Replacing the Gaussian e^{-xr^2} of Theorem 10.1.11 by a t dependent test function gives a linear statistic $\sum_\rho h(\gamma_\rho)$ of the zeros of ξ, whereas the heat trace $\sum_k e^{-xz_k(t)^2/4}$ of the Thorin measure of R_t depends on those zeros nonlinearly:

along the equation of motion of the zeros (Section 4.1.1) its derivative in t is a sum over pairs of zeros. No Euler product for H_t is known either.

17.1.2.4 No GGC multiplier can manufacture positivity

PROPOSITION 17.1.17. *Let V be the Laplace transform of any GGC. Then $V \cdot R$ is the Laplace transform of a GGC if and only if RH holds. For φ only one direction holds for every V: if $V \cdot \varphi$ is a GGC transform then RH holds. The converse depends on V: it fails for $V \equiv 1$ (Theorem 11.1.3) and holds for $V = e^{-c\sqrt{s}}/G$, for which $V\varphi$ is the floor $e^{-c\sqrt{s}}R$ of Theorem 10.2.15.*

Proof. If RH holds, R and V are GGC transforms and so is their product. Conversely, if VR is a GGC transform, then $-(VR)'/(VR)$ and $-V'/V$ are Stieltjes functions, analytic on $\mathbb{C} \setminus (-\infty, 0]$, so $-R'/R$ is analytic there and R has no zeros or poles in $\mathbb{C} \setminus (-\infty, 0]$. By Proposition 10.1.9 this is RH. The same argument applies to $V \cdot \varphi$, since $\varphi = RG$ with G analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. For $V = e^{-c\sqrt{s}}/G$, a GGC transform by Theorem 10.2.12, $V\varphi = e^{-c\sqrt{s}}R$, which is a GGC under RH by Theorem 10.2.15. □

Multipliers can remove obstructions from normalisation, as the floor does, but not the obstruction from the zeros. Thorin measures add under multiplication, so a multiplier can hide negative Thorin mass; but what RH forbids is Thorin mass off the half line, the complex atoms at $-(\rho - \frac{1}{2})^2$ contributed by zeros off the critical line, and no GGC factor can cancel a singularity of R in the cut plane. This is why the floor $e^{-c\sqrt{s}}R$ of Theorem 10.2.15, which uses the best available multiplier $e^{-c\sqrt{s}}/G$, is infinitely divisible but can be shown to be a GGC only by proving RH. Section 14.1 finds the same for the general theory of infinitely divisible laws: each GGC criterion it examines is, for R, equivalent to RH (Section 14.1.5).

17.1.3 NUMERICAL VERIFICATION

The computations of Chapter 10 and of this chapter have three levels of certainty. The inputs to Theorems 11.1.1, 10.2.15, 10.2.17 and 17.1.15 are enclosed in outward rounded interval arithmetic (enclosures E1 to E6). The bounds $233.95 < \operatorname{Re} \kappa(\rho_{213}) < 233.96$ of Corollary 17.1.4 are enclosed in ball arithmetic with Arb [179], and the same computation also encloses the values of S quoted with Theorem 11.1.3. The torus bounds of Tables 17.3 and 17.4 and the second condition of Theorem 17.1.11 are proved by a branch and bound whose final boxes are enclosed in ball arithmetic with Arb (Remark 17.1.13). The constants of Table 17.7 are computed in floating point with `mpmath` at 20 to 30 digits, and the argument principle counts of Section 17.1.1 in floating point with 40 or more digits. The torus minima quoted for $n \geq 40$ after Remark 17.1.13, and the threshold at $n = 40$ in Problem 17.2, are estimated by optimisation and are not certified.

The function w at $x = 10^{-3}, 10^{-2}, 0.1, 1, 10$ is 1.00025, 1.00249, 0.86251, 0.49140, 0.21524; its initial increase rules out self decomposability (Theorem 11.1.3). Talbot inversion of φ gives the density f of the law with Laplace transform φ: $f(0.05) = 0.36222$, $f(1) = 0.13508$, $f(16) = 0.0087321$, and $x^{3/2} f(x) = 0.5589, 0.6382, 0.6908, 0.7220, 0.7392$ at $x = 16, 32, 64, 128, 256$, approaching 0.75773.

Quantity	Value
$c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$	2.6860917096
τ_0, $\theta(\tau_0)$, $\max M = -\theta(\tau_0)/\pi$	6.2898360, −3.5309728, 1.1239436
$M(\gamma_1^{2\,-})$	0.5502528
τ_π with $\theta(\tau_\pi) = -\pi$	3.4362182
x_* with $W_0(x_*) = 0$	0.0032173906
$F(0.003)$, $F(0.0035)$, $F(0.004)$	0.0366850, −0.0442285, −0.1145624
$P(0.004)$	1.99×10^{-13}
$\min S$ over the first 300 ordinates	$S(\gamma_{289}^{-}) = -1.1454808$
$\max S$ over the first 300 ordinates	1.0975638, at $\gamma = 415.4552$

Table 17.7: Constants used in Sections 10.1.4 and 11.1.1.

Three steps remain open, in increasing order of difficulty. First, write the formal Thorin measure of R_t in Vere-Jones form, extending Theorem 10.2.3 with t dependent weights. Second, prove Conjecture S for infinitely many n (Problems 17.1 and 17.2). Third, find a family of positive admissible primal Lévy measures whose sine squared images converge to the image U of $\tilde{\mu}_\xi$ in Proposition 10.2.16.

17.1.4 Open problems

Problem 17.1. Conjecture S. Prove that the symmetrised truncations K_n have all their zeros in $0 < \operatorname{Re} w < \frac{1}{2}$ on the line $\operatorname{Re} w = \frac{1}{4}$ for infinitely many n; by Theorem 17.1.7 this implies RH. Corollaries 17.1.10 and 17.1.12 give its status for $n \le 24$ and $n = 32, 36$. Since Conjecture T and the Hermite Biehler inequality for A_n fail for large n, a proof must establish real rootedness without a one sided inequality, for instance by an interlacing argument in n, or by following the zeros of Z_n continuously in a parameter interpolating between n and $n+1$ and excluding collisions of zeros on the line.

Problem 17.2. Beyond the torus method. Corollary 17.1.12 rests on torus bounds proved by a branch and bound up to $n = 36$; at $n = 40$ the threshold, estimated by optimisation, is already near 4900, and with the band $1.2 \le x \le 1.3$ the method ends between $n = 40$ and $n = 44$, where the torus minimum of P_n in the band reaches zero. Extend Theorem 17.1.11 with narrower bands around $x = \frac{5}{4}$, and find the first n at which P_n has torus zeros with $x \ge \frac{5}{4}$.

Problem 17.3. From finite order to all orders. Theorem 10.2.17 proves that $k = W + c/(2\sqrt{\pi y})$ is n monotone for $n \le 10^{10}$, with the bound on n coming only from the height of verified zeros; the small y branch holds for every n, on $(0, 0.05/n]$. Can the two regimes be made to overlap for all n? The large y branch bounds the zeros above T_0 in absolute value, and the proof applies it where $yT_0^2 > 2n \log T_0$. No bound of this kind works where yT_0^2 is below about n: the peak of $\gamma^{2n} e^{-y\gamma^2}$, at $\gamma^2 = n/y$, then lies above T_0, where the zeros are dense enough that their contributions, taken in absolute value, outweigh the positive term $c\,\Gamma(n+\frac{1}{2}) y^{-n-\frac{1}{2}}/(2\pi)$ and the verified zeros together. So this asks for a prime side estimate for $(-1)^n k^{(n)}$ on a range $y \le y_n$ with y_n growing at least linearly in n; a range that decreases more slowly than $1/n$ raises the order reached but reaches only finitely many orders. In the proof of Theorem 10.2.17 the obstruction is the factor 2^n in the Cauchy estimate for $P^{(n)}$ on the circle of radius $y/2$. A circle of radius close to y, as in the proof of Theorem 8.1.6, loses only a factor of order n (Section 8.1.6), but that proof also stops at a finite order, and Proposition 11.1.2 suggests that at large y the obstruction cannot be removed without the explicit formula.

Problem 17.4. Beyond the Bernstein cone. Theorem 17.1.15 proves Weil positivity for test functions completely monotone in r^2, under an integrability condition. Find the largest cone of test functions, closed under the operations that preserve the Thorin structure, on which positivity can be proved unconditionally, and determine how far it can localise in height.

Problem 17.5. Deforming the binary product. By Theorem 11.1.8 each Euler factor at $\operatorname{Re} s = \sigma$ is $\prod_n (1 + p^{-nu})^{b_n(p^{-\sigma})}$, with integer multiplicities 0 and 1 at $\sigma = 0$ and necklace multiplicities at $\sigma = \frac{1}{2}$. Follow the pseudo spectral trace formula (11.1.3) along the path $q = p^{-\sigma}$, $0 \le \sigma \le \frac{1}{2}$, and determine where and how the positivity of the interference pattern of the progressions can fail.

Problem 17.6. Regularising the autocorrelation. Proposition 10.2.14 factors $\Phi * \Phi$, formally, into the Meixner characteristic function and the positive lattice measure $\mathcal{A}$. The lattice series diverges: truncated at $m, n \le X$ it grows linearly in X, through the square of the pole term of the partial sums of ζ, and the factor $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ does not remove this growth. Find the regularised form of $\operatorname{sech}^{1/2} * \mathcal{A}$ and decide whether it has a Thorin or Lee and Yang structure in v.

Problem 17.7. Unimodality. Proposition 11.1.5 gives numerical evidence that w is unimodal. Prove it, and prove that f is unimodal, with mode near 0.0240, for instance from Steutel's equation (11.1.1) and the unimodality of w. Yamazato's theorem [276], [253, Theorem 53.1], that every self decomposable law is unimodal, shows with no hypothesis that X_R, the variable with Laplace transform R, is unimodal (Corollary 14.1.2 of Section 14.1), but it does not apply to X_φ, whose density is f, since X_φ is infinitely divisible and not self decomposable (Theo-

rem 11.1.3).

Problem 17.8. Infinite divisibility across the strip. Is $w_\sigma > 0$ for every $\sigma \in (0,1)$ and every $x > 0$? For $\frac{1}{2} < \sigma < 1$ the method of Theorem 11.1.1 may apply: in the third form of Theorem 11.1.12 the terms of the zeros below the verified height are Poisson kernels of positive width, so the question reduces to the sign of the explicit pole and archimedean terms at large x, whose limit is governed by $\zeta'(\sigma)/\zeta(\sigma) > 0$. For the completed functions the answer is known: R_σ is infinitely divisible for every $\sigma > \frac{1}{2}$ with no hypothesis (Theorem 12.2.5 and the proof of Proposition 13.1.1). Since φ_σ differs from R_σ by the gamma factor and elementary factors in $\sqrt{s}$, this does not settle the question for w_σ.

Problem 17.9. Meixner Pollaczek truncations. The symmetric expansion of $(\tau^2 + \frac{1}{4})Z(\tau)$ captures the first ordinate at degree 40 but converges slowly. Is there a weighting of its coefficients, or a Jensen type construction from them, whose polynomials are real rooted and whose zeros converge to the ordinates?

Problem 17.10. Nyman Beurling. Theorem 10.1.5 writes $1/\zeta(\frac{1}{2} + \sqrt{s})$ in the span of $\{\operatorname{sech}(\sqrt{s}\log N)\}$ with Möbius weights on $H_{1/2}$. Relate the completeness question of Lai and Vere-Jones for the dilates $\{\operatorname{sech}(\lambda x)\}$ to the Nyman Beurling criterion [219, 23], which Section 13.1.8 sets beside it, and decide whether the sech expansion admits an L^2 version on the boundary $\operatorname{Re}\sqrt{s} = \frac{1}{2}$.

Problem 17.11. Product representations of X_R. By the closure properties behind Corollary 11.1.6, an unconditional representation of X_R built from independent GGC variables by sums, products and weak limits would make R a GGC and prove RH (Proposition 2.1.1); for X_φ the corollary rules out such a representation. The harmonic mean is not available, because Proposition 11.1.7 shows it is not a closure operation for the class. The deconvolution (10.2.1) of Section 10.2.1 gives X_R, formally, as X_φ minus a stable law and a compensated series of the variables Y_{c_n}. Can the compensation be absorbed into a product?

Truncations of the beta lattice and Hermite coefficients

Section 17.1 truncates the lattice of Biane, Pitman and Yor for zeta: the truncated zeros move at the first order rate $\kappa(\rho)/n$ (Proposition 17.1.2), the side of approach is tied to the Riemann Siegel phase and so to Gram's law (Theorem 17.1.3), the zero ρ_{213}, approached from the wrong side, refutes the one sided conjecture (Corollary 17.1.4), and symmetrising by the functional equation leaves a statement that is not refuted (Proposition 17.1.6). We carry this programme to the Dirichlet beta function $L(s,\chi_4)$, with the lattice $\lambda_k = 2\pi^2(k - \frac{1}{2})^2$ and the completed L function as Mellin transform, and add the orders $m \geq 1$ of cumulant matching, an exact mean corrected truncation as a finite sum of incomplete gamma functions, and the drift at all 379 zeros below height 500. The second half is a separate subject: the expansion of Ξ in Hermite functions at a small scale, whose coefficients

sample Ξ at the turning points of the Hermite functions, up to an amplitude factor and an error $O(a^4)$, and have been used to count zeros from the Pólya kernel alone.

The chapter is based on [146, 139]. Where a recomputation differed from a note the recomputed value is printed, and a statement not recomputed is marked unchecked. Three statements of the notes are restated: the drift law needs a hypothesis on the first unmatched cumulant of the remainder (Theorem 17.3.2, Proposition 17.3.4), the exclusion of one sided criteria is conditional on one zero with a negative drift (Corollary 17.4.2), and the asymptotic of the second note is proved in an additive form, uniform in m, the relative form failing at the zeros of Ξ (Theorem 17.9.1, Proposition 17.9.2).

Notation. $\xi_4(s) = \xi(s, \chi_4)$ is that of Section 15.1.7, and Λ is the de Bruijn Newman constant. The variable ω is the exit time of Remark 15.1.23 divided by 2π. ϑ_β and Z_β are the Riemann Siegel and Hardy functions of χ_4, so that ϑ and Z remain those of zeta. Φ is the kernel of (2.1.3), H_t the de Bruijn Newman function of Section 4.1, and H_n with an integer index the Hermite polynomial of Section 21.1. The letter a is the Hermite scale in the second half, and $\lambda_* = 2\pi^2$ in the first.

Floating point computations are called computed and never certified; the zero counts of this chapter are of that kind. The results are these.

1. GRH for χ_4 is equivalent to a local zero free condition on the truncated Mellin transforms (Theorem 17.2.1, proved). The truncated zeros move by $\delta_m n^{-(2m+1)}$ for the hard cutoff and the mean correction, and by a multiple of the first unmatched cumulant difference in general (Theorem 17.3.2, Corollary 17.3.3, proved); the formula for $m \geq 2$ with a generalised gamma convolution remainder is false (Proposition 17.3.4, proved), and the drift of the first zero is computed for three kinds of remainder (Table 17.8).

2. The sign of the real part of the drift is a Gram parity of χ_4 with two corrections (Theorem 17.4.1, proved). Among the 379 zeros below height 500 the mean correction approaches from the left exactly at the five zeros with anomalous Gram parity (computed, Table 17.9). A zero with negative drift excludes a one sided criterion at that order (Corollary 17.4.2, proved); the premise is computed, not proved.

3. The truncations have transient zeros in the left half strip (Table 17.10, computed). The symmetrised truncations satisfy a Hurwitz criterion (Theorem 17.6.1, proved) and keep every simple critical zero on the line locally (Proposition 17.6.2, proved); beyond that only the heights of the computed tables are reached.

4. The Hermite coefficients of Ξ at scale a are the Taylor coefficients of $e^{z^2/2}$ times the backward heat flow of Ξ for the time $2a^2$ (Theorem 17.8.1, proved). They sample Ξ at the turning points $a\sqrt{4m+1}$ with an additive error $O(a^4)$

uniform in m (Theorem 17.9.1, proved), and the first negative coefficient reads off γ_1 (Corollary 17.10.1, proved; Table 17.13, computed).

5. The number of sign changes of the coefficients equals the number of zeros of zeta up to height 1000 at scales up to about 5 to 6, with a lag of order one at the top edge (Remark 17.11.1, computed); what a verified count would prove is Proposition 17.11.2.

17.2 The beta lattice and its truncations

Write $\xi_4(s) = \xi(s,\chi_4) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$ for the completed Dirichlet beta function of Section 15.1.7, entire and symmetric about $s=\frac12$, and let ω be the exit time variable of Remark 15.1.23 divided by 2π. Then

$$\mathbb{E}e^{-\lambda\omega} = \operatorname{sech}\sqrt{\lambda/2} = \prod_{k\ge1}\Big(1+\frac{\lambda}{\lambda_k}\Big)^{-1}, \qquad \lambda_k = \lambda_*\big(k-\tfrac12\big)^2, \quad \lambda_* = 2\pi^2,$$

so that $\omega = \sum_k e_k/\lambda_k$ with independent standard exponential variables e_k, and the Thorin measure of ω is $\sum_k \delta_{\lambda_k}$. By Remark 15.1.23 and the functional equation of ξ_4 the Mellin transform is

$$M(z) = \mathbb{E}\,\omega^{-z} = (2\pi)^z\xi_4(2z), \tag{17.2.1}$$

The nontrivial zeros of $L(s,\chi_4)$ are the points $z=\rho/2$ of the strip $0<\operatorname{Re}z<\frac12$ at which M vanishes, and $g(z)=(2\pi)^{-z}M(z)=\xi_4(2z)$ is symmetric about $\operatorname{Re}z=\frac14$. GRH for χ_4 says that they all lie on $\operatorname{Re}z=\frac14$. This is the lattice of Section 17.1 with the atoms λ_k in place of the squares k^2.

Split $\omega=\omega_n+T_n$ into the first n atoms and the tail $T_n=\sum_{k>n}e_k/\lambda_k$. With $q_j(n)=\sum_{k>n}\lambda_k^{-j}$ the cumulants of the tail are $\kappa_j(T_n)=(j-1)!\,q_j(n)$, and by the midpoint rule (Lemma 17.3.1)

$$\kappa_j(T_n) = \frac{(j-1)!}{\lambda_*^j(2j-1)\,n^{2j-1}}\Big(1-\frac{j(2j-1)}{12\,n^2}+O(n^{-4})\Big). \tag{17.2.2}$$

A truncation replaces T_n by a variable $R_n\ge0$ independent of ω_n and puts $M_n(z)=\mathbb{E}(\omega_n+R_n)^{-z}$. Three choices are used, the third a family.

Hard cutoff $R_n=0$. With $A_k^{(n)}=\prod_{j\le n,\,j\ne k}\lambda_j/(\lambda_j-\lambda_k)$ the density of ω_n is $\sum_{k\le n}A_k^{(n)}\lambda_k e^{-\lambda_k w}$, and

$$M_n^{(0)}(z) = \mathbb{E}\omega_n^{-z} = \Gamma(1-z)\sum_{k\le n}A_k^{(n)}\lambda_k^z,$$

an exponential polynomial times $\Gamma(1-z)$. It is holomorphic for $\operatorname{Re}z<n$ (the density behaves like w^{n-1} at 0); the poles of $\Gamma(1-z)$ at $1,\dots,n-1$ cancel and the pole at n does not, so $M_n^{(0)}$ is not entire.

Mean correction $R_n = c_n = \mathbb{E}T_n = q_1(n) = \psi'(n+\frac{1}{2})/(2\pi^2)$, a constant. Since $\int_0^\infty (w+c)^{-z}\lambda e^{-\lambda w}dw = \lambda^z e^{\lambda c}\Gamma(1-z,\lambda c)$,

$$M_n^{(1)}(z) = \mathbb{E}(\omega_n + c_n)^{-z} = \sum_{k\le n} A_k^{(n)} \lambda_k^z e^{\lambda_k c_n}\, \Gamma(1-z, \lambda_k c_n),$$

a finite sum of upper incomplete gamma functions, entire in z, of the same type as Haglund's approximants of Ξ [94] (Chapters 18 and 19.1.7) but with the weights of a Thorin lattice of $L(s,\chi_4)$.

Matching $R_n \ge 0$ independent of ω_n whose first m cumulants equal those of T_n. For $m = 0$ the hard cutoff is admissible, and c_n is the deterministic choice for $m = 1$; a mean match alone does not force R_n to be constant. The distinction matters for $m \ge 2$ (Section 17.3).

At $n = 40$ and $z = 0.3$ the mean corrected value is $M_{40}^{(1)}(0.3) = 1.70347125$ against $M(0.3) = 1.70347184$, and the hard cutoff at $n = 200$ gives $M_{200}^{(0)}(0.3) = 1.70461$ (computed, by the Laplace integral of the density with no cancellation of the $A_k^{(n)}$; two runs at different precision and step agree to 10^{-34}; more digits: 1.703471255, 1.70347184, 1.704611458).

Theorem 17.2.1 (Local criterion). *Let $R_n \ge 0$ be independent of ω_n with $R_n \to 0$ in probability, as for the hard cutoff and for every nonnegative remainder with $\mathbb{E}R_n = c_n$ (Markov's inequality gives $R_n \to 0$ in probability, since $c_n = O(n^{-1})$). Then $M_n \to M$ locally uniformly on $0 < \operatorname{Re} z < \frac{1}{2}$, and GRH for χ_4 holds if and only if every compact subset of $0 < \operatorname{Re} z < \frac{1}{4}$ is free of zeros of M_n for all sufficiently large n.*

Proof. Let $X_n = \omega_n + R_n$. Coupling the exponential summands gives $\omega_n \uparrow \omega$ almost surely, so $X_n \to \omega$ in probability, and the laws of X_n converge weakly to that of ω. Let K be a compact subset of the strip and choose $b \in (\max_K \operatorname{Re} z, 2)$. For $n \ge 2$, $X_n \ge \omega_2$, so $|X_n^{-z}| \le 1 + \omega_2^{-b}$, which is integrable because the density of ω_2 is $O(w)$ at 0. Weak convergence and this uniform integrability give $\mathbb{E}X_n^{-z} \to \mathbb{E}\omega^{-z}$ for every $z \in K$. The same bound on a slightly larger compact set gives holomorphy and local boundedness, and Vitali's theorem gives local uniform convergence.

If GRH holds, M has no zero on a compact $K \subset \{0 < \operatorname{Re} z < \frac{1}{4}\}$, so $\min_K |M| > 0$ and $M_n \ne 0$ on K for large n. Conversely, if M has a zero in $0 < \operatorname{Re} z < \frac{1}{4}$, a small closed disc about it with zero free boundary lies in the half strip, and Rouché's theorem gives zeros of M_n in the disc for every large n, contradicting the property (M is not identically zero, being positive on the real segment). A zero in $\frac{1}{4} < \operatorname{Re} z < \frac{1}{2}$ is carried to the left half strip by the functional equation $(2\pi)^{-z}M(z) = (2\pi)^{z-\frac{1}{2}}M(\frac{1}{2}-z)$. □

The theorem, the analogue of Theorem 17.1.1 for χ_4, turns GRH into a statement about explicit finite objects, but as an equivalence: the difficulty is to control the zeros of M_n uniformly.

17.3 The drift of the truncated zeros

For $m = 0$ and zeta the drift is Proposition 17.1.2. For χ_4 and every order m one point matters: a remainder matching the first m cumulants of the tail has a first unmatched cumulant of its own, and the drift depends on the difference of the two.

Lemma 17.3.1. *For fixed $j \geq 1$, $q_j(n) = \lambda_*^{-j}\big\{n^{1-2j}/(2j-1) - \frac{j}{12}n^{-2j-1} + O(n^{-2j-3})\big\}$. In particular*

$$\begin{aligned} c_n &= \frac{1}{\lambda_* n} - \frac{1}{12\lambda_* n^3} + O(n^{-5}),\\ v_n = \operatorname{Var} T_n = q_2(n) &= \frac{1}{3\lambda_*^2 n^3} - \frac{1}{6\lambda_*^2 n^5} + O(n^{-7}),\\ \kappa_3(T_n) &= \frac{2}{5\lambda_*^3 n^5} + O(n^{-7}). \end{aligned}$$

Proof. For $p > 1$, expand x^{-p} about the midpoint of $[k-1, k]$ to fourth order: the quadratic term contributes $p(p+1)(k-\frac{1}{2})^{-p-2}/24$ and the remainder sums to $O(n^{-p-3})$. Summing over $k > n$ gives $\sum_{k>n}(k-\frac{1}{2})^{-p} = n^{1-p}/(p-1) - (p/24)n^{-p-1} + O(n^{-p-3})$, using the same estimate without the correction for $\sum_{k>n}(k-\frac{1}{2})^{-p-2}$. Take $p = 2j$. □

Fix $m \geq 0$ and write $r = m+1$, $p = 2r-1$. Let $R_n \geq 0$ be independent of ω_n with $\kappa_j(R_n) = \kappa_j(T_n)$ for $1 \leq j < r$, and assume, for a constant C independent of n and j,

$$\log \mathbb{E}e^{-tR_n} = \sum_{j\geq 1} \frac{(-1)^j \kappa_j(R_n)}{j!} t^j \quad (0 \leq t < n^2/C), \qquad \frac{|\kappa_j(R_n)|}{j!} \leq C^j n^{1-2j}. \tag{17.3.1}$$

This sufficient condition does not follow from matching finitely many cumulants. It holds for the hard cutoff, the constant remainder and the gamma remainders below, and for the tail T_n itself after increasing C, since $\log \mathbb{E}e^{-tT_n} = -\sum_{k>n}\log(1+t/\lambda_k)$ and $q_j(n) \leq (2j-1)^{-1}\lambda_*^{-j}(n-\frac{1}{2})^{1-2j}$. Put $\Delta_{r,n} = \kappa_r(T_n) - \kappa_r(R_n)$.

Theorem 17.3.2 (Drift law). *Under these hypotheses, uniformly for z in a compact subset K of $\operatorname{Re} z > 0$,*

$$M_n(z) - M(z) = -\frac{(-1)^r \Delta_{r,n}}{r!}\,(z)_r\, M(z+r) + O_K\big(n^{-p-2} + n^{-2p}\big), \tag{17.3.2}$$

with $(z)_r = z(z+1)\cdots(z+r-1)$. *Let* $\rho = \frac12 + i\gamma$ *be a simple zero of* ξ_4 *and* $z_0 = \rho/2$. *For all large* n *a fixed small disc about* z_0 *contains exactly one zero* z_n *of* M_n, *it is simple, and*

$$z_n - z_0 = \frac{(-1)^r \Delta_{r,n}}{r!}(z_0)_r \frac{M(z_0+r)}{M'(z_0)} + O\big(n^{-p-2} + n^{-2p}\big), \qquad \frac{M(z_0+r)}{M'(z_0)} = \frac{(2\pi)^r \xi_4(\rho+2r)}{2\,\xi_4'(\rho)}. \tag{17.3.3}$$

If $\Delta_{r,n} = d\,n^{-p} + o(n^{-p})$ *with* $d \neq 0$, *then* $z_n - z_0 = D_r n^{-p} + o(n^{-p})$ *with* $D_r = \dfrac{(-1)^r d\,(2\pi)^r (z_0)_r\, \xi_4(\rho+2r)}{2\,r!\,\xi_4'(\rho)} \neq 0$.

Proof. The Laplace transforms satisfy $\mathbb{E}e^{-t(\omega_n+R_n)} = L(t)e^{D_n(t)}$ with $L(t) = \operatorname{sech}\sqrt{t/2}$ and $D_n = \log \mathbb{E}e^{-tR_n} - \log \mathbb{E}e^{-tT_n}$. The first $r-1$ cumulants cancel, so (17.3.1) gives, for $0 \le t \le n^2/(2C)$, $D_n(t) = -\frac{(-1)^r\Delta_{r,n}}{r!}t^r + O(n^{-p-2}t^{r+1})$ with $\Delta_{r,n} = O(n^{-p})$. On $0 \le t \le s_n = A(\log n)^2$ this is $o(1)$, and the exponential gives $e^{D_n} - 1 = -\frac{(-1)^r\Delta_{r,n}}{r!}t^r + O(n^{-p-2}t^{r+1} + n^{-2p}t^{2r})$. For $t \ge s_n$: with $B = \max_K \operatorname{Re} z$, an integer $Q > B$ and $J_n = \lfloor\sqrt{s_n/\lambda_*}\rfloor$, the first Q factors of $\mathbb{E}e^{-t\omega_n}$ are at most $C_Q t^{-Q}$ and each of the factors with $Q < k \le J_n$ is at most $\frac12$, so $\mathbb{E}e^{-t\omega_n} \le C_Q t^{-Q} 2^{-(J_n - Q)}$. Since $R_n \ge 0$, both $L(t)$ and $\mathbb{E}e^{-t(\omega_n+R_n)}$ are bounded by it, and the Mellin integrals over $[s_n,\infty)$ are $O(n^{-N})$ for every N once A is large. The identity $\mathbb{E}X^{-z} = \Gamma(z)^{-1}\int_0^\infty t^{z-1}\mathbb{E}e^{-tX}dt$ converges absolutely for $\operatorname{Re} z > 0$ and large n, and $\Gamma(z+r)/\Gamma(z) = (z)_r$, so integrating the expansion gives (17.3.2).

By Rouché's theorem a small disc about the simple zero z_0 contains exactly one zero z_n of M_n, and it is simple. Put $E_n = M_n - M$; then $E_n = O(n^{-p})$ and, by Cauchy's estimate on a slightly larger disc, $E_n' = O(n^{-p})$. Thus $z_n - z_0 = O(n^{-p})$, and Taylor's formula $0 = M'(z_0)(z_n - z_0) + E_n(z_0) + O(n^{-2p})$ gives (17.3.3). The ratio follows from (17.2.1): $M'(z_0) = 2(2\pi)^{z_0}\xi_4'(\rho)$ and $M(z_0+r) = (2\pi)^{z_0+r}\xi_4(\rho+2r)$. Finally $(z_0)_r \neq 0$, $\xi_4'(\rho) \neq 0$, and $\xi_4(\rho+2r) \neq 0$ because $\operatorname{Re}(\rho+2r) > 1$, where the Euler product of $L(s,\chi_4)$ has no zero, so $D_r \neq 0$ when $d \neq 0$. □

Corollary 17.3.3 (The two named schemes). *Let* $\rho = \frac12 + i\gamma$ *be a simple zero of* ξ_4. *Then*

$$\delta_m = \frac{(-1)^{m+1}(z_0)_{m+1}\,\xi_4(\rho+2m+2)}{2(m+1)(2m+1)\,\pi^{m+1}\,\xi_4'(\rho)}, \tag{17.3.4}$$

and for the hard cutoff $z_n = z_0 + \delta_0/n + O(n^{-2})$, *for the mean correction* $z_n = z_0 + \delta_1/n^3 + O(n^{-5})$. *For a remainder that satisfies the hypotheses of Theorem 17.3.2 at order* m *and has* $\kappa_{m+1}(R_n) = o(n^{-(2m+1)})$ *the displacement is* $\delta_m n^{-(2m+1)}(1 + o(1))$.

Proof. If $\kappa_r(R_n) = o(n^{-p})$, then $d = (r-1)!/[\lambda_*^r(2r-1)]$ by Lemma 17.3.1, and D_r of Theorem 17.3.2 is δ_m of (17.3.4). For the hard cutoff $r = 1$ and $\Delta_{1,n} = c_n$; for the constant c_n, $r = 2$, $\Delta_{2,n} = v_n$. The error is $O(n^{-3} + n^{-2})$ for $r = 1$ and $O(n^{-5} + n^{-6})$ for $r = 2$. For the hard cutoff the relative error is $O(n^{-1})$, not $O(n^{-2})$: the error term $n^{-2p} = n^{-2}$ of (17.3.2) is the term $\frac{1}{2}c_n^2(z)_2 M(z+2)$ of the expansion of e^{D_n}, which is not cancelled by the variance term $v_n = O(n^{-3})$ and has a nonzero coefficient at z_0. □

The first zero of $L(s, \chi_4)$ is at $\gamma = 6.0209489$. Table 17.8 compares δ_m with the measured $n^{2m+1}(z_n - z_0)$ (computed, precision 40 digits with step $h = 1/4$ against 50 digits with step $h = 1/8$, differences below 10^{-26}). The hard cutoff approaches δ_0 with relative error $(0.27 + 0.51i)/n$, of modulus $0.58/n$, and the mean correction with an error that falls by a factor of 4 at each doubling of n, as the orders $O(n^{-1})$ and $O(n^{-2})$ say; the finite n values agree with the predicted constants to two or three digits, not four. For the hard cutoff the real part of $z_n - \frac{1}{4}$ is 9.44×10^{-3} at $n = 128$ and 4.71×10^{-3} at $n = 256$. For the mean correction it is 5.87×10^{-5}, 3.10×10^{-4}, 4.72×10^{-5}, 6.15×10^{-6}, 7.76×10^{-7}, 9.73×10^{-8} at $n = 4, 8, 16, 32, 64, 128$. The value at $n = 4$ is small by accident (the law $\operatorname{Re}\delta_1/n^3$ gives 3.2×10^{-3}), and the sequence is not monotone. The hard cutoff would need $n \approx \operatorname{Re}\delta_0/(8 \times 10^{-7}) \approx 1.5 \times 10^6$ to reach 8×10^{-7} (from the asymptotic law, not a run).

n	hard cutoff, $n(z_n - z_0)$	mean correction, $n^3(z_n - z_0)$	gamma remainder, $n^5(z_n - z_0)$
16	$1.2424 - 0.5466i$	$0.1934 - 1.0874i$	$-1.1423 - 0.4084i$
32	$1.2234 - 0.5618i$	$0.2015 - 1.0853i$	$-1.1429 - 0.4370i$
64	$1.2137 - 0.5692i$	$0.2035 - 1.0849i$	$-1.1428 - 0.4443i$
128	$1.2089 - 0.5728i$	$0.2040 - 1.0848i$	$-1.1428 - 0.4461i$
limit	$\delta_0 = 1.2041 - 0.5763i$	$\delta_1 = 0.2041 - 1.0848i$	$\frac{4}{9}\delta_2 = -1.1428 - 0.4467i$

Table 17.8: The drift of the first zero of M_n at $\gamma = 6.0209489$ (computed). The last column is the gamma remainder of Proposition 17.3.4, for which $\delta_2 = -2.5713 - 1.0051i$ is not the limit.

17.3.1 Why the matching hypothesis cannot be dropped

The note [146] states (17.3.4) for every m and every remainder matching m cumulants, for instance a generalised gamma convolution with a few atoms. This is false for $m \geq 2$, because the first unmatched cumulant of R_n enters.

Proposition 17.3.4. *Let R_n have the gamma law with shape c_n^2/v_n and scale v_n/c_n, so that $\mathbb{E}R_n = c_n$ and $\operatorname{Var} R_n = v_n$ exactly. It satisfies* (17.3.1)*, and with $r = 3$ the drift law gives $z_n = z_0 + \frac{4}{9}\delta_2\, n^{-5} + O(n^{-7})$. A second remainder $R_n = G_{1,n} + G_{2,n}$, a sum of two independent gamma variables with matched mean and variance and the same natural scale, gives $z_n = z_0 - \frac{1}{9}\delta_2\, n^{-5} + O(n^{-7})$. A*

bounded remainder $R_n = b_n + hB_n$ with B_n a Bernoulli variable, $h > 0$ small and fixed, matched to the mean and variance, has $z_n - z_0 = cn^{-3} + O(n^{-5})$ with $c \neq 0$.

Proof. For the gamma law $\kappa_3(R_n) = 2v_n^2/c_n$, so $\Delta_{3,n} = 2\big(q_3 - q_2^2/q_1\big) = \frac{8}{45\lambda_*^3 n^5} + O(n^{-7}) = (\frac{4}{9} + O(n^{-2}))\kappa_3(T_n)$ by Lemma 17.3.1, and the difference in brackets is positive for every n by the Cauchy Schwarz inequality applied to $\lambda_k^{-1/2}$ and $\lambda_k^{-3/2}$. Theorem 17.3.2 gives the first claim. For two gamma variables with shapes and scales $\big(\frac{8c_n}{5s_n}, \frac{s_n}{2}\big)$ and $\big(\frac{c_n}{15s_n}, 3s_n\big)$, $s_n = v_n/c_n$, the mean is c_n, the variance is v_n and $\kappa_3(R_n) = 4v_n^2/c_n$, so $\Delta_{3,n} = -\frac{2}{45\lambda_*^3 n^5} + O(n^{-7}) = (-\frac{1}{9} + O(n^{-2}))\kappa_3(T_n)$. For the Bernoulli remainder put $p_n = \frac{1}{2}(1 - \sqrt{1 - 4v_n/h^2})$ and $b_n = c_n - hp_n$, so that $R_n \geq 0$ for large n with mean c_n and variance v_n exactly; then $\kappa_3(R_n) = hv_n(1-2p_n)$ is of order n^{-3}. Write $M_h(z) = \mathbb{E}(\omega+h)^{-z}$ and $H_h(z) = h^{-2}\big[M_h(z) - M(z) + hzM(z+1)\big] - \frac{1}{2}(z)_2 M(z+2)$. The method of the proof of Theorem 17.3.2 gives $M_n(z) - M(z) = v_n H_h(z) + O_K(n^{-5})$, and Taylor's theorem gives $H_h(z) = -\frac{h}{6}(z)_3 M(z+3) + O(h^2)$, which does not vanish at z_0 for small h; hence $z_n - z_0 = -H_h(z_0)/(3\lambda_*^2 M'(z_0))\, n^{-3} + O(n^{-5})$ with a nonzero coefficient. □

The gamma column of Table 17.8 agrees: $n^5(z_n - z_0)/\delta_2 = 0.4444$ at $n = 128$, against $\frac{4}{9}$ (computed). No generalised gamma convolution repairs the formula for $m = 2$: for every positive measure U the moments $m_j = \int u^{-j} U(du)$ satisfy $m_1 m_3 \geq m_2^2$, so a generalised gamma convolution remainder with the mean and variance of T_n has $\kappa_3(R_n) \geq 2v_n^2/c_n$, of exact order n^{-5} (proved, by the Cauchy Schwarz inequality), with the one gamma remainder as the extreme case. So δ_m for $m \geq 2$ belongs to a hypothetical remainder with $\kappa_{m+1}(R_n) = o(n^{-(2m+1)})$, not to the generalised gamma convolutions the note proposes.

17.4 The side of approach is a Gram parity

For zeta, Theorem 17.1.3 makes the sign of $\operatorname{Re}\kappa(\rho)$, up to the phase of $\zeta(\frac{5}{2} + i\gamma)$, that of $\sin\theta(\gamma)/Z'(\gamma)$, a Gram parity. The same holds for χ_4 and every m, with one more correction. Put $G(s) = (4/\pi)^{s/2}\Gamma(\frac{s+1}{2})$, so that $\xi_4 = \frac{2}{\sqrt{\pi}} GL$, and let $\vartheta_\beta(x) = \arg G(\frac{1}{2} + ix)$ and $Z_\beta(x) = e^{i\vartheta_\beta(x)} L(\frac{1}{2} + ix, \chi_4)$ be the Riemann Siegel and Hardy functions of χ_4 (ϑ and Z are those of zeta in Section 2.2.1). Then $G(\frac{1}{2} + ix)L(\frac{1}{2} + ix, \chi_4) = |G|\, Z_\beta(x)$ is real, and the Gram points of χ_4 are the solutions of $\vartheta_\beta(x) = j\pi$.

Theorem 17.4.1 (Phase law). *Let $\rho = \frac{1}{2} + i\gamma$, $\gamma > 0$, be a simple zero of ξ_4 and let δ_m be the number* (17.3.4). *Then*

$$\delta_m = \frac{i\, C_m(\gamma)\, e^{i\Theta_m(\gamma)}}{Z'_\beta(\gamma)}, \qquad C_m(\gamma) > 0,$$

$$\Theta_m(\gamma) = \vartheta_\beta(\gamma) + \arg L(\rho+2m+2,\chi_4) - \sum_{j=0}^{m}\Big[\arctan\frac{\frac12+2j}{\gamma} + \arctan\frac{\frac32+2j}{\gamma}\Big],$$

so that $\operatorname{sgn}\operatorname{Re}\delta_m = \operatorname{sgn}\big[-\sin\Theta_m(\gamma)\,Z'_\beta(\gamma)\big]$*, zero included. For an actual remainder satisfying the hypotheses of Theorem 17.3.2 and with* $\eta_m = \lim_n\big[\kappa_{m+1}(T_n) - \kappa_{m+1}(R_n)\big]/\kappa_{m+1}(T_n)$ *the displacement is* $\eta_m\delta_m n^{-(2m+1)}(1+o(1))$ *when the limit exists and* $\eta_m \neq 0$*, and the sign of the real part is* $\operatorname{sgn}\big[-\eta_m\sin\Theta_m(\gamma)Z'_\beta(\gamma)\big]$ *when* $\eta_m\sin\Theta_m \neq 0$*.*

Proof. The gamma factor satisfies $G(s+2) = \frac{4}{\pi}\cdot\frac{s+1}{2}G(s)$, so $\xi_4(\rho+2m+2) = \frac{2}{\sqrt\pi}(\frac{4}{\pi})^{m+1}\prod_{j=0}^m\big(\frac{\rho+1}{2}+j\big)G(\rho)L(\rho+2m+2,\chi_4)$. On the critical line $G(\rho) = |G(\rho)|e^{i\vartheta_\beta(\gamma)}$, and differentiating $G(\frac12+ix)L(\frac12+ix,\chi_4) = |G|Z_\beta(x)$ at a zero gives $i\,GL'(\rho) = |G|Z'_\beta(\gamma)$, that is $\frac{\sqrt\pi}{2}\xi'_4(\rho) = -i|G(\rho)|Z'_\beta(\gamma)$. In (17.3.4) the product $(z_0)_{m+1}\prod_j(\frac{\rho+1}{2}+j) = \prod_{j=0}^m(\frac{\rho}{2}+j)(\frac{\rho+1}{2}+j)$ has factors $\frac14+j+\frac{i\gamma}{2}$ and $\frac34+j+\frac{i\gamma}{2}$ with arguments $\frac{\pi}{2}-\arctan\frac{1/2+2j}{\gamma}$ and $\frac{\pi}{2}-\arctan\frac{3/2+2j}{\gamma}$ for $\gamma>0$. The terms $\frac{\pi}{2}$ total $(m+1)\pi$ and cancel $(-1)^{m+1}$, which leaves $\delta_m = iC_m e^{i\Theta_m}/Z'_\beta(\gamma)$ with $C_m = (4/\pi)^{m+1}|\prod|\,|L(\rho+2m+2,\chi_4)|/[2(m+1)(2m+1)\pi^{m+1}]$. Taking real parts, $\operatorname{Re}(ie^{i\Theta}) = -\sin\Theta$. The last statement is Theorem 17.3.2 with $d = \eta_m\lim_n n^{2m+1}\kappa_{m+1}(T_n)$. □

At two consecutive simple zeros Z'_β has opposite signs, so the sign of the drift alternates with the parity of the zero index unless $\sin\Theta_m$ changes sign, which is the Gram law of χ_4 read through Θ_m. The two corrections to ϑ_β are the arithmetic phase $\arg L(\rho+2m+2,\chi_4)$ and the Pochhammer phase, which is of exact size $-2(m+1)^2/\gamma + O_m(\gamma^{-3})$ (proved, first order sums). The arithmetic phase is bounded, $|L(\sigma+i\gamma,\chi_4)-1| \le \epsilon_\sigma = \sum_{k\ge1}(2k+1)^{-\sigma}$, but it does not tend to zero at fixed $\sigma>1$: the mean square of $\operatorname{Im}L(\sigma+it,\chi_4)$ over $[0,T]$ tends to $\frac12\sum_{n\ge2}\chi_4(n)^2n^{-2\sigma} > 0$ (proved, by orthogonality of distinct frequencies). For zeta the corresponding statement is that $\operatorname{Re}\kappa$ has the sign of $-\cos(\pi S_{\mathrm{mid}})$ up to a shift of the threshold $\frac12$ by about 0.11 (Section 17.1).

17.4.1 All zeros below height 500

The zeros of $L(s,\chi_4)$ below height 500 were located by sign changes of Z_β on two independent grids (steps 0.02 and 0.0137) and refined at two precisions (computed). There are 379 zeros, all simple and on the line as far as the computation sees (the minimal gap is 0.1336, against the step 0.02), in agreement with $\vartheta_\beta(500)/\pi = 379.15$ and with $N(500) = 379.1$ from (15.1.8). This is not a certified count: there is no Turing bound, and the certified count in this book stops at height 300, where $N(300) = 203$ (Section 16.6.1). For each zero the numbers δ_m, $0\le m\le5$, were computed from (17.3.4), and the sign law of Theorem 17.4.1 holds at all 379 zeros for each m (an identity, so a check of the arithmetic).

Measuring the Gram parity by the index $\lfloor \vartheta_\beta(\gamma)/\pi \rfloor$ minus the zero number, 374 zeros have the offset -1 and five do not (two have -2 and three have 0). The five exceptions, at $\gamma = 138.7502, 290.2521, 364.4197, 400.8264, 422.5104$, are listed in Table 17.9. For the mean correction ($m = 1$), $\operatorname{Re}\delta_1 < 0$ exactly at these five zeros and at no other. For the hard cutoff ($m = 0$), $\operatorname{Re}\delta_0 < 0$ at the last four, and at 138.750 the arithmetic phase is large enough to turn the sign. For $m \geq 2$ the Pochhammer phase produces further negative real parts at low height: $\operatorname{Re}\delta_m < 0$ at 8, 11, 19 and 29 of the 379 zeros for $m = 2, 3, 4, 5$ (for $m = 2$ at 6.021, 12.988, 138.750, 236.242, 290.252, 364.420, 400.826 and 422.510, and for $m = 5$ at 22 zeros below height 105). These counts concern the formal δ_m; by Proposition 17.3.4 they give the drift of an actual remainder only after multiplication by η_m.

γ	zero number	gap to the partner	$\operatorname{Re}\delta_0$	$\operatorname{Re}\delta_1$
138.7502	78	0.2929	> 0	< 0
290.2521	195	0.4251	< 0	< 0
364.4197	258	0.3840	< 0	< 0
400.8264	291	0.1336	< 0	< 0
422.5104	309	0.1769	< 0	< 0

Table 17.9: The five zeros of $L(s, \chi_4)$ below height 500 with anomalous Gram parity (computed). The partner is the neighbouring zero at the stated gap (the left neighbour for the zero number 291).

Each of the five lies in a close pair of zeros (gaps recomputed). For zeta the analogue is the list of Section 17.1: three zeros among the first 460 approached from the right, related to $|S_{\mathrm{mid}}| > 0.56$.

Corollary 17.4.2 (No one sided criterion). *Fix $m \geq 0$ and a remainder satisfying the hypotheses of Theorem 17.3.2, for which the limit η_m exists. If some simple zero ρ of ξ_4 on the critical line has $\operatorname{Re}(\eta_m \delta_m) < 0$, then for all large n the function M_n has a zero in $0 < \operatorname{Re} z < \frac{1}{4}$, so no criterion of the form that M_n has no zero in $0 < \operatorname{Re} z < \frac{1}{4}$ for infinitely many n can hold at order m for this remainder.*

Proof. By Theorem 17.3.2, $\operatorname{Re} z_n = \frac{1}{4} + n^{-(2m+1)} \operatorname{Re}(\eta_m \delta_m) + o(n^{-(2m+1)})$, which is below $\frac{1}{4}$ for large n; and the zero lies in the strip $0 < \operatorname{Re} z < \frac{1}{2}$. □

This is the argument of Corollary 17.1.4 in the other direction. Its premise is computed, not proved: for the mean correction ($m = 1$, $\eta_1 = 1$) the zero $\gamma = 138.7502$ has $\operatorname{Re}\delta_1 < 0$ in floating point, and so do the four others. A zero with $\operatorname{Re}\delta_1 < 0$ proved in ball arithmetic would make the corollary unconditional for $m = 1$, exactly as in Corollary 17.1.4. The note states more, that infinitely many Gram anomalies of χ_4 would force infinitely many left approaches at every fixed order m. This does not follow from the Gram parity alone, because the side

is set by $\sin\Theta_m$ and not by $\sin\vartheta_\beta$: at 138.750 the hard cutoff approaches from the right although the zero is anomalous. A precise implication is the following.

PROPOSITION 17.4.3. *Let* $\sigma = 2m + \frac{5}{2}$, $\epsilon_\sigma = \sum_{k\ge1}(2k+1)^{-\sigma} < 1$ *and* $B_m(\gamma) = \arcsin\epsilon_\sigma + \sum_{j=0}^{m}\big[\arctan\frac{1/2+2j}{\gamma} + \arctan\frac{3/2+2j}{\gamma}\big]$. *Then* $|\Theta_m(\gamma) - \vartheta_\beta(\gamma)| \le B_m(\gamma)$. *If infinitely many simple critical zeros satisfy* $-\sin\vartheta_\beta(\gamma)Z'_\beta(\gamma) < 0$ *and* $\operatorname{dist}(\vartheta_\beta(\gamma), \pi\mathbb{Z}) > B_m(\gamma)$, *then infinitely many zeros have* $\operatorname{Re}\delta_m < 0$.

Proof. The disc of centre 1 and radius ϵ_σ contains $L(\sigma+i\gamma,\chi_4)$, and it is seen from the origin under the angle $\arcsin\epsilon_\sigma$, which bounds $|\arg L|$. The interval between $\vartheta_\beta(\gamma)$ and $\Theta_m(\gamma)$ then contains no multiple of π, the two sines have the same nonzero sign, and Theorem 17.4.1 applies. □

The hypotheses of Proposition 17.4.3 on the five ordinates, with the margin $B_m(\gamma)$, are not checked here. Whether Gram anomalies with the margin occur infinitely often for χ_4 is open. For zeta Gram's law fails with positive frequency [267], and the analogy with the five failures below 500 is a reason to expect the same for χ_4 (heuristic). The comparison with the de Bruijn Newman constant, that these are the configurations that make $\Lambda \ge 0$, is an analogy, and the theorem of Rodgers and Tao [250] does not prove a truncation obstruction.

17.5 Transient zeros

Besides the zeros attached to those of L, the truncations have zeros that wander into the left half strip. The counts of Table 17.10 are zeros of M_n in the rectangles $0.02 < \operatorname{Re} z < 0.2499$ (left) and $0.2501 < \operatorname{Re} z < 0.48$ (right) by the argument principle (computed, enclosures of the function values in ball arithmetic but a floating point winding sum, a second run at 1.5 times the precision agrees; not certified).

The zeros of $L(s,\chi_4)$ with $0<\gamma<40$, $40<\gamma<80$ and $80<\gamma<150$ number 14, 23 and 48 (computed, agreeing with (15.1.8)). The note's right counts for the mean correction reach 23 in the band $20 < \operatorname{Im} z < 40$ at $n = 64$ and $n = 128$, and are 14 in the first band at $n = 32$, where the check of this chapter reported 13 (the mean correction counts are computed, see the caption). A count of 13 is not explained by a resolution frontier: the zero of $M_{32}^{(1)}$ near $\rho_1/2$ has $\operatorname{Re} z - \frac14 = 6.15\times10^{-6}$ (Table 17.8) and lies in the band $0.2499 < \operatorname{Re} z < 0.2501$ that the rectangles omit. At $n = 64$ the band $40 < \operatorname{Im} z < 75$ holds 37 captured zeros of the 48 in $80<\gamma<150$ and four zeros in the left half strip (computed, Table 17.10). The note gives their ordinates (heights doubled) as 118.82, 126.01, 138.65 and 142.92, which were not recomputed (unchecked). Each is said to lie between the members of a close pair of zeros of L, namely $(118.538, 119.453)$, $(125.769, 126.299)$, $(138.457, 138.750)$ and $(142.394, 143.329)$; these eight ordinates

truncation	n	$\operatorname{Im} z$	left	right
hard cutoff	32	15 to 60	2	14
hard cutoff	64	15 to 60	2	15
mean correction	32	0.5 to 20	0	13
mean correction	16	20 to 40	3	3
mean correction	32	20 to 40	1	20
mean correction	64	20 to 40	0	23
mean correction	64	40 to 75	4	37
mean correction	128	20 to 40	0	23

Table 17.10: Zeros of M_n in the left and right half strips. The counts are computed at two precisions with identical counts. The source note printed 22 for the hard cutoff at $n = 32$, a blank at $n = 64$ and 14 for the mean correction at $n = 32$, $\operatorname{Im} z$ from 0.5 to 20.

were recomputed and are zeros of L. The third pair contains the first Gram anomaly. The pairs $(44.618, 45.600)$ and $(67.637, 68.366)$ are also pairs of zeros of L (checked); the note reports that at $n = 32$ the hard cutoff has transient zeros between their members, at $0.2356 + 22.5622i$ and $0.2411 + 34.0550i$, the first gone by $n = 256$ and the second at $\operatorname{Re} z = 0.237$ and moving right at $n = 768$ (unchecked: this was not recomputed, since the hard cutoff at $n = 768$ needs about 1500 digits).

The note's picture is that mean correction reduces the transients of the hard cutoff and moves them to a resolution frontier that rises with n. In Table 17.10 the left count in the band $20 < \operatorname{Im} z < 40$ is $3, 1, 0, 0$ for $n = 16, 32, 64, 128$ (computed), consistent with a frontier that rises with n, while the hard cutoff keeps two left zeros at both $n = 32$ and $n = 64$ in $15 < \operatorname{Im} z < 60$. The rectangles of the count do not contain a neighbourhood of the critical line or of the left boundary. A zero $z_n = \frac{1}{4} + \delta_1 n^{-3} + o(n^{-3})$ with $\operatorname{Re} \delta_1 < 0$ eventually lies in $0.2499 < \operatorname{Re} z < \frac{1}{4}$ and is invisible to the count while remaining to the left of the line; so an empty left rectangle at a fixed height does not exclude such zeros, which are consistent with Theorem 17.2.1 because its compact sets stay away from the line. No uniform frontier is proved, and no statement that the transients occur only at close pairs.

17.6 Symmetrised truncations

Impose the functional equation on each truncation: put $g_n(z) = (2\pi)^{-z} M_n(z) = \mathbb{E} e^{-z V_n}$, with $V_n = \log(2\pi(\omega_n + R_n))$, and

$$F_n(z) = g_n(z) + g_n(\tfrac{1}{2} - z).$$

Then $F_n(\frac{1}{2} - z) = F_n(z)$ and $F_n(\bar{z}) = \overline{F_n(z)}$ on the common domain, so F_n is real on $\operatorname{Re} z = \frac{1}{4}$ and its zeros off the line come in quadruples; this is Proposition 17.1.6

for χ_4. If p_n is the density of V_n and $q_n(v) = e^{-v/4}p_n(v)$, then $F_n(\frac{1}{4}+iy) = \int_{\mathbb{R}}[q_n(v)+q_n(-v)]e^{-iyv}dv$, so $F_n(\frac{1}{4}+iy)/F_n(\frac{1}{4})$ is the characteristic function of the positive even density $[q_n(v)+q_n(-v)]/F_n(\frac{1}{4})$ (proved, by the change of variables; for a remainder without a density use measures).

THEOREM 17.6.1 (Hurwitz criterion). *If for infinitely many n all zeros of F_n (for the remainders of Theorem 17.2.1) in $0 < \operatorname{Re} z < \frac{1}{2}$ lie on $\operatorname{Re} z = \frac{1}{4}$, then GRH holds for χ_4.*

Proof. By Theorem 17.2.1 and the functional equation $F_n \to 2\xi_4(2z)$ locally uniformly in the strip. Along the subsequence apply Hurwitz's theorem separately to the two open half strips $0 < \operatorname{Re} z < \frac{1}{4}$ and $\frac{1}{4} < \operatorname{Re} z < \frac{1}{2}$: the nonzero limit $2\xi_4(2z)$ has no zero in either. Hurwitz's theorem is not a theorem on a closed line, so this uses the subsequence and does not apply to one fixed approximant. □

PROPOSITION 17.6.2. *Every simple critical zero of $\xi_4(2z)$ is, for all large n, the limit of exactly one zero of F_n in a small disc about it, and that zero lies on $\operatorname{Re} z = \frac{1}{4}$.*

Proof. Rouché's theorem in a disc centred on the line gives exactly one zero of F_n in it, and the reflection $z \mapsto \frac{1}{2} - \bar{z}$ preserves the disc and the zero set, so the zero lies on the line. This holds even when the corresponding zero of g_n approaches from the left. □

The proposition is local: on a fixed compact region it makes every zero of F_n critical for large n, but says nothing about heights growing with n.

For the hard cutoff, counting all zeros in $0.02 < \operatorname{Re} z < 0.48$, $0.5 < \operatorname{Im} z < 60$ against sign changes on the line gives Table 17.11 (computed, the same at 1.5 times the precision). The count 64 equals the number of zeros of $L(s, \chi_4)$ below height 120 found by the same procedure. For $n = 8$ and 16 every zero of F_n in this rectangle is critical, and the zeros of F_8 already approximate those of L (Table 17.12, computed, heights $\gamma = 2\operatorname{Im} z$).

n	zeros in the strip	on the line	left zeros of g_n
8	64	64	0
16	64	64	0
32	64	60	2
64	64	60	2

Table 17.11: Symmetrised hard cutoff (computed).

REMARK 17.6.3 (Zeros of g_n to the left). For $n = 32$ and 64 the four off line zeros of F_n and the two zeros of g_n to the left of the line are computed counts. The note reads this as a mechanism, that a zero of g_n to the left of the line produces an off

F_8	5.9438	10.3288	12.9300	16.5280	18.1556	21.5568
$L(s,\chi_4)$	6.0209	10.2443	12.9876	16.343	18.292	21.451

Table 17.12: The first zeros of F_8 and of $L(s,\chi_4)$ (computed).

line pair of F_n, in the manner of the Hermite Biehler theorem. That cannot be a general statement. At a zero z_* of g_n one has $F_n(z_*) = g_n(\frac{1}{2} - z_*)$, which need not vanish; and the Hermite Biehler theorem gives real zeros of $E + E^*$ from the absence of zeros of E in a half plane, not the converse. Here is a counterexample with a positive transform. Let $w = z - \frac{1}{4}$, $c = e^{-1/8}$ and $g(z) = (e^w + c)/(e^{-1/4} + c)$, a positive two point Laplace transform with $g(0) = 1$ whose zeros $z = \frac{1}{8} + (2k+1)\pi i$ lie to the left of the line. Its symmetrisation is $F(z) = 2(\cosh w + c)/(e^{-1/4} + c)$, and every solution of $\cosh w = -c$ with $0 < c < 1$ has $\operatorname{Re} w = 0$ (if $w = x + iy$ with $x \neq 0$ then $\sin y = 0$ and $|\cosh w| \geq 1$). So all zeros of F are critical (proved). In the data the off line zeros of F_{32} and F_{64} are four, accompanying two left zeros of g_n; that they coincide is an observation of the note, not recomputed.

The modulus inequality $|g_n(\frac{1}{2} - z)| < |g_n(z)|$ on an open set off the line implies $F_n \neq 0$ there, since $|F_n(z)| \geq |g_n(z)| - |g_n(\frac{1}{2} - z)| > 0$ (proved). It is sufficient and not necessary. The note reports that it fails near the real axis even for $n = 8$ and gives -0.118 at $z = 0.01 + 0.5i$ for the minimum of $\log|g_8(z)| - \log|g_8(\frac{1}{2} - z)|$. The recomputed minimum on the grid $x \in [0.01, 0.25]$, $y \in [0.5, 60.5]$ (steps 0.01 and 0.05) is -0.0282 at the same point (computed): the inequality fails, but with the smaller value. Its failure does not imply an off line zero. The note adds that F_8 has no zeros there because g_8 is positive on the real axis; positivity on the real segment gives a zero free neighbourhood $|\operatorname{Im} z| < c/B$ of it only with a bound $|F_8'| \leq B$ and $c = \min F_8$ on the segment, and neither is given.

With the mean correction, below height 40 ($\gamma < 80$) the function F_{64} has all 37 zeros on the critical line and F_{32} has 35 of 37 (computed, at two precisions with identical counts, in agreement with the note). The note's reading, that the symmetrised mean corrected truncations satisfy the hypothesis below their resolution frontier, rests on two values of n and is an observation.

17.7 Where this leaves the route

Theorem 17.2.1 is the form of Proposition 19.2.1 and of Section 19.2 for χ_4, with the Dirichlet beta lattice in place of the folded Gumbel kernels. Theorem 17.3.2 gives the exact rate of convergence of the truncated zeros for the hard cutoff and the mean correction, and under the matching hypothesis for every order; Theorem 17.4.1 identifies the side as a Gram parity, and to height 500 the drift of the mean correction has negative real part exactly at the five anomalous zeros. What is not proved is that the premise of Corollary 17.4.2 holds (a zero with

$\operatorname{Re}\delta_1 < 0$ in ball arithmetic) and that Gram anomalies with a margin recur. The symmetrised truncations avoid the one sided obstruction by Proposition 17.6.2, and Theorem 17.6.1 reduces GRH for χ_4 to a statement for infinitely many n that is neither proved nor tested beyond the heights of Table 17.11. Nothing in the computation is certified.

17.7.1 Open problems on the truncations

Problem 17.12. Prove that Gram anomalies with the margin $B_m(\gamma)$ of Proposition 17.4.3 occur infinitely often for $L(s,\chi_4)$, adapting the arguments known for ζ [267]. Together with Proposition 17.4.3 this would make Corollary 17.4.2 unconditional at each order m with $\eta_m > 0$.

Problem 17.13. Prove in ball arithmetic that $\operatorname{Re}\delta_1 < 0$ at the zero $\gamma = 138.7502$ of $L(s,\chi_4)$, as is done for zeta in Corollary 17.1.4, and certify the count of 379 zeros below height 500 with a Turing bound.

Problem 17.14. Prove that F_8, the symmetrised hard cutoff with eight atoms, has only critical zeros, by combining the phase of g_8 at low height with the modulus inequality and a Newton polygon bound at large height. The analogues for zeta are Theorem 17.1.9 and Corollary 17.1.10.

Problem 17.15. Determine the largest n for which the symmetrised mean corrected truncation has only critical zeros in the strip, and decide whether the first failure is related to the first Gram anomaly at 138.75.

Problem 17.16. Find a remainder R_n, not a cumulant match, whose drift $\eta\delta_m$ has a sign independent of the Gram parity. By Theorem 17.4.1 this needs a correction whose phase cancels $\vartheta_\beta(\gamma)$, which suggests building R_n from the functional equation and not from the tail.

Problem 17.17. Carry out the analysis of orders $m \geq 1$ for zeta with the lattice of Section 17.1, where $m = 0$ is Theorem 17.1.3, and compare the Gram parity of the corrected phase with the classical failures of Gram's law for zeta (the first near the Gram point $g_{126} \approx 282.455$, where $Z(g_{126}) = -0.0276$).

17.8 Hermite coefficients of Ξ

We expand Ξ in Hermite functions at a small scale, following [139], whose definitions are written out here. The kernel is the Pólya kernel Φ of (2.1.3), even and positive with $\Xi(t) = \int_{\mathbb{R}} \Phi(u) e^{itu} du$; with it the recomputed numbers of the note agree, except where stated, whereas the de Bruijn form Φ_{dBN} of Section 4.1 puts the zeros at 2γ. The kernel satisfies $\int |\Phi(u)|(1+|u|)^N e^{Bu^2+C|u|} du < \infty$ for all $N, B, C \geq 0$ (it decays like $\exp(-\pi e^{2|u|})$), and this is all that is used.

17.8.1 Definitions

Let H_n be the Hermite polynomial of Section 21.1 (physicists' normalisation, the one that gives H_{2d} in the Hermite identity there), and

$$h_n(x) = \pi^{-1/4}(2^n n!)^{-1/2} H_n(x) e^{-x^2/2},$$

the orthonormal Hermite functions. They satisfy $h_n'' + (2n+1-x^2)h_n = 0$, $\int h_n(x) e^{i\xi x} dx = \sqrt{2\pi}\, i^n h_n(\xi)$, and $h_{2m}(0) = (-1)^m \pi^{-1/4}\sqrt{(2m)!}/(2^m m!)$. For a scale $a > 0$ put

$$c_n(a) = \sqrt{a} \int_{\mathbb{R}} \Phi(u) h_n(au)\, du, \qquad t_m = t_m(a) = a\sqrt{4m+1},$$
$$b_m(a) = \sqrt{2\pi}\,(-1)^m c_{2m}(a) = \int_{\mathbb{R}} \Xi(t)\, \frac{h_{2m}(t/a)}{\sqrt{a}}\, dt.$$

The second form of b_m follows from the Fourier relation with $t = ax$. Since Φ is even, $c_n(a) = 0$ for odd n, and $b_m(a)$ is the coefficient of the orthonormal function $h_{2m}(t/a)/\sqrt{a}$ in the expansion of Ξ. The number t_m is the turning point of $h_{2m}(t/a)$: the equation $h_{2m}'' + (4m+1-x^2)h_{2m} = 0$ is oscillatory for $|x| < \sqrt{4m+1}$, that is for $|t| < t_m$. (In Section 21.1 the same polynomials play another role: the Jensen polynomial of degree d is the expectation of a Hermite polynomial of the Brownian variable V.)

With H_t the de Bruijn Newman function of Section 4.1, $8H_t(2\tau) = \int_{\mathbb{R}} \Phi(v) e^{tv^2/4} \cos(\tau v)\, dv$, so the Gaussian smoothing

$$G_a(\tau) = \int_{\mathbb{R}} \Phi(u) e^{-a^2u^2/2} \cos(\tau u)\, du = 8H_{-2a^2}(2\tau)$$
$$= \frac{1}{a\sqrt{2\pi}} \int_{\mathbb{R}} \Xi(s) e^{-(\tau-s)^2/(2a^2)} ds$$

is Ξ smoothed by a Gaussian of variance a^2 ($e^{-a^2u^2/2}$ is the characteristic function of $N(0, a^2)$), the backward heat flow of the kernel for the time $t = -2a^2$.

Theorem 17.8.1 (Hermite coefficients and the heat flow). *For every $a > 0$,*

$$\sum_{m \ge 0} b_m(a) \frac{z^{2m}}{\sqrt{(2m)!}} = \sqrt{2\pi}\, \pi^{-1/4} \sqrt{a}\; e^{z^2/2}\, G_a\big(\sqrt{2}\, az\big),$$

and with $\nu_k(a) = \int \Phi(u) e^{-a^2u^2/2} u^{2k} du > 0$,

$$b_m(a) = \sqrt{2\pi}\, \pi^{-1/4} \sqrt{a}\, \sqrt{(2m)!} \sum_{k=0}^{m} \frac{(-1)^k 2^k a^{2k} \nu_k(a)}{(2k)!\, 2^{m-k} (m-k)!}.$$

Proof. The generating function

$$\sum_n h_n(x) w^n/\sqrt{n!} = \pi^{-1/4} e^{-x^2/2+\sqrt{2}xw-w^2/2}$$

integrated against $\sqrt{a}\,\Phi(u)$ at $x = au$ gives

$$\sum_n c_n(a) w^n/\sqrt{n!} = \pi^{-1/4}\sqrt{a}\, e^{-w^2/2} \int \Phi(u) e^{-a^2u^2/2} e^{\sqrt{2}auw} du.$$

On bounded sets of w the integrand and its derivatives are bounded by a constant times $|\Phi(u)|e^{-a^2u^2/2+C|u|}$, which is integrable, so the coefficients may be extracted termwise. Put $w = iz$: the odd terms vanish by evenness, $(iz)^{2m} = (-1)^m z^{2m}$ supplies the sign in b_m, and $e^{i\sqrt{2}azu}$ integrates to $G_a(\sqrt{2}az)$ by evenness. The moment formula follows by multiplying the series

$$G_a(\sqrt{2}az) = \sum_k (-1)^k 2^k a^{2k} \nu_k z^{2k}/(2k)!$$

by

$$e^{z^2/2} = \sum_j z^{2j}/(2^j j!).$$

Each $\nu_k > 0$ because $\Phi > 0$. □

The moment formula in 60 digit arithmetic agrees with the quadrature value of b_m for $m \le 15$ at $a = 3$ and $a = 1.5$ with relative differences 6.7×10^{-55} and 3.1×10^{-59}, and the two quadratures (steps 1/64 and 1/128) differ by 3.6×10^{-57} and 1.5×10^{-60} (computed). Also $b_3(1.5) = 0.320600426659651$ from Φ and from the integral of Ξ (computed).

Remark 17.8.2. By Rodgers and Tao [250], $\Lambda \ge 0$, so H_t has nonreal zeros for every $t < 0$, and therefore G_a is not real rooted for any $a > 0$, since $t = -2a^2 < 0$ (proved). Since $e^{z^2/2}$ never vanishes, $e^{z^2/2}G_a(\sqrt{2}az)$ has exactly the zeros of $G_a(\sqrt{2}az)$. The note's statement that the coefficients of this product nevertheless recover the real zeros of Ξ is not about zeros of the product; what is true is that the signs of $b_m(a)$ follow those of Ξ at the points t_m away from the zeros as $a \to 0$ (Theorem 17.9.1).

17.9 The turning point asymptotic

The note proposes, as Proposition 1, that $b_m(a) = \sqrt{2a}\, m^{-1/4}\, \Xi(a\sqrt{4m+1})(1 + O(a^2))$ as $a \to 0$ with $t = a\sqrt{4m+1}$ fixed, and derives it by a saddle point argument on the circle $|z| = \sqrt{2m}$ in Theorem 17.8.1. The leading constant is right: the saddles are at $\theta = 0, \pi$ with the exponent $m - 2m\theta^2 + O(m\theta^3)$, the Gaussian in θ has variance $1/(4m)$, the imaginary shift of the argument of G_a has variance

exactly a^2, and $\mathbb{E}G_a(t+iaY)=\Xi(t)$ for $Y\sim N(0,1)$ by Fubini, so the average of G_a over the shifts undoes the smoothing; Stirling's formula and the Gaussian integral give $\sqrt{2a}\,m^{-1/4}$. But the error $O(a^2)$ is not derived (only the shift of the sampling point from $2a\sqrt{m}$ to $a\sqrt{4m+1}$ is), the real displacement $-a\sqrt{m}\,\theta^2$ and the cubic phase are not estimated, and the contribution off the saddles is not bounded. The statement as written is false at the zeros of Ξ (Proposition 17.9.2). The theorem below, proved from the Hermite differential equation, replaces it with an additive estimate, uniform in m, with the exact amplitude.

Put

$$A_m(a)=\sqrt{2\pi a}\,(-1)^m h_{2m}(0)=\sqrt{2}\,\pi^{1/4}\sqrt{a}\,\frac{\sqrt{(2m)!}}{2^m m!}>0,$$
$$M_k(a_0)=\int_{\mathbb{R}}|\Phi(u)||u|^k e^{a_0^2u^2/2}du<\infty.$$

THEOREM 17.9.1 (Additive asymptotic). *For every $a_0>0$, $0<a\le a_0$ and integer $m\ge 0$,*

$$\Big|\frac{b_m(a)}{A_m(a)}-\Xi(t_m)\Big|\le\frac{M_4(a_0)}{12}\,a^4,$$
$$\frac{b_m(a)}{A_m(a)}=\Xi(t_m)+a^4Q(t_m)+R_m(a),\qquad |R_m(a)|\le\frac{M_8(a_0)}{672}\,a^8,$$

with $t_m=a\sqrt{4m+1}$ and the even entire function

$$Q(t)=\frac{\Xi'''(t)}{6t}-\frac{\Xi''(t)}{4t^2}+\frac{\Xi'(t)}{4t^3}\quad(t\neq 0),$$
$$Q(0)=\frac{\Xi^{(4)}(0)}{12}=\frac{1}{12}\int_{\mathbb{R}}\Phi(u)u^4du.$$

Proof. Put $y_m(u)=h_{2m}(au)/h_{2m}(0)$. Then $b_m/A_m=\int\Phi(u)y_m(u)du$, and the Hermite equation gives $y_m''+(t_m^2-a^4u^2)y_m=0$, $y_m(0)=1$, $y_m'(0)=0$. For real t and complex z let $y_{z,t}$ be the even solution of $y''+(t^2-zu^2)y=0$ with these initial values. For $u\ge 0$, $y_{z,t}=\cos tu+z\,L_ty_{z,t}$ with $(L_tf)(u)=\int_0^u\frac{\sin t(u-v)}{t}v^2f(v)dv$ (the kernel is $u-v$ at $t=0$). Since $|\sin(t(u-v))/t|\le u-v$, the Volterra series is majorised by the solution of $w=1+|z|Kw$ with $(Kf)(u)=\int_0^u(u-v)v^2f(v)dv$. The function $v=e^{\sqrt{|z|}u^2/2}$ satisfies $v''=(\sqrt{|z|}+|z|u^2)v\ge|z|u^2v$, so $v\ge 1+|z|Kv$, hence $w\le v$ and

$$|y_{z,t}(u)|\le e^{\sqrt{|z|}u^2/2}\quad(u\in\mathbb{R},\ t\in\mathbb{R}).$$

Iterating the integral equation once and twice, and using $K1=u^4/12$ and $K^21=u^8/672$, gives $|y_{z,t}-\cos tu|\le\frac{|z|u^4}{12}e^{\sqrt{|z|}u^2/2}$ and $|y_{z,t}-\cos tu-zL_t\cos(t\cdot)|\le\frac{|z|^2u^8}{672}e^{\sqrt{|z|}u^2/2}$, with constants independent of t. Take $z=a^4$, $t=t_m$, $a\le a_0$ and integrate against Φ: $\int\Phi\cos(t_mu)du=\Xi(t_m)$, and the first inequality follows.

The function $f_t = (\frac{u^3}{6t} - \frac{u}{4t^3})\sin tu + \frac{u^2}{4t^2}\cos tu$ solves $f'' + t^2 f = u^2 \cos tu$ with $f(0) = f'(0) = 0$, so $f_t = L_t \cos(t\,\cdot)$, and differentiating the Fourier integral of Ξ gives $\int \Phi f_t = Q(t)$. At $t = 0$, $f_0 = u^4/12$. □

For the amplitude, Stirling's expansion gives $A_m(a) = \sqrt{2a}\, m^{-1/4}\big(1 - \frac{1}{16m} + \frac{1}{512m^2} + O(m^{-3})\big)$. Hence for $0 < \varepsilon \le t = t_m \le T$ (so that $m \to \infty$ uniformly as $a \to 0$)

$$\frac{b_m(a)}{\sqrt{2a}\, m^{-1/4}} = \Big(1 - \frac{1}{16m} + O(m^{-2})\Big)\big(\Xi(t) + O(a^4)\big), \tag{17.9.1}$$

an additive statement, uniform even if the interval contains zeros. At a fixed $t > 0$ with $\Xi(t) \neq 0$ one may divide, and then $b_m(a) = \sqrt{2a}\, m^{-1/4}\, \Xi(t)(1+O_t(a^2))$, which is the formula of the note; the leading correction is $-1/(16m) = -a^2/(4t^2) + O(a^4)$ and comes from the amplitude and not from the saddle. A uniform relative error holds on a compact set $K \subset (0,\infty)$ with $\inf_K |\Xi| > 0$. With the sampling point as the variable, $n = m + \frac14 = t^2/(4a^2)$, the amplitude is $A_m = \frac{2a}{\sqrt{t}}\big(1 - \frac{a^4}{8t^4} + O(a^8)\big)$ and

$$b_m(a) = \frac{2a}{\sqrt{t}}\Big\{\Xi(t) + a^4\Big[Q(t) - \frac{\Xi(t)}{8t^4}\Big] + O_{\varepsilon,T}(a^8)\Big\},$$

with no term of order a^2 or a^6; this form is not uniform near $t = 0$, where Theorem 17.9.1 with the exact A_m is the right one. At $m = 0$ one has $t_0 = a$ and $b_0(a) = A_0(a)\int \Phi(u) e^{-a^2u^2/2} du > 0$, and the formula with $m^{-1/4}$ is undefined; at fixed $m \ge 1$, $b_m/A_m \to \Xi(0) > 0$ as $a \to 0$. Even for $m \ge 1$ the relative formula with the prefactor $\sqrt{2a}$ does not extend to $t = 0$: for $m = 1$, $b_1(a)/(\sqrt{2a}\,\Xi(a\sqrt5)) \to \pi^{1/4}/\sqrt2 = 0.9414$ (proved, from $A_1 = \pi^{1/4}\sqrt{a}$).

Proposition 17.9.2. *At every positive real zero γ of Ξ, of any multiplicity, $b_m(a_m) \neq 0$ for all large m, where $a_m = \gamma/\sqrt{4m+1}$. If $Q(\gamma) \neq 0$ then $b_m(a_m) = \frac{2Q(\gamma)}{\sqrt{\gamma}} a_m^5 + O(a_m^9)$. The relative formula of the note therefore fails at the zeros of Ξ, where it would give $b_m(a_m) = 0$.*

Proof. For fixed real t put $F_t(z) = \int \Phi(u) y_{z,t}(u)\, du$. By the bound of the proof of Theorem 17.9.1 it is entire in z, and $F_t(0) = \Xi(t)$. For $t > 0$ and $z = t^4$ the solution is $y = e^{-t^2u^2/2}$, so $F_t(t^4) = \int \Phi e^{-t^2u^2/2} du > 0$. At $t = \gamma$, $F_\gamma(0) = 0$ but $F_\gamma \not\equiv 0$, so $F_\gamma(z) \neq 0$ for small $z \neq 0$; and $b_m(a_m) = A_m(a_m) F_\gamma(a_m^4)$. If $Q(\gamma) \neq 0$ the Taylor expansion at $z = 0$ gives the leading term, with $A_m(a_m) \sim 2a_m/\sqrt{\gamma}$. □

17.10 The first negative coefficient

Let $m^*(a) = \min\{m \ge 0 : b_m(a) < 0\}$, and let $\gamma_1 = 14.134725\ldots$ be the first positive zero of Ξ. It is simple (the first 1.5×10^9 zeros are simple and on the line [271]), so $\Xi > 0$ on $[0, \gamma_1)$ and $\Xi < 0$ just to the right of γ_1.

COROLLARY 17.10.1. *For every $0 < \delta < \gamma_1$ and a small, $b_m(a) > 0$ for all m with $t_m \le \gamma_1 - \delta$, the index $m^*(a)$ exists, and $\lim_{a\to 0} 4a^2 m^*(a) = \gamma_1^2$. If γ_a is the unique zero near γ_1 of $t \mapsto \int \Phi(u) y_{a^4,t}(u)du$, then $\gamma_a = \gamma_1 - a^4 Q(\gamma_1)/\Xi'(\gamma_1) + O(a^8)$ and $m^*(a) = \lfloor (\gamma_a^2/a^2 - 1)/4 \rfloor + 1$ for small a.*

Proof. Let $c_\delta = \min_{[0,\gamma_1-\delta]} \Xi > 0$. By Theorem 17.9.1, $b_m/A_m \ge c_\delta - Ca^4 > 0$ when $t_m \le \gamma_1 - \delta$ and a is small, so $m^*(a)$, where it exists, has $t_{m^*} > \gamma_1 - \delta$ and $4a^2 m^* = t_{m^*}^2 - a^2 > (\gamma_1 - \delta)^2 - a^2$, which gives $\liminf 4a^2 m^* \ge \gamma_1^2$. For the upper bound pick $s > \gamma_1$ with $\Xi(s) < 0$ and $m(a) = \lfloor (s^2/a^2 - 1)/4 \rfloor$; then $t_{m(a)} \to s$, the additive error tends to zero, so $b_{m(a)} < 0$ and $\limsup 4a^2 m^* \le s^2$; let $s \downarrow \gamma_1$. For the exact formula, $F(z,t) = \int \Phi y_{z,t}$ is jointly analytic near $(0, \gamma_1)$ (the kernel bound $|\sin(tw)/t| \le we^{|\operatorname{Im} t| w}$ gives an integrable majorant), $\partial_z F(0,t) = Q(t)$ and $\partial_t F(0,t) = \Xi'(t) \ne 0$, so the analytic implicit function theorem gives a zero $\rho(z) = \gamma_1 - zQ(\gamma_1)/\Xi'(\gamma_1) + O(z^2)$ and $\gamma_a = \rho(a^4)$. For small a the function $F(a^4, \cdot)$ is positive up to γ_a and negative just after it, and the grid has spacing $O(a^2)$ there, so the first negative index is the first grid point strictly after γ_a. □

The note's hypothesis, an error uniform on $[0, \gamma_1 - \delta]$, gives only the lower bound: the limit needs negativity of b_m just above γ_1, so a uniform estimate on an interval reaching beyond γ_1.

The note states the sharper law $m^*(a) = \lceil (\gamma_1^2/a^2 - 1)/4 \rceil$ for $a \le 1.5$. Table 17.13 (computed, second run at step $h/2$ the same) agrees with it at four scales, but it cannot hold for every a. The threshold inside the rounding is $(\gamma_a^2/a^2 - 1)/4$, which differs from $(\gamma_1^2/a^2 - 1)/4$ by $-\gamma_1 Q(\gamma_1) a^2/(2\Xi'(\gamma_1)) + O(a^6)$, and $\gamma_a \ne \gamma_1$ for small a (otherwise F_{γ_1} would vanish identically, against $F_{\gamma_1}(\gamma_1^4) > 0$), so $\gamma_a - \gamma_1 = c\, a^{4p} + o(a^{4p})$ with $c \ne 0$ and $p \ge 1$. If $c > 0$, at $a = \gamma_1/\sqrt{4m+1}$ the ceiling of the note gives m and the first negative coefficient is $m+1$ for large m; if $c < 0$ a slightly smaller a gives the opposite. Such scales exist arbitrarily close to 0, because the displacement of the root is much smaller than the spacing $O(a^2)$ of the grid. At $a = 2$ the formula gives 12.24 and the first negative index is 12, so the ceiling formula, which would give 13, fails there, although $a = 2$ is not in the asymptotic regime; and the reading of γ_1 to about 10^{-2} from $m^*(1/2)$ is only to 0.035 (the value $m^*(\frac{1}{2}) = 200$ places γ_1 in $(14.116, 14.151]$).

a	0.5	0.75	1	1.5	2
$(\gamma_1^2/a^2 - 1)/4$	199.54	88.55	49.70	21.95	12.24
first negative m	200	89	50	22	12

Table 17.13: The first negative Hermite coefficient of Ξ (computed).

17.11 Counting zeros from Φ alone

17.11.1 The ratio table

Table 17.14 gives the ratio $b_m(a)/\big(\sqrt{2a}\,m^{-1/4}\,\Xi(a\sqrt{4m+1})\big)$ at three scales (computed; the note does not say which m, and the indices $m = 9, 36, 81, 144, 182$ at $a = 0.5$, $4, 16, 36, 64, 81$ at $a = 0.75$ and $2, 9, 20, 36, 45$ at $a = 1$ reproduce every printed digit; the sampling points are $t_m = 3.04, 6.02, 9.01, 12.01, 13.50$, $3.09, 6.05, 9.03, 12.02, 13.52$ and $3.00, 6.08, 9.00, 12.04, 13.45$). The last column is the error at $t \approx 9$ divided by a^2, which is 0.0033, 0.0036, 0.0041 (the note printed 0.0032 for the first). The table is explained by the amplitude: the printed values are close to $A_m/(\sqrt{2a}\,m^{-1/4}) = 1 - \frac{1}{16m} + \cdots$ (for example $1 - 1/144 = 0.9931$ at $a = 0.5$, $m = 9$), and the term $a^4Q(t)/\Xi(t)$ of Theorem 17.9.1 is the rest. The coefficients range over four orders of magnitude. At larger a and greater height the ratio is no longer near 1; the note says it is between 1.5 and 9 at $a = 4$ for $100 < t < 200$ and that no H_λ with $\lambda \in [-8, 3]$ fits better, which could not be reproduced (the ratio is not defined near the zeros of Ξ, and it varies in sign over the window).

a	$t \approx 3$	$t \approx 6$	$t \approx 9$	$t \approx 12$	$t \approx 13.5$	error at $t \approx 9$ over a^2
0.50	0.9931	0.9983	0.9992	0.9993	0.9986	0.0033
0.75	0.9847	0.9961	0.9980	0.9976	0.9937	0.0036
1.00	0.9699	0.9931	0.9959	0.9938	0.9830	0.0041

Table 17.14: The ratio of $b_m(a)$ to $\sqrt{2a}\,m^{-1/4}\,\Xi(a\sqrt{4m+1})$ (computed).

17.11.2 Method and the count

The integrand $\Phi(u)h_{2m}(au)$ is not entire: it is analytic in $|\operatorname{Im} u| < \pi/4$ and the lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary ((2.1.3) and the text after it), but it decays like $\exp(-\pi e^{2|u|})$ and the trapezoid rule converges geometrically. For a function analytic in a strip $|\operatorname{Im} u| \le d$ with $M_d = \int(|f(x+id)| + |f(x-id)|)dx < \infty$ the error of the infinite trapezoid rule of step h is at most $M_d/(e^{2\pi d/h} - 1)$ (proved, by shifting the Fourier integral to the edges of the strip and summing two geometric series), with $d < \pi/4$; the constants depend on a and m, so the note's statement that the transform of the integrand is negligible beyond $2T$ and that $h = 2\pi/(3T + 100)$ suffices is not proved. The count was recomputed with this step (the note has about 655 nodes at $T = 500$), the Hermite functions by the three term recurrence for all orders at once, and a working precision of the order of $0.34\,T + 40$ digits (the coefficients near height T have size $e^{-\pi T/4}$), and the positions of the sign changes were compared with a run at half the step and more digits (computed). The note's column of digits does not follow its own rule and is omitted.

T	$N(T)$	coefficients	sign changes of b_m by scale a
100	29	625 at $a=2$	$a=2,3,4,5$: 29, 29, 29, 29
200	79	625 at $a=4$	$a=4,5$: 79, 79; $a=6,7$: 78, 78; $a=8$: 77
500	269	1736 at $a=6$	$a=6$: 269; 7: 268; 8: 267; 9: 265; 10: 263; 14: 245; 18: 192
1000	649	6945 at $a=6$	$a=5,6$: 648, 648 (edge lag)
1004	652	7000 at $a=6$	$a=6$: 651

Table 17.15: Number of sign changes of $b_m(a)$ for $t_m(a) \le T$ against the zero count $N(T)$ (computed). The coefficient counts of the note (627, 1738, 6947, 7002) were two too large.

Remark 17.11.1 (Observation). For a up to about 5 to 6 the number of sign changes of $(b_m(a))_{t_m \le T}$ equals $N(T)$ at $T = 100, 200, 500$, and at $T = 1000$ it equals $N(T)$ except for the last zero within about one unit of the top of the computed range. In every run above the threshold the count was smaller than $N(T)$.

This is an observation (computed, floating point; the positions agree in the second run at $T = 100$, $T = 200$ and $T = 500$, $a = 6$, the other rows being single runs), not a theorem: neither a universal one sided inequality nor an association between sign changes of b_m and zeros of zeta is proved. The zero counts are those of zeta ($N(T) = 29, 79, 269, 649, 652$, mpmath). The equality is a statement about counts and not about signs: at $T = 200$, $a = 4$ the sign of b_m agrees with the sign of $\Xi(t_m)$ at 81.8 percent of the indices ($c = 1$; 81.9 percent at $c = 0$), and sampling at $a\sqrt{4m + c}$ for $c \in [0, 3]$ moves this between 79.7 and 81.9 percent (computed). The sign changes are displaced by a few indices relative to the zeros.

17.11.3 The edge lag at $T = 1000$

At $T = 1000$ both $a = 5$ and $a = 6$ give 648 sign changes against $N(1000) = 649$. The last zero is $\gamma_{649} = 999.79$, 0.3 below the top of the range. Extending the computation at $a = 6$ to $t \le 1004$ the next sign changes occur between consecutive t_m at $(1000.85, 1000.92)$, $(1002.07, 1002.14)$ and $(1003.36, 1003.44)$, for the zeros 999.79, 1001.35 and 1002.40 (computed), and $\gamma_{652} = 1003.27$ again falls inside the lag at the new edge; the note gives the sign changes at 1000.89, 1002.11, 1003.40, which are inside these intervals. The sign changes trail the zeros by about one unit at this height, and the count must be read on $[0, T - \delta]$ with a lag $\delta(a, T)$. The note adds that at low height the turning points sit slightly below the zeros and at $T = 1000$ slightly above, so that δ changes sign with height; this is an observation that Theorem 17.9.1 does not yet explain.

Proposition 17.11.2 (What a verified count proves). *Let $\widetilde{b}_m/A_m$ be computed values of the normalised coefficients with rigorous errors e_m, and let E_m be a rigorous bound for $|b_m/A_m - \Xi(t_m)|$ (for example $M_4(a_0)a^4/12$ from Theorem 17.9.1).*

If $|\widetilde{b}_m/A_m| > e_m + E_m$ *then* $\operatorname{sgn}\widetilde{b}_m = \operatorname{sgn}\Xi(t_m)$. *Suppose there are* S *sign changes at pairs of adjacent sampling points below* T, *all with this property. Then* Ξ *has* S *zeros in disjoint intervals, and if an independent rigorous method gives* $N(T) \le S$, *where* $N(T)$ *is the number of zeros* t *of* Ξ *with* $0 < \operatorname{Re} t < T$, *counted with multiplicity (Turing's method, Section 2.2.1), all those zeros are real and simple.*

Proof. The triangle inequality transfers the signs, the intermediate value theorem gives one zero of Ξ in each of the S disjoint intervals between adjacent sampling points, and the upper bound, counted with multiplicity, leaves no room for another zero or a multiple zero. □

A sign change of b_m certifies nothing by itself, since b_m is an integral of Ξ, not a value. The chain of the note, a rigorous version of the asymptotic, a lower bound for $|\Xi|$ at the sampling points and Turing's method, is Proposition 17.11.2 with Theorem 17.9.1 supplying the first step with explicit constants; the sampling values that are zero or uncertain must be enclosed separately. High precision and agreement after halving the step are diagnostics, not these inequalities, so the counts of Table 17.15 are computed, not verified.

17.12 The best scale and the run lengths

Two quantities compete. The grid $t_m(a)$ has spacing $dt_m/dm = 2a^2/t_m$ (exactly $4a^2/(t_{m+1}+t_m)$ between neighbours), so a small a resolves close zeros; the number of coefficients is $M = T^2/(4a^2) + 1$, so a large a is cheap. The note says that the binding constraint is not resolution, because at the largest successful scale the grid spacing is only 0.12 to 0.17 of the mean gap $2\pi/\log(T/2\pi)$. The recomputed ratios of $2a^2/T$ to the mean gap are 0.22 ($T = 100$, $a = 5$), 0.14 ($T = 200$, $a = 5$), 0.10 ($T = 500$, $a = 6$) and 0.06 ($T = 1000$, $a = 6$) (computed), so the note's range is not reproduced; and the inference does not follow in any case, since two zeros arbitrarily close together can fit between two sampling points whatever the mean gap (the function $(t-c)^2 - \varepsilon^2$ is positive at both ends of a grid interval for small ε). The tables do not separate a missed grid crossing from a distortion of the asymptotic; what is observed is that the counts fall short above $a \approx 5$ to 6 and that the usable threshold rises slowly with T, from at least 5 at $T = 100$ to between 6 and 7 at $T = 500$. With a near the threshold there are about 6.5 coefficients per zero at $T = 500$ and 10.7 at $T = 1000$ (1736/269 and 6945/649), each costing $O(T)$ nodes at $0.34\,T$ digits. As a verification method this is far slower than Odlyzko and Schönhage [221]; its value is structural, since every quantity is an explicit integral of the positive kernel Φ.

The note also reads the sign pattern through Good and Gaskins [86]: with $\Xi(t) = -\frac{1}{2}(t^2+\frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{it}{2})|Z(t)$ for the Hardy function Z and the Riemann Siegel phase $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$, the sign of b_m would rotate by $\theta_m = \vartheta'(t_m)\,dt_m/dm = a^2\log(t_m/2\pi)/t_m$ per step in m, an effective complex ratio, with

mean sign run $\pi/\theta_m = \pi t/(a^2 \log(t/2\pi))$. The rotation law is not derived: the real function Z is not determined by its phase derivative, no recurrence or complex ratio for b_m has been derived, and Theorem 17.9.1 is an asymptotic at bounded height as $a \to 0$ and does not apply at fixed $a = 4$ and increasing height (the attribution to Good and Gaskins is the note's and was not checked). What can be said is an averaged statement.

Proposition 17.12.1 (Average run length). *Let $N(T)$ count the zeros t of Ξ with $0 < \operatorname{Re} t < T$, with multiplicity. Assume the zeros t of Ξ with $T \le \operatorname{Re} t \le T + H$ are real and simple and that the sign changes of $(b_m(a))$ with $t_m \in [T, T + H]$ correspond one to one to them. Then the mean number of coefficient steps per sign change is the number $TH/(2a^2) + O(H^2/a^2 + 1)$ of grid points in the window divided by $N(T + H) - N(T) = \frac{H}{2\pi} \log \frac{T}{2\pi} + O(H^2/T + \log T)$, and for fixed a, $H \to \infty$ and $H = o(T)$ the ratio is asymptotic to $\pi T/(a^2 \log(T/2\pi))$.*

Table 17.16 tests this conditional average, with the predicted values at the window midpoints $60, 100, 140, 180$ and $a = 4$ (arithmetic). The note's observed values 5.29, 7.19, 8.89, 10.50 were not reproduced, because "mean run" is not defined in the note (edge blocks, which window); with the number of indices in the window divided by the number of sign blocks at $T = 200$, $a = 4$, the observed values of the table (computed) are consistent with the average and do not support the rotation law.

window in t ($a = 4$)	40 to 80	80 to 120	120 to 160	160 to 200
predicted $\pi t/(a^2 \log(t/2\pi))$	5.22	7.10	8.86	10.53
observed mean run (indices over sign blocks)	5.00	6.94	8.33	10.23

Table 17.16: Mean sign runs of $b_m(4)$ (computed).

17.13 The Thorin shadow

The note concludes that, by the turning point asymptotic, Hermite coefficients of a real even function sample the function at the points $a\sqrt{4m+1}$, so that sign definiteness of the coefficients is asymptotically equivalent to positivity of the function on the real line, and that a Thorin or GGC positivity statement for the reciprocal kernel has a Hermite form that is no stronger and no weaker. This does not follow and is not true as stated. Let $f(t) = (\sin t/t)^2$, which is even, entire and nonnegative on the real axis, with $f(t) = \frac{1}{4}\int_{-2}^{2}(2 - |u|)e^{itu}du$ and a nonnegative compactly supported kernel. The proof of Theorem 17.9.1 applies to it. At $t = \pi$ one has $f'(\pi) = 0$, $f''(\pi) = 2/\pi^2$, $f'''(\pi) = -12/\pi^3$, so $Q_f(\pi) = -5/(2\pi^4) < 0$, and along $a = \pi/\sqrt{4m+1}$ the normalised coefficient is $a^4 Q_f(\pi) + O(a^8) < 0$ for all large m: a nonnegative function can have negative small scale Hermite coefficients at every large m along that sequence (proved). The correct statements are two.

For a kernel of the class above, if Ξ (or any function of the class) is strictly positive on a compact interval, then every coefficient with t_m in the interval is positive once a is small (the positive minimum and the additive error, with $A_m > 0$). Conversely, if all these coefficients are nonnegative for every small a, the function is nonnegative on the interval (choose grid points converging to any t and pass to the limit in the normalised formula; at $t = 0$ use $m = 0$). Neither characterises the Thorin or GGC property of a reciprocal kernel. The Hermite coefficients add computability; no result saying more is known here.

17.14 Notes and open problems

Nothing computed here is certified in ball arithmetic. The Lehmer pair near $t = 7005$ that the note mentions consists of the zeros 6709 and 6710 at 7005.062866 and 7005.100565, a gap of 0.0377 (computed).

Problem 17.18. Evaluate the constants of Theorem 17.9.1 explicitly (the moments $M_4(a_0)$ and $M_8(a_0)$ and a bound for Q) and use Proposition 17.11.2 with Turing's method to count the zeros of Ξ up to a given height from the coefficients alone, including the edge lag $\delta(a, T)$.

Problem 17.19. Explain the distortion of the counts at large a, and find a modified coefficient sequence, perhaps with $e^{z^2/2}$ replaced by another multiplier in Theorem 17.8.1, that keeps the count exact at larger a and so cuts the number $M = T^2/(4a^2)$ of coefficients.

Problem 17.20. Push the count to $T = 2000$ and to the Lehmer pair near $t = 7005$, where a sign change count is the natural place for a close pair to hide.

Problem 17.21. Find a Hermite statement that is equivalent to a Thorin or GGC positivity property of the reciprocal kernel, or show that none exists, given that sign definiteness of the Hermite coefficients is not equivalent to positivity of the function (the example $(\sin t/t)^2$).

Chapter 18

Friable approximants of Ξ after Miyatake and the cosh ladder

Keep in Riemann's theta series only the integers composed of the primes up to P: the result Ξ_P differs from Ξ by an exponentially small amount yet has infinitely many zeros off the real line. This friable approximant is a test case for any computational criterion for the Riemann Hypothesis. We ask how many real zeros Ξ_P shares with Ξ, why no finite set of primes reaches beyond a height of about $4q^2$, with q the first prime left out (a heuristic formula, checked by computation), where the remaining zeros go, and what the $2\sin^2$ lemma and the Thorin condition can and cannot detect. The last sections read an identity of Miyatake, who worked with the primes up to P inside Pólya's symmetrised kernel, probabilistically (Lévy's area, a Pólya Gamma clock, a logistic partition of unity), and place the bottom of the cosh ladder against the quasi Riemann abscissa.

Here $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(t) = \xi(\frac{1}{2}+it)$ (Chapter 1, Section 2.1). Miyatake [119, 118] approximated ξ using only the integers composed of primes $p \le P$, so that the Dirichlet series becomes the finite Euler product $\zeta_P(s) = \prod_{p\le P}(1-p^{-s})^{-1} = \sum_{n\in\mathcal{F}(P)} n^{-s}$, with $\mathcal{F}(P)$ the set of P friable integers. He worked inside Pólya's symmetrised kernel (Section 18.7); we truncate Riemann's own kernel instead and study the friable approximant Ξ_P of Section 18.1. The incomplete gamma form of truncated theta series, used for all computations below, is also the subject of Haglund [94].

The chapter is organised around the deficit $D_P = \Xi_P - \Xi$ and the negative humps of Ξ. Numbers are computed in floating point at the precision stated unless a certified method is named. The results are these.

1. $|\Xi_P - \Xi| \le |t^2 + \frac{1}{4}|\,\delta_P$ in $|\operatorname{Im} t| \le \frac{3}{2}$ with $\delta_P \approx e^{-\pi q^2}/(\pi q^2)$, and $|\Xi_P - \Xi| \le (1+\delta(a_q))c_P$ on $\mathbb{R}$ with $c_P \approx 4\pi q^2 e^{-\pi q^2}$: the first omitted prime sets the accuracy (Theorems 18.2.1 and 18.3.4, proved).

2. The truncated kernel has a kink of exact size $c_P > 0$, so $\Xi_P \to c_P$ on $\mathbb{R}$; no reweighting of the theta series with $\sum_n |w_n| n^{-\sigma} < \infty$ for every $\sigma > 0$ (this

includes the friable sets) has infinitely many real zeros, and the same holds for any set of omitted integers (Theorems 18.2.3 and 18.2.4, proved).

3. $D_P > 0$ on $\mathbb{R}$ and is, with an explicit uniform error, a sum of Lorentzians (Theorems 18.3.1 and 18.3.4, proved).

4. Real zeros of Ξ_P live in negative humps of Ξ, two per surviving hump, and are lost in pairs by collision; the count is twice the number of humps deeper than D_P, at every height below $3 \cdot 10^{12}$ where the humps are short enough (Theorems 18.4.1 and 18.4.2, proved given the verification of RH to that height [232]). For $P = 7$ the computed count is 264.

5. Ξ_P has more zeros than Ξ: heuristically $N_P(X) = N(X) + y_P(X)/4 + O(1)$; argument principle counts for $P = 2, 3$ fall below this, by 4.86 at $X = 600$ for $P = 2$ (Remark 18.6.1, heuristic and computed).

6. The $2\sin^2$ Lévy density of Ξ_2 agrees with that of Ξ to seven digits for $x \leq 0.1$ (computed) and is positive on the computed range although Ξ_2 has zeros off the line; the support of the Thorin measure can detect them (Section 18.9).

7. Multiplying Riemann's kernel by $\cosh(bt)$ gives, unconditionally for $b \geq \frac{1}{2}$ (Lagarias, Theorem 22.6.1), reciprocals that are Laplace transforms of generalised gamma convolutions, and RH is equivalent to this persisting as $b \downarrow 0$ (Section 18.10).

8. Dividing instead by $\cosh(bt)$, as Miyatake does with $b = \frac{9}{2}$, is smoothing by Lévy's stochastic area, equivalently the de Bruijn Newman flow at a Pólya Gamma random negative time; at $b = 2$ the transform changes character, and for $b > 2$ with $C_b \neq 0$ it is never in the Laguerre Pólya class (Sections 18.11 and 18.12).

9. The bottom of the cosh ladder is bounded by the quasi Riemann abscissa, and in paired form (real and imaginary parts of ξ on vertical lines, a Thorin pair) it lies exactly at it (Section 18.14).

18.1 Riemann's representation, the friable approximant and the kink

18.1.1 Riemann's representation and the kink identity

Let $\psi(x) = \sum_{n \geq 1} e^{-\pi n^2 x}$ and $\Theta_3 = 1 + 2\psi$, so that $\Theta_3(1/x) = \sqrt{x}\,\Theta_3(x)$. Riemann's formula

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2} \int_1^\infty \psi(x)\big(x^{s/2-1} + x^{-(1+s)/2}\big)\,dx$$

becomes, with $s = \frac12 + it$ and $x = e^{2u}$,

$$\Xi(t) = \frac12 - 2\left(t^2+\tfrac14\right)\int_0^\infty \psi(e^{2u})e^{u/2}\cos(tu)\,du$$
$$= \left(t^2+\tfrac14\right)\int_0^\infty k(u)\cos(tu)\,du, \tag{18.1.1}$$

where $k(u) = e^{-u/2} - 2e^{u/2}\psi(e^{2u})$; the pole term $\frac12$ is absorbed by $\int_0^\infty e^{-u/2}\cos(tu)\,du = \frac12/(t^2+\frac14)$. Since $k(u) = 2\cosh(u/2) - e^{u/2}\Theta_3(e^{2u})$ and the theta functional equation makes $e^{u/2}\Theta_3(e^{2u})$ even, k extends to an even real analytic function on $\mathbb{R}$. The classical kernel is that of (2.1.3), $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(tu)\,du$ with Φ positive, even and analytic exactly in $|\operatorname{Im} u| < \pi/4$ (Titchmarsh [264, (2.16.1)]). The two kernels are related by

$$\Phi(u) = 2\sum_{n\ge1}\left(2\pi^2n^4e^{9u/2} - 3\pi n^2e^{5u/2}\right)e^{-\pi n^2e^{2u}} = \tfrac12\left(\tfrac14 k(u) - k''(u)\right), \qquad u \ge 0,$$

which follows from two integrations by parts in (18.1.1): since $k'(0) = 0$ (Lemma 18.1.1 below), $\int_0^\infty(\frac14 k - k'')\cos(tu)\,du = (t^2+\frac14)\int_0^\infty k\cos(tu)\,du = \Xi(t)$, and the terms $e^{-u/2}$ cancel in $\frac14 k - k''$.

Lemma 18.1.1 (Kink identity). $\frac12 + \psi(1) + 4\psi'(1) = 0$; *equivalently* $k'(0) = 0$.

Proof. Differentiate $\Theta_3(1/x) = \sqrt{x}\,\Theta_3(x)$ at $x = 1$ to get $-\Theta_3'(1) = \frac12\Theta_3(1) + \Theta_3'(1)$, that is $\Theta_3(1) + 4\Theta_3'(1) = 0$, and insert $\Theta_3 = 1 + 2\psi$. Then $k'(0) = -\frac12 - \psi(1) - 4\psi'(1) = 0$. □

The folded pole part $e^{-|u|/2}$ has a corner at the origin and the theta part has exactly the opposite corner. Their cancellation, the smoothness of k at 0, is equivalent to the evenness of the continuation of Φ and comes from the functional equation; by Theorem 18.2.4 no reweighting of the theta series with $\sum_n |w_n|n^{-\sigma} < \infty$ for every $\sigma > 0$ keeps it. In terms of the maximum M of a standard Brownian excursion, $k(u) = e^{-u/2}\,\mathbb{E}\min\left(\sqrt{2/\pi}\,M, e^u\right)$ for real u, and Lemma 18.1.1 says $\mathbb{P}(M > \sqrt{\pi/2}) = \frac12 - \psi(1)$ (Remark 21.8.2).

18.1.2 The friable approximant

Definition 18.1.2. For a prime P let $\psi_P(x) = \sum_{n\in\mathcal{F}(P)} e^{-\pi n^2 x}$ and

$$\Xi_P(t) = \frac12 - 2\left(t^2+\tfrac14\right)\int_0^\infty \psi_P(e^{2u})e^{u/2}\cos(tu)\,du, \qquad \xi_P(\tfrac12+it) = \Xi_P(t).$$

Let $q = q(P)$ be the least prime above P.

Ξ_P is even, entire of order one, real on $\mathbb{R}$, and $\xi_P(s) = \xi_P(1-s)$ by construction. Since $\int_0^\infty \psi_P(x) x^{s/2-1} dx = \pi^{-s/2}\Gamma(s/2)\zeta_P(s)$ for $\operatorname{Re} s > 0$, Ξ_P is the Euler product truncation of Riemann's exact kernel.

Proposition 18.1.3 (Incomplete gamma form).

$$\xi_P(s) = \frac{1}{2} + \frac{s(s-1)}{2} \sum_{n \in \mathcal{F}(P)} \Big[(\pi n^2)^{-s/2}\, \Gamma\big(\tfrac{s}{2}, \pi n^2\big) + (\pi n^2)^{-(1-s)/2}\, \Gamma\big(\tfrac{1-s}{2}, \pi n^2\big) \Big].$$

Proof. With $x = e^{2u}$, Definition 18.1.2 is Riemann's formula with ψ_P in place of ψ. The series converges absolutely and uniformly for $x \geq 1$, so it can be integrated term by term, and $\int_1^\infty e^{-\pi n^2 x} x^{z-1} dx = (\pi n^2)^{-z}\Gamma(z, \pi n^2)$ for $z = s/2$ and, since $x^{-(1+s)/2} = x^{(1-s)/2-1}$, for $z = (1-s)/2$. □

All computations in this chapter use this form; with all integers it gives $\xi(\frac{1}{2}+it)$ to working precision (computed: relative difference below 10^{-26} for $t \leq 50$ at 40 digits).

18.1.2.1 The literal Euler product and its poles

Symmetrising the finite Euler product directly gives $F_P(s) = E_P(s) + E_P(1-s)$ with $E_P(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta_P(s)$ and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. Since ζ_P has poles at $s = 2\pi i k/\log p$, F_P has poles on $\operatorname{Re} s = 0$ and $\operatorname{Re} s = 1$.

Proposition 18.1.4. *For $0 < \operatorname{Re} s < 1$, $\xi_P(s) = \frac{1}{2} + F_P(s) - \frac{s(s-1)}{2}[\mathcal{I}_P(s) + \mathcal{I}_P(1-s)]$ with $\mathcal{I}_P(s) = \sum_{n \in \mathcal{F}(P)} (\pi n^2)^{-s/2}\gamma(\frac{s}{2}, \pi n^2)$. The poles of F_P cancel against those of $\mathcal{I}_P$, so the theta form is the pole free regularisation of the symmetrised Euler product.*

Proof. For $0 < \operatorname{Re} s < 1$ the Dirichlet series $\sum_{n \in \mathcal{F}(P)} (\pi n^2)^{-s/2}\Gamma(\frac{s}{2}) = \Gamma_{\mathbb{R}}(s)\zeta_P(s)$ and its reflection converge absolutely, and $\Gamma = \gamma + \Gamma(\cdot, \pi n^2)$ termwise in Proposition 18.1.3 gives the identity. The statement about poles concerns the meromorphic continuations: ξ_P is entire, F_P has the poles above, and on $\operatorname{Re} s = 0$ and $\operatorname{Re} s = 1$ the function $\mathcal{I}_P$ is its continuation, not the series, which diverges there. □

The unregularised F_P pays for its poles with spurious zeros near them. For $P = 2$, near the poles $1 + 2\pi i k/\log 2$, the zeros of F_2 include $1.5151 + 9.1951i$, $1.3184 + 18.0602i$, $1.2268 + 27.3286i$, $1.1815 + 45.4005i$, $0.7885 + 72.7840i$ (inside the critical strip, with its mirror at 0.2115) and $1.1200 + 108.8198i$ (computed, residual below 10^{-26}; this is a selection and not a list of all the zeros of F_2).

PROPOSITION 18.1.5. *Let $P \geq 2$ and let $p \leq P$ be prime. For infinitely many integers k, F_P has a zero s_k with $|s_k - 1 - \frac{2\pi i k}{\log p}| = O(|k|^{-1/2})$, and for $P = 2$ every k with $|k|$ large enough qualifies. So $\operatorname{Re} s_k \to 1$, and the zeros $1 - \bar{s}_k$ have real parts tending to 0 [137].*

Sketch. At $\sigma_k = 1 + \frac{2\pi i k}{\log p}$ the term $E_P(1-s)$ has a simple pole, with residue R_k, while $E_P(s)$ is analytic and nonzero, so by Rouché's theorem F_P has a zero within $2|R_k/E_P(\sigma_k)|$ of $\sigma_k - R_k/E_P(\sigma_k)$ once this ratio is small. The factors $s(s-1)$ in R_k and in $E_P(\sigma_k)$ have equal moduli, Stirling's formula gives $|\Gamma_{\mathbb{R}}(1-\sigma_k)/\Gamma_{\mathbb{R}}(\sigma_k)| \asymp |k|^{-1/2}$, the Euler factors of $\zeta_P(\sigma_k)$ lie on $\operatorname{Re} s = 1$ and are bounded above and below, and the Euler factors $(1 - q^{-(1-s)})^{-1}$ of the residue, $q \leq P$, $q \neq p$, are bounded when every $q^{2\pi i k/\log p}$ stays away from 1. As 1 and the numbers $\log q/\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem gives infinitely many such k; for $P = 2$ there is no such q. The regular part of $E_P(1-s)$ and the variation of $E_P(s)$ are $O(|k|^{-1/2}\log|k|\,|E_P(\sigma_k)|)$ on the disc, by the same estimates and $\Gamma'/\Gamma(s) = O(\log|s|)$. Then $|R_k/E_P(\sigma_k)| = O(|k|^{-1/2})$. The second statement is the symmetry $F_P(s) = F_P(1-s) = \overline{F_P(\bar{s})}$. □

18.2 Approximation, the kink constant and no finite proxy

THEOREM 18.2.1. *Let $\delta_P = \sum_{m \notin \mathcal{F}(P)} e^{-\pi m^2}/(\pi m^2)$. For $|\operatorname{Im} t| \leq \frac{3}{2}$,*

$$|\Xi_P(t) - \Xi(t)| \leq \left|t^2 + \tfrac{1}{4}\right| \delta_P, \qquad \delta_P \leq \frac{e^{-\pi q^2}}{\pi q^2 (1 - e^{-2\pi q})}.$$

Proof. Every $m < q$ is P friable, so every excluded m satisfies $m \geq q$, and $\Xi_P - \Xi = 2(t^2 + \frac{1}{4}) \sum_{m \notin \mathcal{F}(P)} \int_0^\infty e^{-\pi m^2 e^{2u}} e^{u/2} \cos(tu)\, du$. For $t = x + iy$ with $|y| \leq \frac{3}{2}$ we have $|\cos(tu)| e^{u/2} \leq e^{3u/2} e^{u/2} = e^{2u}$, and $\int_0^\infty e^{-\alpha e^{2u}} e^{2u} du = e^{-\alpha}/(2\alpha)$; with $\alpha = \pi m^2$ and the factor 2 this gives the first bound. For the second, $m^2 \geq q^2 + 2q(m-q)$ gives $e^{-\pi m^2}/(\pi m^2) \leq e^{-\pi q^2} e^{-2\pi q(m-q)}/(\pi q^2)$, and the geometric series in $m - q$ is summed. □

Computed values: $\delta_P = 1.85875 \cdot 10^{-14}$, $9.8969 \cdot 10^{-37}$, $9.0827 \cdot 10^{-70}$, $2.1406 \cdot 10^{-168}$ for $P = 2, 3, 5, 7$; on a grid of $|\operatorname{Im} t| \leq \frac{3}{2}$ the ratio $|\Xi_P - \Xi|/(|t^2 + \frac{1}{4}|\delta_P)$ is at most 0.9756, 0.9907, 0.9952, 0.9980, attained at $t = -\frac{3}{2}i$.

COROLLARY 18.2.2 (Certified tracking). *If γ is a simple zero of Ξ and $|\Xi| > |t^2 + \frac{1}{4}|\delta_P$ on $|t - \gamma| = r \leq \frac{3}{2}$, then Ξ_P has exactly one zero in the disc, it is real, and $\gamma_P - \gamma = -(\Xi_P - \Xi)(\gamma)/\Xi'(\gamma)\,(1 + O(\eta))$ with η the ratio of $\max|\Xi_P - \Xi|$ on the disc to $\min|\Xi|$ on the circle.*

Proof. On the circle $|\Xi_P - \Xi| \leq |t^2 + \frac{1}{4}|\delta_P < |\Xi|$ by Theorem 18.2.1, so by Rouché's theorem Ξ_P has one zero in the disc. The disc is symmetric about $\mathbb{R}$ and Ξ_P is real on $\mathbb{R}$, so the conjugate of the zero is a zero in the disc, and uniqueness makes

it real. The first order formula is the Taylor expansion of $\Xi_P = \Xi + D_P$ at γ, whose remainder is controlled on the disc by Cauchy's estimates. □

For $P = 2$ the predicted shifts $-D_2(\gamma_k)/\Xi'(\gamma_k)$ are $+2.487 \times 10^{-9}$, -4.027×10^{-7}, $+7.648 \times 10^{-6}$, -5.539×10^{-4}, $+3.695 \times 10^{-3}$ at the first five zeros, and the computed shifts are $+2.487 \times 10^{-9}$, -4.027×10^{-7}, $+7.648 \times 10^{-6}$, -5.536×10^{-4}, $+3.702 \times 10^{-3}$ (computed at 40 digits); the fourth and fifth agree to three digits, with relative difference $2 \cdot 10^{-3}$ at the fifth (second order effects). The shifts grow like $e^{\pi\gamma/4}$, one decade every $4 \ln 10/\pi = 2.93$ units of height.

THEOREM 18.2.3 (Kink constant). *Folding Ξ_P as in* (18.1.1) *with kernel $k_P(u) = e^{-u/2} - 2e^{u/2}\psi_P(e^{2u})$, put $c_P = -k_P'(0)$. Then*

$$c_P = \sum_{m \notin \mathcal{F}(P)} (4\pi m^2 - 1)e^{-\pi m^2} = 4\pi q^2 e^{-\pi q^2}(1 + O(q^{-2})) > 0,$$

and $\Xi_P(t) = c_P + O_P(t^{-2})$ for real $t \to \infty$. Hence Ξ_P has finitely many real zeros.

Proof. Differentiating, $k_P'(0) = -\frac{1}{2} - \psi_P(1) - 4\psi_P'(1)$; subtracting Lemma 18.1.1 leaves $\sum_{m \notin \mathcal{F}(P)}(1 - 4\pi m^2)e^{-\pi m^2}$. Since k_P is smooth on $[0, \infty)$ with every derivative $O(e^{-u/2})$, two integrations by parts give $\int_0^\infty k_P(u)\cos(tu)\,du = -k_P'(0)t^{-2} + O(t^{-4})$, and multiplying by $t^2 + \frac{1}{4}$ gives the limit. Since $c_P > 0$, $\Xi_P > 0$ for large real t, so its real zeros lie in a bounded set. □

Computed: $c_2 = 5.89126 \cdot 10^{-11}$ against the leading term $5.944 \cdot 10^{-11}$, and $c_3 = 2.43420 \cdot 10^{-32}$ against $2.442 \cdot 10^{-32}$. For $P = 2$ the ratio $\Xi_2(t)/c_2$ is 0.450, 0.775, 0.934, 0.983, 0.9956 at $t = 50, 100, 200, 400, 800$, and $t^2(\Xi_2 - c_2) \to -1.65 \cdot 10^{-7}$, the predicted $c_P/4 + k_P'''(0)$. The limit regime is visible only above the breakdown height τ_P^* of Section 18.4: for $P = 3$ the function Ξ_3 is still dominated by Ξ at $t = 50$ and $t = 100$.

THEOREM 18.2.4 (No summable proxy). *For any real weights $w \neq 0$ with $\sum_n |w_n| n^{-\sigma} < \infty$ for every $\sigma > 0$ (this includes the weights $1_{\mathcal{F}(P)}$, for which the sum is $\zeta_P(\sigma)$, and so Ξ_P), put $\psi_w = \sum w_n e^{-\pi n^2 x}$ and*

$$\Xi_w(t) = \frac{1}{2} - 2\left(t^2 + \tfrac{1}{4}\right)\int_0^\infty \psi_w(e^{2u})e^{u/2}\cos(tu)\,du.$$

Then Ξ_w is even, entire of order at most one, and has finitely many real zeros and infinitely many nonreal zeros.

Proof. Let $k_w(u) = e^{-u/2} - 2e^{u/2}\psi_w(e^{2u})$. The series $\psi_w(x)$ converges absolutely and locally uniformly on $\operatorname{Re} x > 0$ with $|\psi_w(x)| = O_\sigma(x^{-\sigma/2})$ as $x \downarrow 0$ for every $\sigma > 0$ (since $\sup_n n^\sigma e^{-\pi n^2 x} = O_\sigma(x^{-\sigma/2})$), and e^{2u} has positive real part for $|\operatorname{Im} u| < \pi/4$, so k_w is analytic in that strip and real analytic on $\mathbb{R}$; all its derivatives

are $O(e^{-u/2})$ at $+\infty$. Repeated integration by parts gives $\int_0^\infty k_w(u)\cos(tu)\,du = \sum_j (-1)^{j+1} k_w^{(2j+1)}(0) t^{-2j-2} + \dots$, and hence $\Xi_w(t) \sim (-1)^{j_0+1} k_w^{(2j_0+1)}(0)\, t^{-2j_0}$ as $t \to \infty$ on $\mathbb{R}$, for the least j_0 with $k_w^{(2j_0+1)}(0) \neq 0$. Such a j_0 exists. Otherwise the odd part of k_w, which is real analytic on $\mathbb{R}$, would vanish to infinite order at 0, hence identically; then $k_w(-u) = k_w(u) \to 0$ as $u \to \infty$, and this gives $\psi_w(x) \sim \frac{1}{2} x^{-1/2}$ as $x \downarrow 0$, impossible since $|\psi_w(x)| = O_\sigma(x^{-\sigma/2})$ with $\sigma < 1$. So Ξ_w has a nonzero power law limit or decay on $\mathbb{R}$ and finitely many real zeros. For $t = x + iy$ and $s = \frac{|y|}{2} + \frac{1}{4}$, the substitution $v = e^{2u}$ and $\sum_n |w_n| (\pi n^2)^{-s} \leq \pi^{-s} \sum_n |w_n| n^{-1/2}$ give $|\Xi_w(t)| \leq \frac{1}{2} + |t^2 + \frac{1}{4}|\, \pi^{-s} \Gamma(s) \sum_n |w_n| n^{-1/2}$, so the order is at most one. Suppose Ξ_w had finitely many zeros. By Hadamard's theorem it would be $P(t)e^{at}$ with P a polynomial; evenness forces $a = 0$, and boundedness on $\mathbb{R}$ (from the asymptotics) makes P constant. The constant is $\Xi_w(i/2) = \frac{1}{2}$, since $t^2 + \frac{1}{4} = 0$ there. Then the cosine transform of $\psi_w(e^{2u})e^{u/2}$ vanishes for all real t, so $\psi_w = 0$ on $[1, \infty)$; if n is the least index with $w_n \neq 0$, multiplying $\sum_{m \geq n} w_m e^{-\pi m^2 x} = 0$ by $e^{\pi n^2 x}$ and letting $x \to \infty$ gives $w_n = 0$, a contradiction. Hence Ξ_w has infinitely many zeros, and so infinitely many nonreal ones. □

The proof of Theorem 18.2.3 never uses friability: for every nonempty set E of omitted integers, the function built from the integers outside E has $c_E = \sum_{m \in E} (4\pi m^2 - 1) e^{-\pi m^2} > 0$, so it tends to c_E on $\mathbb{R}$ and has finitely many real zeros. Hardy's theorem [95] that Ξ has infinitely many real zeros therefore uses every integer in the theta series: the primes hide in the smoothness of the kernel at the origin. (Weights equal to 1 except at one integer are outside the hypothesis of Theorem 18.2.4, since $\sum n^{-\sigma}$ diverges for $\sigma \leq 1$; for omitted integers the statement is the one just given.)

For $E = \{n > N\}$, that is for the weights $w_n = \mathbf{1}_{n \leq N}$ of Theorem 18.2.4, the constant is $c_E = 1.71806 \cdot 10^{-4}$ for $N = 1$ and $c_E = 5.89126 \cdot 10^{-11}$ for $N = 2$, where Ξ_w and c_E agree with Ξ_2 and c_2 to the digits shown, the integers $n \leq 2$ and the 2 friable integers first differing at $n = 4$. In $0 < t \leq 60$ the function Ξ_w has 2 real zeros for $N = 1$, at 14.1659 and 19.5639, and 6 for $N = 2$, at 14.1347, 21.0220, 25.0109, 30.4243, 32.9388 and 37.4954 (as for Ξ_2, $R_2 = 6$ in Section 18.4), against the 13 zeros of Ξ there. Its nonreal zeros with $0 < \operatorname{Re} t \leq 60$ and $0 < \operatorname{Im} t \leq 12$ are $22.915 + 4.666i$ and $27.930 + 9.028i$ for $N = 1$, and $41.190 + 1.462i$, $45.418 + 4.421i$, $49.549 + 7.364i$ and $53.720 + 10.074i$ for $N = 2$ (computed; argument principle). These functions are not the folded approximations of Section 19.17: by the identity $\Xi_P^{\mathrm{M}} = \Xi_P - c_P$ of Section 18.7, that is two integrations by parts with $k_w'(0) = -c_E$, the transform of the folded sum of the first N terms of (2.1.3) is $\Xi_w - c_E$, so that $8H_N^F(z) = \Xi_w(\frac{z}{2}) - c_E$ in the notation there, and its zeros are the solutions of $\Xi_w = c_E$ [135].

18.3 The deficit: positivity and Lorentzian form

For real t, the termwise form of the definition gives

$$D_P(t) = \left(t^2 + \tfrac{1}{4}\right) \sum_{m \notin \mathcal{F}(P)} C_{a_m}(t), \qquad C_a(t) = \int_0^\infty f_a(w) \cos\left(\tfrac{tw}{2}\right) dw, \quad (18.3.1)$$

where $w = 2u$, $f_a(w) = e^{-ae^w} e^{w/4}$ and $a_m = \pi m^2 \geq 9\pi$. Put $b = a - \frac{1}{4}$ and $\mathcal{L}_a(t) = e^{-a} b/(b^2 + t^2/4)$, the cosine transform of $e^{-a} e^{-bw}$.

Theorem 18.3.1. *$D_P(t) > 0$ for all real t, and $D_P > D_{P'}$ when $P < P'$. Hence $\Xi_P > \Xi_{P'} > \Xi$ on $\mathbb{R}$.*

Proof. For $a \geq \frac{3}{2}$, $f_a' = f_a(\frac{1}{4} - ae^w) < 0$ and $f_a'' = f_a(x^2 - \frac{3}{2}x + \frac{1}{16}) > 0$ with $x = ae^w \geq a$, so f_a is integrable, strictly convex and decreasing to 0. With $K(w) = (1 - \cos \lambda w)/\lambda$, $K' = \sin \lambda w$, two integrations by parts give Pólya's formula $\int_0^\infty f_a(w) \cos(\lambda w)\, dw = \lambda^{-2} \int_0^\infty f_a''(w)(1 - \cos \lambda w)\, dw$ for $\lambda > 0$, which is strictly positive because $f_a'' > 0$ (and the case $\lambda = 0$ is $\int f_a > 0$). So $C_a > 0$. Monotonicity in P holds term by term, since $\mathcal{F}(P) \subset \mathcal{F}(P')$. □

Remark 18.3.2. The proof of Theorem 18.3.1 uses only $a \geq \frac{3}{2}$, so $C_{\pi n^2}(t) > 0$ for every $n \geq 1$ and every real t, with C_a as in (18.3.1), extended to complex t by the same integral. Put

$$W(t) = 4 \int_0^\infty \psi(e^{2u}) e^{u/2} \cos(tu)\, du = 2 \sum_{n \geq 1} C_{\pi n^2}(t),$$

so that $2\Xi(t) = 1 - (t^2 + \frac{1}{4})W(t)$ for every complex t, by (18.1.1). Then $W > 0$ on $\mathbb{R}$, the zeros of Ξ in $\mathbb{C}$ are exactly the solutions of $(t^2 + \frac{1}{4})W(t) = 1$, and RH says that every solution is real. So the transform of no term of the theta series vanishes on the real line, and the zeros of Ξ come from the cancellation between the pole term and the whole series. In terms of laws, $C_{\pi n^2}(t)/C_{\pi n^2}(0)$ is the characteristic function of the symmetric density proportional to $e^{|u|/2} \exp(-\pi n^2 e^{2|u|})$, the law of $\pm\frac{1}{2} \log X$ with an independent fair sign, where X has the density proportional to $x^{-3/4} e^{-\pi n^2 x}$ on $[1, \infty)$. Hence $W(t)/W(0)$ is the characteristic function of the mixture of these laws with the weights $2C_{\pi n^2}(0)/W(0)$, whose density is

$$p_V(u) = \frac{2e^{|u|/2} \psi(e^{2|u|})}{W(0)} = \frac{e^{-|u|/2} - k(u)}{W(0)},$$
$$W(0) = 2 \sum_{n \geq 1} (\pi n^2)^{-1/4}\, \Gamma(\tfrac{1}{4}, \pi n^2) = 0.0230338$$

(computed). None of these laws is ferromagnetic (Proposition 19.1.11). Next, k is bounded, $(\frac{1}{4} - \frac{d^2}{du^2}) e^{-|u|/2} = \delta_0$ in the sense of distributions, and $\Phi = \frac{1}{2}(\frac{1}{4}k - k'')$

holds on all of $\mathbb{R}$, both sides being even. So $k - 2e^{-|\cdot|/2} * \Phi$ is bounded and annihilated by $\frac{1}{4} - \frac{d^2}{du^2}$, whose null solutions $Ae^{u/2} + Be^{-u/2}$ are unbounded unless $A = B = 0$, and

$$k(u) = 2 \int_{\mathbb{R}} e^{-|u-w|/2}\, \Phi(w)\, dw.$$

With the Laplace density $\varkappa(u) = \frac{1}{4} e^{-|u|/2}$, and since $\Phi/\xi(\frac{1}{2})$ is the density of the Pólya law of (2.1.4), this reads

$$\varkappa = (1 - \upsilon)\, \varkappa * \frac{\Phi}{\xi(\frac{1}{2})} + \upsilon\, p_V, \qquad \upsilon = \frac{W(0)}{4} = 1 - 2\xi(\tfrac{1}{2}) = 0.00575844$$

(computed): the Laplace law is the mixture, with weights $1 - \upsilon$ and υ, of the Pólya law plus an independent Laplace variable and of the law p_V. Status: proved [145, 134].

LEMMA 18.3.3. *If $F(0) = F'(0) = 0$ and F, F', F'' are integrable on $[0, \infty)$, then the cosine and sine transforms of F at $\lambda > 0$ are at most $\min(\|F\|_1, \|F''\|_1/\lambda^2)$ in absolute value, and the sine bound needs only $F(0) = 0$. Hence $|\cdot|(\beta^2 + \lambda^2) \le \beta^2 \|F\|_1 + \|F''\|_1$ for every $\beta > 0$.*

Proof. Integrability of F and F' makes $F \to 0$ at infinity, and integrability of F' and F'' makes $F' \to 0$. Two integrations by parts give $\int_0^\infty F \cos \lambda w\, dw = -\lambda^{-1} \int_0^\infty F' \sin \lambda w\, dw = -\lambda^{-2} \int_0^\infty F'' \cos \lambda w\, dw$, where the boundary terms vanish because $F(0) = F'(0) = 0$. For the sine transform, $\int_0^\infty F \sin \lambda w\, dw = \lambda^{-1} \int_0^\infty F' \cos \lambda w\, dw = -\lambda^{-2} \int_0^\infty F'' \sin \lambda w\, dw$, which uses only $F(0) = 0$. The bound by $\|F\|_1$ is trivial, and $\min(A, B/\lambda^2)(\beta^2 + \lambda^2) \le \beta^2 A + B$. □

THEOREM 18.3.4 (Uniform Lorentzian bound). *For $a \ge 9\pi$ and all real t, $|C_a(t) - \mathcal{L}_a(t)| \le \delta(a) \mathcal{L}_a(t)$ with*

$$\delta(a) = \frac{1}{b} \left[\frac{2ab^2}{(b-1)^3} + \frac{2ab}{(b-1)^2} + \frac{2a^2}{(b-2)^3} + \frac{a}{b-1} \right] \le \frac{5.61}{a}.$$

Proof. Write $f_a = e^{-a} e^{-bw} e^{-a\varphi}$ with $\varphi = e^w - 1 - w \ge 0$. Then $C_a - \mathcal{L}_a$ is the cosine transform of $h = e^{-a} e^{-bw} g$ with $g = e^{-a\varphi} - 1$, and $h(0) = h'(0) = 0$. The bounds $|g| \le \frac{a}{2} w^2 e^w$, $|g'| \le awe^w$, $|g''| \le a^2 w^2 e^{2w} + ae^w$ in $h'' = e^{-a} e^{-bw} (b^2 g - 2bg' + g'')$ give $e^a \|h\|_1 \le a/(b-1)^3$ and $e^a \|h''\|_1 \le ab^2/(b-1)^3 + 2ab/(b-1)^2 + 2a^2/(b-2)^3 + a/(b-1)$. Lemma 18.3.3 with $\lambda = t/2$ and $\beta = b$, together with $\mathcal{L}_a(t)(b^2 + t^2/4) = e^{-a} b$, gives the claim. □

The last inequality follows from monotonicity. With $b = a - \frac{1}{4}$ and $a \ge 9\pi$ (so $b > 2$), every term of $\delta(a)$, $a\delta(a)$, $r_1(a)$ and $r_2(a)$ (the last two in Theorem 18.3.5) is a positive constant times a product of factors $a/(a-c)$, $(a-c_1)/(a-c_2)$ with $c_2 > c_1$, and $1/(a-c)$, where $c \ge 0$ and $a > c$; for example $2ab^2/(b(b-1)^3) = 2\,(a/(b-1))(b/(b-1))(1/(b-1))$ and $a^2/(b-2)^2 = (a/(b-2))^2$. Each such factor

is positive and nonincreasing in a, so δ, $a\delta$, $1/(1-\delta)$ (as $\delta(9\pi)=0.198<1$), r_1 and r_2 are nonincreasing on $[9\pi,\infty)$. As an illustration, $a\delta(a)$ equals 5.6067 at $a=9\pi$, 5.204, 5.103, 5.041, 5.0016 and 5.00002 at $a=25\pi$, 49π, 121π, 10^4, 10^6, with limit 5 (the leading terms are $2+2+0+1$ over b), and $\delta(a)\le 5.61/a$ is the value at $a=9\pi$ propagated by the decrease of $a\delta(a)$. By quadrature the true error at $a=9\pi$ is 0.03365 (computed), against the bound 0.198.

Summing, $D_P\approx(t^2+\frac14)\sum_{m\notin\mathcal{F}(P)}\mathcal{L}_{a_m}(t)$. Since $(t^2+\frac14)\mathcal{L}_a(t)=e^{-a}b\,(t^2+\frac14)/(b^2+t^2/4)\le 4e^{-a}b$, Theorem 18.3.4 gives $D_P\le(1+\delta(a_q))c_P\approx 4\pi q^2e^{-\pi q^2}$ on $\mathbb{R}$, and the deficit rises to its limit c_P in the form of the Lorentzian $\mathcal{L}_{a_q}$, whose half width is $2b_q\approx 2\pi q^2$. Near the breakdown height $4q^2$ the deficit is only $D_P(4q^2)/c_P\approx 4/(\pi^2+4)=0.2884$ of its limit; the measured values are 0.2929, 0.2901, 0.2893, 0.2888 for $P=2,3,5,7$ (computed).

Theorem 18.3.5 (Curvature). *For every prime P and $t\ge 14$, $(\log D_P)''(t)\ge -2/t^2-r_2(a_q)-r_1(a_q)^2\ge -0.0239$, where, with $\delta=\delta(a)$,*

$$r_1(a)=\frac{1}{2(1-\delta)b}\Big(1+\frac{a^2}{(b-2)^2}+\frac{a}{(b-1)^2}+\frac{2a}{b-1}\Big),$$
$$r_2(a)=\frac{1}{4(1-\delta)b}\Big(\frac{4}{b}+\frac{2a^2}{(b-2)^3}+\frac{2a}{(b-1)^3}+\frac{4a}{(b-1)^2}\Big)$$

bound $|C_a'|/C_a$ and $|C_a''|/C_a$ uniformly in t.

Proof. Apply Lemma 18.3.3 to f_aw (sine) and f_aw^2 (cosine), with $f_a\le e^{-a}e^{-bw}$, $|f_a'|\le f_aae^w$, $|f_a''|\le f_a(a^2e^{2w}+ae^w)$, and divide by $C_a\ge(1-\delta)\mathcal{L}_a$. Positivity of every C_{a_m} transfers the bounds to the sum $S=\sum_m C_{a_m}$, namely $|S'|\le r_1S$ and $|S''|\le r_2S$, because r_1,r_2 do not increase with a on $[a_q,\infty)\subset[9\pi,\infty)$ (every term of their formulas is of the form treated after Theorem 18.3.4; for illustration $r_1=0.0960$, 0.0281, 0.0136 and $r_2=0.00435$, 0.00045, 0.00011 at $a=9\pi$, 25π, 49π, computed). Then $(\log D_P)''=(\log(t^2+\frac14))''+S''/S-(S'/S)^2\ge -2/t^2-r_2-r_1^2$, because $(\log(t^2+\frac14))''=2(\frac14-t^2)/(t^2+\frac14)^2>-2/t^2$. At $a=9\pi$ one finds $\delta=0.19830$, $r_1=0.095954$, $r_2=0.0043527$ and $2/14^2+r_2+r_1^2=0.023764\le 0.0239$ (computed). □

The value -0.0239 is slightly looser than the computed -0.02376. The computed minima of $(\log D_P)''$ over $t\ge 14$ are -0.010642, -0.010240, -0.010186, -0.010169 for $P=2,3,5,7$, all at $t=14$, well above the bound.

18.4 Humps, pairs and the count of real zeros

Since $\Xi(0)>0$, call the maximal intervals $I_j\subset(0,\infty)$ on which $\Xi<0$ the negative humps, numbered from the left, of length ℓ_j and depth $d_j=\max_{I_j}|\Xi|$; the endpoints of I_j are zeros of Ξ (when the zeros are simple, $I_j=(\gamma_{2j-1},\gamma_{2j})$ for the positive zeros $\gamma_1\le\gamma_2\le\dots$). Let $N(T)$ be the number of zeros $\gamma\in(0,T]$

counted with multiplicity. On $\mathbb{R}$, $\Xi = -AZ$ with Z Hardy's function (Section 2.2) and factor $A(t) = \frac{1}{2}(t^2+\frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{it}{2})| \sim (\pi/2)^{1/4}t^{7/4}e^{-\pi t/4}$ (Stirling). The first hump begins at $\gamma_1 = 14.13\ldots > 14$, so Theorem 18.3.5 applies on every hump.

THEOREM 18.4.1.

(1) Every real zero of Ξ_P lies in a negative hump; $\Xi_P(\gamma_k) = D_P(\gamma_k) > 0$ at the zeros γ_k of Ξ (so no zero of Ξ_P is a zero of Ξ).

(2) Each hump holds an even number of zeros of Ξ_P, counted with multiplicity, and the number R_P of positive real zeros is even.

(3) Assume that every zero of Ξ whose real part lies within $\frac{1}{2}$ of I_j is real (by [232] this holds for every hump below $3\cdot 10^{12}$), and that $\ell_j < 18.29$. Then I_j holds no zero, one double zero or two simple zeros of Ξ_P, and two simple zeros precisely when $\max_{I_j}(|\Xi| - D_P) > 0$.

(4) Surviving humps are nested in P: if I_j carries zeros for Ξ_P, it does for $\Xi_{P'}$, $P' > P$.

Proof. (1) Where $\Xi \geq 0$ we have $\Xi_P = \Xi + D_P > 0$. (2) Ξ_P is positive at both ends of each hump, and $\Xi_P(0) > 0$, $\Xi_P(t) \to c_P > 0$. (3) By the Hadamard product (Equation (2.1.1)), Ξ is of order one and even, so $(\log|\Xi|)''(t) = -\sum_w \mathrm{Re}(t-w)^{-2}$ over all zeros w of Ξ (both signs). A real zero contributes $-(t-w)^{-2} \leq 0$. A nonreal zero $w = \gamma + i\delta'$ has $0 < |\delta'| < \frac{1}{2}$ (the zeros lie in the strip $|\operatorname{Im} w| < \frac{1}{2}$), and contributes $(\delta'^2 - x^2)/(x^2+\delta'^2)^2$ with $x = t-\gamma$, which is negative as soon as $|x| \geq \frac{1}{2}$. For $t \in I_j$ the zeros with $|\operatorname{Re} w - t| < \frac{1}{2}$ are real by hypothesis and all other zeros contribute at most 0. Keeping only the two endpoints $\alpha < \beta$ of I_j, $(\log|\Xi|)'' \leq -(t-\alpha)^{-2} - (\beta-t)^{-2} \leq -8/\ell_j^2$ on I_j, the minimum of the right side being at the midpoint. When $8/\ell_j^2 > 0.0239$, that is $\ell_j < 18.29$, Theorem 18.3.5 makes $g = \log|\Xi| - \log D_P$ strictly concave on I_j, and $g \to -\infty$ at both ends. So g has no zero (if $\max g < 0$), one double zero (if $\max g = 0$) or two simple zeros (if $\max g > 0$). A zero of g is a zero of Ξ_P because $\Xi = -|\Xi|$ on I_j, and the multiplicities agree: $\Xi_P' = -|\Xi|g'$ where $|\Xi| = D_P$. (4) follows from Theorem 18.3.1. □

The hypothesis of (3) is local. The humps used in this chapter lie below 21,400, where the gaps between zeros are of order $2\pi/\log(t/2\pi)$ and $\ell_j < 18.29$ (computed from the tabulated zeros), so for them (3) is an unconditional theorem given [232].

The two zeros of a surviving hump sit near its ends and are pushed inward, which explains the alternating signs of the first order shifts. Real zeros leave the axis only in pairs: deforming by $\Xi + sD_P$, $0 \leq s \leq 1$, a pair of real zeros of $\Xi + sD_P$ collides where $\log|\Xi| - \log(sD_P)$ attains its maximum zero, near but not at the extremum t_j of the hump. Near the collision $\Xi_P \approx D_P(t_j) - d_j + \frac{1}{2}\Xi''(t_j)(t-t_j)^2$,

so the pair is $t_j \pm \sqrt{2(d_j - D_P)/\Xi''(t_j)}$ before and $t_j \pm i\sqrt{2(D_P - d_j)/\Xi''(t_j)}$ after (heuristic).

THEOREM 18.4.2 (Count). *Under the hypotheses of Theorem 18.4.1(3) for every hump meeting* $(0, \tau_P^+)$,

$$R_P = 2\#\{j : \max_{I_j}(|\Xi| - D_P) > 0\},$$

where the count is over all humps. Let τ_P^+ *solve* $D_P(\tau) = 0.63\,A(\tau)\,\tau^{1/6}\log\tau$; *the solution is unique for* $\tau \geq 14$, *and* $\Xi_P > 0$ *on* (τ_P^+, ∞). *Then* $R_P \leq 2\lceil N(\tau_P^+)/2\rceil$.

Proof. The formula is part (3) summed over the humps meeting $(0, \tau_P^+)$; beyond τ_P^+ there is no zero, as follows. Put $\mathcal{R}(t) = D_P(t)/(0.63A(t)t^{1/6}\log t)$. Since $D_P = (t^2 + \frac{1}{4})S$ and $|S'| \leq r_1(a_q)S$ (Theorem 18.3.5), $(\log D_P)' \geq 2t/(t^2 + \frac{1}{4}) - r_1(a_q)$. Also $(\log A)' = 2t/(t^2 + \frac{1}{4}) - \frac{1}{2}\operatorname{Im}\frac{\Gamma'}{\Gamma}(\frac{1}{4} + \frac{it}{2})$. The terms $2t/(t^2 + \frac{1}{4})$ cancel, and with $\operatorname{Im}\frac{\Gamma'}{\Gamma}(\frac{1}{4} + iy) = y\sum_{n\geq 0}\big((n + \frac{1}{4})^2 + y^2\big)^{-1} \geq \frac{\pi}{2} - \frac{1}{4y}$ (compare the sum with an integral) we get, for $t \geq 14$ (so $y = t/2 \geq 7$),

$$(\log\mathcal{R})' \geq \tfrac{1}{2}\Big(\tfrac{\pi}{2} - \tfrac{1}{2t}\Big) - r_1(a_q) - \frac{1}{6t} - \frac{1}{t\log t} > 0.6,$$

using $r_1(a_q) \leq r_1(9\pi) = 0.0960$. With $\mathcal{R}(14) < 1$ (computed: $D_2(14) < 10^{-10}$, and $0.63A(14)14^{1/6}\log 14$ is of order 10^{-3}) and $\mathcal{R} \to \infty$, the equation $\mathcal{R} = 1$ has exactly one solution τ_P^+ for $t \geq 14$ and $\mathcal{R} > 1$ beyond it. By Hiary's estimate $|Z(t)| = |\zeta(\frac{1}{2} + it)| \leq 0.63t^{1/6}\log t$ [98], $|\Xi| = A|Z| < D_P$ for $t > \tau_P^+$, so $\Xi_P > 0$ there. Only humps meeting $(0, \tau_P^+)$ count, and there are at most $\lceil N(\tau_P^+)/2\rceil$ of them, each carrying at most two zeros. □

The monotonicity of r_1 used here is proved (proof of Theorem 18.3.5), so the uniqueness of τ_P^+ and the positivity beyond it are proved.

P	q	τ_P^*	$N(\tau_P^*)$	τ_P^+	$N(\tau_P^+)$	R_P
2	3	39.73	6	41.63	7	6
3	5	104.68	31	107.06	32	32
5	7	201.32	80	203.99	81	80
7	11	490.18	263	493.23	264	264

Table 18.1: Real zero counts (computed in floating point, not certified). τ_P^* solves $A(\tau) = D_P(\tau)$. For $P \leq 5$, R_P is a full sign change scan; for $P = 7$ it is the hump criterion, with a 215 digit check of every hump in $[481, 501]$, the last surviving pair being 489.7932 and 491.0605.

The primes $2, 3, 5, 7$ therefore reproduce the first 264 zeros of Ξ: the computed zeros of Ξ_7 pair with the first 264 zeros γ_k of Ξ, the largest displacement being 0.338 at $k = 264$ (computed); it is 0.764, 0.208, 0.091 for $P = 3, 5, 2$. The counts

R_P of Table 18.1 become theorems once a ball arithmetic certificate of the sign of $\max_{I_j}(\log|\Xi| - \log D_P)$ on every hump below τ_P^+ is supplied. The breakdown height has the heuristic form

$$\tau_P^* \approx 4q^2 + \frac{6}{\pi}\log q + \frac{4}{\pi}\log\frac{\pi^2+4}{4} + 0.03 \approx 4q^2 + \frac{6}{\pi}\log q + 1.61,$$

where $\frac{4}{\pi}\log\frac{\pi^2+4}{4} = 1.583$ comes from the Lorentzian and Stirling estimates and the remaining 0.03 is a fitted constant. The formula gives 39.71, 104.68, 201.33, 490.19 for $q = 3, 5, 7, 11$, against the computed 39.73, 104.68, 201.32, 490.18, and it reproduces all twenty values of Table 18.2 to within 0.05 (computed).

18.5 A survey of the breakdown height up to height 21,000

For every prime $P \le 71$ the hump criterion of Theorem 18.4.2 (computed, with Z evaluated by the Riemann Siegel formula and the zeros located by sign changes on a grid of step 0.02, repeated at step 0.016 with no change in any count) gives $R_P - N(\tau_P^*) \in \{-1, 0, 1, 2, 3\}$, with mean 0.90 and standard error 0.28 (sample standard deviation divided by $\sqrt{20}$) over the twenty primes (Table 18.2); the heights reach 21,326, and no hump was marginal. Compare a heuristic ramp model. Near τ_P^*, $\log A$ falls at rate $\pi/4$ while $\log D_P$ is nearly flat, so hump j survives when $\log\max_{I_j}|Z| > \frac{\pi}{4}(t_j - \tau_P^*)$. Negative humps have density $\log(t/2\pi)/4\pi$ (computed: 0.6475 against 0.6469 in a window of 400 around τ_P^* at $P = 71$) and carry two zeros each, so the model offset is $(2/\pi^2)\log(\tau_P^*/2\pi)\, m_Z(\tau_P^*)$, with $m_Z(T)$ the mean of $\log\max|Z|$ over the negative humps within 200 of T. The model gives the right order of magnitude and no more: its mean over the survey is 0.447 against the observed 0.90, and the rows do not follow it closely.

P	τ_P^*	N	R_P	off	model	P	τ_P^*	N	R_P	off	model
2	39.7	6	6	0	0.16	31	5484	5039	5040	1	0.42
3	104.7	31	32	1	0.23	37	6733	6405	6408	3	0.53
5	201.3	80	80	0	0.29	41	7405	7157	7160	3	0.60
7	490.2	263	264	1	0.37	43	8845	8799	8802	3	0.58
11	682.5	401	400	−1	0.37	47	11245	11616	11616	0	0.54
13	1163	782	782	0	0.37	53	13933	14867	14868	1	0.44
17	1451	1027	1028	1	0.47	59	14893	16050	16052	2	0.52
19	2124	1631	1632	1	0.43	61	17966	19897	19898	1	0.54
23	3372	2838	2840	2	0.41	67	20174	22715	22714	−1	0.56
29	3852	3323	3324	1	0.48	71	21326	24201	24200	−1	0.64

Table 18.2: The survey (computed). $N = N(\tau_P^*)$, off $= R_P - N$, model is the ramp model's expected offset with the window 200 for m_Z.

Conjecture 18.5.1.

$$R_P = N(\tau_P^*) + \frac{2}{\pi^2}\log\frac{\tau_P^*}{2\pi}\, m_Z(\tau_P^*) + O\big((\log q)^{1/2}(\log\log q)^{1/2}\big),$$

with $m_Z(T)$ the mean of $\log\max|Z|$ *over negative humps near T.*

Conjecture 18.5.1 is open. The twenty offsets are consistent with a bounded or slowly growing offset, but they differ from the model by a factor of about two on average, the error term is not motivated by the ramp model, and a growth like $\log q$ cannot be tested at these heights.

18.6 Where the lost zeros go

Beyond τ_P^* the zeros of Ξ_P solve $\Xi = -D_P$. Stirling's formula and the functional equation give $|\Xi(x+iy)| \approx A(x)(x/2\pi)^{|y|/2}$, and matching moduli gives the departure law (heuristic)

$$|y_P(x)| \approx \frac{2\log(D_P(x)/A(x))}{\log(x/2\pi)} \sim \frac{\pi}{2}\,\frac{x-\tau_P^*}{\log(x/2\pi)}, \tag{18.6.1}$$

where the second form follows from $\log(D_P/A) \approx \frac{\pi}{4}(x-\tau_P^*)$. Using $D_P(x)$ rather than its limit c_P matters: near τ_P^*, D_P is only about $0.3c_P$ (computed: 0.336 at $P=2$ and 0.310 at $P=3$), and with D_P the law fits the computed zeros to within 0.104 (Table 18.3). Since $|y_P|$ soon exceeds $\frac{1}{2}$, the departed zeros leave the critical strip altogether.

P	$\operatorname{Re} z$	$\lvert\operatorname{Im} z\rvert$	law with c_P	law (18.6.1)
2	45.418	4.421	5.38	4.46
2	57.879	12.612	13.23	12.65
2	70.174	19.406	19.90	19.51
3	111.384	3.663	4.40	3.64
3	128.300	12.233	12.84	12.24
3	148.311	21.542	22.03	21.56

Table 18.3: Departed zeros of Ξ_P (computed, residuals below 10^{-34}) against the two forms of the departure law. The six zeros are every second zero of the chain of Remark 18.6.1 for $P=2$ and a selection for $P=3$.

Remark 18.6.1 (Zero count, heuristic). With $N_P(X)$ the number of zeros of Ξ_P in $0 < \operatorname{Re} t < X$ (both half planes), and $y_P(X)$ the departure law (18.6.1), the heuristic is

$$N_P(X) = N(X) + \tfrac{1}{4}y_P(X) + O(1).$$

The $O(1)$ is not derived.

Derivation (heuristic). Apply the argument principle to Ξ_P and to Ξ on $[0,X]\times[-Y,Y]$ with $Y \gg y_P(X)$. On the horizontal sides $|\Xi| \gg |D_P|$, and on the left side both functions are real and positive. On the right side $\Xi_P \approx D_P$ for $|y| < y_P$, where $\arg D_P$ is nearly constant, while $\partial_y \arg\Xi = \operatorname{Re}(\Xi'/\Xi) \approx -\pi/4 + y/(2X)$.

The odd term integrates to zero, so the two counts differ by $\frac{\pi}{2}y_P$ radians, that is $y_P/4$ zeros, up to $O(1)$ from the transitions. □

Argument principle counts, evaluated at 20 to 40 digits and equal to an integer within 10^{-15}, give $N_P(X) - N(X)$ equal to 2, 3, 4, 5 for $P = 2$ at $X = 51.5$, 60, 68, 80, against $y_P/4$ equal to 2.17, 3.47, 4.59, 6.12 ($N_P = 12$, 16, 20, 26 against $N = 10$, 13, 16, 21). For $P = 3$ at $X = 120$, 140, 150 they give 2, 4, 6 against 2.03, 4.45, 5.58. Every count was repeated with a second contour height and precision (20 and 28 digits), with the same integer. The difference $N_P - N - y_P/4$ is -0.17, -0.47, -0.59, -1.12 for $P = 2$, and -4.86 at $X = 600$ ($N_P = 384$ at two precisions, $N = 341$, $y_P/4 = 47.86$), so the computed counts do not support a bounded $O(1)$. Tracing the departed zeros of Ξ_2 one by one (each verified by a residual test, in one continuous job) gives a single smooth chain from $41.19+1.46i$ through $45.42+4.42i$ to $601.90 + 191.23i$, with 190 zeros up to 601.90 and 189 with real part below 600. The argument principle on $\operatorname{Re} t < 600$ gives $N_P = 384 = 6 + 2 \cdot 189$, the six real zeros and the chain with its conjugates. (The chain count 189 and the difference $N_P - N = 43$, compared with $y_P/4$ above, are different quantities.) So for $P = 2$ the lost humps release their zeros in pairs, and these, together with about $y_P/4$ new zeros, populate one curve (observed for $P = 2$, and in the counts for $P = 3$). Its density in the upper half plane is, heuristically,

$$\frac{1}{4\pi}\log\frac{x}{2\pi} + \frac{\pi}{16\log(x/2\pi)},$$

larger than the density $\frac{1}{4\pi}\log\frac{x}{2\pi}$ of lost humps by the factor $1+\pi^2/(4\log^2(x/2\pi))$. Integrated from 41.19 to 600 this density gives 197.4 zeros, against the 189 of the chain (four per cent more), and the spacings of the chain, 4.13, 3.72 and 3.39 near $x = 45$, 100 and 150, exceed the reciprocal density there, 3.90, 3.43 and 3.18, by 6 to 9 per cent (over the whole chain the spacings lie between 2.537 and 4.228).

18.7 Miyatake's theorem and the remainder R

Write Ξ in the form used by Miyatake [119], as a combination of two shifted Laplace transforms,

$$\Xi(z) = 2\pi^2\Big\{G_{\mathrm{M}}\big(\tfrac{1}{2}iz - \tfrac{9}{4}\big) + G_{\mathrm{M}}\big(\tfrac{1}{2}iz + \tfrac{9}{4}\big)\Big\}, \qquad G_{\mathrm{M}}(w) = \int_{-\infty}^{\infty} \psi_{\mathrm{M}}(t)e^{wt}\,dt. \tag{18.7.1}$$

Let Φ be the kernel of (2.1.3), so that $\Xi(z) = \int_{\mathbb{R}} \Phi(u)e^{izu}\,du$ and

$$\Phi(t) = 2\pi e^{5t/2}\sum_{n\ge1}(2\pi n^2e^{2t} - 3)\,n^2e^{-\pi n^2e^{2t}}.$$

The identity (18.7.1) holds exactly when

$$8\pi^2\cosh(9t/2)\,\psi_{\mathrm{M}}(2t) = \Phi(t). \tag{18.7.2}$$

Indeed, (18.7.2) gives $\Xi(z) = 8\pi^2 \int \cosh(9t/2)\psi_{\mathrm{M}}(2t)e^{izt}\,dt$, and the substitution $s = 2t$ turns this into $4\pi^2 \int \cosh(9s/4)\psi_{\mathrm{M}}(s)e^{izs/2}\,ds = 2\pi^2\{G_{\mathrm{M}}(\frac{1}{2}iz + \frac{9}{4}) + G_{\mathrm{M}}(\frac{1}{2}iz - \frac{9}{4})\}$. Condition (18.7.2) forces

$$\psi_{\mathrm{M}}(t) = \sum_{n\ge 1} n^4 e^{-2\pi n^2 \cosh t}\bigl(1 - R_{\mathrm{M}}(n^2\pi, t)\bigr), \tag{18.7.3}$$

where

$$1 - R_{\mathrm{M}}(x,t) = \frac{1 - \frac{3}{2x}e^{-t}}{1 + e^{-9t/2}}\, e^{xe^{-t}} \qquad (t > 0),$$

extended evenly. Each term of $\Phi(s/2)/(8\pi^2\cosh(9s/4))$ equals $n^4(1 - \frac{3}{2x}e^{-s})e^{-xe^s}/(1 + e^{-9s/2})$ with $x = \pi n^2$, and $e^{-xe^s} = e^{-2x\cosh s}e^{xe^{-s}}$, which proves (18.7.3). The sum ψ_{M} is even and a single term is not. A numerical check of (18.7.1) and (18.7.2) agrees to thirteen digits at $z = 0$, at $z = 5$ and at the first zero of Ξ. The remainder R_{M} undoes Pólya's three simplifications: $e^{xe^{-t}}$ desymmetrises the exponential, $1/(1 + e^{-9t/2})$ removes the universal factor, $1 - \frac{3}{2x}e^{-t}$ restores the second theta term. Each term of (18.7.3) has a corner at $t = 0$, with one sided logarithmic slopes $+0.0222$, -10.18, -25.97 for $n = 1, 2, 3$ (the slope of term n is $3/(2x - 3) + \frac{9}{4} - x$ with $x = \pi n^2$); only the full sum is smooth, which is Lemma 18.1.1 in Miyatake's coordinates. The ratio $Q = \psi_{\mathrm{M}}/e^{-2\pi\cosh t}$ has $Q(0) = 6.0590692$, so $Q(0) - 1 = 5.059069$, and tends to 1 as $t \to \infty$; numerically it decreases on $[0, 6]$.

Dropping the factors $1 - R_{\mathrm{M}}$ and summing over the P friable integers n (those with every prime factor at most P) gives

$$\xi^{(P)}(z) = 4\pi^2 \sum_{n \text{ friable}} n^4 \Bigl\{ K_{iz/2-9/4}(2\pi n^2) + K_{iz/2+9/4}(2\pi n^2) \Bigr\},$$

the function that Miyatake studies in paper B [119]. His Theorem I states that $\xi^{(P)}$ has all but finitely many zeros real, with $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1)$ of them in $(0, T)$; the proof uses Watson type steepest descent asymptotics for $K_z(a)$, a lower bound $|\zeta_P| \ge \sigma > 0$ for the finite Euler product ζ_P on a strip $\frac{1}{2} - \varepsilon \le x \le A$, the argument principle and Rouché's theorem. Hejhal [96] proved a theorem of this kind for truncated symmetrised kernels. For $P = 2$ we find exactly 79 real zeros of $\xi^{(2)}$ in $(0, 200)$, as many as Ξ, from 8.9929 (against 14.1347 for Ξ) to 198.07. The lower bound σ on $|\zeta_P|$ tends to 0 as $P \to \infty$, so the method does not reach the full sum.

The remainder R_{M} separates two families. Keeping it with the friable integers gives $\Xi_P^{\mathrm{M}} = \Xi_P - c_P$ (two integrations by parts with $k_P'(0) = -c_P$), extremely close to Ξ and, like Ξ_P, with finitely many real zeros (Theorem 18.2.3 for Ξ_P). Dropping it gives functions of the type of $\xi^{(P)}$ and of Pólya's approximant, with almost all zeros real but at a distance from the zeros of Ξ. For $P = 2$ the zeros of Ξ_2 differ from the first five zeros of Ξ by 2.5×10^{-9}, 4.0×10^{-7}, 7.7×10^{-6}, 5.536×10^{-4} and

3.702×10^{-3}, while those of $\xi^{(2)}$ are off by several units already at the first zero. The Pólya shift route to RH through (18.7.1) would need $g(x) = G_{\mathrm{M}}(ix)$ to lie in the Laguerre Pólya class. It does not. Since $\psi_{\mathrm{M}}(t) = \Phi(t/2)/(8\pi^2 \cosh(9t/4))$, ψ_{M} has a pole at $t = 2\pi i/9$, inside the strip where $\Phi(t/2)$ is analytic; hence $g(x) = Ce^{-2\pi x/9}(1 + o(1))$ with $C = 0.0899917116445 > 0$ (Theorem 18.12.1, which proves that g is not Laguerre Pólya, Corollary 18.12.3), and numerically g has no zeros in $(0, 26) \times (-4, 4)$ and none on $(0, 60]$. The pole terms cancel exactly in the shifted sum (18.7.1). In the variable $\cosh t$ this pole is a simple pole at $\cos \frac{2\pi}{9}$, the exponent of the exponential part of Section 18.13 (Proposition 18.13.1). Moreover Q decreases numerically, whereas every universal factor $\mathbb{E} \cosh(tX)$ in the sense of Cardon [34] increases in $|t|$.

Three further facts, all computed except the second half of the third. First, interpolating with factors $1 - \lambda R_{\mathrm{M}}$ gives $\Xi_\lambda = (1 - \lambda)\xi^{(2)} + \lambda \Xi_P^{\mathrm{M}}$, where Ξ_P^{M} keeps R_{M} and has 7 real zeros in $(0, 200)$, the last at 39.53. The real part X of the first nonreal zero of Ξ_λ (the height to which the real zeros survive) satisfies $h(\lambda) \approx h(1) + \frac{4}{\pi} \log \frac{1}{\lambda}$: for $\lambda = 10^{-k}$, $k = 0, \dots, 8$, one finds $h = 44, 45, 50, 51, 56, 57, 62, 63, 67$ on a grid of step 1, with a least squares slope of 2.93 per decade against $\frac{4}{\pi} \log 10 = 2.93$. The survivors are zeros of the Pólya type function, not of Ξ. Second, dividing Riemann's kernel $\Phi(u)$ by $\cosh(bu)$ creates a pole inside its strip of analyticity when $b > 2$, and for $b \le 2$ the transform acquires nonreal zeros at a height that recedes as $b \to 0$, in every case computed: the largest admissible b lies in $(1, 1.25)$ up to height 40, in $(0.5, 0.75)$ up to 100 and in $(0.25, 0.5)$ up to 160. This is the computed analogue, for cosh factors, of the theorem of Rodgers and Tao [250] that the de Bruijn Newman constant is nonnegative (Sections 18.10 and 18.11; the threshold $b = 2$ is explained in Section 18.12). Third, de Bruijn's strip theorem [61] reduces Miyatake's Theorem I to the zeros of $g^{(P)}(w) = 2 \sum_{n \text{ friable}} n^4 K_{iw}(2\pi n^2)$: $\xi^{(P)}(z) = 2\pi^2 \{g^{(P)}(z/2 + 9i/4) + g^{(P)}(z/2 - 9i/4)\}$, so if the zeros of $g^{(P)}$ lie in $|\operatorname{Im} w| \le \frac{9}{4}$, every zero of $\xi^{(P)}$ is real. For $P = 2$ the ten nonreal zeros of $g^{(2)}$ in the upper half plane with real part below 60 have imaginary parts rising to 1.964 (at real part 58.5), and they appear to approach 2, because the governing friable Dirichlet polynomial is geometric with zeros on $\operatorname{Re} s = 0$. For the full sum the condition becomes that partial sums of ζ have zeros only in $\operatorname{Re} s \le \frac{1}{2}$, which is false [212]; and Riemann's own g has, computationally, zeros on a cone starting at $8.96 + 5.20i$ (Section 18.12).

18.8 The pole term, the secant and reweighting

The pole term $\frac{1}{2}/(t^2 + \frac{1}{4})$ is the first term of $\frac{\pi}{2} \operatorname{sech}(\pi t) = \sum_k (-1)^k (k + \frac{1}{2})/((k + \frac{1}{2})^2 + t^2)$, the transform of $\frac{1}{2} \operatorname{sech}(u/2) = \sum_k (-1)^k e^{-(2k+1)|u|/2}$, a kernel that is smooth because its kinks cancel in Abel sum, $\sum_k (-1)^k (2k + 1) = 0$.

Proposition 18.8.1 (Kink budget). *Replace the pole kernel $e^{-u/2}$ in k_P by an*

even kernel κ, smooth on $[0,\infty)$ with derivatives $O(e^{-u/2})$. Then $\lim_{t\to\infty}\Xi_P^\kappa(t) = c_P - \frac12 - \kappa'(0+)$ along $\mathbb{R}$.

Proof. The kernel $k_P^\kappa = k_P + \kappa - e^{-u/2}$ is smooth on $[0,\infty)$ with every derivative $O(e^{-u/2})$, and $\Xi_P^\kappa(t) = (t^2+\frac14)\int_0^\infty k_P^\kappa(u)\cos(tu)\,du$. Two integrations by parts and the Riemann Lebesgue lemma give $(t^2+\frac14)\int_0^\infty k\cos(tu)\,du \to -k'(0+)$. For $k = k_P$ the limit is $c_P = -k_P'(0)$ (Theorem 18.2.3); for $k = k_P^\kappa$ it is $-k_P'(0) - \kappa'(0+) - \frac12 = c_P - \frac12 - \kappa'(0+)$, since $(e^{-u/2})'(0) = -\frac12$. □

So the secant completion, with $\kappa'(0+) = 0$, lowers the limit from c_P to $c_P - \frac12$, which is negative as soon as $c_P < \frac12$, unless the slope $-\frac12$ is restored elsewhere. Miyatake's paper A [118] is on the consistent side: there the pole term becomes $\frac14\operatorname{sech}(u/4)$, with transform $\pi\operatorname{sech}(2\pi z)$ in the variable z of $\Xi(z) = \int\Phi(u)e^{izu}\,du$, while every theta term is symmetrised, so the whole kernel is smooth.

Proposition 18.8.2 (Reweighting). *For weights w put $D_w = \Xi_w - \Xi = (t^2 + \frac14)\sum_n(1-w_n)C_{\pi n^2}$. If $0 \le w_n \le w_n' \le 1$ for every n then $D_w \ge D_{w'} \ge 0$, with $D_{w'} > 0$ unless every $w_n' = 1$. So among weights in $[0,1]$ that vanish on a given set, the indicator of its complement has the smallest deficit; the friable truncation is of this kind.*

Proof. Every $C_{\pi n^2}$ is positive on $\mathbb{R}$ (Remark 18.3.2); compare termwise. □

Doing better requires some $w_n > 1$; then D_w can change sign, and each sign change plants a spurious real zero. For $P = 2$, matching the deficit at the heights $\frac{40}{3}$, $\frac{80}{3}$, 40 with weights on $n \in \{1\},\{1,2\},\{1,2,4\}$ gives 2, 8, 9 real zeros in $(0,80)$, each time with one spurious zero (computed). The weights grow like $e^{\pi n^2}$ ($w_4 - 1 \approx 3.7\times10^9$), because each correction is a Lorentzian of height $e^{-\pi n^2}$ and width $2\pi n^2$ trying to imitate one of width $2\pi q^2$, where q is the least integer outside the friable set.

18.9 The $2\sin^2$ lemma and the Thorin programme

By Hadamard, $\Xi_P(t)/\Xi_P(0) = \prod_{\operatorname{Re}t_0>0}(1 - t^2/t_0^2)$, the product over the zeros t_0 with positive real part, which is complete when Ξ_P has no zero on $i\mathbb{R}$ (Proposition 18.9.1). On the imaginary axis the sine squared lemma in Frullani form (the factor by factor form of (2.1.9)),

$$\log(t_0+iy) + \log(t_0-iy) - 2\log t_0 = 4\int_0^\infty \sin^2\Big(\frac{xy}{2}\Big)e^{-t_0x}\,\frac{dx}{x} \qquad (\operatorname{Re}t_0 > 0),$$

with principal logarithms (for real t_0 the left side is $\log(1 + y^2/t_0^2)$; for complex t_0 it can differ from the principal $\log(1+y^2/t_0^2)$ by $2\pi ik$), applies factor by factor:

$$\log\frac{\Xi_P(iy)}{\Xi_P(0)} = \int_0^\infty 2\sin^2\Big(\frac{xy}{2}\Big)\nu_P(x)\,dx, \tag{18.9.1}$$

where, with $\gamma_j^{(P)}$ the R_P positive real zeros and x_0+iy_0 the departed zeros in the first quadrant,

$$\nu_P(x)=\frac{2}{x}\Big[\sum_{j\le R_P} e^{-\gamma_j^{(P)}x}+2\sum_{\text{departed}} e^{-x_0x}\cos(y_0x)\Big].$$

For Ξ under RH, $\nu(x)=\frac{2}{x}\sum_\gamma e^{-\gamma x}>0$; this positive density is the starting point of the Thorin representation of the reciprocal Ξ function [233]. In ν_P the departed zeros enter through oscillating terms.

PROPOSITION 18.9.1. $\sup_{y\in\mathbb{R}}|\Xi_P(iy)/\Xi(iy)-1|$ *is at most* 6.8×10^{-13} *for* $P=2$ *and at most* 3.4×10^{-35} *for* $P=3$. *In particular* Ξ_P *has no zero on* $i\mathbb{R}$, *and the* $2\sin^2$ *exponents of* Ξ_P *and* Ξ *agree uniformly to that accuracy.*

Proof. We redo the proof of Theorem 18.2.1 on $i\mathbb{R}$. There $\cos(iyu)=\cosh(yu)$, and by the formula in that proof $\Xi_P(iy)-\Xi(iy)=2(y^2-\frac14)\sum_{m\notin\mathcal{F}(P)}\int_0^\infty e^{-\pi m^2e^{2u}}e^{u/2}\cosh(yu)\,du$ (with the sign of $t^2+\frac14=\frac14-y^2$ absorbed in the modulus), whose integrand is positive. Using $\cosh(yu)\le e^{|y|u}$ and the substitution $v=e^{2u}$ gives $|\Xi_P(iy)-\Xi(iy)|\le|y^2-\frac14|\sum_{m\notin\mathcal{F}(P)}(\pi m^2)^{-s}\Gamma(s,\pi m^2)$ with $s=\frac12(|y|+\frac12)$, for every real y, and $\Xi(iy)=\xi(\frac12+y)>0$. The supremum of the quotient of this bound by $\xi(\frac12+y)$ was computed at 30 digits on a grid of step 0.1 in $[0,40]$: it is 6.75×10^{-13} near $y=7.0$ for $P=2$ ($q=3$) and 3.37×10^{-35} near $y=6.9$ for $P=3$ ($q=5$), and it has fallen below 10^{-20} at $y=40$ (the tail decays because $\xi(\frac12+y)$ grows faster than $|y^2-\frac14|\Gamma(s)$). Both suprema are less than 1, so $\Xi_P(iy)\neq0$ (the supremum over the grid is computed, the bound itself is proved). □

The true suprema of the ratio, computed at 40 digits on a grid, are 6.05×10^{-13} at $y=6.85$ for $P=2$ and 3.24×10^{-35} at $y=6.8$ for $P=3$.

x	$\frac{x}{2}\nu_2(x)$	$\frac{x}{2}\nu(x)$	departed part
0.03	6.6432	6.6432	3.8860
0.05	2.6860	2.6860	0.9924
0.1	0.62104	0.62104	0.06521
0.5	8.8375×10^{-4}	8.8375×10^{-4}	1.50×10^{-9}

Table 18.4: The $2\sin^2$ Lévy density of Ξ_2, from its 6 real and 190 traced departed zeros, against that of Ξ. $\nu_2>0$ on the whole computed range $[0.02,3]$.

The departed zeros sit up to 0.34 radians off the real axis (the maximum, 0.343, is at $174.36+62.25i$). The six real zeros of Ξ_2 differ from the first six zeros of Ξ, mainly the sixth (37.495 against 37.586), and together the real zeros of Ξ_2 together with the departed zeros reproduce ν to seven digits for $x\le0.1$ (the difference of $\frac{x}{2}\nu_2$ and $\frac{x}{2}\nu$ is 3.4×10^{-7} at $x=0.03$, a bound set by cutting

the chain of departed zeros at height 602, and at most 2×10^{-12} at $x=0.05$, 0.1, 0.5). The departed part alone reproduces the tail of ν beyond the zeros of Ξ_2 to three digits at $x=0.03$ and to one digit at $x=0.5$. On the computed range, ν_2 is positive with minimum 2.6×10^{-19} at $x=3$.

Remark 18.9.2. Positivity of the $2\sin^2$ Lévy density does not force the zeros of an even entire function of order one to be real. For $f(t)=\cos t\,(1-t^2/(10+i)^2)(1-t^2/(10-i)^2)$ the factor by factor form gives

$$\nu(x)=\frac{2}{x}\Big[\sum_{k\ge0}e^{-(k+\frac12)\pi x}+2e^{-10x}\cos x\Big]>0\qquad(x>0),$$

since $\sum_k e^{-(k+\frac12)\pi x}>e^{-\pi x/2}+e^{-3\pi x/2}\ge2e^{-\pi x}\ge2e^{-10x}\ge-2e^{-10x}\cos x$. On $[0.02,3]$ the friable approximant behaves the same way: $\nu_2>0$ although Ξ_2 has infinitely many nonreal zeros. Whether $\nu_P\ge0$ on all of $(0,\infty)$ is open (Problem 18.2).

What does see the departed zeros is the Thorin structure in $s=y^2$. We have

$$\log\frac{\Xi_P(i\sqrt{s})}{\Xi_P(0)}=\sum_{\operatorname{Re}t_0>0}\log\Big(1+\frac{s}{t_0^2}\Big),$$

which is $\int\log(1+s/u)\,U_P(du)$ with $U_P\ge0$ on $(0,\infty)$ only if every $t_0^2>0$. Equivalently, $\nu_P(x)=\frac{2}{x}\int e^{-x\sqrt{u}}\,U_P(du)$ must come from a positive measure supported on the positive axis. The departed zeros place mass of U_P at the complex points $x_0^2-y_0^2+2ix_0y_0$, that is poles of $1/\xi_P(\frac12+\sqrt{s})$ at $s=-t_0^2\notin(-\infty,0]$, so by Theorem 18.2.4, for every finite P and every $w\neq0$ with $\sum_n|w_n|n^{-\sigma}<\infty$ for all $\sigma>0$ (the weights $1_{\mathcal{F}(P)}$ included), $1/\xi_w(\frac12+\sqrt{s})$ is not the Laplace transform of a generalised gamma convolution. By Remark 18.9.2, a route to RH through the $2\sin^2$ form must use the support of the Thorin measure, not only the sign of the Lévy density.

18.10 The cosh ladder

For real b let

$$F_b(z)=\int_{-\infty}^{\infty}\Phi(u)\cosh(bu)e^{izu}\,du=\tfrac12\big\{\Xi(z+ib)+\Xi(z-ib)\big\},\qquad(18.10.1)$$

which is entire for every b because Φ decays double exponentially, even, real on $\mathbb{R}$, of order one, and satisfies $F_b(0)=2\int_0^\infty\Phi\cosh(bu)\,du>0$. With $\alpha=\frac12+b$ and the variable z real, $F_b(z)=\operatorname{Re}\xi(\alpha+iz)$; in the notation of Section 22.6 it is $A_\alpha(iz)$.

THEOREM 18.10.1 (de Bruijn, Lagarias). *For every $b \geq \frac{1}{2}$ all zeros of F_b are real, and $s \mapsto F_b(0)/F_b(i\sqrt{s})$ is the Laplace transform of a generalised gamma convolution with Thorin measure $\sum_j \delta_{\gamma_j(b)^2}$, where $\pm\gamma_j(b)$ are the zeros of F_b. This holds unconditionally.*

This is Theorem 22.6.1(ii) and Corollary 22.6.2 (the simplicity of the zeros is Lagarias's theorem [186]); the proof is short.

Proof. The zeros of Ξ lie in $|\operatorname{Im} z| < \frac{1}{2}$. By de Bruijn's strip theorem [61], all zeros of $\Xi(z+ib)+\Xi(z-ib)$ then lie in $|\operatorname{Im} z| \leq (\max(\frac{1}{4}-b^2, 0))^{1/2}$, so they are real for $b \geq \frac{1}{2}$. Since F_b is even of order one, Hadamard gives $F_b(z) = F_b(0)\prod_j (1 - z^2/\gamma_j(b)^2)$ with $\sum_j \gamma_j(b)^{-2} < \infty$. Hence $F_b(0)/F_b(i\sqrt{s}) = \prod_j (1 + s/\gamma_j(b)^2)^{-1}$, the Laplace transform of $\sum_j e_j/\gamma_j(b)^2$ with independent standard exponentials e_j. □

Let $B_{\cosh} = \inf\{b \geq 0 :\ F_b \text{ has only real zeros}\}$. Then $B_{\cosh} \leq \frac{1}{2}$ unconditionally, and $B_{\cosh} = 0$ if and only if RH holds: RH gives $B_{\cosh} = 0$ because $F_0 = \Xi$, and conversely if F_{b_n} is real rooted along a sequence $b_n \downarrow 0$, which exists because $B_{\cosh}$ is an infimum, Hurwitz's theorem passes real rootedness to the limit Ξ (the set of real rooted b need not be an interval, and this is not used). So RH says that the ladder of Theorem 18.10.1 reaches the bottom rung. The constant Ω of Chapter 20, defined through ferromagnetic cosine tilts for every $a \geq a_0$, is likewise at most $\frac{1}{2}$ and vanishes under RH.

Below the bottom: dividing by $\cosh(bu)$ instead of multiplying, the transform g_b of Section 18.11 has nonreal zeros at a height that recedes as $b \to 0$, in every case computed (Section 18.7); for Gaussian factors the corresponding statement is the theorem of Rodgers and Tao [250]. If RH holds, $\Lambda = 0$ by their theorem, so along the de Bruijn Newman flow the reciprocal Ξ sits at the edge of the generalised gamma convolution class.

One might hope to lower the bound $B_{\cosh} \leq \frac{1}{2}$ computationally, as the Polymath bound [247] does for the de Bruijn Newman constant. Section 18.14 proves $B_{\cosh} \leq \sigma_{\max} - \frac{1}{2}$, with $\sigma_{\max}$ the supremum of the real parts of the zeros, and that the paired version of the ladder bottoms out exactly at $\sigma_{\max} - \frac{1}{2}$; it conjectures that lowering $B_{\cosh}$ is, at large height, a quasi Riemann hypothesis (Conjecture 18.14.2). Any zero free half plane $\operatorname{Re} s > \sigma_0$ gives $B_{\cosh} \leq \sigma_0 - \frac{1}{2}$; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 and recorded in Chapter 20 would give $B_{\cosh} \leq \frac{3}{8}$.

18.11 Miyatake's identity, read probabilistically

Here and in the next two sections Φ is the kernel of Section 18.7; it is even, positive, analytic in $|\operatorname{Im} u| < \pi/4$ and decays like $\exp(-\pi e^{2|u|})$. For $b > 0$ put

$$g_b(z) = \int_{\mathbb{R}} \Phi(u)\operatorname{sech}(bu)\, e^{izu}\, du.$$

It is entire of order at most one, even and real on $\mathbb{R}$, and Miyatake's transform of ψ_{M} is $g(x) = G_{\mathrm{M}}(ix) = g_{9/2}(2x)/(4\pi^2)$.

PROPOSITION 18.11.1 (Shift identity). *For every $b > 0$ and $z \in \mathbb{C}$, $\Xi(z) = \frac{1}{2}\{g_b(z+ib) + g_b(z-ib)\}$, that is $\Xi = \cos(b\partial) g_b$ with $\partial = d/dz$. Miyatake's identity* (18.7.1) *is the case $b = \frac{9}{2}$.*

Proof. Write $\Phi = \Phi \operatorname{sech}(bu) \cdot \frac{1}{2}(e^{bu} + e^{-bu})$; the integrals converge for all z. □

So $g_b = \sec(b\partial)\Xi$ is Ξ pushed backwards through the universal factor $\cos(b\partial)$, which is why Q decreases while universal factors increase. Let $\mathcal{H}_\lambda(z) = \int_{\mathbb{R}} \Phi(u) e^{\lambda u^2} e^{izu}\, du$ be the de Bruijn Newman family in the normalisation of this chapter; in that of Section 4.1.1, $H_t(z) = \frac{1}{8}\mathcal{H}_{t/4}(z/2)$.

THEOREM 18.11.2 (Lévy area, Pólya Gamma clock, Thorin atoms). *Let $\mathcal{A} = \frac{1}{2}\int_0^1 (W^1 dW^2 - W^2 dW^1)$ be Lévy's stochastic area of a planar Brownian motion (W^1, W^2), with $\mathbb{E}e^{i\lambda\mathcal{A}} = \operatorname{sech}(\lambda/2)$, and let $\varpi \sim \mathrm{PG}(1,0)$ be a Pólya Gamma variable, with $\mathbb{E}e^{-s\varpi} = \operatorname{sech}\sqrt{s/2}$. Then for $x \in \mathbb{R}$*

$$g_b(x) = \mathbb{E}\,\Xi(x + 2b\mathcal{A}) = \int_{\mathbb{R}} \Xi(x-y)\frac{1}{2b}\operatorname{sech}\frac{\pi y}{2b}\, dy = \mathbb{E}_\varpi\big[\mathcal{H}_{-2b^2\varpi}(x)\big],$$

and $2b\mathcal{A} \overset{d}{=} 2b\sqrt{\varpi}\,\mathcal{N} \overset{d}{=} \sum_{k\geq 1} \frac{2b}{(2k-1)\pi} L_k$ with $\mathcal{N}$ standard normal and L_k independent standard Laplace variables. Hence $\Xi \mapsto g_b$ is convolution with an extended generalised gamma convolution whose Thorin measure has atoms at $\pm(2k-1)\pi/(2b)$, the odd multiples of $\pi/9$ for Miyatake.

Proof. $\mathbb{E}e^{iu\,2b\mathcal{A}} = \operatorname{sech}(bu) = \mathbb{E}e^{-2b^2\varpi u^2} = \prod_k (1 + 4b^2u^2/((2k-1)\pi)^2)^{-1}$, the last equality being the product $\cosh x = \prod_k (1 + 4x^2/((2k-1)\pi)^2)$ inverted; insert each form into the definition of g_b and use Fubini. □

So Miyatake's ψ_{M} lives on the negative side of the de Bruijn Newman flow, at a random time with mean $-b^2/2$ (as $\mathbb{E}\varpi = \frac{1}{4}$). The real zero counts of g_b and of $\mathcal{H}_{-b^2/2}$ in $(0, 250)$ track each other: 94 and 96 at $b = 0.5$, 78 and 72 at 0.75, 51 and 51 at 1, 29 and 31 at 1.5, against 108 for Ξ (computed).

In the Mellin variable the identity is a partition of unity. Let $\Upsilon > 0$ have $\log\Upsilon$ with density $2\Phi(u)e^{-u/2}$, so that $\mathbb{E}\Upsilon^s = 2\xi(s)$; this is the square root of the variable V of Biane, Pitman and Yor [26], with $2\xi(s) = \mathbb{E}V^{s/2}$, realised through Brownian excursion and Bessel(3) functionals, and $\xi(s) = \xi(1-s)$ says that Υ^{-1} has the size biased law of Υ.

PROPOSITION 18.11.3 (Logistic partition). *For $c > 0$, $2\xi(s) = \mathbb{E}[\Upsilon^s/(1+\Upsilon^c)] + \mathbb{E}[\Upsilon^{s+c}/(1+\Upsilon^c)]$; with $c = 2b$ and $s = \frac{1}{2} + iz$ the two terms are $g_b(z+ib)$ and $g_b(z-ib)$. The logistic weight is the Pólya Gamma mixture $\Upsilon^c/(1+\Upsilon^c) = \frac{1}{2}\Upsilon^{c/2}\mathbb{E}_\varpi \exp(-\frac{1}{2}\varpi c^2(\log\Upsilon)^2)$ [159].*

Proof. The first identity is $1/(1+\Upsilon^c) + \Upsilon^c/(1+\Upsilon^c) = 1$ under $\mathbb{E}\Upsilon^s = 2\xi(s)$. With $s = \frac12 + iz$ and $c = 2b$ the first term is $\int 2\Phi(u)e^{izu}/(1+e^{2bu})\,du = \int \Phi(u)\operatorname{sech}(bu)e^{-bu}e^{izu}\,du = g_b(z+ib)$, and the second is $g_b(z-ib)$ likewise. The mixture is the case $a = b = 1$ of the identity of Polson, Scott and Windle, $\operatorname{sech}(cL/2) = \mathbb{E}e^{-\varpi c^2 L^2/2}$ with $L = \log\Upsilon$. □

Miyatake's identity is therefore the Pólya Gamma augmentation of a Bernoulli likelihood with log odds $9\log\Upsilon$, the constant 9 being the one that removes the growth factor $e^{9u/2}$ of the leading term of Φ. In the same language each term $n^4e^{-2\pi n^2\cosh t}$ of $\psi_{\mathrm M}$ is a generalised inverse Gaussian density in $r = e^t$, and Pólya's approximant is $8\pi^2 K_{9/4}(2\pi)\,\mathbb{E}\cos(\frac{z}{2}\log\varrho)$ with $\varrho \sim \mathrm{GIG}(\frac94, 2\pi, 2\pi)$, so that $\mathbb{E}\varrho^{iy} = K_{9/4+iy}(2\pi)/K_{9/4}(2\pi)$; the law of ϱ is hyperbolically completely monotone [30].

18.12 The transition at $b = 2$

Theorem 18.12.1 (Exponential tail). *Let $b > 2$, $\pi/(2b) < \tau < \min(\pi/4, 3\pi/(2b))$ and $Y > 0$. Then, uniformly for $\operatorname{Re} z \ge 0$ and $|\operatorname{Im} z| \le Y$,*

$$g_b(z) = C_b e^{-\pi z/(2b)} + O(e^{-\tau \operatorname{Re} z}), \qquad C_b = \frac{2\pi}{b}\,\Phi\Big(\frac{i\pi}{2b}\Big).$$

For $b < 2$ there is no residue and $g_b(x) = O(e^{-\tau x})$ for every $\tau < \pi/4$.

Proof. Shift the contour from $\mathbb{R}$ to $\operatorname{Im} u = \tau$. This is allowed because $\Phi(u + i\sigma)$ decays doubly exponentially in $\operatorname{Re} u$ for $0 \le \sigma \le \tau < \pi/4$ (the factor $\exp(-\pi e^{2|\operatorname{Re} u|}\cos 2\sigma)$ with $\cos 2\sigma > 0$) and Φ is analytic in $|\operatorname{Im} u| < \pi/4$. The only pole of $\operatorname{sech}(bu)$ between the two lines is $u_0 = i\pi/(2b)$, since $\tau < 3\pi/(2b)$, and it lies between them since $\tau > \pi/(2b)$. The residue of $\operatorname{sech}(bu)$ at u_0 is $1/(b\sinh(bu_0)) = 1/(ib)$, so $2\pi i$ times the residue of the integrand is $\frac{2\pi}{b}\Phi(u_0)e^{izu_0} = \frac{2\pi}{b}\Phi(\frac{i\pi}{2b})e^{-\pi z/(2b)}$; here $\Phi(iy)$ is real. On the line $\operatorname{Im} u = \tau$ we have $|e^{izu}| \le e^{-\tau\operatorname{Re} z + Y|\operatorname{Re} u|}$, $\operatorname{sech}(bu)$ is bounded because τ is not an odd multiple of $\pi/(2b)$, and $\Phi(v + i\tau)e^{Y|v|}$ is integrable in v; this gives the error term. For $b < 2$ the pole $i\pi/(2b)$ lies outside the strip $|\operatorname{Im} u| < \pi/4$ of Φ, so the shift crosses no pole and gives $O(e^{-\tau x})$ for every $\tau < \pi/4$. □

The pole of $\operatorname{sech}(bu)$ enters the strip of Φ exactly when the Lévy area kernel $e^{-\pi|y|/(2b)}$ outlasts the decay $|y|^{7/4}e^{-\pi|y|/4}$ of Ξ, so the two thresholds of Section 18.7 coincide. For Miyatake, $C_{9/2} = \frac{4\pi}{9}\Phi(i\pi/9) = 3.5527303732$, and in the scale of g the constant is $C_{9/2}/(4\pi^2) = 0.0899917116445$, matched by $g(x)e^{2\pi x/9}$ at $x = 40, 60, 80$. The sign is proved: on the imaginary axis the nth term of $\Phi(i\pi/9)$ is at most $2(2\pi^2 + 3\pi)n^4e^{-2.4066n^2}$ (the exponent is $\pi\cos(2\pi/9) = 2.4065996$), which sums to less than 2×10^{-6} over $n \ge 3$, while the first two terms have real parts 2.50683 and 0.03762.

THEOREM 18.12.2 (Cusp asymptotics). *With $\delta = \pi(b-2)/(4b)$, as $b \downarrow 2$,*

$$C_b = -\frac{\pi^3}{32b}(2\delta)^{-1/2}\delta^{-4}e^{-\frac{\pi}{8}\cot\delta}\big(1+O(b-2)\big).$$

Proof. Use theta functions with nome $e^{i\pi\tau}$, so that $\vartheta_4(\tau) = \vartheta_3(\tau+1)$ and $\vartheta_4(\tau) = (-i\tau)^{-1/2}\vartheta_2(-1/\tau)$. With Jacobi's $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ of (2.1.3), $\Phi(u) = 2e^{u/2}(2x^2\omega''(x) + 3x\omega'(x))$ with $x = e^{2u}$. Put $u = i\pi/(2b) = i(\frac{\pi}{4} - \delta)$. Then $x = ie^{-2i\delta}$ is near the cusp $x = i$. With $\varepsilon = 1 + ix = 2i\sin\delta\, e^{-i\delta}$ and $w = -i\varepsilon = 2\sin\delta\, e^{-i\delta}$ we have $1 + 2\omega(x) = \vartheta_3(ix) = \vartheta_4(\varepsilon) = w^{-1/2}\vartheta_2(-1/\varepsilon)$ and $1/\varepsilon = \frac{1}{2} - \frac{i}{2}\cot\delta$ exactly. The term $n = 0$ of $\vartheta_2(-1/\varepsilon) = 2\sum_{n\geq 0} e^{-i\pi(n+\frac{1}{2})^2/\varepsilon}$ is $2e^{-i\pi/8}e^{-(\pi/8)\cot\delta}$, and the derivative of its exponent with respect to x is $-\pi/(4\varepsilon^2)$. Hence the leading part of ω'' is $w^{-1/2}e^{-i\pi/8}e^{-(\pi/8)\cot\delta}\pi^2/(16\varepsilon^4)$, and that of $\Phi(u)$ is $2e^{u/2}\cdot 2x^2\omega''$. The phases are $e^{i\pi/8}e^{-i\delta/2}$ (from $e^{u/2}$), $e^{-4i\delta}$ (from x^2 up to the sign -1, as $x^2 = -e^{-4i\delta}$), $e^{i\delta/2}$ (from $w^{-1/2}$), $e^{4i\delta}$ (from ε^{-4}) and $e^{-i\pi/8}$; they multiply to exactly 1, so the leading term is real and negative, equal to $-\frac{\pi^2}{64}(2\delta)^{-1/2}\delta^{-4}e^{-(\pi/8)\cot\delta}(1+O(\delta))$ (moduli: $4\cdot|w|^{-1/2}\cdot\frac{\pi^2}{16}|\varepsilon|^{-4}$ with $|w| = 2\sin\delta$, $|\varepsilon| = 2\sin\delta$). Multiplying by $2\pi/b$ gives the formula. The relative corrections: the second derivative of the exponent and the derivative of $w^{-1/2}$ give $O(\delta)$; the term $3x\omega'$ is relatively $O(\delta^2)$; the terms $n \geq 1$ of ϑ_2 are relatively $O(e^{-\pi\cot\delta})$. Finally $O(\delta) = O(b-2)$; here δ is the distance of the pole $i\pi/(2b)$ from the edge $i\pi/4$ of the strip of Φ. □

The exact modular series (the ϑ_2 side) agrees with the asymptotics to ratios 0.9409, 0.9851, 0.9970, 0.9997 at $b = 2.02, 2.005, 2.001, 2.0001$ (where $C_b \approx -1.58\times 10^{-4324}$); the first ratio is reproduced by the direct series. By Theorem 18.12.2, $C_b < 0$ for b close to 2, and $C_{9/2} > 0$, so C_b has at least one zero in $(2, \frac{9}{2})$. A scan of $(2, 2.5)$ with step 10^{-5} and of $[2.47, 20]$ with step 10^{-3} finds exactly one sign change, at $b_0 = 2.4588641657$, with $C_b < 0$ on $(2, b_0)$ and $C_b > 0$ beyond; this uniqueness is computed on a grid and not certified (Problem 18.11).

COROLLARY 18.12.3. *For $b > 2$ with $C_b \neq 0$, g_b has finitely many real zeros, infinitely many nonreal zeros, and is not in the Laguerre Pólya class. In particular Miyatake's g, for which $b = \frac{9}{2}$ and $C_{9/2} > 0$, is not.*

Proof. Finitely many real zeros follows from Theorem 18.12.1, since $g_b(x) = C_b e^{-\pi x/(2b)}(1+o(1))$ has no sign change for large $|x|$ (g_b is even). A Laguerre Pólya function has only real zeros, so if g_b were in the class it would have finitely many zeros altogether; being of order at most one it would then be $P(z)e^{\alpha z}$ with P a polynomial, evenness forces $e^{2\alpha z} = P(-z)/P(z)$, hence $\alpha = 0$, and a nonzero polynomial cannot tend to 0 on $\mathbb{R}$, while $g_b(0) > 0$ and $g_b(x) \to 0$. The same Hadamard argument shows that g_b cannot have finitely many zeros, so it has infinitely many, all but finitely many nonreal. □

The nonreal zeros escape, heuristically, as follows. Writing $\Xi \approx E(\frac{1}{2} + iz) + E(\frac{1}{2} - iz)$ with $E(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$, the operator $\sec(b\partial)$ damps each term by about $2(x/2\pi)^{-b/2}$, and balancing against $C_b e^{-\pi z/(2b)}$ gives, with $L = \log(x/2\pi)$,

$$|\operatorname{Im} z| \approx b - \tfrac{1}{2} + \frac{2}{L}\Big[\frac{\pi(b-2)}{4b}x - \tfrac{3}{2}\log x + \log C_b - \tfrac{1}{2}\log 4\pi\Big]. \tag{18.12.1}$$

For $b = \frac{9}{2}$ this matches all 31 zeros of $g_{9/2}$ with real part below 160 (from 17.92 to 158.58) to within 0.14 to 0.27, about 0.15 beyond the first four; the first is $17.9168 + 10.3988i$, which in the scale of g is the vertex $8.9584 + 5.1994i$ of the cone of Section 18.7. By (18.12.1) the cone opens like $x/\log x$, so de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ fails already at the first zero (computed), by a margin that grows along the computed zeros (Figure 18.1). The formula (18.12.1) is a heuristic (Problem 18.10).

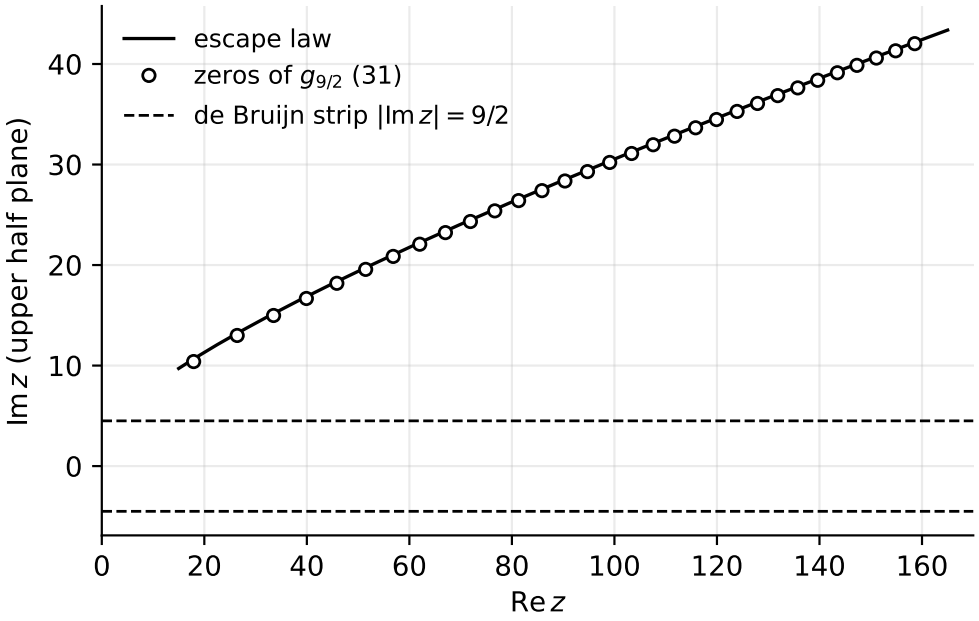


Figure 18.1: The 31 zeros of $g_{9/2}$ with real part below 160 in the upper half plane of z (circles), the escape law (18.12.1) (curve), and the lines $|\operatorname{Im} z| = \frac{9}{2}$ bounding de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ in the scale $w = z/2$ of g (dashed).

For $b < 2$ the root mean square de Bruijn radius of the Pólya Gamma clock, $(\mathbb{E}\, 4b^2\varpi)^{1/2} = b$, equals the strip width that de Bruijn's theorem needs for the shift identity. The strip nevertheless fails: below height 250 the largest imaginary parts of zeros of g_b are 0.367, 1.258, 2.153 for $b = 0.5$, 1, 1.5. Concentrating the

clock does not help. Let $\varpi_h = \mathrm{PG}(h,0)/h$, so that $\mathbb{E}e^{-s\varpi_h} = \cosh(\sqrt{s/2h})^{-h}$ with mean $\frac14$; the corresponding weight at radius b is $\operatorname{sech}(bu/\sqrt{h})^h$, and $h\to\infty$ is the deterministic flow $\mathcal{H}_{-b^2/2}$. At radius 1 the clocks PG(1), PG(4), PG(16) and the deterministic flow $\mathcal{H}_{-1/2}$ give 28 nonreal pairs in $[0,250]\times[-3,3]$ for each (only 27 of the pairs of PG(4) have imaginary part above 0.125) and largest imaginary parts 1.258, 1.331, 1.322, 1.325. The obstruction is the backward direction of the flow, for which de Bruijn's radius has no counterpart.

18.13 The Pólya mixture of ψ_{M}

This section is based on [240]. Write $c=\cosh t$. The map $t\mapsto\cosh t$ takes the strip $|\operatorname{Im} t|<\frac{\pi}{2}$, modulo $t\mapsto -t$, one to one onto the half plane $\operatorname{Re} c>0$, and $\psi_{\mathrm{M}}(t)=\Phi(t/2)/(8\pi^2\cosh(9t/4))$ is even, so ψ_{M} is a function of c there. Without their factors $1-R_{\mathrm{M}}$ the terms of (18.7.3) are $n^4e^{-2\pi n^2c}$, Laplace transforms in c of atoms at $a=2\pi n^2$, and by (18.7.2) they contribute to Ξ the function $\xi^{(P)}$ of Section 18.7 with every n admitted, which we write $\xi^{(\infty)}$. Each of its terms $8\pi^2n^4\operatorname{Re}K_{\frac94+\frac{it}{2}}(2\pi n^2)$, the transform of $8\pi^2n^4\cosh(9u/2)e^{-2\pi n^2\cosh 2u}$, has only real zeros: $\cosh(9u/2)$ is a universal factor [246], and $K_{\frac{it}{2}}(2\pi n^2)$, the transform of $e^{-2\pi n^2\cosh 2u}$, has only real zeros [245]; for $n=1$ this is Ξ^* of Proposition 22.7.1.

Proposition 18.13.1 (The polar part). *As a function of $c=\cosh t$, ψ_{M} is analytic on $\operatorname{Re}c>0$ except for a simple pole at $c_0=\cos\frac{2\pi}{9}$, with residue*

$$r_0=\frac{\sin\frac{2\pi}{9}}{18\pi^2}\,\Phi\Big(\frac{i\pi}{9}\Big)=\frac{\sin\frac{2\pi}{9}}{8\pi^3}\,C_{9/2}>0.$$

Hence, with $\psi_{\mathrm{E}}(t)=r_0\,e^{-2\pi(\cosh t-c_0)}/(\cosh t-c_0)=r_0\int_{2\pi}^\infty e^{c_0a}e^{-a\cosh t}\,da$,

$$\psi_{\mathrm{M}}(t)=\sum_{n\ge1}n^4e^{-2\pi n^2\cosh t}+\psi_{\mathrm{E}}(t)+\psi_{\mathrm{A}}(t),\tag{18.13.1}$$

where ψ_{A} is analytic in c on all of $\operatorname{Re}c>0$.

Proof. $\Phi(t/2)$ is analytic for $|\operatorname{Im}t|<\frac{\pi}{2}$ (Section 18.11), and an even analytic function of t is an analytic function of c near $c=1$, the only critical value of cosh in the strip. In the strip the zeros of $\cosh(9t/4)$ are $t=\pm2\pi i/9$; they are simple, both map to c_0, and $dc/dt=\sinh t=i\sin\frac{2\pi}{9}\neq0$ at $t=2\pi i/9$. The residue in c is therefore $\Phi(i\pi/9)\,i\sin\frac{2\pi}{9}/(8\pi^2\cdot\frac94 i)$, which is the formula, and $C_{9/2}=\frac{4\pi}{9}\Phi(i\pi/9)>0$ by Section 18.12. The atoms are entire in c, and ψ_{E} has the same simple pole with the same residue and no other singularity, so ψ_{A} is analytic on $\operatorname{Re}c>0$. The integral form is $\int_{2\pi}^\infty e^{-a(c-c_0)}\,da=e^{-2\pi(c-c_0)}/(c-c_0)$ for $\operatorname{Re}c>c_0$. □

Here $r_0 = 0.009206406367$, all digits certified by interval arithmetic, and the exponent c_0 of the exponential part depends only on the factor $\cosh(9t/4)$ of (18.7.2) and on the map $c = \cosh t$; Φ enters only through r_0. The lower limit 2π, the position of the first atom, is a choice: any $a_0 > 0$ gives the same pole.

PROPOSITION 18.13.2 (The exponent is forced). *Let $0 < \theta < \frac{\pi}{2}$, $a_0 > 0$, and*

$$\begin{aligned}\mathcal{E}_\theta(z) &= \int_{\mathbb{R}} \frac{\cosh(9u/2)\, e^{-a_0(\cosh 2u - \cos\theta)}}{\cosh 2u - \cos\theta}\, e^{izu}\, du \\ &= \int_{a_0}^{\infty} e^{a\cos\theta}\, \tfrac{1}{2}\Big\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\Big\}\, da,\end{aligned}$$

the exponential mixture with exponent $\cos\theta$ of the Pólya functions $\frac{1}{2}\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\}$ of Proposition 22.7.1. For $\theta/2 < \tau < \frac{\pi}{4}$ and $Y > 0$, uniformly for $\operatorname{Re} z \geq 0$ and $|\operatorname{Im} z| \leq Y$,

$$\mathcal{E}_\theta(z) = \frac{\pi\cos(9\theta/4)}{\sin\theta}\, e^{-\theta z/2} + O(e^{-\tau \operatorname{Re} z}).$$

If $\theta \neq \frac{2\pi}{9}$, then $\mathcal{E}_\theta$ has finitely many real zeros and infinitely many nonreal zeros, and it is not in the Laguerre Pólya class.

Proof. Shift the contour from $\mathbb{R}$ to $\operatorname{Im} u = \tau$ as in the proof of Theorem 18.12.1: on $\operatorname{Im} u = \sigma$, $0 \leq \sigma \leq \tau < \frac{\pi}{4}$, the factor $e^{-a_0\cosh 2u}$ decays doubly exponentially, and the only pole between the two lines is $u_0 = i\theta/2$, where $\cosh 2u_0 = \cos\theta$; the next poles, $\pm i\theta/2 + i\pi\mathbb{Z}$, lie above $\frac{\pi}{4}$. $2\pi i$ times the residue at u_0 is $2\pi i\cos(9\theta/4)e^{-\theta z/2}/(2i\sin\theta)$. For $0 < \theta < \frac{\pi}{2}$, $\cos(9\theta/4) = 0$ only at $\theta = \frac{2\pi}{9}$. The kernel is positive, so $\mathcal{E}_\theta$ is even, real, of order at most one, positive at 0 and tends to 0 on $\mathbb{R}$, and the proof of Corollary 18.12.3 gives the rest. The Bessel form follows from $e^{-a_0(c-\cos\theta)}/(c-\cos\theta) = \int_{a_0}^{\infty} e^{-a(c-\cos\theta)}\, da$ for $c \geq 1$ and $\int_{\mathbb{R}} \cosh(9u/2)e^{-a\cosh 2u}e^{izu}\, du = \frac{1}{2}\{K_{\frac{9}{4}+\frac{iz}{2}}(a) + K_{\frac{9}{4}-\frac{iz}{2}}(a)\}$, which is $\int_{\mathbb{R}} e^{\nu v - a\cosh v}\, dv = 2K_\nu(a)$ after $v = 2u$. □

So among the exponential mixtures with exponent in $(0,1)$ only $c_0 = \cos\frac{2\pi}{9}$ can have only real zeros, and by Proposition 18.13.1 this is the exponent of the polar part of ψ_{M}: with $a_0 = 2\pi$, $\Xi^{\mathrm{E}} = 8\pi^2 r_0\, \mathcal{E}_{2\pi/9}$ is the transform of $8\pi^2\cosh(9u/2)\psi_{\mathrm{E}}(2u)$, whose apparent pole at $u = i\pi/9$ is cancelled by the zero of $\cosh(9u/2)$, and by (18.7.2) and (18.13.1), $\Xi = \xi^{(\infty)} + \Xi^{\mathrm{E}} + \Xi^{\mathrm{A}}$ with Ξ^{A} the transform of $8\pi^2\cosh(9u/2)\psi_{\mathrm{A}}(2u)$.

PROPOSITION 18.13.3 (A Meixner convolution of a Pólya type function). *Let $c_1 = \cos\frac{\pi}{9}$, so that $c_0 = 2c_1^2 - 1$, and*

$$\mathcal{U}(u) = \frac{\cosh(9u/2)}{2(\cosh u - c_1)} = \cosh\tfrac{7u}{2} + 2c_1\cosh\tfrac{5u}{2} + (1+2c_0)\cosh\tfrac{3u}{2} + (1+2c_1)\cosh\tfrac{u}{2}, \tag{18.13.2}$$

so that $\cosh(9u/2)/(\cosh 2u - c_0) = \mathcal{U}(u)/(\cosh u + c_1)$.

(i) $\mathcal{U}$ *is a universal factor, and* $\Xi^{\mathcal{U}}(z) = \int_{\mathbb{R}} \mathcal{U}(u) e^{-2\pi \cosh 2u} e^{izu}\, du = \sum_j m_j \cos(\frac{b_j}{2}\partial) P$, *with* $(b_j, m_j) = (7,1), (5, 2c_1), (3, 1+2c_0), (1, 1+2c_1)$ *and* P *of* (18.1.1), *is in the Laguerre Pólya class.*

(ii) For $0 \le \varphi < \pi$ *the function* $q_\varphi(x) = \frac{(1+\cos\varphi)\sinh(\varphi x)}{\sin\varphi\, \sinh(\pi x)}$, *with* $q_0(x) = 2x/\sinh(\pi x)$, *is a probability density with* $\int_{\mathbb{R}} q_\varphi(x) e^{iux}\, dx = \frac{1+\cos\varphi}{\cosh u + \cos\varphi}$ *and variance* $1/(1+\cos\varphi)$, *and it is a Pólya frequency function. Moreover* $\frac{1+\cos\varphi}{\cosh u + \cos\varphi} = \mathbb{E} e^{-\varsigma_\varphi u^2}$ *with*

$$\varsigma_\varphi = \sum_{k\ge 0} \Big(\frac{e_k}{((2k+1)\pi - \varphi)^2} + \frac{e'_k}{((2k+1)\pi + \varphi)^2} \Big), \qquad \mathbb{E}\varsigma_\varphi = \frac{1}{2(1+\cos\varphi)},$$

e_k, e'_k *independent standard exponential variables, so the transform of* $k(u)(1+\cos\varphi)/(\cosh u + \cos\varphi)$ *is* $q_\varphi * \hat{k}$ *for every kernel* k *for which these integrals converge; for* $k = \Phi$ *it is* $\mathbb{E}\,\mathcal{H}_{-\varsigma_\varphi}$, *a de Bruijn Newman flow at a random negative time, and* $\varphi = \frac{\pi}{2}$ *is* g_b *of Theorem 18.11.2 with* $b = 1$.

(iii) $\Xi^{\mathrm{E}} = \dfrac{8\pi^2 r_0 e^{2\pi c_0}}{1+c_1}\, q_{\pi/9} * \Xi^{\mathcal{U}}$.

Proof. The identity: $\cosh 2u - c_0 = 2(\cosh u - c_1)(\cosh u + c_1)$, and multiplying the four term sum by $2(\cosh u - c_1)$ with $2\cosh u \cosh\frac{bu}{2} = \cosh\frac{(b+2)u}{2} + \cosh\frac{(b-2)u}{2}$ leaves $\cosh\frac{9u}{2}$, the other coefficients vanishing by $2+2c_0 = 4c_1^2$ and $4c_1c_0 = 1+2c_1$, the latter being $4c_1^3 - 3c_1 = \cos\frac{\pi}{3} = \frac{1}{2}$. (i) $\mathcal{U}(iw) = \cos\frac{9w}{2}/(2(\cos w - c_1))$, and the zeros $\pm\frac{\pi}{9} + 2\pi\mathbb{Z}$ of $\cos w - c_1$ are simple zeros of $\cos\frac{9w}{2}$, so $\mathcal{U}(iw)$ is an even real entire function of exponential type whose zeros, the remaining zeros of $\cos\frac{9w}{2}$, are real; it is in the Laguerre Pólya class, and $\mathcal{U}$ is a universal factor by Pólya's theorem [246]. P has only real zeros [245], hence so has $\Xi^{\mathcal{U}}$, which is even of order one, so it is in the class. The shift form is $\int_{\mathbb{R}} k(u)\cosh(bu) e^{izu}\, du = \frac{1}{2}\{\hat{k}(z+ib) + \hat{k}(z-ib)\}$, as in (18.10.1), for $k(u) = e^{-2\pi\cosh 2u}$, whose transform is P, and $b = \frac{b_j}{2}$. (ii) For $0 < \varphi < \pi$ the residue theorem on the rectangle of height 2π, with poles at $i(\pi \mp \varphi)$, gives $\int_{\mathbb{R}} e^{ivx}\, dx/(\cosh x + \cos\varphi) = 2\pi\sinh(\varphi v)/(\sin\varphi\sinh(\pi v))$; Fourier inversion gives the pair, $\varphi \to 0$ gives q_0, the value 1 at $u = 0$ the mass, and the coefficient of u^2 the variance. The bilateral Laplace transform $\int q_\varphi(x) e^{sx}\, dx = (1+\cos\varphi)/(\cos s + \cos\varphi)$, $|\operatorname{Re} s| < \pi - \varphi$, is the reciprocal of an even function of order one with the real zeros $\pm((2k+1)\pi \mp \varphi)$, so q_φ is a Pólya frequency function by Schoenberg's theorem [254] (Theorem 19.5.3). The same zeros give $(\cosh u + \cos\varphi)/(1+\cos\varphi) = \prod_{k\ge 0}(1 + u^2/((2k+1)\pi - \varphi)^2)(1 + u^2/((2k+1)\pi + \varphi)^2)$, and inverting each factor as in the proof of Theorem 18.11.2 gives $\mathbb{E} e^{-\varsigma_\varphi u^2}$, with the mean read off the coefficient of u^2. The convolution form is Fubini's theorem. (iii) Multiply kernels, using (18.13.2) and $1/(\cosh u + c_1) = (1+c_1)^{-1}\int q_{\pi/9}(x) e^{iux}\, dx$. □

For $0 < \varphi < \pi$, q_φ is the law of $(\log V - \log V')/(2\pi)$, where V and V' are independent with the inverted beta density $v^{a-1}/(B(a, 1-a)(1+v))$ on $(0,\infty)$ and $a = \frac{1}{2} + \frac{\varphi}{2\pi}$: by Euler's beta integral and the reflection formula, $\mathbb{E}e^{iu \log V/(2\pi)} = \cos\frac{\varphi}{2} / \cos\frac{\varphi+iu}{2}$, and the product of this with its complex conjugate is $(1+\cos\varphi)/(\cosh u + \cos\varphi)$. Each $\log V/(2\pi)$ is a Meixner variable [90], the logarithm of the local variance in the shrinkage priors of Polson and Scott [155]; for $\varphi = \frac{\pi}{9}$, $a = \frac{5}{9}$.

Three computations bear on Ξ^{E} and the pieces of (18.13.1) (computed). First, Ξ^{E} has 452 zeros in $[0.25, 750] \times [-3, 3]$, all real, with least gap 1.314 at height about 749, against 442.9 for $\frac{T}{2\pi}\log\frac{T}{2\pi e}$ evaluated between 50 and 750, where Ξ^{E} has 443 zeros. Second, the same convolution does not keep the zeros of Ξ real: in $(0, 250]$, $q_\varphi * \Xi$ has 80 real zeros for $\varphi = \frac{\pi}{9}$, 80 for $\varphi = 0$ and 72 for $\varphi = \frac{\pi}{4}$, against 108 for Ξ and 51 for g_b with $b = 1$, the case $\varphi = \frac{\pi}{2}$ (Section 18.11); for $\varphi = \frac{\pi}{9}$ the zeros in $[0.5, 250] \times [-3, 3]$ are real below height 95.5, and the first nonreal one is $95.520 + 0.2361\,i$. For $0 \le \varphi \le \frac{\pi}{2}$, $(\cosh u + \cos\varphi)/(1+\cos\varphi) = \mathbb{E}\cosh(uX)$ with $\Pr(X = 0) = \cos\varphi/(1+\cos\varphi)$ and $\Pr(X = \pm 1) = 1/(2(1+\cos\varphi))$, so these are instances of Problem 18.6, answered in the expected direction below height 250. Third, $\xi^{(\infty)} + \Xi^{\mathrm{E}}$ has 229 real zeros in $(50, 500]$ and 261 zeros in $[50, 500] \times [-3, 3]$, that is 16 conjugate pairs of nonreal zeros, the first at $188.502 + 0.228\,i$, while every term of $\xi^{(\infty)}$ has only real zeros: real zeros of the pieces of (18.13.1) do not pass to their sum. The function $\xi^{(\infty)}$ alone has the same two counts, so Ξ^{E} moves the zeros slightly and changes neither count. The counts come from the argument principle at 160 bits with two quadratures agreeing to 10^{-36}, and interval Newton encloses the two first nonreal zeros, $95.5202007665 + 0.2361064242\,i$ and $188.5020229515 + 0.2278349245\,i$, each as a simple zero up to the quadrature allowance. Heuristically the random negative time of Proposition 18.13.3(ii) pulls the close pairs of zeros of Ξ off the line, as for g_b, while the zeros of $\Xi^{\mathcal{U}}$, like those of Pólya's Ξ^*, are close to evenly spaced.

18.14 The ladder and the quasi Riemann abscissa

The ladder F_b of Section 18.10 is the opposite symmetrisation: $F_b = \cos(b\partial)\Xi$ while $g_b = \sec(b\partial)\Xi$, and in the Mellin variable $F_b(z) = \frac{1}{4}\mathbb{E}[\Upsilon^s(\Upsilon^b + \Upsilon^{-b})]$ with $s = \frac{1}{2} + iz$, the Rademacher Esscher weight in place of the logistic coin. On $\mathbb{R}$, $F_b(x) = \operatorname{Re}\xi(\alpha + ix)$ with $\alpha = \frac{1}{2} + b$. Theorem 18.10.1 is due to de Bruijn and Lagarias, who also showed that on RH all rungs $b > 0$ are real rooted (Theorem 22.6.1). Let $\sigma_{\max} = \sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros of ζ, and let $\gamma_k(b)$ be as in Theorem 18.10.1.

Proposition 18.14.1. $B_{\cosh} \le \sigma_{\max} - \frac{1}{2}$.

Proof. The zeros of Ξ satisfy $|\operatorname{Im} t| \le \Delta = \sigma_{\max} - \frac{1}{2}$. De Bruijn's strip theorem with Δ in place of $\frac{1}{2}$ in the proof of Theorem 18.10.1 shows that the zeros of

F_b satisfy $|\operatorname{Im} z| \le (\max(\Delta^2 - b^2, 0))^{1/2}$, so F_b is real rooted for $b \ge \Delta$ (Theorem 22.6.1(ii)). □

Numerically, below height 100 the number of real zeros of $\operatorname{Re}\xi(\alpha + it)$ is $29 = N(100)$ for every $\alpha \in \{0.5, 0.55, 0.6, 0.7, 0.8, 1, 1.25, 1.5\}$ tested, and the first twelve Thorin atoms $\gamma_k(b)$, which move continuously where the zeros are simple, decrease strictly in b on the grid $\alpha = 0.50, 0.51, \dots, 1.00$ (Figure 18.2); the smallest gap $\gamma_{k+1} - \gamma_k$ is 1.77.

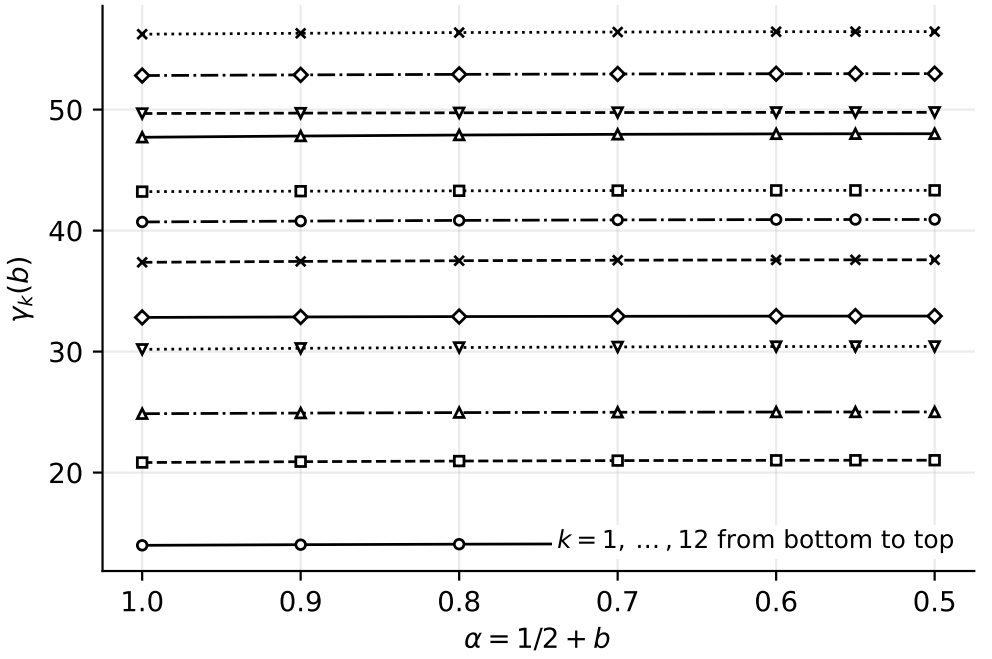


Figure 18.2: The Thorin path: the first twelve atoms $\gamma_k(b)$ against $\alpha = \frac{1}{2} + b$ at the stored values $\alpha = 1, 0.9, 0.8, 0.7, 0.6, 0.55, 0.5$, from the unconditional rung $\alpha = 1$ to the critical line.

Conversely, at height x and $z = x + iy$ with $y > 0$, Stirling and the functional equation give

$$\frac{|\Xi(z + ib)|}{|\Xi(z - ib)|} \approx \Big(\frac{x}{2\pi}\Big)^{y} \frac{|\zeta(\alpha + y + ix)|}{|\zeta(\alpha - y + ix)|},$$

so the two terms of F_b can cancel off the axis only near zeros of ζ with $\beta \approx \alpha + y$, and, heuristically, Rouché near a simple zero with $\beta > \alpha$ produces a zero of F_b near $\gamma + i(\beta - \alpha)$. The cosh shift moves the effective abscissa by the fixed amount b; the heat flow, in the Rodgers Tao approximation, moves it by $\lambda \log(x/2\pi)$, which clears every zero at large height and is why Ki, Kim and Lee [182] have

only finitely many nonreal zeros for every $\lambda > 0$ and the Polymath bound [247] is reachable.

CONJECTURE 18.14.2. *Let* $\sigma_{\max}^{\infty} = \limsup_{\gamma\to\infty} \beta$. *Then* $\sigma_{\max}^{\infty} - \frac{1}{2} \le B_{\cosh} \le \sigma_{\max} - \frac{1}{2}$. *The upper bound is Proposition 18.14.1; the lower bound is conjectured. Under the conjecture, lowering the unconditional bound* $B_{\cosh} \le \frac{1}{2}$ *is equivalent to a quasi Riemann hypothesis whenever* $\sigma_{\max}$ *is approached at large height.*

We test the mechanism on a function with genuine off line zeros. Let χ be the character modulo 5 with $\chi(2) = i$ and $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = 0.2840790438$. The Davenport Heilbronn function [60] is $f(s) = \frac{1-i\kappa}{2} L(s, \chi) + \frac{1+i\kappa}{2} L(s, \bar{\chi})$, completed as $\Lambda(s) = (5/\pi)^{s/2} \Gamma(\frac{s+1}{2}) f(s)$; it satisfies $\Lambda(s) = \Lambda(1 - s)$ (to 2.4×10^{-29} at 30 digits) and is real on the critical line, and its first zero off the line is $0.80851718246 + 85.69934848538i$. The ladder built from Λ has a nonreal pair above this zero for α up to about 0.798, slightly below β. The imaginary part of the pair is 0.2893, 0.2244, 0.1019, 0.0685, 0.0425 at $\alpha = 0.6$, 0.7, 0.78, 0.79, 0.795, and its square falls almost linearly near the collision (the slopes of the square are -0.33, -0.50, -0.57, -0.57 between consecutive values). The pair collides on the axis at $\alpha = 0.798095$, $x = 85.6255$, and is real for $\alpha \ge 0.80$, with the total count of zeros in the box $[83, 88.5] \times [-0.9, 0.9]$ conserved at 4. The small early collision is a finite height effect, since $(x/2\pi)^y$ is only 13.6^y at height 85; it is the reason for $\sigma_{\max}^{\infty}$ rather than $\sigma_{\max}$ in the lower bound.

18.14.1 An exact paired form

The ladder uses only the real part of ξ on the vertical line; pairing it with the imaginary part gives an exact statement. For $\alpha > \frac{1}{2}$ put $E_\alpha(z) = \xi(\alpha - iz)$, $A_\alpha = \frac{1}{2}(E_\alpha + E_\alpha^{\#})$ and $B_\alpha = \frac{i}{2}(E_\alpha - E_\alpha^{\#})$, with $E^{\#}(z) = \overline{E(\bar{z})}$, so that $A_\alpha(t) = \operatorname{Re} \xi(\alpha + it) = F_{\alpha - 1/2}(t)$ and $B_\alpha(t) = \operatorname{Im} \xi(\alpha + it)$ for real t, and $E_\alpha = A_\alpha - iB_\alpha$. Here A_α is the even part of Section 22.6 evaluated at iz.

THEOREM 18.14.3. *For* $\alpha > \frac{1}{2}$ *the following are equivalent: (i)* $\zeta(s) \ne 0$ *for* $\operatorname{Re} s \ge \alpha$; *(ii)* E_α *is Hermite Biehler, that is* $|E_\alpha(z)| > |E_\alpha^{\#}(z)|$ *for* $\operatorname{Im} z > 0$; *(iii)* $\operatorname{Re} \xi(\alpha + it)$ *and* $\operatorname{Im} \xi(\alpha + it)$, *as entire functions of* t, *have only real simple zeros, and these strictly interlace. Hence the paired bottom rung, the infimum of the* $\alpha - \frac{1}{2}$ *for which these hold, is exactly* $\sigma_{\max} - \frac{1}{2}$, *and* RH *holds if and only if (iii) holds for every* $\alpha > \frac{1}{2}$.

Proof. (i)⇒(ii): for $z = x + iy$ with $y > 0$ we have $E_\alpha(z) = \xi(\alpha + y - ix)$ and $E_\alpha^{\#}(z) = \xi(\alpha - y + ix)$. Let $\rho = \beta + i\gamma$ run over the zeros of ξ, a set symmetric under $\rho \mapsto \bar{\rho}$. By the Hadamard product, pairing ρ with $\bar{\rho}$,

$$\frac{|E_\alpha(z)|^2}{|E_\alpha^{\#}(z)|^2} = \prod_\rho \frac{(\alpha + y - \beta)^2 + (x - \gamma)^2}{(\alpha - y - \beta)^2 + (x - \gamma)^2} = \prod_\rho \Big(1 + \frac{4y(\alpha - \beta)}{|\alpha - y + ix - \rho|^2}\Big),$$

and every factor exceeds 1 when $\beta < \alpha$. The product converges absolutely because $\sum \gamma^{-2} < \infty$, and E_α has no real zeros since no β equals α. Hence $|E_\alpha| > |E_\alpha^\#|$ on the upper half plane. (ii)⇒(iii) is the Hermite Biehler theorem in de Branges's form: if $E = A - iB$ is Hermite Biehler, then A and B are real entire with only real simple zeros that strictly interlace. (iii)⇒(i): A_α and B_α are real entire of order one with only real zeros, hence of genus at most one; A_α is even and B_α is odd, so their Hadamard products carry no factor e^{az}: $A_\alpha(z) = \xi(\alpha)\prod_k(1 - z^2/a_k^2)$ and $B_\alpha(z) = \xi'(\alpha)z\prod_k(1 - z^2/b_k^2)$ (without the parity the step fails, since interlacing alone allows an extra factor e^{az}). For strictly interlacing zeros the quotient of these products is, up to sign, a Pick function by Chebotarev's theorem [269]. Since $B_\alpha/A_\alpha \sim (\xi'(\alpha)/\xi(\alpha))z$ at 0, with $\xi(\alpha) > 0$ and $\xi'(\alpha) > 0$ ($\xi'(\alpha) > 0$ holds unconditionally for $\alpha > \frac{1}{2}$, because $\xi'/\xi(\alpha) = \sum_\rho(\alpha - \rho)^{-1}$ is real and each pair ρ, $1 - \rho$ contributes $\operatorname{Re}\left[(\alpha - \rho)^{-1} + (\alpha - 1 + \rho)^{-1}\right] = (2\alpha - 1)(xy + \gamma^2)/((x^2 + \gamma^2)(y^2 + \gamma^2)) > 0$, with $x = \alpha - \beta$, $y = \alpha - 1 + \beta$ and $xy > -\frac{1}{4} > -\gamma^2$), the sign is +: B_α/A_α is a Pick function and $\operatorname{Im}(B_\alpha/A_\alpha) > 0$ on the upper half plane. The identity $|E_\alpha^\#|^2 - |E_\alpha|^2 = -4|A_\alpha|^2 \operatorname{Im}(B_\alpha/A_\alpha)$ then gives $|E_\alpha| > |E_\alpha^\#| \geq 0$ on the upper half plane, and $E_\alpha \neq 0$ on $\mathbb{R}$ because the zeros of A_α and B_α are disjoint. So $E_\alpha(z) = \xi(\alpha - iz)$ has no zeros in the closed upper half plane; since $E_\alpha(z) = 0$ at $z = -\gamma + i(\beta - \alpha)$, this says $\beta < \alpha$ for every zero. (The other sign is excluded by Hardy: $E_\alpha^\#(z) = \xi(\alpha + iz)$ vanishes at $\gamma + i(\alpha - \frac{1}{2})$ for every zero on the line, so it is not Hermite Biehler.) The last two statements follow. □

Its Thorin form is the following. Since A_α is even with $A_\alpha(0) = \xi(\alpha) > 0$ and B_α is odd with $B'_\alpha(0) = \xi'(\alpha) > 0$,

$$\frac{2\xi(\alpha)}{\xi(\alpha + \sqrt{s}) + \xi(\alpha - \sqrt{s})} \quad \text{and} \quad \frac{2\sqrt{s}\,\xi'(\alpha)}{\xi(\alpha + \sqrt{s}) - \xi(\alpha - \sqrt{s})}$$

(these are $\xi(\alpha)/A_\alpha(i\sqrt{s})$ and $B'_\alpha(0)/(B_\alpha(z)/z)$ at $z = i\sqrt{s}$) are Laplace transforms of generalised gamma convolutions with strictly interlacing Thorin atoms exactly when ζ has no zeros with $\operatorname{Re} s \geq \alpha$. Call two such generalised gamma convolutions, whose Thorin atoms a_k^2 and b_k^2 strictly interlace, a Thorin pair: the even and odd parts of ξ about α form a Thorin pair precisely to the right of every zero. For ζ the interlacing holds on $[0, 100]$ at $\alpha = 0.55$, 0.6, 0.7, 1 (and 0.8, 1.25, 1.5), with 29 positive zeros of each function and smallest distance between a zero of A_α and one of B_α equal to 0.50, 0.51, 0.54, 0.69 for $\alpha = 0.55$, 0.6, 0.7, 1. For Davenport Heilbronn the criterion is sharp where the ladder alone is not: at $\alpha = 0.800$ and 0.805 the real part already has only real zeros on $(83, 88.5)$, yet the zeros of the real (A) and imaginary (B) parts come in the order ABAABBAB, while at $\alpha = 0.82$ and 0.85 the order is ABABABAB.

18.14.2 Density one by phase counting

Let $\theta_\alpha(t) = \arg \xi(\alpha + it)$; for $\alpha > \sigma_{\max}$ this is π times the clock U_α of Proposition 22.6.3. Since $\operatorname{Re}(\xi'/\xi)(s) = \sum_\rho \operatorname{Re}(s-\rho)^{-1}$ (Hadamard's product; the same identity is used in the proof of Proposition 22.6.3),

$$\theta'_\alpha(t) = \sum_\rho \frac{\alpha - \beta}{(\alpha-\beta)^2 + (t-\gamma)^2},$$

a sum of Cauchy bumps of mass $+\pi$ for $\beta < \alpha$ and $-\pi$ for $\beta > \alpha$. Write $N^\pm(T)$ for the number of zeros with $|\gamma| \le T$ (so $N^\pm = 2N$ with the usual N counting $0 < \gamma \le T$) and $N^\pm_{>\alpha}(T)$ for those with $\beta > \alpha$.

THEOREM 18.14.4. *Let $\alpha > \frac12$ with $\beta \ne \alpha$ for every zero. Then $\theta_\alpha(T) - \theta_\alpha(-T) = \pi(N^\pm(T) - 2N^\pm_{>\alpha}(T)) + O(\log T)$. Hence $\operatorname{Re}\xi(\alpha+it)$ and $\operatorname{Im}\xi(\alpha+it)$ each have at least $N^\pm(T) - 2N^\pm_{>\alpha}(T) - O(\log T)$ real zeros in $[-T,T]$. If moreover F_b, $b = \alpha - \frac12$, satisfies the Riemann von Mangoldt formula with error $O(\log T)$, its nonreal zeros with $|\operatorname{Re} z| \le T$ number at most $2N^\pm_{>\alpha}(T) + O(\log T)$.*

Proof. Integrating θ'_α over $[-T,T]$, a zero ρ with $|\gamma| \le T$ contributes $\operatorname{sgn}(\alpha - \beta)\,(\arctan \frac{T-\gamma}{|\alpha-\beta|} + \arctan \frac{T+\gamma}{|\alpha-\beta|})$, which is $\pm\pi$ up to a remainder of size $|\alpha-\beta|/(T - |\gamma|)$, and a zero with $|\gamma| > T$ contributes a remainder of size $|\alpha-\beta|/(|\gamma| - T)$, of the opposite sign. Zeros within distance 1 of $\pm T$ number $O(\log T)$ and contribute $O(\log T)$ in all. The remaining remainders sum to $O(\log^2 T)$ if estimated separately; pair ρ with $1-\rho$, which has the same ordinate and combined weight $(\alpha-\beta)+(\alpha-1+\beta) = 2\alpha - 1$, independent of β. Up to $O(\log T)$ the remainders are then $(2\alpha-1)\sum_\gamma \operatorname{sgn}(|\gamma|-T)\big(\frac{1}{||\gamma|-T|} - \frac{1}{|\gamma|+T}\big)$, and partial summation with $N^\pm(t) = \frac{t}{\pi}\log\frac{t}{2\pi e} + O(\log t)$ shows that the contributions from below and from above T cancel to $O(\log T)$. This proves the first formula. A continuous phase θ_α with total change Δ crosses the lattice $\pi\mathbb{Z}$ (zeros of $\operatorname{Im}\xi(\alpha+it)$) and the lattice $\pi\mathbb{Z} + \frac{\pi}{2}$ (zeros of $\operatorname{Re}\xi(\alpha+it)$) at least $|\Delta|/\pi - 1$ times each, with $\Delta = \theta_\alpha(T) - \theta_\alpha(-T)$. Under the Riemann von Mangoldt formula F_b has $N^\pm(T) + O(\log T)$ zeros with $|\operatorname{Re} z| \le T$, of which at least $N^\pm(T) - 2N^\pm_{>\alpha}(T) - O(\log T)$ are real. □

With Ingham's estimate [176], $N^\pm_{>\alpha}(T) = O(T^{3(1-\alpha)/(2-\alpha)}\log^5 T) = o(T)$ for every $\alpha = \frac12 + b > \frac12$, since $3(1-\alpha)/(2-\alpha) < 1$. So, unconditionally, $\operatorname{Re}\xi(\alpha+it)$ has at least $N^\pm(T) - o(T)$ real zeros in $[-T,T]$, against $N^\pm(T) \sim \frac{T}{\pi}\log\frac{T}{2\pi}$. If the rung F_b satisfies the Riemann von Mangoldt formula with error $O(\log T)$, which is open (Problem 18.8), almost all zeros of $\frac12(\xi(s+b) + \xi(s-b))$ lie on the critical line, and as a count the number of conjugate pairs of nonreal zeros up to height T is at most $N^\pm_{>\alpha}(T) + O(\log T)$. For the Davenport Heilbronn function, whose zeros off the line below height 200 have $\beta = 0.8085$, 0.6508, 0.5744, 0.7243 (computed in $0.52 \le \operatorname{Re} s \le 1.5$, $0.1 \le \operatorname{Im} s \le 200$, at heights 85.70, 114.16, 166.48, 176.70), the real part of $\Lambda(\alpha + it)$ has 130, 130, 126, 124 real zeros on

$(0.5, 200)$ at $\alpha = 0.95, 0.85, 0.7, 0.6$, where 0, 0, 2, 3 of those zeros lie to the right of α: the deficits 4 and 6 are two real zeros per zero to the right, as the count predicts. The nonreal zeros of F_b were not counted directly, so that each such zero costs one conjugate pair is established only at the level of these counts. For ζ itself, the error $\theta_\alpha(T) - \theta_\alpha(-T) - \pi N^{\pm}(T)$ at $T = 20, 40, 60, 80, 100$ stays below 1.6 for $\alpha = 0.6, 0.75, 1$ (computed).

18.14.3 Interlacing across rungs

This subsection is based on [241]. Theorem 18.14.3 interlaces A_α with B_α on one rung; for $\alpha > 1$ the Euler product compares different rungs. Normalise $\theta_\alpha(0) = 0$, write $M(\sigma) = -\zeta'(\sigma)/\zeta(\sigma) = \sum_n \Lambda(n) n^{-\sigma}$ for $\sigma > 1$, so that $|(\zeta'/\zeta)(\sigma + it)| \le M(\sigma)$, and take $\arg \zeta(\sigma + it) = \operatorname{Im} \log \zeta(\sigma + it)$ with $\log \zeta(s) = \sum_n \Lambda(n) n^{-s} / \log n$.

Proposition 18.14.5. *Let $1 < \alpha < \alpha'$, $h = \alpha' - \alpha$, $D = \theta_{\alpha'} - \theta_\alpha$,*

$$\mathcal{I} = \int_\alpha^{\alpha'} M(\sigma)\, d\sigma = \log \frac{\zeta(\alpha)}{\zeta(\alpha')}, \qquad J = \sum_p \left(\arctan p^{-\alpha} - \arctan p^{-\alpha'} \right) \le \mathcal{I}.$$

Then $D(t) = \frac{h\pi}{4} + \arg \zeta(\alpha' + it) - \arg \zeta(\alpha + it) + O(1/t)$, so $|D(t) - \frac{h\pi}{4}| \le \mathcal{I} + O(1/t)$.

(i) If $[\frac{h\pi}{4} - \mathcal{I}, \frac{h\pi}{4} + \mathcal{I}] \subset (k\pi, (k+1)\pi)$ for an integer k, the zeros of A_α and $A_{\alpha'}$ alternate for large t, and so do those of B_α and $B_{\alpha'}$.

(ii) If $[\frac{h\pi}{4} - \mathcal{I}, \frac{h\pi}{4} + \mathcal{I}] \subset (k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2})$ for an integer k, the zeros of A_α and $B_{\alpha'}$ alternate for large t, and so do those of B_α and $A_{\alpha'}$.

(iii) If a multiple of π lies in $(\frac{h\pi}{4} - J, \frac{h\pi}{4} + J)$, the zeros of A_α and $A_{\alpha'}$ fail to alternate at arbitrarily large t; if an odd multiple of $\frac{\pi}{2}$ does, so do those of A_α and $B_{\alpha'}$.

Proof. For $\alpha > 1$, $\xi(\alpha + it) \ne 0$ and $\theta'_\alpha > 0$ by the display before Theorem 18.14.4, so $A_\alpha = |\xi| \cos \theta_\alpha$ and $B_\alpha = |\xi| \sin \theta_\alpha$ vanish exactly where θ_α crosses $\frac{\pi}{2} + \pi\mathbb{Z}$ and $\pi\mathbb{Z}$. $D(t)$ is the integral of $\operatorname{Im}(\xi'/\xi)(\sigma + it)$ over $\alpha \le \sigma \le \alpha'$; by Stirling's formula the factor $\frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(\frac{s}{2})$ contributes $\frac{h\pi}{4} + O(1/t)$, and $|\int_\alpha^{\alpha'} \operatorname{Im}(\zeta'/\zeta)(\sigma + it)\, d\sigma| \le \mathcal{I}$. Also $J \le \mathcal{I}$ termwise: $\arctan x - \arctan y \le \log \frac{1-y}{1-x}$ for $0 \le y \le x < 1$, since $\frac{1}{1+u^2} \le \frac{1}{1-u}$, and with $x = p^{-\alpha}$, $y = p^{-\alpha'}$ the right sides sum to $\mathcal{I}$. Let n_1 and n_2 count the zeros in $(0, t]$ of the two functions compared, with phases $\phi_1 = \theta_\alpha$ and $\phi_2 = \theta_{\alpha'}$ in (i), $\phi_2 = \theta_{\alpha'} - \frac{\pi}{2}$ in (ii). Under (i) or (ii), $\phi_2 - \phi_1$ stays in one interval $(k'\pi, (k'+1)\pi)$ for $t \ge t_0$, so $n_2 - n_1$ takes only two adjacent values there, which is alternation. For (iii), $\arg \zeta(\alpha' + it) - \arg \zeta(\alpha + it) = \sum_p \{\arg(1 - p^{-\alpha} w_p) - \arg(1 - p^{-\alpha'} w_p)\}$ with $w_p = p^{-it}$, a sum that converges absolutely and uniformly. By Kronecker's theorem [99, Chapter 23] for the numbers $\log p$ it comes within any $\varepsilon > 0$ of its value J at $w_p = -i$ for all p, and of its value $-J$ at $w_p = i$, at arbitrarily

large t, and D is uniformly continuous, since $|D'| \le M(\alpha) + M(\alpha') + o(1)$. So at arbitrarily large t there are intervals of a fixed length on which $\phi_2 - \phi_1$ lies below the critical multiple of π by ε, followed by intervals on which it lies above it by ε. Since ϕ_1 advances by about $\frac{1}{2}\log\frac{t}{2\pi}$ per unit of t, $n_2 - n_1$ takes some value $m-1$ on the first and $m+1$ on the second; between the last visit to $m-1$ and the first visit to $m+1$ come two zeros of one function with no zero of the other between them. The pairs $(B_\alpha, B_{\alpha'})$ and $(B_\alpha, A_{\alpha'})$ are treated in the same way, with the phases $\theta_\alpha - \frac{\pi}{2}$ and $\theta_{\alpha'} - \frac{\pi}{2}$ or $\theta_{\alpha'}$. □

For $\alpha = 5$, $\alpha' = 11$, $\frac{h\pi}{4} = \frac{3\pi}{2}$ and $\mathcal{I} = 0.035768 < \frac{\pi}{2}$, so A_5 and A_{11} alternate for large t by (i), while A_5 and B_{11} fail to alternate at arbitrarily large t by (iii), which applies because $\frac{h\pi}{4} = \frac{3\pi}{2}$ is itself an odd multiple of $\frac{\pi}{2}$ and $J > 0$; no failure occurs below $t = 1000$ (computed). For $\alpha = 2$, $\alpha' = 3.9$, $\frac{h\pi}{4} = \frac{\pi}{2} - \frac{\pi}{40}$ and $J \ge \arctan\frac{1}{4} - \arctan 2^{-3.9} = 0.17809 > \frac{\pi}{40}$, so A_2 and $B_{3.9}$ fail to alternate at arbitrarily large t; in $(0, 100]$ they have 29 and 29 zeros, which alternate up to height 71.518, and the first failure is the zero 74.163 of $B_{3.9}$, the second of two consecutive zeros of $B_{3.9}$ with no zero of A_2 between them (computed on a grid, not certified). For $\alpha = 2$, (i) applies for $0 < h < 3.4004$ with $k = 0$ and for $4.6198 < h < 7.3682$ with $k = 1$, and (ii) for $0 < h < 1.5161$ with $k = 0$ and for $2.5695 < h < 5.3744$ with $k = 1$ (certified). The lower condition of (i) with $k = 0$, $\mathcal{I} < \frac{h\pi}{4}$, holds for every h once $\alpha \ge \sigma_1 = 1.80732$, where $M(\sigma_1) = \frac{\pi}{4}$.

Proposition 18.14.6. *(i) For $\alpha > 1$, A_α has $N(T) - S(T) + O_\alpha(1)$ zeros in $(0, T]$, where $N(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T)$ counts the zeros of Ξ with $0 < \gamma \le T$. Hence the zeros of A_α do not alternate with the real zeros of Ξ for large t.*

(ii) For $a > 0$, $\operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(a)$ has $\frac{T}{2\pi}(\log\frac{T}{a} - 1) + O_a(1)$ zeros in $(0, T]$. Hence two of these functions with different a do not alternate for large t, and none alternates with the real zeros of Ξ for large t if $a \le 2\pi$, or, under the Riemann hypothesis, if $a > 2\pi$.

Proof. Zero sets that alternate for $t \ge t_0$ have counting functions that differ by $O(1)$. (i) By the proof of Proposition 18.14.5, $\theta_\alpha(t) = \vartheta(t) + \pi + (\alpha - \frac{1}{2})\frac{\pi}{4} + \arg\zeta(\alpha + it) + O(1/t)$ is increasing, $|\arg\zeta(\alpha + it)| \le \log\zeta(\alpha)$, and A_α vanishes where θ_α crosses $\frac{\pi}{2} + \pi\mathbb{Z}$; this gives the count. Alternation with the real zeros of Ξ, at most $N(T)$ in number, would give $S(T) \ge -O(1)$, but S is unbounded in both directions [256]. (ii) For $\nu = \frac{9}{4} + \frac{it}{2}$, $K_\nu(a) = \frac{\pi}{2}(I_{-\nu}(a) - I_\nu(a))/\sin\nu\pi = \frac{1}{2}\Gamma(\nu)(\frac{a}{2})^{-\nu}(1 + O_a(1/t))$, so, as in the proof of Proposition 22.7.2, the zeros for large t are the crossings of $\frac{\pi}{2} + \pi\mathbb{Z}$ by an increasing function $\arg\Gamma(\nu) - \frac{t}{2}\log\frac{a}{2} + O_a(1/t) = \frac{t}{2}\log\frac{t}{a} - \frac{t}{2} + \frac{7\pi}{8} + O_a(1/t)$. The counts of two members differ by $\frac{T}{2\pi}\log\frac{a'}{a} + O(1)$. Against Ξ: for $a < 2\pi$ the count exceeds $N(T)$ by $\frac{T}{2\pi}\log\frac{2\pi}{a} + O(\log T)$; for $a = 2\pi$ it is $N(T) - S(T) + O(1)$ and the argument of (i) applies; for $a > 2\pi$ it falls short of $N(T)$ by $\frac{T}{2\pi}\log\frac{a}{2\pi} + O(\log T)$, which excludes alternation when all zeros of Ξ are real. □

For $a = 2\pi$ the function is $\Xi^*/(8\pi^2)$ of Proposition 22.7.1, whose zeros follow the half Gram points of Proposition 22.7.2.

18.14.4 Universal factors as vertical averages

Every universal factor acts as a vertical average, $\int_{\mathbb{R}} \Phi(u)\mathbb{E}\cosh(uX)e^{izu}\,du = \mathbb{E}\,\Xi(z+iX)$: the ladder is $X = \pm b$ and the de Bruijn Newman flow is $X = \sqrt{2\lambda}\,\mathcal{N}$ with $\mathcal{N}$ standard normal. Since $|\Xi(x+iu)|$ grows like $(x/2\pi)^{|u|/2}$, the average is dominated near the Esscher tilt location

$$u^*(L) = \arg\max_u \big[\tfrac{L}{2}u + \log p_X(u)\big], \qquad L = \log\frac{x}{2\pi},$$

which is λL for the Gaussian, Rodgers and Tao's effective abscissa, and $\operatorname{ess\,sup} X$ for bounded X. Below the asymptotic regime $\lambda L^2 \gg 1$ both flows obey the local model of a pair at $\pm i\eta$, $\eta = \beta - \frac12$: on $z^2+\eta^2$, $\cos(b\partial)$ gives $z^2+\eta^2-b^2$ and $e^{-\lambda\partial^2}$ gives $z^2+\eta^2-2\lambda$, so the model predicts $b_c = \eta$ and $\lambda_c = \eta^2/2$. The heat flow of Λ is the flow by $e^{\lambda u^2}$ of its kernel $\varphi(u) = 2(5/\pi)^{-1/2}e^{3u/2}\sum_{n\ge1} n\,(a\chi(n) + \bar a\bar\chi(n))\exp(-\pi n^2e^{2u}/5)$, $a = \frac{1-i\kappa}{2}$, which is even and gives $\Lambda(\frac12+ix)$ by its Fourier transform. Table 18.5 compares the model with three Davenport Heilbronn zeros.

zero	η	L	b_c	b_c/η	λ_c	$\lambda_c/(\eta^2/2)$
$0.8085 + 85.699i$	0.3085	2.613	0.2981	0.966	0.0441	0.927
$0.6508 + 114.163i$	0.1508	2.900	0.1504	0.997	0.0111	0.978
$0.7243 + 176.702i$	0.2243	3.337	0.2190	0.977	0.0237	0.941

Table 18.5: Critical ladder rung b_c and heat time λ_c at which the pair above each Davenport Heilbronn zero becomes real. In every case $\lambda_c L^2 < 1$ (0.30, 0.09, 0.26), so these heights are pre asymptotic for the heat flow.

For ζ, where an off line zero would lie above 3×10^{12} [232] ($L \approx 27$), the heat flow would absorb a zero at $\lambda \approx \eta/L$, far below the local value $\eta^2/2$ once $\eta > 2/L$ (with $L = 26.89$, $2/L = 0.074$), while the ladder needs $b \approx \eta$. In this local model the cosh ladder, unlike the de Bruijn Newman constant, admits no Polymath style improvement short of a quasi Riemann hypothesis.

Remark 18.14.7 (Mixtures of rungs). For symmetric X, the vertical average $\mathbb{E}\,\Xi(z+iX) = \mathbb{E}F_{|X|}(z)$ is a positive mixture of rungs, and when $\mathbb{E}\cosh(uX) = \prod_k \cosh(\lambda_k u)$, so that $\mathbb{E}X^2 = \sum_k \lambda_k^2$, Proposition 19.1.2, de Bruijn's theorem, puts its zeros in $|\operatorname{Im} z| \le (\max(\frac14 - \mathbb{E}X^2, 0))^{1/2}$. Following [241], each of the following therefore has only real zeros, unconditionally: $\frac12(\Xi + F_b)$ for $b \ge 2^{-1/2}$, from $X = \frac{b}{2}(\varepsilon_1+\varepsilon_2)$ with independent signs ε_k; $\frac34F_b + \frac14F_{3b}$ for $b \ge \frac{1}{2\sqrt3}$, from $X = b(\varepsilon_1+\varepsilon_2+\varepsilon_3)$; $\frac12(F_{b_1}+F_{b_2})$ for $b_1^2+b_2^2 \ge \frac12$, from $\frac12(\cosh b_1u + \cosh b_2u) = \cosh\frac{(b_1+b_2)u}{2}\cosh\frac{(b_2-b_1)u}{2}$; and $\frac{1}{b_2-b_1}\int_{b_1}^{b_2} F_b\,db$ for $b_1^2+b_1b_2+b_2^2 \ge \frac34$, from $\frac{\sinh x}{x} =$

$\prod_{k\ge1}\cosh(2^{-k}x)$. Conversely Ξ is no mixture $\int F_b\,\mu(db)$ with $\mu\ge0$ other than $\mu=\delta_0$: on the kernel side $\int\cosh(bu)\,\mu(db)\equiv1$ forces $\mu=\delta_0$. It is instead the hyperbolic secant smoothing of every rung, $\Xi=p_b*F_b$ with $p_b(x)=\frac{1}{2b}\operatorname{sech}\frac{\pi x}{2b}$, whose characteristic function is $\operatorname{sech}(bu)$; the same smoothing takes Ξ to g_b (Theorem 18.11.2). Beyond products of cosh the variance does not decide. For $\Pr(X=0)=\frac14$ and $\Pr(X=\pm2/\sqrt3)=\frac38$, $\mathbb{E}X^2=1$ and $\mathbb{E}\cosh(uX)=\frac14+\frac34\cosh\frac{2u}{\sqrt3}$ is a universal factor, by Pólya's theorem on universal factors [61, p. 197], since $\frac14+\frac34\cos\frac{2w}{\sqrt3}$ has only real zeros; yet for $p(z)=(z^2+1)^3$, with zeros at height 1,

$$\mathbb{E}\,p(z+iX)=z^6-12z^4+5z^2+\tfrac29$$

has two zeros on the imaginary axis, while the two laws of variance one in the local model above give $\cos(\partial)p=z^4(z^2-12)$ and $e^{-\partial^2/2}p=z^6-12z^4+30z^2-8$, both with only real zeros.

The local model predicts the collision of a pair at $\pm i\eta$ from the variance alone. To the next order the shape of the law enters; [241] tests this numerically, and the following statement makes the test precise.

Proposition 18.14.8 (One pair). *Let G be a real entire function of order less than two, $a\in\mathbb{R}$ with $G(a)\ne0$, $\tau_j=G^{(j)}(a)/G(a)$ and $F_\eta(z)=((z-a)^2+\eta^2)G(z)$. Let ℓ be symmetric, bounded or standard normal, with $\mathbb{E}\ell^2=1$ and $m_4=\mathbb{E}\ell^4$, and put $d_c=\tau_1^2-\frac{m_4-1}{2}\tau_2$. For every $K>|d_c|$ there are $r>0$ and $\eta_0>0$ such that for $0<\eta<\eta_0$ and $|s^2-\eta^2|\le K\eta^4$ the function $\mathbb{E}F_\eta(z+is\ell)$ has exactly two zeros in $|z-a|<r$, and there is s_c with*

$$s_c^2=\eta^2-d_c\,\eta^4+O(\eta^6)$$

such that the two zeros are a nonreal conjugate pair for $s<s_c$, a double real zero at $a+\tau_1\eta^2+O(\eta^4)$ for $s=s_c$, and real and distinct for $s>s_c$.

Proof. Put $z=a+\eta^2x$ and $s^2=\eta^2-\eta^4d$. Since ℓ is symmetric, $\mathbb{E}F_\eta(z+is\ell)$ is an analytic function of z and s^2, and expanding $G(z+is\ell)=G(a)\sum_j\tau_j(z-a+is\ell)^j/j!$ with $\mathbb{E}\ell^2=1$ and $\mathbb{E}\ell^4=m_4$ gives

$$\frac{\mathbb{E}F_\eta(z+is\ell)}{G(a)\,\eta^4}=(x-\tau_1)^2+d-d_c+\eta^2R(x,d,\eta),$$

with R analytic and bounded for bounded x, d and small η; the expectation converges because G has order less than two. As $\eta\to0$ the function tends to $(z-a)^2G(z)$ uniformly near a, so by Hurwitz's theorem it has exactly two zeros in a fixed disc about a, and they lie at bounded x, by Rouché's theorem against $(x-\tau_1)^2$ on $|x-\tau_1|=C$. At $\eta=0$ the system $P=\partial_xP=0$ for the right side has the nondegenerate solution $x=\tau_1$, $d=d_c$, which the implicit function theorem continues to small η. The function is real on $\mathbb{R}$, so the two zeros are real or

conjugate according to the sign of the discriminant; $(x_+ - x_-)^2 = 4(d_c - d) + O(\eta^2)$ is analytic in d with d-derivative $-4 + O(\eta^2)$, so its only zero is $d_c(\eta) = d_c + O(\eta^2)$ (this is where $K > |d_c|$ is used), and its sign is that of $d_c(\eta) - d$. □

With $\ell = (\varepsilon_1 + \cdots + \varepsilon_n)/\sqrt{n}$, the ladder of n independent cosh steps, $m_4 = 3 - \frac{2}{n}$, that is 1, 2 and $\frac{7}{3}$ for one, two and three steps, and $m_4 = 3$ for the Gaussian, the de Bruijn Newman flow. So the variance decides the collision to first order, as in the local model, and the law enters at order η^4 through its kurtosis. Table 18.6 tests the expansion on Ξ itself, with its first zero moved to $\gamma_1 \pm i\eta$ and its mirror to $-\gamma_1 \pm i\eta$, so that $a = \gamma_1$ and $G(z) = \Xi(z)((z+\gamma_1)^2 + \eta^2)/(z^2 - \gamma_1^2)$, which is $\Xi(z)(z+\gamma_1)/(z-\gamma_1)$ times $1 + O(\eta^2)$ near γ_1; this factor changes the expansion in the proof of Proposition 18.14.8 only at order η^6, so the proposition holds with the τ_j of $\Xi(z)(z+\gamma_1)/(z-\gamma_1)$, $\tau_1 = -0.54426$ and $\tau_2 = 0.22728$ (computed).

law of ℓ	m_4	d_c	$\eta = 0.3$	$\eta = 0.15$	$\eta = 0.075$
one cosh step	1	0.2962	0.2827	0.2927	0.2953
two cosh steps	2	0.1826	0.1777	0.1813	0.1823
three cosh steps	$\frac{7}{3}$	0.1447	0.1417	0.1439	0.1445
Gaussian	3	0.0689	0.0684	0.0688	0.0689

Table 18.6: The collision of the moved pair $\gamma_1 \pm i\eta$ of Ξ: $(\eta^2 - s_c^2)/\eta^4$ at three heights η, against its limit $d_c = \tau_1^2 - \frac{m_4 - 1}{2}\tau_2$ of Proposition 18.14.8 (computed; the three cosh cells at $\eta = 0.3$ are certified by a Krawczyk step).

At height T the coefficients τ_1 and τ_2 are typically of size L and L^2, $L = \log \frac{T}{2\pi}$, so the expansion describes pairs with ηL small, the regime $\eta < 2/L$ of the paragraph after Table 18.5, and it says nothing uniform in the height (heuristic).

18.15 Open problems

Problem 18.1. Decide whether the heuristic of Remark 18.6.1 holds with a bounded error.

Problem 18.2. Assuming RH for Ξ, decide whether $\nu_P \geq 0$ on all of $(0, \infty)$ for every P. The angle $y_0/x_0 \sim (\pi/2)/\log x_0$ of the departed chain tends to zero, which suggests yes; the computed range is $[0.02, 3]$.

Problem 18.3. Prove Conjecture 18.5.1, or at least $|R_P - N(\tau_P^*)| = O(\log q)$ under RH and a lower bound for the peaks of Hardy's function Z on negative humps.

Problem 18.4. Find a computable Thorin criterion that detects the departed zeros, for instance the failure of positivity of Hankel or moment matrices built from ν_P beyond a certain order, and use the friable approximants to calibrate it.

PROBLEM 18.5. Prove that all zeros of $g^{(2)}$ lie in $|\operatorname{Im} w| < 2$, and deduce from de Bruijn's theorem that every zero of Miyatake's $\xi^{(2)}$ is real.

PROBLEM 18.6. Prove that no factorisation $\Phi = U\chi$ with $U = \mathbb{E}\cosh(tX)$ a nontrivial universal factor has $\hat{\chi}$ in the Laguerre Pólya class, which would extend the theorem of Rodgers and Tao from Gaussian factors to Cardon's factors.

PROBLEM 18.7. Make the local mechanism of Section 18.14 rigorous at all large heights, proving $B_{\cosh} \geq \sigma^{\infty}_{\max} - \frac{1}{2}$ in Conjecture 18.14.2; Theorem 18.14.3 already settles the paired version. Proposition 18.14.8 proves the mechanism for one pair as $\eta \to 0$ with the other zeros fixed; what is missing is uniformity in the height and in the other zeros.

PROBLEM 18.8. Prove the Riemann von Mangoldt formula for the rungs F_b with error $O(\log T)$, which makes the density one statement after Theorem 18.14.4 unconditional.

PROBLEM 18.9. Prove a Ki, Kim and Lee theorem for the vertical average $\mathbb{E}\,\Xi(z + iX)$ whenever X is unbounded, and show that for bounded X the nonreal zeros at large height sit above zeros of ζ with $\beta > \frac{1}{2} + \operatorname{ess\,sup} X$.

PROBLEM 18.10. Make the escape law (18.12.1) rigorous for $b > 2$, $C_b \neq 0$.

PROBLEM 18.11. Certify, by interval arithmetic, that C_b has exactly one zero b_0 on $(2, \infty)$: an interval Newton enclosure of $b_0 = 2.4588641657$, the sign of $\Phi(i\pi/(2b))$ on $(2, 20]$ by subdivision with a rigorous tail for the series over n, Theorem 18.12.2 with an explicit error near $b = 2$, and a proof that $\Phi(iy) > 0$ for $0 < y \leq \pi/40$ to cover $b > 20$.

PROBLEM 18.12. Decide whether $\mathcal{E}_{2\pi/9}$ of Proposition 18.13.2, equivalently $q_{\pi/9} * \Xi^{\mathcal{U}}$ of Proposition 18.13.3, is in the Laguerre Pólya class, for $a_0 = 2\pi$ or for every $a_0 > 0$. That $q_{\pi/9}$ is a Pólya frequency function does not suffice: convolution with a symmetric probability density of variance v sends $z^2 - r^2$ to $z^2 - r^2 + v$, which has nonreal zeros for $r^2 < v$, and the same convolution gives nonreal zeros for Ξ (Section 18.13). A proof has to use more of $\Xi^{\mathcal{U}}$, for instance the spacing of its zeros.

Notes. The identity (18.7.1) and the function $\xi^{(P)}$ with its Theorem I are Miyatake's [119], and the symmetrised kernel with a secant pole term is that of his paper A [118]; Hejhal [96] treats truncated symmetrised kernels. The ladder is that of de Bruijn and Lagarias, the shift identity and the Brownian and Mellin readings of Section 18.11 use the Biane Pitman Yor identity [26] and the Pólya Gamma mixture of Polson, Scott and Windle [159], the paired form uses Chebotarev's theorem [269], and the counting in Section 18.14.2 uses Ingham's density estimate [176]. The statements about Miyatake's two papers have not been checked against the papers. In Section 18.13 [240] the universal factor theorem is Pólya's [246],

the reality of the zeros of $K_{iz}(a)$ is Pólya's [245], the Pólya frequency criterion is Schoenberg's [254], and the Meixner reading of q_φ uses [90, 155]. Section 18.14.3, Remark 18.14.7 and Table 18.6 follow [241]; the strip bound for products of cosh is de Bruijn's [61], and the unboundedness of $S(T)$ in both directions is Selberg's [256].

Chapter 19

Ferromagnetic and extreme value approximants with certified real zeros

19.1 Ferromagnetic approximations of the Pólya law

By Corollary 20.1.8 and the remark after it, RH holds exactly when the total variation distance δ from the Pólya law to the class of ferromagnetic laws is 0. This section bounds δ above, with no hypothesis, by distances to laws proved ferromagnetic. The distances are enclosed in ball arithmetic with explicit bounds for the discretisation and the tails, so these bounds on δ are proved.

19.1.1 Pólya's approximations

The Gaussian law $N(0, \operatorname{Var} X)$ is ferromagnetic and lies at total variation distance 1.48×10^{-2} from the Pólya law. Pólya kept the leading part of the first term of Φ and made it even:

$$\Phi^*(t) = 8\pi^2 \cosh \tfrac{9t}{2}\, e^{-2\pi \cosh 2t},$$
$$\int \Phi^*(t) e^{i\theta t}\, dt = 4\pi^2 \Big[K_{\frac{9}{4}+\frac{i\theta}{2}}(2\pi) + K_{\frac{9}{4}-\frac{i\theta}{2}}(2\pi) \Big],$$

by $\int e^{-a\cosh u + \nu u}\, du = 2K_\nu(a)$, and he proved that this transform has only real zeros [245]. So the law with density proportional to Φ^* is ferromagnetic. The substitution $g = e^{2t}$ turns $e^{9t/2} e^{-2\pi\cosh 2t}$ into $g^{5/4} e^{-\pi(g+1/g)}$, so this law is that of $\frac{\varepsilon}{2}\log G$ with G generalised inverse Gaussian of that density, and it can be sampled exactly. Keeping the whole first term gives the positive kernel

$$\Psi(t) = \Big(8\pi^2 \cosh \tfrac{9t}{2} - 12\pi \cosh \tfrac{5t}{2} \Big) e^{-2\pi \cosh 2t},$$

and the same theorem of Pólya covers it.

Proposition 19.1.1. *The law with density proportional to Ψ is ferromagnetic.*

Proof. Write $\Psi(t) = \psi(it)f(t)$ with $f(t) = e^{-2\pi\cosh 2t}$ and $\psi(z) = 8\pi^2\cos\frac{9z}{2} - 12\pi\cos\frac{5z}{2}$. In the variable $w = z/2$, ψ is the trigonometric polynomial $8\pi^2\cos 9w - 12\pi\cos 5w$ of degree 9. At $w_j = j\pi/9$, $0 \le j < 18$, the first term is $8\pi^2(-1)^j$ and the second is at most $12\pi < 8\pi^2$ in absolute value, so ψ changes sign at least 18 times in a period, and since a trigonometric polynomial of degree 9 has at most 18 zeros in a period strip, all its zeros are real. So ψ is a real entire function of order 1 with only real zeros. The transform of f is $K_{i\theta/2}(2\pi)$, which has only real zeros, by Pólya's theorem on $\int e^{-\lambda\cosh t}e^{izt}dt$ [61, (1.2)], and f decays faster than $e^{-|t|^b}$ for every b. Pólya's theorem on universal factors [61, p. 197], [34, Proposition 1] then says that the transform of $\psi(it)f(t)$ has only real zeros. The density decays like f, so it has Gaussian moments of all orders. □

As densities both approximations are poor: their total variation distances to p are 0.099 and 0.161, their variances 0.0624 and 0.0726 against $\operatorname{Var} X = 0.0462$, and their values at the origin 1.404 and 1.163 against $p(0) = 1.797$, all enclosed in ball arithmetic. Each transform has 14 zeros in $(0.25, 62.5)$, all real, and an argument principle count in $(0.25, 62.5) \times (-2, 2)$ finds no others; there are also 14 ordinates below 62.5. The positions differ: the first five are 8.99, 19.07, 24.47, 28.97, 33.01 for Φ^* and 7.60, 18.72, 24.28, 28.83, 32.89 for Ψ, against 14.13, 21.02, 25.01, 30.42, 32.94 (computed).

19.1.2 Tilts that shrink the strip

Every zero γ of $\varphi_X = \Xi/\Xi(0)$ has $|\operatorname{Im}\gamma| < \frac{1}{2}$, because $0 < \operatorname{Re}\rho < 1$. Multipliers of the density move the zeros towards the real axis, by an amount de Bruijn measured. For $\lambda_1, \dots, \lambda_N > 0$ and $\mu \ge 0$ put

$$m(t) = e^{\mu t^2}\prod_{k\le N}\cosh(\lambda_k t), \qquad c(m) = 2\mu + \sum_{k\le N}\lambda_k^2.$$

Then $m(t) = \mathbb{E}e^{tY} = \mathbb{E}\cosh(tY)$ for $Y = \sqrt{2\mu}\,Z + \sum_k \lambda_k\varepsilon_k$, and $c(m) = \operatorname{Var} Y$.

Proposition 19.1.2. *The zeros of $\int \Phi(t)m(t)e^{izt}\,dt$ lie in the strip $|\operatorname{Im} z| \le (\max(\frac{1}{4} - c(m), 0))^{1/2}$. The same holds for $N = \infty$ when $\sum_k \lambda_k^2 < \infty$. In particular, with no hypothesis, the law with density proportional to $m\,p$ is ferromagnetic whenever $\operatorname{Var} Y \ge \frac{1}{4}$; for example $\cosh(at)\,p$ for $a \ge \frac{1}{2}$, $\frac{\sinh(ct)}{ct}\,p$ for $c \ge \frac{\sqrt{3}}{2}$, and $e^{\mu t^2}p$ for $\mu \ge \frac{1}{8}$. If no zero of ζ has real part greater than $\frac{7}{8}$, as announced in October 2026 (Section 25.1.2), all of this holds with $\frac{9}{64}$ in place of $\frac{1}{4}$; then $\cosh(at)\,p$ is ferromagnetic for $a \ge \frac{3}{8}$ and $\frac{\sinh(ct)}{ct}\,p$ for $c \ge \frac{3\sqrt{3}}{8}$.*

Proof. For finite N this is de Bruijn's Theorem 13, for the Gaussian factor $e^{\frac{1}{2}(2\mu)t^2}$, followed by his Theorem 12, for the product of cosines [61]; both apply to an even kernel F with $F(t) = O(e^{-|t|^b})$ for some $b > 2$, which Φ and $\Phi e^{\mu t^2}$ are, and both take as input a closed strip containing the zeros, here $|\operatorname{Im} z| \le \frac{1}{2}$. Under the

hypothesis of the last sentence every zero has $\frac{1}{8} \le \operatorname{Re}\rho \le \frac{7}{8}$, by $\xi(s) = \xi(1-s)$, so the zeros of φ_X lie in $|\operatorname{Im} z| \le \frac{3}{8}$, and the same two theorems give the strip with $\frac{9}{64}$ in place of $\frac{1}{4}$. For $N = \infty$ the partial products m_N have $c(m_N) \uparrow c(m)$, and their transforms converge to that of Φm locally uniformly in z, by dominated convergence with $m_N \le m$. By Hurwitz's theorem the zeros of the limit lie in the intersection of the strips. The tilted densities decay like Φ, so they have Gaussian moments of all orders. The three examples have $\operatorname{Var} Y = a^2$, $c^2/3$ (by Viète's product $\frac{\sinh ct}{ct} = \prod_k \cosh \frac{ct}{2^k}$) and 2μ. □

For a single cosine the proof is elementary and shows where the number $\frac{1}{2}$ comes from. The characteristic function of $\cosh(at)p(t)$ is proportional to $\varphi_X(z+ia) + \varphi_X(z-ia)$. Take $z = x + iy$ with $y > 0$ and a pair of zeros $\beta \pm i\eta$ of φ_X, with $c = (x-\beta)^2$. De Bruijn's identity [61, Lemma 1]

$$\big(c + (y+a-\eta)^2\big)\big(c + (y+a+\eta)^2\big) - \big(c + (y-a-\eta)^2\big)\big(c + (y-a+\eta)^2\big) \\ = 8ay\,(c + y^2 + a^2 - \eta^2)$$

shows that the pair contributes a factor larger than 1 to $|\varphi_X(z+ia)|^2/|\varphi_X(z-ia)|^2$ as soon as $a > |\eta|$. With $a \ge \frac{1}{2}$ this holds for every zero, so by the Hadamard product the two terms cannot cancel above the real axis, nor, by symmetry, below it. In the variable $s = \frac{1}{2} + iz$ the transform is $\frac{1}{2}(\xi(s+a) + \xi(s-a))$, and Lagarias proved that its zeros are also simple [186]. For the de Bruijn Newman flow the multiplier is $e^{\lambda t^2/4}$, with $c(m) = \lambda/2$; the proposition gives de Bruijn's bound $\Lambda \le \frac{1}{2}$, and the bound $\Lambda \le 0.2$ of Platt and Trudgian [232] makes $e^{\lambda t^2/4}p$ ferromagnetic for every $\lambda \ge 0.2$, again with no hypothesis.

The three multipliers of the proposition with $\operatorname{Var} Y = \frac{1}{4}$ give total variation distances 2.731×10^{-3}, 2.739×10^{-3} and 2.751×10^{-3} from p, in the order listed. The flow at $\lambda = 0.2$, with $c(m) = 0.1$, gives 1.097×10^{-3}. The law $\cosh(\sqrt{0.1}\,t)\,p$, with the same contraction, gives 1.094×10^{-3}, but whether it is ferromagnetic is not known: the bound of Platt and Trudgian concerns the Gaussian flow, and Proposition 19.1.2 needs $\operatorname{Var} Y \ge \frac{1}{4}$, or $\frac{9}{64}$ if the announced half plane stands, both above 0.1 (distances enclosed in ball arithmetic).

The cosine tilts define a constant in the spirit of Λ. Let Ω be the infimum of the $a_0 \ge 0$ such that $\cosh(at)\,p$ is ferromagnetic for every $a \ge a_0$. Proposition 19.1.2 gives $\Omega \le \frac{1}{2}$, and $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ is free of zeros. Under RH, Proposition 20.1.6(i) gives $\Omega = 0$. Conversely, if $\Omega = 0$, the tilted laws converge to p in total variation as $a \downarrow 0$, and p is ferromagnetic by Theorem 20.1.5. So RH holds if and only if $\Omega = 0$. Unlike the flow, the operators $\varphi \mapsto \varphi(\cdot + ia) + \varphi(\cdot - ia)$ do not form a semigroup, and we do not know whether the set of good values of a is an interval.

Remark 19.1.3. To first order the multipliers of this section, and the factor $1 + t^2/r^2$ of Remark 20.1.7, act on a simple zero in the same way. Let $\varphi(\theta) = L_X(i\theta) = e^{-b\theta^2}\prod_k(1 - \theta^2/c_k^2)$ for a ferromagnetic X, as in (20.1.1), and let c_j be a simple

zero. The factors $\cosh(\lambda t)$, $e^{\mu t^2}$ and $1 + t^2/r^2$ replace φ, up to a constant, by $\varphi - \kappa\varphi'' + O(\kappa^2)$ locally uniformly, with $\kappa = \frac{\lambda^2}{2}$, μ and r^{-2} respectively, so the zero moves to $c_j + \kappa\varphi''(c_j)/\varphi'(c_j) + O(\kappa^2)$, where

$$\frac{\varphi''(c_j)}{\varphi'(c_j)} = \frac{1}{c_j} + 4c_j \sum_{k \neq j} \frac{1}{c_j^2 - c_k^2} - 4bc_j.$$

This is the velocity of the de Bruijn Newman flow of Section 4.1.1, in the variable $\theta = z/2$ and the time $\mu = t/4$: to first order each multiplier runs the flow for time κ. The three differ at the next order, where the coefficients of φ'''' are $\frac{\kappa^2}{6}$, $\frac{\kappa^2}{2}$ and 0. Dilations also preserve the class. This remark is based on [137].

19.1.3 Multipliers on Pólya's kernel

On p the multipliers $m(t) = \mathbb{E}e^{tY}$ of Section 19.1.2 need a hypothesis unless $c(m) \geq \frac{1}{4}$; on Pólya's kernel $e^{-2\pi\cosh 2t}$ they need none, and the kernel itself can be varied. This section is based on [148, 136, 137, 128].

Theorem 19.1.4. *Let $a, \ell > 0$, $\mu \geq 0$ and $\lambda_1, \dots, \lambda_N \geq 0$, and let q be the probability density proportional to $m(t)e^{-a\cosh \ell t}$.*

(i) The law q is ferromagnetic, it has Gaussian moments of all orders, and $b = 0$ in (20.1.1).

(ii) $\int_{\mathbb{R}} m(t)e^{-a\cosh \ell t}\,dt = \frac{2}{\ell}\,\mathbb{E}K_{Y/\ell}(a)$, and $L_q(s) = \mathbb{E}K_{(s+Y)/\ell}(a)/\mathbb{E}K_{Y/\ell}(a)$ for complex s. For $\mu = 0$ the expectations are averages over the 2^N values of Y.

(iii) The clock of q in Theorem 20.1.4 has $\mathbb{E}e^{-uT_q} = \mathbb{E}K_{Y/\ell}(a)/\mathbb{E}K_{(\sqrt{u}+Y)/\ell}(a)$ for $u \geq 0$. For $m \equiv 1$, $T_q = \ell^{-2}\sum_j e_j/\nu_j(a)^2$, where $0 < \nu_1(a) \leq \nu_2(a) \leq \dots$ are the zeros of $\nu \mapsto K_{i\nu}(a)$ on $(0, \infty)$, listed with multiplicity.

For $a = 2\pi$, $\ell = 2$ and $m \equiv 1$, part (iii) is the fifth row of Table 20.1. For $a = 2\pi$, $\ell = 2$ and $m(t) = \cosh\frac{9t}{2}$, q is Pólya's Φ^*, normalised, and (ii) is Proposition 22.7.1. With one field and no tilt the transform is a multiple of $\operatorname{Re} K_{(\lambda_1 + i\theta)/\ell}(a)$, a shifted Bessel kernel, as in Theorem 19.3.3 for integer λ_1/ℓ.

Proof. (i) For $m \equiv 1$, the substitution $v = \ell t$ and $\int_{\mathbb{R}} e^{-a\cosh v + \nu v}\,dv = 2K_\nu(a)$ give the characteristic function $K_{i\theta/\ell}(a)/K_0(a)$. Pólya proved that $z \mapsto K_{iz}(a)$ has only real zeros [245]; here is a short proof. If $K_{i\zeta}(a) = 0$, then $f(x) = K_{i\zeta}(ae^x)$ satisfies $-f'' + a^2e^{2x}f = \zeta^2 f$ on $[0, \infty)$, by Bessel's equation, with $f(0) = 0$, and f and f' decay like $\exp(-ae^x)$ times a power of e^x. Integration by parts against $\bar{f}$ gives

$$\zeta^2 \int_0^\infty |f|^2\,dx = \int_0^\infty \left(|f'|^2 + a^2e^{2x}|f|^2\right) dx > 0,$$

since $f \neq 0$, so ζ is real. Every density in the construction is at most a constant times $\exp(\mu t^2 + \sum_k \lambda_k |t| - \frac{a}{2} e^{\ell |t|})$, so it has Gaussian moments of all orders. Proposition 20.1.6(i), applied once for each field, and Proposition 20.1.6(ii), with any $b_0 > \mu$, give the first statement, and $b = 0$ by the last statement of Theorem 20.1.2. (ii) For real s, Tonelli's theorem gives $\int e^{st} m(t) e^{-a \cosh \ell t}\, dt = \mathbb{E} \int e^{(s+Y)t} e^{-a \cosh \ell t}\, dt = \frac{2}{\ell} \mathbb{E} K_{(s+Y)/\ell}(a)$, which is finite because the left side is; both sides are entire in s, since $|K_\nu(a)| \leq K_{\operatorname{Re} \nu}(a)$. (iii) is Theorem 20.1.4(i) with $b = 0$ and (ii) at $s = \sqrt{u}$. For $m \equiv 1$, $L_q(s) = K_{s/\ell}(a)/K_0(a)$ vanishes exactly at $s = \pm i\ell\nu_j(a)$, so $c_j = \ell \nu_j(a)$. □

Proposition 19.1.5. *For real v let G_v have the generalised inverse Gaussian density $g^{v-1} e^{-\frac{a}{2}(g+1/g)}/(2K_v(a))$ on $(0, \infty)$. In Theorem 19.1.4, let V have the law of Y/ℓ weighted by $K_V(a)$, that is $\mathbb{E}h(V) = \mathbb{E}[h(Y/\ell) K_{Y/\ell}(a)]/\mathbb{E}K_{Y/\ell}(a)$, and given V let G_V have the density of index V. Then q is the law of $\ell^{-1} \log G_V$, and also of $\varepsilon\, \ell^{-1} \log G_{|V|}$, with ε independent of V and $G_{|V|}$. For $\mu = 0$, V takes the values $\ell^{-1} \sum_k \eta_k \lambda_k$, $\eta \in \{\pm 1\}^N$, with probabilities proportional to $K_{\ell^{-1} \sum_k \eta_k \lambda_k}(a)$.*

Proof. The substitution $g = e^{\ell t}$ turns $e^{vt} e^{-a \cosh \ell t}\, dt$ into $\ell^{-1} g^{v/\ell - 1} e^{-\frac{a}{2}(g+1/g)}\, dg$, and $m(t) = \mathbb{E} e^{tY}$. So $e^{\ell X}$, for X with density q, has density proportional to $\mathbb{E}\big[g^{Y/\ell - 1} e^{-\frac{a}{2}(g+1/g)}\big] = \mathbb{E}\big[2K_{Y/\ell}(a) f_{Y/\ell}(g)\big]$, where f_v is the density of G_v; this is the mixture stated. The law of V is symmetric, because that of Y is and $K_{-v} = K_v$, and $1/G_v$ has the law of G_{-v}, which gives the second form. □

With one field the two signs have equal weights, and $a = 2\pi$, $\ell = 2$, $\lambda_1 = \frac{9}{2}$ recover the description of Φ^* in Section 19.1.1. With more fields the weights differ: for $\lambda_1 = \lambda_2 > 0$ the index 0 and the indices $\pm 2\lambda_1/\ell$ have total weights in the ratio $K_0(a) : K_{2\lambda_1/\ell}(a)$, and $K_v(a) = \int_0^\infty e^{-a \cosh u} \cosh(vu)\, du$ increases in $v > 0$. For $\mu = 0$ every law q can be sampled exactly, by drawing V and then G_V.

Remark 19.1.6. The kernel is a product of two Gumbel distribution functions, which builds q from independent variables. Let W_1, W_2 be independent with $\mathbb{P}(W_i \leq x) = e^{-e^{-x}}$, put $c = \log \frac{a}{2}$, $A = (W_1 + c)/\ell$ and $B = -(W_2 + c)/\ell$. Then

$$\mathbb{P}(A \leq t \leq B) = e^{-\frac{a}{2} e^{-\ell t}}\, e^{-\frac{a}{2} e^{\ell t}} = e^{-a \cosh \ell t},$$

which for $a = 2\pi$ and $\ell = 2$ is the two sided Gumbel form of Pólya's kernel in the list after Theorem 19.17.1 (there in the variable $u = t/2$). By Fubini's theorem $\mathbb{E}(B - A)^+ = \int_{\mathbb{R}} e^{-a \cosh \ell t}\, dt = \frac{2}{\ell} K_0(a)$, where $(B - A)^+ = \ell^{-1} \log^+ \big(4e^{-W_1 - W_2}/a^2\big)$. Weight the pair (A, B) by $(B - A)^+ / \mathbb{E}(B - A)^+$ and, given the pair, let U be uniform on $[A, B]$; then U has density $\mathbb{P}(A \leq t \leq B)/\mathbb{E}(B - A)^+$, which is q for $m \equiv 1$. For general m take Y independent of (W_1, W_2), weight by $\mathbf{1}_{\{A < B\}} \int_A^B e^{Ys}\, ds$, and give U the density proportional to e^{Ys} on $[A, B]$; the same computation gives the density proportional to $m(t) e^{-a \cosh \ell t}$. The length bias cannot be dropped: an interval equal

to $[0,1]$ or $[0,2]$ with probability $\frac12$ each puts a uniform point in $[0,1]$ with probability $\frac34$, whereas the density proportional to $\mathbb{P}(t\in[A,B])$ gives $[0,1]$ the mass $\frac23$. In the same way the logistic law of scale $1/\pi$ in the second row of Table 20.1 is the law of $(W_1-W_2)/\pi$, because $\Gamma(1+i\theta)\Gamma(1-i\theta)=\pi\theta/\sinh\pi\theta$.

Remark 19.1.7. The proof of Proposition 18.1.8 holds for $K_{i\nu}(r)$ at every $r>0$, and its Stirling step gives $\vartheta(\nu,r)=\nu\log\frac{2\nu}{er}+\frac{\pi}{4}+O(1/\nu)$. So for $m\equiv1$ the number of zeros of $\varphi_q(\theta)=K_{i\theta/\ell}(a)/K_0(a)$ in $(0,\theta]$ is

$$\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell}+O(1).$$

The zeros of φ_X with real part in $(0,\theta]$ are those with $0<\operatorname{Im}\rho\le\theta$, counted with no hypothesis by the Riemann von Mangoldt formula $\frac{\theta}{2\pi}\log\frac{\theta}{2\pi e}+O(\log\theta)$. The two main terms agree exactly when $\ell=2$ and $a=2\pi$, so among the kernels $e^{-a\cosh\ell t}$ only Pólya's has the main term of Riemann von Mangoldt (compare Section 18.1). Fields keep the main term. If $\mu=0$, a further field λ replaces the transform F by $\operatorname{Re}F(\theta-i\lambda)$ up to a constant; as before Proposition 22.7.2, $\arg F(\theta-i\lambda)$ increases with θ, the zeros on $(0,\infty)$ are its crossings of $(\frac12+\mathbb{Z})\pi$, and integration by parts against the counting function of F shows the count changes by $O(1)$. So for $m(t)=\prod_{k\le N}\cosh(\lambda_k t)$ the count is again $\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell}+O(1)$. A factor $1+t^2/r^2$ (Remark 20.1.7) also changes it by $O(1)$, because the zeros of $\varphi\mp\varphi'/r$ interlace with those of φ, by Rolle's theorem applied to $e^{\mp r\theta}\varphi$. On Pólya's kernel, therefore, $N(\theta)-N_q(\theta)=S(\theta)+O(1)$, where $N(\theta)$ counts the zeros of ξ with $0<\operatorname{Im}\rho\le\theta$ and $N_q(\theta)$ the zeros of φ_q in $(0,\theta]$; Selberg's central limit theorem [256] makes S unbounded above and below, so $\sup_\theta|N(\theta)-N_q(\theta)|=\infty$ for every such q. A Gaussian factor changes the main term: for $\mu>0$ the count is

$$\frac{\theta}{\pi\ell}\log\frac{2\theta}{ea\ell}+\frac{\mu}{\ell^2}\log\theta+O(1),$$

on Pólya's kernel $\vartheta(\theta)/\pi+\frac{\mu}{4}\log\theta+O(1)$, the analogue of the term $\frac{t}{16}\log T$ of Proposition 6.2.7(ii) with $\mu=\frac{t}{4}$. For the proof write $e^{\mu t^2}=\mathbb{E}e^{\sqrt{2\mu}Zt}$, so that $\varphi_q(\theta)$ is proportional to $\mathbb{E}\,F(\theta-i\sqrt{2\mu}Z)$, with F the transform for $\mu=0$. Stirling's formula for $K_{(c+i\theta)/\ell}(a)$, uniform for $|c|\le\theta^{1/3}$, and a saddle point at $\sqrt{2\mu}Z\approx\frac{2\mu}{\ell}\log\theta$ give $\varphi_q(\theta)=A(\theta)\big(\cos\Theta(\theta)+o(1)\big)$ in a strip about the real axis, with $A>0$ and $\Theta(\theta)=\frac{\theta}{\ell}\log\frac{2\theta}{ea\ell}+\frac{\pi\mu}{\ell^2}\log\theta+O(1)$; the tails $|c|>\theta^{1/3}$ are controlled by the bound $|\Gamma(\sigma+i\tau)|\ll|\sigma+i\tau|^{\sigma-1/2}e^{\sigma-\pi|\tau|/2}$ for $\sigma\ge0$. As the zeros are real (Theorem 19.1.4), the count is $\Theta(\theta)/\pi+O(1)$.

Table 19.1 lists members fitted to p by minimising $\mathrm{TV}(q,p)$ numerically. The parameters are exact decimals, so each row is an explicit ferromagnetic law, with its distance enclosed in ball arithmetic as in Section 19.1.6.

The second row is the closest law the search found: neither a second free field nor a tilt with a free kernel lowered the distance. It is ferromagnetic by

$m(t)$	a	ℓ	$\mathrm{TV}(q,p)$
1	2.838657	2.574646	$< 3.2 \times 10^{-4}$
$\cosh(\lambda t)$, $\lambda = 2.146184$	5.483952	2.062866	$< 6.1 \times 10^{-6}$
$\cosh(\lambda t)$, $\lambda = 2.618518$	2π	2	$< 2.0 \times 10^{-3}$
$\cosh^2(\lambda t)$, $\lambda = 1.792548$	2π	2	$< 2.1 \times 10^{-4}$
$e^{\mu t^2}\cosh(\lambda t)$, $\lambda = 2.125355$, $\mu = 0.9417395$	2π	2	$< 8.0 \times 10^{-6}$

Table 19.1: Laws of Theorem 19.1.4 fitted to the Pólya law. The last three rows keep Pólya's kernel $e^{-2\pi\cosh 2t}$ (Remark 19.1.7). The parameters are rounded to seven significant digits from a numerical minimisation and then fixed; the distances are upper bounds enclosed in ball arithmetic and rounded up.

Theorem 19.1.4, with no positivity proof needed, and its distance beats the bound 1.266×10^{-4} of the zero completed law f_{16} (Section 19.1.4, Proposition 19.1.15), though not the bound 1.932×10^{-6} of f_{150}.

A small distance does not pin the zeros. Always $|\varphi_q(\theta) - \varphi_X(\theta)| \le 2\,\mathrm{TV}(q,p)$, but for the second row $|\varphi_X(\theta)| < 2\,\mathrm{TV}(q,p)$ on $[23,100]$ (computed, not certified), so there the bound does not fix the sign of φ_q nor control its zeros. The first zero of φ_q for the second row is 14.133924, against $\gamma_1 = 14.134725$, and over the first twenty zeros the mean and largest distances to $\gamma_1, \dots, \gamma_{20}$ are 0.449 and 1.145; for the fourth row, on Pólya's kernel, they are 0.501 and 1.284. The zeros are located by certified sign changes in ball arithmetic; the absence of further zeros in the strip $|\operatorname{Im}\theta| < 2$ up to the twenty first is computed from the change of phase on a grid, not certified. The clock of the second row has $\mathbb{E}T_q = 0.0231055$ and $\operatorname{Var} T_q = 3.71451 \times 10^{-5}$, against $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = 3.71725993 \times 10^{-5}$.

A sequence of such laws on one kernel cannot approach p, with or without RH.

Proposition 19.1.8. *Let $a_n \to a$ and $\ell_n \to \ell$ with $a, \ell > 0$, and let q_n be the probability density proportional to $m_n(t)e^{-a_n\cosh\ell_n t}$, where $m_n(t) = \psi_n(it)$ for an even function ψ_n of the Laguerre Pólya class with $\psi_n(0) > 0$. Then q_n does not converge in law to p. The multipliers of Theorem 19.1.4 are of this form, and so are $e^{\mu t^2}\prod_{k\ge1}\cosh(\lambda_k t)$ with $\sum_k \lambda_k^2 < \infty$ and the multiplier $\psi(it)$ of Ψ in Proposition 19.1.1.*

Proof. By the Laguerre Pólya theorem [195], $\psi_n(z) = \psi_n(0)e^{-\alpha_n z^2}\prod_j(1 - z^2/z_{n,j}^2)$ with $\alpha_n \ge 0$ and $z_{n,j} > 0$, so $m_n(t) = \psi_n(0)e^{\alpha_n t^2}\prod_j(1 + t^2/z_{n,j}^2)$ is even and nondecreasing in $|t|$, and $|m_n(w)| \le m_n(|w|)$ for complex w, factor by factor. Write $q_n = g_n k_n$ with $k_n(t) = e^{-a_n\cosh\ell_n t}$ and g_n a multiple of m_n. Since k_n decreases in $|t|$, for $U \ge 0$

$$1 \ge \int_U^{U+1} q_n(t)\,dt \ge g_n(U)\,k_n(U+1),$$

and $k_n(U+1) \to e^{-a\cosh\ell(U+1)} > 0$, so the entire functions g_n are bounded on every

disc uniformly in n and form a normal family. If $q_n \to p$ in law, a subsequence of (g_n) converges locally uniformly on $\mathbb{C}$ to an entire function g, so $q_n \to g(t)e^{-a\cosh \ell t}$ locally uniformly on $\mathbb{R}$ along it, and $\int_\alpha^\beta p = \int_\alpha^\beta g(t)e^{-a\cosh \ell t}\,dt$ for all $\alpha < \beta$. Then $\Phi = \Xi(0)\, g(t)e^{-a\cosh \ell t}$ on $\mathbb{R}$ would extend to an entire function, whereas Φ has a natural boundary on the lines $\operatorname{Im} t = \pm\pi/4$ (Section 11.1.3). □

So no choice of fields on one kernel, finitely or infinitely many, gives the Pólya law, which answers the question of [148, 136, 137]. A sequence of laws of Theorem 19.1.4 converging in law to p would prove RH, by Theorem 20.1.5; by the proposition its kernel parameters (a_n, ℓ_n) would have no limit point in $(0,\infty)^2$. Whether such a sequence exists is not known. Spins with these laws can be coupled.

Corollary 19.1.9. *Let $q_1, \dots, q_N$ be laws of Theorem 19.1.4, with parameters a_i, ℓ_i, and let $J = (J_{ij})$ be symmetric with $J_{ij} \ge 0$ and $J_{ii} = 0$. Let $(X_1, \dots, X_N)$ have density proportional to $\exp\big(\sum_{i<j} J_{ij}t_it_j\big)\prod_i q_i(t_i)$ on $\mathbb{R}^N$. Then $M = \sum_i c_iX_i$ is ferromagnetic for all $c_1, \dots, c_N \ge 0$, not all 0.*

Proof. Each $q_i(t)$ is at most a constant times $\exp(-\frac{a_i}{4}e^{\ell_i|t|})$, so the density is integrable and symmetric under $t \mapsto -t$, and $\mathbb{E}e^{bM^2} < \infty$ for every b. By Theorem 19.1.4(i) each q_i has Gaussian moments of all orders and a moment generating function with only imaginary zeros, so by Newman's theorem [214, 196], in its form for fields h_i with $\operatorname{Re} h_i > 0$ for every i, $Z(h) = \mathbb{E}\exp(\sum_i h_iX_i)$ has no zero with $\operatorname{Re} h_i > 0$ for every i. So for $\varepsilon > 0$, $z \mapsto Z(zc + \varepsilon\mathbf{1})$ has no zero in $\operatorname{Re} z > 0$; letting $\varepsilon \downarrow 0$, Hurwitz's theorem and $Z(0) = 1$ show that $L_M(z) = Z(zc)$ has none there, and by symmetry none in $\operatorname{Re} z < 0$. □

For two spins with $J_{12} = J$ and $c_1 = c_2 = 1$, expanding $e^{Jt_1t_2}$ gives the transform of $X_1 + X_2$, up to a constant,

$$Z_J(\theta) = \sum_{n\ge 0} \frac{(-J)^n}{n!}\,\varphi_1^{(n)}(\theta)\,\varphi_2^{(n)}(\theta),$$

with φ_i the characteristic function of q_i. Whether such coupled laws can approach p is not known; Proposition 19.1.8 does not apply to them, since the law of $X_1 + X_2$ is not a multiplier on one kernel.

At $J = 0$ the zero counts add: with every $\mu_i = 0$, the transform $\prod_i \varphi_i(\theta)$ of $X_1 + \cdots + X_N$ has $\sum_i \frac{\theta}{\pi\ell_i}\log\frac{2\theta}{ea_i\ell_i} + O(1)$ zeros in $(0,\theta]$, by Remark 19.1.7, and this has the main terms of the Riemann von Mangoldt formula exactly when

$$\sum_i \frac{1}{\ell_i} = \frac{1}{2}, \qquad \sum_i \frac{\log(a_i\ell_i)}{\ell_i} = \frac{1}{2}\log 4\pi;$$

for $N = 1$ this is Pólya's kernel, and for $\ell_i = 2N$ it asks $\prod_i a_i = (2\pi/N)^N$. Coupling moves the count without bound. For two spins and $J > 0$, Jensen's

inequality under the product of the tilted laws proportional to $q_i(t)e^{-yt}$ gives $\log Z_J(iy) - \log Z_0(iy) \geq J\, m_1(y)m_2(y) + O(1)$, where $-m_i(y)$ is the tilted mean and $m_i(y) \sim \ell_i^{-1} \log y$ by Stirling's formula. Both laws are ferromagnetic with $b = 0$ in (20.1.1) (Theorem 20.1.2), so $\log \big(Z(iy)/Z(0)\big) = \int_0^\infty \log(1+y^2/r^2)\, dN(r)$ for their counting functions, and an $N_J - N_0$ bounded above would make the difference at most $O(\log y)$. So $N_J(\theta) - N_0(\theta)$ is unbounded above. A saddle point computation suggests $N_J(\theta) - N_0(\theta) = \big(\sum_{i<j} J_{ij}/(\ell_i\ell_j)\big) \log\theta + O(1)$ for $X_1 + \cdots + X_N$ (heuristic).

For $N = 2$ and $\ell_1 = \ell_2 = 4$ the condition displayed above is $a_1a_2 = \pi^2$, met by two copies of the law q with $(a, \ell) = (\pi, 4)$ and $m \equiv 1$, for which $\varphi_q(\theta) = K_{i\theta/4}(\pi)/K_0(\pi)$. The Mellin transform $\mathcal{D}$ of the clock of this pair (Section 20.3), and that of a single copy, have no zero in the box $-0.95 < \operatorname{Re} w < -0.05$, $\frac{1}{2} < \operatorname{Im} w < 11$, in which $\mathcal{D}_{\pi U + \pi U'}$ has the two zeros $\frac{\rho_1 - 1}{2}$ and $\frac{\rho_2 - 1}{2}$ (computed; the argument principle on the boundaries of the boxes $\frac{1}{2} < \operatorname{Im} w < \frac{17}{2}$ and $\frac{17}{2} < \operatorname{Im} w < 11$, sampled, not a ball proof, with the uniform pair computed by the same quadrature as a control, which gives winding 1 in each box). The zeros $\pm ic_k$ of this pair have the main term of the Riemann von Mangoldt formula, and those of the uniform pair the count 2θ (Remark 19.1.7). This paragraph is based on [147].

Proposition 19.1.10. *For two spins as above let $J \downarrow 0$.*

(i) If α is a simple zero of φ_1 and $\varphi_2(\alpha) \neq 0$, then Z_J has a zero $\alpha + J\varphi_2'(\alpha)/\varphi_2(\alpha) + O(J^2)$.

(ii) If α is a simple zero of both φ_1 and φ_2, then Z_J has the two real zeros

$$\alpha \pm \sqrt{J} + \frac{J}{4}\Big(\frac{\varphi_1''}{\varphi_1'} + \frac{\varphi_2''}{\varphi_2'}\Big)(\alpha) + O(J^{3/2}),$$

at distance $2\sqrt{J} + O(J^{3/2})$.

Proof. The expansion gives $Z_J = \varphi_1\varphi_2 - J\varphi_1'\varphi_2' + O(J^2)$ locally uniformly, and (i) is the implicit function theorem at the simple zero α of $\varphi_1\varphi_2$. For (ii) put $\theta = \alpha + \sqrt{J}\, v$, $A = \varphi_1'(\alpha)\varphi_2'(\alpha) \neq 0$ and $B = (\varphi_1'\varphi_2'' + \varphi_1''\varphi_2')(\alpha)$; then $Z_J/(AJ) = v^2 - 1 + \sqrt{J}\,\frac{B}{2A}(v^3 - 2v) + O(J)$, and the implicit function theorem at $v = \pm 1$ gives $v = \pm 1 + \sqrt{J}\,\frac{B}{4A} + O(J)$. Both zeros are real because $X_1 + X_2$ is ferromagnetic (Corollary 19.1.9). □

Numerically, for the spins $(a, \ell) = (2\pi, 1)$ and $(2\pi, 1.3)$ with $m \equiv 1$, the transform of $X_1 + X_2$ has 39, 39 and 40 zeros in $(0, 45)$ for $J = 0$, $\frac{1}{8}$ and $\frac{1}{2}$. At $J = \frac{1}{8}$ Proposition 19.1.10(i) predicts the zeros to within 0.084 for the 17 zeros that lie at distance more than $\frac{1}{2}$ from the zeros of the other spin, and to within 0.017 for the 4 beyond 1; the prediction is poor near a near coincidence, at the zeros 43.66 and 43.69 of the two spins, which is the case of a shared zero split by (ii). For two spins

$(2\pi, 1)$ the double zeros of Z_0 split into 24 pairs in $(0, 45)$, and their separations lie between 0.458 and 0.534 (mean 0.486) at $J = \frac{3}{40}$, against $2\sqrt{J} = \sqrt{3/10} = 0.548$, and between 0.580 and 0.919 (mean 0.688) at $J = \frac{1}{4}$, against $2\sqrt{J} = 1$. The separation is below $2\sqrt{J}$ and decreases with height (computed).

PROPOSITION 19.1.11. *For $n \geq 1$ let $r_n(t) = e^{|t|/2} \exp(-\pi n^2 e^{2|t|})$, and consider the laws with densities proportional to r_n and to $\sum_{m\geq 1} r_m$: the components of the theta series in Riemann's formula and their mixture (Remark 18.3.2). Each of these laws is symmetric, has Gaussian moments of all orders and a characteristic function positive on $\mathbb{R}$, and none of them is ferromagnetic: each characteristic function has infinitely many zeros, and none of them is real. Status: proved.*

Proof. Let X have one of these laws, with characteristic function φ. As $\sum_{m\geq 1} e^{-\pi m^2 x} \leq e^{-\pi x}/(1-e^{-3\pi})$ for $x \geq 1$, each density is at most a constant times $e^{|t|/2} \exp(-\pi e^{2|t|})$, which gives the moments, and $\varphi > 0$ on $\mathbb{R}$ by Remark 18.3.2. The same bound gives $\mathbb{E}e^{b|X|} \leq C \int_0^\infty e^{(b+\frac{1}{2})t} e^{-\pi e^{2t}}\, dt \leq C\pi^{-b/2-1/4}\, \Gamma(\frac{b}{2} + \frac{1}{4})$ for $b \geq 0$, so φ is entire and, since $|\varphi(z)| \leq \mathbb{E}e^{|\operatorname{Im} z||X|}$, of order at most one by Stirling's formula. If φ had finitely many zeros, Hadamard's theorem would give $\varphi(z) = P(z)e^{az}$ with a polynomial P, of degree d say, and a constant a, so that $|\varphi(iy)| \leq C'(1 + |y|)^d e^{|a||y|}$. But $\varphi(iy) = \mathbb{E}\cosh(yX) \geq \frac{1}{2} e^{A|y|}\, \mathbb{P}(|X| > A)$ for every $A > 0$, and $\mathbb{P}(|X| > A) > 0$ because the density is positive; with $A > |a|$ this is a contradiction for large $|y|$. So φ has infinitely many zeros, none of them real, and X is not ferromagnetic (Definition 20.1.1). □

The components are laws of $\pm\frac{1}{2}\log X$ with X of gamma type truncated to $[1, \infty)$, whose densities have a corner at the origin; by the same mechanism the cusped tilts of p in Theorem 19.1.16 are not ferromagnetic. The product of two Gumbel distribution functions in Remark 19.1.6 gives instead the smooth kernel $e^{-a\cosh \ell t}$ and the ferromagnetic laws of Theorem 19.1.4. Since $\varphi > 0$ on $\mathbb{R}$, the real zeros of Ξ are zeros of no component: by Remark 18.3.2 they are the points θ where the positive function $(\theta^2 + \frac{1}{4})W(\theta)$ of that remark equals 1. For $n = 1$ the characteristic function has 5 zeros in the box $0 \leq \operatorname{Re}\theta \leq 40$, $0 \leq \operatorname{Im}\theta \leq 30$, the two of least modulus being $13.0885 + 10.8707i$ and $20.1987 + 15.0786i$ (computed; argument principle). The proposition is based on [145].

19.1.4 Laws that keep the first zeros

Let $0 < \gamma_1 \leq \gamma_2 \leq \ldots$ be the ordinates of the zeros on the critical line, with multiplicity, and put $s_K = \sum_{k\leq K} \gamma_k^{-2}$ and $\beta_K = \frac{1}{2}\operatorname{Var} X - s_K$. Keep the first K factors of the Hadamard product and replace the others by the Gaussian factor with the same effect on the variance:

$$\varphi_K(\theta) = e^{-\beta_K \theta^2} \prod_{k\leq K} \Big(1 - \frac{\theta^2}{\gamma_k^2}\Big).$$

A symmetric Maxwell variable M, with density $x^2 e^{-x^2/2}/\sqrt{2\pi}$, has $\mathbb{E}e^{i\theta M} = (1-\theta^2)e^{-\theta^2/2}$ and $\operatorname{Var} M = 3$; it is ferromagnetic, with $L_M(s) = (1+s^2)e^{s^2/2}$. Let $M_1, M_2, \dots$ be independent copies.

THEOREM 19.1.12. *(i) $\beta_K > 0$ for every K. If $3s_K \le \operatorname{Var} X$, then φ_K is the characteristic function of*

$$X_K = \big(\operatorname{Var} X - 3s_K\big)^{1/2} Z + \sum_{k\le K} \frac{M_k}{\gamma_k},$$

whose law f_K is ferromagnetic, with $\operatorname{Var} X_K = \operatorname{Var} X$ and $\kappa_4(X_K) = -12\sum_{k\le K}\gamma_k^{-4}$. The condition holds for $K \le 16$ and fails for $K = 17$.

(ii) For every K, if the inverse Fourier transform f_K of φ_K is nonnegative, then f_K is a ferromagnetic probability density.

(iii) RH holds if and only if $\varphi_K \to \varphi_X$ pointwise as $K \to \infty$. Under RH, $f_K \to p$ in law along every sequence of K for which $f_K \ge 0$.

Proof. With no hypothesis, $\frac{1}{2}\operatorname{Var} X = \sum_k \gamma_k^{-2}$, where the sum runs over one zero of each pair $\pm\gamma$ and complex γ are allowed (Section 20.4.1). A zero off the line contributes, with its conjugate, $2\operatorname{Re}\gamma^{-2} > 0$, since $|\operatorname{Re}\gamma| > 14 > |\operatorname{Im}\gamma|$. So β_K is the sum of γ_k^{-2} over the line zeros with $k > K$ and of these positive terms, and it is positive.

(i) M_k/γ_k has characteristic function $(1-\theta^2/\gamma_k^2)e^{-\theta^2/2\gamma_k^2}$, so the sum over $k \le K$ accounts for every factor of φ_K except $e^{-(\beta_K - s_K/2)\theta^2}$, a Gaussian factor of variance $2\beta_K - s_K = \operatorname{Var} X - 3s_K$. A sum of independent ferromagnetic variables is ferromagnetic. The cumulants come from $\log\varphi_K$. Since s_K increases with K, the last statement is $3s_{16} < \operatorname{Var} X < 3s_{17}$. In ball arithmetic, with the certified zeros of Arb [179] and $\operatorname{Var} X = (\log\xi)''(\frac{1}{2})$ from the Taylor series of $\log\xi$, $3s_{16} = 0.0456803$, $\operatorname{Var} X = 0.0462100$ and $3s_{17} = 0.0463006$.

(ii) f_K has total mass $\varphi_K(0) = 1$, and its moment generating function $e^{\beta_K s^2}\prod_{k\le K}(1+s^2/\gamma_k^2)$ vanishes only on the imaginary axis. By Proposition 19.1.13 below, f_K is a Gaussian density times a polynomial, so it has Gaussian moments.

(iii) As $K \to \infty$, $\beta_K \downarrow \beta_\infty$, the sum of the terms $2\operatorname{Re}\gamma^{-2}$ over the zeros off the line, and $\varphi_K \to \varphi_\infty = e^{-\beta_\infty\theta^2}\prod_k(1-\theta^2/\gamma_k^2)$, a product over the line zeros, whose zeros are all real. If RH holds, $\beta_\infty = 0$ and $\varphi_\infty = \varphi_X$ by the Hadamard product. If RH fails, φ_X vanishes at a nonreal point and φ_∞ does not, so $\varphi_\infty \ne \varphi_X$. Along a sequence with $f_K \ge 0$ the φ_K are characteristic functions, and Lévy's continuity theorem gives convergence in law. □

The law f_{16} is proved ferromagnetic with no hypothesis, and $\operatorname{TV}(f_{16}, p) = 1.27\times10^{-4}$, ten times smaller than for the tilt by the flow at $\lambda = 0.2$. Beyond $K = 16$ the Maxwell construction runs out of Gaussian variance and positivity must

be checked directly, from a closed form in which He_n is the Hermite polynomial with $\mathrm{He}_n(y) = \mathbb{E}(y + iZ)^n$, g_v is the $N(0, v)$ density, and e_j is the jth elementary symmetric function of $\gamma_1^{-2}, \dots, \gamma_K^{-2}$.

PROPOSITION 19.1.13. *For $v > 0$ the function $e^{-v\theta^2/2} \prod_{k\le K}(1 - \theta^2/\gamma_k^2)$ is the Fourier transform of*

$$f_{K,v}(x) = g_v(x)\, Q_{K,v}\big(x/\sqrt{v}\big), \qquad Q_{K,v}(y) = \sum_{j=0}^{K} \frac{e_j}{v^j}\, \mathrm{He}_{2j}(y),$$

and $f_K = f_{K,2\beta_K}$. There is a threshold $v^*(K)$ such that $f_{K,v} \ge 0$ if and only if $v \ge v^*(K)$, and $v^*(K) \le s_K$. Moreover $Q_{K,v}(y) > 0$ for $|y| \ge \sqrt{8K+2}$, so the sign of $f_{K,v}$ is decided on $|x| < \sqrt{(8K+2)v}$.

Proof. Differentiating g_v gives $g_v^{(n)}(x) = (-1)^n v^{-n/2} \mathrm{He}_n(x/\sqrt{v}) g_v(x)$, and the transform of $g_v^{(2j)}$ is $(-1)^j \theta^{2j} e^{-v\theta^2/2}$. Expanding $\prod_k (1 - \theta^2/\gamma_k^2) = \sum_j (-1)^j e_j \theta^{2j}$ gives the formula. For $v < v'$, $f_{K,v'} = f_{K,v} * g_{v'-v}$, so positivity persists as v increases, and the set of good v is closed. At $v = s_K$ the transform is that of $\sum_{k\le K} M_k/\gamma_k$, so $v^*(K) \le s_K$. The zeros of He_n lie in $|y| < \sqrt{4n+2}$, by a Sturm comparison for the Hermite equation, so for $|y| \ge \sqrt{8K+2}$ every He_{2j} with $j \le K$ is positive, and so are the coefficients $e_j v^{-j}$. □

For $K = 1$, $Q_{1,v}(y) = 1 + (y^2 - 1)/(v\gamma_1^2)$, so $v^*(1) = \gamma_1^{-2} = s_1$, the Maxwell value. For larger K the threshold is far smaller; Table 19.2 brackets it for $K \le 800$. The Hermite sum cancels catastrophically, so the computation runs in ball arithmetic [179] at thousands of bits (8210 at $K = 800$), from the certified ordinates $\gamma_1, \dots, \gamma_{800}$ and $\mathrm{Var}\, X = (\log \xi)''(\frac{1}{2})$. The recurrence $\mathrm{He}_{n+1} = y\mathrm{He}_n - n\mathrm{He}_{n-1}$ gives $Q_{K,v}$ in the monomial basis. On $[c - r, c + r]$ its Taylor coefficients T_m at c come exactly from one polynomial product, and $Q_{K,v}(c+h) \ge T_0 - \sum_{m\ge1} |T_m| r^m$ for $|h| \le r$, with no remainder; intervals where this bound is not positive are bisected. A cover of $[0, \sqrt{8K+2}]$, with Proposition 19.1.13 beyond it and evenness, proves $Q_{K,v} > 0$ on the line. This proves $f_K > 0$ for every $1 \le K \le 200$ and for $K = 300$, 400, 600 and 800. For all these K the minimum of $Q = Q_{K,2\beta_K}$ is attained only at $y = 0$: near 0 because $Q(y) - Q(0) \ge y^2\big(b_2 - \sum_{m\ge2} |b_{2m}| y^{2m-2}\big) > 0$ for $0 < |y| \le 2^{-5}$, the b_n being the coefficients of Q, and elsewhere by covers on which the lower bound exceeds $Q(0)$. For the column $v^*(K)$, a complete cover at $v = t_+ \cdot 2\beta_K$ gives $v^*(K) \le t_+ \cdot 2\beta_K$, and a certified negative value of $Q_{K,v}$ at $v = t_- \cdot 2\beta_K$ gives $v^*(K) > t_- \cdot 2\beta_K$, for dyadic rationals $t_- < t_+$ at most 2^{-18} apart.

What matters is the trend of the ratio $v^*(K)/2\beta_K$: while it stays below 1, f_K is a probability density. It increases slowly, to 0.733 at $K = 800$, and its slope in $\log K$ falls from 0.110 between $K = 30$ and 50 to 0.086 between $K = 600$ and 800. A straight line in $\log K$ through the last two points reaches 1 near $K = 1.8 \times 10^4$,

K	$2\beta_K$	s_K	$v^*(K)$	$v^*(K)/2\beta_K$	$\min Q_{K,2\beta_K}$	$\mathrm{TV}(f_K, p)$
0	0.046210	0	0	0	1	1.48×10^{-2}
1	0.036199	0.005005	0.005005	0.138	0.862	3.81×10^{-3}
2	0.031674	0.007268	0.004955	0.156	0.804	2.09×10^{-3}
3	0.028477	0.008867	0.005416	0.190	0.762	1.30×10^{-3}
5	0.024472	0.010869	0.005527	0.226	0.705	6.96×10^{-4}
10	0.019121	0.013544	0.005441	0.285	0.623	2.64×10^{-4}
16	0.015756	0.015227	0.005222	0.331	0.566	1.27×10^{-4}
17	0.015343	0.015434	0.005190	0.338	0.558	1.14×10^{-4}
20	0.014273	0.015969	0.005065	0.355	0.538	8.75×10^{-5}
50	0.009108	0.018551	0.004146	0.455	0.430	1.70×10^{-5}
100	0.006222	0.019994	0.003303	0.531	0.355	4.42×10^{-6}
150	0.004907	0.020652	0.002816	0.574	0.316	1.93×10^{-6}
200	0.004122	0.021044	0.002487	0.603	0.289	
300	0.003199	0.021505	0.002059	0.644	0.255	
400	0.002659	0.021775	0.001784	0.671	0.232	
600	0.002036	0.022087	0.001441	0.708	0.203	
800	0.001677	0.022267	0.001229	0.733	0.184	

Table 19.2: The zero completed laws f_K. For $K \leq 16$, f_K is a ferromagnetic probability density by Theorem 19.1.12(i) ($s_K \leq 2\beta_K$). For every $K \geq 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so f_K is a ferromagnetic probability density by Theorem 19.1.12(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 19.1.13 lies in a certified interval of width at most 1.2×10^{-7}, within 6×10^{-7} of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. Computed from the certified zeros and variance, except $v^*(1) = s_1$.

while the power law $1 - v^*(K)/2\beta_K \approx 1.65K^{-0.27}$, fitted for $K \geq 100$, never does. The data do not decide between them. Either outcome is consistent with RH, since Theorem 19.1.12(iii) asks only for the convergence of φ_K, not for positivity.

19.1.5 The Maxwell series

For $K \leq 16$, Theorem 19.1.12(i) reads as a law the factorisation

$$\varphi_K(\theta) = e^{-(\mathrm{Var}\, X - 3s_K)\theta^2/2}\, \mathbb{E} e^{i\theta \sum_{k \leq K} M_k/\gamma_k}.$$

It holds for every K, though from $K = 17$ on the Gaussian factor has negative variance; moving that factor to the other side and letting $K \to \infty$ gives an identity in law. Let $\mathcal{M} = \sum_k M_k/\gamma_k$ over the zeros on the critical line, which converges almost surely since the terms are independent and centred with summable variances $3\gamma_k^{-2}$.

THEOREM 19.1.14. *Let Z be independent of X. Then RH holds if and only if*

$$X + \left(\tfrac{1}{2}\operatorname{Var} X\right)^{1/2} Z \stackrel{d}{=} \mathcal{M}.$$

Proof. $\mathbb{E}e^{i\theta\mathcal{M}} = \prod_k (1 - \theta^2/\gamma_k^2) e^{-\theta^2/2\gamma_k^2}$, an entire function of θ with only real zeros. Under RH, $\sum_k \gamma_k^{-2} = \frac{1}{2}\operatorname{Var} X$, and the product is $\varphi_X(\theta) e^{-\operatorname{Var} X\, \theta^2/4}$, the characteristic function of the left side. If RH fails, the characteristic function of the left side vanishes at a nonreal point, and that of $\mathcal{M}$ does not. □

A simulation checks the identity numerically (computed, with 10^7 draws). Like Table 20.3, this checks the computation, not RH.

19.1.6 How close

PROPOSITION 19.1.15. *With no hypothesis, the distance δ of Section 20.1.5 is at most* $\mathrm{TV}(f_{150}, p) < 1.932 \times 10^{-6}$, *since f_{150} is a ferromagnetic law by Theorem 19.1.12(ii) and the positivity proof of Section 19.1.4; without that computer assisted proof, f_{16} gives $\delta < 1.266 \times 10^{-4}$, and the second law of Table 19.1, ferromagnetic by Theorem 19.1.4, gives $\delta < 6.024 \times 10^{-6}$.*

Every density here is even, and $\mathrm{TV}(q, p) = \int_0^\infty |q - p|$ is enclosed in ball arithmetic as follows. On $[0, 3]$, cut into intervals $[c - r, c + r]$ with $r = 2^{-12}$, the integral of $|d|$, $d = q - p$, differs from the exact integral of $|d(c) + d'(c)h|$ over $|h| \le r$ by at most $|d''(c)|r^3/3 + M_3 r^4/12$, where M_3 bounds $|d'''|$ by interval evaluation (for the laws f_K, by $\frac{1}{\pi}\int_0^\infty \theta^3 |\varphi_K(\theta)|\, d\theta$); beyond 3 each density is bounded by an explicit function of known logarithmic decay rate. The laws f_K are built from the certified zeros and variance of Section 19.1.4, so the distances are those of the true laws. For the laws of Table 19.1 the normalising constants are Arb's Bessel functions $K_\nu(a)$ at real order, by Theorem 19.1.4(ii), or integrals enclosed by Arb's quadrature when $\mu > 0$, and M_3 comes from interval evaluation of the closed form.

19.1.7 A cusped weight never works

The multipliers of Proposition 19.1.2 are entire. A weight with a cusp at the origin behaves differently: no tilt of p by $e^{-a|t|}$, $a > 0$, is ferromagnetic, however small a is. We write the tilts in the variable $u = t/2$ of the de Bruijn Newman flow (Section 4.1). An even weight $w(u)$ gives the multiplier $m(t) = w(t/2)$, and $8\int_0^\infty w(u)\Phi_{\mathrm{dBN}}(u)\cos(zu)\, du = \int_{\mathbb{R}} \Phi(t) m(t) \cos(\theta t)\, dt$ with $\theta = z/2$, so the zeros of the first function are twice those of the transform of $m\,p$. Thus $e^{-\mu|u|}$ is $e^{-a|t|}$ and $\cosh(\mu u)$ is $\cosh(at)$, with $a = \mu/2$.

THEOREM 19.1.16. *Let $\mu > 0$ and $H^{(\mu)}(z) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$. With C a standard Cauchy variable,*

$$H^{(\mu)}(z) = \mathbb{E}\,H_0(z+\mu C),$$
$$z^2 H^{(\mu)}(z) \to \mu\,\Phi_{\mathrm{dBN}}(0) = 0.4467\mu \quad (z\to\infty,\ z\in\mathbb{R}).$$

Hence $H^{(\mu)}$ is positive beyond some height, has finitely many real zeros and infinitely many non real ones, the law with density proportional to $e^{-\mu|t|/2}p(t)$ is not ferromagnetic, and the tilted clock $R^{(\mu)}(s) = H^{(\mu)}(0)/H^{(\mu)}(2i\sqrt{s})$ is not the Laplace transform of a GGC, for any $\mu > 0$.

Proof. Since $e^{-\mu|u|} = \mathbb{E}\cos(\mu C u)$ and $\mathbb{E}\sin(\mu C u) = 0$, Fubini gives $H^{(\mu)}(z) = \mathbb{E}H_0(z+\mu C) = \int H_0(y)P_\mu(z-y)\,dy$ with $P_\mu(y) = \mu/(\pi(\mu^2+y^2))$. As $H_0(y) = \frac{1}{8}\Xi(y/2)$ decays exponentially and $\int_{\mathbb{R}} H_0(y)\,dy = \pi\Phi_{\mathrm{dBN}}(0)$ by Fourier inversion at $u=0$, dominated convergence, with $z^2P_\mu(z-y) \le 2(\mu + y^2/\mu)/\pi$, gives $z^2H^{(\mu)}(z) \to \mu\Phi_{\mathrm{dBN}}(0)$. The function $H^{(\mu)}$ is even and entire of order one; with finitely many zeros it would be a polynomial times $e^{\alpha z}$, hence a polynomial by evenness, contradicting $H^{(\mu)}(z)\to 0$. So it has infinitely many zeros, almost all non real. It is positive on the imaginary axis, $H^{(\mu)}(iy) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, so the argument of Theorem 9.2.4 applies to $R^{(\mu)}$ verbatim, as it does to R_t in Corollary 9.2.5, and $R^{(\mu)}$ is the Laplace transform of a GGC only if all zeros of $H^{(\mu)}$ are real. By the relation between the variables above, the characteristic function of the tilt $e^{-\mu|t|/2}p(t)$ is $H^{(\mu)}(2\theta)/H^{(\mu)}(0)$, whose non real zeros show that the tilt is not ferromagnetic. □

Numerically the real zeros are lost at once. For $\mu = 1$, $H^{(1)}$ has no sign change on $[0,80]$, where H_0 has six, and the Laguerre inequality $(H')^2 - HH'' \ge 0$, which every real rooted even entire function of order one satisfies, fails for $24 \le z \le 38$ and $43 \le z \le 80$. Even for $\mu = 0.01$ only two sign changes survive on $[0,400]$, against the six of H_0 on $[0,80]$. The cusp of $e^{-\mu|u|}$ at $u=0$ adds a positive term decaying like z^{-2}, which swamps the exponentially small oscillation carrying the zeros of Ξ. The table compares $H^{(\mu)}(z)$ with the tail value $\mu\Phi_{\mathrm{dBN}}(0)/z^2$.

μ	z	$H^{(\mu)}(z)$	$\mu\Phi_{\mathrm{dBN}}(0)/z^2$
1	80	7.237×10^{-5}	6.980×10^{-5}
1	400	2.796×10^{-6}	2.792×10^{-6}
3	400	8.387×10^{-6}	8.376×10^{-6}
0.01	400	2.796×10^{-8}	2.792×10^{-8}

A weight that is smooth at the origin but decays only like u^{-2} is the other natural candidate [154]. For $a > 0$ let

$$H^{[a]}(z) = \int_0^\infty \frac{\Phi_{\mathrm{dBN}}(u)}{1+a^2u^2}\cos(zu)\,du.$$

PROPOSITION 19.1.17. *$H^{[a]}(z) = \mathbb{E}\, H_0(z+aL)$, with L a standard Laplace variable of density $\frac{1}{2}e^{-|y|}$; $H^{[a]} = \int_0^\infty e^{-v} H_{-va^2}\, dv$, an exponential mixture of backward heat flows; and $H_0 = (1 - a^2 \frac{d^2}{dz^2}) H^{[a]}$. Consequently, if $H^{[a]}$ has only real zeros for one $a > 0$, then RH holds.*

Proof. $\mathbb{E}\cos(aLu) = (1 + a^2u^2)^{-1} = \int_0^\infty e^{-v} e^{-va^2u^2} dv$ and $\mathbb{E}\sin(aLu) = 0$, so Fubini gives the first two forms, with $H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du$ as in Section 4.1; the third follows from $-\frac{d^2}{dz^2}\cos(zu) = u^2 \cos(zu)$. The function $H^{[a]}$ is even, real on $\mathbb{R}$ and entire of order one, since its kernel decays like Φ_{dBN}; if its zeros are real it is in the Laguerre Pólya class, and so is $(1 - a\frac{d}{dz})(1 + a\frac{d}{dz}) H^{[a]} = H_0$, because $1 \pm a\frac{d}{dz}$ preserve the class (Laguerre's theorem; see [195]). So H_0 has only real zeros. □

The weight $(1 + a^2u^2)^{-1}$ has poles at $u = \pm i/a$, so it is not one of the entire multipliers of Proposition 19.1.2 and not a universal factor in Pólya's sense; RH is not known to imply that $H^{[a]}$ has real zeros, and whether it has them for some $a > 0$ is open.

Extreme values, hitting times and a path to RH

The ferromagnetic approximants above are laws whose transforms are proved to have only real zeros; the extreme value approximants below are explicit kernels whose real zeros are tracked to a stated height. The Pólya density is an exact superposition of extreme value laws (Section 19.17), and the reciprocal ξ function is the Laplace transform of a sum of independent exponentials with rates γ_k^2, which Kent's theorem reads as the hitting time of a one dimensional diffusion; by Krein's inverse spectral theorem every strictly increasing sequence of positive rates arises this way. Each extreme value approximation therefore defines a candidate hitting time, and real simple zeros become the existence of a string with positive mass. None of this proves RH. It gives a precise target, three routes to it with the place where each stops, the folded Gumbel kernels, whose zeros are tracked rigorously to a stated height, and diagnostics, from Hankel determinants of the kernel moments to threshold fits of the hitting time tail, whose power and limits are measured.

This part is based on [133]; its numbers were recomputed, and values not recomputed are marked unchecked. The kernel Φ here is Φ_{dBN} of Section 4.1 (the normalisation of Rodgers and Tao, with e^{9u} and e^{4u}), not the kernel of (2.1.3); with it $H(z) = \int_0^\infty \Phi(u) \cos(zu)\, du = \frac{1}{8}\Xi(z/2)$, so the zeros of H are at $z = \pm 2\gamma_k$ and the variable of the zeros is $x = z/2$. The eigenvalues of a string are λ_k, the flow time is t and Λ the de Bruijn Newman constant (Section 4.1), and Euler's constant is γ_E.

Results proved elsewhere are cited: Kent's theorem and the reduction of RH to a string with simple spectrum (Propositions 5.4.1 and 5.5.1, Section 5.2.3), the

B spline form of the truncated hitting time (Section 21.2, Proposition 21.2.1), the hazard rate and the Prony extraction of the atoms (Sections 5.2.1 and 5.2.2), the Liouville operator and the diffusion for Pólya's kernel (Proposition 18.1.7, Theorem 18.1.10), the Stieltjes criterion (Theorem 19.6.1), the incomplete gamma form, the kink and Rouché tracking (Proposition 18.1.3, Theorem 18.2.3, Corollary 18.2.2). The results, with their status, are these.

1. Each term of the theta series for Φ is an explicit combination of two Gumbel kernels of fractional order, with Mellin transform $n^{-s}\xi(s)/(4\zeta(s))$ (Theorem 19.17.1, proved). This gives the folded Gumbel kernels, Pólya's two sided kernel and the Dyck chain, whose real zeros to the heights tabulated are computed (Table 19.8). The breakdown height $T_N \approx 4(N+1)^2$ of the folded family is a heuristic balance of the kink against the size of H.

2. The zeros of Ξ are all real and simple if and only if the reciprocal ξ law is the hitting time of a string (Corollary 19.19.1, proved, with the hypotheses of simplicity and of the class of $\widehat{H}$ that the source left out). The tail gives γ_1^2 without any zero, and a Laplace inversion recovers γ_1 to eighteen digits (Proposition 19.19.2, proved under RH, simple zeros and $|\xi'(\rho)| \geq e^{-C|\gamma|}$ for the full series, the exponential excess needing only the two term tail; the digits are computed).

3. The Gumbel pieces have unconditional Hadamard products whose atoms are the poles of the archimedean factor; the atoms γ_k^2 come from the superposition, a sum and not a product (Section 19.20, proved for the pieces, heuristic for the superposition).

4. Pólya's kernel has an explicit string, the Liouville string of Biane, whose first eigenvalues are computed (Section 19.21).

5. For a kernel with infinitely many zeros the zeros are all real if and only if the Hankel determinants of $\mu_j = \sum_k \widehat{\gamma}_k^{-2j}$ are all positive (Proposition 19.22.1, proved). The first eleven pairs are positive for the exact kernel, the two term folded kernel and Pólya's kernel (computed). Finite order positivity cannot see an off line pair above a height that grows with the order (heuristic).

6. Three routes to RH each stop at a known place, and the folded family reduces RH to a height rule for all N (Section 19.2, Proposition 19.2.1, proved). The Rouché inequality behind the rule is certified in ball arithmetic for $N = 1, 2, 3, 4$ at the heights of the table, given the count of zeros of Ξ below the height (Section 19.3); $N = 5$ is computed.

7. The eigenvalue problem for the shifted Bessel kernels is a rational recurrence with real eigenvalues (proved), but the positivity of the extension and its reading as a string are not shown for the tabulated cases; the cotangent

law for fractional shifts, the collisions of the de Bruijn Newman zeros and the masking of the first nonreal pair are in Sections 19.3 and 19.4, with the status stated there.

8. Devroye's alternating series gives a sign test for the reciprocal Ξ law, the mean excess function of the hitting time has an expansion with a rate, and a detection rule for the first nonreal pair in the de Bruijn Newman family is stated and tested (Section 19.5, computed unless stated).

The remaining open problems close the chapter.

19.2 Is there a path to RH?

Corollary 19.19.1 turns RH with simple zeros into the existence of a string. There are three ways one might try to produce it. In this section and the next the variable of the zeros is $x = z/2$ when real and $w = z/2$ when complex, so that the zeros of $H_N^F(2x)$ are the ordinates tracked below; the flow time is t.

19.2.1 Route one: solve the inverse problem for Ξ directly

Find the string whose hitting time has Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ and show its mass is positive. Krein's inverse problem needs the spectral measure, the eigenvalues and their weights, and the zeros alone do not determine the string (Proposition 5.4.1 gives it only up to the weights), so this route is circular without independent access to the mass. A string defines a de Branges space, and de Branges proposed a positivity condition implying RH; Conrey and Li [46] showed that this condition fails for ζ. That does not close the route, but the obvious positivity statements about the string are false.

19.2.2 Route two: a convergent family of explicit strings

Real zeros survive locally uniform limits (Hurwitz). So RH follows from any sequence $\widehat{H}_m \to H$ locally uniformly with each $\widehat{H}_m$ real rooted, and for even real entire functions of order less than two with $\widehat{H}_m(0) > 0$ each has a string if its zeros are simple (Proposition 5.5.1 and Corollary 9.1.2 give the same reduction for hitting times). The candidate families of Section 5.6 stop short in different ways.

- Pólya's kernels have explicit strings (Section 19.21, Proposition 18.1.7) but do not converge to Φ.
- Hejhal's m term kernels converge to Φ, but only almost all their zeros are real [96].
- The de Bruijn Newman family $H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du$ has only real zeros for $t \geq \Lambda$, with $\Lambda \geq 0$ (Rodgers and Tao [250]), $\Lambda \leq 0.22$ (Polymath

[247]) and $\Lambda \le 0.2$ (Platt and Trudgian [232]). The strings exist for $t > \Lambda$, where the zeros are also simple (Section 5.6), and RH is the statement $\Lambda = 0$: the strings survive the limit $t \downarrow 0$.

19.2.3 Route three: the folded family at growing height

Here H_N^F is the cosine transform of the folded kernel $\Phi_N^F = \sum_{n \le N} \phi_n$ on $u > 0$ (Table 19.8), so that $\Phi - \Phi_N^F$ is the omitted set $\{n > N\}$ of Chapter 18; for $N = 0$ the sum is empty and carries no zeros. Proposition 18.1.3, with the integers $n \le N$ in place of $\mathcal{F}(P)$ and its constant $\frac{1}{2}$ replaced by $\frac{1}{2} - c_E$, $c_E = \sum_{m>N}(4\pi m^2 - 1)e^{-\pi m^2}$ (Section 18.2), gives $8H_N^F(z)$ at $s = \frac{1}{2} + \frac{iz}{2}$.

Proposition 19.2.1. *Suppose there is $c > 0$ such that for every $N \ge 1$ all zeros of H_N^F in the disc $|z| < 2c(N+1)^2$ are real. Then RH holds.*

Proof. $H_N^F \to H$ locally uniformly, since $\Phi - \Phi_N^F$ on $u > 0$ is bounded by $\sum_{n>N} |\phi_n|$, which decays like $e^{-\pi(N+1)^2}$ uniformly with all exponential moments. If $H(z_0) = 0$ with $|z_0| < 2c(N_0+1)^2$, Hurwitz's theorem gives zeros $z_N \to z_0$ of H_N^F, $N \ge N_0$, which lie in the disc and so are real. Hence z_0 is real. □

The hypothesis holds for $N = 1, 2, 3, 4$ with $c = 3.6$ (Section 19.3, certified in ball arithmetic given the count of zeros of Ξ below the height of each disc) and for $N = 5$ in floating point (Section 19.4, computed). Each H_N^F is a finite sum of incomplete gamma functions, which makes this the most explicit route, close in spirit to Haglund's conjectures on incomplete gamma approximants [94]. In string language it asks for strings that agree with the true one up to frequency $4(N+1)^2$. The proposition is a reformulation, not a reduction in difficulty: for each N the hypothesis implies RH up to height about $c(N+1)^2$, and nothing in the structure of H_N^F is yet known to force it.

19.2.4 Verdict

There is no proof here. There is a precise target, the positivity of the string mass, and three approaches, each stopping at a known place: the inverse problem needs unavailable data, route two needs the limit $\Lambda = 0$, and route three a height rule equivalent to RH at growing heights. What the extreme value and hitting time viewpoint adds is computational: the bottom of the spectrum is a tail rate estimated by peaks over threshold to eighteen digits without computing zeros; Hankel positivity is a finite order test from the kernel moments that certifies no height (Problem 4); and the folded family turns the height rule into an explicit numerical hypothesis.

19.3 Work on the open problems

This section and the next take up five problems left open by the source, saying for each what is proved, computed and open; this section gives the certified and first computations, Section 19.4 the longer ones.

19.3.1 Problem 1: the height rule for fixed N

Write $R_N = \Phi - \Phi_N^F = \sum_{n>N} \phi_n$ on $u \geq 0$ and $E_N = H - H_N^F$, so $E_N(z) = \int_0^\infty R_N(u)\cos(zu)\,du$. The cosine transform of ϕ_n at complex z is $\frac{1}{2}(F_n(z) + F_n(-z))$, with F_n the incomplete gamma sum; it equals $\operatorname{Re} F_n(z)$ only for real z.

Lemma 19.3.1 (Kink bound). *For real $x \neq 0$,*

$$E_N(x) = -\frac{R_N'(0)}{x^2} - \frac{1}{x^2}\int_0^\infty R_N''(u)\cos(xu)\,du, \qquad |E_N(x)| \leq \frac{B_N}{x^2},$$

with $B_N = |R_N'(0)| + \|R_N''\|_{L^1(0,\infty)}$.

Proof. Integrate by parts twice; R_N and R_N' vanish at infinity and the boundary term $R_N(0)\sin(0)$ is zero. □

This is the integration by parts of Theorem 18.2.3 for the cutoff $\{n > N\}$, with kink $\kappa_N = |R_N'(0)|$. Numerically $B_N = 1.653 \times 10^{-7}$, 2.783×10^{-16} and 5.739×10^{-28} for $N = 2, 3, 4$ (0.079 for $N = 1$). In every case $R_N'' \geq 0$ on $(0, \infty)$, the sign decided in ball arithmetic on every cell of a subdivision, so $\|R_N''\|_{L^1} = |R_N'(0)|$ and $B_N = 2\kappa_N$ exactly (certified, with a rigorous tail bound for the series over n). The leading order $8\pi^3(N+1)^6 e^{-\pi(N+1)^2}$ overstates κ_N by 15%, 8% and 5% for $N = 2, 3, 4$.

Proposition 19.3.2 (Rouché certificate). *Let T lie strictly between two consecutive ordinates, and suppose Ξ has exactly $2\mathcal{N}(T)$ zeros in $|w| < T$, all real and simple. If $|E_N(2Te^{i\vartheta})| < |H(2Te^{i\vartheta})|$ for $\vartheta \in [0, \pi/2]$, and $H_N^F(2x)$ has $\mathcal{N}(T)$ sign changes on $(0, T)$, then every zero of $H_N^F(2x)$ with $|x| < T$ is real.*

Proof. Both functions are even and real on the real axis, so the inequality holds on the whole circle $|z| = 2T$. By Rouché, $H_N^F = H - E_N$ has $2\mathcal{N}(T)$ zeros in the disc, the same number as H. The sign changes supply $2\mathcal{N}(T)$ real ones, so there are no others. □

The inequality on the quarter circle was tested in two ways. It was sampled at 721 points at 250 and 450 bits, with E_N the sum of the omitted incomplete gamma terms (computed). It was also certified: on each of 3200 to 19600 cells in ϑ the bound $|E_N(z)| \leq \int_0^\infty |R_N(u)|\cosh(u \operatorname{Im} z)\,du$ is evaluated in ball arithmetic (a closed form in incomplete gamma functions with a rigorous tail for $n > N + 5$), $|H|$ is the box value of $\frac{1}{8}\Xi(z/2)$, and the signs of $H_N^F(2x)$ are certified on a grid of

step 0.05. Outside ball arithmetic stays only the count of zeros of Ξ in the disc, taken from the book's list of zeros and the verification of RH to this height, which is the hypothesis of the proposition. Table 19.3 gives the results.

N	T	$\mathcal{N}(T)$	sampled $\min \lvert H\rvert/\lvert E_N\rvert$	certified lower bound	$c = T/(N+1)^2$
1	14.4	1	2.45	1.0018	
1	17.5	1	3.65	1.0195	4.38
2	31.7	4	151	3.49	
2	35.3	5	36.3	2.70	3.92
3	51.4	10	2.83×10^5	2555	
3	57.9	12	1965	64.9	3.62
4	86.1	23	1.22×10^6	35921	
4	97.3	28	355	54.2	3.89

Table 19.3: Rouché tests for the folded kernels. The sampled minimum is over 721 points of the quarter circle (computed); the lower bound is over the whole quarter circle (certified in ball arithmetic, and weaker than the sampled minimum because the bound is crude). Every row passes both.

In every row the sampled minimum is attained at $\vartheta = 0$, on the real axis, where Lemma 19.3.1 applies. There the margins $|H(2T)|(2T)^2/B_N$ at the largest heights are 5.77, 262 and 51.4 for $N = 2, 3, 4$, and 0.41 and 0.80 for $N = 1$, so for $N = 1$ the kink bound alone does not suffice and only the certificate on the whole circle does. With Proposition 19.3.2 the rows certify, given the zero count, that every zero of $H_N^F(2x)$ with $|x| < T$ is real, for $N = 1, 2, 3, 4$ and $c = 3.6$. The case $N = 0$ is excluded, since $H_0^F = 0$.

The certificates settle the hypothesis of Proposition 19.2.1 only at these four levels, and they show why the general case is hard. The binding inequality is

$$\tfrac{1}{8}|\Xi(T)| > \frac{B_N}{4T^2}$$

at a point T between consecutive zeros: a lower bound for $|\Xi|$ at Gram type points up to height about $4(N+1)^2$, exactly the information Turing's method uses to verify RH to that height [270]. So the height rule for N contains RH verification up to $c(N+1)^2$, and the folded route can run level with numerical verification of RH but not ahead of it (heuristic).

19.3.2 Problem 2: explicit strings for shifted Bessel kernels

Pólya's kernels involve $\operatorname{Re} K_{\mu+i\nu}(a)$ with $\mu = 9/4$ and $5/4$. For every $\mu \geq 0$ and $a > 0$, $\nu \mapsto \operatorname{Re} K_{\mu+i\nu}(a)$ is a multiple of the characteristic function at $\theta = \nu$ of the law of Theorem 19.1.4 with $\ell = 1$, no tilt and one field $\lambda_1 = \mu$, so its zeros are real. For integer shifts the boundary condition of the eigenvalue problem is explicit. The poles of its rational function are written q_j.

THEOREM 19.3.3. *Let $\mu \geq 0$ be an integer and $a > 0$. By the recurrence $K_{\nu+1}(x) = K_{\nu-1}(x) + (2\nu/x)K_\nu(x)$,*

$$\operatorname{Re} K_{\mu+i\nu}(a) = A_\mu(\nu^2)K_{i\nu}(a) + B_\mu(\nu^2)\, aK'_{i\nu}(a)$$

with real polynomials A_μ, B_μ. Hence, at a point with $B_\mu(\nu^2) \neq 0$, ν is a zero exactly when $g(\tau) = K_{i\nu}(ae^\tau)$ solves the Liouville problem $-g'' + a^2e^{2\tau}g = \lambda g$, $\lambda = \nu^2$, with the spectral boundary condition

$$g'(0) = \beta_\mu(\lambda)\, g(0), \qquad \beta_\mu = -A_\mu/B_\mu.$$

The first cases are Dirichlet for $\mu = 0$, Neumann ($\beta_1 = 0$) for $\mu = 1$,

$$\beta_2 = \tfrac{a^2}{2} - \lambda, \qquad \beta_3 = \frac{4(a^2 - 3\lambda)}{a^2 + 8 - 4\lambda}.$$

The theorem is proved (the recurrence; β_2, β_3 and every entry of Table 19.4 were rederived independently). The Liouville problem is that of Proposition 18.1.7, and the shift lemma for Pólya's kernels is in Section 22.3.

PROPOSITION 19.3.4 (Extended boundary problem). *Suppose $\beta_\mu(\lambda) = \alpha - m\lambda - \sum_{j=1}^{J} \sigma_j/(q_j - \lambda)$ with $m \geq 0$, $\sigma_j > 0$ and q_j real. Then the eigenvalue problem of Theorem 19.3.3, away from the poles, is the eigenvalue problem of a self adjoint operator built from the Liouville operator on $(0, \infty)$ with a point mass m at 0 and J extra coordinates w_j coupled to $g(0)$, with stiffness q_j, unit mass and coupling $-\sqrt{\sigma_j}$. Its eigenvalues are real. If in addition the energy form is nonnegative, for example if every $q_j > 0$ and $\alpha \geq \sum_j \sigma_j/q_j$, they are positive, so that the corresponding ν are real.*

Proof. The boundary condition enters the Rayleigh quotient as $\int_0^\infty (g'^2 + a^2e^{2\tau}g^2)\, d\tau + \beta_\mu(\lambda)g(0)^2$. The term $-m\lambda g(0)^2$ moves to the mass side as a point mass. Each term $-\sigma_j g(0)^2/(q_j - \lambda)$ is the Schur complement of an extra coordinate: eliminating $w_j = \sqrt{\sigma_j}\, g(0)/(q_j - \lambda)$ from $\sum_j [(q_j - \lambda)w_j^2 - 2\sqrt{\sigma_j}\, g(0)w_j]$ leaves $-\sum_j \sigma_j g(0)^2/(q_j - \lambda)$. The energy and mass forms are symmetric and the mass form is positive, so the problem is self adjoint with real eigenvalues. The energy form is the nonnegative integral plus $\alpha g(0)^2 + \sum_j q_j w_j^2 - 2\sum_j \sqrt{\sigma_j}\, g(0)w_j$, which is nonnegative under the stated condition. □

This does not make the extended system a string. A half line with J extra coordinates coupled at 0 carries no Dirichlet condition there, no hitting time is constructed for it, and Kent's theorem is not available. The reading in which $m > 0$ makes the diffusion sticky at 0 and each pole is an extra state visited from 0 is heuristic. What does follow: if the zeros ν_k of $\operatorname{Re} K_{\mu+i\nu}(a)$ are real, positive and simple, Krein's converse (Proposition 5.4.1) gives a string whose hitting time has Laplace transform $\prod_k (1 + s/\nu_k^2)^{-1}$, but it is not identified with the

extended system. Table 19.4 records the sign condition. The zeros of $\operatorname{Re} K_{3+i\nu}(2)$ are 1.41977, 3.84551, 5.74106, 7.37072, 8.84981; for $\mu = 4$, $a = 2\pi$ they are 2.72330, 7.25847, 10.37828, 12.82694, 14.96543; for $\mu = 6$, $a = 2\pi$ they are 1.92369, 5.51737, 8.55690, 11.10450, 13.33385 (computed; the source reports that a discretisation of the extended string with 7000 nodes gives the same lowest five eigenvalues to the digits it prints, which was not rerun).

μ	m	poles q_j at $a = 2\pi$	σ_j at $a = 2\pi$	sign condition at $a = 2\pi$ / $a = 2$
2	1	none	none	yes / yes
3	0	11.87	-3.87	no / yes
4	1/6	7.58	1.98	yes / yes
5	0	5.02, 59.6	-0.165, 270.6	no / yes
6	1/15	3.71, 35.0	0.054, 42.2	yes / yes
7	0	2.88, 26.0, 195.5	0.006, 13.9, 2877	yes / yes
8	1/28	2.34, 20.8, 89.1	0.005, 5.86, 380	yes / yes

Table 19.4: Partial fraction data for β_μ (computed in ball arithmetic). The sign condition of Proposition 19.3.4 is that $m \geq 0$ and all $\sigma_j > 0$; it is a hypothesis of the proposition and does not by itself give a string.

When the sign condition fails, as for $\mu = 3$ and 5 at $a = 2\pi$, the zeros are still real by Pólya's shift lemma (Section 22.3), and Krein's converse still supplies a positive string if they are simple; it is just not this Liouville extension. For Pólya's own orders the recurrence reduces $\mu = 9/4$ and $5/4$ to the base order $1/4$, but $K_{1/4+i\nu}(a)$ is complex for real ν, so the reduction lands on a fractional base string not identified here. The problem is thus reduced to fractional $\mu \in (0, 1)$, taken up in Section 19.4.

19.3.3 Problem 3: the de Bruijn Newman strings

Proposition 19.3.5. *For $H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du$, the mean of the candidate hitting time is*

$$\mu_1(t) = \mathbb{E}T_t = 2\mathbb{E}_t[U^2], \qquad \frac{d\mu_1}{dt} = 2\operatorname{Var}_t(U^2) > 0,$$

where U has density proportional to $e^{tu^2}\Phi(u)$ on $(0,\infty)$. Equivalently $\mu_1(t) = \sum_k \gamma_k(t)^{-2}$, with $\pm 2\gamma_k(t)$ the zeros of H_t counted with multiplicity.

Proof. μ_1 is the ratio of the first two Taylor coefficients of $H_t(2i\sqrt{s})$, namely $4M_2/(2M_0)$ with M_{2m} the moments of the tilted kernel; differentiate in t. □

The moments of the reweighted kernel along the flow are those of Section 4.1.3; what is new is the monotonicity of the mean. Numerically $\mu_1 = 0.0232010$, 0.0231050, 0.0226358, 0.0190909 at $t = 0.2, 0, -1, -10$, and $d\mu_1/dt = 4.78 \times 10^{-4}$

at $t = 0$ (computed). The mean increases strictly through Λ, so it does not see the transition. Nor do low order Hankel determinants: D_n and D'_n are positive for $n \le 5$ on a grid of 13 values of t in $[-10, 0.2]$ (computed), although H_t has nonreal zeros for $t < 0$ [250]. The zeros that leave the line first are high ones, at Lehmer pairs, while low order moments are dominated by $\gamma_1, \gamma_2, \ldots$; the transition at Λ is a high order phenomenon in the string (heuristic).

19.3.4 Problem 4: detecting the first nonreal pair

For the two term folded kernel, the argument principle on the boundary of $[36, 150] \times [-8, -0.05]$ in the w plane gives exactly two zeros,

$$w = 43.13891 - 3.28097i, \qquad w = 47.52276 - 6.25251i,$$

a further 17 in $[36, 150]\times[-40, -8]$, and none with real part below 36 and imaginary part above -40 (computed, floating point; winding numbers 2, 17 and 0). Beyond the breakdown height the zeros peel away into the half plane rather than pairing just off the line (the source's observation, not rerun). The moment Hankel test at 120 digits is positive to order 11, with the same signs at 240 digits (computed), whereas the seven real atoms with the first pair alone already give $D_6 < 0$: the remaining off line atoms mask the first pair (Table 19.7). So a finite order sign is not a local detector. It certifies only that the truncated moment problem has a positive solution, and the order needed to see a nonreal pair is not determined by the number of real atoms below it (heuristic).

19.3.5 Problem 5: higher eigenvalues by joint threshold fits

Replace two point deflation by Prony's method on $2K$ equally spaced values $x_0 + jd$ of the Talbot tail at 80 digits, fitting K exponentials at once: the extraction of Section 5.2.2 with the atoms fitted jointly. Table 19.5 shows the errors reported by the source; it was not rerun, and the spacing d is not stated there.

K	x_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6
3	0.04	4×10^{-14}	2×10^{-7}	7×10^{-4}			
4	0.03	3×10^{-16}	1×10^{-9}	6×10^{-6}	0.084		
5	0.02	8×10^{-16}	3×10^{-10}	7×10^{-7}	0.0075	0.16	
6	0.015	6×10^{-17}	1×10^{-11}	2×10^{-8}	4×10^{-4}	0.017	0.85
6	0.01	2×10^{-10}	1×10^{-6}	3×10^{-4}	0.16	0.27	4.4

Table 19.5: Absolute errors of γ_k recovered by Prony fits of K exponentials to the tail of the hitting time, window starting at threshold x_0 (source values, unchecked).

The pattern is the threshold bias of peaks over threshold fits in exact form. The omitted atoms contribute roughly $e^{-(\gamma_{K+1}^2 - \gamma_k^2)x_0}$ to the error in γ_k, up to conditioning factors (the rule of Section 5.2.2), so raising K allows a lower threshold

(heuristic). Lowering it too far, as in the last row, lets atoms beyond K contaminate the fit and loses accuracy everywhere. The best pair (K, x_0) trades these two biases against the working precision.

19.4 Second round on the open problems

This section returns to the five problems with longer computations. Nothing in it is in interval arithmetic; the status of each number is stated where it is printed.

19.4.1 Problem 1: the circle test at $N = 5$

The test of Proposition 19.3.2 was repeated for the five term folded kernel, with E_5 the sum of the omitted incomplete gamma terms $n = 6, \dots, 14$ (floating point, 30 digits on the circle, 80 digits for the sign changes). Below height 160, $H_5^F(2x)$ has 53 sign changes on a grid of step 0.03 (computed), while Ξ has 58 ordinates there. The source states that each sign change lies within the grid resolution of an ordinate of Ξ; this was not reproduced, since the largest distance from a sign change to the nearest ordinate is 1.04. Sampled at 721 points of the quarter circle, the test gives Table 19.6, each minimum of $|H|/|E_5|$ again on the real axis.

T	$\mathcal{N}(T)$	sampled $\min_\vartheta \lvert H\rvert/\lvert E_5\rvert$	result
128.548	42	6.01×10^6	pass
130.333	43	7.66×10^5	pass
132.293	44	3.55×10^5	pass
134.127	45	3.59×10^4	pass
136.436	46	3.54×10^4	pass
138.926	47	1191	pass
140.430	48	247	pass
142.118	49	157	pass
144.556	50	57.1	pass
146.712	51	3.11	pass
148.738	52	1.36	pass
150.489	53	0.052	fail

Table 19.6: Rouché tests for $N = 5$ (computed: floating point, sampled on the circle, E_5 truncated at $n = 14$).

So, on the sampled circles, every zero of $H_5^F(2x)$ with $|x| < 148.7$ is real, giving $c \geq 148.7/36 = 4.13$ (computed). The margin falls by roughly an order of magnitude every one or two zeros, and the test first fails at $T = 150.5$, just above the predicted breakdown $4 \cdot 6^2 = 144$; a failed circle only stops the test and does not mean a zero has left the line. Across $N = 2, \dots, 5$ the constants are $3.92, 3.62, 3.89, 4.13$, the first three certified (Table 19.3). They are heights at which the test succeeded, not breakdown heights: for $N = 2$ the first nonreal

zero at $w = 43.14 - 3.28i$ (Section 19.3, Problem 4) puts the breakdown at $c \le 43.14/9 = 4.79$. So the rule $T_N \approx 4(N+1)^2$ is consistent with c between 3.6 and 4.2 at every level tested, and nothing more.

19.4.2 Problem 2: fractional shifts and the cotangent law

For noninteger μ there is no finite recurrence, but each zero ν_k of $\operatorname{Re} K_{\mu+i\nu}(a)$ still fixes the Robin ratio of the Liouville eigenfunction,

$$\beta_k = \frac{g'(0)}{g(0)} = \frac{aK'_{i\nu_k}(a)}{K_{i\nu_k}(a)}, \qquad \lambda_k = \nu_k^2.$$

The question is whether these values come from an admissible boundary law $\beta(\lambda) = \alpha - m\lambda - \int \sigma(dq)/(q-\lambda)$ with $m \ge 0$ and $\sigma \ge 0$, which is decreasing between its poles.

Proposition 19.4.1 (Cotangent law). *For fixed $a > 0$ and $\mu > 0$ not an integer, as $k \to \infty$,*

$$\beta_k = \cot\Big(\frac{\pi\mu}{2}\Big)\sqrt{\lambda_k}\,\big(1 + O\big((a/2\nu_k)^{2\mu}\big)\big) + O(1).$$

Sketch. For ν large compared with a, $K_{\mu+i\nu}(a) = \frac{1}{2}\Gamma(\mu+i\nu)(a/2)^{-\mu-i\nu}\big(1 + O((a/2\nu)^{2\mu})\big)$ up to a relative term of order a^2/ν in the leading series, which is absorbed in the $O(1)$, the second term of the small argument expansion being smaller by the stated factor. So the zeros satisfy $\arg\Gamma(\mu+i\nu) - \nu\log(a/2) = \frac{\pi}{2} + k\pi$, and Stirling gives $\arg\Gamma(\mu+i\nu) = \arg\Gamma(i\nu) + \frac{\pi\mu}{2} + O(1/\nu)$. For $\mu = 0$ both terms have equal size and $K_{i\nu}(ae^{\tau}) \approx |\Gamma(i\nu)|\cos(\arg\Gamma(i\nu) - \nu\log(a/2) - \nu\tau)$, whose logarithmic derivative at $\tau = 0$ is $\nu\tan(\arg\Gamma(i\nu) - \nu\log(a/2))$. At the zeros the phase is $\frac{\pi}{2} - \frac{\pi\mu}{2}$ modulo π, giving $\nu\cot(\pi\mu/2)$. □

The phase computation, which gives the leading term, was rederived independently; the error terms were not, so they are heuristic and the statement is proved only for the leading term. The law matches every integer case of Theorem 19.3.3: $\cot(\pi\mu/2) = 0$ for odd μ, where β is bounded, and $\cot(\pi\mu/2) = \infty$ for even μ, where β grows linearly in λ through the point mass. At $a = 2\pi$ the slope of β_k against $\sqrt{\lambda_k}$ over the last zeros below $\nu = 45$ is 1.0007 to 1.0008, 0.4095 to 0.4100 and -0.4178 to -0.4172 for $\mu = \frac{1}{2}, \frac{3}{4}, \frac{5}{4}$, against $\cot(\pi\mu/2) = 1$, 0.4142 and -0.4142 (computed; the range is over windows of the last 6 to 12 zeros, since the source does not define the window, and at least 45 digits are needed because $K_{i\nu}(2\pi)$ is of size 10^{-31} at $\nu = 45$). For $\mu = \frac{1}{4}$ it is 3.04 to 3.06 against 2.41, consistent with the slow correction $(a/2\nu)^{1/2} \approx 0.28$ at $\nu = 40$.

The sign of $\cot(\pi\mu/2)$ splits the problem in two.

- For $\mu \in (1,2)+2\mathbb{Z}$, including Pólya's $\mu = \frac{5}{4}$, β decreases like $-c\sqrt{\lambda}$. Since $\sqrt{\lambda}$ is a Nevanlinna function, this is, heuristically, the growth of an admissible

law with a continuous spread of extra masses, σ carried by the negative half line and so off the spectrum; numerically $\beta_k = 0.37 - 0.417\sqrt{\lambda_k} + 2.43/\sqrt{\lambda_k}$ over the first 24 zeros, with maximal relative error 3.3×10^{-3} (computed; the accuracy 5×10^{-4} stated by the source was not reproduced by a fit of this form in the maximum norm).

- For $\mu \in (0,1)+2\mathbb{Z}$, including $\mu = \frac{9}{4}$, β increases with λ, so an admissible law needs a pole in every gap between consecutive eigenvalues, that is, coupling to a second string with an interlacing spectrum.

Pólya's two term kernel combines one order of each type. Identifying the continuum extension for $\mu = \frac{5}{4}$ and the interlacing string for $\mu = \frac{9}{4}$ is the refined form of the problem (open).

19.4.3 Problem 3: where the de Bruijn Newman zeros collide

For $t < 0$, $H_t(z) = \mathbb{E}H(z+W)$ with W normal of variance $2|t|$, so H_t is a Gaussian convolution of H on a fine real grid (Section 4.1.1 and Theorem 4.1.20 give the same flow). Below $x = 100$, where Ξ has 29 zeros, $H_t(2x)$ has 29 sign changes for $t = -0.25$ and -0.5, then 27, 19, 12 and 8 for $t = -1, -2, -4, -8$ (computed, step 0.05, 60 digits). A bisection of the source places the first collision below height 100 at $t \approx -0.881$, near $x \approx 95.5$, for the pair at 94.651 and 95.871, the closest pair below 100 (gap 1.22); it was not repeated, so the value is unchecked here, and the book's own computation of this collision is in Section 4.1.3. At $t = -4$ a nonreal pair sits at $33.1150940 \pm 0.1505390\,i$ (computed, Section 19.5.4), where, by the source's unchecked reading, the fourth and fifth zeros (30.42 and 32.94 at $t = 0$) have merged.

The moment Hankel test sees these collisions, but late. At 130 digits to order 10, all D_n and D'_n are positive at $t = 0$; at $t = -4$ the first negative value is D'_{10}; at $t = -8$, D_8, D_9, D_{10} and their primed partners are negative while $D_1, \dots, D_7$ and theirs are positive (computed). A nonreal pair of rank about four is thus detected only at order eight to ten, about twice its rank, and the first collision, of rank about 27, is far beyond order ten. A negative determinant excludes a positive measure, hence real zeros, so kernel moments alone give $\Lambda > -4$. This is far weaker than Rodgers and Tao [250], and than the book's Hankel test along the flow (Table 4.3), which gives $t^*_{50} \approx -1.03$ with no failing order up to 50 on $[-1.03, 0]$, so that seeing the collision at -0.881 needs order above 50. What this computation adds is the signs at $t = -4$ and -8 and the shifted determinants D'_n: the Hankel string test does detect a genuine loss of reality when the off line atoms are few, unlike for the folded kernels of Problem 4.

19.4.4 Problem 4: why the first pair is masked

The off line zeros of the two term folded kernel lie on a regular curve. Starting from $43.139 - 3.281i$ and $47.523 - 6.253i$, the source locates 19 zeros w_j in $[36, 150] \times [-40, 0]$, with real parts about 4 apart on average (from the printed end points) and imaginary parts decreasing to -38.6 at real part 114.9, finds by the argument principle 16 more with imaginary part in $[-60, -40]$ and 25 more in $[-100, -60]$, and none with real part above 150 and imaginary part above -40. Only the count $19 = 2 + 17$ in $[36, 150] \times [-40, 0]$ is recomputed (Section 19.3); the geometry and the further counts are unchecked.

The Hankel determinants follow from the Andreief (Heine) identity. With atoms $y_i = 1/\gamma_i^2$ and moments $\mu_{j+1} = \sum_i y_i^{j+1}$,

$$D_n = \sum_{|S|=n} \prod_{i \in S} y_i \prod_{i<j \in S} (y_i - y_j)^2,$$

over n element sets of atoms. A term containing both members of a conjugate pair $y, \bar{y}$ carries the factor $(y - \bar{y})^2 = -4(\operatorname{Im} y)^2$ and is typically negative. Table 19.7 shows D_6 as atoms are added.

atoms used	D_6
7 real atoms only	1.29×10^{-105}
plus the first pair	-1.81×10^{-104}
plus the first two pairs	-2.27×10^{-103}
plus all 19 located pairs	1.09×10^{-100} (unchecked)
kernel moments (all zeros)	8.10×10^{-100}

Table 19.7: The masking of the first nonreal pair in D_6 for the two term folded kernel (computed, 250 digits, except the kernel moments row, which is from 120 and 240 digits, and the row marked unchecked, which is the source's value and needs the 19 located zeros).

The source splits the sum over the 22 atoms of largest modulus by type: all real sets 1.3×10^{-105}, sets containing a full pair -3.4×10^{-102}, and sets containing single members of several pairs $+7.2 \times 10^{-102}$. The first value agrees with the table; the other two were not recomputed (unchecked), and the source does not define the selection of the 22 atoms. If the split is right, masking is a cross term effect: the atoms on the curve have moduli comparable to the real atoms, because $|\gamma|$ along the curve grows only like the real part, and their mixed products $2 \operatorname{Re} \prod$ are positive and larger than the negative full pair terms. A nonreal pair is visible to a finite Hankel test only when the other off line atoms are too small or too spread out to compensate, and the test cannot tell which case it is in (heuristic).

19.4.5 Problem 5: forty digit eigenvalues

Here x is the hitting time variable. At 140 digits, Prony fits of K exponentials to the survival function of the hitting time on windows $x_0 + 0.003j$, $j < 2K$, give the errors in Table 19.8 (computed from the survival series $\sum_k c_k e^{-\lambda_k x}$ over the zeros, where the source used Talbot inversion of degree 160 from $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ alone; the density form of the tail gives larger errors, for example 7×10^{-13} instead of 9×10^{-14} for γ_1 at $K = 6$). The rows were reproduced at 100 and 130 digits to the one digit printed.

K	x_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_8
6	0.02	9×10^{-14}	1×10^{-9}	3×10^{-7}	6×10^{-4}	0.013	0.52	
8	0.02	5×10^{-25}	1×10^{-19}	3×10^{-16}	3×10^{-11}	4×10^{-9}	2×10^{-5}	0.11
10	0.015	8×10^{-34}	1×10^{-27}	1×10^{-23}	7×10^{-18}	4×10^{-15}	2×10^{-10}	5×10^{-5}
10	0.02	2×10^{-39}	7×10^{-33}	2×10^{-28}	5×10^{-22}	6×10^{-19}	2×10^{-13}	4×10^{-7}

Table 19.8: Absolute errors of γ_k from Prony fits to the tail of the reciprocal Ξ hitting time at 140 digits (computed).

With ten atoms the first zero comes out to 39 digits and the eighth to 7. For $K = 10$, the only K with two thresholds, the error falls as the threshold rises from 0.015 to 0.02, so at this precision the threshold should not be below 0.02. The trade off between omitted atom bias and loss of precision shows in the columns: from $K = 6$ to 8 and from 8 to 10 the error of γ_1 falls by 11 and 14 digits, so each added pair of atoms buys ten to fourteen digits in the leading eigenvalues, and the last atom fitted is always the least accurate because the omitted atoms load onto it. For a Lehmer pair with gap δ at height γ the rates differ by about $2\gamma\delta$, and resolving them needs a window long enough for $e^{-2\gamma\delta x}$ to vary appreciably while both terms stay above the working precision, which fixes how many digits a given pair costs (heuristic).

19.5 Devroye's series, the mean excess function and detection

This section tests the extreme value picture through the Laplace transform $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ and a little information about the zeros: Devroye's alternating series, the mean excess function with Schoenberg's criterion, the weak convergence behind them, and detection in the de Bruijn Newman family. Cited, not restated: the residues $R_k = c_k \gamma_k^2$ and the spectral expansion of the density (Section 5.2.1 and Exercise 18.1), the criterion by Pólya frequency functions (Proposition 5.2.2), the weak closure of the GGC class (Corollary 9.1.2) and the digits rule for Prony fits (Section 5.2.2).

19.5.1 Devroye's alternating series and the sign pattern

Devroye's exact samplers for the Jacobi theta laws rest on one device: a density written as a series whose terms alternate in sign and decrease in size is bracketed by its partial sums [59, 62]. The reciprocal Ξ law has such a series under hypotheses. Assume RH, simple zeros and $|\xi'(\rho)| \geq e^{-C|\gamma|}$ as in Theorem 5.3.2; the zeros are $\rho = \frac{1}{2} + i\gamma_k$ with γ_k real, $0 < \gamma_1 < \gamma_2 < \cdots$, and $\pm\gamma_k$ are the zeros of Ξ. Then the density of $X = \sum_k E_k/\gamma_k^2$ is

$$f(x) = \sum_{k \geq 1} c_k \gamma_k^2 e^{-\gamma_k^2 x}, \qquad c_k = \prod_{j \neq k} \Big(1 - \frac{\gamma_k^2}{\gamma_j^2}\Big)^{-1} = \frac{2i\,\xi(\frac{1}{2})}{\gamma_k\,\xi'(\frac{1}{2} + i\gamma_k)}.$$

Without the lower bound for $|\xi'(\rho)|$ only the grouped limit of the residue sums is proved (Theorem 5.3.1), and the partial sums below are not known to converge to f; at a multiple zero c_k is undefined. The two term tail of f needs none of these hypotheses (Proposition 5.3.3).

Proposition 19.5.1 (Sign pattern). *If the atoms $\gamma_1^2 < \gamma_2^2 < \cdots$ are real and simple, then* $\operatorname{sign} c_k = (-1)^{k-1}$.

Proof. Exactly $k-1$ of the factors $1 - \gamma_k^2/\gamma_j^2$ are negative, those with $j < k$. □

Remark 19.5.2. If a pair of atoms $\lambda_k = \gamma_k^2$ is not real, the two coefficients of the pair are complex conjugates, and for every real atom λ_m the pair contributes the factor $|1 - \lambda_m/\lambda_k|^{-2} > 0$ to c_m, so the coefficient of a real atom of rank r among the real atoms has sign $(-1)^{r-1}$. A sign test therefore sees a nonreal pair only if the rank of each atom is known, which is a zero count. The test also reads γ_k and $\xi'(\rho_k)$, not L alone.

Devroye's switch point is the x_k beyond which $|c_{k+1}|\gamma_{k+1}^2 e^{-\gamma_{k+1}^2 x} \leq |c_k|\gamma_k^2 e^{-\gamma_k^2 x}$, namely

$$x_k = \frac{\log\big(|c_{k+1}|\gamma_{k+1}^2/(|c_k|\gamma_k^2)\big)}{\gamma_{k+1}^2 - \gamma_k^2}.$$

Bracketing at a point x needs the terms to alternate and decrease from the first index on, that is $x \geq \sup_k x_k$.

For the reciprocal Ξ law the signs alternate for all 1700 coefficients (computed): $c_1 = 50.871$, $c_2 = -2664.6$, $c_3 = 3.1420 \times 10^4$, $c_4 = -1.3550 \times 10^6$, and so on. Beyond the first switch point the partial sums bracket the density, checked against Talbot inversion at 30 digits.

- At $x = 0.02$, where $f = 80.10420$, the partial sums are 186.93, 16.148, 88.602, 77.170, 80.2026, 80.09549.
- At $x = 0.03$, where $f = 23.433069$, they are 25.352, 23.295, 23.4341, 23.43301, 23.4330693, 23.43306926.

law	signs of c_k	x_1, x_2, x_3
$1/\cosh\sqrt{2s}$	$+-+-+-+-$	.111, .0259, .0114
$\sqrt{2s}/\sinh\sqrt{2s}$	$+-+-+-+-$	.0936, .0329, .0167
reciprocal Ξ	alternating, first 1700 zeros	.01963, .01533, .01385

Table 19.9: Calibration on Devroye's two explicit laws, with atoms $\pi^2(k-\frac{1}{2})^2/2$ and $\pi^2k^2/2$, and the reciprocal Ξ law with atoms γ_k^2. For the two strings $c_k\lambda_k = (-1)^{k-1}\pi(k-\frac{1}{2})$ and $c_k = 2(-1)^{k-1}$ in closed form, so that $x_k = \log\frac{2k+1}{2k-1}/(k\pi^2)$ and $x_k = \log\frac{(k+1)^2}{k^2}/\big((2k+1)\pi^2/2\big)$ (computed, and checked against the product at $K = 4000$ factors). For the reciprocal Ξ law the coefficients come from $\xi'(\rho_k)$ at the first 1700 zeros (computed).

Each sum lies on the opposite side of f from the one before. Over the first 1700 zeros the largest switch point is $x_1 = 0.019627$, and the largest from $k = 4$ on is 0.011656, at $k = 7$ (computed). This gives an exact Devroye sampler for X on $x \geq 0.01963$, and two sided bounds there, conditional on RH with simple zeros, on the lower bound for $|\xi'(\rho)|$, and on $x_k \leq 0.01963$ for every k, which was checked for $k \leq 1699$ only. For large k the switch point depends on the ratios $|\xi'(\rho_k)/\xi'(\rho_{k+1})|$ over the gaps $\gamma_{k+1}^2 - \gamma_k^2$, for which no bound is known. The bounds are computed in floating point, not enclosed.

19.5.2 Schoenberg's theorem and the mean excess function

The mean excess (mean residual life) function of X is $e(u) = \mathbb{E}[X - u \mid X > u]$. It can be computed without zeros, by one Laplace inversion each for numerator and denominator,

$$\int_u^\infty \mathbb{P}(X > x)\,dx = \mathcal{L}^{-1}\Big[\frac{\mathbb{E}X}{s} - \frac{1 - L(s)}{s^2}\Big](u),$$

with

$$\mathbb{E}X = \frac{\xi''(\frac{1}{2})}{2\xi(\frac{1}{2})} = 0.02310499311541897\ldots$$

The criterion by Pólya frequency functions (Proposition 5.2.2) rests on Schoenberg's characterisation [254], restated with its nondegeneracy condition and the Gaussian parameter renamed to avoid a clash with γ_k.

Theorem 19.5.3 (Schoenberg). *An integrable density f is a Pólya frequency function, totally positive of every order, exactly when its bilateral Laplace transform is $1/\Psi(s)$ in a strip around the origin, with $\Psi(s) = Ce^{-\alpha s^2+\delta s}\prod_k(1+\delta_k s)e^{-\delta_k s}$, where $\alpha \geq 0$, δ and δ_k are real, $\sum\delta_k^2 < \infty$ and $0 < \alpha + \sum\delta_k^2$ (Schoenberg 1951).*

Corollary 19.5.4. *RH holds if and only if $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is the Laplace transform of a Pólya frequency density. Under RH that density is log concave, the*

hazard rate of X is increasing, and the mean excess function decreases monotonically,

$$e(0) = \sum_k \gamma_k^{-2} = 0.0231050 \;\searrow\; e(\infty) = \gamma_1^{-2} = 0.00500524.$$

Proof. The function $1/L(s) = \prod_k (1 + s/\gamma_k^2)$ has the form of Theorem 19.5.3 with $\alpha = 0$ and $\delta_k = \gamma_k^{-2}$, which are real exactly when every γ_k is real: an off line zero with γ_k not real gives a nonreal γ_k^2, and a real zero of ξ off the line would lie in $(0,1)$, where, by the classical fact that ζ has no zero in $(0,1)$, there is none. Total positivity of order two is log concavity, which implies an increasing hazard rate, and an increasing hazard rate implies a decreasing mean residual life (Barlow and Proschan [12]). □

So the mean excess function interpolates between the Thorin moment $\sum_k \gamma_k^{-2}$ at $u = 0$ and the first atom at $u = \infty$, and its approach to the limit carries the next atom; the analogous hazard limit, with rate $\gamma_2^2 - \gamma_1^2$, is in Section 5.2.1 and Exercise 18.3.

Proposition 19.5.5 (Expansion and rate). *Under RH with simple zeros and $|\xi'(\rho)| \geq e^{-C|\gamma|}$ as above, and with $\lambda_k = \gamma_k^2$,*

$$e(u) = \frac{\sum_k c_k e^{-\lambda_k u}/\lambda_k}{\sum_k c_k e^{-\lambda_k u}} = \frac{1}{\lambda_1} + A\, e^{-(\lambda_2 - \lambda_1)u} + O\big(e^{-\min(\lambda_3 - \lambda_1,\, 2(\lambda_2 - \lambda_1))u}\big), \qquad A = \frac{c_2}{c_1}\Big(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\Big) > 0.$$

The excess law of X over u converges to the exponential law with rate λ_1 in total variation at rate $e^{-(\lambda_2 - \lambda_1)u}$. Without simple zeros the two term form follows from the two term tail of Proposition 5.3.3, with w_k/λ_k in place of c_k for $k = 1, 2$.

Proof. The first equality is $\int_u^\infty \mathbb{P}(X > x)\,dx / \mathbb{P}(X > u)$ with $\mathbb{P}(X > x) = \sum_k c_k e^{-\lambda_k x}$. Put $\varepsilon = e^{-(\lambda_2 - \lambda_1)u}$. The numerator is $\lambda_1^{-1} c_1 e^{-\lambda_1 u}\big(1 + \frac{c_2 \lambda_1}{c_1 \lambda_2}\varepsilon + O(e^{-(\lambda_3 - \lambda_1)u})\big)$ and the denominator is $c_1 e^{-\lambda_1 u}\big(1 + \frac{c_2}{c_1}\varepsilon + O(e^{-(\lambda_3 - \lambda_1)u})\big)$. Dividing, the first order term is $\frac{c_2}{c_1}\big(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\big)\varepsilon$ and the next is $O(\varepsilon^2)$. Since $c_2/c_1 < 0$ (Proposition 19.5.1) and $1/\lambda_2 < 1/\lambda_1$, $A > 0$. □

The following were computed from the residue series over the first 300 zeros at 50 and 70 digits on the grid $u = 0.005, 0.0075, \ldots, 0.2$ (79 points); the source used Talbot inversion from L alone. Nothing is enclosed in interval arithmetic.

- Log concavity: $ff'' - f'^2 < 0$ at all 79 points (a sampled check). Two of the points, 0.005 and 0.0075, lie below 0.008, where Theorem 5.7.5 starts and where the book's proof is conditional on a saddle point approximation (Proposition 5.7.3).

- Monotonicity: $e(u)$ decreases strictly at every grid step. Beyond $u \approx 0.165$ consecutive values agree with $\gamma_1^{-2} = 0.00500524412359412$ to every printed digit.
- Expansion: Proposition 19.5.5 predicts $A = 0.1436441604$. The observed deviation $e(u) - \gamma_1^{-2}$ against $Ae^{-(\lambda_2-\lambda_1)u}$ is 9.808×10^{-5} against 1.006×10^{-4} at $u = 0.03$, 2.3625×10^{-7} against 2.3637×10^{-7} at $u = 0.055$, and $1.3052007 \times 10^{-12}$ against $1.3052008 \times 10^{-12}$ at $u = 0.105$. From $u = 0.13$ on the agreement is to all eight printed digits; the residual at small u is the λ_3 term.

Read without zeros, as the source does, the expansion gives the first two zeros. An Aitken extrapolation of three values of $e(u)$ estimates $e(\infty) = \gamma_1^{-2}$ and, as $-\log r/h$ for the extrapolated ratio r and the step $h = 0.0075$, the gap $\lambda_2 - \lambda_1$. It gives $\gamma_1 = 14.1347778$, 14.1347266 and 14.1347252 from windows starting at $u = 0.03$, 0.04 and 0.05, and $\gamma_2 = 20.965$, 21.006 and 21.019, against $\gamma_1 = 14.1347251417$ and $\gamma_2 = 21.0220396$. This is weaker than the Prony fits to the tail (Table 19.8); the mean excess is a criterion and a check, not a way of computing zeros.

19.5.3 Weak convergence

Three weak convergence statements sit behind these diagnostics.

1. Pickands, Balkema and de Haan: the excess law over u, rescaled by $e(u)$, converges to a generalised Pareto law [150, 10]. Here the shape is zero and the convergence is exponentially fast (Proposition 19.5.5).
2. Closure: generalised gamma convolutions are closed under weak limits (Bondesson [30]; Corollary 9.1.2), and so are Pólya frequency laws with nondegenerate limits (Karlin [105]). Hence, by Proposition 5.5.1 and Corollary 9.1.2, RH is equivalent to the reciprocal Ξ law being a weak limit of hypoexponential laws $\sum_{k \le K} E_k/\lambda_{k,K}$ with real positive rates, a distributional form of the Hurwitz argument.
3. The exponential B splines of Proposition 21.2.1 (Section 19.20) are such a sequence, but they use the zeros. Zero free sequences exist under RH: the Jensen laws of Section 21.1 are GGC laws for every degree $d \le 2 \cdot 10^{17}$, exist for every d exactly when RH holds, and then converge to the law of X (Proposition 21.1.1). Another candidate is a sequence of approximations $\widehat{L}_N(s) = \widehat{H}_N(0)/\widehat{H}_N(2i\sqrt{s})$ with only real zeros that converge to L. Pólya's kernels give Pólya frequency laws but do not converge. The folded kernels converge, and their zeros were found to be real up to a height of about $c(N+1)^2$ with computed constants c between 3.6 and 4.4 (certified in ball arithmetic for $N \le 4$ given the count of zeros of Ξ in each disc, computed for $N = 5$; Sections 19.3 and 19.4).

19.5.4 Detection in the de Bruijn and Newman family

The family H_t is that of Chapter 4.1, with the variable of the zeros $w = z/2$; its collisions and Hankel test are in Sections 4.1.2 and 4.1.3. At $t = -4$ the fourth and fifth zeros have merged, according to the source (not checked here). By quadrature against the kernel, $H_{-4}(2w)$ is negative at all 41 points of $[28, 38]$ with step 0.25 (computed; the largest value is -6.67×10^{-12} and the smallest modulus is at $w = 33.25$), so it almost touches zero near 33. Root finding locates the pair at

$$w = 33.1150940 \pm 0.1505390\, i, \qquad \gamma^2 = 1096.58679 \pm 9.970228\, i$$

(computed). A short window diagnostic should miss it: the pair adds $e^{-1096.6x}\cos(9.97x)$ to the tail, and over a window of length 0.04 the cosine turns by only 0.4 radians. The source reports, and this was not recomputed, that a six atom Prony fit on such a window returns spurious atoms, one real and a pair, and that the same fit at $t = 0$ returns an equally spurious pair where the true zero is real. Short windows cannot separate a slightly nonreal pair from omitted atom bias.

Remark 19.5.6 (Detection rule, heuristic). To see one radian of oscillation from a nonreal atom $\lambda = a + ib$, a tail window must extend to $x \approx 1/b$. There its term is smaller than the leading term by $e^{-(a-\lambda_1)/b}$, so that the computation needs about $(a - \lambda_1)/(b \ln 10)$ digits beyond the size of the leading term. If the window ends at x_{end} instead, about $(a - \lambda_1)x_{\text{end}}/\ln 10$ digits are needed. Section 5.2.2 states the same kind of rule for real atoms. This is an order of magnitude estimate, not a theorem.

At $t = -4$ the first count is $(1096.59 - 234.42)/(9.970 \ln 10) \approx 37.6$ digits (computed, with $\gamma_1 = 15.31093766$ taken from Table 19.10, which is unchecked). The source's window, below, ends at 0.188, where the pair is smaller than the leading term by about $e^{-162} \approx 10^{-70}$, the count for that window.

The source tested the rule with a window $x_j = 0.08 + 0.012j$, $j = 0, \ldots, 9$, which turns the cosine by about one radian, and five atom Prony fits at 120 digits, with $H_{-4}(2i\sqrt{s})$ computed as the Gaussian average $\mathbb{E}H(2i\sqrt{s} + W)$, W normal with variance 8, by 90 point Gauss and Hermite quadrature on ξ. Table 19.10 is the outcome as the source reports it; the fits were not run for this book, and the only entries recomputed are the pair, the zero of H_{-4} found directly, and the control column, which is $\gamma_1, \ldots, \gamma_5$.

If the table is right, the fit returns $\gamma_1, \ldots, \gamma_5$ for $t = 0$ and for $t = -4$ a conjugate pair with Thorin atom $1096.58679 \pm 9.970227764\, i$, agreeing with the root found directly to all ten digits printed. A threshold analysis of the tail of the inverse Laplace transform of $1/H_{-4}(2i\sqrt{s})$ (at $t = -4$ this need not be a probability density, so it is not a hitting time tail) would then detect a genuine loss of reality from the transform alone, once the window and the precision follow

	$t = 0$ (control)	$t = -4$
$\widehat{\gamma}_1$	14.13472514	15.31093766
$\widehat{\gamma}_2$	21.02203964	22.5998361
$\widehat{\gamma}_3$	25.01085758	26.14438156
$\widehat{\gamma}_4$	30.42487613	$33.11509402 - 0.1505390225\,i$
$\widehat{\gamma}_5$	32.93506159	$33.11509402 + 0.1505390225\,i$

Table 19.10: Long window threshold fits from the Laplace transform alone, as reported by the source (unchecked: the fits were not recomputed; the pair agrees with the zero of H_{-4} found directly, $33.1150940172 \pm 0.150539022\,i$).

Remark 19.5.6. Until the fits are recomputed this is a claim of the source, and the rule itself stays heuristic.

The rule quantifies the limitation of all tail diagnostics. A pair that has only just left the line has small $\operatorname{Im}\gamma^2 = 2\operatorname{Re}\gamma\operatorname{Im}\gamma$, and the digits required grow like $1/\operatorname{Im}\gamma^2$, so it is the hardest kind to see. This is the Lehmer pair phenomenon in tail form (the flow of a Lehmer pair is in Section 4.1.6), and it is why such tests are verification to a finite height and precision rather than proof.

19.6 Remaining open problems

1. Interval arithmetic for the Rouché test at $N = 5$ and for the count of zeros of Ξ in each disc, and the level $N = 6$ near height 196 (Sections 19.3 and 19.4).

2. Identify the continuum extension for $\mu = \frac{5}{4}$ and the interlacing string for $\mu = \frac{9}{4}$, and combine them into a string for Pólya's two term kernel.

3. Give the Hankel order needed to see the de Bruijn and Newman collision at $t \approx -0.881$ near height 95.5. This is answered in part by Table 4.3: no order up to 50 fails on $[-1.03, 0]$, so at the collision time the order needed exceeds 50.

4. Prove that the off line zeros of the folded kernels lie on a curve, with explicit spacing, and decide whether masking persists at all Hankel orders for this family.

5. Turn the Prony scans into an error formula in $(K, x_0, \text{precision})$ and apply it to a Lehmer pair.

6. Find a product structure in the theta superposition of Gumbel pieces that carries the Thorin atoms at γ_k^2 (Section 19.20).

7. Show that the switch points satisfy $x_k \le x_1$ for every k (Section 19.5.1), from properties of L and a lower bound for $|\xi'(\rho)|$, and use Devroye's bounds for

a certified sampler of the reciprocal Ξ law. The alternation itself is proved (Proposition 19.5.1); what is open is the monotonicity of the terms.

8. Higher order total positivity: test PF_3 and beyond for the reciprocal Ξ density from Laplace inversion, since Corollary 19.5.4 makes every order a necessary condition for RH. The ladder of Toeplitz minors of the Pólya density is in Chapter 5.

9. Make the detection rule of Remark 19.5.6 into an algorithm with error control, and estimate its cost for the Lehmer pairs known from verification.

10. One sided approach in the Dyck chain. If $F_n \to 2\xi$ locally uniformly on $\operatorname{Re} s > 0$ and, for every $T > 0$, the Dirichlet polynomial F_n of the Dyck chain (Section 19.17) has no zero in $\frac{1}{2} < \operatorname{Re} s < 1$, $0 < \operatorname{Im} s < T$ for all large n, then RH holds: if RH fails, the functional equation and the symmetry $\xi(\bar{s}) = \overline{\xi(s)}$ give a zero ρ of ξ with $\frac{1}{2} < \operatorname{Re}\rho < 1$ and $\operatorname{Im}\rho > 0$, and by Hurwitz's theorem, for $T > \operatorname{Im}\rho$, F_n has zeros in that rectangle for all large n. Under Conjecture 19.17.2 the hypothesis holds for $T < \gamma_{64} = 169.9119\ldots$ and fails for every $T > \gamma_{64}$. The zeros of ξ below height 170 are $\rho_1, \ldots, \rho_{64}$, simple and on the line [271]. For $T > \gamma_{64}$, the zero $z_n(\rho_{64})$ of Proposition 19.17.3 tends to ρ_{64} and has $\operatorname{Re} z_n(\rho_{64}) = \frac{1}{2} + \delta_n \operatorname{Re} c_{\mathrm{D}}(\rho_{64}) + O(1/n) > \frac{1}{2}$ for large n, since $\operatorname{Re} c_{\mathrm{D}}(\rho_{64}) > 53.08$ (computed in ball arithmetic, Section 19.17). For $T < \gamma_{64}$, $F_n \to 2\xi$ uniformly on the closed rectangle; outside small discs about $\rho_1, \ldots, \rho_{63}$ it carries no zero of ξ, hence none of F_n for large n, and in the disc about ρ_k the only zero is $z_n(\rho_k)$, to the left of the line for large n because $\operatorname{Re} c_{\mathrm{D}}(\rho_k) < 0$ (computed in ball arithmetic). This is the argument of Corollary 17.1.4 for the Dyck chain: conditionally on Conjecture 19.17.2 the hypothesis fails, and one sided approach cannot prove RH along this chain. For $10^5 \le n \le 10^8$ the zero near ρ_{64} lies to the right of the line (computed, Section 19.17). Status: the implication proved, the failure of the hypothesis conditional on Conjecture 19.17.2; open: prove the conjecture, which would also give the locally uniform convergence of the first hypothesis. This item is based on [134, 141].

Chapter 20

The Comtet Majumdar family: damped kernels and a solved model

Comtet and Majumdar [43] proved, in an appendix on the maximum of a random walker, an identity between an integral of the logarithm of $(1 - e^{-k^\mu})/k^\mu$ and $\zeta(1/\mu)$, through a cutoff regularisation. This chapter asks what the identity and the regularisation can do for the programme of approximating Ξ by entire functions whose reciprocals are Laplace transforms of generalised gamma convolutions (GGC) and passing to the limit. The answer has two halves. On the arithmetic side it is negative: the shifted form of the regularisation, and every decreasing damping of the coefficients of the theta series, produce functions outside the Laguerre Pólya class, for structural reasons proved below. On the Thorin side the same shift, read as a tilt of a Lévy measure, gives a positive object: the Thorin density of the tilted uniform law is strictly positive for every positive tilt, and its Mellin transform is the regularised zeta function. The uniform law is thus a solved model of the question asked for ξ in Chapter 12, where the corresponding positivity is proved for every basepoint $\alpha \geq 1$.

The chapter is based on [131]. Its numbers were recomputed; the tables of zeros of the damped kernels (Section 20.5) were not, and are marked unchecked. The kernel Φ is that of (2.1.3), so $\Xi(t) = \int_{\mathbb{R}} \Phi(u) \cos(tu)\, du$, and the sum over n inside Φ carries the terms Φ_n of Section 20.5. The damping parameter is δ. The unnormalised Lorentzian is $\ell_\varepsilon(x) = \varepsilon/(\varepsilon^2 + x^2) = \pi P_\varepsilon(x)$, with P_a the Poisson kernel of Chapter 2. The Thorin density of the uniform law is κ_δ; for ξ the role of κ_α^ξ is played by D_α of Chapter 22. The constant of Comtet and Majumdar is C_μ, the exponential variables are e_k, and the integer part is $\lfloor t \rfloor$. The de Bruijn Newman flow uses the time t and the kernel $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ of Section 4.1, not the parameter λ of the source, whose normalisation differs from the literature (Section 20.6).

Cited, not restated: the Thorin and Lévy forms of the reciprocal ξ function with a basepoint, the compensated density $v_\alpha - b_\alpha$ and its positivity for every $\alpha \geq 1$ (Propositions 12.1.10 and 12.1.12, Theorem 12.1.18, Remark 12.1.20), which re-

place the source's section on ξ, including its threshold $\alpha_1 = 7.7892$ and its first open problem; the equivalence of RH with positivity of the uncompensated density for all $\alpha > \frac{1}{2}$ and Lagarias's criterion (Corollaries 23.1.10 and 23.1.11); the $2\sin^2$ lemma ((2.1.9), Lemma 10.2.4) and its Thorin consequence ((16.5.8), Theorem 16.5.8); the de Bruijn Newman theorem (Theorem 17.1.16); and the kink mechanism for weighted theta sums (Lemma 18.1.1, Theorem 18.2.3, Propositions 18.8.1 and 18.8.2). The results are these.

1. The identity of Comtet and Majumdar holds on the whole strip $0 < \operatorname{Re} s < 1$ and is the functional equation of ζ combined with Spitzer's formula for the maximum of a symmetric stable walk (Propositions 20.1.1 and 20.1.2, proved). The logarithm of the Laplace transform of the uniform law has a regularised signed Lévy representation whose k measure is Lebesgue minus counting measure, with Mellin transform ζ (Proposition 20.1.3, proved as improper integrals with a common cutoff).

2. An even entire function of order less than two has a GGC reciprocal exactly when its zeros are real (Lemma 20.2.1), so a family with GGC reciprocals converging to Ξ proves RH (Corollary 20.2.2, proved).

3. The shifted Comtet Majumdar family ζ_δ has an exact expansion in $\zeta(s+k)$ (Theorem 20.3.2, proved). Its completion has a pole at $s = 3$ for every $\delta \in (0, 2\pi)$, so its reciprocal is not the transform of a GGC nor of any positive measure (Proposition 20.3.3, proved; the source's claim of poles at all odd integers is false and is repaired). Each simple zero moves at an explicit first order rate (Proposition 20.3.3(b), proved), and the first three leave the critical line (computed, Table 20.1). The shifted family is not the literal cutoff of Comtet and Majumdar; the two differ already in sign (Remark 20.3.4, proved).

4. The uniform law tilted by $\delta > 0$ in the sense of the $2\sin^2$ lemma has a strictly positive Thorin density for every $\delta > 0$, by Poisson summation and convexity of the Cauchy kernel (Theorem 20.4.4, proved; the numerical constant of the proof is computed), and the regularised zeta function is its Mellin transform on $1 < \operatorname{Re} s < 3$ (Theorem 20.4.5, proved).

5. Gauss weights on the terms of the theta series give, for every $\delta > 0$, a transform with finitely many real and infinitely many nonreal zeros (Theorem 20.5.1, proved); symmetrising instead produces poles (Section 20.5).

6. The de Bruijn Newman family is a deformation whose GGC property is Theorem 17.1.16; a false convergence criterion for RH in the source and its normalisation of the thresholds are repaired (Section 20.6). The remaining problems are open.

20.1 The identity decoded

Throughout, $s = 1/\mu$ and

$$g(x) = \frac{1}{e^x - 1} - \frac{1}{x}, \qquad x > 0.$$

Then $g(0^+) = -\frac{1}{2}$, $g(x) \sim -1/x$ as $x \to \infty$, and $g < 0$ on $(0, \infty)$.

Proposition 20.1.1 (Mellin form). *For $0 < \operatorname{Re} s < 1$,*

$$I(s) := \frac{1}{\pi} \int_0^\infty x^{-s} g(x)\, dx = \frac{\Gamma(1-s)\zeta(1-s)}{\pi} = \frac{\zeta(s)}{(2\pi)^s \sin(\pi s/2)}. \tag{20.1.1}$$

With $s = 1/\mu$ this is the identity of Comtet and Majumdar [43],

$$\frac{1}{\pi} \int_0^\infty \frac{dk}{k^2} \ln\Bigl[\frac{1 - e^{-k^\mu}}{k^\mu}\Bigr] = \frac{\zeta(1/\mu)}{(2\pi)^{1/\mu} \sin(\pi/(2\mu))}, \qquad 1 < \mu \le 2. \tag{20.1.2}$$

Proof. The first equality is the classical formula $\Gamma(z)\zeta(z) = \int_0^\infty x^{z-1} g(x)\, dx$ for $0 < \operatorname{Re} z < 1$ [264, Equation 2.7.1] with $z = 1 - s$. The second is the functional equation $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$. For (20.1.2), substitute $x = k^\mu$ in the left side and integrate by parts once; the boundary terms vanish because $\mu > 1$. □

Comtet and Majumdar state (20.1.2) for real $\mu \in (1, 2]$, that is $s \in [\frac{1}{2}, 1)$. The proposition holds in the whole strip, in particular on the critical line $s = \frac{1}{2} + it$, which makes it relevant to the zeros; for complex μ only the Mellin form (20.1.1) is proved, the integral in k being formal. The equation numbers of the paper were not checked.

Proposition 20.1.2 (Stable walk reading). *Let S_n be a random walk with symmetric stable steps, $\mathbb{E} e^{ikX} = e^{-|k|^\mu}$, $1 < \mu \le 2$, and $M_n = \max_{0 \le j \le n} S_j$. With $s = 1/\mu$,*

$$\mathbb{E} M_n = \frac{\Gamma(1-s)}{\pi s}\, n^s + \frac{\Gamma(1-s)\zeta(1-s)}{\pi} + O(n^{s-1}).$$

Hence the constant of Comtet and Majumdar is

$$C_\mu = I(s) = \frac{\zeta(s)}{(2\pi)^s \sin(\pi s/2)}.$$

Proof. Spitzer's identity [171] gives $\mathbb{E} M_n = \sum_{k=1}^n \mathbb{E} S_k^+ / k$. By stability $S_k \overset{d}{=} k^s S_1$ and $\mathbb{E} S_1^+ = \frac{1}{2}\mathbb{E}|S_1| = \Gamma(1-s)/\pi$. Hence $\mathbb{E} M_n = \pi^{-1}\Gamma(1-s) \sum_{k \le n} k^{s-1}$, and Euler Maclaurin gives $\sum_{k \le n} k^{s-1} = n^s/s + \zeta(1-s) + \frac{1}{2} n^{s-1} + O(n^{s-2})$. □

So the identity is Spitzer plus Euler Maclaurin plus the functional equation. The map $s \mapsto 1-s$ is $\mu \mapsto \mu/(\mu-1)$, the Hölder conjugate index, and the critical line passes through the self dual point $\mu = 2$, the Gaussian walk. In the normalisation of Proposition 20.1.2, where the Gaussian steps have variance 2, the constant there is $C_2 = \zeta(\frac{1}{2})/\sqrt{\pi} = -0.82392$ (computed); for steps of unit variance it is $\zeta(\frac{1}{2})/\sqrt{2\pi} = -0.58260$ (computed), which is the constant in the paper (their equation 32, not checked). The source printed the second value in the first normalisation. Real indices give real $s \in [\frac{1}{2}, 1)$, where $\zeta < 0$; this is the content of the paper's Appendix A, that there are no zeros on $[\frac{1}{2}, 1)$. The zeros on the critical line are not zeros of a probabilistic offset constant, since a walk with index $\mu = 2/(1+2it)$ is not defined; the factor relating Ξ to C_μ is nonzero and the continuation in s is analytic.

20.1.1 The uniform law and its logarithm

The integrand of (20.1.2) is the logarithm of $\varphi_U(x) = (1-e^{-x})/x = \mathbb{E}e^{-xV}$, $V \sim \mathrm{Uniform}(0,1)$, at $x = k^\mu$.

Proposition 20.1.3. *For $x > 0$,*

$$\ln \frac{1-e^{-x}}{x} = -\lim_{R\to\infty}\Big\{\int_0^R (1-e^{-xt})\frac{dt}{t} - \sum_{1\le n\le R}\frac{1-e^{-xn}}{n}\Big\}, \tag{20.1.3}$$

that is, with the signed k measure $dt - d\lfloor t\rfloor = d\{t\}$ ($\{t\}$ the fractional part), $\ln\varphi_U(x) = -\int_0^\infty (1-e^{-xt})\,t^{-1}(dt - d\lfloor t\rfloor)$ as an improper integral with a common upper cutoff. For $0 < \mathrm{Re}\, z < 1$,

$$\lim_{R\to\infty}\int_0^R t^{-z}\big(d\lfloor t\rfloor - dt\big) = -z\int_0^\infty \{t\}\,t^{-z-1}\,dt = \zeta(z). \tag{20.1.4}$$

Proof. The first integral in the braces is $\int_0^{xR}(1-e^{-v})\,dv/v = \log(xR) + \gamma_E + E_1(xR)$, and the sum is $H_{\lfloor R\rfloor} - \sum_{n\le R} e^{-xn}/n$. Since $H_{\lfloor R\rfloor} - \log R \to \gamma_E$, $E_1(xR)\to 0$ and $\sum_{n\ge1} e^{-xn}/n = -\log(1-e^{-x})$, the braces tend to $\log x - \log(1-e^{-x})$, which is (20.1.3). For (20.1.4), Stieltjes integration by parts gives, at a finite cutoff R, $\int_0^R t^{-z}(d\lfloor t\rfloor - dt) = -\{R\}R^{-z} - z\int_0^R \{t\}t^{-z-1}\,dt$. The boundary term tends to 0, and the last integral converges absolutely because $\{t\} = t$ below 1 and $\{t\} \le 1$ above. The identification with $\zeta(z)$ is Titchmarsh's formula [264, Equation 2.1.5]. □

The common cutoff matters. Neither part is integrable against t^{-1} at infinity, the two parts are mutually singular, and $\int_1^\infty |dt - d\lfloor t\rfloor|/t = \infty$, so the difference is not a signed Lévy measure in the Lebesgue sense and (20.1.3) is a regularised representation. A positive tilt $e^{-\delta t}$ restores integrability, which Section 20.4 uses. The uniform law is not infinitely divisible, and the proposition shows how: what would be its Lévy measure is Lebesgue minus counting. In this language (20.1.1) reads $\pi I(s) = \Gamma(1-s)\int t^{s-1}(d\lfloor t\rfloor - dt)$, a signed Thorin type integral.

20.2 GGC reciprocals and the Laguerre Pólya class

The Laguerre Pólya class $\mathcal{LP}$ consists of the locally uniform limits of real polynomials with only real zeros. The lemma below is known (Propositions 11.1.15 and 2.1.1); it is proved here because the whole chapter uses it.

Lemma 20.2.1. *Let E be even, real on $\mathbb{R}$, entire of order less than two, with $E(0) > 0$. Then $w \mapsto E(0)/E(i\sqrt{w})$ is the Laplace transform of a GGC if and only if all zeros of E are real, that is $E \in \mathcal{LP}$. In that case the Thorin measure is $\sum_k \delta_{\tau_k^2}$, where $\pm\tau_k$ are the zeros of E.*

Proof. An even function of order less than two has the Hadamard product $E(t) = E(0)\prod_k (1 - t^2/\tau_k^2)$ with $\sum_k |\tau_k|^{-2} < \infty$ (order less than two excludes a Gaussian factor e^{-at^2}), and $E(i\sqrt{w})/E(0) = \prod_k (1 + w/\tau_k^2)$ is entire in w. If every τ_k is real, the reciprocal is the Laplace transform of $\sum_k e_k/\tau_k^2$, which converges almost surely, a GGC with the stated Thorin measure. Conversely, a GGC transform $\exp\{-bw - \int \ln(1 + w/x)U(dx)\}$ is analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, so its reciprocal has no zeros there, and the zeros $w = -\tau_k^2$ of the product must lie in $(-\infty, 0)$. □

Corollary 20.2.2. *Let Ξ_δ satisfy the hypotheses of Lemma 20.2.1, let $\Xi_\delta \to \Xi$ locally uniformly as $\delta \to 0$, and let each $\Xi_\delta(0)/\Xi_\delta(i\sqrt{w})$ be a GGC transform. Then RH holds. Conversely, under RH the constant family $\Xi_\delta = \Xi$ qualifies.*

Proof. $\mathcal{LP}$ is closed under locally uniform limits (Hurwitz), so $\Xi \in \mathcal{LP}$. Equivalently the transforms converge pointwise on $(0, \infty)$ and the GGC class is closed under weak limits [30]. □

So the question asked of any approximating family is equivalent to RH, and the realistic aim is a family in which real zeros are visible for structural reasons. The rest of the chapter tests the families the identity suggests.

20.3 The shifted Comtet Majumdar family

Comtet and Majumdar replace $1/(e^x - 1)$ by $1/(e^x - 1 + \epsilon)$ and $1/x$ by $1/(x + \epsilon)$ (their equations B5 to B10, not checked against the paper). With $1 - \epsilon = e^{-\delta}$ one has $1/(e^x - 1 + \epsilon) = e^\delta/(e^{x+\delta} - 1)$. The family studied here is the shift $g(x) \mapsto g(x + \delta)$, a variant of their cutoff with the same limit as $\epsilon \to 0$ that differs from it already at order ϵ^{1-s} (Remark 20.3.4); every result below concerns the shifted family, not the literal cutoff.

Definition 20.3.1. For $\delta > 0$ set $I_\delta(s) = \frac{1}{\pi}\int_0^\infty x^{-s} g(x + \delta)\, dx$ for $0 < \operatorname{Re} s < 1$ and $\zeta_\delta(s) = (2\pi)^s \sin(\pi s/2)\, I_\delta(s)$, so that $\zeta_0 = \zeta$ by (20.1.1), and $\xi_\delta(s) = \frac{1}{2}s(s - 1)\pi^{-s/2}\Gamma(s/2)\zeta_\delta(s)$, continued meromorphically.

THEOREM 20.3.2 (Exact expansion). *For $0 < \delta < 2\pi$,*

$$\pi I_\delta(s) = \Gamma(1-s)F_\delta(s), \qquad F_\delta(s) = \mathrm{Li}_{1-s}(e^{-\delta}) - \Gamma(s)\delta^{-s} = \sum_{k\ge 0} \zeta(1-s-k)\frac{(-\delta)^k}{k!}, \tag{20.3.1}$$

and

$$\begin{aligned} \zeta_\delta(s) &= \sum_{k\ge 0} \Big(\frac{\delta}{2\pi}\Big)^k \frac{(s)_k}{k!}\, \frac{\sin(\pi s/2)}{\sin(\pi(s+k)/2)}\, \zeta(s+k) \\ &= \zeta(s) + \frac{\delta}{2\pi}\, s\tan\frac{\pi s}{2}\, \zeta(s+1) \\ &\quad - \frac{\delta^2}{8\pi^2}\, s(s+1)\zeta(s+2) - \cdots \end{aligned} \tag{20.3.2}$$

The ratio of sines equals $(-1)^j$ for $k = 2j$ and $(-1)^j \tan(\pi s/2)$ for $k = 2j+1$.

Proof. Expanding $1/(e^{x+\delta}-1) = \sum_n e^{-n\delta}e^{-nx}$ gives $\int_0^\infty x^{-s}(e^{x+\delta}-1)^{-1}dx = \Gamma(1-s)\,\mathrm{Li}_{1-s}(e^{-\delta})$, and $\int_0^\infty x^{-s}(x+\delta)^{-1}dx = \delta^{-s}\Gamma(s)\Gamma(1-s)$, both absolutely convergent for $0 < \mathrm{Re}\, s < 1$. The series for F_δ is Lindelöf's expansion $\mathrm{Li}_\sigma(e^m) = \Gamma(1-\sigma)(-m)^{\sigma-1} + \sum_k \zeta(\sigma-k)m^k/k!$, valid for $|m| < 2\pi$, with $\sigma = 1-s$, $m = -\delta$ (the attribution of the expansion was not checked). For (20.3.2) write $A(s) = (2\pi)^s \sin(\pi s/2)/\pi$, so $\zeta(s) = A(s)\Gamma(1-s)\zeta(1-s)$, and substitute $\zeta(1-s-k) = \zeta(s+k)/\{A(s+k)\Gamma(1-s-k)\}$ into each term. Then $\Gamma(1-s)/\Gamma(1-s-k) = (-1)^k(s)_k$ and $A(s)/A(s+k) = (2\pi)^{-k}\sin(\pi s/2)/\sin(\pi(s+k)/2)$. □

The formula (20.3.2) agrees with direct quadrature of I_δ at $s = 0.6$ for $\delta = 0.1$ and 1 (relative difference 0 and 1.0×10^{-21}) and at $s = 0.5 + 5i$ for the same δ (relative differences 1.8×10^{-17} and 3.4×10^{-17}, limited by the 20 digit quadrature), and the form F_δ agrees with the Lindelöf sum to 10^{-21} (computed).

PROPOSITION 20.3.3 (Obstructions). *Let $0 < \delta < 2\pi$.*

(a) $\xi_\delta(s) = \frac{1}{2}s(s-1)\pi^{(s-1)/2}\Gamma\big(\frac{1-s}{2}\big)F_\delta(s)$ is meromorphic and not entire. It has a pole at $s = 3$ for every $\delta \in (0, 2\pi)$ and poles at $s = -1, -2, -3, \ldots$ It has a pole at $s = 2m+1$, $m \ge 2$, exactly when $F_\delta(2m+1) \ne 0$, and $F_\delta(5) = 0$ at $\delta = 4.7215$ and $F_\delta(7) = 0$ at $\delta = 3.0416$ (computed), so there is no pole at $s = 5$ and $s = 7$ for those δ.

(b) Each simple zero $\rho = \frac{1}{2} + i\gamma$ of ζ moves to

$$\rho_\delta = \rho - \frac{\delta}{2\pi}\, \frac{\rho\tan(\pi\rho/2)\,\zeta(\rho+1)}{\zeta'(\rho)} + O(\delta^2), \tag{20.3.3}$$

and $\xi_\delta(s) \ne \xi_\delta(1-s)$.

(c) If $\xi_\delta(\frac12)\neq 0$, the function $w\mapsto \xi_\delta(\frac12)/\xi_\delta(\frac12+\sqrt{w})$ is not the Laplace transform of a GGC, nor of any positive measure.

Proof. (a) From the functional equation, $\pi^{-s/2}\Gamma(s/2)A(s)\Gamma(1-s)=\pi^{(s-1)/2}\Gamma((1-s)/2)$, which gives the formula. $\Gamma((1-s)/2)$ has poles at $s=1,3,5,\dots$, and the factor $s-1$ removes $s=1$. At $s=2m+1$ a pole remains exactly when $F_\delta(2m+1)\neq 0$. Poisson summation for $1/(e^\delta-1)$ gives, for $n\geq 1$,

$$F_\delta(n+1)=n!\sum_{k\neq 0}(\delta-2\pi ik)^{-n-1}.$$

For $n=2$ this is $F_\delta(3)=4\sum_{k\geq 1}\operatorname{Re}(\delta-2\pi ik)^{-3}$, and $\operatorname{Re}(\delta-2\pi ik)^{-3}=|\delta-2\pi ik|^{-3}\cos 3\theta_k$ with $\theta_k=\arctan(2\pi k/\delta)\in(\pi/4,\pi/2)$ because $\delta<2\pi$, so $3\theta_k\in(3\pi/4,3\pi/2)$ and every term is negative. Hence $F_\delta(3)<0$ and the pole at $s=3$ is present. For larger m the term $k=1$ dominates and its real part changes sign near $\delta=2\pi\tan(\pi/(2m+1))$, so no interval $(0,\delta_0)$ avoids the zeros of $F_\delta(2m+1)$; the values 4.7215 and 3.0416 are zeros of $F_\delta(5)$ and $F_\delta(7)$ (computed at forty digits). The source's argument $F_\delta(2m+1)=-\delta\zeta(-2m-1)+O(\delta^3)$ is not uniform in m. At $s=-m$, $m\geq 1$, the factor $\Gamma(s)\delta^{-s}$ has a pole and the other factors are finite and nonzero.

(b) Implicit differentiation of $\zeta_\delta(\rho_\delta)=0$ using the $k=1$ term of (20.3.2). The failure of the functional equation is immediate from (a): a function invariant under $s\mapsto 1-s$ cannot have a pole at -1 and be regular at 2.

(c) The pole of ξ_δ at $s=3$ makes the reciprocal vanish at $w=(5/2)^2=25/4$ (the value $\frac12+\sqrt{w}=3$), whereas the Laplace transform of a nonzero positive measure is strictly positive on $(0,\infty)$. A second reason, for δ small enough that (b) applies: a zero ρ_δ with $\operatorname{Re}\rho_\delta>\frac12$ makes the reciprocal have a pole at $w=(\rho_\delta-\frac12)^2\notin(-\infty,0]$, which a GGC transform cannot have. The source's first reason, that the zero set is not symmetric and so the function is not a function of w, is not an argument: with the principal root $w\mapsto\xi_\delta(\frac12+\sqrt{w})$ is a function on $\mathbb{C}\setminus(-\infty,0]$, which is all that a GGC transform requires. □

Damping applied on the $\zeta(1-s)$ side is carried by the functional equation into the operator series (20.3.2), which mixes $\zeta(s)$ with $\zeta(s+1),\zeta(s+2),\dots$ through the factor $\tan(\pi s/2)$; this is not symmetric and generates poles.

Remark 20.3.4 (The shift is not the literal cutoff). Write $\epsilon=1-e^{-\delta}$. The literal cutoff of Comtet and Majumdar has the integrand $\tilde g_\epsilon(x)=1/(e^x-1+\epsilon)-1/(x+\epsilon)=e^\delta/(e^{x+\delta}-1)-1/(x+\epsilon)$, whereas the shifted family has $g(x+\delta)=1/(e^{x+\delta}-1)-1/(x+\delta)$; both the multiplicative factor e^δ and the second shift differ. In particular $\tilde g_\epsilon(0)=\tilde g_\epsilon'(0)=0$ while $g(\delta)<0$. For $0<\operatorname{Re}s<1$ one finds $\pi\tilde I_\epsilon(s)=\Gamma(1-s)\big[e^\delta\operatorname{Li}_{1-s}(e^{-\delta})-\Gamma(s)\epsilon^{-s}\big]$, and with $\delta=\epsilon+\epsilon^2/2+O(\epsilon^3)$ the two families differ at the order ϵ^{1-s}: $\tilde I_\epsilon(s)=I(s)+\Gamma(1-s)\Gamma(s)(1-s/2)\epsilon^{1-s}/\pi+O(\epsilon)$, while the shifted family's correction is analytic and linear in δ. A sign test shows

	$\gamma_1 = 14.1347$	$\gamma_2 = 21.0220$	$\gamma_3 = 25.0109$
slope, real part of (20.3.3)	1.53709	1.95365	2.14062
$\delta = 0.01$	0.015371	0.019537	0.021406
$\delta = 0.05$	0.076858	0.097694	0.107022
$\delta = 0.10$	0.153723	0.195408	0.214026
$\delta = 0.30$	0.461231	0.586400	0.641850
$\delta = 1.00$	1.537267	1.953476	2.137214

Table 20.1: Displacement $\operatorname{Re}\rho_\delta - \frac{1}{2}$ of the first three zeros of the shifted family, by Newton's method from (20.3.2) at 30 and 45 digits (difference below 5×10^{-25}; computed). The first row is the real part of the complex slope of (20.3.3), whose imaginary parts -0.1565, $+0.0703$, -0.2961 show that $\operatorname{Im}\rho_\delta$ moves too. The displacement is linear in δ only to the digits of the slope for small δ (at $\delta = 1$ the first entry is 1.53727 against 1.53709).

the difference also after continuation. For $x > 0$ one has $e^x - 1 > x$, so $\tilde g_\epsilon(x) < 0$, and $\tilde g_\epsilon(x) = -x^2/(2\epsilon^2) + O(x^3)$ at 0, so the literal integral converges for $0 < \operatorname{Re} s < 3$, and $\tilde\zeta_\epsilon(s) = (2\pi)^s \sin(\pi s/2)\tilde I_\epsilon(s) < 0$ for real $1 < s < 2$, whereas Theorem 20.4.5 below gives $\zeta_\delta(s) > 0$ there. The pole expansions, the zero drift and the Thorin Mellin identity are proved for the shifted family and do not transfer unchanged to the finite ϵ cutoff. (This corrects the source, which called the cutoff exactly the shift; the sign test and the expansion follow from the formulas above and were not run numerically.)

20.4 The uniform law and the $2\sin^2$ lemma

The shifted family does not lie in $\mathcal{LP}$ (Section 20.3). The $2\sin^2$ lemma asks a different question: given a Lévy measure on the primal side, what is the Thorin measure on the dual side? Asked this way the tilt of the uniform law is a positive object for every $\delta > 0$.

20.4.1 The lemma and the Thorin density of a signed Lévy measure

Lemma 20.4.1 (The $2\sin^2$ lemma). *For $x > 0$ and $w \ge 0$,*

$$\frac{e^{-wx} + wx - 1}{x} = \frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{w^2}{y^2}\Big)\big(1 - \cos xy\big)\,dy. \tag{20.4.1}$$

In the variables $u = w^2/2$, $z = y^2/2$ this is $e^{-wx} + wx - 1 = \int_0^\infty \log(1 + u/z)\,\tau_x(z)\,dz$ with $\tau_x(z) = \frac{x}{\pi\sqrt{2z}}\,2\sin^2\big(x\sqrt{z/2}\big) \ge 0$.

This is (2.1.9) (Lemma 10.2.4, Theorem 16.5.8), where it is credited. A short proof: both sides vanish at $w = 0$. Differentiating in w and using $\int_0^\infty (w^2 +$

$y^2)^{-1}dy = \pi/2w$ and $\int_0^\infty \cos(xy)(w^2+y^2)^{-1}dy = \pi e^{-wx}/2w$, the right side gives $\frac{1}{\pi}\int_0^\infty 2w(1-\cos xy)(w^2+y^2)^{-1}dy = 1-e^{-wx}$, the derivative of the left side; the interchange is justified by writing $\log(1+w^2/y^2) = \int_0^w 2v(v^2+y^2)^{-1}dv$ and applying Tonelli to the nonnegative double integral. The change of variable $z = y^2/2$ gives the second form.

The next statement differs from (16.5.8) (Chapter 16) only in its hypothesis, an abstract signed measure M in place of the Lévy measure of ξ.

THEOREM 20.4.2 (Thorin density from a signed Lévy measure). *Let f be analytic and zero free on $\operatorname{Re} s > \alpha_0$ and positive on (α_0, ∞). Let M be a signed measure on $(0,\infty)$ with $\int (1\wedge x)e^{-\alpha x}|M|(dx) < \infty$ for every $\alpha > \alpha_0$, such that*

$$\frac{f'}{f}(s) - \frac{f'}{f}(\alpha) = \int_0^\infty \big(e^{-\alpha x} - e^{-sx}\big) M(dx), \qquad \operatorname{Re} s > \alpha_0,\ \alpha > \alpha_0. \tag{20.4.2}$$

Then for every $\alpha > \alpha_0$ and $w \ge 0$,

$$\log \frac{f(\alpha+w)}{f(\alpha)} - w\,\frac{f'}{f}(\alpha) = \int_0^\infty \log\Big(1+\frac{w^2}{y^2}\Big)\kappa_\alpha(y)\,dy, \tag{20.4.3}$$

$$\kappa_\alpha(y) = \frac{1}{\pi}\int_0^\infty \big(1-\cos xy\big)e^{-\alpha x}M(dx) = \frac{1}{\pi}\Big[\operatorname{Re}\frac{f'}{f}(\alpha+iy) - \frac{f'}{f}(\alpha)\Big]. \tag{20.4.4}$$

If $\kappa_\alpha \ge 0$, then $u \mapsto F_\alpha(u) = \dfrac{f(\alpha)}{f(\alpha+\sqrt{2u})}\exp\big\{\sqrt{2u}\,\frac{f'}{f}(\alpha)\big\}$ is the Laplace transform of a GGC with Thorin density $\tau_\alpha(z) = \kappa_\alpha(\sqrt{2z})/\sqrt{2z}$.

Proof. Differentiating (20.4.2) in s gives $(f'/f)'(s) = \int x e^{-sx}M(dx)$. The function $\int_0^\infty e^{-\alpha x}(e^{-wx}+wx-1)\,M(dx)/x$ has the same second derivative in w as the left side of (20.4.3), and both vanish together with their first derivatives at $w=0$, so they are equal. Now insert (20.4.1). Fubini is justified by Tonelli applied to $|M|$: by (20.4.1) the double integral of the absolute values is $\int e^{-\alpha x}(e^{-wx}+wx-1)|M|(dx)/x \le \int e^{-\alpha x}\min(w^2x/2, w)\,|M|(dx) < \infty$. The second form of κ_α is (20.4.2) at $s = \alpha+iy$, since $\operatorname{Re}(e^{-\alpha x}-e^{-(\alpha+iy)x}) = e^{-\alpha x}(1-\cos xy)$. If $\kappa_\alpha \ge 0$ then $\log F_\alpha(u) = -\int \log(1+u/z)\tau_\alpha(z)\,dz$ with a nonnegative density and a finite integral, which is the definition of a GGC transform [30]. □

When $M \ge 0$, $\kappa_\alpha \ge 0$ automatically. The interesting cases have M signed, counting measure minus a smooth density, with κ_α still positive. Formula (20.4.4) gives a test that needs no Lévy measure: $\kappa_\alpha \ge 0$ if and only if the real part of the logarithmic derivative on the vertical line through α is never below its value on the real axis.

COROLLARY 20.4.3 (Zeros on a line give Lorentzians). *Let a be real and let the hypotheses of Theorem 20.4.2 hold. Suppose $f'/f(s) = a + \sum_k (s-\rho_k)^{-1}$ (symmetric summation) with all $\rho_k = \beta_0 + it_k$ on one vertical line, the set $\{t_k\}$ symmetric*

and $0 \notin \{t_k\}$. *Then for* $\alpha = \beta_0 + \varepsilon$, $\varepsilon > 0$,

$$\kappa_\alpha(y) = \frac{1}{\pi} \sum_k \big[\ell_\varepsilon(y - t_k) - \ell_\varepsilon(t_k)\big], \tag{20.4.5}$$

and if $\sum_k t_k^{-2} < \infty$ *then* $\kappa_\alpha(y)\,dy \to \sum_{t_k>0} \delta_{t_k}(dy)$ *vaguely on* $(0,\infty)$ *as* $\varepsilon \downarrow 0$*; the right side of* (20.4.3) *tends to* $\sum_{t_k>0} \log(1 + w^2/t_k^2)$.

Proof. $\mathrm{Re}(\alpha + iy - \beta_0 - it)^{-1} = \ell_\varepsilon(y - t)$. As $\varepsilon \downarrow 0$, $\pi^{-1}\ell_\varepsilon(\cdot - t) \to \delta_t$; the constant $\pi^{-1}\sum_k \ell_\varepsilon(t_k) \le \pi^{-1}\varepsilon \sum_k t_k^{-2} \to 0$; and peaks at $t_k < 0$ do not reach $y > 0$. For the last limit the integral against $\log(1+w^2/y^2)$ needs the tail bound $\sum_k O(w^2/t_k^2)$ for the dominated convergence, which the hypothesis $\sum t_k^{-2} < \infty$ supplies. □

20.4.2 The tilted uniform law

Take $f = \varphi_U(s) = (1 - e^{-s})/s$. By Proposition 20.1.3 (after the tilt, where the integrals converge absolutely), (20.4.2) holds with $M_U(dx) = dx - d\lfloor x \rfloor$, $f'/f = g$ and $\alpha_0 = 0$, because $g(s) = \sum_{n\ge1} e^{-ns} - 1/s = \int_0^\infty e^{-sx}(d\lfloor x\rfloor - dx)$ and the zeros of φ_U are $2\pi i k$, $k \ne 0$, on the line $\mathrm{Re}\, s = 0$. The tilt is $\alpha = \delta$: it replaces $g(x)$ by $g(x + \delta)$ in the Mellin integral, which is the shifted family of Section 20.3.

Theorem 20.4.4 (Thorin density of the tilted uniform law). *For every* $\delta > 0$,

$$\begin{aligned} \pi\kappa_\delta(y) &= \frac{y^2}{\delta(\delta^2 + y^2)} - \frac{1}{2}\coth\frac{\delta}{2}\,\frac{1 - \cos y}{\cosh\delta - \cos y} \\ &= \sum_{m\ge1} \big[\ell_\delta(y - 2\pi m) + \ell_\delta(y + 2\pi m) \\ &\qquad - 2\ell_\delta(2\pi m)\big], \end{aligned} \tag{20.4.6}$$

and $\kappa_\delta(y) > 0$ *for all* $y > 0$. *Hence* $u \mapsto \dfrac{\varphi_U(\delta)}{\varphi_U(\delta + \sqrt{2u})}\, e^{\sqrt{2u}\, g(\delta)}$ *is the Laplace transform of a GGC with Thorin density* $\tau_\delta(z) = \kappa_\delta(\sqrt{2z})/\sqrt{2z}$. *As* $\delta \downarrow 0$, $\kappa_\delta(y)\,dy \to \sum_{m\ge1} \delta_{2\pi m}$ *vaguely, and the limit of* (20.4.3) *is*

$$\log\varphi_U(w) + \frac{w}{2} = \log\frac{\sinh(w/2)}{w/2} = \sum_{m\ge1} \log\Big(1 + \frac{w^2}{4\pi^2 m^2}\Big),$$

so $(w/2)/\sinh(w/2)$ *is the Laplace transform in* $u = w^2/2$ *of* $\sum_m e_m/(2\pi^2 m^2)$, *a GGC with Thorin measure* $\sum_m \delta_{2\pi^2 m^2}$ *(a law of Biane, Pitman and Yor type [26]; it is* $S_1/4$ *in their notation).*

Proof. The closed form follows from (20.4.4) with $g(s) = \sum_{n\ge1} e^{-ns} - 1/s$: one has $1/\delta - \mathrm{Re}(\delta + iy)^{-1} = y^2/\{\delta(\delta^2 + y^2)\}$ and $\sum_n e^{-n\delta}(1 - \cos ny) =$

$\frac{1}{2}\coth(\delta/2)(1-\cos y)/(\cosh\delta-\cos y)$. The Lorentzian form is Corollary 20.4.3, since $g(s) = -\frac{1}{2} + \sum_{k\neq 0}(s-2\pi ik)^{-1}$. For positivity let $P(y) = \sum_{m\in\mathbb{Z}} \ell_\delta(y-2\pi m) = \frac{1}{2}\sinh\delta/(\cosh\delta-\cos y)$, the periodised Cauchy kernel, so that $\pi\kappa_\delta = [P(y)-\ell_\delta(y)] - [P(0)-\ell_\delta(0)]$.

Case $y \geq \pi$. Here $P(y) \geq P(\pi) = \frac{1}{2}\tanh(\delta/2)$ and $\ell_\delta(y) \leq \ell_\delta(\pi)$, so

$$\pi\kappa_\delta(y) \geq \tfrac{1}{2}\tanh\tfrac{\delta}{2} - \tfrac{1}{2}\coth\tfrac{\delta}{2} + \frac{1}{\delta} - \frac{\delta}{\delta^2+\pi^2} = \frac{\pi^2}{\delta(\delta^2+\pi^2)} - \frac{1}{\sinh\delta} > 0,$$

because $\pi^2\sinh\delta > \pi^2\delta + \pi^2\delta^3/6 > \delta(\delta^2+\pi^2)$, using $\pi^2/6 > 1$.

Case $0 < y < \pi$ and $\delta \leq \sqrt{3}\,\pi$. Since $\ell_\delta''(x) = 2\delta(3x^2-\delta^2)/(\delta^2+x^2)^3$, ℓ_δ is strictly convex on $[\delta/\sqrt{3},\infty) \supset [\pi,\infty)$, and $[2\pi m - y, 2\pi m + y] \subset [\pi,\infty)$. So every bracket in (20.4.6) is positive.

Case $0 < y < \pi$ and $\delta \geq 4.1$. Use $1-\cos y \leq y^2/2$ and $\cosh\delta - \cos y \geq \cosh\delta - 1 = 2\sinh^2(\delta/2)$ in the second term of (20.4.6), and $y^2/\{\delta(\delta^2+y^2)\} \geq y^2/\{\delta(\delta^2+\pi^2)\}$ in the first. Then $\pi\kappa_\delta > 0$ provided $R(\delta) = 8\tanh(\delta/2)\sinh^2(\delta/2)/\{\delta(\delta^2+\pi^2)\} > 1$. The logarithmic derivative of the numerator exceeds $\coth(\delta/2) > 1$, while that of the denominator, $1/\delta + 2\delta/(\delta^2+\pi^2)$, is at most $2/\pi < 1$ for $\delta \geq \pi$. So R is increasing on $[\pi,\infty)$, and $R(4.1) = 1.0322$ (computed, 1.032196804). Since $4.1 < \sqrt{3}\,\pi = 5.4414$, the three cases cover all $\delta > 0$ and $y > 0$. The limit statements are Corollary 20.4.3 together with $\varphi_U(s) = e^{-s/2}\sinh(s/2)/(s/2)$. □

The closed form agrees with the Lorentzian sum to fifteen digits at nine points ($\delta = 0.1, 1, 5$, $y = 0.5, 3, 7$), $\kappa_\delta > 0$ on a grid of step $1/40$ on $(0, 40]$ for ten values of δ from 0.01 to 20, and $R(4) = 0.98037 < 1$, so the threshold 4.1 in the proof is nearly sharp (computed). So the $2\sin^2$ lemma turns the signed measure $dx - d\lfloor x\rfloor$, which is not the Lévy measure of an infinitely divisible law, into a strictly positive Thorin density for every $\delta > 0$. The mechanism is Poisson summation: sum over the integers minus integral is a sum of Lorentzians centred at the dual lattice $2\pi\mathbb{Z}$, and convexity of the Cauchy kernel makes them dominate.

THEOREM 20.4.5 (The shifted zeta as a Thorin Mellin transform). *For* $1 < \operatorname{Re} s < 3$ *and every* $\delta > 0$,

$$\zeta_\delta(s) = (2\pi)^s \int_0^\infty y^{-s}\kappa_\delta(y)\,dy, \tag{20.4.7}$$

where ζ_δ *is the shifted family of Section 20.3, continued analytically (for* $\delta < 2\pi$ *it is the series* (20.3.2)*). At* $\delta = 0$ *the right side is* $(2\pi)^s\sum_m (2\pi m)^{-s} = \zeta(s)$.

Proof. Differentiate (20.4.3) in w: $h(w) := g(\delta+w) - g(\delta) = \int_0^\infty 2w(w^2+y^2)^{-1}\kappa_\delta(y)\,dy$. Since $h(w) = O(w)$ at 0 and $h(w) \to -g(\delta)$ at ∞, the integral $\int_0^\infty w^{-s}h(w)\,dw$ converges for $1 < \operatorname{Re} s < 2$. It is the analytic continuation of $\pi I_\delta(s) = \int w^{-s}g(\delta+w)\,dw$ from $0 < \operatorname{Re} s < 1$: subtracting the value at 0 shifts the Mellin strip. By Tonelli ($\kappa_\delta \geq 0$) and $\int_0^\infty w^{1-s}\,2(w^2+y^2)^{-1}dw = \pi y^{-s}/\sin(\pi s/2)$, we get $I_\delta(s) = \{\sin(\pi s/2)\}^{-1}\int y^{-s}\kappa_\delta(y)\,dy$. Multiply by $(2\pi)^s\sin(\pi s/2)$. The

right side of (20.4.7) converges for $1 < \operatorname{Re} s < 3$, since $\kappa_\delta(y) = O(y^2)$ at 0 and $0 \le \kappa_\delta \le 1/(\pi\delta)$, which gives convergence for $\operatorname{Re} s > 1$ and the continuation to that strip. The restriction $\delta < 2\pi$ belongs to the series (20.3.2) only. □

At $s = 2.5$, $\delta = 0.5$ the right side of (20.4.7) is 1.53343255 (computed with $N = 120$ periods of the lattice, error estimate 5×10^{-8}; $N = 60$ gives 1.5334325) and the series (20.3.2) gives 1.53343255193 (computed). The theorem explains the identity of Comtet and Majumdar: formally (20.1.2) is the Mellin transform of the Thorin measure $\sum_m \delta_{2\pi m}$ of the uniform law, divergent in the strip $0 < s < 1$, and the shift smooths this lattice into the positive density κ_δ. This reading is for the shifted family; for the literal cutoff the Mellin transform is negative at real $s \in (1, 2)$ (Remark 20.3.4).

20.4.3 The Riemann function, by citation

The same construction applies to ξ and is in the book in full. The Lévy measure of ξ is primes minus a smooth density, $M_\xi = \sum_{n\ge2} \Lambda(n)\delta_{\log n} - e^x\,dx + dx/(e^{2x}-1)$, which is $-\sum_\rho e^{\rho x}$ as a distribution by the explicit formula (Proposition 12.1.10; $\Lambda(n)$ is von Mangoldt's function, which this chapter does not otherwise use). Its Thorin density at the basepoint α is π^{-1} times the compensated density $v_\alpha(y) - b_\alpha$ of Chapter 12 (formula (12.1.7)), and Proposition 12.1.12 proves, with no hypothesis on the zeros, that it is positive for every $\alpha \ge 1$ and $y \ne 0$, so the compensated reciprocal is a GGC for every $\alpha \ge 1$; the threshold 7.789, the root 7.789185 of $\frac{1}{8}\zeta(3, \alpha/2 + 1) = (\alpha - 1)^{-3}$ (computed), also appears there. The unconditional GGC property of the uncompensated reciprocal for $\alpha \ge 1$ is Theorem 2.1.2. Positivity of the uncompensated density for every $\alpha > \frac{1}{2}$ is equivalent to RH by Lagarias's criterion (Corollary 23.1.10), and under RH the passage to $\alpha \downarrow \frac{1}{2}$ gives Hadamard's product exactly as Corollary 20.4.3 does for the uniform law (Proposition 12.1.2). Two levels of positivity are distinguished there: level (1), positivity of the uncompensated density, the GGC property of $\xi(\alpha)/\xi(\alpha + \sqrt{s})$; and level (2), positivity of the compensated density, the GGC property of the compensated reciprocal F_α, which strips the stable component $e^{-b\sqrt{s}}$ of index $\frac{1}{2}$ out of level (1). Level (2) implies level (1); no basepoint is known at which (1) holds and (2) fails.

The uniform law is the model in which only level (2) holds. There the uncompensated density is $\pi^{-1} \operatorname{Re} g(\delta + iy) = \pi^{-1}[P(y) - \frac{1}{2} - \ell_\delta(y)]$, which is negative near $y = \pi$ because $P(\pi) = \frac{1}{2}\tanh(\delta/2) < \frac{1}{2}$; the factor $e^{-s/2}$ in φ_U is a drift, and compensation is what makes Theorem 20.4.4 work. For ξ, which has no such factor, level (1) is the natural statement and level (2) is what Proposition 12.1.12 adds.

Remark 20.4.6 (Zeta of the zeros). Under RH, as $\alpha \downarrow \frac{1}{2}$ the Thorin density κ_α^ξ of ξ ((20.4.4) with $f = \xi$) tends vaguely to $\sum_{\gamma>0} \delta_\gamma$, and the argument of Theorem 20.4.5 suggests $\int_0^\infty y^{-s} \kappa_\alpha^\xi(y)\,dy \to \sum_{\gamma>0} \gamma^{-s}$, the zeta function of the

zeros (Voros [204]), for $1 < \operatorname{Re} s < 3$. For the uniform law the same limit is $(2\pi)^{-s}\zeta(s)$ by Theorem 20.4.5. This is heuristic: the limit interchange needs the bounds $|\kappa_\alpha^\xi(y)| \le C\varepsilon y^2$ near the origin and a tail bound of order $\gamma^{-\sigma} + \varepsilon\gamma^{-2}$ for each Lorentzian against $y^{-\sigma}$, summed with the zero counting estimate, which are sketched and not written out. The range $\operatorname{Re} s > 1$ of the source is wrong: the integral diverges at the origin for real $s \ge 3$ because $\kappa_\alpha^\xi(y) = C_\varepsilon y^2 + O(y^4)$ with $C_\varepsilon > 0$ for small ε.

	uniform law, shifted	Riemann ξ
function	$\varphi_U(s) = (1 - e^{-s})/s$	$\xi(s)$
signed k measure M	$dx - d\lfloor x \rfloor$, Lebesgue minus integers	$\sum \Lambda(n)\delta_{\log n} - e^x dx + \frac{dx}{e^{2x}-1}$, primes minus smooth
duality	Poisson summation	explicit formula
zeros	$2\pi i k$, on $\operatorname{Re} s = 0$	ρ; on $\operatorname{Re} s = \frac{1}{2}$ under RH
tilt	$\alpha = \delta$ (the shift)	$\alpha = \frac{1}{2} + \varepsilon$
Thorin density	Lorentzian smoothing of $\sum \delta_{2\pi m}$, positive for all $\delta > 0$ (Theorem 20.4.4); only level (2)	uncompensated and compensated: positive for $\alpha \ge 1$ (Theorem 2.1.2, Proposition 12.1.12); uncompensated for all $\alpha > \frac{1}{2}$ iff RH
limit Thorin measure	$\sum_m \delta_{2\pi m}$	$\sum_{\gamma>0} \delta_\gamma$ counted with multiplicity, under RH
Mellin of Thorin	$(2\pi)^{-s}\zeta(s)$, proved	$\sum_{\gamma>0} \gamma^{-s}$, heuristic

Table 20.2: The uniform law and the Riemann problem under the $2\sin^2$ lemma. Convergence of the Thorin densities is vague convergence. Without RH the limit at $\alpha = \frac{1}{2}$ records only the zeros on the critical line (zeros at $\frac{1}{2} \pm a + i\gamma$ cancel in the real part), so the limit measure is not a test of RH by itself.

20.5 Damping the kernel

The asymmetry of Section 20.3 might be blamed on working with g rather than the symmetric theta representation. This section applies the same damping, with Gauss weights, to the coefficients in Riemann's kernel. The mechanism is that of Chapter 18 (Lemma 18.1.1, Theorem 18.2.3, Propositions 18.8.1 and 18.8.2, with friable weights and the same proof); new here are the Gauss weights, the explicit coefficients and the sign pattern. Write

$$\Phi_n(u) = \bigl(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\bigr)\exp\bigl(-\pi n^2 e^{2u}\bigr), \qquad \Phi = 2\sum_{n\ge 1}\Phi_n,$$

so that $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(ut)\,du$, $\Phi > 0$ and Φ is even; the evenness is the theta transformation formula. For $\delta \ge 0$ let $\Phi_\delta = 2\sum_n e^{-\pi n^2\delta}\Phi_n$ on $[0,\infty)$ and $\Xi_\delta(t) = 2\int_0^\infty \Phi_\delta(u)\cos(ut)\,du$. With $x = e^{2u}$ this is the incomplete gamma form of Ξ (Proposition 18.1.3) with weights $e^{-\pi n^2\delta}$, less the constant $\sum_n(1 -$

$e^{-\pi n^2\delta})(4\pi n^2-1)e^{-\pi n^2}$ (two integrations by parts, as in Section 18.2), the analogue of the factors $(1-\epsilon)^n$ in the paper of Comtet and Majumdar (their B6, not checked). Since Φ_δ decays like $\exp(-\pi e^{2u})$, each Ξ_δ is even, real on $\mathbb{R}$ and entire of order one, $\Xi_\delta \to \Xi$ locally uniformly by dominated convergence, and $\Xi_\delta(0) > 0$ because $\Phi_n > 0$ on $[0,\infty)$; so Ξ_δ satisfies the hypotheses of Lemma 20.2.1. For complex t the transform is $\Xi_\delta(t) = \sum_n e^{-\pi n^2\delta}[B_n(t)+B_n(-t)]$, where $B_n(t) = 2\pi^2 n^4 \Gamma(z_9, \pi n^2)(\pi n^2)^{-z_9} - 3\pi n^2 \Gamma(z_5, \pi n^2)(\pi n^2)^{-z_5}$ with $z_9 = \frac{9}{4}+\frac{it}{2}$, $z_5 = \frac{5}{4}+\frac{it}{2}$ and $\Gamma(\cdot,\cdot)$ the upper incomplete gamma function (this comes from $2\cos(ut) = e^{iut}+e^{-iut}$ before the substitution); on the real axis $B_n(-t) = \overline{B_n(t)}$ and the sum is $2\operatorname{Re} B_n(t)$, the form usually printed, which is not the analytic continuation off the axis.

THEOREM 20.5.1. *For every $\delta > 0$, Ξ_δ has only finitely many real zeros and infinitely many nonreal zeros. Consequently $\Xi_\delta(0)/\Xi_\delta(i\sqrt{w})$ is not the Laplace transform of a GGC.*

Proof. A direct computation gives $\Phi_n'(0) = c_n := \pi n^2 e^{-\pi n^2}\big(15\pi n^2 - \frac{15}{2} - 4\pi^2 n^4\big)$. For $n = 1$ the bracket is $15\pi - 7.5 - 4\pi^2 = 0.1455 > 0$ (computed; the source printed 0.146); for $n \ge 2$ it equals $-n^2(4\pi^2 n^2 - 15\pi) - 7.5 < 0$ because $4\pi^2 n^2 > 15\pi$. Evenness of Φ gives $\sum_n c_n = \Phi'(0)/2 = 0$. Hence for decreasing weights $w_n = e^{-\pi n^2\delta}$,

$$\tfrac{1}{2}\Phi_\delta'(0) = \sum_n w_n c_n = w_1 c_1 + \sum_{n\ge 2} w_n c_n > w_1 \sum_{n\ge 1} c_n = 0,$$

since $w_n < w_1$ and $c_n < 0$ for $n \ge 2$. Two integrations by parts give

$$\int_0^\infty \Phi_\delta(u)\cos(ut)\,du = -\frac{\Phi_\delta'(0)}{t^2} - \frac{1}{t^2}\int_0^\infty \Phi_\delta''(u)\cos(ut)\,du,$$

and the last integral tends to zero by the Riemann Lebesgue lemma. Thus $t^2\Xi_\delta(t) \to -2\Phi_\delta'(0) < 0$, so $\Xi_\delta < 0$ for all large real t and has finitely many real zeros. If Ξ_δ had finitely many zeros in all, Hadamard's theorem would give $\Xi_\delta = Pe^{at}$ with P a polynomial; evenness and reality force $a = 0$, and a nonzero polynomial cannot tend to zero along $\mathbb{R}$. So there are infinitely many zeros, all but finitely many nonreal, and Lemma 20.2.1 applies. □

Damping the n sum destroys the evenness of Φ at $u = 0$, so the even extension $\Phi_\delta(|u|)$ has a kink, which produces a nonoscillating t^{-2} tail that swamps the exponentially small oscillation $|\Xi(t)| \approx t^{7/4}e^{-\pi t/4}$ that carries the zeros. Heuristically the real zeros stop near the height $T(\delta)$ where these balance, $\frac{\pi}{4}T - \frac{15}{4}\ln T \approx \ln(1/\delta)$; for $\delta = 10^{-3}$ this gives $T \approx 24$ (heuristic), and the source reports that the last real zero is at 24.54 (unchecked, not recomputed).

COROLLARY 20.5.2 (Modular constraints). *Let $\Phi_w = 2\sum_n w_n\Phi_n$ with real weights such that $\sum_n |w_n| n^A e^{-\pi n^2} < \infty$ for every $A \ge 0$, and $\Xi_w(t) =$*

$2\int_0^\infty \Phi_w(u)\cos(ut)\,du$. *If* $\Xi_w \in \mathcal{LP}$ *and* Ξ_w *is not of the form* e^{-at^2} *times a polynomial, then* $\sum_n w_n \Phi_n^{(2j+1)}(0) = 0$ *for every* $j \geq 0$.

Proof. If the first nonzero odd derivative is of order $2j+1$, repeated integration by parts gives $\Xi_w(t) \sim 2(-1)^{j+1}\Phi_w^{(2j+1)}(0)\,t^{-2j-2}$, of eventually constant sign, and the argument of Theorem 20.5.1 applies (an element of $\mathcal{LP}$ with finitely many zeros is $e^{-at^2+bt}P$). □

For $w_n \equiv 1$ these linear constraints are the Taylor coefficients at $x = 1$ of Jacobi's identity $\theta(1/x) = \sqrt{x}\,\theta(x)$. The proof of Theorem 20.5.1 shows that the first constraint fails for every weight sequence with $w_1 \geq w_n \geq 0$ for $n \geq 2$ and at least one strict inequality (under the summability condition of Corollary 20.5.2): subtracting $w_1 \sum c_n = 0$ gives $\sum_{n\geq 2}(w_n - w_1)c_n > 0$. This covers the usual Abel, Gauss and Lerch weights; it is not a statement about arbitrary deformations of the coefficients.

20.5.1 Symmetrising does not help

One might restore evenness by defining Φ_δ on the whole line through the n sum and taking the Fourier transform over $\mathbb{R}$. Since $\int_{\mathbb{R}} \Phi_n(u)e^{iut}du = \frac{1}{4}s(s-1)\pi^{-s/2}\Gamma(s/2)\,n^{-s}$ with $s = \frac{1}{2} + it$, this gives

$$\Xi_\delta^\natural(t) = \tfrac{1}{2}\big[\xi_\delta^G(\tfrac{1}{2}+it)+\xi_\delta^G(\tfrac{1}{2}-it)\big], \qquad \xi_\delta^G(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\sum_{n\geq 1} n^{-s}e^{-\pi n^2\delta},$$

the real part of the Gauss damped completed zeta function on the real axis and its continuation elsewhere. Now $\Phi_\delta(u) \sim -6\pi\big(\sum_n n^2 e^{-\pi n^2\delta}\big)e^{5u/2}$ as $u \to -\infty$, only exponential decay, because the double exponential decay of Φ at $-\infty$ is again the theta formula. Correspondingly ξ_δ^G has a pole at $s = -2$ and $\Xi_\delta^\natural$ has poles at $t = \pm\frac{5}{2}i$. The reciprocal $1/\Xi_\delta^\natural(i\sqrt{w})$ then vanishes at $w = 25/4$, exactly as in Proposition 20.3.3(c). Damp on one side and get a kink (Theorem 20.5.1); damp symmetrically and get a pole.

20.5.2 The numerics of the source

The source tabulates zeros of the damped kernels, from mpmath at 18 to 30 digits with sign changes on a grid and an argument principle count on the box $0 < \operatorname{Re} t < T$, $|\operatorname{Im} t| < 4$. Apart from the three values of $\Phi_\delta'(0)$ below, none of it was recomputed and all of it is unchecked. For $\delta = 0.05$ and $\delta = 0.01$ the source finds one real zero in $(0, 60)$ (at 14.1002 and 14.1263) and three zeros in the box, against 13 for Ξ itself; for $\delta = 10^{-3}$ a scan to height 110 finds three real zeros (14.1338, 21.0755, 24.5364); the ratio $t^2\Xi_\delta(t)/\{-2\Phi_\delta'(0)\}$ at $t = 100$ is 0.973 for three values of δ; and $\frac{1}{2}\Phi_\delta'(0) = 1.85\times10^{-4}$, 1.72×10^{-3}, 6.34×10^{-3} at $\delta = 10^{-3}, 10^{-2}, 5\times10^{-2}$, the last three agreeing with a hand recomputation from

$c_1 = 0.019749$, $c_2 = -0.019749$, $c_3 = -4.1 \times 10^{-8}$. For the symmetric damping $e^{-\delta \cosh 2u}$ (the kernel $e^{-2\pi \cosh 2u}$ of Pólya's Bessel function is treated in Section 18.1 and Proposition 18.1.16, and the cosh ladder in Chapter 18) the source finds that all zeros below height 45 stay real for $\delta = 0.5$ and 2, eight and six of them, moving up and thinning out as δ grows (unchecked). If true this separates damping that respects Jacobi's identity from damping that does not, but it concerns low zeros only.

20.6 The de Bruijn Newman family and what the identity contributes

By Corollary 20.5.2 an admissible deformation must keep the even extension of Φ smooth at $u = 0$; multiplying by a smooth even function of u whose derivatives grow at most like e^{Cu^2} does this and keeps the double exponential decay. The classical example is the de Bruijn Newman flow of Section 4.1,

$$H_t(z) = \int_0^\infty e^{tu^2}\, \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du,$$
$$\Phi_{\mathrm{dBN}}(u) = \tfrac{1}{2}\Phi(2u),$$
$$H_0(z) = \tfrac{1}{8}\Xi(z/2).$$

De Bruijn proved that H_t has only real zeros for $t \geq \frac{1}{2}$, the Polymath project for $t \geq 0.22$, Platt and Trudgian for $t \geq 0.2$, and Rodgers and Tao that $\Lambda \geq 0$ (Section 4.1). The source writes the flow with $e^{\lambda u^2}\Phi(u)$ in the variable z, where $H_0 = \Xi$; in that normalisation $H_\lambda(z) = 8H_{4\lambda}(2z)$, the time scales by 4, and the literature thresholds become $\lambda \geq \frac{1}{8}$, 0.055 and 0.05. The source's statement with $\lambda \geq 0.2$ is therefore true (zeros stay real forward in time) but is not the literature result it cites, and the Polymath bound is 0.22, not 0.2. The theorem is Theorem 17.1.16: for $t \geq \Lambda$ the reciprocal $R_t = H_t(0)/H_t(2i\sqrt{s})$ is a GGC transform with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$ where $z_k(t) > 0$ are the zeros of H_t, unconditionally for $t \geq 0.2$, and RH holds if and only if R_t is a GGC for arbitrarily small $t > 0$, that is by Rodgers and Tao if and only if $\Lambda = 0$. The source's statement that the transforms converge as $\lambda \downarrow 0$ if and only if RH holds is false in the direction convergence implies RH: $H_t(i\sqrt{w}) > 0$ for every t, and dominated convergence gives $H_t(0)/H_t(i\sqrt{w}) \to H_0(0)/H_0(i\sqrt{w})$ for every $w > 0$ without any hypothesis. What is equivalent to RH is the GGC property for arbitrarily small t.

The Gaussian multiplier is $\exp(-t\, d^2/dz^2)$ acting on H_0, which preserves real zeros forward in time; a decreasing multiplier is the analogue of $t < 0$, and Rodgers and Tao prove that every $t < 0$ produces nonreal zeros. For the Gaussian family with $t < 0$ and for the coefficient damping of Theorem 20.5.1 this is a theorem. That damping is the wrong direction for every decreasing multiplier, whether applied to n or to u, is a heuristic and is not proved here (the cosh numerics

above, if confirmed, show low zeros staying real), and nothing established in this chapter decides the Laguerre Pólya property of other decreasing multipliers.

What the identity of Comtet and Majumdar contributes is a probabilistic reading of the signed object that must become positive. By Proposition 20.1.3, ζ is the Mellin transform of the negative of the signed k measure $dt - d\lfloor t \rfloor$ of the uniform law (equation (20.1.4)). Under RH, $\Xi(0)/\Xi(i\sqrt{w})$ is the GGC transform with Thorin measure the counting measure of the squared ordinates, and the unnormalised $1/\Xi(i\sqrt{w})$ is only its constant multiple with the same Thorin measure. The identity therefore links two counting measures: integers minus Lebesgue on the arithmetic side, zeros on the spectral side. For real $s \in (0,1)$, $g < 0$ and $\sin(\pi s/2) > 0$ give $\zeta(s) < 0$ by (20.1.1), which is the statement that ζ has no real zeros there.

Problem 20.1. Find the analogue, for the explicit formula, of the convexity and Poisson summation argument of Theorem 20.4.4: a structural reason, not relying on zero locations, why the Lorentzian smoothing of M_ξ against $1 - \cos xy$ stays positive in the strip $\frac{1}{2} < \alpha \leq 1$. (Positivity for $\alpha \geq 1$ is Proposition 12.1.12.)

Problem 20.2. Find a one parameter family of even multipliers $m_\delta(u) \geq 1$, $m_\delta \to 1$, for which the walk interpretation of Proposition 20.1.2 survives (for instance m_δ arising from subordinating the stable walk) and for which real zeros of $\int_{\mathbb{R}} m_\delta \Phi\, e^{iut} du$ can be proved for δ below the known threshold $t = 0.2$ of the flow.

Problem 20.3. The symmetric damping $m_\delta(u) = e^{-\delta \cosh 2u}$ is even and smooth, so every constraint of Corollary 20.5.2 holds for $m_\delta\Phi$, and its improved decay is immediate. Decide whether its transforms lie in $\mathcal{LP}$. It is not the same as weighting θ by the Jacobi invariant weight $e^{-\delta(x+1/x)/2}$: with L the operator taking θ to Φ, $L(W\theta) \neq WL\theta$ because the product rule leaves terms in W' and W'', so $m_\delta\Phi$ is the right object. For $A, B > 0$ one has $\int_{\mathbb{R}} e^{(a+it)u - Ae^{2u} - Be^{-2u}}\, du = (B/A)^{(a+it)/4} K_{(a+it)/2}(2\sqrt{AB})$ (substitute $x = e^{2u}$ in the integral defining K), and each term of $m_\delta\Phi$ has $A = \pi n^2 + \delta/2$, $B = \delta/2$ and $a = \frac{9}{2}$ or $\frac{5}{2}$, which gives Bessel K sums of Chowla Selberg type (compare Chapter 23); this identity gives no conclusion about zeros. A height of order $\exp(c/\delta)$ at which nonreal zeros appear is only a conjecture suggested by the multiplier $e^{-\delta \cosh 2u} \leq 1$, with no theorem or numerical certification, and the low lying numerics are not decisive.

Proposition 12.1.12 settles the source's first open problem, compensated positivity of ξ for $1 < \alpha < 7.789$.

Chapter 21

Approximating clocks: Jensen, Euler product and boundary hitting times

21.1 Jensen hitting times

By Proposition 5.5.1, RH holds if and only if some GGC laws, approximating clocks, converge weakly to the reciprocal Xi law μ, whose Laplace transform is $1/G$ with $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$. This chapter tests the candidates: laws built from the Taylor coefficients of G, from extreme values and smoothed zeros, and from truncated Euler products, together with curved boundaries, Brownian representations and coefficient ladders. Each is asked two questions: is it built from data known without RH, and can it see an off line zero, in particular that of the Davenport Heilbronn function?

Truncated Hadamard products are GGC approximants but circular, being built from the zeros. The Taylor coefficients of G are known unconditionally, and the family built from them is the probabilistic form of the Jensen and Pólya criterion (Proposition 22.4.1).

21.1.1 The family

Write $G(u) = \sum_{k\geq 0} c_k u^k$. The coefficients are positive and computable without any hypothesis: $G(u) = \mathbb{E}\cosh(\sqrt{u}\,V)$ with V the symmetric variable of Section 5.6 (Proposition 21.8.1 identifies it with the logarithm of a size biased range of a Brownian bridge), so that $c_k = \mathbb{E}V^{2k}/(2k)!$. For $d \geq 1$ put

$$J_d(u) = \sum_{k=0}^{d} \binom{d}{k} \frac{k!}{d^k}\, c_k u^k = \sum_{k=0}^{d} c_k \prod_{j<k} \Big(1 - \frac{j}{d}\Big) u^k.$$

Up to the scaling $X = u/d$ and a constant this is the Jensen polynomial of degree d and shift 0 of Griffin, Ono, Rolen and Zagier [89], whose sequence $g(n)$ (not to be confused with the ordinates) is defined by $(-1+4z^2)\Lambda(\frac{1}{2}+z) = \sum g(n) z^{2n}/n!$; since $(-1+4z^2)\Lambda(\frac{1}{2}+z) = 8\xi(\frac{1}{2}+z)$, their $g(k)$ is a constant multiple of $k!\,c_k$.

PROPOSITION 21.1.1. *1. $J_d \to G$ uniformly on compact sets, without any hypothesis.*

2. If J_d has only real zeros, they are negative, $J_d(u) = \prod_{j\le d}(1 + u/\ell_{d,j})$, and $1/J_d$ is the Laplace transform of the GGC law μ_d of $\sum_{j\le d} e_j/\ell_{d,j}$: the time a pure birth chain with rates $\ell_{d,1}, \dots, \ell_{d,d}$ takes to pass through d states.

3. RH holds if and only if μ_d exists for every d, and then $\mu_d \Rightarrow \mu$.

4. μ_d exists for every $d \le 2\cdot 10^{17}$, unconditionally (proved, quoted through [89]).

5. Every μ_d has the same mean $c_1 = p_1$ as μ, and its variance exceeds that of μ by $2c_2/d$.

Proof. For (1), the factors $\prod_{j<k}(1-j/d)$ lie in $[0,1]$ and tend to 1, and $\sum c_k|u|^k < \infty$. For (2), a polynomial with positive coefficients has no zeros in $[0,\infty)$, and a product of linear factors with negative zeros is the transform of a sum of independent exponentials. For (3), if every J_d is hyperbolic then G, as a locally uniform limit of polynomials with only negative zeros, has only negative zeros by Hurwitz's theorem, which is RH; Bondesson's continuity theorem (Corollary 9.1.2) gives the same conclusion directly from $\mu_d \Rightarrow \mu$. Conversely under RH G is in the Laguerre and Pólya class, and the Jensen polynomials of such a function are hyperbolic (Proposition 22.4.1, [89]). Part (4) is the theorem of Chasse on the Jensen polynomials of this sequence, as quoted in a footnote of [89]. Part (5) follows from the factors 1 and $1 - 1/d$ at $k = 1, 2$: the cumulants are $\kappa_1 = c_1$ and $\kappa_2 = c_1^2 - 2c_2(1 - 1/d)$, while $\operatorname{Var}\mu = c_1^2 - 2c_2 = \sum \gamma^{-4}$. □

So the Jensen laws are what Proposition 5.5.1 asks for, built from Brownian moments rather than zeros. The questions are how fast they converge, and what the existence of μ_d at a given degree says about the zeros.

21.1.2 RATE OF CONVERGENCE

Since $\prod_{j<k}(1-j/d) = 1-k(k-1)/(2d)+O(k^4/d^2)$ and $\sum_k k(k-1)c_k u^k = u^2G''(u)$,

$$J_d(u) = G(u) - \frac{u^2 G''(u)}{2d} + O(d^{-2}).$$

At a simple zero $u_n = -\gamma_n^2$ this moves the zero by

$$\frac{\ell_{d,n}}{\gamma_n^2} - 1 = \frac{A_n}{2d} + O(d^{-2}), \qquad A_n = \frac{u_n G''(u_n)}{G'(u_n)}.$$

The coefficients c_k, $k < 600$, were computed from the moments of Pólya's kernel by a trapezoid rule at 260 digits; two step sizes agree to a relative 2×10^{-205} for every $k < 600$, and $c_1 = 0.02310499311541$ and $c_2 = 2.48334053 \times 10^{-4}$ agree with

$(\log\xi)''(\frac12)/2$ and with $(c_1^2-\sum\gamma^{-4})/2$ to twenty and twelve digits (computed). The zeros of J_d near $-\gamma_n^2$ were found by Newton's method at full and half precision, with relative changes below 10^{-110}. Table 21.1 compares the relative shifts with $A_n/(2d)$; here $A_1,\dots,A_6 = -8.69,-16.37,-15.15,-29.57,-15.83,-32.02$ (computed, from the zeros plus a smooth tail they agree to five digits).

d	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$
80	−0.0513	−0.0905	−0.0871	−0.140	−0.104	−0.153
640	−0.00674	−0.0126	−0.0117	−0.0221	−0.0127	−0.0241
10240	−0.000424	−0.000799	−0.000739	−0.00144	−0.000774	−0.00156
$A_n/2d$	−0.000424	−0.000799	−0.000740	−0.00144	−0.000773	−0.00156

Table 21.1: Relative shift $\ell_{d,n}/\gamma_n^2-1$ of the zeros of the Jensen polynomials, against the first order prediction; the last row is $A_n/(2d)$ at $d=10240$ (computed).

The laws converge at the same rate. Table 21.2 gives the largest difference of distribution functions over the grid $y = 0.007 + 0.001\,i$, $i=0,\dots,73$, with $1/(uJ_d(u))$ inverted by Talbot's method (two contour sizes agree to 3×10^{-17}) and μ from its residue expansion. The maximum sits at $y\approx0.016$ to 0.017, and $d\cdot\sup|F_d-F|$ increases towards about 2.76 (computed). On a coarser grid of 18 equispaced points the supremum is smaller by up to fourteen per cent, so the grid has to be named.

d	10	40	160	640	2560
$\sup\lvert F_d-F\rvert$	0.155	0.0560	0.0163	0.00425	0.00107
$d\cdot\sup\lvert F_d-F\rvert$	1.55	2.24	2.60	2.72	2.75

Table 21.2: Distance between the distribution functions of the Jensen law μ_d and of μ on the grid of 74 points of $[0.007, 0.08]$ (computed).

21.1.3 What a degree can see

Near a zero at u the correction $-u^2G''/(2d)$ acts like a heat flow of duration proportional to u^2/d. For a quadratic model $G\approx(u-a)^2+b^2$ of an off line pair with $a=\delta^2-\gamma^2$ and $b=2\delta\gamma$, the correction is $-u^2/d$, and the zeros of J_d are real exactly when $b^2<|u|^2/d$. An off line pair at height γ and displacement δ is therefore invisible to the Jensen polynomials below the degree

$$d^*(\gamma,\delta)\approx\frac{|u|^2}{b^2}\approx\frac{\gamma^2}{4\delta^2}$$

(heuristic: the quadratic model ignores the other zeros).

We tested this on Davenport and Heilbronn. With 600 coefficients of G_{DH} computed in the same way from the positive kernel of Proposition 21.8.3, the

zero of J_d^{DH} starting at $u_\rho = (0.3085 + 85.6993i)^2 = -7344.28 + 52.88i$ was followed as $1/d$ increases from 0 (computed at 240 and 280 digits). Its squared imaginary part decreases linearly in $1/d$, with slope -6.27×10^7 at $1/d = 0$ against $|u_\rho|^2 = 5.39\times10^7$ in the quadratic model, and the pair is real below and complex above the double root $d^* = 20784.9$ of J_d (both coefficient sets give 20784.92): at $d = 20619$ it is real and at $d = 20833$ it has imaginary part 2.44. The model value is $\gamma^2/(4\delta^2) = 19290$. So the Jensen polynomials of Davenport and Heilbronn are not all hyperbolic, as they cannot be, but this zero first breaks them near degree $2\cdot10^4$, far beyond the degrees at which the coefficient ladder of Table 21.8 was tested.

For ξ the same scaling turns Chasse's theorem around. An off line zero at height γ would first make μ_d fail near $d \approx \gamma^2/(4\delta^2)$. Since every zero below $3\cdot10^{12}$ is on the line and $\delta < \frac12$, the heuristic predicts Jensen laws up to degree of order 10^{25}, well beyond the cited $2\cdot10^{17}$, which corresponds to heights near $\sqrt{2\cdot10^{17}} \approx 4.5\cdot10^8$. Conversely, certifying μ_d for all $d \le D$ says about the zeros what a verification up to height about $\sqrt{D}$ says, at a cost growing like $D \approx \gamma^2$ rather than the $\gamma^{1+o(1)}$ of direct verification. This is heuristic: the other zeros change d^* by about 8% for Davenport and Heilbronn (20785/19290 = 1.0775). The unconditional ladder of Section 8.1 trades height against order in a similar way.

21.1.4 A Hermite identity and the de Bruijn flow

Write $u = -z^2$ and use $c_k = \mathbb{E}V^{2k}/(2k)!$. Since $(2k)! = 4^k k!(\frac12)_k$ and $\prod_{j<k}(1 - j/d)\,(-1)^k = (-d)_k/d^k$,

$$J_d(-z^2) = \mathbb{E}\,C_d(zV),$$
$$C_d(x) = {}_1F_1\Big(-d;\tfrac12;\frac{x^2}{4d}\Big) = (-1)^d\frac{d!}{(2d)!}H_{2d}\Big(\frac{x}{2\sqrt{d}}\Big),$$

with H_{2d} the Hermite polynomial (proved). So the Jensen polynomial is the expectation over V of a fixed even polynomial with $2d$ real zeros, which plays the role that cos plays in $G(-z^2) = \mathbb{E}\cos(zV)$. The identity holds to forty digits at the five pairs $(d,z) = (5,0.7),(10,2),(20,3.5),(40,5),(100,7)$ (computed). The Plancherel and Rotach asymptotics of H_{2d} give, for bounded x,

$$C_d(x) = e^{x^2/(8d)}\cos\Big(x\sqrt{1+\tfrac{1}{4d}}\Big) + O(x^3/d^2)$$

with an additive error (heuristic: the error term is quoted from the leading term of Szegő's expansion and was not proved here), which is of the size seen at $x = 7$: the two sides differ by $5.5\cdot10^{-2}$, $3.1\cdot10^{-4}$, $2.9\cdot10^{-6}$ for $d = 10, 100, 1000$ (computed).

With $V = 2u$ in Pólya's variable this is

$$J_d(-z^2) \approx \frac{8}{\Xi(0)} H_t\big(2z(1+\tfrac{1}{8d})\big), \qquad t = \frac{z^2}{2d},$$
$$H_t(x) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(xu)\, \mathrm{d}u,$$

the de Bruijn and Newman function at the height dependent time $t = z^2/(2d)$ (Section 4.1). Near height γ the Jensen truncation of degree d is, to first order, the de Bruijn flow run for time $\gamma^2/(2d)$. Heuristically this gives the resolution law: an off line pair at height γ which the flow merges at time t_c is seen by J_d when

$$d > d^* = \frac{\gamma^2}{2t_c}.$$

For an isolated pair $t_c \approx 2\delta^2$, which recovers $\gamma^2/(4\delta^2)$. For Davenport and Heilbronn the flow H_t of the kernel of G_{DH} was computed: the double root of H_t near $z = 85.6175$ occurs at $t_c = 0.176507$ (steps $h = 1/250$ and $1/300$ give 0.17650714 and 0.17650712; the step $h = 1/200$ gave 0.17639 and is not converged; computed), hence $d^* = 85.699^2/(2 \cdot 0.176507) = 20804.8$, against the directly tracked 20784.9: agreement to 0.1% (computed), which the isolated pair value 19290 does not give.

This answers the converse half of Problem 21.6 at the heuristic level. If RH holds up to height T and $\Lambda \le \Lambda_0$, then near every height $\gamma \ge T$ the Jensen truncation of degree $d \le T^2/(2\Lambda_0)$ is the de Bruijn flow at time at least Λ_0, where all zeros are real, while below T the zeros are real to begin with. So one expects J_d hyperbolic for

$$d \le D(T) = \frac{T^2}{2\Lambda_0}.$$

With de Bruijn's $\Lambda_0 = \frac{1}{2}$ [61] and the height $5.45 \cdot 10^8$ reached by van de Lune, te Riele and Winter [271] this gives $D \approx 3.0 \cdot 10^{17}$, the order of Chasse's $2 \cdot 10^{17}$ (it has not been checked that this is how Chasse's bound was obtained). With $T = 3 \cdot 10^{12}$ and $\Lambda_0 = 0.2$ [232] it gives $D \approx 2.25 \cdot 10^{25}$. Making this rigorous needs a uniform error bound in the Plancherel and Rotach step and control of the height dependence of t, which is the remaining content of Problem 21.6.

21.1.5 Assessment

The Jensen laws are an unconditional, non circular family of hitting times converging to μ, at the rate $2.8/d$ in distribution function on the grid above. They realise Proposition 5.5.1 up to degree $2 \cdot 10^{17}$, and their existence at every degree is RH. But degree d sees an off line pair of displacement δ only at heights up to about $2\delta\sqrt{d}$, an insensitivity of the same δ^2 type as for boundaries in Proposition 21.7.2 (with γ^{-2} here and γ^{-4} there). They do not escape the need to control the zeros at every height; they convert it into finite positivity problems whose size grows like the square of the height.

21.2 Extreme value hitting times

The simplest hitting times with an infinite spectrum are extreme values. If $e_1, \dots, e_n$ are independent standard exponentials, Rényi's representation gives

$$\max_{k \le n} e_k \stackrel{d}{=} \sum_{k=1}^{n} \frac{e_k}{k},$$

the passage time of a pure birth chain with rates $1, 2, \dots, n$, and $\max_k e_k - \log n$ tends to the Gumbel law, whose bilateral Laplace transform is $\Gamma(1+s)$ [162] (compare Chapter 19, Proposition 19.2.1 and Corollary 15.1.10). Since $1/\Gamma(1+s) = e^{\gamma_E s} \prod_k (1+s/k) e^{-s/k}$ is in the Laguerre and Pólya class, the Gumbel density $e^{-x-e^{-x}}$ is a Pólya frequency function of infinite order in Schoenberg's sense [254], and its Hadamard truncations are the centred maxima of n exponentials. Curry and Schoenberg [53] showed that Pólya frequency functions are limits of B splines and used this extreme value example as an illustration. For μ the same structure holds, with the integers replaced by the γ^2.

21.2.1 B splines and Dirichlet means

Proposition 21.2.1. *Let $0 < \lambda_1 < \cdots < \lambda_N$ and let f_N be the density of $\sum_{k \le N} e_k/\lambda_k$. Then*

$$f_N(y) = \Big(\prod_{k \le N} \lambda_k \Big) \frac{y^{N-1}}{(N-1)!} \int M(x; \lambda_1, \dots, \lambda_N) e^{-xy} \, \mathrm{d}x,$$

where $M(\cdot; \lambda_1, \dots, \lambda_N)$ is the Curry and Schoenberg B spline with knots λ_k, normalised to integral one. Equivalently M is the density of the Dirichlet mean $D_N = \sum_k W_k \lambda_k$ with (W_k) uniform on the simplex, and the formula is the case of Dirichlet parameter N of the density formula of James, Roynette and Yor [178, Theorem 2.1].

Proof. By partial fractions $f_N(y) = (\prod \lambda_k)(-1)^{N-1} [\lambda_1, \dots, \lambda_N] e^{-\lambda y}$, a divided difference in λ. The Hermite and Genocchi formula $[\lambda_1, \dots, \lambda_N] g = \mathbb{E}\, g^{(N-1)}(D_N)/(N-1)!$, with $g(\lambda) = e^{-\lambda y}$ and $g^{(N-1)} = (-y)^{N-1} e^{-\lambda y}$, gives the result, and the density of D_N is the B spline [53]. □

For the knots $1, \dots, n$ the B spline is the cardinal one and the formula returns $n e^{-y}(1-e^{-y})^{n-1}$, the density of the maximum. For the knots $\gamma_1^2, \dots, \gamma_N^2$ it returns the Kent density of the truncation μ_N: under RH, μ is the limit of y^{N-1} times Laplace transforms of B splines whose knots are the squared zeros. These densities are positive because B splines with real knots are, and the reality of the knots is what RH asserts. As $N \to \infty$ the Dirichlet mean D_N concentrates near $\frac{1}{N} \sum \gamma_k^2$ and the representation degenerates, as noted in Section 21.8.

21.2.2 The extremal process

The infinite version uses the extremal process of the Gumbel law. Let $\Gamma_1 < \Gamma_2 < \cdots$ be the points of a unit Poisson process on $(0, \infty)$ and $G_k = -\log \Gamma_k$; then $G_1 > G_2 > \cdots$ are the points of the Poisson process with intensity $e^{-x}\,\mathrm{d}x$, the limit of the top order statistics of n exponentials minus $\log n$, and G_1 is standard Gumbel. (These G_k are the extremal points, and have nothing to do with the function G.)

Theorem 21.2.2. *For any $0 < \lambda_1 \le \lambda_2 \le \cdots$ with $\sum 1/\lambda_k < \infty$ and $b_k = k/\lambda_k$,*

$$\sum_k \frac{e_k}{\lambda_k} \stackrel{d}{=} \sum_{k \ge 1} b_k\,(G_k - G_{k+1}),$$

and if moreover $k \log k/\lambda_k \to 0$, the series may be rearranged:

$$\sum_{k \ge 1} b_k\,(G_k - G_{k+1}) = b_1 G_1 + \sum_{k \ge 2} (b_k - b_{k-1}) G_k.$$

In particular, under RH, μ is the law of $\sum_k k\gamma_k^{-2}(G_k - G_{k+1})$, a linear functional of the Gumbel extremal process. For finite N the truncation μ_N is the L statistic $\sum_{k \le N} b_k(e_{(N-k+1)} - e_{(N-k)})$ of N exponentials, with $e_{(0)} = 0$.

Proof. The ratios Γ_k/Γ_{k+1} are independent with laws $\mathrm{Beta}(k, 1)$, so $G_k - G_{k+1} = \log(\Gamma_{k+1}/\Gamma_k)$ are independent exponentials of rates k. Then $b_k(G_k - G_{k+1}) \stackrel{d}{=} e_k/\lambda_k$, which gives the first statement. The partial sums of the rearranged series equal those of the first series plus $b_K G_{K+1}$, and $G_{K+1} \sim -\log K$ almost surely, so the rearrangement converges when $b_K \log K \to 0$; for $\lambda_k = \gamma_k^2$ this holds since $b_k \asymp (\log k)^2/k$. Without the condition on λ_k the second form can fail: the condition $\sum 1/\lambda_k < \infty$ with λ_k increasing gives only $k/\lambda_k \to 0$. The finite statement is Rényi's representation of the spacings. □

The weights $b_k = k/\gamma_k^2$ are 0.0050, 0.0045, 0.0048, 0.0043, 0.0046 for $k \le 5$ (computed), not monotone, and behave like $(\log k)^2/(4\pi^2 k)$ for large k. The top of the extremal process determines the bulk of μ, and the deep points, which accumulate at $-\infty$ like $-\log k$, its lower tail. RH is the statement that these weights are attached to real atoms; the representation itself holds for any positive spectrum.

21.2.3 Smoothed zeros

The extremal form suggests putting the zeros at their expected positions. Let $\hat{t}_k$ solve $\vartheta(\hat{t}_k) = (k - \frac{3}{2})\pi$, with ϑ the Riemann and Siegel theta function, so

that $\hat{t}_k$ is the midpoint of the Gram interval in which Gram's law puts γ_k; then $\hat{t}_1, \hat{t}_2, \hat{t}_3 = 14.52, 20.65, 25.49$ against $14.13, 21.02, 25.01$ (computed). The law

$$\hat{\mu} = \mathcal{L}\Big(\sum_k \frac{e_k}{\hat{t}_k^2} \Big) = \mathcal{L}\Big(\sum_k k \hat{t}_k^{-2} (G_k - G_{k+1}) \Big)$$

is an explicit GGC hitting time defined without any zero. Its Thorin atoms have the counting function of those of μ to leading order, hence the same small time behaviour $-\log \hat{\mu}(0,t] \sim (\log(1/t))^2/(16t)$ (Section 5.2). More generally let $\hat{\mu}^{[N]}$ use the true γ_k for $k \le N$ and $\hat{t}_k$ beyond. Since the first $1.2 \cdot 10^{13}$ zeros are on the line, $\hat{\mu}^{[N]}$ is a GGC unconditionally for every N up to that size. Table 21.3 compares them with μ, using $4 \cdot 10^5$ smoothed zeros plus an integral for the rest, Talbot inversion in double precision, and the residue expansion of μ on the grid $y = 0.007 + 0.001\,i$ of Section 21.1 (computed).

N	$\sup \lvert F_N - F \rvert$	mean error
0	$1.25 \cdot 10^{-2}$	$-2.2 \cdot 10^{-4}$
1	$3.2 \cdot 10^{-3}$	$4.6 \cdot 10^{-5}$
2	$2.7 \cdot 10^{-3}$	$-3.6 \cdot 10^{-5}$
5	$8.6 \cdot 10^{-4}$	$1.1 \cdot 10^{-5}$
10	$2.5 \cdot 10^{-5}$	$-3.2 \cdot 10^{-7}$
30	not resolved	$3.4 \cdot 10^{-8}$
100	not resolved	$-8.4 \cdot 10^{-9}$

Table 21.3: The smoothed zero laws $\hat{\mu}^{[N]}$, with the first N zeros exact and the rest at the Gram midpoints, against μ. The inversion in double precision resolves differences of distribution functions only down to about $2 \cdot 10^{-6}$ for $y \ge 0.011$ (and $1.8 \cdot 10^{-5}$ below), so the supremum for $N = 30$ and $N = 100$ is unchecked; the source printed $2.7 \cdot 10^{-6}$ and $6.8 \cdot 10^{-7}$.

The smoothed law alone is within 1.3% of μ in distribution function, and inserting the first ten zeros brings the error to $2.5 \cdot 10^{-5}$, forty times better than the Jensen law of degree 2560. The error is governed by $\sum_{k>N} |\gamma_k^{-2} - \hat{t}_k^{-2}|$, that is by the fluctuation $S(t)$ of the zero counting function, and enters through the weights on the points G_k with $k > N$ (heuristic). These laws are as unconditional as the zero verification they use, and their convergence to μ as $N \to \infty$ is again RH. What they add is speed: the smooth part carries the archimedean information exactly, and only $S(t)$ needs the zeros. Section 5.2.3 has the Kent approximants with a Weyl tail above a Gram point, which are the continuous counterpart; the two tables are not the same experiment.

21.2.4 The Pólya density

Curry and Schoenberg's theory applies to Pólya frequency functions, those whose bilateral Laplace transform is the reciprocal of a Laguerre and Pólya function.

The Pólya density $p(u) = 4\Phi_{\mathrm{dBN}}(|u|)/\Xi(0)$ of $V/2$ has bilateral transform $\xi(\frac{1}{2} + s/2)/\xi(\frac{1}{2}) = G(s^2/4)$. It is not a Pólya frequency function of infinite order: an entire bilateral transform does not exclude that (the Gaussian, with transform e^{cs^2}, is one), but an entire transform with zeros is not the reciprocal of an entire function. A direct check confirms it. With p normalised by $p(0) = 1$, the Toeplitz determinants $\det[p((i-j)h)]_{i,j\le k}$ are nonnegative for $k \le 4$ at every tested step, but for $k = 5$ they are $-4.0 \cdot 10^{-11}$ and $-1.4 \cdot 10^{-7}$ at $h = 0.03$ and $h = 0.05$ (also $-2.1 \cdot 10^{-20}$ at 0.01 and $-1.8 \cdot 10^{-14}$ at 0.02; the same to eighty digits at 90 and 130 digit precision) and positive for $h \ge 0.07$ among the tested steps (computed). So p appears log concave and PF_4, but it is not PF_5; the $k = 6$ minor is negative at $h = 0.05$ and 0.07 ($-2.3 \cdot 10^{-9}$ and $-5.2 \cdot 10^{-6}$). Curry and Schoenberg apply instead to the dual density f of μ, whose transform $1/G = 1/\mathbb{E}\cosh(\sqrt{u}\,V)$ is the reciprocal of the transform of p in the variable $\sqrt{u}$; f is PF_∞ if and only if RH holds (Proposition 5.2.2). The Pólya side has an extreme value structure of a different form: Biane, Pitman and Yor's variable [26] is the range of a Brownian bridge, and by the Jacobi triple product the alternating theta function

$$\sum_{n\in\mathbb{Z}} (-1)^n q^{n^2} = \prod_{m\ge 1} (1-q^{2m})(1-q^{2m-1})^2, \qquad q = e^{-x},$$

is the distribution function of the maximum of independent exponentials of rates $2m$ and, twice, $2m-1$. With $x = 2a^2$ this is Kolmogorov's law of $\sup|b|$. Its Mellin transform carries the factor $1 - 2^{1-s}$, with zeros on $\mathrm{Re}\, s = 1$; heuristically this is why, independently of RH, a reciprocal law built from this maximum is not a GGC. Only the nonalternating theta function, which is not a product of distribution functions, gives ξ alone.

21.2.5 Assessment

The extreme value view gives the B spline and Dirichlet mean form of every truncation (Proposition 21.2.1), a representation of μ as a linear functional of the Gumbel extremal process (Theorem 21.2.2), and the smoothed zero laws, the fastest converging approximating hitting times found here, at the cost of using the verified zeros. None is non circular in the way the Jensen laws are, and the Jensen laws converge slowly. Either way the location of the zeros has to be put in, through the zeros themselves or through a positivity condition at every degree.

21.3 The prime side hitting times have a universal limit

21.3.1 Truncated Euler products

For $P \ge 2$ let $\zeta_P(z) = \prod_{p\le P}(1-p^{-z})^{-1}$ and let N_P be the random integer with $\Pr(N_P = n) = n^{-\alpha}/\zeta_P(\alpha)$ for n composed of primes up to P. Then $L = \log N_P = \sum_{p\le P}(\log p)G_p$ with independent geometric variables G_p, $\Pr(G_p =$

$k) = (1-p^{-\alpha})p^{-k\alpha}$, so L is infinitely divisible with finite mean $\mathbb{E}L = B_{\alpha,P} = \sum_{p\le P} \log p\, p^{-\alpha}/(1-p^{-\alpha})$, and

$$-\log \mathbb{E}e^{-wL} = \sum_{p\le P}\sum_{k\ge 1} \frac{p^{-k\alpha}}{k}\bigl(1-e^{-wk\log p}\bigr), \qquad \operatorname{Re} w \ge 0.$$

Let T_a denote the first passage time of standard Brownian motion to the level a, so that $\mathbb{E}e^{-uT_a} = e^{-a\sqrt{2u}}$.

Proposition 21.3.1 (Truncated Euler products and first passage to the mean level). *For every $\alpha > 0$ and every P, the function*

$$\Phi_{\alpha,P}(u) = \frac{e^{-B_{\alpha,P}\sqrt{2u}}}{\mathbb{E}e^{-\sqrt{2u}\,L}} = \frac{\zeta_P(\alpha)}{\zeta_P(\alpha+\sqrt{2u})}\, e^{-B_{\alpha,P}\sqrt{2u}}$$

is the Laplace transform of a GGC law Y_P with zero drift, whose Thorin measure has density in the ordinate τ (with atom positions $\tau^2/2$)

$$\frac{1}{\pi}\sum_{p\le P}\sum_{k\ge 1} \log p\, p^{-k\alpha}\bigl(1-\cos(k\tau\log p)\bigr) \;\ge 0.$$

In particular this holds on the critical line $\alpha = \frac12$, unconditionally. If Y_P is independent of N_P and of the Brownian motion, then

$$T_{\mathbb{E}\log N_P} \;\stackrel{d}{=}\; Y_P + T_{\log N_P}.$$

Proof. Put $\tilde\mu = \sum_{p\le P}\sum_{k\ge1} \log p\, p^{-k\alpha}\delta_{k\log p}$, a positive measure with $\int (1\wedge x)\tilde\mu(\mathrm{d}x) < \infty$ because the sum is finite over p and converges over k. With $f(w) = \mathbb{E}e^{-wL}$ the formula above reads $\log f(w)/f(0) - wf'(0)/f(0) = \int (e^{-wx} - 1 + wx)\tilde\mu(\mathrm{d}x)/x$, since $\tilde\mu(\mathrm{d}x)/x$ puts the mass $p^{-k\alpha}/k$ at $k\log p$ and $-f'(0)/f(0) = \mathbb{E}L$. The sine squared lemma (Lemma 10.2.5) therefore gives $\Phi_{\alpha,P}(u) = \exp\bigl(-\int_0^\infty \log(1+2u/\tau^2)U(\mathrm{d}\tau)\bigr)$ with $U(\mathrm{d}\tau) = \pi^{-1}\int(1-\cos\tau x)\tilde\mu(\mathrm{d}x)\,\mathrm{d}\tau$, which is the stated density and is nonnegative termwise; there is no drift because $-\log\Phi_{\alpha,P}(u) = O(\sqrt{u})$. For the second statement, $\mathbb{E}e^{-uT_{\log N_P}} = \mathbb{E}e^{-\sqrt{2u}L}$ and $\mathbb{E}e^{-uT_{\mathbb{E}L}} = e^{-\sqrt{2u}\mathbb{E}L}$, so $\mathbb{E}e^{-uT_{\mathbb{E}L}} = \Phi_{\alpha,P}(u)\,\mathbb{E}e^{-uT_{\log N_P}}$. □

The zeros of $1/\zeta_P(\alpha+w)$ lie on $\operatorname{Re} w = -\alpha$ and cause no difficulty: $w = \sqrt{2u}$ maps $\mathbb{C}\setminus(-\infty,0]$ into $\operatorname{Re} w > 0$, where $\zeta_P(\alpha+w)$ has neither zeros nor poles. The mechanism is a Jensen gap: for any infinitely divisible $L \ge 0$ with finite mean the same argument gives $T_{\mathbb{E}L} \stackrel{d}{=} Y + T_L$ with Y a GGC independent of L, a sharpened form of Jensen's inequality $\mathbb{E}e^{-wL} \ge e^{-w\mathbb{E}L}$; nothing arithmetic is used beyond the infinite divisibility of $\log N_P$. The proposition is the statement of [151] for truncated products, proved here from the sine squared lemma; the statement at

$\alpha > 1$ for the full product is in Remark 21.3.2. The source reports at $\alpha = \frac{1}{2}$, $P = 13$ and $u = 0.7$ the Thorin integral 5.92746438 against $-\log \Phi_{1/2,13}(0.7) = 5.92746429$ (not recomputed).

Remark 21.3.2 (One prime blocks: Möbius, Liouville and the Pólya Gamma law). This remark is based on [154]. Fix a prime p and write $\ell = \log p$, $q = p^{-\alpha}$ with $\alpha > 0$, $w = \sqrt{2u}$ and $\vartheta = \tau\ell$. For a single prime the density of Proposition 21.3.1 sums in closed form,

$$\frac{1}{\pi}\sum_{k\geq 1} \ell\, q^k\big(1-\cos k\vartheta\big) = \frac{1}{\pi}\,\frac{\ell\, q(1+q)(1-\cos\vartheta)}{(1-q)(1-2q\cos\vartheta+q^2)},$$

which vanishes at $\tau = 2\pi k/\ell$, the ordinates of the zeros of $1-p^{-s}$ on $\operatorname{Re} s = 0$, that is of the poles of the Euler factor $e_p(s) = (1-p^{-s})^{-1}$: it is the constant $\frac{\ell}{2\pi}\coth\frac{\alpha\ell}{2}$ minus the periodic row of Cauchy kernels $\pi^{-1}\alpha/(\alpha^2+(\tau-2\pi k/\ell)^2)$ centred there, so as $\alpha \downarrow 0$ its plateau rises like $1/(\pi\alpha)$. The Liouville factor $1/(1+p^{-s})$, the Euler factor of $\sum_n \lambda(n)n^{-s} = \zeta(2s)/\zeta(s)$, gives the odd lattice. Put

$$\Phi^{+}_{p,\alpha}(u) = \frac{1+q}{1+q\,p^{-w}}\exp\Big(-\frac{w\ell q}{1+q}\Big).$$

For every $\alpha > 0$ this is the Laplace transform of a GGC with zero drift, with Thorin density in the ordinate τ

$$\frac{1}{\pi}\,\frac{\ell\, q(1-q)(1-\cos\vartheta)}{(1+q)(1+2q\cos\vartheta+q^2)} \;\geq 0,$$

whose bumps sit at $\tau = \pi(2k-1)/\ell$, the ordinates of the zeros of $1+p^{-s}$ on $\operatorname{Re} s = 0$. Indeed $-\log\Phi^{+}_{p,\alpha}(u) = \sum_{k\geq1}(-1)^{k+1}\frac{q^k}{k}(e^{-wk\ell}-1+wk\ell)$, so the proof of Proposition 21.3.1 applies to the signed measure $\sum_k(-1)^{k+1}\ell q^k\delta_{k\ell}$, whose total variation is that of the Möbius case, and the series $\frac{1}{\pi}\sum_k(-1)^{k+1}\ell q^k(1-\cos k\vartheta)$ sums to the displayed density, which is nonnegative although its terms are not. As $\alpha\downarrow 0$, $\Phi^{+}_{p,\alpha}(u)\to 1/\cosh(\ell\sqrt{u/2})$, the Laplace transform of $\ell^2\omega$ with ω a Pólya Gamma variable $\mathrm{PG}(1,0)$ [159], and the Thorin densities converge vaguely to $\sum_{k\geq1}\delta_{\pi(2k-1)/\ell}$, the Thorin measure of that law (compare Theorem 18.11.2). In the Möbius case the compensator $\ell q/(1-q)$ diverges as $\alpha\downarrow 0$, because e_p has a pole at $s=0$. The even lattice appears instead through $g(s) = (1-p^{-s})/(s\ell) = p^{-s/2}\sinh(s\ell/2)/(s\ell/2)$, the factor $1-p^{-s}$ with its zero at $s=0$ removed, whose zeros $2\pi ik/\ell$, $k\neq 0$, all lie on $\operatorname{Re} s = 0$: at $\alpha = 0$ its compensated reciprocal $e^{wg'(0)/g(0)}g(0)/g(w)$ equals $(w\ell/2)/\sinh(w\ell/2)$, the Laplace transform of $(\ell/2\pi)^2 H$, with H the logistic mixing law $H(1,0)$ of [18], $\mathbb{E}e^{-uH} = \pi\sqrt{2u}/\sinh(\pi\sqrt{2u})$, whose Thorin atoms are the even lattice $\tau = 2\pi k/\ell$, $k\geq 1$.

For $\alpha > 1$ the products over all primes converge and give the GGC transforms

$$\frac{\zeta(\alpha)}{\zeta(\alpha+w)}\, e^{w\frac{\zeta'}{\zeta}(\alpha)} \qquad \text{and} \qquad \frac{\zeta(2\alpha+2w)\,\zeta(\alpha)}{\zeta(\alpha+w)\,\zeta(2\alpha)}\, e^{wG_\lambda(\alpha)},$$

$$G_\lambda(s) = \frac{\zeta'}{\zeta}(s) - 2\frac{\zeta'}{\zeta}(2s),$$

with Thorin densities $\pi^{-1}[\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+i\tau) - \frac{\zeta'}{\zeta}(\alpha)]$ and $\pi^{-1}[\operatorname{Re} G_\lambda(\alpha+i\tau) - G_\lambda(\alpha)]$, the sums of the one prime densities. The exponents carry the sign + because $\sum_p \ell q/(1-q) = -\frac{\zeta'}{\zeta}(\alpha)$ and $\sum_p \ell q/(1+q) = -G_\lambda(\alpha)$. The logarithms of the finite products converge for every u, since $|e^{-x} - 1 + x| \le x$ and $\sum_p \ell q/(1-q) < \infty$, the limits are continuous at $u = 0$, and the class of GGC laws is closed under weak limits [30]. For $0 < \alpha < 1$ neither continued function is the Laplace transform of a probability law (the second is defined only for $\alpha \neq \frac12$, where $\zeta(2\alpha)$ is infinite), whatever the zeros of ζ do: the first vanishes at $u = (1-\alpha)^2/2$, where $\alpha + w = 1$; the second vanishes there when $\frac12 < \alpha < 1$, and has a pole at $u = (\frac12 - \alpha)^2/2$ when $0 < \alpha < \frac12$.

The difficulty is the limit $P \to \infty$. For $\alpha \le 1$ the compensator $B_{\alpha,P}$ diverges, like $P^{1-\alpha}/(1-\alpha)$ when $\alpha < 1$ and like $\log P$ when $\alpha = 1$, faster than $\log \zeta_P(\alpha+w)$ for any $w > 0$. Hence $-\log \Phi_{\alpha,P}(u) \to \infty$ for every $u > 0$: the laws escape to infinity, and Bondesson's closure theorem has nothing to act on.

21.3.2 A Brownian truncation of the Euler product centred at one half

In the strip the primes can be truncated not by size but in Brownian time [154]. Let $W(t) = \sum_{\operatorname{Re}\gamma>0} e^{-t\gamma^2}$ be the heat trace over the zeros, counted with multiplicity, and let T_x be the first passage time of standard Brownian motion to the level $x > 0$, as above.

Theorem 21.3.3 (Brownian truncated Euler product). *Let $U > 0$ and let $s \in \mathbb{C}$ with $\operatorname{Re}(s^2) > -\gamma_1^2$, a region that contains $\mathbb{R}$ and the band $|\operatorname{Im} s| < \gamma_1$, and in which $\operatorname{Re}(\gamma^2 + s^2) > 0$ for every zero. Put*

$$\begin{aligned} B_U(x,s) &= \mathbb{E}\big[1 - e^{-s^2 T_x/2};\, T_x \le U\big] \\ &= \operatorname{erfc}\frac{x}{\sqrt{2U}} - \frac12\Big[e^{-xs}\operatorname{erfc}\frac{x - sU}{\sqrt{2U}} + e^{xs}\operatorname{erfc}\frac{x + sU}{\sqrt{2U}}\Big], \end{aligned}$$

$$k_\infty(u) = e^{u/8} - \frac{\log\pi}{2\sqrt{2\pi u}} + \frac{1}{2\pi}\int_0^\infty e^{-ur^2/2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac14 + \frac{ir}{2}\Big)dr,$$

$$A_U(s) = \int_0^U \big(1 - e^{-s^2 u/2}\big) k_\infty(u)\,\frac{du}{u},$$

$$R_U(s)=\sum_{\operatorname{Re}\gamma>0}\Big[E_1\Big(\frac{\gamma^2U}{2}\Big)-E_1\Big(\frac{(\gamma^2+s^2)U}{2}\Big)\Big],\qquad E_1(z)=\int_1^\infty e^{-zv}\,\frac{dv}{v}.$$

Then, with no hypothesis on the zeros,

$$\frac{\xi(\frac12)}{\xi(\frac12+s)}=e^{-A_U(s)}\prod_p E_{p,U}(s)\;e^{-R_U(s)},$$

$$E_{p,U}(s)=\exp\Big(\sum_{m\ge1}\frac{p^{-m/2}}{m}\,B_U(m\log p,s)\Big),$$

the product converging absolutely, and

$$|R_U(s)|\le\frac{2}{U}\,e^{-m_sU/2}\sum_{\operatorname{Re}\gamma>0}\Big(\frac{1}{\operatorname{Re}\gamma^2}+\frac{1}{\operatorname{Re}(\gamma^2+s^2)}\Big),$$

$$m_s=\gamma_1^2+\min\big(0,\operatorname{Re}(s^2)\big)>0.$$

If moreover $\operatorname{Re}s>0$ *and* $\operatorname{Re}(s^2)\ge0$*, then* $E_{p,U}(s)\to(1-p^{-1/2-s})/(1-p^{-1/2})$ *as* $U\to\infty$*, the Euler factor of* $1/\zeta(\frac12+s)$ *normalised at the centre.*

Proof. For $\operatorname{Re}a,\operatorname{Re}b>0$, Frullani's integral gives $\log(b/a)=\int_0^\infty(e^{-au}-e^{-bu})\frac{du}{u}$ with the principal logarithm. With $a=\gamma^2/2$, $b=(\gamma^2+s^2)/2$ and the paired Hadamard product $\xi(\frac12+s)/\xi(\frac12)=\prod_{\operatorname{Re}\gamma>0}(1+s^2/\gamma^2)$,

$$\log\frac{\xi(\frac12+s)}{\xi(\frac12)}=\int_0^\infty\big(1-e^{-s^2u/2}\big)W\Big(\frac{u}{2}\Big)\frac{du}{u},$$

the sum over the zeros and the integral being exchangeable because the integrals of the absolute values are $O(|s|^2/\operatorname{Re}\gamma^2)$. Every zero has $\operatorname{Re}\gamma^2\ge\gamma_1^2$: on the line trivially, and off it because its ordinate exceeds 3×10^{12} [232] while $|\operatorname{Im}\gamma|<\frac12$. Split the integral at U. On (U,∞), $\int_U^\infty(e^{-au}-e^{-bu})\frac{du}{u}=E_1(aU)-E_1(bU)$ gives $R_U(s)$, and $|E_1(z)|\le e^{-\operatorname{Re}z}/\operatorname{Re}z$ for $\operatorname{Re}z>0$, with $\operatorname{Re}\gamma^2\ge\gamma_1^2$ and $\operatorname{Re}(\gamma^2+s^2)\ge m_s$, gives the bound. On $(0,U]$ insert the arithmetic form (2.1.5) of the heat trace,

$$W\Big(\frac{u}{2}\Big)=k_\infty(u)-\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt n}\,\frac{e^{-(\log n)^2/2u}}{\sqrt{2\pi u}}.$$

Near $u=0$, $k_\infty(u)=O(u^{-1/2}\log(1/u))$ while $1-e^{-s^2u/2}=O(u)$, so the archimedean part converges and gives $A_U(s)$. In the prime part $e^{-x^2/2u}/(u\sqrt{2\pi u})=f_x(u)/x$, where $f_x(u)=x\,e^{-x^2/2u}/\sqrt{2\pi u^3}$ is the density of T_x; since $|1-e^{-s^2u/2}|\le1+e^{|s|^2U/2}$ on $[0,U]$ and $\Pr(T_x\le U)=\operatorname{erfc}(x/\sqrt{2U})\le e^{-x^2/2U}$, the double sum and integral converge absolutely, so they may be exchanged, and the prime power $n=p^m$ contributes

$-\frac{p^{-m/2}}{m}B_U(m\log p,s)$. The closed form of B_U is the reflection principle, $\mathbb{E}[e^{-\lambda T_x};T_x\le U]=\frac12[e^{-x\sqrt{2\lambda}}\operatorname{erfc}(\frac{x}{\sqrt{2U}}-\sqrt{\lambda U})+e^{x\sqrt{2\lambda}}\operatorname{erfc}(\frac{x}{\sqrt{2U}}+\sqrt{\lambda U})]$ with $\sqrt{2\lambda}=s$, an even entire function of s. Exponentiating gives the identity. If $\operatorname{Re}s>0$ and $\operatorname{Re}(s^2)\ge 0$, then $|e^{-s^2T_x/2}|\le 1$, dominated convergence gives $B_U(x,s)\to 1-\mathbb{E}e^{-s^2T_x/2}=1-e^{-xs}$, and $\sum_m\frac{p^{-m/2}}{m}(1-p^{-ms})=\log\frac{1-p^{-1/2-s}}{1-p^{-1/2}}$. $\square$

When $\operatorname{Re}(s^2)<0$ the truncation cannot be removed: the integrand of $B_U(x,s)$ grows exponentially in u, and $B_U(x,s)$ is unbounded as $U\to\infty$. A smoothed Euler product times a smoothed product over the zeros is the structure of the hybrid Euler Hadamard product of Gonek, Hughes and Keating [82]; here the smoothing kernel is the first passage density and the identity is exact for every U. Table 21.4 checks it, with primes up to 10^3 for $U\le 0.25$ and 10^6 for $U=1$ and the first 1700 zeros for R_U, the rest bounded by the theorem (computed).

s	U	$\log\frac{\xi(1/2+s)}{\xi(1/2)}$	$A_U-\sum_p\log E_{p,U}$
0.1	0.05	$2.31048072572\times10^{-4}$	$2.30708738670\times10^{-4}$
0.1	1	$2.31048072572\times10^{-4}$	$2.31048072572\times10^{-4}$
0.49	0.05	$5.54643804846\times10^{-3}$	$5.53831869645\times10^{-3}$
0.49	1	$5.54643804846\times10^{-3}$	$5.54643804846\times10^{-3}$
2	1	$9.21256255914\times10^{-2}$	$9.21256255914\times10^{-2}$
$0.3+3i$	0.25	$-0.207311444533+0.0422063284094i$	$-0.207311444532+0.0422063284090i$

s	U	R_U	bound
0.1	0.05	3.39×10^{-7}	1.25×10^{-2}
0.1	1	2.06×10^{-48}	3.83×10^{-45}
0.49	0.05	8.12×10^{-6}	1.25×10^{-2}
0.49	1	4.68×10^{-47}	3.83×10^{-45}
2	1	3.55×10^{-46}	3.81×10^{-45}
$0.3+3i$	0.25	$(-1.15+0.406i)\times10^{-12}$	1.62×10^{-11}

Table 21.4: Theorem 21.3.3 (computed). Upper block: the two sides of the identity; lower block: the zero remainder R_U and the bound of the theorem; the residual is below 10^{-30} in every row. The rows $s=0.1$, 0.49 and $0.3+3i$ have $\frac12+s$ inside the critical strip. At $U=0.05$ the zero remainder is visible; at $U=1$ it is below the working precision (30 digits). The bound is loose at small U.

21.3.3 The universal limit

The escape of the truncated Euler family of Proposition 21.3.1 can be undone by centring and scaling. At $\alpha=\frac12$ write $L_P=\log N_P=\sum_{p\le P}(\log p)G_p$; its cumulants satisfy $\kappa_j(P)=\sum_{p\le P}(\log p)^j\sum_{m\ge1}m^{j-1}p^{-m/2}\asymp\sqrt{P}(\log P)^{j-1}$. The

Laplace exponent of Y_P is $-\log \mathbb{E}e^{-uY_P} = K_P(\sqrt{2u})$ with

$$K_P(w) = \log \mathbb{E}e^{-w(L_P - \mathbb{E}L_P)} = \sum_{p \le P} \sum_{m \ge 1} \frac{p^{-m/2}}{m} \big(e^{-wm \log p} - 1 + wm \log p\big)$$
$$= \sum_{j \ge 2} \frac{(-1)^j \kappa_j w^j}{j!} \quad (|w| < \tfrac{1}{2};\ \text{the proof does not use the series}).$$

In particular Y_P has mean $\kappa_2(P) = \operatorname{Var} L_P$ and a tail of index $\frac{3}{2}$ governed by $\kappa_3(P)$.

THEOREM 21.3.4 (Universal limit of the Euler hitting times). *As $P \to \infty$,*

$$Z_P = \frac{Y_P - \kappa_2(P)}{\kappa_3(P)^{2/3}} \implies Z, \qquad \log \mathbb{E}e^{-sZ} = \frac{(2s)^{3/2}}{6}, \quad \operatorname{Re} s \ge 0,$$

a spectrally positive stable law of index $\frac{3}{2}$. More precisely, for $\operatorname{Re} s \ge 0$,

$$\Big| \log \mathbb{E}e^{-sZ_P} - \frac{(2s)^{3/2}}{6} \Big| \le \frac{|s|^2}{6} \rho_4(P), \qquad \rho_j(P) = \frac{\kappa_j(P)}{\kappa_3(P)^{j/3}},$$

and $\rho_4(P) \to 0$.

Proof. Put $c = \kappa_3^{2/3}$ and $w = \sqrt{2s/c}$ with the principal root, so $\operatorname{Re} w \ge 0$. Then $\log \mathbb{E}e^{-sZ_P} = s\kappa_2/c - K_P(w)$. For $\operatorname{Re} x \ge 0$ one has $|e^{-x} - 1 + x - x^2/2 + x^3/6| \le |x|^4/24$, since $|e^{-\xi}| \le 1$ on the segment from 0 to x. Apply this with $x = wm \log p$ in the second form of K_P: the terms of degree two and three sum to $\kappa_2 w^2/2 - \kappa_3 w^3/6$, and the rest is at most $\kappa_4 |w|^4/24$. Now $\kappa_2 w^2/2 = \kappa_2 s/c$ cancels the centring, $\kappa_3 w^3/6 = \kappa_3 (2s/c)^{3/2}/6 = (2s)^{3/2}/6$ because $c^{3/2} = \kappa_3$, and $\kappa_4 |w|^4/24 = |s|^2 \rho_4/6$. Finally $\rho_4 \asymp P^{-1/6} (\log P)^{1/3} \to 0$. At $\operatorname{Re} s = 0$ this gives the characteristic functions, so Lévy's continuity theorem gives weak convergence; the limit $\exp((2s)^{3/2}/6)$ has a Laplace exponent $Cs^{3/2}$ with $C > 0$, that of a spectrally positive stable law of index $\frac{3}{2}$. □

Exact computation from the truncated Euler product confirms the limit, with the slow rate $P^{-1/6}$ the cumulant estimate predicts (Table 21.5; computed, primes up to 10^7).

The rate is explicit. One term further,

$$\log \mathbb{E}e^{-sZ_P} = \frac{(2s)^{3/2}}{6} - \rho_4(P) \frac{(2s)^2}{24} + \rho_5(P) \frac{(2s)^{5/2}}{120} - \cdots,$$

and $\rho_4(P) \sim 2^{-1/3} P^{-1/6} (\log P)^{1/3}$. The exact values of $\rho_4(P)$ are 0.54, 0.37, 0.26, 0.19, 0.13 at $P = 10^3, \ldots, 10^7$, while the asymptotic expression gives 0.48, 0.36, 0.26, 0.19, 0.14 (computed). With the ρ_4 and ρ_5 terms the expansion reproduces

P	10^3	10^4	10^5	10^6	10^7	limit
$s=0.5$	0.1466	0.1523	0.1563	0.1591	0.1612	0.1667
$s=1$	0.3946	0.4158	0.4307	0.4416	0.4498	0.4714
$s=2$	1.0443	1.1200	1.1752	1.2168	1.2484	1.3333

Table 21.5: The quantity $\log \mathbb{E}e^{-sZ_P}$ for $P = 10^3, \dots, 10^7$, against its limit $(2s)^{3/2}/6$ (computed).

every entry of Table 21.5 to within $1.8 \cdot 10^{-2}$ at $P = 10^3$ and $2.2 \cdot 10^{-4}$ at $P = 10^7$; at $P = 10^7$ and $s = 1$ it gives 0.44986 against the exact 0.44983 (computed). The bound of the theorem is a rate on the Laplace exponent, which answers the first half of Problem 21.1; the expansion describes the next terms computationally.

The limit depends only on the central limit structure of $\log N_P$. It is the same for every Euler product with the same prime weights, whatever its zeros, and, centred and supported on the whole line, it is not even a hitting time. The renormalisation that rescues the Euler family erases exactly the information RH concerns: the prime side counterpart of the Davenport and Heilbronn test of Section 21.9.

21.4 Thorin rigidity

Theorem 21.4.1 (Thorin rigidity). *If $\mu = \mu_1 * \mu_2$ with μ_1 and μ_2 GGC laws, then RH holds and*

$$U_1 = \sum_{\gamma>0} a_\gamma \delta_{\gamma^2}, \qquad U_2 = \sum_{\gamma>0} (m_\gamma - a_\gamma)\delta_{\gamma^2}, \qquad 0 \le a_\gamma \le m_\gamma,$$

where m_γ is the multiplicity of the zero (so $m_\gamma = 1$ for simple zeros), and μ_1 is the law of $\sum_\gamma g_\gamma/\gamma^2$ with independent gamma variables g_γ of shapes a_γ. No GGC law with drift or with a nonzero absolutely continuous Thorin part is a GGC factor of μ with a GGC cofactor.

Proof. Drifts and Thorin measures add under convolution of GGC laws and are determined by the Laplace transform. So μ is a GGC, which is RH, with drift zero and Thorin measure $\sum m_\gamma \delta_{\gamma^2}$; the nonnegative measures U_1, U_2 summing to it are carried by the points γ^2 with masses in $[0, m_\gamma]$. □

The hypothesis that the cofactor is a GGC cannot be dropped. Assume RH with simple zeros and take $\gamma_1^2 < a < \gamma_2^2$. The function $(1 + u/a)/((1 + u/\gamma_1^2)(1 + u/\gamma_2^2))$ is a mixture of two exponential transforms with weights $(1 - \gamma_1^2/a)/(1 - \gamma_1^2/\gamma_2^2)$ and $(1-\gamma_2^2/a)/(1-\gamma_2^2/\gamma_1^2)$, both positive with sum one, so the exponential law of rate a is an exact factor of μ, with a probability cofactor that is the convolution of this mixture with the remaining exponentials. The same holds for a GGC

whose Thorin density is supported inside (γ_1^2, γ_2^2) with total mass at most one: each cofactor term is a mixture of exponentials, hence infinitely divisible [83, 262], so the exact factor exists with a cofactor that is a probability law (proved). Hence only factorisations into two GGC laws are rigid.

Every prime built law met so far is a GGC with an absolutely continuous Thorin measure: the Euler laws for $\alpha > 1$, the truncated Euler laws Y_P, and the basepoint laws $\mu^{(\alpha)}$ for $\alpha \geq 1$. None is an exact factor of μ with a GGC cofactor, so a proof through such pieces must be a limit in which absolutely continuous Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$. The basepoint descent is a limit of this kind; the Euler family, by Theorem 21.3.4, is not, after the one centring and scaling treated there.

21.5 What the prime side limit would have to be

Remove the compensator $B_{1/2,P}/\pi$ from the Thorin density of Y_P at $\alpha = \frac{1}{2}$ (it is the Thorin density of the stable factor $e^{-B\sqrt{2u}}$) and add the archimedean density $\pi^{-1}A_{1/2}(\tau)$, where $A_{1/2}(\tau) = \operatorname{Re} A(\frac{1}{2} + i\tau)$ and $A(z) = \frac{1}{z} + \frac{1}{z-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{z}{2})$ is the archimedean part of ξ'/ξ. What remains is the formal Thorin density $\pi^{-1}[A_{1/2}(\tau) + \operatorname{Re}\zeta_P'/\zeta_P(\frac{1}{2} + i\tau)]$. Against test functions whose Fourier transforms decay fast enough the prime sums converge, and the explicit formula identifies the limit. In the Gaussian case it is the identity (Weil's explicit formula with the test function e^{-xr^2}, proved)

$$\sum_{\gamma>0} e^{u_\rho x} = e^{x/4} + (\text{archimedean terms}) \\ - \frac{1}{2\sqrt{\pi x}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/(4x)},$$

whose left side is $W(x)$ and in which the pole of ζ appears as $e^{x/4}$. The zeros emerge from the primes as atoms, but only weakly and only after the pole term is added; at no finite stage is the prime side density positive, already failing at $\tau = 0$, where $A_{1/2}(0) = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{1}{4}) = -2.686 < 0$ and $\zeta_P'/\zeta_P(\frac{1}{2}) < 0$. A prime side hitting time proof needs positivity of this limit against every admissible test function, a Weil type criterion at the centre of the strip (compare Chapters 15.12 and 9), the same inequality that is stated at $\alpha = 1$ on windows of one mean zero spacing in [233], now required at $\alpha = \frac{1}{2}$ where the pole term has exponential size.

Remark 21.5.1 (Shape of a hitting time proof, heuristic). A proof of RH through hitting times built from the primes would have to produce GGC laws whose Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$, which is what weak convergence to μ means by the continuity theorem. By Theorem 21.4.1 they cannot be exact factors of μ with GGC cofactors. The truncated Euler family does not do it after centring and scaling (Theorem 21.3.4), which excludes one renormalisation of one family only,

and the positivity of the limit would be a Weil type criterion at the centre. Only the statement about exact factors is a theorem; the others are heuristic.

21.6 Diffusion approximations

Diffusion approximations in the sense of Roberts, Gelman and Gilks [163], in which high dimensional Markov chains, suitably sped up, converge to a Langevin diffusion, offer half of what is needed. (The boundary crossing approach of Roberts and Shortland is a different tool and is treated in Section 21.7.) The hitting time of a level by a one dimensional diffusion started at an entrance or reflecting end is a GGC, by Kent's theorem [181] (Section 5.2.3); from an interior point it need not be. For reflected Brownian motion on $[0,\infty)$ started at $x \in (0,1)$ the time to reach 1 has transform $\cosh(x\sqrt{2u})/\cosh(\sqrt{2u})$, and $-\varphi'/\varphi = \sum_k 1/(u+\lambda_k) - \sum_k 1/(u+\mu_k)$ with $\lambda_k = (k+\frac{1}{2})^2\pi^2/2$ and $\mu_k = \lambda_k/x^2 > \lambda_k$, so the Thorin measure would have negative atoms. Hence any family of processes converging weakly to a one dimensional diffusion started at an entrance or reflecting end, with hitting times converging in law, has a GGC limit: positivity comes free with the diffusion limit. And RH together with the simplicity of the zeros is equivalent to μ being the hitting time of a level by a one dimensional generalised diffusion started at an entrance end.

The other half is identification. The limiting diffusion must have speed measure with spectrum exactly $\{\gamma^2\}$, and diffusion limits of this type are universal, depending only on averaged quantities such as a Fisher information constant. That is the washing out of Theorem 21.3.4; a diffusion scaling of the Euler family is the cruder Gaussian scaling, under which $Y_P/\operatorname{Var} L_P \to 1$ in probability (the fluctuation scale $\kappa_3^{2/3}$ is $o(\kappa_2)$) and every trace of the zeros disappears. A diffusion approximation would supply positivity only if the chains already carried the arithmetic at the scale the limit sees, and by Theorem 21.4.1 through Thorin measures concentrating onto the γ^2, not through exact factors with GGC cofactors. Its natural role is as a tool: if Markov chains built from the primes approximated the Krein string operator, for example through discretisations of the canonical systems of [110], a weak convergence theorem of Roberts type is what would carry nonnegativity to the limit.

21.7 Curved boundaries: the Roberts and Shortland view

Roberts and Shortland [249] compute the law of the first time a Brownian motion crosses a curved, time dependent boundary, the problem behind barrier options with time dependent coefficients. Their question runs from boundary to law; ours runs the other way: for a law ν on $(0,\infty)$, find b such that

$$\tau_b = \inf\{t > 0 : W_t \geq b(t)\}, \qquad W_0 = 0,$$

has law ν. This is the inverse first passage problem. Under mild conditions on ν it has a unique solution, in a suitable viscosity sense [5, 38]. So μ has a boundary whether or not RH holds, and the question is whether its shape sees the zeros. The boundary of μ is computed in Sections 3.1, 12.2 and 15.2; here we add the enclosure at small times, the validation of the solver, and the effect of moving a zero. The two ends of the boundary are fixed by quantities independent of RH (Proposition 21.7.1 and the slope at infinity); the boundary in between is computed to about $2\cdot10^{-4}$; and moving a zero off the line changes it, to first order and heuristically, by δ^2/γ^4 times a profile independent of the zero (Remark 21.7.3).

21.7.1 The small time end

Proposition 21.7.1. *Let b be the boundary of μ, write $F(t)=\mu(0,t]$, and let $m(t)=\inf_{0<s\le t} b(s)$. For every $t>0$,*

$$\sqrt{t}\,\bar\Phi_{\mathrm N}^{-1}\big(F(t)\big)\le b(t),\qquad m(t)\le\sqrt{t}\,\bar\Phi_{\mathrm N}^{-1}\big(\tfrac12 F(t)\big),$$

where $\Phi_{\mathrm N}$ is the standard normal distribution function and $\bar\Phi_{\mathrm N}=1-\Phi_{\mathrm N}$. Moreover $-\log F(t)\sim(\log(1/t))^2/(16t)$ as $t\downarrow0$, so

$$\liminf_{t\downarrow0}\frac{b(t)}{\log(1/t)}\ge\frac{1}{2\sqrt2},\qquad\limsup_{t\downarrow0}\frac{m(t)}{\log(1/t)}\le\frac{1}{2\sqrt2},$$

and if b is nonincreasing near 0 then $b(t)\sim\log(1/t)/(2\sqrt2)$. None of this depends on RH. Status: proved.

Proof. If $W_t\ge b(t)$ then $\tau_b\le t$, so $\bar\Phi_{\mathrm N}(b(t)/\sqrt t)\le F(t)$, which is the first inequality. If $\tau_b\le t$ then $\sup_{s\le t}W_s\ge m(t)$, and by the reflection principle $F(t)\le 2\bar\Phi_{\mathrm N}(m(t)/\sqrt t)$, which is the second. For the asymptotics, Stirling's formula and $\zeta(\sigma)\to1$ give $-\log\mathbb{E}e^{-s\tau}=\log\xi(\frac12+\sqrt s)-\log\xi(\frac12)\sim\frac14\sqrt s\log s$ as $s\to\infty$ through the reals, unconditionally. This is regularly varying of index $\frac12$, and Kasahara's exponential Tauberian theorem [180], [27, Section 4.12] converts it into the Legendre transform asymptotics: the supremum of $\frac14\sqrt s\log s-st$ is attained at $\sqrt s\sim\log(1/t)/(4t)$ and equals $(\log(1/t))^2/(16t)$ to leading order. Since $\bar\Phi_{\mathrm N}^{-1}(p)\sim\sqrt{2\log(1/p)}$ as $p\downarrow0$, both bounds become $\sqrt{2t\cdot(\log(1/t))^2/(16t)}=\log(1/t)/(2\sqrt2)$. When b is nonincreasing, $m=b$ and the two bounds meet. □

The proposition also checks the numerics: for a nonincreasing boundary the two bounds, explicit in F, enclose b. Evaluating the Legendre transform numerically (computed), the ratio $b(t)/\log(1/t)$ is 0.269, 0.328, 0.342, 0.348 at $t=10^{-4},10^{-8},10^{-12},10^{-16}$, approaching $1/(2\sqrt2)=0.354$ slowly, as the log log corrections suggest. Every finite sum of exponentials has a polynomial lower tail and hence $b(0+)=0$. The logarithmic blow up is the trace of infinitely many zeros, through the growth of ξ along the real axis, wherever the zeros lie.

21.7.2 Computing the boundary

Conditioning on the crossing time and using the strong Markov property gives, for every $t > 0$,

$$\bar{\Phi}_{\mathrm{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) = \int_0^t \bar{\Phi}_{\mathrm{N}}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big)\,\nu(\mathrm{d}s),$$

a nonlinear Volterra equation of the first kind for b [229]. We solve it forward with a midpoint rule in s. The difference of the two sides need not be monotone in $b(t)$, and a global root search occasionally jumps to a spurious root; bracketing next to the previous value removes this. The distribution function of μ comes from a contour integral of $\xi(\frac{1}{2})/(s\,\xi(\frac{1}{2}+\sqrt{s}))$ along a vertical line through the saddle point below $t = 0.0064$ and from the residue expansion with 120 zeros above it; the two agree to a relative difference of 10^{-33} where they overlap. Three validations (all computed):

1. On the linear boundary $b(t) = 1 - t/2$, whose crossing law is known in closed form, it recovers b to 10^{-6} for $t \geq 0.05$; the start of the recursion carries a transient of a few 10^{-3} that does not reach the range of interest.

2. Halving the step from $4 \cdot 10^{-5}$ to $2 \cdot 10^{-5}$ changes the boundary of μ by at most $1.5 \cdot 10^{-4}$ on $[0.0045, 0.12]$, at most $8 \cdot 10^{-5}$ for $t \geq 0.006$, and at most $2.2 \cdot 10^{-5}$ for $t \geq 0.01$.

3. The computed boundary is decreasing, and at every grid point it lies inside the enclosure of Proposition 21.7.1, whose width is 0.004 at $t = 0.0045$ and 0.015 at $t = 0.01$. Figure 21.1(b) compares the boundary at small times with the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 21.7.1.

The interval $[0.0045, 0.12]$ carries all but about $2 \cdot 10^{-9}$ of the mass of μ, whose mean is 0.0231.

21.7.3 The large time end and the shape

The tail $\mu(t, \infty) \sim c\,e^{-\gamma_1^2 t}$ forces $b'(t) \to -\sqrt{2}\,\gamma_1 = -19.99$, the slope of a Brownian motion whose drift gives hitting hazard γ_1^2. Matching the survival function of a Brownian motion with drift $c = \sqrt{2}\,\gamma_1$ below a slowly rising level $g(t)$, namely $\mathrm{const} \cdot g\,t^{-3/2} e^{cg - c^2 t/2}$, against $c' e^{-\gamma_1^2 t}$ gives $cg + \log g = \frac{3}{2}\log t + O(1)$, so (heuristic: the matching is quasi static and has no error control)

$$b(t) = -\sqrt{2}\,\gamma_1 t + \frac{3}{2\sqrt{2}\,\gamma_1}\log t - \frac{1}{\sqrt{2}\,\gamma_1}\log\log(\gamma_1^2 t) + O(1),$$
$$t\big(b'(t) + c\big) \to \frac{3}{2c}.$$

We checked this in two ways (computed). For the exponential law, $t(b' + \sqrt{2})$ is 0.80, 0.88, 0.92, 0.94 at $t = 20, 50, 100, 150$, against $3/(2\sqrt{2}) - 1/(\sqrt{2}\log t) = 0.82$,

0.88, 0.91, 0.92. For μ we continued the solution to $t = 2$ with the complementary form of the integral equation, $\Phi_{\mathrm{N}}(b(t)/\sqrt{t}) - \mu(t,\infty) = \int_0^t \Phi_{\mathrm{N}}((b(t) - b(s))/\sqrt{t-s})\,\mu(\mathrm{d}s)$, which keeps full relative accuracy when the survival function is as small as e^{-400}. Then $t(b' + c)$ is 0.0645, 0.0665, 0.0677, 0.0688 at $t = 0.8, 1.2, 1.6, 2$, against 0.0652, 0.0659, 0.0664, 0.0667 from the formula, with $3/(2c) = 0.0750$ as the limit. The limit is not reached on the range computed, and the slow approach on $[0.03, 0.12]$, where the slope is still -19.6 at the right end, is consistent with the $\log t$ correction rather than a numerical artefact. This end sees only the first zero, which is on the line.

Between the ends the shape gives no criterion either. The boundary of the exponential law, the simplest GGC, starts at 0, rises to a maximum of 0.664 at $t = 0.25$, and falls with slope tending to $-\sqrt{2}$; the Gamma law of shape 2 behaves the same way. The boundary of μ falls monotonically from $+\infty$. No simple convexity or monotonicity condition separates GGC boundaries.

21.7.4 Moving a zero off the line

To move a zero off the line we replace one factor $1+s/\gamma_m^2$ of the Hadamard product by the pair $(1 + s/\lambda)(1 + s/\bar{\lambda})$ with $-\lambda = (\delta + i\gamma_m)^2$, the image of the quartet $\frac{1}{2} \pm \delta \pm i\gamma_m$. As a GGC control we use the split real pair $(\gamma_m \pm 0.001)^2$, so that both laws have the same number of factors. The value $\delta = 0.3085$ is the displacement of the Davenport Heilbronn zero of Section 21.9.

zero	δ	$\max \lvert b_{\mathrm{toy}} - b_{\mathrm{GGC}} \rvert$	density negative from	a law
γ_1	0.05	$2.25 \cdot 10^{-4}$	$t = 2.25$	no
γ_1	0.15	$2.0 \cdot 10^{-3}$	$t = 0.77$	no
γ_1	0.3085	$8.8 \cdot 10^{-3}$	$t = 0.39$	no
γ_6	0.3085	$8.2 \cdot 10^{-6}$	never	yes
γ_{20}	0.3085	$6.3 \cdot 10^{-7}$	never	yes
γ_{60}	0.3085	$5.5 \cdot 10^{-8}$	never	yes

Table 21.6: Boundaries with one off line pair against a GGC control on $[0.0045, 0.12]$; for γ_{60} the maximum is over $[0.0045, 0.1]$, since near $t = 0.12$ the solver noise of about $2 \cdot 10^{-7}$ exceeds the signal. "Never" means the density was found positive on a grid of step 0.005 on $(0.0064, 6]$ and at three points below 0.0064, beyond which the tail is governed by γ_1. All entries computed.

When the off line pair is the lowest zero, the pole closest to the origin is complex and the density oscillates in its tail and becomes negative; at $\delta = 0.3085$ this happens at $t = 0.39$, where the remaining mass is about e^{-78}. Beyond that point no boundary exists, and before it the boundary moves by less than 10^{-2}. For ξ this cannot occur, since the lowest zero is on the line. When the pair sits higher the tail is still governed by γ_1, and the toy law is, on the evidence of the computed density, a probability law that is not a GGC (it has complex poles,

Theorem 21.4.1). The size of the change then follows a simple rule, of which two parts are proved.

Proposition 21.7.2. *Moving the pair at height γ off the line by δ multiplies the Laplace transform, relative to the control, by $1+6\delta^2 s/\gamma^4+O(\delta^2 s^2/\gamma^6)$ for $|s| \ll \gamma^2$. To first order this is the shift $\tau \mapsto \tau - 6\delta^2/\gamma^4$ of the mean. Status: proved.*

Proof. The mean of a GGC with atoms λ_j is $\sum 1/\lambda_j$. For the toy pair this contribution is $2\,\mathrm{Re}(1/\lambda) = 2(\gamma^2-\delta^2)/(\gamma^2+\delta^2)^2 = 2/\gamma^2 - 6\delta^2/\gamma^4 + O(\delta^4/\gamma^6)$, and for the control it is $2/\gamma^2 + O(10^{-6}/\gamma^4)$. Higher cumulants change by $O(\delta^2/\gamma^{2k+2})$, and the ratio of the two Laplace transforms is $\big((1+s/\lambda)(1+s/\bar{\lambda})\big)^{-1}$ over the control factor, which expands as stated. □

Remark 21.7.3 (Boundary response, heuristic). If the map from a law to its boundary is linear to first order and the change of the law is, to leading order, a pure shift, then

$$b_{\mathrm{toy}}(t) - b_{\mathrm{GGC}}(t) \approx \frac{6\delta^2}{\gamma^4}\, K(t),$$

where K is the linear response of the boundary to a shift of the law, a profile that does not depend on γ or δ. Linearity, the universality of K and the size of the neglected terms are assumed, not proved. Status: heuristic, checked numerically (Table 21.7).

The step from the law to the boundary is not proved. We computed K directly by solving for the law of $\tau - \Delta$ with $\Delta = 2 \cdot 10^{-6}$; it is -72, -29, -13, -11 at $t = 0.01, 0.02, 0.05, 0.08$, and it is linear in Δ to four digits (the values at $\Delta = 2{\cdot}10^{-6}$ and $4{\cdot}10^{-6}$ agree to 10^{-2}). Table 21.7 compares it with the normalised differences $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$, and Figure 21.1(c) shows the collapse. At $t = 0.02$ the three heights agree with K to 3%, 2% and 0.5%. At small times the agreement sets in only once t exceeds a multiple of $1/\gamma$, where the relevant s are below γ^2.

	$t = 0.01$	$t = 0.02$	$t = 0.05$	$t = 0.08$
γ_6	-11.2	-28.3	-17.0	-14.4
γ_{20}	-36.5	-29.8	-13.7	-11.7
γ_{60}	-60.6	-29.5	-13.1	-11.2
$K(t)$	-71.9	-29.3	-12.9	-11.0

Table 21.7: Normalised boundary change $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$ against the shift response $K(t)$, $\delta = 0.3085$ (computed).

For ξ the consequence is quantitative, to first order and heuristic (Remark 21.7.3). Any zero off the line lies above height $3 \cdot 10^{12}$ [232] and has $\delta < \frac{1}{2}$, so its shift is at most $6 \cdot \frac{1}{4}/(3 \cdot 10^{12})^4 < 2 \cdot 10^{-50}$. The computed response satisfies $|K| \le 215$ on $[0.0045, 0.12]$ (its computed maximum is 195.5 at $t = 0.0045$). For one off line quartet the boundary of μ on the range carrying all but $2 \cdot 10^{-9}$ of the

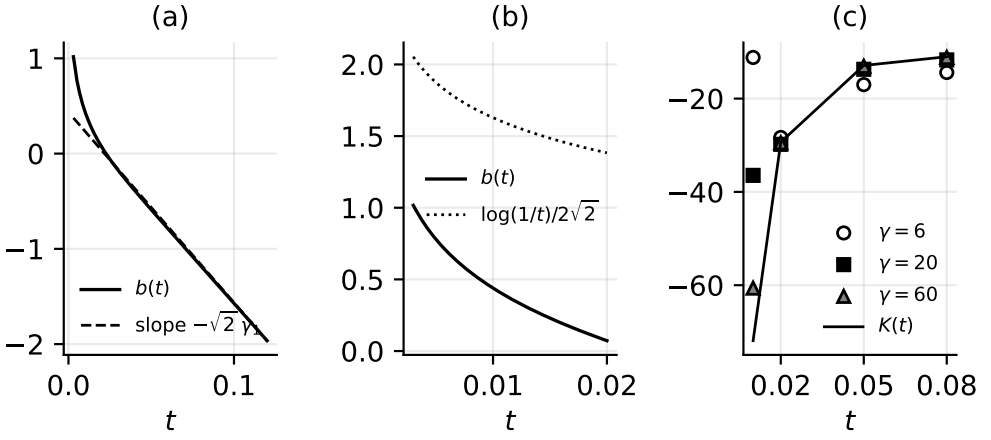


Figure 21.1: (a) The inverse first passage boundary b of the reciprocal Xi law on $[0.003, 0.12]$, with the line of slope $-\sqrt{2}\,\gamma_1$ through its last node. (b) The boundary at small times against the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 21.7.1, which it approaches only slowly. (c) Normalised boundary changes for off line pairs at the zeros 6, 20, 60, against the shift response K (computed).

mass would therefore change, to first order, by less than 10^{-47}. RH failing allows every zero above $3 \cdot 10^{12}$ to be off the line, and then the first order shifts add up to $\frac{3}{2}\sum_{\gamma>T}\gamma^{-4} \approx 8 \cdot 10^{-38}$ at most, so the change is less than 10^{-34} (the computed value of the product with 215 is $1.7 \cdot 10^{-35}$; here $\sum_{\gamma>T}\gamma^{-4} \ll T^{-3}\log T$). The change is larger only at times of order $\log\gamma/\gamma$, where the saddle point s reaches γ^2; by Proposition 21.7.1 the mass there is about $\exp(-\frac{1}{4}\gamma\log\gamma)$.

21.7.5 Assessment

The Roberts and Shortland view is a faithful and cheap encoding of μ: the boundary is computable to about $2 \cdot 10^{-4}$ on the range that carries the mass, with an explicit enclosure at small times. But its ends are fixed by the growth of ξ and by the first zero, and a violation of RH would move it in between, to first order (heuristic, Remark 21.7.3), by about $6\delta^2\gamma^{-4}K(t)$, the same first order shift as in any other representation of μ. A shape test could see RH only if it detected the GGC property itself, through a characterisation of the boundaries whose crossing laws are GGCs. Kent's theorem [181] makes the hitting times of levels by one dimensional diffusions GGCs, but we do not know which curved boundaries for Brownian motion reduce to that case, and we record the question as Problem 21.5. Without such a theorem the method is a computational tool, useful for the prime side laws of Theorem 21.3.4 and the Davenport Heilbronn comparison, with no

positivity of its own.

21.8 Marc Yor's probabilistic Riemann function

Four papers of Marc Yor and his coauthors bear on μ: Biane, Pitman and Yor on Brownian laws whose Mellin transform is ξ [26]; Pitman and Yor on the hyperbolic laws [230]; James, Roynette and Yor on GGCs, Thorin measures and Dirichlet means [178]; and Nikeghbali and Yor on the Barnes G function [218]. Section 18.1 and (12.1.1) already use the first; this section adds the harmonic mean form, the Davenport Heilbronn counterpart and the Wiener and Gamma coupling.

21.8.1 The Biane, Pitman and Yor variable

Let b be a standard Brownian bridge and $Y = \sqrt{2/\pi}\,(\max_u b_u - \min_u b_u)$. Biane, Pitman and Yor prove [26, (4), (5)]

$$\mathbb{E}\, Y^s = 2\xi(s) \qquad (s \in \mathbb{C}).$$

The same law appears in several Brownian guises: $Y^2 \stackrel{d}{=} \frac{\pi}{2}\Sigma_2$ with $\Sigma_2 = \frac{2}{\pi^2}\sum_{n\geq 1}(\varepsilon_n + \hat{\varepsilon}_n)/n^2$, whose Laplace transform is $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$; Σ_2 is the sum of two independent hitting times of 1 by a three dimensional Bessel process; and $\frac{\pi}{2}\sqrt{\Sigma_2}$ is the maximum of a standard Brownian excursion [26, (57)]. The functional equation $\xi(s) = \xi(1-s)$ becomes the size bias identity $\mathbb{E}g(1/Y) = \mathbb{E}\, Y g(Y)$ [26, (20)]. The reciprocal of ξ follows at once.

Proposition 21.8.1. *Let $\mathbb{P}^*$ have density $Y^{1/2}/\mathbb{E}Y^{1/2}$ with respect to $\mathbb{P}$, and let $U = \log Y$. Under $\mathbb{P}^*$ the variable U is symmetric and has the law of the Pólya variable of* (2.1.4)*, with density $\Phi/\Xi(0)$, where $\Xi(z) = \xi(\frac{1}{2} + iz)$ and Φ is the kernel of Riemann and Pólya of* (2.1.3)*; equivalently $U/2$ has density $2\Phi(2u)/\Xi(0)$. Moreover*

$$\mathbb{E}\, e^{-s\tau} = \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{s})} = \frac{1}{\mathbb{E}^* \cosh(\sqrt{s}\, U)} = \frac{1}{\mathbb{E}^*\big[\mathbb{E}[e^{-sT} \,|\, U]^{-1}\big]},$$

where, given U, T is the exit time of a standard Brownian motion from $(-|U|/\sqrt{2}, |U|/\sqrt{2})$. Status: proved.

Proof. From $\mathbb{E}Y^s = 2\xi(s)$, $\mathbb{E}^* e^{wU} = \mathbb{E}\, Y^{1/2+w}/\mathbb{E}\, Y^{1/2} = \xi(\frac{1}{2} + w)/\xi(\frac{1}{2})$. Taking $g(y) = y^{-1/2} f(\log y)$ in the size bias identity gives $\mathbb{E}\, Y^{1/2} f(-\log Y) = \mathbb{E}\, Y^{1/2} f(\log Y)$, so U is symmetric under $\mathbb{P}^*$ and $\mathbb{E}^* e^{wU} = \mathbb{E}^* \cosh(wU)$. Putting $w = iz$ gives $\mathbb{E}^* e^{izU} = \Xi(z)/\Xi(0) = \int_{\mathbb{R}} \Phi(u) e^{izu} \mathrm{d}u/\Xi(0)$, which identifies the density of U (this is (2.1.4)). Finally $\mathbb{E}e^{-sT} = 1/\cosh(a\sqrt{2s})$ for exit from $(-a, a)$, and $a = |U|/\sqrt{2}$. □

So the harmonic mean route of Section 5.6 has an explicit Brownian mixing variable: the logarithm of the normalised range of a Brownian bridge, size biased by its square root. Each conditional exit time is a GGC, and μ is one exactly when the harmonic mean over U preserves that property, which is RH. The problem is restated in Brownian language, not changed.

REMARK 21.8.2. Let M be the maximum of a standard Brownian excursion. Its law is explicit [42, 107]:

$$\mathbb{P}(M \le m) = \sum_{n\in\mathbb{Z}} (1 - 4n^2m^2)e^{-2n^2m^2} \qquad (m > 0),$$

and $Y \overset{d}{=} \sqrt{2/\pi}\, M$, because the range of a Brownian bridge has the law of M, by the path transformation of Vervaat [201]. With $\psi(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ as in Section 18.1.1, and for $y > 0$,

$$\begin{aligned} \mathbb{P}(Y > y) &= 2\sum_{n\ge1} (2\pi n^2y^2 - 1)e^{-\pi n^2y^2}, \\ \mathbb{E}(Y - y)^+ &= 2y\,\psi(y^2), \\ f_Y(y) &= 2\Phi(\log y)\, y^{-3/2}, \end{aligned}$$

where f_Y is the density of Y and Φ the kernel of (2.1.3). The first is the formula above at $m = \sqrt{\pi/2}\, y$. It equals $-\frac{d}{dy}\big(2y\psi(y^2)\big)$, and both sides of the second tend to 0 as $y \to \infty$, which gives the second. The third is $-\frac{d}{dy}$ of the first, compared term by term with (2.1.3) at $u = \log y$. Hence $\mathbb{E}Y^s = 2\int_{\mathbb{R}} \Phi(u)e^{(s-\frac12)u}\,\mathrm{d}u = 2\xi(s)$ for every complex s: the identity of Biane, Pitman and Yor follows from the law of M, term by term. With $y = e^u$ and $\mathbb{E}Y = 2\xi(1) = 1$, the second line gives the folded kernel of (18.1.1) as

$$k(u) = e^{-u/2} - 2e^{u/2}\psi(e^{2u}) = e^{-u/2}\,\mathbb{E}\min(Y, e^u), \qquad u \in \mathbb{R}.$$

Its evenness is the size bias identity with $g(y) = \min(1, e^u/y)$, and the kink identity $k'(0) = 0$ of Lemma 18.1.1 is its case $g = \mathbf{1}_{(0,1)}$, since $k'(0) = \frac12\big(\mathbb{P}(Y > 1) - \mathbb{E}[Y; Y < 1]\big)$. As $\mathbb{E}[Y; Y > 1] - \mathbb{P}(Y > 1) = \mathbb{E}(Y-1)^+ = 2\psi(1)$, the kink identity reads

$$\mathbb{P}\big(M > \sqrt{\pi/2}\big) = \mathbb{P}(Y > 1) = \tfrac12 - \psi(1) = 0.456783$$

(computed). Status: proved. This remark is based on [134].

21.8.2 Davenport and Heilbronn have the same structure

The Biane, Pitman and Yor structure rests on two facts, the positivity and the self duality of the theta function behind ξ, and the Davenport Heilbronn function shares both.

Proposition 21.8.3. *Let a_n be the Davenport Heilbronn coefficients, periodic modulo 5 with $(a_0, \dots, a_4) = (0, 1, \kappa, -\kappa, -1)$ and $\kappa = 0.2840790438$, and let $\theta_{DH}(x) = \sum_{n\geq 1} a_n n\, e^{-\pi n^2 x/5}$. Then $\theta_{DH}(x) > 0$ for every $x > 0$. Consequently there is a probability law on $(0, \infty)$ whose Mellin transform is a constant multiple of ξ_{DH}, a symmetric U_{DH} with $\mathbb{E}^* e^{wU_{DH}} = \xi_{DH}(\frac{1}{2} + w)/\xi_{DH}(\frac{1}{2})$, and a positive Pólya kernel Φ_{DH}. Status: proved.*

Proof. For $x \geq 1$, using $|a_n| \leq 1$,

$$\theta_{DH}(x) \geq e^{-\pi x/5}\Big(1 - \sum_{n\geq 2} n\, e^{-\pi(n^2-1)x/5}\Big) \geq e^{-\pi x/5}(1 - 0.324) > 0.$$

The functional equation of f_{DH} is equivalent to the self duality $\theta_{DH}(1/x) = x^{3/2}\theta_{DH}(x)$, which we also checked numerically (to a relative 10^{-40} at 37 points), and it carries positivity to $0 < x < 1$. Mellin transformation gives $\int_0^\infty \theta_{DH}(x) x^{(s+1)/2}\, \mathrm{d}x/x = \sqrt{5/\pi}\, \xi_{DH}(s)$ (checked to $2 \cdot 10^{-41}$ at $s = \frac{1}{2}$ and $s = 0.8 + i$), so $\theta_{DH}(x)x^{1/2}/x$, normalised, is a probability density with Mellin transform $\xi_{DH}(s)/\xi_{DH}(0)$ in the variable $s/2$. The rest follows as in Proposition 21.8.1. □

The harmonic mean $1/\mathbb{E}^* \cosh(\sqrt{s}\, U_{DH})$ has the complex pole $s = (0.3085 + 85.699i)^2$ coming from the off line zero, so it is not a GGC (Theorem 21.4.1). So the positivity of a self dual kernel, the size bias symmetry and the harmonic mean representation all hold for a function that violates RH. What Davenport and Heilbronn may lack is the Brownian meaning: we do not know whether θ_{DH} is the theta function of any natural Brownian or Bessel functional. That is the only place in the picture where arithmetic could enter (Problem 21.7).

21.8.3 The hyperbolic laws as solvable models

Pitman and Yor [230] study the laws C_1, S_1, T_1 with Laplace transforms $1/\cosh\sqrt{2s}$, $\sqrt{2s}/\sinh\sqrt{2s}$ and $\tanh\sqrt{2s}/\sqrt{2s}$. The first is the exit time of Brownian motion from $(-1, 1)$, and

$$\frac{1}{\cosh\sqrt{2s}} = \prod_{n\geq 1} \Big(1 + \frac{2s}{\pi^2(n - \frac{1}{2})^2}\Big)^{-1}$$

is the exact model of μ under RH, with $\pi^2(n-\frac{1}{2})^2/2$ in place of γ_n^2: a GGC whose Thorin measure is a sum of unit atoms, of infinite total mass [178, Section 1.6]. The law T_1 shows how a GGC fails: its transform has zeros and poles interlacing on the negative axis, so its Thorin measure has atoms of both signs [178]. For μ an off line zero produces instead a pair of complex atoms. Both failures are visible only in the canonical function, not in positivity of the law.

21.8.4 The Wiener and Gamma representation

James, Roynette and Yor [178, Proposition 1.1] show that a positive variable is a GGC if and only if it is a Wiener and Gamma integral $\int_0^\infty h(u)\,\mathrm{d}\gamma_u$ against a standard gamma process, where the Thorin measure is the image of Lebesgue measure under $u \mapsto 1/h(u)$, and h can be taken nonincreasing and is then unique. For μ under RH this gives $h(u) = 1/\gamma^2_{\lceil u \rceil}$ and $\mu = \sum \varepsilon_n/\gamma_n^2$.

The base point laws $\mu^{(\alpha)}$ with transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ fit the same frame, but only when $\alpha \ge \sup_\rho \operatorname{Re}\rho$, which is where $\mu^{(\alpha)}$ is a GGC (Theorem 2.1.2; below that line the Hadamard product has zeros in the slit plane and no Thorin representation exists, see Proposition 14.3.3). For such α the Hadamard product gives

$$\log\frac{\xi(\alpha+\sqrt{s})}{\xi(\alpha)} = \int_0^\infty \log\Bigl(1+\frac{s}{t^2}\Bigr)\,\mathrm{d}N_\alpha(t),$$
$$N'_\alpha(t) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it) = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(t-\gamma)^2},$$

so the Thorin measure in the variable t is $\mathrm{d}N_\alpha$, a Poisson smoothing of the zero counting function at width $\alpha-\beta$. Then $\mu^{(\alpha)}$ is the Wiener and Gamma integral of $h_\alpha(u) = N_\alpha^{-1}(u)^{-2}$. The density N'_α is also defined for $\alpha < \sup_\rho \operatorname{Re}\rho$, and its positivity on every line $\operatorname{Re} s = \alpha > \frac12$ is Lagarias's criterion (Remark 2.1.4), but on a single such line positivity of the density does not make $\mu^{(\alpha)}$ a GGC.

Proposition 21.8.4. *On one gamma process γ, set $X_\alpha = \int_0^\infty h_\alpha(u)\,\mathrm{d}\gamma_u$. Then X_α has law $\mu^{(\alpha)}$ for every $\alpha \ge \sup_\rho \operatorname{Re}\rho$, in particular for every $\alpha \ge 1$. RH holds if and only if N_α is nondecreasing for every $\alpha > \frac12$, and then $N_\alpha \to N$ pointwise at continuity points and $X_\alpha \to X_{1/2} = \sum \varepsilon_n/\gamma_n^2$ in probability as $\alpha \downarrow \frac12$. Status: proved for the representation and the equivalence, the convergence under RH is heuristic (the proof below is a sketch).*

Sketch. The representation is Theorem 2.1.2 together with the Wiener and Gamma characterisation. The equivalence is Lagarias's criterion: each zero with $\beta < \alpha$ contributes a positive Poisson kernel, and a zero with $\beta > \alpha$ contributes a negative kernel of height $1/(\pi(\beta-\alpha))$ at $t=\gamma$, which dominates all other terms as $\alpha \uparrow \beta$, so N_α fails to be monotone for some α close to β. Under RH the kernels tend to unit atoms at the γ_n. The part of $\mathrm{d}N_\alpha$ on $(0,\gamma_1/2)$ has mass $O(\alpha-\frac12)$ and contributes a variable whose transform is $\exp(-O(\alpha-\frac12)\sqrt{s})$, which tends to 1; on the rest the kernels h_α converge in $L^1(\mathrm{d}u)$ to h (this needs $N_\alpha - N = O(\log t)$ on the tail, which we have not written out), and the Wiener and Gamma integrals converge in probability. □

Figure 21.2 shows, against $\alpha = 1, 0.75, 0.6, 0.55$, the minimum of N'_α on $[0,50]$ and the value $N_\alpha(50)$. The densities are positive, with minima 0.0074, 0.0037,

0.0015, 0.0007 (all at $t = 0$), and $N_\alpha(50) = 9.72,\ 9.78,\ 9.88,\ 9.94$ approaches $N(50) = 10$ (computed). The coupling gives a pathwise picture of the base point descent under RH: all the laws $\mu^{(\alpha)}$ are functionals of one gamma process, and an off line zero would appear as a point where N_α stops being monotone as α passes β. Like Proposition 21.8.1, this reformulates RH.

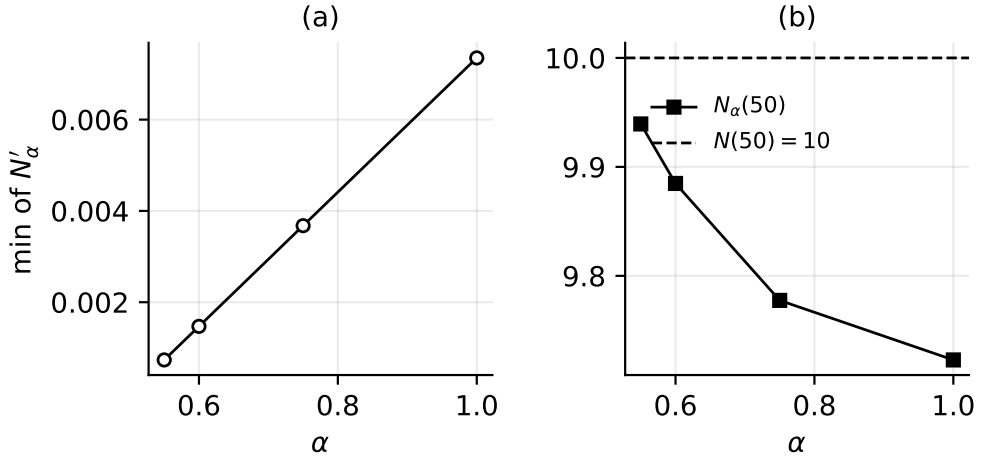


Figure 21.2: (a) The minimum over t of the Thorin density $N'_\alpha(t) = \pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ of the base point laws on $[0, 50]$, attained at $t = 0$, for $\alpha = 1, 0.75, 0.6, 0.55$. (b) The values $N_\alpha(50)$ against $N(50) = 10$. Only these summary values are stored, so the curves on $[0, 50]$ are not drawn (computed).

Dirichlet means, the other main theme of [178], need a Thorin measure of finite total mass θ, in which case the GGC is $\gamma_\theta \cdot D_\theta$ with D_θ a Dirichlet mean [178, (31) to (35)]. The Thorin measures of μ and of every $\mu^{(\alpha)}$ with $\alpha \geq \sup_\rho \operatorname{Re} \rho$ have infinite mass. For the truncation $\mu_N = \sum_{n \leq N} \varepsilon_n / \gamma_n^2$ the representation holds with $\theta = N$, but D_N tends to $\frac{1}{N} \sum_{n \leq N} \gamma_n^{-2} \to 0$ [178, Proposition 1.7], so it degenerates and carries no information about μ.

21.8.5 Moments and the Barnes function

Nikeghbali and Yor [218] express the Barnes G function, and through it the Keating and Snaith moments of characteristic polynomials of random unitary matrices, by sums and products of GGC variables (see Section 14.3.6). This links GGCs to the conjectured moments of ζ on the critical line, but moments of $|\zeta(\frac{1}{2} + it)|$ do not locate zeros, and we see no route from them to the positivity of the Thorin measure of μ.

21.8.6 Assessment

Yor's work supplies three usable things: a Brownian meaning for Pólya's kernel and hence for the harmonic mean form of μ (Proposition 21.8.1), exactly solvable models of μ and of its failure modes in the hyperbolic laws, and a pathwise coupling of the base point laws through the Wiener and Gamma representation (Proposition 21.8.4). It does not supply the arithmetic. Proposition 21.8.3 shows that the positivity and symmetry behind the Biane, Pitman and Yor variable hold for Davenport and Heilbronn too, so they cannot imply RH; only the Brownian interpretation of θ might separate the two.

21.9 The Davenport Heilbronn test

Every proposed argument must fail for the Davenport Heilbronn function f_{DH} [60], which has the functional equation $\xi_{DH}(s) = \xi_{DH}(1-s)$ with $\xi_{DH}(s) = (5/\pi)^{s/2}\Gamma(\frac{s+1}{2})f_{DH}(s)$, no Euler product, and zeros off the line; the zero at $0.808517182456637+85.6993484853776\,i$ is classical [261] (see Section 18.14 for the function and its zeros). We confirmed it by the argument principle (one zero in the box $[0.78, 0.84] \times [85.67, 85.73]$, at two step sizes). Its reciprocal law passes every positivity test we can compute: the Taylor coefficients of $\xi_{DH}(\frac{1}{2}+\sqrt{s})/\xi_{DH}(\frac{1}{2})$ have all contiguous Toeplitz minors positive up to order 10 over the first 100 indices (computed; the check ran to order 14 and index 119 at two precisions, which agree to 80 digits, not an interval proof), indistinguishable from those of ξ, as the angle theorem predicts (Section 21.11). For hitting times the test rules out the de Bruijn and Newman route and the harmonic mean route unless the Euler product enters them, and it rules out any argument that uses only an archimedean factor of this growth and the functional equation.

21.10 The coefficient ladder

The density of the reciprocal Xi law has a Pólya frequency ladder (Section 21.1 and Chapter 22), and so does the canonical function Θ. A third ladder is that of the Taylor coefficients of $F(s) = \xi(\frac{1}{2}+\sqrt{s})/\xi(\frac{1}{2})$; Section 21.11 shows what the three can and cannot see. Both sections are based on [152].

Riemann's integral gives, with the de Bruijn Newman kernel Φ_{dBN} of Section 4.1 (so that $\xi(\frac{1}{2}+w) = 8\int_0^\infty \Phi_{\mathrm{dBN}}(u)\cosh(2wu)\,\mathrm{d}u$),

$$F(s) = \sum_{n\ge 0} c_n s^n, \qquad c_n = \frac{\mathbb{E}V^{2n}}{(2n)!} = \frac{8}{\xi(\frac{1}{2})}\int_0^\infty \Phi_{\mathrm{dBN}}(u)\frac{(2u)^{2n}}{(2n)!}\,\mathrm{d}u > 0,$$

where V is symmetric with the Pólya density $\Phi/\Xi(0)$ of (2.1.3), so that $V/2$ has density proportional to Φ_{dBN}. Then $c_0 = 1$ and, computed at 110 digits from the first 120 zeros and Riemann's integral, $c_1 = \sum_{\mathrm{Im}\,\rho>0}(-s_\rho)^{-1} = 0.023104993$

(with $s_\rho = (\rho - \frac12)^2$), $c_2 = 2.4833405 \times 10^{-4}$ and $c_3 = 1.6743526 \times 10^{-6}$ (status: computed). As a check, $\sum_n c_n = F(1) = \xi(\frac32)/\xi(\frac12) = 1.02335500958$, which the computed series reproduces to the printed digits.

THEOREM 21.10.1 (Coefficient ladder). *(i) RH holds if and only if (c_n) is a Pólya frequency sequence, that is, every minor of the Toeplitz matrix (c_{i-j}) is nonnegative.*

(ii) Unconditionally, (c_n) is PF$_2$: $c_n^2 \geq c_{n-1}c_{n+1}$ for all $n \geq 1$.

Proof. (i) By the theorem of Aissen, Schoenberg and Whitney and of Edrei [7, 70], a sequence with $c_0 = 1$ is PF$_\infty$ if and only if its generating function is $e^{bs}\prod_i(1+\alpha_i s)/\prod_i(1-\beta_i s)$ with $b, \alpha_i, \beta_i \geq 0$ and $\sum_i(\alpha_i+\beta_i) < \infty$. Since F is entire of order one half, a product over its zeros $(\rho-\frac12)^2$, this says exactly that these are real and negative, which is RH.

(ii) Write $b_n = (2n)!\,c_n$. Csordas, Norfolk and Varga proved the Turán inequalities $b_n^2 > \frac{2n-1}{2n+1} b_{n-1}b_{n+1}$ [49], which is the reduction already used in Section 22.4 (the normalisations differ by the factor 4^n, which cancels). Then

$$\frac{c_n^2}{c_{n-1}c_{n+1}} = \frac{b_n^2}{b_{n-1}b_{n+1}} \cdot \frac{(2n-2)!\,(2n+2)!}{(2n)!^2} > \frac{2n-1}{2n+1} \cdot \frac{(2n+2)(2n+1)}{2n(2n-1)} = \frac{n+1}{n} > 1.$$

□

Part (ii) restates the Turán paragraph of Section 22.4 in Toeplitz form; part (i) is the statement not made elsewhere in the book, and puts the coefficients on the footing of the other two objects. The three ladders are

$$\Theta \text{ completely monotone} \iff f \text{ PF}_\infty \iff (c_n) \text{ PF}_\infty \iff \text{RH},$$

with unconditional low rungs for each: $T_k \geq 0$ for $k \leq H^2/80$ for Θ [233, Section 6], in another form Theorem 5.8.1; log concavity of f on $[Y_*, \infty)$ for the density (Theorem 5.7.5, a computer assisted theorem whose enclosed inputs are listed there); and PF$_2$ of (c_n) by Theorem 21.10.1.

The consecutive Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$, normalised by c_n^k, were computed for $k = 2, \dots, 7$ and $k - 1 \leq n \leq 35 - k$ (status: computed). They are all positive and decrease with n, so the smallest sit at the largest n allowed: 4.16×10^{-2}, 1.52×10^{-4}, 7.71×10^{-8}, 7.66×10^{-12}, 1.98×10^{-16} and 1.70×10^{-21}. A different range gives different digits (for $n \leq 36 - k$ the same minima are 4.05×10^{-2}, 1.4×10^{-4}, 6.5×10^{-8}, 5.8×10^{-12}, 1.3×10^{-16} and 9.1×10^{-22}), so the range and the normalisation are part of the statement. Among 399 random minors of orders 3, 4 and 5 (133 of each), none was below -10^{-40}. The higher order Turán inequalities of Dimitrov and Lucas [63] and the hyperbolicity of Jensen

polynomials of Griffin, Ono, Rolen and Zagier [89] are rungs of a closely related ladder on the same coefficients. Of the three ladders, finite rungs of the coefficient ladder are the ones that can be proved, as Theorem 21.11.2 shows next; finite rungs of the density ladder can be certified by computation; and the ladder of Θ is the one that connects to the Thorin measure.

21.11 The angle theorem: what the ladders can and cannot see

The coefficient ladder has a multiplicative structure that the other two lack. It yields an unconditional theorem, a test against a function that violates RH, and a statement of what climbing the ladders can give.

21.11.1 The angle theorem

For a zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\gamma > 0$, let $s_\rho = (\delta + i\gamma)^2$ and let $\theta_\rho \in [0, \pi)$ be the angle between s_ρ and the negative real axis. Since $\tan \theta_\rho = 2|\delta|\gamma/(\gamma^2 - \delta^2)$,

$$\theta_\rho = 2 \arctan \frac{|\delta|}{\gamma}, \tag{21.11.1}$$

and RH is the statement that every θ_ρ vanishes.

Lemma 21.11.1 (Three term sequences). *For $r > 0$ and $0 \le \theta < \pi$, the sequence $(1,\, 2r^{-1}\cos\theta,\, r^{-2})$, the coefficient sequence of $(1 + s/re^{i\theta})(1 + s/re^{-i\theta})$, is PF_k if and only if $\theta \le \pi/(k+1)$.*

Proof. Scale s so that $r = 1$. The contiguous Toeplitz minor of order m is the tridiagonal determinant D_m with $D_m = 2\cos\theta\, D_{m-1} - D_{m-2}$, $D_0 = 1$, $D_1 = 2\cos\theta$, so $D_m = U_m(\cos\theta) = \sin((m+1)\theta)/\sin\theta$. These are positive for $m \le k$ exactly when $(k+1)\theta \le \pi$, which gives necessity. For sufficiency, every minor of a Toeplitz matrix of bandwidth two is a product of contiguous minors and of powers of the end coefficients, which are positive [105]. □

Theorem 21.11.2 (Angle theorem). *If every zero of ζ satisfies $\theta_\rho \le \pi/(k+1)$, then the Taylor coefficients (c_n) of F form a PF_k sequence.*

Proof. Since $\sum |s_\rho|^{-1} < \infty$, F is the product of the factors $1 + s/\gamma^2$ over the zeros on the line and of the real quadratic factors $(1 - s/s_\rho)(1 - s/\overline{s_\rho})$ over the pairs off the line, with multiplicity. The first have PF_∞ coefficient sequences $(1, \gamma^{-2})$; the second have PF_k sequences by Lemma 21.11.1, because $1 - s/s_\rho = 1 + s/(|s_\rho| e^{\pm i\theta_\rho})$. Multiplying generating functions convolves coefficient sequences, which multiplies their Toeplitz matrices, and by the Cauchy Binet formula a product of PF_k matrices is PF_k. The partial products converge coefficientwise, and coefficientwise limits preserve the sign of every minor. □

Corollary 21.11.3 (An unconditional rung count). *Unconditionally, (c_n) is PF_k for every $k \le \lfloor \pi H \rfloor - 1 = 9424777960768$, where $H = 3 \cdot 10^{12}$ is the height to which all zeros are known to lie on the line [232]. More generally, (c_n) is PF_K as soon as every zero with $\gamma \le (K+1)/\pi$ lies on the line.*

Proof. Any zero has $|\delta| < \frac{1}{2}$, so by (21.11.1) $\theta_\rho < 2 \arctan(1/(2\gamma)) < 1/\gamma$. Zeros up to the verified height have $\theta_\rho = 0$; zeros above it have $\theta_\rho < 1/H \le \pi/(k+1)$ for the stated k. In the second statement, a zero off the line has $\gamma > (K+1)/\pi$, so $\theta_\rho < 1/\gamma < \pi/(K+1)$. Apply Theorem 21.11.2. □

The proof uses the exact angle (21.11.1); the cruder $1/(\gamma - 1)$ would fall three short of the stated count. The case $k = 2$ is part (ii) of Theorem 21.10.1, although the Turán inequalities of Csordas, Norfolk and Varga are sharper than PF_2, and the corollary explains the numerics of Section 21.10: no computable rung of the coefficient ladder can fail. The same angle, there of the poles of the factors, drives the unconditional range of the power closure of Theorem 9.3.5 and Corollary 9.3.7.

21.11.2 The Davenport Heilbronn test

The Davenport Heilbronn function $f_{DH}(s) = \sum a_n n^{-s}$ [60], with a_n periodic modulo 5 with values $(1, \kappa, -\kappa, -1, 0)$ and $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1)$, has the completed form $\xi_{DH}(s) = (5/\pi)^{s/2} \Gamma(\frac{s+1}{2}) f_{DH}(s)$ with $\xi_{DH}(s) = \xi_{DH}(1 - s)$ (checked to 60 digits), no Euler product, and zeros off the line. It has the functional equation of ξ and an archimedean term of the same growth, $\frac{1}{2} \log \tau + O(1)$ at height τ, but not the same archimedean factor: its term is $\frac{1}{2} \log(5/\pi) + \frac{1}{2} \psi(\frac{z+1}{2})$ in place of that of ξ. The classical off line zero is $0.8085171824566374 + 85.69934848537759\, i$ [261] (computed again here by root finding at 60 digits).

We computed the first 110 Taylor coefficients of $F_{DH}(s) = \xi_{DH}(\frac{1}{2} + \sqrt{s})/\xi_{DH}(\frac{1}{2})$ by a Cauchy integral on the circle of radius 1000 with 1024 nodes at 220 digits. For ξ itself they agree with the coefficients from Riemann's integral to 3.9×10^{-141} for $n \le 109$, and a second radius (700) agrees with the first to 4.8×10^{-126} for ξ and 2.7×10^{-158} for F_{DH} (status: computed).

order k	2	3	4	5	6	10
ξ: smallest minor	1.47×10^{-2}	6.52×10^{-6}	1.32×10^{-10}	1.66×10^{-16}	1.65×10^{-23}	5.87×10^{-61}
DH: smallest minor	1.58×10^{-2}	7.98×10^{-6}	1.98×10^{-10}	3.26×10^{-16}	4.56×10^{-23}	1.25×10^{-59}

Table 21.8: Smallest contiguous Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$ for $k - 1 \le n \le 104 - k$, normalised by c_n^k, for ξ and for the Davenport Heilbronn function. A further 600 random minors of each of the orders 3, 4 and 6 were nonnegative for both functions, none below -10^{-80}.

The two rows are indistinguishable. The off line zero at height 85.7 has $\theta = 0.0072$ by (21.11.1). If it has the largest angle among the zeros of f_{DH}, Theo-

rem 21.11.2 shows that (c_n^{DH}) is PF_k for every $k \le 435$, because $\pi/\theta - 1 = 435.3$; the condition is not verified here, and zeros of f_{DH} in $\mathrm{Re}\, s > 1$ at low height, which Davenport and Heilbronn showed to exist for such functions, would have larger angles. By the theorem of Aissen, Schoenberg, Whitney and Edrei, (c_n^{DH}) is not PF_∞, since F_{DH} has a zero off the negative axis. So, under that condition, the coefficient ladder cannot tell ξ from a function that violates RH until an order of several hundred, and by Corollary 21.11.3 it cannot tell a hypothetical off line zero of ζ above height H from RH before order 9.4×10^{12}.

21.11.3 How sharp is the angle theorem?

By Theorem 21.11.2 an off line pair at angle θ cannot break the coefficient ladder below order about π/θ. To see whether it breaks near that order, we took the first 30 zeros of ξ, replaced the mth by a pair at angle θ from the negative axis with the same modulus γ_m^2, and found the first order k and index n at which a contiguous Toeplitz minor of the polynomial's coefficients is negative, at 800 bits with the zeros from interval arithmetic (status: computed).

	$m=1$	$m=3$	$m=6$	$m=11$	$m=16$
$\theta=\pi/4$ (bare threshold 4)	6	12	17	21	24
$\theta=\pi/6$ (bare threshold 6)	8	15	24	31	35
$\theta=\pi/10$ (bare threshold 10)	13	20	30	none to 40	none to 40

Table 21.9: First failing order of the coefficient ladder when the mth zero is replaced by a pair at angle θ. The bare threshold is the first failing order for the quadratic factor alone, the least k with $\theta > \pi/(k+1)$. The failing index n is 1, 3, 7, 12 and 17 for $m = 1, 3, 6, 11, 16$ (and 6 at $\theta = \pi/10$, $m = 6$), within one of m; orders above 40 were not tried.

When the pair is the lowest zero, the ladder fails only two or three orders beyond the bare threshold, so the angle theorem is nearly sharp. When the pair sits higher, the zeros below smooth it and the failing order grows: across the table it is $c\,(\pi/\theta)\sqrt{m}$ with c between about 1.2 and 1.7, and the failure appears at index $n \approx m$. This is an observation on the computed entries, not a theorem (Problem 21.8). If the scaling persists, which is heuristic, an off line zero at height γ, position $m \approx N(\gamma)$ and displacement δ first breaks the coefficient ladder at order about $(\pi\gamma/2|\delta|)\sqrt{N(\gamma)}$, of order $\gamma^{3/2}\sqrt{\log\gamma}/|\delta|$. That lies between the angle bound $\gamma/|\delta|$ and the order about $\gamma^2 \log\gamma/|\delta|$ at which the ladder of Θ would be expected to fail. For the Davenport Heilbronn zero at height 85.7, which is about the 44th zero of that function (the main term of the zero count is 43.95 at $T = 85.7$, and 44.8 with the constant $\frac{7}{8}$), the scaling predicts a first failure between about 3500 and 4900 (c from 1.2 to 1.7 times 2894) at index near 44. Minors of that order need thousands of digits, beyond our computations.

21.11.4 The density ladder compared

The same experiment on the density used the first 20 zeros and Toeplitz determinants $\det[f(x_0 + (i-j)h)]$ on a grid of base points x_0 and spacings h, with the arguments computed in exact multiprecision. Table 21.10 is printed as the source note reports it. It has not been reproduced: the note does not define the 79 base points and 5 spacings, and the book does not recompute it. The control, in which the pair is replaced by two real zeros 10^{-6} apart, passes every order to 6 in the same report.

	$m = 1$	$m = 3$	$m = 6$
$\theta = \pi/4$	1	3	5
$\theta = \pi/10$	none to 7	3	6
$\theta = \pi/40$	none to 7	none to 7	6
$\theta = \pi/100,\ \pi/400$	none to 8	none to 8	none to 8 ($m = 6, 11$)

Table 21.10: First failing order of the density ladder for the same kind of perturbation, as reported in the source note (not reproduced). Order 1 means that the density itself becomes negative.

If the table is right, at large angles the density ladder reacts much sooner than the coefficient ladder, at order 3 to 6 where the coefficient ladder needs 12 to 30 (Table 21.9). A plausible reason is that the density sees the pair as an oscillation in time, while the coefficients see it only through a smoothed generating function (heuristic). At angles of order $1/100$ and below, both ladders are blind at every order computed. Such angles are still far larger than any that an off line zero of ζ above the verified height can have, since $\theta_\rho < 1/H < 3.4 \times 10^{-13}$ there by the proof of Corollary 21.11.3. The interval arithmetic repetition of the log concavity theorem, and the open window below Y_*, are in Theorem 5.7.5 and the discussion after it.

21.11.5 Is there a path to RH?

For the coefficient ladder the angle theorem makes the answer precise: rung K is implied by the location of the zeros up to height about K/π, and every off line zero is invisible below order about $\pi\gamma/(2|\delta|)$. The ladder of Θ behaves in the same way with order about γ^2, up to a logarithm, in place of γ [233, Section 6]. The density ladder is, by the computations above if they are confirmed, equally blind at realistic angles, although it reacts sooner when the angle is large. So each ladder reformulates RH so that a finite rung is implied by a finite verification of zeros, and the evidence is that it is no stronger than one. A proof by climbing a ladder would need an argument that controls every rung at once.

The Davenport Heilbronn test sharpens this. The function f_{DH} has the func-

tional equation and an archimedean term of the growth of ξ, violates RH, and passes the low rungs. An argument that applies verbatim to f_{DH} would prove RH where it is false, so it must use a property f_{DH} lacks, evidently the Euler product. Chapter 12 shows where the Euler product gives positivity for free, to the right of 1, where its first moment converges, and Proposition 12.1.13 extends this to the compensated form below 1. Carrying the positivity into the strip is the hard step, and nothing here does it.

21.12 What a hitting time proof would need

Taken together, these results reduce a hitting time proof of RH to one of three constructions, none known.

1. A construction from the primes at basepoint $\alpha = \frac{1}{2}$, finer than central limit scaling, of GGC laws whose Thorin measures converge vaguely to $\sum \delta_{\gamma^2}$. The heuristic Remark 21.5.1 reads its positivity as a Weil type criterion. Equivalently, one would carry the arithmetic positivity of the Euler product at basepoint $\alpha > 1$ into the critical strip. Section 25.1 records what is known about the zero free half planes that this would require: the announced half plane $\operatorname{Re} s > \frac{7}{8}$ is not enough, and the step needed beyond it is itself open.
2. A positivity preserving evolution of the Krein string mass in the de Bruijn and Newman parameter or in the basepoint, sharp at $t = 0$ or $\alpha = \frac{1}{2}$.
3. A proof that the harmonic mean of Brownian exit time transforms given by Riemann's integral is a GGC.

By the Davenport Heilbronn argument of Section 21.11.5, a proof of the second or third kind that does not use the Euler product cannot be expected, which leaves the first. This is heuristic: it excludes arguments that apply verbatim to f_{DH}, not every argument using its functional equation.

21.13 Open problems

Problem 21.1. The rate in Theorem 21.3.4 is explicit, through the cumulant expansion that follows its table. Determine whether any conditioning of Y_P on N_P, or any coupling with the archimedean factor, has a limit that retains the zeros.

Problem 21.2. Compute the Krein string mass functions of the de Bruijn and Newman laws μ_t for $t \geq 0.2$ from their spectral data, and study their evolution as t decreases. Is there a monotone quantity of the string that would force $\Lambda = 0$?

Problem 21.3. Is the harmonic mean $1/\mathbb{E}\cosh(\sqrt{s}V)$ of Brownian exit time transforms a GGC for any explicit class of symmetric V other than those already known to give real zeros through Pólya's theorems?

Problem 21.4. Build Markov chains from the primes whose generators approximate a canonical system of de Branges type for ξ, and prove a diffusion limit theorem for them in the style of Roberts, Gelman and Gilks.

Problem 21.5. Characterise the boundaries b for which the Brownian crossing time τ_b is a GGC. A characterisation that is closed under vague limits of Thorin measures, checked on the computed boundary of μ, would be equivalent to RH; a necessary condition alone would only test it. If the first order response of Section 21.7 is a guide (it is heuristic, Remark 21.7.3), any such condition must be sensitive to changes of order $\delta^2\gamma^{-4}$ in the middle of the boundary, or must act at times of order $\log\gamma/\gamma$.

Problem 21.6. Prove the resolution law of Section 21.1: if F has a non real zero s_ρ with $\operatorname{Im} s_\rho = \omega$ and all other zeros at distance at least η, show that J_d has a non real zero for every $d \geq C|s_\rho|^2/\omega^2$, with C depending on η and the local zero density. Conversely, is there an explicit $D(T)$ such that RH up to height T implies that J_d is hyperbolic for every $d \leq D(T)$?

Problem 21.7. Is the Davenport Heilbronn theta function θ_{DH} the theta function of a natural Brownian or Bessel functional, in the sense in which θ is that of the range of a Brownian bridge? A property shared by all such functionals and failed by θ_{DH} would be a genuinely arithmetic input in probabilistic form.

Problem 21.8. Prove the scaling of Table 21.9: that an off line pair at angle θ and position m first breaks the coefficient ladder at order of size $(\pi/\theta)\sqrt{m}$, at index near m. A local central limit theorem for the coefficients of the product of the m factors below the pair is the natural tool.

Problem 21.9. Explain why the density ladder, if Table 21.10 is confirmed, reacts at low order to pairs at large angles, and find its scaling in θ and m. The question is how fast the advantage over the coefficient ladder disappears as $\theta \to 0$.

Problem 21.10. Enclose the constants M_j of Proposition 5.7.4 rigorously, with a bound on the tail of the integral, completing the computer proof of Theorem 5.7.5, and close the window $(0, Y_*)$ by a rigorous saddle point bound.

21.14 Expansions of the reciprocal Xi function

For reference, the expansions of $1/F(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$, of its law μ and of its canonical function, with region of validity, status and hypotheses; most are proved elsewhere and only pointed to.

Over the zeros

Hadamard product. $1/F(s) = \prod_{\operatorname{Im}\rho>0}(1 - s/s_\rho)^{-1}$, converging absolutely because $\sum |s_\rho|^{-1} < \infty$ (proved); it exhibits μ as a GGC exactly under RH

(Section 2.1).

Basepoint product and Lorentzian Thorin density. Unconditional identities, with a positive Thorin density for $\alpha \geq 1$ (proved; Chapter 12).

Residue expansion. $f(y) = \sum_\rho e^{s_\rho y}/F'(s_\rho)$, unconditional when grouped between suitable circles, with an explicit truncation bound (Theorems 5.3.2 and 5.3.1, Proposition 5.7.4; proved, the constants M_j computed).

Kent's formula. The same series read as the eigenfunction expansion of a Krein string hitting time (Proposition 5.4.1); conditional on RH and simple zeros.

Mittag Leffler series. $1/F(s) = \sum_\rho 1/(F'(s_\rho)(s - s_\rho))$ diverges, because ξ is not in Krein's class (Proposition 5.4.2; proved).

Canonical function over zeros. $\Theta(x) = \sum_{\gamma>0} e^{s_\rho x}$; completely monotone exactly under RH (Section 2.1).

Over the primes

Euler product as a Lévy measure. Valid for $\alpha > 1$, where it gives a GGC after compensation; in the strip the compensator is infinite (proved; Chapter 12 and [151]).

Truncated Euler products. A GGC at every $\alpha > 0$, but the laws escape to infinity as $P \to \infty$ (Theorem 21.3.4).

Möbius series. A signed mixture of first passage times, converging for $\operatorname{Re} z > 1$ (Proposition 5.2.1); inside the strip its convergence is tied to RH.

Explicit formula for Θ. Unconditional for every $x > 0$ (Section 2.1).

From Riemann's integral and theta functions

Cosh transform. $F(s) = \mathbb{E}\cosh(\sqrt{s}\,V)$, hence $1/F(s) \leq \mathbb{E}\operatorname{sech}(\sqrt{s}V)$, a scale mixture of Brownian exit time laws; unconditional and valid for real $s \geq 0$ (Theorem 22.6.1).

Deformations. Pólya's approximations and the de Bruijn Newman flow (Section 4.1); the deformed reciprocals are GGC transforms for $t \geq \Lambda$.

Theta function and Brownian representations. ξ as the Mellin transform of a Jacobi theta law [26], giving the reciprocal through moment and Mellin identities rather than termwise series.

Power series

Taylor series of $1/F$. $1/F(s) = \sum(-1)^n m_n(\mu)s^n/n!$, with m_n the moments of μ and radius of convergence $\gamma_1^2 = 199.79$, set by the first pole (computed: 199.79045).

Cumulant series. $\log F(s) = \sum_{n\geq 1}(-1)^{n+1}\kappa_n s^n/n!$ with $\kappa_n = (n-1)!\sum_{\mathrm{Im}\,\rho>0}(-s_\rho)^{-n}$, which is $(n-1)!\sum_{\gamma>0}\gamma^{-2n}$ under RH (proved).

Taylor coefficients of F. $F(s) = \sum c_n s^n$; PF_∞ exactly under RH and PF_k unconditionally for $k \leq \lfloor \pi H \rfloor - 1$ (Theorem 21.10.1 and Corollary 21.11.3), with the Turán inequalities [49], the higher order Turán inequalities [63] and the hyperbolicity of Jensen polynomials [89].

Asymptotic expansions

Upper tail. $f(y) = w_1 e^{-\gamma_1^2 y} + w_2 e^{-\gamma_2^2 y} + O(e^{-c_3 y})$ with $c_3 = \frac{1}{2}(\gamma_2^2 + \gamma_3^2)$ (Proposition 5.3.3; proved).

Lower tail. $\log f(y) = -\Lambda(y) - \frac{1}{2}\log(2\pi|\psi''(s_y)|) + o(1)$ as $y \to 0$, with Λ the Legendre transform of $\psi = \log F$ (Table 5.8); heuristic, since a rigorous version with derivatives is open.

Only the residue expansion in its grouped form is explicit term by term and valid throughout the critical strip without hypotheses. The prime side expansions converge only to the right of 1, or give Θ rather than the density; the Hadamard product, the Lorentzian sums and Kent's formula give positive objects only when the zeros cooperate.

Part V

Conditional results and synthesis

Chapter 22

The quasi Riemann hypothesis and the Thorin programme

This chapter describes the zero free half planes announced in October 2026, what is claimed, by whom and with what status, how far we could follow the arguments, and what such a half plane would do for the Thorin programme of the two volumes. Nothing in it presents the half plane as established. Every statement that uses it is conditional and says so, and the hypothesis is named as (Q_θ): no Dirichlet L function, ζ included, has a zero with $\operatorname{Re} s > \theta$. The claims are $\theta = \frac{7}{8}$ [227], $\theta = \frac{11}{12}$ [228] and $\theta = 0.874957\ldots$ [198].

The main results are these. Section 22.1 records the claims and their status: announced, formalised in Lean for $\frac{7}{8}$ by the authors of [227], and not refereed. Section 22.2 sets out the structure of the argument as far as we read it: a continuation criterion that turns a power saving for smoothed Möbius sums into a half plane (Proposition 22.2.1, whose short proof we check), Poisson summation in a sextic twist, Kubota's cubic theta function, and a recursion of large sieve type; the steps we did not read are marked. We record the exponent bookkeeping of Step 2 of the outline of [228] (Proposition 22.2.2). Section 22.3 states what (Q_θ) gives in the programme: positivity of the Thorin measure and the minimum at the origin at every level above θ (Corollary 22.3.1, conditional, assembled from Theorems 11.9.1, 11.11.3 and 12.1.18). Section 22.4 argues, as a reading of the manuscripts and not as a theorem, that the $2\sin^2$ lemma (2.1.9) adds no step to the argument (Proposition 22.4.1, proved, gives the sign of the canonical measure): it translates the output of a half plane into the Thorin language, and nothing flows the other way. Section 22.5 describes the step from θ to $\frac{1}{2}$ in the language of the descent in level.

Against the path to RH of Section 1.4 the picture is simple. That path stops at its first step below level one, because positivity at a level $\theta < 1$ is the half plane $\operatorname{Re} s > \theta$. If the announcement stands, the path starts at $\frac{7}{8}$ instead of 1, and the open step is the descent from $\frac{7}{8}$ to $\frac{1}{2}$, a problem of the same kind as before (Section 17.7 and the overture). The argument of [227] does not pass through the Thorin picture, and none of its mechanisms, as far as we can read them, reaches

the critical line.

22.1 The claims and their status

22.1.1 What is claimed and by whom

The repository `https://github.com/openai/math` contains, as family 003 of a catalogue of 721 manuscripts in 373 families produced with an unnamed internal model, three manuscripts on zero free regions [227, 228, 281]. We read the README, the manuscript map and the description of the Lean formalisation of family 003 in the repository, and the PDF files of the first two manuscripts (accessed 28 October 2026).

1. [227], dated 30 September 2026, 199 pages: every finite order Hecke L function over $\mathbb{Q}(\sqrt{-3})$ and every Dirichlet L function is zero free in $\operatorname{Re} s > \frac{7}{8}$, the principal pole at $s = 1$ allowed (Theorem 1.1 there). The proof has two parts: Part I proves the half plane $\operatorname{Re} s > \frac{11}{12}$ (Theorem 3.1 there), and Part II starts from it and reaches $\frac{7}{8}$.

2. [228], dated 5 October 2026, 49 pages: a separate proof of the half plane $\operatorname{Re} s > \frac{11}{12}$ for the same families (Theorem 1.1 there). The README states that this write up was edited by people for readability.

3. [281], dated 1 October 2026: there is an absolute constant $c > 0$ with $(1 - \beta)\log q \geq c$ for every real zero β of every primitive nonprincipal real Dirichlet L function of conductor $q \geq 3$. This uniform exclusion of Landau Siegel zeros also follows from either half plane.

Baiying Liu [198] (arXiv:2610.12234, submitted 8 October 2026, second version 12 October) keeps the construction of [227] and varies three of its parameters. According to the abstract and to the summary of the paper that we obtained, he gets successively the rational bound $B_r = \frac{34999}{40000} = 0.874975$, the algebraic bound $B_{\mathrm{alg}} = (1507 - 2\sqrt{921})/1653 = 0.874957069799\ldots$, and the limiting bound

$$B_{\mathrm{new}} = \frac{18\xi_{\mathrm{new}} + 7}{18\xi_{\mathrm{new}} + 9} = 0.874957019420098946\ldots, \qquad 1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0, \tag{22.1.1}$$

with $\xi_{\mathrm{new}} = 0.388583354266035\ldots$ the root in $(0.38858, 0.38859)$, and he proves that B_{new} is optimal within the system of exponent comparisons he specifies. We checked only the arithmetic: the cubic changes sign on that interval, its root agrees with the trigonometric closed form given by Liu to fifteen digits, (22.1.1) gives the printed value, and $B_{\mathrm{alg}} - B_{\mathrm{new}} = 5.04 \times 10^{-8}$ (computed). Later preprints lower the constant further: Xu and Sang [208], assuming the analytic inputs of [227], give 0.8748867, and Liu [199] claims 0.87489 for finite order Hecke L functions

over every number field; Section 23.1.3 lists them. None of these constants differs from $\frac{7}{8}$ in any way that matters to the Thorin programme, and we write $\frac{7}{8}$ for all of them.

22.1.2 Status

source	half plane	formalised	status
[227]	$\frac{7}{8}$	Lean, by the authors: ζ, all Dirichlet L, Hecke L over $\mathbb{Q}(\sqrt{-3})$	announced 6 October 2026, not refereed
[228]	$\frac{11}{12}$	not listed in the Lean catalogue	announced, human edited, not refereed
[198]	$0.874957\ldots$	Lean files in the author's repository (known to us only from a summary, not built by us)	arXiv preprint, not refereed
[208], [199]	0.8748867, 0.87489	not checked	arXiv preprints, not refereed

The Lean statement for ζ in the repository's comparator challenge is

$$\texttt{riemannZeta_ne_zero_of_seven_eighths_lt_re}\colon \quad \text{for } s \in \mathbb{C} \text{ with } \tfrac{7}{8} < \operatorname{Re} s, \quad \zeta(s) \neq 0,$$

with the permitted axioms `propext`, `Quot.sound` and `Classical.choice`; the Dirichlet statement has the same form with the pole at $s = 1$ of the principal character excluded. The formalisation, according to the repository's description, covers the half plane and the Landau Siegel statement and not the later applications of the paper. Section 25.1.2 reports that the formal proof was rebuilt for this book and accepted by the Lean kernel, and that the comparator confirms that the theorems proved are the published challenge statements. It is not a check of the 199 page manuscript, and it is not refereeing.

A report in Quanta Magazine of 19 October 2026 [282] says that specialists concentrated on the $\frac{11}{12}$ manuscript, judged the $\frac{7}{8}$ manuscript hard to audit, quotes Radziwiłł as finding the argument a surprising synthesis of known ideas and Sarnak as declining to regard it as settled until it has been checked in detail, and records the general expectation that the methods do not reach the critical line. We use this only as a record of the reception, not as evidence for or against the claims.

For the book the status is fixed by the conventions of Section 1.3: the half plane is a hypothesis, named each time.

22.2 The structure of the argument

We read Sections 1 to 3 of [228] in full, Sections 1, 2 and 7.1 of [227], the statement of Proposition 5.1 there, and the parts of Section 10 of [227] that locate the

principal residue. The rest of both manuscripts, about 150 pages of [227] and 35 of [228], we did not read, and nothing below depends on its correctness except where we say so.

22.2.1 From Möbius sums to a half plane

The last step of both manuscripts is classical in kind, a smoothed form of Littlewood's equivalence between bounds for the Mertens function and zero free half planes. In [227, Section 2] it is stated for a variable boundary. Let $F = \mathbb{Q}(\sqrt{-3})$, let β_* be the supremum of $\frac{1}{2}$ and the real parts of the zeros of all primitive finite order Hecke L functions over F in $\frac{1}{2} \le \operatorname{Re} s \le 1$, and for a finite set S of primes let L_F^S be L_F with the Euler factors at S removed.

Proposition 22.2.1 (continuation from a common signal, [227, Proposition 2.1]). *Fix $\sigma_0 \in (\frac{1}{2}, 1)$, suppose $\Delta_0 = \beta_* - \sigma_0 > 0$, let $C(s) = s + c$ with c real, and let $0 < \omega < \Delta_0$ and $\sigma > 0$ be independent of the character. Suppose that for every primitive η there are a finite set S, a function H_η holomorphic on $\operatorname{Re} s > \sigma_0$ with $\sup_{\operatorname{Re} s > \sigma_0} |H_\eta - 1| \le \frac{1}{2}$, and a function J_η with*

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega},$$
$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$
$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds.$$

Then these assumptions contradict $\beta_* > \sigma_0$.

We read the proof in [227] and it is complete as written: the two bounds give $|f_\eta(Z)| \ll Z^{C(\beta_*)-\epsilon_*}$ with $\epsilon_* = \min(\Delta_0 - \omega, \sigma)$, moving the contour to the right gives rapid decay as $Z \downarrow 0$, so the Mellin transform of f_η against $Z^{-C(s)}$ is holomorphic on $\operatorname{Re} s > \beta_* - \epsilon_*$, Fourier inversion on $\operatorname{Re} s = 2$ identifies it with $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s,\eta)$, and since $|H_\eta| \ge \frac{1}{2}$ this continues $1/L_F^S$ across a zero with real part in $(\beta_* - \epsilon_*, \beta_*]$, which exists by the definition of β_*. The uniformity of ϵ_* in η is what makes the supremum β_* usable; the Gaussian $e^{(s-5/6)^2}$ is the device of Section 23.1.3, where its zeta case M_{g} is computed. The two parts of [227] apply the proposition with

$$\sigma_0 = \tfrac{11}{12},\ C(s) = s - \tfrac{2}{3},\ C(\sigma_0) = \tfrac{1}{4}, \qquad \text{and} \qquad \sigma_0 = \tfrac{7}{8},\ C(s) = s - \tfrac{11}{16},\ C(\sigma_0) = \tfrac{3}{16}$$

(exact arithmetic). The passage from Hecke L functions over F to Dirichlet L functions is quadratic base change: away from finitely many Euler factors, $L_F(s, \chi \circ N_{F/\mathbb{Q}}) = L(s,\chi)L(s,\chi\chi_{-3})$, and the possible cancellation at $s = 1$ is excluded by $L(1,\chi_{-3}) \ne 0$ [228, Section 3]. This step we checked.

22.2.2 The outline of the $\frac{11}{12}$ proof

The outline of [228, Section 2] has five steps; we follow it with the notation there.

Step 1. For a finite order Hecke character ν of F and a smooth cutoff W, put $A_1(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\nu(\mathfrak{n})W(N\mathfrak{n}/D)$. A bound $A_1(D) \ll D^{\theta+\varepsilon}$ for every W excludes zeros of $L_F(s,\nu)$ in $\operatorname{Re} s > \theta$: choose $W(y) = y^{-\varrho}\phi(y)$ with $\phi \geq 0$ supported in $(1,2)$, so that the Mellin transform of A_1 equals $\widehat{W}(s)/L_F^S(s,\nu)$ and is holomorphic on $\operatorname{Re} s > \theta$, while $\widehat{W}(\varrho) = \int \phi(y)\,dy/y > 0$, which contradicts $L_F(\varrho,\nu) = 0$ [228, Section 3]. This step we checked.

Step 2. The sum is embedded in the family $A_u(D) = \sum \mu(n)\nu(n)\chi_n(u)W(Nn/D)$, $u \in \mathbb{Z}[\omega]$, with $\chi_n(u) = (u/n)_6$ the sextic residue symbol, and the key estimate is the mean square

$$\sum_{0<Nu\leq H} |A_u(D)|^2 \ll D^{1+\varepsilon}H, \qquad H = D^{1+\vartheta},\ 0 < \vartheta \leq \tfrac{1}{10}, \tag{22.2.1}$$

square root cancellation on average over u [228, (2.3) and Proposition 3.1]. Since $\chi_n(p^6) = 1$ when $p \nmid n$, the rows $u = p^6$ with $Np \approx Y = H^{1/6}$ all equal $A_1(D)$ up to $O(D/Y)$, and there are about $Y/\log Y$ of them. This gives the bound (22.2.3) below and the exponent $\frac{11}{12}$.

Step 3, Poisson summation. The square in (22.2.1) is expanded and Poisson summation is applied in the row variable u. The Fourier transforms of the sextic characters are sextic Gauss sums, and an identity going back to Hasse and Heath-Brown, $\mu(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\alpha(n)\gamma_2(n)$ with $\alpha(n) = n/|n|$ and $G(n)$ a fixed ray class factor, absorbs the Möbius function into normalised cubic Gauss sums $\gamma_2(n)$. The dual problem is a mean square over a new row variable h of sums $B_h(X) = \sum \alpha(n)\gamma_2(n)\xi(n)\chi_n(h)U(Nn/X)$, required to be $\ll D^{2+\varepsilon}$.

Step 4, the cubic theta function. Patterson's formula identifies $\gamma_2(n)$, up to explicit factors, with the Fourier coefficients of Kubota's cubic theta function on hyperbolic three space [285, 284], in the normalisation of Dunn and Radziwiłł [283]. Completing the sum by the indices nb^3 makes the transformation law of the theta function applicable. The transformed sums are twisted by χ_h^3, which is quadratic, and the quadratic large sieve over number fields of Goldmakher and Louvel [286] then bounds their mean square by $(HX)^{\varepsilon}(H + H^2/X)$, without the extra term $(ML)^{2/3}$ of the general large sieve for characters of higher order. That the twist becomes quadratic is the point at which the argument gains.

Step 5, removing the cube factors. The completed sum is undone by Möbius inversion in b. Small cube divisors are handled by Cauchy and Schwarz; large ones by two further Poisson summations, which produce a mean square of the same type with both scales divided by Nb^3 and their ratio unchanged. The resulting recursion contracts the row range by a fixed power of D at each step and terminates after $O_\vartheta(1)$ steps.

The archimedean side of Step 4 is a Mellin Barnes integral, although neither

manuscript uses the word: the transformed weight is

$$V_*^\sharp(x)=\frac{1}{2\pi i}\int_{(0)}\widehat{V}_*(-t)\,\frac{\Gamma(\frac{7}{6}+t)\Gamma(\frac{5}{6}+t)}{\Gamma(\frac{7}{6}-t)\Gamma(\frac{5}{6}-t)}\Big(\frac{(2\pi)^4x}{27}\Big)^{-t}dt$$

[228, (6.6)], a quotient of gamma factors coming from the Mellin transform of the Bessel function $K_{1/3}$ in the cusp expansions of the theta function; [227, Proposition 5.1] has the same kernel $\prod_\pm\Gamma(1+t\pm\frac{1}{6})/\Gamma(1-t\pm\frac{1}{6})$. In Liu's account [198], as we have it, the chain is the same: the continuation criterion, the cubic theta average with its two representations (the theta functional equation giving a direct bound, Poisson summation giving a triple contour integral in (s,w,z) whose residue at $w=1$, $z=\frac{1}{6}$ carries $1/L_F^S(s,\eta)$), a zero detector, and moment estimates.

22.2.3 The passage to $\frac{7}{8}$

Part II of [227] keeps the completed cubic theta sum and modifies it by subtracting selected prime factors, which cancel an unwanted Euler factor, with unequal averaging scales. The principal signal then has $C(s)=s-\frac{11}{16}$. A zero detector produces, for each surviving row, a large inverse polynomial $\sum\mu(\mathfrak{a})\psi(\mathfrak{a})W(N\mathfrak{a}/D)$ and a large plain polynomial with the same character and height, and two separate inductions, a second moment for inverse polynomials with prime factors and a mean square for products of plain polynomials, bound the number of such rows [227, overview, pp. 5 to 7]. In Liu's parametrisation, with $b=l_y-l_x$ the difference of the two length exponents and ℓ the total exponent of the selected primes, $l_x=\frac{1}{2}(1-b-\ell)$, $l_y=\frac{1}{2}(1+b-\ell)$, and matching the direct and principal exponents gives, as reported in a summary of [198] that we have not checked against the paper, the boundary

$$B(\ell)=\frac{11}{12}-\frac{\ell}{4},\tag{22.2.2}$$

so that $\ell=\frac{1}{6}$ gives $\frac{7}{8}$; the moment estimates limit how large ℓ may be, and the moment parameter κ must satisfy $B\le\frac{1}{2}(1+\kappa)$.

22.2.4 What the bookkeeping allows

The exponent of Step 2 is a balance of two terms.

Proposition 22.2.2 (the balance of Step 2). *Suppose that for some $h>0$ the mean square* (22.2.1) *holds with $H=D^h$, and let $Y=H^{1/6}$. Then*

$$|A_1(D)|^2\ll D^{1+\varepsilon}\frac{H}{Y}+\frac{D^2}{Y^2},\qquad A_1(D)\ll D^{e(h)+\varepsilon},\quad e(h)=\max\Big(\frac{1}{2}+\frac{5h}{12},\,1-\frac{h}{6}\Big).\tag{22.2.3}$$

For every $h>0$, $e(h)\ge\frac{6}{7}$, with equality only at $h=\frac{6}{7}$; $e(1)=\frac{11}{12}$ and $e(\frac{11}{10})=\frac{23}{24}$. So, with the trivial bound $O(D/Y)$ for $A_1(D)-A_{p^6}(D)$, no mean square bound of

the form (22.2.1)*, fed into this step with rows of the form* p^6*, gives a half plane to the left of* $\operatorname{Re} s = \frac{6}{7}$*.*

Proof. For the $J \asymp Y/\log Y$ primes with $Y/2 < Np \le Y$ outside S, $|A_1(D) - A_{p^6}(D)| \ll D/Y$, so $|A_1(D)|^2 \le \frac{2}{J}\sum_p |A_{p^6}(D)|^2 + O(D^2/Y^2)$, and the rows p^6 are distinct and of norm at most H; the mean square gives the first term with an extra $\log Y \ll D^{\varepsilon}$. With $H = D^h$ the two terms are $D^{1+5h/6+\varepsilon}$ and $D^{2-h/3}$, whose square roots have the exponents in $e(h)$. The first exponent increases and the second decreases in h; they are equal when $\frac{7h}{12} = \frac{1}{2}$, that is $h = \frac{6}{7}$, where both equal $\frac{6}{7}$. The values at $h = 1$ and $h = \frac{11}{10}$ are exact arithmetic. □

The argument of [228] needs $h = 1 + \vartheta > 1$, which is where the large sieve and the recursion are proved, and lets $\vartheta \downarrow 0$; this is the exponent $\frac{11}{12}$. Bounding the difference of the rows by the same recursion instead removes this floor for every $h > 0$; the constraint that remains is the range $h > 1$ in which (22.2.1) is proved. The sixth powers come from the sextic twist, and a twist of order k with rows p^k would give $e(h) = \max(\frac{1}{2} + \frac{(k-1)h}{2k}, 1 - \frac{h}{k})$, equal to $\frac{2k-1}{2k}$ at $h = 1$ (proved the same way). Liu's formula (22.2.2) says the same of Part II in another form: since $l_x = \frac{1}{2}(1 - b - \ell) \ge 0$ and $l_y = \frac{1}{2}(1 + b - \ell) \ge 0$ force $\ell \le 1$, it gives $B(\ell) \ge \frac{2}{3}$ on its face, and $B = \frac{1}{2}$ would need $\ell = \frac{5}{3}$. This is arithmetic on the formula as reported, not a theorem about the method; Liu's own optimality result, which we have not read, is the precise statement for his system. Neither observation says anything about the correctness of the manuscripts. Both say that the exponents these mechanisms produce are fixed numbers bounded away from $\frac{1}{2}$, which is the reason the specialists quoted in [282] expect the method to stop short of the line.

22.3 What a half plane gives in the Thorin programme

Write $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta + iv)$ for the density at level θ and $b_\theta = (\xi'/\xi)(\theta) = \pi u_\theta(0)$, as in Section 1.3. The two theorems that make the descent in level a single monotone question are Theorem 11.9.1 (positivity is a zero free half plane) and Theorem 12.1.18, whose part (vi) puts the minimum at the origin at every level as the Thorin form of RH; Theorem 11.11.3 proves the same equivalence level by level. Assembled, they give the following.

Corollary 22.3.1 (conditional on (Q_θ)). *Let* $\frac{1}{2} \le \theta < 1$ *and assume that* ζ *has no zero with* $\operatorname{Re} s > \theta$*. Then for every level* $\theta' > \theta$*:*

(i) $u \mapsto \xi(\theta')/\xi(\theta' + \sqrt{u})$ *is the Laplace transform of a GGC* $T_{\theta'}$ *with Thorin density* $u_{\theta'}$ *in the chart* $v = \sqrt{z}$*, and* $u_{\theta'} > 0$*; it is a sum of Cauchy densities of widths in* $[\theta' - \theta, \theta' - 1 + \theta]$*;*

(ii) $u_{\theta'}(v) > u_{\theta'}(0)$ *for every real* $v \neq 0$*, so* $T_{\theta'} = S_{\theta'} + T^{\circ}_{\theta'}$ *with* $S_{\theta'}$ *positive stable of index* $\frac{1}{2}$*,* $\mathbb{E}e^{-uS_{\theta'}} = e^{-b_{\theta'}\sqrt{u}}$*, and* $T^{\circ}_{\theta'}$ *an independent GGC with no drift;*

(iii) the mass deficit of Proposition 11.9.2 vanishes at every height: $M_{\theta'}(T) = M_1(T) + \pi^{-1}\Delta_{\mathrm{top}}(\theta', T)$, and the level mass function of Corollary 11.9.3 is continuous on $(\theta, 1]$;

(iv) the compensated Lévy form of $\log \xi(\theta' + s)/\xi(\theta')$ converges, with error $O(X^{\theta-\theta'+\delta})$ at prime cutoff X, and the $2\sin^2$ lemma converts it into $u_{\theta'}$ (Proposition 12.1.13).

At the level θ itself, (i) holds with an atom at the ordinate of each zero on the line $\operatorname{Re} s = \theta$, if there is one (Theorem 11.1.14). Conversely, (i) for every $\theta' > \theta$, or (ii) for every $\theta' \in (\theta, 1)$, implies (Q_θ).

Proof. (i) is Theorem 2.1.2 with Theorem 11.9.1; the widths are $\theta' - \beta$ with $1 - \theta \le \beta \le \theta$ by the functional equation (Proposition 2.1.3). (ii) for $\theta' \le 1$ is the implication (i)⇒(ii) of Theorem 11.11.3 applied with any level in (θ, θ'), which gives $\ge$, and Theorem 23.1.8 with $\alpha_0 = \theta$, which gives the strict inequality; for $\theta' \ge 1$ it is Proposition 12.1.12, with no hypothesis. The splitting is proved as in Proposition 12.1.12, using $\frac{1}{\pi}\int_0^\infty \log(1 + u/v^2)\,dv = \sqrt{u}$. (iii) is Corollary 11.9.3 with $N(\theta', T) = 0$. (iv) is Proposition 12.1.13 with $\Theta \le \theta$. The converse is Theorem 11.9.1, (i)⇒(ii), and Theorem 11.11.3, (ii)⇒(i), with (ii) at $\theta' = 1$ supplied by Proposition 12.1.12. □

With $\theta = \frac{7}{8}$ the corollary is Theorem 23.1.1(i) and (ii), Theorem 11.11.1 and Corollary 23.1.11, which state it with the constants of the book; its content is that one hypothesis moves every positivity statement of the programme from level one down to θ, and nothing below θ. For the claimed constants the computed values at the boundary are

θ	1	$\frac{11}{12}$	$\frac{7}{8}$	B_{alg}	B_{new}
$b_\theta = (\xi'/\xi)(\theta)$	0.02309570897	0.01924878657	0.01732482643	0.01732284397	0.01732284164
$u_\theta(0) = b_\theta/\pi$	0.007351592492	0.006127079062	0.005514663528	0.005514032492	0.005514031751
$b_\theta/(\theta - \frac{1}{2})$	0.046191418	0.046197088	0.046199537	0.04619954	0.04619954
$2(\theta - \frac{1}{2})^2$	0.5	0.34722222	0.28125	0.28118561	0.28118553

(computed with mpmath at forty digits; the value at $\frac{7}{8}$ agrees with Theorem 11.10.2(ii) and at 1 with the closed form $1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$). The ratio $b_\theta/(\theta - \frac{1}{2})$ increases to $V_\xi = 0.04620998623$ as $\theta \downarrow \frac{1}{2}$ (Theorem 12.1.18(i)); the last row is, under (Q_θ) for $\theta < 1$, the bound $\Lambda \le 2(\theta - \frac{1}{2})^2$ for the de Bruijn Newman constant from de Bruijn's theorem, weaker than $\Lambda \le 0.2$ [232] for every claimed θ.

For the Dirichlet characters χ_4 and χ_{12}, which are within the scope of [227] as stated (every Dirichlet L function), $(\mathrm{Q}_{7/8})$ gives the analogue of part (i) through Proposition 15.1.25 and its χ_{12} counterpart. Part (ii) for them is proved unconditionally only for $\alpha \ge 1$ (Theorem 12.1.18(v)), and the proofs of Theorems 11.11.3 and 23.1.8 use certified ordinates of ζ; between $\frac{7}{8}$ and 1 it is Problem 22.1 below.

22.4 Does the $2\sin^2$ lemma add anything?

The lemma (2.1.9) is an identity: integrated against any real measure L with $\int \min(x, x^2)|L|(dx) < \infty$, it turns a Lévy representation into a Thorin representation whose density is $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\, xL(dx)]$. Positivity of the Thorin density follows when xL is positive. For ξ at a basepoint α the canonical measure is never positive.

Proposition 22.4.1 (the canonical measure is signed). *For every $\alpha > \frac{1}{2}$, the continuous part of the canonical measure $x\Pi_\alpha(dx)$ of Section 12.1,*

$$\Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)dx = e^{-(\alpha-1)x}\Big(\frac{e^{-3x}}{1-e^{-2x}} - 1\Big)dx,$$

is positive on $(0, x_0)$ and negative on (x_0, ∞), where $x_0 = 0.281199574323\ldots$ is the root of $e^{-3x} + e^{-2x} = 1$. Since $x_0 < \log 2$ and the atoms of $x\Pi_\alpha$ sit at $\log n$, $n \geq 2$, the measure $x\Pi_\alpha$ is negative on $(x_0, \log 2)$ for every α, and the sufficient condition $xL \geq 0$ of the $2\sin^2$ lemma holds at no basepoint.

Proof. The factor $g(x) = e^{-3x}/(1 - e^{-2x})$ decreases strictly from $+\infty$ to 0 on $(0, \infty)$, since its numerator decreases and its denominator increases, so $g - 1$ has exactly one zero x_0, where $e^{-3x_0} = 1 - e^{-2x_0}$. At $x = \log 2$, $g = \frac{1}{8}/\frac{3}{4} = \frac{1}{6} < 1$, so $x_0 < \log 2$; the numerical value is computed and gives $g(\log 2) - 1 = -0.8333$. On $(x_0, \log 2)$ there are no prime powers, so the measure there is its continuous part, which is negative. □

So positivity is always a domination question: the pole piece against the archimedean and prime pieces, as in (12.1.7), where for $\alpha > 1$ the negative contribution is the single term $-q_{\alpha-1}(\theta)$. This is why the book proves the minimum at the origin above level one with the primes or the zeros (Propositions 12.1.11 and 12.1.12), never with the lemma alone. Below level one the prime piece and the pole piece each diverge and only their compensated difference has a meaning, and its convergence at α is the bound $\psi(y) - y = O(y^{\Theta+\delta})$ with $\Theta < \alpha$ (Proposition 12.1.13), which is the half plane itself.

Now compare the argument of [227, 228]. Its arithmetic input is the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$ and sextic residue symbols; its analytic output is the continuation of $1/L_F$ (Proposition 22.2.1). The lemma acts on $\log L$, through the von Mangoldt function, and on the shifted reciprocal $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ only through its logarithm. The two meet at one point: a half plane for $1/L$ is the same as a half plane for $\log L$, and through it the bound of Step 1 becomes the convergence statement of Corollary 22.3.1(iv). We conclude, as a matter of fact about the arguments as we read them and not as a theorem, that the lemma adds nothing to the proof: it supplies the Thorin reading of what the proof gives, and no step of Steps 1 to 5 has a counterpart in the lemma. In the other direction a statement of the form "the compensated density is positive

on every line to the right of θ" is equivalent to (Q_θ) (Corollary 23.1.10), so a proof of positivity through the lemma would be a proof of a half plane, and would have to contain an estimate of the strength of (22.2.1) in some form. Whether the smoothed Möbius signal f_η has a Thorin reading of its own is Problem 22.2.

There is one place where the lemma does interact with the half plane, without adding to its proof. Under (Q_θ) the hybrid formula of Section 12.1 for the error of the compensated sum at $\alpha > \theta$ has no growing term, so the compensated $2\sin^2$ density computed from primes up to X converges to u_α at the rate $X^{\theta-\alpha+\delta}$: a computational consequence, conditional, which is Corollary 22.3.1(iv).

22.5 From θ to $\frac{1}{2}$: the descent in level

In the language of Section 1.4 the descent is the monotone family $\theta' \mapsto u_{\theta'}$, $\theta' \in (\frac{1}{2}, 1]$, and RH is the persistence of positivity, equivalently of the minimum at the origin, from $\theta' = 1$ down to $\frac{1}{2}$. Corollary 22.3.1 says that (Q_θ) is exactly this persistence on $(\theta, 1]$. The remaining step has four readings, each a restatement and not a reduction.

As a half plane The step from θ to any $\theta_1 < \theta$ is (Q_{θ_1}). A family of half planes descending to $\frac{1}{2}$ is RH; there is no intermediate statement of positivity type between two levels, because positivity on a single line is weaker than the half plane and is not known to imply it (Remark 2.1.4).

As a deconvolution Under (Q_θ) the densities at levels above θ form a Cauchy semigroup of positive measures, $u_{\theta'+a} = P_a * u_{\theta'}$ (Theorem 8.2.1(a)), and by Theorem 8.2.2 RH is the statement that u_{θ_0}, for any $\theta_0 > \theta$, is the Cauchy smoothing at width $\theta_0 - \frac{1}{2}$ of a positive measure. The hypothesis shortens the width to be undone from $\frac{1}{2}$ to $\theta - \frac{1}{2}$, and the condition number of the discretised inverse, of order $e^{\pi w/h}$ for width w at grid step h (Section 11.1), from $e^{31.4}$ to $e^{23.6}$ at $h = 0.05$ for $\theta = \frac{7}{8}$ (computed); it is still exponential in $1/h$, and the residual of the backward flow at levels below θ remains a diagnostic, not a certificate (Section 11.8). The deconvolution from level one to $\frac{7}{8}$ that the hypothesis makes a Thorin measure has condition number $e^{7.85}$ at the same step (computed), in agreement with Section 11.8.

As a monotone ratio Theorem 12.1.15 gives $v_\alpha(\theta)/(\alpha - \frac{1}{2}) \geq v_{\alpha'}(\theta)/(\alpha' - \frac{1}{2})$ whenever the zeros near height θ that lie further than $(\alpha - \frac{1}{2})/\sqrt{2}$ from the line are well separated from θ. Under (Q_θ) every zero has $\delta = \beta - \frac{1}{2} \leq \theta - \frac{1}{2}$, so by Lemma 12.1.14(b) the hypothesis holds at every height for $\alpha \geq \frac{1}{2} + \sqrt{2}(\theta - \frac{1}{2})$: for $\theta = \frac{7}{8}$ this is $\alpha \geq 1.0303$ (computed), above level one, where the comparison is already unconditional. So the hypothesis does not extend the comparison into

the strip; the ratio argument, like the others, needs δ small relative to the level, which is RH in the limit.

As a condition on ordinates Theorem 11.11.3 reduces the minimum at the origin at every level in $[\theta, 1]$, given (Q_θ), to one inequality $W(v) \geq w^*$ on the ordinates alone, with no level in it, certified on $[13.5, 1418.9]$. The level drops out because the pairing bound and the orbit bound both scale linearly in $2\theta' - 1$. What is left, the hypothesis (Q_θ), is not touched.

In each reading the step from θ to $\frac{1}{2}$ is a zero free half plane for every $\theta_1 > \frac{1}{2}$, uniform in the height. The descent in level is therefore the right frame for the announcement, since the announcement is literally its first step, and the wrong frame for hoping that the announcement makes the rest easier: nothing in the programme turns a half plane at θ into one at a smaller θ_1. The exponents of Section 22.2 say the same from the other side.

22.6 Open problems

Problem 22.1 (the minimum at the origin for χ_4 and χ_{12} under the half plane). Assume $(\mathrm{Q}_{7/8})$ for $L(s, \chi_4)$ and $L(s, \chi_{12})$. Prove $v_\alpha(\theta) > b_\alpha$ for every $\alpha \in (\frac{7}{8}, 1)$ and every $\theta \neq 0$, in the notation of Theorem 12.1.18. The window argument of part (v) there, with the counting formula of [19], should apply above the verified heights; below them the zeros are on the line [231] and the convexity argument of part (ii) needs a margin like that of Table 12.4 at $\alpha < 1$.

Problem 22.2 (a Thorin reading of the signal). The signal of Proposition 22.2.1 is the inverse Mellin transform of $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$, and its zeta analogue M_{g} is computed in Section 23.1.3. Is there a choice of the weight for which the signal is the density of a law whose infinite divisibility, or GGC property, is equivalent to the absence of zeros to the right of a given line, as Theorem 2.1.2 is for the shifted clock? Such a reading would place the direct and the principal estimates of the proof inside the programme.

Problem 22.3 (the exponent of the cubic theta method). Proposition 22.2.2 shows that the step from the mean square to the Möbius bound with the trivial bound for the difference of the rows, cannot pass $\frac{6}{7}$. Is there a corresponding lower bound for the full method of [227], independent of the parametrisation of [198], and is it above $\frac{3}{4}$? In particular, can any argument that bounds the Möbius function through a mean square over a family of twists of fixed order k give an exponent below $\frac{2k-1}{2k}$?

Problem 22.4 (a certificate of the half plane to a height). Theorem 11.9.6 certifies positivity of $u_{7/8}$ up to height 1420 with no hypothesis; part (iii) there, $N(\frac{3}{4}, T^*) = 0$, excludes zeros with $\operatorname{Re} s > \frac{3}{4}$ up to $T^* = 1419.9$. Combine the mass deficit of Proposition 11.9.2 with the zero detector of [227] to give an independent certified

check of the absence of zeros with $\operatorname{Re} s > \frac{7}{8}$ up to a height beyond the verification of RH [232]. This would test the manuscripts, not prove them.

PROBLEM 22.5 (positivity on one line). Does positivity of u_θ on the single line $\operatorname{Re} s = \theta$, for one $\theta \in (\frac{1}{2}, 1)$, imply (Q_θ) for ζ? Remark 2.1.4 shows that it does not for general entire functions of the shape of ξ. A positive answer for ζ would make the descent a question about one line at a time.

Does this give a path to RH? No. If the announced half plane stands, the Thorin programme gains a starting level of $\frac{7}{8}$ in place of 1, with every positivity statement of the book moved down accordingly (Corollary 22.3.1), and nothing below $\frac{7}{8}$. The argument of [227] works with smoothed Möbius sums, sextic twists and the cubic theta function, not with Thorin measures; the $2\sin^2$ lemma only translates its conclusion. The step from $\frac{7}{8}$ to $\frac{1}{2}$ is, in every form the programme gives it, a family of zero free half planes uniform in the height, and the exponents produced by the cubic theta mechanism, as far as we can read them, are fixed numbers bounded away from $\frac{1}{2}$.

Chapter 23

Conditional certificates under a quasi Riemann hypothesis at 7/8

23.1 Consequences of a quasi Riemann hypothesis at 7/8

23.1.1 The claim and its status

Every statement of this section is conditional on the unrefereed half plane

$$(\mathrm{Q}_{7/8}) \quad \zeta(s) \neq 0 \text{ for } \operatorname{Re} s > \tfrac{7}{8}, \text{ so that } \sigma_{\max} \leq \tfrac{7}{8} \text{ and every zero lies in } \tfrac{1}{8} \leq \beta \leq \tfrac{7}{8},$$

unless it says otherwise where it is made. The exceptions are Proposition 23.1.3, Lemma 23.1.6, the direct floating point computations of D_α, and Theorem 23.1.8, Corollary 23.1.10 and the half line result after it, which assume at most a quasi Riemann hypothesis at a general α_0. The half plane was announced on 6 October 2026 by OpenAI [227] (manuscript dated 30 September 2026, with a formal proof in Lean) for every Dirichlet L function, ζ included; a companion manuscript dated 5 October 2026 [228] proves $\operatorname{Re} s > \frac{11}{12}$ separately. Neither is refereed; Section 25.1.2 describes the work. Write $\Delta = \sigma_{\max} - \frac{1}{2} \leq \frac{3}{8}$ for the half width of the strip containing the zeros of Ξ.

23.1.2 What moves from 1 to 7/8

Theorem 23.1.1 (Consequences of $\mathrm{Q}_{7/8}$). *Assume* $(\mathrm{Q}_{7/8})$.

(i) Basepoint laws. For every $\alpha \geq \frac{7}{8}$ the law $\mu^{(\alpha)}$ with Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{s})$ is a GGC. For $\alpha > \frac{7}{8}$ its Thorin measure is absolutely continuous, with density $\pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau)$ in the ordinate variable, a sum of Lorentzians of widths at least $\alpha - \frac{7}{8}$; it has a stable tail $\mu^{(\alpha)}((y, \infty)) \sim b_\alpha / \sqrt{\pi y}$.

(ii) *The clock and Lagarias positivity. For* $\alpha > \frac{7}{8}$ *the clock* U_α *is strictly increasing and* $\operatorname{Re}\frac{\xi'}{\xi} > 0$ *on* $\operatorname{Re} s > \frac{7}{8}$. *In terms of* ζ *this is the one sided bound* $-\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+i\tau) < \operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac{1}{2}\log\frac{\tau}{2\pi} + O(\tau^{-2})$, *with* $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})$, *at every height* τ *and for every* $\alpha > \frac{7}{8}$.

(iii) *Even and odd parts. For* $\alpha \geq \frac{7}{8}$ *the functions* A_α *and* B_α *have only purely imaginary zeros, simple for* $\alpha > \frac{7}{8}$. *Proposition 22.6.3, the exact midpoint formula* (22.11.3) *and, for* $\alpha \leq 1$, *the accounting theorem with* $N(\alpha, T) = 0$ *hold for every* $\alpha > \frac{7}{8}$, *and the even part laws* μ_α^P *are GGCs there.*

(iv) *Heat flow. The de Bruijn Newman constant satisfies* $\Lambda \leq 2\Delta^2 \leq \frac{9}{32} = 0.28125$ *by de Bruijn's theorem alone.*

(v) *Densities.* $N(\sigma, T) = 0$ *for* $\sigma > \frac{7}{8}$, *so in Theorem 22.11.14 the exceptional set at a basepoint* $\alpha \leq \frac{7}{8}$ *is generated only by zeros with* $\frac{1}{2}(\alpha+\frac{1}{2}) < \beta \leq \frac{7}{8}$.

Proof. (i) By Theorems 2.1.2 and 11.1.14, $\mu^{(\alpha)}$ is a GGC when no zero has real part greater than α, and its Thorin measure is the Poisson smoothed zero density, in which a zero of real part $\beta < \alpha$ contributes a Lorentzian of width $\alpha - \beta \geq \alpha - \frac{7}{8}$; a zero with $\beta = \alpha$ gives an atom, hence absolute continuity only for $\alpha > \frac{7}{8}$. For the tail, $1 - \xi(\alpha)/\xi(\alpha+\sqrt{s}) \sim b_\alpha\sqrt{s}$ as $s \downarrow 0$, and Karamata's Tauberian theorem for the survival function, with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, gives $\mu^{(\alpha)}((y,\infty)) \sim b_\alpha/\sqrt{\pi y}$. (ii) For $\alpha > \sigma_{\max}$ every term $\frac{\alpha-\beta}{|\alpha+i\tau-\rho|^2}$ of the Hadamard sum is positive, so $\operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\tau) > 0$ and the clock, whose derivative is $\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$, runs forward. Since $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$, positivity is the stated bound, and $\operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac{1}{2}\log\frac{\tau}{2\pi} + O(\tau^{-2})$ by Stirling's formula for $\frac{\Gamma'}{\Gamma}$. (iii) For A_α this is Theorem 22.6.1(ii) with $\Delta \leq \frac{3}{8}$ and $a = \alpha - \frac{1}{2} \geq \frac{3}{8}$. For B_α it is the Hermite Biehler property of $z \mapsto \xi(\alpha - iz)$, which holds whenever $\sigma_{\max} \leq \alpha$ (Remark 22.6.4; the pairing of ρ with $1 - \bar{\rho}$ proves it). Simplicity follows from the strict monotonicity of the clock in (ii), since the zeros of $A_\alpha(it)$ are transversal crossings of half integer levels. The remaining statements are the propositions cited with $\alpha > \sigma_{\max}$. (iv) In the normalisation $H_0(z) = \frac{1}{8}\Xi(\frac{z}{2})$ the zeros of H_0 lie in $|\operatorname{Im} z| \leq 2\Delta$, and de Bruijn's theorem [61] makes H_t real rooted for $t \geq \frac{1}{2}(2\Delta)^2$. (v) is immediate. □

The bound $\Lambda \leq \frac{9}{32}$ is weaker than the bound 0.2 of Platt and Trudgian [232], which sharpens the bound 0.22 of the Polymath project [247] (Table 4.8), both obtained by the barrier method and extensive computation; a smaller a priori Δ might lower them, but only by redoing the computation. The stable tails of (i) show that the laws at basepoints between $\frac{7}{8}$ and 1 have heavy tails of nearly the weight of the law at 1; only the law at $\frac{1}{2}$ has a finite mean. At the edge the density of (i) is computable from ξ without zeros: on $0 \leq \tau \leq 60$, $\operatorname{Re}\frac{\xi'}{\xi}(\frac{7}{8}+i\tau)$ is smallest at $\tau = 0$, where it equals $\frac{\xi'}{\xi}(\frac{7}{8}) = 0.0173248$, and near each ordinate it

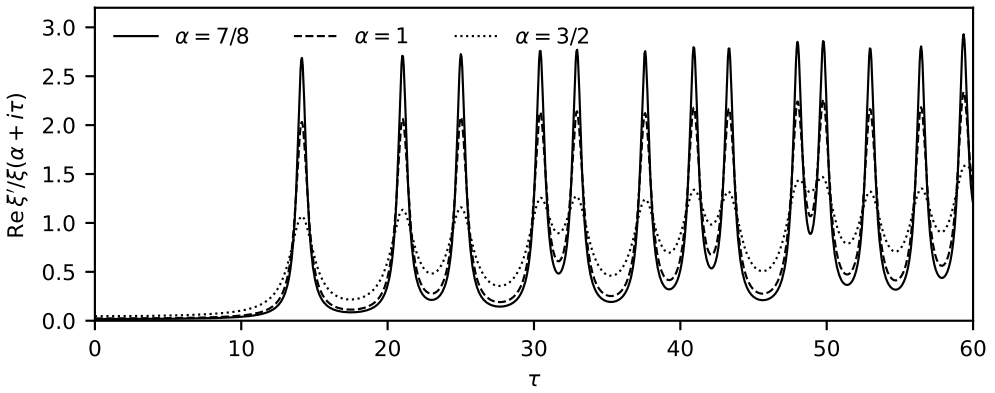


Figure 23.1: The function $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$, which is π times the Thorin density of Theorem 23.1.1(i) in the ordinate variable, at $\alpha=\frac{7}{8}$ (solid), 1 (dashed) and $\frac{3}{2}$ (dotted) for $0\le\tau\le 60$, computed from ξ with no zeros. The zeros in this range lie on the critical line, and each contributes a Lorentzian of width $\alpha-\frac{1}{2}$ and height $1/(\alpha-\frac{1}{2})$, that is $\frac{8}{3}$, 2 and 1; at $\alpha=\frac{7}{8}$ the computed peaks are 2.69 to 2.93. On this range the minimum is at $\tau=0$, where the value is $\frac{\xi'}{\xi}(\alpha)=0.0173$, 0.0231 and 0.0461. Positivity at every height for $\alpha>\frac{7}{8}$ is Theorem 23.1.1(ii), under $(\mathrm{Q}_{7/8})$, and not a consequence of the figure.

has a peak of height at least about $\frac{8}{3}$ and width $\frac{3}{4}$, narrower than the smallest gap 1.77 between ordinates there, so the Thorin measure at $\alpha=\frac{7}{8}$ already separates the zeros (Figure 23.1).

Elsewhere the half plane acts as Table 23.1 lists; every entry is conditional on [227] and proved where cited, except the heuristic count in the last row.

Proposition 23.1.2 (Poles of the reciprocal under $\mathrm{Q}_{7/8}$). *Assume* $(\mathrm{Q}_{7/8})$. *Every pole* $(\rho-\frac{1}{2})^2$ *of* $1/\xi(\frac{1}{2}+\sqrt{w})$ *lies in the closed region* $(\operatorname{Im}w)^2\le 4\Delta^2(\Delta^2-\operatorname{Re}w)\le\frac{9}{16}(\frac{9}{64}-\operatorname{Re}w)$, *bounded by a parabola with focus* 0 *and vertex* $\Delta^2\le\frac{9}{64}$, *against the vertex* $\frac{1}{4}$ *with no hypothesis. Theorem 9.3.9 holds with the sliver* $\mathcal{E}_{T_0}$ *replaced by the thinner set*

$$\mathcal{E}'_{T_0}=\Big\{x+iy:\ x<-T_0^2+\tfrac{3}{4}T_0+\tfrac{9}{64},\ y^2\le\tfrac{9}{16}\big(\tfrac{9}{32}-x\big)\Big\},$$

whose half width at the abscissa $-X^2$ *is about* $\frac{3}{4}X$, *against about* X *for* $\mathcal{E}_{T_0}$.

Proof. In the notation of the proof of Theorem 9.3.9 a zero is $\gamma=a+ib$ with $0\le b\le\Delta\le\frac{3}{8}$, and its pole $w=-\gamma^2$ has $\operatorname{Re}w=b^2-a^2$ and $(\operatorname{Im}w)^2=4a^2b^2=4b^2(b^2-\operatorname{Re}w)$, which increases in b. In that proof, $b\le\frac{3}{8}$ in place of $b<\frac{1}{2}$ gives $x\le 2b^2-(a-b)^2<-T_0^2+\frac{3}{4}T_0+\frac{9}{64}$ and $y^2\le 4b^2(2b^2-x)\le\frac{9}{16}(\frac{9}{32}-x)$ on D_γ, and the rest is unchanged. □

Chapter	What $(\mathrm{Q}_{7/8})$ gives there
12	the compensated Lévy form for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ (Proposition 12.1.13)
13	$R(x) = O(x^{7/16+\delta})$ (Theorem 13.2.7); the range $0 < y < \frac{4\pi}{3}$ of Remark 13.1.9
25	step 7 of the path, with $\alpha = \frac{7}{8}$ (Section 25.1.2)
20	$\cosh(at)\,p$ ferromagnetic for $a \geq \frac{3}{8}$; $\Omega \leq \frac{3}{8}$ (Section 19.1.2)
6, 8	$\Lambda \leq \frac{9}{32}$, weaker than 0.2 (Corollary 6.3.3)
16	the bound on ζ'/ζ behind $v_\alpha > 0$ is Theorem 23.1.1(ii) (Section 16.5)
9	equivalent to the damped form being nonnegative; $\lambda_{\min}(\frac{7}{8}, a) \geq 0$ (Theorem 9.1.2)
16	with the half plane for every Dirichlet L function, Theorem 16.6.5 for every $q \geq 3$; the heuristic count of Section 16.1.2 does not lower K

Table 23.1: Where the half plane $\operatorname{Re} s > \frac{7}{8}$ of [227], which has not been refereed, acts outside this section.

23.1.3 The Gaussian smoothed Möbius sum

The proof of [227] bounds smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$ (Section 25.1.2), and its last step is a continuation: a bound on the growth of such a sum makes the reciprocal L function analytic in a half plane. Liu [198] claims an optimisation of its exponents to the half plane $\operatorname{Re} s > 0.874957\ldots$, and in a second preprint [199], the half plane $\operatorname{Re} s > B_\flat = 0.8748866470\ldots$, with $B_\flat$ the root near 0.8749 of $133488B^4 - 376596B^3 + 384534B^2 - 167541B + 26231$, for every Hecke L function of finite order over every number field K, obtained over $K(\zeta_{12})$ and carried down to K. Xu and Sang [208], assuming the analytic inputs of [227], give $8748867/10^7 \approx 0.8748867$, which agrees with $B_\flat$ to six digits. All these claims rest on the argument of [227]; none is refereed, and the Lean proof covers $\frac{7}{8}$ only. We checked only that $B_\flat$ is a root of the quartic. Nothing below uses them: $B_\flat$ lies 1.13×10^{-4} below $\frac{7}{8}$ and would change the bound of Theorem 23.1.1(iv) from 0.28125 to $2(B_\flat - \frac{1}{2})^2 = 0.28108$. For ζ alone, with a Gaussian weight in the Mellin variable, the sum and its explicit formula can be written down and computed; nothing in the rest of this subsection depends on $(\mathrm{Q}_{7/8})$. For $x > 0$ put

$$M_{\mathrm{g}}(x) = \frac{1}{2\pi i}\int_{(2)} x^{v-2/3}\,\frac{e^{(v-5/6)^2}}{\zeta(v)}\,dv = \frac{x^{-2/3}}{2\sqrt{\pi}}\sum_{n\geq 1}\mu(n)\Big(\frac{x}{n}\Big)^{5/6}\exp\Big(-\frac{1}{4}\log^2\frac{x}{n}\Big), \tag{23.1.1}$$

where the second form follows from $1/\zeta(v) = \sum_n \mu(n)n^{-v}$ and

$$\frac{1}{2\pi i}\int_{(c)} e^{vw}e^{(v-c)^2}dv = \frac{e^{cw-w^2/4}}{2\sqrt{\pi}},$$

a Möbius sum smoothed by a Gaussian of standard deviation $\sqrt{2}$ in $\log n$ around $\log x$.

PROPOSITION 23.1.3. *Let* $\sigma_{\max} = \sup \operatorname{Re} \rho$.

(i) For every $K \geq 1$ *there are heights* $T_m \in [m, m+1]$ *such that for every* $x > 0$

$$M_{\mathrm{g}}(x) = \lim_{m\to\infty} \sum_{|\gamma|<T_m} \operatorname*{Res}_{v=\rho} \frac{x^{v-2/3} e^{(v-5/6)^2}}{\zeta(v)} + \sum_{k=1}^{K} \frac{e^{(2k+5/6)^2}}{\zeta'(-2k)}\, x^{-2k-2/3} + I_K(x),$$

with

$$I_K(x) = \frac{1}{2\pi i} \int_{(-2K-1)} x^{v-2/3} e^{(v-5/6)^2} \zeta(v)^{-1} dv = O(x^{-2K-5/3}).$$

At a simple zero the residue is $e^{(\rho-5/6)^2} x^{\rho-2/3}/\zeta'(\rho)$. *As* $K \to \infty$ *the trivial zero series diverges: its kth term has modulus*

$$\frac{2(2\pi)^{2k} e^{(2k+5/6)^2} x^{-2k-2/3}}{(2k)!\,\zeta(2k+1)}.$$

(ii) $\sigma_{\max} \leq \sigma$ *if and only if* $M_{\mathrm{g}}(x) = O(x^{\sigma-2/3+\varepsilon})$ *as* $x \to \infty$ *for every* $\varepsilon > 0$. *RH is the exponent* $-\frac{1}{6}$ *and* $(\mathrm{Q}_{7/8})$ *the exponent* $\frac{5}{24}$.

(iii) $M_{\mathrm{g}}(x) = c_1 x^{-8/3} + E(x) + O(x^{-11/3})$, *with* $c_1 = e^{(17/6)^2}/\zeta'(-2) = -1.00659 \times 10^5$ *and* E *the sum over the zeros in (i), and* $E(x)$ *is not* $O(x^{\sigma-2/3})$ *for any* $\sigma < \sigma_{\max}$. *So* $c_1 x^{-8/3}$ *is the leading term of* M_{g} *on a finite range of* x *only, not its asymptotic.*

Proof. (i) Move the line of integration from $\operatorname{Re} v = 2$ to $\operatorname{Re} v = -2K-1$ inside the rectangles $|\operatorname{Im} v| < T_m$. On the horizontal sides $|e^{(v-5/6)^2}| = e^{(\operatorname{Re} v - 5/6)^2 - T_m^2}$, and $T_m \in [m, m+1]$ can be chosen with $|\zeta(\sigma \pm iT_m)|^{-1} \leq e^{C(\log T_m)^2}$ for $-2K-1 \leq \sigma \leq 2$ ([264, Chapter 9] for $-1 \leq \sigma \leq 2$, the functional equation to the left of -1), so the horizontal integrals tend to zero. The residues at $v = -2k$ use $\zeta'(-2k) = (-1)^k (2k)!\,\zeta(2k+1)/(2(2\pi)^{2k})$; at $v = 1$ the integrand vanishes. (ii) If $M_{\mathrm{g}}(x) = O(x^{\sigma-2/3+\varepsilon})$, then $\int_0^\infty M_{\mathrm{g}}(x) x^{2/3-v}\, dx/x$ converges absolutely for $\operatorname{Re} v > \sigma + \varepsilon$, since near $x = 0$ the series in (23.1.1) gives $M_{\mathrm{g}}(x) x^{2/3} = O(x^{5/6} e^{-\log^2 x/4})$. By Mellin inversion it equals $e^{(v-5/6)^2}/\zeta(v)$ for $\operatorname{Re} v > 1$, hence for $\operatorname{Re} v > \sigma + \varepsilon$ by continuation, so ζ has no zero there. Conversely, if $\sigma_{\max} \leq \sigma$, then $1/\zeta(v) = O(|\operatorname{Im} v|^{\varepsilon})$ on $\operatorname{Re} v \geq \sigma + \varepsilon$, $|\operatorname{Im} v| \geq 1$, by the Borel and Carathéodory argument of [264, Section 14.2], which uses only the absence of zeros to the right of σ; moving the line to $\operatorname{Re} v = \sigma + \varepsilon$ gives the bound. (iii) Take $K = 1$ in (i). If $E(x) = O(x^{\sigma-2/3})$ with $\sigma < \sigma_{\max}$, then $M_{\mathrm{g}}(x) = O(x^{\sigma'-2/3})$ with $\sigma' = \max(\sigma, -2) < \sigma_{\max}$, and (ii) gives $\sigma_{\max} \leq \sigma'$, a contradiction. □

Computed three ways, from the Möbius sum over $n \le 4\times10^7$ and from the integral on $\operatorname{Re} v = 2$ and on $\operatorname{Re} v = 0.6$, between which no zero lies below height 3×10^{12} [232], far beyond the reach of the Gaussian factor, $M_{\mathrm g}(10^3) = -5.53094\times10^{-4}$; the ratio $M_{\mathrm g}(x)/(c_1x^{-8/3})$ is 0.5495, 0.9470, 0.99856 and 0.999983 at $x = 10^3$, 10^4, 10^5 and 10^6 (the Möbius sum is accurate only while its window $n \lesssim xe^8$ fits below 4×10^7). The first pair of zeros has coefficient $|e^{(\rho_1-5/6)^2}/\zeta'(\rho_1)| = 2.404\times10^{-87}$ and overtakes $c_1x^{-8/3}$ only at $x \approx 10^{36.5}$. Below that the sum is numerically its trivial zero term, and whether its exponent is $-\frac16$ or larger cannot be seen at any computable x. The continuation step does not need to see it: it reads the exponent, which the rightmost zero sets however small its coefficient. Reading an exponent is no obstruction in itself; what stops a method of this kind is the size of the exponents it can prove.

$M_{\mathrm g}$ is the Mellin counterpart of the density f of the clock T (Section 5.2.1): both invert a reciprocal, $e^{(v-5/6)^2}/\zeta(v)$ on a vertical line and $\xi(\frac12)/\xi(\frac12+\sqrt{u})$ on a Bromwich line, and both have two expansions. On the prime side each is a Möbius series with no hypothesis, (23.1.1) here and Proposition 5.2.1 there. On the zero side each is a residue series, Proposition 23.1.3(i) here and, under RH with simple zeros, $f(t) = \sum_k R_k e^{-\gamma_k^2 t}$ there. They differ in the archimedean factor. The chosen Gaussian puts the factor $e^{-\gamma^2}$ into the weight of each zero. The clock carries the gamma factor of ξ, which puts $e^{-\pi\gamma_k/4}$ into $|\xi'(\rho_k)|$, so that the residues R_k grow like $e^{\pi\gamma_k/4}$ up to powers of γ_k and the factor $1/|\zeta'(\rho_k)|$. That is why the spectral expansion of $P(T>t)$ at $t = 0.008$ is far off with 20 zeros and within 10^{-8} with 50, while $M_{\mathrm g}$ shows no zero below $x = 10^{36}$.

Proposition 23.1.4 (Mertens function under $\mathrm{Q}_{7/8}$). *Assume* ($\mathrm{Q}_{7/8}$). *Then* $M(x) = \sum_{n\le x}\mu(n) = O(x^{7/8+\varepsilon})$ *for every* $\varepsilon>0$, *and* $\sum_n \mu(n)n^{-s}$ *converges to* $1/\zeta(s)$ *on* $\operatorname{Re} s > \frac78$. *So the Möbius series of* $1/\zeta(\frac12+\sqrt{s})$ *converges on* $H_{3/8} = \{\operatorname{Re}\sqrt{s} > \frac38\} = \{u+iy:\ u > \frac{9}{64} - \frac{16}{9}y^2\}$, *which contains the region* $H_{1/2}$ *of absolute convergence of Section 10.1.*

Proof. With no zero in $\operatorname{Re} s > \frac78$, $\log\zeta$ is analytic there, and the argument of [264, Section 14.2], with $\frac78$ in place of $\frac12$, gives $1/\zeta(s) = O(|t|^{\varepsilon})$ uniformly on $\operatorname{Re} s \ge \frac78+\varepsilon$, $|t|\ge 2$. Perron's formula with the contour on $\operatorname{Re} s = \frac78+\varepsilon$, as in [264, Section 14.25], gives the bound on M. Partial summation makes the series converge locally uniformly on $\operatorname{Re} s > \frac78$, and its sum, analytic there, equals $1/\zeta$ on $\operatorname{Re} s > 1$. With $\sqrt{s} = c+id$, $s = c^2-d^2+2icd$, and $c = \frac38$ traces $x = \frac{9}{64} - \frac{16}{9}y^2$. □

This is the unsmoothed form of Proposition 23.1.3(ii), whose exponent $\frac{5}{24}$ is $\frac78 - \frac23$.

Remark 23.1.5 (Least prime in a progression). Assume, as [227] claims, that no Dirichlet L function vanishes in $\operatorname{Re} s > \frac78$, so that every nontrivial zero has $\frac18 \le \beta \le \frac78$ and $|\rho| \ge \frac18$. The explicit formula for $\psi(x,\chi)$ [54, Chapters 19 and 20] with $T = x$, and $\sum_{|\gamma|\le x}|\rho|^{-1} = O(\log^2(qx))$, give $\psi(x;q,a) = x/\varphi(q) + O(x^{7/8}\log^2(qx))$

uniformly in $(a,q)=1$, so the least prime $p \equiv a \pmod q$ is $O(q^{8+\varepsilon})$ (7.9928 with the unrefereed $B_\flat$). This is weaker than the unconditional exponent 5.2 of Xylouris [209], and it is not the Linnik constant K of Chapter 16.

23.1.4 The compensated clock condition at every height

Lemma 23.1.6 (Window count). *For $d>0$ and $r>d+e$, the number of zeros, counted with multiplicity, with ordinates in $[r-d,r+d]$ is at least*

$$\nu(r,d)=\frac{\vartheta(r+d)-\vartheta(r-d)}{\pi}-s(r+d)-s(r-d),$$

where ϑ is the Riemann Siegel theta function, $s(t)=2.5167$ for $e\le t\le X_0=30\,610\,046\,000$, and $s(t)=0.1095\log t+0.2042\log\log t+3.0305$ for $t>X_0$.

Proof. $N(t)=\frac{1}{\pi}\vartheta(t)+1+S(t)$ (Section 2.2.1), and $|S(t)|\le s(t)$ for $t\ge e$ by (1.7) and Corollary 1.4 of Hasanalizade, Shen and Wong [100], as in the proof of Proposition 12.1.11. The bound of Trudgian [268], which is often quoted, has an incomplete proof [19, 100] and is not used. □

Theorem 23.1.7 (Compensated clock under $\mathrm{Q}_{7/8}$, counting form). *Assume $(\mathrm{Q}_{7/8})$ and let $\alpha>\frac78$. Put $c_{\min}=\alpha-\frac78$, $c_{\max}=\alpha-\frac18$, $f(c,x)=c/(c^2+x^2)$ and $m(d)=\min\{f(c_{\min},d),f(c_{\max},d)\}$. If for some $d>0$ and some $T_0>d+e$ one has $\nu(r,d)\,m(d)>b_\alpha$ for all $r\ge T_0$, then*

$$D_\alpha(r)=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)-\frac{\xi'}{\xi}(\alpha)>0\qquad\text{for all } r\ge T_0.$$

For $r+d\le H=3\times10^{12}$ the conclusion also holds if $m(d)$ is replaced by $f(\alpha-\frac12,d)$.

Proof. By the Hadamard product, $D_\alpha(r)=\sum_\rho f(\alpha-\beta_\rho,r-\gamma_\rho)-b_\alpha$, summed symmetrically over all zeros $\rho=\beta_\rho+i\gamma_\rho$ counted with multiplicity. Under $(\mathrm{Q}_{7/8})$ every width $c_\rho=\alpha-\beta_\rho$ lies in $[c_{\min},c_{\max}]$, because $\frac18\le\beta_\rho\le\frac78$, so every term is positive. For $|r-\gamma_\rho|\le d$ we have $f(c_\rho,r-\gamma_\rho)\ge f(c_\rho,d)$, and $f(\cdot,d)$ increases on $(0,d)$ and decreases on (d,∞), so its minimum on an interval is attained at an endpoint and $f(c_\rho,d)\ge m(d)$. Lemma 23.1.6 counts the terms, whence $D_\alpha(r)\ge\nu(r,d)\,m(d)-b_\alpha$. For $r+d\le H$ every zero with ordinate in $[r-d,r+d]$ lies on the critical line by the verification of [232], so $c_\rho=\alpha-\frac12$ and $f(c_\rho,d)=f(\alpha-\frac12,d)$. □

Table 23.2 gives the heights. For each α and half width d put $k=\lfloor b_\alpha/m(d)\rfloor+1$; if the window holds at least k zeros then $k\,m(d)>b_\alpha$, and by Lemma 23.1.6 it does as soon as $\nu(r,d)\ge k$. The table lists, for the d that makes the last height smallest, the height T_0 beyond which $\nu(r,d)\ge k$. Under $(\mathrm{Q}_{7/8})$ the off line threshold is finite for every $\alpha>\frac78$, because the window count grows like $(\frac{d}{\pi}-0.219)\log r$ for $d>0.69$ while $m(d)$ is fixed. For α close to $\frac78$, however, it exceeds the verified height H,

and the off line range $r \geq T_0$ and the on line range $r + d \leq H$ leave the heights between H and T_0 uncovered. The two ranges overlap from $\alpha_* = 0.8888$ on: at $\alpha = 0.8888$ the count gives $T_0 = 2.30 \times 10^{12}$ with $d = 3.5$ and sixteen zeros, whereas at $\alpha = 0.8887$ it gives $3.19 \times 10^{12} > H$, and the smallest α for which the off line threshold T_0 is at most H is 0.88875. For the eight basepoints of the table the ranges overlap, and the count covers every height beyond the on line threshold. The scan in α is on a grid; we do not prove that the thresholds are monotone in α, and Theorem 23.1.8 makes this unnecessary.

α	b_α	T_0 off line	(d, k)	T_0 on line	(d, k)
0.900	0.01848	1.9×10^6	$(3, 7)$	147	$(13, 8)$
0.925	0.01963	5.2×10^4	$(3.5, 5)$	147	$(13, 8)$
0.950	0.02079	1.3×10^4	$(5, 7)$	147	$(13, 8)$
1.000	0.02310	3.5×10^3	$(6.5, 8)$	147	$(13, 8)$
1.500	0.04614	324	$(8, 5)$	147	$(13, 8)$
2.000	0.06907	231	$(10.5, 7)$	147	$(13, 8)$
5.000	0.20166	191	$(12, 8)$	167	$(12.5, 8)$
7.800	0.31322	238	$(13, 10)$	214	$(12.5, 9)$

Table 23.2: Heights beyond which Theorem 23.1.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width d of the window and the number k of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8} \leq \beta \leq \frac{7}{8}$; "on line" uses $\beta = \frac{1}{2}$, valid for $r + d \leq H$. Computed with the bound of Lemma 23.1.6.

The count loses information in one place: a zero at distance $c_{\min} = \alpha - \frac{7}{8}$ from the line $\operatorname{Re} s = \alpha$ contributes only about $c_{\min}/d^2$ to D_α, which tends to 0 with $\alpha - \frac{7}{8}$, whereas b_α stays near 0.0173; this is why the method stops at $\alpha_* = 0.8888$ rather than at $\frac{7}{8}$, and why any extension of the verified height only lowers α_* towards $\frac{7}{8}$.

Theorem 23.1.8 (Compensated clock under a quasi Riemann hypothesis). *Let $\frac{1}{2} \leq \alpha_0 < 1$ and assume that ζ has no zero with $\operatorname{Re} s > \alpha_0$. Then $D_\alpha(r) > 0$ for every α with $\alpha_0 < \alpha \leq 1$ and every real $r \neq 0$. Together with Proposition 12.1.12, which has no hypothesis, $D_\alpha(r) > 0$ for every $\alpha > \alpha_0$ and every real $r \neq 0$.*

Proof. Fix α and put $a = \alpha - \frac{1}{2}$ and $c_{\max} = \alpha + \alpha_0 - 1$, which is less than 1 because $\alpha \leq 1$ and $\alpha_0 < 1$. As D_α is even we take $r > 0$. By the Hadamard product, as in the proof of Theorem 23.1.7, $D_\alpha(r) = \sum_\rho f(\alpha - \beta_\rho, r - \gamma_\rho) - b_\alpha$ with $f(c, x) = c/(c^2 + x^2)$. By hypothesis and the functional equation, $1 - \alpha_0 \leq \beta_\rho \leq \alpha_0 < \alpha$ for every zero, so every width $c_\rho = \alpha - \beta_\rho$ lies in $(0, c_{\max}]$ and every term is positive.

Window estimate. Fix $d > 0$ and let W be the multiset of zeros with ordinate in $[r - d, r + d]$, with n elements. The map $\rho \mapsto 1 - \bar{\rho}$ preserves the ordinate and the multiplicity and fixes exactly the zeros on the critical line, so W consists of zeros

on the line, of width a, and of pairs $\{\rho, 1-\bar\rho\}$ of the same ordinate whose widths c, c' satisfy $c + c' = 2a$ however far from the line the zero lies. For $0 < c \le c_{\max}$ and $|x| \le d$ we have $c^2 + x^2 \le c_{\max}^2 + d^2$, hence $f(c,x) \ge c/(c_{\max}^2 + d^2)$; summing over W, in which the widths add up to na,

$$D_\alpha(r) \;\ge\; \frac{na}{c_{\max}^2 + d^2} - b_\alpha \;\ge\; a\Big(\frac{n}{c_{\max}^2 + d^2} - 0.04621\Big),$$

by Lemma 22.11.13. With $d = 4$ and $c_{\max} < 1$ we have $c_{\max}^2 + d^2 < 17$ and $0.04621 \cdot 17 = 0.7856 < 1$, so $n \ge 1$ suffices: if the window $[r-4, r+4]$ contains the ordinate of at least one zero, then $D_\alpha(r) > a\big(\frac{1}{17} - 0.04621\big) > 0.0126\,a$.

It remains to show, at every $r > 0$, that the window contains an ordinate or that $D_\alpha(r) > 0$ for another reason. There are three ranges.

(a) $0 < r \le \theta_0 = \gamma_1 - \alpha/\sqrt3$, where $\theta_0 \ge \gamma_1 - 1/\sqrt3 = 13.557$. Every zero has $|r - \gamma_\rho| \ge \gamma_1 - r \ge \alpha/\sqrt3 > c_\rho/\sqrt3$, because $c_\rho < \alpha$, and $x \mapsto f(c,x)$ is strictly convex where $|x| > c/\sqrt3$, since $\partial_x^2 f = 2c(3x^2 - c^2)/(c^2 + x^2)^3$. So D_α is strictly convex on $[-\theta_0, \theta_0]$ (the series of second derivatives converges uniformly there), even, and zero at the origin, hence positive for $0 < |r| \le \theta_0$; this is the argument of Proposition 12.1.11.

(b) $\theta_0 < r \le \gamma_{1700} + 4 = 2201.26$. The first 1700 ordinates, computed in ball arithmetic, have consecutive differences at most 6.8873 (the largest, between $\gamma_1 = 14.1347$ and $\gamma_2 = 21.0220$; the next is 5.4140), so the windows $[\gamma_k - 4, \gamma_k + 4]$ cover $[\gamma_1 - 4, \gamma_{1700} + 4]$, and $\gamma_1 - 4 = 10.13 < \theta_0$. Every r of the range lies in one of them, that is, its window contains an ordinate.

(c) $r \ge 2200$. By Lemma 23.1.6 with $d = 4$ the window holds at least $\nu(r,4)$ zeros, and we show $\nu(r,4) \ge 2.42$. Put $I(r) = (\vartheta(r+4) - \vartheta(r-4))/\pi$. Since $\vartheta''(t) = \frac14 \sum_{n\ge0} (n+\frac14) t \,\big((n+\frac14)^2 + \frac{t^2}{4}\big)^{-2} > 0$, I increases, and on $[2200, X_0 - 4]$ we have $\nu(r,4) = I(r) - 2 \cdot 2.5167 \ge \nu(2200,4) = 7.4591 - 5.0334 = 2.4257$. For $r \ge X_0 - 4$ Stirling's formula gives $I(r) = \frac{4}{\pi} \log \frac{r}{2\pi} + O(r^{-2})$ with an error below 10^{-9}, and s is nondecreasing and, at $t = r + 4 \ge X_0$, at most $s_{\log}(t) = 0.1095 \log t + 0.2042 \log\log t + 3.0305$, so $\nu(r,4) \ge \frac{4}{\pi} \log \frac{r}{2\pi} - 2 s_{\log}(r+4) - 10^{-9}$. The derivative of the right side is at least $\big(\frac{4}{\pi} - 2{\cdot}0.1095 - \frac{2{\cdot}0.2042}{\log X_0}\big)/(r+4) > 1/(r+4)$, so it increases, and at $r = X_0 - 4$ it equals $28.4018 - 12.6490 - 10^{-9} > 15.75$. Hence $\nu(r,4) \ge 2.42 > 0$ for all $r \ge 2200$, and the window contains a zero. Ranges (b) and (c) overlap in $[2200, 2201.26]$. □

For $\alpha_0 = \frac12$ the theorem is the direction RH ⇒ compensated positivity of Theorem 12.1.18(vi), and the window argument is that of the proof of part (v) of Theorem 12.1.18.

Remark 23.1.9 (One block for each zero). Under the hypothesis of Theorem 23.1.8 every width $c = \alpha - \beta_\rho$ is positive, and the Hadamard product splits the compensated reciprocal into one factor for each pair $\rho, \bar\rho$ with $\gamma = \gamma_\rho > 0$

[154]:

$$\Phi_{\alpha,\rho}\Big(\frac{w^2}{2}\Big) = \frac{c^2+\gamma^2}{(c+w)^2+\gamma^2}\, e^{2cw/(c^2+\gamma^2)}.$$

By Frullani's integral $-\log \Phi_{\alpha,\rho}(w^2/2) = \int_0^\infty (e^{-wy} + wy - 1)\, \varrho_\rho(dy)$, with the damped cosine wave $\varrho_\rho(dy) = -2y^{-1}e^{-cy}\cos(\gamma y)\, dy$, a signed measure with $\int (y \wedge y^2)\, |\varrho_\rho|(dy) < \infty$. The $2\sin^2$ lemma, applied to its positive and negative parts, gives the density $\pi^{-1}[f(c, r-\gamma) + f(c, r+\gamma) - 2f(c,\gamma)]$ in the ordinate r, a recentred pair of Cauchy bumps. Summed over the zeros these are $\pi^{-1}D_\alpha(r)$, as in the proof of Theorem 23.1.8, the constants $2f(c_\rho, \gamma_\rho)$ adding up to b_α. No single block is a GGC, nor even a Laplace transform, since $\Phi_{\alpha,\rho}(w^2/2) \to \infty$ as $w \to \infty$: positivity is a property of the sum, in which the bumps of the zeros near height r outweigh the accumulated constants. As $c \downarrow 0$ the block becomes $(1 + w^2/\gamma^2)^{-1}$, an exponential law in $u = w^2/2$ whose Thorin measure is the unit atom at $r = \gamma$; this is Proposition 10.2.16, zero by zero.

Corollary 23.1.10. *For $\frac{1}{2} \le \alpha_0 < 1$ the following are equivalent: (a) ζ has no zero with $\operatorname{Re} s > \alpha_0$; (b) $D_\alpha(r) \ge 0$ for every $\alpha \in (\alpha_0, 1)$ and every real r at which D_α is defined. When they hold, $D_\alpha(r) > 0$ for every $\alpha \in (\alpha_0, 1)$ and every real $r \neq 0$.*

Proof. (a) implies (b) by Theorem 23.1.8, and (b) implies (a) by Proposition 22.11.10, since $D_\alpha \ge 0$ for $\alpha \ge 1$ by Proposition 12.1.12. □

For $\alpha_0 = \frac{1}{2}$ this is part (vi) of Theorem 12.1.18.

The basepoints at which the compensated reciprocal is a GGC form a closed half line [154]. Let $\Theta = \sup \operatorname{Re} \rho$, the $\sigma_{\max}$ of Theorem 22.11.7. For $\alpha < \Theta$ a zero with $\operatorname{Re}\rho > \alpha$ gives the transform a pole at $u = (\rho - \alpha)^2/2$, off the negative axis, so it is not a GGC, whose transform is holomorphic off $(-\infty, 0]$ [30]. For $\alpha > \Theta$ it is one: by Theorem 23.1.8 with $\alpha_0 = \Theta$ and Theorem 22.11.7 with $\delta = \alpha - \Theta$ when $\alpha < 1$, and by Proposition 12.1.12 when $\alpha \ge 1$. At $\alpha = \Theta$, as $\alpha \downarrow \Theta$ the transforms $e^{\sqrt{2u}\, b_\alpha}\xi(\alpha)/\xi(\alpha + \sqrt{2u})$ converge for every $u \ge 0$ to the transform at Θ, which is continuous at $u = 0$; so the laws converge weakly and the limit is a GGC, the class being closed under weak limits. So the set is exactly $[\Theta, \infty)$; for $\Theta = \frac{1}{2}$ this is RH. A zero on the line $\operatorname{Re} s = \Theta$ gives the Thorin measure at Θ an atom at its ordinate, of mass its multiplicity, the vague limit of its Cauchy kernel of width $\alpha - \Theta$; if Θ is not attained there is no atom.

Corollary 23.1.11. *Assume $(\mathrm{Q}_{7/8})$ and let $\alpha > \frac{7}{8}$. Then the compensated reciprocal $e^{wb_\alpha}\xi(\alpha)/\xi(\alpha + w)$ is the Laplace transform in $u = w^2/2$ of a GGC, with the Thorin density (22.11.6) of the $2\sin^2$ lemma, positive at every height. For $\alpha \ge 1$ this holds with no hypothesis, by the factorisation of Proposition 12.1.12.*

Proof. The hypotheses of Theorem 22.11.7 hold for ξ with $\delta = \alpha - \frac{7}{8}$, and its condition (22.11.7) is $D_\alpha \ge 0$, which holds by Theorem 23.1.8 for $\frac{7}{8} < \alpha \le 1$ and

by Proposition 12.1.12 for $\alpha \geq 1$. □

So under $(\mathrm{Q}_{7/8})$ the conclusion of the $2\sin^2$ lemma, positivity of the Thorin density of the compensated reciprocal, holds on every line to the right of $\frac{7}{8}$, inside the critical strip for $\frac{7}{8} < \alpha < 1$, although its hypothesis, positivity of the primal measure, cannot even be stated there, because the Euler product diverges.

Numerics. We computed the constants of the proof of Theorem 23.1.8 for $\alpha_0 = \frac{7}{8}$. The margin $a/(c_{\max}^2 + 16) - b_\alpha$ for one zero in the window, with b_α computed by mpmath at sixty values of α in $(\frac{7}{8}, 1]$, is at least 0.0053, with the minimum at $\alpha = 0.8751$, and the ratio b_α/a is at most 0.04620; the inequality $f(c,x) + f(2a - c, x) \geq 2a/(c_{\max}^2 + d^2)$ for a pair holds on grids of α, c and x with smallest ratio 1.0000047. The quantity $\nu(r, 4)$ was recomputed with the Stirling series for ϑ and with mpmath at ten heights up to 10^{16}, agreeing to 7.1×10^{-15}; it never exceeds the number of ordinates in $[r - 4, r + 4]$ for r from 40 to 2190.

The direct computation of D_α, which assumes nothing, agrees. A computation finds $D_\alpha(r) > 0$ on $(0, 600]$ at step 0.01 (60 000 heights) at $\alpha = 0.9$, 0.95, 1, 1.001, 2, 4 and 7.8, and on $(0, 200]$ at the fifteen equally spaced basepoints from 1 to 7.8 and at twelve further ones (1.001, 1.01, 1.1, 1.2, 1.5, 2, 3, 4, 5, 6, 7 and 7.789), with the smallest value of D_α at the first height of the grid, where D_α grows like r^2 and $D_\alpha(r)/r^2$ is 1.1×10^{-4} at $\alpha = 1$ and 9.0×10^{-4} at $\alpha = 7.8$. At heights above 2 the minimum over $(2, 600]$ is 3.7×10^{-4}, 4.2×10^{-4} and 4.6×10^{-4} at $\alpha = 0.9$, 0.95 and 1, attained just above $r = 2$. For $\alpha < 1$ these computations are floating point evidence, not a proof.

23.1.5 What it does not give, and what to look for in the proof

The claimed theorem is a fixed half plane; the routes of the book need more.

1. Bondesson's closure gives RH from GGC basepoint laws along a sequence $\alpha_n \downarrow \frac{1}{2}$ (Corollary 9.1.2). $(\mathrm{Q}_{7/8})$ supplies them for $\alpha \geq \frac{7}{8}$ only; the open strip is $\frac{1}{2} < \alpha < \frac{7}{8}$.

2. The even part laws and the accounting theorem need no zero to the right of α; below $\frac{7}{8}$ the term $2N(\alpha, T)$ of Theorem 22.11.16 remains and is controlled only by density estimates.

The structural point is Corollary 23.1.10: a quasi Riemann hypothesis at α_0 is equivalent to the positivity of the compensated Thorin densities of the $2\sin^2$ lemma on every line to the right of α_0. The proof announced in [227] does not go through the Thorin picture (Section 25.1.2), but by the equivalence whatever it proves has a Thorin reading. If it contains an explicit lower bound for $\operatorname{Re} \frac{\zeta'}{\zeta}$ or for a mollified Dirichlet polynomial on $\operatorname{Re} s > \frac{7}{8}$, then a bound at least as strong as $-\operatorname{Re} \mathcal{A}$ is, by part (ii) of Theorem 23.1.1, positivity of the clock, a Thorin

statement in disguise, and the natural next step towards $\alpha_0 < \frac{7}{8}$ is to transfer it directly to the clock: the identity (22.11.9) at $\alpha < 1$, with the prime sum replaced by whatever the proof controls.

Chapter 24

What would close the gap: open problems of the computational programme

Every chapter of this volume ends with its open problems, and this chapter does not restate them. It sorts them by what they would accomplish, so that a reader looking for the next step can see which problems bear on the Riemann Hypothesis itself and which would sharpen a computation. The sorting follows the path of Section 1.4 of Volume I and its three constraints.

24.1 Problems on the descent itself

These are the problems whose solution would be progress on RH, because they concern the step from level one to the critical line for all heights at once.

1. What a proof by deconvolution would still need, and why every local computation stops short of it: Section 13.10.

2. Whether the backward flow is a path at all, and the problems of the level one measure: Sections 10.6 and 10.7.

3. The questions on the explicit formula read as a deconvolution, including a rigorous matrix pencil (Problem 12.1): Section 12.9.

4. The Balazard, Saias and Yor criterion at every level, and the limits of its Monte Carlo solution: Sections 14.8 and 14.9.

5. Where the uniform step of the phase diagram could come from, the de Bruijn Newman flow being the only known source of an analytic tail below the basepoint one (below $\frac{7}{8}$ if the announced half plane is confirmed): Section 6.6.

Each of these meets the first constraint of Section 1.4 of Volume I: a computation in a window cannot settle a statement for all heights (Section 1.4). A solution has to be an argument, and the Euler product has to enter it.

24.2 Problems on windows of the explicit formula

Weil positivity in compact windows is certified to finite support lengths, and the barriers of Chapters 15 and 16 say how far such certificates can see: Sections 15.11, 15.12, 15.23, 16.6 and 16.7. These problems would extend what is certified. A window certificate is a statement about finitely many support lengths, so the first constraint applies to it as it does to every computation.

24.3 Problems on approximants

The truncations, the friable approximants, the extreme value approximants and the approximating clocks each agree with Ξ to high accuracy and yet can have zeros off the line. Their open problems ask where the zeros of the approximants go and what a computational criterion must see to tell them from Ξ: Sections 17.7, 17.14, 18.15, 19.2, 19.6, 21.12 and 21.13. They test the second and third constraints of Section 1.4 of Volume I: an approximant that keeps the functional equation but not the Euler product, or a property that survives a Gaussian tilt, is where a false criterion would be caught.

24.4 Problems on cost and certification

The remaining problems concern the computation as such: whether the level one density can be evaluated in time polynomial in both the logarithm of the height and the number of digits (Problem 11.3), a posteriori bounds for the accelerated flow, and a calibrated Bayesian inversion of the backward flow (Problem 10.1): Sections 11.11 and 11.12. They would make the computations of this volume faster and more rigorous, and they bear on RH only through the windows they make reachable.

Appendix A

The Pólya evaluators

A.1 The Pólya evaluators

Ball arithmetic is that of Arb [179], through its Python interface; reference values in floating point use `mpmath` [213].

The evaluator of Section 2.1 on the real u axis works in ball arithmetic: it computes the node values $\Phi_j = \Phi(jh)$, $0 \le j \le J$, from the theta series (2.1.1), and from them $P(E)+P(E^{-1})$ and $Q(E)-Q(E^{-1})$ with $E = e^{sh}$ and the polynomials P and Q of Section 2.1.1, the trapezoid approximations to $\xi(\frac{1}{2}+s)$ and $\xi'(\frac{1}{2}+s)$. The defaults are 240 bits, $h = 1/200$ and $|u| \le 3$, so that $J = 600$. Most runs use 420 bits, the parameters of Section 2.1.1; the Keiper Li coefficients of Section 23.6 use 600 bits, and the Selberg computation of Section 5.1 uses 300.

The tilted evaluator implements Proposition 2.1.1. With the nodes $u_j = x_j + i\eta$, $x_j = jh$, $|j| \le J$, where $J = \lceil x_{\max}/h \rceil$ for the cut $|x| \le x_{\max}$, it returns the trapezoid sums of the two integrals of the proposition. The production runs evaluate each exponential directly, not by Horner's rule, whose ball radii grow (Section 2.1.5). The Lehmer pair and its de Bruijn Newman flow use $\delta = 0.01$, 256 bits, $h = 6 \times 10^{-4}$ and $|x| \le 3.9$; the count at height 1000 in Section 2.2.3.2 uses $\delta = 0.1$, 200 bits, $h = 0.004$ and $|x| \le 4.2$; the Laplace transform of the clock for Sections 5.2 and 3.1 switches to the tilted evaluator with $\delta = 0.1$, 320 bits, $h = 0.003$ and $|x| \le 4.5$ when $|\operatorname{Im} s| \ge 120$; the runs of Table 2.2 use other δ, precision, h, cut and heights.

In both evaluators the radii of the returned balls bound only the rounding errors of the finite sums. Three errors lie outside them: the discretisation error of the trapezoid rule, bounded in Section 2.1.1; the cut of the axis at $|u| \le 3$, where $\Phi < 10^{-500}$, or at $|x| \le x_{\max}$ on the tilted line; and the truncation of the theta series, whose loop stops at the first term with midpoint below 2^{-b-20} at working precision b and does not add the tail to the radius. The stopping test compares double precision numbers, so for $b \ge 1055$ its threshold underflows to zero and the test is never met. The untilted route needs more than 1054 bits for thirty digits above height about 840 (Table 2.3), and there the test has to be made in ball

arithmetic. The runs in this book use at most 600 bits and are not affected.

Appendix B

Notation

The tables list the notation used in more than one section; the first five cover the book and record second meanings that some letters carry locally. Sums and products over k run over the pairs $\{\rho, 1-\rho\}$.

symbol	meaning	defined in
ξ, Ξ	the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$	Section 2.1
ρ, γ_k	a zero $\rho = \beta + i\gamma$ of ξ, that is a nontrivial zero of ζ; the ordinates $0 < \gamma_1 \le \gamma_2 \le \cdots$ of the zeros with $\gamma > 0$	Section 2.1
a_k	$-(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$, equal to γ_k^2 on the line (in Section 23.6, the Taylor coefficients of $\log \xi(1+w)$)	(2.1.1)
Φ, p, X	the Pólya kernel in Titchmarsh's normalisation; the Pólya density $p = \Phi/\Xi(0)$; a random variable with density p (in Section 10.2, p is the Meixner density; in Section 9.3, X is the clock variable; in Sections 12.1 and 15.2, X is the prime cutoff)	(2.1.3), (2.1.4)
S_2, Y, J	$S_2 = \sum_{k\ge1} \frac{2}{\pi^2 k^2}(e_k + e_k')$ with independent standard exponential variables, the sum of two independent copies of the first passage time J of the level 1 by a three dimensional Bessel process started at 0; $Y = \frac{1}{2}\log(\pi S_2/2)$, with $\mathbb{E}e^{sY} = 2\xi(s)$; X is Y tilted by $e^{Y/2}$	Section 2.1, Section 12.1.1
u, z, θ	the variable $u = s^2$ of the clock (also the variable of Φ); the variable of Thorin measures; the chart $\theta = \sqrt{z}$, which is also the ordinate (the Kent approximants of Section 5.2 write w for u)	Section 1.1
$G(u)$	$\xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$; the clock is $1/G$ (in Section 11.1.3, G is a Laguerre Pólya function)	Section 1.1

symbol	meaning	defined in
$\mathcal{G}(u,t)$	the Gaussian smoothing $\sum e^{-2t(\tilde{\gamma}-u)^2}$ of the zeros at resolution $1/(2\sqrt{t})$, with $\tilde{\gamma}=\gamma-i\delta$ for a zero $\frac{1}{2}+\delta+i\gamma$ with $\gamma>0$	Section 8.1.5
T, e_γ	the clock variable, with $\mathbb{E}e^{-uT}=1/G(u)$, which exists with no hypothesis and is the centre clock of Chapters 12 to 14; under RH, $T=\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (T is also a height, as in $N(T)$)	Section 2.1, Proposition 2.1.1
U, U_α	the Thorin measure of a GGC H, with $\mathbb{E}e^{-uH}=\exp\{-du-\int\log(1+u/z)\,U(dz)\}$ and drift d; the signed measure $U_\alpha(dz)=v_\alpha(\sqrt{z})\,dz/(2\pi\sqrt{z})$ at a basepoint α; for the clock, the Thorin functional $\sum_k \delta_{a_k}$	Section 2.1, (2.1.6), Section 8.2.1
k	the canonical function of an infinitely divisible law on $[0,\infty)$ with Lévy density $k(t)/t$ ($k(y)$ in Sections 8.1 and 10.2)	Section 4.1

symbol	meaning	defined in
U_{RH}, U_{line}, $U^{\pm}_{\mathrm{RH}}$	$\sum_{\gamma>0}\delta_{\gamma^2}$; the same sum over the zeros on the line; $\sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$	(1.1.1), Proposition 2.1.3
$\mathcal{W}$, $T_{\mathcal{W}}$, $U_{\mathcal{W}}$, U	the principal branch of Lambert's function, $\mathcal{W}(u)e^{\mathcal{W}(u)}=u$ (not the heat trace W); the Lambert law, with Laplace transform $\mathcal{W}(u)/u=e^{-\mathcal{W}(u)}$; its Thorin measure; a variable uniform on $(0,1)$ (not the Thorin functional U)	Sections 14.4.1, 14.4.3
φ_α, T_α	$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, the clock at a basepoint $\alpha\geq\frac{1}{2}$, with $\varphi_{1/2}=1/G$; the variable with $\mathbb{E}e^{-uT_\alpha}=\varphi_\alpha(u)$, which exists for every $\alpha>\frac{1}{2}$ and is a GGC if and only if no zero has $\beta>\alpha$, in particular for every $\alpha\geq 1$; $T_{1/2}=T$	Theorems 2.1.2, 12.2.5
$\widetilde{T}$, $\widetilde{T}_\alpha$, $\sigma^{(\alpha)}$	the centre clock tilted by $e^{-T/4}$, and by $e^{-(\alpha-\frac{1}{2})^2T}$, so that $\widetilde{T}=\widetilde{T}_1$; the subordinator with Laplace exponent $u+2(\alpha-\frac{1}{2})\sqrt{u}$, with $T_\alpha\stackrel{d}{=}\sigma^{(\alpha)}_{\widetilde{T}_\alpha}$	Theorems 12.2.2, 12.2.5
H_a, $H_a^{(c)}$	the first passage time of B_t+ct to the level $a\geq 0$, for a standard Brownian motion B, and $H_a=H_a^{(0)}$; unrelated to H_t	Section 12.2

symbol	meaning	defined in
Y_α, Z	the normal scale mixture $Y_\alpha = \sqrt{2T_\alpha}\, Z$, with Z standard normal and independent of T_α, whose characteristic function is $\xi(\alpha)/\xi(\alpha + \lvert u\rvert)$ (in Section 14.1, Z is the subordinator with $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$)	Proposition 13.1.1
$W(t)$, $c_k(t)$	the heat trace $\sum_k e^{-a_k t}$, also written $\int e^{-tz} U(dz)$; the signed derivatives $(-1)^k W^{(k)}(t)$ (in Chapters 10 and 11 the time is x, or y where x is a Lévy variable, as in the proof of Lemma 10.2.5 and from Proposition 10.2.16 to the end of Section 10.2; the numbers $c_k(u)$ of Section 9.1 are local to it)	Sections 2.1, 8.1
m_n, $\mathcal{H}_N$, $\mathcal{H}'_N$, ι_N, ι'_N	the Stieltjes moments $\sum_k a_k^{-n-1}$, with $m_0 = \mathbb{E}T$; the Hankel matrices $[m_{i+j}]_{i,j<N}$ and $[m_{i+j+1}]_{i,j<N}$; the numbers of their negative eigenvalues, which count the zeros off the line (Section 4.1 uses the same symbols for the moments $m_n(t)$ of H_t)	Section 8.1, Theorem 8.1.7
$W^{[j]}$, β_j, B_j	the ladder functions $(-t\frac{d}{dt})^j W$, with $W^{[0]} = W$ (along the de Bruijn Newman flow $W^{[j]}(x;t)$, with x the time of the heat trace and t that of the flow); the polynomials with $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)e^{-at}$; the Touchard polynomials, $\beta_j(u) = (-1)^j B_j(-u)$	Section 14.4.4
L_n, L_∞	Urbanik's classes of multiply self decomposable laws, with L_0 the self decomposable laws; unrelated to the Post Widder functionals $L_n[W]$ of Section 17.1.2	Section 14.4.4.1
$Z^{[j]}$	the jth driving process of the centre clock, $Z^{[1]} = Z$, with Laplace exponent $(u\frac{d}{du})^j \log G$	Lemma 14.4.7
g, g_k, H, λ^*	the candidate generating exponent $\Psi_Z \circ H$ and its Taylor coefficients, with H the inverse of $\log G$; the branch point $\log\lvert G(u^*)\rvert + i\pi$ of H	Section 14.4.4.4
Φ_t, $\widehat{g}$, R_g	the damped Pólya kernel $e^{tv^2/4}\Phi(v)$; the cosine transform $\int g(v)\cos(\theta v)\, dv$ of an even kernel; $\widehat{g}(0)/\widehat{g}(i\sqrt{s})$, with $R_{\Phi_t} = R_t$	Section 14.4.5.1
X_t, p_t, Q_t, ϱ_t, h_t	the variable with density $p_t = \Phi_t/\Xi_t(0)$, $X_0 = X$; the radius of Khinchine's mixture, $X_t = UQ_t$; the radius of the arcsine mixture, $X_t = \varrho_t \cos\Theta$, and its density	Section 14.4.5.2
$\mathcal{B}$, f, $\mathcal{B}_g$, $\mathcal{B}_t$	$\frac{\pi}{2}S_2$, with $\mathbb{E}\mathcal{B}^{s/2} = 2\xi(s)$; its density $b^{-5/4}\Phi(\frac{1}{2}\log b)$; the variable with density proportional to $b^{-5/4} g(\frac{1}{2}\log b)$; $\mathcal{B}_{\Phi_t}$	Section 14.4.5.3
ℓ, $\kappa_k(\lambda)$, $c_k(\lambda)$, σ_λ	a Laplace transform $\mathbb{E}e^{-\lambda\mathcal{V}}$; the cumulants $(-1)^k(\log\ell)^{(k)}(\lambda)$; $\kappa_{k+1}(\lambda)/k!$, the moments of σ_λ for a GGC (not the $c_k(t)$ of the heat trace)	(14.4.9)

symbol	meaning	defined in
$\mathcal{K}_n(\lambda)$, $\mathcal{K}'_n(\lambda)$	the Hankel matrices $[c_{i+j}(\lambda)]_{i,j<n}$ and $[c_{i+j+1}(\lambda)]_{i,j<n}$, positive semidefinite for a GGC	Section 14.4.5.3
ℓ_t, $\dot{\ell}$, $\mathcal{J}$	$\mathbb{E}e^{-\lambda\mathcal{B}_t}$; its derivative in t at $t=0$, up to a term in ℓ_0; the first order change of the Thorin measure in a gap, in the proof of Proposition 14.4.15	Section 14.4.5.3
ϑ_a, $\xi(s,a)$, K_a, $\mathcal{B}_a$	the theta series $\sum n^p a_n e^{-\pi n^2 x/q}$ of a periodic sequence; its completed Dirichlet series; the kernel $2e^{(p+1/2)v}\vartheta_a(e^{2v})$; $\mathcal{B}_{K_a}$ (for $a=\chi_{-4}$, $1/\mathcal{B}_a$ is the variable Z of Remark 15.1.23)	Section 14.4.5.4
χ_d, S_a, Σ_a	the real primitive character of discriminant d ($\chi_{-4}=\chi_4$); the positivity sum of Lemma 14.4.17; the numerator of the reflected transform (14.4.10)	Section 14.4.5.4
f_{DH}, $\varkappa$, f_r	the function of Davenport and Heilbronn; its coefficient 0.28407904…; the self dual family modulo 7	Section 14.4.5.4
k_n	the inverse Laplace transform of Li's φ_n, $\lambda_n = \int k_n W$	Problem 14.4

symbol	meaning	defined in
ε	the resolution; the basepoint is $\frac{1}{2}+\varepsilon$	Section 1.1
$v_\alpha(r)$	$\mathrm{Re}(\xi'/\xi)(\alpha+ir)$, which is π times the density of U_α in the chart $\theta=\sqrt{z}$ (Chapters 12 to 14 write $v_\alpha(\theta)$)	Section 2.1
b_α, Π_α	$b_\alpha = \frac{\xi'}{\xi}(\alpha) = v_\alpha(0)$, the drift of the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx}-1+sx)\,\Pi_\alpha(dx)$ (for $\alpha\geq 1$, and for $\frac{1}{2}<\alpha<1$ when $\Theta<\alpha$), with $b_1 = \sum_\rho \rho^{-1} = \lambda_1$; its signed Lévy measure, with atoms $\Lambda(n)/(n^\alpha \log n)$ at $\log n$	Proposition 12.1.10, Section 12.1.5
A, A_α, Q, ν_1	the gamma piece $A(\theta) = \frac{1}{2}[\mathrm{Re}\,\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})]$ and the compensated prime piece $Q(\theta) = \lim_{X\to\infty}\int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ of $v_1 = b_1 + A + Q$; the gamma piece A_α at α; the prime measure $\nu_1 = \sum_n \Lambda(n)n^{-1}\delta_{\log n} - dx$ on $(0,\infty)$, compensated by Lebesgue measure and unrelated to $\nu(\sigma,T)$ (Section 15.2 uses A, A_α and Q for $L(s,\chi_{12})$)	(12.1.6), Proposition 12.1.3

symbol	meaning	defined in
S, T_1°; S_α, T_α°	the positive stable variable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$; the GGC with no drift, independent of S, whose Thorin density in the chart $\theta = \sqrt{z}$ is $(A+Q)/\pi$, so that $T_1 \stackrel{d}{=} S + T_1^\circ$ (unrelated to $S(t)$ and S_2); the same at a basepoint $\alpha \geq 1$, with b_α and $(v_\alpha - b_\alpha)/\pi$, so that $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$	Propositions 12.1.11, 12.1.12
V_α, V	the variable with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, whose density $e^{(\alpha-\frac{1}{2})y}p(y)\,\xi(\frac{1}{2})/\xi(\alpha)$ is a tilt of the Pólya density, with no hypothesis; $V_{1/2} = X$ and $V = V_1$	Section 2.1, Section 12.1.1
$\rho_\varepsilon(\theta)$	the Thorin density at basepoint $\frac{1}{2}+\varepsilon$ in the chart $\theta = \sqrt{z}$, $\pi^{-1}v_{1/2+\varepsilon}(\theta)$	(2.1.7)
μ_ε	the signed measure $U_{1/2+\varepsilon}$, with density $\rho_\varepsilon(\theta)\,d\theta$ in the chart θ; the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2}+\varepsilon > \operatorname{Re}\rho$ for every zero, and called the Thorin measure at resolution ε in general ($\mu(n)$ is the Möbius function)	Section 2.1

symbol	meaning	defined in
P_a	the Poisson (Cauchy) kernel $a/(\pi(a^2+x^2))$, with $P_{-a} = -P_a$	Section 2.1
$d_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * U_{\mathrm{RH}}^{\pm}\|_{\mathrm{TV}}$, the distance to the RH measure smoothed at the same resolution, which vanishes under RH	Section 3.1
$\tilde{d}_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, the matched distance computed against the model $\widetilde{U}$ of the zero list (the first 1700 ordinates and the density $\pi^{-1}\vartheta'$ beyond them, each with its reflection)	Section 3.1
κ_{cell}	the cell constant $\pi^{-1}\sum_k \lvert\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\rvert$ of the 79 cells $(b_{k-1}, b_k]$ below height 200, so that $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$	Section 3.1
$M_\varepsilon(\theta)$	the cumulative mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$, with the branch of the argument that vanishes at $\theta = 0$ (the Mertens function $M(t) = \sum_{d\leq t}\mu(d)$ in Section 13.3)	Section 2.1.2, Section 3.3
$\tau_\varepsilon(T)$	π^{-1} times the change of $\arg\xi$ along the segment from $\frac{1}{2}+\varepsilon+iT$ to $\frac{1}{2}+iT$, so that $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac{1}{2}+\varepsilon, T)$ ($\tau(n)$ is Ramanujan's function in Section 15.1)	Proposition 3.3.2

symbol	meaning	defined in
$N(T)$, $\nu(\sigma,T)$	the number of zeros with $0<\operatorname{Im}\rho\le T$; the number of those with $\operatorname{Re}\rho>\sigma$ (Sections 3.3 and 14.3 write $N(\sigma,T)$ for the number with $\operatorname{Re}\rho\ge\sigma$)	Section 1.1
$S(t)$, $S_\varepsilon(t)$	$\pi^{-1}\arg\zeta(\frac12+it)$, with the argument continued along $2\to 2+it\to\frac12+it$; $\pi^{-1}\arg\zeta(\frac12+\varepsilon+it)$, continued along the line $\operatorname{Re}s=\frac12+\varepsilon$ from $S_\varepsilon(0+)=-1$ if $0<\varepsilon<\frac12$ and from 0 if $\varepsilon>\frac12$	Sections 2.2.1, 3.3.1
$\vartheta(t)$	the Riemann Siegel theta function $\operatorname{Im}\log\Gamma(\frac14+\frac{it}{2})-\frac{t}{2}\log\pi$	Section 2.2
$Z(t)$	Hardy's function $e^{i\vartheta(t)}\zeta(\frac12+it)$, real for real t	Section 2.2
Θ	a height, the top of a range of ordinates, with $\Theta=200$ for the window of Chapters 2 to 8; in Chapters 10 and 11 and Sections 12.1 and 13.2, $\sup\operatorname{Re}\rho$	Section 2.1.2, Section 10.1
T_0	the height 3×10^{12} below which RH has been verified [232]	Section 1.1
λ_n	the Keiper Li coefficients $\sum_\rho[1-(1-1/\rho)^n]$	Section 23.6
H_t, Λ, Φ_{dBN}	the de Bruijn Newman flow, normalised by $H_0(z)=\frac18\Xi(z/2)$; the de Bruijn Newman constant; the kernel $\Phi_{\mathrm{dBN}}(u)=\frac12\Phi(2u)$ of the flow ($\Lambda(n)$ is von Mangoldt's function)	Section 4.1
$P(\theta)$, ν_n, A_P, K^P	Pólya's Bessel function $P(\theta)=K_{i\theta/2}(2\pi)$; the zeros of $\nu\mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta=\pm2\nu_n$; the operator $-\frac{d^2}{dx^2}+e^{2x}$ on $[\log2\pi,\infty)$ with a Dirichlet condition, whose eigenvalues are the ν_n^2; the trace $K^P(y)=\sum_n e^{-2\nu_n y}$	Section 18.1

symbol	meaning	defined in
δ	the displacement of an off line zero $\frac12\pm\delta+i\gamma$, written d in Sections 1.4, 8.2, 8.3, 23.6, 4.1 and 25.2; in Chapter 2 and where its parameters are quoted, the tilt margin, with $0<\delta<\pi/4$; in Section 9.3, the angle of a zero	Propositions 2.1.3, 2.1.1
η	the tilted height $\eta=\pi/4-\delta$ (Dedekind's eta function in Sections 15.1, 9.3 and 15.2; Dirichlet's eta function $\eta(s)=(1-2^{1-s})\zeta(s)$ in Section 13.1 and, as $\eta(1+v)=(1-2^{-v})\zeta(1+v)$, in Chapters 10 and 11)	Proposition 2.1.1
$K_\eta(\varepsilon)$	the kernel constant $\int_{\mathbb{R}}\lvert\Phi(x\pm i\eta)\rvert\,e^{\varepsilon\lvert x\rvert}\,dx$ of the trapezoid bound, with $K_\eta=K_\eta(0)$	Section 2.1.1

symbol	meaning	defined in
ψ	the digamma function Γ'/Γ (Chebyshev's function $\sum_{n\le y}\Lambda(n)$ at real $y \ge 1$ in Sections 11.1, 12.1, 13.2 and 14.3, and $\psi(y,\chi_{12})$ in Section 15.2)	Section 2.1
γ_E	Euler's constant	Section 8.1
$\mathcal{N}$	the standard normal distribution function, with $\overline{\mathcal{N}} = 1 - \mathcal{N}$	Section 3.1
$\xi(s,\chi_{12})$, K	the completed L function $(12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12}) = \int_0^\infty \eta(ix)x^{s/2}\,dx/x$; its kernel $K(u) = 2e^{u/2}\eta(ie^{2u})$, with $\xi(\frac12+s,\chi_{12}) = \int_{\mathbb{R}} K(u)e^{su}\,du$ (a variable Y with density $K/\xi(\frac12,\chi_{12})$ plays the role of X)	Section 15.1.1
$\xi(s,\chi_4)$, K_4, Y_4	the completed Dirichlet beta function $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$, equal to $\int_0^\infty \eta(ix)^3 x^{(s+1)/2}\,dx/x$; its kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$, with $\xi(\frac12+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\,du$; the variable with density $K_4/\xi(\frac12,\chi_4)$	Section 15.1.7.1
$R(x)$, $R_h(x)$	Riesz's function $x\sum_{n\ge1}\mu(n)n^{-2}e^{-x/n^2}$; its twist $x\sum_{n\ge1}\mu(n)n^{-2}(x/n^2)^{ih}e^{-x/n^2}$, with $R_0 = R$	Sections 13.1.8, 13.2.1
$\xi_K(s)$, C_K	the completed Dedekind zeta function $s(s-1)\lvert d_K\rvert^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ of a number field K, with $\xi_{\mathbb{Q}} = 2\xi$; the constant of $\sum_{N(\mathfrak{a})\le x}\mu(\mathfrak{a})^2/N(\mathfrak{a})$, with $C = C_{\mathbb{Q}} = 1.7171377$	Sections 13.3.1, 13.3.3
$\lVert\cdot\rVert_{\mathrm{TV}}$	the total variation $\lvert\cdot\rvert(E)$ of a signed measure on the window E	Section 1.1
GGC, HCM, ID, SD, BO, EGGC	generalised gamma convolution; hyperbolically completely monotone; infinitely divisible; self decomposable; the Bondesson class; the extended class of generalised gamma convolutions, the weak limits of convolutions of gamma laws and their reflections	Section 4.1, Section 4.1.2

symbol	meaning	defined in

Sections 9.2 and 9.3 use the following notation.

symbol	meaning	defined in
s, $\varphi(s)$	the variable u of the clock; the clock $\varphi_{1/2}(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$	Section 9.2
W	the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace	Section 9.2
T_α	Bosch's variable $(Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with independent positive α stable $Z_\alpha, \tilde{Z}_\alpha$; here α is a stability index, not a basepoint	Section 9.2
X, p, ε_h, δ, θ, ϑ	under RH, the clock variable T; an exponent, not the Pólya density; an elasticity, not a resolution; the angle of a zero; angles, not the ordinate and the Riemann Siegel function	Section 9.3
ψ, S_{on}, R_{off}	$-\varphi'/\varphi = \sum_\gamma (s + \gamma^2)^{-1}$, the function of the Stieltjes form; its sums over the real and over the nonreal γ (in Section 14.3.7, the parts of G'/G in the variable u)	Theorems 9.2.4, 9.3.9
D_γ, $\mathcal{E}_{T_0}$, θ_0	for a nonreal zero $\gamma = a + ib$, the closed disc $\lvert s - (b^2 - a^2)\rvert \le 2ab$ with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$; the parabolic sliver $\{x + iy :\ x < -T_0^2 + T_0 + \frac{1}{4},\ y^2 < \frac{1}{2} - x\}$ that contains every D_γ; its angle at the tip, seen from the origin	Theorem 9.3.9

The numerical checks behind Section 9.3 were computed for this book. Chapters 10 and 11 use the following notation.

symbol	meaning	defined in
s, t, x	the variable u of the clock; the variable z of Thorin measures; the time of the heat trace $W(x)$	Section 10.1
τ, z	the ordinate; the argument of ζ and ξ	Section 10.1
G, R, φ	the gamma factor $\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$; the reciprocal Xi function $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the clock $1/G$ of Section 2.1; the normalised reciprocal zeta function $\varphi(s) = \zeta(\frac{1}{2})/\{(1 - 4s)\zeta(\frac{1}{2} + \sqrt{s})\}$, with $\varphi = RG$	(10.1.4)

symbol	meaning	defined in
R_σ, φ_σ, R_t	$\xi(\sigma)/\xi(\sigma+\sqrt{s})$, which the rest of the book writes φ_σ; the normalised reciprocal zeta function $\zeta(\sigma)/\{(1-s/(1-\sigma)^2)\zeta(\sigma+\sqrt{s})\}$ at σ, with $\varphi_{1/2}=\varphi$, not the φ_σ of the rest of the book; $H_t(0)/H_t(2i\sqrt{s})$, with $R_0=R$ (also in Section 4.1, which writes u for s)	Sections 10.1, 17.1
θ, c, c_n	the Riemann Siegel theta function (ϑ elsewhere); $c=-\theta'(0)=\zeta'(\frac12)/\zeta(\frac12)=2.6860917$ (also quoted in Sections 8.1 and 15.1); $c_n=2n+\frac12$	Section 10.1, (10.1.6)
$W_0(x)$, $w(x)$, $M(t)$	the archimedean part $\frac{1}{\pi}\int_0^\infty e^{-x\tau^2}\theta'(\tau)\,d\tau$ of $W(x)$; the canonical function $w=W-W_0$ of φ; $N(\sqrt{t})-\theta(\sqrt{t})/\pi=1+S(\sqrt{t})$	Section 10.1
X_φ, X_R	random variables with Laplace transforms φ and R, so that X_R has the law of the clock variable T	Section 10.2
κ, Y_κ	a Poisson width; the variable with $\mathbb{E}e^{-sY_\kappa}=\kappa/(\kappa+\sqrt{s})$, whose Thorin density is a Poisson kernel of width κ in $\sqrt{t}$	Section 10.2
$\eta(1+v)$, χ_4, $\mathcal{S}_n$	Dirichlet's eta function; the nontrivial character modulo 4; the truncated lattice variables $\mathcal{S}_n=\sum_{k\le n}k^{-2}(e_k+e'_k)$, with $\mathcal{S}=\frac{\pi^2}{2}S_2$ and tails $\mathcal{T}_n=\mathcal{S}-\mathcal{S}_n$ (also in Section 14.1)	Sections 10.1, 17.1
B_N, h_λ	the weights $B_N=\sum_{n^m=N}d_m a_n$ of the Vere-Jones representation, with $a_n=\Lambda(n)/(\sqrt{n}\log n)$ and $\sum_n d_n n^{-v}=1/\eta(1+v)$, which vanish unless N is a prime power p^n and are then sums of necklace polynomials at $p^{-1/2}$; the heat trace of the sawtooth measure U_λ, a Jacobi theta function	Section 10.1, Theorem 11.1.8

List of Figures

1.1 Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with W replaced by its sum over the first 1700 zeros. The change is at most 4×10^{-11}, near $t = 7 \times 10^{-7}$, and falls below 10^{-1000} by $t = 10^{-3}$. The sum over 1700 zeros is smaller than W, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with W from the explicit formula (2.1.5) the maximum is 3.66×10^{-11}, at $t = 7.2 \times 10^{-7}$. 9

2.1 The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2} + \varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 2.1.3). . 22

2.2 The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed. 29

2.3 The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of γ_1, the kernel of γ_2 and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction. 31

3.1 Cumulative Thorin mass $\mu_\varepsilon((0, \theta^2]) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of U_{RH}, on the whole window (left) and on $28 \le \theta \le 52$ (right). 38

3.2 The three signatures of Proposition 3.2.1 as functions of δ/ε: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1. 40

6.1 The Thorin phase diagram. Solid curves are the boundaries of certified GGC regions carried along the flow by Theorem 6.3.2(i), and the dashed curve is that of the region conditional on the half plane of Corollary 6.3.3(d); the dotted curves are the level sets of $t + 2(\alpha - \frac{1}{2})^2$, labelled by the bound on Λ that a point on them gives. RH is the point $(0, \frac{1}{2})$. 103
6.2 $v_t(\alpha, \theta) = \operatorname{Re} \xi_t'/\xi_t(\alpha + i\theta)$, which is π times the Thorin density of the shifted time t clock when $(t, \alpha) \in \mathcal{G}$, on the lines $\operatorname{Re} s = 0.6$ and 0.75, at $t = 0$ (solid) and $t = 0.2$ (dashed), from the Pólya integral. Dotted lines mark the ordinates of the zeros of ζ. 105
6.3 Left: $\lambda_{\min}$ of the Weil Gram matrix against the support length L for two resolutions. Right: a pair of zeros at height g moved to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$, signed logarithm of $\lambda_{\min}$ for $w = 0.05$. The pair at $g \approx 17.6$ is detected at every L, the pair at $g \approx 54.7$ only at $L = 4$, and the pair at $g \approx 146.7$ not at all within $L \leq 4$. 112
6.4 Left: the minimum normalised gap and $\min_k \Delta_k^2 g_k$ along the flow, both monotone on the computed range; the second crosses the Lehmer threshold $\frac{4}{5}$ near $t = 0.17$. Right: D^2 for the closest pair below height 200, at ordinates 184.87 and 185.60, against the isolated pair law $D_0^2 + 8t$. 114
6.5 The bound $\Lambda \leq 0.22$ of Polymath15 and the twelve rows of their table of conditional bounds [247] against the barrier X (verification height $X/2$), with the fits $\Lambda = c/\log X$ and $\log X = a/\Lambda + b$. Dashed lines mark $\Lambda = 0.1$ and 0.05. 115

8.1 The Thorin density ρ_ε across the gap at height 9878 (gap 0.872) with a hypothetical quadruple at distance $\delta = 0.45$ inserted at its midpoint, at four levels $\varepsilon = 0.05$, 0.15, 0.25 and 0.4. It is positive at $\varepsilon = 0.05$ and $\varepsilon = 0.15$ and negative near the midpoint at $\varepsilon = 0.25$ and $\varepsilon = 0.4$. The vertical scale is symmetric logarithmic, linear within ± 0.1. 146
8.2 The visibility region $\varepsilon < \delta < \delta_{\det}(\varepsilon)$ (shaded band) at the midpoint of the 10000th gap. Solid: $\delta_{\det}$ from the actual zeros through ξ'/ξ; dashed: the midpoint lattice of Proposition 8.3.2 with the same gap; dotted: $\delta = \varepsilon$, the lower edge of the band. The marked point is $\varepsilon = 0.1895$, $\delta = 0.45$. 148

9.1 The converged ladder on a logarithmic scale: $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ against the window half length a at $a = 0.25$, 0.5, 0.75, 1, 1.5, 2, 3, 4, for $\varepsilon = 0.5$, 0.375, 0.2, 0.1, 0.05 (five marker styles in black and grey), each with the value $2b_{1/2+\varepsilon}$, the limit where the hypotheses of Theorem 9.2.1 hold, as a dashed line labelled m, and for $\varepsilon = 0$ at $a = 0.25$ and 0.5 only (heavy line with stars). The values are those of Table 9.1; the legend and the axis write y and $\lambda(a, y)$ for ε and $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$. 161

10.1 Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. The zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement δ of the fifth zero, the δ^2 law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$. 184

14.1 The Davenport and Heilbronn Lévy density. Left: K_{DH} by the explicit formula and by the sum over zeros, with the off line terms in absolute value. Right: $K_{\mathrm{DH}}(u)e^{\gamma_1 u} - 1$ on a logarithmic scale; the quantity $K_{\mathrm{DH}}(u)e^{\gamma_1 u}$ tends to 1, from above for large u since the first zero is simple and the second lies on the line. Beyond $u = 9$ the right panel shows rounding noise. 327

14.2 Vertical curves at $\theta = \frac{1}{2}$ (computed, floating point). Left: $P(f > \lambda)$ (black) and $P(f < -\lambda)$ (grey) under the Cauchy law centred at t_0, for ζ at $t_0 = 0$ (solid) and Davenport and Heilbronn at $t_0 = 85.7$ (dashed), $\lambda \le 10$, logarithmic vertical axis. Right: the running balance $\mathbb{E}[\mathrm{clip}(f, -L, L)] - \log|h(1 + it_0)|$, which converges to the defect $D_{1/2}(t_0)$, for ζ at $t_0 = 0$ and 85.7 and for Davenport and Heilbronn at $t_0 = 85.7$, with the prediction 1.4407 dotted; window $|\tau - t_0| \le 200$, neglected Cauchy mass 0.0016. 336

14.3 Root mean square error against N (from the 400 replications of Table 14.4), drawn for $\theta = \frac{1}{2}$ and $\frac{7}{8}$ for plain sampling, YKS and YKS with control, with the fitted slopes in the legend. The dotted lines have slope -2 (anchored at the first YKS point), the rate of the theorem of Yakowitz, Krimmel and Szidarovszky for integrands in $C^2[0, 1]$, and slope $-\frac{1}{2}$ (anchored at the first plain point). 338

14.4 Root mean square error against N for the half disc identity (data of Table 14.5). The dotted lines have slopes $-\frac{1}{2}$, -2 and $-\frac{5}{2}$. . . . 343

14.5 The integrand $\log|f|\,\Delta\varphi$ on the box (121×121 points, one grey scale for both functions). The cross marks the Davenport and Heilbronn zero. The large values are at the left and right edges of the box, where $\Delta\varphi$ is largest and $\log|f|$ is large; the picture does not show the zero, which is why the two look alike. Their integrals are 0 and 0.632. 351

14.6 Root mean square error of $I(\varphi)$ against N in two dimensions, 20 replicates, for ζ (left) and Davenport and Heilbronn (right), with the plain, Voronoi, Delaunay, jittered, grid and (Davenport and Heilbronn) deflated grid rules, the deflating zero being off by 1.4×10^{-4}. The dotted lines have slopes $-\frac{1}{2}$ and $-\frac{3}{2}$. 352

15.1 The modulated heat trace $\mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$ against the window centre u at $x = 0.02$ (solid), 0.1 (dashed) and 0.5 (dotted), from the first 1700 zeros; dotted verticals mark the ordinates of the zeros. 395

15.2 The least value of D over windows of length 0.4 on $[0, 60]$, for the actual zeros (solid, the first 1700 zeros with the smooth tail; the least window minimum is 0.0427, in the window centred at $x = 25.5$) and with one hypothetical off line pair added at height γ and displacement δ, $(\gamma, \delta) = (100, 0.3)$ (dashed), $(100, 0.2)$ (dash dotted) and $(1000, 0.3)$ (dotted). Vertical dotted lines mark the predicted levels $x_* = 18.1$, 27.2 and 33.5 of Remark 15.18.3. 398

15.3 Left: the excess of the least eigenvalue of the damped Yoshida form over its floor, $\lambda_{\min}(\alpha, a)/(2b_\alpha) - 1$, against the window half length a, for $\alpha = 0.55$, 0.6, 0.75, 1 (finite sections of $\max(96, \lceil 24a \rceil)$ modes, logarithmic scales); it tends to zero, roughly like a^{-2}. Right: $\log_{10}$ of the finite section value at $\alpha = \frac{1}{2}$ for $N = \max(24, 24a)$ and $N = \max(12, 12a)$ modes, with the level of double precision and the curve $\exp(-10ae^{2a})$. 407

15.4 Left: the symbol $v_\alpha(\tau) = \operatorname{Re} \xi'/\xi(\alpha + i\tau)$ on $[0, 60]$ for $\alpha = 1$, 0.75, 0.6, 0.55 (logarithmic scale), with b_α as thin horizontal lines. Right: the smallest value of v_α in windows of length 10 on $[0, 2000]$, with the minimum above $\tau = 20$ circled. Above $\tau = 20$ the minima are 0.1901, 0.0968, 0.0389 and 0.0195 for $\alpha = 1$, 0.75, 0.6, 0.55, which are 8.2, 8.4, 8.4 and 8.4 times b_α, all at $\tau = 27.67$, between the third and the fourth ordinate (the largest gap above the second); the lowest local minimum other than the one at the origin is lower, $4.90\, b_1$ at $\tau = 17.49$ for $\alpha = 1$ and $4.98\, b_{0.55}$ for $\alpha = 0.55$, in the gap between the first two ordinates. 408

16.1 Detection thresholds of Table 16.3 and of the displacements $\delta = 0.001$ and 0.003 of the text against the height γ of the hypothetical quadruple (logarithmic axis), for $\delta = 0.001$, 0.003, 0.01, 0.05, 0.2, 0.45 (points of equal δ joined), with the horizon $\frac{1}{2}\log(\gamma/2\pi)$ dashed; the vertical axis is the half length a of the window. Left: criterion (i). Right: criterion (ii), with upper bounds at the heights 1000 and 2000. The points are the thresholds refined by bisection to 0.005, so that they lie up to 0.05 below the entries of the table. 422

17.1 Left: real parts of the zeros of F_n near $\rho_j/2$, in the variable $u = w + 1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against n, converging to $-\operatorname{Re}\kappa(\rho_j)$. 436

18.1 The 31 zeros of $g_{9/2}$ with real part below 160 in the upper half plane of z (circles), the escape law (18.12.1) (curve), and the lines $|\operatorname{Im} z| = \frac{9}{2}$ bounding de Bruijn's strip $|\operatorname{Im} w| \le \frac{9}{4}$ in the scale $w = z/2$ of g (dashed). 503

18.2 The Thorin path: the first twelve atoms $\gamma_k(b)$ against $\alpha = \frac{1}{2} + b$ at the stored values $\alpha = 1, 0.9, 0.8, 0.7, 0.6, 0.55, 0.5$, from the unconditional rung $\alpha = 1$ to the critical line. 508

21.1 (a) The inverse first passage boundary b of the reciprocal Xi law on $[0.003, 0.12]$, with the line of slope $-\sqrt{2}\,\gamma_1$ through its last node. (b) The boundary at small times against the asymptote $\log(1/t)/(2\sqrt{2})$ of Proposition 21.7.1, which it approaches only slowly. (c) Normalised boundary changes for off line pairs at the zeros 6, 20, 60, against the shift response K (computed). 595

21.2 (a) The minimum over t of the Thorin density $N'_\alpha(t) = \pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ of the base point laws on $[0, 50]$, attained at $t = 0$, for $\alpha = 1, 0.75, 0.6, 0.55$. (b) The values $N_\alpha(50)$ against $N(50) = 10$. Only these summary values are stored, so the curves on $[0, 50]$ are not drawn (computed). 600

23.1 The function $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau)$, which is π times the Thorin density of Theorem 23.1.1(i) in the ordinate variable, at $\alpha = \frac{7}{8}$ (solid), 1 (dashed) and $\frac{3}{2}$ (dotted) for $0 \le \tau \le 60$, computed from ξ with no zeros. The zeros in this range lie on the critical line, and each contributes a Lorentzian of width $\alpha - \frac{1}{2}$ and height $1/(\alpha - \frac{1}{2})$, that is $\frac{8}{3}$, 2 and 1; at $\alpha = \frac{7}{8}$ the computed peaks are 2.69 to 2.93. On this range the minimum is at $\tau = 0$, where the value is $\frac{\xi'}{\xi}(\alpha) =$ 0.0173, 0.0231 and 0.0461. Positivity at every height for $\alpha > \frac{7}{8}$ is Theorem 23.1.1(ii), under $(\mathrm{Q}_{7/8})$, and not a consequence of the figure. 627

List of Tables

1.1 The four pieces of the sine squared decomposition (1.3.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond 2×10^6 (1.3×10^{-3} at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece. 6

1.2 The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, g the midpoint of the pair. k^* is the least order at which $c_k^F(t) < 0$ for some $t > 0$, found by the scan above (computed, not certified); $c_{k^*}^F(t^*) < 0$ is certified in ball arithmetic. The last column is $c_{k^*}^F(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$. 10

1.3 What finite tests establish and what they cannot exclude. 12

1.4 Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [71, Chapters 6 to 8]. 17

2.1 Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts. 23

2.2 The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of ξ'/ξ, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height. 24

2.3 Working precision needed by the untilted and the tilted Pólya route for D target digits. The untilted bits are $(D+\pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D+\delta\theta/\log 10+5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8\times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices. . . 24
2.4 The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached. 27

3.1 Distances between the Pólya computed Thorin measure μ_ε and U_{RH} on $0 < z \leq 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.1.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \leq 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, L^1 in θ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row. 37
3.2 Proposition 3.3.2 at $T = 200$ by the Pólya route (computed, not certified). 44

4.1 The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; t is the de Bruijn Newman time. 50
4.2 Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \ldots, m_{59}(t)$; at $t = 0$ they are γ_1, γ_2, γ_3, γ_5 and γ_{10}. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \ldots, \theta_{12}(t)$ to within 3.1×10^{-7} of the zeros that Newton's method finds on H_t, and $\theta_{13}(t)$ only to within 5.2×10^{-4}. 51
4.3 The first failing order $n_{\mathrm{fail}}(t) \leq 50$ of the Hankel test along the flow on the grid of step 0.01 and, at one time on each step, the least order at which the pair that starts at (γ_k, γ_{k+1}) breaks the test alone (real: the pair has not yet collided; > 50: no order up to 50 fails). The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone. 53
4.4 The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros γ_k, γ_{k+1} it starts from, and its position θ; the number of real zeros below it; the number of collisions so far among $\gamma_1, \ldots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\mathrm{fail}}(t)$ and $n'_{\mathrm{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30. . . 55

4.5 The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time t_{coll} from the census of Section 4.1.3, against predictions computed from H_0: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms τ_c and τ_B, and the frozen predictor of Section 4.1.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of H_t. . 58

4.6 Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13, 14)$ and $(19, 20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 4.1.7 and 4.1.9; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of H_t. 66

4.7 Two sided brackets for the collision time of seven pairs, in the units of H_t: the lower bound $1/m(0)$ of Corollary 4.1.13, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound s^* of Corollary 4.1.15, with ρ the largest value of Q and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 4.1.12 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. 66

4.8 Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [250, Table 1], upper bounds from [61], [182], [247] and [232]. Odlyzko's bound is the one computed in [223, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication. 69

5.1 The barrier $\mathcal{B} = [0.6, 1] \times [497.4, 497.9]$, $X = 994.8$, $y_0 = 0.2$, at six of the 21 grid times (computed, floating point at 214 digits; the rows $t = 0.05$ to 0.19 are not recomputed in ball arithmetic). Taylor ratio: the largest over the walk of $\sum_{k \le 8} \frac{D^k}{k!} |\partial_t^k \xi_t| / |\xi_t|$. 89

6.1 The three routes of the book that reach every height. All other routes are finite information. 95

6.2 The value at the origin and at the minimum of a selection of the valleys of $v_t(\alpha, \cdot)$ between consecutive zeros. The ordinate θ is that of the minimum at $t = 0$ and $\alpha = 0.6$; at $t = 0.2$ the minimum lies 0.06 to 0.08 to the left, and at $\alpha = 0.75$ it moves by less than 0.002. 106

6.3 Constants for the certificate at $t = 0.2$, in floating point. 109

6.4 The Weil Gram matrix on hat functions. The last column splits $\lambda_{\min} = v^* G v$ for the minimising unit vector into the pole, prime (with its sign) and archimedean pieces. 111

6.5 The smallest eigenvalue from the arithmetic side against the sum of squares of $\hat{g}_{\min}$ over the first 1500 zeros, with the zeros (by index) that carry the largest shares. 112

7.1 Stability between levels, computed. The source is a sum of 40 Lorentzians of width $w = 0.05$ on a period 400, the data are at width $a = \frac{1}{2}$ with white noise of size ε at every frequency, and $P_b * \mu$ is recovered by an oracle spectral cutoff, with ε from 10^{-12} to 10^{-3}. The fitted slope is that of log of the error in $P_b * \mu$ against $\log \varepsilon$; the predicted slope is the exponent $(b+w)/(a+w)$ of Proposition 7.3.2. Nineteen values of ε (the error does not depend on the noise draw). All four fitted slopes are within 0.004 of the prediction. Status: computed, illustration. 130

8.1 Mean modulus of $E(N, s)$ of (8.2.2) for χ_4 over $s = \alpha + 10i$ and $s = \alpha + 25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10} N$ fitted by least squares on the four values of N. Four values of N and two values of b illustrate the GRH statement of the proposition and do not verify it. 143

8.2 The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate γ_{n_0} of the n_0th zero. Replacing a zero: the 20 zeros γ_n, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of γ_n and γ_{n+1}, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros γ_n and γ_{n+1} are removed as well. Median and least are over the 20 values of δ_c, and the ratio is the median over the 20 points of δ_c divided by the local gap. 145

8.3 Levels ε at which ρ_ε is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)). 146

8.4 The number K of greedy levels of Corollary 8.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$. 149

9.1 The ladder $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, by power law extrapolation from $N = 384, 768, 1536$. The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2}+\varepsilon)$ of Theorem 9.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed. At $\varepsilon = 0$ the entries for $a \ge 0.75$ are omitted: in double precision they are below the rounding level of about 10^{-13}, and the upper bounds from extended precision are given in the text. 159

9.2 The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is 1.3293×10^{-3}. 160

9.3 The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of 10^{-4}, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 9.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 9.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times a^2, is at most 4×10^{-5} in these units, an error estimate and not a bound. 162

9.4 Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of 10^{-4}: the lower bound of the proof of Theorem 9.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1 - \ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 9.3, extended to $a = 6$), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound. 163

10.1 Spectral peaks of the level one Thorin measure (computed). Every peak lies within 10^{-4} of the logarithm of a prime or prime power. The heights, relative to that of $n = 3$, depend on the window and follow the relative value of $\Lambda(n)/n$ to within 0.04. 171

10.2 The Fourier backward flow from samples of u_1 on $\operatorname{Re} s = 1$ (computed). The error is the largest distance between a zero in $100 < v < 300$ and the nearest peak. At $\Omega = 60$ the sharp cutoff has 213 peaks of height at least a quarter of the maximum, 109 of them at the zeros, and the maximum is +26.6. 173

10.3 EM inversion of u_1 on $[100, 300]$ (computed). The source printed 4.1×10^{-3} in the last row; the recomputation gives 4.05×10^{-3}, and the grid sensitivity of the residual is of that size. 177

10.4 Quartet injection test with controls (computed); total residual after 10000 EM steps. The last row is the boundary case $\sigma = \theta'$. 178
10.5 The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero γ_k in the data window, the centre of mass c_k of its cluster, the error of the centre and the total mass of the cluster. 184
10.6 Nonnegative deconvolution of $\rho_{0.1}$ with the signature g of a displaced fifth zero added to the data. Here r is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$. 185

11.1 The mass deficit at $T = 100$, where $M_1(100) = 29.1112$ and $N(100) = 29$ (the author's computation, not rerun). 211
11.2 The certificates of Theorem 11.9.6, as recomputed. 218

12.1 The matrix pencil at height 200, level one data, taper $\ell = 10$: errors in the position and in the real part of the recovered zeros, by distance from the centre (computed by the author, not rerun). . . 253

13.1 Particle filter posterior for the zeros of ζ in the data window $[78, 94]$ after all levels down to $\theta = 0.6$ (computed by the author). The true ordinates are 79.3374, 82.9104, 84.7355, 87.4253, 88.8091 and 92.4919. 280
13.2 The backward Cauchy flow as a filter: the posterior of $\delta^* = \max|\sigma_j - \frac{1}{2}|$ over the zeros of ζ in $[80, 92]$ as levels are added, with cumulative cost (computed by the author). The bounds measure the displacement of one unpaired zero from the line, which the functional equation forbids. 280
13.3 Posterior standard deviation of σ_j for the six zeros of ζ in $[78, 94]$, against the first order Fisher information formula ($\varepsilon = 0.01$, $\Delta v = 0.04$). The predicted row is computed; the filter row is the author's run. 282
13.4 Contour moments, computed in ball arithmetic at 128 and 192 bits; the two precisions agree, and the integrals are enclosed, so that each M_0 is an enclosure containing one integer. The note printed 7×10^{-22} and 0.006 for the first two entries, floating point quadrature residuals (the second from zeros at distance 0.05 from the edge); the exact values are 0. The zero of the last row is the Newton root with $K = 1$ and M_1 as an interval. 283
13.5 Coverage of the GBC posterior quantiles of δ^* on held out simulations (computed by the author). The middle quantiles undercover, an approximation and calibration error of the quantile network; at an atom exact quantiles overcover. The upper tail is slightly short and is the place to add conformal calibration. 284

13.6 Davenport and Heilbronn windows flagged by GBC (computed by the author; the zeros are those of Section 13.2). The four zeros off the line below height 200 are flagged, and located to within 0.1 in the four windows that target them. The window at 180 sees the zero at 176.70 near its edge, outside its target interval $[177, 183]$, and overstates δ: a boundary false alarm. No other window of either function is flagged; for all 35 windows of ζ the posterior median is 0 and the 95% quantile is at most 0.007. 285
13.7 Costs of reported runs (computed by the author). The two runs for ζ differ in window ($[78, 94]$ with six zeros against $[80, 92]$ with four), in zeros and in data, so their ratio of about 115 in evaluations is a comparison of two different runs and not a matched speedup. For Davenport and Heilbronn the GBC initialised filter succeeds where the plain filter stalls (at tempering exponent $\lambda \approx 0.003$), but the posterior is strongly non Gaussian because the level $\theta = 0.8$ lies within 0.0085 of $\sigma = 0.8085$, so the bridge from the Gaussian proposal still needs many steps. 286
13.8 GBC posterior medians of δ. Mean absolute error 0.0096 with the rigid simulator and 0.0034 with the sine kernel simulator. 287
13.9 Monte Carlo spread over twelve runs, computed in the source note. Least squares log log slopes of the printed values are -0.44, -0.62 and -0.47, against $-1/2$ from Theorem 13.5.1; with twelve runs per cell each standard deviation has a relative error of about 20%, so the slopes carry about ± 0.15. 291
13.10 Negative mass of u_θ for the Davenport and Heilbronn function on intervals of length 3 centred at its two lowest zeros off the line (computed). The bound is the number of zeros with $\sigma_\rho > \theta$ at the centre. 297
13.11 Negative mass of u_θ on the left half disc around the zeros of the Davenport and Heilbronn function off the line, on a 40×81 grid (computed by the author, not recomputed). 300

13.12 Zero free constants $1 - \sigma_\rho \geq c/\log \gamma_\rho$ from admissible pairs, in the scheme of Corollary 13.8.6 (computed). The optimal polynomial of degree 4 has $a \approx (1, 1.705, 1.044, 0.425, 0.089)$, and the single level rows to degree 8 are over squares with $b_j \geq 0$ (with only $p \geq 0$ and $a_k \geq 0$ the value grows by about 1% at degree 8); the best level for the derivative family at degree 4 is $x = 0.89$. The rows with derivatives run to $\deg q = 5$ and are optima on discrete constraint grids (computed); at $\deg q = 4$ and 5 the kernels are not certified admissible (at $\deg q = 5$ a dense audit of positive definiteness gives -1.6×10^{-6}). The cap of the source note, 0.432 for the degree 4 polynomial, is replaced by the sharper cap of Proposition 13.8.7. The last row is outside the class of Lemma 13.8.5. 306

13.13 Ceiling of the scheme of Corollary 13.8.6 over nonnegative positive definite smoothings, by discretised linear programming (computed, not certified). Refining the frequency grid near the origin moves the value for $3 + 4\cos\omega + \cos 2\omega$ from 0.0460 to 0.0450; on the stated uniform grid, changing the number of points or $\lambda_{\max}$ moves the third digit. The rows with the optimised polynomial are lower estimates, since the alternating searches end at local optima (the degree 4 gain over the fixed polynomial is below 10^{-4}). 308

13.14 Nested sampling for ζ on bands of length 10^4, with N live heights, M moves per replacement and K iterations (computed). In every band the score stays positive. The column min s is the smallest value found by the run, not the band minimum. The scores at the printed heights and the normalised gaps, measured between consecutive zeros found as sign changes of Hardy's Z function, are recomputed with enclosure radius below 6×10^{-12}; the run parameters and $\log X$ are the source's. The smallest values sit in the largest normalised gaps. 312

14.1 The dictionary between the two levels. 322

14.2 Moments of $Z = \log h_\theta(\theta + i\tau)$ under $P_{1-\theta}(t_0 - \cdot)$, computed in floating point: Poisson weighted panels between the zeros with tanh sinh nodes, midpoint cells beyond distance 1000 out to $|\tau - t_0| \leq 13000$, and the mass beyond the window (2.45×10^{-5} at $\theta = \frac{1}{2}$, 1.22×10^{-5} at $\theta = \frac{3}{4}$) estimated from the sample moments of the outer window. Entries are from the finer of two meshes; the larger of the difference between the meshes and half the tail correction is at most 2.1×10^{-5} in every entry. For ζ the equal variance and zero covariance identities of Theorem 14.5.2 hold to that accuracy; for Davenport and Heilbronn (DH), with $h = f$, they fail at both levels. 330

14.3 Local defects $D_\theta(t_0)$ (computed, floating point). Davenport and Heilbronn is abbreviated DH. The quadrature columns are the integral over $|\tau - t_0| \le W$ minus $\log|h_\theta(1 + it_0)|$, with the zeros as breakpoints: they are realised truncation errors of the window, not accuracies, and change with W. Outside the window the Cauchy mass is 0.0053 at $\theta = \frac{1}{2}$, 0.0027 at $\theta = \frac{3}{4}$ and 0.0013 at $\theta = \frac{7}{8}$ for $W = 60$. Monte Carlo (computed) uses $n = 20{,}000$ evaluations for ζ and 4,000 for DH. Plain Monte Carlo uses exact Cauchy draws, with standard error $\mathrm{sd}/\sqrt{n}$. The vertical sampler takes a pilot mean c from $n/20$ draws, $\epsilon = 0.05$ times the standard deviation of the pilot, an even mixture of Cauchy independence proposals and Gaussian steps of scale $(1-\theta)/2$, and a burn in of $n/20$ steps, with batch means standard errors (50 batches). The Davenport and Heilbronn prediction is the sum of the Green potentials of the zeros found below height 700; for $\theta = \frac{7}{8}$ none has $\beta > \frac{7}{8}$, but f has zeros with $\beta > 1$ [60], so the exact defect is positive. 335

14.4 Root mean square error of the Cauchy average of $\log|\zeta(\theta+i\tau)|$, $t_0 = 0$, over 400 replications (computed, floating point; the sampling error of each entry is about 3.5% and larger for the heavy tailed plain estimator). The control uses $M = 100$ at $\theta = \frac{1}{2}$ and $M = 1000$ at $\theta = \frac{7}{8}$. Gain is the ratio of plain to YKS with control. Fitted slopes over $500 \le N \le 16{,}000$ at $\theta = \frac{1}{2}$ and $\frac{7}{8}$: plain -0.49 and -0.50, YKS -0.66 and -0.60, YKS with control -0.58 and -0.52. 338

14.5 Root mean square error at $N = 32{,}768$ for the half disc identity of Theorem 14.6.6, $R = 1$, $t_0 = 85.7$, 200 replications, with the slopes fitted for $N \ge 1024$ in brackets (computed, floating point). The target is the right side of the identity; for the Davenport and Heilbronn function (DH) it includes the Green term $G_H(z, \rho)$. . . . 343

14.6 Truncated Cauchy average of $\log|\zeta(\theta + i\tau)|$ over $|\tau - t_0| \le T$ at $t_0 = 85.7$, minus $\log|h_\theta(1 + it_0)|$; as in Remark 14.2.2, the zeros above 3×10^{12} move the whole line average away from $\log|h_\theta(1+it_0)|$ by less than 10^{-12}, with no hypothesis. Mean error and spread over 10 random shifts of the grid (computed, floating point). At $\theta = \frac{1}{2}$ the zeros with $|\gamma - t_0| \le T+1$ (the zero range convention, counted in the last column) are deflated and their window means added back (the unbiased rule). The shifted grid is unbiased for the window integral, so the mean error is minus the Cauchy tail beyond T, up to the spread. In this range $T^2 \cdot |\text{error}|$ stays below 0.27; the bounds of [16] allow slow growth. 345

14.7 Root mean square error of the weighted rules at independent uniform nodes on $Q = [-1, 1]^d$, 200 replications in one dimension and 30 in two (about 5 percent noise in each root mean square error in one dimension and about 13 percent in two), the largest N in brackets, slopes fitted for $N \geq 4000$ (computed, floating point). The bump $\prod_i b(x_i/0.9)$ vanishes near ∂Q (case (a) of Theorem 14.7.3); the exponentials do not, and their error is the bias of Remark 14.7.4. The slopes are consistent with the last column, the exponent of Theorem 14.7.3, of Lemma 14.6.8 for the trapezoid row and of the Remark for the last row. 350

14.8 Root mean square error of $I(\varphi)$ (computed, floating point, 20 replicates; the relative standard error of an entry is 16%). Jittered is Delaunay on jittered nodes; deflated grid uses an estimate of the zero that is off by 1.4×10^{-4}, along $(1 + i)/\sqrt{2}$. 352

14.9 Deflated shifted grid for Davenport and Heilbronn, root mean square error over 10 shifts, as a function of the error ε in the deflating zero (computed, floating point; the relative standard error of an entry is 22%). The last row is the grid without deflation. A synthetic test of the two regimes of Proposition 14.7.5 shows the error falling to $N = 2^{22}$ for every $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, with slope near -0.6 while $\varepsilon\sqrt{N} \leq 1$ and near -1.1 afterwards; the constant depends on the bump and the box: at $\varepsilon = 10^{-4}$, $N = 262{,}144$ the ratio of the error to ε is 0.030 here and 0.0050 in the synthetic test, a factor 6. 354

15.1 The explicit formula for the heat trace, agreeing to all printed digits (computed). 364

15.2 Archimedean part, prime part and heat trace (computed). Digits cancelled is $\log_{10}(\mathcal{A}/W)$, which differs from the measure of the caption of Table 8.4. 367

15.3 The first sign change t_n of $(-1)^n \mathcal{A}^{(n)}$ against n/λ_*, $\lambda_* = r_*^2 = 39.5620$ (computed). 369

15.4 Zeros recovered by Prony's method from the heat trace computed from Γ and the primes (computed, 110 digits; the source printed errors of 10^{-30} for $k \leq 3$, which was the floor of its reference, and the digits shown are those of the source). 370

15.5 End of the positivity window of order n when only the primes in S are kept (computed). The entries > 0.8 mean no failure up to the end of the scan. 373

15.6 V_δ for the primes, the GUE prediction, the Cramér model and the Poisson level (computed). The source printed $10.19, 7.69, 5.34, 4.06, 3.23, 1.10$ for the primes (all within two standard errors of these) and Cramér values 14.19 to 15.22 with a last entry 10.09, which the recomputation does not reproduce. . . . 375
15.7 Windows for $W'' > 0$ suggested by the bound of Proposition 15.9.2(2) from the primes up to N_0 with the Rosser and Schoenfeld tail (computed in floating point on a grid; not certified). 378
15.8 Energy of the normalised prime error on the log scale (computed). The source printed 0.0506 for the first window and 7.54, 9.78, 9.14, 7.15 for a Cramér realisation; the Cramér values are heavy tailed and one realisation does not support a growth rate. 380
15.9 The tent value $D(x)$ from the primes (Proposition 15.18.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x = \log 2$ and $x = \log 5$ differ by Landau's term, see the text. 399
15.10 The least eigenvalue $\lambda_{\min}(\alpha, a)$ of the damped Yoshida form on the window $[-a, a]$, from the matrices of the form on the modes χ_n, $|n| \le N$. For $\alpha > \frac{1}{2}$ the entries are converged values, from Richardson extrapolation of $N = 96$, 192, 384, with an uncertainty of at most 3×10^{-7} at $a = 0.35$ and 1.1×10^{-8} for $a \ge 0.8$ (an independent Legendre basis agrees to 1.5×10^{-7} at most). For $\alpha = \frac{1}{2}$ the entries are the finite section values at $N = \max(24, 24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a = 4$ ($N = 96$) was computed with 640 digits. The last row is the floor $2b_\alpha = 2\xi'(\alpha)/\xi(\alpha)$. Computed. 406

16.1 Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order N shown. The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling. 417
16.2 The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number K of atoms killed and the even exponent m, with the optimal K and m, the last ordinate γ_K killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 16.4, and T_* itself. 419

16.3 Smallest window half length a at which Yoshida's form on $[-a, a]$ detects a hypothetical off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05. The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\leq$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90. . . 421

16.4 The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem. 424

16.5 The comb constant $A_a(\alpha)$ of (16.5.1) and the barrier height $T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha = \frac{1}{2}$ is Zhu's [279], who prints $A_{0.8}(\frac{1}{2}) = 2.9420$ and $T_1 = 119$ at $a = 0.8$, 1.0×10^7 at $a = 1.6$ and 2.4×10^{11} at $a = 2$; the digits here are 2.94197, 119, 1.04×10^7 and 2.44×10^{11}. For $\alpha > 1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha) = 18.88$, 3.01, 1.14 at $\alpha = 1.1$, 1.5, 2. . . 425

16.6 The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 16.5.2 for several values of $\lambda = \lambda_{\min}^{\text{even}}(\frac{1}{2}, a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices, not bounds; Zhu's reference values of his Table 1, 9.34×10^{-7} and 1.65×10^{-17}, give 1.01×10^{-5} and 1.79×10^{-16}. 428

17.1 Argument principle counts for Conjecture T: rectangles searched and zeros of Q_n found with $\operatorname{Re} u > \frac{5}{4}$. 434

17.2 Zeros of the symmetrised truncations K_n in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of ξ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for K_8 exceeds that for ξ by one; see the text. 439

17.3 Bounds for the torus extremes over the band $1.2 \leq x \leq 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 17.1.11 holds with these values. The bounds are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 17.1.13); the programs are in `verification_reciprocal_zeta/torus/`. 442

17.4 As Table 17.3, for the single band variant: bounds for the extremes over the band $1 \leq x \leq \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values. 443

17.5 Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none. 446
17.6 Smallest order n^* at which an inserted off line quadruple at height γ and distance δ makes $\min_t L_n[W](t)$ negative. 448
17.7 Constants used in Sections 10.1.4 and 11.1.1. 451
17.8 The drift of the first zero of M_n at $\gamma = 6.0209489$ (computed). The last column is the gamma remainder of Proposition 17.3.4, for which $\delta_2 = -2.5713 - 1.0051i$ is not the limit. 459
17.9 The five zeros of $L(s, \chi_4)$ below height 500 with anomalous Gram parity (computed). The partner is the neighbouring zero at the stated gap (the left neighbour for the zero number 291). 462
17.10 Zeros of M_n in the left and right half strips. The counts are computed at two precisions with identical counts. The source note printed 22 for the hard cutoff at $n = 32$, a blank at $n = 64$ and 14 for the mean correction at $n = 32$, $\operatorname{Im} z$ from 0.5 to 20. 464
17.11 Symmetrised hard cutoff (computed). 465
17.12 The first zeros of F_8 and of $L(s, \chi_4)$ (computed). 466
17.13 The first negative Hermite coefficient of Ξ (computed). 472
17.14 The ratio of $b_m(a)$ to $\sqrt{2a}\, m^{-1/4}\, \Xi(a\sqrt{4m+1})$ (computed). 473
17.15 Number of sign changes of $b_m(a)$ for $t_m(a) \le T$ against the zero count $N(T)$ (computed). The coefficient counts of the note (627, 1738, 6947, 7002) were two too large. 474
17.16 Mean sign runs of $b_m(4)$ (computed). 476

18.1 Real zero counts (computed in floating point, not certified). τ_P^* solves $A(\tau) = D_P(\tau)$. For $P \le 5$, R_P is a full sign change scan; for $P = 7$ it is the hump criterion, with a 215 digit check of every hump in $[481, 501]$, the last surviving pair being 489.7932 and 491.0605. . 490
18.2 The survey (computed). $N = N(\tau_P^*)$, off $= R_P - N$, model is the ramp model's expected offset with the window 200 for m_Z. 491
18.3 Departed zeros of Ξ_P (computed, residuals below 10^{-34}) against the two forms of the departure law. The six zeros are every second zero of the chain of Remark 18.6.1 for $P = 2$ and a selection for $P = 3$. 492
18.4 The $2\sin^2$ Lévy density of Ξ_2, from its 6 real and 190 traced departed zeros, against that of Ξ. $\nu_2 > 0$ on the whole computed range $[0.02, 3]$. 497
18.5 Critical ladder rung b_c and heat time λ_c at which the pair above each Davenport Heilbronn zero becomes real. In every case $\lambda_c L^2 < 1$ (0.30, 0.09, 0.26), so these heights are pre asymptotic for the heat flow. 514

18.6 The collision of the moved pair $\gamma_1 \pm i\eta$ of Ξ: $(\eta^2 - s_c^2)/\eta^4$ at three heights η, against its limit $d_c = \tau_1^2 - \frac{m_4-1}{2}\tau_2$ of Proposition 18.14.8 (computed; the three cosh cells at $\eta = 0.3$ are certified by a Krawczyk step). 516

19.1 Laws of Theorem 19.1.4 fitted to the Pólya law. The last three rows keep Pólya's kernel $e^{-2\pi\cosh 2t}$ (Remark 19.1.7). The parameters are rounded to seven significant digits from a numerical minimisation and then fixed; the distances are upper bounds enclosed in ball arithmetic and rounded up. 525

19.2 The zero completed laws f_K. For $K \le 16$, f_K is a ferromagnetic probability density by Theorem 19.1.12(i) ($s_K \le 2\beta_K$). For every $K \ge 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so f_K is a ferromagnetic probability density by Theorem 19.1.12(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 19.1.13 lies in a certified interval of width at most 1.2×10^{-7}, within 6×10^{-7} of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. Computed from the certified zeros and variance, except $v^*(1) = s_1$. 531

19.3 Rouché tests for the folded kernels. The sampled minimum is over 721 points of the quarter circle (computed); the lower bound is over the whole quarter circle (certified in ball arithmetic, and weaker than the sampled minimum because the bound is crude). Every row passes both. 539

19.4 Partial fraction data for β_μ (computed in ball arithmetic). The sign condition of Proposition 19.3.4 is that $m \ge 0$ and all $\sigma_j > 0$; it is a hypothesis of the proposition and does not by itself give a string. . 541

19.5 Absolute errors of γ_k recovered by Prony fits of K exponentials to the tail of the hitting time, window starting at threshold x_0 (source values, unchecked). 542

19.6 Rouché tests for $N = 5$ (computed: floating point, sampled on the circle, E_5 truncated at $n = 14$). 543

19.7 The masking of the first nonreal pair in D_6 for the two term folded kernel (computed, 250 digits, except the kernel moments row, which is from 120 and 240 digits, and the row marked unchecked, which is the source's value and needs the 19 located zeros). 546

19.8 Absolute errors of γ_k from Prony fits to the tail of the reciprocal Ξ hitting time at 140 digits (computed). 547

19.9 Calibration on Devroye's two explicit laws, with atoms $\pi^2(k-\frac{1}{2})^2/2$ and $\pi^2k^2/2$, and the reciprocal Ξ law with atoms γ_k^2. For the two strings $c_k\lambda_k = (-1)^{k-1}\pi(k-\frac{1}{2})$ and $c_k = 2(-1)^{k-1}$ in closed form, so that $x_k = \log\frac{2k+1}{2k-1}/(k\pi^2)$ and $x_k = \log\frac{(k+1)^2}{k^2}/\big((2k+1)\pi^2/2\big)$ (computed, and checked against the product at $K = 4000$ factors). For the reciprocal Ξ law the coefficients come from $\xi'(\rho_k)$ at the first 1700 zeros (computed). 549

19.10 Long window threshold fits from the Laplace transform alone, as reported by the source (unchecked: the fits were not recomputed; the pair agrees with the zero of H_{-4} found directly, $33.1150940172 \pm 0.150539022\,i$). 553

20.1 Displacement $\operatorname{Re}\rho_\delta - \frac{1}{2}$ of the first three zeros of the shifted family, by Newton's method from (20.3.2) at 30 and 45 digits (difference below 5×10^{-25}; computed). The first row is the real part of the complex slope of (20.3.3), whose imaginary parts -0.1565, $+0.0703$, -0.2961 show that $\operatorname{Im}\rho_\delta$ moves too. The displacement is linear in δ only to the digits of the slope for small δ (at $\delta = 1$ the first entry is 1.53727 against 1.53709). 562

20.2 The uniform law and the Riemann problem under the $2\sin^2$ lemma. Convergence of the Thorin densities is vague convergence. Without RH the limit at $\alpha = \frac{1}{2}$ records only the zeros on the critical line (zeros at $\frac{1}{2} \pm a + i\gamma$ cancel in the real part), so the limit measure is not a test of RH by itself. 567

21.1 Relative shift $\ell_{d,n}/\gamma_n^2 - 1$ of the zeros of the Jensen polynomials, against the first order prediction; the last row is $A_n/(2d)$ at $d = 10240$ (computed). 575

21.2 Distance between the distribution functions of the Jensen law μ_d and of μ on the grid of 74 points of $[0.007, 0.08]$ (computed). 575

21.3 The smoothed zero laws $\hat{\mu}^{[N]}$, with the first N zeros exact and the rest at the Gram midpoints, against μ. The inversion in double precision resolves differences of distribution functions only down to about $2\cdot10^{-6}$ for $y \geq 0.011$ (and $1.8\cdot10^{-5}$ below), so the supremum for $N = 30$ and $N = 100$ is unchecked; the source printed $2.7\cdot10^{-6}$ and $6.8\cdot10^{-7}$. 580

21.4 Theorem 21.3.3 (computed). Upper block: the two sides of the identity; lower block: the zero remainder R_U and the bound of the theorem; the residual is below 10^{-30} in every row. The rows $s = 0.1$, 0.49 and $0.3+3i$ have $\frac{1}{2}+s$ inside the critical strip. At $U = 0.05$ the zero remainder is visible; at $U = 1$ it is below the working precision (30 digits). The bound is loose at small U. 586

21.5 The quantity $\log \mathbb{E}e^{-sZ_P}$ for $P = 10^3, \dots, 10^7$, against its limit $(2s)^{3/2}/6$ (computed). 588
21.6 Boundaries with one off line pair against a GGC control on $[0.0045, 0.12]$; for γ_{60} the maximum is over $[0.0045, 0.1]$, since near $t = 0.12$ the solver noise of about $2 \cdot 10^{-7}$ exceeds the signal. "Never" means the density was found positive on a grid of step 0.005 on $(0.0064, 6]$ and at three points below 0.0064, beyond which the tail is governed by γ_1. All entries computed. 593
21.7 Normalised boundary change $(b_{\mathrm{toy}} - b_{\mathrm{GGC}})\gamma^4/6\delta^2$ against the shift response $K(t)$, $\delta = 0.3085$ (computed). 594
21.8 Smallest contiguous Toeplitz minors $\det[c_{n+i-j}]_{i,j<k}$ for $k-1 \le n \le 104 - k$, normalised by c_n^k, for ξ and for the Davenport Heilbronn function. A further 600 random minors of each of the orders 3, 4 and 6 were nonnegative for both functions, none below -10^{-80}. . . 604
21.9 First failing order of the coefficient ladder when the mth zero is replaced by a pair at angle θ. The bare threshold is the first failing order for the quadratic factor alone, the least k with $\theta > \pi/(k+1)$. The failing index n is 1, 3, 7, 12 and 17 for $m = 1, 3, 6, 11, 16$ (and 6 at $\theta = \pi/10$, $m = 6$), within one of m; orders above 40 were not tried. 605
21.10 First failing order of the density ladder for the same kind of perturbation, as reported in the source note (not reproduced). Order 1 means that the density itself becomes negative. 606

23.1 Where the half plane $\operatorname{Re} s > \frac{7}{8}$ of [227], which has not been refereed, acts outside this section. 628
23.2 Heights beyond which Theorem 23.1.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width d of the window and the number k of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8} \le \beta \le \frac{7}{8}$; "on line" uses $\beta = \frac{1}{2}$, valid for $r + d \le H$. Computed with the bound of Lemma 23.1.6. 632

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